UM

HPM5300

32-bit High-performance Microcontroller based on RISC-V Core Rev 1.1

HPM5300 Series High-performance Microcontroller User Manual

Applicable to the following products of Shanghai HPMicro Semiconductor Co., Ltd.:

HPM5361ICB1、HPM5361ICF1、HPM5361IEG1
HPM5331ICB1、HPM5331ICF1、HPM5331IEG1
HPM5321ICB1、HPM5321ICF1、HPM5321IEG1
HPM5301IEG1

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1 Product Overview

1.1 System Block Diagram

The system block diagram of this product is shown in Figure 1.

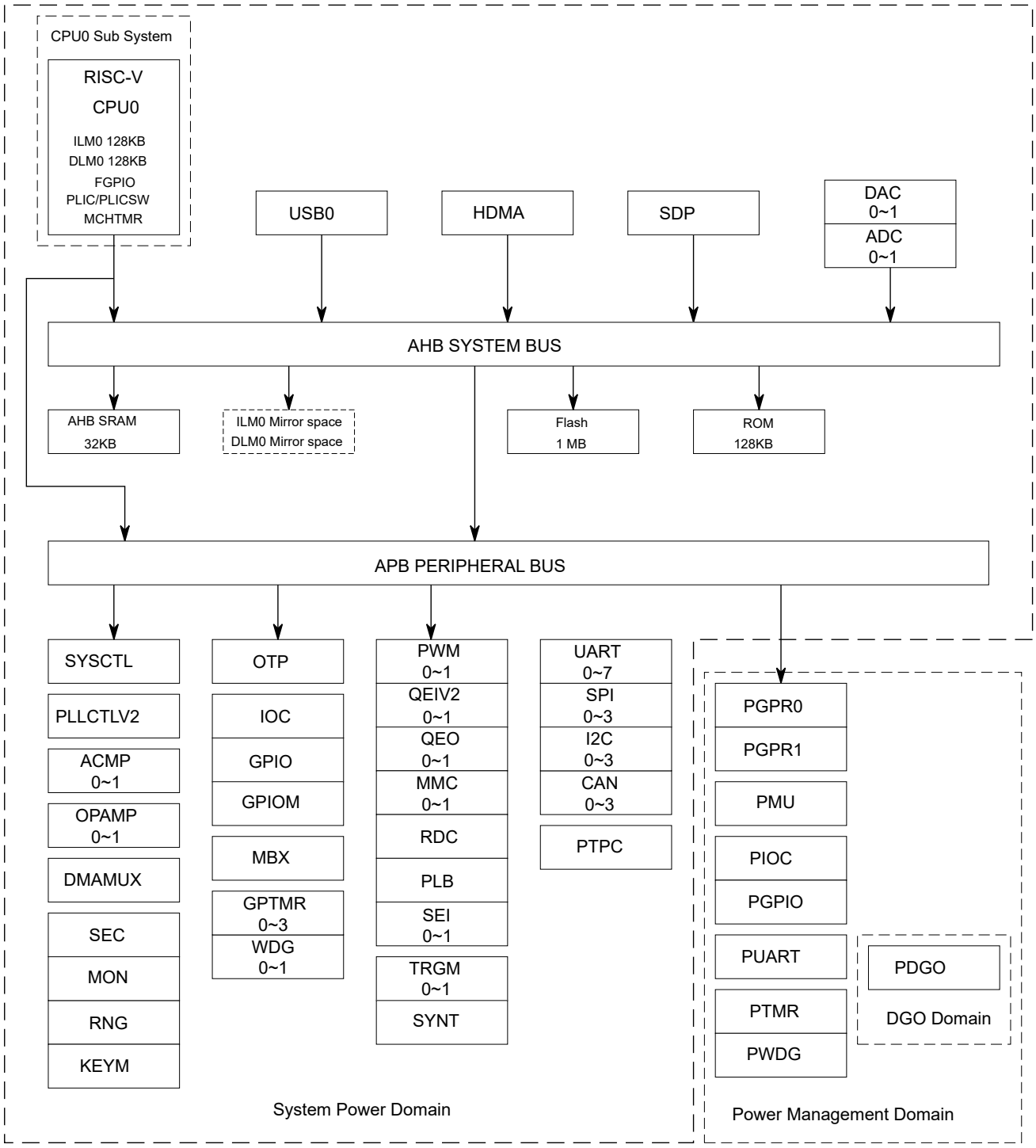


Figure 1: System Architecture Block Diagram

Table 1 summarizes the definitions of all peripheral abbreviations in Figure 1.

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Abbreviation	Description
CPU0 Subsystem	Subsystem including RISC-V CPU0, its local memory and private peripherals
HART	Hardware Thread. As defined in the RISC-V specification, a HART is a unit that implements the complete RISC-V architecture and can execute instructions independently. In this manual, HART is equivalent to the RISC-V core.
ILM	Instruction Local Memory
DLM	Data Local Memory
FGPIO	Fast General Purpose Input Output Controller
USB	Universal Serial Bus
SDP	Secure Data Processor
HDMA	AHB Peripheral Bus DMA Controller
AHB SRAM	AHB Bus SRAM
EXIP	Encrypted Execution-In-Place Module
ADC	Analog-to-Digital Converter
DAC	Digital-to-Analog Converter
SYSCTL	System Control Module
PLLCTL	PLL Controller
ACMP	Analog Comparator
MBX	Mailbox
DMAMUX	DMA Request Router
IOC	Input Output Controller
PIOC	Power Management Domain IO Controller
GPIO	General Purpose Input Output Controller
PGPIO	Power Management Domain GPIO Controller
GPIOM	GPIO Manager
OTP	One Time Programmable Memory
PWM	Pulse Width Modulation Timer
QEIv2	Quadrature Encoder Input
QEO	Quadrature Encoder Output
SEI	Serial Encoder Interface
MMC	Motion Management Controller
RDC	Resolver Decoder
PLB	Programmable Logic Block
TRGM	Trigger Manager
SYNT	Sync Timer
GPTMR	General Purpose Timer
PTMR	General Purpose Timer in Power Management Domain
EWDG	Watchdog
PWDG	Watchdog in Power Management Domain

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Abbreviation	Description
UART	Universal Asynchronous Receiver and Transmitter
PUART	Universal Asynchronous Receiver and Transmitter in Power Management Domain
SPI	Serial Peripheral Interface
I2C	Inter-Integrated Circuit Bus
CAN	Controller Area Network
PTPC	Precise Time Protocol Module
RNG	Random Number Generator
KEYM	Key Manager
PGPR0/1	Power Management Domain General Purpose Registers 0/1
PCFG	Power Management Domain Configuration Module
PDGO	Power Management Domain Power On/Off Module
SEC	Security Manager
MON	Security Monitor
System Power Domain	In this manual, system power domain specifically refers to the logic and memory circuits powered by VDD_SOC
Power Management Domain	In this manual, power management domain specifically refers to the logic and memory circuits powered by VPMC

Table 1: Peripheral Abbreviation Summary

1.2 Feature Summary

This section describes the main features of this product.

1.2.1 Core and System

32-bit RISC-V processor with the following features:

- RV32-IMAFDCBP instruction set
  - Integer instruction set
  - Multiplication instruction set
  - Atomic instruction set
  - Single-precision floating-point instruction set
  - Double-precision floating-point instruction set
  - Compressed instruction set
  - Bit operation instruction set
  - DSP unit supporting SIMD and DSP instructions, compatible with P extension instruction set
- Performance up to 3.57 CoreMark / MHz
- Privilege modes support Machine mode and User mode
- Supports 16 Physical Memory Protection (PMP) regions
- Supports 16KB L1 instruction cache and 16KB L1 data cache

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- Supports 128 KB Instruction Local Memory (ILM) and 128 KB Data Local Memory (DLM)

The processor is equipped with 1 Platform Level Interrupt Controller (PLIC) for managing RISC-V external interrupts

- Supports multiple interrupt sources
- Supports 8 programmable interrupt priority levels
- Interrupt nesting extension and interrupt vector extension

The processor core is equipped with 1 software interrupt controller PLICSW to manage RISC-V software interrupts

- Generates RISC-V software interrupts

The processor core is equipped with 1 machine timer MCHTMR to manage RISC-V timer interrupts

- Generates RISC-V timer interrupts

DMA Controllers:

- HDMA, supporting 32 channels, for low-latency data transfer between peripheral registers and memory, also applicable for data transfer between memories
- Supports DMA request routing assigned to any DMA controller

Includes 1 Mailbox (MBX) supporting communication between different processes of the processor:

- Supports independent information sending and receiving interfaces
- Supports interrupt generation

1.2.2 Internal Memory

Internal memories include:

- 288 KB on-chip SRAM
  - ILM0, 128KB instruction local memory for RISC-V CPU0
  - DLM0, 128KB data local memory for RISC-V CPU0
  - AHB SRAM, 32KB, suitable for low-latency access by HDMA
- General purpose registers
  - Power management domain general purpose registers PGPR, 2 groups of 64 bytes each (128 bytes total), can retain data when the system power domain is powered off
  - DGO general purpose registers DGO_GPR, 16 bytes capacity, can retain data when the system power domain and power management domain are powered off
- Internal Read-Only Memory (ROM), 128KB capacity, stores the boot code of this product, Flashloader and part of peripheral driver programs
- One-Time Programmable memory (OTP), 4096 bits, can be used to store part of the chip's factory information, user keys, security configurations, boot configurations and other data

1.2.3 Power Management

This product integrates a complete power management system:

- Multiple on-chip power supplies
  - DCDC voltage converter, providing 0.9~1.3V output to power circuits in the system power domain. The DCDC output is adjustable to support Dynamic Voltage Frequency Scaling (DVFS)

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for optimizing power consumption during operation
  - LDOPMC, linear regulator with typical 1.1V output, powers circuits in the power management domain
  - LDOOTP, linear regulator with typical 2.5V output, powers OTP, can only be turned on during OTP programming
- Operation modes and low-power modes: Wait mode, Stop mode, Sleep mode and Shutdown mode
- The chip integrates power-on reset circuit
- The chip integrates low-voltage detection circuit

1.2.4 Clock

The clock management system of this product supports multiple clock sources and clock low-power management:

- External clock sources:
  - 24MHz on-chip oscillator OSC24M, supports 24MHz crystal, also supports external 24MHz active clock input through pins
- Internal clock sources:
  - Internal RC oscillator RC24M, 24MHz frequency, can be configured as alternate clock source for PLL
  - Internal 32KHz RC oscillator RC32K
- 2 Phase-Locked Loops (PLL), supporting fractional frequency division and spread spectrum
- Supports low-power management, supports automatic clock gating

1.2.5 Reset

Global reset can reset the entire chip including power management domain and system power domain. Reset sources include:

- RESETN pin reset (RESETN)

System power domain reset can reset the system power domain. Reset sources include:

- VPMC pin brownout reset (VPMC Brownout)
- Debug reset (DEBUG RST)
- Watchdog reset (WDOGx RST)
- Software reset (SW RST)

1.2.6 Boot

BootROM is the first program executed after the chip is powered on, it supports the following functions:

- Boot from serial NOR FLASH
- UART/USB boot
- In-System Programming (ISP)
- Secure boot
- Low-power wakeup
- Multiple ROM APIs

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1.2.7 External Memory

External memory interfaces include:

- 1 serial bus controller XPI, which can connect various off-chip SPI serial storage devices and serial bus compatible devices. Each XPI:
  - Supports 1/2/4-bit data mode, supports 2 CS chip select signals
  - Supports SDR and DDR, maximum 166MHz
  - Supports Quad-SPI and Octal-SPI serial NOR Flash
  - Supports serial NAND Flash
  - Supports HyperBus, HyperRAM and HyperFlash
  - Supports Quad/Oct SPI PSRAM

1.2.8 Motion Control System

Motion control system includes:

- 2 8-channel PWM timers, PWM modulation accuracy up to 3.0ns, supports complementary PWM output, dead time insertion and fault protection
- 2 Quadrature Encoder Input (QEIv2)
- 2 Quadrature Encoder Output (QEO)
- 2 Motion Management Controllers (MMC)
- 1 Resolver Decoder (RDC)
- 1 Programmable Logic Block (PLB)
- 2 Serial Encoder Interfaces (SEI)
- 1 Trigger Manager (TRGM), each module can interact with modules inside or outside the motion control system through TRGM
- 1 sync timer for synchronization

1.2.9 Timers

Timers include:

- 5 groups of 32-bit general purpose timers, one group (PTMR) is located in the power management domain, supports low-power wakeup. Each group of general purpose timers includes 4 32-bit counters
- 3 watchdogs, one (PWDG) is located in the power management domain

1.2.10 Communication Peripherals

Supports a rich set of communication peripherals, including:

- 9 Universal Asynchronous Receiver/Transmitters (UART), one (PUART) is located in the power management domain, supports low-power wakeup
- 4 Serial Peripheral Interfaces (SPI)
- 4 Inter-Integrated Circuit (I2C) buses, supporting Standard (100kbps), Fast (400kbps) and Fast+ (1 Mbps) modes
- 4 Controller Area Networks (CAN), supporting CAN_FD
  - Supports CAN 2.0B standard, 1Mbps
  - Supports CAN FD, 8 Mbps

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  - Supports timestamp
- 1 Precise Time Protocol Module (PTPC), supporting 2 groups of timestamp modules, each group includes a 64-bit counter, connected to the CAN module, the CAN module can read timestamp information from the port at any time
- 1 USB OTG controller, integrated with 1 high-speed USB-PHY
  - Complies with *Universal Serial Bus Specification Rev. 2.0*

1.2.11 Analog Peripherals

Analog peripherals include:

- 2 16-bit Analog-to-Digital Converters (ADC)
  - 16-bit successive approximation ADC
  - Supports 16 input channels
  - 2M sampling rate, 4M sampling rate when conversion accuracy is set to 12-bit
- 2 high-speed comparators
  - Operating voltage 3.0 ~ 3.6V, supports rail-to-rail input
  - Built-in 8-bit DAC
- 2 Digital-to-Analog Converters (DAC)
  - 12-bit accuracy, 1MSPS, supports output buffer
- 2 operational amplifiers (OPAMP)
  - Supports PGA mode

1.2.12 Input and Output

- Provides 8 groups from PA to PY with up to 56 GPIO function multiplexing pins
- IO supports 3V voltage mode, group power supply
- IO supports open-drain control, internal pull-up/down, drive strength adjustment, built-in Schmitt trigger
- GPIO controller
  - Supports reading input of any IO or controlling output of IO
  - Supports IO input trigger interrupt
- Fast GPIO controller (FGPIO), as a private IO fast access interface for the processor
- Provides a GPIO manager to manage IO control permissions of each GPIO controller
- Power management domain dedicated IO PYxx has exclusive GPIO controller and IO configuration module, supports state retention in low-power mode

1.2.13 Information Security System

Information security modules include:

- Secure Data Processor (SDP), the on-chip encryption and decryption algorithm engine:
  - Supports AES-128/256, SM4, supports ECB mode and CBC mode
  - Supports SHA-1/SHA-256, SM3
- Encrypted Execution-In-Place module (EXIP):
  - Tightly coupled with serial bus controller XPI, supports online decryption of external NOR Flash
  - AES-128 CTR mode, zero-wait cycle decryption
  - Supports RFC3394 key unwrapping, protects Data Encryption Key (DEK) through Key Encryp-

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tion Key (KEK)

- Key Manager (KEYM):
  - Supports loading keys from OTP key area through independent data path
  - Supports key obfuscation
  - Supports loading random keys from True Random Number Generator (RNG)
  - Supports generating Session Key
  - Supports independent data path to transmit keys to Secure Data Processor (SDP)
- Key area in OTP, supports storage and protection of:
  - Related keys for SDP and EXIP
  - Related keys for secure boot
  - Related keys for secure debugging
  - Product life cycle configuration
- True Random Number Generator (RNG):
  - 3 independent entropy sources are internal analog noise sources
- Security Manager (SEC):
  - Monitors product life cycle
  - Configures system security status
  - Formulates security rules and monitors security rule violation events
  - Associated with Monitor (MON), monitors VPMC power supply and OSC24M clock
- Secure boot mechanism based on BOOT ROM, supports encrypted boot, supports trusted execution environment

1.2.14 System Debugging

System debugging modules include:

- Supports JTAG interface
  - Complies with *RISC-V External Debug Support V0.13* specification
  - Supports IEEE1149.1
  - Accesses RISC-V core registers, CSRs and memory
- Debug port locking function
  - Open mode: debugging function is enabled
  - Locked mode: debugging function is disabled, can be unlocked with debug key
  - Disabled mode: debugging function is permanently disabled

1.3 Document Conventions

1.3.1 List of Register-related Abbreviations

The following abbreviations are used in the register descriptions of this manual:

Abbreviation	Description
ro (Read Only)	Software can only read this bit.
rc (Read Only, Clear on Read)	Software can only read this bit, and the bit is cleared after being read.

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Abbreviation	Description
wo (Write Only)	Software can only write to this bit.
w1c (Write 1 to Clear)	Software can clear this bit by writing 1. Writing 0 has no effect on the value of this bit.
w1s (Write 1 to Set)	Software can set this bit to 1 by writing 1. Writing 0 has no effect on the value of this bit.
wc (Write to Clear)	Software can clear this bit by writing any value.
ws (Write to Set)	Software can set this bit to 1 by writing any value.
rw (Read/Write)	Software can read and write this bit.
rw1c (Readable/Write 1 to Clear)	Software can read this bit, and can also clear this bit by writing 1. Writing 0 has no effect on the value of this bit.
rw1s (Readable/Write 1 to Set)	Software can read this bit, and can also set this bit to 1 by writing 1. Writing 0 has no effect on the value of this bit.
rwc (Readable/Write to Clear)	Software can read this bit, and can also clear this bit by writing any value.
rws (Readable/Write to Set)	Software can read this bit, and can also set this bit to 1 by writing any value.

Table 2: List of Abbreviations for Register Descriptions

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2 Processor Core

This product integrates a high-performance RISC-V processor as the CPU, which complies with the following RISC-V specifications:

- ***The RISC-V Instruction Set Manual, Volume I: User-Level ISA, Version 2.2***
- ***The RISC-V Instruction Set Manual Volume II: Privileged Architecture Version 1.11***
- ***The RISC-V Debug Specification, Version 0.13***

2.1 Central Processing Unit

32-bit high-performance RISC-V embedded processor, supporting the following instruction sets:

- RISC-V, RV32I: Base integer instruction set
- RISC-V, M extension: Multiplication and division instruction set
- RISC-V, A extension: Atomic instruction set
- RISC-V, F extension: Single-precision floating-point instruction set
- RISC-V, D extension: Double-precision floating-point instruction set
- RISC-V, C extension: Compressed instruction set
- RISC-V, B extension: Bit manipulation instruction set

It also supports extended instruction sets:

- RISC-V, P extension: SIMD and DSP extended instruction set

Other features:

- Dynamic branch prediction
- Processor performance monitor
- Unaligned memory access

2.2 Bus and Memory Interface

The RISC-V processor is configured to support the following bus interfaces:

- Bus master interface, through which the processor accesses on-chip memory and other resources.
- Bus slave interface, through which other on-chip bus master devices can access the processor's data and instruction local memories
- Instruction local memory interface, through which the processor accesses high-speed instruction local memory
- Data local memory interface, through which the processor accesses high-speed data local memory

2.3 TRAP

According to the RISC-V specification, the control flow transfer of processor instruction execution caused by exceptions or interrupts is called trap. Exceptions are triggered by the processor's own instruction execution, while interrupts are generated by internal and external interrupt sources of the processor. When a trap occurs, the processor will interrupt the current instruction execution flow, disable interrupts, save required kernel general-purpose registers to the stack, and then execute the corresponding trap service routine.

For details of the RISC-V processor TRAP, please refer to Chapter 4.

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For all peripherals of this product that support interrupt generation and their interrupt vector table, please refer to Section 4.4.

2.4 Machine Timer MCHTMR

The machine timer is a 64-bit timer compliant with the RISC-V specification. It runs on a fixed clock, provides a fixed time base, and can generate Machine Timer Interrupt. The MCHTMR includes a 64-bit counter mtime and a 64-bit comparator mtimecmpx. When mtime >= mtimecmpx, an interrupt is generated.

2.5 Hardware Performance Monitor

The hardware performance monitor complies with the RISC-V specification, including:

- mcycle / mcycleh CSR: 64-bit counter that counts the number of clock cycles the processor has run since a specific time point.
- minstret / minstreth CSR: 64-bit counter that counts the number of retired instructions executed by the processor since a specific time point.
- hpmcounter3~6 / hpmcounterh3~6 CSR: 4 64-bit event counters. By configuring CSR mhpmevent3~6, the event counters can be set to count the following events since a specific time point:
  - Execution count of specific instructions, including integer instructions such as LOAD, STORE and multiplication, floating-point related instructions, and program flow control instructions such as jump and return
  - Number of accesses to processor local memories ILM and DLM
  - Number of cache-related events: such as instruction or data cache hits, misses, etc.
  - Number of mispredictions by the branch prediction module, etc.

2.6 Privilege Modes

This product complies with *The RISC-V Instruction Set Manual Volume II: Privileged Architecture Version 1.11* specification.

- Supports Machine Mode (M-mode)
- Supports User Mode (U-mode)

Supports Physical Memory Protection (PMP) with 16 regions.

2.7 Physical Memory Protection

As defined in the RISC-V specification, RISC-V processors can support the PMP (Physical Memory Protection) module. PMP provides read, write and execute protection for specified address ranges in the memory map. PMP permission checks are performed based on the CPU's privilege mode, that is, users can configure PMP to grant software running in User Mode read, write or execute permissions for specified address ranges.

By configuring PMP, users can also revoke read, write or execute permissions for specified address ranges from software running in Machine Mode.

The RISC-V processor of this product supports 16 PMP entries, which can be configured through 4 PMP CFG CSRs and 16 PMP ADDR CSRs.

Users can configure the base address and length of the specified memory area through the PMP ADDR

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CSR.

Users can configure the PMP CFG CSR to specify whether software in User Mode has read, write, or execute permissions for the specified address range.

Users can lock the PMP CSR configuration. Once locked, even if the CPU is in Machine Mode, the PMP related CSRs cannot be modified until the next reset. Meanwhile, once locked, the PMP configuration takes effect not only for User Mode, but also for Machine Mode.

2.8 Related Documents

For more information about the RISC-V open source instruction set, please visit the official RISC-V website at https://riscv.org/.

For RISC-V instruction set related specifications, please visit https://riscv.org/technical/specifications/.

For information about the DSP extension instructions of the RISC-V CPU of this product, please visit http://www.andestech.com/en/products-solutions/product-documentation/ and download *AndeStar V5 DSP ISA Extension Specification*.

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3 Control and Status Registers of RISC-V Processor

This chapter lists the control and status register group (control and status register, abbreviated as CSR) of the RISC-V processor core of this product.

CSRs are private registers of the RISC-V CPU. The RISC-V specification defines a series of standard CSRs, and processor manufacturers can also implement CSRs in the CPU to support non-standard functions.

Main functions of CSRs include:

- Inherent processor information related: such as processor manufacturer information, architecture information, number of cores, etc.;
- Interrupt related: such as interrupt enable/disable, interrupt entry and other information;
- Interrupt response related: such as interrupt cause, interrupt return address and other information;
- Memory protection related: set memory access attributes for different address spaces, such as readable, writable, executable, etc.;
- Performance statistics and debug interface related

The RISC-V instruction set defines a series of instructions for accessing CSRs, through which users can operate CSR registers, for example:

```
csrr rd, csr  //Read CSR
csrw csr, rs  //Write CSR
```

Users can refer to relevant RISC-V specifications for more related information.

3.1 Control Status Register (CSR) Description

The list of control status registers (CSRs) is as follows. Note that the address offset in the list represents the address in the CSR coding space of the RISC-V CPU:

Offset	Register Name	Description	Type	Reset Value
0x000	USTATUS	User Status	standard read/write	0x00000000
0x004	UIE	User Interrupt Enable	standard read/write	0x00000000
0x005	UTVEC	User Trap Vector Base Address	standard read/write	0x00000000
0x040	USCRATCH	User Scratch Register	standard read/write	0x00000000
0x041	UEPC	User Exception Program Counter	standard read/write	0x00000000
0x042	UCAUSE	User Cause Register	standard read/write	0x00000000
0x043	UTVAL	User Trap Value	standard read/write	0x00000000
0x044	UIP	User Interrupt Pending	standard read/write	0x00000000
0x300	MSTATUS	Machine Status	standard read/write	0x00001800
0x301	MISA	Machine ISA Register	standard read/write	-
0x304	MIE	Machine Interrupt Enable	standard read/write	0x00000000
0x305	MTVEC	Machine Trap Vector Base Address	standard read/write	0x00000000

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Offset	Register Name	Description	Type	Reset Value
0x306	MCOUNTEREN	Machine Counter Enable	standard read/write	0x00000000
0x320	MCOUNTINHIBIT	Machine Counter-Inhibit	non-standard read/write	0x00000000
0x323	MHPMEVENT3	Machine Performance Monitoring Event Selector	standard read/write	0x00000000
0x324	MHPMEVENT4	Machine Performance Monitoring Event Selector	standard read/write	0x00000000
0x325	MHPMEVENT5	Machine Performance Monitoring Event Selector	standard read/write	0x00000000
0x326	MHPMEVENT6	Machine Performance Monitoring Event Selector	standard read/write	0x00000000
0x340	MSCRATCH	Machine Scratch Register	standard read/write	0x00000000
0x341	MEPC	Machine Exception Program Counter	standard read/write	0x00000000
0x342	MCAUSE	Machine Cause Register	standard read/write	0x00000000
0x343	MTVAL	Machine Trap Value	standard read/write	0x00000000
0x344	MIP	Machine Interrupt Pending	standard read/write	0x00000000
0x3A0	PMPCFG0	PMP Configuration Registers	standard read/write	0x00000000
0x3A1	PMPCFG1	PMP Configuration Registers	standard read/write	0x00000000
0x3A2	PMPCFG2	PMP Configuration Registers	standard read/write	0x00000000
0x3A3	PMPCFG3	PMP Configuration Registers	standard read/write	0x00000000
0x3B0	PMPADDR[PMPADDR0]	PMP Address Register	standard read/write	-
0x3B1	PMPADDR[PMPADDR1]	PMP Address Register	standard read/write	-
0x3B2	PMPADDR[PMPADDR2]	PMP Address Register	standard read/write	-
0x3B3	PMPADDR[PMPADDR3]	PMP Address Register	standard read/write	-
0x3B4	PMPADDR[PMPADDR4]	PMP Address Register	standard read/write	-
0x3B5	PMPADDR[PMPADDR5]	PMP Address Register	standard read/write	-
0x3B6	PMPADDR[PMPADDR6]	PMP Address Register	standard read/write	-
0x3B7	PMPADDR[PMPADDR7]	PMP Address Register	standard read/write	-
0x3B8	PMPADDR[PMPADDR8]	PMP Address Register	standard read/write	-
0x3B9	PMPADDR[PMPADDR9]	PMP Address Register	standard read/write	-
0x3BA	PMPADDR[PMPADDR10]	PMP Address Register	standard read/write	-
0x3BB	PMPADDR[PMPADDR11]	PMP Address Register	standard read/write	-
0x3BC	PMPADDR[PMPADDR12]	PMP Address Register	standard read/write	-
0x3BD	PMPADDR[PMPADDR13]	PMP Address Register	standard read/write	-
0x3BE	PMPADDR[PMPADDR14]	PMP Address Register	standard read/write	-
0x3BF	PMPADDR[PMPADDR15]	PMP Address Register	standard read/write	-
0x7A0	TSELECT	Trigger Select	standard read/write	0x00000000
0x7A1	TDATA1	Trigger Data 1	standard read/write	0x20000000
0x7A1	MCONTROL	Match Control	standard read/write	0x22400000

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Offset	Register Name	Description	Type	Reset Value
0x7A1	ICOUNT	Instruction Count	standard read/write	0x20000400
0x7A1	ITRIGGER	Interrupt Trigger	standard read/write	0x20000000
0x7A1	ETRIGGER	Exception Trigger	standard read/write	0x20000000
0x7A2	TDATA2	Trigger Data 2	standard read/write	0x00000000
0x7A3	TDATA3	Trigger Data 3	standard read/write	0x00000000
0x7A3	TEXTRA	Trigger Extra	standard read/write	0x00000000
0x7A4	TINFO	Trigger Info	standard read/write	-
0x7A5	TCONTROL	Trigger Control	standard read/write	0x00000000
0x7A8	MCONTEXT	Machine Context	standard read/write	0x00000000
0x7AA	SCONTEXT	Supervisor Context	standard read/write	0x00000000
0x7B0	DCSR	Debug Control and Status Register	debug-mode-only	0x40000603
0x7B1	DPC	Debug Program Counter	debug-mode-only	0x00000000
0x7B2	DSCRATCH0	Debug Scratch Register 0	debug-mode-only	0x00000000
0x7B3	DSCRATCH1	Debug Scratch Register 1	debug-mode-only	0x00000000
0x7C0	MILMB	Instruction Local Memory Base Register	non-standard read/write	-
0x7C1	MDLMB	Data Local Memory Base Register	non-standard read/write	-
0x7C2	MECC_CODE	ECC Code Register	non-standard read/write	0x00000001
0x7C3	MNVEC	NMI Vector Base Address Register	non-standard read/write	-
0x7C4	MXSTATUS	Machine Extended Status	non-standard read/write	0x00000000
0x7C5	MPFT_CTL	Performance Throttling Control Register	non-standard read/write	0x00000000
0x7C6	MHSP_CTL	Machine Hardware Stack Protection Control	non-standard read/write	0x00000000
0x7C7	MSP_BOUND	Machine SP Bound Register	non-standard read/write	0xFFFFFFFF
0x7C8	MSP_BASE	Machine SP Base Register	non-standard read/write	0xFFFFFFFF
0x7C9	MDCAUSE	Machine Detailed Trap Cause	non-standard read/write	0x00000000
0x7CA	MCACHE_CTL	Cache Control Register	non-standard read/write	0x00000800
0x7CB	MCCTLBEGINADDR	Machine CCTL Begin Address	non-standard read/write	0x00000000
0x7CC	MCCTLCOMMAND	Machine CCTL Command	non-standard read/write	0x00000000

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Offset	Register Name	Description	Type	Reset Value
0x7CD	MCCTLDATA	Machine CCTL Data	non-standard read/write	0x00000000
0x7CE	MCOUNTERWEN	Machine Counter Write Enable	non-standard read/write	0x00000000
0x7CF	MCOUNTERINTEN	Machine Counter Interrupt Enable	non-standard read/write	0x00000000
0x7D0	MMISC_CTL	Machine Miscellaneous Control Register	non-standard read/write	0x00000048
0x7D1	MCOUNTERMASK_M	Machine Counter Mask for Machine Mode	non-standard read/write	0x00000000
0x7D2	MCOUNTERMASK_S	Machine Counter Mask for Supervisor Mode	non-standard read/write	0x00000000
0x7D3	MCOUNTERMASK_U	Machine Counter Mask for User Mode	non-standard read/write	0x00000000
0x7D4	MCOUNTEROVF	Machine Counter Overflow Status	non-standard read/write	0x00000000
0x7E0	DEXC2DBG	Exception Redirection Register	non-standard read/write	0x00000000
0x7E1	DDCAUSE	Debug Detailed Cause	non-standard read/write	0x00000000
0x800	UITB	Instruction Table Base Address Register	non-standard read/write	0x00000000
0x801	UCODE	Code Register	non-standard read/write	0x00000000
0x809	UDCAUSE	User Detailed Trap Cause	non-standard read/write	0x00000000
0x80B	UCCTLBEGINADDR	User CCTL Begin Address	non-standard read/write	0x00000000
0x80C	UCCTLCOMMAND	User CCTL Command	non-standard read/write	0x00000000
0xB00	MCYCLE	Machine Cycle Counter	standard read/write	0x00000000
0xB02	MINSTRET	Machine Instruction-Retired Counter	standard read/write	0x00000000
0xB03	MHPMCOUNTER3	Machine Performance Monitoring Counter	standard read/write	0x00000000
0xB04	MHPMCOUNTER4	Machine Performance Monitoring Counter	standard read/write	0x00000000
0xB05	MHPMCOUNTER5	Machine Performance Monitoring Counter	standard read/write	0x00000000
0xB06	MHPMCOUNTER6	Machine Performance Monitoring Counter	standard read/write	0x00000000

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Offset	Register Name	Description	Type	Reset Value
0xB80	MCYCLEH	Machine Cycle Counter	standard read/write	0x00000000
0xB82	MINSTRETH	Machine Instruction-Retired Counter	standard read/write	0x00000000
0xB83	MHPMCOUNTER3H	Machine Performance Monitoring Counter	standard read/write	0x00000000
0xB84	MHPMCOUNTER4H	Machine Performance Monitoring Counter	standard read/write	0x00000000
0xB85	MHPMCOUNTER5H	Machine Performance Monitoring Counter	standard read/write	0x00000000
0xB86	MHPMCOUNTER6H	Machine Performance Monitoring Counter	standard read/write	0x00000000
0xC00	CYCLE	CYCLE Register		0x00000000
0xC80	CYCLEH	CYCLE High 32-bit register		0x00000000
0xF11	MVENDORID	Machine Vendor ID Register	standard read only	0x0000031E
0xF12	MARCHID	Machine Architecture ID Register	standard read only	0x00000025
0xF13	MIMPID	Machine Implementation ID Register	standard read only	-
0xF14	MHARTID	Hart ID Register	standard read only	0x00000000

Table 3: CSR Register List

3.2 Control Status Register (CSR) Detailed Information

The detailed description of control status registers (CSRs) is as follows:

3.2.1 USTATUS (0x0) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																											UPIE	RSVD				UIE
N/A																											RW	N/A				RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	0

USTATUS [31:0]

Bits	Name	Description
4	UPIE	UPIE holds the value of the UIE bit prior to a trap.
0	UIE	U mode interrupt enable bit. 0:Disabled 1:Enabled

USTATUS Bits

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3.2.2 UIE (0x4) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																							UEIE	RSVD			UTIE	RSVD			USIE
N/A																							RW	N/A			RW	N/A			RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	0	x	x	x	0

UIE [31:0]

Bits	Name	Description
8	UEIE	U mode external interrupt enable bit 0:Disabled 1:Enabled
4	UTIE	U mode timer interrupt enable bit. 0:Disabled 1:Enabled
0	USIE	U mode software interrupt enable bit. 0:Disabled 1:Enabled

UIE Bits

3.2.3 UTVEC (0x5) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BASE_31_2																														RSVD	
RW																														N/A	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x

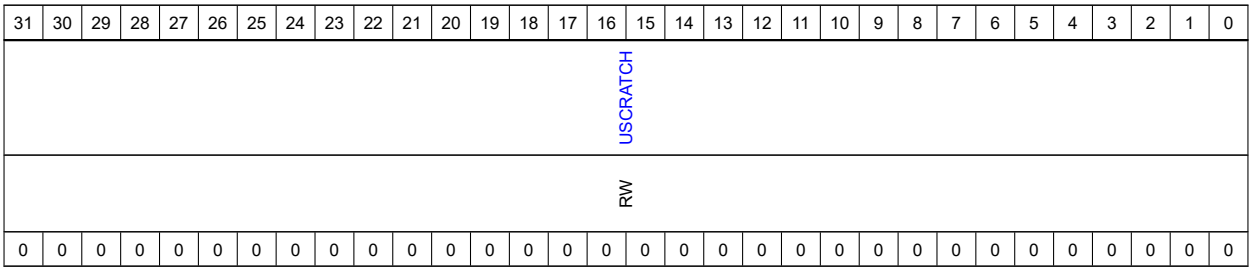
UTVEC [31:0]

Bits	Name	Description
31-2	BASE_31_2	Base address for interrupt and exception handlers. See description above for alignment requirements when PLIC is in the vector mode.

UTVEC Bits

3.2.4 USCRATCH (0x40) (standard read/write)

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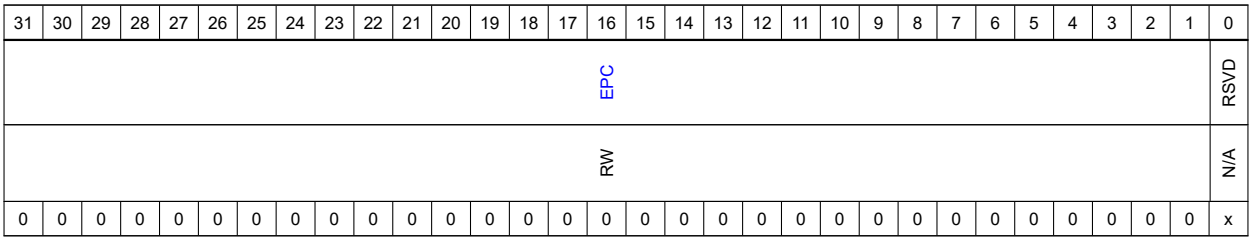


USCRATCH [31:0]

Bits	Name	Description
31-0	USCRATCH	Scratch register storage.

USCRATCH Bits

3.2.5 UEPC (0x41) (standard read/write)

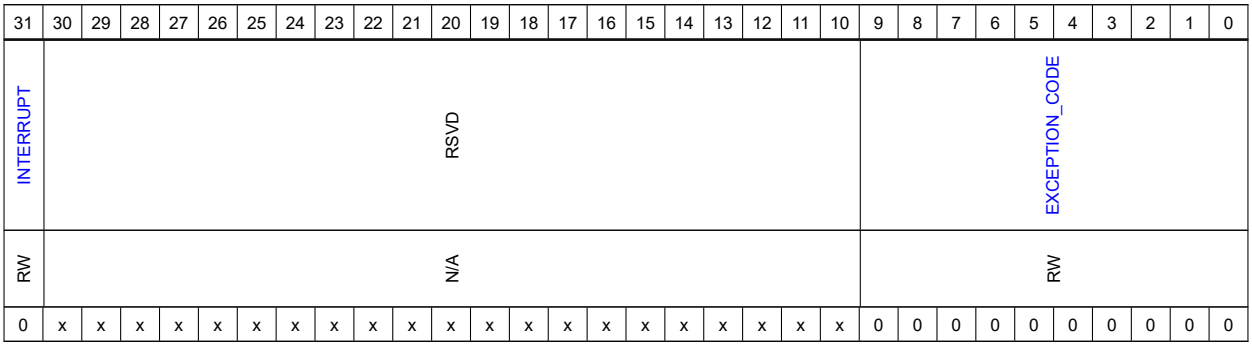


UEPC [31:0]

Bits	Name	Description
31-1	EPC	Exception program counter.

UEPC Bits

3.2.6 UCAUSE (0x42) (standard read/write)



UCAUSE [31:0]

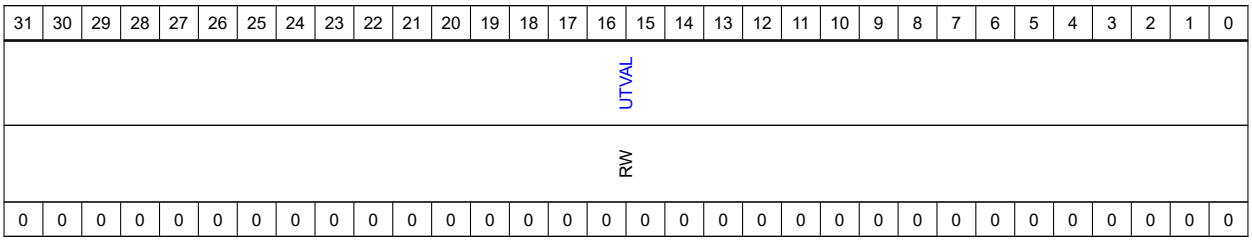
Bits	Name	Description
31	INTERRUPT	Interrupt.

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Bits	Name	Description
9-0	EXCEPTION_CODE	Exception Code. When interrupt is 1: 0:User software interrupt 4:User timer interrupt 8:User external interrupt When interrupt is 0: 0:Instruction address misaligned 1:Instruction access fault 2:Illegal instruction 3:Breakpoint 4:Load address misaligned 5:Load access fault 6:Store/AMO address misaligned 7:Store/AMO access fault 8:Environment call from U-mode 9-11:Reserved 12:Instruction page fault 13:Load page fault 14:Reserved 15:Store/AMO page fault 32：Stack overflow exception 33:Stack underflow exception 40-47:Reserved

UCAUSE Bits

3.2.7 UTVAL (0x43) (standard read/write)



UTVAL [31:0]

Bits	Name	Description
31-0	UTVAL	Exception-specific information for software trap handling.

UTVAL Bits

3.2.8 UIP (0x44) (standard read/write)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																							UEIP	RSVD			UTIP	RSVD			USIP
N/A																							RW	N/A			RW	N/A			RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	0	x	x	x	0

UIP [31:0]

Bits	Name	Description
8	UEIP	U mode external interrupt pending bit. 0:Not pending 1:Pending
4	UTIP	U mode timer interrupt pending bit. 0:Not pending 1:Pending
0	USIP	U mode software interrupt pending bit. 0:Not pending 1:Pending

UIP Bits

3.2.9 MSTATUS (0x300) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SD	RSVD											MXR	RSVD	MPRV	XS		FS		MPP		RSVD			MPIE	RSVD		UPIE	MIE	RSVD		UIE
RO	N/A											RW	N/A	RW	RO		RW		RW		N/A			RW	N/A		RW	RW	N/A		RW
0	x	x	x	x	x	x	x	x	x	x	x	0	x	0	0	0	0	0	1	1	x	x	x	0	x	x	0	0	x	x	0

MSTATUS [31:0]

Bits	Name	Description
31	SD	SD summarizes whether either the FS field or XS field is dirty.
19	MXR	MXR controls whether execute-only pages are readable. It has no effect when page-based virtual memory is not in effect 0:Execute-only pages are not readable 1:Execute-only pages are readable
17	MPRV	When the MPRV bit is set, the memory access privilege for load and store are specified by the MPP field. When U-mode is not available, this field is hardwired to 0.

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Bits	Name	Description
16-15	XS	XS holds the status of the architectural states (ACE registers) of ACE instructions. The value of this field is zero if ACE extension is not configured. This field is primarily managed by software. The processor hardware assists the state managements in two regards: Illegal instruction exceptions are triggered when XS is Off. XS is updated to the Dirty state with the execution of ACE instructions when XS is not Off. Changing the setting of this field has no effect on the contents of ACE states. In particular, setting XS to Off does not destroy the states, nor does setting XS to Initial clear the contents. 0:Off 1:Initial 2:Clean 3:Dirty
14-13	FS	FS holds the status of the architectural states of the floating-point unit, including the fcsr CSR and f0 –f31 floating-point data registers. The value of this field is zero and read-only if the processor does not have FPU. This field is primarily managed by software. The processor hardware assists the state managements in two regards: Attempts to access fcsr or any f register raise an illegal-instruction exception when FS is Off. FS is updated to the Dirty state with the execution of any instruction that updates fcsr or any f register when FS is Initial or Clean. Changing the setting of this field has no effect on the contents of the floating-point register states. In particular, setting FS to Off does not destroy the states, nor does setting FS to Initial clear the contents. 0:Off 1:Initial 2:Clean 3:Dirty
12-11	MPP	MPP holds the privilege mode prior to a trap. Encoding for privilege mode is described in Table5. When U-mode is not available, this field is hardwired to 3.
7	MPIE	MPIE holds the value of the MIE bit prior to a trap.
4	UPIE	UPIE holds the value of the UIE bit prior to a trap.
3	MIE	M mode interrupt enable bit. 0: Disabled 1: Enabled

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Bits	Name	Description
0	UIE	U mode interrupt enable bit. 0: Disabled 1: Enabled

MSTATUS Bits

3.2.10 MISA (0x301) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BASE		RSVD				Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
RO		N/A				RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO
0	1	x	x	x	x	0	0	1	0	0	*	0	*	0	0	0	0	*	1	0	0	0	1	0	0	*	0	*	1	0	*

MISA [31:0]

Bits	Name	Description
31-30	BASE	The general-purpose register width of the native base integer ISA. 0:Reserved 1:32 2:64 3:128
25	Z	Reserved
24	Y	Reserved
23	X	Non-standard extensions present
22	W	Reserved
21	V	Tentatively reserved for Vector extension
20	U	User mode implemented 0:Machine 1:Machine + User / Machine + Supervisor + User
19	T	Tentatively reserved for Transactional Memory extension
18	S	Supervisor mode implemented 0:Machine / Machine + User 1:Machine + Supervisor + User
17	R	Reserved
16	Q	Quad-precision floating-point extension
15	P	Tentatively reserved for Packed-SIMD extension
14	O	Reserved
13	N	User-level interrupts supported 0:no 1:yes

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Bits	Name	Description
12	M	Integer Multiply/Divide extension
11	L	Tentatively reserved for Decimal Floating-Point extension
10	K	Reserved
9	J	Tentatively reserved for Dynamically Translated Languages extension
8	I	RV32I/64I/128I base ISA
7	H	Reserved
6	G	Additional standard extensions present
5	F	Single-precision floating-point extension 0:none 1:double+single precision / single precision
4	E	RV32E base ISA
3	D	Double-precision floating-point extension 0:single precision / none 1:double+single precision
2	C	Compressed extension
1	B	Tentatively reserved for Bit operations extension
0	A	Atomic extension 0:no 1:yes

MISA Bits

3.2.11 MIE (0x304) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD													PMOVI	BWEI	IMECCI	RSVD				MEIE	RSVD		UEIE	MTIE	RSVD		UTIE	MSIE	RSVD		USIE
N/A													RW	RW	RW	N/A				RW	N/A		RW	RW	N/A		RW	RW	N/A		RW
x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	x	x	x	x	0	x	x	0	0	x	x	0	0	x	x	0

MIE [31:0]

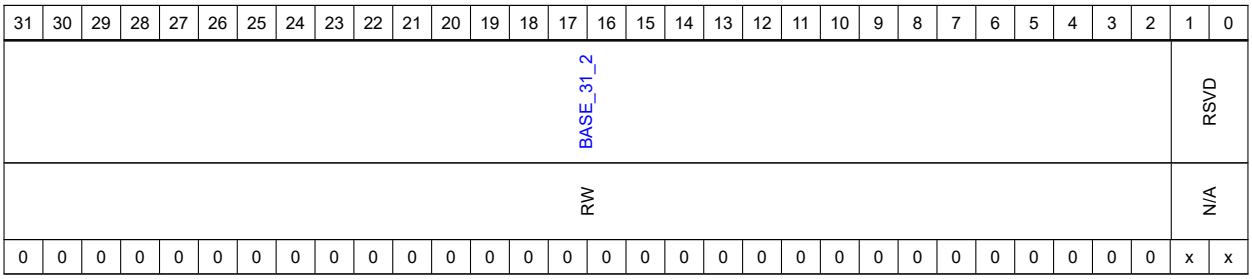
Bits	Name	Description
18	PMOVI	Performance monitor overflow local interrupt enable bit 0:Disabled 1:Enabled
17	BWEI	Bus read/write transaction error local interrupt enable bit. The processor may receive bus errors on load/store instructions or cache writebacks. 0:Disabled 1:Enabled

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Bits	Name	Description
16	IMECCI	Imprecise ECC error local interrupt enable bit. The processor may receive imprecise ECC errors on slave port accesses or cache writebacks. 0:Disabled 1:Enabled
11	MEIE	M mode external interrupt enable bit 0:Disabled 1:Enabled
8	UEIE	U mode external interrupt enable bit 0:Disabled 1:Enabled
7	MTIE	M mode timer interrupt enable bit. 0:Disabled 1:Enabled
4	UTIE	U mode timer interrupt enable bit. 0:Disabled 1:Enabled
3	MSIE	M mode software interrupt enable bit 0:Disabled 1:Enabled
0	USIE	U mode software interrupt enable bit. 0:Disabled 1:Enabled

MIE Bits

3.2.12 MTVEC (0x305) (standard read/write)



MTVEC [31:0]

Bits	Name	Description
31-2	BASE_31_2	Base address for interrupt and exception handlers. See description above for alignment requirements when PLIC is in the vector mode

MTVEC Bits

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3.2.13 MCOUNTEREN (0x306) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																									HPM6	HPM5	HPM4	HPM3	IR	TM	CY
N/A																									RW	RW	RW	RW	RW	RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0

MCOUNTEREN [31:0]

Bits	Name	Description
6	HPM6	See register description
5	HPM5	See register description
4	HPM4	See register description
3	HPM3	See register description
2	IR	See register description
1	TM	See register description
0	CY	See register description

MCOUNTEREN Bits

3.2.14 MCOUNTINHIBIT (0x320) (non-standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																									HPM6	HPM5	HPM4	HPM3	IR	TM	CY
N/A																									RW	RW	RW	RW	RW	RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0

MCOUNTINHIBIT [31:0]

Bits	Name	Description
6	HPM6	See register description.
5	HPM5	See register description.
4	HPM4	See register description.
3	HPM3	See register description.
2	IR	See register description.
1	TM	See register description.
0	CY	See register description.

MCOUNTINHIBIT Bits

3.2.15 MHPMEVENT3 (0x323) (standard read/write)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																							SEL					TYPE			
N/A																							RW					RW			
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0

MHPMEVENT3 [31:0]

Bits	Name	Description
8-4	SEL	Event Selectors table
3-0	TYPE	Event Selectors table

MHPMEVENT3 Bits

TYPE	SEL	事件名称	解释
0	1	Cycle count	Number of elapsed processor clock cycles
0	2	Retired instruction count	Number of retired instructions
0	3	Integer load instruction count	Number of retired load instructions (including LR).
0	4	Integer store instruction count	Number of retired store instructions (including SC).
0	5	Atomic instruction count	Number of retired atomic instructions (not including LR and SC).
0	6	System instruction count	Number of retired SYSTEM instructions(instructions with major opcode equal to 0b1110011).
0	7	Integer computational instruction count	Number of retired integer computational instructions.
0	8	Conditional branch instruction count	Number of retired conditional branch instructions.
0	9	Taken conditional branch instruction count	Number of retired conditional branch instructions that are taken.
0	10	JAL instruction count	Number of retired JAL instructions.
0	11	JALR instruction count	Number of retired JALR instructions.This event selector also counts the events monitored by the return instruction count event selector defined in the next row.
0	12	Return instruction count	Number of retired return instructions. Return instructions are JALR instructions with zero immediate offset and the following operands: • (rd != x1/x5) and (rs1 == x1/x5) • rd == x1 and rs1 == x5 • rd == x5 and rs1 == x1
0	13	Control transfer instruction count	Number of retired unconditional jumps (JAL and JALR) and conditional branch instructions.
0	14	EXEC.IT instruction count	Number of retired EXEC.IT instructions.

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TYPE	SEL	事件名称	解释
0	15	Integer multiplication instruction count	Number of retired integer multiplication instructions.
0	16	Integer division instruction count	Number of retired integer division/remainder instructions.
0	17	Floating-point load instruction count	Number of retired floating-point load instructions.
0	18	Floating-point store instruction count	Number of retired floating-point store instructions.
0	19	Floating-point addition instruction count	Number of retired floating-point addition/subtraction instructions.
0	20	Floating-point multiplication instruction count	Number of retired floating-point multiplication instructions.
0	21	Floating-point fused multiply-add instruction count	Number of retired floating-point fused multiply-add/subtraction instructions (FMADD, FMSUB, FNMSUB, FNMADD).
0	22	Floating-point division or square-root instruction count	Number of retired floating-point division/square-root instructions.
0	23	Other floating-point instruction count	Number of retired floating-point instructions not counted by the previous floating-point instruction event selectors
0	24	Integer multiplication and add/sub instruction count	Number of retired integer multiplication and add/sub instructions.
0	25	Retired operation count	Number of retired operations. a floating-point multiply-add instruction is counted as 2 operations. other instructions are counted as 1 operation
1	0	ILM access	Number of ILM transfers, including speculative instruction fetch, load/store accesses, ECC repair and slave port accesses.
1	1	DLM access	Number of DLM transfers, including speculative load/store accesses, ECC repair and slave port accesses.
1	2	I-Cache access	Number of completed I-Cache fetch access.
1	3	I-Cache miss	Number of I-Cache fetch miss.
1	4	D-Cache access	Number of completed D-Cache load-and-store access. Misaligned load/store accesses might increase this counter by either one or two, depending on access sizes and alignments. Only misaligned accesses crossing two cache lines are guaranteed to result in increment of two.

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TYPE	SEL	事件名称	解释
1	5	D-Cache miss	The event counts the number of D-Cache load-and-store miss. Misaligned load/store accesses might increase this counter by either zero, one or two, depending on access sizes, alignments and whether the accessed lines are in D-Cache.
1	6	D-Cache load access	Number of completed D-Cache load access. See the D-Cache access count event selector for the handling of misaligned load accesses.
1	7	D-Cache load miss	Number of D-Cache load miss. See the D-Cache miss count event selector for the handling of misaligned load accesses.
1	8	D-Cache store access	Number of completed D-Cache store access. See the D-Cache access count event selector for the handling of misaligned load accesses.
1	9	D-Cache store miss	Number of D-Cache store miss. See the D-Cache miss count event selector for the handling of misaligned load accesses.
1	10	D-Cache writeback	Number of D-Cache writeback.
1	11	Cycles waiting for I-Cache fill data	Number of cycles waiting for the return of the critical word of I-Cache misses from the system bus. This event selector does not monitor accesses to I/O regions or accesses to cacheable regions when I-Cache is turned off.
1	12	Cycles waiting for D-Cache fill data	Number of cycles waiting for the return of the critical word of D-Cache misses from the system bus. This event selector does not monitor accesses to I/O regions or accesses to cacheable regions when D-Cache is turned off.
1	13	Uncached fetch data access from bus	Number of accesses of uncached instruction data returning from the system bus. This event selector monitors accesses to I/O regions or accesses to cacheable regions when I-Cache is not configured or off.
1	14	Uncached load data access from bus	Number of accesses of uncached load data returning from the system bus. This event selector monitors accesses to I/O regions or accesses to cacheable regions when D-Cache is not configured or off.

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TYPE	SEL	事件名称	解释
1	15	Cycles waiting for uncached fetch data from bus	Number of cycles waiting for the instruction data to return from the system bus. This event selector monitors accesses to I/O regions or accesses to cacheable regions when I-Cache is not configured or off.
1	16	Cycles waiting for uncached load data from bus	Number of cycles waiting for the load data to return from the system bus. This event selector monitors accesses to I/O regions or accesses to cacheable regions when D-Cache is not configured or off.
1	23	Hardware prefetch bus access	This event counts the bus accesses generated by the hardware data prefetcher.
2	0	Misprediction of conditional branches (direction)	Number of misprediction of committed conditional branches.
2	1	Misprediction of taken conditional branches (direction)	Number of misprediction of committed taken conditional branches.
2	2	Misprediction of targets of Return instructions	Number of misprediction of committed Return instruction.

Table 4: Event Selectors

3.2.16 MHPMEVENT4 (0x324) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																							SEL					TYPE			
N/A																							RW					RW			
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0

MHPMEVENT4 [31:0]

Bits	Name	Description
8-4	SEL	Event Selectors table
3-0	TYPE	Event Selectors table

MHPMEVENT4 Bits

3.2.17 MHPMEVENT5 (0x325) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																							SEL					TYPE			
N/A																							RW					RW			

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0

MHPMEVENT5 [31:0]

Bits	Name	Description
8-4	SEL	Event Selectors table
3-0	TYPE	Event Selectors table

MHPMEVENT5 Bits

3.2.18 MHPMEVENT6 (0x326) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																							SEL					TYPE			
N/A																							RW					RW			
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0

MHPMEVENT6 [31:0]

Bits	Name	Description
8-4	SEL	Event Selectors table
3-0	TYPE	Event Selectors table

MHPMEVENT6 Bits

3.2.19 MSCRATCH (0x340) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MSCRATCH																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MSCRATCH [31:0]

Bits	Name	Description
31-0	MSCRATCH	Scratch register storage.

MSCRATCH Bits

3.2.20 MEPC (0x341) (standard read/write)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EPC																															RSVD
RW																															N/A
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x

MEPC [31:0]

Bits	Name	Description
31-1	EPC	Exception program counter.

MEPC Bits

3.2.21 MCAUSE (0x342) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INTERRUPT	RSVD																			EXCEPTION_CODE											
RW	N/A																			RW											
0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0

MCAUSE [31:0]

Bits	Name	Description
31	INTERRUPT	Interrupt

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Bits	Name	Description
11-0	EXCEPTION_CODE	Exception code When interrupt is 1, the value means: 0:User software interrupt 1:Supervisor software interrupt 3:Machine software interrupt 4:User timer interrupt 5:Supervisor timer interrupt 7:Machine timer interrupt 8:User external interrupt 9:Supervisor external interrupt 11:Machine external interrupt 16:Imprecise ECC error interrupt (slave port accesses, D-Cache evictions, and nonblocking load/stores) (M-mode) 17:Bus read/write transaction error interrupt (M-mode) 18:Performance monitor overflow interrupt (M-mode) 256+16:Imprecise ECC error interrupt (slave port accesses, D-Cache evictions, and nonblocking load/stores) (S-mode) 256+17:Bus write transaction error interrupt (S-mode) 256+18:Performance monitor overflow interrupt (S-mode) When interrupt bit is 0, the value means: 0:Instruction address misaligned 1:Instruction access fault 2:Illegal instruction 3:Breakpoint 4:Load address misaligned 5:Load access fault 6:Store/AMO address misaligned 7:Store/AMO access fault 8:Environment call from U-mode 9:Environment call from S-mode 11:Environment call from M-mode 32:Stack overflow exception 33:Stack underflow exception 40-47:Reserved

MCAUSE Bits

3.2.22 MTVAL (0x343) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MTVAL																															

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MTVAL [31:0]

Bits	Name	Description
31-0	MTVAL	Exception-specific information for software trap handling.

MTVAL Bits

3.2.23 MIP (0x344) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD													PMOVI	BWEI	IMECCI	RSVD				MEIP	RSVD	SEIP	UEIP	MTIP	RSVD	STIP	UTIP	MSIP	RSVD	SSIP	USIP
N/A													RW	RW	RW	N/A				RW	N/A	RW	RW	RW	N/A	RW	RW	RW	N/A	RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	x	x	x	x	0	x	0	0	0	x	0	0	0	x	0	0

MIP [31:0]

Bits	Name	Description
18	PMOVI	Performance monitor overflow local interrupt pending bit. 0:Not pending 1:Pending
17	BWEI	Bus read/write transaction error local interrupt pending bit. The processor may receive bus errors on load/store instructions or cache writebacks. 0:Not pending 1:Pending
16	IMECCI	Imprecise ECC error local interrupt enable bit. The processor may receive imprecise ECC errors on slave port accesses or cache writebacks. 0:Not pending 1:Pending
11	MEIP	M mode external interrupt pending bit. 0:Not pending 1:Pending
9	SEIP	S mode external interrupt pending bit. 0:Not pending 1:Pending

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Bits	Name	Description
8	UEIP	U mode external interrupt pending bit. 0:Not pending 1:Pending
7	MTIP	M mode timer interrupt pending bit. 0:Not pending 1:Pending
5	STIP	S mode timer interrupt pending bit. 0:Not pending 1:Pending
4	UTIP	U mode timer interrupt pending bit 0:Not pending 1:Pending
3	MSIP	M mode software interrupt pending bit. 0:Not pending 1:Pending
1	SSIP	S mode software interrupt pending bit. 0:Not pending 1:Pending
0	USIP	U mode software interrupt pending bit. 0:Not pending 1:Pending

MIP Bits

3.2.24 PMPCFG0 (0x3A0) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PMP3CFG								PMP2CFG								PMP1CFG								PMP0CFG							
RW								RW								RW								RW							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PMPCFG0 [31:0]

Bits	Name	Description
31-24	PMP3CFG	See PMPCFG register
23-16	PMP2CFG	See PMPCFG register
15-8	PMP1CFG	See PMPCFG register
7-0	PMP0CFG	See PMPCFG register

PMPCFG0 Bits

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3.2.25 PMPCFG1 (0x3A1) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PMP7CFG								PMP6CFG								PMP5CFG								PMP4CFG							
RW								RW								RW								RW							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PMPCFG1 [31:0]

Bits	Name	Description
31-24	PMP7CFG	See PMPCFG register
23-16	PMP6CFG	See PMPCFG register
15-8	PMP5CFG	See PMPCFG register
7-0	PMP4CFG	See PMPCFG register

PMPCFG1 Bits

3.2.26 PMPCFG2 (0x3A2) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PMP11CFG								PMP10CFG								PMP9CFG								PMP8CFG							
RW								RW								RW								RW							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PMPCFG2 [31:0]

Bits	Name	Description
31-24	PMP11CFG	See PMPCFG register
23-16	PMP10CFG	See PMPCFG register
15-8	PMP9CFG	See PMPCFG register
7-0	PMP8CFG	See PMPCFG register

PMPCFG2 Bits

3.2.27 PMPCFG3 (0x3A3) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PMP15CFG								PMP14CFG								PMP13CFG								PMP12CFG							
RW								RW								RW								RW							

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PMPCFG3 [31:0]

Bits	Name	Description
31-24	PMP15CFG	See PMPCFG register
23-16	PMP14CFG	See PMPCFG register
15-8	PMP13CFG	See PMPCFG register
7-0	PMP12CFG	See PMPCFG register

PMPCFG3 Bits

7	6	5	4	3	2	1	0
L	RSVD		A		X	W	R
W1S*	N/A		RW		RW	RW	RW
0	x	x	0	0	0	0	0

Table 5: PMPCFG [7:0]

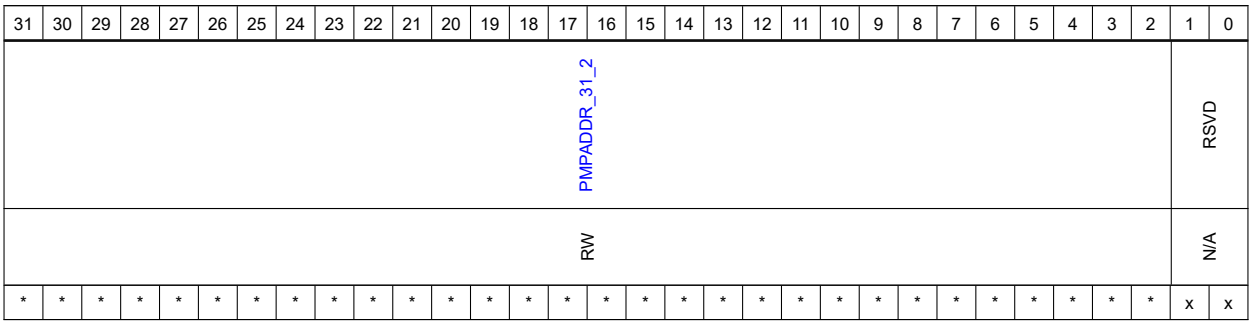
位域	名称	描述
7	L	Write lock and permission enforcement bit for Machine mode 0:Machine mode writes to PMP entry registers are allowed. R/W/X permissions apply to S and U modes. 1:For PMP entry i, writes to PMPiCFG and PMPADDRi are ignored. Additionally, if PMPiCFG.A is set to TOR, writes to pmpaddri-1 are ignored as well. As for permission enforcement, R/W/X permissions apply to all modes. This bit can only be cleared to 0 with a system reset.
4-3	A	Address matching mode. 0:OFF: Null region. 1:TOR: Top of range. For PMP entry 0, it matches any address A < pmpaddr0. For PMP entry i, it matches any address A such that pmpaddri > A >= pmpaddri-1. But the 4-byte range is not supported. 2:Reserved. 3:NAPOT: Naturally aligned power-of-2 region. This mode makes use of the low-order bits of the associated address register to encode the size of the range. The minimal size of NAPOT regions must be 8 bytes.
2	X	Instruction execution control. 0:Instruction execution is not allowed 1:Instruction execution is allowed.

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位域	名称	描述
1	W	Write access control. 0:Write accesses are not allowed. 1:Write accesses are allowed.
0	R	Read access control. 0:Read accesses are not allowed. 1:Read accesses are allowed.

Table 6: PMPCFG 位域

3.2.28 PMPADDR[PMPADDR0] (0x3B0 + 0x1 * n) (standard read/write)

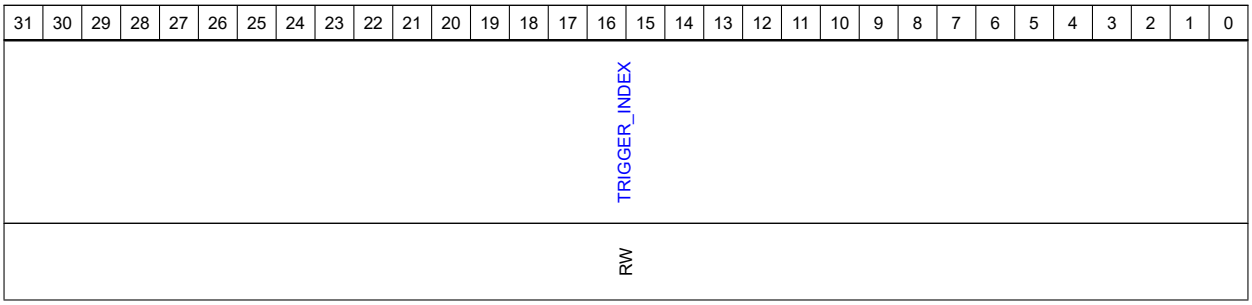


PMPADDR [31:0]

Bits	Name	Description
31-2	PMPADDR_31_2	Register Content : Match Size(Byte) aaaa. . . aaa0 8 aaaa. . . aa01 16 aaaa. . . a011 32 . . . . . . aa01. . . 1111 2^{XLEN} a011. . . 1111 2^{XLEN+1} 0111. . . 1111 2^{XLEN+2} 1111. . . 1111 2^{XLEN+3*1}

PMPADDR Bits

3.2.29 TSELECT (0x7A0) (standard read/write)



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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TSELECT [31:0]

Bits	Name	Description
31-0	TRIGGER_INDEX	This register determines which trigger is accessible through other trigger registers.

TSELECT Bits

3.2.30 TDATA1 (0x7A1) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TYPE				DMODE	DATA																										
RW				RW	RW																										
0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TDATA1 [31:0]

Bits	Name	Description
31-28	TYPE	Indicates the trigger type. 0:The selected trigger is invalid. 2:The selected trigger is an address/data match trigger. 3:The selected trigger is an instruction count trigger 4:The selected trigger is an interrupt trigger. 5:The selected trigger is an exception trigger.
27	DMODE	Setting this field to indicate the trigger is used by Debug Mode. 0:Both Debug-mode and M-mode can write the currently selected trigger registers. 1:Only Debug Mode can write the currently selected trigger registers. Writes from M-mode is ignored.
26-0	DATA	Trigger-specific data

TDATA1 Bits

3.2.31 MCONTROL (0x7A1) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TYPE				DMODE	MASKMAX						RSVD					ACTION				CHAIN	MATCH				M	RSVD		U	EXECUTE	STORE	LOAD
RW				RW	RO						N/A					RW				RW	RW				RW	N/A		RW	RW	RW	RW
0	0	1	0	0	0	1	0	0	1	0	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	x	x	0	0	0	0

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

MCONTROL [31:0]

Bits	Name	Description
31-28	TYPE	Indicates the trigger type. 0:The selected trigger is invalid. 2:The selected trigger is an address/data match trigger.
27	DMODE	Setting this field to indicate the trigger is used by Debug Mode. 0:Both Debug-mode and M-mode can write the currently selected trigger registers 1:Only Debug Mode can write the currently selected trigger registers. Writes from M-mode is ignored.
26-21	MASKMAX	Indicates the largest naturally aligned range supported by the hardware is 2^12 bytes.
15-12	ACTION	Setting this field to select what happens when this trigger matches. 0:Raise a breakpoint exception 1:Enter Debug Mode. (Only supported when DMODE is 1.)
11	CHAIN	Setting this field to enable trigger chain. 0:When this trigger matches, the configured action is taken. 1:While this trigger does not match, it prevents the trigger with the next index from matching. If Number of Triggers is 2, this field is hardwired to 0 on trigger 1 (tselect = 1). If Number of Triggers is 4, this field is hardwired to 0 on trigger 3 (tselect = 3). If Number of Triggers is 8, this field is hardwired to 0 on trigger 3 and trigger 7 (tselect = 3 or 7).
10-7	MATCH	Setting this field to select the matching scheme. 0:Matches when the value equals tdata2. 1:Matches when the top M bits of the value match the top M bits of tdata2. M is 31 minus the index of the least-significant bit containing 0 in tdata2. 2:Matches when the value is greater than (unsigned) or equal to tdata2. 3:Matches when the value is less than (unsigned) tdata2
6	M	Setting this field to enable this trigger in M-mode.
3	U	Setting this field to enable this trigger in U-mode.
2	EXECUTE	Setting this field to enable this trigger to compare virtual address of an instruction.
1	STORE	Setting this field to enable this trigger to compare virtual address of a store.

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Bits	Name	Description
0	LOAD	Setting this field to enable this trigger to compare virtual address of a load.

MCONTROL Bits

3.2.32 ICOUNT (0x7A1) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TYPE				DMODE	RSVD																COUNT	M	RSVD		U	ACTION						
RW				RW	N/A																RO	RW	N/A		RW	RW						
0	0	1	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	1	0	x	x	0	0	0	0	0	0	0

ICOUNT [31:0]

Bits	Name	Description
31-28	TYPE	The selected trigger is an instruction count trigger.
27	DMODE	Setting this field to indicate the trigger is used by Debug Mode. 0:Both Debug-mode and M-mode can write the currently selected trigger registers. 1:Only Debug Mode can write the currently selected trigger registers. Writes from M-mode is ignored.
10	COUNT	This field is hardwired to 1 for single-stepping support
9	M	Setting this field to enable this trigger in M-mode.
6	U	Setting this field to enable this trigger in U-mode.
5-0	ACTION	Setting this field to select what happens when this trigger matches. 0:Raise a breakpoint exception 1:Enter Debug Mode. (Only supported when DMODE is 1.)

ICOUNT Bits

3.2.33 ITRIGGER (0x7A1) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TYPE				DMODE	RSVD																	M	RSVD		U	ACTION						
RW				RW	N/A																	RW	N/A		RW	RW						
0	0	1	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	0	0	0	0	0	0	0

ITRIGGER [31:0]

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Bits	Name	Description
31-28	TYPE	The selected trigger is an interrupt trigger.
27	DMODE	Setting this field to indicate the trigger is used by Debug Mode. 0:Both Debug-mode and M-mode can write the currently selected trigger registers. 1:Only Debug Mode can write the currently selected trigger registers. Writes from M-mode is ignored.
9	M	Setting this field to enable this trigger in M-mode.
6	U	Setting this field to enable this trigger in U-mode.
5-0	ACTION	Setting this field to select what happens when this trigger matches. 0:Raise a breakpoint exception. 1:Enter Debug Mode. (Only supported when DMODE is 1.)

ITRIGGER Bits

3.2.34 ETRIGGER (0x7A1) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TYPE				DMODE	RSVD																NMI	M	RSVD		U	ACTION						
RW				RW	N/A																RW	RW	N/A		RW	RW						
0	0	1	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	x	x	0	0	0	0	0	0	0

ETRIGGER [31:0]

Bits	Name	Description
31-28	TYPE	The selected trigger is an exception trigger.
27	DMODE	Setting this field to indicate the trigger is used by Debug Mode. 0:Both Debug-mode and M-mode can write the currently selected trigger registers. 1:Only Debug Mode can write the currently selected trigger registers. Writes from M-mode is ignored.
10	NMI	Setting this field to enable this trigger in non-maskable interrupts, regardless of the values of s, u, and m.
9	M	Setting this field to enable this trigger in M-mode.
6	U	Setting this field to enable this trigger in U-mode.
5-0	ACTION	Setting this field to select what happens when this trigger matches. 0:Raise a breakpoint exception 1:Enter Debug Mode. (Only supported when DMODE is 1.)

ETRIGGER Bits

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3.2.35 TDATA2 (0x7A2) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TDATA2 [31:0]

Bits	Name	Description
31-0	DATA	This register provides accesses to the tdata2 register of the currently selected trigger registers selected by the tselect register, and it holds trigger-specific data..

TDATA2 Bits

3.2.36 TDATA3 (0x7A3) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TDATA3 [31:0]

Bits	Name	Description
31-0	DATA	This register provides accesses to the tdata3 register of the currently selected trigger registers selected by the tselect register, and it holds trigger-specific data..

TDATA3 Bits

3.2.37 TEXTRA (0x7A3) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MVALUE						MSELECT	RSVD														SVALUE									SSELECT	
RW						RW	N/A														RW									RW	
0	0	0	0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0

TEXTRA [31:0]

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Bits	Name	Description
31-26	MVALUE	Data used together with MSELECT.
25	MSELECT	0:Ignore MVALUE. 1:This trigger will only match if the lower bits of mcontext equal MVALUE.
10-2	SVALUE	Data used together with SSELECT.
1-0	SSELECT	0:Ignore MVALUE 1:This trigger will only match if the lower bits of scontext equal SVALUE 2This trigger will only match if satp.ASID equals SVALUE.

TEXTRA Bits

3.2.38 TINFO (0x7A4) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																INFO															
N/A																RO															
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*

TINFO [31:0]

Bits	Name	Description
15-0	INFO	One bit for each possible type in tdata1. Bit N corresponds to type N. If the bit is set, then that type is supported by the currently selected trigger. If the currently selected trigger does not exist, this field contains 1. 0:When this bit is set, there is no trigger at this tselect 1:Reserved and hardwired to 0. 2：When this bit is set, the selected trigger supports type of address/data match trigger 3:When this bit is set, the selected trigger supports type of instruction count trigger. 4:When this bit is set, the selected trigger supports type of interrupt trigger 5:When this bit is set, the selected trigger supports type of exception trigger 15:When this bit is set, the selected trigger exists (so enumeration shouldn’t terminate), but is not currently available. Others:Reserved for future use.

TINFO Bits

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3.2.39 TCONTROL (0x7A5) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																								MPTE	RSVD			MTE	RSVD		
N/A																								RW	N/A			RW	N/A		
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	0	x	x	x

TCONTROL [31:0]

Bits	Name	Description
7	MPTE	M-mode previous trigger enable field. When a trap into M-mode is taken, MPTE is set to the value of MTE.
3	MTE	M-mode trigger enable field. When a trap into M-mode is taken, MTE is set to 0. When the MRET instruction is executed, MTE is set to the value of MPTE. 0:Triggers do not match/fire while the hart is in M-mode. 1:Triggers do match/fire while the hart is in M-mode.

TCONTROL Bits

3.2.40 MCONTEXT (0x7A8) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																										MCONTEXT					
N/A																										RW					
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0

MCONTEXT [31:0]

Bits	Name	Description
5-0	MCONTEXT	Machine mode software can write a context number to this register, which can be used to set triggers that only fire in that specific context.

MCONTEXT Bits

3.2.41 SCONTEXT (0x7AA) (standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																											SCONTEXT					

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
N/A																							RW								
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0

SCONTEXT [31:0]

Bits	Name	Description
8-0	SCONTEXT	Machine mode software can write a context number to this register, which can be used to set triggers that only fire in that specific context.

SCONTEXT Bits

3.2.42 DCSR (0x7B0) (debug-mode-only)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
XDEBUGVER				RSVD												EBREAKM	RSVD		EBREAKU	STEPIE	STOPCOUNT	STOPTIME	CAUSE			RSVD	MPRVEN	NMIP	STEP	PRV	
RO				N/A												RW	N/A		RW	RW	RW	RW	RO			N/A	RW	RO	RW	RW	
0	1	0	0	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	0	0	1	1	0	0	0	x	0	0	0	1	1

DCSR [31:0]

Bits	Name	Description
31-28	XDEBUGVER	Version of the external debugger. 0 indicates that no external debugger exists and 4 indicates that the external debugger conforms to the RISC-V External Debug Support (TD003) V0.13
15	EBREAKM	This bit controls the behavior of EBREAK instructions in Machine Mode 0:Generate a regular breakpoint exception 1:Enter Debug Mode
12	EBREAKU	This bit controls the behavior of EBREAK instructions in User/Application Mode 0:Generate a regular breakpoint exception 1:Enter Debug Mode
11	STEPIE	This bit controls whether interrupts are enabled during single stepping 0:Disable interrupts during single stepping 1:Allow interrupts in single stepping

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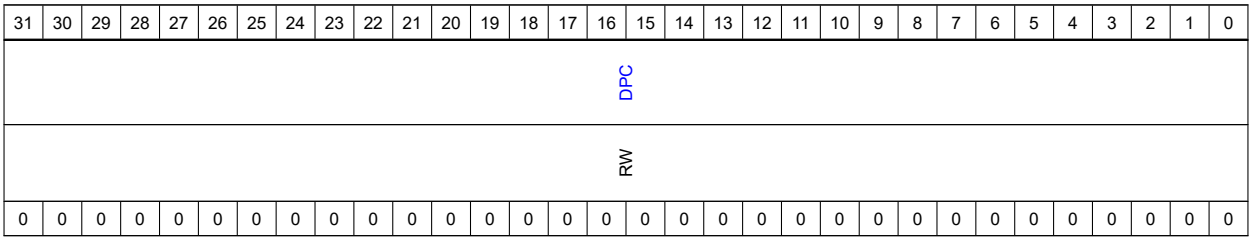
Bits	Name	Description
10	STOPCOUNT	This bit controls whether performance counters are stopped in Debug Mode. 0:Do not stop counters in Debug Mode 1:Stop counters in Debug Mode
9	STOPTIME	This bit controls whether timers are stopped in Debug Mode. The processor only drives its stoptime output pin to 1 if it is in Debug Mode and this bit is set. Integration effort is required to make timers in the platform observe this pin to really stop them. 0:Do not stop timers in Debug Mode 1:Stop timers in Debug Mode
8-6	CAUSE	Reason why Debug Mode was entered. When there are multiple reasons to enter Debug Mode, the priority to determine the CAUSE value will be: trigger module > EBREAK > halt-on-reset > halt request > single step. Halt requests are requests issued by the external debugger 0:Reserved 1:EBREAK 2:Trigger module 3:Halt request 4:Single step 5:Halt-on-reset 6-7:Reserved
4	MPRVEN	This bit controls whether mstatus.MPRV takes effect in Debug Mode. 0:MPRV in mstatus is ignored in Debug Mode. 1:MPRV in mstatus takes effect in Debug Mode.
3	NMIP	When this bit is set, there is a Non-Maskable-Interrupt (NMI) pending for the hart. Since an NMI can indicate a hardware error condition, reliable debugging may no longer be possible once this bit becomes set.
2	STEP	This bit controls whether non-Debug Mode instruction execution is in the single step mode. When set, the hart returns to Debug Mode after a single instruction execution. If the instruction does not complete due to an exception, the hart will immediately enter Debug Mode before executing the trap handler, with appropriate exception registers set. 0:Single Step Mode is off 1:Single Step Mode is on

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Bits	Name	Description
1-0	PRV	The privilege level that the hart was operating in when Debug Mode was entered. The external debugger can modify this value to change the hart’s privilege level when exiting Debug Mode. 0:User/Application 1:Supervisor 2:Reserved 3:Machine

DCSR Bits

3.2.43 DPC (0x7B1) (debug-mode-only)

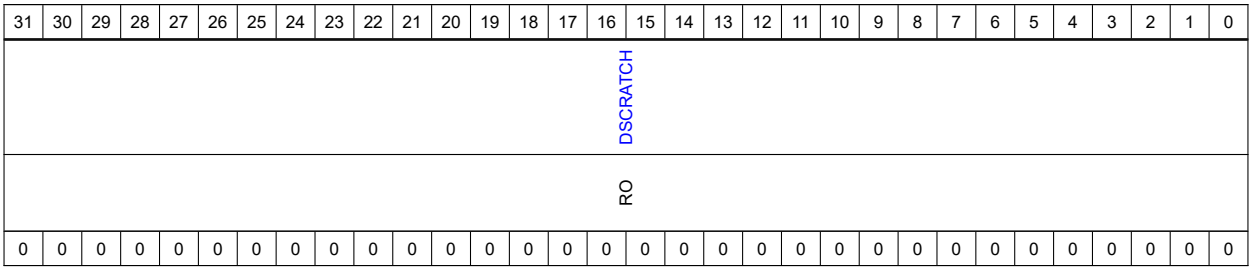


DPC [31:0]

Bits	Name	Description
31-0	DPC	Debug Program Counter. Bit 0 is hardwired to 0.

DPC Bits

3.2.44 DSCRATCH0 (0x7B2) (debug-mode-only)



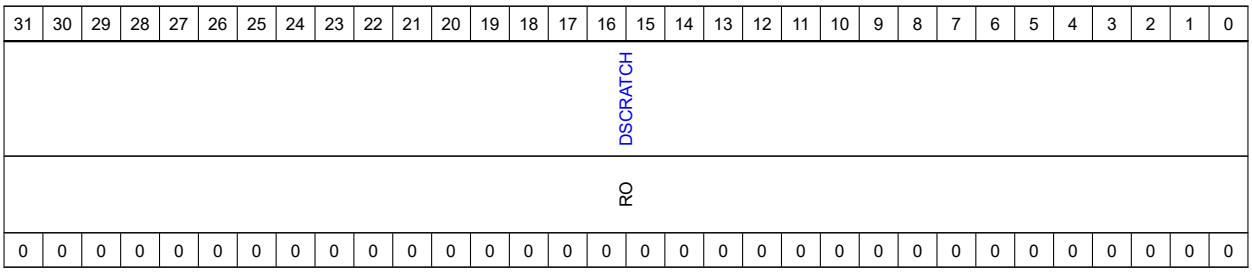
DSCRATCH0 [31:0]

Bits	Name	Description
31-0	DSCRATCH	A scratch register that is reserved for use by Debug Module.

DSCRATCH0 Bits

3.2.45 DSCRATCH1 (0x7B3) (debug-mode-only)

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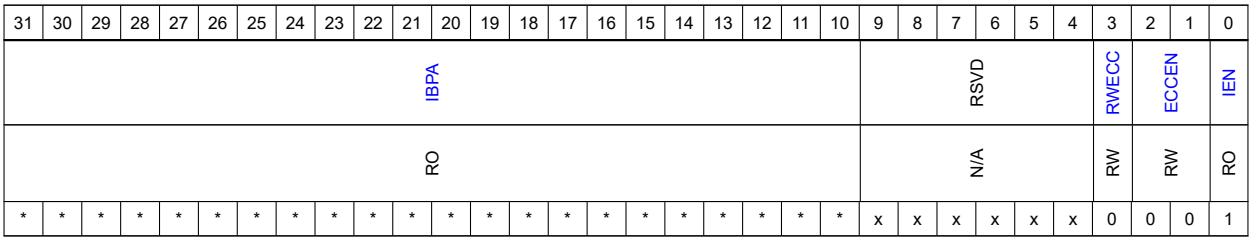


DSCRATCH1 [31:0]

Bits	Name	Description
31-0	DSCRATCH	A scratch register that is reserved for use by Debug Module.

DSCRATCH1 Bits

3.2.46 MILMB (0x7C0) (non-standard read/write)



MILMB [31:0]

Bits	Name	Description
31-10	IBPA	The base physical address of ILM. It has to be an integer multiple of the ILM size
3	RWECC	Controls diagnostic accesses of ECC codes of the ILM RAMs. When set, load/store to ILM reads/writes ECC codes to the mecc_code register. This bit can be set for injecting ECC errors to test the ECC handler. 0:Disable diagnostic accesses of ECC codes 1:Enable diagnostic accesses of ECC codes
2-1	ECCEN	Parity/ECC enable control: 0:Disable parity/ECC 1:Reserved 2:Generate exceptions only on uncorrectable parity/ECC errors 3:Generate exceptions on any type of parity/ECC errors
0	IEN	ILM enable control: 0:ILM is disabled 1:ILM is enabled

MILMB Bits

3.2.47 MDLMB (0x7C1) (non-standard read/write)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DBPA																						RSVD						RWECC	ECCEN		DEN
RO																						N/A						RW	RW		RO
*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	x	x	x	x	x	x	0	0	0	1

MDLMB [31:0]

Bits	Name	Description
31-10	DBPA	The base physical address of DLM. It has to be an integer multiple of the DLM size
3	RWECC	Controls diagnostic accesses of ECC codes of the DLM RAMs. When set, load/store to DLM reads/writes ECC codes to the mecc_code register. This bit can be set for injecting ECC errors to test the ECC handler. 0:Disable diagnostic accesses of ECC codes 1:Enable diagnostic accesses of ECC codes
2-1	ECCEN	Parity/ECC enable control: 0:Disable parity/ECC 1:Reserved 2:Generate exceptions only on uncorrectable parity/ECC errors 3:Generate exceptions on any type of parity/ECC errors
0	DEN	DLM enable control: 0:DLM is disabled 1:DLM is enabled

MDLMB Bits

3.2.48 MECC_CODE (0x7C2) (non-standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD									INSN	RAMID				P	C	RSVD									CODE							
N/A									RO	RO				RO	RO	N/A									RW							
x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	1

MECC_CODE [31:0]

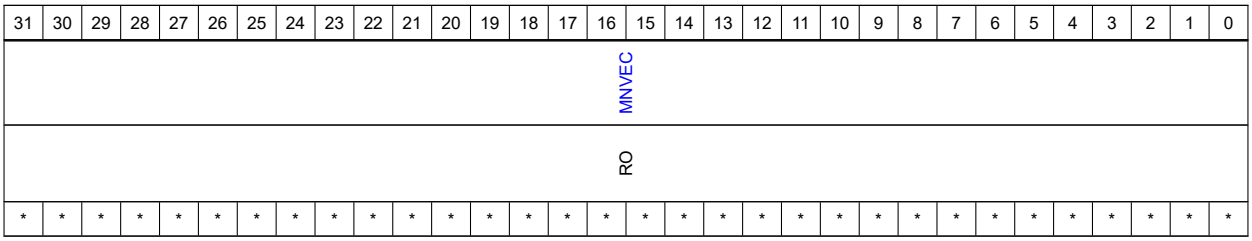
Bits	Name	Description
22	INSN	Indicates if the parity/ECC error is caused by instruction fetch or data access. 0:Data access 1:Instruction fetch

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Bits	Name	Description
21-18	RAMID	The ID of RAM that caused parity/ECC errors. This bit is updated on parity/ECC error exceptions. 0-1:Reserved 2:Tag RAM of I-Cache 3:Data RAM of I-Cache 4:Tag RAM of D-Cache 5:Data RAM of D-Cache 6:Tag RAM of TLB 7:Data RAM of TLB 8:ILM 9:DLM 10-15:Reserved
17	P	Precise error. This bit is updated on parity/ECC error exceptions. 0:Imprecise error 1:Precise error
16	C	Correctable error. This bit is updated on parity/ECC error exceptions. 0:Uncorrectable error 1:Correctable error
6-0	CODE	This field records the ECC value on ECC error exceptions. This field is also used to read/write the ECC codes when diagnostic access of ECC codes are enabled (milmb.RWECC or mdlmb.RWECC is 1).

MECC_CODE Bits

3.2.49 MNVEC (0x7C3) (non-standard read/write)



MNVEC [31:0]

Bits	Name	Description
31-0	MNVEC	Base address of the NMI handler. Its value is the zero-extended value of the reset_vector.

MNVEC Bits

3.2.50 MXSTATUS (0x7C4) (non-standard read/write)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																										PDME	DME	PIME	IME	PPFT_EN	PFT_EN
N/A																										RW	RW	RW	RW	RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0

MXSTATUS [31:0]

Bits	Name	Description
5	PDME	For saving previous DME state on entering a trap. This field is hardwired to 0 if data cache and data local memory are not supported.
4	DME	Data Machine Error flag. It indicates an exception occurred at the data cache or data local memory (DLM). Load/store accesses will bypass D-Cache when this bit is set. The exception handler should clear this bit after the machine error has been dealt with.
3	PIME	When mcause is imprecise exception (in the form of an interrupt), the PM field records the privileged mode of the instruction that caused the imprecise exception. The PM field encoding is defined as follows: 0: User mode 1: Supervisor mode 2: Reserved 3: Machine mode
2	IME	When mcause is imprecise exception (in the form of an interrupt), the PM field records the privileged mode of the instruction that caused the imprecise exception. The PM field encoding is defined as follows: 0: User mode 1: Supervisor mode 2: Reserved 3: Machine mode
1	PPFT_EN	When mcause is imprecise exception (in the form of an interrupt), the PM field records the privileged mode of the instruction that caused the imprecise exception. The PM field encoding is defined as follows: 0: User mode 1: Supervisor mode 2: Reserved 3: Machine mode

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Bits	Name	Description
0	PFT_EN	Enable performance throttling. When throttling is enabled, the processor executes instructions at the performance level specified in mpft_ctl.T_LEVEL. On entering a trap: PPFT_EN <= PFT_EN; PFT_EN <= mpft_ctl.FAST_INT ? 0 :PFT_EN; On executing an MRET instruction: PFT_EN <= PPFT_EN; This field is hardwired to 0 if the PowerBrake feature is not supported.

MXSTATUS Bits

3.2.51 MPFT_CTL (0x7C5) (non-standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																							FAST_INT	T_LEVEL				RSVD			
N/A																							RW	RW				N/A			
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	x	x	x	x

MPFT_CTL [31:0]

Bits	Name	Description
8	FAST_INT	Fast interrupt response. If this field is set, mxstatus.PFT_EN will be automatically cleared when the processor enters an interrupt handler.
7-4	T_LEVEL	Throttling Level. The processor has the highest performance at throttling level 0 and the lowest performance at throttling level 15. 0:Level 0 (the highest performance) 1-14:Level 1-14 15:Level 15 (the lowest performance)

MPFT_CTL Bits

3.2.52 MHSP_CTL (0x7C6) (non-standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																										M	S	U	SCHM	UDF_EN	OVF_EN
N/A																										RW	RW	RW	RW	RW	RW

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0

MHSP_CTL [31:0]

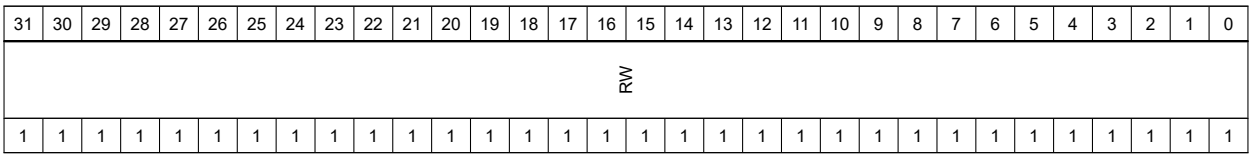
Bits	Name	Description
5	M	Enables the SP protection and recording mechanism in Machine mode 0:The mechanism is disabled in Machine mode. 1: The mechanism is enabled in Machine mode.
4	S	Enables the SP protection and recording mechanism in Supervisor mode 0:The mechanism is disabled in Supervisor mode 1:The mechanism is enabled in Supervisor mode
3	U	Enables the SP protection and recording mechanism in User mode 0:The mechanism is disabled in User mode 1:The mechanism is enabled in User mode.
2	SCHM	Selects the operating scheme of the stack protection and recording mechanism 0:Stack overflow/underflow detection 1:Top-of-stack recording
1	UDF_EN	Enable bit for the stack underflow protection mechanism. This bit will be cleared to 0 automatically by hardware when a stack protection (overflow or underflow) exception is taken. 0:The stack underflow protection is disabled 1:The stack underflow protection is enabled.
0	OVF_EN	Enable bit for the stack overflow protection and recording mechanism. This bit will be cleared to 0 automatically by hardware when a stack protection (overflow or underflow) exception is taken. 0:The stack overflow protection and recording mechanism are disabled. 1:The stack overflow protection and recording mechanism are enabled.

MHSP_CTL Bits

3.2.53 MSP_BOUND (0x7C7) (non-standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MSP_BOUND																															

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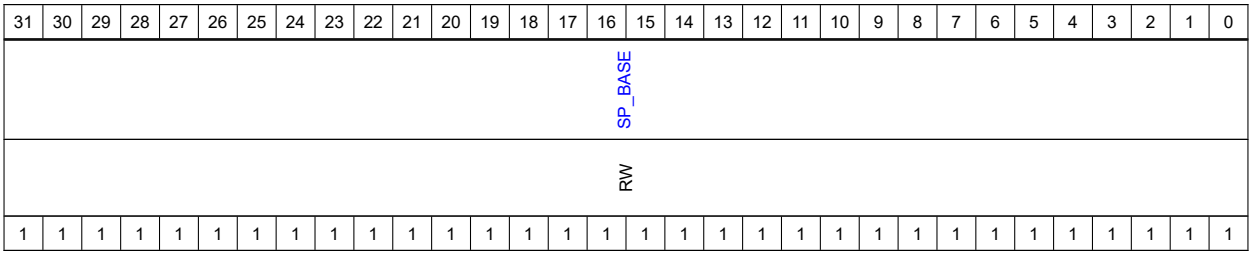


MSP_BOUND [31:0]

Bits	Name	Description
31-0	MSP_BOUND	Machine SP Bound

MSP_BOUND Bits

3.2.54 MSP_BASE (0x7C8) (non-standard read/write)

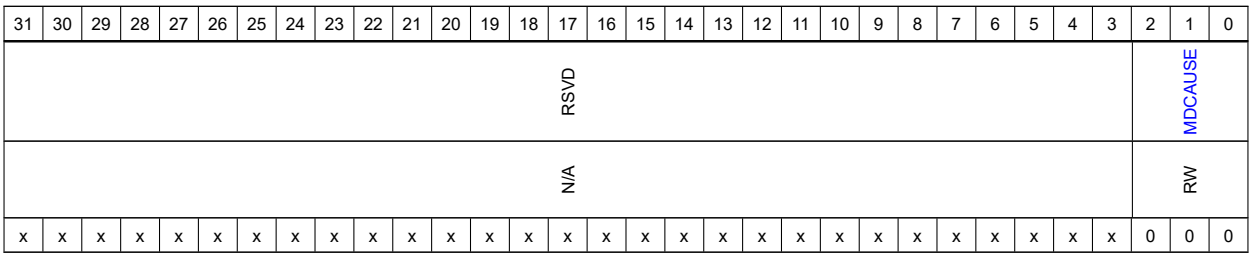


MSP_BASE [31:0]

Bits	Name	Description
31-0	SP_BASE	Machine SP base

MSP_BASE Bits

3.2.55 MDCAUSE (0x7C9) (non-standard read/write)



MDCAUSE [31:0]

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Bits	Name	Description
2-0	MDCAUSE	This register further disambiguates causes of traps recorded in the mcause register. The value of MDCAUSE for precise exception: When mcause == 1 (Instruction access fault): 0:Reserved; 1:ECC/Parity error; 2:PMP instruction access violation; 3:Bus error; 4:PMA empty hole access When mcause == 2 (Illegal instruction): 0:Please parse the mtval CSR; 1:FP disabled exception; 2:ACE disabled exception When mcause == 5 (Load access fault): 0:Reserved; 1:ECC/Parity error; 2:PMP load access violation; 3:Bus error; 4:Misaligned address; 5:PMA empty hole access; 6:PMA attribute inconsistency; 7:PMA NAMO exception When mcause == 7 (Store access fault): 0:Reserved; 1:ECC/Parity error; 2:PMP load access violation; 3:Bus error; 4:Misaligned address; 5:PMA empty hole access; 6:PMA attribute inconsistency; 7:PMA NAMO exception The value of MDCAUSE for imprecise exception: When mcause == Local Interrupt 16 or Local Interrupt 272 (16 + 256) (ECC error local interrupt) 0:Reserved; 1:LM slave port ECC/Parity error; 2:Imprecise store ECC/Parity error; 3:Imprecise load ECC/Parity error When mcause == Local Interrupt 17 or Local Interrupt 273 (17 + 256) (Bus read/write transaction error local interrupt) 0:Reserved; 1:Bus read error; 2:Bus write error; 3:PMP error caused by load instructions; 4:PMP error caused by store instructions; 5:PMA error caused by load instructions; 6:PMA error caused by store instructions

MDCAUSE Bits

3.2.56 MCACHE_CTL (0x7CA) (non-standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																				IC_FIRST_WORD	RSVD		CCTL_SUEN	DC_RWECC	IC_RWECC	DC_ECCEN		IC_ECCEN		DC_EN	IC_EN
N/A																				RO	N/A		RW	RW	RW	RW		RW		RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	1	x	x	0	0	0	0	0	0	0	0	0

MCACHE_CTL [31:0]

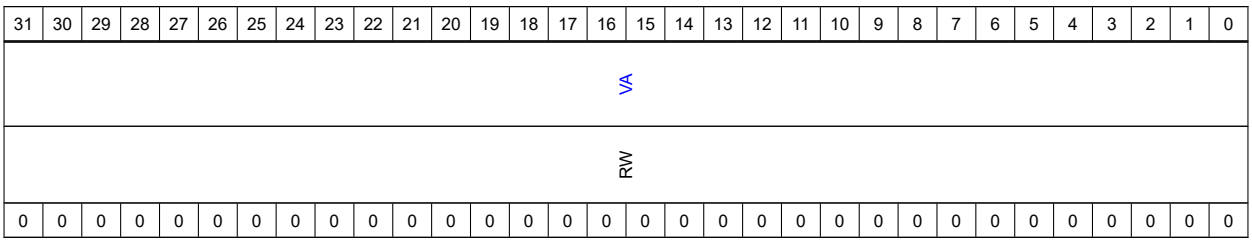
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Bits	Name	Description
11	IC_FIRST_WORD	Cache miss allocation filling policy 0:Cache line data is returned critical (double) word first 1:Cache line data is returned the lowest address (double) word first
8	CCTL_SUEN	Enable bit for Superuser-mode and User-mode software to access ucctlbeginaddr and ucctlcommand CSRs 0:Disable ucctlbeginaddr and ucctlcommand accesses in S/U mode 1:Enable ucctlbeginaddr and ucctlcommand accesses in S/U mode
7	DC_RWECC	Controls diagnostic accesses of ECC codes of the data cache RAMs. It is set to enable CCTL operations to access the ECC codes. This bit can be set for injecting ECC errors to test the ECC handler 0:Disable diagnostic accesses of ECC codes 1:Enable diagnostic accesses of ECC codes
6	IC_RWECC	Controls diagnostic accesses of ECC codes of the instruction cache RAMs. It is set to enable CCTL operations to access the ECC codes . This bit can be set for injecting ECC errors to test the ECC handler. 0:Disable diagnostic accesses of ECC codes 1:Enable diagnostic accesses of ECC codes
5-4	DC_ECCEN	Parity/ECC error checking enable control for the data cache. 0:Disable parity/ECC 1:Reserved 2:Generate exceptions only on uncorrectable parity/ECC errors 3:Generate exceptions on any type of parity/ECC errors
3-2	IC_ECCEN	Parity/ECC error checking enable control for the instruction cache 0:Disable parity/ECC 1:Reserved 2:Generate exceptions only on uncorrectable parity/ECC errors 3:Generate exceptions on any type of parity/ECC errors
1	DC_EN	Controls if the data cache is enabled or not. 0:D-Cache is disabled 1:D-Cache is enabled
0	IC_EN	Controls if the instruction cache is enabled or not. 0:I-Cache is disabled 1:I-Cache is enabled

MCACHE_CTL Bits

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3.2.57 MCCTLBEGINADDR (0x7CB) (non-standard read/write)

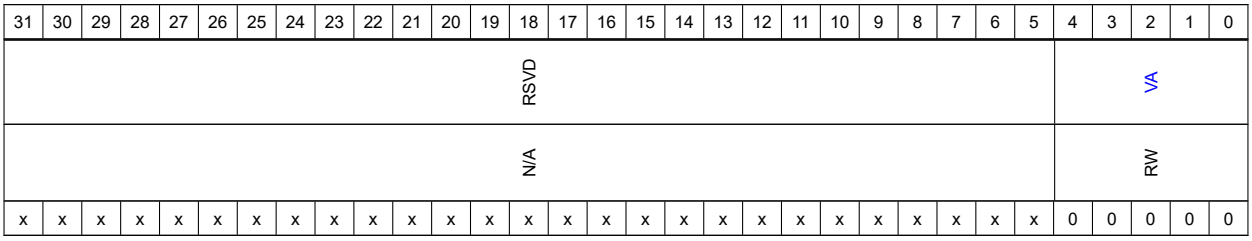


MCCTLBEGINADDR [31:0]

Bits	Name	Description
31-0	VA	This register holds the address information required by CCTL operations

MCCTLBEGINADDR Bits

3.2.58 MCCTLCOMMAND (0x7CC) (non-standard read/write)



MCCTLCOMMAND [31:0]

Bits	Name	Description
4-0	VA	See CCTL Command Definition Table

MCCTLCOMMAND Bits

值 (DEC)	值 (BIN)	命令	类型
0	0b00_000	L1D_VA_INVAL	VA
1	0b00_001	L1D_VA_WB	VA
2	0b00_010	L1D_VA_WBINVAL	VA
3	0b00_011	L1D_VA_LOCK	VA
4	0b00_100	L1D_VA_UNLOCK	VA
6	0b00_110	L1D_WBINVAL_ALL	-
7	0b00_111	L1D_WB_ALL	-
8	0b01_000	L1I_VA_INVAL	VA
11	0b01_011	L1I_VA_LOCK	VA
12	0b01_100	L1I_VA_UNLOCK	VA

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值 (DEC)	值 (BIN)	命令	类型
16	0b10_000	L1D_IX_INVAL	Index
17	0b10_001	L1D_IX_WB	Index
18	0b10_010	L1D_IX_WBINVAL	Index
19	0b10_011	L1D_IX_RTAG	Index
20	0b10_100	L1D_IX_RDATA	Index
21	0b10_101	L1D_IX_WTAG	Index
22	0b10_110	L1D_IX_WDATA	Index
23	0b10_111	L1D_INVAL_ALL	-
24	0b11_000	L1I_IX_INVAL	Index
27	0b11_011	L1I_IX_RTAG	Index
28	0b11_100	L1I_IX_RDATA	Index
29	0b11_101	L1I_IX_WTAG	Index
30	0b11_110	L1I_IX_WDATA	Index

Table 7: CCTL Command Definition

3.2.59 MCCTLDATA (0x7CD) (non-standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																											VA				
N/A																											RW				
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0

MCCTLDATA [31:0]

Bits	Name	Description
4-0	VA	See CCTL Commands Which Access mcctldata Table

MCCTLDATA Bits

值 (DEC)	值 (BIN)	命令	类型
3	0b00_011	L1D_VA_LOCK	VA
11	0b01_011	L1I_VA_LOCK	VA
19	0b10_011	L1D_IX_RTAG	Index
20	0b10_100	L1D_IX_RDATA	Index
21	0b10_101	L1D_IX_WTAG	Index
22	0b10_110	L1D_IX_WDATA	Index
27	0b11_011	L1I_IX_RTAG	Index
28	0b11_100	L1I_IX_RDATA	Index
29	0b11_101	L1I_IX_WTAG	Index
30	0b11_110	L1I_IX_WDATA	Index

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值 (DEC)	值 (BIN)	命令	类型

Table 8: CCTL Commands Which Access mcctldata

3.2.60 MCOUNTERWEN (0x7CE) (non-standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																									HPM6	HPM5	HPM4	HPM3	IR	RSVD	CY
N/A																									RW	RW	RW	RW	RW	N/A	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	x	0

MCOUNTERWEN [31:0]

Bits	Name	Description
6	HPM6	See register description
5	HPM5	See register description
4	HPM4	See register description
3	HPM3	See register description
2	IR	See register description
0	CY	See register description

MCOUNTERWEN Bits

3.2.61 MCOUNTERINTEN (0x7CF) (non-standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																									HPM6	HPM5	HPM4	HPM3	IR	RSVD	CY
N/A																									RW	RW	RW	RW	RW	N/A	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	x	0

MCOUNTERINTEN [31:0]

Bits	Name	Description
6	HPM6	See register description
5	HPM5	See register description
4	HPM4	See register description
3	HPM3	See register description
2	IR	See register description
0	CY	See register description

MCOUNTERINTEN Bits

3.2.62 MMISC_CTL (0x7D0) (non-standard read/write)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																									MSA_UNA	RSVD		BRPE	RVCOMPM	VEC_PLIC	RSVD
N/A																									RW	N/A		RW	RW	RW	N/A
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	1	x	x	1	0	0	x

MMISC_CTL [31:0]

Bits	Name	Description
6	MSA_UNA	This field controls whether the load/store instructions can access misaligned memory locations without generating Address Misaligned exceptions. Supported instructions: LW/LH/LHU/SW/SH 0:Misaligned accesses generate Address Misaligned exceptions. 1:Misaligned accesses generate Address Misaligned exceptions.
3	BRPE	Branch prediction enable bit. This bit controls all branch prediction structures. 0:Disabled 1:Enabled This bit is hardwired to 0 if branch prediction structure is not supported.
2	RVCOMPM	RISC-V compatibility mode enable bit. If the compatibility mode is turned on, all specific instructions become reserved instructions 0:Disabled 1:Enabled
1	VEC_PLIC	Selects the operation mode of PLIC: 0:Regular mode 1:Vector mode Please note that both this bit and the vector mode enable bit (VECTORED) of the Feature Enable Register in NCEPLIC100 should be turned on for the vectored interrupt support to work correctly. This bit is hardwired to 0 if the vectored PLIC feature is not supported.

MMISC_CTL Bits

3.2.63 MCOUNTERMASK_M (0x7D1) (non-standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																									HPM6	HPM5	HPM4	HPM3	IR	RSVD	CY

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
N/A																									RW	RW	RW	RW	RW	N/A	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	x	0

MCOUNTERMASK_M [31:0]

Bits	Name	Description
6	HPM6	See register description
5	HPM5	See register description
4	HPM4	See register description
3	HPM3	See register description
2	IR	See register description
0	CY	See register description

MCOUNTERMASK_M Bits

3.2.64 MCOUNTERMASK_S (0x7D2) (non-standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																									HPM6	HPM5	HPM4	HPM3	IR	RSVD	CY
N/A																									RW	RW	RW	RW	RW	N/A	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	x	0

MCOUNTERMASK_S [31:0]

Bits	Name	Description
6	HPM6	See register description
5	HPM5	See register description
4	HPM4	See register description
3	HPM3	See register description
2	IR	See register description
0	CY	See register description

MCOUNTERMASK_S Bits

3.2.65 MCOUNTERMASK_U (0x7D3) (non-standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																									HPM6	HPM5	HPM4	HPM3	IR	RSVD	CY
N/A																									RW	RW	RW	RW	RW	N/A	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	x	0

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

MCOUNTERMASK_U [31:0]

Bits	Name	Description
6	HPM6	See register description
5	HPM5	See register description
4	HPM4	See register description
3	HPM3	See register description
2	IR	See register description
0	CY	See register description

MCOUNTERMASK_U Bits

3.2.66 MCOUNTEROVF (0x7D4) (non-standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																									HPM6	HPM5	HPM4	HPM3	IR	RSVD	CY
N/A																									RW	RW	RW	RW	RW	N/A	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	x	0

MCOUNTEROVF [31:0]

Bits	Name	Description
6	HPM6	See register description
5	HPM5	See register description
4	HPM4	See register description
3	HPM3	See register description
2	IR	See register description
0	CY	See register description

MCOUNTEROVF Bits

3.2.67 DEXC2DBG (0x7E0) (non-standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD												PMOV	RSVD			BWE	SLPECC	ACE	HSP	MEC	RSVD		UEC	SAF	SAM	LAF	LAM	NMI	II	IAF	IAM
N/A												RW	N/A			RW	RW	RW	RW	RW	N/A		RW	RW	RW	RW	RW	RW	RW	RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	0	0	0	0	0	x	x	0	0	0	0	0	0	0	0	0

DEXC2DBG [31:0]

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Bits	Name	Description
19	PMOV	Indicates whether performance counter overflow interrupts are redirected to enter Debug Mode 0:Do not redirect 1:Redirect
15	BWE	Indicates whether Bus-write Transaction Error local interrupts are redirected to enter Debug Mode 0:Do not redirect 1:Redirect
14	SLPECC	Indicates whether local memory slave port ECC Error local interrupts are redirected to enter Debug Mode 0:Do not redirect 1:Redirect
13	ACE	Indicates whether ACE-related exceptions are redirected to enter Debug Mode. This bit is present only when mmsc_cfg.ACE is set 0:Do not redirect 1:Redirect
12	HSP	Indicates whether Stack Protection exceptions are redirected to enter Debug Mode. This bit is present only when mmsc_cfg.HSP is set. 0:Do not redirect 1:Redirect
11	MEC	Indicates whether M-mode Environment Call exceptions are redirected to enter Debug Mode 0:Do not redirect 1:Redirect
8	UEC	Indicates whether U-mode Environment Call exceptions are redirected to enter Debug Mode. 0:Do not redirect 1:Redirect
7	SAF	Indicates whether Store Access Fault exceptions are redirected to enter Debug Mode. 0:Do not redirect 1:Redirect
6	SAM	Indicates whether Store Access Misaligned exceptions are redirected to enter Debug Mode. 0:Do not redirect 1:Redirect
5	LAF	Indicates whether Load Access Fault exceptions are redirected to enter Debug Mode. 0:Do not redirect 1:Redirect

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Bits	Name	Description
4	LAM	Indicates whether Load Access Misaligned exceptions are redirected to enter Debug Mode 0:Do not redirect 1:Redirect
3	NMI	Indicates whether Non-Maskable Interrupt 0:Do not redirect 1:Redirect
2	II	Indicates whether Illegal Instruction exceptions are redirected to enter Debug Mode. 0:Do not redirect 1:Redirect
1	IAF	Indicates whether Instruction Access Fault exceptions are redirected to enter Debug Mode 0:Do not redirect 1:Redirect
0	IAM	Indicates whether Instruction Access Misaligned exceptions are redirected to enter Debug Mode. 0:Do not redirect 1:Redirect

DEXC2DBG Bits

3.2.68 DDCAUSE (0x7E1) (non-standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																SUBTYPE								MAINTYPE							
N/A																RO								RO							
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DDCAUSE [31:0]

Bits	Name	Description
15-8	SUBTYPE	Subtypes for main type. The table below lists the subtypes for DCSR.CAUSE==1 and DDCAUSE.MAINTYPE==3. 0:Illegal instruction 1:Privileged instruction 2:Non-existent CSR 3:Privilege CSR access 4:Read-only CSR update

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Bits	Name	Description
7-0	MAINTYPE	Cause for redirection to Debug Mode. 0:Software Breakpoint (EBREAK) 1:Instruction Access Misaligned (IAM) 2:Instruction Access Fault (IAF) 3:Illegal Instruction (II) 4:Non-Maskable Interrupt (NMI) 5:Load Access Misaligned (LAM) 6:Load Access Fault (LAF) 7:Store Access Misaligned (SAM) 8:Store Access Fault (SAF) 9:U-mode Environment Call (UEC) 10:S-mode Environment Call (SEC) 11:Instruction page fault 12:M-mode Environment Call (MEC) 13:Load page fault 14:Reserved 15:Store/AMO page fault 16:Imprecise ECC error 17;Bus write transaction error 18:Performance Counter overflow 19–31:Reserved 32:Stack overflow exception 33:Stack underflow exception 34:ACE disabled exception 35–39:Reserved 40–47:ACE exception ≥48:Reserved

DDCAUSE Bits

3.2.69 UITB (0x800) (non-standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDR																														RSVD	HW
RW																														N/A	RO
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	0

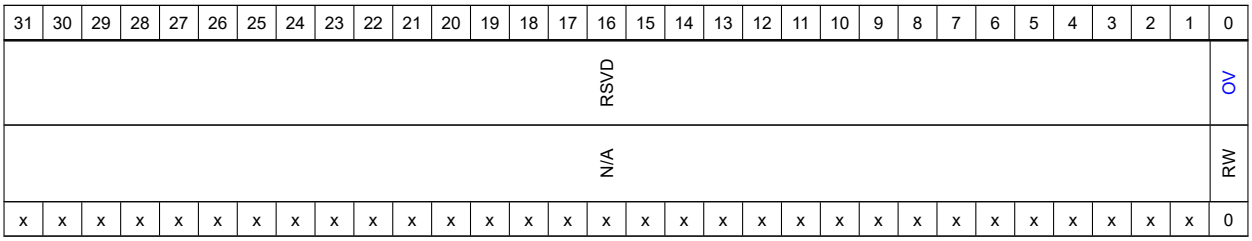
UITB [31:0]

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Bits	Name	Description
31-2	ADDR	The base address of the CoDense instruction table. This field is reserved if uitb.HW == 1.
0	HW	This bit specifies if the CoDense instruction table is hardwired. 0:The instruction table is located in memory. uitb.ADDR should be initialized to point to the table before using the CoDense instructions. 1:The instruction table is hardwired. Initialization of uitb.ADDR is not needed before using the CoDense instructions.

UITB Bits

3.2.70 UCODE (0x801) (non-standard read/write)

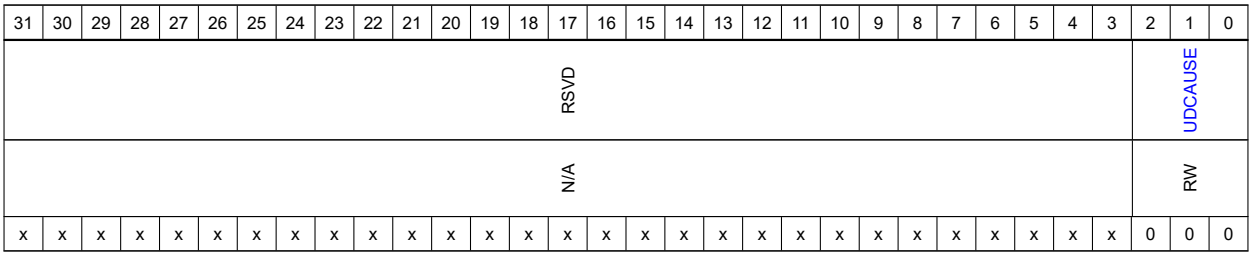


UCODE [31:0]

Bits	Name	Description
0	OV	Overflow flag. It will be set by DSP instructions with a saturated result. 0:A saturated result is not generated 1:A saturated result is generated

UCODE Bits

3.2.71 UDCAUSE (0x809) (non-standard read/write)



UDCAUSE [31:0]

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Bits	Name	Description
2-0	UDCAUSE	This register further disambiguates causes of traps recorded in the ucause register. See the list below for details. The value of UDCAUSE for precise exception: When ucause == 1 (Instruction access fault) 0:Reserved 1:ECC/Parity error 2:PMP instruction access violation 3:Bus error 4:PMA empty hole access When ucause == 2 (Illegal instruction) 0:Please parse the utval CSR 1:FP disabled exception 2:ACE disabled exception When ucause == 5 (Load access fault) 0:Reserved 1:ECC/Parity error 2:PMP load access violation 3:Bus error 4:Misaligned address 5:PMA empty hole access 6:PMA attribute inconsistency 7:PMA NAMO exception When ucause == 7 (Store access fault) 0:Reserved 1:ECC/Parity error 2:PMP store access violation 3:Bus error 4:Misaligned address 5:PMA empty hole access 6:PMA attribute inconsistency 7:PMA NAMO exception

UDCAUSE Bits

3.2.72 UCCTLBEGINADDR (0x80B) (non-standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VA																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
UCCTLBEGINADDR [31:0]																															

Bits	Name	Description
31-0	VA	It is an alias to the mcctlbeginaddr register and it is only accessible to Supervisor-mode and User-mode software when mcache_ctl.CCTL_SUEN is 1. Otherwise illegal instruction exceptions will be triggered.

UCCTLBEGINADDR Bits

3.2.73 UCCTLCOMMAND (0x80C) (non-standard read/write)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																											VA				
N/A																											RW				
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0

UCCTLCOMMAND [31:0]

Bits	Name	Description
4-0	VA	See User CCTL Command Definition Table

UCCTLCOMMAND Bits

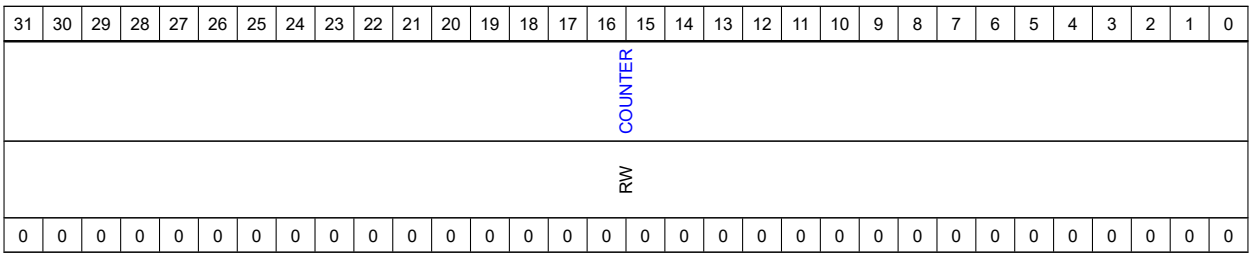
值 (DEC)	值 (BIN)	命令	类型	用户模式允许
0	0b00_000	L1D_VA_INVAL	VA	1
1	0b00_001	L1D_VA_WB	VA	1
2	0b00_010	L1D_VA_WBINVAL	VA	1
3	0b00_011	L1D_VA_LOCK	VA	0
4	0b00_100	L1D_VA_UNLOCK	VA	0
6	0b00_110	L1D_WBINVAL_ALL	-	0
7	0b00_111	L1D_WB_ALL	-	0
8	0b01_000	L1I_VA_INVAL	VA	1
11	0b01_011	L1I_VA_LOCK	VA	0
12	0b01_100	L1I_VA_UNLOCK	VA	0
16	0b10_000	L1D_IX_INVAL	Index	0
17	0b10_001	L1D_IX_WB	Index	0
18	0b10_010	L1D_IX_WBINVAL	Index	0
19	0b10_011	L1D_IX_RTAG	Index	0
20	0b10_100	L1D_IX_RDATA	Index	0
21	0b10_101	L1D_IX_WTAG	Index	0

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值 (DEC)	值 (BIN)	命令	类型	用户模式允许
22	0b10_110	L1D_IX_WDATA	Index	0
23	0b10_111	L1D_INVAL_ALL	-	0
24	0b11_000	L1I_IX_INVAL	Index	0
27	0b11_011	L1I_IX_RTAG	Index	0
28	0b11_100	L1I_IX_RDATA	Index	0
29	0b11_101	L1I_IX_WTAG	Index	0
30	0b11_110	L1I_IX_WDATA	Index	0

Table 9: CCTL Command Definition

3.2.74 MCYCLE (0xB00) (standard read/write)

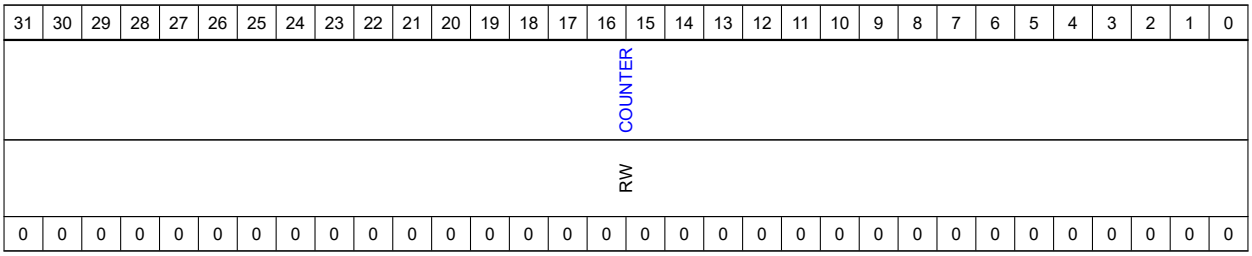


MCYCLE [31:0]

Bits	Name	Description
31-0	COUNTER	the lower 32 bits of Machine Cycle Counter

MCYCLE Bits

3.2.75 MINSTRET (0xB02) (standard read/write)



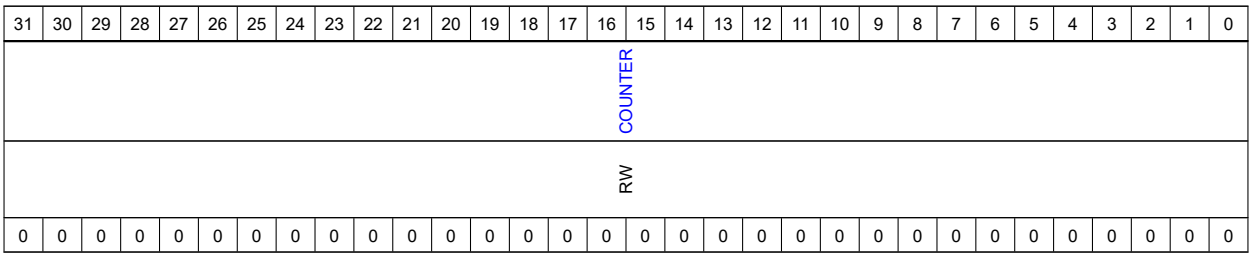
MINSTRET [31:0]

Bits	Name	Description
31-0	COUNTER	the lower 32 bits of Machine Instruction-Retired Counter

MINSTRET Bits

3.2.76 MHPMCOUNTER3 (0xB03) (standard read/write)

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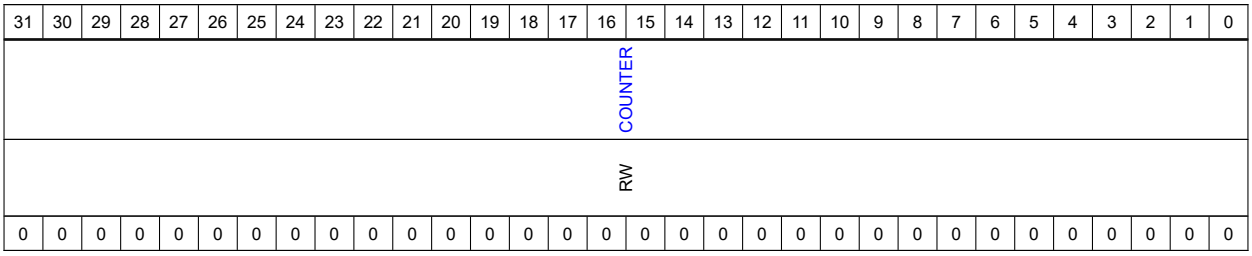


MHPMCOUNTER3 [31:0]

Bits	Name	Description
31-0	COUNTER	count the num- ber of events selected by mhpmevent3

MHPMCOUNTER3 Bits

3.2.77 MHPMCOUNTER4 (0xB04) (standard read/write)

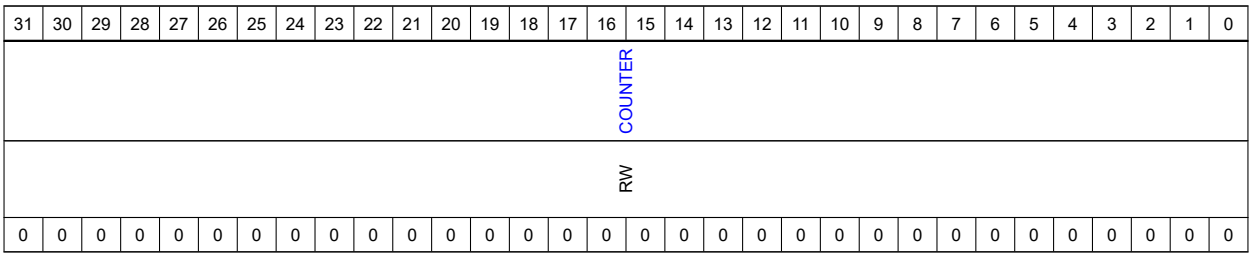


MHPMCOUNTER4 [31:0]

Bits	Name	Description
31-0	COUNTER	count the num- ber of events selected by mhpmevent4

MHPMCOUNTER4 Bits

3.2.78 MHPMCOUNTER5 (0xB05) (standard read/write)



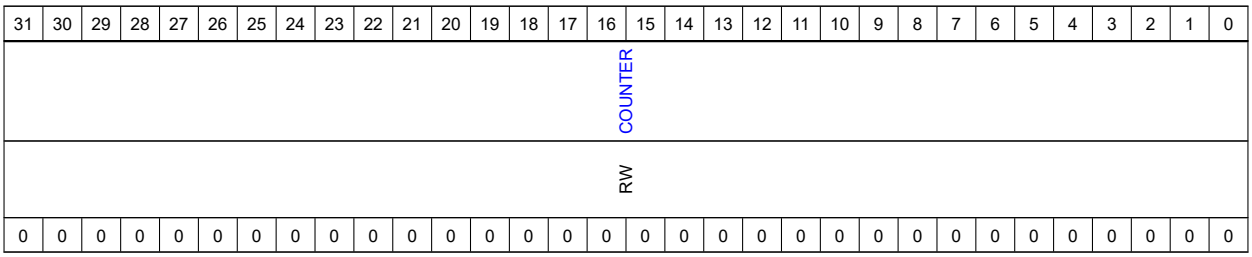
MHPMCOUNTER5 [31:0]

Bits	Name	Description
31-0	COUNTER	count the num- ber of events selected by mhpmevent5

MHPMCOUNTER5 Bits

3.2.79 MHPMCOUNTER6 (0xB06) (standard read/write)

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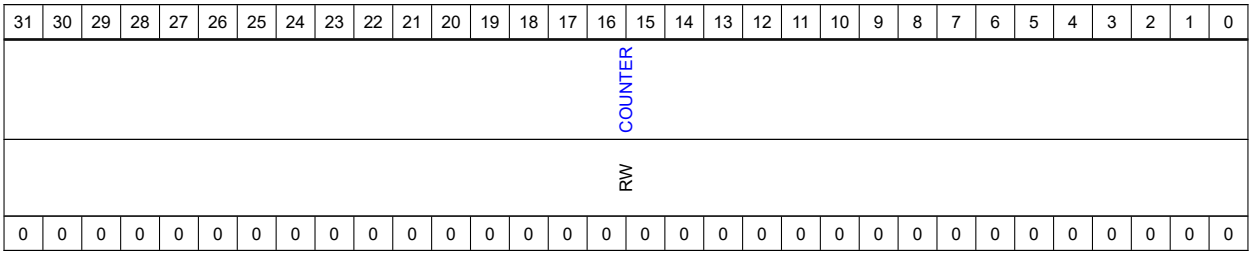


MHPMCOUNTER6 [31:0]

Bits	Name	Description
31-0	COUNTER	count the num- ber of events selected by mhpmevent6

MHPMCOUNTER6 Bits

3.2.80 MCYCLEH (0xB80) (standard read/write)

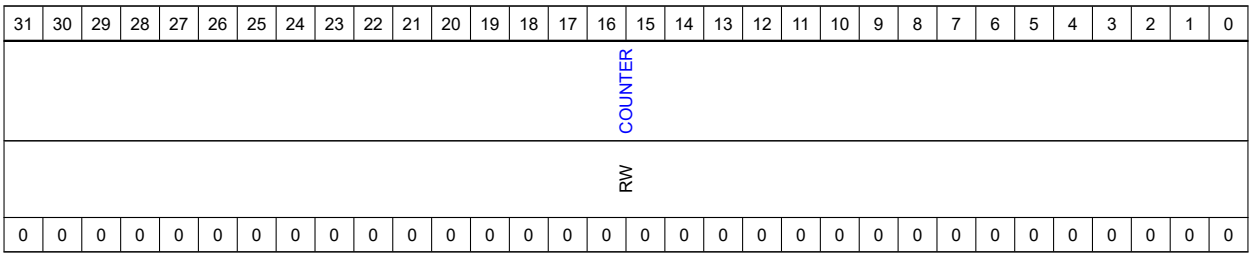


MCYCLEH [31:0]

Bits	Name	Description
31-0	COUNTER	the higher 32 bits of Machine Cycle Counter

MCYCLEH Bits

3.2.81 MINSTRETH (0xB82) (standard read/write)



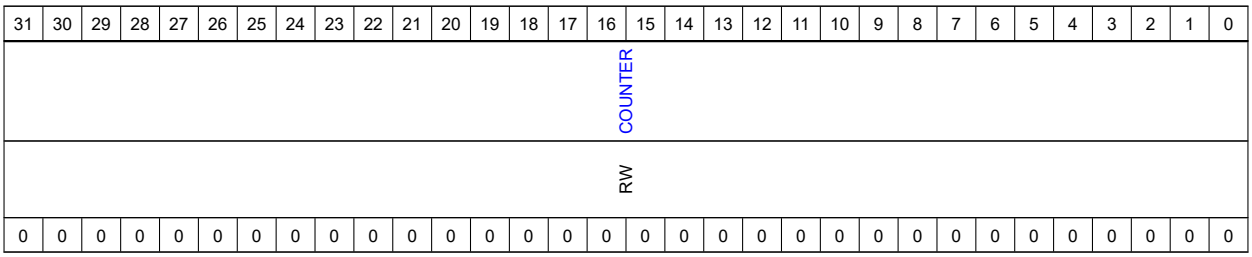
MINSTRETH [31:0]

Bits	Name	Description
31-0	COUNTER	the higher 32 bits of Machine Instruction-Retired Counter

MINSTRETH Bits

3.2.82 MHPMCOUNTER3H (0xB83) (standard read/write)

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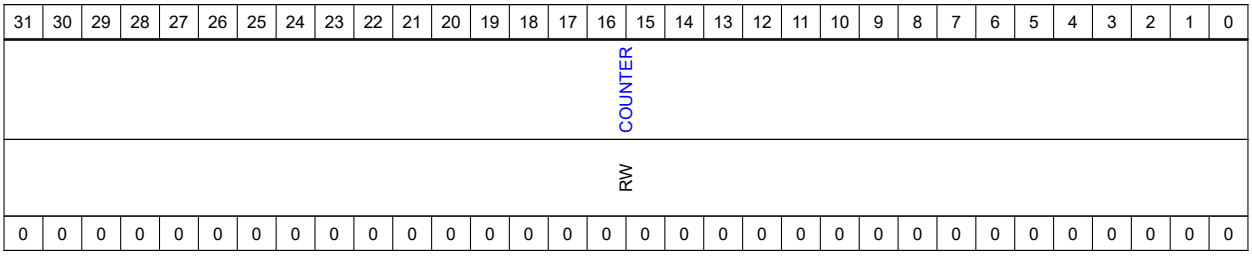


MHPMCOUNTER3H [31:0]

Bits	Name	Description
31-0	COUNTER	count the num- ber of events selected by mhpmevent3

MHPMCOUNTER3H Bits

3.2.83 MHPMCOUNTER4H (0xB84) (standard read/write)

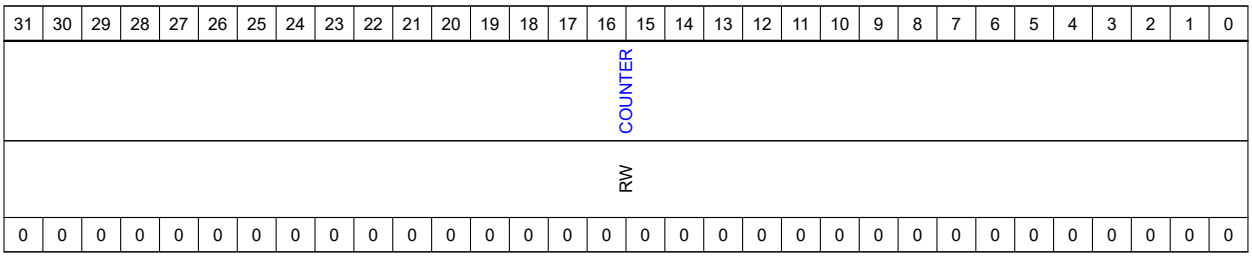


MHPMCOUNTER4H [31:0]

Bits	Name	Description
31-0	COUNTER	count the num- ber of events selected by mhpmevent4

MHPMCOUNTER4H Bits

3.2.84 MHPMCOUNTER5H (0xB85) (standard read/write)



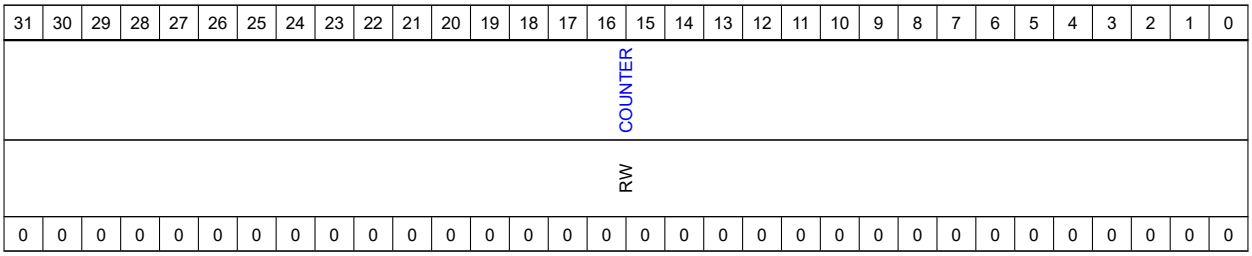
MHPMCOUNTER5H [31:0]

Bits	Name	Description
31-0	COUNTER	count the num- ber of events selected by mhpmevent5

MHPMCOUNTER5H Bits

3.2.85 MHPMCOUNTER6H (0xB86) (standard read/write)

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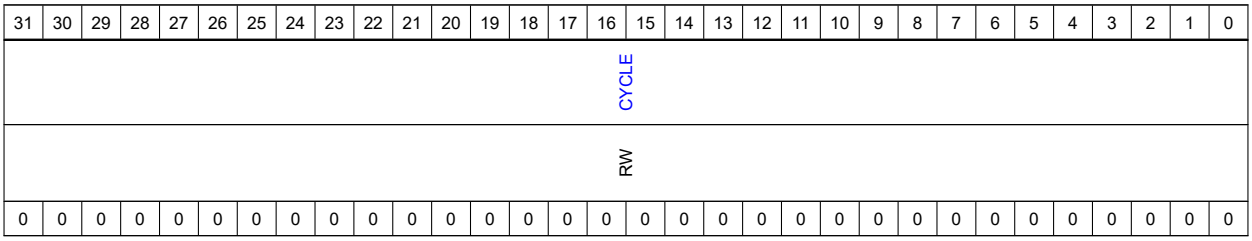


MHPMCOUNTER6H [31:0]

Bits	Name	Description
31-0	COUNTER	count the num- ber of events selected by mhpmevent6

MHPMCOUNTER6H Bits

3.2.86 CYCLE (0xC00) ()

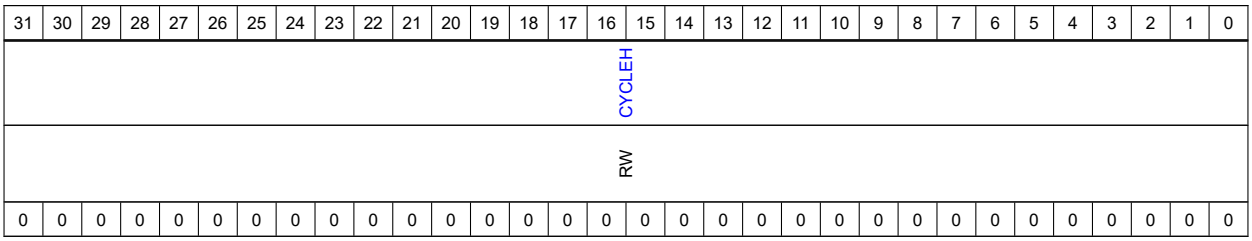


CYCLE [31:0]

Bits	Name	Description
31-0	CYCLE	Cycle Counter

CYCLE Bits

3.2.87 CYCLEH (0xC80) ()



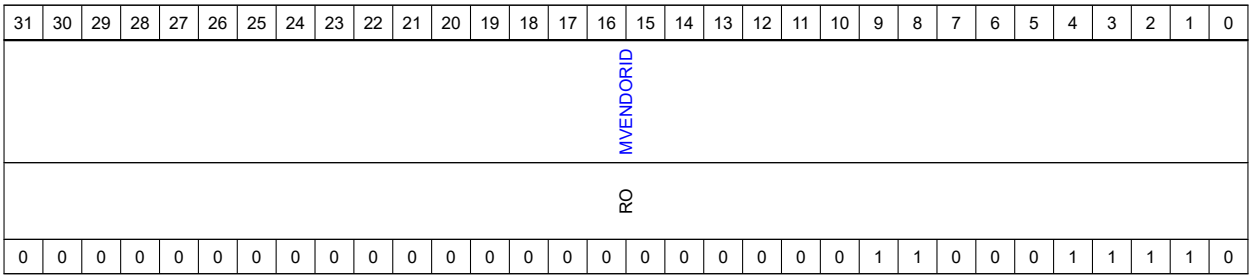
CYCLEH [31:0]

Bits	Name	Description
31-0	CYCLEH	Cycle Counter Higher 32-bit

CYCLEH Bits

3.2.88 MVENDORID (0xF11) (standard read only)

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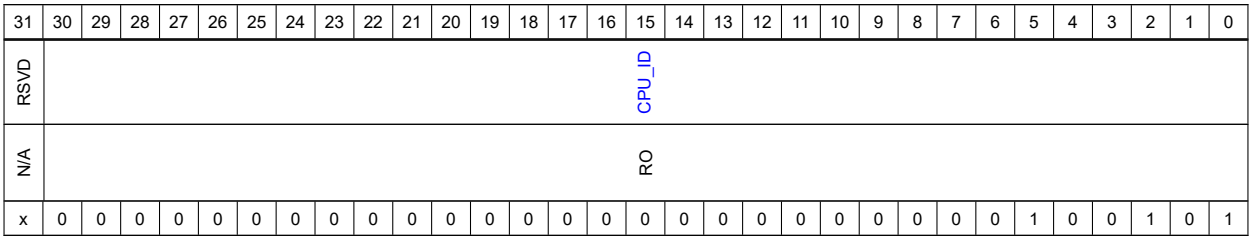


MVENDORID [31:0]

Bits	Name	Description
31-0	MVENDORID	The manufacturer ID

MVENDORID Bits

3.2.89 MARCHID (0xF12) (standard read only)

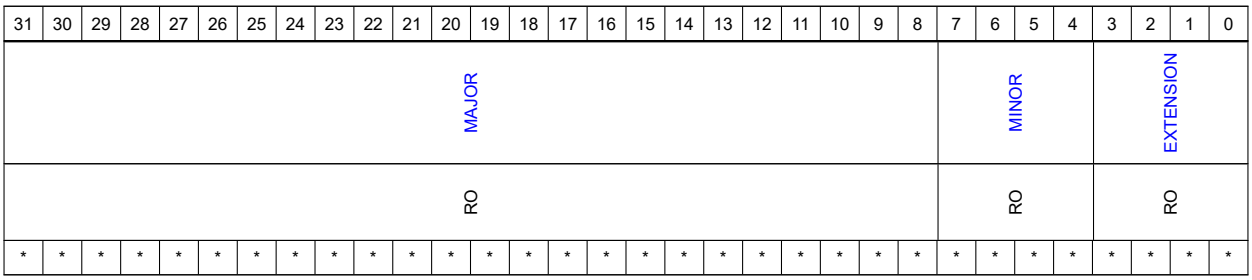


MARCHID [31:0]

Bits	Name	Description
30-0	CPU_ID	CPU ID

MARCHID Bits

3.2.90 MIMPID (0xF13) (standard read only)



MIMPID [31:0]

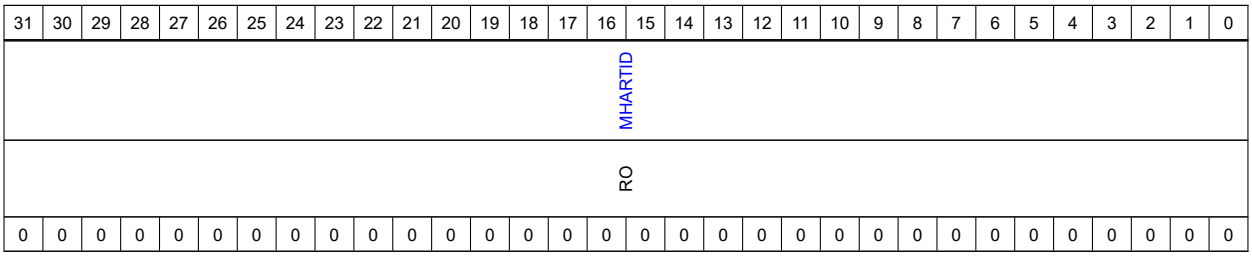
Bits	Name	Description
31-8	MAJOR	Revision major
7-4	MINOR	Revision minor
3-0	EXTENSION	Revision extension

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Bits	Name	Description

MIMPID Bits

3.2.91 MHARTID (0xF14) (standard read only)



MHARTID [31:0]

Bits	Name	Description
31-0	MHARTID	Hart ID

MHARTID Bits

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4 Exceptions and Interrupts of TRAP Processors

This chapter describes the TRAP of RISC-V processors, namely processor exceptions and interrupts.

For all peripherals of this product that support interrupt generation and their interrupt vector tables, please refer to Section 4.4.

4.1 Overview of RISC-V Processor TRAP

This product uses a high-performance RISC-V processor as the central processing unit, which complies with the RISC-V specification. RISC-V defines all program execution flow transfers caused by the processor itself or peripherals as trap. When a trap occurs, the processor will interrupt the execution of the current program and switch to respond to the trap service program (handler).

According to the RISC-V specification, traps are divided into two categories: exceptions and interrupts. An exception refers to a TRAP triggered by the processor itself due to abnormal instruction execution. An interrupt refers to a request sent by various peripherals to the processor to require a response.

Interrupts can be divided into local interrupts and external interrupts. Local interrupts are mainly generated by the processor itself or its private devices, such as software interrupts and timer interrupts. Local interrupts can be managed by accessing the processor's CSR (Control and Status Registers). Various interrupts generated by most peripherals of this product are managed by the Platform-Level Interrupt Controller (PLIC), whose interrupt request output is connected to the Machine External Interrupt of the RISC-V processor.

The PLIC manages all on-chip peripheral interrupts. The RISC-V specification defines all functions of the PLIC. The PLIC on this product is compatible with the RISC-V specification and supports enhanced features: interrupt nesting extension and interrupt vector extension.

4.1.1 Exceptions

RISC-V processors support the following exceptions:

- Instruction address misaligned
  - When the instruction jumps to a non-aligned address
- Instruction access fault
  - Bus error during instruction fetch
  - PMP error during instruction fetch
  - PMA error during instruction fetch
- Illegal instruction
  - Execution of illegal instructions
  - Access to illegal CSRs
- Breakpoint
- Load address misaligned
- Load access fault
  - Bus error when the Load instruction loads data
  - PMP error when the Load instruction loads data
  - PMA error when the Load instruction loads data
- Store/AMO address misaligned

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- Store/AMO access fault
  - Bus error when the Store instruction stores data
  - PMP error when the Store instruction stores data
  - PMA error when the Store instruction stores data
- Environment call from U-mode
- Stack overflow exception
- Stack underflow exception

4.1.2 Interrupts

Interrupts supported by the RISC-V processor can be divided into local interrupts and external interrupts:

Local interrupts are generated by the processor's own devices, including:

- M-Mode software interrupt, generated by the software interrupt controller PLICSW
- M-Mode timer interrupt, generated by the machine timer MCHTMR
- M-mode Bus read/write transaction error interrupt
- M-mode Performance monitor overflow interrupt

External interrupts refer to Machine External Interrupts. External interrupts are managed by the PLIC. The PLIC arbitrates all interrupts according to priority and sends eligible interrupts to the processor.

4.1.3 Platform-Level Interrupt Controller (PLIC)

The Platform-Level Interrupt Controller of this product manages all interrupts generated by on-chip peripherals.

PLIC has the following features:

- The PLIC outputs interrupts to the external interrupt interface of the processor
- Supports 32 priority levels.
- Supports configurable interrupt edge trigger or level trigger
- Supports interrupt nesting, allowing high-priority interrupts to preempt ongoing low-priority interrupts
- Supports interrupt vectors. The interrupt vector mode can simplify the interrupt declaration program defined by the RISC-V specification and reduce interrupt response latency

4.1.4 Software Interrupt Controller (PLICSW)

The machine software interrupt of the RISC-V processor in this product is managed by the software interrupt controller PLICSW. The programming model of PLICSW is similar to that of PLIC, but it only contains one interrupt source. Users can generate a machine software interrupt by setting the interrupt pending bit of PLICSW.

4.1.5 Machine Timer (MCHTMR)

The RISC-V specification defines the machine timer. The machine timer is required to operate at a fixed clock frequency and generate timer interrupts. The machine timer interrupt of the RISC-V processor in this product is generated by the machine timer MCH_TMR. The machine timer supports a 64-bit counter and a 64-bit comparator. When the counter value reaches or exceeds the comparator value, a Machine Timer Interrupt is generated.

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4.2 TRAP and Processor Privilege Modes

The RISC-V processor used in this product supports two privilege modes defined by the RISC-V specification: Machine Mode and User Mode.

When the RISC-V processor is running, it is always in one of the privilege modes. Machine Mode has the highest privilege, while User Mode has the lowest privilege. Privilege modes can be used to isolate software and on-chip resources allocated to the software. When a HART runs in a lower privilege mode, it is not allowed to access resources exclusive to higher privilege modes (such as a certain memory segment, a certain peripheral, etc.). Users can thus establish a trusted execution environment to protect sensitive information or resources from unauthorized access.

The RISC-V specification stipulates that when any TRAP occurs, the processor mode will switch to Machine Mode to process the TRAP service routine. Note that even if some interrupts are designed for lower privilege modes, when these interrupts occur, the processor will still enter Machine Mode to process their interrupt service routines.

In this product, the PLIC supports 1 interrupt request output, which is connected to the Machine External Interrupt of the RISC-V processor. After an interrupt occurs, the processor will respond to the interrupt in Machine Mode.

In this product, the interrupt request output of the PLIC is connected to the Machine External Interrupt of the RISC-V processor, Users can run software in User Mode, but when an interrupt or exception occurs, the processor privilege mode will switch to Machine Mode.

If users do not need to differentiate software through the processor's privilege modes and assign different resource access permissions to the software, only the Machine External Interrupt request output of the PLIC can be used. In this case, both regular programs and interrupt service routines run in Machine Mode.

4.3 Interrupt Nesting

The interrupt nesting extension means that this product allows the processor to suspend low-priority interrupt service routines and switch to respond to high-priority interrupt service routines. The PLIC of this product adds this extended function on the basis of the PLIC defined by the RISC-V specification, enabling the processor to better meet the requirements of real-time system applications.

Users can enable interrupt nesting by setting the PREEMPT bit in the PLIC's Feature Enable Register to 1. When the processor is responding to an interrupt service routine, if the MIE bit in the mstatus CSR is set to 1, a higher-priority interrupt can preempt the current interrupt service routine. After the processor responds to the high-priority interrupt, it will continue to process the interrupted interrupt service routine.

During the execution of an interrupt service routine, the PLIC will not respond to interrupts of the same or lower priority.

The specific implementation process of interrupt nesting is as follows:

- When an interrupt occurs (interrupt claim), the PLIC will save the current value of the Priority Threshold Register, and assign the priority of the new interrupt to the Priority Threshold Register.
- When the interrupt ends (interrupt completion), the PLIC will restore the Priority Threshold Register to its previous value.

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In this product, the PLIC manages almost all peripheral interrupts and outputs interrupt requests to the processor's Machine External Interrupt. Therefore, the interrupt priority settings of the PLIC are only valid for peripheral interrupts managed by the PLIC.

When the software sets the MIE bit of the mstatus CSR to 1 in the interrupt service routine, the PLIC allows higher-priority interrupts as new Machine External Interrupts to preempt the current interrupt. In addition, processor exceptions and other local interrupts, such as software interrupts or timer interrupts, can also preempt the current interrupt. The priorities of these exceptions and other interrupts are not affected by the PLIC's priority settings.

4.4 Interrupt Vectors

When a trap occurs, the processor will jump to the address pointed to by the mtvec CSR (Control and Status Register) to execute the interrupt service routine. The software can determine the source of the trap (interrupt or exception) through the mcause CSR. If the trap is an external interrupt (Machine External Interrupt), the software needs to further determine the source of the interrupt through the PLIC's interrupt pending register.

The interrupt vector extension is an enhanced feature of the PLIC, which can reduce the interrupt response latency of the peripherals of this product. The interrupt vector extension means that when a trap is triggered by an external interrupt, the processor will read the address pointer of the interrupt service routine from the interrupt vector table and assign it to the processor's Program Counter (PC). In this way, the processor directly jumps to the corresponding interrupt service routine, and the software does not need to first read the mcause CSR and then the PLIC's claim/complete register to determine the source of the interrupt.

Users can enable the interrupt vector mode by setting the VECTORED bit in the PLIC's Feature Enable Register to 1 and the VEC_PLIC bit in the MMISC CSR to 1.

The RISC-V specification stipulates that by default, all traps are handled by Machine Mode. In this case, the base address of the interrupt vector table is stored in the processor's mtvec (Machine Trap Vector Base Address Register) CSR.

For Machine External Interrupts (output by PLIC Target 0, which sets the MEIP bit of the mip CSR to 1):

- The base address is mtvec, and the address of the interrupt vector with sequence number N is: base address + 4 × N.
- The base address is mtvec + 1024, and the address of the interrupt vector with sequence number N is: base address + 4 × N
- The address of the interrupt vector with sequence number N is: base address + 4 × N

The interrupt vector allocation of this product is as follows:

Interrupt Number	Interrupt Name	Description
1	GPIO0_A	
2	GPIO0_B	
3	GPIO0_X	
4	GPIO0_Y	
5	GPTMR0	

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Interrupt Number	Interrupt Name	Description
6	GPTMR1	
7	GPTMR2	
8	GPTMR3	
13	UART0	
14	UART1	
15	UART2	
16	UART3	
17	UART4	
18	UART5	
19	UART6	
20	UART7	
21	I2C0	
22	I2C1	
23	I2C2	
24	I2C3	
25	SPI0	
26	SPI1	
27	SPI2	
28	SPI3	
29	TSNS	
30	MBX0A	
31	MBX0B	
32	EWDG0	
33	EWDG1	
34	HDMA	
35	MCAN0	
36	MCAN1	
37	MCAN2	
38	MCAN3	
39	PTPC	
40	PWM0	
41	QEI0	
42	SEI0[0]	
43	MMC0	
44	TRGMUX0	
45	PWM1	
46	QEI1	
47	SEI0[1]	
48	MMC1	

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Interrupt Number	Interrupt Name	Description
49	TRGMUX1	
50	RDC	
51	USB0	
52	XPI0	
53	SDP	
54	PSEC	
55	SECMON	
56	RNG	
57	FUSE	
58	ADC0	
59	ADC1	
60	DAC0	
61	DAC1	
62	ACMP_0	
63	ACMP_1	
64	SYSCTL	
65	PGPIO	
66	PTMR	
67	PUART	
68	PEWDG	
69	BROWNOUT	
70	PAD_WAKEUP	
71	DEBUG0	
72	DEBUG1	

Table 10: IRQ List

4.5 Interrupt Response Flow

Take the regular interrupt service routine flow in Machine Mode (M-Mode) as an example, the processor hardware will perform the following actions:

- Update the value of the mepc CSR to the current PC (Program Counter)
- Update the value of the mstatus CSR:
  - Update the MPP bit field to the current privilege mode
  - Update the MPIE value to MIE
  - Set the MIE bit to 0
- Update the value of the mcause CSR, which contains the current interrupt source information
- Switch the processor privilege mode to Machine Mode
- Update the PC (Program Counter) according to the value of the mtvec CSR:
  - If interrupt vectors are not supported, the value of the mtvec CSR is assigned to the PC
  - If interrupt vectors are supported, the value of base address + 4 * N is assigned to the PC,

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where N is the interrupt ID number of the PLIC

.

After that, the software starts to execute the interrupt service routine, and the software should:

- Save the necessary processor integer registers and push them to the stack
- If interrupt vectors are not supported, or the occurred TRAP is an exception or local interrupt, save the mcause CSR
- If interrupt nesting is required:
  - Save the mepc CSR
  - Save the mstatus CSR
  - Save other necessary CSRs as required, such as CSRs related to floating-point/DSP extensions, etc.
  - Set the MIE bit of the mstatus CSR to 1, which allows new interrupts to be generated
- Start executing the interrupt service routine:
  - If interrupt vectors are not supported, or the occurred TRAP is an exception or local interrupt, it is necessary to determine the cause of the TRAP according to mcause, and execute the corresponding processing routine. If it is an external interrupt sent by the PLIC to the processor, it is necessary to read the PLIC's Claim/Complete register to complete the interrupt claim, obtain the interrupt ID number to determine the interrupt source, and then execute the corresponding interrupt service routine.
  - If interrupt vectors are supported, directly read the interrupt flag bit of the corresponding peripheral, determine the cause of the interrupt, and execute the interrupt service routine

After completing the TRAP service routine, the processing flow is as follows:

- Write the interrupt ID to the PLIC's Claim/Complete register to indicate that the interrupt is completed.
- If interrupt nesting is enabled:
  - Set the MIE bit of the mstatus CSR to 0 to stop responding to new interrupts
  - Insert a FENCE instruction to ensure that the previous write operation to the PLIC's Claim/-Complete register is completed
  - Restore the mepc CSR
  - Restore the mstatus CSR
  - Restore other previously saved CSRs, such as CSRs related to floating-point/DSP extensions, etc.
- The software restores the processor integer registers previously pushed to the stack
- The software executes the MRET instruction to return from Machine Mode.

Once the MRET instruction is executed, the processor hardware will perform the following operations:

- Restore the value of the mepc CSR to the PC (Program Counter)
- Restore the value of the mstatus CSR:
  - Restore the processor's privilege mode according to the value of the MPP bit field
  - Restore the value of MPIE to MIE

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5 Machine Timer MCHTMR

This chapter introduces the functions and features of the machine timer MCHTMR.

5.1 Feature Summary

This chapter describes the main features of the machine timer MCHTMR:

- 64-bit real-time counter
- 64-bit count comparator
- Supports 32-bit and 64-bit register access
- Supports sending machine timer interrupts to the RISC-V core

The block diagram of MCHTMR is shown in Figure 2.

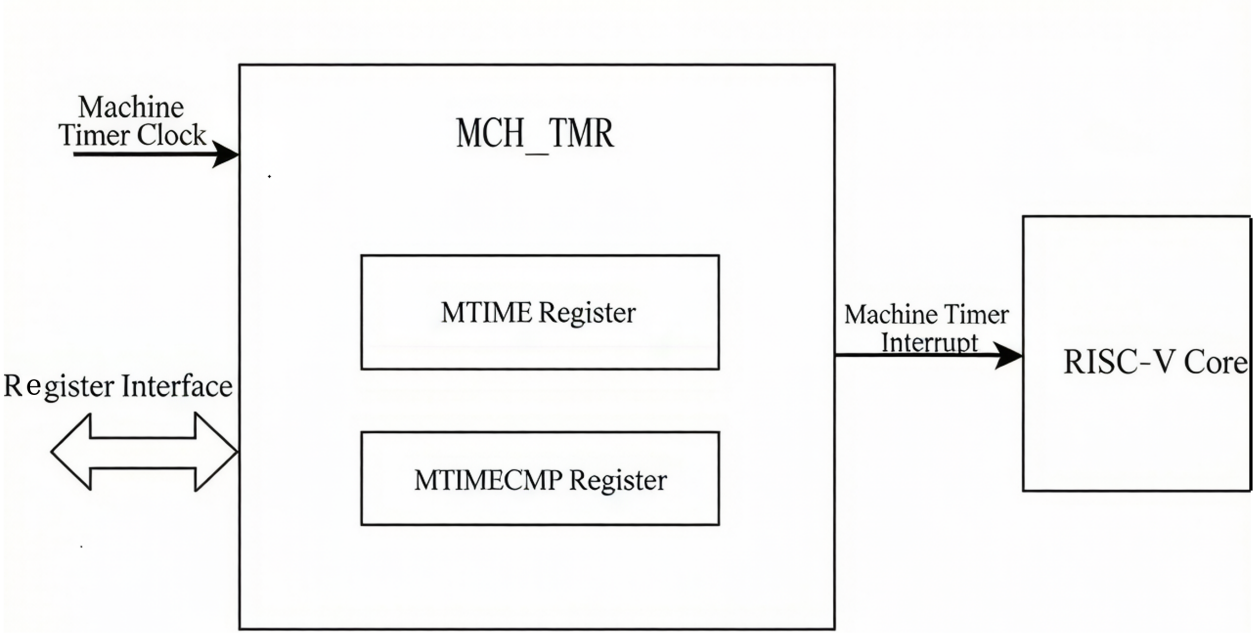


Figure 2: MCHTMR Block Diagram

5.2 Function Description

This chapter describes the functions of the machine timer MCHTMR.

5.2.1 Timekeeping and Timing Functions

MCHTMR has a built-in 64-bit counter, and the counter value increases by 1 each time the machine timer clock toggles. The current count value can be obtained by reading the MTIME register.

The timing function is implemented by setting the MTIMECMP register. When the counter value is greater than or equal to the value of the MTIMECMP register, MCHTMR sends a machine timer interrupt to the RISC-V core.

When the machine timer interrupt is active, a write operation to the MTIMECMP register will clear the interrupt.

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5.2.2 Timer Reset

The internal counter starts counting from 1. Rewriting the MTIME register can reset the counter to a specific value, and modifying the MTIME register must follow certain rules:

The MTIME register is divided into two 32-bit registers: MTIME[63:32] and MTIME[31:0]. When modifying the MTIME register, you must operate on the two 32-bit registers separately in 32-bit mode.

- If bits 31 to 29 of the target MTIME value to be set are 7, i.e. MTIME[31:29]=7, the configuration process is as follows:
    1. Set MTIME[31:0] to 0
    2. Set the target value of MTIME[63:32]
    3. Set the target value of MTIME[31:0]
- For other general cases:
    1. Set the target value of MTIME[31:0]
    2. Set the target value of MTIME[63:32]

5.3 MCHTMR Register Description

The register list of MCHTMR is as follows:

MCHTMR base address: 0xE6000000

Offset	Register Name	Description	Reset Value
0x0000	MTIME	Machine Time	0x0000000000020210
0x0008	MTIMECMP	Machine Time Compare	0x0000000000020210

Table 11: MCHTMR Register List

5.4 MCHTMR Register Details

5.4.1 MTIME (0x0)

63	62	61	60	59	58	57	56	55	54	53	52	51	50	49	48	47	46	45	44	43	42	41	40	39	38	37	36	35	34	33	32
MTIME																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0

MTIME [63:32]

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MTIME																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0

MTIME [31:0]

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Bits	Name	Description
63-0	MTIME	Machine time

MTIME Bits

5.4.2 MTIMECMP (0x8)

63	62	61	60	59	58	57	56	55	54	53	52	51	50	49	48	47	46	45	44	43	42	41	40	39	38	37	36	35	34	33	32
MTIMECMP																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0

MTIMECMP [63:32]

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MTIMECMP																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0

MTIMECMP [31:0]

Bits	Name	Description
63-0	MTIMECMP	Machine time compare

MTIMECMP Bits

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6 Platform-Level Interrupt Controller (PLIC)

This chapter describes the functions and features of the Platform-Level Interrupt Controller (PLIC).

6.1 Feature Summary

This chapter describes the main features of the PLIC:

- Configurable interrupt trigger modes
- Supports vectored interrupts
- Supports 32 priority levels
- Supports interrupt priority preemption

6.2 Functional Description

This chapter describes the functions of the PLIC.

The PLIC collects interrupt requests from all interrupt sources into the interrupt gateway, performs priority processing according to the configuration, and then forwards the interrupt requests to the RISC-V core.

Interrupt configuration includes enable, priority and other settings. This PLIC supports 8 interrupt priority levels, the higher the value set in the priority register, the higher the interrupt priority. Note that software should not modify the interrupt configuration when there are unprocessed interrupts.

6.2.1 Vectored Interrupt Extension

The RISC-V standard requires that the PLIC notify the RISC-V core when an external interrupt is received. The RISC-V core reads the CLAIM_COMPLETE register of the PLIC to obtain the interrupt ID, and this read operation is called interrupt claim. After obtaining the interrupt ID, the RISC-V core executes the corresponding interrupt handler, and then writes the ID back to the CLAIM_COMPLETE register of the PLIC. This write operation is called interrupt complete.

The PLIC of this product also supports vectored interrupt extension, that is, when the PLIC notifies the RISC-V core of an interrupt request, it will send the interrupt ID at the same time. The RISC-V core sends the interrupt claim to the PLIC via hardware, and the software does not need to read the PLIC to obtain the interrupt ID and perform the interrupt claim, so as to speed up interrupt processing.

The vectored interrupt extension is enabled by configuring the FEATURE_ENABLE.VECTORED register.

6.2.2 Interrupt Priority Preemption

The PLIC supports interrupt preemption priority. The priority of each interrupt ID can be configured in the PRIORITY register. The maximum valid priority value is 31, and a larger value indicates a higher interrupt priority. When priority preemption is enabled, if the RISC-V core has claimed an interrupt but has not yet completed it, and a new higher-priority interrupt occurs, the PLIC will send a new interrupt request to the RISC-V core again.

Setting the FEATURE.PREEMPT register to 1 enables the priority preemption function.

In the interrupt handler, global interrupts are enabled by setting the [MIE] bit field of the Machine Status Register (MSTATUS) in the Control and Status Register (CSR) module of the RISC-V processor , to allow higher-priority interrupts to preempt the currently executing interrupt handler.

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In non-vectored mode, the RISC-V core should enable interrupt preemption after saving the context and completing the interrupt claim. Since both interrupt claim and interrupt complete are load/store instructions executed on the device region, their order is automatically guaranteed, so there is no need to additionally disable global interrupts, nor to insert a FENCE instruction after issuing the interrupt complete operation.

The recommended non-vectored mode interrupt handler is as follows:

1. Save registers and CSRs to the stack
2. Send an interrupt claim operation to the PLIC
3. Set the [MIE] bit field of the Machine Status Register (MSTATUS) to 1 to enable global interrupts
4. Process the interrupt
5. Send an interrupt complete operation to the PLIC
6. Restore registers and CSRs
7. Return from the interrupt handler

In vectored mode, the RISC-V core can enable interrupt preemption after saving the context. To avoid conflict with the next interrupt claim, global interrupt enable must be disabled before executing interrupt complete. In addition, a FENCE I/O instruction should be executed before exiting the interrupt handler to ensure that the interrupt complete operation has reached the PLIC.

The recommended vectored mode interrupt handler is as follows:

1. Save registers and CSRs to the stack
2. Set the [MIE] bit field of the Machine Status Register (MSTATUS) to 1 to enable global interrupts
3. Process the interrupt
4. Set the [MIE] bit field of the Machine Status Register (MSTATUS) to 0 to disable global interrupts
5. Send an interrupt complete operation to the PLIC
6. Restore registers and CSRs
7. Execute the FENCE I/O instruction to ensure that the interrupt complete operation has been delivered to the PLIC
8. Return from the interrupt handler

The THRESHOLD register can be set to an interrupt priority threshold, and only interrupts with a priority higher than this threshold will be notified to the RISC-V core. When an interrupt has been claimed but not completed, the THRESHOLD register will be updated in real time to the priority of the currently processed interrupt, so as to mask lower-priority interrupts and allow higher-priority interrupt preemption.

6.3 PLIC Register Description

The PLIC register list is as follows:

PLIC base address: 0xE4000000

Offset	Register Name	Description	Reset Value
0x0000	FEATURE	Feature enable register	0x00000000
0x0004	PRIORITY[PRIORITY1]	Source priority	0x00000001
0x0008	PRIORITY[PRIORITY2]	Source priority	0x00000001
0x000C	PRIORITY[PRIORITY3]	Source priority	0x00000001
0x0010	PRIORITY[PRIORITY4]	Source priority	0x00000001

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Offset	Register Name	Description	Reset Value
0x0014	PRIORITY[PRIORITY5]	Source priority	0x00000001
0x0018	PRIORITY[PRIORITY6]	Source priority	0x00000001
0x001C	PRIORITY[PRIORITY7]	Source priority	0x00000001
0x0020	PRIORITY[PRIORITY8]	Source priority	0x00000001
0x0024	PRIORITY[PRIORITY9]	Source priority	0x00000001
0x0028	PRIORITY[PRIORITY10]	Source priority	0x00000001
0x002C	PRIORITY[PRIORITY11]	Source priority	0x00000001
0x0030	PRIORITY[PRIORITY12]	Source priority	0x00000001
0x0034	PRIORITY[PRIORITY13]	Source priority	0x00000001
0x0038	PRIORITY[PRIORITY14]	Source priority	0x00000001
0x003C	PRIORITY[PRIORITY15]	Source priority	0x00000001
0x0040	PRIORITY[PRIORITY16]	Source priority	0x00000001
0x0044	PRIORITY[PRIORITY17]	Source priority	0x00000001
0x0048	PRIORITY[PRIORITY18]	Source priority	0x00000001
0x004C	PRIORITY[PRIORITY19]	Source priority	0x00000001
0x0050	PRIORITY[PRIORITY20]	Source priority	0x00000001
0x0054	PRIORITY[PRIORITY21]	Source priority	0x00000001
0x0058	PRIORITY[PRIORITY22]	Source priority	0x00000001
0x005C	PRIORITY[PRIORITY23]	Source priority	0x00000001
0x0060	PRIORITY[PRIORITY24]	Source priority	0x00000001
0x0064	PRIORITY[PRIORITY25]	Source priority	0x00000001
0x0068	PRIORITY[PRIORITY26]	Source priority	0x00000001
0x006C	PRIORITY[PRIORITY27]	Source priority	0x00000001
0x0070	PRIORITY[PRIORITY28]	Source priority	0x00000001
0x0074	PRIORITY[PRIORITY29]	Source priority	0x00000001
0x0078	PRIORITY[PRIORITY30]	Source priority	0x00000001
0x007C	PRIORITY[PRIORITY31]	Source priority	0x00000001
0x0080	PRIORITY[PRIORITY32]	Source priority	0x00000001
0x0084	PRIORITY[PRIORITY33]	Source priority	0x00000001
0x0088	PRIORITY[PRIORITY34]	Source priority	0x00000001
0x008C	PRIORITY[PRIORITY35]	Source priority	0x00000001
0x0090	PRIORITY[PRIORITY36]	Source priority	0x00000001
0x0094	PRIORITY[PRIORITY37]	Source priority	0x00000001
0x0098	PRIORITY[PRIORITY38]	Source priority	0x00000001
0x009C	PRIORITY[PRIORITY39]	Source priority	0x00000001
0x00A0	PRIORITY[PRIORITY40]	Source priority	0x00000001
0x00A4	PRIORITY[PRIORITY41]	Source priority	0x00000001
0x00A8	PRIORITY[PRIORITY42]	Source priority	0x00000001
0x00AC	PRIORITY[PRIORITY43]	Source priority	0x00000001
0x00B0	PRIORITY[PRIORITY44]	Source priority	0x00000001

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Offset	Register Name	Description	Reset Value
0x00B4	PRIORITY[PRIORITY45]	Source priority	0x00000001
0x00B8	PRIORITY[PRIORITY46]	Source priority	0x00000001
0x00BC	PRIORITY[PRIORITY47]	Source priority	0x00000001
0x00C0	PRIORITY[PRIORITY48]	Source priority	0x00000001
0x00C4	PRIORITY[PRIORITY49]	Source priority	0x00000001
0x00C8	PRIORITY[PRIORITY50]	Source priority	0x00000001
0x00CC	PRIORITY[PRIORITY51]	Source priority	0x00000001
0x00D0	PRIORITY[PRIORITY52]	Source priority	0x00000001
0x00D4	PRIORITY[PRIORITY53]	Source priority	0x00000001
0x00D8	PRIORITY[PRIORITY54]	Source priority	0x00000001
0x00DC	PRIORITY[PRIORITY55]	Source priority	0x00000001
0x00E0	PRIORITY[PRIORITY56]	Source priority	0x00000001
0x00E4	PRIORITY[PRIORITY57]	Source priority	0x00000001
0x00E8	PRIORITY[PRIORITY58]	Source priority	0x00000001
0x00EC	PRIORITY[PRIORITY59]	Source priority	0x00000001
0x00F0	PRIORITY[PRIORITY60]	Source priority	0x00000001
0x00F4	PRIORITY[PRIORITY61]	Source priority	0x00000001
0x00F8	PRIORITY[PRIORITY62]	Source priority	0x00000001
0x00FC	PRIORITY[PRIORITY63]	Source priority	0x00000001
0x0100	PRIORITY[PRIORITY64]	Source priority	0x00000001
0x0104	PRIORITY[PRIORITY65]	Source priority	0x00000001
0x0108	PRIORITY[PRIORITY66]	Source priority	0x00000001
0x010C	PRIORITY[PRIORITY67]	Source priority	0x00000001
0x0110	PRIORITY[PRIORITY68]	Source priority	0x00000001
0x0114	PRIORITY[PRIORITY69]	Source priority	0x00000001
0x0118	PRIORITY[PRIORITY70]	Source priority	0x00000001
0x011C	PRIORITY[PRIORITY71]	Source priority	0x00000001
0x0120	PRIORITY[PRIORITY72]	Source priority	0x00000001
0x1000	PENDING[PENDING0]	Pending status	0x00000000
0x1004	PENDING[PENDING1]	Pending status	0x00000000
0x1008	PENDING[PENDING2]	Pending status	0x00000000
0x100C	PENDING[PENDING3]	Pending status	0x00000000
0x1080	TRIGGER[TRIGGER0]	Trigger type	0x00000000
0x1084	TRIGGER[TRIGGER1]	Trigger type	0x00000000
0x1088	TRIGGER[TRIGGER2]	Trigger type	0x00000000
0x108C	TRIGGER[TRIGGER3]	Trigger type	0x00000000
0x1100	NUMBER	Number of supported interrupt sources and targets	-
0x1104	INFO	Version and the maximum priority	-

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Offset	Register Name	Description	Reset Value
0x2000	TARGETINT[TARGET0][INTEN][INTEN0]	machine interrupt enable	0x00000000
0x2004	TARGETINT[TARGET0][INTEN][INTEN1]	machine interrupt enable	0x00000000
0x2008	TARGETINT[TARGET0][INTEN][INTEN2]	machine interrupt enable	0x00000000
0x200C	TARGETINT[TARGET0][INTEN][INTEN3]	machine interrupt enable	0x00000000
0x2010	TARGETINT[TARGET0][INTEN][INTEN4]	machine interrupt enable	0x00000000
0x2014	TARGETINT[TARGET0][INTEN][INTEN5]	machine interrupt enable	0x00000000
0x200000	TARGETCONFIG[TARGET0][THRESHOLD]	Target0 priority threshold	0x00000000
0x200004	TARGETCONFIG[TARGET0][CLAIM]	Target claim and complete	0x00000000
0x200400	TARGETCONFIG[TARGET0][PPS]	Preempted priority stack	0x00000000

Table 12: PLIC Register List

6.4 PLIC Register Details

The detailed description of PLIC registers is as follows:

6.4.1 FEATURE (0x0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																														VECTORED	PREEMPT
N/A																														RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0

FEATURE [31:0]

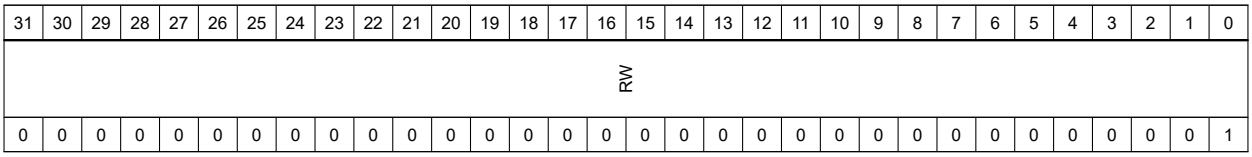
Bits	Name	Description
1	VECTORED	Vector mode enable 0: Disabled 1: Enabled
0	PREEMPT	Preemptive priority interrupt enable 0: Disabled 1: Enabled

FEATURE Bits

6.4.2 PRIORITY (0x4 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PRIORITY																															

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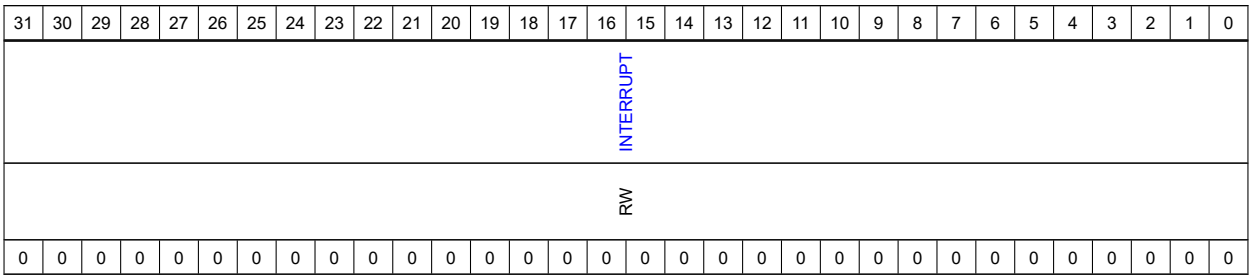


PRIORITY [31:0]

Bits	Name	Description
31-0	PRIORITY	Interrupt source priority. The valid range of this field is 0-31. 0: Never interrupt 1-31: Interrupt source priority. The larger the value, the higher the priority.

PRIORITY Bits

6.4.3 PENDING (0x1000 + 0x4 * n)

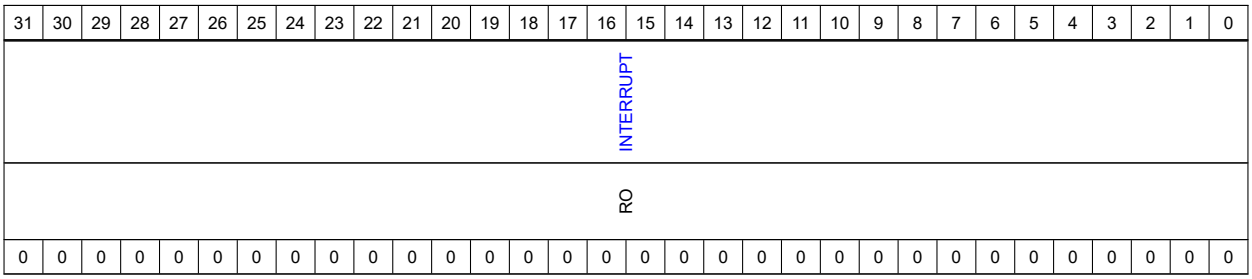


PENDING [31:0]

Bits	Name	Description
31-0	INTERRUPT	The interrupt pending status of inpterrupt sources. Every interrupt source occupies 1 bit.

PENDING Bits

6.4.4 TRIGGER (0x1080 + 0x4 * n)



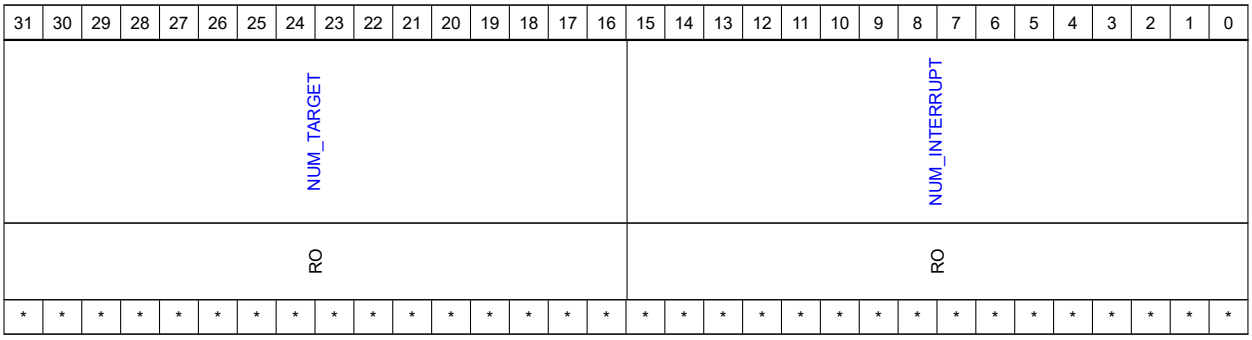
TRIGGER [31:0]

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Bits	Name	Description
31-0	INTERRUPT	The interrupt trigger type of interrupt sources. Every interrupt source occupies 1 bit. 0: Level-triggered interrupt 1: Edge-triggered interrupt

TRIGGER Bits

6.4.5 NUMBER (0x1100)

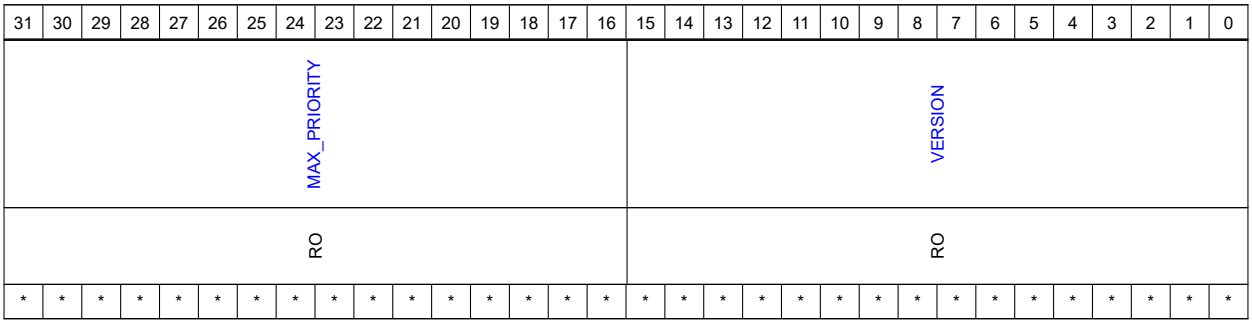


NUMBER [31:0]

Bits	Name	Description
31-16	NUM_TARGET	The number of supported targets
15-0	NUM_INTERRUPT	The number of supported interrupt sources

NUMBER Bits

6.4.6 INFO (0x1104)



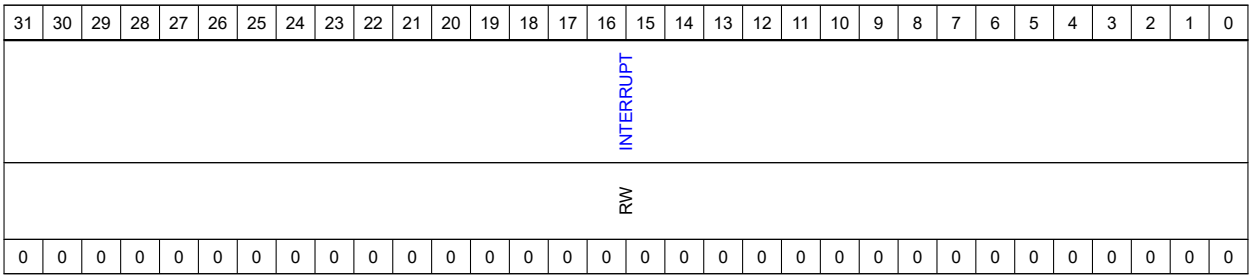
INFO [31:0]

Bits	Name	Description
31-16	MAX_PRIORITY	The maximum priority supported
15-0	VERSION	The version of the PLIC design

INFO Bits

6.4.7 TARGETINT[INTEN] (0x2000 + 0x80 * n + 0x4 * m)

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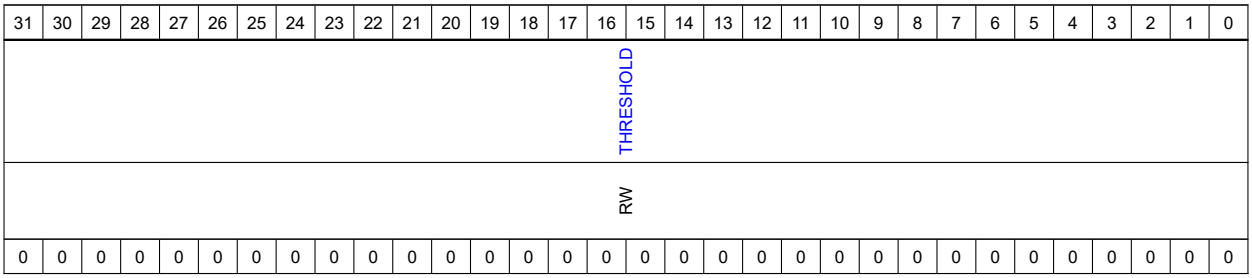


TARGETINT[INTEN] [31:0]

Bits	Name	Description
31-0	INTERRUPT	The interrupt enable bit for interrupt. Every interrupt source occupies 1 bit.

TARGETINT[INTEN] Bits

6.4.8 TARGETCONFIG[THRESHOLD] (0x200000 + 0x1000 * n)

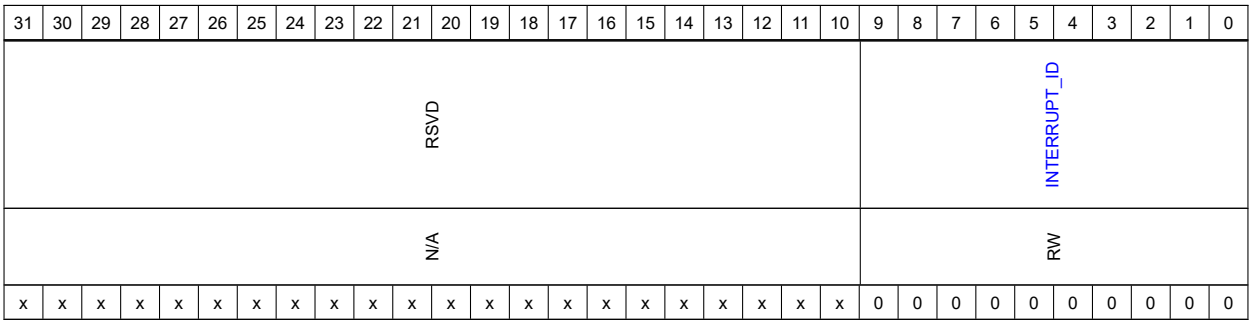


TARGETCONFIG[THRESHOLD] [31:0]

Bits	Name	Description
31-0	THRESHOLD	Interrupt priority threshold.

TARGETCONFIG[THRESHOLD] Bits

6.4.9 TARGETCONFIG[CLAIM] (0x200004 + 0x1000 * n)



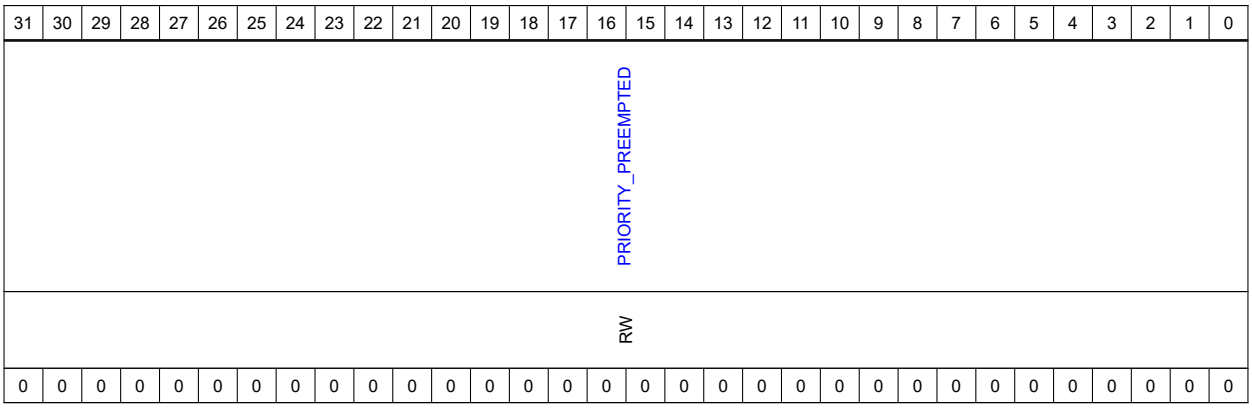
TARGETCONFIG[CLAIM] [31:0]

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Bits	Name	Description
9-0	INTERRUPT_ID	On reads, indicating the interrupt source that has being claimed. On writes, indicating the interrupt source that has been handled (completed).

TARGETCONFIG[CLAIM] Bits

6.4.10 TARGETCONFIG[PPS] (0x200400 + 0x1000 * n)



TARGETCONFIG[PPS] [31:0]

Bits	Name	Description
31-0	PRIORITY_PREEMPTED	Each bit indicates if the corresponding priority level has been preempted by a higher-priority interrupt.

TARGETCONFIG[PPS] Bits

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7 Platform-Level Software Interrupt Controller PLICSW

This chapter describes the functions and features of the platform-level software interrupt controller PLICSW.

7.1 Feature Summary

This chapter describes the main features of the platform-level interrupt controller PLIC:

- Supports generating interrupts through software programming

7.2 Functional Description

This chapter describes the functions of the platform-level interrupt controller PLIC.

The RISC-V core can trigger a software interrupt by writing 1 to the PENDING register. After receiving the interrupt, the RISC-V core needs to read the CLAIM register to complete the interrupt claim. After the interrupt processing is completed, it needs to write to the CLAIM register to perform the interrupt completion operation.

7.3 PLICSW Register Description

The register list of PLIC is as follows:

PLICSW base address: 0xE6400000

Offset	Register Name	Description	Reset Value
0x1000	PENDING	Pending status	0x00000000
0x2000	INTEN	Interrupt enable	0x00000000
0x200004	CLAIM	Claim and complete.	0x00000000

Table 13: PLIC_SW Register List

7.3.1 Detailed PLICSW Register Information

The detailed description of PLIC registers is as follows:

7.3.2 PENDING (0x1000)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																														INTERRUPT	RSVD
N/A																														RW	N/A
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x

PENDING [31:0]

Bits	Name	Description
1	INTERRUPT	writing 1 to trigger software interrupt

PENDING Bits

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7.3.3 INTEN (0x2000)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																															INTERRUPT
N/A																															RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

INTEN [31:0]

Bits	Name	Description
0	INTERRUPT	enable software interrupt

INTEN Bits

7.3.4 CLAIM (0x200004)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																															INTERRUPT_ID
N/A																															RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

CLAIM [31:0]

Bits	Name	Description
0	INTERRUPT_ID	On reads, indicating the interrupt source that has being claimed. On writes, indicating the interrupt source that has been handled (completed).

CLAIM Bits

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8 Power Management

This chapter describes the power management system.

8.1 Power System Architecture

This product is divided into 2 main power domains in terms of power structure, as shown in Figure 3:

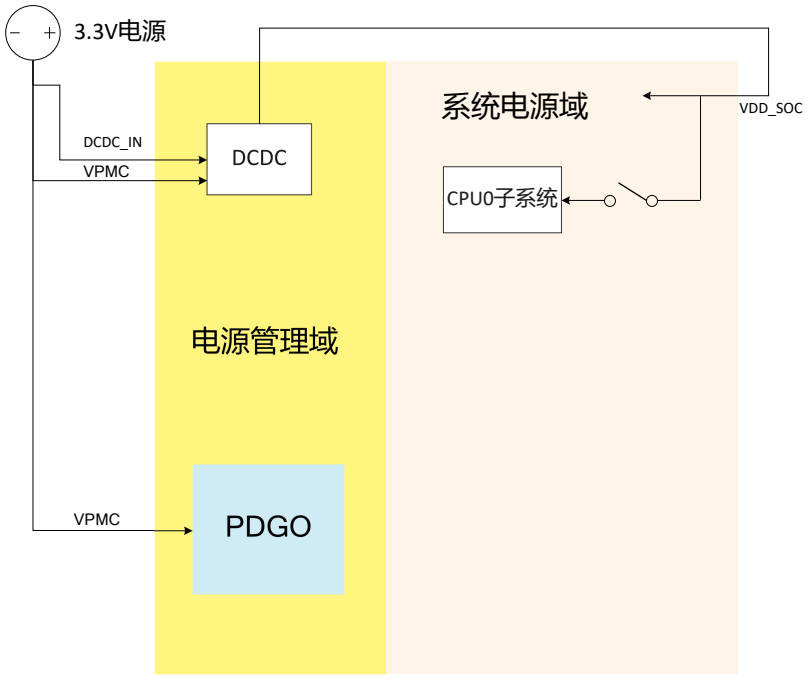


Figure 3: Power System Block Diagram

The 2 power domains include:

- The system power domain contains most of the functional modules in the chip, including 1 subsystem (CPU0) with independent power switch control, and several other functional modules
- The power management domain integrates a DCDC to supply power to the system power domain, and several wake-up capable functional modules that can maintain low-power operation when the system power domain is turned off. The PDGO module in the power management domain can operate at ultra-low power consumption and retain necessary data when other modules in the power management domain are turned off

The main functional modules in each power domain can be viewed in Figure 1.

In typical application scenarios, the chip can operate with a single 3.3V power supply.

8.2 Power Supply System

The chip integrates 2 channels of LDO linear regulators and 1 channel of DCDC switching power regulator to support single 3.3V power supply applications.

The block diagram of the power supply system is shown in Figure 4:

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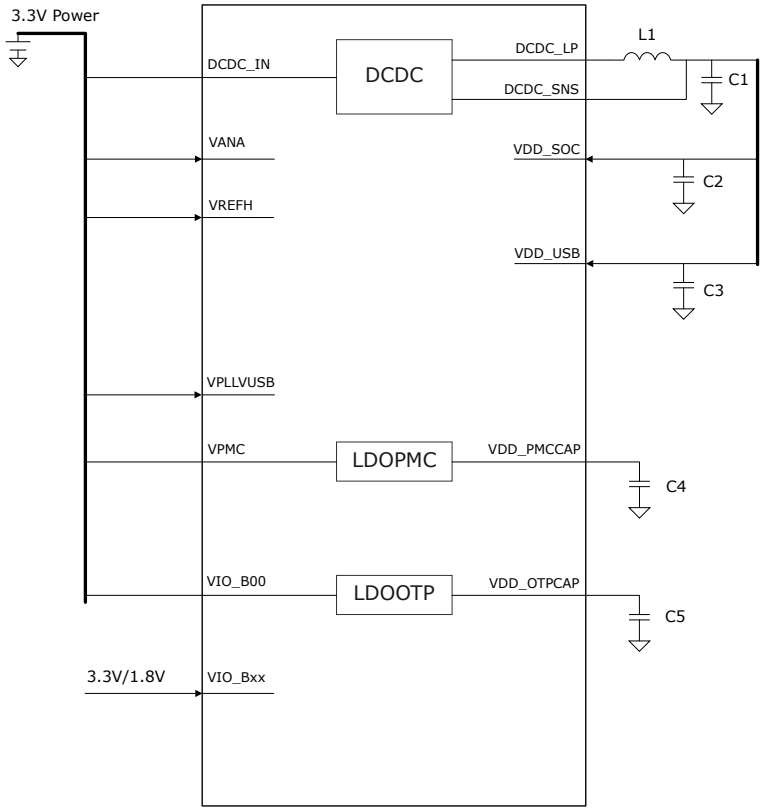


Figure 4: Block Diagram of Power Supply System with Built-in DCDC

Note: When using the built-in DCDC, the accuracy of the 16-bit ADC will degrade to a certain extent. For precautions for on-chip ADC usage, please refer to the ADC-related application documents at: https://www.hpmicro.com/design-resources/chip-information.

The chip supports powering the system power domain with an external power supply instead of the built-in DCDC, as shown in Figure 5:

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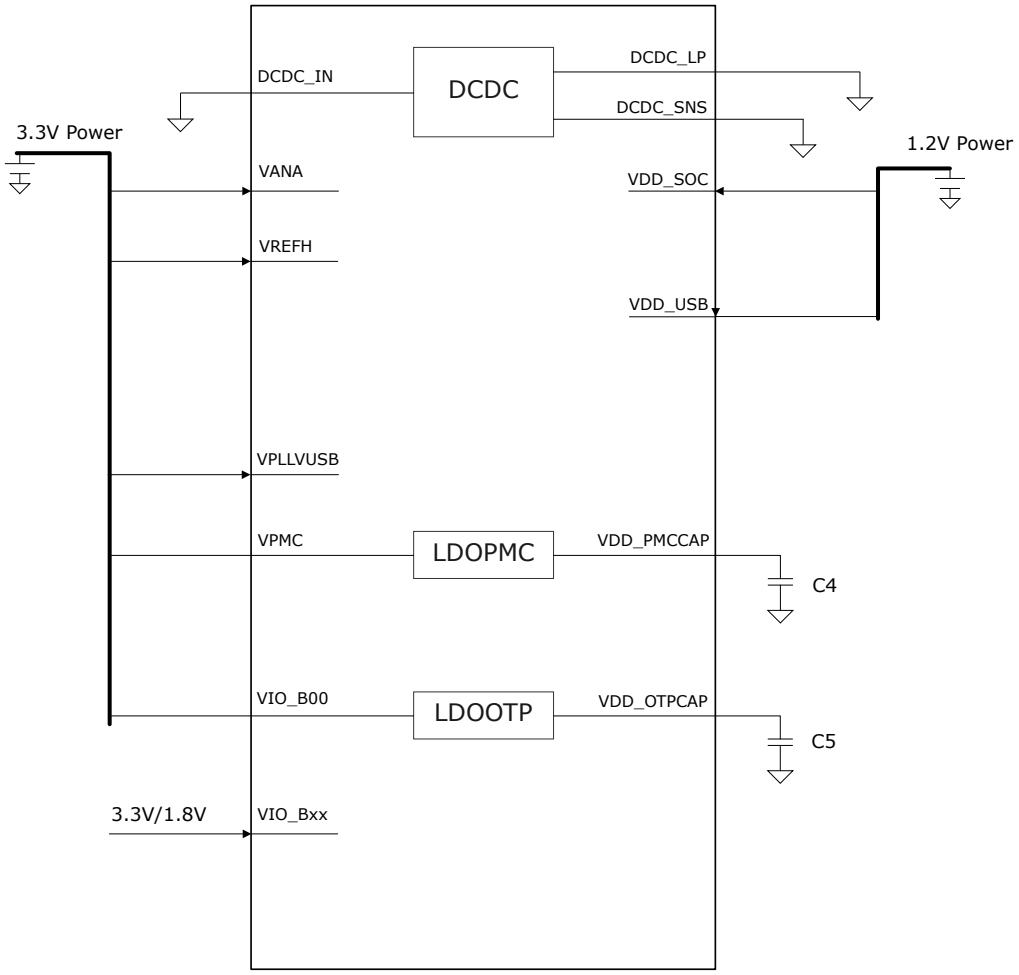


Figure 5: Block Diagram of Power Supply System without Built-in DCDC

8.2.1 Switching Power Regulator DCDC

The DCDC is located in the power management domain. It converts the 3.3V external power supply to a typical 1.1V output to power the circuits in the system power domain.

The output voltage of the DCDC is adjustable and supports dynamic voltage and frequency scaling.

DCDC_IN is the power input pin of the DCDC. DCDC_LP is the power output pin of the DCDC, which should be connected to a 2.2~10uH inductor. DCDC_SNS is the feedback pin of the DCDC, which should be connected to the power output of the DCDC.

When the user chooses to use an external power supply instead of the on-chip DCDC to power the system power domain, it is recommended to connect the relevant DCDC pins DCDC_IN, DCDC_LP and DCDC_SNS to ground through a 10kΩ resistor.

8.2.2 Power Management Domain Linear Regulator LDOPMC

LDOPMC is located in the power management domain. It converts the 3.3V external power supply to a typical 1.1V output to power the circuits in the power management domain except the PDGO module. The PDGO module is powered by the 3.3V VPMC supply.

LDOPMC uses the VPMC pin as its power input, and a 4.7uF decoupling capacitor needs to be connected

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to the VDD_PMCCAP pin.

8.2.3 OTP Linear Regulator LDOOTP

LDOOTP is located in the power management domain. It converts the 3.3V external power supply to a typical 2.5V output to power the OTP programming circuit.

LDOOTP should only be enabled during OTP programming operations, and should remain turned off at other times.

LDOOTP uses the VPMC pin as its power input, and a 4.7uF decoupling capacitor needs to be connected to the VDD_OTPCAP pin.

8.3 IO Power Supply

The GPIO pins of the chip are divided into multiple groups (Banks). Each IO group has an independent IO power supply pin VIO_B*, and GPIOs support 3.3V voltage.

The IO power supply pin used by each IO can be found in Chapter 20.

The power management domain GPIOs PY* are fixedly powered by VPMC, so they can only operate in 3V mode.

8.4 Power Pin Description

Power Pin	Voltage Range	Description
VPMC	3.0V ~ 3.6V	Power input for LDOPMC, power input for LDOOTP, power supply for analog circuits in power management domain and PDG module of digital circuits
VDD_PMCCAP	-	Decoupling capacitor connection pin for LDOPMC
VANA	3.0V ~ 3.6V	Power supply for PLL, ADC, ACMP analog circuits
VREFH	2.4V ~ VANA	ADC high-side reference voltage ACMP built-in DAC high-side reference voltage DAC high-side reference voltage
DCDC_IN	3.0V ~ 3.6V	DCDC power input
DCDC_LP	-	DCDC power output, can supply power to digital circuits in the system power domain
DCDC_SNS	-	DCDC power feedback pin, should be connected to DCDC_LP
VDD_SOC	0.925V ~ 1.26V	Power supply pin for digital circuits in the system power domain
VUSB	3.0V ~ 3.6V	Power supply for USB analog circuits, which is shorted to VIO_B00 on this product.
VDD_USB	0.925V ~ 1.26V	Power supply for USB digital circuits, usually shorted to VDD_SOC
VIO_B00 ~ VIO_B01	3.3V	IO power supply pins

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Power Pin	Voltage Range	Description
DCDC_GND	-	DCDC ground pin
VSS	-	Ground pin
VREFL	-	ADC low-side reference voltage, must be grounded

Table 14: Power and Ground Pin Description

8.5 Low Power Overview

This product supports multiple power consumption modes, sorted by power consumption from high to low and wake-up time from short to long as follows:

- Run mode (RUN)
- Wait mode (WAIT)
- Stop mode (STOP)
- Standby mode (STANBY)
- Shutdown mode (SHUTDOWN)

In Run mode, the CPU executes instructions normally, and all required functional modules operate normally. Unneeded functional modules can be turned off, and the CPU clock frequency and supply voltage can be adjusted by software.

In Wait mode, the CPU core clock is turned off triggered by the WFI instruction, other functional modules maintain the state in Run mode, and the CPU can resume operation and process interrupts immediately when an interrupt occurs.

Stop mode is triggered by the CPU's WFI instruction. Through pre-configuration, each subsystem and module in the system power domain, including the CPU's own power supply, can be flexibly turned off or maintained under the control of the SYSCTL module.

In Standby mode, the entire system power domain is in reset or power-down state. This mode can be triggered by the CPU's WFI instruction, or by software operation.

In Shutdown mode, all parts of the system power domain and power management domain except the PDGO module are in reset or power-down state, only the PDGO remains operational. This mode can be triggered by software operation.

Special note: The CPU local memories ILM and DLM of this product support memory retention function. When the system power domain is powered (VDD_SOC is powered) and the CPU0 subsystem is powered down (VDD_CPU is powered down), the memory core of the CPU local memories ILM and DLM remains powered, and the stored content can be retained to facilitate subsequent power-up recovery. This function does not require configuration and is not a low-power mode. It is automatically implemented in the above modes as long as the system power domain is powered and the CPU0 subsystem is powered down.

8.6 Power Management Function Related IOs

The PDGO module in the power management domain is the basis of the chip's power management system, and its IOs can be used for power management related functions:

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Pin	Default Function	Description
RESETN	RESETN	Global reset input pin, which can reset the entire system and can also be used to exit shutdown mode, see **??** for details
WAKEUP	WAKEUP	Power wake-up input pin, which can be used to exit shutdown mode and other low-power modes, see **??** for details

Table 15: Power Management Related IOs

8.7 VPMC Undervoltage Detection

This product supports VPMC voltage undervoltage detection (Brownout BOR) function. When the VPMC voltage drops to a certain level, an undervoltage warning (BOR Warning) can be generated. When the VPMC voltage drops further, an undervoltage reset (Brownout Reset) can be generated. Users can refer to the product datasheet for the voltage values for undervoltage warning assertion and de-assertion, as well as the voltage values for undervoltage reset assertion and de-assertion.

The undervoltage warning can generate an interrupt, and the user needs to correctly enable the undervoltage warning interrupt in the platform interrupt controller. When an undervoltage condition occurs, bit 0 of the RESET_FLAG register of the PPOR module will be set to 1.

If the user wishes to use the undervoltage warning interrupt, it is necessary to confirm that the DCDC_PROT [DISABLE_POWER_LOSS] bit of the PCFG module is set to 1.

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9 System Reset

This chapter describes the system reset of this product.

9.1 Reset Overview

The reset system supports resetting different ranges of the chip, as shown in Figure 6:

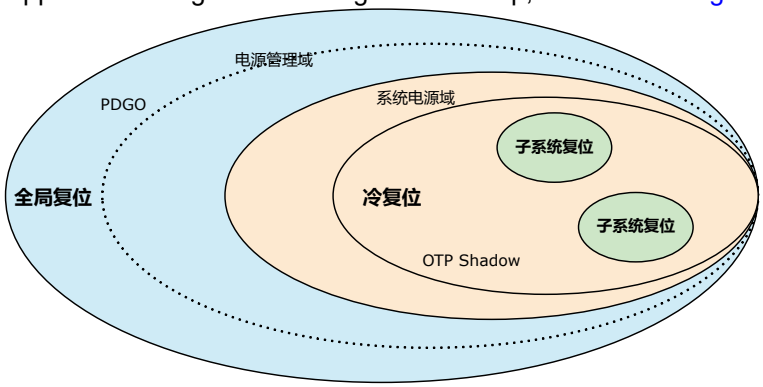


Figure 6: Reset System

The relationship between the range of each reset in the figure and the power domain structure is as follows:

- The PDGO module in the power management domain triggers a reset of the entire chip, namely global reset
- Reset of parts other than PDGO in the power management domain does not affect the PDGO module, but will trigger reset of the system power domain
- Reset of the system power domain does not affect the power management domain, but will reset all subsystems

9.2 Global Reset

Global reset resets the entire chip, and it is triggered in the following two scenarios:

- Power-up of the VPMC power pin
- The RESETN pin is pulled low by an external circuit

The RESETN pin is a dedicated IO.

The RESETN pin generates a global reset by default. You can modify its reset range by configuring the BDGO module, so that the state of PDGO in the power management domain is not affected. For details, see Section 11.3.

Power-up of the VPMC pin will reset the entire chip.

9.3 System Power Domain Reset

System power domain reset:

- Cold Reset: The system power domain retains the value of OTP shadow registers, and does not reset debug-related circuits.

The system power domain has multiple reset sources, including:

- VPMC supply voltage is lower than approximately 2.7V

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- VPMC supply voltage is lower than approximately 2.5V, this reset will also reset the PDGO area
- Debug Reset: Reset request issued by an external debugger
- Security Violation
- Watchdog timeout
- Software reset

In addition to resets triggered by the above reset sources, the software can also reset a specific subsystem individually by controlling the SYSCTL module. For details, see the relevant chapters of SYSCTL.

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10 Clock System

This chapter describes the clock system of this product.

10.1 Clock System Overview

The clock system is mainly composed of clock sources and functional clocks, and its structure is shown in Figure 7.

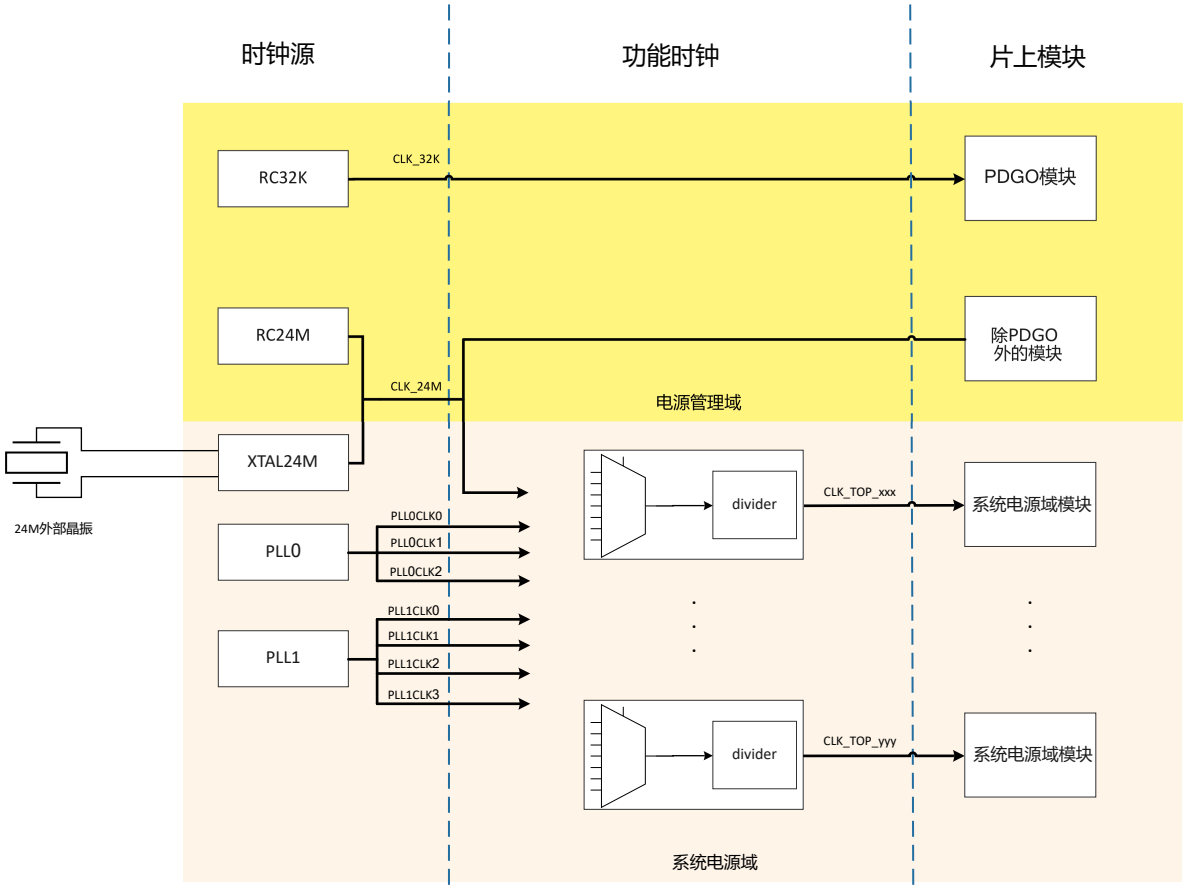


Figure 7: Clock System Block Diagram

Clock sources include external crystal oscillator XTAL, internal RC oscillator, Phase-Locked Loop (PLL), etc. Clock sources can generate clocks with different frequencies and different accuracies.

The functional clock CLK_TOP is the clock obtained after selecting and dividing the frequency of the clock source.

10.2 Clock Sources

10.2.1 32KHz Clock Source CLK_32K

Modules that generate CLK_32K clock include:

- RC32K - Internal 32KHz RC oscillator

CLK_32K directly provides clock for the PDGO module in the power management domain.

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10.2.2 24MHz Clock Source CLK_24M

Modules that generate the 24MHz clock source include:

- XTAL24M - 24MHz high-speed oscillator, supporting 24MHz external crystal, and also supporting external input of 24 MHz active clock through pins
- RC24M - Internal 24MHz RC oscillator

XTAL24M and RC24M together form the CLK_24M clock source. When XTAL24M is enabled and the frequency is stable, CLK_24M comes from the output clock of XTAL24M. When XTAL24M is turned off or not stable yet, CLK_24M comes from the output clock of RC24M. The switching between the two clocks is performed automatically.

XTAL24M is automatically enabled by default. For its on-off control in low-power mode, please refer to Chapter 13.

CLK_24M directly provides clocks for modules in the power management domain.

For some packages, the ROM will not start the XTAL24M external crystal, but directly use the internal RC24M as the PLL reference clock. Users are advised to note that they need to manually start the external crystal and switch the PLL reference clock through software according to their own situation later. For the PLL reference clock selection configuration register, please refer to Chapter 14.

10.2.3 Phase-Locked Loop (PLL)

The product has 2 built-in PLLs to generate various frequency clocks required by the system. Among them, PLL0 has 3 channels, and PLL1 has 4 independently divided clock outputs, so there are 7 PLL clock sources in total.

The PLL uses the XTAL24M clock as the reference clock by default. Each PLL has been preset with a default operating frequency, as shown in Table 16:

For the modification method of PLL frequency and frequency division coefficient, please refer to Chapter 14.

10.3 Functional Clock

The functional clock can be selected from multiple clock sources and divided according to the configuration, and then provides operating clocks for each module in the system power domain.

There are 8 clock sources in the system, so each functional clock has an 8-to-1 multiplexer for clock source selection. The correspondence between the selection value and the clock source is shown in Table 16.

Selection Value	Clock Source	Default Frequency	Description
0	CLK_24M	24MHz	24MHz clock source
1	PLL0CLK0	720MHz	Default clock source for the processor
2	PLL0CLK1	600MHz	
3	PLL0CLK2	400MHz	
4	PLL1CLK0	800MHz	
5	PLL1CLK1	666MHz	

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Selection Value	Clock Source	Default Frequency	Description
6	PLL1CLK2	500MHz	
7	PLL1CLK3	266MHz	

Table 16: Clock Source Selection and Default Frequency

Each functional clock can be set with a separate frequency division coefficient to divide the selected clock source. The frequency division coefficient ranges from any integer from 1 to 256.

For ease of use, each functional clock has been preset with clock source selection and frequency division coefficients. For details, see Table 17. For the modification method of clock source selection and frequency division coefficient, please refer to Section 13.2.

Functional Clock Name	Default Clock Source	Default Division Coefficient	Default Frequency	Application
CLK_TOP_HART0	PLL0CLK0	2	360MHz	CPU0 core/ ILM0/DLM0/FGPIO0
CLK_TOP_MCHTMR0	CLK24M	1	24MHz	MCHTMR0 timing
CLK_TOP_AHB	CLK_TOP_HART0	2	180MHz	Peripheral bus/HDMA/AHB SRAM/ Motor system/ACMP/GPIO
CLK_TOP_CAN0	PLL1CLK0	10	80MHz	CAN0 interface
CLK_TOP_CAN1	PLL1CLK0	10	80MHz	CAN1 interface
CLK_TOP_CAN2	PLL1CLK0	10	80MHz	CAN2 interface
CLK_TOP_CAN3	PLL1CLK0	10	80MHz	CAN3 interface
CLK_TOP_GPTMR0	PLL1CLK0	8	100MHz	GPTMR0 timing
CLK_TOP_GPTMR1	PLL1CLK0	8	100MHz	GPTMR1 timing
CLK_TOP_GPTMR2	PLL1CLK0	8	100MHz	GPTMR2 timing
CLK_TOP_GPTMR3	PLL1CLK0	8	100MHz	GPTMR3 timing
CLK_TOP_I2C0	PLL1CLK0	10	80MHz	I2C0 interface
CLK_TOP_I2C1	PLL1CLK0	10	80MHz	I2C1 interface
CLK_TOP_I2C2	PLL1CLK0	10	80MHz	I2C2 interface
CLK_TOP_I2C3	PLL1CLK0	10	80MHz	I2C3 interface
CLK_TOP_SPI0	PLL1CLK0	10	80MHz	SPI0 interface
CLK_TOP_SPI1	PLL1CLK0	10	80MHz	SPI1 interface
CLK_TOP_SPI2	PLL1CLK0	10	80MHz	SPI2 interface
CLK_TOP_SPI3	PLL1CLK0	10	80MHz	SPI3 interface
CLK_TOP_UART0	PLL1CLK0	10	80MHz	UART0 interface
CLK_TOP_UART1	PLL1CLK0	10	80MHz	UART1 interface

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Functional Clock Name	Default Clock Source	Default Division Coefficient	Default Frequency	Application
CLK_TOP_UART2	PLL1CLK0	10	80MHz	UART2 interface
CLK_TOP_UART3	PLL1CLK0	10	80MHz	UART3 interface
CLK_TOP_UART4	PLL1CLK0	10	80MHz	UART4 interface
CLK_TOP_UART5	PLL1CLK0	10	80MHz	UART5 interface
CLK_TOP_UART6	PLL1CLK0	10	80MHz	UART6 interface
CLK_TOP_UART7	PLL1CLK0	10	80MHz	UART7 interface
CLK_TOP_XPI0	PLL1CLK1	2	333MHz*	XPI0 interface
CLK_TOP_ANA0	PLL1CLK0	4	200MHz	ADC0 alternative clock
CLK_TOP_ANA1	PLL1CLK0	4	200MHz	ADC1 alternative clock
CLK_TOP_ANA2	PLL1CLK0	4	200MHz	DAC0 alternative clock
CLK_TOP_ANA3	PLL1CLK0	4	200MHz	DAC1 alternative clock
CLK_TOP_REF0	PLL1CLK2	10	50MHz	REF0 alternative clock
CLK_TOP_REF1	PLL1CLK2	10	50MHz	REF1 alternative clock
CLK_TOP_ADC0	CLK_TOP_AHB		180MHz	ADC0
CLK_TOP_ADC1	CLK_TOP_AHB		180MHz	ADC1
CLK_TOP_DAC0	CLK_TOP_AHB		180MHz	DAC0
CLK_TOP_DAC1	CLK_TOP_AHB		180MHz	DAC1

Table 17: Functional Clock Summary

*The actual frequency is related to the boot configuration.

Note that the functional clocks of ADC and DAC adopt a two-stage multiplexer structure to support any number of ADCs or DACs working synchronously or asynchronously.

The functional clock structure of ADC and DAC is shown in Figure 8.

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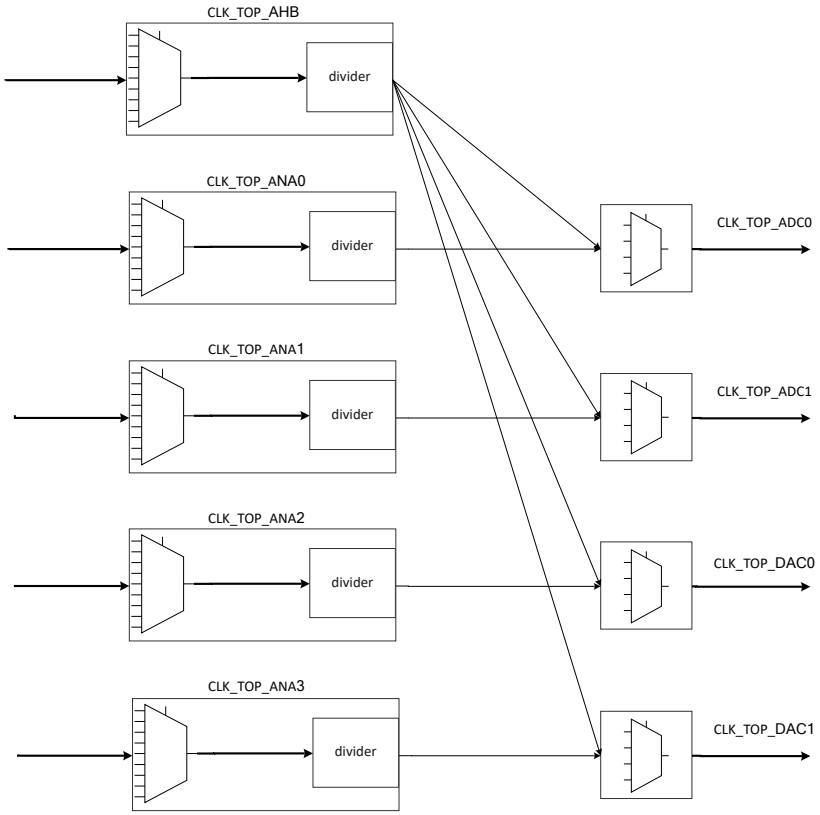


Figure 8: ADC Functional Clock Structure

The previous-stage functional clock selected by the multiplexer on CLK_TOP_ADCx and CLK_TOP_DACx is shown in Table 18.

Selection Value	**CLK_TOP_ADC0 Clock Selection**
0	CLK_TOP_AHB
1	CLK_TOP_ANA0
Selection Value	**CLK_TOP_ADC1 Clock Selection**
0	CLK_TOP_AHB
1	CLK_TOP_ANA1
Selection Value	**CLK_TOP_DAC0 Clock Selection**
0	CLK_TOP_AHB
1	CLK_TOP_ANA2
Selection Value	**CLK_TOP_DAC1 Clock Selection**
0	CLK_TOP_AHB
1	CLK_TOP_ANA3

Table 18: Two-Stage Functional Clock Selection

CLK_TOP_ADCx and CLK_TOP_DACx select CLK_TOP_AHB by default in sequence.

This structure enables any number of ADCs or DACs to work with the same clock or work at independent clock frequencies respectively.

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10.4 Modules Using Clock Sources Directly

Some modules in the system do not use functional clocks but are directly connected to clock sources, including all modules in the power management domain and some timing modules in the system power domain, which are listed in Table 19.

Module	Clock Source
Power Management Domain Modules	
PDGO	CLK_32K
PTMR	CLK_24M
PWDG	CLK_24M and CLK_32K
PUART	CLK_24M
PGPIO	CLK_24M
System Power Domain Modules	
WDG0~WDG1	CLK_32K and CLK_TOP_AHB
USB0	CLK_24M

Table 19: Module Clock List

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11 Power Management Domain Power On/Off Module (PDGO)

This chapter describes the main functions and usage of the PDGO module in the power management domain.

11.1 Main Features

PDGO is powered by 3.3V VPMC, and its main functions include:

- 32KHz clock source management
- Global reset pin RESETN control
- Power management domain power on/off control
- Supports software wakeup function: single wakeup or multiple wakeups
- Supports 4 groups of 32-bit general-purpose registers

11.2 CLK_32K Clock Source Management

There is an IRC32K clock oscillator module in the power management domain. After the power management domain is powered on, PDGO uses the IRC32K clock as the CLK_32K clock source to operate.

The accuracy of IRC32K can be calibrated via Fuse, and users can also calibrate the accuracy of IRC32K by setting the DGO_RC32K_CFG register.

11.3 Global Reset Pin RESETN Control

The global reset pin RESETN is a dedicated pin. Users can set the RETENTION bit of the DGO_CTR0 register to 1, so that the pin reset only resets the system power domain and the parts of the power management domain except PDGO, without affecting the state of PDGO.

In addition, the global reset pin can also be used as a wakeup source to exit the shutdown mode, see Section 11.4.2 for details.

11.4 Power Management Domain Power On/Off Control

PDGO can control the entry and exit of shutdown mode. In shutdown mode, the entire system power domain and the parts of the power management domain except the PDGO module are in reset or power-down state, and only PDGO remains working. The system has extremely low power consumption in this mode.

11.4.1 Entering Shutdown Mode

Entering shutdown mode is triggered by software operation. Write a non-zero count value to the COUNTER bit of the DGO_TURNOFF register, then the counter decrements by 1 every 24M clock cycle until the counter reaches 1, and the system enters shutdown mode. After entering shutdown mode, this register will be reset.

11.4.2 Exiting Shutdown Mode

The ways to exit shutdown mode are as follows:

- Software countdown single wakeup
- Software countdown multiple wakeups
- Wakeup pin WAKEUP

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- Reset pin RESETN

Software countdown single wakeup: Before the shutdown operation, users need to set the AUTO_SYS_WAKEUP bit of the DGO_CTRL1 register to 0, and write a non-zero count value to the WAKEUP_COUNTER bit of the DGO_CTRL3 register. The counter starts decrementing by 1 every 32KHz clock cycle from the moment WAKEUP_COUNTER is a non-zero value, until the counter reaches 2, a wakeup event is generated, and the hardware automatically clears the WAKEUP_COUNTER bit of the DGO_CTRL3 register.

Note that the wakeup countdown does not start after shutdown, so users need to control the time difference between the shutdown countdown and the wakeup countdown properly.

Software countdown multiple wakeups: Before the shutdown operation, users need to set the AUTO_SYS_WAKEUP bit of the DGO_CTRL1 register to 1, and write a non-zero count value to the WAKEUP_COUNTER bit of the DGO_CTRL3 register. The counter starts decrementing by 1 every 32KHz clock cycle from the moment WAKEUP_COUNTER is a non-zero value, until the counter reaches 2, a wakeup event is generated. When the counter reaches 0, it starts decrementing by 1 every 32KHz clock cycle from WAKEUP_COUNTER again, until the counter reaches 2, another wakeup event is generated. This cycles until the user manually clears the AUTO_SYS_WAKEUP bit of the DGO_CTRL1 register to stop generating wakeup events, and the hardware automatically clears the WAKEUP_COUNTER bit of the DGO_CTRL3 register at the same time.

Wakeup pin WAKEUP: When the system is in shutdown mode, a valid high level detected on the wakeup pin WAKEUP can generate a wakeup event to exit shutdown mode.

Reset pin RESETN: When the system is in shutdown mode, a valid low level detected on the reset pin RESETN can generate a wakeup event to exit shutdown mode.

Note: Wakeup events only take effect after the system enters shutdown mode.

11.5 General-Purpose Registers

PDGO has 4 groups of 32-bit general-purpose GPR registers, which support saving user data when RESETN reset occurs. In this case, the RETENTION bit field of the DGO_CTR0 register needs to be set to 1 in advance. Note that it is also necessary to ensure that PDGO is powered on.

11.6 PDGO Register Description

The register list of PDGO is as follows:

PDGO base address: 0xF4134000

Offset	Register Name	Description	Reset Value
0x0000	DGO_TURNOFF	trunoff control	0x00000000
0x0004	DGO_RC32K_CFG	RC32K CLOCK	0x00000000
0x0600	DGO_GPR00	Generic control 0	0x00000000
0x0604	DGO_GPR01	Generic control 1	0x00000000
0x0608	DGO_GPR02	Generic control 2	0x00000000
0x060C	DGO_GPR03	Generic control 3	0x00000000
0x0700	DGO_CTR0	control register 0	0x00000000
0x0704	DGO_CTR1	control register 1	0x00000000

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Offset	Register Name	Description	Reset Value
0x0708	DGO_CTR2	control register 2	0x00000000
0x070C	DGO_CTR3	control register 3	0x00000000
0x0710	DGO_CTR4	control register 4	0x00000000

Table 20: PDGO Register List

11.7 PDGO Register Detailed Information

The detailed description of PDGO registers is as follows:

11.7.1 DGO_TURNOFF (0x0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
COUNTER																															
WO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DGO_TURNOFF [31:0]

Bits	Name	Description
31-0	COUNTER	trunoff counter, counter stops when it counts down to 0, the trunoff occurs when the counter value is 1.

DGO_TURNOFF Bits

11.7.2 DGO_RC32K_CFG (0x4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IRC_TRIMMED	RSVD							CAPEX7_TRIM	CAPEX6_TRIM	RSVD													CAP_TRIM								
RW	N/A							RW	RW	N/A													RW								
0	x	x	x	x	x	x	x	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0

DGO_RC32K_CFG [31:0]

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Bits	Name	Description
31	IRC_TRIMMED	IRC32K trim happened, this bit set by hardware after trim value loaded, and stop load, write 0 will clear this bit and reload trim value 0: irc is not trimmed 1: irc is trimmed
23	CAPEX7_TRIM	IRC32K bit 7
22	CAPEX6_TRIM	IRC32K bit 6
8-0	CAP_TRIM	capacitor trim bits

DGO_RC32K_CFG Bits

11.7.3 DGO_GPR00 (0x600)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GPR																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DGO_GPR00 [31:0]

Bits	Name	Description
31-0	GPR	Generic control

DGO_GPR00 Bits

11.7.4 DGO_GPR01 (0x604)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GPR																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

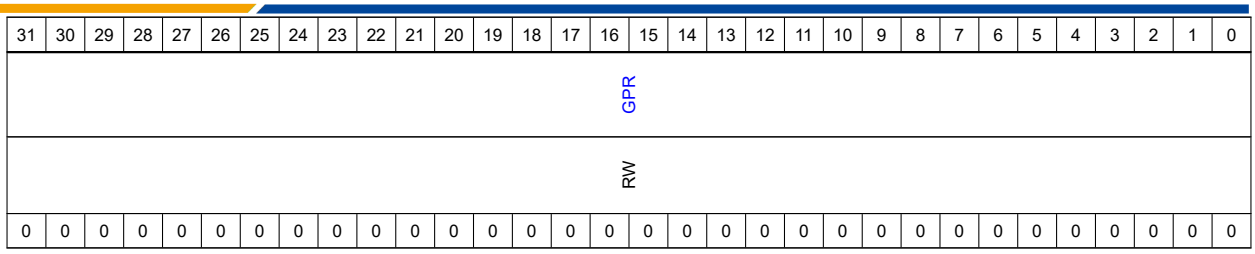
DGO_GPR01 [31:0]

Bits	Name	Description
31-0	GPR	Generic control

DGO_GPR01 Bits

11.7.5 DGO_GPR02 (0x608)

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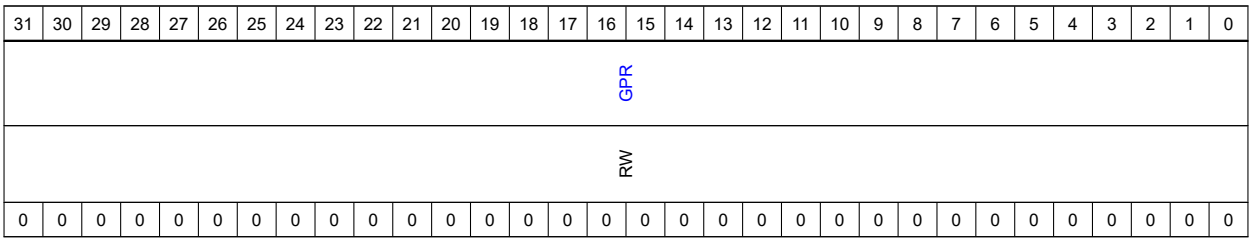


DGO_GPR02 [31:0]

Bits	Name	Description
31-0	GPR	Generic control

DGO_GPR02 Bits

11.7.6 DGO_GPR03 (0x60C)

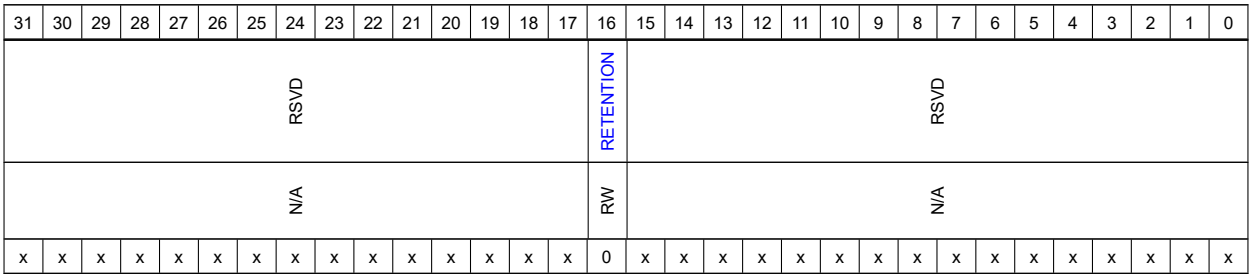


DGO_GPR03 [31:0]

Bits	Name	Description
31-0	GPR	Generic control

DGO_GPR03 Bits

11.7.7 DGO_CTR0 (0x700)



DGO_CTR0 [31:0]

Bits	Name	Description
16	RETENTION	dgo register status retenion

DGO_CTR0 Bits

11.7.8 DGO_CTR1 (0x704)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AOTO_SYS_WAKEUP	RSVD														WAKEUP_EN	RSVD															PIN_WAKEUP_STATUS
RW	N/A														RW	N/A															RO
0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

DGO_CTR1 [31:0]

Bits	Name	Description
31	AOTO_SYS_WAKEUP	software wakeup：0：wakeup once；1：auto wakeup Continuously
16	WAKEUP_EN	permit wakeup pin or software wakeup
0	PIN_WAKEUP_STATUS	wakeup pin status

DGO_CTR1 Bits

11.7.9 DGO_CTR2 (0x708)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD							RESETN_PULLUP_DISABLE	RSVD							WAKEUP_PULLDN_DISABLE	RSVD															
N/A							RW	N/A							RW	N/A															
x	x	x	x	x	x	x	0	x	x	x	x	x	x	x	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

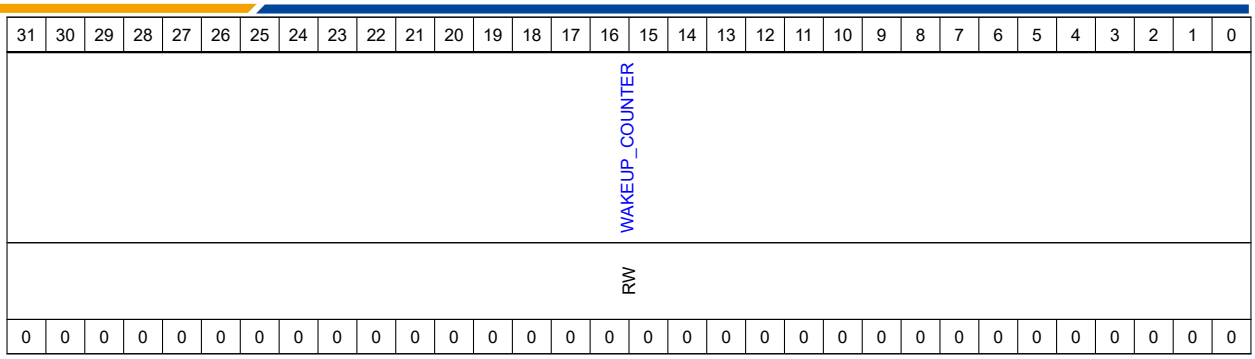
DGO_CTR2 [31:0]

Bits	Name	Description
24	RESETN_PULLUP_DISABLE	resetn pin pull up disable
16	WAKEUP_PULLDN_DISABLE	wakeup pin pull down disable

DGO_CTR2 Bits

11.7.10 DGO_CTR3 (0x70C)

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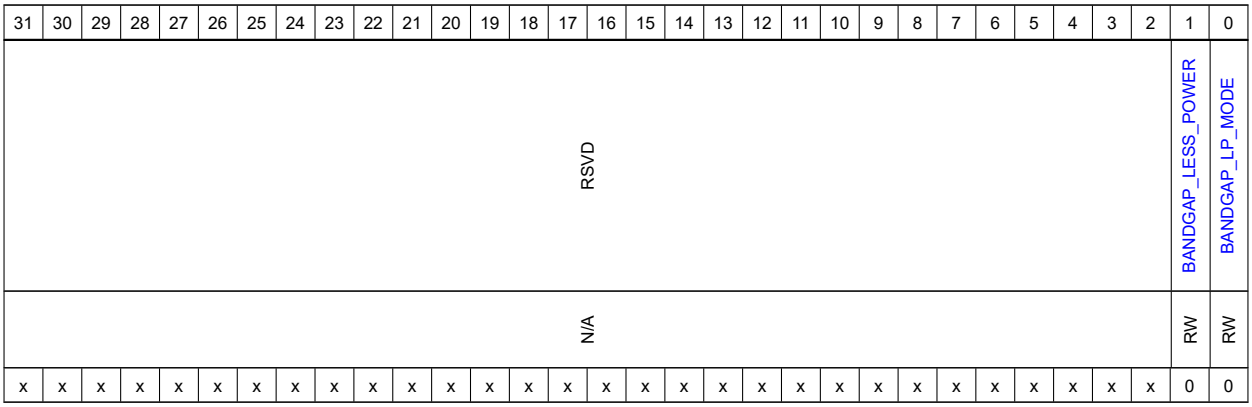


DGO_CTR3 [31:0]

Bits	Name	Description
31-0	WAKEUP_COUNTER	software wakeup counter

DGO_CTR3 Bits

11.7.11 DGO_CTR4 (0x710)



DGO_CTR4 [31:0]

Bits	Name	Description
1	BANDGAP_LESS_POWER	Banggap work in power save mode, banggap function normally 0: banggap works in high performance mode 1: banggap works in power saving mode
0	BANDGAP_LP_MODE	Banggap work in low power mode, banggap function limited 0: banggap works in normal mode 1: banggap works in low power mode

DGO_CTR4 Bits

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12 Power Management Domain Configuration Module PCFG

This section describes the main features of the power domain manager PCFG.

12.1 Feature Summary

PCFG is an integrated control module for power supplies, clocks and analog peripherals in the power management domain. Users can implement the following functions through PCFG:

- Linear regulator LDOPMC control
- Linear regulator LDOOTP control
- Switching power supply DCDC configuration
- Power management domain gated clock management

12.2 Functional Description

This section describes the functions of PCFG.

12.2.1 Linear Regulator LDOPMC Configuration

The linear regulator LDOPMC supplies power to all modules in the power management domain except PDGO. It is enabled by default, and supports software forced shutdown or automatic hardware shutdown in shutdown mode.

12.2.2 Linear Regulator LDOOTP Configuration

The linear regulator LDOOTP supplies power to the on-chip one-time programmable memory OTP. Before programming OTP, you must first enable the ENABLE bit of the LDO2P5 register and wait for the READY bit to be valid. After OTP programming is completed, the ENABLE bit must be cleared to turn off this LDO.

12.2.3 Switching Power Supply DCDC Configuration

The switching power supply DCDC can be used to supply power to the system power domain. The output voltage of DCDC is set through the VOLT bit of the DCDC_MODE register, with a default voltage of 1.15V.

DCDC is enabled by default. When the chip uses an external power supply to directly power VDD_SOC, set the MODE bit of the DCDC_MODE register to 0 to turn off DCDC.

When the system enters sleep mode (see Section 15.4 for details), DCDC can also be turned off or enter a low-power state accordingly. The related control is implemented through the TRAP and RETENTION bits of the POWER_TRAP register, and their combination relationship is shown in Table 21:

TRAP Bit	RETENTION Bit	Setting
0	-	DCDC working state remains unchanged when the system is in sleep mode
1	0	DCDC is turned off when the system is in sleep mode

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TRAP Bit	RETENTION Bit	Setting
1	1	DCDC enters low-power mode when the system is in sleep mode

Table 21: DCDC Low Power State Control

When DCDC enters low-power mode, the output voltage is set in the STBY_VOLT bit of the DCDC_LPMODE register. A lower voltage can be set according to the application, for example, the voltage value for maintaining the content of the memory in the system power domain.

DCDC supports obtaining the current current intensity. After enabling the current measurement function, it can be read from the LEVEL bit of the DCDC_CURRENT register.

12.2.4 Clock Gating of Power Management Domain Modules

Users can turn on or off the clock of some modules in the power management domain through the SCG_CTRL register.

- PGPIO
- PIOC
- PTMR
- PWDG
- PUART

12.2.5 System Power Domain Wakeup

The wakeup sources in the power management domain are:

- WAKEUP input pin
- PTMR interrupt
- PUART interrupt
- PWDG interrupt
- PGPIO interrupt

The above wakeup sources can be allowed to wake up the system through the WAKE_MASK register of the PCFG module, and the wakeup source status can be read and cleared in the WAKE_CAUSE register.

12.2.6 Power Management Domain Reset Control Module PPOR

PPOR manages the reset of the system power domain.

Users can enable or disable these resets through the RESET$_{ENABLE}$ register.

Users can view the source of the last reset through the RESET$_{FLAG}$ register.

The RESET$_{COLD}$ register is used to control whether each reset source generates a cold reset or a warm reset.

Users can configure the software reset countdown counter through the COUNTER bit field of SOFTWARE_RESET. Users can configure COUNTER arbitrarily. When the counter counts down to 2, a software reset is generated. Writing 0 to COUNTER can turn off the software reset and cancel the ongoing software reset countdown.

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12.3 PCFG Register Description

The register list of PCFG is as follows:

PCFG base address: 0xF4104000

Offset	Register Name	Description	Reset Value
0x0000	BANDGAP	BANGGAP control	0x00101010
0x0004	LDO1P1	1V LDO config	0x0000044C
0x0008	LDO2P5	2.5V LDO config	0x000009C4
0x0010	DCDC_MODE	DCDC mode select	0x0001047E
0x0014	DCDC_LPMODE	DCDC low power mode	0x00000384
0x0018	DCDC_PROT	DCDC protection	0x00000010
0x001C	DCDC_CURRENT	DCDC current estimation	0x00000000
0x0020	DCDC_ADVMODE	DCDC advance setting	0x03120040
0x0024	DCDC_ADVPARAM	DCDC advance parameter	0x00006E1C
0x0028	DCDC_MISC	DCDC misc parameter	0x00070100
0x002C	DCDC_DEBUG	DCDC Debug	0x00005DBF
0x0030	DCDC_START_TIME	DCDC ramp time	0x0001193F
0x0034	DCDC_RESUME_TIME	DCDC resume time	0x00008C9F
0x0040	POWER_TRAP	SOC power trap	0x00000000
0x0044	WAKE_CAUSE	Wake up source	0x00000000
0x0048	WAKE_MASK	Wake up mask	0x00000000
0x004C	SCG_CTRL	Clock gate control in PMIC	0xFFFFFFFF
0x0060	RC24M	RC 24M config	0x00000310
0x0064	RC24M_TRACK	RC 24M track mode	0x00000000
0x0068	TRACK_TARGET	RC 24M track target	0x00000000
0x006C	STATUS	RC 24M track status	0x00000000

Table 22: PCFG Register List

12.4 PCFG Register Detailed Information

The detailed description of PCFG registers is as follows:

12.4.1 BANDGAP (0x0)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VBG_TRIMMED	RSVD										VBG_1P0_TRIM					RSVD			VBG_P65_TRIM					RSVD			VBG_P50_TRIM				
RW	N/A										RW					N/A			RW					N/A			RW				
0	x	x	x	x	x	x	x	x	x	x	1	0	0	0	0	x	x	x	1	0	0	0	0	x	x	x	1	0	0	0	0

BANDGAP [31:0]

Bits	Name	Description
31	VBG_TRIMMED	Bandgap trim happened, this bit set by hardware after trim value loaded, and stop load, write 0 will clear this bit and reload trim value 0: bandgap is not trimmed 1: bandgap is trimmed
20-16	VBG_1P0_TRIM	Banggap 1.0V output trim value
12-8	VBG_P65_TRIM	Banggap 1.0V output trim value
4-0	VBG_P50_TRIM	Banggap 1.0V output trim value

BANDGAP Bits

12.4.2 LDO1P1 (0x4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																				VOLT											
N/A																				RW											
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	1	0	0	0	1	0	0	1	1	0	0

LDO1P1 [31:0]

Bits	Name	Description
11-0	VOLT	LDO output voltage in mV, value valid through 700-1320, , step 20mV. Hardware select voltage no less than target if not on valid steps, with maximum 1320mV. 700: 700mV 720: 720mV . . . 1320:1320mV

LDO1P1 Bits

12.4.3 LDO2P5 (0x8)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD			READY	RSVD											ENABLE	RSVD				VOLT											
N/A			RO	N/A											RW	N/A				RW											
x	x	x	0	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	x	1	0	0	1	1	1	0	0	0	1	0	0

LDO2P5 [31:0]

Bits	Name	Description
28	READY	Ready flag, will set 1ms after enabled or voltage change 0: LDO is not ready for use 1: LDO is ready
16	ENABLE	LDO enable 0: turn off LDO 1: turn on LDO
11-0	VOLT	LDO output voltage in mV, value valid through 2125-2900, step 25mV. Hardware select voltage no less than target if not on valid steps, with maximum 2900mV. 2125: 2125mV 2150: 2150mV . . . 2900:2900mV

LDO2P5 Bits

12.4.4 DCDC_MODE (0x10)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD			READY	RSVD									MODE			RSVD				VOLT											
N/A			RO	N/A									RW			N/A				RW											
x	x	x	0	x	x	x	x	x	x	x	x	x	0	0	1	x	x	x	x	0	1	0	0	0	1	1	1	1	1	1	0

DCDC_MODE [31:0]

Bits	Name	Description
28	READY	Ready flag 0: DCDC is applying new change 1: DCDC is ready

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Bits	Name	Description
18-16	MODE	DCDC work mode XX0: trun off 001: basic mode 011: generic mode 101: automatic mode 111: expert mode
11-0	VOLT	DCDC voltage in mV in normal mode, value valid through 600-1375, , step 25mV. Hardware select voltage no less than target if not on valid steps, with maximum 1375mV. 600: 600mV 625: 625mV . . . 1375:1375mV

DCDC_MODE Bits

12.4.5 DCDC_LPMODE (0x14)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																				STBY_VOLT											
N/A																				RW											
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	1	1	1	0	0	0	0	1	0	0

DCDC_LPMODE [31:0]

Bits	Name	Description
11-0	STBY_VOLT	DCDC voltage in mV in standby mode, , value valid through 600-1375, , step 25mV. Hardware select voltage no less than target if not on valid steps, with maximum 1375mV. 600: 600mV 625: 625mV . . . 1375:1375mV

DCDC_LPMODE Bits

12.4.6 DCDC_PROT (0x18)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD			ILIMIT_LP	RSVD			OVERLOAD_LP	RSVD							POWER_LOSS_FLAG	DISABLE_OVERVOLTAGE	RSVD						OVERVOLT_FLAG	DISABLE_SHORT	RSVD		SHORT_CURRENT	RSVD			SHORT_FLAG
N/A			RW	N/A			RO	N/A							RO	RW	N/A						RO	RW	N/A		RW	N/A			RO
x	x	x	0	x	x	x	0	x	x	x	x	x	x	x	0	0	x	x	x	x	x	x	0	0	x	x	1	x	x	x	0

DCDC_PROT [31:0]

Bits	Name	Description
28	ILIMIT_LP	over current setting for low power mode 0:250mA 1:200mA
24	OVERLOAD_LP	over current in low power mode 0: current is below setting 1: overcurrent happened in low power mode
16	POWER_LOSS_FLAG	power loss 0: input power is good 1: input power is too low
15	DISABLE_OVERVOLTAGE	ouput over voltage protection 0: protection enabled, DCDC will shut down is output voltage is unexpected high 1: protection disabled, DCDC continue to adjust output voltage
8	OVERVOLT_FLAG	output over voltage flag 0: output is normal 1: output is unexpected high
7	DISABLE_SHORT	disable output short circuit protection 0: short circuits protection enabled, DCDC shut down if short circuit on ouput detected 1: short circuit protection disabled
4	SHORT_CURRENT	short circuit current setting 0: 2.0A, 1: 1.3A
0	SHORT_FLAG	short circuit flag 0: current is within limit 1: short circuits detected

DCDC_PROT Bits

12.4.7 DCDC_CURRENT (0x1C)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																ESTI_EN	RSVD						VALID	RSVD			LEVEL				
N/A																RW	N/A						RO	N/A			RO				
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	x	x	x	0	x	x	x	0	0	0	0	0

DCDC_CURRENT [31:0]

Bits	Name	Description
15	ESTI_EN	enable current measure
8	VALID	Current level valid 0: data is invalid 1: data is valid
4-0	LEVEL	DCDC current level, current level is num * 50mA

DCDC_CURRENT Bits

12.4.8 DCDC_ADVMODE (0x20)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD					EN_RCSCALE			RSVD		DC_C		DC_R				RSVD									EN_FF_DET	EN_FF_LOOP	EN_AUTOLP	EN_DCM_EXIT	EN_SKIP	EN_IDLE	EN_DCM
N/A					RW			N/A		RW		RW				N/A									RW	RW	RW	RW	RW	RW	RW
x	x	x	x	x	0	1	1	x	x	0	1	0	0	1	0	x	x	x	x	x	x	x	x	x	1	0	0	0	0	0	0

DCDC_ADVMODE [31:0]

Bits	Name	Description
26-24	EN_RCSCALE	Enable RC scale
21-20	DC_C	Loop C number
19-16	DC_R	Loop R number
6	EN_FF_DET	enable feed forward detect 0: feed forward detect is disabled 1: feed forward detect is enabled
5	EN_FF_LOOP	enable feed forward loop 0: feed forward loop is disabled 1: feed forward loop is enabled
4	EN_AUTOLP	enable auto enter low power mode 0: do not enter low power mode 1: enter low power mode if current is detected low

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Bits	Name	Description
3	EN_DCM_EXIT	avoid over voltage 0: stay in DCM mode when voltage excess 1: change to CCM mode when voltage excess
2	EN_SKIP	enable skip on narrow pulse 0: do not skip narrow pulse 1: skip narrow pulse
1	EN_IDLE	enable skip when voltage is higher than threshold 0: do not skip 1: skip if voltage is excess
0	EN_DCM	DCM mode 0: CCM mode 1: DCM mode

DCDC_ADVMODE Bits

12.4.9 DCDC_ADVPARAM (0x24)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																	MIN_DUT							RSVD	MAX_DUT						
N/A																	RW							N/A	RW						
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	1	1	0	1	1	1	0	x	0	0	1	1	1	0	0

DCDC_ADVPARAM [31:0]

Bits	Name	Description
14-8	MIN_DUT	minimum duty cycle
6-0	MAX_DUT	maximum duty cycle

DCDC_ADVPARAM Bits

12.4.10 DCDC_MISC (0x28)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD			EN_HYST	RSVD		HYST_SIGN	HYST_THRS	RSVD			RC_SCALE	RSVD	DC_FF			RSVD						OL_THRE		RSVD			OL_HYST	RSVD	DELAY	CLK_SEL	EN_STEP
N/A			RW	N/A		RW	RW	N/A			RW	N/A	RW			N/A						RW		N/A			RW	N/A	RW	RW	RW
x	x	x	0	x	x	0	0	x	x	x	0	x	1	1	1	x	x	x	x	x	x	0	1	x	x	x	0	x	0	0	0

DCDC_MISC [31:0]

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Bits	Name	Description
28	EN_HYST	hysteres enable
25	HYST_SIGN	hysteres sign
24	HYST_THRS	hysteres threshold
20	RC_SCALE	Loop RC scale threshold
18-16	DC_FF	Loop feed forward number
9-8	OL_THRE	overload threshold in low power mode
4	OL_HYST	voltage ripple threshold in low power mode 0: 12.5mV 1: 25mV
2	DELAY	enable delay 0: delay disabled, 1: delay enabled
1	CLK_SEL	clock selection 0: select DCDC internal oscillator 1: select RC24M oscillator
0	EN_STEP	enable stepping in voltage change 0: stepping disabled 1: steping enabled

DCDC_MISC Bits

12.4.11 DCDC_DEBUG (0x2C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD												UPDATE_TIME																			
N/A												RW																			
x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	1	0	1	1	1	0	1	1	0	1	1	1	1	1	1

DCDC_DEBUG [31:0]

Bits	Name	Description
19-0	UPDATE_TIME	DCDC voltage change time in 24M clock cycles, default value is 1mS

DCDC_DEBUG Bits

12.4.12 DCDC_START_TIME (0x30)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD												START_TIME																			
N/A												RW																			
x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	1	0	0	0	1	1	0	0	1	0	0	1	1	1	1	1	1

DCDC_START_TIME [31:0]

Bits	Name	Description
19-0	START_TIME	Start delay for DCDC to turn on, in 24M clock cycles, default value is 3mS

DCDC_START_TIME Bits

12.4.13 DCDC_RESUME_TIME (0x34)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD												RESUME_TIME																			
N/A												RW																			
x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	1	0	0	0	1	1	0	0	1	0	0	1	1	1	1	1

DCDC_RESUME_TIME [31:0]

Bits	Name	Description
19-0	RESUME_TIME	Resume delay for DCDC to recover from low power mode, in 24M clock cycles, default value is 1.5mS

DCDC_RESUME_TIME Bits

12.4.14 POWER_TRAP (0x40)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TRIGGERED	RSVD														RETENTION	RSVD															TRAP
RW	N/A														RW	N/A															RW
0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

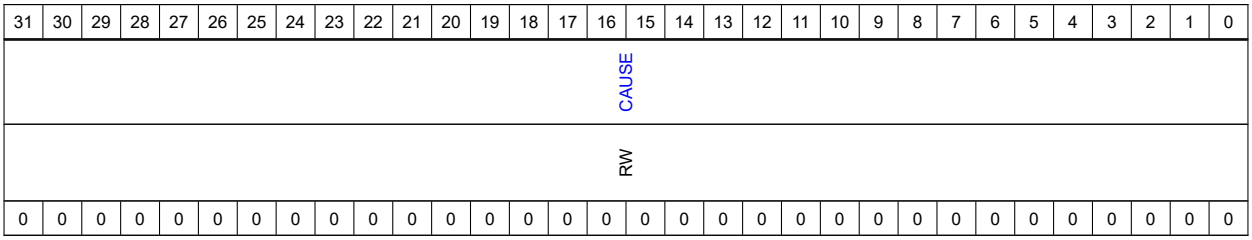
POWER_TRAP [31:0]

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Bits	Name	Description
31	TRIGGERED	Low power trap status, thit bit will set when power related low power flow triggered, write 1 to clear this flag. 0: low power trap is not triggered 1: low power trap triggered
16	RETENTION	DCDC enter standby mode, which will reduce voltage for memory content retention 0: Shutdown DCDC 1: reduce DCDC voltage
0	TRAP	Enable trap of SOC power supply, trap is used to hold SOC in low power mode for DCDC to enter further low power mode, this bit will self-clear when power related low pwer flow triggered 0: trap not enabled, pmic side low power function disabled 1: trap enabled, STOP operation leads to PMIC low power flow if SOC is not retentioned.

POWER_TRAP Bits

12.4.15 WAKE_CAUSE (0x44)



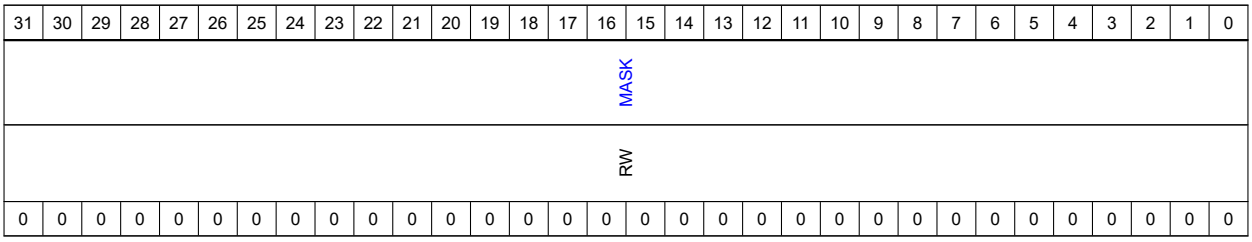
WAKE_CAUSE [31:0]

Bits	Name	Description
31-0	CAUSE	wake up cause, each bit represents one wake up source, write 1 to clear the register bit 0: wake up source is not active during last wakeup 1: wake up source is active furing last wakeup bit 0: pmic_enable bit 7: UART interrupt bit 8: TMR interrupt bit 9: WDG interrupt bit10: GPIO in PMIC interrupt bit31: pin wakeup

WAKE_CAUSE Bits

12.4.16 WAKE_MASK (0x48)

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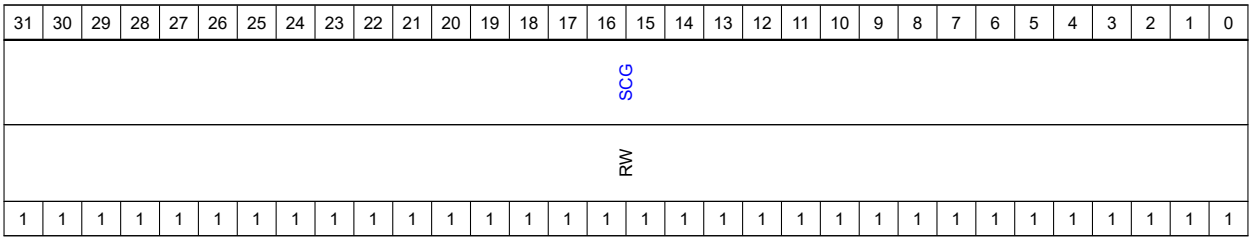


WAKE_MASK [31:0]

Bits	Name	Description
31-0	MASK	mask for wake up sources, each bit represents one wakeup source 0: allow source to wake up system 1: disallow source to wakeup system bit 0: pmic_enable bit 7: UART interrupt bit 8: TMR interrupt bit 9: WDG interrupt bit10: GPIO in PMIC interrupt bit31: pin wakeup

WAKE_MASK Bits

12.4.17 SCG_CTRL (0x4C)



SCG_CTRL [31:0]

Bits	Name	Description
31-0	SCG	control whether clock being gated during PMIC low power flow, 2 bits for each peripheral 00,01: reserved 10: clock is always off 11: clock is always on bit6-7:gpio bit8-9:ioc bit10-11: timer bit12-13:wdog bit14-15:uart

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Bits	Name	Description

SCG_CTRL Bits

12.4.18 RC24M (0x60)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RC_TRIMMED	RSVD																				TRIM_C			RSVD			TRIM_F				
RW	N/A																				RW			N/A			RW				
0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	1	1	x	x	x	1	0	0	0	0

RC24M [31:0]

Bits	Name	Description
31	RC_TRIMMED	RC24M trim happened, this bit set by hardware after trim value loaded, and stop load, write 0 will clear this bit and reload trim value 0: RC is not trimmed 1: RC is trimmed
10-8	TRIM_C	Coarse trim for RC24M, bigger value means faster
4-0	TRIM_F	Fine trim for RC24M, bigger value means faster

RC24M Bits

12.4.19 RC24M_TRACK (0x64)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD															SEL24M	RSVD											RETURN	RSVD			TRACK
N/A															RW	N/A											RW	N/A			RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	0

RC24M_TRACK [31:0]

Bits	Name	Description
16	SEL24M	Select track reference 0: select 32K as reference 1: select 24M XTAL as reference
4	RETURN	Retrun default value when XTAL loss 0: remain last tracking value 1: switch to default value

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Bits	Name	Description
0	TRACK	track mode 0: RC24M free running 1: track RC24M to external XTAL

RC24M_TRACK Bits

12.4.20 TRACK_TARGET (0x68)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PRE_DIV																TARGET															
RW																RW															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TRACK_TARGET [31:0]

Bits	Name	Description
31-16	PRE_DIV	Divider for reference source
15-0	TARGET	Target frequency multiplier of divided source

TRACK_TARGET Bits

12.4.21 STATUS (0x6C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD											SEL32K	RSVD			SEL24M	EN_TRIM	RSVD				TRIM_C		RSVD			TRIM_F				
N/A											RO	N/A			RO	RO	N/A				RO		N/A			RO				
x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	0	0	x	x	x	x	0	0	0	x	x	x	0	0	0	0	0

STATUS [31:0]

Bits	Name	Description
20	SEL32K	track is using XTAL32K 0: track is not using XTAL32K 1: track is using XTAL32K
16	SEL24M	track is using XTAL24M 0: track is not using XTAL24M 1: track is using XTAL24M
15	EN_TRIM	default value takes effect 0: default value is invalid 1: default value is valid

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Bits	Name	Description
10-8	TRIM_C	default coarse trim value
4-0	TRIM_F	default fine trim value

STATUS Bits

12.5 PPOR Register Description

The register list of PPOR is as follows:

PPOR base address: 0xF4100000

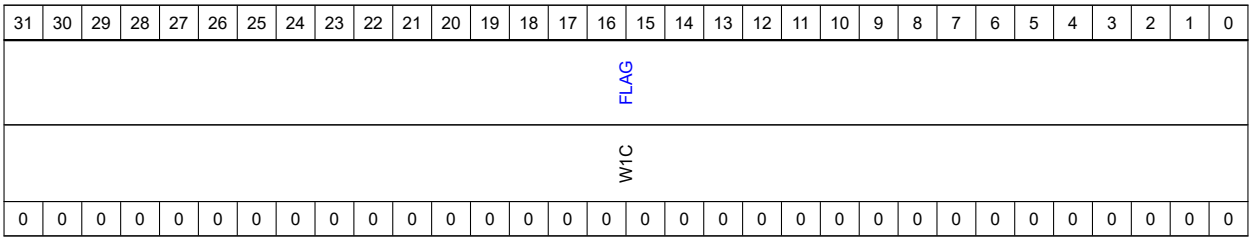
Offset	Register Name	Description	Reset Value
0x0000	RESET_FLAG	flag indicate reset source	0x00000000
0x0004	RESET_STATUS	reset source status	0x00000000
0x0008	RESET_HOLD	reset hold attribute	0x00000000
0x000C	RESET_ENABLE	reset source enable	0xFFFFFFFF
0x0010	RESET_TYPE	reset type triggered by reset	0x00000000
0x001C	SOFTWARE_RESET	Software reset counter	0x00000000

Table 23: PPOR Register List

12.6 PPOR Register Detailed Information

The detailed description of PPOR registers is as follows:

12.6.1 RESET_FLAG (0x0)



RESET_FLAG [31:0]

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Bits	Name	Description
31-0	FLAG	reset reason of last hard reset, write 1 to clear each bit 0: brownout 1: temperature 4: debug reset 5: jtag soft reset 8: cpu0 lockup(not available) 9: cpu1 lockup(not available) 10: cpu0 request(not available) 11: cpu1 request(not available) 16: watch dog 0 17: watch dog 1 18: watch dog 2(not available) 19: watch dog 3(not available) 24: pmic watch dog 30: jtag ieee reset 31: software

RESET_FLAG Bits

12.6.2 RESET_STATUS (0x4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
STATUS																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

RESET_STATUS [31:0]

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Bits	Name	Description
31-0	STATUS	current status of reset sources 0: brownout 1: temperature 4: debug reset 5: jtag soft reset 8: cpu0 lockup(not available) 9: cpu1 lockup(not available) 10: cpu0 request(not available) 11: cpu1 request(not available) 16: watch dog 0 17: watch dog 1 18: watch dog 2(not available) 19: watch dog 3(not available) 24: pmic watch dog 30: jtag ieee reset 31: software

RESET_STATUS Bits

12.6.3 RESET_HOLD (0x8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HOLD																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

RESET_HOLD [31:0]

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Bits	Name	Description
31-0	HOLD	hold arrtibute, when set, SOC keep in reset status until reset source release, or, reset will be released after SOC enter reset status 0: brownout 1: temperature 4: debug reset 5: jtag soft reset 8: cpu0 lockup(not available) 9: cpu1 lockup(not available) 10: cpu0 request(not available) 11: cpu1 request(not available) 16: watch dog 0 17: watch dog 1 18: watch dog 2(not available) 19: watch dog 3(not available) 24: pmic watch dog 30: jtag ieee reset 31: software

RESET_HOLD Bits

12.6.4 RESET_ENABLE (0xC)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ENABLE																															
RW																															
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

RESET_ENABLE [31:0]

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Bits	Name	Description
31-0	ENABLE	enable of reset sources 0: brownout 1: temperature 4: debug reset 5: jtag soft reset 8: cpu0 lockup(not available) 9: cpu1 lockup(not available) 10: cpu0 request(not available) 11: cpu1 request(not available) 16: watch dog 0 17: watch dog 1 18: watch dog 2(not available) 19: watch dog 3(not available) 24: pmic watch dog 30: jtag ieee reset 31: software

RESET_ENABLE Bits

12.6.5 RESET_TYPE (0x10)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TYPE																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

RESET_TYPE [31:0]

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Bits	Name	Description
31-0	TYPE	reset type of reset sources, 0 for cold reset, all system control setting cleared except debug/fuse/ioc; 1 for hot reset, keep system control setting and debug/fuse/ioc setting, only clear some subsystem 0: brownout 1: temperature 4: debug reset 5: jtag soft reset 8: cpu0 lockup(not available) 9: cpu1 lockup(not available) 10: cpu0 request(not available) 11: cpu1 request(not available) 16: watch dog 0 17: watch dog 1 18: watch dog 2(not available) 19: watch dog 3(not available) 24: pmic watch dog 30: jtag ieee reset 31: software

RESET_TYPE Bits

12.6.6 SOFTWARE_RESET (0x1C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
COUNTER																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

SOFTWARE_RESET [31:0]

Bits	Name	Description
31-0	COUNTER	counter decrease in 24MHz and stop at 0, trigger reset when value reach 2, software can write 0 to cancel reset

SOFTWARE_RESET Bits

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13 System Control Module SYSCTL

This chapter introduces the main functions and usage of the system control module SYSCTL.

13.1 Feature Summary

The system control module SYSCTL is the management module of the system power domain. Its main functions are:

- Functional clock configuration
- Processor boot management
- System power domain resource management
- Low power management
- Clock measurement

13.2 Functional Clock Configuration

The configuration of the functional clock is implemented in the SYSCTL module. Users can configure the functional clock by modifying the CLOCK_CPU[x] and CLOCK[x] registers. The MUX bit is used to select the clock source, and the clock source list is shown in Table 16. The DIV bit is used to set the frequency division coefficient. The valid configurable value of the register is 0 255, and the actual frequency division coefficient is the register value +1. For example, if the DIV bit is set to 3, the selected clock source will be divided by 4 and output.

When the LOC_BUSY bit is 0, it indicates that the functional clock is ready for use under the current configuration.

Both the frequency division coefficient and clock source selection support on-the-fly modification.

13.3 CPU Boot Management

CPU0 is the only core, so the system always boots from CPU0.

13.3.1 Boot Management

The boot of CPU0 supports fast program jump after wake-up from low power, see Section 19.5 for details. SYSCTL provides the following support in this process:

- The RESET/SLEEP/WAKE bits of the CPU[CPU0][LP] register are used to identify event records
- The CPU[CPU0][GPR0] register is used to store the fast program jump entry
- The CPU[CPU0][GPRx] registers are used to implement program verification for fast jumps

13.4 System Power Domain Resource Management

The system control module SYSCTL manages the clock, power switch and reset of the entire system power domain.

This section will introduce the switch control mechanism of various resources in the system power domain in detail. For a simple description of the programming model, you can directly jump to Section 13.4.6 to view.

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13.4.1 Resource Nodes

In the system power domain, various nodes that can be turned on or off are called resource nodes, including each functional module, functional clock, subsystem power supply and reset, clock source, etc.

The RESOURCE[x] register is used to configure resource nodes, and the MODE bit is used to configure the control mode of the resource node. The software can directly modify the MODE bit to directly control the enable and disable of a resource.

For functional modules, set the MODE bit of their RESOURCE[*] register to 1 to enable the module, and set it to 2 to disable the module.

For CPU subsystems, set the MODE bit of their RESOURCE[POW_*] register to 1 to turn on the power switch of the subsystem, and set it to 2 to turn off the power switch.

For CPU and SOC subsystems, set the MODE bit of their RESOURCE[RST_*] register to 2 to keep the subsystem in reset, and set it to 1 to exit reset.

For functional clocks and clock sources, set the MODE bit of their RESOURCE[CLK_TOP_*] and RESOURCE[CLK_SRC_*] registers to 1 to enable the clock, and set it to 2 to disable the clock.

When the LOC_BUSY bit is 0, it indicates that the resource node is ready for use.

In practical applications, the control mode of the resource node is usually configured to the default value 0, that is, automatic mode. The specific working mechanism is detailed in the next subsection.

13.4.2 Chain Structure of Resource Nodes

There are dependencies between resource nodes. For example, the normal operation of a functional module depends on the correct output of the functional clock, which in turn depends on the frequency stability of clock sources such as PLL. This dependency constitutes a chain structure. The downstream of the chain is each terminal functional module, the midstream is resources that directly serve the functional modules such as functional clocks, power supplies and resets, and the upstream is mainly each clock source.

Figure 9 takes the ENET0 and SDXC0 modules in the CONN subsystem as an example to describe the chain relationship of resource nodes.

*Only for explaining the structural principle, it does not mean that the product must have ENET0/SDCX0/CONN subsystems

The normal operation of the downstream resource node ENET0 depends on the interface clock provided by CLK_TOP_ENET0, the bus clock provided by CLK_TOP_AXI, and also requires the CONN subsystem where it is located to be in a non-reset state and the power switch to be turned on.

The midstream resource node CLK_TOP_ENET0 depends on the clock source PLL2CLK1 by default, CLK_TOP_AXI depends on the clock source PLL1CLK1 by default, and the reset exit of the CONN subsystem depends on the power switch of the CONN subsystem being turned on.

The upstream resource node PLL1CLK1 clock source depends on PLL1 being enabled and frequency stabilized, and PLL1 in turn depends on XTAL24M providing a valid reference clock.

All resource nodes are uniformly managed by the SYSCTL module, and the dependencies between resource nodes are automatically implemented in hardware, so users do not need to pay attention to the specific dependencies. On the resource node chain, when a downstream node needs to be turned on, it will automatically request all midstream and upstream resource nodes it depends on to be turned on. If the MODE bit in the

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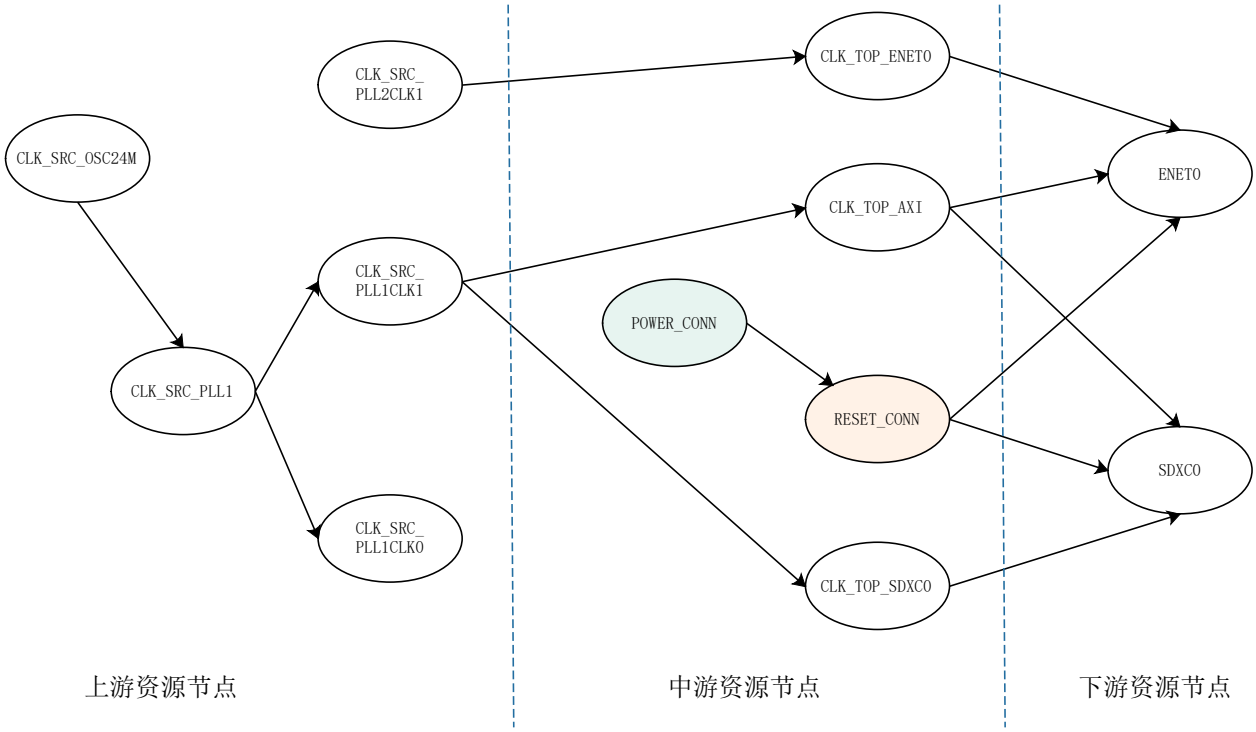


Figure 9: Chain Relationship of Resource Nodes

RESOURCE[] register of the midstream and upstream related resource nodes is 0, that is, in automatic mode, the node will automatically turn on after receiving the request, and at the same time request the more upstream resource nodes to turn on. The resource nodes work in automatic mode by default.

Take the ENET0 module as an example, when all midstream and upstream resource nodes work in the default automatic mode, modifying the MODE bit of the RESOURCE[ENET0] register to 1 is to enable the ENET0 module, the LOC_BUSY bit of the RESOURCE[ENET0] register will be set to 1, and at the same time through chain conduction, all related resource nodes will be automatically enabled, until XTAL24M oscillates stably, PLL1 clock is locked, CLK_TOP_AXI and CLK_TOP_ENET0 clocks are enabled, and the CONN subsystem power is turned on and reset is completed, the LOC_BUSY bit of the RESOURCE[ENET0] register will be set to 0, indicating that the ENET0 module is ready for use.

13.4.3 Automatic Shutdown Mechanism of Resource Nodes

Resource nodes on the chain structure can be automatically turned on according to requests, and can also be automatically turned off under certain conditions.

After a resource node is turned off, it will send a shutdown request to the resource nodes it depends on. A resource node is allowed to be turned off only when all downstream nodes connected to it have sent shutdown requests.

Assuming that Figure 9 shows the complete dependency relationship of the relevant resource nodes, when all midstream and upstream resource nodes work in the default automatic mode, if the MODE bit of the RESOURCE[ENET0] register is modified to 2, that is, to shut down the ENET0 module, first the module clock of ENET0 will be turned off, and then it will request to shut down all resource nodes it depends on. Since CLK_TOP_ENET0 only serves one downstream resource node, ENET0, after receiving the shutdown request, it

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will perform the shutdown operation and continue to send a shutdown request to its dependent upstream resource node PLL2CLK1.

CLK_TOP_AXI will also receive the shutdown request from ENET0, but since SDXC0 connected to its downstream is still in working state, CLK_TOP_AXI is not allowed to be turned off, and the same is true for the RESET_CONN resource node.

13.4.4 Resource Node Keep Enabled

The automatic shutdown mechanism of resource nodes can automatically transmit the shutdown request to the most upstream of each chain, such as XTAL24M and PLL, under permitted conditions. The re-enable of these resources will consume a certain amount of time. If you want the shutdown request to stop propagating when it reaches a certain resource node, you only need to set the MODE bit of the RESOURCE[] register of this node to 1 to force it to remain enabled.

Common scenarios include: keeping the PLL from being turned off or XTAL24M working all the time, and keeping the power of the subsystem.

For subsystems with independent power switches, in low power mode, there are several possible working states:

- Clock retention: Keep some or all module resource nodes in the subsystem enabled, so that these modules can start working at any time. This requires setting the MODE bit of the corresponding RESOURCE[*] register to 1
- Register retention: Allow all module resource nodes in the subsystem to be turned off, but do not reset the subsystem and retain power, so that the states of all registers and logic circuits in the subsystem are saved. This requires setting the MODE bits of the RESOURCE[RST_*] and RESOURCE[POW_*] registers of the subsystem to 1
- Memory retention: Allow the subsystem to enter the reset state, but retain power, so that the data in the memory in the subsystem can be retained. This requires setting the MODE bit of the RESOURCE[POW_*] register of the subsystem to 1
- Subsystem power off

There is another way to set the subsystem state, see Section 13.5.2 for details.

13.4.5 Connection between Functional Modules and Processors

When midstream and upstream resource nodes work in automatic mode, they will automatically turn on and off according to the requests of their downstream nodes. When downstream functional module resource nodes work in automatic mode, they can turn on and off according to the state of the processor.

SYSCTL provides a connection group (link group 0) to connect functional modules with the processor, and group0 is for CPU0 use.

The connection between functional modules and groups is set in the GROUP[*] register. Each bit of the register corresponds to a functional module. Setting the bit to 1 means connecting and binding the module to this group. Table 24 lists the correspondence between the bits of the GROUP register and the functional modules.

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GROUPx[0] Register		GROUPx[1] Register	
bit	**Resource Node**	**bit**	**Resource Node**
0	AHB/APB Peripheral Bus	0	UART7
1	ILM/DLM	1	WDG0
2	MCHTMR	2	WDG1
3	ROM	3	MBX
4	CAN0	4	TSNS
5	CAN1	5	CRC
6	CAN2	6	ADC0
7	CAN3	7	ADC1
8	PTPC	8	DAC0
9	Reserved	9	DAC1
10	Reserved	10	ACMP
11	Reserved	11	OPA0
12	Reserved	12	OPA1
13	GPTMR0	13	MOTOR
14	GPTMR1	14	RNG
15	GPTMR2	15	SDP
16	GPTMR3	16	KEYM
17	I2C0	17	GPIO
18	I2C1	18	HDMA
19	I2C2	19	XPI
20	I2C3	20	USB
21	SPI0	21	
22	SPI1	22	
23	SPI2	23	
24	SPI3	24	
25	UART0	25	
26	UART1	26	
27	UART2	27	
28	UART3	28	
29	UART4	29	
30	UART5	30	
31	UART6	31	

Table 24: Function Module Connection Bit Table

To ensure the basic system can work, SYSCTL has made default configurations for some resource nodes, for example, CPU0 is connected and bound to group0 by default, and the resource nodes AHB/APB peripheral bus and ROM are connected to group0 by default.

The connection between the CPU and the group is configured through the AFFILIATE[x] register, and the default value of the LINK bit of the AFFILIATE[CPU0] register is 0b1, that is, CPU0 has been connected to group0.

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When the CPU and the functional module resource nodes are connected to the same group, and the resource nodes work in automatic mode, these resource nodes will automatically turn on or off as the CPU state changes. For example, when the CPU connects a resource node to the group bound to the CPU itself by modifying the GROUP register, the functional module of this node will be automatically enabled, and a series of resources dependent on the module will be automatically turned on together. This is also the most convenient way to enable functional modules recommended by the system.

Accordingly, when the CPU enters sleep, the resource nodes connected to it will also be automatically turned off. For detailed behavior management after CPU sleep, please refer to Section 13.5.

After introducing the connection function between group and CPU, Figure 9 can be expanded to the form of Figure 10, forming a complete chain structure.

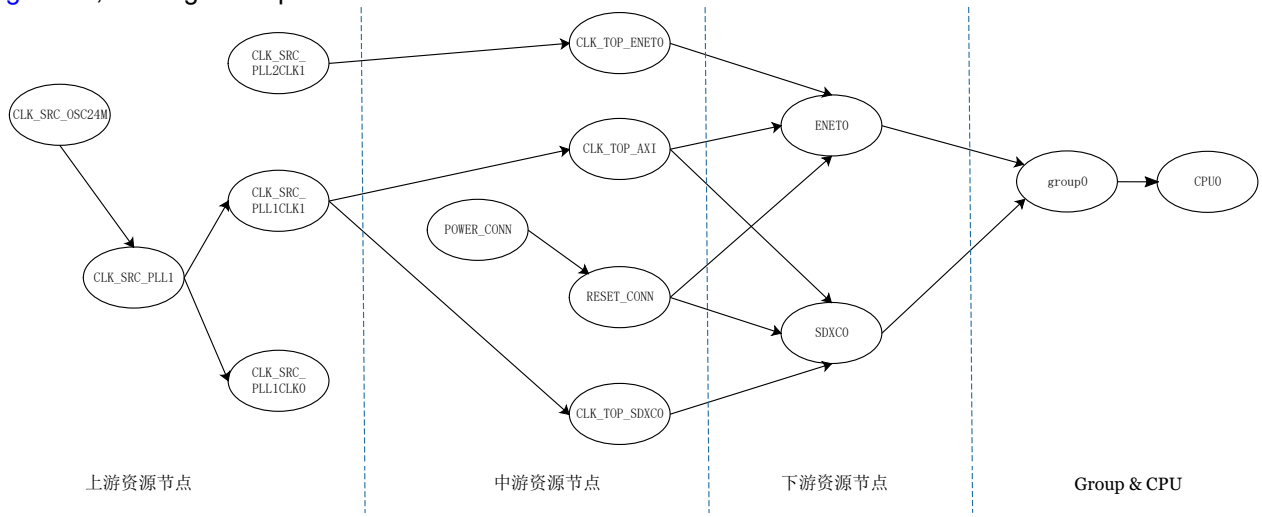


Figure 10: Chain Relationship between Resource Nodes and Group/CPU

*Only for explaining the structural principle, it does not mean that the product must have ENET0/SDCX0/CONN subsystems

In the figure, ENET0 and SDXC0 are connected to group0, indicating that CPU0 needs to use these two modules, which also forms the dependency relationship of the resource node chain.

When CPU0 enters sleep, it will send a shutdown request to both ENET0 and SDXC0, and ENET0 and SDXC0 will shut down and transmit the shutdown request to the upstream of the chain.

13.4.6 Usage of Resource Nodes

All functional modules in the system power domain are resource nodes, which are listed in Table 24.

Except for modules such as AHB/APB peripheral bus and ROM which are enabled by default, most functional modules are disabled by default.

If the software on CPU0 needs to use a functional module, it only needs to set the corresponding bit of the module in the GROUP0 register to 1, and then wait until the GLB_BUSY bit of the RESOURCE[*] register corresponding to the module is 0, and all necessary resources will be automatically turned on.

13.5 Low Power Management

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13.5.1 Processor Low Power Modes and Wake-up

Executing the WFI instruction is the sign that the processor switches from running mode to low power mode. The system behavior after entering low power mode is set by the MODE bit of the CPU[CPUx][LP] register.

- MODE bit is b00 - WAIT Mode: Only the processor core clock is turned off
- MODE bit is b01 - STOP Mode: Turn off the processor core clock, and transmit the shutdown request upstream on the resource node chain, allowing more resource nodes to be turned off
- MODE bit is b10 - RUN Mode: Keep the processor core clock running continuously
- MODE bit is b11 - Invalid setting

The processor will only trigger SYSCTL to perform the shutdown operation on the resource node chain when entering stop mode.

Wake-up from stop mode is generally triggered by an interrupt. Which interrupt sources can wake up the system is configured in the CPU[CPUx][WAKEUP_ENABLE] register of SYSCTL. Setting the bit to 1 means the corresponding interrupt source can wake up the CPU.

The bit allocation of the CPU[CPUx][WAKEUP_ENABLE] register is consistent with the interrupt number defined in the product interrupt vector table.

System wake-up aims at the CPU exiting sleep and being able to handle interrupts. Therefore, when an interrupt that can wake up the CPU occurs, the SYSCTL module will automatically turn on all nodes on the resource node chain where the CPU is located in sequence.

13.5.2 State Retention of Resource Nodes in Low Power Mode

For subsystems with independent power switches, if their power and reset resource nodes work in automatic mode, the processor sleep may turn off their power, resulting in the loss of all register and memory information in the subsystem.

For resource nodes such as XTAL24M and PLL, if they work in automatic mode, processor sleep may cause them to automatically shut down, resulting in a lot of time waiting for their frequency to stabilize when the system wakes up.

To separately control the behavior of these resource nodes in low power mode, in addition to directly modifying their RESOURCE[] registers to change the MODE from automatic mode to enable mode, it is more recommended to use the RETENTION[CPUx] register to control the behavior of some midstream and upstream resource nodes.

Each bit in the RETENTION register corresponds to a resource node. Setting the bit to 1 can control the resource to ignore downstream shutdown requests and remain enabled.

SYSCTL provides 1 RETENTION register, which is allocated to CPU0. When the software on CPU0 wants some resource nodes to remain enabled in low power mode, it only needs to modify the RETENTION register and set the corresponding bit to 1.

For a controllable resource node, as long as the RETENTION register requires it to remain enabled, it will remain working in low power mode.

Note that the RETENTION configuration is independent of the resource node chain and group connection, and can be understood as a forced enable switch added to some midstream and upstream resource nodes.

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The specific bit allocation in the RETENTION register is shown in Table 25:

bit	Resource Node Kept Enabled
0	System power domain power supply
1	System power domain reset
2	CPU0 subsystem power supply
3	CPU0 subsystem reset
4	XTAL24M
5	PLL0
6	PLL1

Table 25: RETENTION Register Bit Table

13.6 Clock Measurement Module

The clock measurement module allows users to select on-chip clocks for measurement and save the current results in the result register.

There are 4 clock measurement units SLICE0 3 in SYSCTL. Users can select the clock to be measured through the SELECTION bit of the MONITOR[SLICEx][CONTROL] register. The measurement clock corresponding to the SELECTION value is shown in Table 26.

Select the measurement reference to be XTAL32K or XTAL24M through the REFERENCE bit of the MONITOR[SLICEx][CONTROL] register, and select the accuracy to be 1KHz or 1Hz through the ACCURAY bit of the MONITOR[SLICEx][CONTROL] register.

SELECTION	Measurement Clock Selection
1	clk_32k
2	clk_rc24m
3	clk_xtal24m
4	clk_usb0_phy
20	clk_24m
21	pll0clk0
22	pll0clk1
23	pll0clk2
24	pll1clk0
25	pll1clk1
26	pll1clk2
27	pll1clk3
128	clk_top_cpu0
129	clk_top_mct0
130	clk_top_can0
131	clk_top_can1
132	clk_top_can2

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SELECTION	Measurement Clock Selection
133	clk_top_can3
134	Reserved
135	Reserved
136	Reserved
137	Reserved
138	clk_top_gptmr0
139	clk_top_gptmr1
140	clk_top_gptmr2
141	clk_top_gptmr3
142	clk_top_i2c0
143	clk_top_i2c1
144	clk_top_i2c2
145	clk_top_i2c3
146	clk_top_spi0
147	clk_top_spi1
148	clk_top_spi2
149	clk_top_spi3
150	clk_top_uart0
151	clk_top_uart1
152	clk_top_uart2
153	clk_top_uart3
154	clk_top_uart4
155	clk_top_uart5
156	clk_top_uart6
157	clk_top_uart7
158	clk_top_xpi0
159	clk_top_ana0
160	clk_top_ana1
161	clk_top_ana2
162	clk_top_ana3
163	clk_top_ref0
164	clk_top_ref1

Table 26: Measurement Clock Selection Table

After the configuration is completed, write 1 to the START bit of the MONITOR[SLICEx][CONTROL] register to start the clock measurement. After waiting for the VALID flag bit of the MONITOR[SLICEx][CONTROL] register to be set to 1, the user can read the clock frequency information from the MONITOR[SLICEx][CURRENT] register.

Note: The clock measurement function is only applicable to clock frequencies below 400MHz.

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13.7 SYSCTL Register Description

The register list of SYSCTL is as follows:

SYSCTL base address: 0xF4000000

Offset	Register Name	Description	Reset Value
0x0000	RESOURCE[CPU0]	Resource control register for cpu0_core	0x00000000
0x0004	RESOURCE[CPX0]	Resource control register for cpu0_subsys	0x00000000
0x0054	RESOURCE[POW_CPU0]	Resource control register for pow_cpu0	0x00000000
0x0058	RESOURCE[RST_SOC]	Resource control register for rst_soc	0x00000000
0x005C	RESOURCE[RST_CPU0]	Resource control register for rst_cpu0	0x00000000
0x0080	RESOURCE[CLK_SRC_XTAL]	Resource control register for xtal	0x00000000
0x0084	RESOURCE[CLK_SRC_PLL0]	Resource control register for pll0	0x00000000
0x0088	RESOURCE[CLK_SRC_CLK0_PLL0]	Resource control register for clk0_pll0	0x00000000
0x008C	RESOURCE[CLK_SRC_CLK1_PLL0]	Resource control register for clk1_pll0	0x00000000
0x0090	RESOURCE[CLK_SRC_CLK2_PLL0]	Resource control register for clk2_pll0	0x00000000
0x0094	RESOURCE[CLK_SRC_PLL1]	Resource control register for pll1	0x00000000
0x0098	RESOURCE[CLK_SRC_CLK0_PLL1]	Resource control register for clk0_pll1	0x00000000
0x009C	RESOURCE[CLK_SRC_CLK1_PLL1]	Resource control register for clk1_pll1	0x00000000
0x00A0	RESOURCE[CLK_SRC_CLK2_PLL1]	Resource control register for clk2_pll1	0x00000000
0x00A4	RESOURCE[CLK_SRC_CLK3_PLL1]	Resource control register for clk3_pll1	0x00000000
0x00A8	RESOURCE[CLK_SRC_PLL0_REF]	Resource control register for pll0 ref clock	0x00000000
0x00AC	RESOURCE[CLK_SRC_PLL1_REF]	Resource control register for pll1 ref clock	0x00000000
0x0100	RESOURCE[CLK_TOP_CPU0]	Resource control register for clk_top_cpu0	0x00000000

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Offset	Register Name	Description	Reset Value
0x0104	RESOURCE[CLK_TOP_MCT0]	Resource control register for clk_top_mct0	0x00000000
0x0108	RESOURCE[CLK_TOP_CAN0]	Resource control register for clk_top_can0	0x00000000
0x010C	RESOURCE[CLK_TOP_CAN1]	Resource control register for clk_top_can1	0x00000000
0x0110	RESOURCE[CLK_TOP_CAN2]	Resource control register for clk_top_can2	0x00000000
0x0114	RESOURCE[CLK_TOP_CAN3]	Resource control register for clk_top_can3	0x00000000
0x0128	RESOURCE[CLK_TOP_TMR0]	Resource control register for clk_top_tmr0	0x00000000
0x012C	RESOURCE[CLK_TOP_TMR1]	Resource control register for clk_top_tmr1	0x00000000
0x0130	RESOURCE[CLK_TOP_TMR2]	Resource control register for clk_top_tmr2	0x00000000
0x0134	RESOURCE[CLK_TOP_TMR3]	Resource control register for clk_top_tmr3	0x00000000
0x0138	RESOURCE[CLK_TOP_I2C0]	Resource control register for clk_top_i2c0	0x00000000
0x013C	RESOURCE[CLK_TOP_I2C1]	Resource control register for clk_top_i2c1	0x00000000
0x0140	RESOURCE[CLK_TOP_I2C2]	Resource control register for clk_top_i2c2	0x00000000
0x0144	RESOURCE[CLK_TOP_I2C3]	Resource control register for clk_top_i2c3	0x00000000
0x0148	RESOURCE[CLK_TOP_SPI0]	Resource control register for clk_top_spi0	0x00000000
0x014C	RESOURCE[CLK_TOP_SPI1]	Resource control register for clk_top_spi1	0x00000000
0x0150	RESOURCE[CLK_TOP_SPI2]	Resource control register for clk_top_spi2	0x00000000
0x0154	RESOURCE[CLK_TOP_SPI3]	Resource control register for clk_top_spi3	0x00000000
0x0158	RESOURCE[CLK_TOP_URT0]	Resource control register for clk_top_urt0	0x00000000
0x015C	RESOURCE[CLK_TOP_URT1]	Resource control register for clk_top_urt1	0x00000000
0x0160	RESOURCE[CLK_TOP_URT2]	Resource control register for clk_top_urt2	0x00000000

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Offset	Register Name	Description	Reset Value
0x0164	RESOURCE[CLK_TOP_URT3]	Resource control register for clk_top_urt3	0x00000000
0x0168	RESOURCE[CLK_TOP_URT4]	Resource control register for clk_top_urt4	0x00000000
0x016C	RESOURCE[CLK_TOP_URT5]	Resource control register for clk_top_urt5	0x00000000
0x0170	RESOURCE[CLK_TOP_URT6]	Resource control register for clk_top_urt6	0x00000000
0x0174	RESOURCE[CLK_TOP_URT7]	Resource control register for clk_top_urt7	0x00000000
0x0178	RESOURCE[CLK_TOP_XPI0]	Resource control register for clk_top_xpi0	0x00000000
0x017C	RESOURCE[CLK_TOP_ANA0]	Resource control register for clk_top_ana0	0x00000000
0x0180	RESOURCE[CLK_TOP_ANA1]	Resource control register for clk_top_ana1	0x00000000
0x0184	RESOURCE[CLK_TOP_ANA2]	Resource control register for clk_top_ana2	0x00000000
0x0188	RESOURCE[CLK_TOP_ANA3]	Resource control register for clk_top_ana3	0x00000000
0x018C	RESOURCE[CLK_TOP_REF0]	Resource control register for clk_top_ref0	0x00000000
0x0190	RESOURCE[CLK_TOP_REF1]	Resource control register for clk_top_ref1	0x00000000
0x0194	RESOURCE[CLK_TOP_ADC0]	Resource control register for clk_top_adc0	0x00000000
0x0198	RESOURCE[CLK_TOP_ADC1]	Resource control register for clk_top_adc1	0x00000000
0x019C	RESOURCE[CLK_TOP_DAC0]	Resource control register for clk_top_dac0	0x00000000
0x01A0	RESOURCE[CLK_TOP_DAC1]	Resource control register for clk_top_dac1	0x00000000
0x0400	RESOURCE[AHB0]	Resource control register for soc_bus	0x00000000
0x0404	RESOURCE[LMM0]	Resource control register for lmm0	0x00000000
0x0408	RESOURCE[MCT0]	Resource control register for mchtmr0	0x00000000
0x040C	RESOURCE[ROM0]	Resource control register for rom	0x00000000

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Offset	Register Name	Description	Reset Value
0x0410	RESOURCE[CAN0]	Resource control register for can0	0x00000000
0x0414	RESOURCE[CAN1]	Resource control register for can1	0x00000000
0x0418	RESOURCE[CAN2]	Resource control register for can2	0x00000000
0x041C	RESOURCE[CAN3]	Resource control register for can3	0x00000000
0x0420	RESOURCE[PTPC]	Resource control register for ptpc	0x00000000
0x0434	RESOURCE[TMR0]	Resource control register for tmr0	0x00000000
0x0438	RESOURCE[TMR1]	Resource control register for tmr1	0x00000000
0x043C	RESOURCE[TMR2]	Resource control register for tmr2	0x00000000
0x0440	RESOURCE[TMR3]	Resource control register for tmr3	0x00000000
0x0444	RESOURCE[I2C0]	Resource control register for i2c0	0x00000000
0x0448	RESOURCE[I2C1]	Resource control register for i2c1	0x00000000
0x044C	RESOURCE[I2C2]	Resource control register for i2c2	0x00000000
0x0450	RESOURCE[I2C3]	Resource control register for i2c3	0x00000000
0x0454	RESOURCE[SPI0]	Resource control register for spi0	0x00000000
0x0458	RESOURCE[SPI1]	Resource control register for spi1	0x00000000
0x045C	RESOURCE[SPI2]	Resource control register for spi2	0x00000000
0x0460	RESOURCE[SPI3]	Resource control register for spi3	0x00000000
0x0464	RESOURCE[URT0]	Resource control register for uart0	0x00000000
0x0468	RESOURCE[URT1]	Resource control register for uart1	0x00000000
0x046C	RESOURCE[URT2]	Resource control register for uart2	0x00000000

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Offset	Register Name	Description	Reset Value
0x0470	RESOURCE[URT3]	Resource control register for uart3	0x00000000
0x0474	RESOURCE[URT4]	Resource control register for uart4	0x00000000
0x0478	RESOURCE[URT5]	Resource control register for uart5	0x00000000
0x047C	RESOURCE[URT6]	Resource control register for uart6	0x00000000
0x0480	RESOURCE[URT7]	Resource control register for uart7	0x00000000
0x0484	RESOURCE[WDG0]	Resource control register for wdg0	0x00000000
0x0488	RESOURCE[WDG1]	Resource control register for wdg1	0x00000000
0x048C	RESOURCE[MBX0]	Resource control register for mbx0	0x00000000
0x0490	RESOURCE[TSNS]	Resource control register for tsns	0x00000000
0x0494	RESOURCE[CRC0]	Resource control register for crc	0x00000000
0x0498	RESOURCE[ADC0]	Resource control register for adc0	0x00000000
0x049C	RESOURCE[ADC1]	Resource control register for adc1	0x00000000
0x04A0	RESOURCE[DAC0]	Resource control register for dac0	0x00000000
0x04A4	RESOURCE[DAC1]	Resource control register for dac1	0x00000000
0x04A8	RESOURCE[ACMP]	Resource control register for acmp	0x00000000
0x04AC	RESOURCE[OPA0]	Resource control register for opamp0	0x00000000
0x04B0	RESOURCE[OPA1]	Resource control register for opamp1	0x00000000
0x04B4	RESOURCE[MOT0]	Resource control register for motor	0x00000000
0x04B8	RESOURCE[RNG0]	Resource control register for rng	0x00000000
0x04BC	RESOURCE[SDP0]	Resource control register for sdp	0x00000000
0x04C0	RESOURCE[KMAN]	Resource control register for key_manager	0x00000000

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Offset	Register Name	Description	Reset Value
0x04C4	RESOURCE[GPIO]	Resource control register for gpio	0x00000000
0x04C8	RESOURCE[HDMA]	Resource control register for hdma	0x00000000
0x04CC	RESOURCE[XPI0]	Resource control register for xpi0	0x00000000
0x04D0	RESOURCE[USB0]	Resource control register for usb	0x00000000
0x04D4	RESOURCE[REF0]	Resource control register for ref0	0x00000000
0x04D8	RESOURCE[REF1]	Resource control register for ref1	0x00000000
0x0800	GROUP0[LINK0][VALUE]	Group setting	0x00000000
0x0804	GROUP0[LINK0][SET]	Group setting	0x00000000
0x0808	GROUP0[LINK0][CLEAR]	Group setting	0x00000000
0x080C	GROUP0[LINK0][TOGGLE]	Group setting	0x00000000
0x0810	GROUP0[LINK1][VALUE]	Group setting	0x00000000
0x0814	GROUP0[LINK1][SET]	Group setting	0x00000000
0x0818	GROUP0[LINK1][CLEAR]	Group setting	0x00000000
0x081C	GROUP0[LINK1][TOGGLE]	Group setting	0x00000000
0x0900	AFFILIATE[CPU0][VALUE]	Affiliate of Group	0x00000000
0x0904	AFFILIATE[CPU0][SET]	Affiliate of Group	0x00000000
0x0908	AFFILIATE[CPU0][CLEAR]	Affiliate of Group	0x00000000
0x090C	AFFILIATE[CPU0][TOGGLE]	Affiliate of Group	0x00000000
0x0920	RETENTION[CPU0][VALUE]	Retention Control	0x00000000
0x0924	RETENTION[CPU0][SET]	Retention Control	0x00000000
0x0928	RETENTION[CPU0][CLEAR]	Retention Control	0x00000000
0x092C	RETENTION[CPU0][TOGGLE]	Retention Control	0x00000000
0x1000	POWER[CPU0][STATUS]	Power Setting	0x80000000
0x1004	POWER[CPU0][LF_WAIT]	Power Setting	0x000000FF
0x100C	POWER[CPU0][OFF_WAIT]	Power Setting	0x0000000F
0x1010	POWER[CPU0][RET_WAIT]	Power Setting	0x0000000F
0x1400	RESET[SOC][CONTROL]	Reset Setting	0x80000000
0x1404	RESET[SOC][CONFIG]	Reset Setting	0x00402003
0x140C	RESET[SOC][COUNTER]	Reset Setting	0x00000000
0x1410	RESET[CPU0][CONTROL]	Reset Setting	0x80000000
0x1414	RESET[CPU0][CONFIG]	Reset Setting	0x00402003
0x141C	RESET[CPU0][COUNTER]	Reset Setting	0x00000000
0x1800	CLOCK_CPU[CLK_TOP_CPU0]	Clock setting	0x00000000
0x1804	CLOCK[CLK_TOP_MCT0]	Clock setting	0x00000000

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Offset	Register Name	Description	Reset Value
0x1808	CLOCK[CLK_TOP_CAN0]	Clock setting	0x00000000
0x180C	CLOCK[CLK_TOP_CAN1]	Clock setting	0x00000000
0x1810	CLOCK[CLK_TOP_CAN2]	Clock setting	0x00000000
0x1814	CLOCK[CLK_TOP_CAN3]	Clock setting	0x00000000
0x1828	CLOCK[CLK_TOP_TMR0]	Clock setting	0x00000000
0x182C	CLOCK[CLK_TOP_TMR1]	Clock setting	0x00000000
0x1830	CLOCK[CLK_TOP_TMR2]	Clock setting	0x00000000
0x1834	CLOCK[CLK_TOP_TMR3]	Clock setting	0x00000000
0x1838	CLOCK[CLK_TOP_I2C0]	Clock setting	0x00000000
0x183C	CLOCK[CLK_TOP_I2C1]	Clock setting	0x00000000
0x1840	CLOCK[CLK_TOP_I2C2]	Clock setting	0x00000000
0x1844	CLOCK[CLK_TOP_I2C3]	Clock setting	0x00000000
0x1848	CLOCK[CLK_TOP_SPI0]	Clock setting	0x00000000
0x184C	CLOCK[CLK_TOP_SPI1]	Clock setting	0x00000000
0x1850	CLOCK[CLK_TOP_SPI2]	Clock setting	0x00000000
0x1854	CLOCK[CLK_TOP_SPI3]	Clock setting	0x00000000
0x1858	CLOCK[CLK_TOP_URT0]	Clock setting	0x00000000
0x185C	CLOCK[CLK_TOP_URT1]	Clock setting	0x00000000
0x1860	CLOCK[CLK_TOP_URT2]	Clock setting	0x00000000
0x1864	CLOCK[CLK_TOP_URT3]	Clock setting	0x00000000
0x1868	CLOCK[CLK_TOP_URT4]	Clock setting	0x00000000
0x186C	CLOCK[CLK_TOP_URT5]	Clock setting	0x00000000
0x1870	CLOCK[CLK_TOP_URT6]	Clock setting	0x00000000
0x1874	CLOCK[CLK_TOP_URT7]	Clock setting	0x00000000
0x1878	CLOCK[CLK_TOP_XPI0]	Clock setting	0x00000000
0x187C	CLOCK[CLK_TOP_ANA0]	Clock setting	0x00000000
0x1880	CLOCK[CLK_TOP_ANA1]	Clock setting	0x00000000
0x1884	CLOCK[CLK_TOP_ANA2]	Clock setting	0x00000000
0x1888	CLOCK[CLK_TOP_ANA3]	Clock setting	0x00000000
0x188C	CLOCK[CLK_TOP_REF0]	Clock setting	0x00000000
0x1890	CLOCK[CLK_TOP_REF1]	Clock setting	0x00000000
0x1C00	ADCCLK[CLK_TOP_ADC0]	Clock setting	0x00000000
0x1C04	ADCCLK[CLK_TOP_ADC1]	Clock setting	0x00000000
0x1C08	DACCLK[CLK_TOP_DAC0]	Clock setting	0x00000000
0x1C0C	DACCLK[CLK_TOP_DAC1]	Clock setting	0x00000000
0x2000	GLOBAL00	Clock senario	0x00000000
0x2400	MONITOR[SLICE0][CONTROL]	Clock measure and monitor control	0x00000000
0x2404	MONITOR[SLICE0][CURRENT]	Clock measure result	0x00000000

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Offset	Register Name	Description	Reset Value
0x2408	MONITOR[SLICE0][LOW_LIMIT]	Clock lower limit	0xFFFFFFFF
0x240C	MONITOR[SLICE0][HIGH_LIMIT]	Clock upper limit	0x00000000
0x2420	MONITOR[SLICE1][CONTROL]	Clock measure and monitor control	0x00000000
0x2424	MONITOR[SLICE1][CURRENT]	Clock measure result	0x00000000
0x2428	MONITOR[SLICE1][LOW_LIMIT]	Clock lower limit	0xFFFFFFFF
0x242C	MONITOR[SLICE1][HIGH_LIMIT]	Clock upper limit	0x00000000
0x2440	MONITOR[SLICE2][CONTROL]	Clock measure and monitor control	0x00000000
0x2444	MONITOR[SLICE2][CURRENT]	Clock measure result	0x00000000
0x2448	MONITOR[SLICE2][LOW_LIMIT]	Clock lower limit	0xFFFFFFFF
0x244C	MONITOR[SLICE2][HIGH_LIMIT]	Clock upper limit	0x00000000
0x2460	MONITOR[SLICE3][CONTROL]	Clock measure and monitor control	0x00000000
0x2464	MONITOR[SLICE3][CURRENT]	Clock measure result	0x00000000
0x2468	MONITOR[SLICE3][LOW_LIMIT]	Clock lower limit	0xFFFFFFFF
0x246C	MONITOR[SLICE3][HIGH_LIMIT]	Clock upper limit	0x00000000
0x2800	CPU[CPU0][LP]	CPU0 LP control	0x00001000
0x2804	CPU[CPU0][LOCK]	CPU0 Lock GPR	0x00000000
0x2808	CPU[CPU0][GPR][GPR0]	CPU0 GPR0	0x00000000
0x280C	CPU[CPU0][GPR][GPR1]	CPU0 GPR1	0x00000000
0x2810	CPU[CPU0][GPR][GPR2]	CPU0 GPR2	0x00000000
0x2814	CPU[CPU0][GPR][GPR3]	CPU0 GPR3	0x00000000
0x2818	CPU[CPU0][GPR][GPR4]	CPU0 GPR4	0x00000000
0x281C	CPU[CPU0][GPR][GPR5]	CPU0 GPR5	0x00000000
0x2820	CPU[CPU0][GPR][GPR6]	CPU0 GPR6	0x00000000
0x2824	CPU[CPU0][GPR][GPR7]	CPU0 GPR7	0x00000000
0x2828	CPU[CPU0][GPR][GPR8]	CPU0 GPR8	0x00000000
0x282C	CPU[CPU0][GPR][GPR9]	CPU0 GPR9	0x00000000
0x2830	CPU[CPU0][GPR][GPR10]	CPU0 GPR10	0x00000000
0x2834	CPU[CPU0][GPR][GPR11]	CPU0 GPR11	0x00000000
0x2838	CPU[CPU0][GPR][GPR12]	CPU0 GPR12	0x00000000
0x283C	CPU[CPU0][GPR][GPR13]	CPU0 GPR13	0x00000000
0x2840	CPU[CPU0][WAKEUP_STATUS][STATUS0]	CPU0 wakeup IRQ status	0x00000000
0x2844	CPU[CPU0][WAKEUP_STATUS][STATUS1]	CPU0 wakeup IRQ status	0x00000000
0x2848	CPU[CPU0][WAKEUP_STATUS][STATUS2]	CPU0 wakeup IRQ status	0x00000000

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Offset	Register Name	Description	Reset Value
0x284C	CPU[CPU0][WAKEUP_STATUS][STATUS3]	CPU0 wakeup IRQ status	0x00000000
0x2880	CPU[CPU0][WAKEUP_ENABLE][ENABLE0]	CPU0 wakeup IRQ enable	0x00000000
0x2884	CPU[CPU0][WAKEUP_ENABLE][ENABLE1]	CPU0 wakeup IRQ enable	0x00000000
0x2888	CPU[CPU0][WAKEUP_ENABLE][ENABLE2]	CPU0 wakeup IRQ enable	0x00000000
0x288C	CPU[CPU0][WAKEUP_ENABLE][ENABLE3]	CPU0 wakeup IRQ enable	0x00000000

Table 27: SYSCTL Register List

Note: The boot code of this product's ROM will configure this module, so the register value read by the software may be inconsistent with the reset value described in this chapter.

13.8 Detailed Register Information

The detailed description of SYSCTL registers is as follows:

13.8.1 RESOURCE (0x0 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GLB_BUSY	LOC_BUSY	RSVD																												MODE	
RO	RO	N/A																												RW	
0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0

RESOURCE [31:0]

Bits	Name	Description
31	GLB_BUSY	global busy 0: no changes pending to any nodes 1: any of nodes is changing status
30	LOC_BUSY	local busy 0: no change is pending for current node 1: current node is changing status
1-0	MODE	resource work mode 0:auto turn on and off as system required(recommended) 1:always on 2:always off 3:reserved

RESOURCE Bits

13.8.2 GROUP0[VALUE] (0x800 + 0x10 * n)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LINK																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

GROUP0[VALUE] [31:0]

Bits	Name	Description
31-0	LINK	denpendency on peripherals, index count from resource ahbp(0x400), each bit represents a peripheral 0: peripheral is not needed 1: periphera is needed

GROUP0[VALUE] Bits

13.8.3 GROUP0[SET] (0x804 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LINK																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

GROUP0[SET] [31:0]

Bits	Name	Description
31-0	LINK	denpendency on peripherals, index count from resource ahbp(0x400), each bit represents a peripheral 0: no effect 1: add periphera into this group，periphera is needed

GROUP0[SET] Bits

13.8.4 GROUP0[CLEAR] (0x808 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LINK																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

GROUP0[CLEAR] [31:0]

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Bits	Name	Description
31-0	LINK	denpendency on peripherals, index count from resource ahbp(0x400), each bit represents a peripheral 0: no effect 1: delete periphera in this group，periphera is not needed

GROUP0[CLEAR] Bits

13.8.5 GROUP0[TOGGLE] (0x80C + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LINK																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

GROUP0[TOGGLE] [31:0]

Bits	Name	Description
31-0	LINK	denpendency on peripherals, index count from resource ahbp(0x400), each bit represents a peripheral 0: no effect 1: toggle the result that whether periphera is needed before

GROUP0[TOGGLE] Bits

13.8.6 AFFILIATE[VALUE] (0x900 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																												LINK			
N/A																												RW			
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0

AFFILIATE[VALUE] [31:0]

Bits	Name	Description
3-0	LINK	Affiliate groups of cpu0, each bit represents a group bit0: cpu0 depends on group0 bit1: cpu0 depends on group1 bit2: cpu0 depends on group2 bit3: cpu0 depends on group3

AFFILIATE[VALUE] Bits

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13.8.7 AFFILIATE[SET] (0x904 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																												LINK			
N/A																												RW			
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0

AFFILIATE[SET] [31:0]

Bits	Name	Description
3-0	LINK	Affiliate groups of cpu0，each bit represents a group 0: no effect 1: the group is assigned to CPU0

AFFILIATE[SET] Bits

13.8.8 AFFILIATE[CLEAR] (0x908 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																												LINK			
N/A																												RW			
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0

AFFILIATE[CLEAR] [31:0]

Bits	Name	Description
3-0	LINK	Affiliate groups of cpu0, each bit represents a group 0: no effect 1: the group is not assigned to CPU0

AFFILIATE[CLEAR] Bits

13.8.9 AFFILIATE[TOGGLE] (0x90C + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																												LINK			
N/A																												RW			
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0

AFFILIATE[TOGGLE] [31:0]

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Bits	Name	Description
3-0	LINK	Affiliate groups of cpu0, each bit represents a group 0: no effect 1: toggle the result that whether the group is assigned to CPU0 before

AFFILIATE[TOGGLE] Bits

13.8.10 RETENTION[VALUE] (0x920 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																	LINK														
N/A																	RW													
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

RETENTION[VALUE] [31:0]

Bits	Name	Description
14-0	LINK	retention setting while CPU0 enter stop mode, each bit represents a resource bit00: soc_mem is kept on while cpu0 stop bit01: soc_ctx is kept on while cpu0 stop bit02: cpu0_mem is kept on while cpu0 stop bit03: cpu0_ctx is kept on while cpu0 stop bit04: xtal_hold is kept on while cpu0 stop bit05: pll0_hold is kept on while cpu0 stop bit06: pll1_hold is kept on while cpu0 stop

RETENTION[VALUE] Bits

13.8.11 RETENTION[SET] (0x924 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																	LINK														
N/A																	RW													
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

RETENTION[SET] [31:0]

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Bits	Name	Description
14-0	LINK	retention setting while CPU0 enter stop mode, each bit represents a resource 0: no effect 1: keep

RETENTION[SET] Bits

13.8.12 RETENTION[CLEAR] (0x928 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																	LINK														
N/A																	RW													
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

RETENTION[CLEAR] [31:0]

Bits	Name	Description
14-0	LINK	retention setting while CPU0 enter stop mode, each bit represents a resource 0: no effect 1: no keep

RETENTION[CLEAR] Bits

13.8.13 RETENTION[TOGGLE] (0x92C + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																	LINK														
N/A																	RW													
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

RETENTION[TOGGLE] [31:0]

Bits	Name	Description
14-0	LINK	retention setting while CPU0 enter stop mode, each bit represents a resource 0: no effect 1: toggle the result that whether the resource is kept on while CPU0 stop before

RETENTION[TOGGLE] Bits

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13.8.14 POWER[STATUS] (0x1000 + 0x14 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FLAG	FLAG_WAKE	RSVD												MEM_RET_N	MEM_RET_P	RSVD			LF_DISABLE	RSVD			LF_ACK	RSVD							
RW	RW	N/A												RO	RO	N/A			RO	N/A			RO	N/A							
1	0	x	x	x	x	x	x	x	x	x	x	x	x	0	0	x	x	x	0	x	x	x	0	x	x	x	x	x	x	x	x

POWER[STATUS] [31:0]

Bits	Name	Description
31	FLAG	flag represents power cycle happened from last clear of this bit 0: power domain did not edurance power cycle since last clear of this bit 1: power domain enduranced power cycle since last clear of this bit
30	FLAG_WAKE	flag represents wakeup power cycle happened from last clear of this bit 0: power domain did not edurance wakeup power cycle since last clear of this bit 1: power domain enduranced wakeup power cycle since last clear of this bit
17	MEM_RET_N	memory info retention control signal 0: memory enter retention mode 1: memory exit retention mode
16	MEM_RET_P	memory info retention control signal 0: memory not enterexitretention mode 1: memory enter retention mode
12	LF_DISABLE	low fanout power switch disable 0: low fanout power switches are turned on 1: low fanout power switches are truned off
8	LF_ACK	low fanout power switch feedback 0: low fanout power switches are turned on 1: low fanout power switches are truned off

POWER[STATUS] Bits

13.8.15 POWER[LF_WAIT] (0x1004 + 0x14 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD												WAIT																			

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
N/A												RW																			
x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1

POWER[LF_WAIT] [31:0]

Bits	Name	Description
19-0	WAIT	wait time for low fan out power switch turn on, default value is 255 0: 0 clock cycle 1: 1 clock cycles . . . clock cycles count on 24MHz

POWER[LF_WAIT] Bits

13.8.16 POWER[OFF_WAIT] (0x100C + 0x14 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD												WAIT																			
N/A												RW																			
x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1

POWER[OFF_WAIT] [31:0]

Bits	Name	Description
19-0	WAIT	wait time for power switch turn off, default value is 15 0: 0 clock cycle 1: 1 clock cycles . . . clock cycles count on 24MHz

POWER[OFF_WAIT] Bits

13.8.17 POWER[RET_WAIT] (0x1010 + 0x14 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD												WAIT																			
N/A												RW																			
x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1

POWER[RET_WAIT] [31:0]

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Bits	Name	Description
19-0	WAIT	wait time for memory retention mode transition, default value is 15 0: 0 clock cycle 1: 1 clock cycles . . . clock cycles count on 24MHz

POWER[RET_WAIT] Bits

13.8.18 RESET[CONTROL] (0x1400 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FLAG	FLAG_WAKE	RSVD																									HOLD	RSVD			RESET
RW	RW	N/A																									RW	N/A			RW
1	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	0

RESET[CONTROL] [31:0]

Bits	Name	Description
31	FLAG	flag represents reset happened from last clear of this bit 0: domain did not edurance reset cycle since last clear of this bit 1: domain enduranced reset cycle since last clear of this bit
30	FLAG_WAKE	flag represents wakeup reset happened from last clear of this bit 0: domain did not edurance wakeup reset cycle since last clear of this bit 1: domain enduranced wakeup reset cycle since last clear of this bit
4	HOLD	perform reset and hold in reset, until ths bit cleared by software 0: reset is released for function 1: reset is assert and hold
0	RESET	perform reset and release imediately 0: reset is released 1 reset is asserted and will release automatically

RESET[CONTROL] Bits

13.8.19 RESET[CONFIG] (0x1404 + 0x10 * n)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD								PRE_WAIT								RSTCLK_NUM								POST_WAIT							
N/A								RW								RW								RW							
x	x	x	x	x	x	x	x	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	1

RESET[CONFIG] [31:0]

Bits	Name	Description
23-16	PRE_WAIT	wait cycle numbers before assert reset 0: wait 0 cycle 1: wait 1 cycles . . . Note, clock cycle is base on 24M
15-8	RSTCLK_NUM	reset clock number(must be even number) 0: 0 cycle 1: 0 cycles 2: 2 cycles 3: 2 cycles . . . Note, clock cycle is base on 24M
7-0	POST_WAIT	time guard band for reset release 0: wait 0 cycle 1: wait 1 cycles . . . Note, clock cycle is base on 24M

RESET[CONFIG] Bits

13.8.20 RESET[COUNTER] (0x140C + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD												COUNTER																			
N/A												RW																			
x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

RESET[COUNTER] [31:0]

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Bits	Name	Description
19-0	COUNTER	self clear trigger counter, reset triggered when counter value is 1, write 0 will cancel reset 0: wait 0 cycle 1: wait 1 cycles . . . Note, clock cycle is base on 24M

RESET[COUNTER] Bits

13.8.21 CLOCK_CPU (0x1800 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GLB_BUSY	LOC_BUSY	RSVD	PRESERVE	RSVD								SUB0_DIV				RSVD					MUX			DIV							
RO	RO	N/A	RW	N/A								RW				N/A					RW			RW							
0	0	x	0	x	x	x	x	x	x	x	x	0	0	0	0	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0

CLOCK_CPU [31:0]

Bits	Name	Description
31	GLB_BUSY	global busy 0: no changes pending to any clock 1: any of nodes is changing status
30	LOC_BUSY	local busy 0: a change is pending for current node 1: current node is changing status
28	PRESERVE	preserve function against global select 0: select global clock setting 1: not select global clock setting
19-16	SUB0_DIV	ahb bus divider, the bus clock is generated by cpu_clock/div 0: divider by 1 1: divider by 2 ...

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Bits	Name	Description
10-8	MUX	current mux in clock component 0:osc0_clk0 1:pll0_clk0 2:pll0_clk1 3:pll0_clk2 4:pll1_clk0 5:pll1_clk1 6:pll1_clk2 7:pll1_clk3
7-0	DIV	clock divider 0: divider by 1 1: divider by 2 2: divider by 3 . . . 255: divider by 256

CLOCK_CPU Bits

13.8.22 CLOCK (0x1804 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GLB_BUSY	LOC_BUSY	RSVD	PRESERVE	RSVD																	MUX			DIV							
RO	RO	N/A	RW	N/A																	RW			RW							
0	0	x	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0

CLOCK [31:0]

Bits	Name	Description
31	GLB_BUSY	global busy 0: no changes pending to any clock 1: any of nodes is changing status
30	LOC_BUSY	local busy 0: a change is pending for current node 1: current node is changing status
28	PRESERVE	preserve function against global select 0: select global clock setting 1: not select global clock setting

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Bits	Name	Description
10-8	MUX	current mux in clock component 0:osc0_clk0 1:pll0_clk0 2:pll0_clk1 3:pll0_clk2 4:pll1_clk0 5:pll1_clk1 6:pll1_clk2 7:pll1_clk3
7-0	DIV	clock divider 0: divider by 1 1: divider by 2 2: divider by 3 . . . 255: divider by 256

CLOCK Bits

13.8.23 ADCCLK (0x1C00 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GLB_BUSY	LOC_BUSY	RSVD	PRESERVE	RSVD																			MUX	RSVD							
RO	RO	N/A	RW	N/A																			RW	N/A							
0	0	x	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	x	x	x	x	x

ADCCLK [31:0]

Bits	Name	Description
31	GLB_BUSY	global busy 0: no changes pending to any clock 1: any of nodes is changing status
30	LOC_BUSY	local busy 0: a change is pending for current node 1: current node is changing status
28	PRESERVE	preserve function against global select 0: select global clock setting 1: not select global clock setting
8	MUX	current mux 0: ahb0 clock 1: ana clock N

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Bits	Name	Description

ADCCLK Bits

13.8.24 DACCLK (0x1C08 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GLB_BUSY	LOC_BUSY	RSVD	PRESERVE	RSVD																			MUX	RSVD							
RO	RO	N/A	RW	N/A																			RW	N/A							
0	0	x	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	x	x	x	x	x

DACCLK [31:0]

Bits	Name	Description
31	GLB_BUSY	global busy 0: no changes pending to any clock 1: any of nodes is changing status
30	LOC_BUSY	local busy 0: a change is pending for current node 1: current node is changing status
28	PRESERVE	preserve function against global select 0: select global clock setting 1: not select global clock setting
8	MUX	current mux 0: ahb0 clock 1: ana clock N

DACCLK Bits

13.8.25 GLOBAL00 (0x2000)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																								MUX							
N/A																								RW							
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0

GLOBAL00 [31:0]

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Bits	Name	Description
7-0	MUX	global clock override request bit0: override to preset0 bit1: override to preset1 bit2: override to preset2 bit3: override to preset3 bit4: override to preset4 bit5: override to preset5 bit6: override to preset6 bit7: override to preset7

GLOBAL00 Bits

13.8.26 MONITOR[CONTROL] (0x2400 + 0x20 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VALID	RSVD			DIV_BUSY	RSVD		OUTEN	DIV								HIGH	LOW	RSVD	START	RSVD	MODE	ACCURACY	REFERENCE	SELECTION							
RW	N/A			RO	N/A		RW	RW								RW	RW	N/A	RW	N/A	RW	RW	RW	RW							
0	x	x	x	0	x	x	0	0	0	0	0	0	0	0	0	0	0	x	0	x	0	0	0	0	0	0	0	0	0	0	0

MONITOR[CONTROL] [31:0]

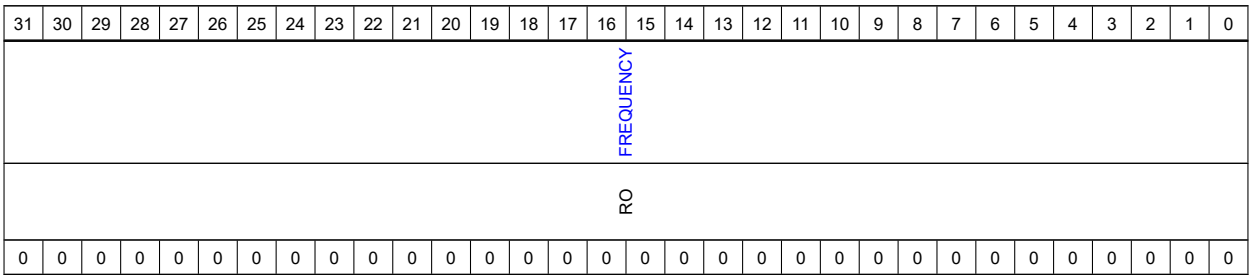
Bits	Name	Description
31	VALID	result is ready for read 0: not ready 1: result is ready
27	DIV_BUSY	divider is applying new setting
24	OUTEN	enable clock output
23-16	DIV	output divider
15	HIGH	clock frequency higher than upper limit
14	LOW	clock frequency lower than lower limit
12	START	start measurement
10	MODE	work mode, 0: register value will be compared to measurement 1: upper and lower value will be recordered in register
9	ACCURACY	measurement accuracy, 0: resolution is 1kHz 1: resolution is 1Hz
8	REFERENCE	reference clock selection, 0: 32k 1: 24M

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Bits	Name	Description
7-0	SELECTION	clock measurement selection

MONITOR[CONTROL] Bits

13.8.27 MONITOR[CURRENT] (0x2404 + 0x20 * n)

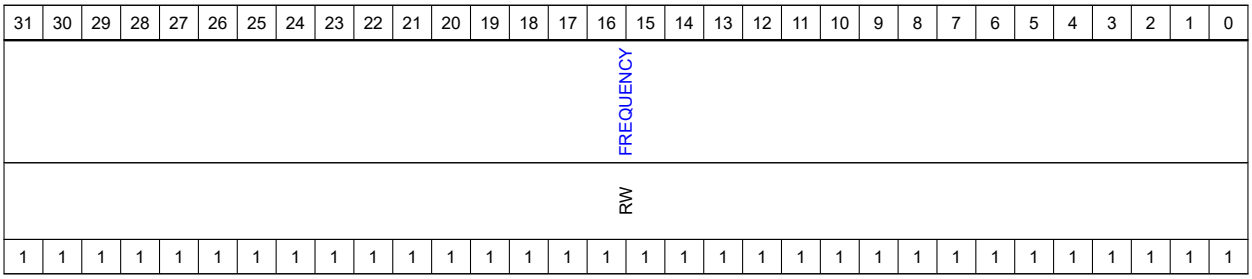


MONITOR[CURRENT] [31:0]

Bits	Name	Description
31-0	FREQUENCY	self updating measure result

MONITOR[CURRENT] Bits

13.8.28 MONITOR[LOW_LIMIT] (0x2408 + 0x20 * n)

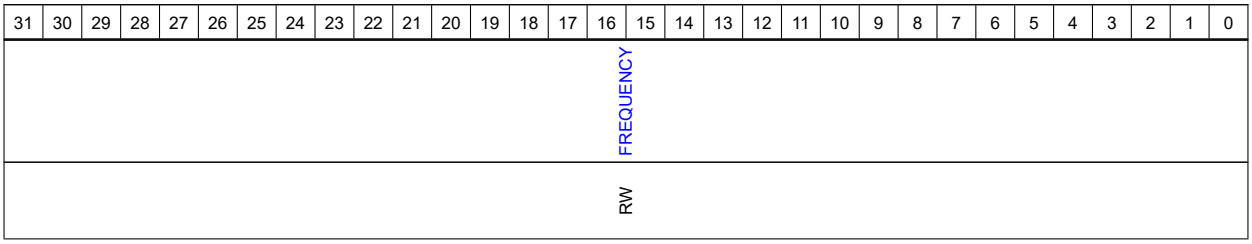


MONITOR[LOW_LIMIT] [31:0]

Bits	Name	Description
31-0	FREQUENCY	lower frequency

MONITOR[LOW_LIMIT] Bits

13.8.29 MONITOR[HIGH_LIMIT] (0x240C + 0x20 * n)



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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

MONITOR[HIGH_LIMIT] [31:0]

Bits	Name	Description
31-0	FREQUENCY	upper frequency

MONITOR[HIGH_LIMIT] Bits

13.8.30 CPU[LP] (0x2800 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WAKE_CNT								RSVD							HALT	RSVD		WAKE	EXEC	RSVD	WAKE_FLAG	SLEEP_FLAG	RESET_FLAG	RSVD							MODE	
RW								N/A							RW	N/A		RO	RO	N/A	RW	RW	RW	N/A							RW	
0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	0	x	x	0	1	x	0	0	0	x	x	x	x	x	x	0	0

CPU[LP] [31:0]

Bits	Name	Description
31-24	WAKE_CNT	CPU0 wake up counter, counter satuated at 255, write 0x00 to clear
16	HALT	halt request for CPU0, 0: CPU0 will start to execute after reset or receive wakeup request 1: CPU0 will not start after reset, or wakeup after WFI
13	WAKE	CPU0 is waking up 0: CPU0 wake up not asserted 1: CPU0 wake up asserted
12	EXEC	CPU0 is executing 0: CPU0 is not executing 1: CPU0 is executing
10	WAKE_FLAG	CPU0 wakeup flag, indicate a wakeup event got active, write 1 to clear this bit 0: CPU0 wakeup not happened 1: CPU0 wake up happened
9	SLEEP_FLAG	CPU0 sleep flag, indicate a sleep event got active, write 1 to clear this bit 0: CPU0 sleep not happened 1: CPU0 sleep happened

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Bits	Name	Description
8	RESET_FLAG	CPU0 reset flag, indicate a reset event got active, write 1 to clear this bit 0: CPU0 reset not happened 1: CPU0 reset happened
1-0	MODE	Low power mode, system behavior after WFI 00: CPU clock stop after WFI 01: System enter low power mode after WFI 10: Keep running after WFI 11: reserved

CPU[LP] Bits

13.8.31 CPU[LOCK] (0x2804 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																GPR														LOCK	RSVD
N/A																RW														RW	N/A
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x

CPU[LOCK] [31:0]

Bits	Name	Description
15-2	GPR	Lock bit for CPU_DATA0 to CPU_DATA13, once set, this bit will not clear untile next reset
1	LOCK	Lock bit for CPU_LOCK

CPU[LOCK] Bits

13.8.32 CPU[GPR] (0x2808 + 0x400 * n + 0x4 * m)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GPR																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CPU[GPR] [31:0]

Bits	Name	Description
31-0	GPR	register for software to handle resume, can save resume address or status

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Bits	Name	Description

CPU[GPR] Bits

13.8.33 CPU[WAKEUP_STATUS] (0x2840 + 0x400 * n + 0x4 * m)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
STATUS																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CPU[WAKEUP_STATUS] [31:0]

Bits	Name	Description
31-0	STATUS	IRQ values

CPU[WAKEUP_STATUS] Bits

13.8.34 CPU[WAKEUP_ENABLE] (0x2880 + 0x400 * n + 0x4 * m)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ENABLE																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CPU[WAKEUP_ENABLE] [31:0]

Bits	Name	Description
31-0	ENABLE	IRQ wakeup enable

CPU[WAKEUP_ENABLE] Bits

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14 Phase-Locked Loop Controller PLLCTL

This section describes the main functions and features of the phase-locked loop controller PLLCTL.

14.1 Feature Summary

This product integrates a phase-locked loop controller PLLCTL for configuring the on-chip PLL. Main features are as follows:

- XTAL oscillator configuration
- On-chip PLL working mode configuration
  - Configure PLL reference clock
  - Configure PLL operating frequency
  - Configure PLL spread spectrum function
- Output frequency division configuration
- Support hardware automatic control
- Support frequency modification during operation

14.2 Functional Description

This section describes the main functions of the phase-locked loop controller PLLCTL.

14.2.1 XTAL Oscillator

XTAL oscillator You can read the working status of XTAL, users can query the ENABLE flag bit of the XTAL register, a value of 1 indicates that the crystal oscillator as the reference clock source is turned on, query the RESPONSE flag bit of the XTAL register, when set to 1, it indicates that the reference clock source is in stable status. You can set the time required for XTAL startup, the default value of this time is 3ms, this setting affects the entry and exit time of turning off XTAL low-power mode.

14.2.2 PLL Reference Clock Setting

The PLL reference clock can be selected through REFSEL:

- 0: Select XTAL
- 1: Select RC24M

This selection can be modified when the PLL is working, modification during operation will cause short-term fluctuation of the PLL output clock frequency.

14.2.3 PLL Operating Frequency Configuration

The output frequency of the PLL voltage-controlled oscillator is as follows:

$$F_vco = F_ref \times (MFI + (MFN \div MFD))$$

The default value of MFD is 240000000, and the value 1 of MFN represents 0.1Hz.

The MFI and MFN fields of the PLL register support modification during operation, after modification, you can read the BUSY flag to judge whether the PLL works stably under the new settings. The PLL will not stop

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clock output when switching to the new settings. MFD does not support modification during operation, if modified, it will take effect after the PLL is turned off and turned on again.

14.2.4 PLL Spread Spectrum Mode

The spread spectrum mode of PLL can be enabled through SPREAD:

- 0: Disable spread spectrum
- 1: Enable spread spectrum

Spread spectrum can be enabled or disabled when the PLL is working. The PLL adopts down spread spectrum, when the PLL spread spectrum is enabled, the VCO oscillation frequency sweeps between the upper and lower limits of spread spectrum, increasing or decreasing

$$F_ref \times (STEP \div MFD)$$

every 24M clock cycle.

The upper limit of VCO oscillation frequency is:

$$F_vco = F_ref \times (MFI + (MFN \div MFD))$$

The lower limit of VCO oscillation frequency is:

$$F_vco = F_ref \times (MFI + ((MFN - STOP) \div MFD))$$

When the PLL works in spread spectrum mode, STEP and STOP do not support setting modification during operation. If modified at this time, the new settings will take effect after spread spectrum is disabled and re-enabled, or the PLL is turned off and turned on again.

14.2.5 PLL Output Frequency Divider

The VCO clock is divided and output to the clock module to generate the system clock. This divider is a fractional divider, and the division coefficient steps by 0.2.

$$F_out = F_vco \div (1 + 0.2 \times DIV)$$

DIV supports modification of operating frequency during operation.

14.3 PLLCTL Registers

14.3.1 Register Description

The register list of PLLCTLv2 is as follows:

PLLCTLV2 base address: 0xF40C0000

Offset	Register Name	Description	Reset Value
0x0000	XTAL	OSC configuration	0x0001FFFF

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Offset	Register Name	Description	Reset Value
0x0080	PLL[PLL0][MFI]	PLL0 multiple register	0x00000010
0x0084	PLL[PLL0][MFN]	PLL0 fraction numerator register	0x09896800
0x0088	PLL[PLL0][MFD]	PLL0 fraction demoninator register	0x0E4E1C00
0x008C	PLL[PLL0][SS_STEP]	PLL0 spread spectrum step register	0x00000000
0x0090	PLL[PLL0][SS_STOP]	PLL0 spread spectrum stop register	0x00000000
0x0094	PLL[PLL0][CONFIG]	PLL0 confguration register	0x00000000
0x0098	PLL[PLL0][LOCKTIME]	PLL0 lock time register	0x000009C4
0x009C	PLL[PLL0][STEPTIME]	PLL0 step time register	0x000009C4
0x00A0	PLL[PLL0][ADVANCED]	PLL0 advance configuration register	0x00000000
0x00C0	PLL[PLL0][DIV][DIV0]	PLL0 divider output 0 configuration register	0x00000000
0x00C4	PLL[PLL0][DIV][DIV1]	PLL0 divider output 1 configuration register	0x00000000
0x00C8	PLL[PLL0][DIV][DIV2]	PLL0 divider output 2 configuration register	0x00000000
0x0100	PLL[PLL1][MFI]	PLL0 multiple register	0x00000010
0x0104	PLL[PLL1][MFN]	PLL0 fraction numerator register	0x09896800
0x0108	PLL[PLL1][MFD]	PLL0 fraction demoninator register	0x0E4E1C00
0x010C	PLL[PLL1][SS_STEP]	PLL0 spread spectrum step register	0x00000000
0x0110	PLL[PLL1][SS_STOP]	PLL0 spread spectrum stop register	0x00000000
0x0114	PLL[PLL1][CONFIG]	PLL0 confguration register	0x00000000
0x0118	PLL[PLL1][LOCKTIME]	PLL0 lock time register	0x000009C4
0x011C	PLL[PLL1][STEPTIME]	PLL0 step time register	0x000009C4
0x0120	PLL[PLL1][ADVANCED]	PLL0 advance configuration register	0x00000000
0x0140	PLL[PLL1][DIV][DIV0]	PLL0 divider output 0 configuration register	0x00000000
0x0144	PLL[PLL1][DIV][DIV1]	PLL0 divider output 1 configuration register	0x00000000
0x0148	PLL[PLL1][DIV][DIV2]	PLL0 divider output 2 configuration register	0x00000000

Table 28: PLLCTLV2 Register List

The detailed description of PLLCTLv2 registers is as follows:

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14.3.2 XTAL (0x0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BUSY	RSVD	RESPONSE	ENABLE	RSVD								RAMP_TIME																			
RO	N/A	RO	RO	N/A								RW																			
0	x	0	0	x	x	x	x	x	x	x	x	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

XTAL [31:0]

Bits	Name	Description
31	BUSY	Busy flag 0: Oscillator is working or shutdown 1: Oscillator is changing status
29	RESPONSE	Crystal oscillator status 0: Oscillator is not stable 1: Oscillator is stable for use
28	ENABLE	Crystal oscillator enable status 0: Oscillator is off 1: Oscillator is on
19-0	RAMP_TIME	Rampup time of XTAL oscillator in cycles of RC24M clock 0: 0 cycle 1: 1 cycle 2: 2 cycle 1048575: 1048575 cycles

XTAL Bits

14.3.3 PLL[MFI] (0x80 + 0x80 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BUSY	RSVD	RESPONSE	ENABLE	RSVD																					MFI						
RO	N/A	RO	RO	N/A																					RW						
0	x	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	1	0	0	0	0

PLL[MFI] [31:0]

Bits	Name	Description
31	BUSY	Busy flag 0: PLL is stable or shutdown 1: PLL is changing status

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Bits	Name	Description
29	RESPONSE	PLL status 0: PLL is not stable 1: PLL is stable for use
28	ENABLE	PLL enable status 0: PLL is off 1: PLL is on
6-0	MFI	loop back divider of PLL, support from 13 to 42, f=fref*(mfi + mfn/mfd) 0-15: invalid 16: divide by 16 17: divide by17 . . . 42: divide by 42 43~:invalid

PLL[MFI] Bits

14.3.4 PLL[MFN] (0x84 + 0x80 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD		MFN																													
N/A		RW																													
x	x	0	0	1	0	0	1	1	0	0	0	1	0	0	1	0	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0

PLL[MFN] [31:0]

Bits	Name	Description
29-0	MFN	Numeratorof fractional part,f=fref*(mfi + mfn/mfd). This field supports changing while running.

PLL[MFN] Bits

14.3.5 PLL[MFD] (0x88 + 0x80 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD		MFD																													
N/A		RW																													
x	x	0	0	1	1	1	0	0	1	0	0	1	1	1	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0

PLL[MFD] [31:0]

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Bits	Name	Description
29-0	MFD	Demoninator of fraction part,f=fref*(mfi + mfn/mfd). This field should not be changed during PLL enabled. If changed, change will take efftect when PLL re-enabled.

PLL[MFD] Bits

14.3.6 PLL[SS_STEP] (0x8C + 0x80 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																STEP															
N/A																RW															
x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PLL[SS_STEP] [31:0]

Bits	Name	Description
29-0	STEP	Step of spread spectrum modulator. This register should not be changed during PLL and spread spectrum enabled. If changed, new value will take effect when PLL disabled or spread spectrum disabled.

PLL[SS_STEP] Bits

14.3.7 PLL[SS_STOP] (0x90 + 0x80 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																STOP															
N/A																RW															
x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PLL[SS_STOP] [31:0]

Bits	Name	Description
29-0	STOP	Stop point of spread spectrum modulator This register should not be changed during PLL and spread spectrum enabled. If changed, new value will take effect when PLL disabled or spread spectrum disabled.

PLL[SS_STOP] Bits

14.3.8 PLL[CONFIG] (0x94 + 0x80 * n)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																							SPREAD	RSVD							REFSEL
N/A																							RW	N/A							RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	x	x	x	x	0

PLL[CONFIG] [31:0]

Bits	Name	Description
8	SPREAD	Enable spread spectrum function. This field supports changing during PLL running.
0	REFSEL	Select reference clock, This filed support changing while running, but application must take frequency error and jitter into consideration. And if MFN changed before reference switch, application need make sure time is enough for MFN updating. 0: XTAL24M 1: IRC24M

PLL[CONFIG] Bits

14.3.9 PLL[LOCKTIME] (0x98 + 0x80 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																LOCKTIME															
N/A																RW															
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	1	0	0	1	1	1	0	0	0	1	0	0

PLL[LOCKTIME] [31:0]

Bits	Name	Description
15-0	LOCKTIME	Lock time of PLL in 24M clock cycles, typical value is 2500. If MFI changed during PLL startup, PLL lock time may be longer than this setting.

PLL[LOCKTIME] Bits

14.3.10 PLL[STEPTIME] (0x9C + 0x80 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																STEPTIME															

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
N/A																RW															
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	1	0	0	1	1	1	0	0	0	1	0	0

PLL[STEPTIME] [31:0]

Bits	Name	Description
15-0	STEPTIME	Step time for MFI on-the-fly change in 24M clock cycles, typical value is 2500.

PLL[STEPTIME] Bits

14.3.11 PLL[ADVANCED] (0xA0 + 0x80 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD			SLOW	RSVD			DITHER	RSVD																							
N/A			RW	N/A			RW	N/A																							
x	x	x	0	x	x	x	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

PLL[ADVANCED] [31:0]

Bits	Name	Description
28	SLOW	Use slow lock flow, PLL lock expendite is disabled. This mode might be stabler. And software need config LOCKTIME field accordingly. 0: fast lock enabled, lock time is 100us 1: fast lock disabled, lock time is 400us
24	DITHER	Enable dither function

PLL[ADVANCED] Bits

14.3.12 PLL[DIV] (0xC0 + 0x80 * n + 0x4 * m)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BUSY	RSVD	RESPONSE	ENABLE	RSVD																						DIV					
RO	N/A	RO	RO	N/A																						RW					
0	x	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0

PLL[DIV] [31:0]

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Bits	Name	Description
31	BUSY	Busy flag 0: divider is working 1: divider is changing status
29	RESPONSE	Divider response status 0: Divider is not stable 1: Divider is stable for use
28	ENABLE	Divider enable status 0: Divider is off 1: Divider is on
5-0	DIV	Divider factor, divider factor is DIV/5 + 1 0: divide by 1 1: divide by 1.2 2: divide by 1.4 . . . 63: divide by 13.6

PLL[DIV] Bits

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15 Low Power Management

This chapter introduces the system-level low power management and applications.

15.1 RUN Mode

The processor is in RUN mode when running programs normally. The following power reduction methods are supported in RUN mode:

- Configure the GROUPx registers of SYSCTL correctly to enable only necessary functional modules, and unused modules will be automatically turned off to save power
- Appropriately reduce the clock frequency of CPU and functional modules by configuring FBDIV, POSTDIV of PLL and the frequency division coefficients of functional clocks in SYSCTL
- If the system power domain is powered by DCDC, configure the PCFG module to appropriately reduce the DCDC output voltage as required

15.2 WAIT Mode

If the MODE bit of SYSCTL register CPU[CPUx][LP] is set to b00, the CPU will enter WAIT mode after executing the WFI instruction, and the CPU core clock will be turned off. But the CPU can wake up immediately to process interrupts when receiving an interrupt.

15.3 STOP Mode

If the MODE bit of SYSCTL register CPU[CPUx][LP] is set to b01, the CPU will enter STOP mode after executing the WFI instruction. In STOP mode, the SYSCTL module will turn off part of the resource nodes in the system power domain according to the configuration.

Compared with WAIT mode, STOP mode allows more resources to be turned off and provides flexible configuration options. Users need to perform reasonable configuration in the SYSCTL module according to the requirements of power consumption and wake-up time.

Consider the simplest application scenario 1: The system only uses CPU0, only necessary functional modules are connected and enabled through the GROUP0 register of SYSCTL, and the RETENTION[CPU0] register is modified to all 0s. First, unused resource nodes have been automatically turned off in RUN mode. After the CPU executes WFI and enters STOP mode, the clock of CPU0, the power supply of CPU0 subsystem, the functional modules used by CPU0, functional clocks, power supplies of other subsystems, PLL and 24M crystal oscillator will be turned off in sequence, only the function of SYSCTL is reserved to wait for wake-up.

Application scenario 2: The system only uses CPU0. If it is required that the register state of CPU0 will not be lost after entering STOP mode, that is, the CPU0 subsystem will not be powered down or reset, and the 24M crystal oscillator and PLL will not be turned off, so as to achieve faster wake-up speed, the RETENTION[CPU0] register of SYSCTL needs to be set, and the bits corresponding to CPU0 registers, 24M crystal oscillator and PLL should be set to 1.

Application scenario 3: The system only uses CPU0. If it is required that a functional module such as GPTMR0 keeps running after CPU0 enters STOP mode, and the module is not within the management range of the RETENTION register, the RESOURCE[GPTMR0] register of SYSCTL needs to be set, and the MODE bit should be set to 1 to keep it enabled all the time.

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Note that the default value of the RETENTION[CPU0] register reserves the power supply of the CPU0 subsystem and the system power domain.

Interrupts can wake up the system from STOP mode. These interrupts can come from functional modules that remain working in the system power domain, or from the power management domain. It is necessary to ensure that the expected wake-up interrupt source has been enabled in the SYSCTL register CPU[CPUx][WAKEUP_ENABLE][E

In conclusion, the steps involved in configuring STOP mode are as follows:

- Configure the MODE bit of SYSCTL register CPU[CPUx][LP] to b01
- Configure SYSCTL register RETENTION[x] to keep part of the resource nodes enabled
- Configure SYSCTL register RESOURCE[x] to keep more resource nodes enabled
- Configure SYSCTL register CPU[CPUx][WAKEUP_ENABLE][ENABLEx] to specify the wake-up interrupt source
- Configure the module used to generate wake-up interrupts, set the interrupt trigger conditions and enable the interrupt
- Execute WFI instruction

15.4 STANDBY Mode

In STANDBY mode, the entire system power domain no longer operates, and it may be in reset state or power-down state. If there is no resource node that needs to keep running in the RETENTION register in the SYSCTL configuration, and there is no resource node that is forcibly enabled in the RESOURCE register, then when all CPUs enter STOP mode, the entire system power domain has no operation requirements. After turning off all resource nodes, the SYSCTL module will send a request to the power management domain to turn off the system power domain. After the request is issued, the system power domain will be cold reset first, and then the PCFG module will perform operations according to the settings of the POWER_TRAP register:

If DCDC is used to supply power to the system power domain, the DCDC will switch states according to Table 21: normal operation, turn off, or enter low power mode, where DCDC can be configured to provide a lower output voltage in low power mode.

After entering STANDBY mode, the system can only be woken up by the interrupt of the module in the power management domain. For details of the wake-up sources, see Section 12.2.5.

In addition, the unused modules in the power management domain can turn off their clocks by setting the SCG_CTRL register of PCFG, so as to further reduce the system power consumption in STANDBY mode. For details, see Section 12.2.4.

In conclusion, the steps involved in configuring STANDBY mode are as follows:

- Configure the MODE bit of SYSCTL register CPU[CPUx][LP] to b01
- Configure SYSCTL register RETENTION[x] to all 0s
- Configure the MODE bit in all SYSCTL register RESOURCE[x] to be not 1
- Configure the TRIGGERED bit and RETENTION bit of the POWER_TRAP register of the PCFG module to set the holding, turning off or voltage reduction of the system power domain power supply
- If an external power supply is used to supply power to the system power domain and pins are used to control its behavior, the function selection of PZ10 or PZ11 needs to be configured in BIOC
- Configure the WAKE_MASK register of the PCFG module to enable the wake-up source

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- Configure the interrupt trigger conditions of the wake-up source module in the power management domain and enable the interrupt
- Execute WFI instruction

15.5 SHUTDOWN Mode

In SHUTDOWN mode, only the PDGO module of the power management domain can work normally, and other modules of the power management domain except PDGO and the system power domain are in reset or power-down state.

The way to enter SHUTDOWN mode is software register writing:

- Write a non-zero countdown value to the countdown shutdown register DGO_TURNOFF, and the system enters SHUTDOWN mode after the countdown ends

Consider the following scenarios according to the external power supply structure of the chip:

Scenario 1: Single 3.3V power supply, and DCDC is used to supply power to the system power domain. When entering SHUTDOWN mode, the modules of the power management domain except DGO enter the reset state, and the DCDC is turned off so that the system power domain is powered down.

Scenario 2: Single 3.3V power supply, and an external power supply is used to supply power to the system power domain. When entering SHUTDOWN mode, the modules of the power management domain except DGO and the system power domain enter the reset state.

In SHUTDOWN mode, the system can only be woken up from the PDGO module. For details of how to exit SHUTDOWN mode, see **??**.

15.6 CPU Local Memory Retention

The CPU local memories ILM and DLM of this product support memory retention function. When the system power domain is powered (VDD_SOC is powered on) and the CPU0 subsystem is powered down (VDD_CPU is powered off), the memory core of ILM and DLM of the CPU local memory remains powered, and the stored content can be retained, which is convenient for future power-on recovery. This function does not require configuration, nor is it a low-power mode. In the above modes, as long as the system power domain is powered and the CPU0 subsystem is powered down, this function can be automatically implemented.

15.7 Low Power Summary

Figure 11 describes the control flow and scope of each low-power mode.

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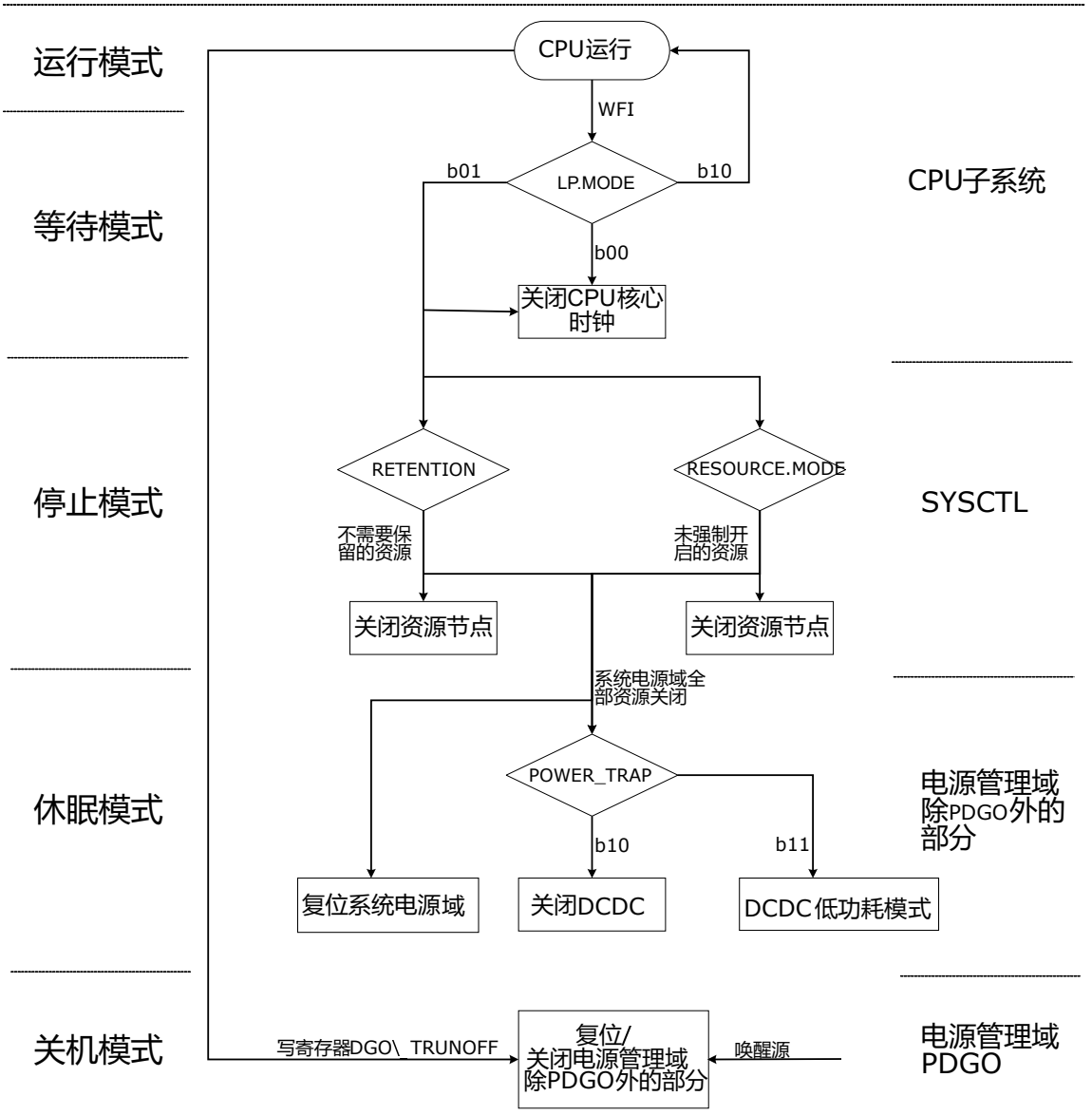


Figure 11: Low Power Mode Flow Chart

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16 System Memory Mapping

This chapter describes the system memory mapping of this product.

16.1 System Memory Map

The system memory mapping table of this product is as follows:

Start Address	End Address	Size	Name	Description
0x00000000	0x0001FFFF	128 KBytes	CPU0_ILM_SLV	INSTRUCTION LOCAL MEMORY
0x00060000	0x0007FFFF	128 KBytes	CPU0_ILM_SLV_ALIAS	
0x00080000	0x0009FFFF	128 KBytes	CPU0_DLM_SLV	DATA LOCAL MEMORY
0x000A0000	0x000BFFFF	128 KBytes	CPU0_DLM_SLV_ALIAS	
0x000C0000	0x000FFFFF	256 KBytes	FGPIO	FAST GPIO CONTROLLER
0x01040000	0x0105FFFF	128 KBytes	CPU0_ILM_SLV_ALS	CPU ILM MIRROR
0x01060000	0x0107FFFF	128 KBytes	CPU0_DLM_SLV_ALS	CPU DLM MIRROR
0x20000000	0x2001FFFF	128 KBytes	ROM	READ-ONLY MEMORY (ROM)
0x30000000	0x300FFFFF	1 MBytes	DM	DEBUG MODULE
0x80000000	0x8FEFFFFF	255 MBytes	XPI0	XPI SERIAL BUS CONTROLLER MEMORY SPACE
0xE4000000	0xE43FFFFF	4 MBytes	PLIC	PLATFORM INTERRUPT CONTROLLER (PLIC)
0xE6000000	0xE60FFFFF	1 MBytes	MCHTMR	MACHINE TIMER
0xE6400000	0xE67FFFFF	4 MBytes	PLICSW	PLATFORM SOFTWARE INTERRUPT CONTROLLER
0xF0000000	0xF0003FFF	16 KBytes	GPTMR0	TIMER 0
0xF0004000	0xF0007FFF	16 KBytes	GPTMR1	TIMER 1
0xF0008000	0xF000BFFF	16 KBytes	GPTMR2	TIMER 2
0xF000C000	0xF000FFFF	16 KBytes	GPTMR3	TIMER 3
0xF0040000	0xF0043FFF	16 KBytes	UART0	UNIVERSAL ASYNCHRONOUS RE-CEIVER/TRANSMITTER UART0

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Start Address	End Address	Size	Name	Description
0xF0044000	0xF0047FFF	16 KBytes	UART1	UNIVERSAL ASYNCHRONOUS RECEIVER/TRANSMITTER UART1
0xF0048000	0xF004BFFF	16 KBytes	UART2	UNIVERSAL ASYNCHRONOUS RECEIVER/TRANSMITTER UART2
0xF004C000	0xF004FFFF	16 KBytes	UART3	UNIVERSAL ASYNCHRONOUS RECEIVER/TRANSMITTER UART3
0xF0050000	0xF0053FFF	16 KBytes	UART4	UNIVERSAL ASYNCHRONOUS RECEIVER/TRANSMITTER UART4
0xF0054000	0xF0057FFF	16 KBytes	UART5	UNIVERSAL ASYNCHRONOUS RECEIVER/TRANSMITTER UART5
0xF0058000	0xF005BFFF	16 KBytes	UART6	UNIVERSAL ASYNCHRONOUS RECEIVER/TRANSMITTER UART6
0xF005C000	0xF005FFFF	16 KBytes	UART7	UNIVERSAL ASYNCHRONOUS RECEIVER/TRANSMITTER UART7
0xF0060000	0xF0063FFF	16 KBytes	I2C0	INTER-INTEGRATED CIRCUIT BUS I2C0
0xF0064000	0xF0067FFF	16 KBytes	I2C1	INTER-INTEGRATED CIRCUIT BUS I2C1
0xF0068000	0xF006BFFF	16 KBytes	I2C2	INTER-INTEGRATED CIRCUIT BUS I2C2
0xF006C000	0xF006FFFF	16 KBytes	I2C3	INTER-INTEGRATED CIRCUIT BUS I2C3
0xF0070000	0xF0073FFF	16 KBytes	SPI0	SERIAL PERIPHERAL INTERFACE SPI0
0xF0074000	0xF0077FFF	16 KBytes	SPI1	SERIAL PERIPHERAL INTERFACE SPI1

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Start Address	End Address	Size	Name	Description
0xF0078000	0xF007BFFF	16 KBytes	SPI2	SERIAL PERIPHERAL INTERFACE SPI2
0xF007C000	0xF007FFFF	16 KBytes	SPI3	SERIAL PERIPHERAL INTERFACE SPI3
0xF0080000	0xF0083FFF	16 KBytes	CRC	CYCLIC REDUNDANCY CHECK (CRC)
0xF0090000	0xF0093FFF	16 KBytes	TSNS	TEMPERATURE SENSOR
0xF00A0000	0xF00A3FFF	16 KBytes	MBX0A	MAILBOX MBX0A
0xF00A4000	0xF00A7FFF	16 KBytes	MBX0B	MAILBOX MBX0B
0xF00B0000	0xF00B3FFF	16 KBytes	EWDG0	WATCHDOG WDG0
0xF00B4000	0xF00B7FFF	16 KBytes	EWDG1	WATCHDOG WDG1
0xF00C0000	0xF00C3FFF	16 KBytes	MISC	MISCELLANEOUS CONTROL REGISTERS
0xF00C4000	0xF00C7FFF	16 KBytes	DMAMUX	DMA REQUEST ROUTER (DMAMUX)
0xF00C8000	0xF00CBFFF	16 KBytes	HDMA	DMA CONTROLLER (HDMA)
0xF00D0000	0xF00D3FFF	16 KBytes	GPIO0	GPIO CONTROLLER GPIO0
0xF00D8000	0xF00DBFFF	16 KBytes	GPIOM	GPIO MANAGER
0xF0280000	0xF0283FFF	16 KBytes	MCAN0	CONTROLLER AREA NETWORK CAN0
0xF0284000	0xF0287FFF	16 KBytes	MCAN1	CONTROLLER AREA NETWORK CAN1
0xF0288000	0xF028BFFF	16 KBytes	MCAN2	CONTROLLER AREA NETWORK CAN2
0xF028C000	0xF028FFFF	16 KBytes	MCAN3	CONTROLLER AREA NETWORK CAN3
0xF02FC000	0xF02FFFFF	16 KBytes	PTPC	PRECISION TIME PROTOCOL CONTROLLER (PTPC)
0xF0300000	0xF0303FFF	16 KBytes	QEI0	QUADRATURE ENCODER INPUT QEI0
0xF0304000	0xF0307FFF	16 KBytes	QEI1	QUADRATURE ENCODER INPUT QEI1
0xF0308000	0xF030BFFF	16 KBytes	QEO0	QUADRATURE ENCODER OUTPUT QEO0

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Start Address	End Address	Size	Name	Description
0xF030C000	0xF030FFFF	16 KBytes	QEO1	QUADRATURE ENCODER OUTPUT QEO1
0xF0310000	0xF0313FFF	16 KBytes	MMC0	MOTION MANAGEMENT CONTROLLER MMC0
0xF0314000	0xF0317FFF	16 KBytes	MMC1	MOTION MANAGEMENT CONTROLLER MMC1
0xF0318000	0xF031BFFF	16 KBytes	PWM0	PWM TIMER PWM0
0xF031C000	0xF031FFFF	16 KBytes	PWM1	PWM TIMER PWM1
0xF0320000	0xF0323FFF	16 KBytes	RDC	ROTATIONAL DECODER (RDC)
0xF0324000	0xF0327FFF	16 KBytes	PLB	PROGRAMMABLE LOGIC BLOCK (PLB)
0xF0328000	0xF032BFFF	16 KBytes	SYNT	SYNCHRONIZATION TIMER (SYNT)
0xF032C000	0xF032FFFF	16 KBytes	SEI	SERIAL ENCODER INTERFACE (SEI)
0xF033C000	0xF033FFFF	16 KBytes	TRGM0	MOTOR GENERAL-PURPOSE REGISTER (GPR)
0xF0400000	0xF0407FFF	32 KBytes	HRAM	AHB SRAM
0xF3000000	0xF3003FFF	16 KBytes	XPI0	XPI SERIAL BUS CONTROLLER
0xF300C000	0xF300FFFF	16 KBytes	USB0	UNIVERSAL SERIAL BUS (USB)
0xF3040000	0xF3043FFF	16 KBytes	SDP	SECURE DATA PROCESSOR (SDP)
0xF3044000	0xF3047FFF	16 KBytes	SEC	SECURITY MANAGER
0xF3048000	0xF304BFFF	16 KBytes	MON	SECURITY MONITOR
0xF304C000	0xF304FFFF	16 KBytes	RNG	RANDOM NUMBER GENERATOR (RNG)
0xF3050000	0xF3053FFF	16 KBytes	OTP	FUSE (OTP)
0xF3054000	0xF3057FFF	16 KBytes	KEYM	KEY MANAGER (KMAN)
0xF3080000	0xF3083FFF	16 KBytes	ADC0	ANALOG-TO-DIGITAL CONVERTER ADC0
0xF3084000	0xF3087FFF	16 KBytes	ADC1	ANALOG-TO-DIGITAL CONVERTER ADC1
0xF3090000	0xF3093FFF	16 KBytes	DAC0	DIGITAL-TO-ANALOG CONVERTER DAC0
0xF3094000	0xF3097FFF	16 KBytes	DAC1	DIGITAL-TO-ANALOG CONVERTER DAC1

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Start Address	End Address	Size	Name	Description
0xF30A0000	0xF30A3FFF	16 KBytes	OPAMP0	OPERATIONAL AMPLIFIER OPAMP0
0xF30A4000	0xF30A7FFF	16 KBytes	OPAMP1	OPERATIONAL AMPLIFIER OPAMP1
0xF30B0000	0xF30B3FFF	16 KBytes	ACMP	ANALOG COMPARATOR (ACMP)
0xF4000000	0xF403FFFF	256 KBytes	SYSCTL	SYSTEM CONTROL MODULE
0xF4040000	0xF407FFFF	256 KBytes	IOC	IO CONTROLLER
0xF40C0000	0xF40FFFFF	256 KBytes	PLLCTLV2	PLL CONTROLLER
0xF4100000	0xF4103FFF	16 KBytes	PPOR	POWER MANAGEMENT DOMAIN RESET MODULE
0xF4104000	0xF4107FFF	16 KBytes	PCFG	POWER MANAGEMENT DOMAIN CONFIGURATION REGISTERS
0xF4110000	0xF4113FFF	16 KBytes	PGPR0	POWER MANAGEMENT DOMAIN GPR 0
0xF4114000	0xF4117FFF	16 KBytes	PGPR1	POWER MANAGEMENT DOMAIN GPR 1
0xF4118000	0xF411BFFF	16 KBytes	PIOC	POWER MANAGEMENT DOMAIN IO CONTROLLER
0xF411C000	0xF411FFFF	16 KBytes	PGPIO	POWER MANAGEMENT DOMAIN GPIO CONTROLLER
0xF4120000	0xF4123FFF	16 KBytes	PTMR	POWER MANAGEMENT DOMAIN TIMER
0xF4124000	0xF4127FFF	16 KBytes	PUART	POWER MANAGEMENT DOMAIN UART
0xF4128000	0xF412BFFF	16 KBytes	PEWDG	POWER MANAGEMENT DOMAIN WATCHDOG
0xF4134000	0xF4137FFF	16 KBytes	PDGO	POWER MANAGEMENT DOMAIN DGO REGISTERS

Table 29: Address space allocation

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17 OTP Mapping Table

The OTP mapping table of this product is as follows:

Offset	Name	Position	Function	Type
0	HARD_LOCK	[0:31]	Fuse value lock	Security
1	LIFECYCLE_B	[28:31]	Lifecycle	Security
1	TCU_DISABLE	[19:19]	Disable test	Security
1	DEBUG_DISABLE	[17:17]	Disable CPU debug	Security
1	JTAG_DISABLE	[16:16]	Disable JTAG port	Security
1	PUK_REVOKE	[8:15]	Public key revoke	Security
1	LIFECYCLE_A	[0:3]	Lifecycle	Security
3	SW_VER	[0:31]	Software version	Security
8	DIE_TRACE	[0:127]	Die trace data	Identity
12	DEBUG_KEY	[0:127]	Debug key	Secret
21	TSNS_BASE	[0:11]	Temperature sensor baseline	General
21	TSNS_SLOPE	[12:23]	Temperature sensor slope	General
64	CHIP_ID	[0:31]	Chip ID code	General
68	USB_PID	[0:15]	USB product ID	General
68	USB_VID	[16:31]	USB vendor ID	General
80	PUBLIC_KEY_HASH	[0:255]	Public key hash	Identity
88	UUID	[0:127]	Unique ID	Identity
96	EXIP0_KEY	[0:255]	EXIP key 0	Secret
112	MASTER_KEY	[0:255]	Symmetric root key	Secret

Table 30: OTP List

For this product, users can program OTP Word 69~79 by themselves to store application data.

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18 Bus

This chapter introduces the bus structure of this product.

18.1 Bus Structure Overview

The bus system of this product consists of multiple system bus matrices. All on-chip modules are connected through these system bus matrices, enabling bus masters such as processors and DMA to access bus slaves such as memory and peripherals. Since each master and slave device is connected to different positions of different bus matrices, the access efficiency of different master devices to different slave devices will also vary.

This product includes the following system bus matrices:

- AHB system bus, which is the central bus. Most master devices on the system access on-chip resources through it;
- APB peripheral bus, through which the registers of on-chip peripherals can be accessed;

18.2 Bus Configuration

This chapter introduces the connection and configuration of each bus.

18.2.1 AHB System Bus

The AHB system bus is the central bus matrix.

The master devices connected to it are:

- RISC-V CPU
- HDMA
- Debug module (DM)
- USB
- XPISLV
- Encryption and decryption module
- ADCx&DACx

The slave devices accessible through it are:

- On-chip read-only memory ROM
- XPI controller and supporting online decryption module
- General peripheral register interface
- CAN bus register interface and motor control peripheral register interface
- HRAM
- Security module register interface, analog peripheral register interface, USB, XPI, EXIP registers, etc.
- Power management domain peripheral register interface, system control module register interface, phase-locked loop register, etc.
- ILM and ILM mirror, DLM and DLM mirror, etc.

18.2.2 APB Peripheral Bus

The APB peripheral bus connects most of the on-chip peripherals. RISC-V processor and DMA access peripheral registers through it.

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19 BootROM

This chapter details the features and applications related to BootROM.

19.1 BootROM Overview

BootROM is the first program executed after the chip powers on, it supports the following features:

- Boot from Serial NOR FLASH
  - Support for code execution in place on FLASH or copying to internal RAM for execution
  - Support for AES-CTR mode in-place decryption execution (EXIP)
- Serial Boot
  - Boot from UART0 or USB0, support streaming code download to system RAM for execution
- In-System Programming (ISP)
  - Support UART0 and USB0 (USB-HID protocol)
  - Support programming NOR FLASH connected to XPI0
  - Support programming on-chip OTP
- Secure Boot
  - Support ECDSA-P256 + SHA256 secure boot
  - Support AES-CBC + SHA256 secure encrypted boot
  - Support SM2+SM3 secure boot
  - Support SM4-CBC + SM3 secure encrypted boot
- Low Power Wakeup
  - Support fast wakeup via BootROM in CPU low power down mode
- Debug Port Permission Management
- Rich ROM API

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19.2 Boot Flow

This chapter describes BootROM boot-related processes.

BootROM supports the following boot modes:

- Primary boot mode (XPI NOR boot mode)
- Serial boot mode (UART, USB-HID)
- In-System Programming mode (In-System-Programming)

19.2.1 Boot Flow Diagram

This section shows the BootROM boot flow diagram.

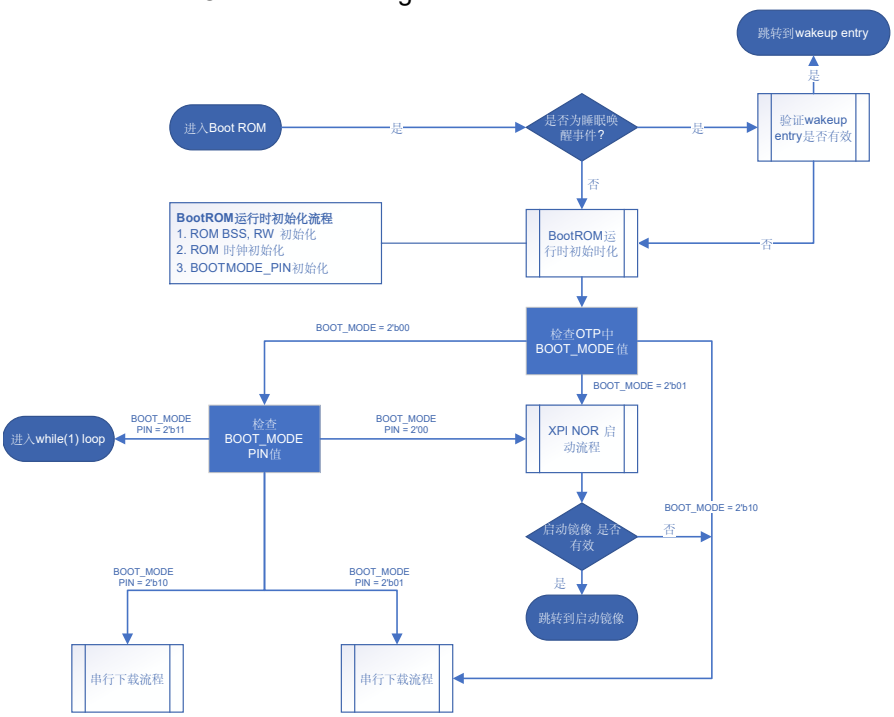


Figure 12: BootROM Boot Flow Diagram

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19.2.2 XPI NOR Boot

19.2.2.1 Supported Serial NOR Devices

Boot ROM supports booting from mainstream Serial NOR FLASH, including but not limited to:

- QSPI NOR FLASH (Winbond, GigaDevice, ISSI, etc.)

19.2.2.2 FLASH Connection Method

BootROM connects to 4-wire FLASH through the following typical connection.

QSPI FLASH Connected via CA

The SoC connects to 4-wire FLASH through the XPI CA port. Most 4-wire FLASH on the market do not provide a DQS pin, in which case XPIn.CA_DQS should be left floating, and this pin cannot be used for other purposes (if the XPI info section has special statements, follow those statements).



Figure 13: XPI QSPI NOR FLASH CA Port Connection

FLASH Connection Method Specification

BootROM determines the corresponding FLASH connection based on the following OTP fields:

- Default XPI pin group selected based on XPI_PIN_GROUP

For detailed XPI pin assignment, see XPI info section.

FLASH Configuration Information Specification

BootROM determines the corresponding FLASH configuration information based on the following OTP fields:

- Default FLASH configuration information source determined based on XPI_NOR_CFG_SRC: Serial NOR FLASH or OTP
- Default FLASH access command after power-on determined based on XPI_DEFAULT_READ
- Decide whether to enable encrypted execute-in-place based on ENCRYPT_XIP

When FLASH configuration information comes from OTP:

- FLASH read operation clock frequency determined based on XPI_FREQ_OPTION
- FLASH type determined based on PROBE_TYPE
- XPI IO drive strength determined based on XPI_DRIVE_STRENGTH
- dummy cycle value for read determined based on DUMMY_CYCLE

When FLASH configuration information comes from FLASH, determine the FLASH read command to use based on XPI_DEFAULT_READ, and read FLASH configuration information from FLASH address 0x400.

Note: For detailed description of FLASH configuration information, see XPI NOR Configuration Option

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19.2.2.3 XPI NOR Boot Flow

When users select to boot from XPI NOR, after the device powers on, it will execute XPI NOR boot according to the flow shown in Figure (14).

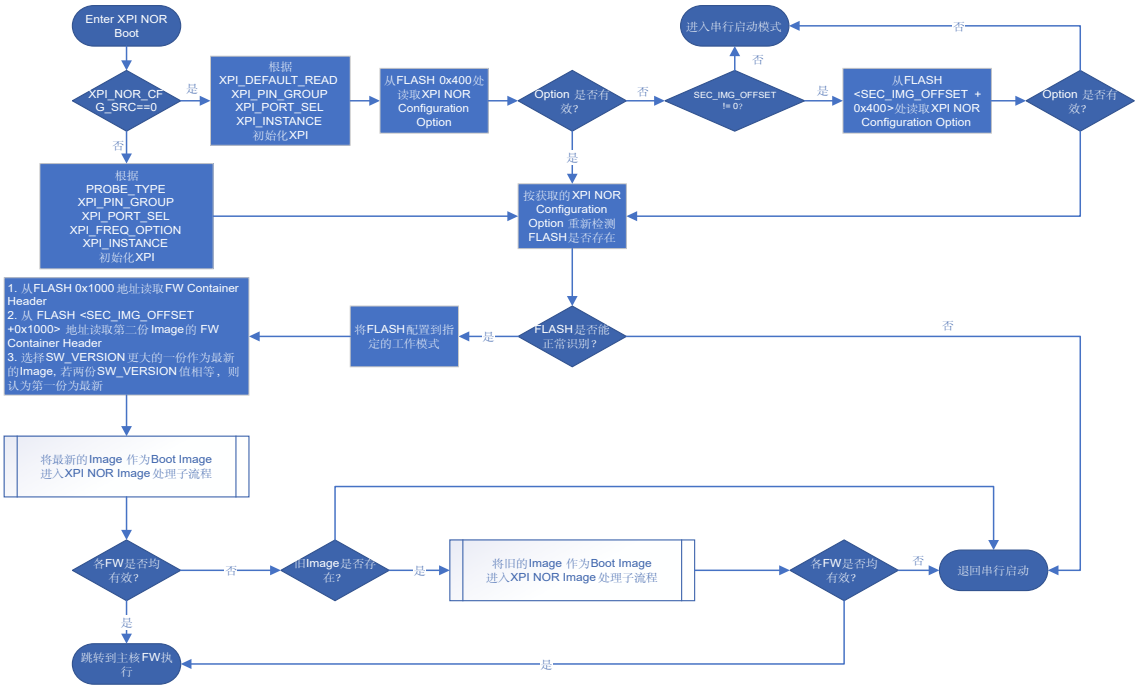


Figure 14: XPI NOR Boot Flow

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19.2.2.4 XPI NOR Boot Image Layout

The XPI NOR boot image layout in FLASH is shown in Figure (15):



Figure 15: XPI NOR Boot Image Layout

19.2.2.5 Execute-in-Place Decryption

BootROM supports execute-in-place decryption via XPI NOR. When the ENCRYPT_XIP field in OTP is set to 1, BootROM will force enable EXIP during XPI NOR boot and attempt to decrypt the EXIP Blob using EXIP0_KEK (XPI_INSTANCE value is 0)/EXIP1_KEK (XPI_INSTANCE value is 1). After the EXIP Blob is decrypted correctly, ROM will configure the corresponding decryption information (decryption region, initial vector, etc.) based on the information in the EXIP BLOB. After EXIP configuration is complete, ROM can perform execute-in-place decryption.

Note: The boot image header cannot be encrypted, otherwise BootROM cannot boot the image

Execute-in-place decryption is divided into two steps:

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1. Decrypt the encrypted EXIP BLOB using $EXIP_KEK_n$ from OTP. The EXIP BLOB uses the RFC3394 standard for encryption.
2. Configure the EXIP module using information from the EXIP BLOB to prepare the execute-in-place decryption environment.

19.2.3 In-System Programming (In-System-Programming)

In ISP mode, ROM supports the following interfaces:

- UART
- USB (USB-HID)

In ISP mode, ROM supports the following functions:

- Query runtime information (query-rte)
- Configure runtime information (config-rte)
- Configure memory (configure-memory)
- Write memory device (write-memory)
- Read memory device (read-memory)
- Load image (load-image)
- Erase memory device (erase)
- Reset device (reset)
- Generate firmware Blob (gen-fwblob)

19.2.3.1 Command Protocol

BootROM supports the following commands:

- Query runtime environment (query-rte)
- Configure runtime environment (config-rte)
- Configure memory device (configure-memory)
- Write memory device (write-memory)
- Read memory device (read-memory)
- Load image (load-image)
- Erase memory device (erase)
- Reset SoC (reset)
- Generate firmware Blob (gen-fwblob)

Communication Process

Under the ROM communication protocol, each command includes 4 phases as shown in the following figure.

1. Phase 1: Host command and data transmission phase. In this phase, if the command only needs to transmit simple parameters, the command data transmission only includes one data packet (such as reset/erase); if the command includes data transmission, the data transmission includes one mixed packet (command + data) and several pure data packets.
2. Phase 2: Slave sends ACK/NAK/Abort phase. For each data packet in phase 1, the slave needs to give corresponding confirmation. When data processing has no exceptions, reply ACK; when data verification error occurs, reply NAK; when data processing error occurs, reply Abort.

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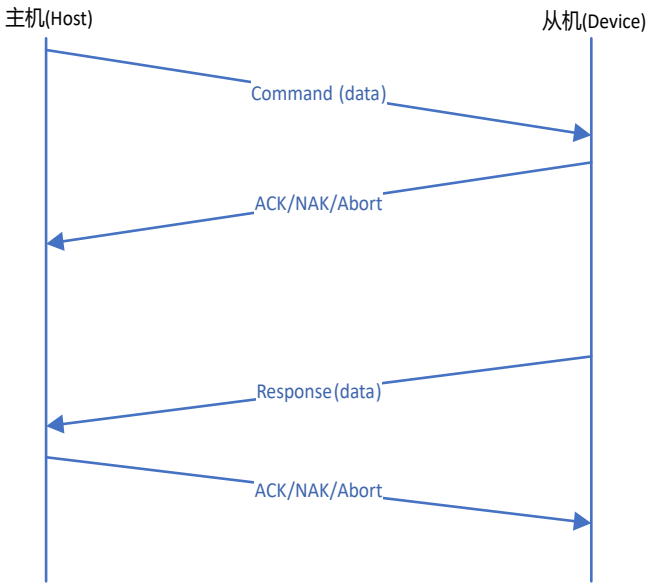


Figure 16: Serial Communication Protocol

3. Phase 3: Slave sends response phase. After phase 1 data transmission is completed or Device replies Abort, communication moves to phase 3. In this phase, if the command only requires the slave to reply simple status, the data packet only has one response packet (such as reset/erase); if the slave needs to reply data, the slave will first reply the command response frame, then reply the remaining data to be sent. After each frame transmission ends, the slave needs to wait for the host's confirmation frame. After receiving the Abort sent by the host, communication ends.
4. Phase 4: Host sends confirmation for slave response. When the host receives the response packet from the slave, it needs to reply with ACK/NAK/Abort. When the data packet has no exceptions, reply ACK; when data packet verification error occurs, reply NAK; when the slave sends more data than expected, reply Abort.

Command Data Structure

Each command uniformly includes a 4-byte command header, detailed definitions are shown in the following table:

Offset (bytes)	Field	Description
0	Command Identifier	1 - Query runtime environment (query-rte) 2 - Configure runtime environment (config-rte) 3 - Configure memory (configure-memory) 4 - Write memory (write-memory) 5 - Read memory (read-memory) 6 - Load image (load-image) 7 - Erase (erase) 8 - Reset (reset) 9 - Generate firmware Blob (gen-fwblob)

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Offset (bytes)	Field	Description
1	Command Parameter Count	-
2	Command Type	0 - Command + Data 1 - Data only 2 - Response only
3	Reserved	-

Table 31: Command Data Structure

Query Runtime Environment (query-rte)

This command can be used to query the following status:

- ROM parameters
- Active communication interface parameters
- Last boot status
- Memory attributes

The detailed command data structure is shown in the following table:

Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	1 - query-rte
1	1	Command Parameter Count	1 / 2
2	1	Command Type	0 - Command + Data
3	1	Reserved	-
4	4	Runtime Environment Identifier	0 - ROM Parameters 1 - Current Active Peripheral Info 3 - Last Boot Status 4 - Memory Attributes

Table 32: Query Runtime Environment (query-rte) Command Data Structure

When ROM receives this command, if the parameters are valid, ROM will reply with the corresponding response packet. The response packet parameters vary depending on different commands.

ROM Parameter Response Packet

Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	1 - query-rte

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<table>
<tr><th>Offset (bytes)</th><th>Width (bytes)</th><th>Field</th><th>Description</th></tr>
<tr><td>1</td><td>1</td><td>Command Parameter Count</td><td>5</td></tr>
<tr><td>2</td><td>1</td><td>Command Type</td><td>2 - Response only</td></tr>
<tr><td>3</td><td>1</td><td>Reserved</td><td>-</td></tr>
<tr><td>4</td><td>4</td><td>Command Status</td><td>0 - Success</td></tr>
<tr><td>8</td><td>4</td><td>ROM Version</td><td>Bit[31:24] ASCII: V(0x56)
Bit[23:16] Major version: 0x01
Bit[15:8] Minor version: 0x00
Bit[7:0] Bug fix: 0x00</td></tr>
<tr><td>12</td><td>4</td><td>SoC Version</td><td>Consistent with ROM version number definition</td></tr>
<tr><td>16</td><td>4</td><td>Miscellaneous</td><td>Bit[7:0] - Lifecycle
Bit[31:8] - Reserved</td></tr>
<tr><td>20</td><td>4</td><td>ROM Feature Info</td><td>
<table>
<tr><td>Bit[31:27]</td><td>Reserved</td></tr>
<tr><td>Bit[26:26]</td><td>Support SM4$_{CCM}$</td></tr>
<tr><td>Bit[25:25]</td><td>Support AES$_{CCM}(256-bitkey)$</td></tr>
<tr><td>Bit[24:24]</td><td>Support AES$_{CCM}(128-bitkey)$</td></tr>
<tr><td>Bit[23:20]</td><td>Reserved</td></tr>
<tr><td>Bit[19:19]</td><td>Support SM2</td></tr>
<tr><td>Bit[18:18]</td><td>Support ECDSA P521</td></tr>
<tr><td>Bit[17:17]</td><td>Support ECDSA P384</td></tr>
<tr><td>Bit[16:16]</td><td>Support ECDSA P256</td></tr>
<tr><td>Bit[15:12]</td><td>Reserved</td></tr>
<tr><td>Bit[11:11]</td><td>Support Device Configuration Block</td></tr>
<tr><td>Bit[10:10]</td><td>Support Command Container</td></tr>
<tr><td>Bit[9:9]</td><td>Support Encrypted Boot</td></tr>
<tr><td>Bit[8:8]</td><td>Support Secure Boot</td></tr>
<tr><td>Bit[7:4]</td><td>Reserved</td></tr>
<tr><td>Bit[3:3]</td><td>Support In-System Programming (ISP)</td></tr>
<tr><td>Bit[2:2]</td><td>Support Serial Boot</td></tr>
<tr><td>Bit[1:1]</td><td>Support Recovery Boot</td></tr>
<tr><td>Bit[0:0]</td><td>Support Primary Boot</td></tr>
</table>
</td></tr>
</table>

Table 33: ROM Parameter Response Packet

Active Peripheral Info Packet

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Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	1 - query-rte
1	1	Command Parameter Count	If parameters are invalid, this value is 0, otherwise it's "word" bytes +1
2	1	Command Type	2 - Response only
3	1	Reserved	-
4	4	Command Status	0 - Success 2 - Invalid Parameters
8	4	General Info	Bit[7:0] Word count (in 32-bit units) Bit[31:8] - Reserved
12	4	Peripheral Mask	0x01 - UART 0x10 - USB-HID
16	4	Speed	For UART, this field represents baud rate; for USB-HID, this field is meaningless
20	8	Reserved	-
28	4	Maximum Packet Length	Maximum data packet length (not including packet header and CRC check)

Table 34: Active Peripheral Info Packet

Last ROM Boot Status Packet

Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	1 - query-rte
1	1	Command Parameter Count	If parameters are invalid, this value is 0, otherwise it's "word" bytes +1
2	1	Command Type	2 - Response only
3	1	Reserved	-
4	4	Command Status	0 - Success

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Offset (bytes)	Width (bytes)	Field	Description
8	4	Boot Parameters	Boot Parameters Bit[31:16] Reserved Bit[15:8] Boot status word count (in 32-bit units) Bit[7:0] Boot Mode

Table 35: Last ROM Boot Status

Memory Device Attribute Packet

Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	1 - query-rte
1	1	Command Parameter Count	If parameters are invalid, this value is 0, otherwise it's "word" bytes +1
2	1	Command Type	2 - Response only
3	1	Reserved	-
4	4	Command Status	0 - Success
8	4	General Info	Bit[7:0] Memory device attribute word count (in 32-bit units) Bit[15:8] Memory size unit: 0 - 1 byte, 1 - 2 bytes 2 - 4 bytes Others - Reserved
12	8	Memory Device Identifier	TBD
16	4	Start Address	Memory device start address
20	4	End Address	Memory device end address
24	4	Memory Device Capacity	Memory device capacity
28	4	Page Size (Optional)	Only valid for FLASH devices
32	4	Sector Size (Optional)	Only valid for FLASH devices

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Offset (bytes)	Width (bytes)	Field	Description
32	4	Block Size (Optional)	Only valid for FLASH devices

Table 36: Memory Device Attribute Response Packet

Configure Runtime Environment (configure-rte)

This command can configure active peripheral parameters, currently only UART baud rate supports configuration. Detailed command data structure:

Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	2 - configure-rte
1	1	Command Parameter Count	4
2	1	Command Type	0 - Command + Data
3	1	Reserved	-
4	4	Runtime Environment Identifier	1 - Active Peripheral Info
8	4	General Info	Bit[7:0] Parameter word count (in 32-bit units) Others - Reserved
12	4	Peripheral Mask	1 - UART
16	4	Speed	UART baud rate

Table 37: Configure Runtime Environment (configure-rte)

The response frame data structure for this command:

Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	2 - configure-rte
1	1	Command Parameter Count	1
2	1	Command Type	2 - Response only
3	1	Reserved	-

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Offset (bytes)	Width (bytes)	Field	Description
4	4	Command Status	0 - Success 2 - Invalid Parameters

Table 38: Configure Runtime Environment Response Packet

Configure Memory (configure-memory)

This command is used to enable support for external memory devices through memory device configuration information stored at the specified address, such as enabling support for Serial NOR FLASH. Detailed command data structure:

Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	3 - configure-memory
1	1	Command Parameter Count	2
2	1	Command Type	0 - Command + Data
3	1	Reserved	-
4	Memory Device Identifier	0x10000 - NOR FLASH mounted on XPI0 0x10001 - NOR FLASH mounted on XPI1	
8	4	Memory Configura-tion Block Address	Memory configuration block address (only supports obtaining from on-chip RAM)

Table 39: Configure Memory (configure-memory)

The response frame data structure for this command:

Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	3 - configure-memory

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Offset (bytes)	Width (bytes)	Field	Description
1	1	Command Parameter Count	1
2	1	Command Type	2 - Response only
3	1	Reserved	-
4	4	Command Status	0 - Success 2 - Invalid Parameters

Table 40: configure-memory Response Frame Data Structure

Write Memory Device (write-memory)

This command is used to write data to on-chip SRAM, on-chip OTP, and external XPI NOR. This command can consist of a single data packet or multiple data packets. The data structure of the first packet is defined as follows:

Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	4 - write-memory
1	1	Command Parameter Count	3
2	1	Command Type	0 - Command + Data
3	1	Reserved	-
4	4	Address to Write	Device address to write to
8	4	Length	Data length to write
12	4	Memory Device Identifier	0 - On-chip RAM 0x10000 - NOR FLASH mounted on XPI0 0x10001 - NOR FLASH mounted on XPI1 0x20000 - OTP
16	Variable length	Data	Data to write to memory device

Table 41: Configure Memory (write-memory) - Command + Data

The data structure from the second packet to the last packet:

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Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	4 - write-memory
1	1	Command Parameter Count	0
2	1	Command Type	1- Data only
3	1	Reserved	-
4	Variable length	Data	Data to write to memory device

Table 42: Configure Memory (write-memory) - Data Only

When data can be sent in one packet, only the first type of data packet is needed; when data needs to be sent in multiple packets, the second type of data packets are needed. The length of the Data part in the data packet is obtained from the data packet length received in various communication interfaces.

If an error occurs during write-memory execution, the slave will abort transmission and notify the host of the error code in the response packet.

When write-memory successfully completes execution, report the success code to the host in the response packet.

The write-memory response packet data structure is defined as follows:

Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	4 - configure-memory
1	1	Command Parameter Count	1
2	1	Command Type	2 - Response only
3	1	Reserved	-
4	4	Command Status	0 - Success Others - Error Code

Table 43: write-memory Response Frame Data Structure

Read Memory (read-memory)

This command is used to read data from on-chip SRAM, on-chip OTP, and external XPI NOR. This command consists of one command packet and multiple response packets.

The command packet data structure:

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Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	5 - read-memory
1	1	Command Parameter Count	3
2	1	Command Type	0 - Command + Data
3	1	Reserved	-
4	4	Address to Read	Device address to read from
8	4	Length	Data length to read
12	4	Memory Device Identifier	0 - On-chip RAM 0x10000 - NOR FLASH mounted on XPI0 0x10001 - NOR FLASH mounted on XPI1 0x20000 - OTP

Table 44: read-memory Command Data Frame

When the slave receives this command, it will first check the parameters. If parameter check fails, it will directly reply Abort to the host in phase 2 of the command, then notify the error code in the response packet. After receiving the phase 4 confirmation from the host, this command ends. If parameter check passes, the slave replies ACK to the host in phase 2, then the slave will first notify the host that parameter check succeeded in the first response packet, and then send the requested data to the host sequentially through data packets.

The first type of response frame replied by the slave:

Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	5 - read-memory
1	1	Command Parameter Count	1
2	1	Command Type	2 - Response only
3	1	Reserved	-
4	4	Command Status	0 - Success Others - Error Code

Table 45: read-memory Response Frame Data Structure

The slave's pure data packet frame:

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Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	5 - read-memory
1	1	Command Parameter Count	0
2	1	Command Type	1 - Data only
3	1	Reserved	-
4	Variable length	Data	Data read from memory device

Table 46: read-memory Response Frame Data Structure

Load Image (load-image)

This command is used to load and execute executable images, or load and parse command containers. This command consists of pure data packets, one command consists of several data packets.

The data packet structure:

Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	6 - load-image
1	1	Command Parameter Count	0
2	1	Command Type	1- Data only
3	1	Reserved	-
4	Variable length	Data	Image data

Table 47: load-image Data Frame Structure

After data transmission is complete, the slave will reply with one response packet, whose data structure is:

Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	6 - load-image
1	1	Command Parameter Count	1

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Offset (bytes)	Width (bytes)	Field	Description
2	1	Command Type	2 - Response only
3	1	Reserved	-
4	4	Command Status	0 - Success Others - Error Code

Table 48: load-image Response Frame Data Structure

Erase Memory Device (erase)

This command can be used to erase part or all of the sectors of external XPI NOR, its data structure:

Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	7 - erase
1	1	Command Parameter Count	2 - Chip erase 4 - Sector erase
2	1	Command Type	1- Data only
3	1	Reserved	-
4	4	Erase Type	0 - Sector erase 1 - Chip erase
8	4	Memory Device Identifier	0x10000 - NOR FLASH on XPI0 0x10001 - NOR FLASH on XPI1
12	4	Erase Address	Only valid when erase type is sector erase
16	4	Erase Length	Only valid when erase type is sector erase

Table 49: erase Data Frame Structure

The command response frame:

Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	7 - erase
1	1	Command Parameter Count	1

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Offset (bytes)	Width (bytes)	Field	Description
2	1	Command Type	2 - Response only
3	1	Reserved	-
4	4	Command Status	0 - Success Others - Error Code

Table 50: erase Response Frame Data Structure

Reset (reset)

This command can reset the SoC. Its data structure:

Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	8 - reset
1	1	Command Parameter Count	1
2	1	Command Type	1- Data only
3	1	Reserved	-
4	4	Reset Type	Reserved in this SoC

Table 51: reset Data Frame Structure

The response frame for this command:

Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	8 - reset
1	1	Command Parameter Count	1
2	1	Command Type	2 - Response only
4	4	Command Status	0 - Success Others - Error Code

Table 52: reset Response Frame Data Structure

Generate Firmware Blob (gen-fwblob)

This command can generate the firmware BLOB required for encrypted images.

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Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	9 - gen-fwblob
1	1	Command Parameter Count	1 + Key bytes /4 (calculated starting after key parameter)
2	1	Command Type	1- Data only
3	1	Reserved	-
4	4	Key Parameter	Key parameter **Byte 0 - alg (algorithm)** 0x33 - AES, 0x55 - SM4 **Byte 1 - mode** 0x11 - CBC-MAC **Byte 2 - key**$_{s}rc(symmetrickeysource)$ $1-FMK, 2-ZMK, 3-OTPKEY0, 4-OTPKEY1$
8	4	Reserved	-
12	16/32	Symmetric Key	Key for encrypting firmware

Table 53: Generate Firmware Blob (gen-fwblob) Data Frame Structure

When the slave receives this command, it will first check the parameters. If parameter check fails, it will directly reply Abort to the host in phase 2 of the command, then notify the error code in the response packet. After receiving the phase 4 confirmation from the host, this command ends. If parameter check passes, the slave replies ACK to the host in phase 2, then the slave will first notify the host that parameter check succeeded in the first response packet, and then send the requested data to the host sequentially through data packets.

The first type of response frame replied by the slave:

Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	9 - gen-fwblob
1	1	Command Parameter Count	1
2	1	Command Type	2 - Response only
3	1	Reserved	-

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Offset (bytes)	Width (bytes)	Field	Description
4	4	Command Status	0 - Success Others - Error Code

Table 54: gen-fwblob Response Frame Data Structure

The slave's pure data packet frame:

Offset (bytes)	Width (bytes)	Field	Description
0	1	Command Identifier	9 - gen-fwblob
1	1	Command Parameter Count	0
2	1	Command Type	1 - Data only
3	1	Reserved	-
4	Variable length	Data	Return complete firmware Blob data

Table 55: gen-fwblob Reply Data Frame

19.2.4 UART Communication

ROM supports UART communication with default parameters 115200, 8-N-1, baud rate can be switched through configure-rte command.

The UART communication protocol data packets are of two types:

- Payload packet
- Acknowledgment packet

19.2.4.1 Payload Packet

The payload frame consists of 4 parts: packet header, packet type, payload, and CRC check. Detailed data packet structure:

Offset (bytes)	Width (bytes)	Field	Description
0	1	Start Byte	Must be 0x5A
1	1	Packet Type	0xA5 - Payload packet
2	2	Payload Length	Payload length, cannot exceed 512 bytes
4	<Payload$_{Length}$>	Data Payload	Data payload, used to carry data in the command protocol

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Offset (bytes)	Width (bytes)	Field	Description
4 + <Payload$_{Length}$>	2	CRC Check Value	CRC check value of the first <4+Payload$_{Length}$> $bytes of the entire packet$

Table 56: Payload Packet Data Structure

The CRC calculation range is from the start byte to the last byte of the data payload.

The CRC algorithm is a variant of XMODEM. Its C language implementation:

```
uint16_t crc16_update(const uint8_t *src, uint32_t lengthInBytes)
{
    uint32_t crc = 0;
    uint32_t j;
    for (j = 0; j < lengthInBytes; ++j)
    {
        uint32_t i;
        uint32_t byte = src[j];
        crc ^= byte << 8;
        for (i = 0; i < 8; ++i)
        {
            uint32_t temp = crc << 1;
            if (crc & 0x8000)
            {
                temp ^= 0x1021;
            }
            crc = temp;
        }
    }
    return (uint16_t)crc;
}
```

19.2.4.2 Acknowledgment Packet (ACK Packet)

The acknowledgment packet is used to implement ACK, NAK, or Abort for payload packets. Detailed data packet structure:

Offset (bytes)	Width (bytes)	Field	Description
0	1	Start Byte	Must be 0x5A
1	1	Packet Type	0xA1 - ACK 0xA2 - NAK 0xA3 - Abort

Table 57: Acknowledgment Packet

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The sender of each payload packet needs to wait for the receiver's acknowledgment packet before proceeding with subsequent data transmission.

The payload part of the data packet can carry various commands or data introduced above.

19.2.5 USB-HID Communication

BootROM supports communication via USB-HID to implement functions such as FLASH programming, EFUSE programming, loading boot images, etc.

19.2.5.1 USB-HID Descriptor Information

The USB descriptor configuration in BootROM is shown in the following table:

Descriptor Identifier	Content/Value
VID	0x34b7
PID	0x0005
Vendor Descriptor	HPMICRO Semiconductor Co., Ltd
Product Descriptor	Boot ROM Open (factory settings) Boot ROM Closed (mass production settings) Boot ROM Return (factory return settings)
HID Packet Length	When connected to High-Speed HUB: 516 bytes When connected to Full-Speed HUB: 64 bytes
HID Interrupt Transfer Interval	When connected to High-Speed HUB: 125us When connected to Full-Speed HUB: 1ms

Table 58: USB Descriptor

19.2.5.2 USB Endpoint Information

BootROM uses the following three endpoints:

- Control (Endpoint 0) - Used to obtain device descriptors
- OUT (Endpoint 1) - Used for Host to send data to Device
- IN (Endpoint 2) - Used for Device to send data to Host

19.2.5.3 USB-HID Communication Protocol

The USB-HID communication protocol is similar to UART, both include payload packets and acknowledgment packets.

USB-HID Payload Packet

The USB-HID payload packet includes four fields: packet header, packet type, payload length, and data payload. Detailed data structure is shown below:

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Offset (bytes)	Width (bytes)	Field	Description
0	1	Start Byte	Host to Device is 0x1, Device to Host is 0x2
1	1	Packet Type	0xA5 - Payload packet
2	2	Payload Length	Payload length, cannot exceed 512 bytes
4	<Payload$_{Length}$ >	Data Payload	Data payload, used to carry data in the command protocol

Table 59: USB-HID Payload Packet Data Structure

USB-HID Acknowledgment Packet

The USB-HID acknowledgment packet includes three fields: packet header, packet type, payload length. Detailed acknowledgment packet data structure is shown below:

Offset (bytes)	Width (bytes)	Field	Description
0	1	Start Byte	Host to Device is 0x1, Device to Host is 0x2
1	1	Packet Type	0xA1 - ACK 0xA2 - NAK 0xA3 - Abort
2	2	Payload Length	Fixed to 0

Table 60: USB-HID Acknowledgment Packet Data Structure

19.3 Boot Image

The boot image supported by this device consists of two parts: Firmware Container (FW Container) and Firmware (Firmware). The FW Container consists of FW Container Header, FW Info Table, Device Configuration Block, and Signature Block. The boot image layout is shown in the figure.

19.3.1 Firmware Container Header (FW Container Header)

The detailed data structure of the firmware container header is shown below:

Table 61: Firmware Container Header

Offset (bytes)	Field	Width (bytes)	Description
0x00	tag	1	Boot image tag, must be 0xBF
0x01	version	1	Boot image format version bit[7:4] - Major version bit[3:0] - Minor version This value must be 0x10 on this SoC

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Table 61: Firmware Container Header

<table>
<tr><th>Offset (bytes)</th><th>Field</th><th>Width (bytes)</th><th>Description</th></tr>
<tr><td>0x02</td><td>length</td><td>2</td><td>Firmware header length
Includes firmware container header, firmware info table, device configuration block, and signature block</td></tr>
<tr><td>0x04</td><td>flags</td><td>4</td><td>Corresponding flags
<table>
<tr><td>Bit[3:0]</td><td>SRK set index</td></tr>
<tr><td>Bit[7:4]</td><td>SRK index</td></tr>
<tr><td>Bit[11:8]</td><td>SRK_REVOKE_MASK
Each bit represents revoking the corresponding position’s SRK. When this byte is non-zero, BootROM will attempt to revoke the corresponding SRK according to this field’s value and burn PUK_REVOKE in OTP to the same value</td></tr>
<tr><td>Bit[19:16]</td><td>Signature Type
1 - ECDSA P256
4 - SM3</td></tr>
<tr><td>Bit[23:20]</td><td>Firmware BLOB Location
0 - BLOB is in signature block
1 - BLOB is at the firmware’s location in storage medium</td></tr>
<tr><td>Bit[27:24]</td><td>Lifecycle
0 - No change to lifecycle
Others - BootROM will switch lifecycle to this field’s value</td></tr>
<tr><td>Bit[31:28]</td><td>Reserved</td></tr>
</table></td></tr>
<tr><td>0x08</td><td>sw_version</td><td>2</td><td>Software version number, defines the current image’s version. When there are multiple images in storage medium, ROM determines which is newer based on this field</td></tr>
<tr><td>0x0A</td><td>fuse_version</td><td>1</td><td>This field is used for image rollback prevention, this field’s value must not be less than the SW_VER field in OTP
Note: The number of 1 bits in SW_VER represents the version number (e.g., 0xffff represents version number 15)</td></tr>
<tr><td>0x0B</td><td>number_of_fw</td><td>1</td><td>Specifies the number of firmware included in the current image, this SoC supports a maximum of two</td></tr>
</table>

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Table 61: Firmware Container Header

Offset (bytes)	Field	Width (bytes)	Description
0x0C	device_config_block_offset	2	Device configuration block offset relative to firmware container header, device configuration block is optional, if current image doesn't have this block, this field must be 0
0x0E	signature_block_offset	2	Signature block offset relative to firmware container header, if signature block doesn't exist, this field must be 0

Table 61: Firmware Container Header

19.3.2 Firmware Information Table (FW Info Table)

The firmware information table mainly provides firmware execution address, length, offset in storage medium relative to firmware container header, firmware hash value, etc. Its detailed data structure is shown below:

Offset (bytes)	Field	Width (bytes)	Description
0x00	offset	4	Firmware offset relative to firmware container header
0x04	size	4	Firmware size
0x08	flags	4	Corresponding flags Bit[3:0]: fw_type - Firmware type 0 - Executable image 1 - Command container Bit[7:4]: core_id - CPU ID number 0 - CPU0 1 - CPU1 Bit[11:8]: hash_type - Hash type 0 - None 1 - SHA256 4 - SM3 Bit[15:12]: is_encrypted - Whether encrypted 0 - Not encrypted 1 - Firmware encrypted Others: Reserved
0x0C	reserved	4	Reserved for future use
0x10	load_addr	4	Firmware load address
0x14	reserved	4	Reserved for future use
0x18	entry_point	4	Firmware entry point
0x1C	reserved	4	Reserved for future use

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Offset (bytes)	Field	Width (bytes)	Description
0x20	hash	64	Firmware hash value, currently only uses 32 bytes When firmware type is executable file, this field is the hash value of the entire FW When firmware type is command container, this field is the hash value of the command container header packet. When firmware is encrypted firmware, the first 128 bits of this field are used as initial vector
0x60	reserved	32	Reserved for future use

Table 62: Firmware Container Header

19.3.3 Device Configuration Information Block (Device Configuration Info Block)

The device configuration block can support the following 4 types of configuration information:

- XPI NOR configuration information
- Wakeup entry check information
- OTP UNLOCK information

The device configuration information block consists of the following parts:

- Device Configuration Information Block Header (Device Configuration Block Header)
- Device Configuration Information
  - Device Configuration Information Header
  - Device Configuration Information Content

The device configuration information header data structure:

Bit[31:16]	Bit[15:8]	Bit[7:0]
Configuration info block length	Version number, currently 0x10	Configuration info type 0xC0 - Total device configuration info

Table 63: Configuration Info Block Header (Configure Info Block Header)

Each configuration info has its own header, defined as follows:

Bit[31:16]	Bit[15:8]	Bit[7:0]
Configuration info block length	Parameter For XPI NOR, XPI RAM represents XPI instance number	Configuration info type 0xC1 - XPI NOR configuration info 0xC2 - XPI RAM configuration info 0xC3 - SDRAM configuration info 0xC4 - General register info 0xC5 - Wakeup verification configuration info 0xC6 - OTP unlock info

Table 64: Configuration Info Header (Configure Info Header)

Following the configuration info header is the corresponding configuration information, detailed definitions

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Figure 17: Boot Image Layout

of each configuration info are shown in subsequent sections.

19.3.3.1 XPI NOR Configuration Information

XPI NOR configuration information is divided into two types: one is simplified XPI NOR configuration information, the other is complete XPI NOR configuration block.

XPI NOR Configuration Option

Offset (bytes)	Field	Description
0x00	Header	Bit[31:12] - tag: Tag, must be 0xfcf90 Bit[11:4] - Reserved Bit[3:0] - words: Configuration option word count (not including Header itself)

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Offset (bytes)	Field	Description
0x04	Option0	**Bit[31:28] - Probe Type** 0 - SFDP SDR, 1 - SFDP DDR 2 - 1-4-4 Read 0xEB, 3 - 1-2-2 Read 0xBB 4 - HyperBus 1V8, 5 - HyperBus 3V0 6 - OctaBus DDR, 8 - Xccela DDR 10 - EcoXiP DDR **Bit[27:24] - Data pin count for commands sent after power-on reset** 0 - SPI, 1 - DPI, 2 - QPI, 3 - OPI **Bit[23:20] - Data pin count for commands sent after FLASH configuration is complete** 0 - SPI, 1 - DPI, 2 - QPI, 3 - OPI **Bit[19:16] - Sequence to enable 1-4-4 mode** 0 - Not needed or automatic 1 - QE bit is at bit 6 in Status Register1 2 - QE bit is at bit1 in Status Register2 3 - QE bit is at bit7 in Status Register2 4 - QE bit is at bit1 in Status Register2 and should be programmed by 0x31 **Bit[15:8] - dummy cycles** 0 - Auto detect / Default value Others - User-specified value, for DTR read, this value should be actual datasheet dummy cycle × 2 **Bit[7:4] - Miscellaneous** 0 - None 1 - SPI mode 2 - Internal loopback mode 3 - Force external DQS **Bit[3:0] - Frequency option** 0 - Don’t change existing configuration Others - SoC corresponding value

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Offset (bytes)	Field	Description
0x08	Option1	This field is valid when ”words” field value > 1 Bit[31:20] - Reserved **Bit[19:16] - IO Voltage** 0 - IO is 3V, 1 - IO is 1.8V **Bit[15:12] - Pin group selection** 0 - First group, 1 - Second group **Bit[11:8] - Connection mode selection** 0 - PORTA_CS0, 1 - PORTB_CS0, 2 - PORTA_CS0 + PORTB_CS0, 3 - PORTA_CS0 + PORTA_CS1, 4 - PORTB_CS0 + PORTB_CS1 **Bit[7:0] - IO drive strength** 0 - Default value (maximum drive strength) Others - See (TBD)
0x0C	Option2	This field is valid when ”words” > 2 Bit[31:16] - Reserved **Bit[15:12] - Erase sector command option** 0 - Erase 4KB, 1 - Erase 32KB, 2 - Erase 64KB, 3 - Erase 256KB **Bit[11:8] - Sector size option** 0 - 4KB, 1 - 32KB, 2 - 64KB, 3 - 256KB **Bit[7:0] - FLASH size option** 0 - 4MB, 1 - 8MB, 2 - 16MB

Table 65: XPI NOR Configuration Option (XPI NOR Configuration Option)

XPI NOR Configuration Block

The XPI NOR configuration block data structure is shown in the following table:

Offset (bytes)	Field	Width (bytes)	Description
0x000	tag	4	Configuration block tag, fixed at 0x524f4e58
0x004	reserved	4	-
0x008	rxclk_src	1	Sample clock source 0 - Internal loopback 1 - DQS internal loopback 3 - External DQS input
0x009	clk_freq	1	Clock frequency option, see 19.8.2.1
0x00A	drive_strength	1	Drive strength 0 - Default value (maximum drive strength) Others - See (TBD)

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Offset (bytes)	Field	Width (bytes)	Description
0x00B	column_addr_width	1	Column address width Only valid for HyperFLASH, other FLASH must be set to 0
0x00C	rxclk_src_for_init	1	Value of rxclk_src during FLASH initialization
0x00D	config_in_progress	1	Indicates whether FLASH is currently being initialized 1 - Initializing in progress, Others - Not initializing
0x00E	reserved	2	-
0x010	port_info	16	- Connected port info This field is divided into 4 groups, respectively representing CA_CS0, CA_CS1, CB_CS0, CB_CS1 Each group's 4 bytes are defined as follows: Byte 0 - Enable: Enable bit, indicates this connection is enabled Byte 1 - Group: Pin group: 0 indicates first group, 1 indicates second group Byte 2, Byte 3 - Reserved for future use
0x020	device_info	0x50	Detailed definition see 67
0x070	instruction_set	0x90	Standard XPI NOR instruction set, divided into 9 groups, each 16 bytes represents a complete instruction sequence, grouping info as follows: 0 - FLASH Read sequence 1 - FLASH Write sequence 2 - FLASH Read Status sequence 3 - FLASH Read Status XPI sequence (QPI/OPI mode) 4 - FLASH Write Enable sequence 5 - FLASH Write Enable XPI sequence (QPI/OPI mode) 6 - FLASH Erase Sector sequence 7 - FLASH Erase Block sequence 8 - FLASH Erase Chip sequence **Note: For HyperFLASH, the above definitions don't apply, BootROM supports dedicated FLASH operations**

Table 66: XPI NOR Configuration Block

The XPI NOR device info table data structure is shown below:

Offset (bytes)	Field	Width (bytes)	Description
0x00	size_in_kbytes	4	FLASH size (in KB)
0x04	page_size	2	Page size
0x06	sector_size_kbytes	2	Sector size (in KB)
0x08	block_size_kbytes	2	Block size (in KB)

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Offset (bytes)	Field	Width (bytes)	Description
0x0A	busy_offset	1	Busy bit offset (in FLASH Status register)
0x0B	busy_polarity	1	Busy bit polarity: 0 - 1 indicates busy 1 - 0 indicates busy
0x0C	data_pads	1	Number of data pins 0 - 1 wire, 1 - 2 wires, 2 - 4 wires, 3 - 8 wires
0x0D	en_ddr_mode	1	Enable DDR/DTR mode
0x0E	clk_freq_for_dev_cfg	1	Clock frequency used during FLASH configuration Clock frequency definition see 19.8.2.1
0x0F	working_mode_por	1	FLASH working mode after power-on reset 0 - Extended SPI 1 - XPI (DPI/QPI/OPI) 2 - HyperBus
0x10	working_mode	1	FLASH working mode after initialization 0 - Extended SPI 1 - XPI (DPI/QPI/OPI) 2 - HyperBus
0x11	en_diff_clk	1	Enable differential clock Only 1.8V HyperFLASH needs to set this option to 1, other FLASH must set to 0
0x12	data_valid_time	1	-
0x13	en_half_clk_for_ non_read_cmd	1	Enable reducing clock frequency by half for non-read commands
0x14	clk_freq_for_ non_read_cmd		Set clock frequency for non-read commands Clock frequency definition see 19.8.2.1
0x15	reserved	3	-
0x18	cs_hold_time	1	Set CS hold clock cycles
0x19	cs_setup_time	1	Set CS setup clock cycles
0x1A	cs_interval	1	Interval clock cycles between two commands
0x1B	en_dev_mode_cfg	1	Enable device mode configuration
0x1C	flash_state_ctx	4	FLASH state context Context definition see (TBD)
0x20	mode_cfg_list	4	Includes two groups, each 2 bytes, each group's data structure is as follows: Byte 0 - Configuration command type Byte 1 - Parameter size
0x24	mode_cfg_param	8	Includes 2 groups, each 4 bytes, corresponds one-to-one with elements in the above list
0x2C	reserved	4	-

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Offset (bytes)	Field	Width (bytes)	Description
0x30	cfg_instr_seq	32	Includes 2 groups, each 16 bytes is a command sequence, corresponds one-to-one with elements in the above list Detailed command sequence definition see (TBD)

Table 67: XPI NOR Configuration Block

19.3.3.2 Wakeup Verification Information

Wakeup verification information provides relevant parameters used to verify the integrity and legitimacy of the wakeup program in wakeup mode, including wakeup program entry address, wakeup program code start address, length, and corresponding SHA256 hash value. This information is verified by BootROM with signature when secure boot is enabled.

Detailed data structure:

Offset (bytes)	Field	Width (bytes)	Description
0x00	Start	4	Wakeup code start address
0x04	wake_up_entry	4	Wakeup code entry address Note: This address must be within the address range specified by start and length
0x08	Length	4	Wakeup code length
0x0C	Reserved	4	-
0x10	wake_up_entry_hash	32	Hash value of code within start and length specified range

Table 68: Wakeup Verification Information (Wake-up Check Info)

19.3.3.3 OTP Unlock Information

The OTP unlock operation data structure:

Bit[31:24]	Bit[23:16]	Bit[15:8]	Bit[7:0]
Tag, must be 0x55	Reserved	Reserved	Unlock mask When Lifecycle is Closed or higher, BootROM will disable write permissions for corresponding OTP fields by default. When BootROM detects that the following bits are set to 1 in OTP unlock information, it will unlock corresponding write permissions Bit 0 - HARD_LOCK Bit 1 - LIFECYCLE, PUBKEY_REVOKE, etc. Bit 2 - Boot configuration info part Bit 3 - OTP KEY0 (for encrypt boot) Bit 4 - OTP KEY1 (for encrypt boot)

Table 69: OTP Unlock Information (OTP Unlock Info)

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19.3.4 Signature Block (Signature Block)

The complete signature block provides the following information:

- Signature Block Header (Signature Block Header)
- Super Root Key Table (SRK Table) - This part includes Root Key Table Header and 4 Root Key Table entries (SRK0 - SRK3)
- Signature (Signature)
- Certificate (Certificate) - This part is optional
- Firmware Blob (FW Blob)

The layout of each part in the signature block in target storage is shown in the figure:

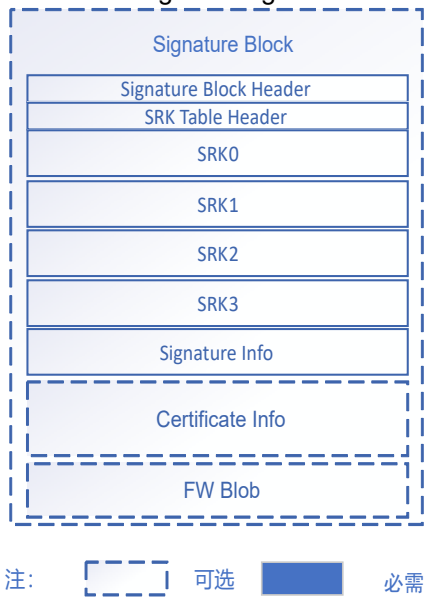


Figure 18: Signature Block Layout

19.3.4.1 Signature Block Header (Signature Block Header)

The signature block header includes the following information:

- Signature block length
- Signature block version number
- Signature tag
- Root key table (SRK Table) relative offset
- Certificate relative offset
- Signature relative offset
- FW Blob relative offset (if FW Blob is in Container)

Detailed data structure is shown in the following table:

Offset (bytes)	Field	Width (bytes)	Description

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<table>
<tr><th>Offset (bytes)</th><th>Field</th><th>Width (bytes)</th><th>Description</th></tr>
<tr><td>0x00</td><td>Header</td><td>4</td><td>Signature block Header
<table>
<tr><td>Bit[31:16]</td><td>Length - Length of entire signature block</td></tr>
<tr><td>Bit [15:8]</td><td>Version - Signature block version, fixed at 0x10 for this SoC</td></tr>
<tr><td>Bit [7:0]</td><td>Tag - Signature block tag, fixed at 0xD0</td></tr>
</table></td></tr>
<tr><td>0x04</td><td>Flags</td><td>4</td><td>Signature block flags
Bit0 Key identifier 1 - OEM Key
Others - Reserved bits</td></tr>
<tr><td>0x08</td><td>Member_Offset</td><td>8</td><td>Relative offset of each part in signature
<table>
<tr><td>Bit[15:0]</td><td>srk_table_offset
SRK table offset relative to signature block first address</td></tr>
<tr><td>Bit[31:16]</td><td>signature_offset
Signature offset relative to signature block first address</td></tr>
<tr><td>Bit[47:32]</td><td>certificate_offset (optional)
Certificate offset relative to signature block first address
Only valid when secondary certificate is used</td></tr>
<tr><td>Bit[63:48]</td><td>blob_offset (optional)
Firmware BLOB offset relative to signature block first address
Only valid when encrypted firmware exists and firmware BLOB is in signature block</td></tr>
</table></td></tr>
</table>

Table 70: Signature Block Header (Signature Block Header)

Note: To ensure CPU access efficiency to each part in the signature block, the start addresses of SRK table, signature, certificate, and BLOB must all be aligned to 8 bytes

19.3.5 Root Key Table (SRK Table)

The SRK Table consists of Root Key Table Header (SRK Table Header) and Root Key Table entries (SRK Item).

19.3.5.1 Root Key Table Header (SRK Table Header)

The root key table header consists of 4 bytes:

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Bit[31:16]	Bit[15:8]	Bit[7:0]
SRK table length (stored in little-endian)	SRK table version, currently 0x10	SRK table tag, fixed at 0xD1

Table 71: Root Key Table Header (SRK Table Header)

19.3.5.2 Root Key Table Entry (SRK Table Item)

The root key table entry includes public key related information, detailed data structure:

Offset (bytes)	Field	Width (bytes)	Description
0x00	Header	4	Bit[31:16] Root key table entry length (stored in little-endian) Bit[15:8] Key type, 0x21 - Public key Tag - Must be 0xE1
0x04	Flags	4	Bit[31-16] - Reserved Bit[15:8] Curve type: 1 - ECDSA P256, 4 - SM2 Bit[7:0] Hash type: 1 - SHA256, 4 - SM3
0x08	Key Info	4	Bit[7:0] - <X_Length>: Big number X length Bit[23:16] - <Y_Length>: Big number Y length Others - Reserved bits
0x0C	Big Number X	<X_Length>	Big number X
0x0C + <X_Length>	Big Number Y	<Y_Length>	Big number Y

Table 72: Root Key Table Entry (SRK Table Item)

19.3.6 Signature Information (Signature Info)

The signature information data structure is shown below:

Offset (bytes)	Field	Width (bytes)	Description
0x00	Header	4	Bit[31:16] - <Sig_Length>: Signature info length (little-endian) Bit[15:8] - Signature info version, currently fixed at 0x10 Bit[7:0] - Tag, fixed at 0xD2
0x04 0x08	reserved Signature Data	4 <Sig_Length>	Reserved for future use Signature data

Table 73: Signature Information (Signature Info)

19.3.7 Certificate (Certificate)

The certificate data structure is shown below:

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Offset (bytes)	Field	Width (bytes)	Description
0x00	Header	4	Bit[31:16] - <Sig_Length>: Certificate length (little-endian) Bit[15:8] - Certificate version, currently fixed at 0x10 Bit[7:0] - Tag, fixed at 0xD3
0x04	Offset	4	Signature info offset relative to certificate header
0x08	SRK Table Item	<SRK_Table_Item_Length>	Key table entry, detailed info see (19.3.5.2)
0x08 + <SRK_Table_Item_Length>	Signature Info	<Signature_Info_Length>	Signature info, detailed info see (19.3.6)

Table 74: Certificate Header (Certificate Header)

19.3.8 Firmware BLOB (FW Blob)

Firmware Blob is an encrypted packaging format that encapsulates the FW decryption key, defined as follows:

Offset (bytes)	Field	Width (bytes)	Description
0x00	Header	4	Bit[31:16] - <FWBLOB_Length>: Firmware BLOB length (little-endian) Bit[15:8] - Version, currently fixed at 0x10 Bit[7:0] - Tag, fixed at 0xD4
0x04	Flags	4	Flags Byte 0 - Key Source Symmetric key source: 01 - FMK, 2 - ZMK, 3 - OTP KEY0, 4 - OTP KEY1 Byte 1 - Key Length Key length: 0x10 - 128bit, 0x20 - 256bit Byte 2 - Encryption Algorithm Encryption algorithm: 0x33 - AES, 0x55 - SM4 Byte 3 - Mode Encryption mode: 0x11-CBC-MAC
0x08	IV	16	Initial vector - Used to decrypt encrypted Blob key - Used to decrypt encapsulated KEK

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<table>
<tr><th>Offset (bytes)</th><th>Field</th><th>Width (bytes)</th><th>Description</th></tr>
<tr><td>0x18</td><td>Blob_content</td><td><Blob_Length></td><td>Blob content
<table>
<tr><th>Offset</th><th>Width (bytes)</th><th>Field</th><th>Description</th></tr>
<tr><td>0x00</td><td>0x10 / 0x20</td><td>Encrypted Blob Key</td><td>Encrypted Blob key
This key is used to extract the encrypted packaged DEK (DEK is used for encrypted firmware decryption)</td></tr>
<tr><td>0x10 / 0x20</td><td>0x10 / 0x20</td><td>Wrapped DEK</td><td>Encrypted packaged DEK</td></tr>
<tr><td>0x20 / 0x40</td><td>16</td><td>MAC</td><td>Message authentication code</td></tr>
</table></td></tr>
</table>

Table 75: Firmware Blob (FW Blob)

KEK is encapsulated by AES_CCM or SM4_CCM algorithm, where:

- AAD is empty
- NONCE is the first 13 bytes of the IV field
- MAC length is 16 bytes

BootROM extracts the Blob key from Encrypted Blob Key according to the Key specified by Key source and the selected algorithm (AES/SM4), and extracts DEK from Blob, then decrypts the firmware combining the IV in FW and DEK, and performs hash verification on the decrypted firmware.

19.3.9 Command Container (Command Container)

The command container is a package encapsulation of several commands supported by ROM, this format supports encryption (AES-CBC/SM4-CBC) or plaintext, users can choose the corresponding packaging method according to their system's security requirements.

The command container data packets are packaged in blocks, the block length is variable, indicated by the next_block_info field at the end of the previous block. The first block is fixed as the command container header, with fixed length. The last block doesn't include next_block_info.

Offset (bytes)	Field	Width (bytes)	Description
0x00	Payload	<Data_Bytes>	Data payload part
<Data_Bytes>	next_block_info	16	Next block information Byte 0-1: Next block size Byte 2-3: Whether it is the last block Byte 4-15: Reserved
<Data_Bytes> + 16	next_block_hash	16	Next block hash value. The hash value length is determined by hash_type in 62

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Offset (bytes)	Field	Width (bytes)	Description

Table 76: Command Container (Command Container)

19.3.9.1 Command Container Header (Command Container Header)

Offset (bytes)	Field	Width (bytes)	Description
0x00	Tag	4	Command container tag: 0x33764343 (ASCII: CCv1)
4	Version	4	Command container format version, currently 0x56010000 (V1.0.0)
8	Flags	4	Reserved for future use
12	Reserved	48	Reserved for future use
64	Description	48	Simple description of command container, e.g., (xpi nor boot image)

Table 77: Command Container Header (Command Container Header)

19.3.9.2 Commands Supported by Command Container

The command container supports the following commands:

- configure-memory
- write-memory
- erase
- reset

Configure-memory data block structure is as follows:

Offset (bytes)	Field	Width (bytes)	Description
0x00	Command Tag	1	Command tag - 0x03
0x01	Reserved	1	Reserved
0x02	Command Misc.	1	Command miscellaneous - Fixed at 0
0x03	Reserved	1	Reserved
0x4	Memory Identifier	4	Memory device identifier 0x10000 - XPI0 0x10001 - XPI1
0x08	Reserved	4	Reserved
12	Configuration Block Data Bytes	4	Configuration block data bytes

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Offset (bytes)	Field	Width (bytes)	Description
16	Configuration Block Data	<Configuration Block Data Bytes>	Configuration block data When data length is less than 16 bytes, must pad with 0 to align to 16 bytes

Table 78: Configure Memory Device Data Block

Write-memory data block structure is as follows:

Offset (bytes)	Field	Width (bytes)	Description
0x00	Command Tag	1	Command tag - 0x04
0x01	Reserved	1	Reserved
0x02	Command Misc.	1	Command miscellaneous - Fixed at 0
0x03	Reserved	1	Reserved
0x4	Memory Identifier	4	Memory device identifier 0x10000 - XPI0 0x10001 - XPI1 0x20000 - OTP
0x08	Start	4	Start address
12	Data Bytes	4	Number of bytes to write
16	Data Block	<Data Bytes>	Data to write (maximum 256 bytes) When data length is less than 16 bytes, must pad with 0 to align to 16 bytes

Table 79: Write Memory Device Data Block

Erase data block structure is as follows:

Offset (bytes)	Field	Width (bytes)	Description
0x00	Command Tag	1	Command tag - 0x07
0x01	Reserved	1	Reserved
0x02	Command Misc.	1	Command miscellaneous: 0x0 - Sector Erase 0x1 - Chip Erase
0x03	Reserved	1	Reserved
0x4	Memory Identifier	4	Memory device identifier 0x10000 - XPI0 0x10001 - XPI1
0x08	Start	4	Start address

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Offset (bytes)	Field	Width (bytes)	Description
12	Length	4	Number of bytes to erase (ignored for Chip Erase)

Table 80: Erase Memory Device Data Block

Reset data block structure is as follows:

Offset (bytes)	Field	Width (bytes)	Description
0x00	Command Tag	1	Command tag - 0x08
0x01	Reserved	1	Reserved
0x02	Command Misc.	1	Command miscellaneous, fixed at 0
0x03	Reserved	1	Reserved
0x4	Reset Type	4	Reset type, reserved field on this SoC
0x08	Reserved	8	Reserved bits

Table 81: Reset Data Block

19.3.9.3 Encrypted Command Container (Encrypted Command Container)

The command container supports encryption in AES-CBC/SM4-CBC mode. The IV is the first 128 bits of the current block's hash value. The command container encryption key is packaged in the FW Blob. The IV of the command container header is the first 128 bits of the hash value in the firmware information table.

19.4 Secure Boot

19.4.1 Secure Boot Introduction

This device supports secure boot.

Secure boot supports signature and authentication based on digital certificates derived from Public Key Infrastructure (PKI).

The secure boot of this device supports the following two certificate signature and authentication modes:

1. Direct signature authentication based on the primary certificate generated by CA, where the primary certificate supports up to 4.
2. Signature authentication based on secondary certificates issued by primary certificates, adopting a chain of trust approach to authenticate level by level, where the primary certificate supports up to 4 and the secondary certificate supports up to 1.

This device supports ECDSA P256 and SM2 signature authentication.

To simplify the entire secure boot process and reduce unrelated format conversion and parsing, BootROM uses fixed encoding for the following parts:

- Public key encoding - See Root Key Table section
- Digital signature encoding - See Signature Info section
- Certificate - See Certificate section

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19.4.2 Secure Boot Process

To shorten the entire secure boot time, the secure boot design model is as follows:

1. Only authenticate the FW Container header, where the data to be signed/authenticated is shown in the following figure:

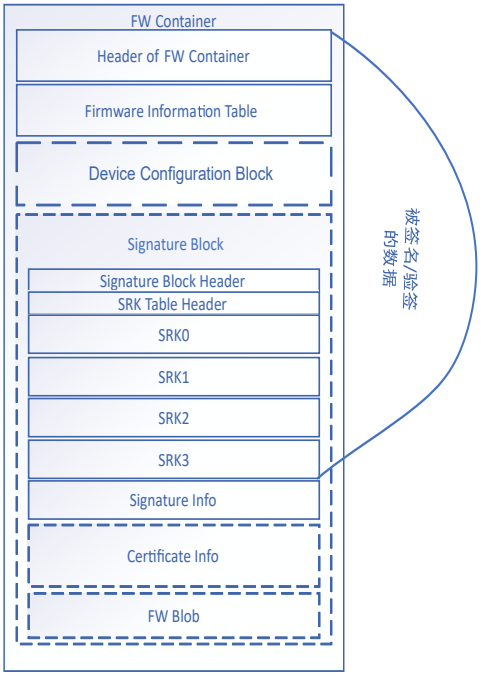


Figure 19: Firmware Header Part to Be Signed

2. Adopt hash comparison method for FW, where the FW hash value is included in the firmware info table as shown in the figure above. So the hash value itself has been authenticated, this design ensures signature uniqueness.
3. After FW Container header authentication passes, BootROM parses the firmware table and processes each firmware's verification separately according to the verification method specified in this table. This device supports two verification methods:
    - Plaintext hash verification based on SHA256 or SM3
    - Decryption verification based on AES-CCM or SM4-CCM
4. After FW Container header verification and each firmware verification pass, the entire boot image security authentication passes, BootROM supports on-demand boot or processing corresponding firmware. If any of the above verification steps fail, the boot image is considered invalid.

19.4.3 Firmware Container Authentication

For secure boot supporting primary certificates, the process is as follows:

1. BootROM parses FW Container, obtains signature block, and extracts SRK table from signature block, then calculates SRK table hash value
2. BootROM obtains SRK table hash value from OTP (determines SRK table hash value source according to $SRK_{SET in Container header)}, compare swith the hash value calculated in the previous step, if matches, SRK is valid Calculate$
3. Uses SRK specified in FW Container header to authenticate signature, if authentication passes, FW Container header is valid and not tampered

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For secure boot supporting secondary certificates, the process is as follows:

1. BootROM parses FW Container, obtains signature block, and extracts SRK table from signature block, then calculates SRK table hash value
2. BootROM obtains SRK table hash value from $SRK_{HASHfieldinOTP(determinesSRKtablehashvaluesourceaccordingto}$
3. Uses SRK specified in FW Container header to authenticate signature, if authentication passes, secondary certificate is valid, otherwise secure boot fails
4. Calculates hash value of data from FW Container header to SRK table end as data to be signed
5. Uses secondary certificate to authenticate the data from the previous step. If authentication passes, FW Container header is valid and not tampered

19.4.4 Firmware Authentication

After FW Container authentication passes, if device configuration block exists, BootROM completes device configuration according to device configuration block parameters, then enters firmware info processing phase. ROM supports two firmware authentication methods:

- Plaintext-based hash verification, BootROM completes FW hash value calculation according to address and length specified in firmware info table, and compares result with value in firmware info table.

  When FW Container authentication public key type is ECDSA P256, firmware hash value is based on SHA256 algorithm

  When FW Container authentication public key type is SM2, firmware hash value is based on SM3 algorithm or SHA256 algorithm
- CBC-based (AES-CBC/SM4-CBC) decryption and hash verification. BootROM adopts encryption/decryption while verifying approach according to load address and length specified in firmware info table.

  When FW Container authentication public key type is ECDSA P256, firmware uses AES-CBC + SHA256 to decrypt and verify hash value.

  When FW Container authentication public key type is SM2, firmware uses SM4-CBC + SM3 / AES-CBC + SHA256 to decrypt and verify hash value.

19.4.5 Public Key Revocation

As product usage time increases, currently used public keys may face expiration or private key leakage risks. When this happens, currently used public keys need to be revoked. BootROM doesn't provide automatic revocation of public keys, user programs need to revoke corresponding public keys as needed. BootROM provides two methods to revoke public keys.

- Revoke via FW Container.
  Set the bit corresponding to current key in SRK_REVOKE in Firmware Container Header to 1, sign new image with new private key, burn new image into storage medium or load via serial boot mode. After new image loading completes, BootROM will automatically complete revocation of specified public key.
- Revoke via user program.
  Use device configuration info block in image, and unlock write permission of PUK_REVOKE. Sign new image with new private key, burn new image into storage medium or load via serial boot mode. After new image is successfully booted, users can write PUK_REVOKE via OTP driver to complete public key revocation.

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19.5 Low Power Wakeup

In specific low power modes, the CPU core is powered down, and only necessary modules are powered on. Upon specific events (such as pressing a wakeup key), the CPU core can be woken up and powered on again to execute BootROM code, which then provides subsequent wakeup support. ROM supports the following low power wakeup options:

- Fast Jump: Directly jump to the user-specified wakeup address, only performing basic address range validity checks
- Strong Verification Jump: Perform integrity verification (SHA256) on the user-specified range
- Disable Low Power Wakeup: Force cold boot, ignore low power wakeup options

19.5.1 Fast Jump

Users need to fill in a valid wakeup address at SYSCTL.CPU0_GPR0 and configure the correct wakeup source. If the address is invalid, BootROM will perform a complete boot process.

Example code:

```
HPM_SYSCTL->CPU0_LP |= SYSCTL_CPU0_LP_STOP_MASK;
HPM_SYSCTL->RETENTION[0].VALUE = 0;
for (uint32_t i = 0; i < 4; i++) {
    HPM_SYSCTL->CPU0_WAKEUP_ENABLE[i] = ~0;
}
HPM_PMIC_WDG->WREN = 0x5AA5UL;
HPM_PMIC_WDG->CTRL = WDT_CTRL_INTTIME(2) | WDT_CTRL_RSTEN(0) | WDT_CTRL_INTEN(1) \
| WDT_CTRL_CLKSEL(1) | WDT_CTRL_EN(1);
HPM_SYSCTL->CPU0_GPR0 = (uint32_t)&lowpower_wakeup_test;
__asm("wfi");
```

19.5.2 Strong Verification Jump

When the FORCE_WAKEUP_ENTRY_CHK OTP bit is set, BootROM will force integrity verification of the wakeup code. Only after the integrity verification passes will BootROM execute the wakeup code; otherwise, BootROM will perform a complete boot process. In addition to fast jump configuration, users also need to fill in the program start address and length in SYSCTL.CPU0_PARAM[0] and SYSCTL.CPU0_PARAM[1], and fill in the SHA256 hash of the entire wake-up code in SYSCTL.CPU0_DATA[0] to SYSCTL.CPU0_DATA[7]. The address entry in SYSCTL.CPU_GPR0 must be within the address range specified by CPU0_GPR1 and CPU0_GPR2.

When CPU0 is woken up, ROM will calculate the SHA256 hash value based on the address range in SYSCTL.CPU0_PARAM[0] and SYSCTL.CPU0_PARAM[1] and compare it with the values in SYSCTL.CPU0_DATA[0] to SYSCTL.CPU0_DATA[7]. Only when the results match will BootROM jump to the wakeup address entry.

To ensure security, users can lock the above registers through SYSCTL.CPU0_LOCK.

Enabling methods:

- Configure in user code as described above
- Configure in the device configuration information block device configuration information block, see wakeup verification information section for details. When users need BootROM to ensure wakeup code

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is also authenticated by secure boot, this method is recommended. In this mode, BootROM will lock the above registers before jumping to the user program.

19.5.3 Disable Low Power Wakeup

When the FORCE_COLD_BOOT OTP bit is set, BootROM will ignore wakeup requests and perform a complete secure boot process.

19.6 Debug Port Permission Management

Access to the SoC debug port is managed jointly by SoC hardware and ROM.

- When the device is in CREATE mode, the Debug port is enabled by default, and BootROM execution can be interrupted and taken over at any time.
- When the device leaves the factory, the Debug port is disabled by default, and debug port control is taken over by BootROM. When the SoC leaves BootROM, BootROM will enable access permissions to the DEBUG port.
  - In ISP boot mode, when BootROM detects a DEBUG request, it will open DEBUG port access permissions and enter a while(1) loop.
  - When there is a normal boot image, BootROM will open DEBUG port access permissions before jumping to the APP and continue executing the user application.

19.7 ROM API

This section details all API interfaces supported by this SoC.

19.7.1 Introduction

BootROM exports common FLASH operation API interfaces and security-related API interfaces. User programs can directly call these interfaces for fast FLASH and security-related programming to save development and verification time. APIs are categorized by module, the following are the API categories supported by this SoC:

- run_bootloader interface
- XPI NOR driver interface
- XPI low-level driver interface
- OTP API interface
- Secure boot API interface
- SDP API interface
- EXIP API interface

On this SoC, the APIs provided by BootROM are organized based on the API root list at 0x2001_FF00. The API root data structure is as follows:

```
typedef struct {
    const uint32_t version;                          /*!< offset: 0x00 */
    const char *copyright;                           /*!< offset: 0x04 */
    hpm_stat_t (*const run_bootloader)(void *arg);   /*!< offset: 0x08 */
    const otp_driver_interface_t *otp_driver_if;     /*!< offset: 0x0c */
```

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```
    const xpi_driver_interface_t *xpi_driver_if;          /*!< offset: 0x10 */
    const xpi_nor_driver_interface_t *xpi_nor_driver_if; /*!< offset: 0x14 */
    const xpi_ram_driver_interface_t *xpi_ram_driver_if; /*!< offset: 0x18 */
    const sdp_driver_interface_t *sdp_driver_if;          /*!< offset: 0x1c */
    const sec_boot_api_interface_t *sec_boot_api_if;      /*!< offset: 0x20 */
} bootloader_api_table_t;

#define ROM_API_TABLE_ROOT  (const bootloader_api_table_t*)0x2001FF00U
```

19.7.2 run_bootloader API

User programs can call this API to jump to specified boot mode (such as entering ISP mode or selecting to boot another image). Its prototype definition is:

```
typedef struct {
    uint32_t index:8;
    uint32_t peripheral:8;
    uint32_t src:8;
    uint32_t tag:8;
}api_boot_arg_t;

#define API_BOOT_TAG  (0xEBU)
#define API_BOOT_SRC_FUSE (0U)
#define API_BOOT_SRC_PRIMARY (1U)
#define API_BOOT_SRC_SERIAL_BOOT (2U)
#define API_BOOT_SRC_ISP (3U)
#define API_BOOT_SRC_MAX    (API_BOOT_SRC_ISP)
#define API_BOOT_PERIPH_AUTO (0U)
#define API_BOOT_PERIPH_UART (1U)
#define API_BOOT_PERIPH_USBHID (2U)
#define API_BOOT_PERIPH_MASK (API_BOOT_PERIPH_USBHID)

void run_bootloader(void *arg);
```

19.7.2.1 Example:

Users want to enter in-system programming mode in application program, can enter as follows:

```
api_boot_arg_t boot_arg = {.index = 0,
            .peripheral = API_BOOT_PERIPH_AUTO,
            .src = API_BOOT_SRC_ISP,
            .tag = API_BOOT_TAG};
ROM_API_TABLE_ROOT->run_bootloader(&boot_arg);
```

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19.7.3 OTP API

OTP API provides OTP read and write operations. Its prototype definition is as follows:

```
typedef struct {
    uint32_t version;
    void (*init)(void);
    uint32_t reserved;
    uint32_t (*read_from_shadow)(uint32_t addr);
    uint32_t (*read_from_ip)(uint32_t addr);
    hpm_stat_t (*program)(uint32_t addr, const uint32_t *src, uint32_t num_of_words);
    hpm_stat_t (*reload)(efuse_region_t region);
    hpm_stat_t (*lock_otp)(uint32_t addr, otp_lock_option_t lock_option);
    hpm_stat_t (*lock_shadow)(uint32_t addr, otp_lock_option_t lock_option);
    hpm_stat_t (*set_configurable_region)(uint32_t start, uint32_t num_of_words);
    hpm_stat_t (*write_shadow_register)(uint32_t addr, uint32_t data);
} otp_driver_interface_t;
```

19.7.3.1 init

This API is used to initialize OTP driver, confirm this API has been called before calling OTP read and write operations.

19.7.3.2 read_from_shadow

This API is used to read specified OTP word from OTP Shadow register.

19.7.3.3 read_from_ip

This API is used to read specified OTP word from OTP using IP read method.

19.7.3.4 program

This API is used to write data to specified OTP address range. This API's parameter definition is as follows:

- addr - Specified write address
- src - Pointer to data to write
- num_of_words - Number of words to write (in 32-bit units)

19.7.3.5 reload

This API is used to reload shadow register values for specified region. The parameter definition is as follows:

```
typedef enum {
    efuse_region0_mask = 1U,   /*!< Address range: [0, 7]   */
    efuse_region1_mask = 2U,   /*!< Address range: [8, 15]  */
    efuse_region2_mask = 4U,   /*!< Address range: [16, 127] */
    efuse_region3_mask = 8U,   /*!< Address range: user defined */
}otp_region_t;
```

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19.7.3.6 lock_otp

This API can lock corresponding permissions of specified OTP word. Its parameter definition is as follows:

```
typedef enum {
    efuse_no_lock = 0,
    efuse_read_only = 1,
    efuse_permanent_no_lock = 2,
    efuse_disable_access = 3,
    efuse_lock_option_max = efuse_disable_access,
}otp_lock_option_t;
```

19.7.3.7 set_configurable_region

This API is used to set user-specified configurable region.

19.7.3.8 write_shadow_register

This API can directly write specified value to specified shadow register.

19.7.3.9 Example

If users want to read the 26th word of OTP. Can obtain via following two methods: Method 1: Obtain via Shadow register:

```
uint32_t fuse_word = ROM_API_TABLE_ROOT->otp_driver_if->read_from_shadow(26);
```

Method 2: Obtain via IP read:

```
uint32_t fuse_word = ROM_API_TABLE_ROOT->otp_driver_if->read_from_ip(26);
```

19.7.4 XPI Low-Level Driver API

XPI low-level API provides XPI module specific function configuration and data transfer, its API interface prototype definition is as follows:

```
typedef struct {
    uint32_t version;
    hpm_stat_t (*get_default_config)(xpi_config_t *xpi_config);
    hpm_stat_t (*get_default_device_config)(xpi_device_config_t *dev_config);
    hpm_stat_t (*init)(XPI_Type *base, xpi_config_t *xpi_config);
    hpm_stat_t (*config_ahb_buffer)(XPI_Type *base, xpi_ahb_buffer_cfg_t *ahb_buf_cfg);
    hpm_stat_t (*config_device)(XPI_Type *base, xpi_device_config_t *dev_cfg,
            xpi_port_t port);
    hpm_stat_t (*update_instr_table)(XPI_Type *base, const uint32_t *inst_base,
            uint32_t seq_idx, uint32_t num);
    hpm_stat_t (*transfer_blocking)(XPI_Type *base, xpi_xfer_ctx_t *xfer);
    void (*software_reset)(XPI_Type *base);
    bool (*is_idle)(XPI_Type *base);
```

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```
    void (*update_dllcr)(XPI_Type *base, uint32_t serial_root_clk_freq,
                uint32_t data_valid_time, xpi_port_t port, uint32_t dly_target);

    hpm_stat_t (*get_abs_apb_xfer_addr)(XPI_Type *base, xpi_xfer_port_t port,
                uint32_t in_addr, uint32_t *out_addr);
    hpm_stat_t (*iomux_init)(XPI_Type *base, const xpi_io_config_t *io_cfg);
    hpm_stat_t (*clock_init)(XPI_Type *base, const xpi_clk_config_t *clk_cfg);
} xpi_driver_interface_t;
```

19.7.4.1 get_default_config

This API is used to get XPI default configuration parameters. xpi_config_t definition see SDK.

Example:

```
xpi_config_t xpi_config;
hpm_stat_t status = ROM_API_TABLE_ROOT->xpi_driver_if->get_default_config(&xpi_config);
```

19.7.4.2 get_default_device_config

This API is used to get device default configuration parameters. xpi_device_config_t definition see SDK.

Example:

```
xpi_device_config_t device_config;
status = ROM_API_TABLE_ROOT->xpi_driver_if->get_default_device_config(&device_config);
```

19.7.4.3 init

Configure XPI controller according to xpi_config parameter.

Example:

```
status = ROM_API_TABLE_ROOT->xpi_driver_if->init(HPM_XPI0, &xpi_config);
```

19.7.4.4 config_ahb_buffer

Configure AHB Buffer information. xpi_ahb_buffer_cfg_t definition is as follows:

```
typedef struct
{
    struct
    {
        uint8_t priority;           /* Offset: 0x00 */
        uint8_t master_idx;         /* Offset: 0x01 */
        uint8_t buf_size_in_dword;  /* Offset: 0x02 */
        bool enable_prefetch;       /* Offset: 0x03 */
    }entry[XPI_AHB_BUF_CR_COUNT];
}xpi_ahb_buffer_cfg_t;
```

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Example:

```
xpi_ahb_buffer_cfg_t ahb_cfg;
for(uint32_t i=0; i<XPI_AHB_BUF_CR_COUNT; i++) {
    ahb_cfg.entry[i].enable_prefetch = false;
}
ahb_cfg.entry[0].priority = 0;
ahb_cfg.entry[0].master_idx = 0;
ahb_cfg.entry[0].buf_size_in_dword = 64;
ahb_cfg.entry[0].enable_prefetch = true;
status = ROM_API_TABLE_ROOT->xpi_driver_if->config_ahb_buffer(HPM_XPI0, &ahb_cfg);
```

19.7.4.5 config_device

Configure device information to XPI controller.

Example:

```
    status = ROM_API_TABLE_ROOT->xpi_driver_if->config_device(HPM_XPI0, &device_config);
```

19.7.4.6 update_instr_table

Update instruction table to XPI controller.

Example:

```
    uint32_t lut[4] = {XPI_INSTR_SEQ(XPI_PHASE_CMD_SDR, XPI_1PAD, 0x03,
                            XPI_PHASE_RADDR_SDR, XPI_1PAD, 0x18),
                       XPI_INSTR_SEQ(XPI_PHASE_READ_SDR, XPI_1PAD, 0x04,
                            XPI_PHASE_STOP, XPI_1PAD, 0x00),
                       0, 0};
    status = ROM_API_TABLE_ROOT->xpi_driver_if->update_instr_table(HPM_XPI0,
                                                &lut[0], 0, 1);
```

19.7.4.7 transfer_blocking

Complete blocking transfer (read/write/send command). Example:

```
    uint32_t buffer[128];
    xpi_xfer_ctx_t xfer = {0};
    xfer.cmd_type = xpi_apb_xfer_type_read;
    xfer.addr = 0x1000;
    xfer.seq_idx = 0;
    xfer.seq_num = 1;
    xfer.port = xpi_xfer_port_auto;
    xfer.buf = &buffer[0];
    xfer.xfer_size = sizeof(buffer);
    status = ROM_API_TABLE_ROOT->xpi_driver_if->transfer_blocking(HPM_XPI0, &xfer);
```

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19.7.4.8 software_reset

Perform software reset on XPI module. Example:

```
ROM_API_TABLE_ROOT->xpi_driver_if->software_reset(HPM_XPI0);
```

19.7.4.9 is_idle

Check if XPI module is idle. Example:

```
bool is_idle = ROM_API_TABLE_ROOT->xpi_driver_if->is_idle(HPM_XPI0);
```

19.7.4.10 update_dllcr

Update delay line (delay line) settings. Example:

```
uint32_t serial_root_clk_freq = 133000000UL;
uint32_t data_valid_time = 100;
xpi_port_t port = xpi_port_a1;
uint32_t delay_target = 8;
ROM_API_TABLE_ROOT->xpi_driver_if->update_dllcr(HPM_XPI0, serial_root_clk_freq,
                                                data_valid_time, port, delay_target);
```

19.7.5 XPI NOR API

XPI NOR API provides support for most FLASH devices on the market, can easily perform FLASH read and write operations, its API interface prototype definition is as follows:

```
typedef struct {
    uint32_t version;
    hpm_stat_t (*get_config)(XPI_Type *base,
               xpi_nor_config_t *nor_cfg,
               xpi_nor_config_option_t *cfg_option);
    hpm_stat_t (*init)(XPI_Type *base, xpi_nor_config_t *nor_config);
    hpm_stat_t (*enable_write)(XPI_Type *base,
               xpi_xfer_port_t port,
               const xpi_nor_config_t *nor_config, uint32_t addr);
    hpm_stat_t (*get_status)(XPI_Type *base, xpi_xfer_port_t port,
               const xpi_nor_config_t *nor_config, uint32_t addr,
               uint16_t *out_status);
    hpm_stat_t (*wait_busy)(XPI_Type *base, xpi_xfer_port_t port,
               const xpi_nor_config_t *nor_config, uint32_t addr);
    hpm_stat_t (*erase)(XPI_Type *base, xpi_xfer_port_t port,
               const xpi_nor_config_t *nor_config, uint32_t start,
               uint32_t length);
    hpm_stat_t (*erase_chip)(XPI_Type *base, xpi_xfer_port_t port,
               const xpi_nor_config_t *nor_config);
    hpm_stat_t (*erase_sector)(XPI_Type *base, xpi_xfer_port_t port,
```

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```
                const xpi_nor_config_t *nor_config, uint32_t addr);
    hpm_stat_t (*erase_block)(XPI_Type *base, xpi_xfer_port_t port,
                const xpi_nor_config_t *nor_config, uint32_t addr);
    hpm_stat_t (*program)(XPI_Type *base, xpi_xfer_port_t port,
                const xpi_nor_config_t *nor_config, const uint32_t *src,
                uint32_t dst_addr, uint32_t length);
    hpm_stat_t (*read)(XPI_Type *base, xpi_xfer_port_t port,
                const xpi_nor_config_t *nor_config, uint32_t *dst,
                uint32_t start, uint32_t length);
    hpm_stat_t (*page_program_nonblocking)(XPI_Type *base, xpi_xfer_port_t port,
                const xpi_nor_config_t *nor_config, const uint32_t *src,
                uint32_t dst_addr, uint32_t length);
    hpm_stat_t (*erase_sector_nonblocking)(XPI_Type *base, xpi_xfer_port_t port,
                const xpi_nor_config_t *nor_config, uint32_t addr);
    hpm_stat_t (*erase_block_nonblocking)(XPI_Type *base, xpi_xfer_port_t port,
                const xpi_nor_config_t *nor_config, uint32_t addr);
    hpm_stat_t (*erase_chip_nonblocking)(XPI_Type *base, xpi_xfer_port_t port,
                const xpi_nor_config_t *nor_config);
    hpm_stat_t (*restore_spi_protocol)(XPI_Type *base, xpi_xfer_port_t port,
                xpi_nor_config_t *config, flash_run_context_t *run_ctx);
    hpm_stat_t (*write_persistent)(const uint32_t data);
    hpm_stat_t (*read_persistent)(uint32_t *data);
    hpm_stat_t (*auto_config)(XPI_Type *base, xpi_nor_config_t *nor_cfg,
                xpi_nor_config_option_t *cfg_option);
    hpm_stat_t (*get_property)(XPI_Type *base, xpi_nor_config_t *nor_cfg,
                uint32_t property_id, uint32_t *value);
    hpm_stat_t (*set_property)(XPI_Type *base, xpi_nor_config_t *nor_cfg,
                uint32_t property_id, uint32_t value);
} xpi_nor_driver_interface_t;
```

19.7.5.1 get_config

Get FLASH configuration according to FLASH configuration options. This API supports most 4-wire and 8-wire serial NOR FLASH on the market. Detailed definition see XPI NOR Configuration Information.

Example:

```
    xpi_nor_config_option_t flash_option = {0};
    xpi_nor_config_t flash_config;
    flash_option.header.U = 0xfcf90001;
    flash_option.option0.U = 0x00000007;

    hpm_stat_t status = ROM_API_TABLE_ROOT->xpi_nor_driver_if->get_config(HPM_XPI0,
                                            &flash_config, &flash_option);
```

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19.7.5.2 init

Initialize XPI according to FLASH configuration. Example:

```
status = ROM_API_TABLE_ROOT->xpi_nor_driver_if->init(HPM_XPI0, &flash_config);
```

19.7.5.3 enable_write

Enable FLASH write operation. Example:

```
status = ROM_API_TABLE_ROOT->xpi_nor_driver_if->enable_write(HPM_XPI0,
                                        xpi_xfer_port_auto, &flash_config);
```

19.7.5.4 get_status

Get FLASH current status. Example:

```
uint16_t flash_status;
status = ROM_API_TABLE_ROOT->xpi_nor_driver_if->get_status(HPM_XPI0,
                            xpi_xfer_port_auto, &flash_config, 0, &flash_status);
```

19.7.5.5 wait_busy

FLASH busy wait operation. API will block waiting while FLASH is busy. Example:

```
status = ROM_API_TABLE_ROOT->xpi_nor_driver_if->wait_busy(HPM_XPI0,
                        xpi_xfer_port_auto, &flash_config, 0);
```

19.7.5.6 erase

Blocking erase specified FLASH range, this command sequentially executes FLASH erase and busy wait operations. Example:

```
status = ROM_API_TABLE_ROOT->xpi_nor_driver_if->erase(HPM_XPI0,
                            xpi_xfer_port_auto, &flash_config, 0, 0x2000);
```

19.7.5.7 erase_chip

Blocking erase entire FLASH. This command sequentially executes FLASH erase and busy wait operations. Example:

```
status = ROM_API_TABLE_ROOT->xpi_nor_driver_if->erase_chip(HPM_XPI0,
                            xpi_xfer_port_auto, &flash_config);
```

19.7.5.8 erase_sector

Blocking erase specified FLASH sector. This command sequentially executes FLASH erase and busy wait operations. Example:

```
status = ROM_API_TABLE_ROOT->xpi_nor_driver_if->erase_sector(HPM_XPI0,
                                xpi_xfer_port_auto, &flash_config, 0x4000);
```

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19.7.5.9 erase_block

Blocking erase specified FLASH block. This command sequentially executes FLASH erase and busy wait operations. Example:

```
status = ROM_API_TABLE_ROOT->xpi_nor_driver_if->erase_block(HPM_XPI0,
                                xpi_xfer_port_auto, &flash_config, 0x4000);
```

19.7.5.10 program

Blocking program data to specified FLASH area. This command sequentially executes FLASH write operation and busy wait operation. Example:

```
uint32_t buffer[64];
uint8_t *buf_8 = (uint8_t*)&buffer[0];
for(uint32_t i=0; i<256; i++) {
    buf_8[i] = (uint8_t)i & 0xFF;
}
status = ROM_API_TABLE_ROOT->xpi_nor_driver_if->program(HPM_XPI0,
            xpi_xfer_port_auto, &flash_config, buffer, 0, sizeof(buffer));
```

19.7.5.11 read

Blocking read specified FLASH area. Example:

```
uint32_t buffer[64];
status = ROM_API_TABLE_ROOT->xpi_nor_driver_if->read(HPM_XPI0,
                    xpi_xfer_port_auto, &flash_config, buffer, 0, sizeof(buffer));
```

19.7.5.12 page_program_nonblocking

Non-blocking write data to specified FLASH page. This API only completes PAGE Program command sending, user programs need to complete subsequent busy wait operations. Example:

```
uint32_t buffer[64];
uint8_t *buf_8 = (uint8_t*)&buffer[0];
for(uint32_t i=0; i<256; i++) {
    buf_8[i] = (uint8_t)i & 0xFF;
}
status = ROM_API_TABLE_ROOT->xpi_nor_driver_if->page_program_nonblocking(HPM_XPI0,
            xpi_xfer_port_auto, &flash_config, buffer, 0, sizeof(buffer));
```

19.7.5.13 erase_sector_nonblocking

Non-blocking erase specified FLASH sector, this API only completes Sector Erase command sending, user programs need to complete subsequent busy wait operations. Example:

```
status = ROM_API_TABLE_ROOT->xpi_nor_driver_if->erase_sector_nonblocking(HPM_XPI0,
                    xpi_xfer_port_auto, &flash_config, 0x4000);
```

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19.7.5.14 erase_block_nonblocking

Non-blocking erase specified FLASH block, this API only completes Block Erase command sending, user programs need to complete subsequent busy wait operations. Example:

```
status = ROM_API_TABLE_ROOT->xpi_nor_driver_if->erase_block_nonblocking(HPM_XPI0,
        xpi_xfer_port_auto, &flash_config, 0x4000);
```

19.7.5.15 erase_chip_nonblocking

Non-blocking erase entire FLASH space, this API only completes Chip Erase command sending, user programs need to complete subsequent busy wait operations. Example:

```
status = ROM_API_TABLE_ROOT->xpi_nor_driver_if->erase_chip_nonblocking(HPM_XPI0,
        xpi_xfer_port_auto, &flash_config, 0x4000);
```

19.7.6 Secure Boot API

Secure boot API provides security authentication support for data compatible with BootROM boot images. Its API interface prototype definition is as follows:

```
typedef struct {
    uint32_t version;
    secure_status_t (*init)(rom_core_ctx_t *ctx, uint32_t core_id);
    secure_status_t (*auth_img_hdr)(rom_core_ctx_t *ctx,
                    const boot_image_hdr_t *img_hdr,
                    const uint8_t *blob,
                    const uint8_t *srk_hash);
    secure_status_t (*auth_fw)(rom_core_ctx_t *ctx,
                    const firmware_info_t *fw_info);
    secure_status_t (*hdl_img)(rom_core_ctx_t *ctx,
                    const image_buf_t *img_hdr_buf,
                    const boot_interface_t *boot_if,
                    uint32_t img_offset);
} sec_boot_api_interface_t;
```

Detailed structure definition see examples in SDK.

19.7.7 SDP API

Detailed structure definition see examples in SDK

19.7.8 EXIP API

Detailed structure definition see hpm_romapi.h header file in SDK

19.8 SoC-Specific Information

19.8.1 OTP Memory Map

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OTP Word Index	Bit Off-set	Field Width (Bits)	Field	Description
1	0	4	LIFE_CYCLE_A	Life Cycle
	8	8	PUBK_REVOKE	Public Key REVOKE Lower 4 bits are valid, bit_n represents revoking the nth public key
	28	4	LIFE_CYCLE_B	Duplicate Life Cycle Actual life cycle value is LIFE_CYCLE_A|LIFE_CYCLE_B
3	0	32	SW_VER	Software minimum version number
24	0	4	XPI_FREQ_OPTION	XPI Frequency Option See XPI Frequency Option
	5	1	XPI_PIN_GROUP	XPI Pin Group 0 - First group 1 - Second group
	8	4	PROBE_TYPE	Serial NOR Probe Type 0 - Based on SFDP (SDR mode) 1 - Based on SFDP (DDR/DTR mode) 2 - 1-4-4 Read (command 0xEB, 24-bit address) 3 - 1-2-2 Read (command 0xBB, 24-bit address) Others - Reserved
	12	1	ENCRYPT_XIP	Encrypted Execute-in-Place Enable 0 - Do not enable encrypted execute-in-place 1 - Enable encrypted execute-in-place
	13	1	XPI_NOR_CFG_SRC	Select XPI NOR Configuration Option Location 0 - Look for XPI NOR CFG OPTION from FLASH offset 0x400 1 - Look for XPI NOR CFG OPTION from OTP
	14	2	XPI_DEFAULT_READ	XPI Default Read Mode 0 - 1-1-1 Read, command 0x03, 24-bit address Other - Reserved
	16	4	BOOT_MODE	Boot Mode Select 0 - Boot mode determined by $BOOT_{MODE}pins$ • 0 - Boot from XPI NOR • 1 - In-system programming mode • 2 - In-system programming mode • Others - Reserved $1 - Boot from XPI NOR$ $2 - Serial boot mode$ $4 - In - system programming mode$ $Others - Reserved$

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OTP Word Index	Bit Off-set	Field Width (Bits)	Field	Description
	20	4	DRIVE_STRENGTH	IO Drive Strength Selection 0 - BootROM default configuration Others - See DS field definition in IOC PAD_CTL
	24	7	DUMMY_CYCLE	Serial NOR FLASH DUMMY CYCLE Value 0 - DUMMY CYCLE value automatically detected by ROM / use ROM preset value Others - Use DUMMY CYCLE value defined in this field
25	0	16	SEC_IMG_OFFSET	Second Boot Image Address Offset 0 - Second boot image does not exist Others - SEC_IMG_OFFSET×256KB
	16	16	MAX_IMG_LEN	Maximum Boot Image Length 0 - SEC_IMG_OFFSET×256KB Others - MAX_IMG_LEN×256KB
26	0	2	FORCE_COLD_BOOT	Force Cold Boot 0 - Support low-power wakeup Others - Force cold boot
	2	2	FORCE_WAKEUP_ENTRY_CHK	Force Wakeup Entry Integrity Check 0 - No need to check wakeup entry integrity 1 - Force detect wakeup entry integrity (based on SHA256)
	4	1	HIGH_SPEED_BOOT	High-Speed Boot Mode Enable 0 - Normal boot mode (CPU CORE frequency approx. 325MHz) 1 - High-speed boot mode (CPU CORE frequency approx. 650MHz)
	5	27	Reserved	-
27	0	32	Reserved	-
80	0	32	SRK_HASH[31:0]	SRK_HASH stored in little-endian format in OTP If SRK_HASH is represented in bytes as (b0, b1, b2, b3, ..., b30, b31), then in OTP it is stored sequentially as: 0xb3b2b1b0, 0xb7b6b5b4 ..., 0xb31b30b29b28
81	0	32	SRK_HASH[63:32]	
82	0	32	SRK_HASH[95:64]	
83	0	32	SRK_HASH[127:96]	
84	0	32	SRK_HASH[159:128]	
85	0	32	SRK_HASH[191:160]	
86	0	32	SRK_HASH[223:192]	
87	0	32	SRK_HASH[255:224]	
96	0	32	EXIP0_KEK[31:0]	KEK stored in big-endian format in OTP. If KEK is represented in bytes as (b0, b1, b2, ..., b14, b15) then in OTP it is stored as 0xb15b14b13b12, ..., 0xb3b2b1b0
97	0	32	EXIP0_KEK[63:32]	
98	0	32	EXIP0_KEK[95:64]	

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OTP Word Index	Bit Off-set	Field Width (Bits)	Field	Description
99	0	32	EXIP0_KEK[127:96]	
100	0	32	OTP_KEK0[31:0]	-
101	0	32	OTP_KEK0[63:32]	
102	0	32	OTP_KEK0[95:64]	
103	0	32	OTP_KEK0[127:96]	

Table 82: OTP Memory Map Related to BootROM

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19.8.2 BootROM-Supported Peripheral Information

19.8.2.1 XPI Information

XPI0 Pin Groups

Pin Group	Pin	Function	Description
1	PA31	**XPI0.CA_CS0**	When FLASH device supports SPI mode, XPI accesses FLASH through **bold** pins by default
	PA24	XPI0.CA_CS1	
	PA27	**XPI0.CA_SCLK**	
	PA28	**XPI0.CA_D[0]**	
	PA30	**XPI0.CA_D[1]**	
	PA29	XPI0.CA_D[2]	
	PA26	XPI0.CA_D[3]	
	PA25	XPI0.CA_DQS	
2	PX02	XPI0.CA_CS0	-
	PX06	XPI0.CA_SCLK	
	PX05	XPI0.CA_D[0]	
	PX01	XPI0.CA_D[1]	
	PX00	XPI0.CA_D[2]	
	PX07	XPI0.CA_D[3]	
	PX03	XPI0.CA_DQS	

Table 83: XPI0 Pin Information

Note:
- When FLASH without DQS pin support is connected to XPI0 CA pins, CA_DQS pin can be used for other purposes
- Second group XPI pins are only valid for SiP package chips

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XPI Frequency Option

XPI NOR frequencies supported by BootROM are as follows:

- 1 - 31MHz (Note: In SDR mode, this value is 29.6MHz, in DTR/DDR mode, this value is 28.6MHz)
- 2 - 50MHz
- 3 - 66MHz
- 4 - 80MHz
- 5 - 102MHz
- 6 - 120MHz (Note: In SDR mode, this value is 114MHz, in DTR/DDR mode, this value is 125MHz)
- 7 - 133MHz
- 8 - 166MHz

Note: The above frequencies are derived from SoC default PRESET clock dividers. If the corresponding clock source (such as PLL1CLK1) frequency changes, the above frequencies will change. Users need to confirm beforehand that changes to PLL clock affect XPI frequency.

19.8.2.2 UART Pins

Chip Pin	Function	Function Description
PA00	UART0.TXD	UART0 Pins
PA01	UART0.RXD	

Table 84: UART0 Pins

19.8.2.3 USB Pins

Chip Pin	Function	Function Description
USB0.DP	USB_DP	USB0 Pins
USB0.DM	USB_DM	

Table 85: USB0 Pins

Note: The above USB pins are only valid for LQFP100 package

19.8.2.4 Boot Mode (BOOT_MODE) Pins

Chip Pin	Function	Function Description
PA02	BOOT_MODE[0]	Boot Mode Pins
PA03	BOOT_MODE[1]	

Table 86: BOOT_MODE Pins

19.8.3 BootROM Pre-Occupied Register Information

Register	Function	Description/Remarks
SYSCTL.CPU0.GPR0	CPU0 Wakeup Entry	CPU0 wakeup program entry

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SYSCTL.CPU0.PARAM0	CPU0 Wakeup Code Start Address	CPU0 wakeup program start address (for integrity check)
SYSCTL.CPU0.PARAM1	CPU0 Wakeup Code Length	CPU0 wakeup program length (for integrity check)
SYSCTL.CPU0_DATA0 - SYSCTL.CPU0_DATA7	CPU0 Wakeup Code Hash Value	After FORCE_WAKEUP_ENTRY_CHK is set to 1, BootROM needs to perform SHA256 calculation based on the address range specified by SYSCTL.CPU0_PARAM0 and SYSCTL.CPU0_PARAM1, and compare with the values in PMIC.GPR4-PMIC.GPR11. Only when the results match is jumping allowed, otherwise execute complete boot process
SYSCTL.CPU0_LOCK	Lock Register for Above Registers	-
PMIC.GPR0	run_bootloader Parameter	See ROM API Introduction
PMIC.GPR1	XPI NOR FLASH Status Context	Saves XPI NOR running context, detailed definition see
PMIC.GPR2	ROM Specific Marker	-

Table 87: Registers Occupied by BootROM

19.8.4 Register Range Supported by General Register Configuration

The following modules support configuration in General Register Information:

- SYSCTL
- IOC
- XPI0

19.8.5 BootROM Memory Map

Address Range	Function
0x2000_0000 - 0x2001_FFFF	BootROM code and data
0x0009_7000 - 0x0009_FFFF	BSS, RW, Stack, etc.

Table 88: BootROM Memory Map

19.8.6 BootROM Lifecycle

Encoding	Lifecycle	Description
4'b0000	CREATE	In this mode BootROM does not control DEBUG interface

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4’b0001	NONSEC	In this mode DEBUG port is controlled by BootROM, BootROM supports booting both unsigned and signed boot images
4’b0011	SECURE	In this mode DEBUG port is controlled by BootROM, BootROM only supports booting signed boot images
4’b0111	RETURN	In this mode chip security-related information is disabled, BootROM can boot unsigned boot images
4’b1011	NONRET	This mode behaves consistently with closed/deployed mode, the only difference is that OEM manufacturers can only switch to discard mode in this mode
4’b1111	SCRIBE	Discard mode, BootROM cannot execute normal boot operations in this mode

Table 89: BootROM Lifecycle Encoding

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20 Pin Configuration and Functions (PINMUX)

20.1 IO Function Assignment

The pin configuration and functions of this product are as follows:

Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
25	20	16	12	PA00	GPIO_A_00(ALT0) GPTMR1_COMP_0(ALT1) UART0_TXD(ALT2) MCAN0_TXD(ALT7) PWM0_FAULT_0(ALT16) PWM1_P_0(ALT17) TRGM0_P_00(ALT18) PWM1_FAULT_0(ALT19) SYSCTL_CLK_OBS_0(ALT24)	-	B00
24	19	15	11	PA01	GPIO_A_01(ALT0) GPTMR1_CAPT_0(ALT1) UART0_RXD(ALT2) MCAN0_RXD(ALT7) PWM0_FAULT_1(ALT16) PWM1_P_1(ALT17) TRGM0_P_01(ALT18) ACMP_COMP_0(ALT19) SYSCTL_CLK_OBS_1(ALT24)	-	B00

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
19	14	11	8	PA02	GPIO_A_02(ALT0) GPTMR1_COMP_1(ALT1) UART0_DE(ALT2) UART0_RTS(ALT3) I2C0_SCL(ALT4) MCAN0_STBY(ALT7) ACMP_COMP_0(ALT16) PWM1_P_2(ALT17) TRGM0_P_02(ALT18) ACMP_COMP_1(ALT19) QEI1_F(ALT20) SYSCTL_CLK_OBS_2(ALT24)	-	B00
18	13	10	7	PA03	GPIO_A_03(ALT0) GPTMR1_CAPT_1(ALT1) UART0_CTS(ALT3) I2C0_SDA(ALT4) SPI3_CS_3(ALT5) MCAN1_STBY(ALT7) ACMP_COMP_1(ALT16) PWM1_P_3(ALT17) TRGM0_P_03(ALT18) PWM1_FAULT_1(ALT19) QEI1_H1(ALT20) SYSCTL_CLK_OBS_3(ALT24)	-	B00

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
2	80	64	1	PA04	GPIO_A_04(ALT0) UART1_CTS(ALT3) SPI0_CS_0(ALT5) MCAN1_RXD(ALT7) PWM0_P_0(ALT16) PWM1_P_4(ALT17) TRGM0_P_04(ALT18) RDC0_EXC_P(ALT19) QEI1_A(ALT20) QEO1_A(ALT21) SEI1_DE(ALT22) JTAG_TDO(ALT24)	-	B00
1	79	63	48	PA05	GPIO_A_05(ALT0) GPTMR1_COMP_2(ALT1) UART1_DE(ALT2) UART1_RTS(ALT3) SPI0_SCLK(ALT5) MCAN1_TXD(ALT7) PWM0_P_1(ALT16) PWM1_P_5(ALT17) TRGM0_P_05(ALT18) RDC0_EXC_N(ALT19) QEI1_B(ALT20) QEO1_B(ALT21) SEI1_CK(ALT22) JTAG_TDI(ALT24)	-	B00

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
100	78	62	47	PA06	GPIO_A_06(ALT0) GPTMR0_CAPT_0(ALT1) UART1_RXD(ALT2) I2C1_SDA(ALT4) SPI0_MISO(ALT5) PWM0_P_2(ALT16) PWM1_P_6(ALT17) TRGM0_P_06(ALT18) QEI1_Z(ALT20) QEO1_Z(ALT21) SEI1_TX(ALT22) JTAG_TCK(ALT24)	-	B00
99	77	61	46	PA07	GPIO_A_07(ALT0) GPTMR0_COMP_0(ALT1) UART1_TXD(ALT2) I2C1_SCL(ALT4) SPI0_MOSI(ALT5) PWM0_P_3(ALT16) PWM1_P_7(ALT17) TRGM0_P_07(ALT18) QEI1_H0(ALT20) SEI1_RX(ALT22) JTAG_TMS(ALT24)	-	B00
3	1	1	2	PA08	GPIO_A_08(ALT0) GPTMR0_COMP_1(ALT1) UART2_TXD(ALT2) I2C2_SCL(ALT4) SPI3_CS_2(ALT5) MCAN2_TXD(ALT7) PWM0_P_4(ALT16) PWM0_FAULT_0(ALT18) JTAG_TRST(ALT24)	-	B00

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
4	2	2	3	PA09	GPIO_A_09(ALT0) GPTMR0_CAPT_1(ALT1) UART2_RXD(ALT2) I2C2_SDA(ALT4) SPI3_CS_1(ALT5) MCAN2_RXD(ALT7) PWM0_P_5(ALT16) PWM0_FAULT_1(ALT18) SOC_REF0(ALT24)	-	B00
5	3	3	4	PA10	GPIO_A_10(ALT0) GPTMR0_COMP_2(ALT1) UART2_DE(ALT2) UART2_RTS(ALT3) SPI3_CS_0(ALT5) MCAN2_STBY(ALT7) PWM0_P_6(ALT16) PWM1_FAULT_0(ALT17) ACMP_COMP_0(ALT18) QEI1_A(ALT20) QEO0_A(ALT21) SEI1_DE(ALT22)	-	B00
6	4	4	-	PA11	GPIO_A_11(ALT0) UART2_CTS(ALT3) SPI3_SCLK(ALT5) PWM0_P_7(ALT16) PWM1_FAULT_1(ALT17) ACMP_COMP_1(ALT18) QEI1_B(ALT20) QEO0_B(ALT21) SEI1_CK(ALT22) EWDG0_RST(ALT24)	-	B00

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
11	7	8	-	PA12	GPIO_A_12(ALT0) UART3_CTS(ALT3) I2C3_SDA(ALT4) SPI3_MISO(ALT5) PWM0_P_0(ALT16) PWM1_FAULT_0(ALT17) PWM0_FAULT_0(ALT18) RDC0_EXC_P(ALT19) QEI1_Z(ALT20) QEO0_Z(ALT21) SEI1_TX(ALT22)	-	B00
12	8	9	-	PA13	GPIO_A_13(ALT0) GPTMR1_COMP_3(ALT1) UART3_DE(ALT2) UART3_RTS(ALT3) I2C3_SCL(ALT4) SPI3_MOSI(ALT5) MCAN3_STBY(ALT7) PWM0_P_1(ALT16) PWM1_FAULT_1(ALT17) PWM0_FAULT_1(ALT18) RDC0_EXC_N(ALT19) QEI1_H0(ALT20) SEI1_RX(ALT22)	-	B00
13	9	-	-	PA14	GPIO_A_14(ALT0) UART3_RXD(ALT2) SPI3_DAT2(ALT5) MCAN3_RXD(ALT7) PWM0_P_2(ALT16) ACMP_COMP_0(ALT18) QEI1_H1(ALT20) EWDG1_RST(ALT24)	-	B00

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
14	10	-	-	PA15	GPIO_A_15(ALT0) GPTMR0_COMP_3(ALT1) UART3_TXD(ALT2) SPI3_DAT3(ALT5) MCAN3_TXD(ALT7) PWM0_P_3(ALT16) ACMP_COMP_1(ALT18) QEI1_F(ALT20) SOC_REF0(ALT24)	-	B00
98	76	-	-	PA16	GPIO_A_16(ALT0) GPTMR3_COMP_0(ALT1) UART4_TXD(ALT2) MCAN0_TXD(ALT7) PWM0_P_4(ALT16) PWM1_P_0(ALT17) TRGM0_P_04(ALT18) QEO0_A(ALT21) SEI1_DE(ALT22)	-	B00
97	75	-	-	PA17	GPIO_A_17(ALT0) GPTMR3_CAPT_0(ALT1) UART4_RXD(ALT2) MCAN0_RXD(ALT7) PWM0_P_5(ALT16) PWM1_P_1(ALT17) TRGM0_P_05(ALT18) QEO0_B(ALT21) SEI1_CK(ALT22)	-	B00

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
93	72	-	-	PA18	GPIO_A_18(ALT0) GPTMR3_COMP_1(ALT1) UART4_DE(ALT2) UART4_RTS(ALT3) I2C0_SCL(ALT4) MCAN0_STBY(ALT7) PWM0_P_6(ALT16) PWM1_P_2(ALT17) TRGM0_P_06(ALT18) QEO0_Z(ALT21) SEI1_TX(ALT22)	-	B00
92	71	-	-	PA19	GPIO_A_19(ALT0) GPTMR3_CAPT_1(ALT1) UART4_CTS(ALT3) I2C0_SDA(ALT4) SPI1_CS_3(ALT5) MCAN1_STBY(ALT7) PWM0_P_7(ALT16) PWM1_P_3(ALT17) TRGM0_P_07(ALT18) SEI1_RX(ALT22)	-	B00
91	-	-	-	PA20	GPIO_A_20(ALT0) UART5_CTS(ALT3) SPI2_CS_0(ALT5) MCAN1_RXD(ALT7) PWM0_FAULT_0(ALT16) PWM1_P_4(ALT17) TRGM0_P_00(ALT18) QEI0_A(ALT20) QEO0_A(ALT21) SEI0_DE(ALT22)	-	B00

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
90	-	-	-	PA21	GPIO_A_21(ALT0) GPTMR3_COMP_2(ALT1) UART5_DE(ALT2) UART5_RTS(ALT3) SPI2_SCLK(ALT5) MCAN1_TXD(ALT7) PWM0_FAULT_1(ALT16) PWM1_P_5(ALT17) TRGM0_P_01(ALT18) QEI0_B(ALT20) QEO0_B(ALT21) SEI0_CK(ALT22)	-	B00
89	-	-	-	PA22	GPIO_A_22(ALT0) GPTMR2_CAPT_0(ALT1) UART5_RXD(ALT2) I2C1_SDA(ALT4) SPI2_MISO(ALT5) PWM1_P_6(ALT17) TRGM0_P_02(ALT18) PWM1_FAULT_0(ALT19) QEI0_Z(ALT20) QEO0_Z(ALT21) SEI0_TX(ALT22)	-	B00

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
88	-	-	-	PA23	GPIO_A_23(ALT0) GPTMR2_COMP_0(ALT1) UART5_TXD(ALT2) I2C1_SCL(ALT4) SPI2_MOSI(ALT5) PWM1_P_7(ALT17) TRGM0_P_03(ALT18) PWM1_FAULT_1(ALT19) QEI0_H0(ALT20) SEI0_RX(ALT22)	-	B00
87	70	57	44	PA24	GPIO_A_24(ALT0) GPTMR2_COMP_1(ALT1) UART6_TXD(ALT2) I2C2_SCL(ALT4) SPI1_CS_2(ALT5) MCAN2_TXD(ALT7) XPI0_CA_CS1(ALT14) PWM0_P_0(ALT16) PWM1_P_0(ALT17) TRGM0_P_00(ALT18) QEI0_H1(ALT20)	-	B00
86	69	56	43	PA25	GPIO_A_25(ALT0) GPTMR2_CAPT_1(ALT1) UART6_RXD(ALT2) I2C2_SDA(ALT4) SPI1_CS_1(ALT5) MCAN2_RXD(ALT7) XPI0_CA_DQS(ALT14) PWM0_P_1(ALT16) PWM1_P_1(ALT17) TRGM0_P_01(ALT18) QEI0_F(ALT20)	-	B00

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
82	66	55	42	PA26	GPIO_A_26(ALT0) GPTMR2_COMP_2(ALT1) UART6_DE(ALT2) UART6_RTS(ALT3) SPI1_CS_0(ALT5) MCAN2_STBY(ALT7) XPI0_CA_D_3(ALT14) PWM0_P_2(ALT16) PWM1_P_2(ALT17) TRGM0_P_02(ALT18) QEI0_A(ALT20) QEO0_A(ALT21) SEI0_DE(ALT22) SYSCTL_CLK_OBS_0(ALT24)	-	B00
81	65	54	41	PA27	GPIO_A_27(ALT0) UART6_CTS(ALT3) SPI1_SCLK(ALT5) XPI0_CA_SCLK(ALT14) PWM0_P_3(ALT16) PWM1_P_3(ALT17) TRGM0_P_03(ALT18) QEI0_B(ALT20) QEO0_B(ALT21) SEI0_CK(ALT22) SYSCTL_CLK_OBS_1(ALT24)	-	B00

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
80	64	53	40	PA28	GPIO_A_28(ALT0) UART7_CTS(ALT3) I2C3_SDA(ALT4) SPI1_MISO(ALT5) XPI0_CA_D_0(ALT14) PWM0_P_4(ALT16) PWM1_P_4(ALT17) TRGM0_P_04(ALT18) RDC0_EXC_P(ALT19) QEI0_Z(ALT20) QEO0_Z(ALT21) SEI0_TX(ALT22) SYSCTL_CLK_OBS_2(ALT24)	-	B00
79	63	52	39	PA29	GPIO_A_29(ALT0) GPTMR3_COMP_3(ALT1) UART7_DE(ALT2) UART7_RTS(ALT3) I2C3_SCL(ALT4) SPI1_MOSI(ALT5) MCAN3_STBY(ALT7) XPI0_CA_D_2(ALT14) PWM0_P_5(ALT16) PWM1_P_5(ALT17) TRGM0_P_05(ALT18) RDC0_EXC_N(ALT19) QEI0_H0(ALT20) SEI0_RX(ALT22) SYSCTL_CLK_OBS_3(ALT24) USB0_OC(ALT25)	-	B00

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
78	62	51	38	PA30	GPIO_A_30(ALT0) UART7_RXD(ALT2) SPI1_DAT2(ALT5) MCAN3_RXD(ALT7) XPI0_CA_D_1(ALT14) PWM0_P_6(ALT16) PWM1_P_6(ALT17) TRGM0_P_06(ALT18) QEI0_H1(ALT20) SOC_REF0(ALT24) USB0_PWR(ALT25)	-	B00
77	61	50	37	PA31	GPIO_A_31(ALT0) GPTMR2_COMP_3(ALT1) UART7_TXD(ALT2) SPI1_DAT3(ALT5) MCAN3_TXD(ALT7) XPI0_CA_CS0(ALT14) PWM0_P_7(ALT16) PWM1_P_7(ALT17) TRGM0_P_07(ALT18) QEI0_F(ALT20) USB0_ID(ALT25)	-	B00
63	49	41	-	PB00	GPIO_B_00(ALT0) GPTMR1_COMP_0(ALT1) UART0_TXD(ALT2) MCAN0_TXD(ALT7) PWM0_P_0(ALT16) PWM1_FAULT_0(ALT17) TRGM0_P_04(ALT18) ACMP_COMP_0(ALT19)	ADC0_IN15 ADC1_IN15 ACMP_CMP0_INN7 OPA0_OUT	B01

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
62	48	40	-	PB01	GPIO_B_01(ALT0) GPTMR1_CAPT_0(ALT1) UART0_RXD(ALT2) MCAN0_RXD(ALT7) PWM0_P_1(ALT16) PWM1_FAULT_1(ALT17) TRGM0_P_05(ALT18) ACMP_COMP_1(ALT19)	ADC0_IN14 ADC1_IN14 ACMP_CMP1_INN7 OPA1_OUT	B01
61	-	-	-	PB02	GPIO_B_02(ALT0) GPTMR1_COMP_1(ALT1) UART0_DE(ALT2) UART0_RTS(ALT3) I2C0_SCL(ALT4) MCAN0_STBY(ALT7) PWM0_P_2(ALT16) ACMP_COMP_1(ALT17) TRGM0_P_06(ALT18) PWM0_FAULT_0(ALT19)	ADC0_IN12 ADC1_IN12 ACMP_CMP0_INP7 OPA0_EXT	B01
60	-	-	-	PB03	GPIO_B_03(ALT0) GPTMR1_CAPT_1(ALT1) UART0_CTS(ALT3) I2C0_SDA(ALT4) SPI2_CS_3(ALT5) MCAN1_STBY(ALT7) PWM0_P_3(ALT16) ACMP_COMP_0(ALT17) TRGM0_P_07(ALT18) PWM0_FAULT_1(ALT19)	ADC0_IN8 ADC1_IN8 ACMP_CMP1_INP7 OPA1_EXT	B01

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
59	-	-	-	PB04	GPIO_B_04(ALT0) UART1_CTS(ALT3) SPI3_CS_0(ALT5) MCAN1_RXD(ALT7) PWM0_P_4(ALT16) PWM1_P_0(ALT17) TRGM0_P_00(ALT18) QEI1_A(ALT20) QEO1_A(ALT21) SEI0_DE(ALT22)	ADC0_IN0 ADC1_IN0 ACMP_CMP0_INN5 ACMP_CMP1_INN5 OPA0_INP0	B01
58	-	-	-	PB05	GPIO_B_05(ALT0) GPTMR1_COMP_2(ALT1) UART1_DE(ALT2) UART1_RTS(ALT3) SPI3_SCLK(ALT5) MCAN1_TXD(ALT7) PWM0_P_5(ALT16) PWM1_P_1(ALT17) TRGM0_P_01(ALT18) QEI1_B(ALT20) QEO1_B(ALT21) SEI0_CK(ALT22)	ADC0_IN13 ADC1_IN13 ACMP_CMP0_INN3 ACMP_CMP1_INN3 OPA0_INN0	B01
54	-	-	-	PB06	GPIO_B_06(ALT0) GPTMR0_CAPT_0(ALT1) UART1_RXD(ALT2) I2C1_SDA(ALT4) SPI3_MISO(ALT5) PWM0_P_6(ALT16) PWM1_P_2(ALT17) TRGM0_P_02(ALT18) RDC0_EXC_P(ALT19) QEI1_Z(ALT20) QEO1_Z(ALT21) SEI0_TX(ALT22)	ADC0_IN9 ADC1_IN9 ACMP_CMP0_INP5 ACMP_CMP1_INP5 OPA0_INP1	B01

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
53	-	-	-	PB07	GPIO_B_07(ALT0) GPTMR0_COMP_0(ALT1) UART1_TXD(ALT2) I2C1_SCL(ALT4) SPI3_MOSI(ALT5) PWM0_P_7(ALT16) PWM1_P_3(ALT17) TRGM0_P_03(ALT18) RDC0_EXC_N(ALT19) QEI1_H0(ALT20) SEI0_RX(ALT22)	ADC0_IN10 ADC1_IN10 ACMP_CMP0_INP3 ACMP_CMP1_INP3 OPA0_INN1	B01
52	44	36	27	PB08	GPIO_B_08(ALT0) GPTMR0_COMP_1(ALT1) UART2_TXD(ALT2) I2C2_SCL(ALT4) SPI2_CS_2(ALT5) MCAN2_TXD(ALT7) ACMP_COMP_0(ALT16) PWM1_P_4(ALT17) QEI1_H1(ALT20) QEO1_A(ALT21) SEI1_DE(ALT22) USB0_ID(ALT25)	ADC0_IN11 ADC1_IN11 DAC0_OUT ACMP_CMP0_INN6 ACMP_CMP1_INN6 OPA0_INP2 OPA1_INP2	B01
51	43	35	26	PB09	GPIO_B_09(ALT0) GPTMR0_CAPT_1(ALT1) UART2_RXD(ALT2) I2C2_SDA(ALT4) SPI2_CS_1(ALT5) MCAN2_RXD(ALT7) ACMP_COMP_1(ALT16) PWM1_P_5(ALT17) QEI1_F(ALT20) QEO1_B(ALT21) SEI1_CK(ALT22) USB0_OC(ALT25)	ADC0_IN1 ADC1_IN1 DAC1_OUT ACMP_CMP0_INP6 ACMP_CMP1_INP6 OPA0_INN2 OPA1_INN2	B01

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
50	42	34	25	PB10	GPIO_B_10(ALT0) GPTMR0_COMP_2(ALT1) UART2_DE(ALT2) UART2_RTS(ALT3) SPI2_CS_0(ALT5) MCAN2_STBY(ALT7) ACMP_COMP_0(ALT16) PWM1_P_6(ALT17) QEI0_H1(ALT20) QEO1_Z(ALT21) SEI1_TX(ALT22) USB0_PWR(ALT25)	ADC0_IN2 ADC1_IN2 ACMP_CMP0_INP4 ACMP_CMP1_INP4 OPA0_INP3 OPA1_INP3	B01
49	41	33	24	PB11	GPIO_B_11(ALT0) UART2_CTS(ALT3) SPI2_SCLK(ALT5) ACMP_COMP_1(ALT16) PWM1_P_7(ALT17) QEI0_F(ALT20) SEI1_RX(ALT22)	ADC0_IN3 ADC1_IN3 ACMP_CMP0_INN4 ACMP_CMP1_INN4 OPA0_INN3 OPA1_INN3	B01
48	40	32	23	PB12	GPIO_B_12(ALT0) UART3_CTS(ALT3) I2C3_SDA(ALT4) SPI2_MISO(ALT5) PWM1_FAULT_0(ALT16) PWM1_P_0(ALT17) TRGM0_P_00(ALT18) QEI0_A(ALT20) QEO1_A(ALT21) SEI0_DE(ALT22)	ADC0_IN4 ADC1_IN4 ACMP_CMP0_INN2 ACMP_CMP1_INN2 OPA1_INP0	B01

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
47	39	31	22	PB13	GPIO_B_13(ALT0) GPTMR1_COMP_3(ALT1) UART3_DE(ALT2) UART3_RTS(ALT3) I2C3_SCL(ALT4) SPI2_MOSI(ALT5) MCAN3_STBY(ALT7) PWM1_FAULT_1(ALT16) PWM1_P_1(ALT17) TRGM0_P_01(ALT18) QEI0_B(ALT20) QEO1_B(ALT21) SEI0_CK(ALT22)	ADC0_IN5 ADC1_IN5 ACMP_CMP0_INP2 ACMP_CMP1_INP2 OPA1_INN0	B01
46	38	30	21	PB14	GPIO_B_14(ALT0) UART3_RXD(ALT2) SPI2_DAT2(ALT5) MCAN3_RXD(ALT7) PWM0_FAULT_0(ALT16) PWM1_P_2(ALT17) TRGM0_P_02(ALT18) RDC0_EXC_P(ALT19) QEI0_Z(ALT20) QEO1_Z(ALT21) SEI0_TX(ALT22)	ADC0_IN6 ADC1_IN6 ACMP_CMP0_INN1 ACMP_CMP1_INN1 OPA1_INP1	B01

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
45	37	29	20	PB15	GPIO_B_15(ALT0) GPTMR0_COMP_3(ALT1) UART3_TXD(ALT2) SPI2_DAT3(ALT5) MCAN3_TXD(ALT7) PWM0_FAULT_1(ALT16) PWM1_P_3(ALT17) TRGM0_P_03(ALT18) RDC0_EXC_N(ALT19) QEI0_H0(ALT20) SEI0_RX(ALT22)	ADC0_IN7 ADC1_IN7 ACMP_CMP0_INP1 ACMP_CMP1_INP1 OPA1_INN1	B01
-	-	-	-	PX00	GPIO_X_00(ALT0) GPTMR2_COMP_0(ALT1) UART4_TXD(ALT2) MCAN0_TXD(ALT7) XPI0_CA_D_2(ALT14)	-	B00
-	-	-	-	PX01	GPIO_X_01(ALT0) GPTMR2_CAPT_0(ALT1) UART4_RXD(ALT2) MCAN0_RXD(ALT7) XPI0_CA_D_1(ALT14)	-	B00
-	-	-	-	PX02	GPIO_X_02(ALT0) GPTMR2_COMP_1(ALT1) UART4_DE(ALT2) UART4_RTS(ALT3) I2C0_SCL(ALT4) MCAN0_STBY(ALT7) XPI0_CA_CS0(ALT14)	-	B00

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
-	-	-	-	PX03	GPIO_X_03(ALT0) GPTMR2_CAPT_1(ALT1) UART4_CTS(ALT3) I2C0_SDA(ALT4) MCAN1_STBY(ALT7) XPI0_CA_DQS(ALT14)	-	B00
-	-	-	-	PX04	GPIO_X_04(ALT0) UART5_CTS(ALT3) SPI1_CS_0(ALT5) MCAN1_RXD(ALT7) XPI0_CA_CS1(ALT14)	-	B00
-	-	-	-	PX05	GPIO_X_05(ALT0) GPTMR2_COMP_2(ALT1) UART5_DE(ALT2) UART5_RTS(ALT3) SPI1_SCLK(ALT5) MCAN1_TXD(ALT7) XPI0_CA_D_0(ALT14)	-	B00
-	-	-	-	PX06	GPIO_X_06(ALT0) GPTMR3_CAPT_0(ALT1) UART5_RXD(ALT2) I2C1_SDA(ALT4) SPI1_MISO(ALT5) XPI0_CA_SCLK(ALT14)	-	B00
-	-	-	-	PX07	GPIO_X_07(ALT0) GPTMR3_COMP_0(ALT1) UART5_TXD(ALT2) I2C1_SCL(ALT4) SPI1_MOSI(ALT5) XPI0_CA_D_3(ALT14)	-	B00

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
41	34	28	19	PY00	GPIO_Y_00(ALT0) GPTMR3_COMP_0(ALT1) UART0_TXD(ALT2) MCAN2_TXD(ALT7) PWM0_P_0(ALT16) PWM1_P_4(ALT17) PWM0_FAULT_0(ALT18) USB0_ID(ALT25)	-	VPMC
40	33	27	18	PY01	GPIO_Y_01(ALT0) GPTMR3_CAPT_0(ALT1) UART0_RXD(ALT2) MCAN2_RXD(ALT7) PWM0_P_1(ALT16) PWM1_P_5(ALT17) PWM0_FAULT_1(ALT18) EWDG0_RST(ALT24) USB0_OC(ALT25)	-	VPMC
39	32	26	-	PY02	GPIO_Y_02(ALT0) GPTMR3_COMP_1(ALT1) UART0_DE(ALT2) UART0_RTS(ALT3) I2C2_SCL(ALT4) MCAN2_STBY(ALT7) PWM0_P_2(ALT16) PWM1_P_6(ALT17) ACMP_COMP_0(ALT18) PWM1_FAULT_0(ALT19) EWDG1_RST(ALT24) USB0_PWR(ALT25)	-	VPMC

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
38	31	25	-	PY03	GPIO_Y_03(ALT0) GPTMR3_CAPT_1(ALT1) UART0_CTS(ALT3) I2C2_SDA(ALT4) MCAN3_STBY(ALT7) PWM0_P_3(ALT16) PWM1_P_7(ALT17) ACMP_COMP_1(ALT18) PWM1_FAULT_1(ALT19)	-	VPMC
37	30	-	-	PY04	GPIO_Y_04(ALT0) UART1_CTS(ALT3) SPI2_CS_0(ALT5) MCAN3_RXD(ALT7) PWM0_P_4(ALT16) TRGM0_P_04(ALT18)	-	VPMC
36	29	-	-	PY05	GPIO_Y_05(ALT0) GPTMR3_COMP_2(ALT1) UART1_DE(ALT2) UART1_RTS(ALT3) SPI2_SCLK(ALT5) MCAN3_TXD(ALT7) PWM0_P_5(ALT16) TRGM0_P_05(ALT18) EWDG0_RST(ALT24)	-	VPMC
35	-	-	-	PY06	GPIO_Y_06(ALT0) GPTMR2_CAPT_0(ALT1) UART1_RXD(ALT2) I2C3_SDA(ALT4) SPI2_MISO(ALT5) PWM0_P_6(ALT16) TRGM0_P_06(ALT18) EWDG1_RST(ALT24)	-	VPMC

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
34	-	-	-	PY07	GPIO_Y_07(ALT0) GPTMR2_COMP_0(ALT1) UART1_TXD(ALT2) I2C3_SCL(ALT4) SPI2_MOSI(ALT5) PWM0_P_7(ALT16) TRGM0_P_07(ALT18)	-	VPMC
7,17,20,44,57,69,83,94	5,15,36,47,54,73	5,12,39,45,58	5,9,28,33,45	VDD_SOC	-	-	-
8,10,16,22,32,43,56,66,70,76,84,95	6,12,17,27,35,46,55,68,74	6,13,23,38,46,59	-	VSS	-	-	-
9,15,85,96	11,67	7,47,60	6,34	VIO_B00	-	-	-
21	16	-	-	DCDC_SNS	-	-	-
23	18	14	10	VDD_OTPCAP	-	-	-
26	21	17	13	RESETN	-	-	-
27	22	18	14	WAKEUP	-	-	-
28	23	19	-	DCDC_GND	-	-	-
29	24	20	15	DCDC_LP	-	-	-
30	25	21	-	DCDC_IN	-	-	-

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Package				PIN NAME	Digital Function	Analog Functi	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48				
31	26	22	16	VPMC	-	-	-
33	28	24	17	VDD_PMCCAP	-	-	-
42,55	45	37	29	VIO_B01	-	-	-
64	50	42	30	VREFL	-	-	-
65	51	43	31	VREFH	-	-	-
67	52	44	32	VANA	-	-	-
68	53	-	-	USBVBUS	-	-	-
71	56	-	-	VPLL	-	-	-
72	57	-	-	USB_DP	-	-	-
73	58	-	-	USB_DM	-	-	-
74	59	48	35	XTALI	-	-	-
75	60	49	36	XTALO	-	-	-

Table 90: SOC IOMUX

20.2 IO Function Assignment for Power Management Domain

The IO pin configuration and functions of the power management domain of this product are as follows:

Package				PIN NAME	Digital Function	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48			
41	34	28	19	PY00	PGPIO_Y_00(ALT0) PUART_TXD(ALT1) PTMR_COMP_0(ALT2) SOC_GPIO_Y_00(ALT3)	VPMC
40	33	27	18	PY01	PGPIO_Y_01(ALT0) PUART_RXD(ALT1) PTMR_COMP_1(ALT2) SOC_GPIO_Y_01(ALT3)	VPMC

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Package				PIN NAME	Digital Function	Power Group
LQFP_100	LQFP_80	LQFP_64	QFN_48			
39	32	26	-	PY02	PGPIO_Y_02(ALT0) PUART_RTS(ALT1) PTMR_COMP_2(ALT2) SOC_GPIO_Y_02(ALT3)	VPMC
38	31	25	-	PY03	PGPIO_Y_03(ALT0) PUART_CTS(ALT1) PTMR_COMP_3(ALT2) SOC_GPIO_Y_03(ALT3)	VPMC
37	30	-	-	PY04	PGPIO_Y_04(ALT0) PTMR_COMP_0(ALT2) SOC_GPIO_Y_04(ALT3)	VPMC
36	29	-	-	PY05	PGPIO_Y_05(ALT0) PEWDG_RST(ALT1) PTMR_CAPT_0(ALT2) SOC_GPIO_Y_05(ALT3)	VPMC
35	-	-	-	PY06	PGPIO_Y_06(ALT0) PTMR_COMP_1(ALT2) SOC_GPIO_Y_06(ALT3)	VPMC
34	-	-	-	PY07	PGPIO_Y_07(ALT0) PTMR_CAPT_1(ALT2) SOC_GPIO_Y_07(ALT3)	VPMC

Table 91: PMIC IOMUX

20.3 Peripheral Pin Assignment for System Power Domain

The pin assignment summary of peripherals in the system power domain of this product by different modules is as follows:

ACMP signal name	Pin
CMP0_INN1	PB14
CMP0_INN2	PB12
CMP0_INN3	PB05
CMP0_INN4	PB11
CMP0_INN5	PB04
CMP0_INN6	PB08
CMP0_INN7	PB00
CMP0_INP1	PB15
CMP0_INP2	PB13

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ACMP signal name	Pin
CMP0_INP3	PB07
CMP0_INP4	PB10
CMP0_INP5	PB06
CMP0_INP6	PB09
CMP0_INP7	PB02
CMP1_INN1	PB14
CMP1_INN2	PB12
CMP1_INN3	PB05
CMP1_INN4	PB11
CMP1_INN5	PB04
CMP1_INN6	PB08
CMP1_INN7	PB01
CMP1_INP1	PB15
CMP1_INP2	PB13
CMP1_INP3	PB07
CMP1_INP4	PB10
CMP1_INP5	PB06
CMP1_INP6	PB09
CMP1_INP7	PB03
COMP_0	PA01 PA02 PA10 PA14 PB00 PB03 PB08 PB10 PY02
COMP_1	PA02 PA03 PA11 PA15 PB01 PB02 PB09 PB11 PY03

Table 92: ACMP SIGNAL PIN

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ADC0 signal name	Pin
IN0	PB04
IN1	PB09
IN2	PB10
IN3	PB11
IN4	PB12
IN5	PB13
IN6	PB14
IN7	PB15
IN8	PB03
IN9	PB06
IN10	PB07
IN11	PB08
IN12	PB02
IN13	PB05
IN14	PB01
IN15	PB00

Table 93: ADC0 SIGNAL PIN

ADC1 signal name	Pin
IN0	PB04
IN1	PB09
IN2	PB10
IN3	PB11
IN4	PB12
IN5	PB13
IN6	PB14
IN7	PB15
IN8	PB03
IN9	PB06
IN10	PB07
IN11	PB08
IN12	PB02
IN13	PB05
IN14	PB01
IN15	PB00

Table 94: ADC1 SIGNAL PIN

DAC0 signal name	Pin
OUT	PB08

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DAC0 signal name	Pin

Table 95: DAC0 SIGNAL PIN

DAC1 signal name	Pin
OUT	PB09

Table 96: DAC1 SIGNAL PIN

EWDG0 signal name	Pin
RST	PA11 PY01 PY05

Table 97: EWDG0 SIGNAL PIN

EWDG1 signal name	Pin
RST	PA14 PY02 PY06

Table 98: EWDG1 SIGNAL PIN

GPIO signal name	Pin
A_00	PA00
A_01	PA01
A_02	PA02
A_03	PA03
A_04	PA04
A_05	PA05
A_06	PA06
A_07	PA07
A_08	PA08
A_09	PA09
A_10	PA10
A_11	PA11
A_12	PA12
A_13	PA13
A_14	PA14
A_15	PA15
A_16	PA16
A_17	PA17
A_18	PA18

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GPIO signal name	Pin
A_19	PA19
A_20	PA20
A_21	PA21
A_22	PA22
A_23	PA23
A_24	PA24
A_25	PA25
A_26	PA26
A_27	PA27
A_28	PA28
A_29	PA29
A_30	PA30
A_31	PA31
B_00	PB00
B_01	PB01
B_02	PB02
B_03	PB03
B_04	PB04
B_05	PB05
B_06	PB06
B_07	PB07
B_08	PB08
B_09	PB09
B_10	PB10
B_11	PB11
B_12	PB12
B_13	PB13
B_14	PB14
B_15	PB15
X_00	PX00
X_01	PX01
X_02	PX02
X_03	PX03
X_04	PX04
X_05	PX05
X_06	PX06
X_07	PX07
Y_00	PY00
Y_01	PY01
Y_02	PY02

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GPIO signal name	Pin
Y_03	PY03
Y_04	PY04
Y_05	PY05
Y_06	PY06
Y_07	PY07

Table 99: GPIO SIGNAL PIN

GPTMR0 signal name	Pin
CAPT_0	PA06 PB06
CAPT_1	PA09 PB09
COMP_0	PA07 PB07
COMP_1	PA08 PB08
COMP_2	PA10 PB10
COMP_3	PA15 PB15

Table 100: GPTMR0 SIGNAL PIN

GPTMR1 signal name	Pin
CAPT_0	PA01 PB01
CAPT_1	PA03 PB03
COMP_0	PA00 PB00
COMP_1	PA02 PB02
COMP_2	PA05 PB05
COMP_3	PA13 PB13

Table 101: GPTMR1 SIGNAL PIN

GPTMR2 signal name	Pin
CAPT_0	PA22

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GPTMR2 signal name	Pin
	PX01 PY06
CAPT_1	PA25 PX03
COMP_0	PA23 PX00 PY07
COMP_1	PA24 PX02
COMP_2	PA26 PX05
COMP_3	PA31

Table 102: GPTMR2 SIGNAL PIN

GPTMR3 signal name	Pin
CAPT_0	PA17 PX06 PY01
CAPT_1	PA19 PY03
COMP_0	PA16 PX07 PY00
COMP_1	PA18 PY02
COMP_2	PA21 PY05
COMP_3	PA29

Table 103: GPTMR3 SIGNAL PIN

I2C0 signal name	Pin
SCL	PA02 PA18 PB02 PX02
SDA	PA03 PA19 PB03 PX03

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I2C0 signal name	Pin

Table 104: I2C0 SIGNAL PIN

I2C1 signal name	Pin
SCL	PA07 PA23 PB07 PX07
SDA	PA06 PA22 PB06 PX06

Table 105: I2C1 SIGNAL PIN

I2C2 signal name	Pin
SCL	PA08 PA24 PB08 PY02
SDA	PA09 PA25 PB09 PY03

Table 106: I2C2 SIGNAL PIN

I2C3 signal name	Pin
SCL	PA13 PA29 PB13 PY07
SDA	PA12 PA28 PB12 PY06

Table 107: I2C3 SIGNAL PIN

JTAG signal name	Pin
TCK	PA06
TDI	PA05
TDO	PA04

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JTAG signal name	Pin
TMS	PA07
TRST	PA08

Table 108: JTAG SIGNAL PIN

MCAN0 signal name	Pin
RXD	PA01 PA17 PB01 PX01
STBY	PA02 PA18 PB02 PX02
TXD	PA00 PA16 PB00 PX00

Table 109: MCAN0 SIGNAL PIN

MCAN1 signal name	Pin
RXD	PA04 PA20 PB04 PX04
STBY	PA03 PA19 PB03 PX03
TXD	PA05 PA21 PB05 PX05

Table 110: MCAN1 SIGNAL PIN

MCAN2 signal name	Pin
RXD	PA09 PA25 PB09 PY01
STBY	PA10

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MCAN2 signal name	Pin
	PA26 PB10 PY02
TXD	PA08 PA24 PB08 PY00

Table 111: MCAN2 SIGNAL PIN

MCAN3 signal name	Pin
RXD	PA14 PA30 PB14 PY04
STBY	PA13 PA29 PB13 PY03
TXD	PA15 PA31 PB15 PY05

Table 112: MCAN3 SIGNAL PIN

OPA0 signal name	Pin
EXT	PB02
INN0	PB05
INN1	PB07
INN2	PB09
INN3	PB11
INP0	PB04
INP1	PB06
INP2	PB08
INP3	PB10
OUT	PB00

Table 113: OPA0 SIGNAL PIN

OPA1 signal name	Pin
EXT	PB03

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OPA1 signal name	Pin
INN0	PB13
INN1	PB15
INN2	PB09
INN3	PB11
INP0	PB12
INP1	PB14
INP2	PB08
INP3	PB10
OUT	PB01

Table 114: OPA1 SIGNAL PIN

PWM0 High Speed Group	High speed signal name	Pin
A	P_0	PA04
	P_1	PA05
	P_2	PA06
	P_3	PA07
	P_4	PA08
	P_5	PA09
	P_6	PA10
	P_7	PA11
B	P_0	PA12
	P_1	PA13
	P_2	PA14
	P_3	PA15
	P_4	PA16
	P_5	PA17
	P_6	PA18
	P_7	PA19
C	P_0	PA24
	P_1	PA25
	P_2	PA26
	P_3	PA27
	P_4	PA28
	P_5	PA29
	P_6	PA30
	P_7	PA31
D	P_0	PB00
	P_1	PB01
	P_2	PB02
	P_3	PB03

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PWM0 High Speed Group	High speed signal name	Pin
	P_4	PB04
	P_5	PB05
	P_6	PB06
	P_7	PB07
E	P_0	PY00
	P_1	PY01
	P_2	PY02
	P_3	PY03
	P_4	PY04
	P_5	PY05
	P_6	PY06
	P_7	PY07

Table 115: PWM0 HIGH SPEED SIGNAL PIN

PWM0 Non-high speed signal name	Pin
FAULT_0	PA00 PA08 PA12 PA20 PB02 PB14 PY00
FAULT_1	PA01 PA09 PA13 PA21 PB03 PB15 PY01

Table 116: PWM0 NON-HIGH SPEED SIGNAL PIN

PWM1 High Speed Group	High speed signal name	Pin
A	P_0	PA00
	P_1	PA01
	P_2	PA02
	P_3	PA03
	P_4	PA04

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PWM1 High Speed Group	High speed signal name	Pin
	P_5	PA05
	P_6	PA06
	P_7	PA07
B	P_0	PA16
	P_1	PA17
	P_2	PA18
	P_3	PA19
	P_4	PA20
	P_5	PA21
	P_6	PA22
	P_7	PA23
C	P_0	PA24
	P_1	PA25
	P_2	PA26
	P_3	PA27
	P_4	PA28
	P_5	PA29
	P_6	PA30
	P_7	PA31
D	P_0	PB04
	P_1	PB05
	P_2	PB06
	P_3	PB07
	P_4	PB08
	P_5	PB09
	P_6	PB10
	P_7	PB11
E	P_0	PB12
	P_1	PB13
	P_2	PB14
	P_3	PB15
	P_4	PY00
	P_5	PY01
	P_6	PY02
	P_7	PY03

Table 117: PWM1 HIGH SPEED SIGNAL PIN

PWM1 Non-high speed signal name	Pin
	PA00

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PWM1 Non-high speed signal name	Pin
	PA10 PA12 PA22 PB00 PB12 PY02
FAULT_1	PA03 PA11 PA13 PA23 PB01 PB13 PY03

Table 118: PWM1 NON-HIGH SPEED SIGNAL PIN

QEI0 signal name	Pin
A	PA20 PA26 PB12
B	PA21 PA27 PB13
F	PA25 PA31 PB11
H0	PA23 PA29 PB15
H1	PA24 PA30 PB10
Z	PA22 PA28 PB14

Table 119: QEI0 SIGNAL PIN

QEI1 signal name	Pin
A	PA04 PA10

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QEI1 signal name	Pin
	PB04
B	PA05 PA11 PB05
F	PA02 PA15 PB09
H0	PA07 PA13 PB07
H1	PA03 PA14 PB08
Z	PA06 PA12 PB06

Table 120: QEI1 SIGNAL PIN

QEO0 signal name	Pin
A	PA10 PA16 PA20 PA26
B	PA11 PA17 PA21 PA27
Z	PA12 PA18 PA22 PA28

Table 121: QEO0 SIGNAL PIN

QEO1 signal name	Pin
A	PA04 PB04 PB08 PB12
B	PA05 PB05

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QEO1 signal name	Pin
	PB09 PB13
Z	PA06 PB06 PB10 PB14

Table 122: QEO1 SIGNAL PIN

RDC0 signal name	Pin
EXC_N	PA05 PA13 PA29 PB07 PB15
EXC_P	PA04 PA12 PA28 PB06 PB14

Table 123: RDC0 SIGNAL PIN

SEI0 signal name	Pin
CK	PA21 PA27 PB05 PB13
DE	PA20 PA26 PB04 PB12
RX	PA23 PA29 PB07 PB15
TX	PA22 PA28 PB06 PB14

Table 124: SEI0 SIGNAL PIN

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SEI1 signal name	Pin
CK	PA05 PA11 PA17 PB09
DE	PA04 PA10 PA16 PB08
RX	PA07 PA13 PA19 PB11
TX	PA06 PA12 PA18 PB10

Table 125: SEI1 SIGNAL PIN

SOC signal name	Pin
REF0	PA09 PA15 PA30

Table 126: SOC SIGNAL PIN

SPI0 High Speed Group	High speed signal name	Pin
A	CS_0	PA04
	MISO	PA06
	MOSI	PA07
	SCLK	PA05

Table 127: SPI0 HIGH SPEED SIGNAL PIN

SPI1 High Speed Group	High speed signal name	Pin
A	CS_0	PA26
	CS_1	PA25
	CS_2	PA24
	CS_3	PA19
	DAT2	PA30
	DAT3	PA31

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SPI1 High Speed Group	High speed signal name	Pin
	MISO	PA28
	MOSI	PA29
	SCLK	PA27
B	CS_0	PX04
	MISO	PX06
	MOSI	PX07
	SCLK	PX05

Table 128: SPI1 HIGH SPEED SIGNAL PIN

SPI2 High Speed Group	High speed signal name	Pin
A	CS_0	PB10
	CS_1	PB09
	CS_2	PB08
	CS_3	PB03
	DAT2	PB14
	DAT3	PB15
	MISO	PB12
	MOSI	PB13
	SCLK	PB11
B	CS_0	PA20
	MISO	PA22
	MOSI	PA23
	SCLK	PA21
C	CS_0	PY04
	MISO	PY06
	MOSI	PY07
	SCLK	PY05

Table 129: SPI2 HIGH SPEED SIGNAL PIN

SPI3 High Speed Group	High speed signal name	Pin
A	CS_0	PA10
	CS_1	PA09
	CS_2	PA08
	CS_3	PA03
	DAT2	PA14
	DAT3	PA15
	MISO	PA12

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SPI3 High Speed Group	High speed signal name	Pin
	MOSI	PA13
	SCLK	PA11
B	CS_0	PB04
	MISO	PB06
	MOSI	PB07
	SCLK	PB05

Table 130: SPI3 HIGH SPEED SIGNAL PIN

SYSCTL signal name	Pin
CLK_OBS_0	PA00 PA26
CLK_OBS_1	PA01 PA27
CLK_OBS_2	PA02 PA28
CLK_OBS_3	PA03 PA29

Table 131: SYSCTL SIGNAL PIN

TRGM0 signal name	Pin
P_00	PA00 PA20 PA24 PB04 PB12
P_01	PA01 PA21 PA25 PB05 PB13
P_02	PA02 PA22 PA26 PB06 PB14
P_03	PA03 PA23 PA27 PB07

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TRGM0 signal name	Pin
	PB15
P_04	PA04 PA16 PA28 PB00 PY04
P_05	PA05 PA17 PA29 PB01 PY05
P_06	PA06 PA18 PA30 PB02 PY06
P_07	PA07 PA19 PA31 PB03 PY07

Table 132: TRGM0 SIGNAL PIN

UART0 signal name	Pin
CTS	PA03 PB03 PY03
DE	PA02 PB02 PY02
RTS	PA02 PB02 PY02
RXD	PA01 PB01 PY01
TXD	PA00 PB00 PY00

Table 133: UART0 SIGNAL PIN

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UART1 signal name	Pin
CTS	PA04 PB04 PY04
DE	PA05 PB05 PY05
RTS	PA05 PB05 PY05
RXD	PA06 PB06 PY06
TXD	PA07 PB07 PY07

Table 134: UART1 SIGNAL PIN

UART2 signal name	Pin
CTS	PA11 PB11
DE	PA10 PB10
RTS	PA10 PB10
RXD	PA09 PB09
TXD	PA08 PB08

Table 135: UART2 SIGNAL PIN

UART3 signal name	Pin
CTS	PA12 PB12
DE	PA13 PB13
RTS	PA13 PB13
RXD	PA14 PB14
TXD	PA15

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UART3 signal name	Pin
	PB15

Table 136: UART3 SIGNAL PIN

UART4 signal name	Pin
CTS	PA19 PX03
DE	PA18 PX02
RTS	PA18 PX02
RXD	PA17 PX01
TXD	PA16 PX00

Table 137: UART4 SIGNAL PIN

UART5 signal name	Pin
CTS	PA20 PX04
DE	PA21 PX05
RTS	PA21 PX05
RXD	PA22 PX06
TXD	PA23 PX07

Table 138: UART5 SIGNAL PIN

UART6 signal name	Pin
CTS	PA27
DE	PA26
RTS	PA26
RXD	PA25
TXD	PA24

Table 139: UART6 SIGNAL PIN

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UART7 signal name	Pin
CTS	PA28
DE	PA29
RTS	PA29
RXD	PA30
TXD	PA31

Table 140: UART7 SIGNAL PIN

USB0 signal name	Pin
ID	PA31 PB08 PY00
OC	PA29 PB09 PY01
PWR	PA30 PB10 PY02

Table 141: USB0 SIGNAL PIN

XPI0 High Speed Group	High speed signal name	Pin
A	CA_CS0	PA31
	CA_CS1	PA24
	CA_DQS	PA25
	CA_D_0	PA28
	CA_D_1	PA30
	CA_D_2	PA29
	CA_D_3	PA26
	CA_SCLK	PA27
B	CA_CS0	PX02
	CA_CS1	PX04
	CA_DQS	PX03
	CA_D_0	PX05
	CA_D_1	PX01
	CA_D_2	PX00
	CA_D_3	PX07
	CA_SCLK	PX06
C	CA_CS0	PA31
	CA_CS1	PA24
	CA_DQS	PX03

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XPI0 High Speed Group	High speed signal name	Pin
	CA_D_0	PA28
	CA_D_1	PA30
	CA_D_2	PA29
	CA_D_3	PA26
	CA_SCLK	PA27

Table 142: XPI0 HIGH SPEED SIGNAL PIN

20.4 Peripheral Pin Assignment for Power Management Domain

The pin assignment summary of peripherals in the power management domain of this product by different modules is as follows:

PEWDG signal name	Pin
RST	PY05

Table 143: PEWDG SIGNAL PIN

PGPIO signal name	Pin
Y_00	PY00
Y_01	PY01
Y_02	PY02
Y_03	PY03
Y_04	PY04
Y_05	PY05
Y_06	PY06
Y_07	PY07

Table 144: PGPIO SIGNAL PIN

PTMR signal name	Pin
CAPT_0	PY05
CAPT_1	PY07
COMP_0	PY00 PY04
COMP_1	PY01 PY06
COMP_2	PY02
COMP_3	PY03

Table 145: PTMR SIGNAL PIN

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PUART signal name	Pin
CTS	PY03
RTS	PY02
RXD	PY01
TXD	PY00

Table 146: PUART SIGNAL PIN

SOC signal name	Pin
GPIO_Y_00	PY00
GPIO_Y_01	PY01
GPIO_Y_02	PY02
GPIO_Y_03	PY03
GPIO_Y_04	PY04
GPIO_Y_05	PY05
GPIO_Y_06	PY06
GPIO_Y_07	PY07

Table 147: SOC SIGNAL PIN

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20.5 Special Function Pins

By default, the chip selects three different boot modes via the BOOT_MODE[1:0]=[PA03:PA02] pins. The boot configuration is shown in Table 148, and other special pin configurations are shown in Table 149.

Boot Mode Selection Pins		Boot Mode	Description
BOOT_MODE1	BOOT_MODE0		
0	0	XPI NOR boot	boot from serial NOR FLASH connected to XPI0
0	1	In-System Programming (ISP)/Serial Boot	Burn firmware from UART0/USB0, OTP, or boot from UART0/USB0
1	0	In-System Programming (ISP)/Serial Boot	Burn firmware from UART0/USB0, OTP, or boot from UART0/USB0
1	1	Retention Mode	Retention Mode

Table 148: Startup configuration table

Pin Name	Description	Recommended Usage
XTALI	24MHz clock input	Connect to 24MHz crystal or active clock
XTALO	24MHz clock output	Connect to 24MHz crystal or leave floating

Table 149: Special Function Pin Configuration

_auto_part=0 Note: For LQFP64 and QFN48 packages of this product:

- USB_DP is multiplexed with PA24. When used for USB function, PA24 shall be configured as analog function (set the ANALOG bit of PAD[FUNC_CTL] to 1)
- USB_DM is multiplexed with PA25. When used for USB function, PA25 shall be configured as analog function (set the ANALOG bit of PAD[FUNC_CTL] to 1)
- When PA24 and PA25 are not used for USB, configure the USB PHY_CTRL0 register (PHY_CTRL0 |= 0x001000E0u) to disable the pull-down resistors of DP and DM.
- In In-System Programming (ISP) mode, only burning firmware from UART0 or booting from UART0 is supported

20.6 IO Reset Status

Table 150 summarizes the status of all IOs of this product after system reset:

Name	Status after Reset	Function after Reset
PA04	High-impedance	JTAG.TDO

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Name	Status after Reset	Function after Reset
PA05	Input with internal pull-up	JTAG.TDI
PA06	Input with internal pull-down	JTAG.TCK
PA07	Input with internal pull-up	JTAG.TMS
PA08	Input with internal pull-up	JTAG.TRST
PY00	Output high level	PUART.TXD
PY01	Input with internal pull-up	PUART.RXD
PY02	Output high level	PUART.RTS
PY03	Input with internal pull-up	PUART.CTS
PY05	Output high level	PWDG.RST
Other IOs	Input state holder	GPIO

Table 150: IO Reset Status Table

20.7 IO Power Supply Pins

For this product, the power supply pin for IOs of GPIO Port A (PA00~PA31) is VIO_B00.

The power supply pin for IOs of GPIO Port B (PB00~PB15) is VIO_B01.

The power supply pin for IOs of GPIO Port Y (PY00~PY07) is VPMC.

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21 Overview of Input/Output Module

This chapter describes the input/output (IO) related modules of this product. The IO related control modules of this product include general-purpose IO control module (IOC), power management domain IO control module (PIOC), GPIO controller, fast GPIO control, GPIO manager and power management domain GPIO controller (PGPIO).

21.1 IO Controller

The IO controller module includes general-purpose IO controller (IOC) and power management domain IO controller (PIOC).

The general-purpose IO controller (IOC) can control general-purpose IOs (PA, PB, PX).

The power management domain IO controller (PIOC) can control power management domain IOs (PY). It has the same functions as the general-purpose IOC, and can configure the basic attributes and peripheral functions of power management domain IOs.

The PIOC can map one or more IOs in the power management domain IOs (PY) to the system power domain. After that, these IOs can be controlled by the IOC.

The IO controller supports configuration of any IO, such as open-drain control, internal pull-up/pull-down control, Schmitt trigger, slew rate, driving capability, etc. It can also configure the peripheral multiplexing function mapping, analog input and IO status monitoring function for each IO. The peripheral multiplexing function of general-purpose IOs is configured by the IOC, as shown in Figure 20.

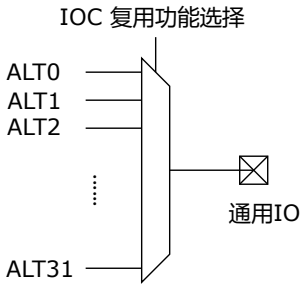


Figure 20: General-purpose IO Peripheral Multiplexing Function Selection

The peripheral multiplexing function of power management domain IOs (PY) is configured by the PIOC and IOC, as shown in Figure 21.

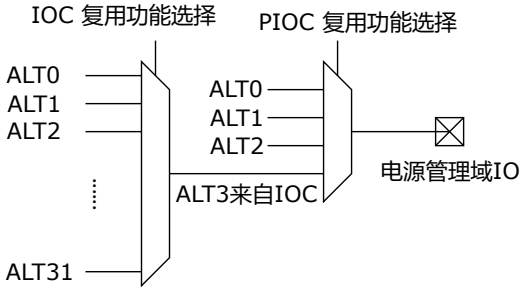


Figure 21: Power Management Domain IO Peripheral Multiplexing Function Selection

When the peripheral multiplexing function of the power management domain IO (PY) is set to ALT3 by the

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PIOC, the IOC configuration for this IO takes effect, while the PIOC configuration no longer takes effect.

21.2 GPIO Controller

GPIO controllers include: GPIO controller (GPIO0), fast GPIO controller (FGPIO0), and power management domain GPIO controller (PGPIO).

GPIO0 and FGPIO0 can control general-purpose IOs (PA, PB, PX). For any IO, the GPIO manager (GPIOM) configuration determines which controller takes effect.

The power management domain GPIO controller (PGPIO) can control power management domain IOs (PY).

The PIOC can map one or more IOs in the power management domain IOs (PY) to the system power domain. After that, these IOs can be controlled by GPIO0 or FGPIO0.

The GPIO controller GPIO0 and fast GPIO controller FGPIO0 can control on-chip general-purpose IOs (PA, PB, PX).

The GPIO control of general-purpose IOs is shown in Figure 22.

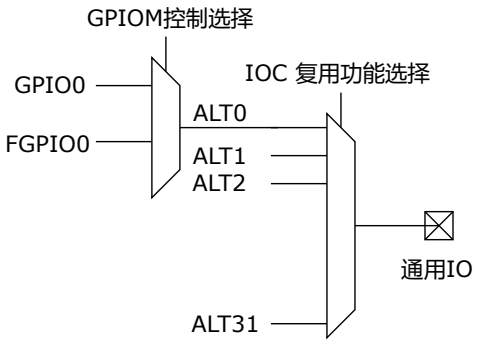


Figure 22: General-purpose IO GPIO Control Selection

With proper PIOC and IOC configuration, 2 GPIO controllers (GPIO0, GPIO1) and 2 fast GPIO controllers (FGPIO0, FGPIO1) can control power management domain IOs (PY).

The GPIO control of power management domain IOs is shown in Figure 23.

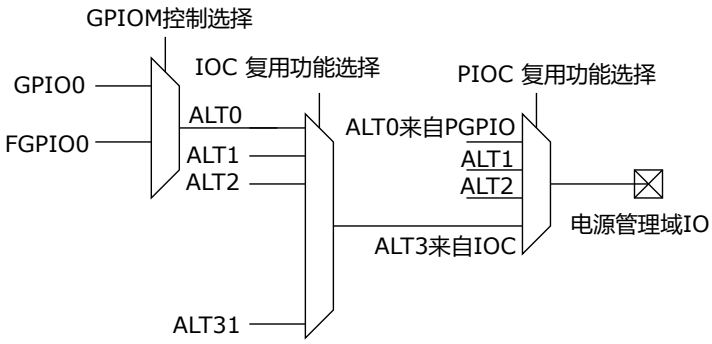


Figure 23: Power Management Domain IO GPIO Control Selection

The fast GPIO controller (FGPIO0) is a private peripheral of processor CPU0, and can only be accessed by the RISC-V processor. The processor supports zero-wait-cycle access to its own fast GPIO controller.

The GPIO controller and the fast GPIO controller have basically the same functions, they can read inputs by IO port, configure IO direction (input or output), set IO output, or simultaneously set one or more IO outputs

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to high, low or toggle.

The GPIO controller supports configuring and generating GPIO interrupts. The fast GPIO controller does not support interrupts.

The power management domain GPIO controller (PGPIO) is the GPIO controller for power management domain IOs (PY), it has the same functions of reading IO inputs and configuring IO outputs as the GPIO controller. In addition, as a power management domain peripheral, it can keep working when the system power domain is powered off, and the PGPIO interrupt can wake up the system when the system power domain is powered off.

21.3 GPIO Manager (GPIOM)

The GPIO manager is used to manage all IOs. Its main function is to configure the management authority of IOs. The GPIO manager selects either the fast GPIO controller or the GPIO controller for each IO as the controller of this IO.

The GPIO manager supports configuring the visibility of any IO input, that is, whether the input registers of the fast GPIO controller and the GPIO controller can read the IO input.

After reset, the GPIO manager defaults to GPIO0 controlling all general-purpose IOs. All GPIO controllers and fast GPIO controllers can read the IO input.

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22 IO Controllers IOC, PIOC and BIOC

This chapter describes the main features and functions of the IO controller IOC.

22.1 Feature Summary

The main features of the IO controller IOC are as follows:

- Peripheral multiplexing function mapping
- Output loopback control
- Analog function configuration
- Slew rate configuration
- Open-drain setting
- Schmitt trigger
- Pull-up/pull-down configuration
- Driving capability configuration

22.2 Functional Description

This chapter describes the functions of the IO controller IOC.

22.2.1 Basic IO Configuration

The IO controller IOC can be used to configure the basic attributes of IOs, including open-drain selection, Schmitt trigger on/off, slew rate, internal pull-up/pull-down resistors, and driving strength. Users can configure the basic attributes of any IO through the IOC_X_PAD_CTL register.

The IOs of this product support two operating voltages: 3.3 V and 1.8 V, among which IOs in the battery domain only operate at 3.3 V.

The IO open-drain selection means that users can configure the IO as open-drain output. If the IO is configured as open-drain output, it outputs normally when outputting low level; when outputting high level, the IO will not drive high level but output high impedance, and an external pull-up resistor needs to be configured by the user.

The IO Schmitt trigger means that users can enable the input Schmitt trigger of the IO, that is, enable input hysteresis, to improve anti-interference capability.

The IO can also enable internal pull-up/pull-down resistors, that is, it can be configured as internal pull-up or pull-down.

The IO supports slew rate configuration.

The IO supports driving capability configuration.

22.2.2 IO Peripheral Function Configuration

Users can configure the peripheral functions of the IO through the IOC_X_FUNC_CTL register of IOC, including output loopback function, analog function and peripheral function mapping.

After enabling the output loopback function of the IO, users can read the output signal at the input end.

After enabling the analog function of the IO, this IO can be used as a pin for analog peripherals, such as

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ADC, ACMP, etc. It should be noted that when the pin is selected for analog function through the configuration of the ANALOG bit field of the IOC_X_FUNC_CTL register, the digital input and output functions of the pin will be automatically masked, including disabling the pin's digital input and output enable (OE), pull-up/pull-down enable (PE), and hold capability enable (KE), to reduce the impact on the input and output of the analog module. Of course, when it is necessary to use the digital output as the analog input on the same pin (for example, measuring the GPIO output voltage value of the pin through the ADC module), or use the analog output as the digital input (for example, GPIO reads the approximate output of the DAC module, i.e. high/low level), the ANALOG bit field of the IOC_X_FUNC_CTL register does not need to be enabled, but the digital function of the pin needs to be configured in advance to avoid simultaneous output from digital and analog modules.

Users can select the peripheral function mapped to this IO through the IO's peripheral function multiplexer. For the mapping table of IOs and peripheral functions, please refer to the relevant chapters of PINMUX.

22.3 IOC Registers

22.3.1 Register Description

The register list of IOC is as follows:

IOC base address: 0xF4040000

PIOC base address: 0xF4118000

Offset	Register Name	Description	Reset Value
0x0000	PAD[PA00][FUNC_CTL]	ALT SELECT	0x00000000
0x0004	PAD[PA00][PAD_CTL]	PAD SETTINGS	0x01010056
0x0008	PAD[PA01][FUNC_CTL]	ALT SELECT	0x00000000
0x000C	PAD[PA01][PAD_CTL]	PAD SETTINGS	0x01010056
0x0010	PAD[PA02][FUNC_CTL]	ALT SELECT	0x00000000
0x0014	PAD[PA02][PAD_CTL]	PAD SETTINGS	0x01010056
0x0018	PAD[PA03][FUNC_CTL]	ALT SELECT	0x00000000
0x001C	PAD[PA03][PAD_CTL]	PAD SETTINGS	0x01010056
0x0020	PAD[PA04][FUNC_CTL]	ALT SELECT	0x00000000
0x0024	PAD[PA04][PAD_CTL]	PAD SETTINGS	0x01010056
0x0028	PAD[PA05][FUNC_CTL]	ALT SELECT	0x00000000
0x002C	PAD[PA05][PAD_CTL]	PAD SETTINGS	0x01010056
0x0030	PAD[PA06][FUNC_CTL]	ALT SELECT	0x00000000
0x0034	PAD[PA06][PAD_CTL]	PAD SETTINGS	0x01010056
0x0038	PAD[PA07][FUNC_CTL]	ALT SELECT	0x00000000
0x003C	PAD[PA07][PAD_CTL]	PAD SETTINGS	0x01010056
0x0040	PAD[PA08][FUNC_CTL]	ALT SELECT	0x00000000
0x0044	PAD[PA08][PAD_CTL]	PAD SETTINGS	0x01010056
0x0048	PAD[PA09][FUNC_CTL]	ALT SELECT	0x00000000
0x004C	PAD[PA09][PAD_CTL]	PAD SETTINGS	0x01010056
0x0050	PAD[PA10][FUNC_CTL]	ALT SELECT	0x00000000
0x0054	PAD[PA10][PAD_CTL]	PAD SETTINGS	0x01010056

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Offset	Register Name	Description	Reset Value
0x0058	PAD[PA11][FUNC_CTL]	ALT SELECT	0x00000000
0x005C	PAD[PA11][PAD_CTL]	PAD SETTINGS	0x01010056
0x0060	PAD[PA12][FUNC_CTL]	ALT SELECT	0x00000000
0x0064	PAD[PA12][PAD_CTL]	PAD SETTINGS	0x01010056
0x0068	PAD[PA13][FUNC_CTL]	ALT SELECT	0x00000000
0x006C	PAD[PA13][PAD_CTL]	PAD SETTINGS	0x01010056
0x0070	PAD[PA14][FUNC_CTL]	ALT SELECT	0x00000000
0x0074	PAD[PA14][PAD_CTL]	PAD SETTINGS	0x01010056
0x0078	PAD[PA15][FUNC_CTL]	ALT SELECT	0x00000000
0x007C	PAD[PA15][PAD_CTL]	PAD SETTINGS	0x01010056
0x0080	PAD[PA16][FUNC_CTL]	ALT SELECT	0x00000000
0x0084	PAD[PA16][PAD_CTL]	PAD SETTINGS	0x01010056
0x0088	PAD[PA17][FUNC_CTL]	ALT SELECT	0x00000000
0x008C	PAD[PA17][PAD_CTL]	PAD SETTINGS	0x01010056
0x0090	PAD[PA18][FUNC_CTL]	ALT SELECT	0x00000000
0x0094	PAD[PA18][PAD_CTL]	PAD SETTINGS	0x01010056
0x0098	PAD[PA19][FUNC_CTL]	ALT SELECT	0x00000000
0x009C	PAD[PA19][PAD_CTL]	PAD SETTINGS	0x01010056
0x00A0	PAD[PA20][FUNC_CTL]	ALT SELECT	0x00000000
0x00A4	PAD[PA20][PAD_CTL]	PAD SETTINGS	0x01010056
0x00A8	PAD[PA21][FUNC_CTL]	ALT SELECT	0x00000000
0x00AC	PAD[PA21][PAD_CTL]	PAD SETTINGS	0x01010056
0x00B0	PAD[PA22][FUNC_CTL]	ALT SELECT	0x00000000
0x00B4	PAD[PA22][PAD_CTL]	PAD SETTINGS	0x01010056
0x00B8	PAD[PA23][FUNC_CTL]	ALT SELECT	0x00000000
0x00BC	PAD[PA23][PAD_CTL]	PAD SETTINGS	0x01010056
0x00C0	PAD[PA24][FUNC_CTL]	ALT SELECT	0x00000000
0x00C4	PAD[PA24][PAD_CTL]	PAD SETTINGS	0x01010056
0x00C8	PAD[PA25][FUNC_CTL]	ALT SELECT	0x00000000
0x00CC	PAD[PA25][PAD_CTL]	PAD SETTINGS	0x01010056
0x00D0	PAD[PA26][FUNC_CTL]	ALT SELECT	0x00000000
0x00D4	PAD[PA26][PAD_CTL]	PAD SETTINGS	0x01010056
0x00D8	PAD[PA27][FUNC_CTL]	ALT SELECT	0x00000000
0x00DC	PAD[PA27][PAD_CTL]	PAD SETTINGS	0x01010056
0x00E0	PAD[PA28][FUNC_CTL]	ALT SELECT	0x00000000
0x00E4	PAD[PA28][PAD_CTL]	PAD SETTINGS	0x01010056
0x00E8	PAD[PA29][FUNC_CTL]	ALT SELECT	0x00000000
0x00EC	PAD[PA29][PAD_CTL]	PAD SETTINGS	0x01010056
0x00F0	PAD[PA30][FUNC_CTL]	ALT SELECT	0x00000000
0x00F4	PAD[PA30][PAD_CTL]	PAD SETTINGS	0x01010056

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Offset	Register Name	Description	Reset Value
0x00F8	PAD[PA31][FUNC_CTL]	ALT SELECT	0x00000000
0x00FC	PAD[PA31][PAD_CTL]	PAD SETTINGS	0x01010056
0x0100	PAD[PB00][FUNC_CTL]	ALT SELECT	0x00000000
0x0104	PAD[PB00][PAD_CTL]	PAD SETTINGS	0x01010056
0x0108	PAD[PB01][FUNC_CTL]	ALT SELECT	0x00000000
0x010C	PAD[PB01][PAD_CTL]	PAD SETTINGS	0x01010056
0x0110	PAD[PB02][FUNC_CTL]	ALT SELECT	0x00000000
0x0114	PAD[PB02][PAD_CTL]	PAD SETTINGS	0x01010056
0x0118	PAD[PB03][FUNC_CTL]	ALT SELECT	0x00000000
0x011C	PAD[PB03][PAD_CTL]	PAD SETTINGS	0x01010056
0x0120	PAD[PB04][FUNC_CTL]	ALT SELECT	0x00000000
0x0124	PAD[PB04][PAD_CTL]	PAD SETTINGS	0x01010056
0x0128	PAD[PB05][FUNC_CTL]	ALT SELECT	0x00000000
0x012C	PAD[PB05][PAD_CTL]	PAD SETTINGS	0x01010056
0x0130	PAD[PB06][FUNC_CTL]	ALT SELECT	0x00000000
0x0134	PAD[PB06][PAD_CTL]	PAD SETTINGS	0x01010056
0x0138	PAD[PB07][FUNC_CTL]	ALT SELECT	0x00000000
0x013C	PAD[PB07][PAD_CTL]	PAD SETTINGS	0x01010056
0x0140	PAD[PB08][FUNC_CTL]	ALT SELECT	0x00000000
0x0144	PAD[PB08][PAD_CTL]	PAD SETTINGS	0x01010056
0x0148	PAD[PB09][FUNC_CTL]	ALT SELECT	0x00000000
0x014C	PAD[PB09][PAD_CTL]	PAD SETTINGS	0x01010056
0x0150	PAD[PB10][FUNC_CTL]	ALT SELECT	0x00000000
0x0154	PAD[PB10][PAD_CTL]	PAD SETTINGS	0x01010056
0x0158	PAD[PB11][FUNC_CTL]	ALT SELECT	0x00000000
0x015C	PAD[PB11][PAD_CTL]	PAD SETTINGS	0x01010056
0x0160	PAD[PB12][FUNC_CTL]	ALT SELECT	0x00000000
0x0164	PAD[PB12][PAD_CTL]	PAD SETTINGS	0x01010056
0x0168	PAD[PB13][FUNC_CTL]	ALT SELECT	0x00000000
0x016C	PAD[PB13][PAD_CTL]	PAD SETTINGS	0x01010056
0x0170	PAD[PB14][FUNC_CTL]	ALT SELECT	0x00000000
0x0174	PAD[PB14][PAD_CTL]	PAD SETTINGS	0x01010056
0x0178	PAD[PB15][FUNC_CTL]	ALT SELECT	0x00000000
0x017C	PAD[PB15][PAD_CTL]	PAD SETTINGS	0x01010056
0x0D00	PAD[PX00][FUNC_CTL]	ALT SELECT	0x00000000
0x0D04	PAD[PX00][PAD_CTL]	PAD SETTINGS	0x01010056
0x0D08	PAD[PX01][FUNC_CTL]	ALT SELECT	0x00000000
0x0D0C	PAD[PX01][PAD_CTL]	PAD SETTINGS	0x01010056
0x0D10	PAD[PX02][FUNC_CTL]	ALT SELECT	0x00000000
0x0D14	PAD[PX02][PAD_CTL]	PAD SETTINGS	0x01010056

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Offset	Register Name	Description	Reset Value
0x0D18	PAD[PX03][FUNC_CTL]	ALT SELECT	0x00000000
0x0D1C	PAD[PX03][PAD_CTL]	PAD SETTINGS	0x01010056
0x0D20	PAD[PX04][FUNC_CTL]	ALT SELECT	0x00000000
0x0D24	PAD[PX04][PAD_CTL]	PAD SETTINGS	0x01010056
0x0D28	PAD[PX05][FUNC_CTL]	ALT SELECT	0x00000000
0x0D2C	PAD[PX05][PAD_CTL]	PAD SETTINGS	0x01010056
0x0D30	PAD[PX06][FUNC_CTL]	ALT SELECT	0x00000000
0x0D34	PAD[PX06][PAD_CTL]	PAD SETTINGS	0x01010056
0x0D38	PAD[PX07][FUNC_CTL]	ALT SELECT	0x00000000
0x0D3C	PAD[PX07][PAD_CTL]	PAD SETTINGS	0x01010056
0x0E00	PAD[PY00][FUNC_CTL]	ALT SELECT	0x00000000
0x0E04	PAD[PY00][PAD_CTL]	PAD SETTINGS	0x01010056
0x0E08	PAD[PY01][FUNC_CTL]	ALT SELECT	0x00000000
0x0E0C	PAD[PY01][PAD_CTL]	PAD SETTINGS	0x01010056
0x0E10	PAD[PY02][FUNC_CTL]	ALT SELECT	0x00000000
0x0E14	PAD[PY02][PAD_CTL]	PAD SETTINGS	0x01010056
0x0E18	PAD[PY03][FUNC_CTL]	ALT SELECT	0x00000000
0x0E1C	PAD[PY03][PAD_CTL]	PAD SETTINGS	0x01010056
0x0E20	PAD[PY04][FUNC_CTL]	ALT SELECT	0x00000000
0x0E24	PAD[PY04][PAD_CTL]	PAD SETTINGS	0x01010056
0x0E28	PAD[PY05][FUNC_CTL]	ALT SELECT	0x00000000
0x0E2C	PAD[PY05][PAD_CTL]	PAD SETTINGS	0x01010056
0x0E30	PAD[PY06][FUNC_CTL]	ALT SELECT	0x00000000
0x0E34	PAD[PY06][PAD_CTL]	PAD SETTINGS	0x01010056
0x0E38	PAD[PY07][FUNC_CTL]	ALT SELECT	0x00000000
0x0E3C	PAD[PY07][PAD_CTL]	PAD SETTINGS	0x01010056

Table 151: IOC Register List

The detailed description of IOC registers is as follows:

22.3.2 PAD[FUNC_CTL] (0x0 + 0x8 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD															LOOP_BACK	RSVD							ANALOG	RSVD			ALT_SELECT				
N/A															RW	N/A							RW	N/A			RW				
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	x	x	x	x	0	x	x	x	0	0	0	0	0

PAD[FUNC_CTL] [31:0]

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Bits	Name	Description
16	LOOP_BACK	force input on 0: disable 1: enable
8	ANALOG	select analog pin in pad 0: disable 1: enable
4-0	ALT_SELECT	alt select 0: ALT0 1: ALT1 ... 31:ALT31

PAD[FUNC_CTL] Bits

22.3.3 PAD[PAD_CTL] (0x4 + 0x8 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD							HYS	RSVD		PRS		RSVD	PS	PE	KE	RSVD							OD	RSVD	SR	SPD		RSVD	DS		
N/A							RW	N/A		RW		N/A	RW	RW	RW	N/A							RW	N/A	RW	RW		N/A	RW		
x	x	x	x	x	x	x	1	x	x	0	0	x	0	0	1	x	x	x	x	x	x	x	0	x	1	0	1	x	1	1	0

PAD[PAD_CTL] [31:0]

Bits	Name	Description
24	HYS	schmitt trigger enable 0: disable 1: enable
21-20	PRS	select pull up/down internal resistance strength: For pull down, only have 100 Kohm resistance For pull up: 00: 100 KOhm 01: 47 KOhm 10: 22 KOhm 11: 22 KOhm
18	PS	pull select 0: pull down 1: pull up
17	PE	pull enable 0: pull disable 1: pull enable

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Bits	Name	Description
16	KE	keeper capability enable 0: keeper disable 1: keeper enable
8	OD	open drain 0: open drain disable 1: open drain enable
6	SR	slew rate 0: Slow slew rate 1: Fast slew rate
5-4	SPD	additional 2-bit slew rate to select IO cell operation frequency range with reduced switching noise 00: Slow frequency slew rate(50Mhz) 01: Medium frequency slew rate(100 Mhz) 10: Fast frequency slew rate(150 Mhz) 11: Max frequency slew rate(200Mhz)
2-0	DS	drive strength 1.8V Mode: 000: 260 Ohm 001: 260 Ohm 010: 130 Ohm 011: 88 Ohm 100: 65 Ohm 101: 52 Ohm 110: 43 Ohm 111: 37 Ohm 3.3V Mode: 000: 157 Ohm 001: 157 Ohm 010: 78 Ohm 011: 53 Ohm 100: 39 Ohm 101: 32 Ohm 110: 26 Ohm 111: 23 Ohm

PAD[PAD_CTL] Bits

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23 GPIO Controller

This section describes the main features and functions of the GPIO controller. GPIO controller types include:

- GPIO Controller: GPIO0
- Fast GPIO Controller: FGPIO0 (CPU0 fast access)
- Power Management Domain GPIO Controller: PGPIO

23.1 Feature Summary

This section describes the key features of the GPIO controller:

- Configure IOs as input or output
- Read IO input status
- Set IO output
- Atomic operations to set IO output high, output low, and toggle

23.2 Functional Description

23.2.1 GPIO Control

The GPIO controller supports input/output direction control. In the OE register, each IO has a corresponding DIRECTION control bit. The user can set this bit to 1 to configure the corresponding IO as GPIO output, or clear it to 0 to configure it as GPIO input.

In output mode, the user can configure the GPIO output state through the DO register of the GPIO controller.

GPIO output supports atomic operation registers:

- Output high register SET: setting the corresponding bit to 1 in this register drives the corresponding IO output high; setting it to 0 leaves the IO output unchanged.
- Output low register CLEAR: setting the corresponding bit to 1 in this register drives the corresponding IO output low; setting it to 0 leaves the IO output unchanged.
- Toggle register TOGGLE: setting the corresponding bit to 1 in this register toggles the corresponding IO output; setting it to 0 leaves the IO output unchanged.

The user can read the IO level status through the DI register of the GPIO controller.

Note: The DI register enables IO monitoring. That is, regardless of whether the GPIO is configured as input or output, or whether the corresponding IO controller IOC maps the IO function to GPIO (in which case the output loopback must be enabled via IOC), the user can always read the IO status from DI.

23.2.2 GPIO Interrupt

In the GPIO controller, the following controllers support generating interrupts: GPIO0 , PGPIO

In the GPIO controller, the following controllers do not support generating interrupts: FGPIO0

The user can specify the interrupt type through the TP register of the GPIO controller. Setting the corresponding bit to 1 means the interrupt is edge-triggered; clearing the corresponding bit to 0 means the interrupt is level-triggered.

The user can specify the interrupt polarity through the PL register of the GPIO controller. Setting the corresponding bit to 1 means the interrupt is triggered by a falling edge or low level; clearing the corresponding bit to

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0 means the interrupt is triggered by a rising edge or high level.

GPIO supports generating asynchronous interrupts. The user can select synchronous or asynchronous interrupts through the AS register. Asynchronous interrupts allow interrupt generation when the system clock is abnormal or absent.

The PD register can be used to select whether to enable dual-edge triggered interrupts. Setting the corresponding bit to 1 enables both rising and falling edge interrupts; clearing the corresponding bit to 0 allows only single-edge triggered interrupts, with the specific edge determined by the PL register configuration.

After the interrupt options are configured, the user can enable GPIO interrupts through the IE register of the GPIO controller. Setting the corresponding bit to 1 in the IE register enables the interrupt for the corresponding IO. When an interrupt event is triggered, the user can query the interrupt status through the IF register of the GPIO controller. Setting the corresponding flag bit to 1 indicates that the corresponding IO has a pending interrupt. Writing 1 to the flag bit can clear this flag bit.

23.3 GPIO Register List

The GPIO register list is as follows:

FGPIO base address: 0x000C0000

GPIO0 base address: 0xF00D0000

PGPIO base address: 0xF411C000

Offset	Register Name	Description	Reset Value
0x0000	DI[GPIOA][VALUE]	GPIO input value	0x00000000
0x0010	DI[GPIOB][VALUE]	GPIO input value	0x00000000
0x00D0	DI[GPIOX][VALUE]	GPIO input value	0x00000000
0x00E0	DI[GPIOY][VALUE]	GPIO input value	0x00000000
0x0100	DO[GPIOA][VALUE]	GPIO output value	0x00000000
0x0104	DO[GPIOA][SET]	GPIO output set	0x00000000
0x0108	DO[GPIOA][CLEAR]	GPIO output clear	0x00000000
0x010C	DO[GPIOA][TOGGLE]	GPIO output toggle	0x00000000
0x0110	DO[GPIOB][VALUE]	GPIO output value	0x00000000
0x0114	DO[GPIOB][SET]	GPIO output set	0x00000000
0x0118	DO[GPIOB][CLEAR]	GPIO output clear	0x00000000
0x011C	DO[GPIOB][TOGGLE]	GPIO output toggle	0x00000000
0x01D0	DO[GPIOX][VALUE]	GPIO output value	0x00000000
0x01D4	DO[GPIOX][SET]	GPIO output set	0x00000000
0x01D8	DO[GPIOX][CLEAR]	GPIO output clear	0x00000000
0x01DC	DO[GPIOX][TOGGLE]	GPIO output toggle	0x00000000
0x01E0	DO[GPIOY][VALUE]	GPIO output value	0x00000000
0x01E4	DO[GPIOY][SET]	GPIO output set	0x00000000
0x01E8	DO[GPIOY][CLEAR]	GPIO output clear	0x00000000
0x01EC	DO[GPIOY][TOGGLE]	GPIO output toggle	0x00000000

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Offset	Register Name	Description	Reset Value
0x0200	OE[GPIOA][VALUE]	GPIO direction value	0x00000000
0x0204	OE[GPIOA][SET]	GPIO direction set	0x00000000
0x0208	OE[GPIOA][CLEAR]	GPIO direction clear	0x00000000
0x020C	OE[GPIOA][TOGGLE]	GPIO direction toggle	0x00000000
0x0210	OE[GPIOB][VALUE]	GPIO direction value	0x00000000
0x0214	OE[GPIOB][SET]	GPIO direction set	0x00000000
0x0218	OE[GPIOB][CLEAR]	GPIO direction clear	0x00000000
0x021C	OE[GPIOB][TOGGLE]	GPIO direction toggle	0x00000000
0x02D0	OE[GPIOX][VALUE]	GPIO direction value	0x00000000
0x02D4	OE[GPIOX][SET]	GPIO direction set	0x00000000
0x02D8	OE[GPIOX][CLEAR]	GPIO direction clear	0x00000000
0x02DC	OE[GPIOX][TOGGLE]	GPIO direction toggle	0x00000000
0x02E0	OE[GPIOY][VALUE]	GPIO direction value	0x00000000
0x02E4	OE[GPIOY][SET]	GPIO direction set	0x00000000
0x02E8	OE[GPIOY][CLEAR]	GPIO direction clear	0x00000000
0x02EC	OE[GPIOY][TOGGLE]	GPIO direction toggle	0x00000000
0x0300	IF[GPIOA][VALUE]	GPIO interrupt flag value	0x00000000
0x0310	IF[GPIOB][VALUE]	GPIO interrupt flag value	0x00000000
0x03D0	IF[GPIOX][VALUE]	GPIO interrupt flag value	0x00000000
0x03E0	IF[GPIOY][VALUE]	GPIO interrupt flag value	0x00000000
0x0400	IE[GPIOA][VALUE]	GPIO interrupt enable value	0x00000000
0x0404	IE[GPIOA][SET]	GPIO interrupt enable set	0x00000000
0x0408	IE[GPIOA][CLEAR]	GPIO interrupt enable clear	0x00000000
0x040C	IE[GPIOA][TOGGLE]	GPIO interrupt enable toggle	0x00000000
0x0410	IE[GPIOB][VALUE]	GPIO interrupt enable value	0x00000000
0x0414	IE[GPIOB][SET]	GPIO interrupt enable set	0x00000000
0x0418	IE[GPIOB][CLEAR]	GPIO interrupt enable clear	0x00000000
0x041C	IE[GPIOB][TOGGLE]	GPIO interrupt enable toggle	0x00000000
0x04D0	IE[GPIOX][VALUE]	GPIO interrupt enable value	0x00000000
0x04D4	IE[GPIOX][SET]	GPIO interrupt enable set	0x00000000
0x04D8	IE[GPIOX][CLEAR]	GPIO interrupt enable clear	0x00000000
0x04DC	IE[GPIOX][TOGGLE]	GPIO interrupt enable toggle	0x00000000
0x04E0	IE[GPIOY][VALUE]	GPIO interrupt enable value	0x00000000
0x04E4	IE[GPIOY][SET]	GPIO interrupt enable set	0x00000000
0x04E8	IE[GPIOY][CLEAR]	GPIO interrupt enable clear	0x00000000
0x04EC	IE[GPIOY][TOGGLE]	GPIO interrupt enable toggle	0x00000000
0x0500	PL[GPIOA][VALUE]	GPIO interrupt polarity value	0x00000000
0x0504	PL[GPIOA][SET]	GPIO interrupt polarity set	0x00000000
0x0508	PL[GPIOA][CLEAR]	GPIO interrupt polarity clear	0x00000000
0x050C	PL[GPIOA][TOGGLE]	GPIO interrupt polarity toggle	0x00000000

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Offset	Register Name	Description	Reset Value
0x0510	PL[GPIOB][VALUE]	GPIO interrupt polarity value	0x00000000
0x0514	PL[GPIOB][SET]	GPIO interrupt polarity set	0x00000000
0x0518	PL[GPIOB][CLEAR]	GPIO interrupt polarity clear	0x00000000
0x051C	PL[GPIOB][TOGGLE]	GPIO interrupt polarity toggle	0x00000000
0x05D0	PL[GPIOX][VALUE]	GPIO interrupt polarity value	0x00000000
0x05D4	PL[GPIOX][SET]	GPIO interrupt polarity set	0x00000000
0x05D8	PL[GPIOX][CLEAR]	GPIO interrupt polarity clear	0x00000000
0x05DC	PL[GPIOX][TOGGLE]	GPIO interrupt polarity toggle	0x00000000
0x05E0	PL[GPIOY][VALUE]	GPIO interrupt polarity value	0x00000000
0x05E4	PL[GPIOY][SET]	GPIO interrupt polarity set	0x00000000
0x05E8	PL[GPIOY][CLEAR]	GPIO interrupt polarity clear	0x00000000
0x05EC	PL[GPIOY][TOGGLE]	GPIO interrupt polarity toggle	0x00000000
0x0600	TP[GPIOA][VALUE]	GPIO interrupt type value	0x00000000
0x0604	TP[GPIOA][SET]	GPIO interrupt type set	0x00000000
0x0608	TP[GPIOA][CLEAR]	GPIO interrupt type clear	0x00000000
0x060C	TP[GPIOA][TOGGLE]	GPIO interrupt type toggle	0x00000000
0x0610	TP[GPIOB][VALUE]	GPIO interrupt type value	0x00000000
0x0614	TP[GPIOB][SET]	GPIO interrupt type set	0x00000000
0x0618	TP[GPIOB][CLEAR]	GPIO interrupt type clear	0x00000000
0x061C	TP[GPIOB][TOGGLE]	GPIO interrupt type toggle	0x00000000
0x06D0	TP[GPIOX][VALUE]	GPIO interrupt type value	0x00000000
0x06D4	TP[GPIOX][SET]	GPIO interrupt type set	0x00000000
0x06D8	TP[GPIOX][CLEAR]	GPIO interrupt type clear	0x00000000
0x06DC	TP[GPIOX][TOGGLE]	GPIO interrupt type toggle	0x00000000
0x06E0	TP[GPIOY][VALUE]	GPIO interrupt type value	0x00000000
0x06E4	TP[GPIOY][SET]	GPIO interrupt type set	0x00000000
0x06E8	TP[GPIOY][CLEAR]	GPIO interrupt type clear	0x00000000
0x06EC	TP[GPIOY][TOGGLE]	GPIO interrupt type toggle	0x00000000
0x0700	AS[GPIOA][VALUE]	GPIO interrupt asynchronous value	0x00000000
0x0704	AS[GPIOA][SET]	GPIO interrupt asynchronous set	0x00000000
0x0708	AS[GPIOA][CLEAR]	GPIO interrupt asynchronous clear	0x00000000
0x070C	AS[GPIOA][TOGGLE]	GPIO interrupt asynchronous toggle	0x00000000
0x0710	AS[GPIOB][VALUE]	GPIO interrupt asynchronous value	0x00000000
0x0714	AS[GPIOB][SET]	GPIO interrupt asynchronous set	0x00000000

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Offset	Register Name	Description	Reset Value
0x0718	AS[GPIOB][CLEAR]	GPIO interrupt asynchronous clear	0x00000000
0x071C	AS[GPIOB][TOGGLE]	GPIO interrupt asynchronous toggle	0x00000000
0x07D0	AS[GPIOX][VALUE]	GPIO interrupt asynchronous value	0x00000000
0x07D4	AS[GPIOX][SET]	GPIO interrupt asynchronous set	0x00000000
0x07D8	AS[GPIOX][CLEAR]	GPIO interrupt asynchronous clear	0x00000000
0x07DC	AS[GPIOX][TOGGLE]	GPIO interrupt asynchronous toggle	0x00000000
0x07E0	AS[GPIOY][VALUE]	GPIO interrupt asynchronous value	0x00000000
0x07E4	AS[GPIOY][SET]	GPIO interrupt asynchronous set	0x00000000
0x07E8	AS[GPIOY][CLEAR]	GPIO interrupt asynchronous clear	0x00000000
0x07EC	AS[GPIOY][TOGGLE]	GPIO interrupt asynchronous toggle	0x00000000
0x0800	PD[GPIOA][VALUE]	GPIO dual edge interrupt enable value	0x00000000
0x0804	PD[GPIOA][SET]	GPIO dual edge interrupt enable set	0x00000000
0x0808	PD[GPIOA][CLEAR]	GPIO dual edge interrupt enable clear	0x00000000
0x080C	PD[GPIOA][TOGGLE]	GPIO dual edge interrupt enable toggle	0x00000000
0x0810	PD[GPIOB][VALUE]	GPIO dual edge interrupt enable value	0x00000000
0x0814	PD[GPIOB][SET]	GPIO dual edge interrupt enable set	0x00000000
0x0818	PD[GPIOB][CLEAR]	GPIO dual edge interrupt enable clear	0x00000000
0x081C	PD[GPIOB][TOGGLE]	GPIO dual edge interrupt enable toggle	0x00000000
0x08D0	PD[GPIOX][VALUE]	GPIO dual edge interrupt enable value	0x00000000
0x08D4	PD[GPIOX][SET]	GPIO dual edge interrupt enable set	0x00000000

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Offset	Register Name	Description	Reset Value
0x08D8	PD[GPIOX][CLEAR]	GPIO dual edge interrupt enable clear	0x00000000
0x08DC	PD[GPIOX][TOGGLE]	GPIO dual edge interrupt enable toggle	0x00000000
0x08E0	PD[GPIOY][VALUE]	GPIO dual edge interrupt enable value	0x00000000
0x08E4	PD[GPIOY][SET]	GPIO dual edge interrupt enable set	0x00000000
0x08E8	PD[GPIOY][CLEAR]	GPIO dual edge interrupt enable clear	0x00000000
0x08EC	PD[GPIOY][TOGGLE]	GPIO dual edge interrupt enable toggle	0x00000000

Table 152: GPIO Register List

23.4 GPIO Register Description

The detailed description of GPIO registers is as follows:

23.4.1 DI[VALUE] (0x0 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INPUT																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DI[VALUE] [31:0]

Bits	Name	Description
31-0	INPUT	GPIO input bus value, each bit represents a bus bit 0: low level presents on chip pin 1: high level presents on chip pin

DI[VALUE] Bits

23.4.2 DO[VALUE] (0x100 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OUTPUT																															

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DO[VALUE] [31:0]

Bits	Name	Description
31-0	OUTPUT	GPIO output register value, each bit represents a bus bit 0: chip pin output low level when direction is output 1: chip pin output high level when direction is output

DO[VALUE] Bits

23.4.3 DO[SET] (0x104 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OUTPUT																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DO[SET] [31:0]

Bits	Name	Description
31-0	OUTPUT	GPIO output register value, each bit represents a bus bit 0: chip pin output low level when direction is output 1: chip pin output high level when direction is output

DO[SET] Bits

23.4.4 DO[CLEAR] (0x108 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OUTPUT																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DO[CLEAR] [31:0]

Bits	Name	Description
31-0	OUTPUT	GPIO output register value, each bit represents a bus bit 0: chip pin output low level when direction is output 1: chip pin output high level when direction is output

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Bits	Name	Description

DO[CLEAR] Bits

23.4.5 DO[TOGGLE] (0x10C + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OUTPUT																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DO[TOGGLE] [31:0]

Bits	Name	Description
31-0	OUTPUT	GPIO output register value, each bit represents a bus bit 0: chip pin output low level when direction is output 1: chip pin output high level when direction is output

DO[TOGGLE] Bits

23.4.6 OE[VALUE] (0x200 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DIRECTION																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

OE[VALUE] [31:0]

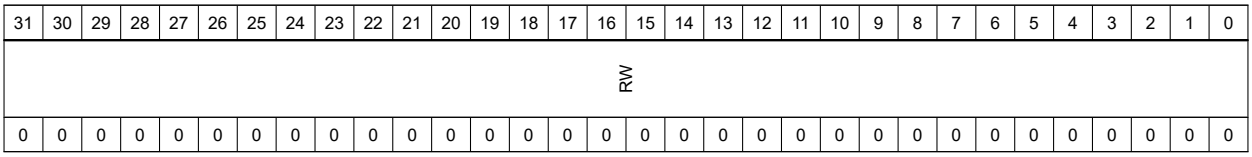
Bits	Name	Description
31-0	DIRECTION	GPIO direction, each bit represents a bus bit 0: input 1: output

OE[VALUE] Bits

23.4.7 OE[SET] (0x204 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DIRECTION																															

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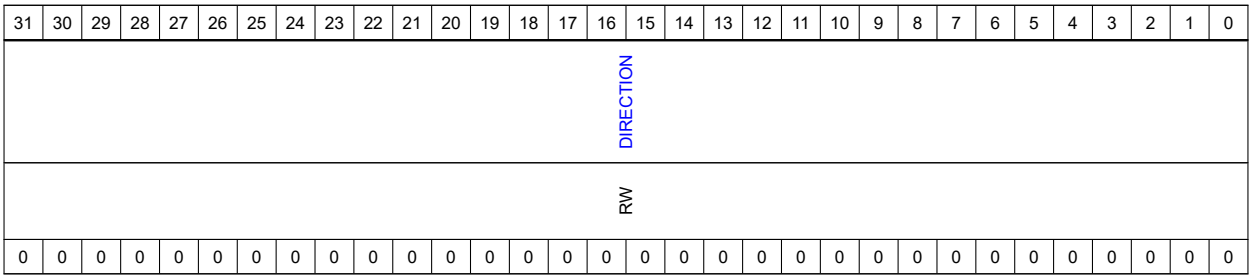


OE[SET] [31:0]

Bits	Name	Description
31-0	DIRECTION	GPIO direction, each bit represents a bus bit 0: input 1: output

OE[SET] Bits

23.4.8 OE[CLEAR] (0x208 + 0x10 * n)

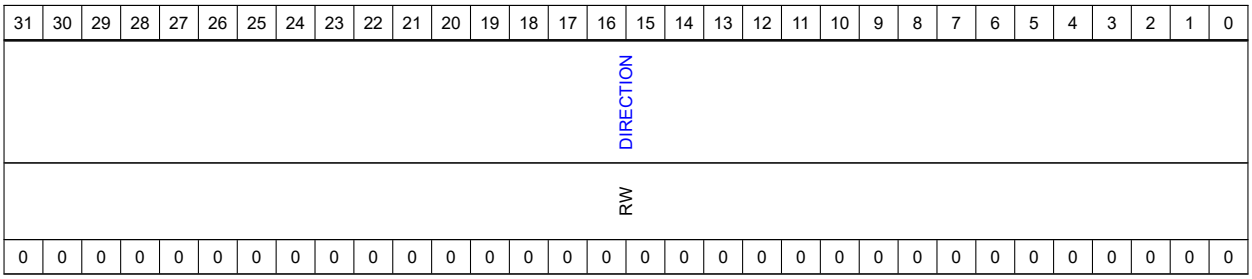


OE[CLEAR] [31:0]

Bits	Name	Description
31-0	DIRECTION	GPIO direction, each bit represents a bus bit 0: input 1: output

OE[CLEAR] Bits

23.4.9 OE[TOGGLE] (0x20C + 0x10 * n)



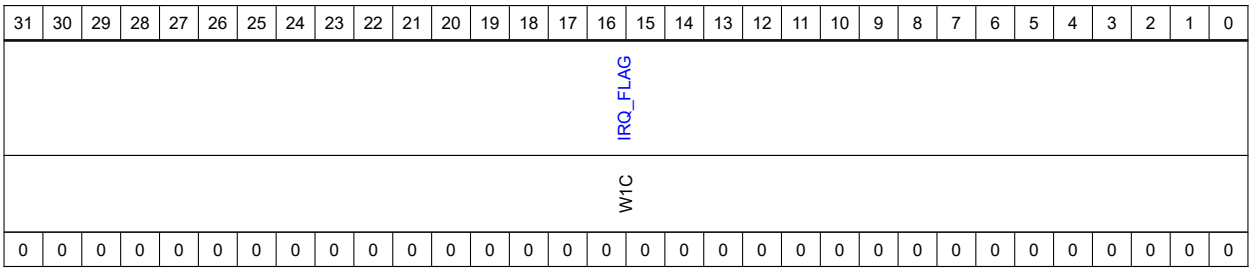
OE[TOGGLE] [31:0]

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Bits	Name	Description
31-0	DIRECTION	GPIO direction, each bit represents a bus bit 0: input 1: output

OE[TOGGLE] Bits

23.4.10 IF[VALUE] (0x300 + 0x10 * n)

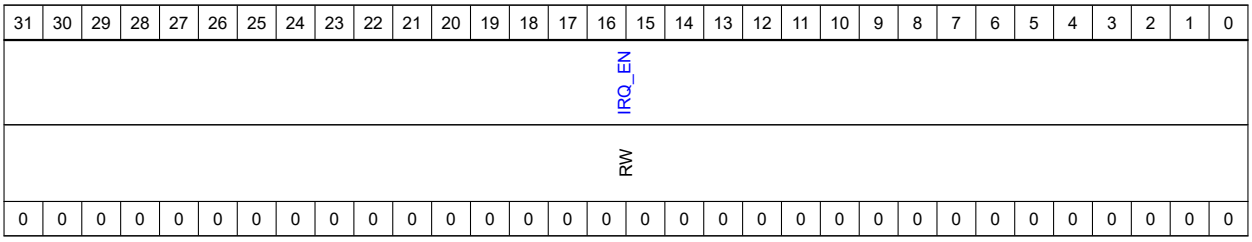


IF[VALUE] [31:0]

Bits	Name	Description
31-0	IRQ_FLAG	GPIO interrupt flag, write 1 to clear this flag 0: no irq 1: irq pending

IF[VALUE] Bits

23.4.11 IE[VALUE] (0x400 + 0x10 * n)



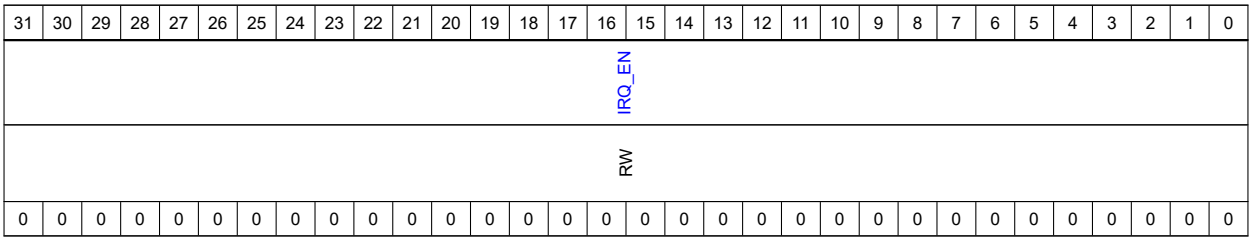
IE[VALUE] [31:0]

Bits	Name	Description
31-0	IRQ_EN	GPIO interrupt enable, each bit represents a bus bit 0: irq is disabled 1: irq is enable

IE[VALUE] Bits

23.4.12 IE[SET] (0x404 + 0x10 * n)

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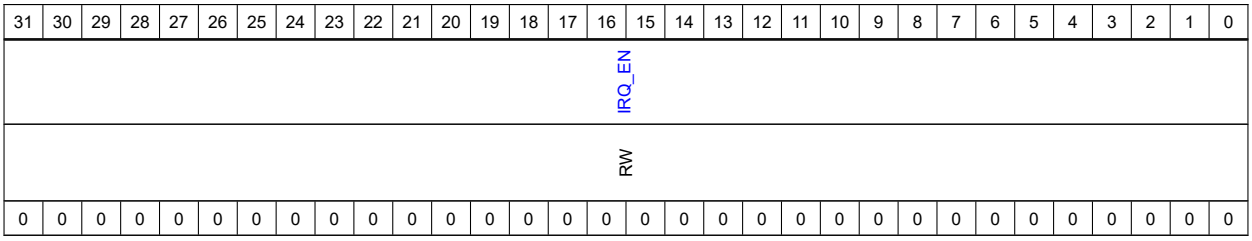


IE[SET] [31:0]

Bits	Name	Description
31-0	IRQ_EN	GPIO interrupt enable, each bit represents a bus bit 0: irq is disabled 1: irq is enable

IE[SET] Bits

23.4.13 IE[CLEAR] (0x408 + 0x10 * n)

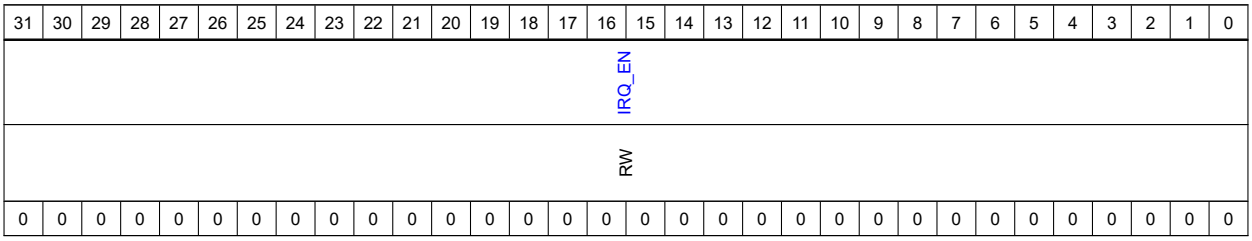


IE[CLEAR] [31:0]

Bits	Name	Description
31-0	IRQ_EN	GPIO interrupt enable, each bit represents a bus bit 0: irq is disabled 1: irq is enable

IE[CLEAR] Bits

23.4.14 IE[TOGGLE] (0x40C + 0x10 * n)



IE[TOGGLE] [31:0]

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Bits	Name	Description
31-0	IRQ_EN	GPIO interrupt enable, each bit represents a bus bit 0: irq is disabled 1: irq is enable

IE[TOGGLE] Bits

23.4.15 PL[VALUE] (0x500 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IRQ_POL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PL[VALUE] [31:0]

Bits	Name	Description
31-0	IRQ_POL	GPIO interrupt polarity, each bit represents a bus bit 0: irq is high level or rising edge 1: irq is low level or falling edge

PL[VALUE] Bits

23.4.16 PL[SET] (0x504 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IRQ_POL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PL[SET] [31:0]

Bits	Name	Description
31-0	IRQ_POL	GPIO interrupt polarity, each bit represents a bus bit 0: irq is high level or rising edge 1: irq is low level or falling edge

PL[SET] Bits

23.4.17 PL[CLEAR] (0x508 + 0x10 * n)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IRQ_POL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PL[CLEAR] [31:0]

Bits	Name	Description
31-0	IRQ_POL	GPIO interrupt polarity, each bit represents a bus bit 0: irq is high level or rising edge 1: irq is low level or falling edge

PL[CLEAR] Bits

23.4.18 PL[TOGGLE] (0x50C + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IRQ_POL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PL[TOGGLE] [31:0]

Bits	Name	Description
31-0	IRQ_POL	GPIO interrupt polarity, each bit represents a bus bit 0: irq is high level or rising edge 1: irq is low level or falling edge

PL[TOGGLE] Bits

23.4.19 TP[VALUE] (0x600 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IRQ_TYPE																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

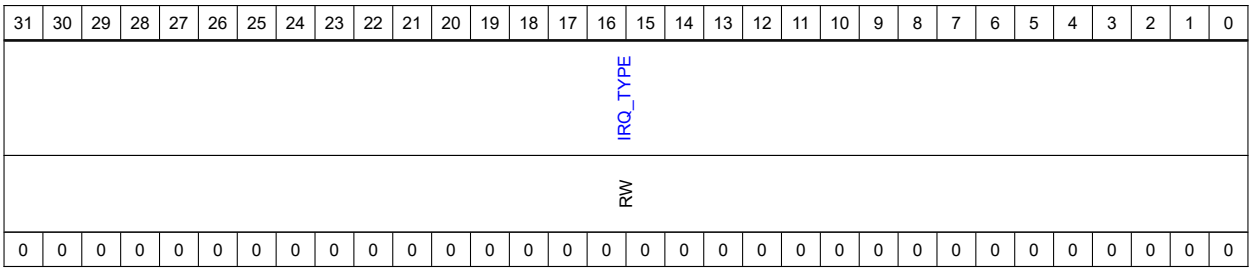
TP[VALUE] [31:0]

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Bits	Name	Description
31-0	IRQ_TYPE	GPIO interrupt type, each bit represents a bus bit 0: irq is triggered by level 1: irq is triggered by edge

TP[VALUE] Bits

23.4.20 TP[SET] (0x604 + 0x10 * n)

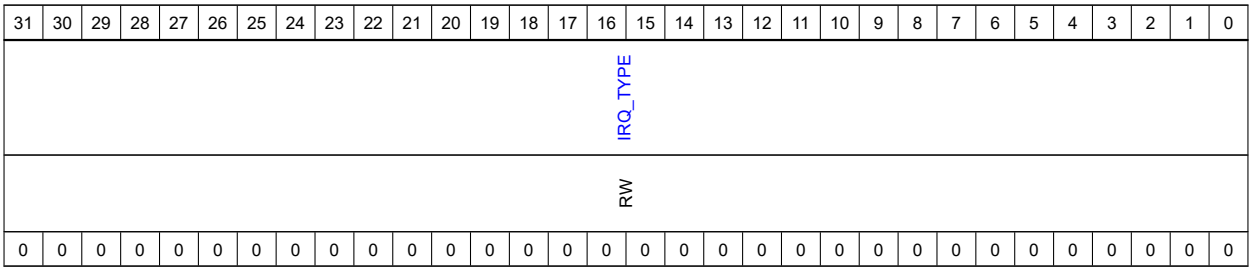


TP[SET] [31:0]

Bits	Name	Description
31-0	IRQ_TYPE	GPIO interrupt type, each bit represents a bus bit 0: irq is triggered by level 1: irq is triggered by edge

TP[SET] Bits

23.4.21 TP[CLEAR] (0x608 + 0x10 * n)



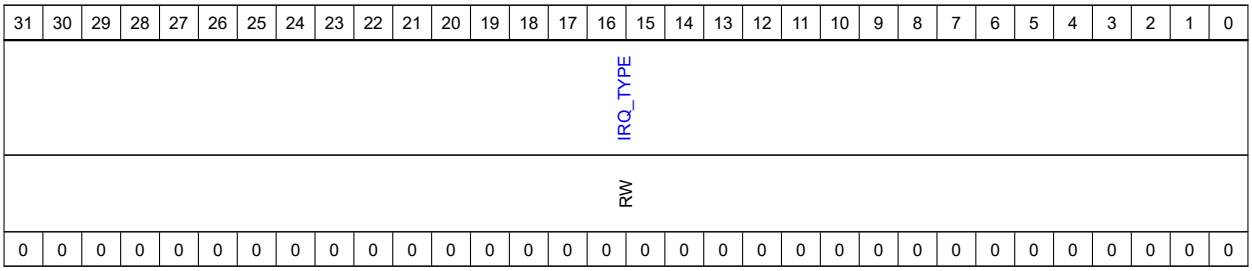
TP[CLEAR] [31:0]

Bits	Name	Description
31-0	IRQ_TYPE	GPIO interrupt type, each bit represents a bus bit 0: irq is triggered by level 1: irq is triggered by edge

TP[CLEAR] Bits

23.4.22 TP[TOGGLE] (0x60C + 0x10 * n)

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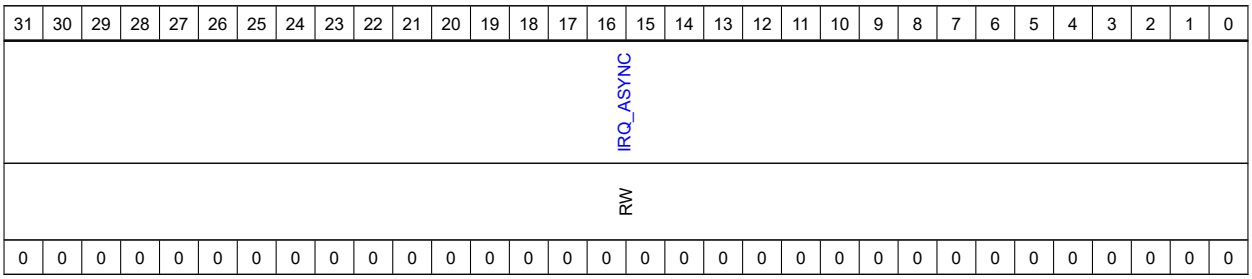


TP[TOGGLE] [31:0]

Bits	Name	Description
31-0	IRQ_TYPE	GPIO interrupt type, each bit represents a bus bit 0: irq is triggered by level 1: irq is triggered by edge

TP[TOGGLE] Bits

23.4.23 AS[VALUE] (0x700 + 0x10 * n)

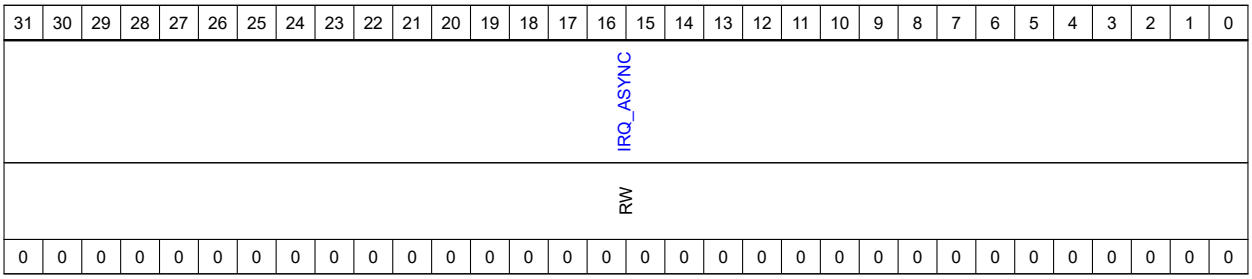


AS[VALUE] [31:0]

Bits	Name	Description
31-0	IRQ_ASYNC	GPIO interrupt asynchronous, each bit represents a bus bit 0: irq is triggered base on system clock 1: irq is triggered combinational Note: combinational interrupt is sensitive to environment noise

AS[VALUE] Bits

23.4.24 AS[SET] (0x704 + 0x10 * n)



AS[SET] [31:0]

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Bits	Name	Description
31-0	IRQ_ASYNC	GPIO interrupt asynchronous, each bit represents a bus bit 0: irq is triggered base on system clock 1: irq is triggered combinational Note: combinational interrupt is sensitive to environment noise

AS[SET] Bits

23.4.25 AS[CLEAR] (0x708 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
																IRQ_ASYNC															
																RW															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

AS[CLEAR] [31:0]

Bits	Name	Description
31-0	IRQ_ASYNC	GPIO interrupt asynchronous, each bit represents a bus bit 0: irq is triggered base on system clock 1: irq is triggered combinational Note: combinational interrupt is sensitive to environment noise

AS[CLEAR] Bits

23.4.26 AS[TOGGLE] (0x70C + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
																IRQ_ASYNC															
																RW															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

AS[TOGGLE] [31:0]

Bits	Name	Description
31-0	IRQ_ASYNC	GPIO interrupt asynchronous, each bit represents a bus bit 0: irq is triggered base on system clock 1: irq is triggered combinational Note: combinational interrupt is sensitive to environment noise

AS[TOGGLE] Bits

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23.4.27 PD[VALUE] (0x800 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																															IRQ_DUAL
N/A																															RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

PD[VALUE] [31:0]

Bits	Name	Description
0	IRQ_DUAL	GPIO dual edge interrupt enable 0: single edge interrupt 1: dual edge interrupt enable

PD[VALUE] Bits

23.4.28 PD[SET] (0x804 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																															IRQ_DUAL
N/A																															RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

PD[SET] [31:0]

Bits	Name	Description
0	IRQ_DUAL	GPIO dual edge interrupt enable set 0: keep original edge interrupt type 1: dual edge interrupt enable

PD[SET] Bits

23.4.29 PD[CLEAR] (0x808 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																															IRQ_DUAL
N/A																															RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

PD[CLEAR] [31:0]

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Bits	Name	Description
0	IRQ_DUAL	GPIO dual edge interrupt enable clear 0: keep original edge interrupt type 1: single edge interrupt enable

PD[CLEAR] Bits

23.4.30 PD[TOGGLE] (0x80C + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																															IRQ_DUAL
N/A																															RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

PD[TOGGLE] [31:0]

Bits	Name	Description
0	IRQ_DUAL	GPIO dual edge interrupt enable toggle 0: keep original edge interrupt type 1: change original edge interrupt type to another one.

PD[TOGGLE] Bits

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24 GPIO Manager (GPIOM)

This chapter describes the main features and functions of the GPIO Manager (GPIOM). The main function of the GPIOM is to specify the module for which the GPIO configuration takes effect for any IO. Each IO can be individually assigned to be controlled by either the GPIO Controller (GPIO0) or the Fast GPIO Controller (FGPIO0).

24.1 Feature Summary

This chapter introduces the main features of the GPIOM:

- Assign IOs to the specified GPIO controller
- Configure whether the IO input is visible to a specific GPIO controller
- GPIOM register access control
- GPIOM register lock

24.2 Functional Description

This chapter describes the functions of the GPIOM.

24.2.1 GPIO Assignment

This product supports multiple GPIO controllers, which is designed to facilitate GPIO resource isolation between multi-core and multi-task environments. The main function of the GPIOM is to manage all IOs. Each IO can be individually assigned to be controlled by a specific GPIO controller, and other GPIO controllers cannot control this IO.

Users can manage the y pin of Px through the GPIOxASSIGNy register. The SELECT bit field of this register can select which GPIO controller controls this IO:

- 0'b0, GPIO0
- 0'b1, FGPIO0

The HIDE bit field of the GPIOxASSIGNy register consists of 2 HIDE bits. When each bit is set to 1, it means that the input of this IO is not visible in the DI register of the corresponding GPIO controller, that is, the GPIO controller cannot read the input of this IO.

- HIDE[0], set to 1 means the IO is invisible to GPIO0
- HIDE[1], set to 1 means the IO is invisible to FGPIO0

24.2.2 Access Control

The LOCK bit of the GPIOxASSIGNy register can be used to lock the corresponding register of this IO. Once set to 1, the configuration of this register cannot be changed until the next reset.

The NON_SEC bit of the GPIOxASSIGNy register can control access permissions according to the system security status. Once set to 1, the register can only be accessed in the secure state. Please refer to the relevant chapters of system security for detailed information about the system security status.

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24.3 GPIOM Register List

The GPIOM register list is as follows:

GPIOM base address: 0xF00D8000

Offset	Register Name	Description	Reset Value
0x0000	ASSIGN[GPIOA][PIN][PIN00]	GPIO mananger	0x00000000
0x0004	ASSIGN[GPIOA][PIN][PIN01]	GPIO mananger	0x00000000
0x0008	ASSIGN[GPIOA][PIN][PIN02]	GPIO mananger	0x00000000
0x000C	ASSIGN[GPIOA][PIN][PIN03]	GPIO mananger	0x00000000
0x0010	ASSIGN[GPIOA][PIN][PIN04]	GPIO mananger	0x00000000
0x0014	ASSIGN[GPIOA][PIN][PIN05]	GPIO mananger	0x00000000
0x0018	ASSIGN[GPIOA][PIN][PIN06]	GPIO mananger	0x00000000
0x001C	ASSIGN[GPIOA][PIN][PIN07]	GPIO mananger	0x00000000
0x0020	ASSIGN[GPIOA][PIN][PIN08]	GPIO mananger	0x00000000
0x0024	ASSIGN[GPIOA][PIN][PIN09]	GPIO mananger	0x00000000
0x0028	ASSIGN[GPIOA][PIN][PIN10]	GPIO mananger	0x00000000
0x002C	ASSIGN[GPIOA][PIN][PIN11]	GPIO mananger	0x00000000
0x0030	ASSIGN[GPIOA][PIN][PIN12]	GPIO mananger	0x00000000
0x0034	ASSIGN[GPIOA][PIN][PIN13]	GPIO mananger	0x00000000
0x0038	ASSIGN[GPIOA][PIN][PIN14]	GPIO mananger	0x00000000
0x003C	ASSIGN[GPIOA][PIN][PIN15]	GPIO mananger	0x00000000
0x0040	ASSIGN[GPIOA][PIN][PIN16]	GPIO mananger	0x00000000
0x0044	ASSIGN[GPIOA][PIN][PIN17]	GPIO mananger	0x00000000
0x0048	ASSIGN[GPIOA][PIN][PIN18]	GPIO mananger	0x00000000
0x004C	ASSIGN[GPIOA][PIN][PIN19]	GPIO mananger	0x00000000
0x0050	ASSIGN[GPIOA][PIN][PIN20]	GPIO mananger	0x00000000
0x0054	ASSIGN[GPIOA][PIN][PIN21]	GPIO mananger	0x00000000
0x0058	ASSIGN[GPIOA][PIN][PIN22]	GPIO mananger	0x00000000
0x005C	ASSIGN[GPIOA][PIN][PIN23]	GPIO mananger	0x00000000
0x0060	ASSIGN[GPIOA][PIN][PIN24]	GPIO mananger	0x00000000
0x0064	ASSIGN[GPIOA][PIN][PIN25]	GPIO mananger	0x00000000
0x0068	ASSIGN[GPIOA][PIN][PIN26]	GPIO mananger	0x00000000
0x006C	ASSIGN[GPIOA][PIN][PIN27]	GPIO mananger	0x00000000
0x0070	ASSIGN[GPIOA][PIN][PIN28]	GPIO mananger	0x00000000
0x0074	ASSIGN[GPIOA][PIN][PIN29]	GPIO mananger	0x00000000
0x0078	ASSIGN[GPIOA][PIN][PIN30]	GPIO mananger	0x00000000
0x007C	ASSIGN[GPIOA][PIN][PIN31]	GPIO mananger	0x00000000
0x0080	ASSIGN[GPIOB][PIN][PIN00]	GPIO mananger	0x00000000
0x0084	ASSIGN[GPIOB][PIN][PIN01]	GPIO mananger	0x00000000
0x0088	ASSIGN[GPIOB][PIN][PIN02]	GPIO mananger	0x00000000
0x008C	ASSIGN[GPIOB][PIN][PIN03]	GPIO mananger	0x00000000

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Offset	Register Name	Description	Reset Value
0x0090	ASSIGN[GPIOB][PIN][PIN04]	GPIO mananger	0x00000000
0x0094	ASSIGN[GPIOB][PIN][PIN05]	GPIO mananger	0x00000000
0x0098	ASSIGN[GPIOB][PIN][PIN06]	GPIO mananger	0x00000000
0x009C	ASSIGN[GPIOB][PIN][PIN07]	GPIO mananger	0x00000000
0x00A0	ASSIGN[GPIOB][PIN][PIN08]	GPIO mananger	0x00000000
0x00A4	ASSIGN[GPIOB][PIN][PIN09]	GPIO mananger	0x00000000
0x00A8	ASSIGN[GPIOB][PIN][PIN10]	GPIO mananger	0x00000000
0x00AC	ASSIGN[GPIOB][PIN][PIN11]	GPIO mananger	0x00000000
0x00B0	ASSIGN[GPIOB][PIN][PIN12]	GPIO mananger	0x00000000
0x00B4	ASSIGN[GPIOB][PIN][PIN13]	GPIO mananger	0x00000000
0x00B8	ASSIGN[GPIOB][PIN][PIN14]	GPIO mananger	0x00000000
0x00BC	ASSIGN[GPIOB][PIN][PIN15]	GPIO mananger	0x00000000
0x00C0	ASSIGN[GPIOB][PIN][PIN16]	GPIO mananger	0x00000000
0x00C4	ASSIGN[GPIOB][PIN][PIN17]	GPIO mananger	0x00000000
0x00C8	ASSIGN[GPIOB][PIN][PIN18]	GPIO mananger	0x00000000
0x00CC	ASSIGN[GPIOB][PIN][PIN19]	GPIO mananger	0x00000000
0x00D0	ASSIGN[GPIOB][PIN][PIN20]	GPIO mananger	0x00000000
0x00D4	ASSIGN[GPIOB][PIN][PIN21]	GPIO mananger	0x00000000
0x00D8	ASSIGN[GPIOB][PIN][PIN22]	GPIO mananger	0x00000000
0x00DC	ASSIGN[GPIOB][PIN][PIN23]	GPIO mananger	0x00000000
0x00E0	ASSIGN[GPIOB][PIN][PIN24]	GPIO mananger	0x00000000
0x00E4	ASSIGN[GPIOB][PIN][PIN25]	GPIO mananger	0x00000000
0x00E8	ASSIGN[GPIOB][PIN][PIN26]	GPIO mananger	0x00000000
0x00EC	ASSIGN[GPIOB][PIN][PIN27]	GPIO mananger	0x00000000
0x00F0	ASSIGN[GPIOB][PIN][PIN28]	GPIO mananger	0x00000000
0x00F4	ASSIGN[GPIOB][PIN][PIN29]	GPIO mananger	0x00000000
0x00F8	ASSIGN[GPIOB][PIN][PIN30]	GPIO mananger	0x00000000
0x00FC	ASSIGN[GPIOB][PIN][PIN31]	GPIO mananger	0x00000000
0x0680	ASSIGN[GPIOX][PIN][PIN00]	GPIO mananger	0x00000000
0x0684	ASSIGN[GPIOX][PIN][PIN01]	GPIO mananger	0x00000000
0x0688	ASSIGN[GPIOX][PIN][PIN02]	GPIO mananger	0x00000000
0x068C	ASSIGN[GPIOX][PIN][PIN03]	GPIO mananger	0x00000000
0x0690	ASSIGN[GPIOX][PIN][PIN04]	GPIO mananger	0x00000000
0x0694	ASSIGN[GPIOX][PIN][PIN05]	GPIO mananger	0x00000000
0x0698	ASSIGN[GPIOX][PIN][PIN06]	GPIO mananger	0x00000000
0x069C	ASSIGN[GPIOX][PIN][PIN07]	GPIO mananger	0x00000000
0x06A0	ASSIGN[GPIOX][PIN][PIN08]	GPIO mananger	0x00000000
0x06A4	ASSIGN[GPIOX][PIN][PIN09]	GPIO mananger	0x00000000
0x06A8	ASSIGN[GPIOX][PIN][PIN10]	GPIO mananger	0x00000000
0x06AC	ASSIGN[GPIOX][PIN][PIN11]	GPIO mananger	0x00000000

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Offset	Register Name	Description	Reset Value
0x06B0	ASSIGN[GPIOX][PIN][PIN12]	GPIO mananger	0x00000000
0x06B4	ASSIGN[GPIOX][PIN][PIN13]	GPIO mananger	0x00000000
0x06B8	ASSIGN[GPIOX][PIN][PIN14]	GPIO mananger	0x00000000
0x06BC	ASSIGN[GPIOX][PIN][PIN15]	GPIO mananger	0x00000000
0x06C0	ASSIGN[GPIOX][PIN][PIN16]	GPIO mananger	0x00000000
0x06C4	ASSIGN[GPIOX][PIN][PIN17]	GPIO mananger	0x00000000
0x06C8	ASSIGN[GPIOX][PIN][PIN18]	GPIO mananger	0x00000000
0x06CC	ASSIGN[GPIOX][PIN][PIN19]	GPIO mananger	0x00000000
0x06D0	ASSIGN[GPIOX][PIN][PIN20]	GPIO mananger	0x00000000
0x06D4	ASSIGN[GPIOX][PIN][PIN21]	GPIO mananger	0x00000000
0x06D8	ASSIGN[GPIOX][PIN][PIN22]	GPIO mananger	0x00000000
0x06DC	ASSIGN[GPIOX][PIN][PIN23]	GPIO mananger	0x00000000
0x06E0	ASSIGN[GPIOX][PIN][PIN24]	GPIO mananger	0x00000000
0x06E4	ASSIGN[GPIOX][PIN][PIN25]	GPIO mananger	0x00000000
0x06E8	ASSIGN[GPIOX][PIN][PIN26]	GPIO mananger	0x00000000
0x06EC	ASSIGN[GPIOX][PIN][PIN27]	GPIO mananger	0x00000000
0x06F0	ASSIGN[GPIOX][PIN][PIN28]	GPIO mananger	0x00000000
0x06F4	ASSIGN[GPIOX][PIN][PIN29]	GPIO mananger	0x00000000
0x06F8	ASSIGN[GPIOX][PIN][PIN30]	GPIO mananger	0x00000000
0x06FC	ASSIGN[GPIOX][PIN][PIN31]	GPIO mananger	0x00000000
0x0700	ASSIGN[GPIOY][PIN][PIN00]	GPIO mananger	0x00000000
0x0704	ASSIGN[GPIOY][PIN][PIN01]	GPIO mananger	0x00000000
0x0708	ASSIGN[GPIOY][PIN][PIN02]	GPIO mananger	0x00000000
0x070C	ASSIGN[GPIOY][PIN][PIN03]	GPIO mananger	0x00000000
0x0710	ASSIGN[GPIOY][PIN][PIN04]	GPIO mananger	0x00000000
0x0714	ASSIGN[GPIOY][PIN][PIN05]	GPIO mananger	0x00000000
0x0718	ASSIGN[GPIOY][PIN][PIN06]	GPIO mananger	0x00000000
0x071C	ASSIGN[GPIOY][PIN][PIN07]	GPIO mananger	0x00000000
0x0720	ASSIGN[GPIOY][PIN][PIN08]	GPIO mananger	0x00000000
0x0724	ASSIGN[GPIOY][PIN][PIN09]	GPIO mananger	0x00000000
0x0728	ASSIGN[GPIOY][PIN][PIN10]	GPIO mananger	0x00000000
0x072C	ASSIGN[GPIOY][PIN][PIN11]	GPIO mananger	0x00000000
0x0730	ASSIGN[GPIOY][PIN][PIN12]	GPIO mananger	0x00000000
0x0734	ASSIGN[GPIOY][PIN][PIN13]	GPIO mananger	0x00000000
0x0738	ASSIGN[GPIOY][PIN][PIN14]	GPIO mananger	0x00000000
0x073C	ASSIGN[GPIOY][PIN][PIN15]	GPIO mananger	0x00000000
0x0740	ASSIGN[GPIOY][PIN][PIN16]	GPIO mananger	0x00000000
0x0744	ASSIGN[GPIOY][PIN][PIN17]	GPIO mananger	0x00000000
0x0748	ASSIGN[GPIOY][PIN][PIN18]	GPIO mananger	0x00000000
0x074C	ASSIGN[GPIOY][PIN][PIN19]	GPIO mananger	0x00000000

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Offset	Register Name	Description	Reset Value
0x0750	ASSIGN[GPIOY][PIN][PIN20]	GPIO mananger	0x00000000
0x0754	ASSIGN[GPIOY][PIN][PIN21]	GPIO mananger	0x00000000
0x0758	ASSIGN[GPIOY][PIN][PIN22]	GPIO mananger	0x00000000
0x075C	ASSIGN[GPIOY][PIN][PIN23]	GPIO mananger	0x00000000
0x0760	ASSIGN[GPIOY][PIN][PIN24]	GPIO mananger	0x00000000
0x0764	ASSIGN[GPIOY][PIN][PIN25]	GPIO mananger	0x00000000
0x0768	ASSIGN[GPIOY][PIN][PIN26]	GPIO mananger	0x00000000
0x076C	ASSIGN[GPIOY][PIN][PIN27]	GPIO mananger	0x00000000
0x0770	ASSIGN[GPIOY][PIN][PIN28]	GPIO mananger	0x00000000
0x0774	ASSIGN[GPIOY][PIN][PIN29]	GPIO mananger	0x00000000
0x0778	ASSIGN[GPIOY][PIN][PIN30]	GPIO mananger	0x00000000
0x077C	ASSIGN[GPIOY][PIN][PIN31]	GPIO mananger	0x00000000

Table 153: GPIOM Register List

24.4 GPIOM Register Description

The detailed description of GPIOM registers is as follows:

24.4.1 ASSIGN[PIN] (0x0 + 0x80 * n + 0x4 * m)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LOCK	RSVD																			HIDE				RSVD						SELECT	
RW	N/A																			RW				N/A						RW	
0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	x	x	x	x	x	x	0	0

ASSIGN[PIN] [31:0]

Bits	Name	Description
31	LOCK	lock fields in this register, lock can only be cleared by soc reset 0: fields can be changed 1: fields locked to current value, not changeable
11-8	HIDE	pin value visibility to gpios, bit0: 1, invisible to soc gpio0; 0: visible to soc gpio0 bit2: 1, invisible to cpu0 fast gpio; 0: visible to cpu0 fast gpio
1-0	SELECT	select which gpio controls chip pin, 0: soc gpio0; 2: cpu0 fastgpio

ASSIGN[PIN] Bits

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25 Memory Overview

This chapter introduces the relevant interfaces of the internal memory and external storage peripherals of this product.

25.1 Internal SRAM

This product integrates a total of 288 KB internal RAM, including processor local memory and general-purpose memory.

25.1.1 Local Memory

Local Memory (LM) includes Instruction Local Memory (ILM) and Data Local Memory (DLM), which refers to the memory that works at the same clock frequency as the core and supports zero-wait cycle access for the core. The RISC-V CPU0 on this product is configured with 256 KB local memory. The local memory LM can be directly accessed by the processor, and can also be accessed by other on-chip master devices through the processor's bus slave interface, corresponding to LM and LM_SLV on the system address mapping table.

The allocation of ILD and DLM is as follows:

- ILM0, instruction local memory of RISC-V CPU0, 128KB;
- DLM0, data local memory of RISC-V CPU0, 128KB;

25.1.2 General-purpose Memory

This product supports the following general-purpose memory:

- AHB SRAM, 32KB, located on the AHB bus;

25.2 General Purpose Registers

This product contains a certain capacity of General Purpose Register (GPR) for users to store data for special purposes.

- Power management domain general purpose register PGPR, with a capacity of 128 bytes, located in the power management domain, can retain data when the system power domain is powered off, and is not affected by the reset of the system power domain. Individual registers may be occupied by ROM, please refer to the ROM chapter for specific occupation details.
- DGO domain general purpose register DGO_GPR, with a capacity of 16 bytes, located in the DGO domain, can retain data when the system power domain and power management domain are powered off.

25.3 Serial Bus Controller XPI

The serial bus controller XPI supports connecting various serially accessible external memories, and can also interconnect with some devices that support serial buses.

The external memories supported by XPI include serial NOR Flash, serial NAND Flash, serial PSRAM, etc., and also support HyperBus devices such as HyperRAM and HyperFlash.

When the CA port of XPI0 is connected to the on-chip 1MB NOR Flash, it supports execute-in-place (XIP).

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And the online decryption of code and data can be realized through the online decryption module EXIP, that is, data and code can be stored in the on-chip NOR Flash in ciphertext form.

25.4 Read-only Memory ROM

This product supports 128KB internal read-only memory ROM.

ROM stores the boot code, Flashloader and part of the peripheral driver of this product.

25.5 One-Time Programmable Memory OTP

This product supports 4096 bit fuses as on-chip one-time programmable memory OTP.

OTP is used to store the following information:

- Unique ID of the product
- Boot configuration information of the product
- Security configuration information of the product
- Part of the security keys
- Calibration data of on-chip analog modules stored when the product leaves the factory
- General data for users

The management of the one-time programmable memory OTP is implemented by the OTP controller. Users can read and program the OTP through the OTP controller. In addition, the OTP controller is also responsible for implementing read protection and write protection of OTP.

For this product, users need to turn on LDOOTP when programming the OTP fuse array, and must turn off LDOOTP after programming is completed.

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26 OTP and OTP Controller

This chapter introduces the main functions and features of the on-chip one-time programmable memory OTP and OTP controller.

26.1 Feature Summary

The features of the on-chip one-time programmable memory OTP are as follows:

- OTP is implemented based on a fuse array, each bit in the fuse array can be programmed once, and cannot be changed after programming
- OTP fuse array contains 128 words, each word contains 32 bits of fuses, with a total capacity of 4096 bits
- OTP words are divided into the following different types according to functions and features:
  - IDENTITY: Chip factory information, UUID and other information are classified as IDENTITY type. This type of information is programmed by the original manufacturer, and the public key HASH for secure boot is also classified as IDENTITY type, which can be programmed by customers
  - SECURE: OTP words such as chip life cycle and OTP hardware lock bits are classified as SECURE type
  - SECRET: Various keys, such as Debug Key, EXIP KEK, key manager FMK, etc. are classified as SECRET type
  - GENERAL: General data that users can freely program is classified as GENERAL type. Control OTP words reserved for BOOT ROM are also classified as GENERAL type
- Supports 32-bit OTP hardware lock bits, each lock bit corresponds to 4 OTP words, combined with the shadow lock bits and fuse lock bits of the OTP controller, to realize OTP read and program protection

26.1.1 OTP Controller Feature Summary

The management of the one-time programmable memory OTP of this product is implemented by the OTP controller. Users can realize read and program operations on OTP through the OTP controller. In addition, the OTP controller is also responsible for implementing OTP read protection and write protection.

The OTP controller is divided into 2 parts in the system address mapping:

- Body part (named OTP in the memory mapping table)
- Shadow part (named OTPSHW in the memory mapping table)

The features of the OTP controller, as well as the different configurations of the body part (OTP) and the shadow part (OTPSHW) are as follows:

- Supports 128 shadow registers, corresponding to 128 words of OTP
  - The body part (OTP) contains shadow registers 0~15, corresponding to OTP words 0~15
  - The shadow part (OTPSHW) contains shadow registers 16~127, corresponding to OTP words 16~127
- Shadow lock register, each OTP word corresponds to 2 lock bits, the shadow lock register can configure read and write protection of shadow registers
  - The body part (OTP) contains shadow lock register 0, corresponding to OTP words 0~15

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  - The shadow part (OTPSHW) contains shadow lock registers 1~7, corresponding to OTP words 16~127
- Supports 128 fuse registers, corresponding to 128 words of OTP, the fuse register is the direct read and write interface of OTP
  - The body part (OTP) contains fuse registers 0~127, corresponding to OTP words 0~127
  - The shadow part (OTPSHW) does not support fuse registers
- Fuse lock register, each OTP word corresponds to 2 lock bits, the fuse lock register can configure read and write protection of the fuse array
  - The body part (OTP) contains fuse lock registers 0~7, corresponding to OTP words 0~127
  - The shadow part (OTPSHW) does not support fuse lock registers
- Supports OTP operation engine, supports OTP read, write, and OTP shadow register reload, supports interrupt generation
  - The operation engine of the body part (OTP) supports OTP words 0~127
  - The shadow part (OTPSHW) does not support the operation engine

26.2 OTP Controller Functional Description

26.2.1 OTP Shadow Registers

The OTP controller contains 128 shadow registers, each shadow register corresponds to a word of OTP. The OTP controller will load all data of the OTP fuse array into the shadow registers when the chip is released from reset.

Users can also manually reload specific OTP words to the corresponding shadow registers through the OTP operation engine. For details, please refer to Section 26.4.

If a shadow register is not configured with read or write protection, users can read and modify the value of the shadow register arbitrarily. Note that some system configuration related control bits stored in OTP actually take effect after being loaded into the shadow register. Therefore, modifying the OTP shadow register can directly modify the corresponding configuration without programming OTP. For example, setting the DEBUG_DISABLE bit of the OTP shadow register to 1 can immediately close the debug port.

OTP shadow registers include a set of shadow lock registers. The lock registers are used to configure read protection, write protection of shadow registers and whether to allow shadow registers to reload the corresponding word values from the OTP fuse array.

Each OTP shadow register (32-bit) corresponds to 2 lock control bits. The definition of the LOCK bit field is as follows:

- 2'b00, unlocked, no restriction on user reading and writing shadow registers
- 2'b01, single lock, users can read shadow registers, but are not allowed to write shadow registers, and are not allowed to modify the LOCK bit itself
- 2'b10, unlocked, no restriction on user reading and writing shadow registers, and modification of the LOCK bit itself is not allowed
- 2'b11, double lock, users are not allowed to read or write shadow registers, and modification of the LOCK bit itself is not allowed

For shadow registers corresponding to different types of OTP words, the actual read protection, write, and

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reload protection effects are affected by both the shadow lock register and the OTP hardware lock bit. For specific effects, please refer to Section 26.5.

26.3 OTP Fuse Registers

The OTP controller contains 128 fuse registers, each fuse register corresponds to a word of OTP. The fuse register is essentially the read and write interface of the OTP fuse array. Reading the fuse register can directly read the value of the corresponding word of the OTP fuse array bypassing the OTP shadow register.

To prevent user misoperation, the fuse register includes an unlock register UNLOCK. Writing 0x4F50454E (corresponding to ASCII string ”OPEN”) to the unlock register can unlock the write protection of the fuse register. After unlocking, writing to the fuse register can directly start programming the corresponding OTP word. If the user writes any value other than 0x4F50454E to the unlock register, the write protection of the fuse register will be activated immediately.

OTP fuse registers include a set of fuse lock registers. The lock registers can configure read protection and program protection of the corresponding words of the fuse array.

Each OTP word (32-bit) corresponds to 2 lock control bits. The definition of the LOCK bit field is as follows:

- 2’b00, unlocked, no restriction on user reading and writing fuse registers
- 2’b01, single lock, users can read fuses, but are not allowed to program fuses, and are not allowed to modify the LOCK bit itself
- 2’b10, unlocked, no restriction on user reading and writing fuse registers, and modification of the LOCK bit itself is not allowed
- 2’b11, double lock, users are not allowed to read or program fuses, and modification of the LOCK bit itself is not allowed

For different types of words in the OTP fuse array, the actual read protection and program protection effects are affected by both the fuse lock register and the OTP hardware lock bit. For specific effects, please refer to Section 26.5.

26.4 OTP Operation Engine

The OTP controller supports direct reading and writing of the OTP fuse array by reading and writing fuse registers. However, if the processor directly reads and writes the fuse register, the read and write delay of the fuse array itself is long, which may block the processor. Therefore, the OTP controller supports a set of operation engines. Users can realize reading and programming of the OTP fuse array through the operation engine, and can also reload the OTP fuse array into the corresponding shadow register.

The steps to read OTP through the operation engine are as follows:

- Write the sub-address of the target word to the ADDR register, that is, 0 represents Word 0, 1 represents Word 1, and so on.
- Write 0x52454144 (corresponding to ASCII string ”READ”) to the CMD register
- Wait for the INT_FLAG[READ] flag bit to be set to 1
- Read the value of the target word from the DATA register

The steps to program OTP through the operation engine are as follows:

- Write the sub-address of the target word to the ADDR register, that is, 0 represents Word 0, 1 represents

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Word 1, and so on.

- Write the program value of the target word to the DATA register
- Write 0x424C4F57 (corresponding to ASCII string ”BLOW”) to the CMD register
- Wait for the INT_FLAG[WRITE] flag bit to be set to 1

Users can reload OTP shadow registers through the operation engine, which supports 4 reload regions:

- REGION[LOAD_REGION0], reload OTP Word 0 ~ Word7 to SHAWD0 ~ SHADOW7
- REGION[LOAD_REGION1], reload OTP Word 8 ~ Word15 to SHAWD7 ~ SHADOW15
- REGION[LOAD_REGION2], reload OTP Word 16 ~ Word127 to SHAWD16 ~ SHADOW127
- REGION[LOAD_REGION3], user-defined start and end word addresses of OTP words

Users can set bits 0~3 of the LOAD_REQ register to 1 to start the reload of LOAD_REGION0~3.

Note: Before starting REGION[LOAD_REGION3], users need to configure the START bit field of the REGION[LOAD_REGION3] register and the STOP bit field of the REGION[LOAD_REGION3] register to specify a continuous segment of OTP words. The START bit field defines the first OTP word address to be reloaded, the STOP bit field defines the end word address of the reload, that is, reload OTP Word (START) ~ OTP Word (STOP -1) to SHAWD (START) ~ SHADOW(STOP -1).

Users can query the reload process through the LOAD$_COMP$ register, and the flag bit is set to 1 when the reload of the corresponding reload region is completed.

Note that the read, write and reload operations of OTP by the OTP operation engine are restricted by OTP hardware lock bits, shadow register lock bits and fuse register lock bits. If the corresponding OTP word or OTP shadow register is protected by read or write protection, the read, write and reload operations of the OTP operation engine will not succeed.

26.5 OTP Read and Write Protection

The OTP controller supports configuring different read and write protection policies for different types of words in OTP. The actual read and write protection effects of the OTP fuse array, fuse registers, and shadow registers are affected by OTP hardware lock bits, shadow register lock bits and fuse register lock bits at the same time.

Word 0 of the OTP fuse array is the 32-bit hardware lock bit HARD_LOCK[31:0]. Programming HARD_LOCK can apply read or write protection to the specified OTP area. Programming several bits in HARD_LOCK to 1 will apply hardware lock to the corresponding OTP area.

Note that OTP supports 32-bit hardware lock bits HARD_LOCK[31:0], each bit corresponds to 4 words of OTP. The lock bits of OTP fuse registers and shadow registers, each 2 bits correspond to one OTP word or one 32-bit shadow register.

All OTP data can be divided into the following 4 types by word. For different types of OTP words, after setting the corresponding hardware lock bit, the effect of read protection or write protection is different:

- IDENTITY
- SECURE
- SECRET
- GENERAL

Users can refer to the OTP mapping table to understand the type attribution of each OTP word.

The OTP read and write protection policies are summarized as follows:

- Effect of shadow register lock bit

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  - When the lock control bit of the shadow register is set to unlocked, the shadow register can be read and written freely, or the value of the OTP array can be reloaded into the shadow register
  - When the lock control bit of the shadow register is set to single lock, the shadow register is write-protected, and it is not allowed to reload the value of the OTP array into the shadow register
  - When the lock control bit of the shadow register is set to double lock, the shadow register is read-protected and write-protected, and it is not allowed to reload the value of the OTP array into the shadow register
- Effect of fuse register lock bit
  - When the lock control bit of the fuse register is set to unlocked, the fuse register can be read and written freely
  - When the lock control bit of the fuse register is set to single lock, the fuse array is write-protected, that is, it is not allowed to program the fuse array by writing to the fuse register or the OTP operation engine
  - When the lock control bit of the fuse register is set to double lock, the fuse array is read-protected and write-protected, that is, it is not allowed to read or program the fuse array through the fuse register or the OTP operation engine
- Effect of OTP hardware lock bit
  - For IDENTITY type OTP words, when the corresponding hardware lock bit is 1, only write protection is applied, that is, it is allowed to read the OTP word through the fuse register or the OTP operation engine, and it is also allowed to read the shadow register corresponding to the OTP word. But it is not allowed to program the OTP word through the fuse register or the OTP operation engine, nor is it allowed to write or reload the shadow register corresponding to the OTP word.
  - For SECURE type OTP words, when the corresponding hardware lock bit is 1, only write protection is applied, that is, it is allowed to read the OTP word through the fuse register or the OTP operation engine, and it is also allowed to read the shadow register corresponding to the OTP word. But it is not allowed to program the OTP word through the fuse register or the OTP operation engine, nor is it allowed to write or reload the shadow register corresponding to the OTP word. Note: The shadow register corresponding to the product life cycle in the SECURE type OTP word is an exception, users are allowed to write this shadow register to advance the current life cycle (set the product life cycle bit from 0 to 1), but are not allowed to roll back the life cycle (clear the product life cycle bit from 1 to 0)
  - For SECRET type OTP words, when the corresponding hardware lock bit is 1, both read protection and write protection are applied, that is, it is not allowed to read the OTP word through the fuse register or the OTP operation engine, nor is it allowed to read the shadow register corresponding to the OTP word. It is also not allowed to program the OTP word through the fuse register or the OTP operation engine, nor is it allowed to write or reload the shadow register corresponding to the OTP word.
  - For GENERAL type OTP words, when the corresponding hardware lock bit is 1, only write protection is applied to the fuse array, that is, it is allowed to read the OTP word through the fuse register or the OTP operation engine, and it is also allowed to read the shadow register corresponding to the OTP word. It is not allowed to program the OTP word through the fuse

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register or the OTP operation engine, but it is allowed to write or reload the shadow register corresponding to the OTP word.

- For a word in the OTP array, if the read protection or program protection configured by the hardware lock bit and the fuse lock bit conflict, the protection always takes effect. Similarly, for an OTP shadow register, if the read protection or write protection configured by the hardware lock bit and the shadow lock bit conflict, the protection always takes effect.

The following table summarizes the read and write protection effects of different types of OTP words:

IDENTITY Type	Hardware Lock	Fuse Single Lock	Fuse Double Lock	Shadow Single Lock	Shadow Double Lock
Read Fuse	Y	Y	N	Y	Y
Write Fuse	N	N	N	Y	Y
Shadow Register Reload	Y	Y	Y	N	N
Read Shadow Register	Y	Y	Y	Y	N
Write Shadow Register	N	Y	Y	N	N

Table 154: Summary of Read and Write Protection for IDENTITY Type OTP Words

SECURE Type	Hardware Lock	Fuse Single Lock	Fuse Double Lock	Shadow Single Lock	Shadow Double Lock
Read Fuse	Y	Y	N	Y	Y
Write Fuse	N	N	N	Y	Y
Shadow Register Reload	Y	Y	Y	N	N
Read Shadow Register	Y	Y	Y	Y	N
Write Shadow Register	N	Y	Y	N	N

Table 155: Summary of Read and Write Protection for SECURE Type OTP Words

SECRET Type	Hardware Lock	Fuse Single Lock	Fuse Double Lock	Shadow Single Lock	Shadow Double Lock
Read Fuse	N	Y	N	Y	Y
Write Fuse	N	N	N	Y	Y
Shadow Register Reload	Y	Y	Y	N	N
Read Shadow Register	N	Y	Y	Y	N
Write Shadow Register	N	Y	Y	N	N

Table 156: Summary of Read and Write Protection for SECRET Type OTP Words

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GENERAL Type	Hardware Lock	Fuse Single Lock	Fuse Double Lock	Shadow Single Lock	Shadow Double Lock
Read Fuse	Y	Y	N	Y	Y
Write Fuse	N	N	N	Y	Y
Shadow Register Reload	Y	Y	Y	N	N
Read Shadow Register	Y	Y	Y	Y	N
Write Shadow Register	Y	Y	Y	N	N

Table 157: Summary of Read and Write Protection for GENERAL Type OTP Words

26.6 OTP Controller Registers

The OTP register list is as follows:

OTP base address: 0xF3050000

Offset	Register Name	Description	Reset Value
0x0000	SHADOW[SHADOW000]	Fuse shadow registers	0x00000000
0x0004	SHADOW[SHADOW001]	Fuse shadow registers	0x00000000
0x0008	SHADOW[SHADOW002]	Fuse shadow registers	0x00000000
0x000C	SHADOW[SHADOW003]	Fuse shadow registers	0x00000000
0x0010	SHADOW[SHADOW004]	Fuse shadow registers	0x00000000
0x0014	SHADOW[SHADOW005]	Fuse shadow registers	0x00000000
0x0018	SHADOW[SHADOW006]	Fuse shadow registers	0x00000000
0x001C	SHADOW[SHADOW007]	Fuse shadow registers	0x00000000
0x0020	SHADOW[SHADOW008]	Fuse shadow registers	0x00000000
0x0024	SHADOW[SHADOW009]	Fuse shadow registers	0x00000000
0x0028	SHADOW[SHADOW010]	Fuse shadow registers	0x00000000
0x002C	SHADOW[SHADOW011]	Fuse shadow registers	0x00000000
0x0030	SHADOW[SHADOW012]	Fuse shadow registers	0x00000000
0x0034	SHADOW[SHADOW013]	Fuse shadow registers	0x00000000
0x0038	SHADOW[SHADOW014]	Fuse shadow registers	0x00000000
0x003C	SHADOW[SHADOW015]	Fuse shadow registers	0x00000000
0x0040	SHADOW[SHADOW016]	Fuse shadow registers	0x00000000
0x0044	SHADOW[SHADOW017]	Fuse shadow registers	0x00000000
0x0048	SHADOW[SHADOW018]	Fuse shadow registers	0x00000000
0x004C	SHADOW[SHADOW019]	Fuse shadow registers	0x00000000
0x0050	SHADOW[SHADOW020]	Fuse shadow registers	0x00000000
0x0054	SHADOW[SHADOW021]	Fuse shadow registers	0x00000000
0x0058	SHADOW[SHADOW022]	Fuse shadow registers	0x00000000

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Offset	Register Name	Description	Reset Value
0x005C	SHADOW[SHADOW023]	Fuse shadow registers	0x00000000
0x0060	SHADOW[SHADOW024]	Fuse shadow registers	0x00000000
0x0064	SHADOW[SHADOW025]	Fuse shadow registers	0x00000000
0x0068	SHADOW[SHADOW026]	Fuse shadow registers	0x00000000
0x006C	SHADOW[SHADOW027]	Fuse shadow registers	0x00000000
0x0070	SHADOW[SHADOW028]	Fuse shadow registers	0x00000000
0x0074	SHADOW[SHADOW029]	Fuse shadow registers	0x00000000
0x0078	SHADOW[SHADOW030]	Fuse shadow registers	0x00000000
0x007C	SHADOW[SHADOW031]	Fuse shadow registers	0x00000000
0x0080	SHADOW[SHADOW032]	Fuse shadow registers	0x00000000
0x0084	SHADOW[SHADOW033]	Fuse shadow registers	0x00000000
0x0088	SHADOW[SHADOW034]	Fuse shadow registers	0x00000000
0x008C	SHADOW[SHADOW035]	Fuse shadow registers	0x00000000
0x0090	SHADOW[SHADOW036]	Fuse shadow registers	0x00000000
0x0094	SHADOW[SHADOW037]	Fuse shadow registers	0x00000000
0x0098	SHADOW[SHADOW038]	Fuse shadow registers	0x00000000
0x009C	SHADOW[SHADOW039]	Fuse shadow registers	0x00000000
0x00A0	SHADOW[SHADOW040]	Fuse shadow registers	0x00000000
0x00A4	SHADOW[SHADOW041]	Fuse shadow registers	0x00000000
0x00A8	SHADOW[SHADOW042]	Fuse shadow registers	0x00000000
0x00AC	SHADOW[SHADOW043]	Fuse shadow registers	0x00000000
0x00B0	SHADOW[SHADOW044]	Fuse shadow registers	0x00000000
0x00B4	SHADOW[SHADOW045]	Fuse shadow registers	0x00000000
0x00B8	SHADOW[SHADOW046]	Fuse shadow registers	0x00000000
0x00BC	SHADOW[SHADOW047]	Fuse shadow registers	0x00000000
0x00C0	SHADOW[SHADOW048]	Fuse shadow registers	0x00000000
0x00C4	SHADOW[SHADOW049]	Fuse shadow registers	0x00000000
0x00C8	SHADOW[SHADOW050]	Fuse shadow registers	0x00000000
0x00CC	SHADOW[SHADOW051]	Fuse shadow registers	0x00000000
0x00D0	SHADOW[SHADOW052]	Fuse shadow registers	0x00000000
0x00D4	SHADOW[SHADOW053]	Fuse shadow registers	0x00000000
0x00D8	SHADOW[SHADOW054]	Fuse shadow registers	0x00000000
0x00DC	SHADOW[SHADOW055]	Fuse shadow registers	0x00000000
0x00E0	SHADOW[SHADOW056]	Fuse shadow registers	0x00000000
0x00E4	SHADOW[SHADOW057]	Fuse shadow registers	0x00000000
0x00E8	SHADOW[SHADOW058]	Fuse shadow registers	0x00000000
0x00EC	SHADOW[SHADOW059]	Fuse shadow registers	0x00000000
0x00F0	SHADOW[SHADOW060]	Fuse shadow registers	0x00000000
0x00F4	SHADOW[SHADOW061]	Fuse shadow registers	0x00000000
0x00F8	SHADOW[SHADOW062]	Fuse shadow registers	0x00000000

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Offset	Register Name	Description	Reset Value
0x00FC	SHADOW[SHADOW063]	Fuse shadow registers	0x00000000
0x0100	SHADOW[SHADOW064]	Fuse shadow registers	0x00000000
0x0104	SHADOW[SHADOW065]	Fuse shadow registers	0x00000000
0x0108	SHADOW[SHADOW066]	Fuse shadow registers	0x00000000
0x010C	SHADOW[SHADOW067]	Fuse shadow registers	0x00000000
0x0110	SHADOW[SHADOW068]	Fuse shadow registers	0x00000000
0x0114	SHADOW[SHADOW069]	Fuse shadow registers	0x00000000
0x0118	SHADOW[SHADOW070]	Fuse shadow registers	0x00000000
0x011C	SHADOW[SHADOW071]	Fuse shadow registers	0x00000000
0x0120	SHADOW[SHADOW072]	Fuse shadow registers	0x00000000
0x0124	SHADOW[SHADOW073]	Fuse shadow registers	0x00000000
0x0128	SHADOW[SHADOW074]	Fuse shadow registers	0x00000000
0x012C	SHADOW[SHADOW075]	Fuse shadow registers	0x00000000
0x0130	SHADOW[SHADOW076]	Fuse shadow registers	0x00000000
0x0134	SHADOW[SHADOW077]	Fuse shadow registers	0x00000000
0x0138	SHADOW[SHADOW078]	Fuse shadow registers	0x00000000
0x013C	SHADOW[SHADOW079]	Fuse shadow registers	0x00000000
0x0140	SHADOW[SHADOW080]	Fuse shadow registers	0x00000000
0x0144	SHADOW[SHADOW081]	Fuse shadow registers	0x00000000
0x0148	SHADOW[SHADOW082]	Fuse shadow registers	0x00000000
0x014C	SHADOW[SHADOW083]	Fuse shadow registers	0x00000000
0x0150	SHADOW[SHADOW084]	Fuse shadow registers	0x00000000
0x0154	SHADOW[SHADOW085]	Fuse shadow registers	0x00000000
0x0158	SHADOW[SHADOW086]	Fuse shadow registers	0x00000000
0x015C	SHADOW[SHADOW087]	Fuse shadow registers	0x00000000
0x0160	SHADOW[SHADOW088]	Fuse shadow registers	0x00000000
0x0164	SHADOW[SHADOW089]	Fuse shadow registers	0x00000000
0x0168	SHADOW[SHADOW090]	Fuse shadow registers	0x00000000
0x016C	SHADOW[SHADOW091]	Fuse shadow registers	0x00000000
0x0170	SHADOW[SHADOW092]	Fuse shadow registers	0x00000000
0x0174	SHADOW[SHADOW093]	Fuse shadow registers	0x00000000
0x0178	SHADOW[SHADOW094]	Fuse shadow registers	0x00000000
0x017C	SHADOW[SHADOW095]	Fuse shadow registers	0x00000000
0x0180	SHADOW[SHADOW096]	Fuse shadow registers	0x00000000
0x0184	SHADOW[SHADOW097]	Fuse shadow registers	0x00000000
0x0188	SHADOW[SHADOW098]	Fuse shadow registers	0x00000000
0x018C	SHADOW[SHADOW099]	Fuse shadow registers	0x00000000
0x0190	SHADOW[SHADOW100]	Fuse shadow registers	0x00000000
0x0194	SHADOW[SHADOW101]	Fuse shadow registers	0x00000000
0x0198	SHADOW[SHADOW102]	Fuse shadow registers	0x00000000

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Offset	Register Name	Description	Reset Value
0x019C	SHADOW[SHADOW103]	Fuse shadow registers	0x00000000
0x01A0	SHADOW[SHADOW104]	Fuse shadow registers	0x00000000
0x01A4	SHADOW[SHADOW105]	Fuse shadow registers	0x00000000
0x01A8	SHADOW[SHADOW106]	Fuse shadow registers	0x00000000
0x01AC	SHADOW[SHADOW107]	Fuse shadow registers	0x00000000
0x01B0	SHADOW[SHADOW108]	Fuse shadow registers	0x00000000
0x01B4	SHADOW[SHADOW109]	Fuse shadow registers	0x00000000
0x01B8	SHADOW[SHADOW110]	Fuse shadow registers	0x00000000
0x01BC	SHADOW[SHADOW111]	Fuse shadow registers	0x00000000
0x01C0	SHADOW[SHADOW112]	Fuse shadow registers	0x00000000
0x01C4	SHADOW[SHADOW113]	Fuse shadow registers	0x00000000
0x01C8	SHADOW[SHADOW114]	Fuse shadow registers	0x00000000
0x01CC	SHADOW[SHADOW115]	Fuse shadow registers	0x00000000
0x01D0	SHADOW[SHADOW116]	Fuse shadow registers	0x00000000
0x01D4	SHADOW[SHADOW117]	Fuse shadow registers	0x00000000
0x01D8	SHADOW[SHADOW118]	Fuse shadow registers	0x00000000
0x01DC	SHADOW[SHADOW119]	Fuse shadow registers	0x00000000
0x01E0	SHADOW[SHADOW120]	Fuse shadow registers	0x00000000
0x01E4	SHADOW[SHADOW121]	Fuse shadow registers	0x00000000
0x01E8	SHADOW[SHADOW122]	Fuse shadow registers	0x00000000
0x01EC	SHADOW[SHADOW123]	Fuse shadow registers	0x00000000
0x01F0	SHADOW[SHADOW124]	Fuse shadow registers	0x00000000
0x01F4	SHADOW[SHADOW125]	Fuse shadow registers	0x00000000
0x01F8	SHADOW[SHADOW126]	Fuse shadow registers	0x00000000
0x01FC	SHADOW[SHADOW127]	Fuse shadow registers	0x00000000
0x0200	SHADOW_LOCK[LOCK00]	Fuse shadow lock	0x00000000
0x0204	SHADOW_LOCK[LOCK01]	Fuse shadow lock	0x00000000
0x0208	SHADOW_LOCK[LOCK02]	Fuse shadow lock	0x00000000
0x020C	SHADOW_LOCK[LOCK03]	Fuse shadow lock	0x00000000
0x0210	SHADOW_LOCK[LOCK04]	Fuse shadow lock	0x00000000
0x0214	SHADOW_LOCK[LOCK05]	Fuse shadow lock	0x00000000
0x0218	SHADOW_LOCK[LOCK06]	Fuse shadow lock	0x00000000
0x021C	SHADOW_LOCK[LOCK07]	Fuse shadow lock	0x00000000
0x0400	FUSE[FUSE000]	Fuse Array	0x00000000
0x0404	FUSE[FUSE001]	Fuse Array	0x00000000
0x0408	FUSE[FUSE002]	Fuse Array	0x00000000
0x040C	FUSE[FUSE003]	Fuse Array	0x00000000
0x0410	FUSE[FUSE004]	Fuse Array	0x00000000
0x0414	FUSE[FUSE005]	Fuse Array	0x00000000
0x0418	FUSE[FUSE006]	Fuse Array	0x00000000

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Offset	Register Name	Description	Reset Value
0x041C	FUSE[FUSE007]	Fuse Array	0x00000000
0x0420	FUSE[FUSE008]	Fuse Array	0x00000000
0x0424	FUSE[FUSE009]	Fuse Array	0x00000000
0x0428	FUSE[FUSE010]	Fuse Array	0x00000000
0x042C	FUSE[FUSE011]	Fuse Array	0x00000000
0x0430	FUSE[FUSE012]	Fuse Array	0x00000000
0x0434	FUSE[FUSE013]	Fuse Array	0x00000000
0x0438	FUSE[FUSE014]	Fuse Array	0x00000000
0x043C	FUSE[FUSE015]	Fuse Array	0x00000000
0x0440	FUSE[FUSE016]	Fuse Array	0x00000000
0x0444	FUSE[FUSE017]	Fuse Array	0x00000000
0x0448	FUSE[FUSE018]	Fuse Array	0x00000000
0x044C	FUSE[FUSE019]	Fuse Array	0x00000000
0x0450	FUSE[FUSE020]	Fuse Array	0x00000000
0x0454	FUSE[FUSE021]	Fuse Array	0x00000000
0x0458	FUSE[FUSE022]	Fuse Array	0x00000000
0x045C	FUSE[FUSE023]	Fuse Array	0x00000000
0x0460	FUSE[FUSE024]	Fuse Array	0x00000000
0x0464	FUSE[FUSE025]	Fuse Array	0x00000000
0x0468	FUSE[FUSE026]	Fuse Array	0x00000000
0x046C	FUSE[FUSE027]	Fuse Array	0x00000000
0x0470	FUSE[FUSE028]	Fuse Array	0x00000000
0x0474	FUSE[FUSE029]	Fuse Array	0x00000000
0x0478	FUSE[FUSE030]	Fuse Array	0x00000000
0x047C	FUSE[FUSE031]	Fuse Array	0x00000000
0x0480	FUSE[FUSE032]	Fuse Array	0x00000000
0x0484	FUSE[FUSE033]	Fuse Array	0x00000000
0x0488	FUSE[FUSE034]	Fuse Array	0x00000000
0x048C	FUSE[FUSE035]	Fuse Array	0x00000000
0x0490	FUSE[FUSE036]	Fuse Array	0x00000000
0x0494	FUSE[FUSE037]	Fuse Array	0x00000000
0x0498	FUSE[FUSE038]	Fuse Array	0x00000000
0x049C	FUSE[FUSE039]	Fuse Array	0x00000000
0x04A0	FUSE[FUSE040]	Fuse Array	0x00000000
0x04A4	FUSE[FUSE041]	Fuse Array	0x00000000
0x04A8	FUSE[FUSE042]	Fuse Array	0x00000000
0x04AC	FUSE[FUSE043]	Fuse Array	0x00000000
0x04B0	FUSE[FUSE044]	Fuse Array	0x00000000
0x04B4	FUSE[FUSE045]	Fuse Array	0x00000000
0x04B8	FUSE[FUSE046]	Fuse Array	0x00000000

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Offset	Register Name	Description	Reset Value
0x04BC	FUSE[FUSE047]	Fuse Array	0x00000000
0x04C0	FUSE[FUSE048]	Fuse Array	0x00000000
0x04C4	FUSE[FUSE049]	Fuse Array	0x00000000
0x04C8	FUSE[FUSE050]	Fuse Array	0x00000000
0x04CC	FUSE[FUSE051]	Fuse Array	0x00000000
0x04D0	FUSE[FUSE052]	Fuse Array	0x00000000
0x04D4	FUSE[FUSE053]	Fuse Array	0x00000000
0x04D8	FUSE[FUSE054]	Fuse Array	0x00000000
0x04DC	FUSE[FUSE055]	Fuse Array	0x00000000
0x04E0	FUSE[FUSE056]	Fuse Array	0x00000000
0x04E4	FUSE[FUSE057]	Fuse Array	0x00000000
0x04E8	FUSE[FUSE058]	Fuse Array	0x00000000
0x04EC	FUSE[FUSE059]	Fuse Array	0x00000000
0x04F0	FUSE[FUSE060]	Fuse Array	0x00000000
0x04F4	FUSE[FUSE061]	Fuse Array	0x00000000
0x04F8	FUSE[FUSE062]	Fuse Array	0x00000000
0x04FC	FUSE[FUSE063]	Fuse Array	0x00000000
0x0500	FUSE[FUSE064]	Fuse Array	0x00000000
0x0504	FUSE[FUSE065]	Fuse Array	0x00000000
0x0508	FUSE[FUSE066]	Fuse Array	0x00000000
0x050C	FUSE[FUSE067]	Fuse Array	0x00000000
0x0510	FUSE[FUSE068]	Fuse Array	0x00000000
0x0514	FUSE[FUSE069]	Fuse Array	0x00000000
0x0518	FUSE[FUSE070]	Fuse Array	0x00000000
0x051C	FUSE[FUSE071]	Fuse Array	0x00000000
0x0520	FUSE[FUSE072]	Fuse Array	0x00000000
0x0524	FUSE[FUSE073]	Fuse Array	0x00000000
0x0528	FUSE[FUSE074]	Fuse Array	0x00000000
0x052C	FUSE[FUSE075]	Fuse Array	0x00000000
0x0530	FUSE[FUSE076]	Fuse Array	0x00000000
0x0534	FUSE[FUSE077]	Fuse Array	0x00000000
0x0538	FUSE[FUSE078]	Fuse Array	0x00000000
0x053C	FUSE[FUSE079]	Fuse Array	0x00000000
0x0540	FUSE[FUSE080]	Fuse Array	0x00000000
0x0544	FUSE[FUSE081]	Fuse Array	0x00000000
0x0548	FUSE[FUSE082]	Fuse Array	0x00000000
0x054C	FUSE[FUSE083]	Fuse Array	0x00000000
0x0550	FUSE[FUSE084]	Fuse Array	0x00000000
0x0554	FUSE[FUSE085]	Fuse Array	0x00000000
0x0558	FUSE[FUSE086]	Fuse Array	0x00000000

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Offset	Register Name	Description	Reset Value
0x055C	FUSE[FUSE087]	Fuse Array	0x00000000
0x0560	FUSE[FUSE088]	Fuse Array	0x00000000
0x0564	FUSE[FUSE089]	Fuse Array	0x00000000
0x0568	FUSE[FUSE090]	Fuse Array	0x00000000
0x056C	FUSE[FUSE091]	Fuse Array	0x00000000
0x0570	FUSE[FUSE092]	Fuse Array	0x00000000
0x0574	FUSE[FUSE093]	Fuse Array	0x00000000
0x0578	FUSE[FUSE094]	Fuse Array	0x00000000
0x057C	FUSE[FUSE095]	Fuse Array	0x00000000
0x0580	FUSE[FUSE096]	Fuse Array	0x00000000
0x0584	FUSE[FUSE097]	Fuse Array	0x00000000
0x0588	FUSE[FUSE098]	Fuse Array	0x00000000
0x058C	FUSE[FUSE099]	Fuse Array	0x00000000
0x0590	FUSE[FUSE100]	Fuse Array	0x00000000
0x0594	FUSE[FUSE101]	Fuse Array	0x00000000
0x0598	FUSE[FUSE102]	Fuse Array	0x00000000
0x059C	FUSE[FUSE103]	Fuse Array	0x00000000
0x05A0	FUSE[FUSE104]	Fuse Array	0x00000000
0x05A4	FUSE[FUSE105]	Fuse Array	0x00000000
0x05A8	FUSE[FUSE106]	Fuse Array	0x00000000
0x05AC	FUSE[FUSE107]	Fuse Array	0x00000000
0x05B0	FUSE[FUSE108]	Fuse Array	0x00000000
0x05B4	FUSE[FUSE109]	Fuse Array	0x00000000
0x05B8	FUSE[FUSE110]	Fuse Array	0x00000000
0x05BC	FUSE[FUSE111]	Fuse Array	0x00000000
0x05C0	FUSE[FUSE112]	Fuse Array	0x00000000
0x05C4	FUSE[FUSE113]	Fuse Array	0x00000000
0x05C8	FUSE[FUSE114]	Fuse Array	0x00000000
0x05CC	FUSE[FUSE115]	Fuse Array	0x00000000
0x05D0	FUSE[FUSE116]	Fuse Array	0x00000000
0x05D4	FUSE[FUSE117]	Fuse Array	0x00000000
0x05D8	FUSE[FUSE118]	Fuse Array	0x00000000
0x05DC	FUSE[FUSE119]	Fuse Array	0x00000000
0x05E0	FUSE[FUSE120]	Fuse Array	0x00000000
0x05E4	FUSE[FUSE121]	Fuse Array	0x00000000
0x05E8	FUSE[FUSE122]	Fuse Array	0x00000000
0x05EC	FUSE[FUSE123]	Fuse Array	0x00000000
0x05F0	FUSE[FUSE124]	Fuse Array	0x00000000
0x05F4	FUSE[FUSE125]	Fuse Array	0x00000000
0x05F8	FUSE[FUSE126]	Fuse Array	0x00000000

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Offset	Register Name	Description	Reset Value
0x05FC	FUSE[FUSE127]	Fuse Array	0x00000000
0x0600	FUSE_LOCK[LOCK00]	Fuse lock	0x00000000
0x0604	FUSE_LOCK[LOCK01]	Fuse lock	0x00000000
0x0608	FUSE_LOCK[LOCK02]	Fuse lock	0x00000000
0x060C	FUSE_LOCK[LOCK03]	Fuse lock	0x00000000
0x0610	FUSE_LOCK[LOCK04]	Fuse lock	0x00000000
0x0614	FUSE_LOCK[LOCK05]	Fuse lock	0x00000000
0x0618	FUSE_LOCK[LOCK06]	Fuse lock	0x00000000
0x061C	FUSE_LOCK[LOCK07]	Fuse lock	0x00000000
0x0800	UNLOCK	UNLOCK	0x00000000
0x0804	DATA	DATA	0x00000000
0x0808	ADDR	ADDR	0x00000000
0x080C	CMD	CMD	0x00000000
0x0A00	LOAD_REQ	LOAD Request	0x00000007
0x0A04	LOAD_COMP	LOAD complete	0x00000007
0x0A20	REGION[LOAD_REGION0]	LOAD region	0x00000800
0x0A24	REGION[LOAD_REGION1]	LOAD region	0x00001008
0x0A28	REGION[LOAD_REGION2]	LOAD region	0x00000010
0x0A2C	REGION[LOAD_REGION3]	LOAD region	0x00000000
0x0C00	INT_FLAG	interrupt flag	0x00000000
0x0C04	INT_EN	interrupt enable	0x00000000

Table 158: OTP Register List

26.7 OTP Controller Register Details

The detailed description of OTP registers is as follows:

26.7.1 SHADOW (0x0 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SHADOW																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

SHADOW [31:0]

Bits	Name	Description
31-0	SHADOW	shadow register of fuse for pmic area for PMIC, index valid for 0-15, for SOC index valid for 16-128

SHADOW Bits

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26.7.2 SHADOW_LOCK (0x200 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LOCK																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

SHADOW_LOCK [31:0]

Bits	Name	Description
31-0	LOCK	lock for pmic part shadow registers, 2 bits per 32 bit word, lock behavior is different between different fuse types 00: not locked 01: soft locked 10: not locked, and cannot lock in furture 11: double locked

SHADOW_LOCK Bits

26.7.3 FUSE (0x400 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FUSE																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

FUSE [31:0]

Bits	Name	Description
31-0	FUSE	fuse array, valid in PMIC part only read operation will read out value in fuse array write operation will update fuse array value(please make sure fuse is unlocked and 2.5V power is ready)

FUSE Bits

26.7.4 FUSE_LOCK (0x600 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LOCK																															
RW																															

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

FUSE_LOCK [31:0]

Bits	Name	Description
31-0	LOCK	lock for fuse array, 2 bits per 32 bit word, lock behavior is different between different fuse types 00: not locked 01: soft locked 10: not locked, and cannot lock in furture 11: double locked

FUSE_LOCK Bits

26.7.5 UNLOCK (0x800)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
UNLOCK																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

UNLOCK [31:0]

Bits	Name	Description
31-0	UNLOCK	unlock word for fuse array operation write ”OPEN” to unlock fuse array, write any other value will lock write to fuse. Please make sure 24M crystal is running and 2.5V LDO working properly

UNLOCK Bits

26.7.6 DATA (0x804)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DATA [31:0]

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Bits	Name	Description
31-0	DATA	data register for non-blocking access this register hold dat read from fuse array or data to by programmed to fuse array

DATA Bits

26.7.7 ADDR (0x808)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																									ADDR						
N/A																									RW						
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0

ADDR [31:0]

Bits	Name	Description
6-0	ADDR	word address to be read or write

ADDR Bits

26.7.8 CMD (0x80C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMD																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CMD [31:0]

Bits	Name	Description
31-0	CMD	command to access fure array ”BLOW” will update fuse word at ADDR to value hold in DATA ”READ” will fetch fuse value in at ADDR to DATA register

CMD Bits

26.7.9 LOAD_REQ (0xA00)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																												REQUEST			

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
N/A																												RW			
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	1	1	1

LOAD_REQ [31:0]

Bits	Name	Description
3-0	REQUEST	reload request for 4 regions bit0: region0 bit1: region1 bit2: region2 bit3: region3

LOAD_REQ Bits

26.7.10 LOAD_COMP (0xA04)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																												COMPLETE			
N/A																												RW			
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	1	1	1

LOAD_COMP [31:0]

Bits	Name	Description
3-0	COMPLETE	reload complete sign for 4 regions bit0: region 0 bit1: region1 bit2: region2 bit3: region3

LOAD_COMP Bits

26.7.11 REGION (0xA20 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																	STOP							RSVD	START						
N/A																	RW							N/A	RW						
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	x	0	0	0	0	0	0	0

REGION [31:0]

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Bits	Name	Description
14-8	STOP	stop address of load region, fuse word at end address will NOT be reloaded region0: fixed at 8 region1: fixed at 16 region2: fixed at 0, region3: usrer configurable
6-0	START	start address of load region, fuse word at start address will be reloaded region0: fixed at 0 region1: fixed at 8 region2: fixed at 16, region3: usrer configurable

REGION Bits

26.7.12 INT_FLAG (0xC00)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																													WRITE	READ	LOAD
N/A																													RW	RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0

INT_FLAG [31:0]

Bits	Name	Description
2	WRITE	fuse write flag, write 1 to clear 0: fuse is not written or writing 1: value in DATA register is programmed into fuse
1	READ	fuse read flag, write 1 to clear 0: fuse is not read or reading 1: fuse value is put in DATA register
0	LOAD	fuse load flag, write 1 to clear 0: fuse is not loaded or loading 1: fuse loaded

INT_FLAG Bits

26.7.13 INT_EN (0xC04)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																													WRITE	READ	LOAD

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
N/A																													RW	RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0

INT_EN [31:0]

Bits	Name	Description
2	WRITE	fuse write interrupt enable 0: fuse write interrupt is not enable 1: fuse write interrupt is enable
1	READ	fuse read interrupt enable 0: fuse read interrupt is not enable 1: fuse read interrupt is enable
0	LOAD	fuse load interrupt enable 0: fuse load interrupt is not enable 1: fuse load interrupt is enable

INT_EN Bits

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27 Overview of DMA, DMAMUX and Mailbox

This chapter describes the DMA controller, DMAMUX and communication mailbox MBX.

27.1 DMA Controller

This product integrates 1 DMA controller.

- HDMA, which is connected to the AHB peripheral bus as a master device. It enables real-time data transfer between peripheral registers and memory, as well as between memories.

HDMA has 32 DMA request input signals, and these DMA request sources come from the DMA Request Multiplexer (DMAMUX).

27.2 DMA Request Multiplexer

DMA Request Multiplexer (DMAMUX) allocates DMA requests from each peripheral module to 32 output signals, which serve as the DMA transfer request sources for HDMA.

Users can connect DMA requests from specific peripherals to the 32 channels of HDMA by configuring the DMAMUX registers.

The specific allocation of DMAMUX inputs is as follows:

DMA MUX Index	DMA MUX SOURCE Name	Description
0	GPTMR0_0	GPTMR0 Channel 0
1	GPTMR0_1	GPTMR0 Channel 1
2	GPTMR0_2	GPTMR0 Channel 2
3	GPTMR0_3	GPTMR0 Channel 3
4	GPTMR1_0	GPTMR1 Channel 0
5	GPTMR1_1	GPTMR1 Channel 1
6	GPTMR1_2	GPTMR1 Channel 2
7	GPTMR1_3	GPTMR1 Channel 3
8	GPTMR2_0	GPTMR2 channel 0
9	GPTMR2_1	GPTMR2 Channel 1
10	GPTMR2_2	GPTMR2 Channel 2
11	GPTMR2_3	GPTMR2 Channel 3
12	GPTMR3_0	GPTMR3 channel 0
13	GPTMR3_1	GPTMR3 Channel 1
14	GPTMR3_2	GPTMR3 Channel 2
15	GPTMR3_3	GPTMR3 Channel 3
20	UART0_RX	UART0 Receive
21	UART0_TX	UART0 Transmit
22	UART1_RX	UART1 Receive
23	UART1_TX	UART1 Transmit
24	UART2_RX	UART2 Receive
25	UART2_TX	UART2 Transmit

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DMA MUX Index	DMA MUX SOURCE Name	Description
26	UART3_RX	UART3 Receive
27	UART3_TX	UART3 Transmit
28	UART4_RX	UART4 Receive
29	UART4_TX	UART4 Transmit
30	UART5_RX	UART5 Receive
31	UART5_TX	UART5 Transmit
32	UART6_RX	UART6 Receive
33	UART6_TX	UART6 Transmit
34	UART7_RX	UART7 Receive
35	UART7_TX	UART7 Transmit
36	I2C0	I2C0
37	I2C1	I2C1
38	I2C2	I2C2
39	I2C3	I2C3
40	SPI0_RX	SPI0 Receive
41	SPI0_TX	SPI0 Transmit
42	SPI1_RX	SPI1 Receive
43	SPI1_TX	SPI1 Transmit
44	SPI2_RX	SPI2 Receive
45	SPI2_TX	SPI2 Transmit
46	SPI3_RX	SPI3 Receive
47	SPI3_TX	SPI3 Transmit
48	MCAN0	CAN0
49	MCAN1	CAN1
50	MCAN2	CAN2
51	MCAN3	CAN3
52	MOT_0	TRGM0 DMA request 0 (contains DMA requests for PWM0, PWM1, QEI0, QEI1, MMC0, MMC1, SEI0, SEI1)
53	MOT_1	TRGM0 DMA request 1 (contains DMA requests for PWM0, PWM1, QEI0, QEI1, MMC0, MMC1, SEI0, SEI1)
54	MOT_2	TRGM0 DMA request 2 (contains DMA requests for PWM0, PWM1, QEI0, QEI1, MMC0, MMC1, SEI0, SEI1)
55	MOT_3	TRGM0 DMA request 3 (contains DMA requests for PWM0, PWM1, QEI0, QEI1, MMC0, MMC1, SEI0, SEI1)
56	MOT_4	TRGM0 DMA request 4 (contains DMA requests for PWM0, PWM1, QEI0, QEI1, MMC0, MMC1, SEI0, SEI1)

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DMA MUX Index	DMA MUX SOURCE Name	Description
57	MOT_5	TRGM0 DMA request 5 (contains DMA requests for PWM0, PWM1, QEI0, QEI1, MMC0, MMC1, SEI0, SEI1)
58	MOT_6	TRGM0 DMA request 6 (contains DMA requests for PWM0, PWM1, QEI0, QEI1, MMC0, MMC1, SEI0, SEI1)
59	MOT_7	TRGM0 DMA request 7 (contains DMA requests for PWM0, PWM1, QEI0, QEI1, MMC0, MMC1, SEI0, SEI1)
60	XPI0_RX	XPI0 Receive
61	XPI0_TX	XPI0 Transmit
62	DAC0	DAC0
63	DAC1	DAC1
64	ACMP0	ACMP0
65	ACMP1	ACMP1

Table 159: DMA MUX List

27.3 Communication Mailbox MBX

This product supports 1 communication mailbox MBX for inter-process communication. Processors can send data to each other using MBX, which supports interrupt generation.

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28 DMA Controller

This chapter introduces the functions and features of the DMA controller.

28.1 Overview

The main features of the DMA controller are as follows:

- Supports 32 configurable channels
- Channels support 2-level priority configuration
- Channels with the same priority use Round-Robin arbitration
- Supports chained connection of multiple DMA tasks
- Supports infinite loop

28.2 DMA Architecture Block Diagram

The structure diagram of this module is shown in Figure 24:

In the DMA module, dma_apb_csr is the control register module, the dma_ahb_ahbmst module implements

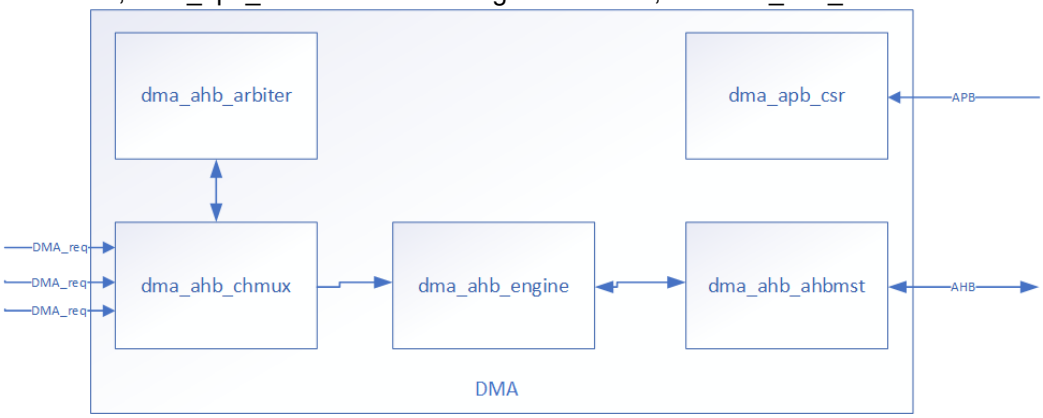


Figure 24: DMA Structure Block Diagram

the AHB master interface, the DMA_ahb_engine module implements the DMA operation engine state machine, and the dma_ahb_arbiter and dma_ahb_chmux modules implement arbitration and selection of multiple DMA requests.

28.3 Pin Description

Table 160: DMA Pin Description

Pin	Direction	Function Description
dma_req[31:0]	Input	Input DMA request signal
dma_ack[31:0]	Output	Output DMA response signal

28.4 Function Description

The DMA controller supports 32 channels, and each channel can independently configure data transmission parameters. Users can use multiple DMA channels at the same time, and the DMA controller will alternately perform data transmission tasks of different channels, instead of waiting for one channel to complete transmission

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before responding to the transmission request of the next channel. That is to say, the DMA controller may complete the data transmission of a single channel in multiple times, and after all tasks of this channel are completed, it will set the corresponding flag bit in the status register.

The data transmission of a DMA channel may be interrupted due to a data transmission error, or it may be aborted by software. In this case, the DMA controller will stop transmitting data and set the corresponding flag bit in the status register.

28.4.1 DMA Hardware Handshake

The DMA controller supports 32 pairs of request-response handshake signals. The request signal comes from the functional module and is routed to the DMA controller via DMAMUX. The DMA controller uses a specified channel to execute the request according to the configuration, and sends a response signal to DMAMUX after completing the data transmission with the length of SrcBurstSize*SrcWidth to complete the hardware handshake.

28.4.2 DMA Chained Transmission

DMA supports chained transmission, which can continuously complete multiple transmission tasks with different configurations without processor intervention.

To use the chained transmission function of DMA, users need to pre-construct a series of interconnected DMA task descriptor lists in memory. The first element of the list is the control register of the DMA channel, and the remaining task descriptors are stored in memory. The ChnLLPointer control word in the DMA task descriptor is the pointer to the next task descriptor of DMA channel n. If the value of ChnLLPointer is non-zero, the DMA controller will fetch the next task descriptor from the address pointed to by ChnLLPointer after completing the corresponding task of the current task descriptor. If the value of ChnLLPointer is 0, it means that this task descriptor is the last one in the list and the chain is terminated.

Table 161 lists the format of the task descriptor. The storage address of the descriptor needs to be 8-byte aligned, and the descriptor itself should not cross the 4K boundary.

28.4.3 Infinite Loop

DMA supports infinite loop mode. Before the hardware starts transmission, it will record the currently configured source address/destination address/transmission length. After the current transmission length is completed, it will reset the source address/destination address/transmission length and continue transmission.

Note: Chained transmission is not supported in infinite loop mode.

28.4.4 Data Order

The DMA controller supports 3 address modes: increment mode, decrement mode and fixed mode.

- Increment mode: After each DMA access to the source or destination address, the next access will target the address after the current source or destination address
- Decrement mode: After each DMA access to the source or destination address, the next access will target the address before the current source or destination address
- Fixed mode: After each DMA access to the source or destination address, the source or destination address for the next access remains unchanged

When the mode configuration of the source and destination addresses is the same, the DMA will keep the

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Table 161: DMA Chained Task Descriptor

Name	Address Offset	Description
Ctrl	0x00	Channel control, the format is the same as the CTRL register
TranSize	0x04	Total transmission length, the format is the same as the TRANSIZE register
SrcAddrL	0x08	Source address L, the format is the same as the SRCADDRL register
SrcAddrH	0x0C	Source address H, the format is the same as the SRCADDRH register
DstAddrL	0x10	Destination address L, the format is the same as the DSTADDRL register
DstAddrH	0x14	Destination address H, the format is the same as the DSTADDRH register
LLPointerL	0x18	Linked list pointer L, the format is the same as the LLPOINTERL register
LLPointerH	0x1C	Linked list pointer H, the format is the same as the LLPOINTERH register

byte order of the data in the source and destination areas unchanged. However, if the address configurations of the source and destination are opposite, the byte order of the data between the source and destination addresses will also be reversed. The byte order in fixed mode is the same as that in increment mode.

28.5 DMA Registers

The DMA register list is as follows:

HDMA base address: 0xF00C8000

Offset	Register Name	Description	Reset Value
0x0004	IDMISC	ID Misc	0x00000000
0x0010	DMACFG	DMAC Configuration Register	0x00000000
0x0014	DMACTRL	DMAC Control Register	0x00000000
0x0018	CHABORT	Channel Abort Register	0x00000000
0x0024	INTHALFSTS	Harlf Complete Interrupt Status	0x00000000
0x0028	INTTCSTS	Trans Complete Interrupt Status Register	0x00000000
0x002C	INTABORTSTS	Abort Interrupt Status Register	0x00000000
0x0030	INTERRSTS	Error Interrupt Status Register	0x00000000
0x0034	CHEN	Channel Enable Register	0x00000000
0x0040	CHCTRL[CH0][CTRL]	Channel ch0 Control Register	0x00000000
0x0044	CHCTRL[CH0][TRANSIZE]	Channel ch0Transfer Size Register	0x00000000

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Offset	Register Name	Description	Reset Value
0x0048	CHCTRL[CH0][SRCADDR]	Channel ch0 Source Address Low Part Register	0x00000000
0x004C	CHCTRL[CH0][CHANREQCTRL]	Channel ch0 DMA Request Control Register	0x00000000
0x0050	CHCTRL[CH0][DSTADDR]	Channel ch0 Destination Address Low Part Register	0x00000000
0x0058	CHCTRL[CH0][LLPOINTER]	Channel ch0 Linked List Pointer Low Part Register	0x00000000
0x0060	CHCTRL[CH1][CTRL]	Channel ch1 Control Register	0x00000000
0x0064	CHCTRL[CH1][TRANSIZE]	Channel ch1Transfer Size Register	0x00000000
0x0068	CHCTRL[CH1][SRCADDR]	Channel ch1 Source Address Low Part Register	0x00000000
0x006C	CHCTRL[CH1][CHANREQCTRL]	Channel ch1 DMA Request Control Register	0x00000000
0x0070	CHCTRL[CH1][DSTADDR]	Channel ch1 Destination Address Low Part Register	0x00000000
0x0078	CHCTRL[CH1][LLPOINTER]	Channel ch1 Linked List Pointer Low Part Register	0x00000000
0x0080	CHCTRL[CH2][CTRL]	Channel ch2 Control Register	0x00000000
0x0084	CHCTRL[CH2][TRANSIZE]	Channel ch2Transfer Size Register	0x00000000
0x0088	CHCTRL[CH2][SRCADDR]	Channel ch2 Source Address Low Part Register	0x00000000
0x008C	CHCTRL[CH2][CHANREQCTRL]	Channel ch2 DMA Request Control Register	0x00000000
0x0090	CHCTRL[CH2][DSTADDR]	Channel ch2 Destination Address Low Part Register	0x00000000
0x0098	CHCTRL[CH2][LLPOINTER]	Channel ch2 Linked List Pointer Low Part Register	0x00000000
0x00A0	CHCTRL[CH3][CTRL]	Channel ch3 Control Register	0x00000000
0x00A4	CHCTRL[CH3][TRANSIZE]	Channel ch3Transfer Size Register	0x00000000
0x00A8	CHCTRL[CH3][SRCADDR]	Channel ch3 Source Address Low Part Register	0x00000000
0x00AC	CHCTRL[CH3][CHANREQCTRL]	Channel ch3 DMA Request Control Register	0x00000000
0x00B0	CHCTRL[CH3][DSTADDR]	Channel ch3 Destination Address Low Part Register	0x00000000
0x00B8	CHCTRL[CH3][LLPOINTER]	Channel ch3 Linked List Pointer Low Part Register	0x00000000

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Offset	Register Name	Description	Reset Value
0x00C0	CHCTRL[CH4][CTRL]	Channel ch4 Control Register	0x00000000
0x00C4	CHCTRL[CH4][TRANSIZE]	Channel ch4Transfer Size Register	0x00000000
0x00C8	CHCTRL[CH4][SRCADDR]	Channel ch4 Source Address Low Part Register	0x00000000
0x00CC	CHCTRL[CH4][CHANREQCTRL]	Channel ch4 DMA Request Control Register	0x00000000
0x00D0	CHCTRL[CH4][DSTADDR]	Channel ch4 Destination Address Low Part Register	0x00000000
0x00D8	CHCTRL[CH4][LLPOINTER]	Channel ch4 Linked List Pointer Low Part Register	0x00000000
0x00E0	CHCTRL[CH5][CTRL]	Channel ch5 Control Register	0x00000000
0x00E4	CHCTRL[CH5][TRANSIZE]	Channel ch5Transfer Size Register	0x00000000
0x00E8	CHCTRL[CH5][SRCADDR]	Channel ch5 Source Address Low Part Register	0x00000000
0x00EC	CHCTRL[CH5][CHANREQCTRL]	Channel ch5 DMA Request Control Register	0x00000000
0x00F0	CHCTRL[CH5][DSTADDR]	Channel ch5 Destination Address Low Part Register	0x00000000
0x00F8	CHCTRL[CH5][LLPOINTER]	Channel ch5 Linked List Pointer Low Part Register	0x00000000
0x0100	CHCTRL[CH6][CTRL]	Channel ch6 Control Register	0x00000000
0x0104	CHCTRL[CH6][TRANSIZE]	Channel ch6Transfer Size Register	0x00000000
0x0108	CHCTRL[CH6][SRCADDR]	Channel ch6 Source Address Low Part Register	0x00000000
0x010C	CHCTRL[CH6][CHANREQCTRL]	Channel ch6 DMA Request Control Register	0x00000000
0x0110	CHCTRL[CH6][DSTADDR]	Channel ch6 Destination Address Low Part Register	0x00000000
0x0118	CHCTRL[CH6][LLPOINTER]	Channel ch6 Linked List Pointer Low Part Register	0x00000000
0x0120	CHCTRL[CH7][CTRL]	Channel ch7 Control Register	0x00000000
0x0124	CHCTRL[CH7][TRANSIZE]	Channel ch7Transfer Size Register	0x00000000
0x0128	CHCTRL[CH7][SRCADDR]	Channel ch7 Source Address Low Part Register	0x00000000
0x012C	CHCTRL[CH7][CHANREQCTRL]	Channel ch7 DMA Request Control Register	0x00000000

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Offset	Register Name	Description	Reset Value
0x0130	CHCTRL[CH7][DSTADDR]	Channel ch7 Destination Address Low Part Register	0x00000000
0x0138	CHCTRL[CH7][LLPOINTER]	Channel ch7 Linked List Pointer Low Part Register	0x00000000
0x0140	CHCTRL[CH8][CTRL]	Channel ch8 Control Register	0x00000000
0x0144	CHCTRL[CH8][TRANSIZE]	Channel ch8Transfer Size Register	0x00000000
0x0148	CHCTRL[CH8][SRCADDR]	Channel ch8 Source Address Low Part Register	0x00000000
0x014C	CHCTRL[CH8][CHANREQCTRL]	Channel ch8 DMA Request Control Register	0x00000000
0x0150	CHCTRL[CH8][DSTADDR]	Channel ch8 Destination Address Low Part Register	0x00000000
0x0158	CHCTRL[CH8][LLPOINTER]	Channel ch8 Linked List Pointer Low Part Register	0x00000000
0x0160	CHCTRL[CH9][CTRL]	Channel ch9 Control Register	0x00000000
0x0164	CHCTRL[CH9][TRANSIZE]	Channel ch9Transfer Size Register	0x00000000
0x0168	CHCTRL[CH9][SRCADDR]	Channel ch9 Source Address Low Part Register	0x00000000
0x016C	CHCTRL[CH9][CHANREQCTRL]	Channel ch9 DMA Request Control Register	0x00000000
0x0170	CHCTRL[CH9][DSTADDR]	Channel ch9 Destination Address Low Part Register	0x00000000
0x0178	CHCTRL[CH9][LLPOINTER]	Channel ch9 Linked List Pointer Low Part Register	0x00000000
0x0180	CHCTRL[CH10][CTRL]	Channel ch10 Control Register	0x00000000
0x0184	CHCTRL[CH10][TRANSIZE]	Channel ch10Transfer Size Register	0x00000000
0x0188	CHCTRL[CH10][SRCADDR]	Channel ch10 Source Address Low Part Register	0x00000000
0x018C	CHCTRL[CH10][CHANREQCTRL]	Channel ch10 DMA Request Control Register	0x00000000
0x0190	CHCTRL[CH10][DSTADDR]	Channel ch10 Destination Address Low Part Register	0x00000000
0x0198	CHCTRL[CH10][LLPOINTER]	Channel ch10 Linked List Pointer Low Part Register	0x00000000
0x01A0	CHCTRL[CH11][CTRL]	Channel ch11 Control Register	0x00000000
0x01A4	CHCTRL[CH11][TRANSIZE]	Channel ch11Transfer Size Register	0x00000000

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Offset	Register Name	Description	Reset Value
0x01A8	CHCTRL[CH11][SRCADDR]	Channel ch11 Source Address Low Part Register	0x00000000
0x01AC	CHCTRL[CH11][CHANREQCTRL]	Channel ch11 DMA Request Control Register	0x00000000
0x01B0	CHCTRL[CH11][DSTADDR]	Channel ch11 Destination Address Low Part Register	0x00000000
0x01B8	CHCTRL[CH11][LLPOINTER]	Channel ch11 Linked List Pointer Low Part Register	0x00000000
0x01C0	CHCTRL[CH12][CTRL]	Channel ch12 Control Register	0x00000000
0x01C4	CHCTRL[CH12][TRANSIZE]	Channel ch12Transfer Size Register	0x00000000
0x01C8	CHCTRL[CH12][SRCADDR]	Channel ch12 Source Address Low Part Register	0x00000000
0x01CC	CHCTRL[CH12][CHANREQCTRL]	Channel ch12 DMA Request Control Register	0x00000000
0x01D0	CHCTRL[CH12][DSTADDR]	Channel ch12 Destination Address Low Part Register	0x00000000
0x01D8	CHCTRL[CH12][LLPOINTER]	Channel ch12 Linked List Pointer Low Part Register	0x00000000
0x01E0	CHCTRL[CH13][CTRL]	Channel ch13 Control Register	0x00000000
0x01E4	CHCTRL[CH13][TRANSIZE]	Channel ch13Transfer Size Register	0x00000000
0x01E8	CHCTRL[CH13][SRCADDR]	Channel ch13 Source Address Low Part Register	0x00000000
0x01EC	CHCTRL[CH13][CHANREQCTRL]	Channel ch13 DMA Request Control Register	0x00000000
0x01F0	CHCTRL[CH13][DSTADDR]	Channel ch13 Destination Address Low Part Register	0x00000000
0x01F8	CHCTRL[CH13][LLPOINTER]	Channel ch13 Linked List Pointer Low Part Register	0x00000000
0x0200	CHCTRL[CH14][CTRL]	Channel ch14 Control Register	0x00000000
0x0204	CHCTRL[CH14][TRANSIZE]	Channel ch14Transfer Size Register	0x00000000
0x0208	CHCTRL[CH14][SRCADDR]	Channel ch14 Source Address Low Part Register	0x00000000
0x020C	CHCTRL[CH14][CHANREQCTRL]	Channel ch14 DMA Request Control Register	0x00000000
0x0210	CHCTRL[CH14][DSTADDR]	Channel ch14 Destination Address Low Part Register	0x00000000
0x0218	CHCTRL[CH14][LLPOINTER]	Channel ch14 Linked List Pointer Low Part Register	0x00000000

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Offset	Register Name	Description	Reset Value
0x0220	CHCTRL[CH15][CTRL]	Channel ch15 Control Register	0x00000000
0x0224	CHCTRL[CH15][TRANSIZE]	Channel ch15Transfer Size Register	0x00000000
0x0228	CHCTRL[CH15][SRCADDR]	Channel ch15 Source Address Low Part Register	0x00000000
0x022C	CHCTRL[CH15][CHANREQCTRL]	Channel ch15 DMA Request Control Register	0x00000000
0x0230	CHCTRL[CH15][DSTADDR]	Channel ch15 Destination Address Low Part Register	0x00000000
0x0238	CHCTRL[CH15][LLPOINTER]	Channel ch15 Linked List Pointer Low Part Register	0x00000000
0x0240	CHCTRL[CH16][CTRL]	Channel ch16 Control Register	0x00000000
0x0244	CHCTRL[CH16][TRANSIZE]	Channel ch16Transfer Size Register	0x00000000
0x0248	CHCTRL[CH16][SRCADDR]	Channel ch16 Source Address Low Part Register	0x00000000
0x024C	CHCTRL[CH16][CHANREQCTRL]	Channel ch16 DMA Request Control Register	0x00000000
0x0250	CHCTRL[CH16][DSTADDR]	Channel ch16 Destination Address Low Part Register	0x00000000
0x0258	CHCTRL[CH16][LLPOINTER]	Channel ch16 Linked List Pointer Low Part Register	0x00000000
0x0260	CHCTRL[CH17][CTRL]	Channel ch17 Control Register	0x00000000
0x0264	CHCTRL[CH17][TRANSIZE]	Channel ch17Transfer Size Register	0x00000000
0x0268	CHCTRL[CH17][SRCADDR]	Channel ch17 Source Address Low Part Register	0x00000000
0x026C	CHCTRL[CH17][CHANREQCTRL]	Channel ch17 DMA Request Control Register	0x00000000
0x0270	CHCTRL[CH17][DSTADDR]	Channel ch17 Destination Address Low Part Register	0x00000000
0x0278	CHCTRL[CH17][LLPOINTER]	Channel ch17 Linked List Pointer Low Part Register	0x00000000
0x0280	CHCTRL[CH18][CTRL]	Channel ch18 Control Register	0x00000000
0x0284	CHCTRL[CH18][TRANSIZE]	Channel ch18Transfer Size Register	0x00000000
0x0288	CHCTRL[CH18][SRCADDR]	Channel ch18 Source Address Low Part Register	0x00000000
0x028C	CHCTRL[CH18][CHANREQCTRL]	Channel ch18 DMA Request Control Register	0x00000000

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Offset	Register Name	Description	Reset Value
0x0290	CHCTRL[CH18][DSTADDR]	Channel ch18 Destination Address Low Part Register	0x00000000
0x0298	CHCTRL[CH18][LLPOINTER]	Channel ch18 Linked List Pointer Low Part Register	0x00000000
0x02A0	CHCTRL[CH19][CTRL]	Channel ch19 Control Register	0x00000000
0x02A4	CHCTRL[CH19][TRANSIZE]	Channel ch19Transfer Size Register	0x00000000
0x02A8	CHCTRL[CH19][SRCADDR]	Channel ch19 Source Address Low Part Register	0x00000000
0x02AC	CHCTRL[CH19][CHANREQCTRL]	Channel ch19 DMA Request Control Register	0x00000000
0x02B0	CHCTRL[CH19][DSTADDR]	Channel ch19 Destination Address Low Part Register	0x00000000
0x02B8	CHCTRL[CH19][LLPOINTER]	Channel ch19 Linked List Pointer Low Part Register	0x00000000
0x02C0	CHCTRL[CH20][CTRL]	Channel ch20 Control Register	0x00000000
0x02C4	CHCTRL[CH20][TRANSIZE]	Channel ch20Transfer Size Register	0x00000000
0x02C8	CHCTRL[CH20][SRCADDR]	Channel ch20 Source Address Low Part Register	0x00000000
0x02CC	CHCTRL[CH20][CHANREQCTRL]	Channel ch20 DMA Request Control Register	0x00000000
0x02D0	CHCTRL[CH20][DSTADDR]	Channel ch20 Destination Address Low Part Register	0x00000000
0x02D8	CHCTRL[CH20][LLPOINTER]	Channel ch20 Linked List Pointer Low Part Register	0x00000000
0x02E0	CHCTRL[CH21][CTRL]	Channel ch21 Control Register	0x00000000
0x02E4	CHCTRL[CH21][TRANSIZE]	Channel ch21Transfer Size Register	0x00000000
0x02E8	CHCTRL[CH21][SRCADDR]	Channel ch21 Source Address Low Part Register	0x00000000
0x02EC	CHCTRL[CH21][CHANREQCTRL]	Channel ch21 DMA Request Control Register	0x00000000
0x02F0	CHCTRL[CH21][DSTADDR]	Channel ch21 Destination Address Low Part Register	0x00000000
0x02F8	CHCTRL[CH21][LLPOINTER]	Channel ch21 Linked List Pointer Low Part Register	0x00000000
0x0300	CHCTRL[CH22][CTRL]	Channel ch22 Control Register	0x00000000
0x0304	CHCTRL[CH22][TRANSIZE]	Channel ch22Transfer Size Register	0x00000000

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Offset	Register Name	Description	Reset Value
0x0308	CHCTRL[CH22][SRCADDR]	Channel ch22 Source Address Low Part Register	0x00000000
0x030C	CHCTRL[CH22][CHANREQCTRL]	Channel ch22 DMA Request Control Register	0x00000000
0x0310	CHCTRL[CH22][DSTADDR]	Channel ch22 Destination Address Low Part Register	0x00000000
0x0318	CHCTRL[CH22][LLPOINTER]	Channel ch22 Linked List Pointer Low Part Register	0x00000000
0x0320	CHCTRL[CH23][CTRL]	Channel ch23 Control Register	0x00000000
0x0324	CHCTRL[CH23][TRANSIZE]	Channel ch23Transfer Size Register	0x00000000
0x0328	CHCTRL[CH23][SRCADDR]	Channel ch23 Source Address Low Part Register	0x00000000
0x032C	CHCTRL[CH23][CHANREQCTRL]	Channel ch23 DMA Request Control Register	0x00000000
0x0330	CHCTRL[CH23][DSTADDR]	Channel ch23 Destination Address Low Part Register	0x00000000
0x0338	CHCTRL[CH23][LLPOINTER]	Channel ch23 Linked List Pointer Low Part Register	0x00000000
0x0340	CHCTRL[CH24][CTRL]	Channel ch24 Control Register	0x00000000
0x0344	CHCTRL[CH24][TRANSIZE]	Channel ch24Transfer Size Register	0x00000000
0x0348	CHCTRL[CH24][SRCADDR]	Channel ch24 Source Address Low Part Register	0x00000000
0x034C	CHCTRL[CH24][CHANREQCTRL]	Channel ch24 DMA Request Control Register	0x00000000
0x0350	CHCTRL[CH24][DSTADDR]	Channel ch24 Destination Address Low Part Register	0x00000000
0x0358	CHCTRL[CH24][LLPOINTER]	Channel ch24 Linked List Pointer Low Part Register	0x00000000
0x0360	CHCTRL[CH25][CTRL]	Channel ch25 Control Register	0x00000000
0x0364	CHCTRL[CH25][TRANSIZE]	Channel ch25Transfer Size Register	0x00000000
0x0368	CHCTRL[CH25][SRCADDR]	Channel ch25 Source Address Low Part Register	0x00000000
0x036C	CHCTRL[CH25][CHANREQCTRL]	Channel ch25 DMA Request Control Register	0x00000000
0x0370	CHCTRL[CH25][DSTADDR]	Channel ch25 Destination Address Low Part Register	0x00000000
0x0378	CHCTRL[CH25][LLPOINTER]	Channel ch25 Linked List Pointer Low Part Register	0x00000000

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Offset	Register Name	Description	Reset Value
0x0380	CHCTRL[CH26][CTRL]	Channel ch26 Control Register	0x00000000
0x0384	CHCTRL[CH26][TRANSIZE]	Channel ch26Transfer Size Register	0x00000000
0x0388	CHCTRL[CH26][SRCADDR]	Channel ch26 Source Address Low Part Register	0x00000000
0x038C	CHCTRL[CH26][CHANREQCTRL]	Channel ch26 DMA Request Control Register	0x00000000
0x0390	CHCTRL[CH26][DSTADDR]	Channel ch26 Destination Address Low Part Register	0x00000000
0x0398	CHCTRL[CH26][LLPOINTER]	Channel ch26 Linked List Pointer Low Part Register	0x00000000
0x03A0	CHCTRL[CH27][CTRL]	Channel ch27 Control Register	0x00000000
0x03A4	CHCTRL[CH27][TRANSIZE]	Channel ch27Transfer Size Register	0x00000000
0x03A8	CHCTRL[CH27][SRCADDR]	Channel ch27 Source Address Low Part Register	0x00000000
0x03AC	CHCTRL[CH27][CHANREQCTRL]	Channel ch27 DMA Request Control Register	0x00000000
0x03B0	CHCTRL[CH27][DSTADDR]	Channel ch27 Destination Address Low Part Register	0x00000000
0x03B8	CHCTRL[CH27][LLPOINTER]	Channel ch27 Linked List Pointer Low Part Register	0x00000000
0x03C0	CHCTRL[CH28][CTRL]	Channel ch28 Control Register	0x00000000
0x03C4	CHCTRL[CH28][TRANSIZE]	Channel ch28Transfer Size Register	0x00000000
0x03C8	CHCTRL[CH28][SRCADDR]	Channel ch28 Source Address Low Part Register	0x00000000
0x03CC	CHCTRL[CH28][CHANREQCTRL]	Channel ch28 DMA Request Control Register	0x00000000
0x03D0	CHCTRL[CH28][DSTADDR]	Channel ch28 Destination Address Low Part Register	0x00000000
0x03D8	CHCTRL[CH28][LLPOINTER]	Channel ch28 Linked List Pointer Low Part Register	0x00000000
0x03E0	CHCTRL[CH29][CTRL]	Channel ch29 Control Register	0x00000000
0x03E4	CHCTRL[CH29][TRANSIZE]	Channel ch29Transfer Size Register	0x00000000
0x03E8	CHCTRL[CH29][SRCADDR]	Channel ch29 Source Address Low Part Register	0x00000000
0x03EC	CHCTRL[CH29][CHANREQCTRL]	Channel ch29 DMA Request Control Register	0x00000000

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Offset	Register Name	Description	Reset Value
0x03F0	CHCTRL[CH29][DSTADDR]	Channel ch29 Destination Address Low Part Register	0x00000000
0x03F8	CHCTRL[CH29][LLPOINTER]	Channel ch29 Linked List Pointer Low Part Register	0x00000000
0x0400	CHCTRL[CH30][CTRL]	Channel ch30 Control Register	0x00000000
0x0404	CHCTRL[CH30][TRANSIZE]	Channel ch30Transfer Size Register	0x00000000
0x0408	CHCTRL[CH30][SRCADDR]	Channel ch30 Source Address Low Part Register	0x00000000
0x040C	CHCTRL[CH30][CHANREQCTRL]	Channel ch30 DMA Request Control Register	0x00000000
0x0410	CHCTRL[CH30][DSTADDR]	Channel ch30 Destination Address Low Part Register	0x00000000
0x0418	CHCTRL[CH30][LLPOINTER]	Channel ch30 Linked List Pointer Low Part Register	0x00000000
0x0420	CHCTRL[CH31][CTRL]	Channel ch31 Control Register	0x00000000
0x0424	CHCTRL[CH31][TRANSIZE]	Channel ch31Transfer Size Register	0x00000000
0x0428	CHCTRL[CH31][SRCADDR]	Channel ch31 Source Address Low Part Register	0x00000000
0x042C	CHCTRL[CH31][CHANREQCTRL]	Channel ch31 DMA Request Control Register	0x00000000
0x0430	CHCTRL[CH31][DSTADDR]	Channel ch31 Destination Address Low Part Register	0x00000000
0x0438	CHCTRL[CH31][LLPOINTER]	Channel ch31 Linked List Pointer Low Part Register	0x00000000

Table 162: DMAV2 Register List

28.6 DMA Register Details

The detailed description of DMA registers is as follows:

28.6.1 IDMISC (0x4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																DMASTATE			CURCHAN					RSVD							
N/A																RO			RO					N/A							
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x

IDMISC [31:0]

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Bits	Name	Description

15-13	DMASTATE	DMA state machine
		localparam ST_IDLE = 3'b000;
		localparam ST_READ = 3'b001;
		localparam ST_READ_ACK = 3'b010;
		localparam ST_WRITE = 3'b011;
		localparam ST_WRITE_ACK = 3'b100;
		localparam ST_LL = 3'b101;
		localparam ST_END = 3'b110;
		localparam ST_END_WAIT = 3'b111;

12-8	CURCHAN	current channel in used

IDMISC Bits

28.6.2 DMACFG (0x10)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CHAINXFR	REQSYNC	RSVD				DATAWIDTH		ADDRWIDTH							CORENUM	BUSNUM	REQNUM					FIFODEPTH						CHANNELNUM			
RO	RO	N/A				RO		RO							RO	RO	RO					RO						RO			
0	0	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DMACFG [31:0]

Bits	Name	Description
31	CHAINXFR	Chain transfer 0x0: Chain transfer is not configured 0x1: Chain transfer is configured
30	REQSYNC	DMA request synchronization. The DMA request synchronization should be configured to avoid signal integrity problems when the request signal is not clocked by the system bus clock, which the DMA control logic operates in. If the request synchronization is not configured, the request signal is sampled directly without synchronization. 0x0: Request synchronization is not configured 0x1: Request synchronization is configured
25-24	DATAWIDTH	AXI bus data width 0x0: 32 bits 0x1: 64 bits 0x2: 128 bits 0x3: 256 bits

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Bits	Name	Description
23-17	ADDRWIDTH	AXI bus address width 0x18: 24 bits 0x19: 25 bits ... 0x40: 64 bits Others: Invalid
16	CORENUM	DMA core number 0x0: 1 core 0x1: 2 cores
15	BUSNUM	AXI bus interface number 0x0: 1 AXI bus 0x1: 2 AXI busses
14-10	REQNUM	Request/acknowledge pair number 0x0: 0 pair 0x1: 1 pair 0x2: 2 pairs ... 0x10: 16 pairs
9-4	FIFODEPTH	FIFO depth 0x4: 4 entries 0x8: 8 entries 0x10: 16 entries 0x20: 32 entries Others: Invalid
3-0	CHANNELNUM	Channel number 0x1: 1 channel 0x2: 2 channels ... 0x8: 8 channels Others: Invalid

DMACFG Bits

28.6.3 DMACTRL (0x14)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																															RESET
N/A																															WO
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

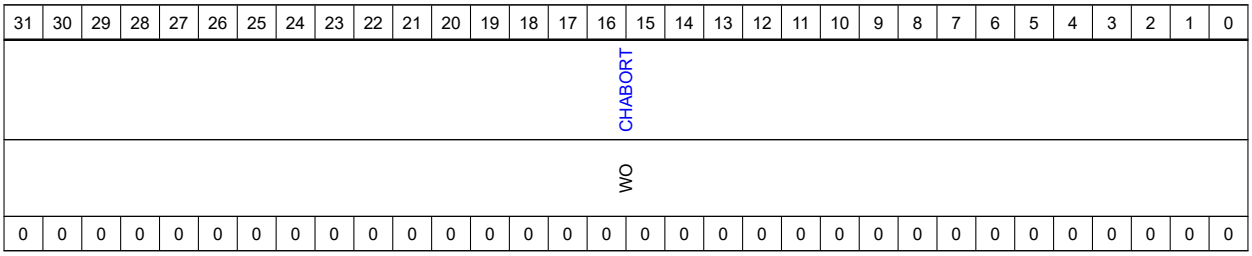
DMACTRL [31:0]

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Bits	Name	Description
0	RESET	Software reset control. Write 1 to this bit to reset the DMA core and disable all channels. Note: The software reset may cause the in-completion of AXI transaction.

DMACTRL Bits

28.6.4 CHABORT (0x18)

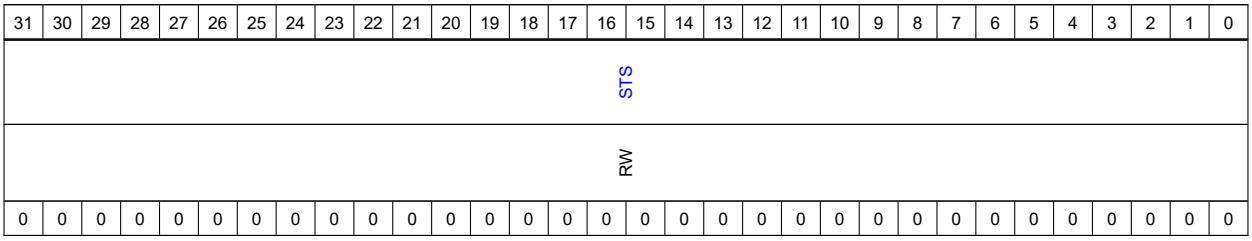


CHABORT [31:0]

Bits	Name	Description
31-0	CHABORT	Write 1 to bit n to abort channel n. The bits should only be set when the corresponding channels are enabled. Otherwise, the writes will be ignored for channels that are not enabled. (N: Number of channels)

CHABORT Bits

28.6.5 INTHALFSTS (0x24)



INTHALFSTS [31:0]

Bits	Name	Description
31-0	STS	half transfer done irq status

INTHALFSTS Bits

28.6.6 INTTCSTS (0x28)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
STS																															
W1C																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

INTTCSTS [31:0]

Bits	Name	Description
31-0	STS	The terminal count status, one bit per channel. The terminal count status is set when a channel transfer finishes without the abort or error event. 0x0: Channel n has no terminal count status 0x1: Channel n has terminal count status

INTTCSTS Bits

28.6.7 INTABORTSTS (0x2C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
STS																															
W1C																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

INTABORTSTS [31:0]

Bits	Name	Description
31-0	STS	The abort status of channel, one bit per channel. The abort status is set when a channel transfer is aborted. 0x0: Channel n has no abort status 0x1: Channel n has abort status

INTABORTSTS Bits

28.6.8 INTERRSTS (0x30)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
STS																															
W1C																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

INTERRSTS [31:0]

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Bits	Name	Description
31-0	STS	The error status, one bit per channel. The error status is set when a channel transfer encounters the following error events: - Bus error - Unaligned address - Unaligned transfer width - Reserved configuration 0x0: Channel n has no error status 0x1: Channel n has error status

INTERRSTS Bits

28.6.9 CHEN (0x34)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CHEN																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CHEN [31:0]

Bits	Name	Description
31-0	CHEN	Alias of the Enable field of all ChnCtrl registers

CHEN Bits

28.6.10 CHCTRL[CTRL] (0x40 + 0x20 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INFINITELOOP	HANDSHAKEOPT	PRIORITY	BURSTOPT	SRCBURSTSIZE				SRCWIDTH			DSTWIDTH			SRCMODE	DSTMODE	SRCADDRCTRL		DSTADDRCTRL		RSVD							INTHALFCNTMASK	INTABTMASK	INTERRMASK	INTTCMASK	ENABLE
RW	RW	RW	RW	RW				RW			RW			RW	RW	RW		RW		N/A							RW	RW	RW	RW	RW
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	0	0	0	0	0

CHCTRL[CTRL] [31:0]

Bits	Name	Description
31	INFINITELOOP	set to loop current config infinitely
30	HANDSHAKEOPT	0: one request to transfer one burst 1: one request to transfer all the data defined in ch_tts

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Bits	Name	Description
29	PRIORITY	Channel priority level 0x0: Lower priority 0x1: Higher priority
28	BURSTOPT	set to change burst_size definition
27-24	SRCBURSTSIZE	Source burst size. This field indicates the number of transfers before DMA channel re-arbitration. The burst transfer byte number is (SrcBurstSize * SrcWidth). 0x0: 1 transfer 0x1: 2 transfers 0x2: 4 transfers 0x3: 8 transfers 0x4: 16 transfers 0x5: 32 transfers 0x6: 64 transfers 0x7: 128 transfers 0x8: 256 transfers 0x9:512 transfers 0xa: 1024 transfers 0xb - 0xf: Reserved, setting this field with a reserved value triggers the error exception
23-21	SRCWIDTH	Source transfer width 0x0: Byte transfer 0x1: Half-word transfer 0x2: Word transfer 0x3: Double word transfer 0x4: Quad word transfer 0x5: Eight word transfer 0x6 - 0x7: Reserved, setting this field with a reserved value triggers the error exception

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Bits	Name	Description
20-18	DSTWIDTH	Destination transfer width. Both the total transfer byte number and the burst transfer byte number should be aligned to the destination transfer width; otherwise the error event will be triggered. For example, destination transfer width should be set as byte transfer if total transfer byte is not aligned to half-word. See field SrcBurstSize above for the definition of burst transfer byte number and section 3.2.8 for the definition of the total transfer byte number. 0x0: Byte transfer 0x1: Half-word transfer 0x2: Word transfer 0x3: Double word transfer 0x4: Quad word transfer 0x5: Eight word transfer 0x6 - 0x7: Reserved, setting this field with a reserved value triggers the error exception
17	SRCMODE	Source DMA handshake mode 0x0: Normal mode 0x1: Handshake mode Normal mode is enabled and started by software set Enable bit; Handshake mode is enabled by software set Enable bit, started by hardware dma request from DMAMUX block
16	DSTMODE	Destination DMA handshake mode 0x0: Normal mode 0x1: Handshake mode the difference bewteen Source/Destination handshake mode is: the dma block will response hardware request after read in Source handshake mode; the dma block will response hardware request after write in Destination handshake mode; NOTE: can't set SrcMode and DstMode at same time, otherwise result unknown.
15-14	SRCADDRCTRL	Source address control 0x0: Increment address 0x1: Decrement address 0x2: Fixed address 0x3: Reserved, setting the field with this value triggers the error exception

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Bits	Name	Description
13-12	DSTADDRCTRL	Destination address control 0x0: Increment address 0x1: Decrement address 0x2: Fixed address 0x3: Reserved, setting the field with this value triggers the error exception
4	INTHALFCNTMASK	Channel half interrupt mask 0x0: Allow the half interrupt to be triggered 0x1: Disable the half interrupt
3	INTABTMASK	Channel abort interrupt mask 0x0: Allow the abort interrupt to be triggered 0x1: Disable the abort interrupt
2	INTERRMASK	Channel error interrupt mask 0x0: Allow the error interrupt to be triggered 0x1: Disable the error interrupt
1	INTTCMASK	Channel terminal count interrupt mask 0x0: Allow the terminal count interrupt to be triggered 0x1: Disable the terminal count interrupt
0	ENABLE	Channel enable bit 0x0: Disable 0x1: Enable

CHCTRL[CTRL] Bits

28.6.11 CHCTRL[TRANSIZE] (0x44 + 0x20 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD				TRANSIZE																											
N/A				RW																											
x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CHCTRL[TRANSIZE] [31:0]

Bits	Name	Description
27-0	TRANSIZE	Total transfer size from source. The total number of transferred bytes is (TranSize * SrcWidth). This register is cleared when the DMA transfer is done. If a channel is enabled with zero total transfer size, the error event will be triggered and the transfer will be terminated.

CHCTRL[TRANSIZE] Bits

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28.6.12 CHCTRL[SRCADDR] (0x48 + 0x20 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SRCADDRL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CHCTRL[SRCADDR] [31:0]

Bits	Name	Description
31-0	SRCADDRL	Low part of the source starting address. When the transfer completes, the value of {SrcAddrH,SrcAddrL} is updated to the ending address. This address must be aligned to the source transfer size; otherwise, an error event will be triggered.

CHCTRL[SRCADDR] Bits

28.6.13 CHCTRL[CHANREQCTRL] (0x4C + 0x20 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD			SRCREQSEL					RSVD			DSTREQSEL					RSVD															
N/A			RW					N/A			RW					N/A															
x	x	x	0	0	0	0	0	x	x	x	0	0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

CHCTRL[CHANREQCTRL] [31:0]

Bits	Name	Description
28-24	SRCREQSEL	Source DMA request select. Select the request/ack handshake pair that the source device is connected to.
20-16	DSTREQSEL	Destination DMA request select. Select the request/ack handshake pair that the destination device is connected to.

CHCTRL[CHANREQCTRL] Bits

28.6.14 CHCTRL[DSTADDR] (0x50 + 0x20 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DSTADDRL																															

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CHCTRL[DSTADDR] [31:0]

Bits	Name	Description
31-0	DSTADDRL	Low part of the destination starting address. When the transfer completes, the value of {DstAddrH,DstAddrL} is updated to the ending address. This address must be aligned to the destination transfer size; otherwise the error event will be triggered.

CHCTRL[DSTADDR] Bits

28.6.15 CHCTRL[LLPOINTER] (0x58 + 0x20 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LLPOINTERL																											RSVD			
RW																											N/A			
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x

CHCTRL[LLPOINTER] [31:0]

Bits	Name	Description
31-5	LLPOINTERL	Low part of the pointer to the next descriptor. The pointer must be 32byte aligned.

CHCTRL[LLPOINTER] Bits

28.7 DMA Configuration and Usage Instructions

Taking starting a DMA handshake mode transmission as an example, the required steps are:

1. Configure the register transize of the corresponding channel, the total data volume is transize x sr-cwidth;
2. Configure the register srcaddr of the corresponding channel, which points to the starting address of the source data;
3. Configure the register chanreqctl of the corresponding channel, including source data DMA request selection srcreqsel or destination data DMA request selection dstreqsel;
4. Configure the register dstaddr of the corresponding channel, which points to the starting address of the destination data;
5. If it is non-chained transmission, configure llpointer to 0x0, otherwise configure it to the starting address of the next chained data structure;

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6. Configure the register ctrl of the corresponding channel, including the priority field, source burst length srcburstsize field, source data bit width srcwidth, destination data bit width dstwidth field, and channel enable field enable;
7. Configure DMAMUX to route srcreqsel or dstreqsel to the corresponding channel;
8. Read the register inttcsts to determine whether the transmission of the corresponding channel is completed;

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29 DMA Request Router DMAMUX

This chapter introduces the functions and features of the DMA request router DMAMUX.

29.1 Overview

This chapter describes the main features of the DMA request router DMAMUX:

- Supports up to 128 peripheral DMA requests
- Supports 32 output channels

For all peripherals of this product that support generating DMA requests and their allocation details on the DMA request router DMAMUX, please refer to Section 27.2.

29.2 DMAMUX Architecture Diagram

The structure diagram of this module is shown in Figure 25.

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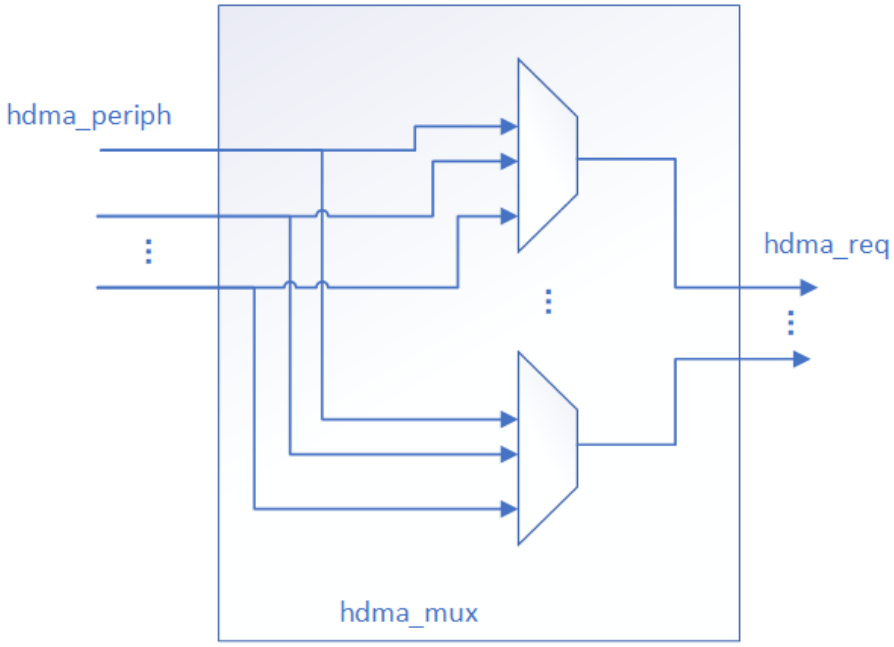


Figure 25: DMAMUX Structure Block Diagram

29.3 Pin Description

Table 163: DMAMUX Pin Description

Pin	Direction	Function Description
hdma_req[31:0]	Output	DMA request signal output to the DMA controller
hdma_ack[31:0]	Input	Input DMA acknowledge signal
hdma_req_periph[127:0]	Input	Input DMA request signals from each peripheral
hdma_ack_periph[127:0]	Output	DMA acknowledge signals output to each peripheral

29.4 Function Description

The function of the DMA request router is to realize flexible and dynamic mapping of peripheral DMA requests to each channel of the DMA controller, enabling any peripheral to send DMA requests to any channel of any DMA controller.

The steps to configure DMAMUX through the configuration register CHCFGx of each DMAMUX channel are as follows:

- Select the peripheral DMA request through the SOURCE bit of MUXCFGx
- Set the ENABLE bit of MUXCFGx to 1 to enable channel x of DMAMUX

29.5 DMAMUX Registers

The list of DMAMUX registers is as follows:

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DMAMUX base address: 0xF00C4000

Offset	Register Name	Description	Reset Value
0x0000	MUXCFG[HDMA_MUX0]	HDMA MUX0 Configuration	0x00000000
0x0004	MUXCFG[HDMA_MUX1]	HDMA MUX1 Configuration	0x00000000
0x0008	MUXCFG[HDMA_MUX2]	HDMA MUX2 Configuration	0x00000000
0x000C	MUXCFG[HDMA_MUX3]	HDMA MUX3 Configuration	0x00000000
0x0010	MUXCFG[HDMA_MUX4]	HDMA MUX4 Configuration	0x00000000
0x0014	MUXCFG[HDMA_MUX5]	HDMA MUX5 Configuration	0x00000000
0x0018	MUXCFG[HDMA_MUX6]	HDMA MUX6 Configuration	0x00000000
0x001C	MUXCFG[HDMA_MUX7]	HDMA MUX7 Configuration	0x00000000
0x0020	MUXCFG[HDMA_MUX8]	HDMA MUX8 Configuration	0x00000000
0x0024	MUXCFG[HDMA_MUX9]	HDMA MUX9 Configuration	0x00000000
0x0028	MUXCFG[HDMA_MUX10]	HDMA MUX10 Configuration	0x00000000
0x002C	MUXCFG[HDMA_MUX11]	HDMA MUX11 Configuration	0x00000000
0x0030	MUXCFG[HDMA_MUX12]	HDMA MUX12 Configuration	0x00000000
0x0034	MUXCFG[HDMA_MUX13]	HDMA MUX13 Configuration	0x00000000
0x0038	MUXCFG[HDMA_MUX14]	HDMA MUX14 Configuration	0x00000000
0x003C	MUXCFG[HDMA_MUX15]	HDMA MUX15 Configuration	0x00000000
0x0040	MUXCFG[HDMA_MUX16]	HDMA MUX16 Configuration	0x00000000
0x0044	MUXCFG[HDMA_MUX17]	HDMA MUX17 Configuration	0x00000000
0x0048	MUXCFG[HDMA_MUX18]	HDMA MUX18 Configuration	0x00000000
0x004C	MUXCFG[HDMA_MUX19]	HDMA MUX19 Configuration	0x00000000
0x0050	MUXCFG[HDMA_MUX20]	HDMA MUX20 Configuration	0x00000000
0x0054	MUXCFG[HDMA_MUX21]	HDMA MUX21 Configuration	0x00000000
0x0058	MUXCFG[HDMA_MUX22]	HDMA MUX22 Configuration	0x00000000
0x005C	MUXCFG[HDMA_MUX23]	HDMA MUX23 Configuration	0x00000000
0x0060	MUXCFG[HDMA_MUX24]	HDMA MUX24 Configuration	0x00000000
0x0064	MUXCFG[HDMA_MUX25]	HDMA MUX25 Configuration	0x00000000
0x0068	MUXCFG[HDMA_MUX26]	HDMA MUX26 Configuration	0x00000000
0x006C	MUXCFG[HDMA_MUX27]	HDMA MUX27 Configuration	0x00000000
0x0070	MUXCFG[HDMA_MUX28]	HDMA MUX28 Configuration	0x00000000
0x0074	MUXCFG[HDMA_MUX29]	HDMA MUX29 Configuration	0x00000000
0x0078	MUXCFG[HDMA_MUX30]	HDMA MUX30 Configuration	0x00000000
0x007C	MUXCFG[HDMA_MUX31]	HDMA MUX31 Configuration	0x00000000

Table 164: DMAMUX Register List

29.6 DMAMUX Register Details

The detailed description of DMAMUX registers is as follows:

29.6.1 MUXCFG (0x0 + 0x4 * n)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ENABLE	RSVD																								SOURCE						
WO	N/A																								WO						
0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0

MUXCFG [31:0]

Bits	Name	Description
31	ENABLE	DMA Mux Channel Enable Enables the channel for DMA Mux. The DMA has separate channel enables/disables, which should be used to disable or reconfigure a DMA channel. 0b - DMA Mux channel is disabled 1b - DMA Mux channel is enabled
6-0	SOURCE	DMA Channel Source Specifies which DMA source, if any, is routed to a particular DMA channel. See the ”DMA MUX Mapping”

MUXCFG Bits

29.7 DMAMUX Configuration and Usage Instructions

The steps required to configure DMAMUX are as follows:

1. Configure the MUXCFG.SOURCE field of the corresponding channel register to select one of the 128 peripheral DMA requests;
2. Meanwhile, set the MUXCFG.ENABLE bit of the corresponding channel register to 1;

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30 Communication Mailbox (MBX)

This chapter introduces the functions and features of the communication mailbox MBX.

30.1 Feature Summary

The communication mailbox MBX supports inter-core communication between processor cores and inter-process communication within a processor. Its main features are as follows:

- Supports 2 register access interfaces
- Each interface is equipped with TX FIFO and RX FIFO
- Supports flag bits to reflect the status of TX FIFO and RX FIFO
- Supports interrupt generation

30.2 Function Description

This chapter describes the functions of the communication mailbox MBX.

30.2.1 MBX Register Access Interface

The communication mailbox MBX has 2 sets of register access interfaces, Interface A and Interface B. Both Interface A and Interface B are equipped with a set of TX FIFO registers, RX FIFO registers, control registers and status registers.

Data sent by the user from the TX (transmit) end of Interface A can be received at the RX (receive) end of Interface B. Similarly, the RX end of Interface A can receive data sent from the TX end of Interface B.

30.2.2 FIFO Management

The TX FIFO depth of the MBX transmit end is 4 words (4×32 bits), and the write interface of TX FIFO is the TXREG register. MBX supports indicating various statuses of TX FIFO, including whether TX FIFO is empty, whether TX FIFO is full, and the number of empty words in TX FIFO.

The RX FIFO depth of the MBX receive end is 4 words (4×32 bits). The read interface of RX FIFO is the RXREG register. MBX supports indicating various statuses of RX FIFO, including whether RX FIFO is empty, whether RX FIFO is full, and the number of valid words in RX FIFO.

30.2.3 Interrupt

MBX supports generating the following interrupts:

- An interrupt can be triggered when TX FIFO is empty
- An interrupt can be triggered when TX FIFO is not full
- An interrupt can be triggered when RX FIFO is not empty
- An interrupt can be triggered when RX FIFO is full
- An interrupt can be triggered when the content of the sent message word is empty, that is, it has been read
- An interrupt can be triggered when the content of the received message word is valid

MBX can generate two interrupts, corresponding to Port A and Port B respectively. They can be allocated to different cores in a multi-core system or to different processes in a single-core system.

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30.3 MBX Registers

The list of MBX registers is as follows:

MBX0A base address: 0xF00A0000

MBX0B base address: 0xF00A4000

Offset	Register Name	Description	Reset Value
0x0000	CR	Command Registers	0x00000000
0x0004	SR	Status Registers	0x000000E2
0x0008	TXREG	Transmit word message to other core.	0x00000000
0x000C	RXREG	Receive word message from other core.	0x00000000
0x0010	TXWRD[TXFIFO0]	TXFIFO for sending message to other core	0x00000000
0x0020	RXWRD[RXFIFO0]	RXFIFO for receiving message from other core	0x00000000

Table 165: MBX Register List

30.4 MBX Register Details

The detailed description of MBX registers is as follows:

30.4.1 CR (0x0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TXRESET	RSVD															BARCTL		RSVD					BEIE	TFMAIE	TFMEIE	RFMAIE	RFMFIE	RSVD		TWMEIE	RWMVIE
RW	N/A															RW		N/A					RW	RW	RW	RW	RW	N/A		RW	RW
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CR [31:0]

Bits	Name	Description
31	TXRESET	Reset TX Fifo and word.

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Bits	Name	Description
15-14	BARCTL	Bus Access Response Control, when bit 15:14= 00: no bus error will be generated, no wait for fifo write when fifo full and no wait for word/fifo read when word message invalid or fifo empty; or when write to word/fifo message will be ignored. 01: bus error will be generated when: 1, access invalid address; 2, write to ready only addr; 3, write to fulled fifo or valid message; 4, read from a emptied fifo/word message. 10: no error will be generated, but bus will wait when 1, write to fulled fifo/reg message; 2, read from a emptied fifo/reg message; write to word message will overwrite the existing reg value enven word message are still valid; read from invalid word message will read out last read out message data.happen. 11: reserved.
8	BEIE	Bus Error Interrupt Enable, will enable the interrupt for any bus error as described in the SR bit 13 to bit 8. 1, enable the bus access error interrupt. 0, disable the bus access error interrupt.
7	TFMAIE	TX FIFO message available interrupt enable. 1, enable the TX FIFO massage available interrupt. 0, disable the TX FIFO message available interrupt.
6	TFMEIE	TX FIFO message empty interrupt enable. 1, enable the TX FIFO massage empty interrupt. 0, disable the TX FIFO message empty interrupt.
5	RFMAIE	RX FIFO message available interrupt enable. 1, enable the RX FIFO massage available interrupt. 0, disable the RX FIFO message available interrupt.
4	RFMFIE	RX fifo message full interrupt enable. 1, enable the RX fifo message full interrupt. 0, disable the RX fifo message full interrupt.
1	TWMEIE	TX word message empty interrupt enable. 1, enable the TX word massage empty interrupt. 0, disable the TX word message empty interrupt.
0	RWMVIE	RX word message valid interrupt enable. 1, enable the RX word massage valid interrupt. 0, disable the RX word message valid interrupt.

CR Bits

30.4.2 SR (0x4)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD								RFVC				TFEC				RSVD		ERRRE	EWTRF	ERRFE	EWTFF	EAIVA	EW2RO	TFMA	TFME	RFMA	RFMF	RSVD		TWME	RWMV
N/A								RO				RO				N/A		W1C	W1C	W1C	W1C	W1C	W1C	RW	RW	RO	RO	N/A		RO	RO
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	0	0	0	0	0	0	1	1	1	0	0	0	1	0

SR [31:0]

Bits	Name	Description
23-20	RFVC	RX FIFO valid message count
19-16	TFEC	TX FIFO empty message word count
13	ERRRE	bus Error for read when rx word message are still invalid, this bit is W1C bit. 1, read from word message when the word message are still invalid will cause this error bit set. 0, nothis kind of bus error; write this bit to 1 will clear this bit when this kind of error happen.
12	EWTRF	bus Error for write when tx word message are still valid, this bit is W1C bit. 1, write to word message when the word message are still valid will cause this error bit set. 0, nothis kind of bus error; write this bit to 1 will clear this bit when this kind of error happen.
11	ERRFE	bus Error for read when rx fifo empty, this bit is W1C bit. 1, read from a empty rx fifo will cause this error bit set. 0, nothis kind of bus error; write this bit to 1 will clear this bit when this kind of error happen.
10	EWTFF	bus Error for write when tx fifo full, this bit is W1C bit. 1, write to a fulled tx fifo will cause this error bit set. 0, nothis kind of bus error; write this bit to 1 will clear this bit when this kind of error happen.
9	EAIVA	bus Error for Accessing Invalid Address; this bit is W1C bit. 1, read and write to invalid address in the bus of this block, will set this bit. 0, nothis kind of bus error; write this bit to 1 will clear this bit when this kind of error happen.
8	EW2RO	bus Error for Write to Read Only address; this bit is W1C bit. 1, write to read only address happened in the bus of this block. 0, nothis kind of bus error; write this bit to 1 will clear this bit when this kind of error happen.

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Bits	Name	Description
7	TFMA	TX FIFO Message slot available, the 4x32 TX FIFO message buffer to the other core full, will not trigger any interrupt. 1, TXFIFO message buffer has slot available 0, no slot available (fifo full)
6	TFME	TX FIFO Message Empty, no any data in the message FIFO buffer from other core, will not trigger any interrupt.message from other core. 1, no any message data in TXFIFO from other core. 0, there are some data in the 4x32 TX FIFO from other core yet.
5	RFMA	RX FIFO Message Available, available data in the 4x32 TX FIFO message buffer to the other core, will trigger interrupt if the related interrupt enable bit set in the control (CR) registrer. 1, no any data in the 4x32 TXFIFO message buffer. 0, there are some data in the the 4x32 TXFIFO message buffer already.
4	RFMF	RX FIFO Message Full, message from other core; will trigger interrupt if the related interrupt enable bit set in the control (CR) registrer. 1, the other core had written 4x32 message in the RXFIFO. 0, no 4x32 RX FIFO message from other core yet.
1	TWME	TX word message empty, will trigger interrupt if the related interrupt enable bit set in the control (CR) registrer. 1, means this core had write word message to TXREG. 0, means no valid word message in the TXREG yet.
0	RWMV	RX word message valid, will trigger interrupt if the related interrupt enable bit set in the control (CR) registrer. 1, the other core had written word message in the RXREG. 0, no valid word message yet in the RXREG.

SR Bits

30.4.3 TXREG (0x8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TXREG																															
WO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

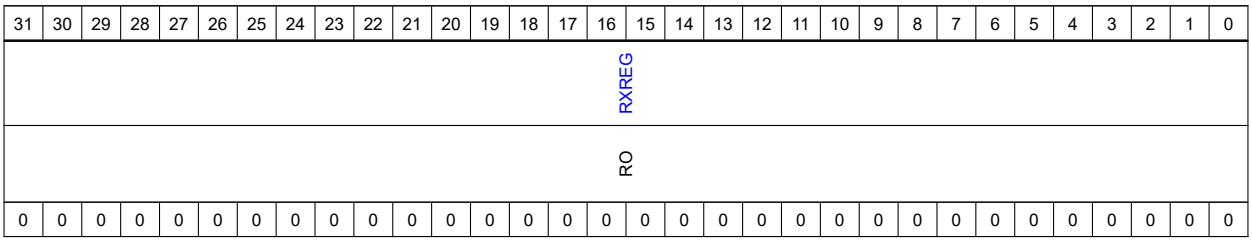
TXREG [31:0]

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Bits	Name	Description
31-0	TXREG	Transmit word message to other core.

TXREG Bits

30.4.4 RXREG (0xC)

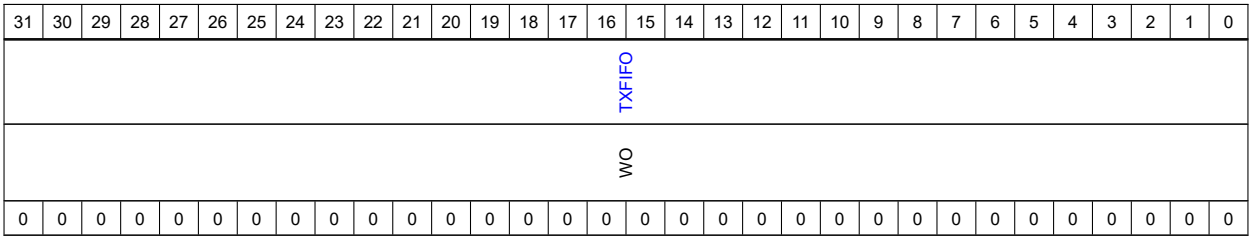


RXREG [31:0]

Bits	Name	Description
31-0	RXREG	Receive word message from other core.

RXREG Bits

30.4.5 TXWRD (0x10 + 0x4 * n)

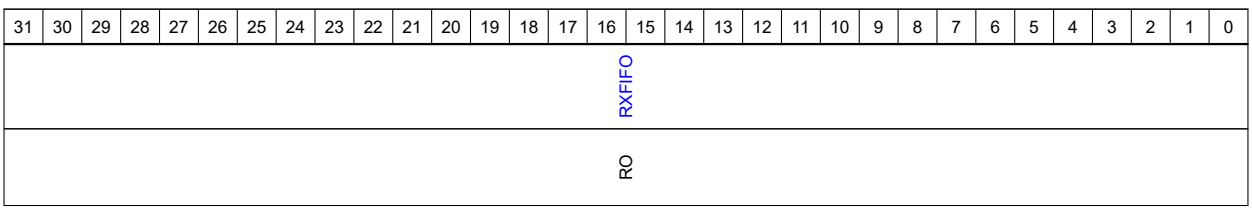


TXWRD [31:0]

Bits	Name	Description
31-0	TXFIFO	TXFIFO for sending message to other core, FIFO size, 4x32 can write one of the word address to push data to the FIFO; can also use 4x32 burst write from 0x010 to push 4 words to the FIFO.

TXWRD Bits

30.4.6 RXWRD (0x20 + 0x4 * n)



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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

RXWRD [31:0]

Bits	Name	Description
31-0	RXFIFO	RXFIFO for receiving message from other core, FIFO size, 4x32 can read one of the word address to pop data to the FIFO; can also use 4x32 burst read from 0x020 to read 4 words from the FIFO.

RXWRD Bits

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31 Cyclic Redundancy Check (CRC)

This chapter introduces the functions and features of Cyclic Redundancy Check (CRC).

31.1 Feature Summary

The main features of CRC are as follows:

- Supports independent calculation for 8 channels
- Supports programmable CRC algorithms (supports all CRC algorithms with the highest power no higher than 32)
- Supports byte, half-word and full-word transmission
- Supports quick configuration of 15 common CRC algorithms

31.2 Functional Description

As a slave device on the system APB bus, the CRC module allows users to complete configuration, bitstream input and CRC check code reading through the bus. For the CRC module, after inputting the bitstream to be checked into the CHN[n][DATA] register according to the user-defined CRC algorithm, you can get the corresponding CRC check code from CHN[n][result]. The CRC module provides 8 independent calculation channels, each of which can independently configure its CRC algorithm and complete the calculation independently. The input bitstream supports byte, half-word and full-word transmission, and also supports configuring the bitstream to flip the order by byte or by bit. Each channel supports fast pre-configuration of 15 common CRC algorithms.

CRC algorithm configuration is required before using CRC, which can be completed quickly by configuring each algorithm register in detail or configuring the pre-configuration register. It supports discontinuous bitstream transmission, and the calculation result of discontinuous bitstream transmission is consistent with that of continuous transmission. The calculation result in CHN[n][result] is always the check code of the currently accumulated transmitted bitstream.

31.3 CRC Algorithm Configuration

- CHN[n][POLY]: Polynomial expression of the CRC algorithm, for example, the polynomial corresponding to CRC-32 is expressed as 0x04C11DB7. If the CRC polynomial is less than 32 bits, write 0 to the high bits of this configuration register.
- CHN[n][INIT_DATA]: CRC initial value.
- CHN[n][XOROUT]: XOR value of the CRC output value, that is, the final check code is obtained after the CRC calculation result is XORed with this parameter.
- CHN[n][MISC_SETTING]: Miscellaneous configuration of CRC, including: POLY_WIDTH (length of CRC check code), REV_IN (whether to flip the bitstream by bit), REV_OUT (whether to flip the output result by bit), BYTE_REV (whether to flip the input bitstream by byte).
- CHN[n][PRE_SET]: CRC fast pre-configuration register. The CRC algorithm corresponding to the pre-configuration register has higher priority than the configuration of the above detailed algorithm parameter registers, that is, after the user configures the pre-configuration register, the previously configured algorithm parameters (if any) will be overwritten by the corresponding parameters of the pre-configuration.

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31.4 CRC Registers

The list of CRC registers is as follows:

CRC base address: 0xF0080000

Offset	Register Name	Description	Reset Value
0x0000	CHN[0][PRE_SET]	0 pre set for crc setting	0x00000000
0x0004	CHN[0][CLR]	chn0 clear crc result and setting	0x00000000
0x0008	CHN[0][POLY]	chn0 poly	0x00000000
0x000C	CHN[0][INIT_DATA]	chn0 init_data	0x00000000
0x0010	CHN[0][XOROUT]	chn0 xorout	0x00000000
0x0014	CHN[0][MISC_SETTING]	chn0 misc_setting	0x00000000
0x0018	CHN[0][DATA]	chn0 data	0x00000000
0x001C	CHN[0][RESULT]	chn0 result	0x00000000
0x0040	CHN[1][PRE_SET]	1 pre set for crc setting	0x00000000
0x0044	CHN[1][CLR]	chn1 clear crc result and setting	0x00000000
0x0048	CHN[1][POLY]	chn1 poly	0x00000000
0x004C	CHN[1][INIT_DATA]	chn1 init_data	0x00000000
0x0050	CHN[1][XOROUT]	chn1 xorout	0x00000000
0x0054	CHN[1][MISC_SETTING]	chn1 misc_setting	0x00000000
0x0058	CHN[1][DATA]	chn1 data	0x00000000
0x005C	CHN[1][RESULT]	chn1 result	0x00000000
0x0080	CHN[2][PRE_SET]	2 pre set for crc setting	0x00000000
0x0084	CHN[2][CLR]	chn2 clear crc result and setting	0x00000000
0x0088	CHN[2][POLY]	chn2 poly	0x00000000
0x008C	CHN[2][INIT_DATA]	chn2 init_data	0x00000000
0x0090	CHN[2][XOROUT]	chn2 xorout	0x00000000
0x0094	CHN[2][MISC_SETTING]	chn2 misc_setting	0x00000000
0x0098	CHN[2][DATA]	chn2 data	0x00000000
0x009C	CHN[2][RESULT]	chn2 result	0x00000000
0x00C0	CHN[3][PRE_SET]	3 pre set for crc setting	0x00000000
0x00C4	CHN[3][CLR]	chn3 clear crc result and setting	0x00000000
0x00C8	CHN[3][POLY]	chn3 poly	0x00000000
0x00CC	CHN[3][INIT_DATA]	chn3 init_data	0x00000000
0x00D0	CHN[3][XOROUT]	chn3 xorout	0x00000000
0x00D4	CHN[3][MISC_SETTING]	chn3 misc_setting	0x00000000
0x00D8	CHN[3][DATA]	chn3 data	0x00000000
0x00DC	CHN[3][RESULT]	chn3 result	0x00000000
0x0100	CHN[4][PRE_SET]	4 pre set for crc setting	0x00000000
0x0104	CHN[4][CLR]	chn4 clear crc result and setting	0x00000000
0x0108	CHN[4][POLY]	chn4 poly	0x00000000
0x010C	CHN[4][INIT_DATA]	chn4 init_data	0x00000000

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Offset	Register Name	Description	Reset Value
0x0110	CHN[4][XOROUT]	chn4 xorout	0x00000000
0x0114	CHN[4][MISC_SETTING]	chn4 misc_setting	0x00000000
0x0118	CHN[4][DATA]	chn4 data	0x00000000
0x011C	CHN[4][RESULT]	chn4 result	0x00000000
0x0140	CHN[5][PRE_SET]	5 pre set for crc setting	0x00000000
0x0144	CHN[5][CLR]	chn5 clear crc result and setting	0x00000000
0x0148	CHN[5][POLY]	chn5 poly	0x00000000
0x014C	CHN[5][INIT_DATA]	chn5 init_data	0x00000000
0x0150	CHN[5][XOROUT]	chn5 xorout	0x00000000
0x0154	CHN[5][MISC_SETTING]	chn5 misc_setting	0x00000000
0x0158	CHN[5][DATA]	chn5 data	0x00000000
0x015C	CHN[5][RESULT]	chn5 result	0x00000000
0x0180	CHN[6][PRE_SET]	6 pre set for crc setting	0x00000000
0x0184	CHN[6][CLR]	chn6 clear crc result and setting	0x00000000
0x0188	CHN[6][POLY]	chn6 poly	0x00000000
0x018C	CHN[6][INIT_DATA]	chn6 init_data	0x00000000
0x0190	CHN[6][XOROUT]	chn6 xorout	0x00000000
0x0194	CHN[6][MISC_SETTING]	chn6 misc_setting	0x00000000
0x0198	CHN[6][DATA]	chn6 data	0x00000000
0x019C	CHN[6][RESULT]	chn6 result	0x00000000
0x01C0	CHN[7][PRE_SET]	7 pre set for crc setting	0x00000000
0x01C4	CHN[7][CLR]	chn7 clear crc result and setting	0x00000000
0x01C8	CHN[7][POLY]	chn7 poly	0x00000000
0x01CC	CHN[7][INIT_DATA]	chn7 init_data	0x00000000
0x01D0	CHN[7][XOROUT]	chn7 xorout	0x00000000
0x01D4	CHN[7][MISC_SETTING]	chn7 misc_setting	0x00000000
0x01D8	CHN[7][DATA]	chn7 data	0x00000000
0x01DC	CHN[7][RESULT]	chn7 result	0x00000000

Table 166: CRC Register List

31.5 CRC Register Details

The detailed description of CRC registers is as follows:

31.5.1 CHN[PRE_SET] (0x0 + 0x40 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																								PRE_SET							
N/A																								RW							

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0

CHN[PRE_SET] [31:0]

Bits	Name	Description
7-0	PRE_SET	0: no pre set 1: CRC32 2: CRC32-AUTOSAR 3: CRC16-CCITT 4: CRC16-XMODEM 5: CRC16-MODBUS 1: CRC32 2: CRC32-autosar 3: CRC16-ccitt 4: CRC16-xmodem 5: CRC16-modbus 6: crc16_dnp 7: crc16_x25 8: crc16_usb 9: crc16_maxim 10: crc16_ibm 11: crc8_maxim 12: crc8_rohc 13: crc8_itu 14: crc8 15: crc5_usb

CHN[PRE_SET] Bits

31.5.2 CHN[CLR] (0x4 + 0x40 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																															CLR
N/A																															RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

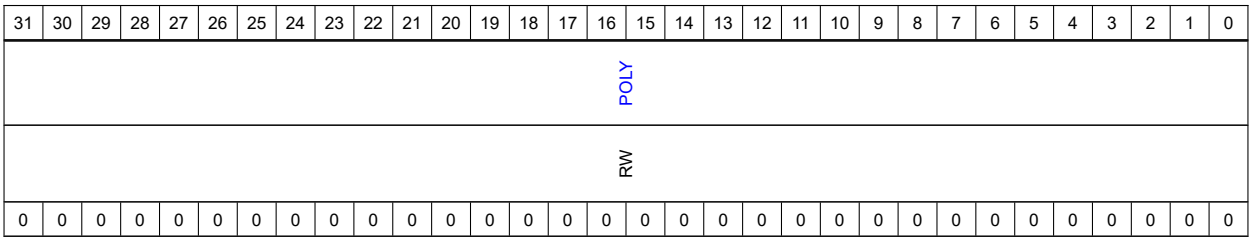
CHN[CLR] [31:0]

Bits	Name	Description
0	CLR	write 1 to clr crc setting and result for its channel. always read 0.

CHN[CLR] Bits

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31.5.3 CHN[POLY] (0x8 + 0x40 * n)

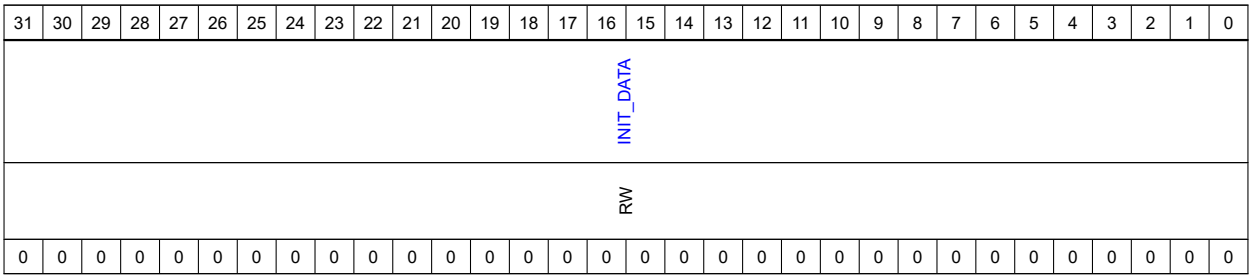


CHN[POLY] [31:0]

Bits	Name	Description
31-0	POLY	poly setting

CHN[POLY] Bits

31.5.4 CHN[INIT_DATA] (0xC + 0x40 * n)

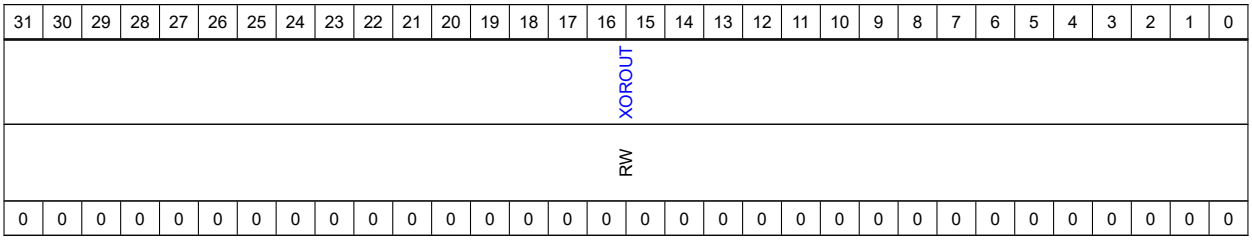


CHN[INIT_DATA] [31:0]

Bits	Name	Description
31-0	INIT_DATA	initial data of CRC

CHN[INIT_DATA] Bits

31.5.5 CHN[XOROUT] (0x10 + 0x40 * n)



CHN[XOROUT] [31:0]

Bits	Name	Description
31-0	XOROUT	XOR for CRC result

CHN[XOROUT] Bits

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31.5.6 CHN[MISC_SETTING] (0x14 + 0x40 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD							BYTE_REV	RSVD							REV_OUT	RSVD							REV_IN	RSVD		POLY_WIDTH					
N/A							RW	N/A							RW	N/A							RW	N/A		RW					
x	x	x	x	x	x	x	0	x	x	x	x	x	x	x	0	x	x	x	x	x	x	x	0	x	x	0	0	0	0	0	0

CHN[MISC_SETTING] [31:0]

Bits	Name	Description
24	BYTE_REV	0: no wrap input byte order 1: wrap input byte order
16	REV_OUT	0: no wrap output bit order 1: wrap output bit order
8	REV_IN	0: no wrap input bit order 1: wrap input bit order
5-0	POLY_WIDTH	crc data length

CHN[MISC_SETTING] Bits

31.5.7 CHN[DATA] (0x18 + 0x40 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CHN[DATA] [31:0]

Bits	Name	Description
31-0	DATA	data for crc

CHN[DATA] Bits

31.5.8 CHN[RESULT] (0x1C + 0x40 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RESULT																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

CHN[RESULT] [31:0]

Bits	Name	Description
31-0	RESULT	crc result

CHN[RESULT] Bits

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32 Motion Control System Overview

This chapter introduces the motion control system of this product. The motion control system of this product includes 2 PWM timers, 2 quadrature encoder interfaces, 2 quadrature encoder outputs, 2 motion management controllers, 2 serial encoder controllers, 1 resolver demodulator and 1 interconnection manager. It can be used in conjunction with on-chip analog peripherals such as analog-to-digital converter ADC, comparator ACMP, etc. This product includes a global timer to provide timestamps for the motion control system.

All motion control units and synchronous timers operate on the external bus clock (CLK_TOP_AHB) to achieve minimum latency.

The motor control system of this product is shown in Figure 26.

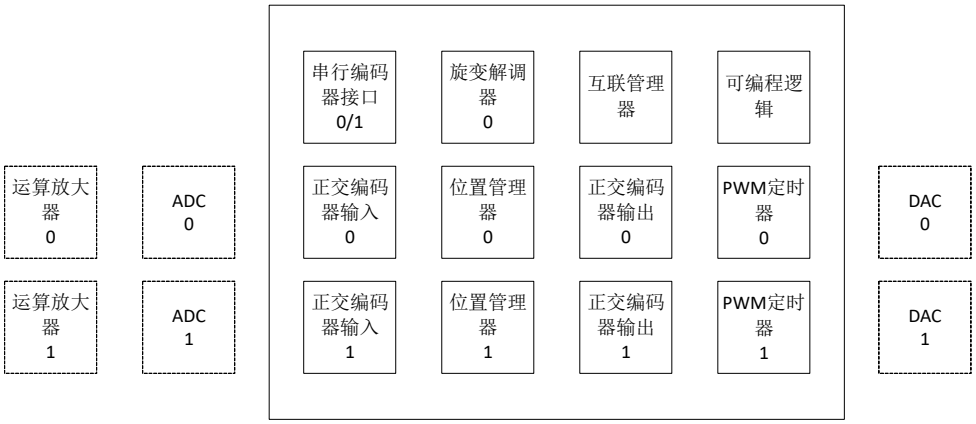


Figure 26: Motor System Block Diagram

The partial connection relationship of the motor control system of this product is shown in Figure 27.

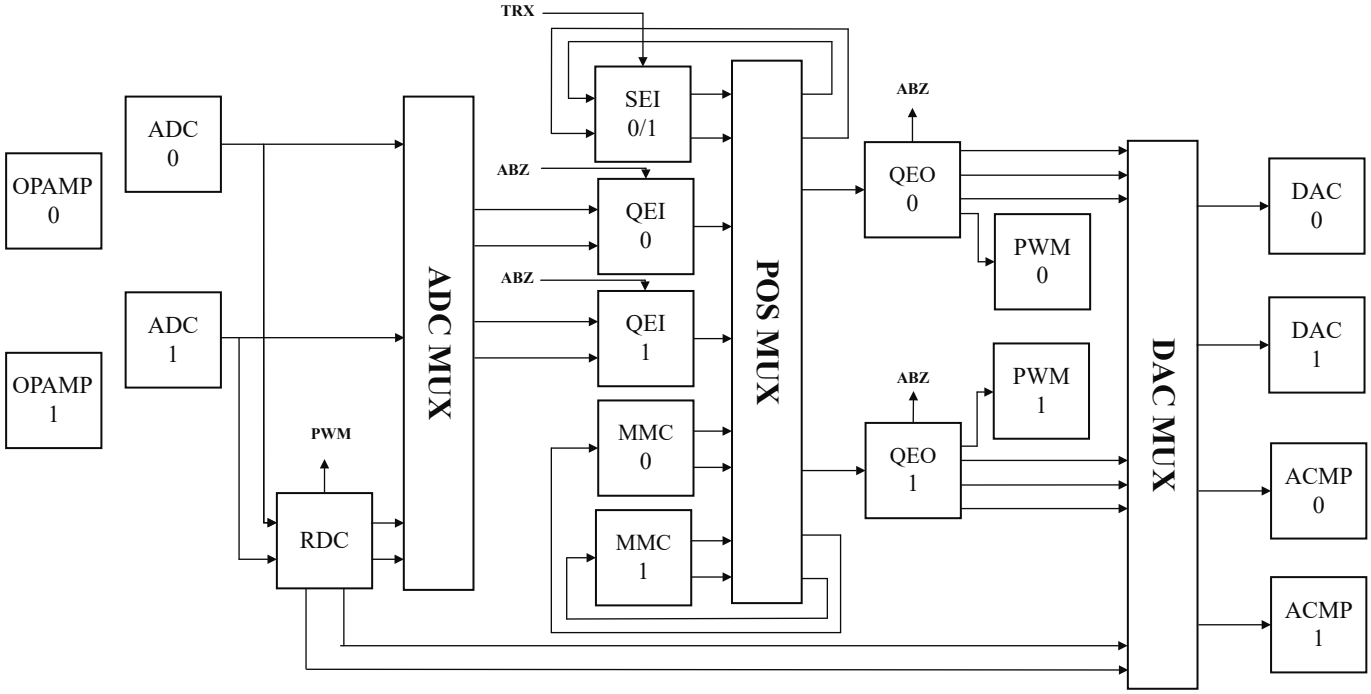


Figure 27: Partial Connection Block Diagram of Motor System

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32.1 PWM Timer

This product supports 2 PWM timers. A single PWM supports up to 28-bit counter and 24 channels, of which channels 0~7 are used to generate 8 independent or 4 pairs of complementary PWM outputs, supporting dead zone control and fault protection. The remaining 16 channels support output comparison, which can be output to IO or other on-chip modules through the interconnection manager. The PWM timer supports input capture.

On this product, the signal connection of a single PWM timer is as follows:

- Outputs of channels 0~7 are connected to IO through PWM control logic
- Outputs of channels 8~23 are connected to the interconnection manager in the motor control unit
- Input captures 0~7 are from IO
- Input captures 8~23 are from the interconnection manager in the motor control unit
- Counter synchronization trigger input SYNCI is from the interconnection manager in the motor control unit
- Forced output enable input FRCI is from the interconnection manager in the motor control unit
- Trigger input SHSYNCI for forced output shadow register to take effect is from the interconnection manager in the motor control unit
- External fault protection inputs FAULTE0~1 are from IO, which can realize fault protection without clock
- Internal fault protection inputs FAULTI0~3 are from the interconnection manager in the motor control unit, which need to realize fault protection when the motor system clock works normally

For the connection between each PWM module signal and the signal in the interconnection manager of the motor control unit, please refer to Section 34.3.5.

32.2 Quadrature Encoder Interface QEI

This product supports 2 quadrature encoder interfaces QEI. QEI supports various types of external quadrature encoders, and also supports digital Hall signal input to sense the position information of the motor, and provide motor steering and speed information.

On this product, the signal connection of a single quadrature encoder interface QEI is as follows:

- A/U phase input APH, from IO
- B/V phase input BPH, from IO
- Z/W phase input ZPH, from IO
- H signal input HPH, also known as HOME, from IO
- F signal input FPH, also known as FAULT, from IO
- Position output, output to position repeater
- Sine ADC input, from ADC repeater
- Cosine ADC input, from ADC repeater
- Snapshot input SNAPI, from the interconnection manager in the motor control unit
- Trigger output TRGO, connected to the interconnection manager in the motor control unit

Each signal of the quadrature encoder interface QEI supports flexible allocation through the interconnection manager in the motor control unit, and can be connected to each IO of TRGM or other on-chip peripherals. For specific signal connections, please refer to Section 34.3.5.

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32.3 Quadrature Encoder Output QEO

This product supports 2 quadrature encoder outputs QEO. QEO supports sending quadrature coded signals to the outside.

On this product, the signal connection of a single quadrature encoder output QEO is as follows:

- A/U phase output, output through IO
- B/V phase output, output through IO
- Z/W phase output, output through IO
- Position input, from position repeater
- A-phase DAC output, output to DAC repeater
- B-phase DAC output, output to DAC repeater
- C-phase DAC output, output to DAC repeater
- Commutation signal output, for commutation of stepper motors and brushless DC motors, connected to the interconnection manager in the motor control unit

Each signal of the quadrature encoder output QEO supports flexible allocation through the interconnection manager in the motor control unit, and can be connected to each IO of TRGM or other on-chip peripherals. For specific signal connections, please refer to Section 34.3.5.

32.4 Serial Encoder Interface SEI

This product supports 2 serial encoder interfaces SEI. SEI supports synchronous and asynchronous encoders, and can be used as host and encoder.

On this product, the signal connection of the serial encoder interface SEI is as follows:

- CK bidirectional clock, from IO
- TX output, output through IO
- RX input, from IO
- DE output enable, output through IO
- Position input, from position repeater
- Position output, output to position repeater
- Trigger input, from the interconnection manager in the motor control unit
- Trigger output, connected to the interconnection manager in the motor control unit

Each signal of the serial encoder interface SEI supports flexible allocation through the interconnection manager in the motor control unit, and can be connected to each IO of TRGM or other on-chip peripherals. For specific signal connections, please refer to Section 34.3.5.

32.5 Position Manager MMC

This product supports 2 position managers MMC. MMC supports generation, analysis, compensation and prediction output of motion signals.

On this product, the signal connection of the position manager MMC is as follows:

- Position input, from position repeater
- Position output 0, output to position repeater
- Position output 1, output to position repeater

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- Trigger input, from the interconnection manager in the motor control unit
- Trigger output, connected to the interconnection manager in the motor control unit

Each signal of the position manager MMC supports flexible allocation through the interconnection manager in the motor control unit, and can be connected to each IO of TRGM or other on-chip peripherals. For specific signal connections, please refer to Section 34.3.5.

32.6 Resolver Demodulator RDC

This product supports 1 resolver demodulator RDC. RDC supports outputting resolver excitation signals and demodulating the orthogonal signals sampled by ADC.

On this product, the signal connection of the resolver demodulator RDC is as follows:

- PWM excitation signal output, output through IO
- Sine excitation signal output, output to DAC repeater
- Sine ADC input, from ADC repeater
- Cosine ADC input, from ADC repeater
- Sine ADC output, output to ADC repeater
- Cosine ADC output, output to ADC repeater
- Trigger input, from the interconnection manager in the motor control unit
- Trigger output, connected to the interconnection manager in the motor control unit

Each signal of the resolver demodulator RDC supports flexible allocation through the interconnection manager in the motor control unit, and can be connected to each IO of TRGM or other on-chip peripherals. For specific signal connections, please refer to Section 34.3.5.

32.7 Programmable Logic Unit PLB

This product supports 1 programmable logic unit PLB. PLB supports combinational logic operations and counting operations.

On this product, the signal connection of the programmable logic unit PLB is as follows:

- Trigger input, from the interconnection manager in the motor control unit
- Trigger output, connected to the interconnection manager in the motor control unit

Each signal of the programmable logic unit PLB supports flexible allocation through the interconnection manager in the motor control unit, and can be connected to each IO of TRGM or other on-chip peripherals. For specific signal connections, please refer to Section 34.3.5.

32.8 Interconnection Manager TRGM

The interconnection manager TRGM of this product supports interconnection between signals of various devices inside and outside the motor control unit, which can integrate various on-chip peripherals to realize mutual synchronization and cooperation between peripherals.

The interconnection manager supports multiple inputs, which come from IO, various peripherals inside and outside the motor control unit, and interconnection managers of other motor control units. For the input signals of each interconnection manager, please refer to Section 34.3.5.

The interconnection manager supports multiple outputs, and users can select one of the many input signals

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of the interconnection manager to connect to any output. For the output signal allocation of each interconnection manager, please refer to Section 34.3.6.

Note: When connecting the internal signal of the motor system to the outside of the motor system, the clock frequency and valid signal length need to be considered.

The following signals have only one motor system clock cycle, which can be connected arbitrarily inside the motor system, and it is not recommended to go out of the motor system:

Quadrature decoder n trigger output; Hall sensor n interface trigger output; Synchronous timer channel n.

The interconnection manager supports managing DMA requests of peripherals inside and outside the motor control unit. On this product, DMA requests of devices in the motor control unit are not directly connected to DMAMUX, but are transferred through TRGM. For DMA requests managed by each TRGM, please refer to Section 34.3.4.

The interconnection manager supports multiple digital filters, which can filter specific input signals of TRGM. For the input signals configured with digital filters in each TRGM, please refer to Section 34.3.3.

32.9 Synchronous Timer SYNT

This product supports 1 synchronous timer SYNT. SYNT supports 32-bit counter and 4 output comparison channels, and the output signal is connected to the input of TRGM of 2 motor control units on this product. It is used for synchronization of peripherals inside and outside each motor control unit.

For the output signal connection of the synchronous timer SYNCT, please refer to Section 34.3.5.

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33 PWM Timer PWM

This chapter introduces the main functions and features of the PWM timer of this product.

33.1 Feature Summary

This section describes the main features of the PWM controller:

- 28 (24 +4) bit resolution counter, supports up-counting mode
- Supports counter synchronization
- Up to 24 comparators, can be used for output comparison or input capture
- Up to 24 channels, among which channels 0~7 can be used for PWM output
  - Supports 8 independent or 4 pairs of complementary PWM outputs
  - Complementary PWM supports dead zone insertion, supports independent configuration of bilateral dead zone width
  - Supports forcing PWM output to a specified state
  - Supports fault protection input, and the state of each PWM output channel can be configured individually when an error occurs (such as when fault protection input is triggered)
- Supports flexible allocation of different numbers of comparators for each output channel, flexibly controls output signals, and generates such as edge-aligned PWM, left-right asymmetric center-aligned PWM and more complex output signals
- Supports generating various DMA requests and interrupt requests
- Some registers are equipped with shadow registers, which support flexible update/effective timing of new register values

The block diagram of PWM is shown in Figure 28.

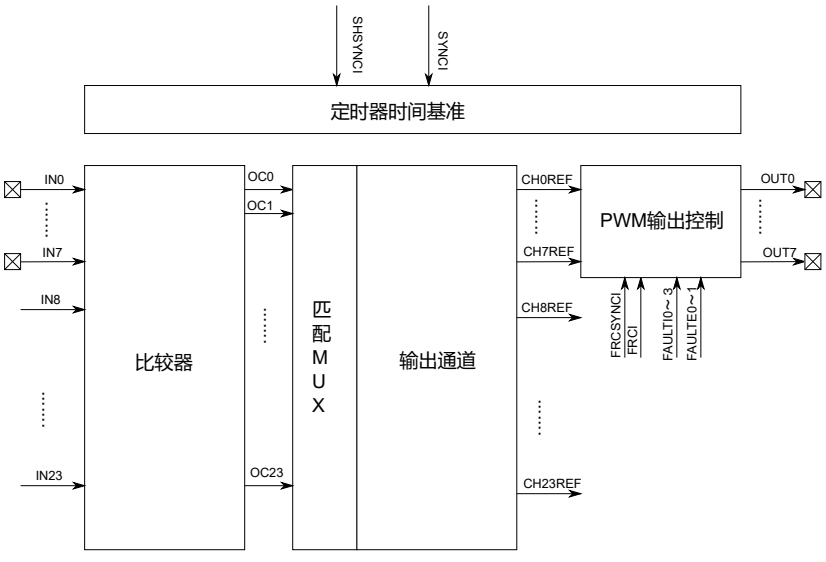


Figure 28: Block Diagram of PWM Timer

33.2 Functional Description

This section describes the functions of each sub-module of the PWM timer.

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33.2.1 Timer Time Base

The function of the timer time base module is to determine the running time and period of the PWM timer.

It consists of the following parts:

- Counter: The counter includes a counter part and an extended counting part. The counter is 24-bit, and the extended part is 4-bit, which can be combined into a 28-bit counter for use
- Start register: You can set the counter start value and extended start value
- Reload register: You can set the counter reload value and extended reload value

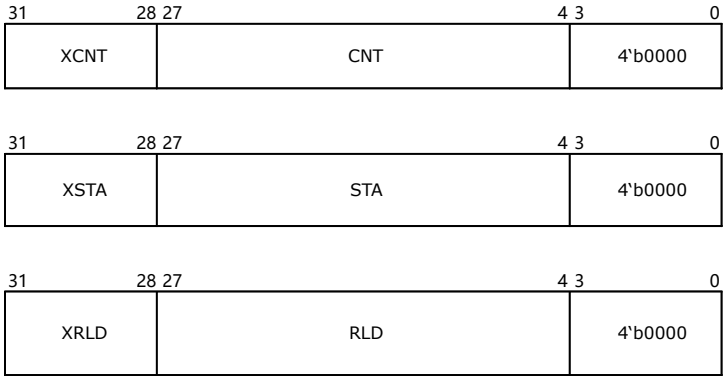


Figure 29: Time Base Module of PWM

The counter is composed of a 24-bit counter (CNT) and a 4-bit extended counter (XCNT), which can be expanded into a 28-bit counter.

The start register STA can set the start value of the 24-bit counter (STA) and the start value of the 4-bit extended counter (XSTA).

The reload register RLD can set the reload value of the 24-bit counter (RLD) and the reload value of the 4-bit extended counter (XRLD).

The counter, start register and reload register each occupy 32-bit register space, with the lower 4 bits being 0.

The timer supports up-counting.

After the counter is enabled (bit CEN of GCR is set to 1), it always starts counting from the start value. When the timer value (CNT) counts to the reload value (RLD), the reload flag bit RLDF is set to 1. At this time, the extended counter value increases by 1, and the counter value returns to the start value (STA). When the extended counter value (XCNT) also counts to the extended reload value (XRLD), the extended reload flag bit XRLDF is set to 1, and the extended counter returns to the start value.

If the counter start value is set to 0 and the reload value is set to x, the actual timing period of the timer is x+1 clock cycles. If the counter start value STA is set to 24'h000000, the reload value RLD is set to 24'hFFFFFF, the extended start value XSTA is set to 4'h0, and the extended reload value XRLD is set to 4'hF, the effect is equivalent to expanding the counter from 24 bits to 28 bits.

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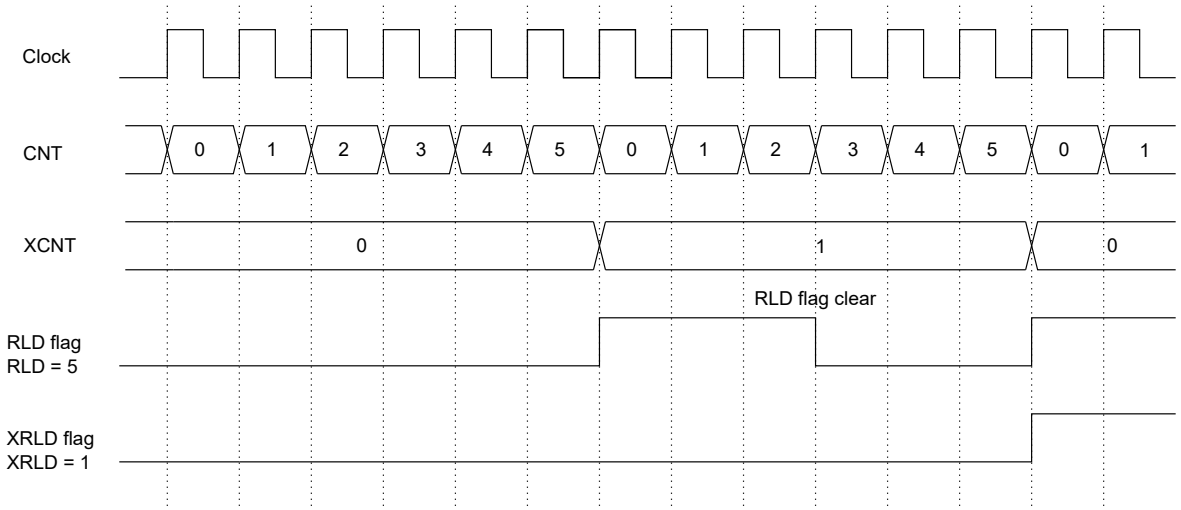


Figure 30: Schematic Diagram of PWM Counter Counting, Reload and Extended Reload Flag Setting

After the user configures the STA and RLD registers, set the CEN bit of GCR to 1, and the counter CNT starts counting.

Users can also use the synchronization trigger input (SYNCI) of the PWM timer to synchronize the timing when the counter starts counting. Users can configure the RLDSYNCEN bit and XRLDSYNCEN bit of GCR to select whether to enable external synchronization. When the RLDSYNCEN bit of GCR is set to 1, the synchronization trigger input (SYNCI) can reset the counter to the start value (CNT = STA), and the reload flag bit RLDF is set to 1. If the RLDSYNCEN bit of GCR is set to 1, the extended bit of the counter is reset to the extended bit of the start value (XCNT = XSTA), and the extended reload flag bit XRLDF is also set to 1. It should be noted that the synchronization trigger input (SYNCI) of the PWM timer takes effect when a high level is captured. When the user configures the TRGOCFG[PWM_SYNCI] register of TRGM to select the trigger signal, the rising or falling edge of the trigger signal can also be converted into a pulse signal according to actual needs, so as to achieve the effect of taking effect when a specific edge is captured.

For multi-motor cooperative work, the synchronization timer SYNT can be used to generate synchronization pulses, so that multiple PWMs start counting at the same time, or start counting at a specified interval.

The counter module also supports a half-reload flag bit HALFRLDF, which is set to 1 when the counter counts to half of (RLD - STA) (note that XRLD and XSTA are not included).

The control module manages the following flag bits:

- RLDF: This bit is set to 1 when the counter (excluding extended counting bits) counts to the value of the reload register, or when the counter is reset by SYNCI
- XRLDF: This bit is set to 1 when the extended counting bits of the extended counter count to the extended reload counting bits, or when the counter is reset by SYNCI
- HALFRLDF: This bit is set to 1 when the counter counts to the middle of the start value and the reload value

33.2.2 PWM Generation

PWM generation requires the use of comparators and channels. Its main function is to generate output reference signals in units of channels using a combination of comparators.

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As shown in the figure below, the PWM generation module includes 24 comparators and 24 output channels.

Among them, channels 0 to 7 are PWM output channels, and the channel outputs CH0REF ~ CH7REF are connected to PWM control logic, such as complementary control, dead zone generation, fault protection and other modules. The final output signals OUT0~OUT7 are sent to the chip pins.

Channels 8 and later are general-purpose output channels. The output reference signals of these channels do not pass through the PWM control logic, and can also output signals for internal and external use.

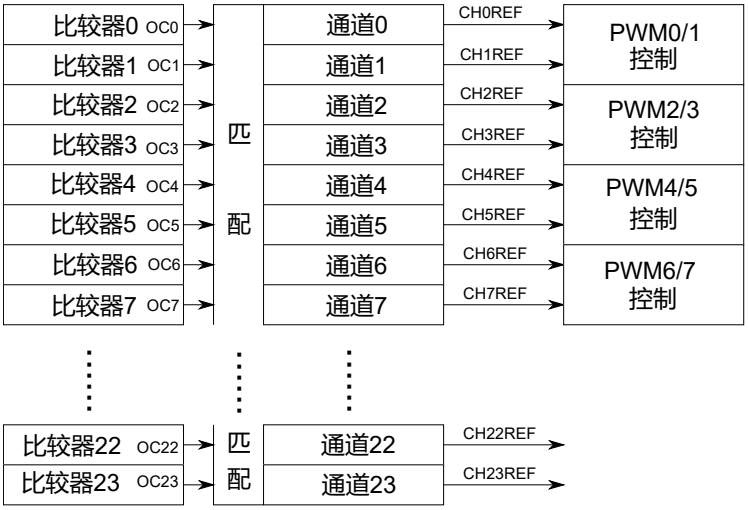


Figure 31: PWM Generation Module of PWM Timer

When the 24 comparators of the PWM timer are used for PWM generation, they need to be configured to output comparison mode (set the CMPMODE bit of CMPCFGx to 0). At this time, when the counter value is equal to the value of the comparison register, a matching event is generated. Note that the comparator, like the timer, includes a 24-bit comparison bit (CMP) and a 4-bit extended comparison bit (XCMP). You can set the XCNTCMPEN bit of CMPCFGx to select whether the 4-bit extended bits participate in the comparison.

- When all bits of XCNTCMPEN[3:0] are set to 0, the 4-bit extended comparison bits do not participate in the comparison. When the 24-bit counter counts to the 24-bit comparison bit, that is, CNT == CMPx, a matching event is generated
- When one or more bits of XCNTCMPEN[3:0] are set to 1, the corresponding extended comparison bits will be compared with the corresponding extended counting bits of the timer. For example, when XCNTCMPEN[3:0] = 4'b1111, a matching event is generated when the 4-bit extended counting bits reach the 4-bit extended comparison bits and the 24-bit counter counts to the 24-bit comparison bit, that is, XCNT == XCMP CNT == CMP.

The comparator register also contains 4 bits for fractional comparison (CMPHLF and CMPJIT). Setting these bits can make the accuracy of the matching event generated by the comparison register less than one PWM timer clock cycle. CMPHLF is a half-cycle comparison bit, which is implemented by the lower edge of the PWM timer clock. Setting this bit to 1 can delay the generation time of the matching event by 1/2 clock cycle. CMPJIT is a jitter comparison bit, which is implemented by 4, 8, 16 cycle edge jitter, and achieves a delay effect with accuracy of 1/4, 1/8 and 1/16 clock cycle after averaging.

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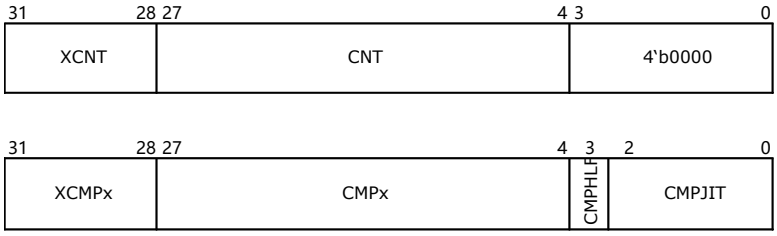


Figure 32: Bit Field Diagram of PWM Comparator Corresponding to Counter

Users can assign one or more consecutive comparators to a channel to achieve flexible and complex output. Users set the CHCFGx [CMPSELBEG] bit to select the start number of the comparison register assigned to channel x, and set the CHCFGx [CMPSELEND] bit to select the end number of the comparison register assigned to channel x.

One comparator can be assigned to multiple channels at the same time.

For example, the following is the configuration of an application, with 3 PWM outputs and two control signal outputs:

Set CHCFG[0][CMPSELBEG] = 5'b00000, CHCFG[0][CMPSELEND] = 5'b00010, which means that comparison registers 0~2 are assigned to channel 0.

Set CHCFG[1][CMPSELBEG] = 5'b00011, CHCFG[1][CMPSELEND] = 5'b00011, which means that comparison register 3 is assigned to channel 1.

Set CHCFG[5][CMPSELBEG] = 5'b00100, CHCFG[5][CMPSELEND] = 5'b00111, which means that comparison registers 4~7 are assigned to channel 5.

Set CHCFG[8][CMPSELBEG] = 5'b00100, CHCFG[8][CMPSELEND] = 5'b00101, which means that comparison registers 4~5 are assigned to channel 8.

Set CHCFG[9][CMPSELBEG] = 5'b00111, CHCFG[9][CMPSELEND] = 5'b00111, which means that comparison register 7 is assigned to channel 9.

When the counter CNT counts to the CMP or XCMP configured in the comparator CMPx, a matching event is generated, and the OCx output is set to logic 1. When the counter CNT value reaches the reload register and a reload event occurs, the output is reset to logic 0.

If the comparator value CMPx is equal to the reload value RLD, OCx will keep outputting logic 1. Therefore, setting CMPx = RLD can be used to generate a PWM output with 100% duty cycle.

When the comparator value is greater than the reload value, that is, CMPx > RLD, since the counter value CNT never reaches CMPx, the comparator output OCx will remain logic 0.

Note: The output OCx of the comparator is a PWM signal. The duty cycle of the PWM is determined by CMPx and RLD together. Setting CMPx == RLD can obtain a PWM with 100% duty cycle, and setting CMPx > RLD can obtain a PWM with 0% duty cycle.

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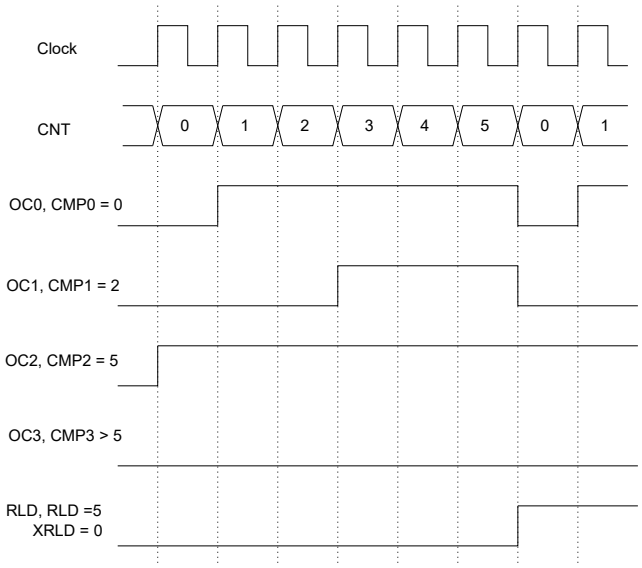


Figure 33: Schematic Diagram of PWM Counter Counting, Reload and Extended Reload Flag Setting

The output reference signal CHxREF of channel x is obtained by XOR of all comparator outputs OC_BEG ~ OC_END assigned to it, that is:

$$CHxREF = OC_BEG \oplus OC_BEG + 1 \oplus OC_BEG + 2 \oplus \cdots \oplus OC_END$$

33.2.3 PWM Generation Examples

This section provides several examples of generating specific PWM outputs by configuring comparators in combination with output channels.

The following is an example of generating edge-aligned PWM on the outputs of channel 0 (CH0REF) and channel 1 (CH1REF).

Assign comparator 0 to channel 0, that is, CHCFG0[CMPSELBEG] = 0, CHCFG0[CMPSELEND] = 0

Assign comparator 1 to channel 1, that is, CHCFG1[CMPSELBEG] = 1, CHCFG1[CMPSELEND] = 1

Set comparators 0~1 to level output mode. Set the values of comparators 0~1 as $STA < CMP0 < CMP1 < RLD$. In this way, the outputs of CH0REF and CH1REF are obtained as shown in Figure 34:

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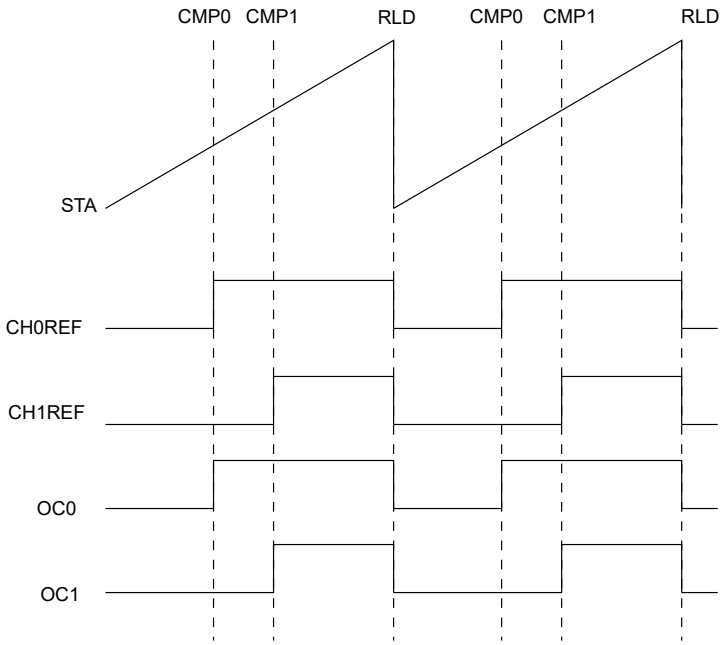


Figure 34: Edge-Aligned PWM Generation Example Diagram

The following is an example of generating center-aligned phase-shifted PWM on the outputs of channel 0 (CH0REF) and channel 1 (CH1REF).

Assign comparators 0 and 1 to channel 0, that is, CHCFG0[CMPSELBEG] = 0, CHCFG0[CMPSELEND] = 1

Assign comparators 2 and 3 to channel 1, that is, CHCFG1[CMPSELBEG] = 2, CHCFG1[CMPSELEND] = 3

Set comparators 0~3 to level output mode.

Set the values of comparators 0~3 as $STA < CMP2 < CMP0 < CMP3 < CMP1 < RLD$. In this way, the outputs of CH0REF and CH1REF are obtained as shown in Figure 35:

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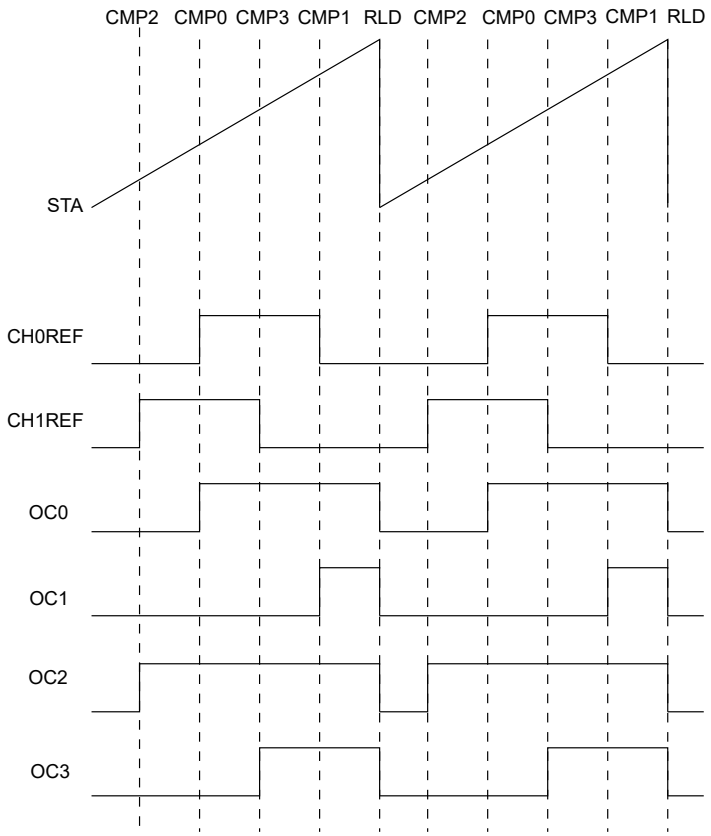


Figure 35: Center-Aligned Phase-Shifted PWM Generation Example Diagram

The following is an example of generating double-toggle PWM on the output of channel 0 (CH0REF).

Assign comparators 0 ~ 3 to channel 0, that is, CHCFG0[CMPSELBEG] = 0, CHCFG0[CMPSELEND] = 3.

Set comparators 0~3 to level output mode.

Set the values of comparators 0~3 as $STA < CMP0 < CMP1 < CMP2 < CMP3 < RLD$.

In this way, the output of CH0REF is obtained as shown in Figure 36:

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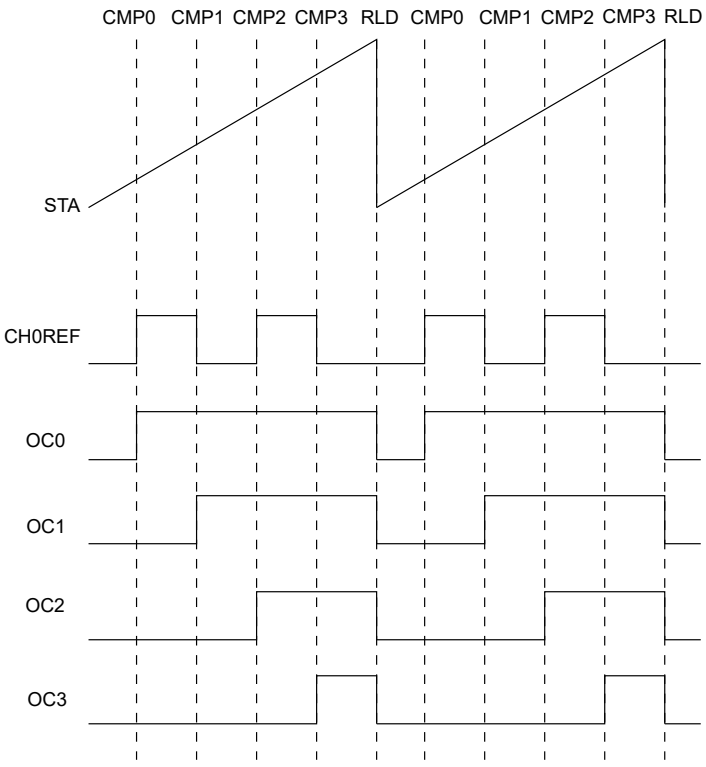


Figure 36: Double-Toggle PWM Generation Example Diagram

33.2.4 Overview of PWM Output Control

The reference signals (CH0REF~CH7REF) of the PWM output channels (channels 0~7) undergo subsequent complementary control, dead zone insertion, inversion control, forced output, and fault protection to form output signals (OUT0~OUT7) to the IO. These PWM control logics can be configured through the control bits in the PWMCFG[x] and CHCFG[x] registers.

Taking a pair of adjacent PWM outputs as an example, the PWM control logic is shown in Figure 37.

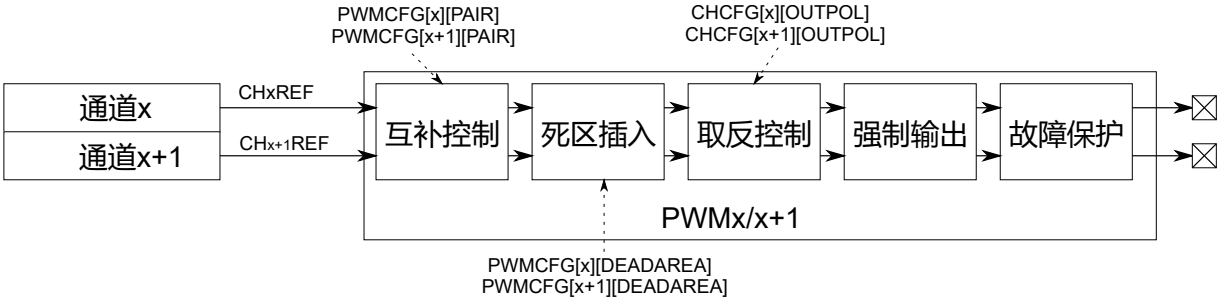


Figure 37: PWM Output Control Example Diagram

33.2.5 PWM Complementary Control

The PWM timer supports generating paired complementary PWM outputs. When the PWMCFGx[PAIR] bit is set to 1, PWM0 and PWM1, PWM2 and PWM3, PWM4 and PWM5, PWM6 and PWM7 can be set as complementary PWM outputs. Note that the complementary output will take effect only when the PAIR bits of the paired PWMCFGx registers are all set to 1. Once set to PWM complementary output, the configuration of the

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even-numbered PWM channel takes effect, and the output of the odd-numbered PWM channel is the inverted output of the even-numbered channel.

For example, when PWM0 and PWM1 are configured as complementary outputs, the PWM0 output (OUT0) is the output reference signal (CH0REF) of channel 0, and PWM1 (OUT1) is the inverted signal of PWM0.

33.2.6 Dead Zone Control

When the PWM timer is configured to output a pair of complementary PWM outputs, the user can insert a certain delay between the toggles of the paired output reference signals through the dead zone control module to avoid simultaneous conduction of external switches controlled by the PWM output.

The PWM timer allows users to insert different delays on the 2 complementary reference signals, so that users can optimize the dead zone time according to the characteristics of off-chip switching devices.

Users can configure the dead zone length through the PWMCFGx[DEADAREA] bit. The registers of paired channel x and channel x+1 must be configured at the same time to correctly generate the dead zone.

For example, when PWM output 0 and output 1 are configured as complementary PWM outputs, you need to configure: PWMCFG0[DEADAREA] can delay each rising edge of the channel 0 reference signal by several clock cycles. PWMCFG1[DEADAREA] can delay each rising edge of the channel 1 reference signal by several clock cycles.

Note: The unit of the dead zone length configured by PWMCFGx[DEADAREA] is 0.5 PWM timer clock cycles. The minimum supported dead zone length is 1 clock cycle.
That is: PWMCFGx[DEADAREA] = 20'h00003, indicating that the dead zone length is 1.5 clock cycles
PWMCFGx[DEADAREA] = 20'h000A0, indicating that the dead zone length is 80 clock cycles

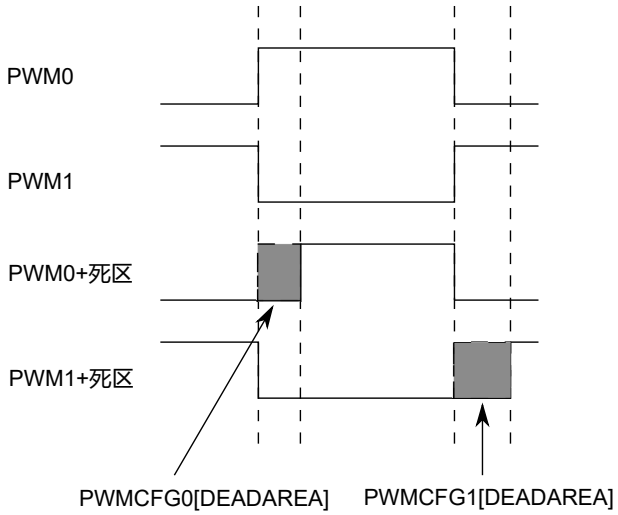


Figure 38: Schematic Diagram of PWM Dead Zone Control

33.2.7 Output Inversion

The level required to turn on and off external devices can be high or low. To adapt to various external devices, the output of all channels of the PWM timer can be configured with output polarity by setting the CHCFGx[OP] bit:

- When set to 0, the output is CHxREF without change
- When set to 1, the output is the inverted CHxREF

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33.2.8 Forced Output Control

The forced output control module can act on the reference signals of channels 0~7, allowing users to configure the reference signals to a specified state. This module can easily generate the 6-step commutation PWM output required to drive BLDC motors.

By configuring the PWMCFGx[FRCSRCSEL] bit, you can freely select whether the forced debugging of PWM outputs 0~7 is triggered by hardware or software:

- When this bit is set to 0, it means that the forced output of PWM output x is controlled by the external input FRCI. The active polarity of FRCI can be selected through the GCR[FRCPOL] bit. GCR[FRCPOL] = 1'b0 means that FRCI logic 1 is active, otherwise logic 0 is active.
- When this bit is set to 1, and the software sets the GCR[SWFRC] bit to 1, the forced output is active.

By configuring the register GCR[FRCTIME], users can select the timing when the output takes effect:

- 2'b00: Take effect immediately
- 2'b01: The forced output takes effect when the counter reload flag bit RLD is set to 1 after being enabled
- 2'b10: The forced output takes effect when a rising edge is captured on the synchronous input trigger (FRCSYNCI) after being enabled
- 2'b11: Forced output is not active

Note: GCR[FRCTIME] also controls the timing when the forced output becomes invalid. That is, after the forced output is turned off, the timing when the output becomes invalid is also controlled by it.

By configuring the register FRCMD[MODEx], you can configure the state of the outputs of channels 0~7 when the forced output takes effect:

- 2'b00: Force output logic 0
- 2'b01: Force output logic 1
- 2'b10: Turn off the output, the pin becomes high impedance HiZ
- 2'b11: Do not perform forced output, that is, the output remains the reference signal CHxREF of channel x unchanged

Note: The FRCMD register is equipped with a shadow register. Writing directly to the FRCMD register does not necessarily take effect immediately. If the user wants to change the forced output state of a PWM output, for example, from forced output logic 0 to output logic 1, the effective time is determined by FRCSHDWUPT of PWMCFG[x]. Please refer to Section 33.2.12 for details.

33.2.9 Fault Protection

The PWM timer allows users to cut off the PWM output signal quickly in real time in emergency situations.

The PWM timer supports a total of 6 internal and external fault protection input signals, including 4 internal fault input signals (FAULTI0~FAULTI3) and 2 external fault input signals (FAULTE0, FAULTE1).

The external fault input signals (FAULTE0, FAULTE1) of the PWM timer can also work when the PWM timer clock is lost. Users can set the GCR[FAULTExPOL] bit to select whether the external fault input signal is active at logic 1 or logic 0.

- 1'b0: FAULTEx is active when logic 1
- 1'b1: FAULTEx is active when logic 0

Users can turn on or off the external fault input of the PWM timer by configuring the GCR[FAULTExEN] bit:

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- 1'b0: Turn off fault protection, PWM output does not respond to external fault input signal x (FAULTEx)
- 1'b1: Enable fault protection, and it takes effect when the fault input signal (FAULTIx) is logic 1 or logic 0 according to the setting of the GCR[FAULTExPOL] bit

Users can turn on or off the internal fault input of the PWM timer by configuring the GCR[FAULTIxEN] bit.

- 1'b0: Turn off fault protection, PWM output does not respond to internal fault input signal x (FAULTIx)
- 1'b1: Enable fault protection, internal fault input is active when logic 1

When the fault input is active, the PWM timer can force the outputs (OUT0 ~ OUT7) of channels 0~7 to a certain safe predetermined state.

Users can choose to enable one or more or even all fault input signals. When any of the enabled fault inputs is active, the fault protection is valid.

By configuring PWMCFGx[FAULTMODE], all PWM0~7 outputs can be independently configured as:

- 2'b0x: When a fault occurs, turn off the output, the pin becomes high impedance HiZ
- 2'b10: When a fault occurs, the output becomes logic 0
- 2'b11: When a fault occurs, the output becomes logic 1

Once a fault occurs, the PWM output will not recover until the system state changes.

Users can select the timing of PWM recovery by setting PWMCFGx[FAULTRECTIME]:

- 2'b00: After the fault is recovered, the output recovers immediately.
- 2'b01: After the fault is recovered, the output recovers when the counter reload flag bit RLD is set to 1.
- 2'b10: After the fault is recovered, the output recovers after the counter matches a certain CMP. You can select one of the 24 comparators as the fault recovery timing through the GCR [FAULTRECHWSEL] bit.
- 2'b11: After the fault is recovered, the output recovers after the software sets the GCR[FAULTCLR] bit to 1. Note that it is recommended to clear FAULTCLR to 0 after the fault is eliminated, otherwise the PWM will automatically recover after the next fault occurs and recovers because the FAULTCLR bit is set to 1.

33.2.10 Debug Mode Support

The PWM timer allows users to disconnect the PWM output when entering the debug mode, so as to protect external power devices.

You can set the GCR[DEBUGFAULT] bit. When this bit is set to 1, once the chip enters the debug mode, the PWM output state will be forced to the specified state just like fault protection. The specific state of the pin is determined by the setting of the PWMCFGx[FAULTMODE] bit, same as fault protection.

33.2.11 Input Capture Module

Each comparator of the PWM timer can also be used in input capture mode.

Note: Once a comparator is configured to input capture mode, it can no longer be assigned to a channel for output.

Users can configure the CMPCFGx[CMPMODE] bit to select the working mode of the comparator:

- 1'b0: Output comparison mode, the comparator can be assigned to a channel for output
- 1'b1: Input capture mode. At this time, the comparator is used to capture the toggle of the input signal and save the counter values (CNT and XCNT) when toggling.

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After configuring the comparator to input capture mode, the user can read the counter value captured by CMPx on the rising edge of the signal (INx) from the CAPPOSx register; read the counter value captured by CMPx on the falling edge from the CAPNEGx register. The CAPPOSx and CAPNEGx registers will save the 24-bit counter (CNT) value of the counter register and the 4-bit extended counter (XCNT) value.

Note: The values of the CAPPOSx register and CAPNEGx register will be refreshed when a new edge is captured, and the previously captured values will be lost.

33.2.12 Shadow Register

Some registers of the PWM timer support shadow registers. The function of the shadow register is to provide a read-write interface for some key registers of the timer. When the processor accesses the register, it actually changes its shadow register, and the new value does not take effect immediately. Only at the time specified by the user, the value of the shadow register is updated to the register, thereby changing the working state of the PWM timer.

The following three groups of registers of the PWM timer support shadow registers:

- Counter start register STA (including STA/XSTA) and reload register RLD (including RLD/XRLD)
- Comparator CMPx, when it is used for output comparison
- FRCMD[FRCMD] bit in the control register, that is, the forced output control mode bits of channels 0~7 (PWM channels)

When users write to these registers and register bits, they will not take effect immediately. They need to wait for a specified time to take effect uniformly.

For the counter start register STA and counter reload register RLD, the effective time of their shadow register values (including STA, XSTA, RLD, XRLD) can be set through the SHCR[CNTSHDWUPT] bit:

- 2'b00: Take effect after the software sets the SHLK [SHLK] bit to 1
- 2'b01: Take effect in real time, within one cycle after the register is written
- 2'b10: Take effect after a certain CMP of the timer matches. Users can select one of comparators 0~23 through SHCR [CNTSHDWSEL]. The match can be the output comparison of the comparator or input capture. Users can set the value of the selected comparator CMPx to be equal to RLD or xRLD, so as to achieve the purpose of updating the shadow register after a complete PWM cycle.
- 2'b11: Take effect when a high level is captured on the shadow register reload trigger input SHRLDSYNCI. When configuring the TRGOCFG[PWM_SHRLDSYNCI] register of TRGM to select the trigger signal, the user can also convert the rising or falling edge of the input signal into a valid pulse signal according to actual needs, so as to achieve effect when a specific edge is captured.

For the comparator CMPx of the PWM timer, the effective time of their shadow register values can be set through the CMPCFGx[CMPSHDWUPT] bit:

- 2'b00: Take effect after the software sets the SHLK [SHLK] bit to 1
- 2'b01: Take effect in real time, within one cycle after the register is written
- 2'b10: Take effect after a certain CMP of the timer matches. Users can select one of comparators 0~23 through GCR [CMPSHDWSEL]. The match can be the output comparison of the comparator or input capture. Users can set the value of the selected comparator CMPx to be equal to RLD or xRLD, so as to achieve the purpose of updating the shadow register after a complete PWM cycle.
- 2'b11: Take effect when a high level is captured on the shadow register reload trigger input SHRLDSYNCI.

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When configuring the TRGOCFG[PWM_SHRLDSYNCI] register of TRGM to select the trigger signal, the user can also convert the rising or falling edge of the input signal into a valid pulse signal according to actual needs, so as to achieve effect when a specific edge is captured.

Note: During the operation of the PWM timer, if the duty cycle of the PWM output waveform is changed in real time by updating the comparator CMPx, it is recommended that the user configure the CMPCFGx[CMPSHDWUPT] bit of CMPx to 2'b10 or 2'b11, that is, the effective time of the shadow register of CMPx is specified by hardware. Avoid software changing the configuration at an inappropriate time, resulting in abnormal output waveform.

The effective time of the shadow register bits of the forced output mode bits of channels 0~7 (PWM channels) can be set through the PWMCFGx[FRCSHUPT] bit:

- 2'b00: Take effect after the software sets the SHLK[SHLK] bit to 1
- 2'b01: Take effect in real time, within one cycle after the register is written
- 2'b10: Take effect after a certain CMP of the timer matches. Users can select one of comparators 0~23 through SHCR [FRCSHDWSEL]. The match can be the output comparison of the comparator or input capture. Users can set the value of the selected comparator CMPx to be equal to RLD or xRLD, so as to achieve the purpose of updating the shadow register after a complete PWM cycle.
- 2'b11: Take effect when a high level is captured on the shadow register reload trigger input SHRLDSYNCI. When configuring the TRGOCFG[PWM_SHRLDSYNCI] register of TRGM to select the trigger signal, the user can also convert the rising or falling edge of the input signal into a valid pulse signal according to actual needs, so as to achieve effect when a specific edge is captured.

In addition, the shadow register supports write protection. Users can set the SHCR[SHLKEN] bit to 1, and then set the SHclk[SHLK] bit to 1. In this way, any write operation to the shadow register by the software is invalid.

Users can unlock the shadow register by writing the magic code 0xB0382607 to the UNLK register. After unlocking, the write operation to all shadow registers can be resumed.

During the process of updating multiple shadow registers by the user, it is possible that the software or DMA only updates part of the shadow registers, and the effective time of the shadow registers has arrived, resulting in errors. Users can avoid this situation through the following operation steps:

- Set the SHCR[SHLKEN] bit to 1, then set the SHLK[SHLK] bit to 1 to enable shadow register write protection
- Write the magic code 0xB0382607 to the UNLK register to unlock the write protection of the shadow register
- Update all shadow registers that need to be updated
- Set the SHCR[SHLK] bit to 1 again, and then the shadow register write protection is re-locked.
- Then, when the next shadow register effective event occurs, the new register value takes effect. Note that if the effective timing control bit is set to 2'b00, that is, it takes effect after the software sets the SHLK[SHLK] bit to 1, then the user needs to set the SHLK[SHLK] bit to 1 again to make the new value of the shadow register take effect.

33.2.13 Interrupt and DMA

The PWM timer supports generating the following interrupts:

- RLD: When a reload event occurs, that is, the counter counts to the value of the reload register, or the

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counter reload is triggered by the synchronous trigger input SYNCI

- XRLD: When the extended counting bits of the extended counter count to the extended reload counting bits, or when the counter is reset by SYNCI
- HALFRLD: When the counter counts to the middle of the start register and the reload register
- When the comparator is set to output comparison and a matching event occurs
- When the comparator is set to input capture and a specified edge jump is captured on the input channel
- When fault protection takes effect

The PWM timer supports generating the following DMA requests:

- RLD: When a reload event occurs, that is, the counter counts to the value of the reload register, or the counter reload is triggered by the synchronous trigger input SYNCI
- XRLD: When the extended counting bits of the extended counter count to the extended reload counting bits, or when the counter is reset by SYNCI
- HALFRLD: When the counter counts to the middle of the start register and the reload register
- When the comparator is set to output comparison and a matching event occurs
- When the comparator is set to input capture and a specified edge jump is captured on the input channel

33.3 PWM Registers

The register list of PWM is as follows:

PWM0 base address: 0xF0318000

PWM1 base address: 0xF031C000

Offset	Register Name	Description	Reset Value
0x0000	UNLK	Shadow registers unlock register	0x00000000
0x0004	STA	Counter start register	0x00000000
0x0008	RLD	Counter reload register	0x00000000
0x000C	CMP[0]	Comparator register	0x00000000
0x0010	CMP[1]	Comparator register	0x00000000
0x0014	CMP[2]	Comparator register	0x00000000
0x0018	CMP[3]	Comparator register	0x00000000
0x001C	CMP[4]	Comparator register	0x00000000
0x0020	CMP[5]	Comparator register	0x00000000
0x0024	CMP[6]	Comparator register	0x00000000
0x0028	CMP[7]	Comparator register	0x00000000
0x002C	CMP[8]	Comparator register	0x00000000
0x0030	CMP[9]	Comparator register	0x00000000
0x0034	CMP[10]	Comparator register	0x00000000
0x0038	CMP[11]	Comparator register	0x00000000
0x003C	CMP[12]	Comparator register	0x00000000
0x0040	CMP[13]	Comparator register	0x00000000
0x0044	CMP[14]	Comparator register	0x00000000

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Offset	Register Name	Description	Reset Value
0x0048	CMP[15]	Comparator register	0x00000000
0x004C	CMP[16]	Comparator register	0x00000000
0x0050	CMP[17]	Comparator register	0x00000000
0x0054	CMP[18]	Comparator register	0x00000000
0x0058	CMP[19]	Comparator register	0x00000000
0x005C	CMP[20]	Comparator register	0x00000000
0x0060	CMP[21]	Comparator register	0x00000000
0x0064	CMP[22]	Comparator register	0x00000000
0x0068	CMP[23]	Comparator register	0x00000000
0x0078	FRCMD	Force output mode register	0x00000000
0x007C	SHLK	Shadow registers lock register	0x00000000
0x0080	CHCFG[0]	Output channel configure register	0x00000000
0x0084	CHCFG[1]	Output channel configure register	0x00000000
0x0088	CHCFG[2]	Output channel configure register	0x00000000
0x008C	CHCFG[3]	Output channel configure register	0x00000000
0x0090	CHCFG[4]	Output channel configure register	0x00000000
0x0094	CHCFG[5]	Output channel configure register	0x00000000
0x0098	CHCFG[6]	Output channel configure register	0x00000000
0x009C	CHCFG[7]	Output channel configure register	0x00000000
0x00A0	CHCFG[8]	Output channel configure register	0x00000000
0x00A4	CHCFG[9]	Output channel configure register	0x00000000
0x00A8	CHCFG[10]	Output channel configure register	0x00000000
0x00AC	CHCFG[11]	Output channel configure register	0x00000000
0x00B0	CHCFG[12]	Output channel configure register	0x00000000
0x00B4	CHCFG[13]	Output channel configure register	0x00000000
0x00B8	CHCFG[14]	Output channel configure register	0x00000000

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Offset	Register Name	Description	Reset Value
0x00BC	CHCFG[15]	Output channel configure register	0x00000000
0x00C0	CHCFG[16]	Output channel configure register	0x00000000
0x00C4	CHCFG[17]	Output channel configure register	0x00000000
0x00C8	CHCFG[18]	Output channel configure register	0x00000000
0x00CC	CHCFG[19]	Output channel configure register	0x00000000
0x00D0	CHCFG[20]	Output channel configure register	0x00000000
0x00D4	CHCFG[21]	Output channel configure register	0x00000000
0x00D8	CHCFG[22]	Output channel configure register	0x00000000
0x00DC	CHCFG[23]	Output channel configure register	0x00000000
0x00F0	GCR	Global control register	0x00000000
0x00F4	SHCR	Shadow register control register	0x00000000
0x0100	CAPPOS[0]	Capture rising edge register	0x00000000
0x0104	CAPPOS[1]	Capture rising edge register	0x00000000
0x0108	CAPPOS[2]	Capture rising edge register	0x00000000
0x010C	CAPPOS[3]	Capture rising edge register	0x00000000
0x0110	CAPPOS[4]	Capture rising edge register	0x00000000
0x0114	CAPPOS[5]	Capture rising edge register	0x00000000
0x0118	CAPPOS[6]	Capture rising edge register	0x00000000
0x011C	CAPPOS[7]	Capture rising edge register	0x00000000
0x0120	CAPPOS[8]	Capture rising edge register	0x00000000
0x0124	CAPPOS[9]	Capture rising edge register	0x00000000
0x0128	CAPPOS[10]	Capture rising edge register	0x00000000
0x012C	CAPPOS[11]	Capture rising edge register	0x00000000
0x0130	CAPPOS[12]	Capture rising edge register	0x00000000
0x0134	CAPPOS[13]	Capture rising edge register	0x00000000
0x0138	CAPPOS[14]	Capture rising edge register	0x00000000
0x013C	CAPPOS[15]	Capture rising edge register	0x00000000
0x0140	CAPPOS[16]	Capture rising edge register	0x00000000
0x0144	CAPPOS[17]	Capture rising edge register	0x00000000
0x0148	CAPPOS[18]	Capture rising edge register	0x00000000
0x014C	CAPPOS[19]	Capture rising edge register	0x00000000

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Offset	Register Name	Description	Reset Value
0x0150	CAPPOS[20]	Capture rising edge register	0x00000000
0x0154	CAPPOS[21]	Capture rising edge register	0x00000000
0x0158	CAPPOS[22]	Capture rising edge register	0x00000000
0x015C	CAPPOS[23]	Capture rising edge register	0x00000000
0x0170	CNT	Counter	0x00000000
0x0180	CAPNEG[0]	Capture falling edge register	0x00000000
0x0184	CAPNEG[1]	Capture falling edge register	0x00000000
0x0188	CAPNEG[2]	Capture falling edge register	0x00000000
0x018C	CAPNEG[3]	Capture falling edge register	0x00000000
0x0190	CAPNEG[4]	Capture falling edge register	0x00000000
0x0194	CAPNEG[5]	Capture falling edge register	0x00000000
0x0198	CAPNEG[6]	Capture falling edge register	0x00000000
0x019C	CAPNEG[7]	Capture falling edge register	0x00000000
0x01A0	CAPNEG[8]	Capture falling edge register	0x00000000
0x01A4	CAPNEG[9]	Capture falling edge register	0x00000000
0x01A8	CAPNEG[10]	Capture falling edge register	0x00000000
0x01AC	CAPNEG[11]	Capture falling edge register	0x00000000
0x01B0	CAPNEG[12]	Capture falling edge register	0x00000000
0x01B4	CAPNEG[13]	Capture falling edge register	0x00000000
0x01B8	CAPNEG[14]	Capture falling edge register	0x00000000
0x01BC	CAPNEG[15]	Capture falling edge register	0x00000000
0x01C0	CAPNEG[16]	Capture falling edge register	0x00000000
0x01C4	CAPNEG[17]	Capture falling edge register	0x00000000
0x01C8	CAPNEG[18]	Capture falling edge register	0x00000000
0x01CC	CAPNEG[19]	Capture falling edge register	0x00000000
0x01D0	CAPNEG[20]	Capture falling edge register	0x00000000
0x01D4	CAPNEG[21]	Capture falling edge register	0x00000000
0x01D8	CAPNEG[22]	Capture falling edge register	0x00000000
0x01DC	CAPNEG[23]	Capture falling edge register	0x00000000
0x01F0	CNTCOPY	Counter copy	0x00000000
0x0200	PWMCFG[0]	PWM channel configure register	0x00000000
0x0204	PWMCFG[1]	PWM channel configure register	0x00000000
0x0208	PWMCFG[2]	PWM channel configure register	0x00000000
0x020C	PWMCFG[3]	PWM channel configure register	0x00000000
0x0210	PWMCFG[4]	PWM channel configure register	0x00000000
0x0214	PWMCFG[5]	PWM channel configure register	0x00000000
0x0218	PWMCFG[6]	PWM channel configure register	0x00000000
0x021C	PWMCFG[7]	PWM channel configure register	0x00000000
0x0220	SR	Status register	0x00000000
0x0224	IRQEN	Interrupt request enable register	0x00000000

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Offset	Register Name	Description	Reset Value
0x0228	CAP_EN_NEG	Capture irq mode	0x00000000
0x022C	DMAEN	DMA request enable register	0x00000000
0x0230	CMPCFG[CMPCFG0]	Comparator configure register	0x00000000
0x0234	CMPCFG[1]	Comparator configure register	0x00000000
0x0238	CMPCFG[2]	Comparator configure register	0x00000000
0x023C	CMPCFG[3]	Comparator configure register	0x00000000
0x0240	CMPCFG[4]	Comparator configure register	0x00000000
0x0244	CMPCFG[5]	Comparator configure register	0x00000000
0x0248	CMPCFG[6]	Comparator configure register	0x00000000
0x024C	CMPCFG[7]	Comparator configure register	0x00000000
0x0250	CMPCFG[8]	Comparator configure register	0x00000000
0x0254	CMPCFG[9]	Comparator configure register	0x00000000
0x0258	CMPCFG[10]	Comparator configure register	0x00000000
0x025C	CMPCFG[11]	Comparator configure register	0x00000000
0x0260	CMPCFG[12]	Comparator configure register	0x00000000
0x0264	CMPCFG[13]	Comparator configure register	0x00000000
0x0268	CMPCFG[14]	Comparator configure register	0x00000000
0x026C	CMPCFG[15]	Comparator configure register	0x00000000
0x0270	CMPCFG[16]	Comparator configure register	0x00000000
0x0274	CMPCFG[17]	Comparator configure register	0x00000000
0x0278	CMPCFG[18]	Comparator configure register	0x00000000
0x027C	CMPCFG[19]	Comparator configure register	0x00000000
0x0280	CMPCFG[20]	Comparator configure register	0x00000000
0x0284	CMPCFG[21]	Comparator configure register	0x00000000
0x0288	CMPCFG[22]	Comparator configure register	0x00000000
0x028C	CMPCFG[23]	Comparator configure register	0x00000000

Table 167: PWM Register List

33.4 Detailed PWM Register Information

The detailed description of PWM registers is as follows:

33.4.1 UNLK (0x0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
																SHUNLK															
																RW															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

UNLK [31:0]

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Bits	Name	Description
31-0	SHUNLK	write 0xB0382607 to unlock the shadow registers of register offset from 0x04 to 0x78, otherwise the shadow registers can not be written.

UNLK Bits

33.4.2 STA (0x4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
XSTA				STA																								RSVD			
RW				RW																								N/A			
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

STA [31:0]

Bits	Name	Description
31-28	XSTA	pwm timer counter extended start point, should back to this value after reach xrld
27-4	STA	pwm timer counter start value sta/rld will be loaded from shadow register to work register at main counter reload time, or software write unlk.shunlk

STA Bits

33.4.3 RLD (0x8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
XRLD				RLD																								RSVD			
RW				RW																								N/A			
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

RLD [31:0]

Bits	Name	Description
31-28	XRLD	timeout counter extended reload point, counter will reload to xsta after reach this point
27-4	RLD	pwm timer counter reload value

RLD Bits

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33.4.4 CMP (0xC + 0x4 * m)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
XCMP				CMP																								CMPHLF	CMPJIT		
RW				RW																								RW	RW		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CMP [31:0]

Bits	Name	Description
31-28	XCMP	extended counter compare value
27-4	CMP	clock counter compare value, the compare output is 0 at default, set to 1 when compare value meet, and clr to 0 when timer reload. Software can invert the output by setting chan_cfg.out_polarity.
3	CMPHLF	half clock counter compare value
2-0	CMPJIT	jitter counter compare value

CMP Bits

33.4.5 FRCMD (0x78)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																FRCMD															
N/A																RW															
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

FRCMD [31:0]

Bits	Name	Description
15-0	FRCMD	2bit for each PWM output channel (0-7); 00: force output 0 01: force output 1 10: output highz 11: no force

FRCMD Bits

33.4.6 SHLK (0x7C)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SHLK	RSVD																														
RW	N/A																														
0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

SHLK [31:0]

Bits	Name	Description
31	SHLK	write 1 to lock all shawdow register, write access is not permitted

SHLK Bits

33.4.7 CHCFG (0x80 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD			CMPSELEND					RSVD			CMPSELBEG					RSVD														OUTPOL	RSVD
N/A			RW					N/A			RW					N/A														RW	N/A
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0

CHCFG [31:0]

Bits	Name	Description
28-24	CMPSELEND	assign the last comparator for this output channel
20-16	CMPSELBEG	assign the first comparator for this output channel
1	OUTPOL	output polarity, set to 1 will invert the output

CHCFG Bits

33.4.8 GCR (0xF0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FAULTI3EN	FAULTI2EN	FAULTI1EN	FAULTI0EN	DEBUGFAULT	FRCPOL	RSVD	HWSHDWEDG	CMPSHDWSEL					FAULTRECEDG	FAULTRECHWSEL					FAULTE1EN	FAULTE0EN	FAULTEXPOL		RLDSYNCEN	CEN	FAULTCLR	XRLDSYNCEN	RSVD	TIMERRESET	FRCTIME		SWFRC
RW	RW	RW	RW	RW	RW	N/A	RW	RW					RW	RW					RW	RW	RW		RW	RW	RW	RW	N/A	RW	W/O		RW
0	0	0	0	0	0	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	0	0	0	0

GCR [31:0]

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Bits	Name	Description
31	FAULTI3EN	1- enable the internal fault input 3
30	FAULTI2EN	1- enable the internal fault input 2
29	FAULTI1EN	1- enable the internal fault input 1
28	FAULTI0EN	1- enable the internal fault input 0
27	DEBUGFAULT	1- enable debug mode output protection
26	FRCPOL	polarity of input pwm_force, 1- active low 0- active high
24	HWSHDWEDG	When hardware event is selected as shawdow register effective time and the select comparator is configured as input capture mode. This bit assign its which edge is used as compare shadow register hardware load event. 1- Falling edge 0- Rising edge
23-19	CMPSHDWSEL	This bitfield select one of the comparators as hardware event time to load comparator shadow registers
18	FAULTRECEDG	When hardware load is selected as output fault recover trigger and the selected channel is capture mode. This bit assign its effective edge of fault recover trigger. 1- Falling edge 0- Rising edge
17-13	FAULTRECHWSEL	Selec one of the 24 comparators as fault output recover trigger.
12	FAULTE1EN	1- enable the external fault input 1
11	FAULTE0EN	1- enable the external fault input 0
10-9	FAULTEXPOL	external fault polarity 1-active low 0-active high
8	RLDSYNCEN	1- pwm timer counter reset to reload value (rld) by synci is enabled
7	CEN	1- enable the pwm timer counter 0- stop the pwm timer counter
6	FAULTCLR	1- Write 1 to clear the fault condition. The output will recover if FAULTRECTIME is set to 2b’11. User should write 1 to this bit after the active FAULT signal de-assert and before it re-assert again.
5	XRLDSYNCEN	1- pwm timer extended counter (xcnt) reset to extended reload value (xrld) by synci is enabled
3	TIMERRESET	set to clear current timer(total 28bit, main counter and tmout_count). Auto clear

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Bits	Name	Description
2-1	FRCTIME	This bit field select the force effective time 00: force immediately 01: force at main counter reload time 10: force at FRCSYNCI 11: no force
0	SWFRC	1- write 1 to enable software force, if the frcsrcsel is set to 0, force will take effect

GCR Bits

33.4.9 SHCR (0xF4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																			FRCSHDWSEL					CNTSHDWSEL						CNTSHDWUPT		SHLKEN
N/A																			RW					RW						RW		RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0

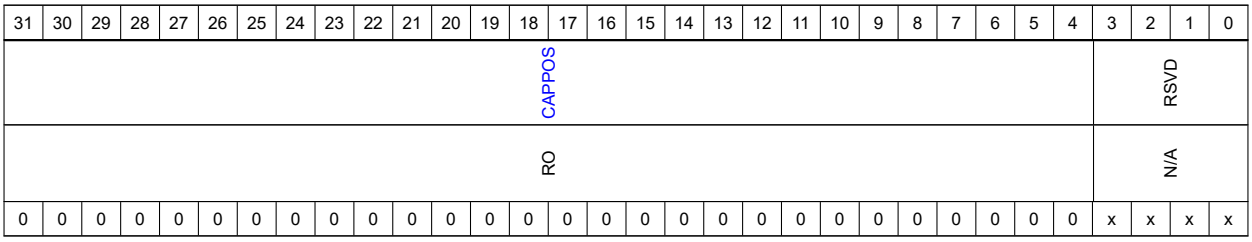
SHCR [31:0]

Bits	Name	Description
12-8	FRCSHDWSEL	This bitfield select one of the comparators as hardware event time to load FRCMD shadow registers
7-3	CNTSHDWSEL	This bitfield select one of the comparators as hardware event time to load the counter related shadow registers (STA and RLD)
2-1	CNTSHDWUPT	This bitfield select when the counter related shadow registers (STA and RLD) will be loaded to its work register 00: after software set shlk bit of shlk register 01: immediately after the register being modified 10: after hardware event assert, user can select one of the comparators to generate this hardware event. The comparator can be either output compare mode or input capture mode. 11: after SHSYNCI assert
0	SHLKEN	1- enable shadow registers lock feature, 0- disable shadow registers lock, shlk bit will always be 0

SHCR Bits

33.4.10 CAPPOS (0x100 + 0x4 * n)

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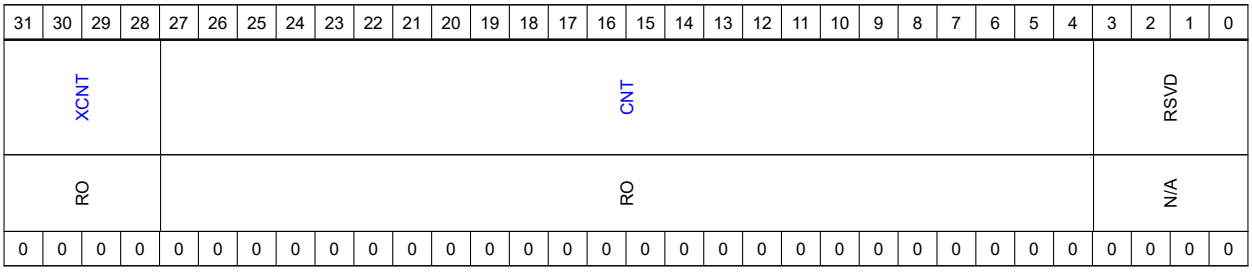


CAPPOS [31:0]

Bits	Name	Description
31-4	CAPPOS	counter value captured at input posedge

CAPPOS Bits

33.4.11 CNT (0x170)

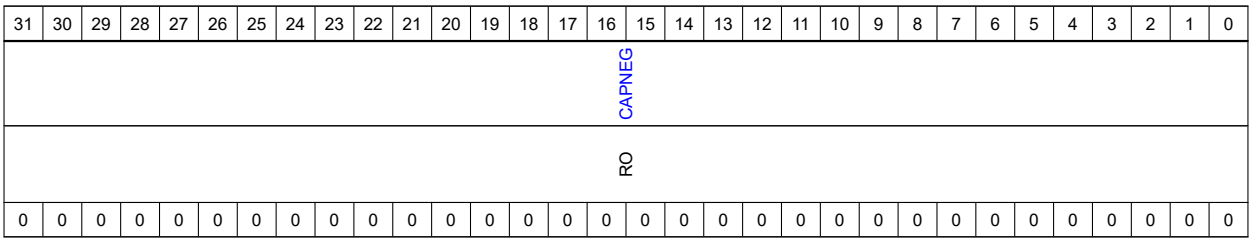


CNT [31:0]

Bits	Name	Description
31-28	XCNT	current extended counter value
27-4	CNT	current clock counter value

CNT Bits

33.4.12 CAPNEG (0x180 + 0x4 * n)



CAPNEG [31:0]

Bits	Name	Description
31-0	CAPNEG	counter value captured at input signal falling edge

CAPNEG Bits

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33.4.13 CNTCOPY (0x1F0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
XCNT				CNT																								RSVD			
RO				RO																								N/A			
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CNTCOPY [31:0]

Bits	Name	Description
31-28	XCNT	current extended counter value
27-4	CNT	current clock counter value

CNTCOPY Bits

33.4.14 PWMCFG (0x200 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD			OEN	FRCSHDWUPT		FAULTMODE		FAULTRECTIME		FRCSRCSEL	PAIR	DEADAREA																			
N/A			RW	RW		RW		RW		RW	RW	RW																			
x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PWMCFG [31:0]

Bits	Name	Description
28	OEN	PWM output enable 1- output is enabled 0- output is disabled
27-26	FRCSHDWUPT	This bitfield select when the FRCMD shadow register will be loaded to its work register 00: after software set shlk bit of shlk register 01: immediately after the register being modified 10: after hardware event assert, user can select one of the comparators to generate this hardware event. The comparator can be either output compare mode or input capture mode. 11: after SHSYNCI assert

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Bits	Name	Description
25-24	FAULTMODE	This bitfield defines the PWM output status when fault condition happen 00: force output 0 01: force output 1 1x: output highz
23-22	FAULTRECTIME	This bitfield select when to recover PWM output after fault condition removed. 00: immediately 01: after pwm timer counter reload time 10: after hardware event assert, user can select one of the comparators to generate this hardware event. The comparator can be either output compare mode or input capture mode. 11: after software write faultclr bit in GCR register
21	FRCSRCSEL	Select sources for force output 0- force output is enabled when FRCI assert 1- force output is enabled by software write swfrc to 1
20	PAIR	1- PWM output is in pair mode. Note the two PWM outputs need to be both set to pair mode. 0- PWM output is in independant mode.
19-0	DEADAREA	This bitfield define the PWM pair deadarea length. The unit is 0.5 cycle. The minimum length of deadarea is 1 cycle. Note: user should configure pair bit and this bitfield before PWM output is enabled.

PWMCFG Bits

33.4.15 SR (0x220)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD				FAULTF	XRLDF	HALFRLDF	RLDF	CMPFX																							
N/A				W1C	W1C	W1C	W1C	W1C																							
x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

SR [31:0]

Bits	Name	Description
27	FAULTF	fault condition flag
26	XRLDF	extended reload flag, this flag set when xcnt count to xrld value or when SYNCI assert

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Bits	Name	Description
25	HALFRLDF	half reload flag, this flag set when cnt count to rld/2
24	RLDF	reload flag, this flag set when cnt count to rld value or when SYNCI assert
23-0	CMPFX	comparator output compare or input capture flag

SR Bits

33.4.16 IRQEN (0x224)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD				FAULTIRQE	XRLDIRQE	HALFRLDIRQE	RLDIRQE	CMPIRQEX																							
N/A				RW	RW	RW	RW	RW																							
x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

IRQEN [31:0]

Bits	Name	Description
27	FAULTIRQE	fault condition interrupt enable
26	XRLDIRQE	extended reload flag interrupt enable
25	HALFRLDIRQE	half reload flag interrupt enable
24	RLDIRQE	reload flag interrupt enable
23-0	CMPIRQEX	comparator output compare or input capture flag interrupt enable

IRQEN Bits

33.4.17 CAP_EN_NEG (0x228)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD								CAP_EN_NEG																							
N/A								RW																							
x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CAP_EN_NEG [31:0]

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Bits	Name	Description
23-0	CAP_EN_NEG	In capture mode, those bits setting for generate capture irq when capture falling edge. Default those bits is 0, generate capture irq when capture rising edge.

CAP_EN_NEG Bits

33.4.18 DMAEN (0x22C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD				FAULTEN	XRLDEN	HALFRLDEN	RLDEN	CMPENX																							
N/A				RW	RW	RW	RW	RW																							
x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DMAEN [31:0]

Bits	Name	Description
27	FAULTEN	fault condition DMA request enable
26	XRLDEN	extended reload flag DMA request enable
25	HALFRLDEN	half reload flag DMA request enable
24	RLDEN	reload flag DMA request enable
23-0	CMPENX	comparator output compare or input capture flag DMA request enable

DMAEN Bits

33.4.19 CMPCFG (0x230 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																								XCNTCMPEN				CMPSHDWUPT		CMPMODE	RSVD
N/A																								RW				RW		RW	N/A
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0

CMPCFG [31:0]

Bits	Name	Description
7-4	XCNTCMPEN	This bitfield enable the comparator to compare xcmp with xcnt.

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Bits	Name	Description
3-2	CMPSHDWUPT	This bitfield select when the comparator shadow register will be loaded to its work register 00: after software set shlk bit of shlk register 01: immediately after the register being modified 10: after hardware event assert, user can select one of the comparators to generate this hardware event. The comparator can be either output compare mode or input capture mode. 11: after SHSYNCI assert
1	CMPMODE	comparator mode 0- output compare mode 1- input capture mode

CMPCFG Bits

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34 Trigger Manager TRGM

The Trigger Manager supports a number of input and output (IO) ports, which can be allocated to peripherals connected to the Trigger Manager as their inputs or outputs.

On this product, the input and output signals of multiple peripherals can be interconnected to enable multiple peripherals to work in coordination with each other. The Trigger Manager can configure these connections.

For the input signal assignment of the Trigger Manager on this product, please refer to Section 34.3.5.

For the output signal assignment of the Trigger Manager on this product, please refer to Section 34.3.6.

The inputs of the Trigger Manager are equipped with digital filters, which can filter some signals.

For the digital filter assignment of the Trigger Manager on this product, please refer to Section 34.3.3.

The Trigger Manager also manages DMA requests from peripherals and sends selected DMA requests to the DMA controller.

For the DMA request signal assignment of the Trigger Manager on this product, please refer to Section 34.3.4.

34.1 Feature Summary

This section describes the main features of the Trigger Manager TRGM (Trigger Manager):

- Multiplexer (MUX) array
- Supports multiple inputs and multiple outputs
- Each output can be configured independently to select from a variety of inputs
- Input signal digital filter
- Output signal polarity inversion
- Signal edge-to-pulse conversion
- Trigger Manager DMA requests and interrupt requests, managing DMA requests from PWM, QEI and HALL

34.2 Block Diagram

The system block diagram of TRGM is shown in Figure 39.

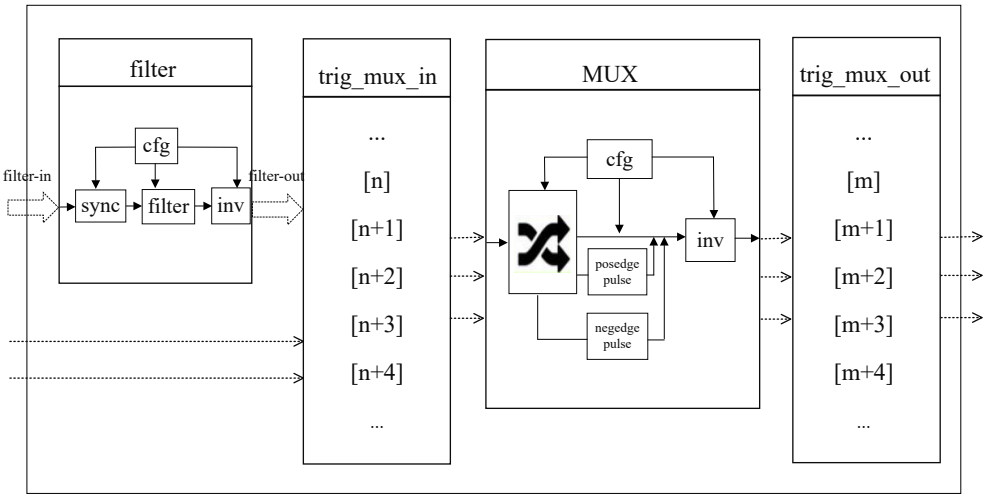


Figure 39: TRGM System Block Diagram

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34.3 Functional Description

This section describes the functions of the Trigger Manager TRGM.

34.3.1 Manager Signal Input/Output Multiplexer

Users can connect any input signal of TRGM to the output of the Trigger Manager by configuring the Trigger Manager registers, and configure the TRGOCFGx[OCFG] bit to select one of the TRGM inputs as the output of OUTx.

34.3.2 Output Configuration

For the output of the Trigger Manager, after selecting the signal input, users can also invert the output by configuring the TRGOCFGx[OPOL] bit. When the TRGOCFGx[OPOL] bit is set to 0, the output signal remains unchanged; when set to 1, the output signal is inverted.

In addition, the output configuration of the Trigger Manager supports edge-to-pulse conversion for signals. When the TRGOCFGx[REDG2PEN] bit is set to 1, each rising edge of the input signal will output a pulse with a width of one Trigger Manager clock cycle.

When the TRGOCFGx[FEDG2PEN] bit is set to 1, each falling edge of the input signal will output a pulse with a width of one Trigger Manager clock cycle.

34.3.3 Trigger Manager Digital Filter and Its Assignment

The Trigger Manager TRGM provides digital filters for some input signals. Users can configure the working mode of the filter through the FILTCFGx[MODE] register:

- 3'b000: Bypass mode, digital filter is disabled
- 3'b100: Glitch filtering mode: After the filter input flips, the output will also flip immediately, and then the filter input will be ignored for a certain period of time. In this mode, the filter output follows the input immediately while avoiding glitches in the output signal.
- 3'b101: Delay filter: The filter input needs to remain flipped for a certain period of time before the filter output flips
- 3'b110: Peak filtering mode: After the filter input is set to logic 0, it needs to remain for a certain period of time before the filter output is set to logic 0; when the filter input is set to 1, the filter output will be set to 1 immediately. This mode is designed to filter out input wave peaks that are not long enough.
- 3'b111: Valley filtering mode: After the filter input is set to logic 1, it needs to remain for a certain period of time before the filter output is set to logic 1; when the filter input is set to 0, the filter output will be set to 0 immediately. This mode is designed to filter out input wave valleys that are not long enough.

Users can configure the filter width through the FILTCFGx[FILTLEN] register bit, in units of the Trigger Manager clock cycle.

Users can invert the filter output by setting the FILTCFGx[OUTINV] bit to 1.

The Trigger Manager supports multiple digital filters to filter specific input signals of TRGM.

This section describes the correspondence between the digital filters of the Trigger Manager and the input signals.

Digital filter assignment of the Trigger Manager:

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TRGM0_FILTER MUX Index	TRGM0_FILTER MUX SOURCE Name	Description
0	PWM0_IN0	PWM timer 0 capture Input 0
1	PWM0_IN1	PWM timer 0 capture Input 1
2	PWM0_IN2	PWM timer 0 capture Input 2
3	PWM0_IN3	PWM timer 0 capture Input 3
4	PWM0_IN4	PWM timer 0 capture Input 4
5	PWM0_IN5	PWM timer 0 capture Input 5
6	PWM0_IN6	PWM timer 0 capture Input 6
7	PWM0_IN7	PWM timer 0 capture Input 7
8	PWM1_IN0	PWM timer 1 capture Input 0
9	PWM1_IN1	PWM timer 1 capture Input 1
10	PWM1_IN2	PWM timer 1 capture Input 2
11	PWM1_IN3	PWM timer 1 capture Input 3
12	PWM1_IN4	PWM timer 1 capture Input 4
13	PWM1_IN5	PWM timer 1 capture Input 5
14	PWM1_IN6	PWM timer 1 capture Input 6
15	PWM1_IN7	PWM timer 1 capture Input 7
16	TRGM_P00	TRGM input 0 from IO
17	TRGM_P01	TRGM input 1 from IO
18	TRGM_P02	TRGM input 2 from IO
19	TRGM_P03	TRGM input 3 from IO
20	TRGM_P04	TRGM input 4 from IO
21	TRGM_P05	TRGM input 5 from IO
22	TRGM_P06	TRGM input 6 from IO
23	TRGM_P07	TRGM input 7 from IO
24	PWM0_FAULT0	PWM timer 0 Input fault0
25	PWM0_FAULT1	PWM timer 0 Input fault1
26	PWM1_FAULT0	PWM timer 1 Input fault0
27	PWM1_FAULT1	PWM timer 1 Input fault1

Table 168: TRGM0_FILTER MUX List

34.3.4 Trigger Manager DMA Requests and Interrupt Requests

The Trigger Manager supports managing DMA requests from some peripherals interconnected with it, and these DMA requests are forwarded to the DMAMUX through the Trigger Manager.

The Trigger Manager supports 8 DMA requests. Users can configure DMACFG0 ~ DMACFG7 registers to select the DMA requests output to the DMA controller from the table below.

In addition, the Trigger Manager supports 2 interrupt requests, and TRGM input signals can be used as signals to trigger interrupt events. Similarly, TRGM input signals can also be used as signals to trigger DMA requests.

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DMA request assignment of the Trigger Manager:

Index	Source	Description
0	PWM0_CMP0	The capture input or matches output of PWM timer 0 comparator 0
1	PWM0_CMP1	The capture input or matches output of PWM timer 0 comparator 1
2	PWM0_CMP2	The capture input or matches output of PWM timer 0 comparator 2
3	PWM0_CMP3	The capture input or matches output of PWM timer 0 comparator 3
4	PWM0_CMP4	The capture input or matches output of PWM timer 0 comparator 4
5	PWM0_CMP5	The capture input or matches output of PWM timer 0 comparator 5
6	PWM0_CMP6	The capture input or matches output of PWM timer 0 comparator 6
7	PWM0_CMP7	The capture input or matches output of PWM timer 0 comparator 7
8	PWM0_CMP8	The capture input or matches output of PWM timer 0 comparator 8
9	PWM0_CMP9	The capture input or matches output of PWM timer 0 comparator 9
10	PWM0_CMP10	The capture input or matches output of PWM timer 0 comparator 10
11	PWM0_CMP11	The capture input or matches output of PWM timer 0 comparator 11
12	PWM0_CMP12	The capture input or matches output of PWM timer 0 comparator 12
13	PWM0_CMP13	The capture input or matches output of PWM timer 0 comparator 13
14	PWM0_CMP14	The capture input or matches output of PWM timer 0 comparator 14
15	PWM0_CMP15	The capture input or matches output of PWM timer 0 comparator 15
16	PWM0_CMP16	The capture input or matches output of PWM timer 0 comparator 16
17	PWM0_CMP17	The capture input or matches output of PWM timer 0 comparator 17
18	PWM0_CMP18	The capture input or matches output of PWM timer 0 comparator 18

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Index	Source	Description
19	PWM0_CMP19	The capture input or matches output of PWM timer 0 comparator 19
20	PWM0_CMP20	The capture input or matches output of PWM timer 0 comparator 20
21	PWM0_CMP21	The capture input or matches output of PWM timer 0 comparator 21
22	PWM0_CMP22	The capture input or matches output of PWM timer 0 comparator 22
23	PWM0_CMP23	The capture input or matches output of PWM timer 0 comparator 23
24	PWM0_RLD	PWM timer 0 counter reload
25	PWM0_HALFRLD	PWM timer 0 half cycle reload
26	PWM0_XRLD	PWM timer 0 extended counter reload
27	PWM1_CMP0	The capture input or matches output of PWM timer 1 comparator 0
28	PWM1_CMP1	The capture input or matches output of PWM timer 1 comparator 1
29	PWM1_CMP2	The capture input or matches output of PWM timer 1 comparator 2
30	PWM1_CMP3	The capture input or matches output of PWM timer 1 comparator 3
31	PWM1_CMP4	The capture input or matches output of PWM timer 1 comparator 4
32	PWM1_CMP5	The capture input or matches output of PWM timer 1 comparator 5
33	PWM1_CMP6	The capture input or matches output of PWM timer 1 comparator 6
34	PWM1_CMP7	The capture input or matches output of PWM timer 1 comparator 7
35	PWM1_CMP8	The capture input or matches output of PWM timer 1 comparator 8
36	PWM1_CMP9	The capture input or matches output of PWM timer 1 comparator 9
37	PWM1_CMP10	The capture input or matches output of PWM timer 1 comparator 10
38	PWM1_CMP11	The capture input or matches output of PWM timer 1 comparator 11
39	PWM1_CMP12	The capture input or matches output of PWM timer 1 comparator 12
40	PWM1_CMP13	The capture input or matches output of PWM timer 1 comparator 13

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Index	Source	Description
41	PWM1_CMP14	The capture input or matches output of PWM timer 1 comparator 14
42	PWM1_CMP15	The capture input or matches output of PWM timer 1 comparator 15
43	PWM1_CMP16	The capture input or matches output of PWM timer 1 comparator 16
44	PWM1_CMP17	The capture input or matches output of PWM timer 1 comparator 17
45	PWM1_CMP18	The capture input or matches output of PWM timer 1 comparator 18
46	PWM1_CMP19	The capture input or matches output of PWM timer 1 comparator 19
47	PWM1_CMP20	The capture input or matches output of PWM timer 1 comparator 20
48	PWM1_CMP21	The capture input or matches output of PWM timer 1 comparator 21
49	PWM1_CMP22	The capture input or matches output of PWM timer 1 comparator 22
50	PWM1_CMP23	The capture input or matches output of PWM timer 1 comparator 23
51	PWM1_RLD	PWM timer 1 Counter reload
52	PWM1_HALFRLD	PWM timer 1 half cycle reload
53	PWM1_XRLD	PWM timer 1 Extended counter reload
54	QEI0	DMA request for QEI0
55	QEI1	DMA request for QEI1
56	MMC0	DMA request for MMC0
57	MMC1	DMA request for MMC1
58	SEI0	DMA request 0 for SEI0
59	SEI1	DMA request 0 for SEI1
60	TRGM0	DMA request 0 for TRGM (from TRGM output 135)
61	TRGM1	DMA request 1 for TRGM (from TRGM output 136)

Table 169: TRGM DMAMUX List

34.3.5 Trigger Manager Input Assignment

This section describes the input signal assignment of the Trigger Manager.

Users can configure the lower 2 bits to output a fixed value of 0 or 1.

Input signal assignment of the Trigger Manager:

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Index	Source	Description
0	VSS	Low level
1	VDD	High level
2	DEBUG_FLAG	the flag bit of debug mode enters
3	USB0_SOF	USB0 frame start
4	PTPC_CMP0	PTPC output comparison 0
5	PTPC_CMP1	PTPC output comparison 1
6	ACMP0_OUT	Comparator 0 output
7	ACMP1_OUT	Comparator 1 output
8	GPTMR0_OUT2	GPTMR0 channel 2
9	GPTMR0_OUT3	GPTMR0 channel 3
10	GPTMR1_OUT2	GPTMR1 channel 2
11	GPTMR1_OUT3	GPTMR1 channel 3
12	GPTMR2_OUT2	GPTMR2 channel 2
13	GPTMR2_OUT3	GPTMR2 channel 3
14	GPTMR3_OUT2	GPTMR3 channel 2
15	GPTMR3_OUT3	GPTMR3 channel 3
16	TRGM0_P0	TRGM0 Input 0 (from IO)
17	TRGM0_P1	TRGM0 Input 1 (from IO)
18	TRGM0_P2	TRGM0 Input 2 (from IO)
19	TRGM0_P3	TRGM0 Input 3 (from IO)
20	TRGM0_P4	TRGM0 Input 4 (from IO)
21	TRGM0_P5	TRGM0 Input 5 (from IO)
22	TRGM0_P6	TRGM0 Input 6 (from IO)
23	TRGM0_P7	TRGM0 Input 7 (from IO)
24	SYNT0_CH0	SYNT0 Channel 0
25	SYNT0_CH1	SYNT0 Channel 1
26	SYNT0_CH2	SYNT0 Channel 2
27	SYNT0_CH3	SYNT0 channel 3
28	MMC0_TRGO_0	MMC0 triggers output 0
29	MMC0_TRGO_1	MMC0 triggers output 1
30	MMC1_TRGO_0	MMC1 triggers output 0
31	MMC1_TRGO_1	MMC1 triggers output 1
32	QEO0_TRGO_0	QEO0 triggers output 0
33	QEO0_TRGO_1	QEO0 triggers output 1
34	QEO0_TRGO_2	QEO0 triggers output 2
35	QEO0_TRGO_3	QEO0 triggers output 3
36	QEO0_TRGO_4	QEO0 triggers output 4
37	QEO0_TRGO_5	QEO0 triggers output 5
38	QEO0_TRGO_6	QEO0 triggers output 6
39	QEO0_TRGO_7	QEO0 triggers output 7

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Index	Source	Description
40	QEO1_TRGO_0	QEO1 triggers output 0
41	QEO1_TRGO_1	QEO1 triggers output 1
42	QEO1_TRGO_2	QEO1 triggers output 2
43	QEO1_TRGO_3	QEO1 triggers output 3
44	QEO1_TRGO_4	QEO1 triggers output 4
45	QEO1_TRGO_5	QEO1 triggers output 5
46	QEO1_TRGO_6	QEO1 triggers output 6
47	QEO1_TRGO_7	QEO1 triggers output 7
48	PWM0_CH8REF	PWM timer 0 channel 8 reference output
49	PWM0_CH9REF	PWM timer 0 channel 9 reference output
50	PWM0_CH10REF	PWM timer 0 channel 10 reference output
51	PWM0_CH11REF	PWM timer 0 channel 11 reference output
52	PWM0_CH12REF	PWM timer 0 channel 12 reference output
53	PWM0_CH13REF	PWM timer 0 channel 13 reference output
54	PWM0_CH14REF	PWM timer 0 channel 14 reference output
55	PWM0_CH15REF	PWM timer 0 channel 15 reference output
56	PWM1_CH8REF	PWM timer 1 channel 8 reference output
57	PWM1_CH9REF	PWM timer 1 channel 9 reference output
58	PWM1_CH10REF	PWM timer 1 channel 10 reference output
59	PWM1_CH11REF	PWM timer 1 channel 11 reference output
60	PWM1_CH12REF	PWM timer 1 channel 12 reference output
61	PWM1_CH13REF	PWM timer 1 channel 13 reference output
62	PWM1_CH14REF	PWM timer 1 channel 14 reference output
63	PWM1_CH15REF	PWM timer 1 channel 15 reference output
64	PLB_OUT00	PLB module output 0

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Index	Source	Description
65	PLB_OUT01	PLB module output 1
66	PLB_OUT02	PLB module output 2
67	PLB_OUT03	PLB module output 3
68	PLB_OUT04	PLB module output 4
69	PLB_OUT05	PLB module output 5
70	PLB_OUT06	PLB module output 6
71	PLB_OUT07	PLB module output 7
72	PLB_OUT08	PLB module output 8
73	PLB_OUT09	PLB module output 9
74	PLB_OUT10	PLB module output 10
75	PLB_OUT11	PLB module output 11
76	PLB_OUT12	PLB module output 12
77	PLB_OUT13	PLB module output 13
78	PLB_OUT14	PLB module output 14
79	PLB_OUT15	PLB module output 15
80	PLB_OUT16	PLB module output 16
81	PLB_OUT17	PLB module output 17
82	PLB_OUT18	PLB module output 18
83	PLB_OUT19	PLB module output 19
84	PLB_OUT20	PLB module output 20
85	PLB_OUT21	PLB module output 21
86	PLB_OUT22	PLB module output 22
87	PLB_OUT23	PLB module output 23
88	PLB_OUT24	PLB module output 24
89	PLB_OUT25	PLB module output 25
90	PLB_OUT26	PLB module output 26
91	PLB_OUT27	PLB module output 27
92	PLB_OUT28	PLB module output 28
93	PLB_OUT29	PLB module output 29
94	PLB_OUT30	PLB module output 30
95	PLB_OUT31	PLB module output 31
96	RDC_TRGO_0	RDC triggers output 0
97	RDC_TRGO_1	RDC triggers output 1
98	QEI1_TRGO	QEI1 triggers output
99	QEI0_TRGO	QEI0 triggers output
100	SEI_TRGO_0	SEI triggers output 0
101	SEI_TRGO_1	SEI triggers output 1
102	SEI_TRGO_2	SEI triggers output 2
103	SEI_TRGO_3	SEI triggers output 3
104	SEI_TRGO_4	SEI triggers output 4

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Index	Source	Description
105	SEI_TRGO_5	SEI triggers output 5
106	SEI_TRGO_6	SEI triggers output 6
107	SEI_TRGO_7	SEI triggers output 7
108	PWM0_FAULT0	PWM0 fault Input 0 (from IO)
109	PWM0_FAULT1	PWM0 fault Input 1 (from IO)
110	PWM1_FAULT0	PWM1 fault Input 0 (from IO)
111	PWM1_FAULT1	PWM1 fault Input 1 (from IO)
112	PWM0_CAPIN0	PWM0 Input 0 (from IO)
113	PWM0_CAPIN1	PWM0 Input 1 (from IO)
114	PWM0_CAPIN2	PWM0 Input 2 (from IO)
115	PWM0_CAPIN3	PWM0 Input 3 (from IO)
116	PWM0_CAPIN4	PWM0 Input 4 (from IO)
117	PWM0_CAPIN5	PWM0 Input 5 (from IO)
118	PWM0_CAPIN6	PWM0 Input 6 (from IO)
119	PWM0_CAPIN7	PWM0 Input 7 (from IO)
120	PWM1_CAPIN0	PWM1 Input 0 (from IO)
121	PWM1_CAPIN1	PWM1 Input 1 (from IO)
122	PWM1_CAPIN2	PWM1 Input 2 (from IO)
123	PWM1_CAPIN3	PWM1 Input 3 (from IO)
124	PWM1_CAPIN4	PWM1 Input 4 (from IO)
125	PWM1_CAPIN5	PWM1 Input 5 (from IO)
126	PWM1_CAPIN6	PWM1 Input 6 (from IO)
127	PWM1_CAPIN7	PWM1 Input 7 (from IO)

Table 170: TRGM TRIGMUX INPUT List

34.3.6 Trigger Manager Output Assignment

This section describes the output signal assignment of the Trigger Manager.

Output signal assignment of the Trigger Manager:

Index	Destination	Description
0	MOT2OPAMP0_0	OPAMP0 triggers input 0
1	MOT2OPAMP0_1	OPAMP0 triggers input 1
2	MOT2OPAMP0_2	OPAMP0 triggers input 2
3	MOT2OPAMP0_3	OPAMP0 triggers input 3
4	MOT2OPAMP0_4	OPAMP0 triggers input 4
5	MOT2OPAMP0_5	OPAMP0 triggers input 5
6	MOT2OPAMP0_6	OPAMP0 triggers input 6
7	MOT2OPAMP0_7	OPAMP0 triggers input 7
8	MOT2OPAMP1_0	OPAMP1 triggers input 0

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Index	Destination	Description
9	MOT2OPAMP1_1	OPAMP1 triggers input 1
10	MOT2OPAMP1_2	OPAMP1 triggers input 2
11	MOT2OPAMP1_3	OPAMP1 triggers input 3
12	MOT2OPAMP1_4	OPAMP1 triggers input 4
13	MOT2OPAMP1_5	OPAMP1 triggers input 5
14	MOT2OPAMP1_6	OPAMP1 triggers input 6
15	MOT2OPAMP1_7	OPAMP1 triggers input 7
16	GPTMR0_IN2	GPTMR0 channel 2 input
17	GPTMR0_IN3	GPTMR0 channel 3 input
18	GPTMR0_SYNCI	GPTMR0 counter synchronous input
19	GPTMR1_IN2	GPTMR1 channel 2 input
20	GPTMR1_IN3	GPTMR1 channel 3 input
21	GPTMR1_SYNCI	GPTMR1 counter synchronous input
22	GPTMR2_IN2	GPTMR2 channel 2 input; This bit is also activated as DAC0 ladder mode 1
23	GPTMR2_IN3	GPTMR2 channel 3 input; This bit is also activated as DAC0 ladder mode 2
24	GPTMR2_SYNCI	GPTMR2 counter synchronous input; This bit is also activated as DAC0 ladder mode 3
25	GPTMR3_IN2	GPTMR3 channel 2 input; This bit is also activated as DAC1 ladder mode 1
26	GPTMR3_IN3	GPTMR3 channel 3 input; This bit is also activated as DAC1 ladder mode 2
27	GPTMR3_SYNCI	GPTMR3 counter synchronous input; This bit is also activated as DAC1 ladder mode 3
28	ACMP0_WIN	Comparator 0 window mode input
29	ACMP1_WIN	Comparator 1 window mode input
30	DAC0_BUFTRG	DAC0 buffer mode starts to trigger; This bit is also activated as DAC0 ladder mode 0
31	DAC1_BUFTRG	DAC1 buffer mode starts to trigger; This bit is also activated as DAC1 ladder mode 0
32	ADC0_STRGI	The sequence conversion of ADC0 triggers input
33	ADC1_STRGI	The sequence conversion of ADC1 triggers input
34	ADCX_PTRGI0A	The preemption conversion of ADC0, 1 triggers input 0A

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Index	Destination	Description
35	ADCX_PTRGI0B	The preemption conversion of ADC0, 1 triggers input 0B
36	ADCX_PTRGI0C	The preemption conversion of ADC0, 1 triggers input 0C
37	ADCX_PTRGI1A	The preemption conversion of ADC0, 1 triggers input 1A
38	ADCX_PTRGI1B	The preemption conversion of ADC0, 1 triggers input 1B
39	ADCX_PTRGI1C	The preemption conversion of ADC0, 1 triggers input 1C
40	ADCX_PTRGI2A	The preemption conversion of ADC0, 1 triggers input 2A
41	ADCX_PTRGI2B	The preemption conversion of ADC0, 1 triggers input 2B
42	ADCX_PTRGI2C	The preemption conversion of ADC0, 1 triggers input 2C
43	ADCX_PTRGI3A	The preemption conversion of ADC0, 1 triggers input 3A
44	ADCX_PTRGI3B	The preemption conversion of ADC0, 1 triggers input 3B
45	ADCX_PTRGI3C	The preemption conversion of ADC0, 1 triggers input 3C
46	MCAN_PTPC0_CAP	PTPC input capture 0
47	MCAN_PTPC1_CAP	PTPC input capture 1
48	QEO0_TRIG_IN0	QEO0 triggers input 0
49	QEO0_TRIG_IN1	QEO0 triggers input 1
50	QEO1_TRIG_IN0	QEO1 triggers input 0
51	QEO1_TRIG_IN1	QEO1 triggers input 1
52	SEI_TRIG_IN0	SEI triggers input 0
53	SEI_TRIG_IN1	SEI triggers input 1
54	SEI_TRIG_IN2	SEI triggers input 2
55	SEI_TRIG_IN3	SEI triggers input 3
56	SEI_TRIG_IN4	SEI triggers input 4
57	SEI_TRIG_IN5	SEI triggers input 5
58	SEI_TRIG_IN6	SEI triggers input 6
59	SEI_TRIG_IN7	SEI triggers input 7
60	MMC0_TRIG_IN0	MMC0 triggers input 0
61	MMC0_TRIG_IN1	MMC0 triggers input 1
62	MMC1_TRIG_IN0	MMC1 triggers input 0
63	MMC1_TRIG_IN1	MMC1 triggers input 1
64	PLB_IN_00	PLB module input 00

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Index	Destination	Description
65	PLB_IN_01	PLB module input 01
66	PLB_IN_02	PLB module input 02
67	PLB_IN_03	PLB module input 03
68	PLB_IN_04	PLB module input 04
69	PLB_IN_05	PLB module input 05
70	PLB_IN_06	PLB module input 06
71	PLB_IN_07	PLB module input 07
72	PLB_IN_08	PLB module input 08
73	PLB_IN_09	PLB module input 09
74	PLB_IN_10	PLB module input 10
75	PLB_IN_11	PLB module input 11
76	PLB_IN_12	PLB module input 12
77	PLB_IN_13	PLB module input 13
78	PLB_IN_14	PLB module input 14
79	PLB_IN_15	PLB module input 15
80	PLB_IN_16	PLB module input 16
81	PLB_IN_17	PLB module input 17
82	PLB_IN_18	PLB module input 18
83	PLB_IN_19	PLB module input 19
84	PLB_IN_20	PLB module input 20
85	PLB_IN_21	PLB module input 21
86	PLB_IN_22	PLB module input 22
87	PLB_IN_23	PLB module input 23
88	PLB_IN_24	PLB module input 24
89	PLB_IN_25	PLB module input 25
90	PLB_IN_26	PLB module input 26
91	PLB_IN_27	PLB module input 27
92	PLB_IN_28	PLB module input 28
93	PLB_IN_29	PLB module input 29
94	PLB_IN_30	PLB module input 30
95	PLB_IN_31	PLB module input 31
96	TRGM0_P0	TRGM Output 0 (output to IO)
97	TRGM0_P1	TRGM Output 1 (output to IO)
98	TRGM0_P2	TRGM Output 2 (output to IO)
99	TRGM0_P3	TRGM Output 3 (output to IO)
100	TRGM0_P4	TRGM Output 4 (output to IO)
101	TRGM0_P5	TRGM Output 5 (output to IO)
102	TRGM0_P6	TRGM Output 6 (output to IO)
103	TRGM0_P7	TRGM Output 7 (output to IO)
104	PWM_IN8	PWM timer x capture input 8

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Index	Destination	Description
105	PWM_IN9	PWM timer x capture input 9
106	PWM_IN10	PWM timer x capture input 10
107	PWM_IN11	PWM timer x capture input 11
108	PWM_IN12	PWM timer x capture input 12
109	PWM_IN13	PWM timer x capture input 13
110	PWM_IN14	PWM timer x capture input 14
111	PWM_IN15	PWM timer x capture input 15
112	PWM0_FRCI	the input value for PWM timer 0 forces control
113	PWM0_FRCSYNCI	the synchronous input for PWM timer 0 forces control
114	PWM0_SYNCI	PWM timer 0 counter synchronously triggers input
115	PWM0_SHRLDSYNCI	PWM timer 0 Shadow register to activate trigger input
116	PWM0_FAULTI0	PWM timer 0 Fault protection input 0
117	PWM0_FAULTI1	PWM timer 0 Fault protection input 1
118	PWM1_FRCI	the input value for PWM timer 1 forces control
119	PWM1_FRCSYNCI	the synchronous input for PWM timer 1 forces control
120	PWM1_SYNCI	PWM timer 1 Counter synchronizes trigger input
121	PWM1_SHRLDSYNCI	PWM timer 1 Shadow register kicks in trigger input
122	PWM1_FAULTI0	PWM timer 1 Fault protection input 0
123	PWM1_FAULTI1	PWM timer 1 Fault protection Input 1
124	RDC_TRIG_IN0	RDC triggers input 0
125	RDC_TRIG_IN1	RDC triggers input 1
126	SYNCTIMER_TRIG	SYNT triggers input
127	QEI0_TRIG_IN	QEI0 triggers input
128	QEI1_TRIG_IN	QEI1 triggers input
129	QEI0_PAUSE	QEI0 Pause input
130	QEI1_PAUSE	QEI1 Pause input
131	UART_TRIG0	UART0/1/2/3 Trigger signal
132	UART_TRIG1	UART4/5/6/7 Trigger signal
133	TRGM_IRQ0	TRGM interrupt signal 0
134	TRGM_IRQ1	TRGM interrupt signal 1
135	TRGM_DMA0	TRGM DMA request 0
136	TRGM_DMA1	TRGM DMA request 1

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Index	Destination	Description

Table 171: TRGM TRIGMUX OUTPUT List

34.4 TRGM Register List

The register list of TRGM is as follows:

TRGM0 base address: 0xF033C000

Offset	Register Name	Description	Reset Value
0x0000	FILTCFG[PWM0_IN0]	Filter configure register	0x00000000
0x0004	FILTCFG[PWM0_IN1]	Filter configure register	0x00000000
0x0008	FILTCFG[PWM0_IN2]	Filter configure register	0x00000000
0x000C	FILTCFG[PWM0_IN3]	Filter configure register	0x00000000
0x0010	FILTCFG[PWM0_IN4]	Filter configure register	0x00000000
0x0014	FILTCFG[PWM0_IN5]	Filter configure register	0x00000000
0x0018	FILTCFG[PWM0_IN6]	Filter configure register	0x00000000
0x001C	FILTCFG[PWM0_IN7]	Filter configure register	0x00000000
0x0020	FILTCFG[PWM1_IN0]	Filter configure register	0x00000000
0x0024	FILTCFG[PWM1_IN1]	Filter configure register	0x00000000
0x0028	FILTCFG[PWM1_IN2]	Filter configure register	0x00000000
0x002C	FILTCFG[PWM1_IN3]	Filter configure register	0x00000000
0x0030	FILTCFG[PWM1_IN4]	Filter configure register	0x00000000
0x0034	FILTCFG[PWM1_IN5]	Filter configure register	0x00000000
0x0038	FILTCFG[PWM1_IN6]	Filter configure register	0x00000000
0x003C	FILTCFG[PWM1_IN7]	Filter configure register	0x00000000
0x0040	FILTCFG[TRGM_P_00]	TRGM IO Filter configure register	0x00000000
0x0044	FILTCFG[TRGM_P_01]	TRGM IO Filter configure register	0x00000000
0x0048	FILTCFG[TRGM_P_02]	TRGM IO Filter configure register	0x00000000
0x004C	FILTCFG[TRGM_P_03]	TRGM IO Filter configure register	0x00000000
0x0050	FILTCFG[TRGM_P_04]	TRGM IO Filter configure register	0x00000000
0x0054	FILTCFG[TRGM_P_05]	TRGM IO Filter configure register	0x00000000
0x0058	FILTCFG[TRGM_P_06]	TRGM IO Filter configure register	0x00000000
0x005C	FILTCFG[TRGM_P_07]	TRGM IO Filter configure register	0x00000000

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Offset	Register Name	Description	Reset Value
0x0060	FILTCFG[PWM0_FAULT0]	Filter configure register	0x00000000
0x0064	FILTCFG[PWM0_FAULT1]	Filter configure register	0x00000000
0x0068	FILTCFG[PWM1_FAULT0]	Filter configure register	0x00000000
0x006C	FILTCFG[PWM1_FAULT1]	Filter configure register	0x00000000
0x0100	TRGOCFG[MOT2OPAMP0_0]	Trigger manager output configure register	0x00000000
0x0104	TRGOCFG[MOT2OPAMP0_1]	Trigger manager output configure register	0x00000000
0x0108	TRGOCFG[MOT2OPAMP0_2]	Trigger manager output configure register	0x00000000
0x010C	TRGOCFG[MOT2OPAMP0_3]	Trigger manager output configure register	0x00000000
0x0110	TRGOCFG[MOT2OPAMP0_4]	Trigger manager output configure register	0x00000000
0x0114	TRGOCFG[MOT2OPAMP0_5]	Trigger manager output configure register	0x00000000
0x0118	TRGOCFG[MOT2OPAMP0_6]	Trigger manager output configure register	0x00000000
0x011C	TRGOCFG[MOT2OPAMP0_7]	Trigger manager output configure register	0x00000000
0x0120	TRGOCFG[MOT2OPAMP1_0]	Trigger manager output configure register	0x00000000
0x0124	TRGOCFG[MOT2OPAMP1_1]	Trigger manager output configure register	0x00000000
0x0128	TRGOCFG[MOT2OPAMP1_2]	Trigger manager output configure register	0x00000000
0x012C	TRGOCFG[MOT2OPAMP1_3]	Trigger manager output configure register	0x00000000
0x0130	TRGOCFG[MOT2OPAMP1_4]	Trigger manager output configure register	0x00000000
0x0134	TRGOCFG[MOT2OPAMP1_5]	Trigger manager output configure register	0x00000000
0x0138	TRGOCFG[MOT2OPAMP1_6]	Trigger manager output configure register	0x00000000
0x013C	TRGOCFG[MOT2OPAMP1_7]	Trigger manager output configure register	0x00000000
0x0140	TRGOCFG[GPTMR0_IN2]	Trigger manager output configure register	0x00000000
0x0144	TRGOCFG[GPTMR0_IN3]	Trigger manager output configure register	0x00000000

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Offset	Register Name	Description	Reset Value
0x0148	TRGOCFG[GPTMR0_SYNCI]	Trigger manager output configure register	0x00000000
0x014C	TRGOCFG[GPTMR1_IN2]	Trigger manager output configure register	0x00000000
0x0150	TRGOCFG[GPTMR1_IN3]	Trigger manager output configure register	0x00000000
0x0154	TRGOCFG[GPTMR1_SYNCI]	Trigger manager output configure register	0x00000000
0x0158	TRGOCFG[GPTMR2_IN2]	also as DAC0_STPTRG1	0x00000000
0x015C	TRGOCFG[GPTMR2_IN3]	also as DAC0_STPTRG2	0x00000000
0x0160	TRGOCFG[GPTMR2_SYNCI]	also as DAC0_STPTRG3	0x00000000
0x0164	TRGOCFG[GPTMR3_IN2]	also as DAC1_STPTRG1	0x00000000
0x0168	TRGOCFG[GPTMR3_IN3]	also as DAC1_STPTRG2	0x00000000
0x016C	TRGOCFG[GPTMR3_SYNCI]	also as DAC1_STPTRG3	0x00000000
0x0170	TRGOCFG[ACMP_CH0_WIN]	Trigger manager output configure register	0x00000000
0x0174	TRGOCFG[ACMP_CH1_WIN]	Trigger manager output configure register	0x00000000
0x0178	TRGOCFG[DAC0_BUFTRG]	also as DAC0_STPTRG0	0x00000000
0x017C	TRGOCFG[DAC1_BUFTRG]	also as DAC1_STPTRG0	0x00000000
0x0180	TRGOCFG[ADC0_STRGI]	Trigger manager output configure register	0x00000000
0x0184	TRGOCFG[ADC1_STRGI]	Trigger manager output configure register	0x00000000
0x0188	TRGOCFG[ADCX_PTRGI0A]	Trigger manager output configure register	0x00000000
0x018C	TRGOCFG[ADCX_PTRGI0B]	Trigger manager output configure register	0x00000000
0x0190	TRGOCFG[ADCX_PTRGI0C]	Trigger manager output configure register	0x00000000
0x0194	TRGOCFG[ADCX_PTRGI1A]	Trigger manager output configure register	0x00000000
0x0198	TRGOCFG[ADCX_PTRGI1B]	Trigger manager output configure register	0x00000000
0x019C	TRGOCFG[ADCX_PTRGI1C]	Trigger manager output configure register	0x00000000
0x01A0	TRGOCFG[ADCX_PTRGI2A]	Trigger manager output configure register	0x00000000
0x01A4	TRGOCFG[ADCX_PTRGI2B]	Trigger manager output configure register	0x00000000

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Offset	Register Name	Description	Reset Value
0x01A8	TRGOCFG[ADCX_PTRGI2C]	Trigger manager output configure register	0x00000000
0x01AC	TRGOCFG[ADCX_PTRGI3A]	Trigger manager output configure register	0x00000000
0x01B0	TRGOCFG[ADCX_PTRGI3B]	Trigger manager output configure register	0x00000000
0x01B4	TRGOCFG[ADCX_PTRGI3C]	Trigger manager output configure register	0x00000000
0x01B8	TRGOCFG[CAN_PTPC0_CAP]	Trigger manager output configure register	0x00000000
0x01BC	TRGOCFG[CAN_PTPC1_CAP]	Trigger manager output configure register	0x00000000
0x01C0	TRGOCFG[QEO0_TRIG_IN0]	Trigger manager output configure register	0x00000000
0x01C4	TRGOCFG[QEO0_TRIG_IN1]	Trigger manager output configure register	0x00000000
0x01C8	TRGOCFG[QEO1_TRIG_IN0]	Trigger manager output configure register	0x00000000
0x01CC	TRGOCFG[QEO1_TRIG_IN1]	Trigger manager output configure register	0x00000000
0x01D0	TRGOCFG[SEI_TRIG_IN0]	Trigger manager output configure register	0x00000000
0x01D4	TRGOCFG[SEI_TRIG_IN1]	Trigger manager output configure register	0x00000000
0x01D8	TRGOCFG[SEI_TRIG_IN2]	Trigger manager output configure register	0x00000000
0x01DC	TRGOCFG[SEI_TRIG_IN3]	Trigger manager output configure register	0x00000000
0x01E0	TRGOCFG[SEI_TRIG_IN4]	Trigger manager output configure register	0x00000000
0x01E4	TRGOCFG[SEI_TRIG_IN5]	Trigger manager output configure register	0x00000000
0x01E8	TRGOCFG[SEI_TRIG_IN6]	Trigger manager output configure register	0x00000000
0x01EC	TRGOCFG[SEI_TRIG_IN7]	Trigger manager output configure register	0x00000000
0x01F0	TRGOCFG[MMC0_TRIG_IN0]	Trigger manager output configure register	0x00000000
0x01F4	TRGOCFG[MMC0_TRIG_IN1]	Trigger manager output configure register	0x00000000

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Offset	Register Name	Description	Reset Value
0x01F8	TRGOCFG[MMC1_TRIG_IN0]	Trigger manager output configure register	0x00000000
0x01FC	TRGOCFG[MMC1_TRIG_IN1]	Trigger manager output configure register	0x00000000
0x0200	TRGOCFG[PLB_IN_00]	Trigger manager output configure register	0x00000000
0x0204	TRGOCFG[PLB_IN_01]	Trigger manager output configure register	0x00000000
0x0208	TRGOCFG[PLB_IN_02]	Trigger manager output configure register	0x00000000
0x020C	TRGOCFG[PLB_IN_03]	Trigger manager output configure register	0x00000000
0x0210	TRGOCFG[PLB_IN_04]	Trigger manager output configure register	0x00000000
0x0214	TRGOCFG[PLB_IN_05]	Trigger manager output configure register	0x00000000
0x0218	TRGOCFG[PLB_IN_06]	Trigger manager output configure register	0x00000000
0x021C	TRGOCFG[PLB_IN_07]	Trigger manager output configure register	0x00000000
0x0220	TRGOCFG[PLB_IN_08]	Trigger manager output configure register	0x00000000
0x0224	TRGOCFG[PLB_IN_09]	Trigger manager output configure register	0x00000000
0x0228	TRGOCFG[PLB_IN_10]	Trigger manager output configure register	0x00000000
0x022C	TRGOCFG[PLB_IN_11]	Trigger manager output configure register	0x00000000
0x0230	TRGOCFG[PLB_IN_12]	Trigger manager output configure register	0x00000000
0x0234	TRGOCFG[PLB_IN_13]	Trigger manager output configure register	0x00000000
0x0238	TRGOCFG[PLB_IN_14]	Trigger manager output configure register	0x00000000
0x023C	TRGOCFG[PLB_IN_15]	Trigger manager output configure register	0x00000000
0x0240	TRGOCFG[PLB_IN_16]	Trigger manager output configure register	0x00000000
0x0244	TRGOCFG[PLB_IN_17]	Trigger manager output configure register	0x00000000

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Offset	Register Name	Description	Reset Value
0x0248	TRGOCFG[PLB_IN_18]	Trigger manager output configure register	0x00000000
0x024C	TRGOCFG[PLB_IN_19]	Trigger manager output configure register	0x00000000
0x0250	TRGOCFG[PLB_IN_20]	Trigger manager output configure register	0x00000000
0x0254	TRGOCFG[PLB_IN_21]	Trigger manager output configure register	0x00000000
0x0258	TRGOCFG[PLB_IN_22]	Trigger manager output configure register	0x00000000
0x025C	TRGOCFG[PLB_IN_23]	Trigger manager output configure register	0x00000000
0x0260	TRGOCFG[PLB_IN_24]	Trigger manager output configure register	0x00000000
0x0264	TRGOCFG[PLB_IN_25]	Trigger manager output configure register	0x00000000
0x0268	TRGOCFG[PLB_IN_26]	Trigger manager output configure register	0x00000000
0x026C	TRGOCFG[PLB_IN_27]	Trigger manager output configure register	0x00000000
0x0270	TRGOCFG[PLB_IN_28]	Trigger manager output configure register	0x00000000
0x0274	TRGOCFG[PLB_IN_29]	Trigger manager output configure register	0x00000000
0x0278	TRGOCFG[PLB_IN_30]	Trigger manager output configure register	0x00000000
0x027C	TRGOCFG[PLB_IN_31]	Trigger manager output configure register	0x00000000
0x0280	TRGOCFG[TRGM_P_00]	Trigger manager output configure register for TRGM IO	0x00000000
0x0284	TRGOCFG[TRGM_P_01]	Trigger manager output configure register for TRGM IO	0x00000000
0x0288	TRGOCFG[TRGM_P_02]	Trigger manager output configure register for TRGM IO	0x00000000
0x028C	TRGOCFG[TRGM_P_03]	Trigger manager output configure register for TRGM IO	0x00000000
0x0290	TRGOCFG[TRGM_P_04]	Trigger manager output configure register for TRGM IO	0x00000000
0x0294	TRGOCFG[TRGM_P_05]	Trigger manager output configure register for TRGM IO	0x00000000

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Offset	Register Name	Description	Reset Value
0x0298	TRGOCFG[TRGM_P_06]	Trigger manager output configure register for TRGM IO	0x00000000
0x029C	TRGOCFG[TRGM_P_07]	Trigger manager output configure register for TRGM IO	0x00000000
0x02A0	TRGOCFG[PWM_IN8]	Trigger manager output configure register	0x00000000
0x02A4	TRGOCFG[PWM_IN9]	Trigger manager output configure register	0x00000000
0x02A8	TRGOCFG[PWM_IN10]	Trigger manager output configure register	0x00000000
0x02AC	TRGOCFG[PWM_IN11]	Trigger manager output configure register	0x00000000
0x02B0	TRGOCFG[PWM_IN12]	Trigger manager output configure register	0x00000000
0x02B4	TRGOCFG[PWM_IN13]	Trigger manager output configure register	0x00000000
0x02B8	TRGOCFG[PWM_IN14]	Trigger manager output configure register	0x00000000
0x02BC	TRGOCFG[PWM_IN15]	Trigger manager output configure register	0x00000000
0x02C0	TRGOCFG[PWM0_FRCI]	Trigger manager output configure register	0x00000000
0x02C4	TRGOCFG[PWM0_FRCSYNCI]	Trigger manager output configure register	0x00000000
0x02C8	TRGOCFG[PWM0_SYNCI]	Trigger manager output configure register	0x00000000
0x02CC	TRGOCFG[PWM0_SHRLDSYNCI]	Trigger manager output configure register	0x00000000
0x02D0	TRGOCFG[PWM0_FAULTI0]	Trigger manager output configure register	0x00000000
0x02D4	TRGOCFG[PWM0_FAULTI1]	Trigger manager output configure register	0x00000000
0x02D8	TRGOCFG[PWM1_FRCI]	Trigger manager output configure register	0x00000000
0x02DC	TRGOCFG[PWM1_FRCSYNCI]	Trigger manager output configure register	0x00000000
0x02E0	TRGOCFG[PWM1_SYNCI]	Trigger manager output configure register	0x00000000
0x02E4	TRGOCFG[PWM1_SHRLDSYNCI]	Trigger manager output configure register	0x00000000

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Offset	Register Name	Description	Reset Value
0x02E8	TRGOCFG[PWM1_FAULTI0]	Trigger manager output configure register	0x00000000
0x02EC	TRGOCFG[PWM1_FAULTI1]	Trigger manager output configure register	0x00000000
0x02F0	TRGOCFG[RDC_TRIG_IN0]	Trigger manager output configure register	0x00000000
0x02F4	TRGOCFG[RDC_TRIG_IN1]	Trigger manager output configure register	0x00000000
0x02F8	TRGOCFG[SYNCTIMER_TRIG]	Trigger manager output configure register	0x00000000
0x02FC	TRGOCFG[QEI0_TRIG_IN]	Trigger manager output configure register	0x00000000
0x0300	TRGOCFG[QEI1_TRIG_IN]	Trigger manager output configure register	0x00000000
0x0304	TRGOCFG[QEI0_PAUSE]	Trigger manager output configure register	0x00000000
0x0308	TRGOCFG[QEI1_PAUSE]	Trigger manager output configure register	0x00000000
0x030C	TRGOCFG[UART_TRIG0]	Trigger manager output configure register	0x00000000
0x0310	TRGOCFG[UART_TRIG1]	Trigger manager output configure register	0x00000000
0x0314	TRGOCFG[TRGM_IRQ0]	Trigger manager output configure register	0x00000000
0x0318	TRGOCFG[TRGM_IRQ1]	Trigger manager output configure register	0x00000000
0x031C	TRGOCFG[TRGM_DMA0]	Trigger manager output configure register	0x00000000
0x0320	TRGOCFG[TRGM_DMA1]	Trigger manager output configure register	0x00000000
0x0400	DMACFG[0]	DMA request configure register	0x00000000
0x0404	DMACFG[1]	DMA request configure register	0x00000000
0x0408	DMACFG[2]	DMA request configure register	0x00000000
0x040C	DMACFG[3]	DMA request configure register	0x00000000
0x0410	DMACFG[4]	DMA request configure register	0x00000000
0x0414	DMACFG[5]	DMA request configure register	0x00000000
0x0418	DMACFG[6]	DMA request configure register	0x00000000
0x041C	DMACFG[7]	DMA request configure register	0x00000000
0x0500	GCR	General Control Register	0x00000000
0x0510	ADC_MATRIX_SEL	adc matrix select register	0x00000000
0x0514	DAC_MATRIX_SEL	dac matrix select register	0x00000000

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Offset	Register Name	Description	Reset Value
0x0518	POS_MATRIX_SEL0	position matrix select register0	0x00000000
0x051C	POS_MATRIX_SEL1	position matrix select register1	0x00000000
0x0600	TRGM_IN[0]	trigmux input read register0	0x00000000
0x0604	TRGM_IN[1]	trigmux input read register1	0x00000000
0x0608	TRGM_IN[2]	trigmux input read register2	0x00000000
0x060C	TRGM_IN[3]	trigmux input read register3	0x00000000
0x0620	TRGM_OUT[0]	trigmux output read register0	0x00000000
0x0624	TRGM_OUT[1]	trigmux output read register1	0x00000000
0x0628	TRGM_OUT[2]	trigmux output read register2	0x00000000
0x062C	TRGM_OUT[3]	trigmux output read register3	0x00000000
0x0630	TRGM_OUT[4]	trigmux output read register4	0x00000000

Table 172: TRGM Register List

34.5 TRGM Register Description

The detailed description of TRGM registers is as follows:

34.5.1 FILTCFG (0x0 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD															OUTINV	MODE			SYNCEN	FILTLEN_SHIFT			FILTLEN_BASE								
N/A															RW	RW			RW	RW			RW								
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

FILTCFG [31:0]

Bits	Name	Description
16	OUTINV	1- Filter will invert the output 0- Filter will not invert the output
15-13	MODE	This bitfields defines the filter mode 000-bypass; 100-rapid change mode; 101-delay filter mode; 110-stalbe low mode; 111-stable high mode
12	SYNCEN	set to enable sychronization input signal with TRGM clock
11-9	FILTLEN_SHIFT	
8-0	FILTLEN_BASE	This bitfields defines the filter counter length.

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Bits	Name	Description

FILTCFG Bits

34.5.2 TRGOCFG (0x100 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																				OUTINV	FEDG2PEN	REDG2PEN	RSVD		TRIGOSEL							
N/A																				RW	RW	RW	N/A		RW							
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	x	x	0	0	0	0	0	0	0

TRGOCFG [31:0]

Bits	Name	Description
11	OUTINV	1- Invert the output
10	FEDG2PEN	1- The selected input signal falling edge will be convert to an pulse on output. The output pulse can be stably used within the motor control system. When connecting the signal outside the motor system, due to the asynchronization of the clock systems, the clock frequency and signal active length need to be considered.
9	REDG2PEN	1- The selected input signal rising edge will be convert to an pulse on output. The output pulse can be stably used within the motor control system. When connecting the signal outside the motor system, due to the asynchronization of the clock systems, the clock frequency and signal active length need to be considered.
6-0	TRIGOSEL	This bitfield selects one of the TRGM inputs as output.

TRGOCFG Bits

34.5.3 DMACFG (0x400 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DMAMUX_EN	RSVD																									DMASRCSEL					
RW	N/A																									RW					
0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0

DMACFG [31:0]

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Bits	Name	Description
31	DMAMUX_EN	
5-0	DMASRCSEL	This field selects one of the DMA requests as the DMA request output.

DMACFG Bits

34.5.4 GCR (0x500)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																								TRGOPEN							
N/A																								RW							
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0

GCR [31:0]

Bits	Name	Description
7-0	TRGOPEN	The bitfield enable the TRGM outputs.

GCR Bits

34.5.5 ADC_MATRIX_SEL (0x510)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
QEI1_ADC1_SEL								QEI1_ADC0_SEL								QEI0_ADC1_SEL								QEI0_ADC0_SEL							
RW								RW								RW								RW							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ADC_MATRIX_SEL [31:0]

Bits	Name	Description
31-24	QEI1_ADC1_SEL	0-adc0; 1-adc1; 2-rdc_adc0; 3-rdc_adc1; bit7 is used to invert adc_value; others reserved
23-16	QEI1_ADC0_SEL	
15-8	QEI0_ADC1_SEL	
7-0	QEI0_ADC0_SEL	

ADC_MATRIX_SEL Bits

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34.5.6 DAC_MATRIX_SEL (0x514)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DAC1_DAC_SEL								DAC0_DAC_SEL								ACMP1_DAC_SEL								ACMP0_DAC_SEL							
RW								RW								RW								RW							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DAC_MATRIX_SEL [31:0]

Bits	Name	Description
31-24	DAC1_DAC_SEL	0-qeo0_dac0; 1-qeo0_dac1; 2-qeo0_dac2; 3-qeo1_dac0; 4-qeo1_dac1; 5-qeo1_dac2; 6-rdc_dac0; 7-rdc_dac1; bit7 is used to invert dac_value; others reserved
23-16	DAC0_DAC_SEL	
15-8	ACMP1_DAC_SEL	
7-0	ACMP0_DAC_SEL	

DAC_MATRIX_SEL Bits

34.5.7 POS_MATRIX_SEL0 (0x518)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MMC1_POSIN_SEL								MMC0_POSIN_SEL								SEI_POSIN1_SEL								SEI_POSIN0_SEL							
RW								RW								RW								RW							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

POS_MATRIX_SEL0 [31:0]

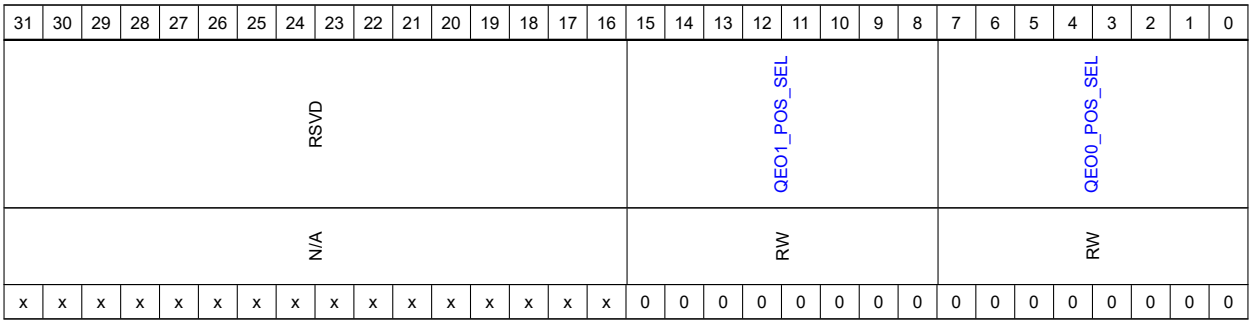
Bits	Name	Description
31-24	MMC1_POSIN_SEL	0-sei_pos_out0; 1-sei_pos_out1; 2-qei0_pos; 3-qei1_pos; 4-mmc0_pos_out0; 5-mmc0_pos_out1; 6-mmc1_pos_out0; 7-mmc1_pos_out1; bit7 is used to invert position value; others reserved

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Bits	Name	Description
23-16	MMC0_POSIN_SEL	
15-8	SEI_POSIN1_SEL	
7-0	SEI_POSIN0_SEL	

POS_MATRIX_SEL0 Bits

34.5.8 POS_MATRIX_SEL1 (0x51C)

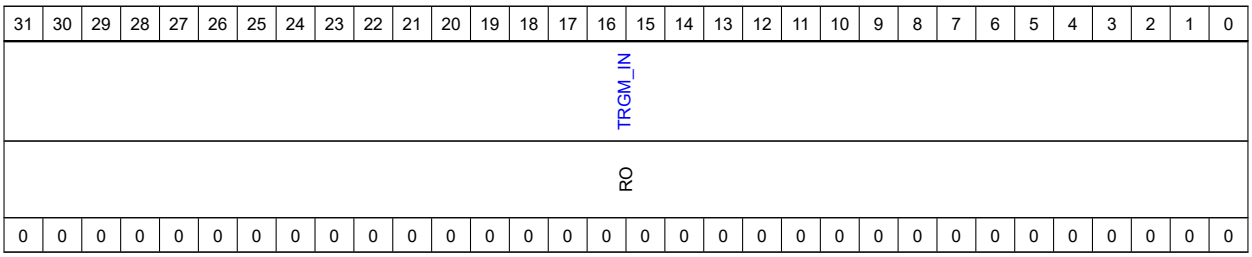


POS_MATRIX_SEL1 [31:0]

Bits	Name	Description
15-8	QEO1_POS_SEL	
7-0	QEO0_POS_SEL	

POS_MATRIX_SEL1 Bits

34.5.9 TRGM_IN (0x600 + 0x4 * n)



TRGM_IN [31:0]

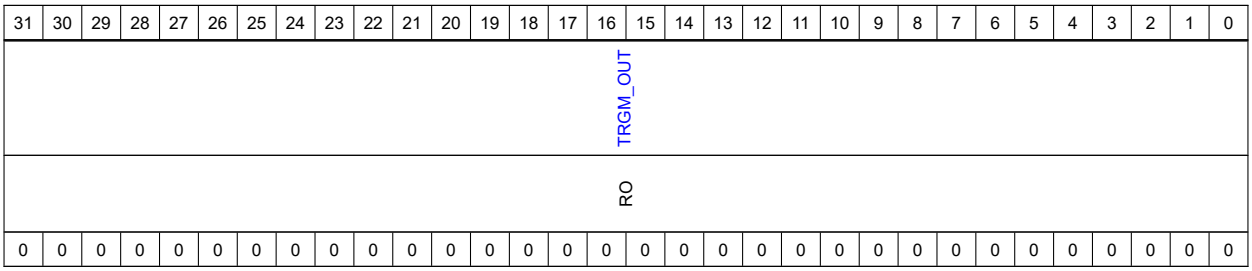
Bits	Name	Description
31-0	TRGM_IN	mmc1_trig_out[1:0], mmc0_trig_out[1:0],sync_pulse[3:0],moto_gpio_in_sync[7:0],//31:16 gtmr3_to_motor_sync[1:0],gtmr2_to_motor_sync[1:0],gtmr1_to_motor_sync[1:0],g //15:8 acmp_out_sync[1:0],can2mot_event_sync[1:0],usb0_sof_tog_sync,pwm_debug,1 //7:0

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Bits	Name	Description

TRGM_IN Bits

34.5.10 TRGM_OUT (0x620 + 0x4 * n)



TRGM_OUT [31:0]

Bits	Name	Description

31-0	TRGM_OUT	motor_to_opamp0[7:0] = trig_mux_out[7:0];
		motor_to_opamp1[7:0] = trig_mux_out[15:8];
		motor_to_gtmr0_capt[1:0] = trig_mux_out[17:16];
		motor_to_gtmr0_sync = trig_mux_out[18];
		motor_to_gtmr1_capt[1:0] = trig_mux_out[20:19];
		motor_to_gtmr1_sync = trig_mux_out[21];
		motor_to_gtmr2_capt[1:0] = trig_mux_out[23:22];
		motor_to_gtmr2_sync = trig_mux_out[24];
		motor_to_gtmr3_capt[1:0] = trig_mux_out[26:25];
		motor_to_gtmr3_sync = trig_mux_out[27];
		acmp_window[1:0] = trig_mux_out[29:28];
		dac0_buf_trigger = trig_mux_out[30];
		dac1_buf_trigger = trig_mux_out[31];
		dac0_step_trigger[3:0] = {trig_mux_out[24:22],trig_mux_out[30]};//use same buf_trig, and gtmr2
		dac1_step_trigger[3:0] = {trig_mux_out[27:25],trig_mux_out[31]}; //use same buf_trig, and gtmr3

TRGM_OUT Bits

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35 Synchronization Timer (SYNT)

This chapter describes the main functions and features of the Synchronization Timer (SYNT).

35.1 Feature Summary

This section describes the main features of the SYNT:

- Supports synchronization event output
  - 32-bit counter and reload register
  - Supports 4 channels, each channel is equipped with one compare register
- Supports output of unified timestamp to the motion control system
  - 32-bit counter for timestamp generation
  - Supports recording timestamps when events are triggered
  - Supports software-based timestamp update

35.2 Functional Description

There are two independent 32-bit counters inside the SYNT, which are used for synchronization event generation and timestamp generation respectively.

35.2.1 Synchronization Event Output

The SYNT uses a 32-bit counter to generate synchronization events, and users can configure the reload value of this counter through the reload register RLD.

During operation, set the GCR[CEN] bit to 1 to enable the counter. After being enabled, the counter always starts counting from 0. When the counter value reaches the value in the RLD register, the counter resets to 0 and continues counting.

Users can reset the counter to 0 by setting the GCR[CRST] bit to 1, and GCR[CRST] needs to be cleared manually after the reset is completed. If the GCR[COUNTER_DEBUG_EN] bit field is enabled, the counter will stop counting in debug mode.

The SYNT supports 4 channels, each configured with a 32-bit compare register CMPx. When the counter count reaches the value of the compare register, a compare event with a duration of one SYNT clock pulse is output. This event can be connected to other modules of the motor system, such as PWM, QEI, etc., through the interconnection manager for synchronization purposes.

For example, by connecting to the synchronous trigger input (SYNCI) of the PWM, 4 motor systems can start operating precisely in the same clock cycle, and the phase difference between the 4 motor systems can also be adjusted accurately.

Note: The SYNT is designed for synchronization between on-chip motor systems. Its operating clock is synchronized with the motor system, so it cannot be used for peripherals in other clock domains.

The SYNT does not support generating interrupt or DMA requests.

35.2.2 Timestamp Output

The SYNT can output a 32-bit timestamp to other modules in the motion control system (motor system), providing a unified time reference for collaborative work between different modules of the motion control system.

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This timestamp is generated by a 32-bit counter.

During operation, set the GCR[TIMESTAMP_ENABLE] bit to 1 to enable the counter. After being enabled, the counter always starts counting from 0. When the counter value reaches 0xFFFF_FFFF, the counter automatically resets to 0 and continues counting.

Users can reset the counter to 0 by setting the GCR[TIMESTAMP_RESET] bit to 1, and GCR[TIMESTAMP_RESET] is automatically cleared after being written with 1. If the GCR[TIMESTAMP_DEBUG_EN] bit field is enabled, the counter will stop counting in debug mode.

Users can update the value of this counter through software: Set GCR[TIMESTAMP_SET_NEW] to 1 to directly update the counter value (timestamp) to the value written in the TIMESTAMP_NEW register; Set GCR[TIMESTAMP_DEC_NEW] to 1 to update the counter value to the current counter value minus the value written in the TIMESTAMP_NEW register; Set GCR[TIMESTAMP_INC_NEW] to 1 to update the counter value to the current counter value plus the value written in the TIMESTAMP_NEW register.

The SYNT also supports recording timestamps when events are triggered. By configuring the interconnection manager, events from the motion control system are selected as the trigger signal for the SYNT, which triggers the SYNT to save the current counter timestamp to the TIMESTAMP_SAV register.

35.3 SYNT Register List

The register list of SYNT is as follows:

SYNT base address: 0xF0328000

Offset	Register Name	Description	Reset Value
0x0000	GCR	Global control register	0x00000000
0x0004	RLD	Counter reload register	0x00000000
0x0008	TIMESTAMP_NEW	timestamp new value register	0x00000000
0x000C	CNT	Counter	0x00000000
0x0010	TIMESTAMP_SAV	timestamp trig save value	0x00000000
0x0014	TIMESTAMP_CUR	timestamp read value	0x00000000
0x0020	CMP[0]	Comparator	0x00000000
0x0024	CMP[1]	Comparator	0x00000000
0x0028	CMP[2]	Comparator	0x00000000
0x002C	CMP[3]	Comparator	0x00000000

Table 173: SYNT Register List

35.4 SYNT Register Details

The detailed description of SYNT registers is as follows:

35.4.1 GCR (0x0)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIMESTAMP_INC_NEW	TIMESTAMP_DEC_NEW	TIMESTAMP_SET_NEW	TIMESTAMP_RESET	RSVD																						TIMESTAMP_DEBUG_EN	TIMESTAMP_ENABLE	RSVD	COUNTER_DEBUG_EN	CRST	CEN
WO	WO	WO	WO	N/A																						RW	RW	N/A	RW	RW	RW
0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	x	0	0	0

GCR [31:0]

Bits	Name	Description
31	TIMESTAMP_INC_NEW	set to increase the timesamp with new value, auto clr
30	TIMESTAMP_DEC_NEW	set to decrease the timesamp with new value, auto clr
29	TIMESTAMP_SET_NEW	set the timesamp to new value, auto clr
28	TIMESTAMP_RESET	reset timesamp to 0, auto clr
5	TIMESTAMP_DEBUG_EN	set to enable cpu_debug_mode to stop the timesamp
4	TIMESTAMP_ENABLE	set to enable the timesamp , clr to stop
2	COUNTER_DEBUG_EN	set to enable cpu_debug_mode to stop the counter
1	CRST	1- Reset counter
0	CEN	1- Enable counter

GCR Bits

35.4.2 RLD (0x4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RLD																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

RLD [31:0]

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Bits	Name	Description
31-0	RLD	counter reload value

RLD Bits

35.4.3 TIMESTAMP_NEW (0x8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VALUE																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TIMESTAMP_NEW [31:0]

Bits	Name	Description
31-0	VALUE	new value for timesamp , can be used as set/inc/dec

TIMESTAMP_NEW Bits

35.4.4 CNT (0xC)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CNT [31:0]

Bits	Name	Description
31-0	CNT	counter

CNT Bits

35.4.5 TIMESTAMP_SAV (0x10)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VALUE																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

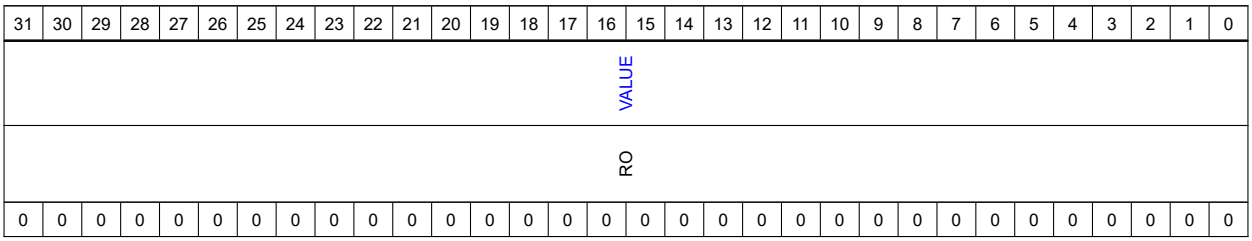
TIMESTAMP_SAV [31:0]

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Bits	Name	Description
31-0	VALUE	use the trigger to save timesamp here

TIMESTAMP_SAV Bits

35.4.6 TIMESTAMP_CUR (0x14)

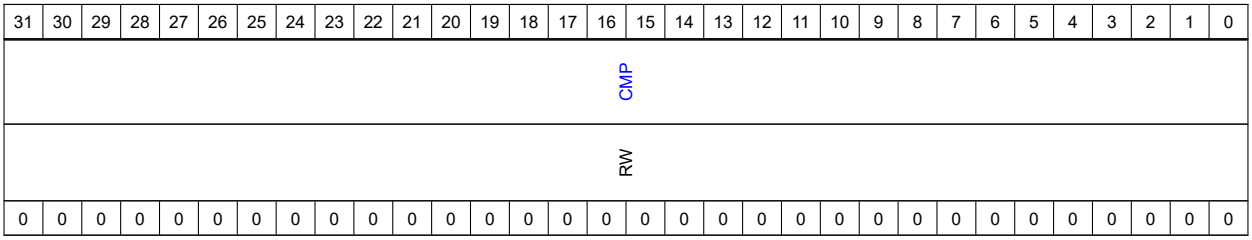


TIMESTAMP_CUR [31:0]

Bits	Name	Description
31-0	VALUE	current timesamp value

TIMESTAMP_CUR Bits

35.4.7 CMP (0x20 + 0x4 * n)



CMP [31:0]

Bits	Name	Description
31-0	CMP	comparator value, the output will assert when counter count to this value

CMP Bits

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36 Quadrature Encoder Interface QEIV2

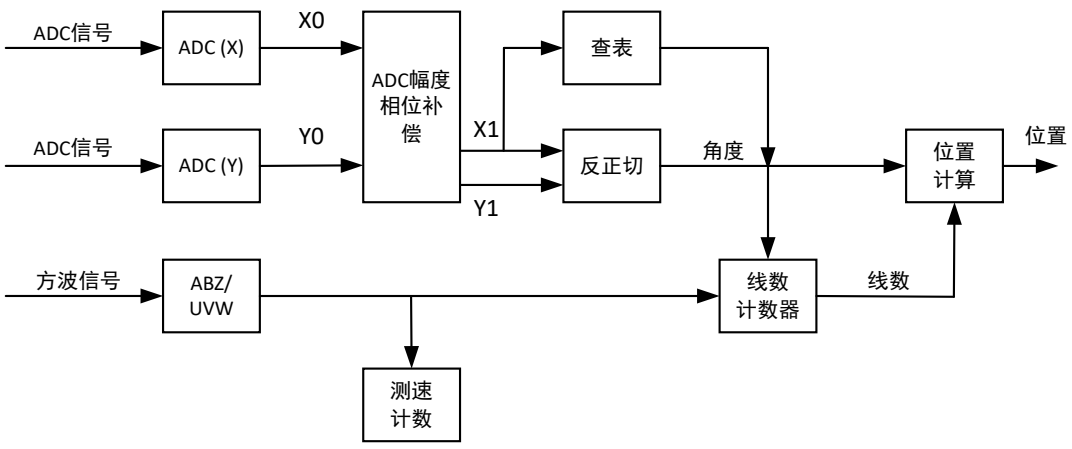
Measures speed and generates position information based on input signals.

36.1 Feature Summary

This section describes the main features of the Quadrature Encoder Interface QEIV2:

- Supports single-phase, two-phase (quadrature/up-down/direction pulse), three-phase (120° phase difference) square wave input;
- Supports index signal INDEX/Z, which can generate interrupts, reset the line count counter, and modify the revolution count counter
- Supports two homing signals HOME, which only generate interrupts
- Supports single-phase, two-phase (quadrature) sine wave input (via ADC harness)
- Supports amplitude, center value and included angle correction for sine wave signals
- Supports line count up to 32 bits
- Supports two preset position trigger interrupts, with configurable specified revolution count, line count, direction and position
- Supports software trigger latch, event trigger latch, and two external trigger latches (including revolution count, line count, direction, angle and position information)
- Supports two pulse count based speed measurement counters, two speed measurement results are stored, and speed measurement time is recorded
- Supports two duration based speed measurement counters, two speed measurement results are stored, and speed measurement time is recorded
- Supports converting position to binary position, which is output to the position manager through the position harness (including position acquisition time)
- Allows software to modify position, line count counter and revolution count counter (only absolute value mode is supported), supports incremental and absolute value modes
- Supports watchdog and stall detection
- All interrupt signals can also be configured to generate DMA requests, trigger latches, or trigger output signals

36.2 Architecture Diagram



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Figure 40: QEI Architecture Diagram

36.3 Pin Description

Pin	Direction	Function Description
QEI_A	Input	Phase A signal, can also be used as the U-phase signal of UVW.
QEI_B	Input	Phase B signal, can also be used as the V-phase signal of UVW.
QEI_Z	Input	Phase Z signal, can also be used as the W-phase signal of UVW.
QEI_F	Input	Fault signal, which can cut off position information output when active, and can generate interrupts.
QEI_H0	Input	Home0 signal, can generate interrupts.
QEI_H1	Input	Home1 signal, can generate interrupts.

Table 174: QEI Pin Description

36.4 Functional Description

36.4.1 QEIV2 Decoding Modes

The quadrature encoder interface QEIV2 module (hereinafter referred to as QEI module) square wave encoding has three input signals: phase A, B and Z. The encoding logic parses the rotation information of the motor rotor based on these three-phase input signals.

The quadrature encoder interface supports single-phase, two-phase (quadrature/up-down/direction pulse), three-phase (120° phase difference) square wave, and sine-cosine wave decoding modes. Users can select the corresponding decoding mode in the [ENCTYP] bit field of the control register CR according to actual requirements.

36.4.1.1 Single-phase Digital Mode

For single-phase digital mode, only phase A signal input is required, but users must know the rotation direction of the motor rotor in advance.

Users need to configure the [ENCTYP] bit field of the control register CR to 4, and configure the [ZCNTCFG] bit field of the control register CR according to the situation of phase Z. In addition, it is also necessary to configure the number of motor lines and phase calculation parameters.

36.4.1.2 ABZ Mode

For AB two-phase quadrature signals, when phase A leads phase B, it indicates that the motor rotation direction is forward; when phase A lags phase B, it indicates that the motor rotation direction is reverse. When the quadrature encoder interface receives a forward rotating square wave, the AB phase counter (COUNT[CURRENT][PH] register, offset address 0x0034) increments by 1 when the signal flips, and the AB phase counter (PHASE_CNT)

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also increments by 1; if a reverse rotating square wave is received, the AB phase counter decrements by 1 when the signal flips, and the AB phase counter (PHASE_CNT) also decrements by 1. The process is shown in the figures below. The rotation direction of the motor and the input status of AB phase can be obtained through the [DIR] bit field of the speed register (COUNT[CURRENT][SPD]).

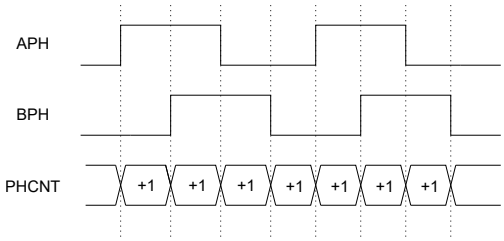


Figure 41: Schematic Diagram of Forward Rotation Phase Counting

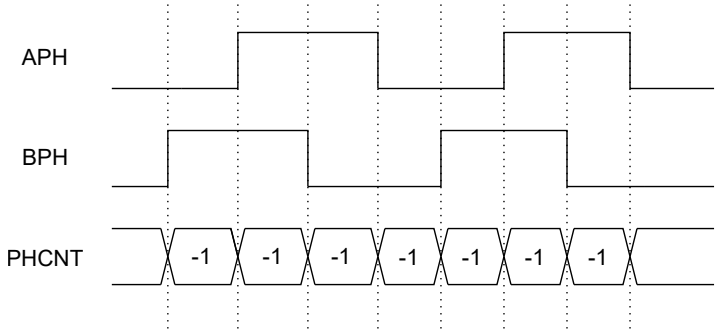


Figure 42: Schematic Diagram of Reverse Rotation Phase Counting

Users can set the phase configuration register (PHCFG[PHASE]) according to the motor and quadrature encoder, which represents the total number of PHASE counts for one revolution of the motor.

Whenever the AB phase counter counts forward to PHMAX, the AB phase counter will be reset to 0.

Whenever the AB phase counter counts in reverse to 0, the AB phase counter will be reset to the user-configured motor line count value (i.e., the [PHMAX value] of the phase configuration register PHCFG).

Phase Z is usually called the index phase. When AB phase indicates forward rotation, after capturing the rising edge of phase Z, the Z phase counter increments by 1; when AB phase indicates reverse rotation, after capturing the falling edge of phase Z, the Z phase counter decrements by 1. The motor outputs one pulse on phase Z per revolution, so this signal can be used to locate the zero point of each revolution of the motor. The quadrature decoder provides 2 counting modes for the Z phase counter, which users can configure through the PHCFG[ZCNTCFG] bit.

To ensure that the motor always locates to the same edge of the phase Z signal regardless of forward or reverse rotation, the internal logic of the quadrature decoder synchronizes the phase Z signal with the A/B phase signals, as follows:

- When the motor rotates forward, phase Z signal capture is always synchronized with the rising edge of phase A
- When the motor rotates in reverse, phase Z signal capture is always synchronized with the falling edge of phase A

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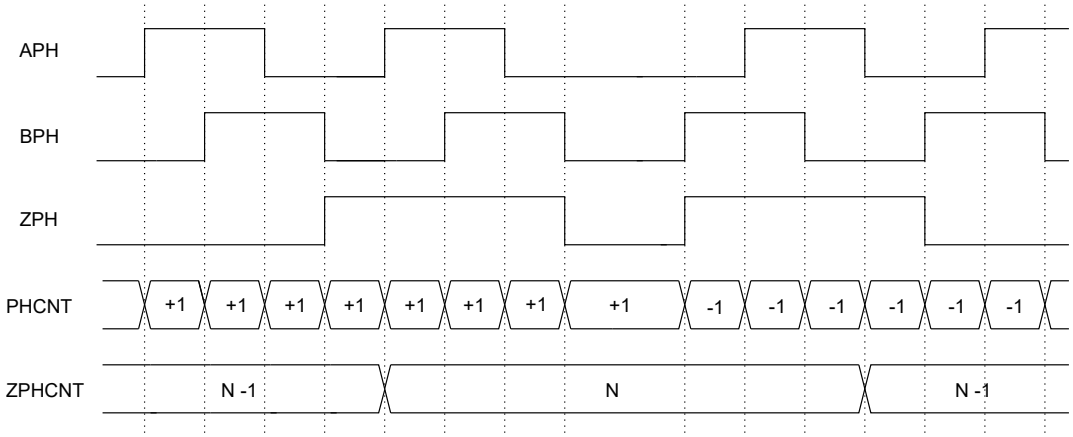


Figure 43: Schematic Diagram of Z-phase Input Synchronization

Z-phase Calibration Function

In addition, phase Z signal can be used to calibrate the phase counter. Users need to set the value of the PHCFG register, which represents the Z-phase correction value. When this register is set to 0, it indicates that the phase Z signal is the 0 point of motor rotation. It can also be set to any non-zero value, indicating that the phase Z signal has an offset relative to the 0 position.

- When the motor rotates forward, the rising edge of phase Z is captured, reset PHCNT = PHIDX;
- When the motor rotates in reverse, the falling edge of phase Z is captured, reset PHCNT = PHIDX - 1. Note that if PHIDX is set to 0, PHCNT will be reset to the phase overflow value PHMAX;

36.4.1.3 PD Mode

For PD mode, the phase A input pin is used to input P-phase pulse signals, and the phase B input pin is used to input D-phase pulse signals. The D phase indicates the rotation direction of the motor: high level indicates forward rotation, low level indicates reverse rotation. During forward rotation, each pulse of P phase increments the phase counter by 1; during reverse rotation, each pulse of P phase decrements the phase counter by 1.

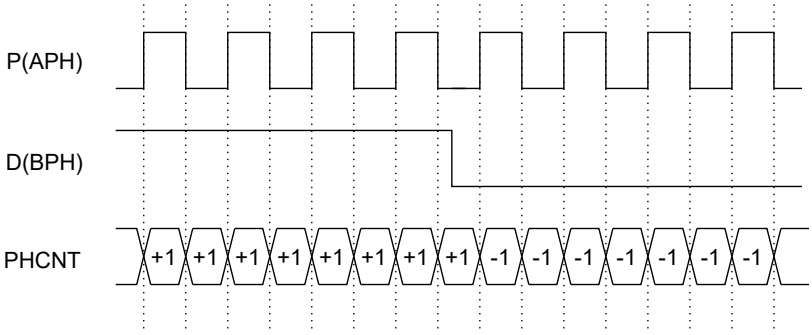


Figure 44: Schematic Diagram of PD Mode

36.4.1.4 UD Mode

For UD mode, the phase A input pin is used to input U-phase pulse signals, and the phase B input pin is used to input D-phase pulse signals. Each pulse input to U phase indicates forward rotation of the motor, which

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increments the phase counter by 1. Each pulse input to D phase indicates reverse rotation of the motor, which decrements the phase counter by 1.

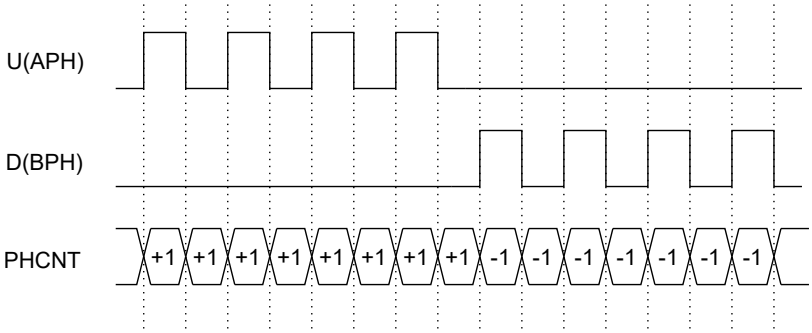


Figure 45: Schematic Diagram of Up-Down (UD) Decoding Mode

36.4.1.5 UVW Mode

For UVW mode, the phase A input pin is used to input U-phase pulse signals, the phase B input pin is used to input V-phase pulse signals, and the phase Z input pin is used to input W-phase pulse signals.

One cycle of any of the three input signals corresponds to one revolution of the motor rotor. The figure below shows a schematic diagram of forward rotation with 120° phase difference between the three signals. According to the three input signals, they can be divided into 6 phases. When this mode is enabled, the configuration of registers including UVW_POS_CFG[UVW_POS0_CFG], UVW_POS_CFG[UVW_POS1_CFG], UVW_POS_CFG[UVW_POS2_CFG], UVW_POS_CFG[UVW_POS3_CFG], UVW_POS_CFG[UVW_POS4_CFG], UVW_POS_CFG[UVW_POS5_CFG] is performed based on the six phase levels of the UVW three-phase signal and the value of the [UVW_POS_OPT0] bit field of the QEI configuration register (QEI_CFG). Note that this module requires 120° phase difference between the three signals.

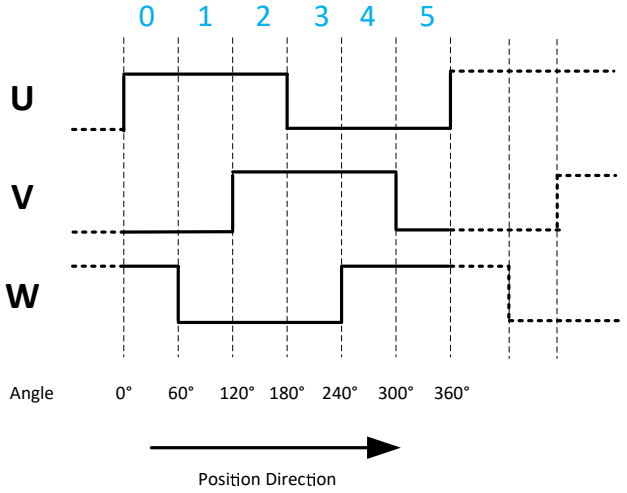


Figure 46: Schematic Diagram of UVW Three-phase Signal Position Encoding

36.4.1.6 Quadrature Sine-Cosine Wave Mode

The QEI module supports sine-cosine wave input from ADC. When sine wave input is used, the signal comes from the on-chip ADC. Users can configure the ADC source in X-channel ADC configuration register 0

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(ADCX_CFG0) and Y-channel ADC configuration register 0 (ADCY_CFG0), i.e., specify which ADC and which channel the sin and cos channels come from. Note that if both sin and cos come from the same ADC, the signal delay between the two channels will cause large deviation in the calculated angle.

Since ADC inputs are all unsigned numbers, it is necessary to configure the offset in X-channel ADC configuration register 2 (ADCX_CFG2) and Y-channel ADC configuration register 2 (ADCY_CFG2). The ADC value will subtract the offset to convert the input unsigned number into a signed number.

If the input sin and cos signals are not completely orthogonal or have different amplitudes, you can configure parameters in X-channel ADC configuration register 1 (ADCX_CFG1) or Y-channel ADC configuration register 1 (ADCY_CFG1) for correction and compensation.

If the sine wave signal phase offset is B, i.e., the input is Y0=sin(A+B), X0=cosA, parameters need to be configured to obtain sinA and cosA.

$$sinA = sin((A+B)-B)$$

$$= sin(A+B)*cosB - cos(A+B)*sinB$$

$$= sin(A+B)*cosB - (cosA*cosB*sinB - sinA*sinB*sinB)$$

After sorting: $sinA*(1-(sinB)^2) = sin(A+B)*cosB - cosA*cosB*sinB$

That is $sinA*(cosB)^2 = sin(A+B)*cosB - cosA*cosB*sinB$

We can get $sinA = sin(A+B)/cosB + cosA*(-tanB)$

Therefore, the configuration of adcx_cfg1 and adcy_cfg1 is as follows:

x_param0 = 1(0x4000)

x_param1 = -tanB

y_param0 = 0

y_param1 = 1/cosB

This way, X1 remains cosA unchanged, and Y1 obtains sinA.

The calculation principle of parameter correction and compensation is shown in the figure below.

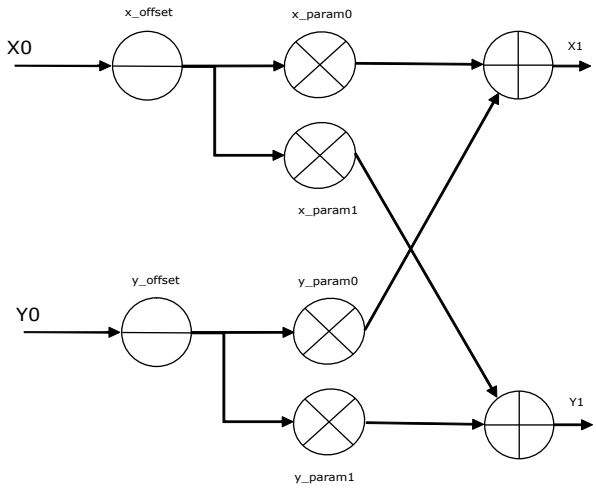


Figure 47: ADC Amplitude and Phase Compensation

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x_param0 and y_param0 are multiplied by a coefficient to scale the value of x1, and x_param1 and y_param1 are multiplied by a coefficient to scale the value of y1.

Special Note: Too small ADC amplitude will lead to inaccurate position calculation results. The closer the maximum positive amplitude is to 0xFFFFFFFF, the more accurate the result will be

36.4.1.7 Single-phase Sine Wave Mode

If single-phase sine wave input is used, the sine wave signal can only be input from the X channel. At this time, the input sine wave signal does not need to be distinguished as sine or cosine. The QEI module derives another sine wave value from the input single-phase sine wave signal through the internal lookup table, so as to calculate the position information. If the ADC amplitude of the single-phase sine wave is too small, the calculated position result will be inaccurate. The closer the maximum positive amplitude is to 0xFFFFFFFF, the more accurate the result will be.

36.4.2 H-phase and F-phase Functions

H-phase

H-phase input, also known as HOME phase, represents the zero position on some motors. The quadrature decoder supports setting the SR[HOME2FEN] flag bit to 1 in the following cases:

- Rising edge of H is detected during forward rotation of the motor, and CR[HRDIR0] bit is set to 1
- Falling edge of H is detected during forward rotation of the motor, and CR[HFDIR0] bit is set to 1
- Rising edge of H is detected during reverse rotation of the motor, and CR[HRDIR1] bit is set to 1
- Falling edge of H is detected during reverse rotation of the motor, and CR[HFDIR1] bit is set to 1

When the SR[HOMEF] flag bit is set to 1, an interrupt can be generated. Users can also choose to use this flag bit to reset each counter of the decoder, i.e., when the SR[HOME] flag bit is set to 1, reset the ZCNT counter of the quadrature decoder (when [HRSTZ] bit of CR register is set to 1).

F-phase

F-phase is called FAULT phase. Users can set [FAULTPOS] bit of CR register to 1. When the motor system fails, input a rising edge to F-phase, so that the QEI module stops outputting position valid signal after the failure occurs. Users can configure [FAULTE] bit of IRQEN register to 1, enable interrupt generation after detecting the rising edge of F-phase.

36.4.3 Simple Speed Measurement

For square wave signals, the QEI module provides two speed measurement methods: one based on pulse count, one based on duration (clock count). Each method has two counters that can be measured separately based on different parameters.

36.4.3.1 Based on Pulse Count

Select the required edge from the input square wave signal to generate pulses (you can select one or more rising edges, falling edges, or both rising and falling edges from A, B, Z signals; combinations like A rising edge and B falling edge are not supported).

Two different pulse counts can be configured (only one set of circuits is drawn in the figure). When the rising

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or falling edge of the input signal is detected, the pulse counter starts working, incrementing by 1 for each edge, and the cycle counter also starts working, incrementing by 1 for each clock cycle.

When the pulse counter equals the configured pulse count, a trigger signal is generated, the current cycle counter value is stored in cycle snapshot 0, and the previous cycle snapshot 0 is stored in cycle snapshot 1. The cycle counter and pulse counter are cleared and restart to work.

Software can read the latest two speed measurement results in the two snapshot registers.

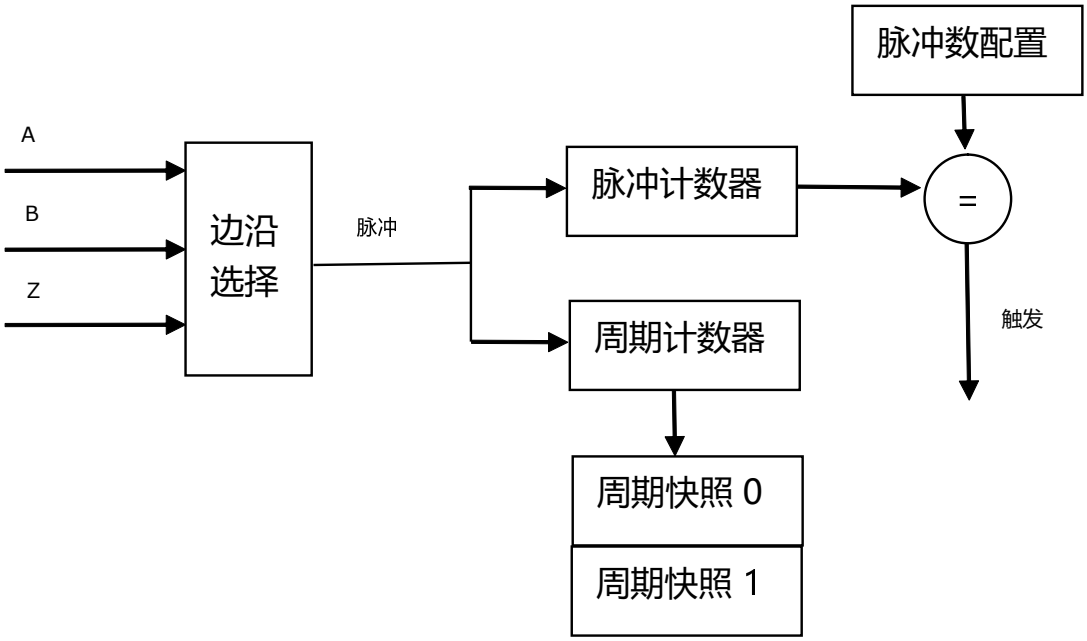


Figure 48: Pulse-based Speed Measurement

36.4.3.2 Based on Duration

When the pulse signal is detected, the pulse counter and cycle counter start working. Each time the pulse counter increments by 1, the current cycle counter value is stored in the cycle count.

When the cycle counter equals the configured cycle count, a trigger signal is generated, the current pulse counter value and cycle count are stored in snapshot 0, and the previous snapshot 0 is stored in snapshot 1. The cycle counter and pulse counter are cleared and restart to work.

Software can read the latest two speed measurement results in the two snapshot registers.

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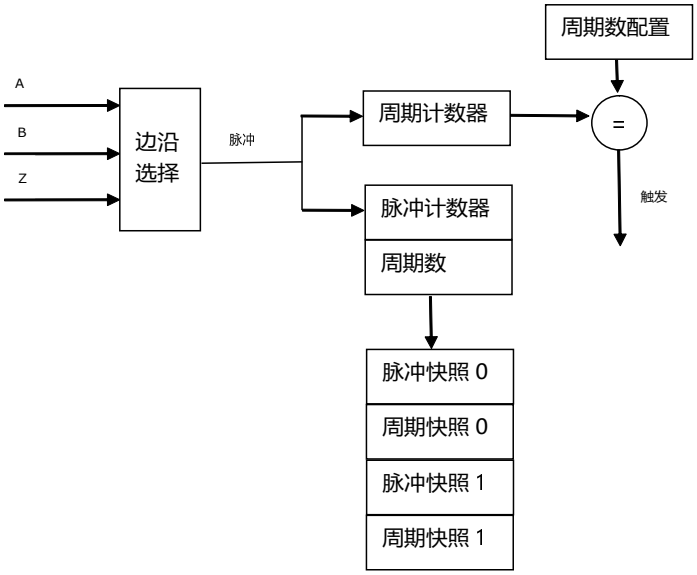


Figure 49: Duration-based Speed Measurement

36.4.4 Watchdog Counter

The watchdog counter is a timer running with the quadrature encoder interface clock, accumulating in each clock cycle. The watchdog counter is cleared when any flip of phase A or phase B input signal is detected. Users can set the WDGCFG register arbitrarily to configure the timeout period. The function of the watchdog counter is to provide an alarm for users if the motor stops rotating for more than a specified period of time.

Users set the WDGCFG[WDGEN] bit to 1 to enable the watchdog counter. The timeout length of the watchdog counter can be set through the WDGCFG[WDGTO] bit. When the watchdog count reaches the timeout, the SR[WDGF] flag bit is set.

When the user clears the flag bit by writing 1 to SR[WDGF], the watchdog counter will also be cleared accordingly.

36.4.5 Interrupt, DMA and Trigger Output

The interrupt events supported by the QEI module include watchdog count timeout, missing phase Z, phase Z flag generation, HOME phase flag generation, direction change flag generation, etc. Users can enable the required interrupt events in the Interrupt Request Enable Register (IRQEN) according to actual requirements. Please refer to the corresponding registers for details.

The DMA requests supported by the QEI module include watchdog count timeout, missing phase Z, phase Z flag generation, HOME phase flag generation, direction change flag generation, etc. Users can enable the required DMA requests in the DMA Request Enable Register (DMAEN) according to actual requirements. Please refer to the corresponding registers for details.

The QEI module supports sending a trigger output signal TRGO with a length of one pulse in the following cases: watchdog count timeout, missing phase Z, phase Z flag generation, HOME phase flag generation, direction change flag generation, etc. Users can enable the required trigger events in the Trigger Output Enable Register (TRGOEN) according to actual requirements. Please refer to the corresponding registers for details.

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36.5 QEI Register Configuration Step Guide

1. ABZ Mode, PD Mode or UD Mode

1. Configure PHZCNT on the Control Register (CR), configure ZCNTCFG according to the situation of phase Z (**Special Note: If there is no phase Z signal, do not configure the phase Z pin**), and select the required decoding mode in the [ENCTYP] bit;

2. Configure the maximum number of motor lines on the Phase Configuration Register (PHCFG) according to requirements;

3. Configure the reset value of PHZCNT on the Phase Index Register (PHIDX) according to requirements;

4. Configure the phase calculation parameters on the register with offset address 0x230 (PHASE_PARAM) according to requirements;

5. Configure the QEI Configuration Register (QEI_CFG) according to requirements: configure [NEGEDGE_EN], [POSIDGE_EN] bit fields to select enable rising and falling edges of the signal; set [SIGA_EN], [SIGB_EN] and [SIGZ_EN] bit fields to 1 to enable ABZ three-phase signals;

6. Configure the position output threshold in the Position Change Threshold Register (POS_THRESHOLD) according to requirements;

7. Configure the Interrupt Request Enable Register (IRQEN) or Trigger Output Enable Register (TRGOEN) according to requirements to enable the required interrupt function or trig_out event;

2. UVW Mode

1. Configure the Z-phase counting method in the [ZCNTCFG] bit field of the Control Register (CR) according to requirements, and select the UVW decoding mode in the [ENCTYP] bit;

2. Configure the value of the Phase Configuration Register (PHIDX) to 5;

3. Configure the QEI Configuration Register (QEI_CFG) according to requirements: configure [NEGEDGE_EN], [POSIDGE_EN] bit fields to select enable rising and falling edges of the signal; set [SIGA_EN], [SIGB_EN] and [SIGZ_EN] bit fields to 1 to enable ABZ three-phase signals;

4. Configure the values of UVM Position 0 Register (UVW_POS_CFG[UVW_POS0_CFG]), UVM Position 1 Register (UVW_POS_CFG[UVW_POS1_CFG]), UVM Position 2 Register (UVW_POS_CFG[UVW_POS2_CFG]), UVM Position 3 Register (UVW_POS_CFG[UVW_POS3_CFG]), UVM Position 4 Register (UVW_POS_CFG[UVW_POS4_CF and UVM Position 5 Register (UVW_POS_CFG[UVW_POS5_CFG]) according to the setting of [UVW_POS_OPT0] bit field of the QEI Configuration Register (QEI_CFG) and the six phase levels of the UVW three-phase signal;

5. Configure the values of UVW Position 0 Angle Register (UVW_POS[UVW_POS0]), UVW Position 1 Angle Register (UVW_POS[UVW_POS1]), UVW Position 2 Angle Register (UVW_POS[UVW_POS2]), UVW Position 3 Angle Register (UVW_POS[UVW_POS3]), UVW Position 4 Angle Register (UVW_POS[UVW_POS4]), UVW Position 5 Angle Register (UVW_POS[UVW_POS5]) according to the output positions of the six UVW phases;

6. Configure the Interrupt Request Enable Register (IRQEN) or Trigger Output Enable Register (TRGOEN) according to requirements to enable the required interrupt function or trig_out event;

3. Quadrature Sine-Cosine Wave Mode

1. Configure the decoding mode to quadrature sine-cosine wave mode, i.e., configure the [ENCTYP] bit field of the Control Register CR to value 6;

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2. Configure the revolution count method according to requirements, and configure the value of [ZCNTCFG] bit field of the Control Register CR according to the situation of phase Z;

3. Configure the ADC source of the X channel according to the source of the sine-cosine wave, the value of X-channel ADC configuration register (ADCX_CFG0); at the same time, configure the ADC source of the Y channel, the value of Y-channel ADC configuration register (ADCY_CFG0);

4. Configure the number of lines (PHCNT_MAX) according to the number of motor lines, i.e., set the value of the Phase Configuration Register (PHCFG); at the same time, configure the Phase Calculation Parameter Register (PHASE_PARAM) according to the number of motor lines;

5. Configure the Interrupt Request Enable Register (IRQEN) or Trigger Output Enable Register (TRGOEN) according to requirements to enable the required interrupt function or trig_out event;

4. Single-phase Sine Wave Mode

1. Configure the [ENCTYP] bit field of the Control Register CR to value 5, i.e., configure to single-phase sine wave mode;

2. Configure the value of [ZCNTCFG] bit field of the Control Register CR according to the situation of phase Z;

3. Configure the ADC source of the X channel in the X-channel ADC configuration register 0 (ADCX_CFG0) according to requirements;

4. Configure the X parameters in the X-channel ADC configuration register 1 (ADCX_CFG1) according to requirements;

5. Configure the offset value of the ADC value in the X-channel ADC configuration register 2 (ADCX_CFG2) according to requirements;

6. Set the Phase Configuration Register (PHCFG) according to the number of motor lines, and configure the value of the Phase Calculation Parameter Register (PHASE_PARAM) at the same time;

7. Configure the Interrupt Request Enable Register (IRQEN) or Trigger Output Enable Register (TRGOEN) according to requirements to enable the required interrupt function or trig_out event;

36.6 QEIV2 Register List

The QEIV2 register list is as follows:

QEI0 base address: 0xF0300000

QEI1 base address: 0xF0304000

Offset	Register Name	Description	Reset Value
0x0000	CR	Control register	0x00000000
0x0004	PHCFG	Phase configure register	0xFFFFFFFF
0x0008	WDGCFG	Watchdog configure register	0x00000000
0x000C	PHIDX	Phase index register	0x00000000
0x0010	TRGOEN	Tigger output enable register	0x00000000
0x0014	READEN	Read event enable register	0x00000000

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Offset	Register Name	Description	Reset Value
0x0018	ZCMP	Z comparator	0x80000000
0x001C	PHCMP	Phase comparator	0x80000000
0x0020	SPDCMP	Speed comparator	0x80000000
0x0024	DMAEN	DMA request enable register	0x00000000
0x0028	SR	Status register	0x00000000
0x002C	IRQEN	Interrupt request register	0x00000000
0x0030	COUNT[CURRENT][Z]	Z counter	0x00000000
0x0034	COUNT[CURRENT][PH]	Phase counter	0x00000000
0x0038	COUNT[CURRENT][SPD]	Speed counter	0x00000000
0x003C	COUNT[CURRENT][TMR]	Timer counter	0x00000000
0x0040	COUNT[READ][Z]	Z counter	0x00000000
0x0044	COUNT[READ][PH]	Phase counter	0x00000000
0x0048	COUNT[READ][SPD]	Speed counter	0x00000000
0x004C	COUNT[READ][TMR]	Timer counter	0x00000000
0x0050	COUNT[SNAP0][Z]	Z snap register	0x00000000
0x0054	COUNT[SNAP0][PH]	Phase snap register	0x00000000
0x0058	COUNT[SNAP0][SPD]	Speed snap register	0x00000000
0x005C	COUNT[SNAP0][TMR]	Timer snap register	0x00000000
0x0060	COUNT[SNAP1][Z]	Z snap register 1	0x00000000
0x0064	COUNT[SNAP1][PH]	Phase snap register 1	0x00000000
0x0068	COUNT[SNAP1][SPD]	Speed snap register 1	0x00000000
0x006C	COUNT[SNAP1][TMR]	Timer snap register 1	0x00000000
0x0080	ZCMP2	Z comparator	0x80000000
0x0084	PHCMP2	Phase comparator	0x80000000
0x0088	SPDCMP2	Speed comparator	0x80000000
0x008C	MATCH_CFG		0x00000000
0x0090	FILT_CFG[FILT_CFG_A]	A signal filter config	0x00001000
0x0094	FILT_CFG[FILT_CFG_B]	B signal filter config	0x00001000
0x0098	FILT_CFG[FILT_CFG_Z]	Z signal filter config	0x00001000
0x009C	FILT_CFG[FILT_CFG_H]	H signal filter config	0x00001000
0x00A0	FILT_CFG[FILT_CFG_H2]	H2 signal filter config	0x00001000
0x00A4	FILT_CFG[FILT_CFG_F]	F signal filter config	0x00001000
0x0100	QEI_CFG	qei config register	0x00000000
0x0110	PULSE0_NUM	pulse0_num	0x00000000
0x0114	PULSE1_NUM	pulse1_num	0x00000000
0x0118	CYCLE0_CNT	cycle0_cnt	0x00000000
0x011C	CYCLE0PULSE_CNT	cycle0pulse_cnt	0x00000000
0x0120	CYCLE1_CNT	cycle1_cnt	0x00000000
0x0124	CYCLE1PULSE_CNT	cycle1pulse_cnt	0x00000000
0x0128	CYCLE0_SNAP0	cycle0_snap0	0x00000000

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Offset	Register Name	Description	Reset Value
0x012C	CYCLE0_SNAP1	cycle0_snap1	0x00000000
0x0130	CYCLE1_SNAP0	cycle1_snap0	0x00000000
0x0134	CYCLE1_SNAP1	cycle1_snap1	0x00000000
0x0140	CYCLE0_NUM	cycle0_num	0x00000000
0x0144	CYCLE1_NUM	cycle1_num	0x00000000
0x0148	PULSE0_CNT	pulse0_cnt	0x00000000
0x014C	PULSE0CYCLE_CNT	pulse0cycle_cnt	0x00000000
0x0150	PULSE1_CNT	pulse1_cnt	0x00000000
0x0154	PULSE1CYCLE_CNT	pulse1cycle_cnt	0x00000000
0x0158	PULSE0_SNAP0	pulse0_snap0	0x00000000
0x015C	PULSE0CYCLE_SNAP0	pulse0cycle_snap0	0x00000000
0x0160	PULSE0_SNAP1	pulse0_snap1	0x00000000
0x0164	PULSE0CYCLE_SNAP1	pulse0cycle_snap1	0x00000000
0x0168	PULSE1_SNAP0	pulse1_snap0	0x00000000
0x016C	PULSE1CYCLE_SNAP0	pulse1cycle_snap0	0x00000000
0x0170	PULSE1_SNAP1	pulse1_snap1	0x00000000
0x0174	PULSE1CYCLE_SNAP1	pulse1cycle_snap1	0x00000000
0x0200	ADCX_CFG0	adcx_cfg0	0x00000000
0x0204	ADCX_CFG1	adcx_cfg1	0x00004000
0x0208	ADCX_CFG2	adcx_cfg2	0x80000000
0x0210	ADCY_CFG0	adcy_cfg0	0x00000000
0x0214	ADCY_CFG1	adcy_cfg1	0x40000000
0x0218	ADCY_CFG2	adcy_cfg2	0x80000000
0x0220	CAL_CFG	cal_cfg	0x00000100
0x0230	PHASE_PARAM	phase_param	0xFFFFFFFF
0x0238	POS_THRESHOLD	pos_threshold	0x80000000
0x0240	UVW_POS[UVW_POS0]	uvw_pos0	0x00000000
0x0244	UVW_POS[UVW_POS1]	uvw_pos1	0x00000000
0x0248	UVW_POS[UVW_POS2]	uvw_pos2	0x00000000
0x024C	UVW_POS[UVW_POS3]	uvw_pos3	0x00000000
0x0250	UVW_POS[UVW_POS4]	uvw_pos4	0x00000000
0x0254	UVW_POS[UVW_POS5]	uvw_pos5	0x00000000
0x0258	UVW_POS_CFG[UVW_POS0_CFG]	uvw_pos0_cfg	0x00000000
0x025C	UVW_POS_CFG[UVW_POS1_CFG]	uvw_pos1_cfg	0x00000000
0x0260	UVW_POS_CFG[UVW_POS2_CFG]	uvw_pos2_cfg	0x00000000
0x0264	UVW_POS_CFG[UVW_POS3_CFG]	uvw_pos3_cfg	0x00000000
0x0268	UVW_POS_CFG[UVW_POS4_CFG]	uvw_pos4_cfg	0x00000000
0x026C	UVW_POS_CFG[UVW_POS5_CFG]	uvw_pos5_cfg	0x00000000
0x0280	PHASE_CNT	phase_cnt	0x00000000

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Offset	Register Name	Description	Reset Value
0x0284	PHASE_UPDATE	phase_update	0x00000000
0x0288	POSITION	position	0x00000000
0x028C	POSITION_UPDATE	position_update	0x00000000
0x0290	ANGLE	angle	0x00000000
0x0294	POS_TIMEOUT	pos_timeout	0x7FFFFFFF

Table 175: QEIV2 Register List

36.7 QEIV2 Register Detailed Information

The detailed description of QEIV2 registers is as follows:

36.7.1 CR (0x0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
READ	RSVD								ZCNTCFG	PHCALIZ	Z_ONLY_EN	H2FDIR0	H2FDIR1	H2RDIR0	H2RDIR1	PAUSEPOS	PAUSESPD	PAUSEPH	PAUSEZ	HFDIR0	HFDIR1	HRDIR0	HRDIR1	RSVD	FAULTPOS	SNAPEN	RSTCNT	RD_SEL	ENCTYP		
WO	N/A								RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	N/A	RW	RW	RW	RW	RW		
0	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	0	0	0	0	0	0	0

CR [31:0]

Bits	Name	Description
31	READ	1- load phcnt, zcnt, spdcnt and tmrcnt into their read registers. Hardware auto-clear; read as 0
22	ZCNTCFG	1- zcnt will increment when phcnt upcount to phmax, decrement when phcnt downcount to 0 0- zcnt will increment or decrement when Z input assert
21	PHCALIZ	1- phcnt will set to phidx when Z input assert(for abz digital signsl)
20	Z_ONLY_EN	1- phcnt will set to phidx when Z input assert(only using z phase, ignore ab phase signals)
19	H2FDIR0	
18	H2FDIR1	
17	H2RDIR0	
16	H2RDIR1	
15	PAUSEPOS	1- pause position output valid when PAUSE assert
14	PAUSESPD	1- pause spdcnt when PAUSE assert
13	PAUSEPH	1- pause phcnt when PAUSE assert
12	PAUSEZ	1- pause zcnt when PAUSE assert

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Bits	Name	Description
11	HFDIR0	1- HOMEF will set at H rising edge when dir == 1 (negative rotation direction)
10	HFDIR1	1- HOMEF will set at H rising edge when dir == 0 (positive rotation direction)
9	HRDIR0	1- HOMEF will set at H falling edge when dir == 1 (negative rotation direction)
8	HRDIR1	1- HOMEF will set at H falling edge when dir == 1 (positive rotation direction)
6	FAULTPOS	
5	SNAPEN	1- load phcnt, zcnt, spdcnt and tmrcnt into their snap registers when snapi input assert
4	RSTCNT	1- reset zcnt, spdcnt and tmrcnt to 0. reset phcnt to phidx
3	RD_SEL	define the width/counter value(affect width_match, width_match2, width_cur, timer_cur, width_read, timer_read, width_snap0,width_snap1, timer_snap0, timer_snap1) 0 : same as hpm1000/500/500s; 1: use width for position; use timer for angle
2-0	ENCTYP	000-abz; 001-pd; 010-ud; 011-UVW(hal) 100-single A; 101-single sin; 110: sin&cos

CR Bits

36.7.2 PHCFG (0x4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PHMAX																															
RW																															
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

PHCFG [31:0]

Bits	Name	Description
31-0	PHMAX	maximum phcnt number, phcnt will rollover to 0 when it upcount to phmax

PHCFG Bits

36.7.3 WDGCFG (0x8)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WDGEN	WDOG_CFG			WDGTO																											
RW	RW			RW																											
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

WDGCFG [31:0]

Bits	Name	Description
31	WDGEN	1- enable wdog counter
30-28	WDOG_CFG	define as stop if phase_cnt change is less than it if 0, then each change of phase_cnt will clear wdog counter; if 2, then phase_cnt change larger than 2 will clear wdog counter
27-0	WDGTO	watch dog timeout value

WDGCFG Bits

36.7.4 PHIDX (0xC)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PHIDX																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PHIDX [31:0]

Bits	Name	Description
31-0	PHIDX	phcnt reset value, phcnt will reset to phidx when phcaliz set to 1

PHIDX Bits

36.7.5 TRGOEN (0x10)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WDGFEN	HOMEFEN	POSCMPFEN	ZPHFEN	ZMISSFEN	WIDTHTMFEN	POS2CMPFEN	DIRCHGFEN	CYCLE0FEN	CYCLE1FEN	PULSE0FEN	PULSE1FEN	HOME2FEN	FAULTFEN	RSVD																	
RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	N/A																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

TRGOEN [31:0]

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Bits	Name	Description
31	WDGFEN	1- enable trigger output when wdg flag set
30	HOMEFEN	1- enable trigger output when homef flag set
29	POSCMPFEN	1- enable trigger output when poscmpf flag set
28	ZPHFEN	1- enable trigger output when zphf flag set
27	ZMISSFEN	
26	WIDTHTMFEN	
25	POS2CMPFEN	
24	DIRCHGFEN	
23	CYCLE0FEN	
22	CYCLE1FEN	
21	PULSE0FEN	
20	PULSE1FEN	
19	HOME2FEN	
18	FAULTFEN	

TRGOEN Bits

36.7.6 READEN (0x14)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WDGFEN	HOMEFEN	POSCMPFEN	ZPHFEN	ZMISSFEN	WIDTHTMFEN	POS2CMPFEN	DIRCHGFEN	CYCLE0FEN	CYCLE1FEN	PULSE0FEN	PULSE1FEN	HOME2FEN	FAULTFEN	RSVD																	
RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	N/A																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

READEN [31:0]

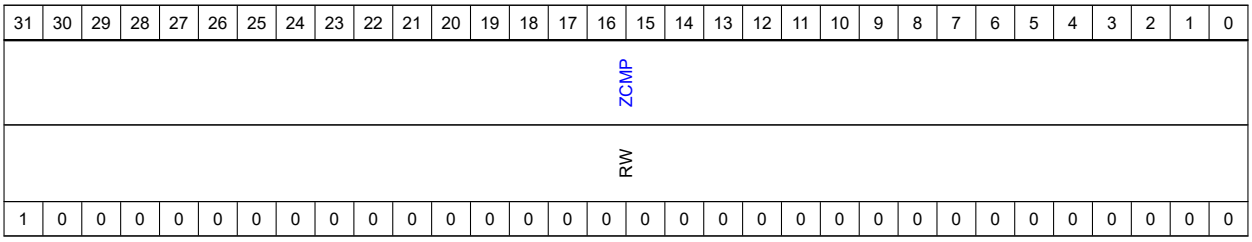
Bits	Name	Description
31	WDGFEN	1- load counters to their read registers when wdg flag set
30	HOMEFEN	1- load counters to their read registers when homef flag set
29	POSCMPFEN	1- load counters to their read registers when poscmpf flag set
28	ZPHFEN	1- load counters to their read registers when zphf flag set
27	ZMISSFEN	
26	WIDTHTMFEN	
25	POS2CMPFEN	
24	DIRCHGFEN	
23	CYCLE0FEN	
22	CYCLE1FEN	
21	PULSE0FEN	
20	PULSE1FEN	

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Bits	Name	Description
19	HOME2FEN	
18	FAULTFEN	

READEN Bits

36.7.7 ZCMP (0x18)

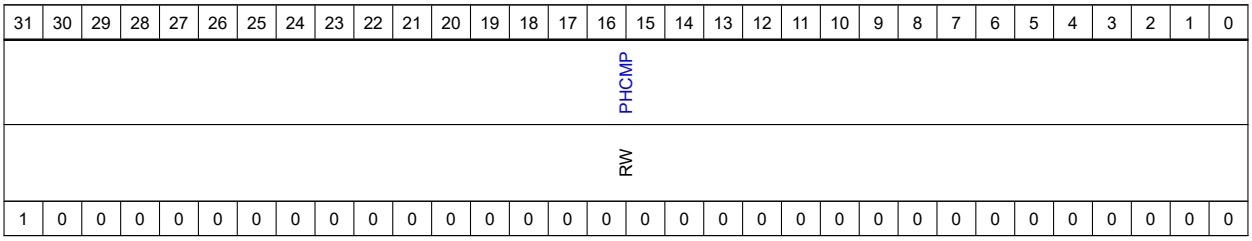


ZCMP [31:0]

Bits	Name	Description
31-0	ZCMP	zcnt postion compare value

ZCMP Bits

36.7.8 PHCMP (0x1C)

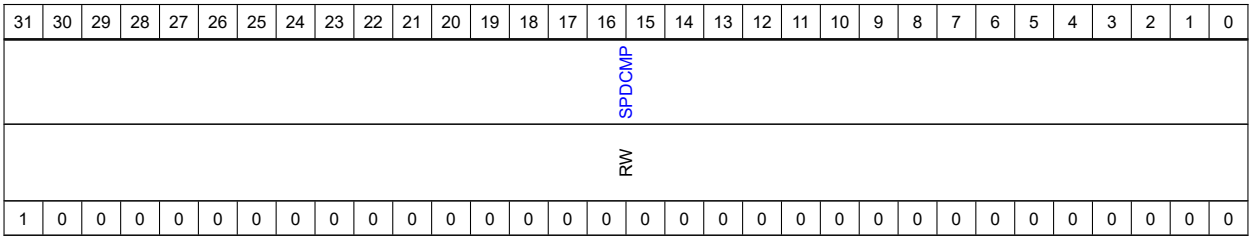


PHCMP [31:0]

Bits	Name	Description
31-0	PHCMP	phcnt position compare value

PHCMP Bits

36.7.9 SPDCMP (0x20)



SPDCMP [31:0]

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Bits	Name	Description
31-0	SPDCMP	spdcnt position compare value

SPDCMP Bits

36.7.10 DMAEN (0x24)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WDGFEN	HOMEFEN	POSCMPFEN	ZPHFEN	ZMISSFEN	WIDTHTMFEN	POS2CMPFEN	DIRCHGFEN	CYCLE0FEN	CYCLE1FEN	PULSE0FEN	PULSE1FEN	HOME2FEN	FAULTFEN	RSVD																	
RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	N/A																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

DMAEN [31:0]

Bits	Name	Description
31	WDGFEN	1- generate dma request when wdg flag set
30	HOMEFEN	1- generate dma request when homef flag set
29	POSCMPFEN	1- generate dma request when poscmpf flag set
28	ZPHFEN	1- generate dma request when zphf flag set
27	ZMISSFEN	
26	WIDTHTMFEN	
25	POS2CMPFEN	
24	DIRCHGFEN	
23	CYCLE0FEN	
22	CYCLE1FEN	
21	PULSE0FEN	
20	PULSE1FEN	
19	HOME2FEN	
18	FAULTFEN	

DMAEN Bits

36.7.11 SR (0x28)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WDGF	HOMEF	POSCMPF	ZPHF	ZMISSF	WIDTHTMF	POS2CMPF	DIRCHGF	CYCLE0F	CYCLE1F	PULSE0F	PULSE1F	HOME2F	FAULTF	RSVD																	
RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	N/A																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

SR [31:0]

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Bits	Name	Description
31	WDGF	watchdog flag
30	HOMEF	home flag
29	POSCMPF	postion compare match flag
28	ZPHF	z input flag
27	ZMISSF	
26	WIDTHTMF	
25	POS2CMPF	
24	DIRCHGF	
23	CYCLE0F	
22	CYCLE1F	
21	PULSE0F	
20	PULSE1F	
19	HOME2F	
18	FAULTF	

SR Bits

36.7.12 IRQEN (0x2C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WDGIE	HOMEIE	POSCMPIE	ZPHIE	ZMISSE	WIDTHTME	POS2CMPE	DIRCHGE	CYCLE0E	CYCLE1E	PULSE0E	PULSE1E	HOME2E	FAULTE	RSVD																	
RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	N/A																	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

IRQEN [31:0]

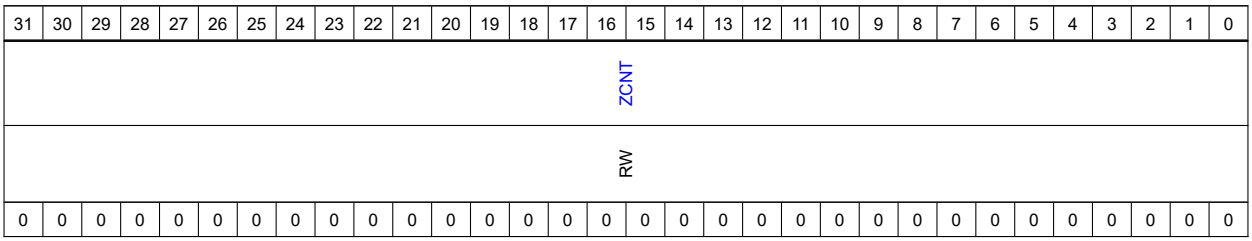
Bits	Name	Description
31	WDGIE	1- generate interrupt when wdg flag set
30	HOMEIE	1- generate interrupt when homef flag set
29	POSCMPIE	1- generate interrupt when poscmpf flag set
28	ZPHIE	1- generate interrupt when zphf flag set
27	ZMISSE	
26	WIDTHTME	
25	POS2CMPE	
24	DIRCHGE	
23	CYCLE0E	
22	CYCLE1E	
21	PULSE0E	
20	PULSE1E	
19	HOME2E	

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Bits	Name	Description
18	FAULTE	

IRQEN Bits

36.7.13 COUNT[Z] (0x30 + 0x10 * n)

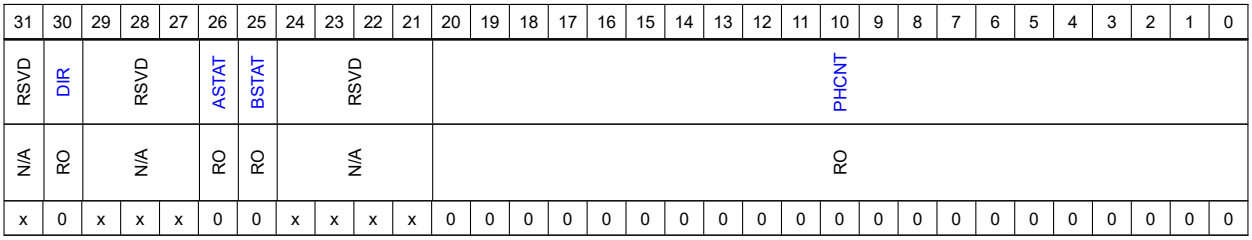


COUNT[Z] [31:0]

Bits	Name	Description
31-0	ZCNT	zcnt value

COUNT[Z] Bits

36.7.14 COUNT[PH] (0x34 + 0x10 * n)



COUNT[PH] [31:0]

Bits	Name	Description
30	DIR	1- reverse rotation 0- forward rotation
26	ASTAT	1- a input is high 0- a input is low
25	BSTAT	1- b input is high 0- b input is low
20-0	PHCNT	phcnt value

COUNT[PH] Bits

36.7.15 COUNT[SPD] (0x38 + 0x10 * n)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DIR	ASTAT	BSTAT	RSVD	SPDCNT																											
RO	RO	RW	N/A	RO																											
0	0	0	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

COUNT[SPD] [31:0]

Bits	Name	Description
31	DIR	1- reverse rotation 0- forward rotation
30	ASTAT	1- a input is high 0- a input is low
29	BSTAT	1- b input is high 0- b input is low
27-0	SPDCNT	spdcnt value

COUNT[SPD] Bits

36.7.16 COUNT[TMR] (0x3C + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TMRCNT																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

COUNT[TMR] [31:0]

Bits	Name	Description
31-0	TMRCNT	32 bit free run timer

COUNT[TMR] Bits

36.7.17 ZCMP2 (0x80)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ZCMP2																															
RW																															
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

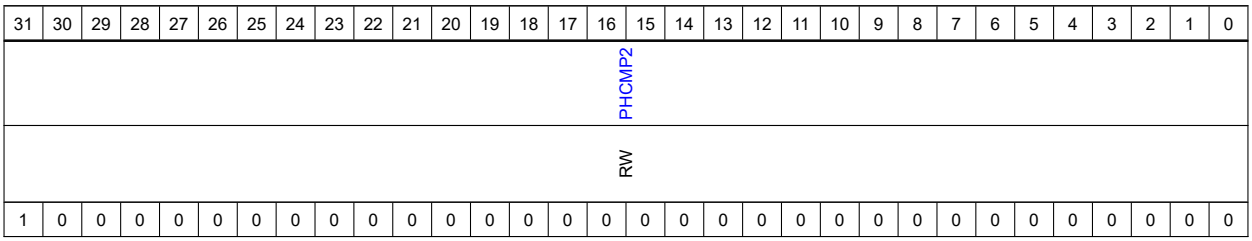
ZCMP2 [31:0]

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Bits	Name	Description
31-0	ZCMP2	

ZCMP2 Bits

36.7.18 PHCMP2 (0x84)

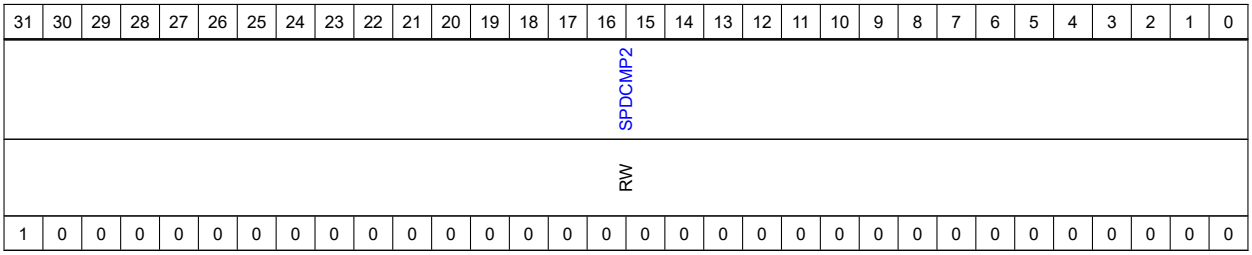


PHCMP2 [31:0]

Bits	Name	Description
31-0	PHCMP2	

PHCMP2 Bits

36.7.19 SPDCMP2 (0x88)

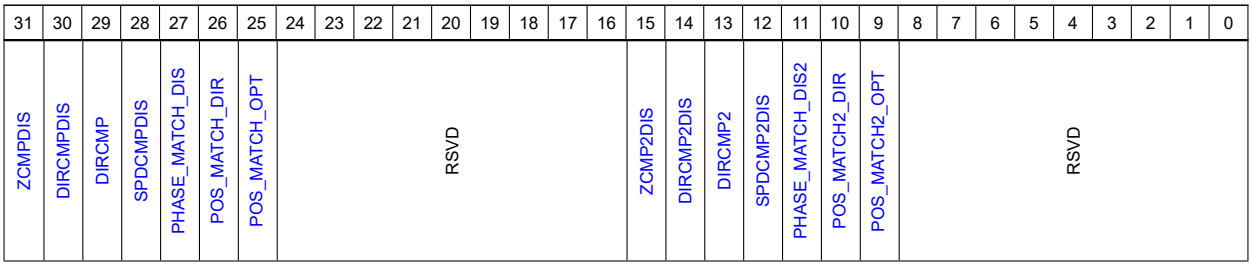


SPDCMP2 [31:0]

Bits	Name	Description
31-0	SPDCMP2	

SPDCMP2 Bits

36.7.20 MATCH_CFG (0x8C)



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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RW	RW	RW	RW	RW	RW	RW	N/A									RW	RW	RW	RW	RW	RW	RW	N/A								
0	0	0	0	0	0	0	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x	x

MATCH_CFG [31:0]

Bits	Name	Description
31	ZCMPDIS	1- postion compare not include zcnt
30	DIRCMPDIS	1- postion compare not include rotation direction
29	DIRCMP	0- position compare need positive rotation 1- position compare need negative rotation
28	SPDCMPDIS	
27	PHASE_MATCH_DIS	
26	POS_MATCH_DIR	
25	POS_MATCH_OPT	
15	ZCMP2DIS	
14	DIRCMP2DIS	
13	DIRCMP2	
12	SPDCMP2DIS	
11	PHASE_MATCH_DIS2	
10	POS_MATCH2_DIR	
9	POS_MATCH2_OPT	

MATCH_CFG Bits

36.7.21 FILT_CFG (0x90 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD															OUTINV	MODE			SYNCEN	FILTLEN											
N/A															RW	RW			RW	RW											
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0

FILT_CFG [31:0]

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Bits	Name	Description
16	OUTINV	1- Filter will invert the output 0- Filter will not invert the output
15-13	MODE	This bitfields defines the filter mode 000-bypass; 100-rapid change mode; 101-delay filter mode; 110-stable low mode; 111-stable high mode
12	SYNCEN	set to enable sychronization input signal with TRGM clock
11-0	FILTLEN	This bitfields defines the filter counter length.

FILT_CFG Bits

36.7.22 QEI_CFG (0x100)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																			SPEED_DIR_CHG_EN	RSVD						UVW_POS_OPT0	NEGEDGE_EN	POSIDGE_EN	SIGZ_EN	SIGB_EN	SIGA_EN
N/A																			RW	N/A						RW	RW	RW	RW	RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	x	x	x	0	0	0	0	0	0

QEI_CFG [31:0]

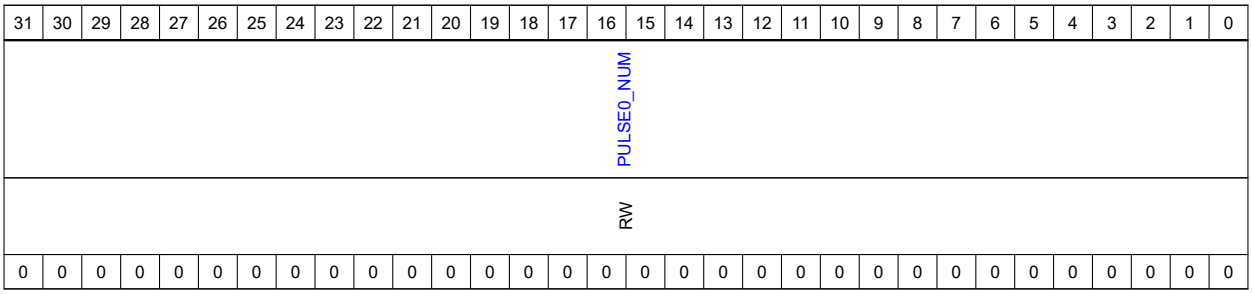
Bits	Name	Description
12	SPEED_DIR_CHG_EN	clear counter if detect direction change
5	UVW_POS_OPT0	set to output next area position for QEO use; clr to output exact point position for MMC use
4	NEGEDGE_EN	bit4: negedge enable bit3: posedge enable bit2: W in hal enable bit1: signal b(or V in hal) enable bit0: signal a(or U in hal) enable such as: 01001: use posedge A 11010: use both edge of signal B 11111: use both edge of all HAL siganls
3	POSIDGE_EN	
2	SIGZ_EN	

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Bits	Name	Description
1	SIGB_EN	
0	SIGA_EN	

QEI_CFG Bits

36.7.23 PULSE0_NUM (0x110)

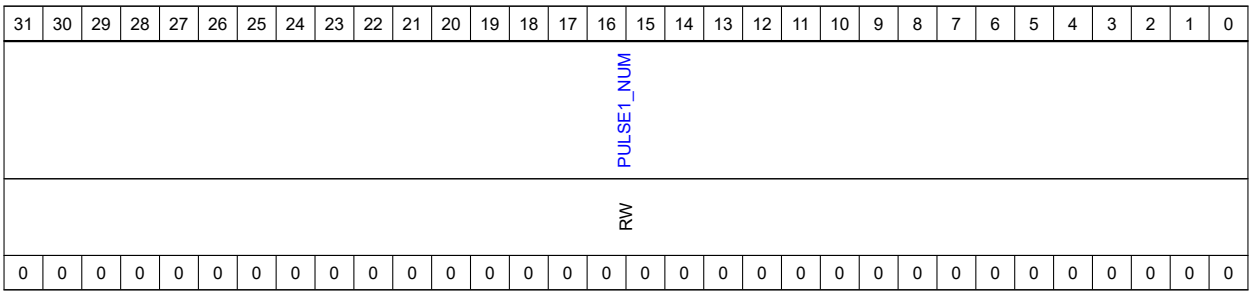


PULSE0_NUM [31:0]

Bits	Name	Description
31-0	PULSE0_NUM	for speed detection, will count the cycle number for configed pulse_num

PULSE0_NUM Bits

36.7.24 PULSE1_NUM (0x114)



PULSE1_NUM [31:0]

Bits	Name	Description
31-0	PULSE1_NUM	

PULSE1_NUM Bits

36.7.25 CYCLE0_CNT (0x118)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CYCLE0_CNT																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CYCLE0_CNT [31:0]

Bits	Name	Description
31-0	CYCLE0_CNT	

CYCLE0_CNT Bits

36.7.26 CYCLE0PULSE_CNT (0x11C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CYCLE0PULSE_CNT																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CYCLE0PULSE_CNT [31:0]

Bits	Name	Description
31-0	CYCLE0PULSE_CNT	

CYCLE0PULSE_CNT Bits

36.7.27 CYCLE1_CNT (0x120)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CYCLE1_CNT																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

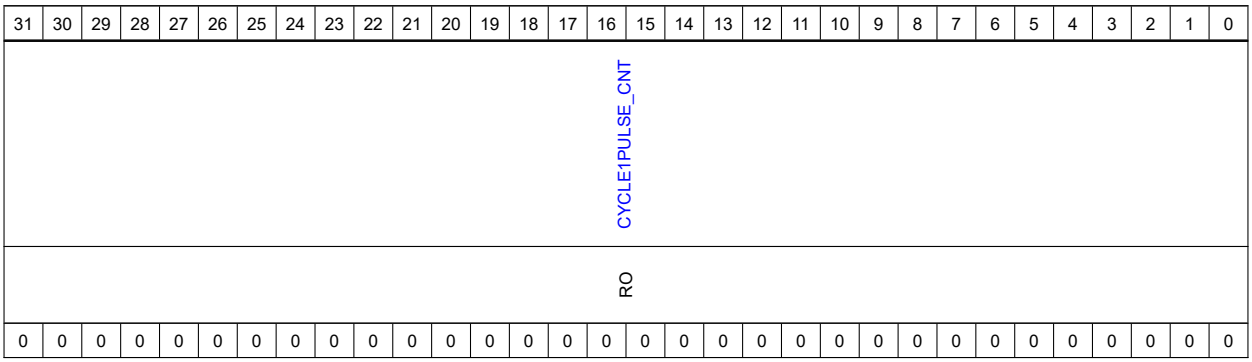
CYCLE1_CNT [31:0]

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Bits	Name	Description
31-0	CYCLE1_CNT	

CYCLE1_CNT Bits

36.7.28 CYCLE1PULSE_CNT (0x124)

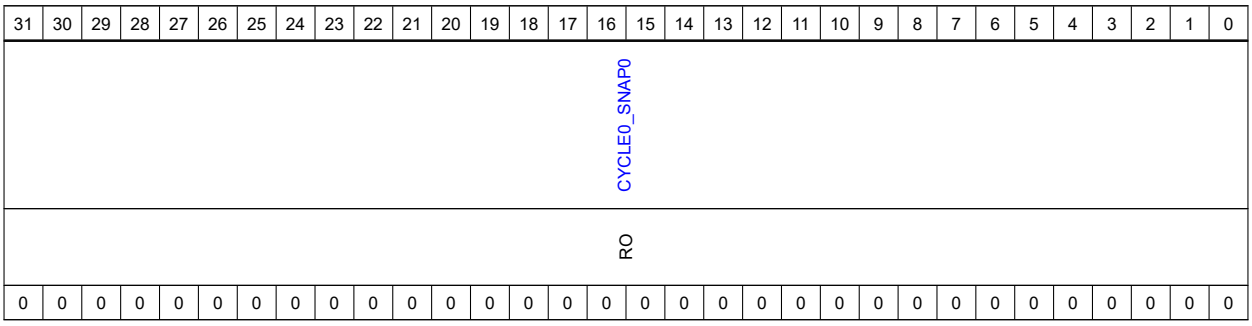


CYCLE1PULSE_CNT [31:0]

Bits	Name	Description
31-0	CYCLE1PULSE_CNT	

CYCLE1PULSE_CNT Bits

36.7.29 CYCLE0_SNAP0 (0x128)



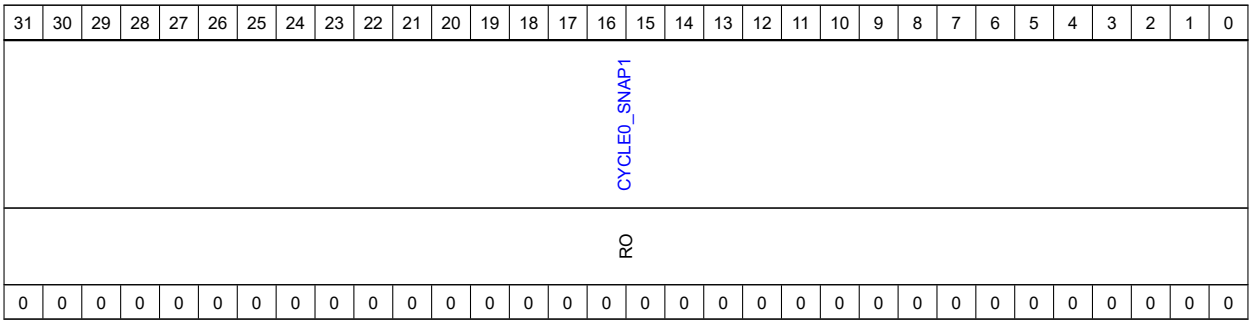
CYCLE0_SNAP0 [31:0]

Bits	Name	Description
31-0	CYCLE0_SNAP0	

CYCLE0_SNAP0 Bits

36.7.30 CYCLE0_SNAP1 (0x12C)

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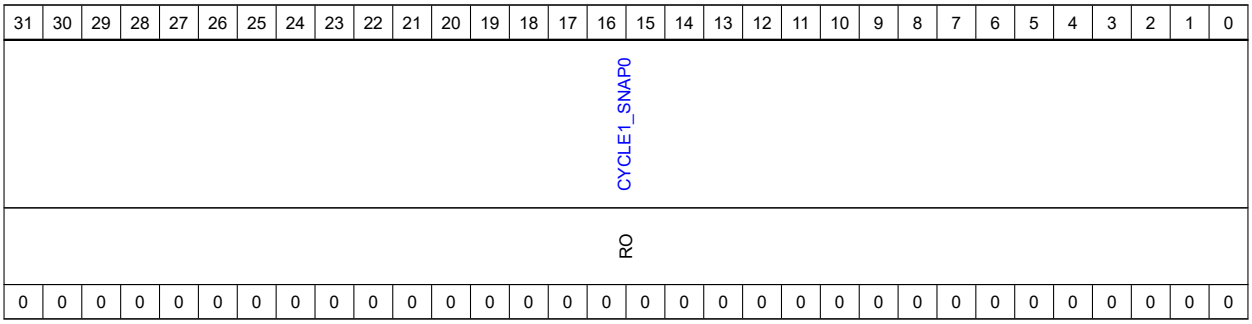


CYCLE0_SNAP1 [31:0]

Bits	Name	Description
31-0	CYCLE0_SNAP1	

CYCLE0_SNAP1 Bits

36.7.31 CYCLE1_SNAP0 (0x130)

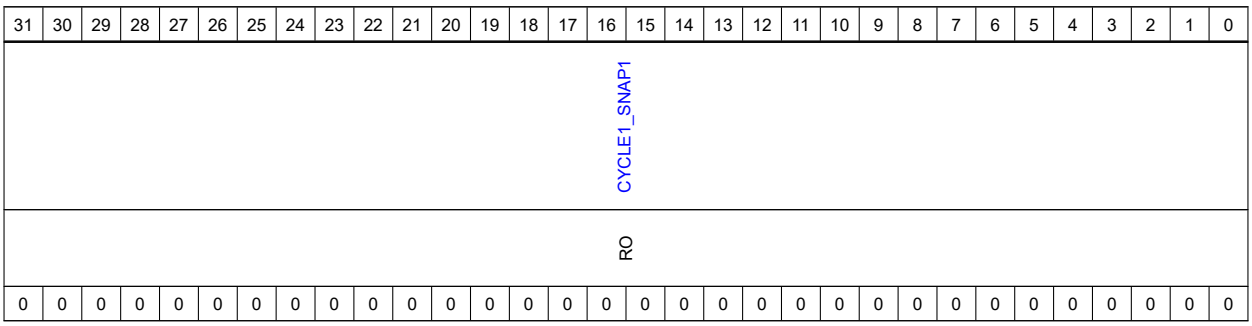


CYCLE1_SNAP0 [31:0]

Bits	Name	Description
31-0	CYCLE1_SNAP0	

CYCLE1_SNAP0 Bits

36.7.32 CYCLE1_SNAP1 (0x134)



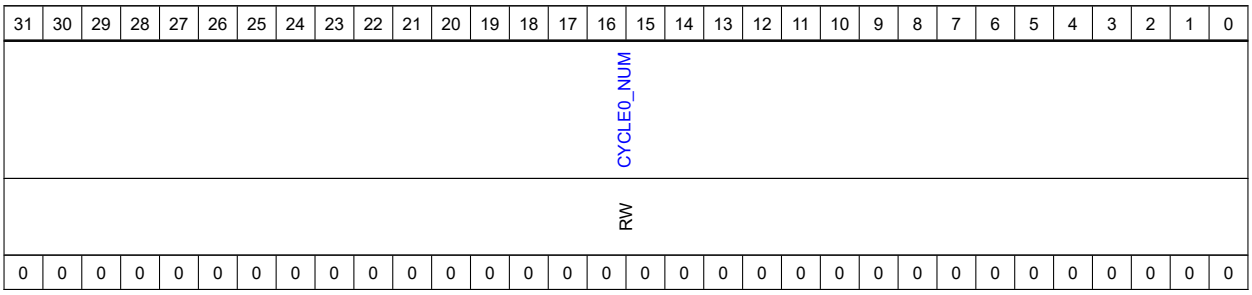
CYCLE1_SNAP1 [31:0]

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Bits	Name	Description
31-0	CYCLE1_SNAP1	

CYCLE1_SNAP1 Bits

36.7.33 CYCLE0_NUM (0x140)

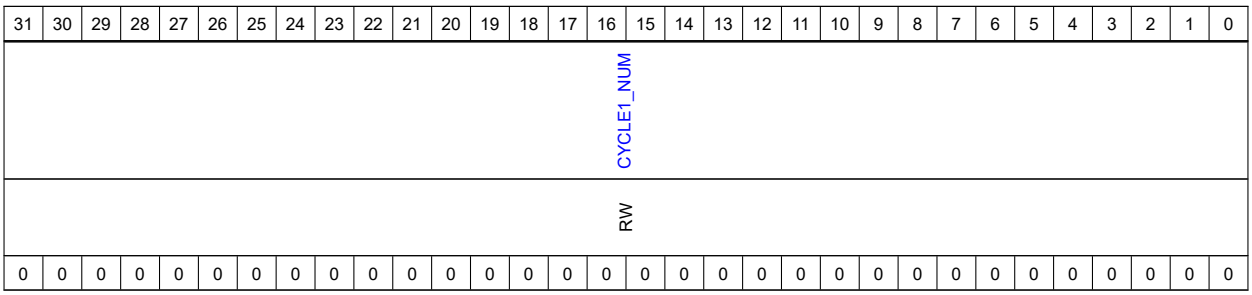


CYCLE0_NUM [31:0]

Bits	Name	Description
31-0	CYCLE0_NUM	

CYCLE0_NUM Bits

36.7.34 CYCLE1_NUM (0x144)

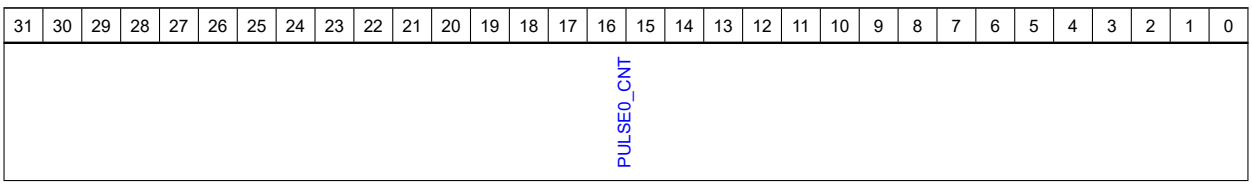


CYCLE1_NUM [31:0]

Bits	Name	Description
31-0	CYCLE1_NUM	

CYCLE1_NUM Bits

36.7.35 PULSE0_CNT (0x148)



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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PULSE0_CNT [31:0]

Bits	Name	Description
31-0	PULSE0_CNT	

PULSE0_CNT Bits

36.7.36 PULSE0CYCLE_CNT (0x14C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PULSE0CYCLE_CNT																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PULSE0CYCLE_CNT [31:0]

Bits	Name	Description
31-0	PULSE0CYCLE_CNT	

PULSE0CYCLE_CNT Bits

36.7.37 PULSE1_CNT (0x150)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PULSE1_CNT																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PULSE1_CNT [31:0]

Bits	Name	Description
31-0	PULSE1_CNT	

PULSE1_CNT Bits

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36.7.38 PULSE1CYCLE_CNT (0x154)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PULSE1CYCLE_CNT																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PULSE1CYCLE_CNT [31:0]

Bits	Name	Description
31-0	PULSE1CYCLE_CNT	

PULSE1CYCLE_CNT Bits

36.7.39 PULSE0_SNAP0 (0x158)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PULSE0_SNAP0																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PULSE0_SNAP0 [31:0]

Bits	Name	Description
31-0	PULSE0_SNAP0	

PULSE0_SNAP0 Bits

36.7.40 PULSE0CYCLE_SNAP0 (0x15C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PULSE0CYCLE_SNAP0																															

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PULSE0CYCLE_SNAP0 [31:0]

Bits	Name	Description
31-0	PULSE0CYCLE_SNAP0	

PULSE0CYCLE_SNAP0 Bits

36.7.41 PULSE0_SNAP1 (0x160)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PULSE0_SNAP1																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PULSE0_SNAP1 [31:0]

Bits	Name	Description
31-0	PULSE0_SNAP1	

PULSE0_SNAP1 Bits

36.7.42 PULSE0CYCLE_SNAP1 (0x164)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PULSE0CYCLE_SNAP1																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

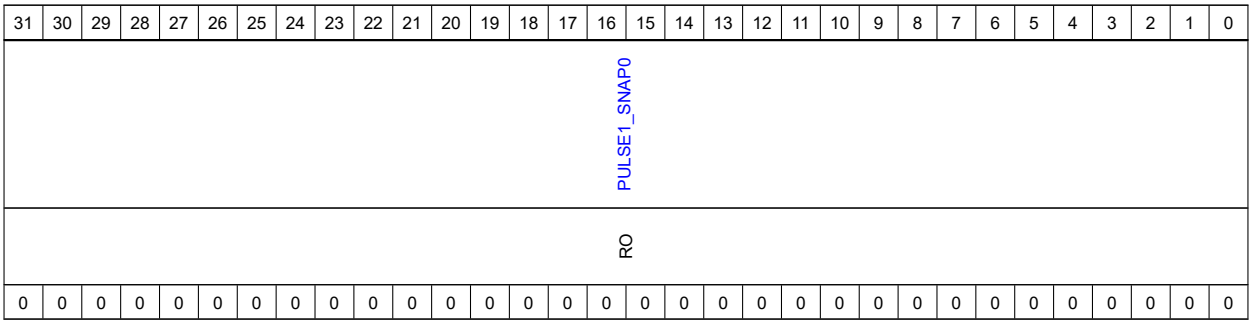
PULSE0CYCLE_SNAP1 [31:0]

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Bits	Name	Description
31-0	PULSE0CYCLE_SNAP1	

PULSE0CYCLE_SNAP1 Bits

36.7.43 PULSE1_SNAP0 (0x168)

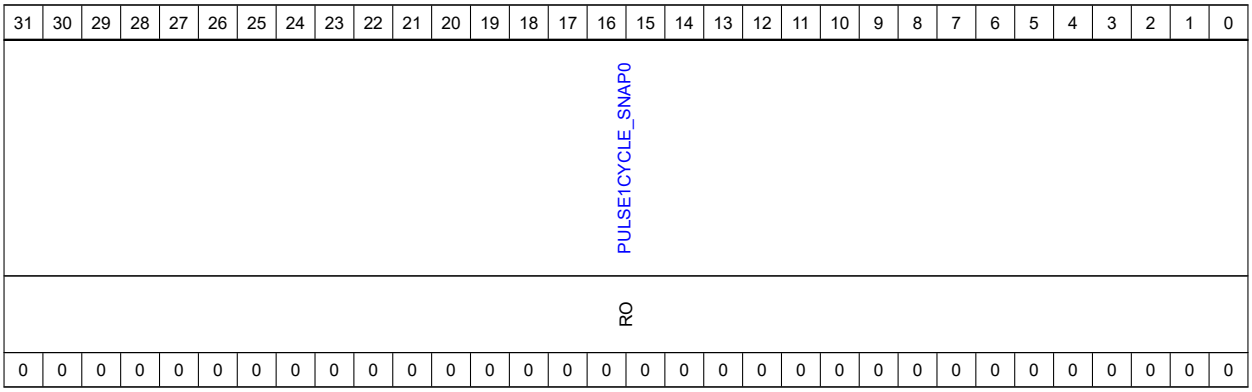


PULSE1_SNAP0 [31:0]

Bits	Name	Description
31-0	PULSE1_SNAP0	

PULSE1_SNAP0 Bits

36.7.44 PULSE1CYCLE_SNAP0 (0x16C)



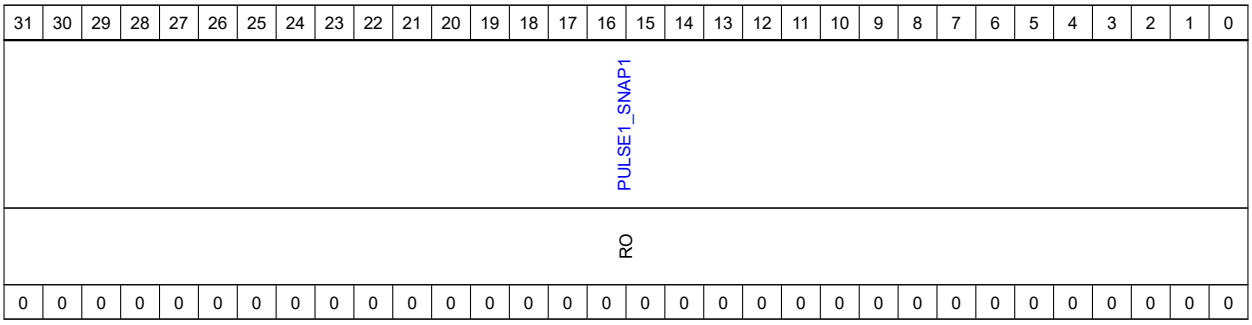
PULSE1CYCLE_SNAP0 [31:0]

Bits	Name	Description
31-0	PULSE1CYCLE_SNAP0	

PULSE1CYCLE_SNAP0 Bits

36.7.45 PULSE1_SNAP1 (0x170)

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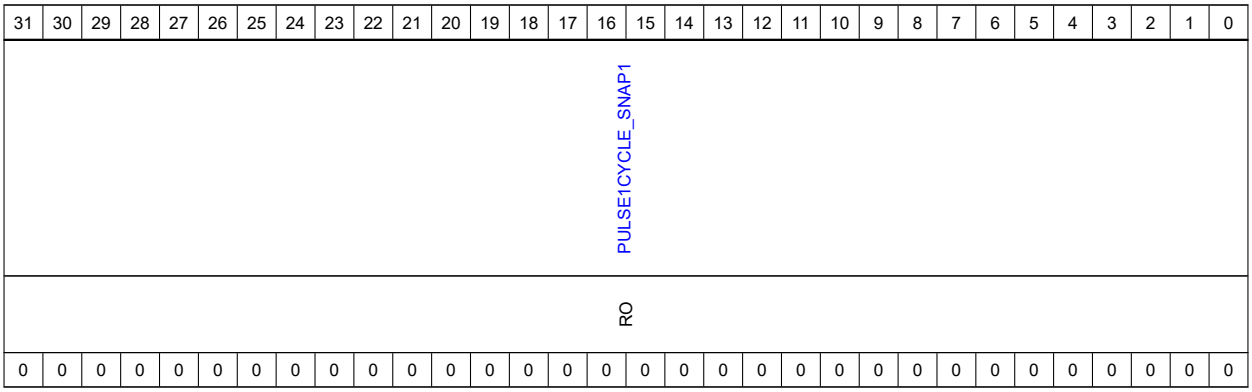


PULSE1_SNAP1 [31:0]

Bits	Name	Description
31-0	PULSE1_SNAP1	

PULSE1_SNAP1 Bits

36.7.46 PULSE1CYCLE_SNAP1 (0x174)

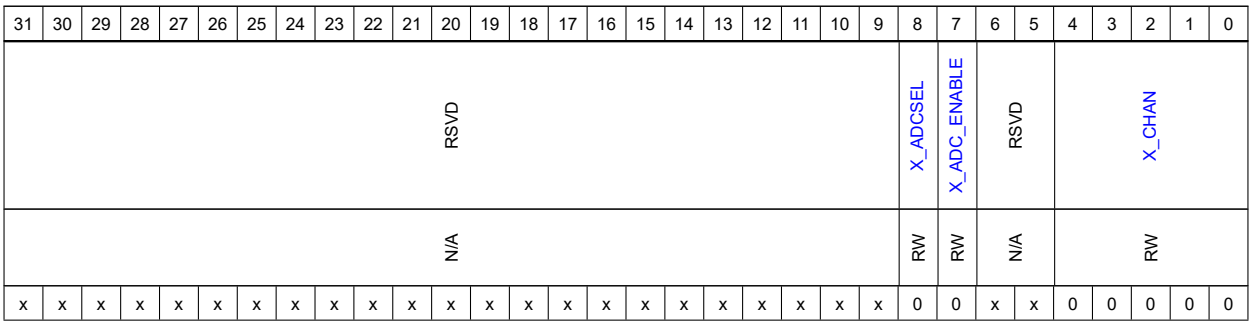


PULSE1CYCLE_SNAP1 [31:0]

Bits	Name	Description
31-0	PULSE1CYCLE_SNAP1	

PULSE1CYCLE_SNAP1 Bits

36.7.47 ADCX_CFG0 (0x200)



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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

ADCX_CFG0 [31:0]

Bits	Name	Description
8	X_ADCSEL	
7	X_ADC_ENABLE	
4-0	X_CHAN	

ADCX_CFG0 Bits

36.7.48 ADCX_CFG1 (0x204)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
X_PARAM1																X_PARAM0															
RW																RW															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ADCX_CFG1 [31:0]

Bits	Name	Description
31-16	X_PARAM1	
15-0	X_PARAM0	

ADCX_CFG1 Bits

36.7.49 ADCX_CFG2 (0x208)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
X_OFFSET																															
RW																															
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ADCX_CFG2 [31:0]

Bits	Name	Description
31-0	X_OFFSET	

ADCX_CFG2 Bits

36.7.50 ADCY_CFG0 (0x210)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																							Y_ADCSEL	Y_ADC_ENABLE	RSVD		Y_CHAN				
N/A																							RW	RW	N/A		RW				
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	x	x	0	0	0	0	0

ADCY_CFG0 [31:0]

Bits	Name	Description
8	Y_ADCSEL	
7	Y_ADC_ENABLE	
4-0	Y_CHAN	

ADCY_CFG0 Bits

36.7.51 ADCY_CFG1 (0x214)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Y_PARAM1																Y_PARAM0															
RW																RW															
0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ADCY_CFG1 [31:0]

Bits	Name	Description
31-16	Y_PARAM1	
15-0	Y_PARAM0	

ADCY_CFG1 Bits

36.7.52 ADCY_CFG2 (0x218)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Y_OFFSET																															
RW																															
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

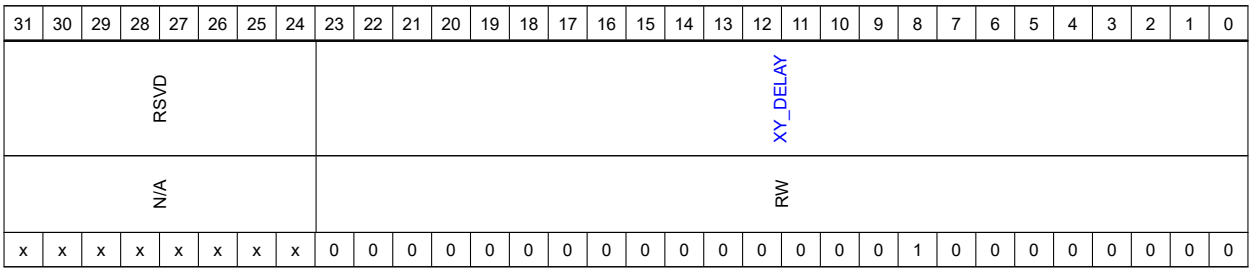
ADCY_CFG2 [31:0]

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Bits	Name	Description
31-0	Y_OFFSET	

ADCY_CFG2 Bits

36.7.53 CAL_CFG (0x220)

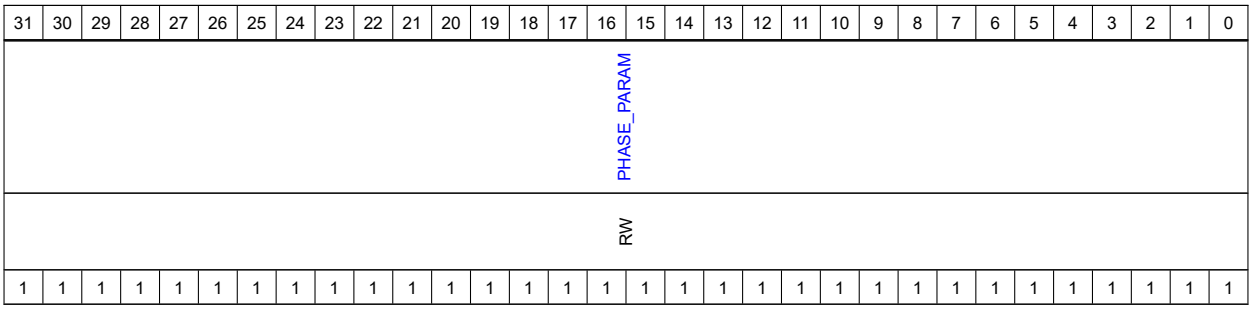


CAL_CFG [31:0]

Bits	Name	Description
23-0	XY_DELAY	valid x/y delay, larger than this delay will be treated as invalid data. Default 1.25us@200MHz; max 80ms;

CAL_CFG Bits

36.7.54 PHASE_PARAM (0x230)



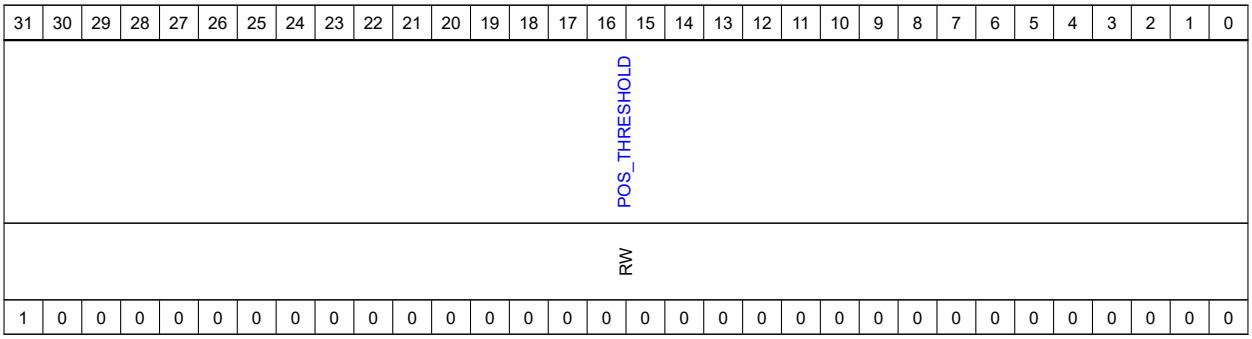
PHASE_PARAM [31:0]

Bits	Name	Description
31-0	PHASE_PARAM	

PHASE_PARAM Bits

36.7.55 POS_THRESHOLD (0x238)

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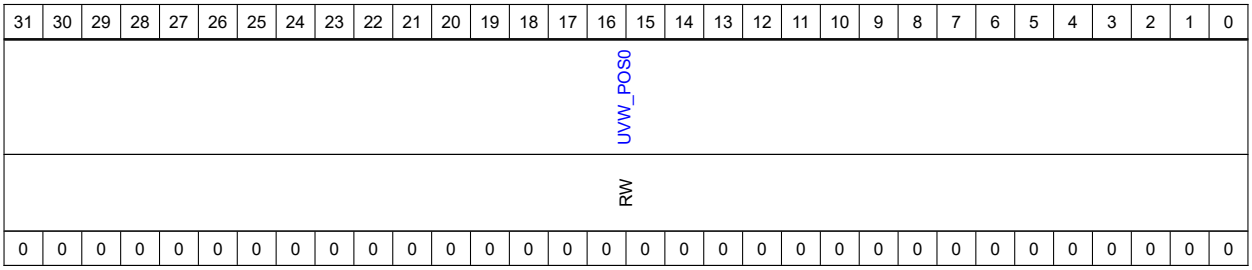


POS_THRESHOLD [31:0]

Bits	Name	Description
31-0	POS_THRESHOLD	

POS_THRESHOLD Bits

36.7.56 UVW_POS (0x240 + 0x4 * n)



UVW_POS [31:0]

Bits	Name	Description
31-0	UVW_POS0	

UVW_POS Bits

36.7.57 UVW_POS_CFG (0x258 + 0x4 * n)



UVW_POS_CFG [31:0]

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Bits	Name	Description
6	POS_EN	
5-4	U_POS_SEL	
3-2	V_POS_SEL	
1-0	W_POS_SEL	

UVW_POS_CFG Bits

36.7.58 PHASE_CNT (0x280)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PHASE_CNT																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PHASE_CNT [31:0]

Bits	Name	Description
31-0	PHASE_CNT	

PHASE_CNT Bits

36.7.59 PHASE_UPDATE (0x284)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INC	DEC	VALUE																													
WO	WO	WO																													
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

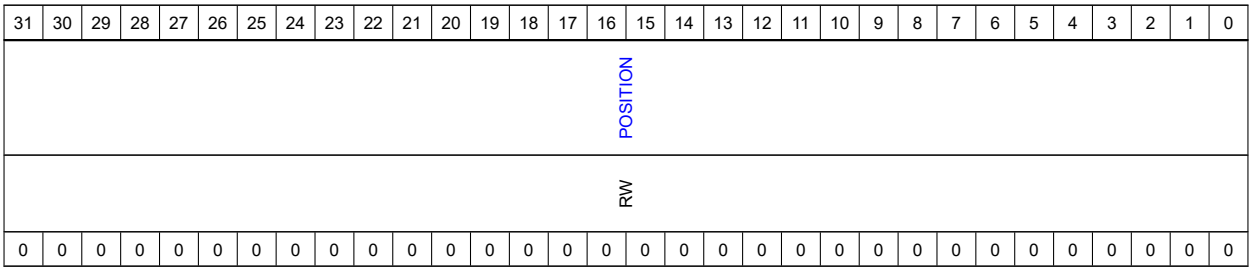
PHASE_UPDATE [31:0]

Bits	Name	Description
31	INC	set to add value to phase_cnt
30	DEC	set to minus value from phase_cnt(set inc and dec same time willl act inc)
29-0	VALUE	value to be added or minus from phase_cnt. only valid when inc or dec is set in one 32bit write operation

PHASE_UPDATE Bits

36.7.60 POSITION (0x288)

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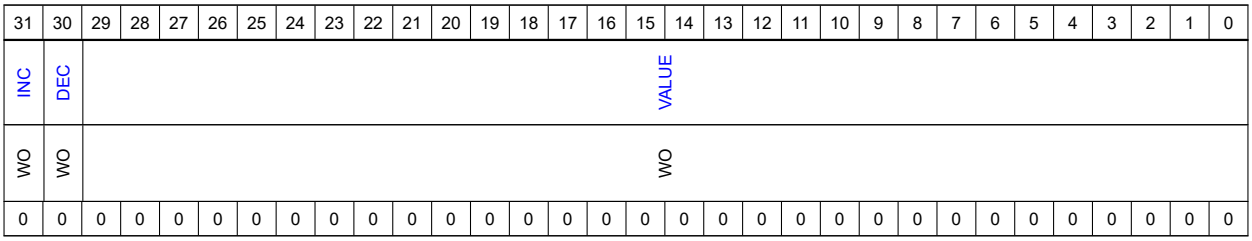


POSITION [31:0]

Bits	Name	Description
31-0	POSITION	

POSITION Bits

36.7.61 POSITION_UPDATE (0x28C)

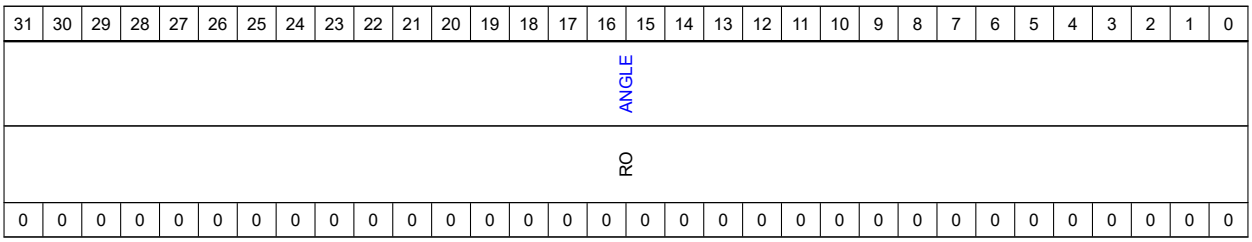


POSITION_UPDATE [31:0]

Bits	Name	Description
31	INC	set to add value to position
30	DEC	set to minus value from position(set inc and dec same time willl act inc)
29-0	VALUE	value to be added or minus from position. only valid when inc or dec is set in one 32bit write operation

POSITION_UPDATE Bits

36.7.62 ANGLE (0x290)



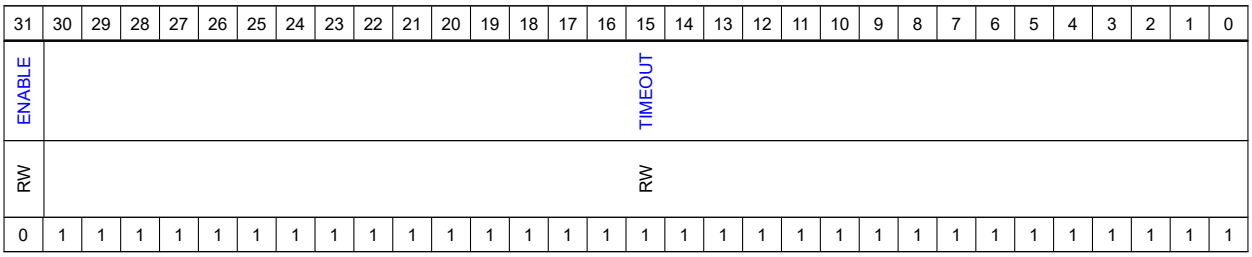
ANGLE [31:0]

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Bits	Name	Description
31-0	ANGLE	

ANGLE Bits

36.7.63 POS_TIMEOUT (0x294)



POS_TIMEOUT [31:0]

Bits	Name	Description
31	ENABLE	enable position timeout feature, if timeout, send valid again
30-0	TIMEOUT	postion timeout value

POS_TIMEOUT Bits

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37 Quadrature Encoder Output (QEO)

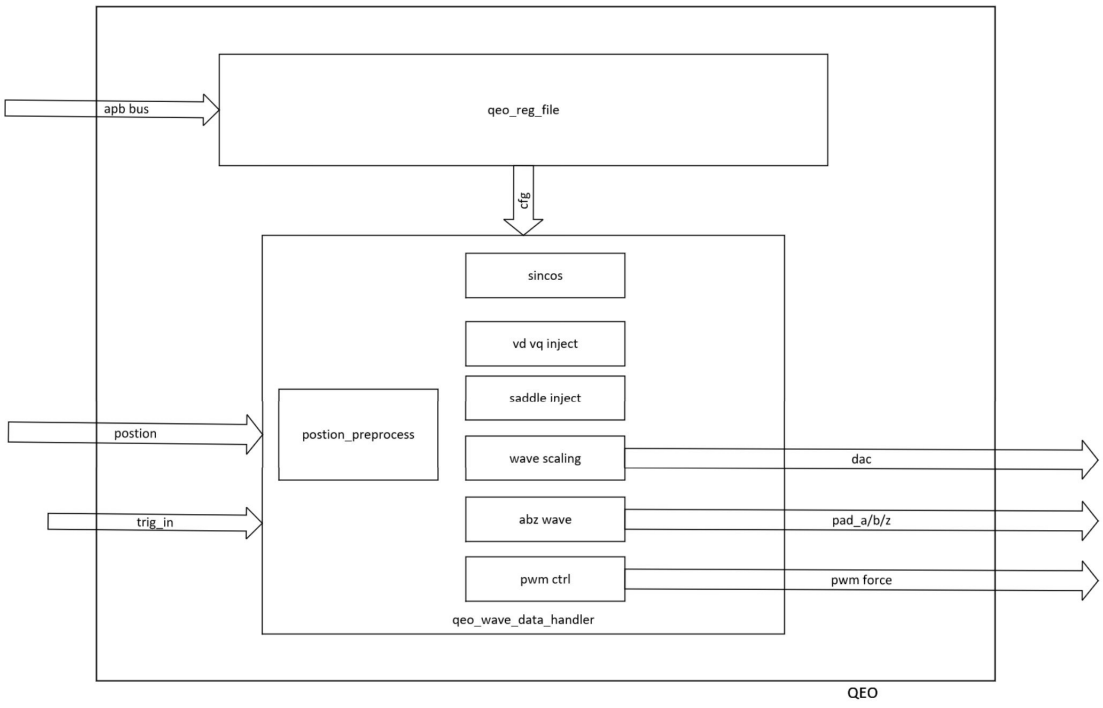
This chapter describes the main functions and features of the Quadrature Encoder Output (QEO).

37.1 Feature Summary

This section introduces the main features of QEO:

- Obtain position data from the system position manager
- Support position compensation
- Support line count conversion
- Support single-phase, two-phase (quadrature, up/down, pulse & direction), three-phase digital pulse output
- Support INDEX/Z signal output
- Configurable maximum and minimum pulse width
- Support analog signal output from DAC
- Support PWM output for three-phase BLDC and two-phase stepper motor control
- Support direct output for three-phase BLDC and two-phase stepper motor
- Support sine wave / full-wave rectified sine (hump) wave / saddle wave / sawtooth wave output
- Support signal adjustment such as amplification and offset
- Specifiable duty cycle for over-limit signals

37.2 Block Diagram



37.3 Functional Description

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37.3.1 Position Preprocessing

QEO has two types of position sources:

1) System position manager;

2) Position injected by software via writing to the POSTION_SOFTWARE register.

The valid bit width of the position input to QEO is 32 bits. The transition from 0x00000000 to 0xFFFFFFFF corresponds to 1 line count, equal to 2π radians or 360 degrees.

Each valid position input to QEO will be preprocessed first, which includes two steps:

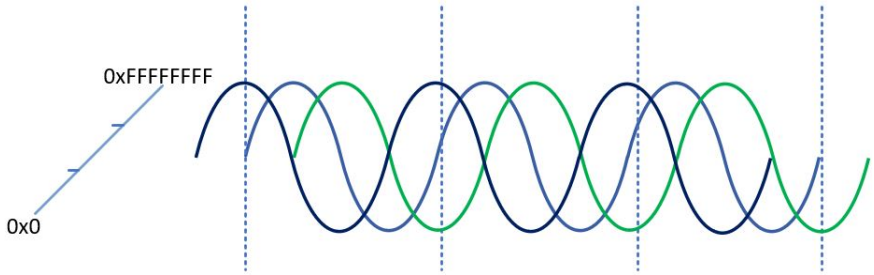
1) Line count conversion;

2) Position compensation.

Line count conversion means that the upstream position input originally corresponds to 1 line count when transitioning from 0x00000000 to 0xFFFFFFFF, after passing through the line count conversion module, the same position transition corresponds to n line counts, where n is the value of the WAVE/ABZ/PWM[RESOLUTION] register.

Position compensation refers to applying phase offset to the position after line count conversion.

Sine wave, ABZ and PWM FORCE modules have independent configuration registers for line count conversion and position compensation.



Taking sine wave output as an example in the figure above, if line count conversion is set to 3, and the three channels are configured with a 120° phase difference between each other, the output will generate 3 sine wave cycles when the input transitions from 0x0 to 0xFFFFFFFF.

37.3.2 Sine Wave Output

Sine wave output supports 4 modes:

1) Cosine wave; 2) Full-wave rectified sine (hump) wave; 3) Saddle wave; 4) Sawtooth wave.

Each output mode is equipped with 3 output channels, and the phase offset can be configured independently via WAVE[PHASH_SHIFT][WAVEn].

37.3.2.1 VD/VQ Control for Sine Wave

For all 3 output channels of sine wave, software can configure the WAVE[VD_VQ_INJECT] register to control the VD/VQ value corresponding to each sine wave.

$$V = \cos(angle) * VD - \sin(angle) * VQ \quad (1)$$

V represents the output value after inverse Park transform, and angle is the radian value corresponding to the

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preprocessed QEO input position.

37.3.2.2 Saddle Wave Mode in Sine Wave Output

When saddle wave output is selected, there are two saddle wave modes, which can be selected by software via configuring WAVE[MODE].SADDLE_TYPE:

1) Standard saddle wave, which can be output normally only when the phase difference between the three output channels is 120°.

2) Saddle wave synthesized by triangular wave superposition, which has no requirement on the phase difference of the three output channels.

37.3.2.3 Scaling and Limiting of Sine Wave

Each channel of sine wave output supports independent scaling and limiting. The scaling factor is controlled by the WAVE[AMPLITUDE] register, which can be configured as n * (1/4096), where n can be any integer from 0 to 65535, coming from WAVE[AMPLITUDE][15:0].

The scaled output can be adjusted for midpoint via WAVE[MID_POINT][WAVEn]. The part exceeding the sine wave output range can be limited by configuring WAVE[MODE][WAVEn_ABOVE_MAX_LIMIT] and WAVE[MODE][WAVEn_BELOW_MIN_LIMIT]. Limiting control is also supported within the range, please refer to WAVE[LIMIT][WAVEn][MAX] and WAVE[LIMIT][WAVEn][MIN] for details.

37.3.2.4 Dead Zone Compensation of Sine Wave

After limiting, each channel of sine wave output can also be compensated for dead zone prevention. Compensation can be performed by configuring WAVE[DEADZONE_SHIFT][WAVEn].

37.3.3 ABZ Output

ABZ is digital pulse output, which supports:

1) A/B two-phase quadrature output with internal position tracking;

2) High-speed Pulse & Revise output;

3) Up & Down output;

4) A/B two-phase quadrature output without internal position tracking.

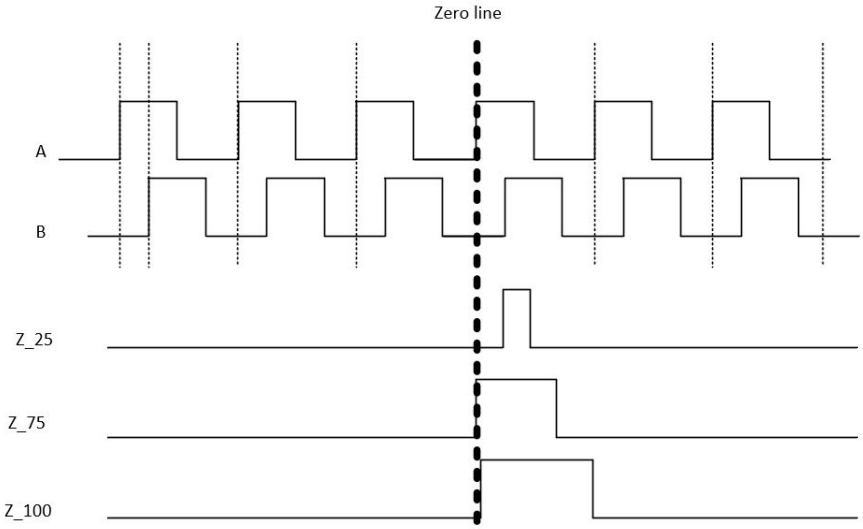
37.3.3.1 A/B Two-Phase Quadrature Output with Internal Position Tracking

The ABZ output module of QEO has a built-in ABZ internal position manager. After system reset, the internal position is reset to zero. When QEO receives a new input position, the internal position tracks the input position with the minimum period set by ABZ[LINE_WIDTH], and stops tracking when the internal position matches the input position. The minimum period corresponds to the period of 1/4 line count of A and B outputs.

Software can configure ABZ[POSTION_SYNC] to use the next valid position received by QEO as the position of the internal position manager. This function can be used to configure the ABZ starting position at startup to prevent incorrect ABZ tracking.

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When ABZ is set to A/B two-phase quadrature output with internal position tracking, the outputs of A, B and Z are determined by the internal position value.



Z_25, Z_75 and Z_100 in the figure above are three output modes of Z-phase, which are controlled by the ABZ[MODE].Z_TYPE register.

37.3.3.2 High-Speed Pulse & Revise Output

The high-speed pulse output mode also uses the ABZ internal position.

Phase A outputs clock pulses. When the internal position tracks the input position in the forward direction, phase B maintains high level output; when the internal position tracks in the reverse direction, phase B maintains low level output.

37.3.3.3 Up & Down Output

The Up & Down output mode also uses the ABZ internal position.

When the internal position tracks the input position in the forward direction, phase A outputs clock pulses; when the internal position tracks in the reverse direction, phase B outputs clock pulses.

37.3.3.4 A/B Two-Phase Quadrature Output without Internal Position Tracking

In this mode, the outputs of A/B/Z are determined by the real-time input position of QEO. The minimum period and watchdog configuration are invalid for this mode.

37.3.3.5 Watchdog for Internal Position Manager

To prevent users from being unable to judge whether the system is in normal mode when there is no position update for a long time, software can configure ABZ[MODE].EN_WDOG and ABZ[WDOG_WIDTH] to enable the watchdog function. After enabling the watchdog function, if the QEO input position does not change within the set number of cycles, the internal position will automatically step forward by 1/4 line count and then step backward by 1/4 line count, or step backward by 1/4 line count first and then step forward by 1/4 line count.

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37.3.4 PWM FORCE Output

QEO provides 8 groups of PWM FORCE control signals. Each group of PWM FORCE corresponds to 2-bit control signals. The PWM FORCE control signals are determined by looking up the PWM commutation table according to the input position.

The input position comes from the QEO position preprocessing module. Software can configure PWM[MODE].PHASE_N to determine the number of commutation steps, with a maximum of 12 commutation steps. Then fill the commutation table into PWM [PHASE_TABLE] [POSEDGEn] or PWM [PHASE_TABLE] [NEGEDGEn], the number of n depends on the number of commutation steps configured by PWM[MODE].PHASE_NUM. Whether to use PWM [PHASE_TABLE] [POSEDGEn] or PWM [PHASE_TABLE] [NEGEDGEn] is determined by the PWM[MODE].REVISE_UP_DN register.

This module also provides a PWM_FORCE lookup table for safe mode. The lookup table is determined by the PWM[MODE].PWMn_SAFETY register. Two methods are supported to enter safe mode:

1) Configure QEO.TRIGGER_IN[1] via TRIG_MUX, and enter safe mode when the source is high.

2) Enter safe mode by configuring PWM[MODE].PWM_ENTER_SAFETY_MODE.

37.4 QEO Register List

The QEO register list is as follows:

QEO0 base address: 0xF0308000

QEO1 base address: 0xF030C000

Offset	Register Name	Description	Reset Value
0x0000	WAVE[MODE]	analog waves mode	0x00000000
0x0004	WAVE[RESOLUTION]	resolution of wave0/1/2	0x00000000
0x0008	WAVE[PHASE_SHIFT][WAVE0]	wave0 phase shifter	0x00000000
0x000C	WAVE[PHASE_SHIFT][WAVE1]	wave1 phase shifter	0x00000000
0x0010	WAVE[PHASE_SHIFT][WAVE2]	wave2 phase shifter	0x00000000
0x0014	WAVE[VD_VQ_INJECT][WAVE0]	wave0 vd vq inject value	0x00000000
0x0018	WAVE[VD_VQ_INJECT][WAVE1]	wave1 vd vq inject value	0x00000000
0x001C	WAVE[VD_VQ_INJECT][WAVE2]	wave2 vd vq inject value	0x00000000
0x0020	WAVE[VD_VQ_LOAD]	load wave0/1/2 vd vq value	0x00000000
0x0024	WAVE[AMPLITUDE][WAVE0]	wave0 amplitude	0x00000000
0x0028	WAVE[AMPLITUDE][WAVE1]	wave1 amplitude	0x00000000
0x002C	WAVE[AMPLITUDE][WAVE2]	wave2 amplitude	0x00000000
0x0030	WAVE[MID_POINT][WAVE0]	wave0 output middle point offset	0x00000000
0x0034	WAVE[MID_POINT][WAVE1]	wave1 output middle point offset	0x00000000
0x0038	WAVE[MID_POINT][WAVE2]	wave2 output middle point offset	0x00000000
0x003C	WAVE[LIMIT][WAVE0][MIN]	wave0 low area limit value	0x00000000
0x0040	WAVE[LIMIT][WAVE0][MAX]	wave0 high area limit value	0xFFFFFFFF
0x0044	WAVE[LIMIT][WAVE1][MIN]	wave1 low area limit value	0x00000000

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Offset	Register Name	Description	Reset Value
0x0048	WAVE[LIMIT][WAVE1][MAX]	wave1 high area limit value	0xFFFFFFFF
0x004C	WAVE[LIMIT][WAVE2][MIN]	wave2 low area limit value	0x00000000
0x0050	WAVE[LIMIT][WAVE2][MAX]	wave2 high area limit value	0xFFFFFFFF
0x0054	WAVE[DEADZONE_SHIFT][WAVE0]	deadzone_shifter_wave0	0x00000000
0x0058	WAVE[DEADZONE_SHIFT][WAVE1]	deadzone_shifter_wave1	0x00000000
0x005C	WAVE[DEADZONE_SHIFT][WAVE2]	deadzone_shifter_wave2	0x00000000
0x0060	ABZ[MODE]	wave_a/b/z output mode	0x00000000
0x0064	ABZ[RESOLUTION]	resolution of wave_a/b/z	0x00000000
0x0068	ABZ[PHASE_SHIFT][A]	wave_a phase shifter	0x00000000
0x006C	ABZ[PHASE_SHIFT][B]	wave_b phase shifter	0x00000000
0x0070	ABZ[PHASE_SHIFT][Z]	wave_z phase shifter	0x00000000
0x0074	ABZ[LINE_WIDTH]	Two-phase orthogonality wave 1/4 period	0x00000000
0x0078	ABZ[WDOG_WIDTH]	wdog width of qeo	0x00000000
0x007C	ABZ[POSTION_SYNC]	sync abz owned postion	0x00000000
0x0080	PWM[MODE]	pwm mode	0x00000000
0x0084	PWM[RESOLUTION]	resolution of pwm	0x00000000
0x0088	PWM[PHASE_SHIFT][A]	pwm_a phase shifter	0x00000000
0x008C	PWM[PHASE_SHIFT][B]	pwm_b phase shifter	0x00000000
0x0090	PWM[PHASE_SHIFT][C]	pwm_c phase shifter	0x00000000
0x0094	PWM[PHASE_SHIFT][D]	pwm_d phase shifter	0x00000000
0x0098	PWM[PHASE_TABLE][POSEDGE0]	pwm_phase_table 0	0x00000000
0x009C	PWM[PHASE_TABLE][POSEDGE1]	pwm_phase_table 1	0x00000000
0x00A0	PWM[PHASE_TABLE][POSEDGE2]	pwm_phase_table 2	0x00000000
0x00A4	PWM[PHASE_TABLE][POSEDGE3]	pwm_phase_table 3	0x00000000
0x00A8	PWM[PHASE_TABLE][POSEDGE4]	pwm_phase_table 4	0x00000000
0x00AC	PWM[PHASE_TABLE][POSEDGE5]	pwm_phase_table 5	0x00000000
0x00B0	PWM[PHASE_TABLE][POSEDGE6]	pwm_phase_table 6	0x00000000
0x00B4	PWM[PHASE_TABLE][POSEDGE7]	pwm_phase_table 7	0x00000000
0x00B8	PWM[PHASE_TABLE][POSEDGE8]	pwm_phase_table 8	0x00000000
0x00BC	PWM[PHASE_TABLE][POSEDGE9]	pwm_phase_table 9	0x00000000
0x00C0	PWM[PHASE_TABLE][POSEDGE10]	pwm_phase_table 10	0x00000000
0x00C4	PWM[PHASE_TABLE][POSEDGE11]	pwm_phase_table 11	0x00000000
0x00C8	PWM[PHASE_TABLE][NEGEDGE0]	pwm_phase_table 0	0x00000000
0x00CC	PWM[PHASE_TABLE][NEGEDGE1]	pwm_phase_table 1	0x00000000
0x00D0	PWM[PHASE_TABLE][NEGEDGE2]	pwm_phase_table 2	0x00000000
0x00D4	PWM[PHASE_TABLE][NEGEDGE3]	pwm_phase_table 3	0x00000000
0x00D8	PWM[PHASE_TABLE][NEGEDGE4]	pwm_phase_table 4	0x00000000

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Offset	Register Name	Description	Reset Value
0x00DC	PWM[PHASE_TABLE][NEGEDGE5]	pwm_phase_table 5	0x00000000
0x00E0	PWM[PHASE_TABLE][NEGEDGE6]	pwm_phase_table 6	0x00000000
0x00E4	PWM[PHASE_TABLE][NEGEDGE7]	pwm_phase_table 7	0x00000000
0x00E8	PWM[PHASE_TABLE][NEGEDGE8]	pwm_phase_table 8	0x00000000
0x00EC	PWM[PHASE_TABLE][NEGEDGE9]	pwm_phase_table 9	0x00000000
0x00F0	PWM[PHASE_TABLE][NEGEDGE10]	pwm_phase_table 10	0x00000000
0x00F4	PWM[PHASE_TABLE][NEGEDGE11]	pwm_phase_table 11	0x00000000
0x00F8	POSTION_SOFTWARE	softwave inject postion	0x00000000
0x00FC	POSTION_SEL	select softwave inject postion	0x00000000
0x0100	STATUS	qeo status	0x00000000
0x0104	DEBUG0	qeo debug 0	0x00000000
0x0108	DEBUG1	qeo debug 1	0x00000000
0x010C	DEBUG2	qeo debug 2	0x00000000
0x0110	DEBUG3	qeo debug 3	0x00000000

Table 176: QEO Register List

37.5 QEO Register Details

The detailed description of QEO registers is as follows:

37.5.1 WAVE[MODE] (0x0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WAVE2_ABOVE_MAX_LIMIT		WAVE2_HIGH_AREA1_LIMIT	WAVE2_HIGH_AREA0_LIMIT	WAVE2_LOW_AREA1_LIMIT	WAVE2_LOW_AREA0_LIMIT	WAVE2_BELOW_MIN_LIMIT		WAVE1_ABOVE_MAX_LIMIT		WAVE1_HIGH_AREA1_LIMIT	WAVE1_HIGH_AREA0_LIMIT	WAVE1_LOW_AREA1_LIMIT	WAVE1_LOW_AREA0_LIMIT	WAVE1_BELOW_MIN_LIMIT		WAVE0_ABOVE_MAX_LIMIT		WAVE0_HIGH_AREA1_LIMIT	WAVE0_HIGH_AREA0_LIMIT	WAVE0_LOW_AREA1_LIMIT	WAVE0_LOW_AREA0_LIMIT	WAVE0_BELOW_MIN_LIMIT		SADDLE_TYPE	EN_WAVE2_VD_VQ_INJECT	EN_WAVE1_VD_VQ_INJECT	EN_WAVE0_VD_VQ_INJECT	RSVD		WAVES_OUTPUT_TYPE	
RW		RW	RW	RW	RW	RW		RW		RW	RW	RW	RW	RW		RW		RW	RW	RW	RW	RW		RW	RW	RW	RW	N/A		RW	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	0	0

WAVE[MODE] [31:0]

Bits	Name	Description
31-30	WAVE2_ABOVE_MAX_LIMIT	wave2 above max limit mode. 0: output 0xffff. 1: output 0x0. 2: output as level_max_limit2.level0_max_limit

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Bits	Name	Description
29	WAVE2_HIGH_AREA1_LIMIT	wave2 high area1 limit mode. 0: output 0xffff. 1: output as level_max_limit2.level0_max_limit
28	WAVE2_HIGH_AREA0_LIMIT	wave2 high area0 limit mode. 0: output 0xffff. 1: output as level_max_limit2.level0_max_limit
27	WAVE2_LOW_AREA1_LIMIT	wave2 low area1 limit mode. 0: output 0. 1: output as level_min_limit2.level1_min_limit
26	WAVE2_LOW_AREA0_LIMIT	wave2 low area0 limit mode. 0: output 0. 1: output as level_min_limit2.level1_min_limit
25-24	WAVE2_BELOW_MIN_LIMIT	wave2 below min limit mode. 0: output 0. 1: output 0xffff. 2: output as level_min_limit2.level1_min_limit
23-22	WAVE1_ABOVE_MAX_LIMIT	wave1 above max limit mode. 0: output 0xffff. 1: output 0x0. 2: output as level_max_limit1.level0_max_limit
21	WAVE1_HIGH_AREA1_LIMIT	wave1 high area1 limit mode. 0: output 0xffff. 1: output as level_max_limit1.level0_max_limit
20	WAVE1_HIGH_AREA0_LIMIT	wave1 high area0 limit mode. 0: output 0xffff. 1: output as level_max_limit1.level0_max_limit
19	WAVE1_LOW_AREA1_LIMIT	wave1 low area1 limit mode. 0: output 0. 1: output as level_min_limit1.level1_min_limit
18	WAVE1_LOW_AREA0_LIMIT	wave1 low area0 limit mode. 0: output 0. 1: output as level_min_limit1.level1_min_limit
17-16	WAVE1_BELOW_MIN_LIMIT	wave1 below min limit mode. 0: output 0. 1: output 0xffff. 2: output as level_min_limit1.level1_min_limit
15-14	WAVE0_ABOVE_MAX_LIMIT	wave0 above max limit mode. 0: output 0xffff. 1: output 0x0. 2: output as level_max_limit0.level0_max_limit

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Bits	Name	Description
13	WAVE0_HIGH_AREA1_LIMIT	wave0 high area1 limit mode. 0: output 0xffff. 1: output as level_max_limit0.level0_max_limit
12	WAVE0_HIGH_AREA0_LIMIT	wave0 high area0 limit mode. 0: output 0xffff. 1: output as level_max_limit0.level0_max_limit
11	WAVE0_LOW_AREA1_LIMIT	wave0 low area1 limit mode. 0: output 0. 1: output as level_min_limit0.level1_min_limit
10	WAVE0_LOW_AREA0_LIMIT	wave0 low area0 limit mode. 0: output 0. 1: output as level_min_limit0.level1_min_limit
9-8	WAVE0_BELOW_MIN_LIMIT	wave0 below min limit mode. 0: output 0. 1: output 0xffff. 2: output as level_min_limit0.level1_min_limit
7	SADDLE_TYPE	saddle type seclect; 0:standard saddle. 1: triple-cos saddle.
6	EN_WAVE2_VD_VQ_INJECT	wave2 VdVq inject enable. 0: disable VdVq inject. 1: enable VdVq inject.
5	EN_WAVE1_VD_VQ_INJECT	wave1 VdVq inject enable. 0: disable VdVq inject. 1: enable VdVq inject.
4	EN_WAVE0_VD_VQ_INJECT	wave0 VdVq inject enable. 0: disable VdVq inject. 1: enable VdVq inject.
1-0	WAVES_OUTPUT_TYPE	wave0/1/2 output mode. 0: cosine wave. 1: saddle wave. 2. abs cosine wave. 3. saw wave

WAVE[MODE] Bits

37.5.2 WAVE[RESOLUTION] (0x4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LINES																															

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

WAVE[RESOLUTION] [31:0]

Bits	Name	Description
31-0	LINES	wave0/1/2 resolution

WAVE[RESOLUTION] Bits

37.5.3 WAVE[PHASE_SHIFT] (0x8 + 0x4 * m)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																VAL															
N/A																RW															
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

WAVE[PHASE_SHIFT] [31:0]

Bits	Name	Description
15-0	VAL	wave0 phase shifter value, default is 0x0. write other value will shift phase early as (cfg_value/2^16) period

WAVE[PHASE_SHIFT] Bits

37.5.4 WAVE[VD_VQ_INJECT] (0x14 + 0x4 * m)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VQ_VAL																VD_VAL															
RW																RW															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

WAVE[VD_VQ_INJECT] [31:0]

Bits	Name	Description
31-16	VQ_VAL	Vq inject value
15-0	VD_VAL	Vd inject value

WAVE[VD_VQ_INJECT] Bits

37.5.5 WAVE[VD_VQ_LOAD] (0x20)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																															LOAD
N/A																															WO
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

WAVE[VD_VQ_LOAD] [31:0]

Bits	Name	Description
0	LOAD	load wave0/1/2 vd vq value. always read 0 0: vd vq keep previous value. 1: load wave0/1/2 vd vq value at sametime.

WAVE[VD_VQ_LOAD] Bits

37.5.6 WAVE[AMPLITUDE] (0x24 + 0x4 * m)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD															EN_SCAL	AMP_VAL															
N/A															RW	RW															
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

WAVE[AMPLITUDE] [31:0]

Bits	Name	Description
16	EN_SCAL	enable wave amplitude scaling. 0: disable; 1: enable
15-0	AMP_VAL	amplitude scaling value. bit15-12 are integer part value. bit11-0 are fraction value.

WAVE[AMPLITUDE] Bits

37.5.7 WAVE[MID_POINT] (0x30 + 0x4 * m)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

WAVE[MID_POINT] [31:0]

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Bits	Name	Description
31-0	VAL	wave0 output middle point, use this value as 32 bit signed value. bit 31 is signed bit. bit30-27 is integer part value. bit26-0 is fraction value.

WAVE[MID_POINT] Bits

37.5.8 WAVE[LIMIT][MIN] (0x3C + 0x8 * m)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LIMIT1																LIMIT0															
RW																RW															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

WAVE[LIMIT][MIN] [31:0]

Bits	Name	Description
31-16	LIMIT1	low area limit level1
15-0	LIMIT0	low area limit level0

WAVE[LIMIT][MIN] Bits

37.5.9 WAVE[LIMIT][MAX] (0x40 + 0x8 * m)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LIMIT1																LIMIT0															
RW																RW															
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

WAVE[LIMIT][MAX] [31:0]

Bits	Name	Description
31-16	LIMIT1	high area limit level1
15-0	LIMIT0	high area limit level0

WAVE[LIMIT][MAX] Bits

37.5.10 WAVE[DEADZONE_SHIFT] (0x54 + 0x4 * m)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																VAL															

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
N/A																RW															
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

WAVE[DEADZONE_SHIFT] [31:0]

Bits	Name	Description
15-0	VAL	wave0 deadzone shifter value

WAVE[DEADZONE_SHIFT] Bits

37.5.11 ABZ[MODE] (0x60)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD			REVERSE_EDGE_TYPE	RSVD			EN_WDOG	RSVD			Z_POLARITY	RSVD			B_POLARITY	RSVD			A_POLARITY	RSVD		Z_TYPE		RSVD		B_TYPE		RSVD		A_TYPE	
N/A			RW	N/A			RW	N/A			RW	N/A			RW	N/A			RW	N/A		RW		N/A		RW		N/A		RW	
x	x	x	0	x	x	x	0	x	x	x	0	x	x	x	0	x	x	x	0	x	x	0	0	x	x	0	0	x	x	0	0

ABZ[MODE] [31:0]

Bits	Name	Description
28	REVERSE_EDGE_TYPE	pulse reverse wave，reverse edge point: 0: between pulse's posedge and negedge, min period dedicated by the num line_width 1: edge change point flow pulse's negedge.
24	EN_WDOG	enable abz wdog: 0: disable abz wdog. 1: enable abz wdog.
20	Z_POLARITY	wave_z polarity. 0: normal output. 1: invert normal output
16	B_POLARITY	wave_b polarity. 0: normal output. 1: invert normal output
12	A_POLARITY	wave_a polarity. 0: normal output. 1: invert normal output

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Bits	Name	Description
9-8	Z_TYPE	wave_z type: 0: zero pulse and output high at both wave_a and wave_b are high. mantain about 25% period. 1: zero pulse output high about 75% period. start from 0 to 75% period. 2: zero pulse output high about 100% period. 3: wave_z output as tree-phase wave same as wave_a/wave_b
5-4	B_TYPE	wave_b type: 0: Two-phase orthogonality wave_b. 1: reverse wave of pulse/reverse type. 2: down wave of up/down type. 3: Three-phase orthogonality wave_b.
1-0	A_TYPE	wave_a type: 0: Two-phase orthogonality wave_a. 1: pulse wave of pulse/reverse type. 2: up wave of up/down type. 3: Three-phase orthogonality wave_a.

ABZ[MODE] Bits

37.5.12 ABZ[RESOLUTION] (0x64)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LINES																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ABZ[RESOLUTION] [31:0]

Bits	Name	Description
31-0	LINES	wave_a/b/z resolution

ABZ[RESOLUTION] Bits

37.5.13 ABZ[PHASE_SHIFT] (0x68 + 0x4 * m)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																VAL															
N/A																RW															
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

ABZ[PHASE_SHIFT] [31:0]

Bits	Name	Description
15-0	VAL	wave_a phase shifter value, default is 0x0. write other value will shift phase early as (cfg_value/2^16) period.

ABZ[PHASE_SHIFT] Bits

37.5.14 ABZ[LINE_WIDTH] (0x74)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LINE																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ABZ[LINE_WIDTH] [31:0]

Bits	Name	Description
31-0	LINE	the num of system clk by 1/4 period when using as Two-phase orthogonality.

ABZ[LINE_WIDTH] Bits

37.5.15 ABZ[WDOG_WIDTH] (0x78)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WIDTH																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

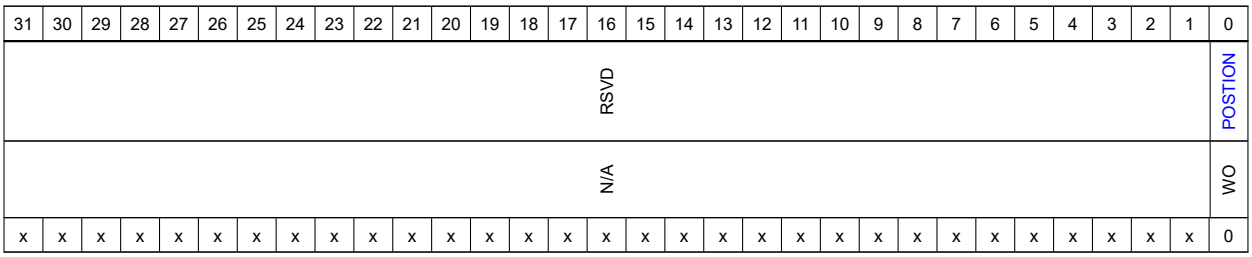
ABZ[WDOG_WIDTH] [31:0]

Bits	Name	Description
31-0	WIDTH	wave will step 1/4 line to reminder user QEO still in controlled if QEO has no any toggle after the num of wdog_width sys clk.

ABZ[WDOG_WIDTH] Bits

37.5.16 ABZ[POSTION_SYNC] (0x7C)

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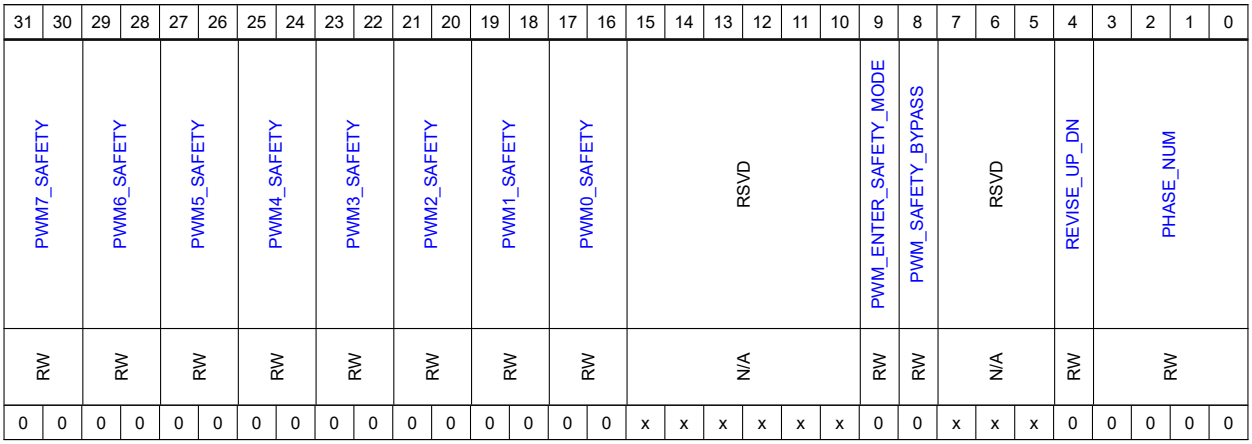


ABZ[POSTION_SYNC] [31:0]

Bits	Name	Description
0	POSTION	load next valid postion into abz owned postion. always read 0 0: sync abz owned postion with next valid postion. 1: not sync.

ABZ[POSTION_SYNC] Bits

37.5.17 PWM[MODE] (0x80)



PWM[MODE] [31:0]

Bits	Name	Description
31-30	PWM7_SAFETY	PWM safety mode phase table
29-28	PWM6_SAFETY	PWM safety mode phase table
27-26	PWM5_SAFETY	PWM safety mode phase table
25-24	PWM4_SAFETY	PWM safety mode phase table
23-22	PWM3_SAFETY	PWM safety mode phase table
21-20	PWM2_SAFETY	PWM safety mode phase table
19-18	PWM1_SAFETY	PWM safety mode phase table
17-16	PWM0_SAFETY	PWM safety mode phase table
9	PWM_ENTER_SAFETY_MODE	PWM enter safety mode 0: not enter 1: enter

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Bits	Name	Description
8	PWM_SAFETY_BYPASS	PWM safety mode bypass 0: not bypass 1: bypass
4	REVISE_UP_DN	exchange PWM pairs’ output 0: not exchange. 1: exchange.
3-0	PHASE_NUM	pwm force phase number.

PWM[MODE] Bits

37.5.18 PWM[RESOLUTION] (0x84)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LINES																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PWM[RESOLUTION] [31:0]

Bits	Name	Description
31-0	LINES	pwm resolution

PWM[RESOLUTION] Bits

37.5.19 PWM[PHASE_SHIFT] (0x88 + 0x4 * m)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																VAL															
N/A																RW															
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PWM[PHASE_SHIFT] [31:0]

Bits	Name	Description
15-0	VAL	pwm_a phase shifter value, default is 0x0. write other value will shift phase early as (cfg_value/2^16) period

PWM[PHASE_SHIFT] Bits

37.5.20 PWM[PHASE_TABLE] (0x98 + 0x4 * m)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																PWM7		PWM6		PWM5		PWM4		PWM3		PWM2		PWM1		PWM0	
N/A																RW		RW		RW		RW		RW		RW		RW		RW	
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PWM[PHASE_TABLE] [31:0]

Bits	Name	Description
15-14	PWM7	pwm phase table value
13-12	PWM6	pwm phase table value
11-10	PWM5	pwm phase table value
9-8	PWM4	pwm phase table value
7-6	PWM3	pwm phase table value
5-4	PWM2	pwm phase table value
3-2	PWM1	pwm phase table value
1-0	PWM0	pwm phase table value

PWM[PHASE_TABLE] Bits

37.5.21 POSTION_SOFTWARE (0xF8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
POSTION_SOFTWAVE																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

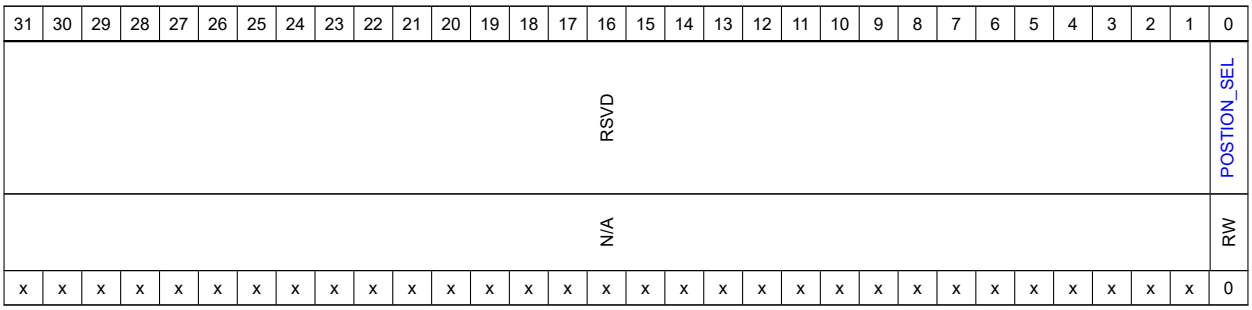
POSTION_SOFTWARE [31:0]

Bits	Name	Description
31-0	POSTION_SOFTWAVE	softwave inject postion

POSTION_SOFTWARE Bits

37.5.22 POSTION_SEL (0xFC)

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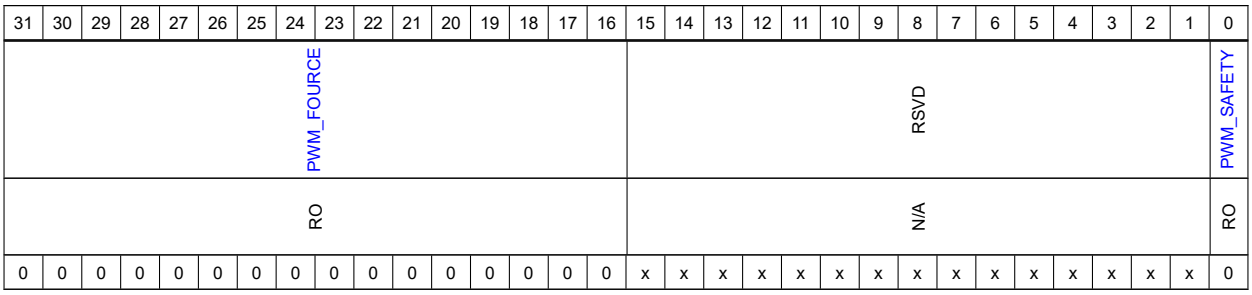


POSTION_SEL [31:0]

Bits	Name	Description
0	POSTION_SEL	enable softwave inject postion. 0: disable. 1: enable.

POSTION_SEL Bits

37.5.23 STATUS (0x100)

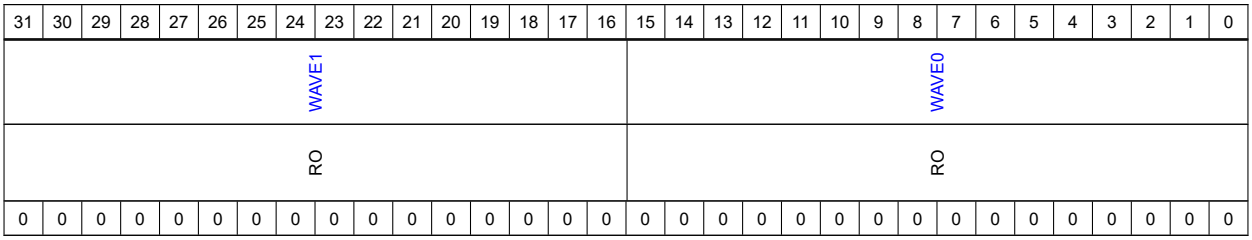


STATUS [31:0]

Bits	Name	Description
31-16	PWM_FOURCE	qeo_pwm_force observe
0	PWM_SAFETY	pwm_fault status

STATUS Bits

37.5.24 DEBUG0 (0x104)



DEBUG0 [31:0]

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Bits	Name	Description
31-16	WAVE1	wave1 observe
15-0	WAVE0	wave0 observe

DEBUG0 Bits

37.5.25 DEBUG1 (0x108)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD			QEO_FINISH	RSVD			WAVE_Z	RSVD			WAVE_B	RSVD			WAVE_A	WAVE2															
N/A			RO	N/A			RO	N/A			RO	N/A			RO	RO															
x	x	x	0	x	x	x	0	x	x	x	0	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DEBUG1 [31:0]

Bits	Name	Description
28	QEO_FINISH	qeo finish observe
24	WAVE_Z	wave_z observe
20	WAVE_B	wave_b observe
16	WAVE_A	wave_a observe
15-0	WAVE2	wave2 observe

DEBUG1 Bits

37.5.26 DEBUG2 (0x10C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ABZ_OWN_POSTION																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

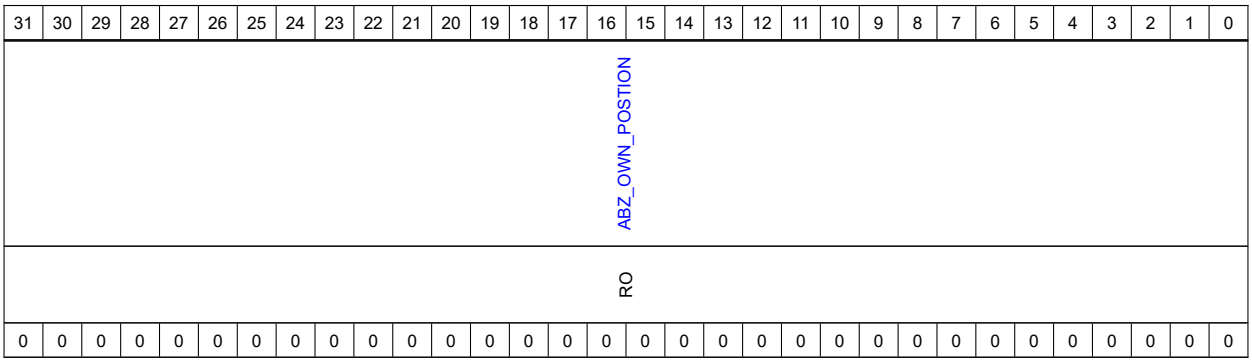
DEBUG2 [31:0]

Bits	Name	Description
31-0	ABZ_OWN_POSTION	abz_own_postion observe

DEBUG2 Bits

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37.5.27 DEBUG3 (0x110)



DEBUG3 [31:0]

Bits	Name	Description
31-0	ABZ_OWN_POSTION	abz_own_postion observe

DEBUG3 Bits

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38 Motion Processing Unit MMC

38.1 Overview

This chapter introduces the main functions and features of MMC.

MMC is also called Motion Processing Unit. It receives rotor position and timestamp from other modules, predicts and compensates the rotor position at future time through built-in algorithm and outputs the result, while providing speed and acceleration estimation. Currently supported features are shown in Table 177:

Function	Main Features
During Initialization	· Configure initial position, speed and acceleration. · Configure increments of initial position, speed and acceleration.
During Operation	· Dynamically configure initial position, speed and acceleration. · Dynamically configure increments of initial position, speed and acceleration.
Open-loop Mode	· After setting initial position, speed and acceleration, the system evolves continuously with time. Only the prediction module works in this mode.
Trigger Mode	· Configurable trigger sources including position, time and external events. Support dynamic modification of initial position, speed, acceleration and their increments. Currently only rising edge detection is supported for external events.
Interrupt	· Support configuration completion interrupt, trigger interrupt and exception interrupt.

Table 177: Main Features of MMC

38.2 MMC Architecture Diagram

The structure diagram of this module is shown in Figure 50. The in-cycle position and number of turns are collectively referred to as position.

38.3 Function Description

38.3.1 General Description of MMC

Its main data flow is as follows: measured data is sent to the tracker to train relevant parameters in the tracker. Training is performed once for each incoming measured data. After each training, the trained parameters are sent to the shadow register, and the two predictors will output position, speed, acceleration and other information at the predicted time according to their respective predicted timestamps and the latest content in the shadow register.

38.3.2 Generation of Predictor Timestamp

Each predictor has an independent timestamp generation mechanism. The following lists several optional modes. In the diagram:

- Black time t_n is the time to be predicted
- Red time t_n' is the time when calculation starts
- Yellow time t_tn is the arrival time of the trigger signal

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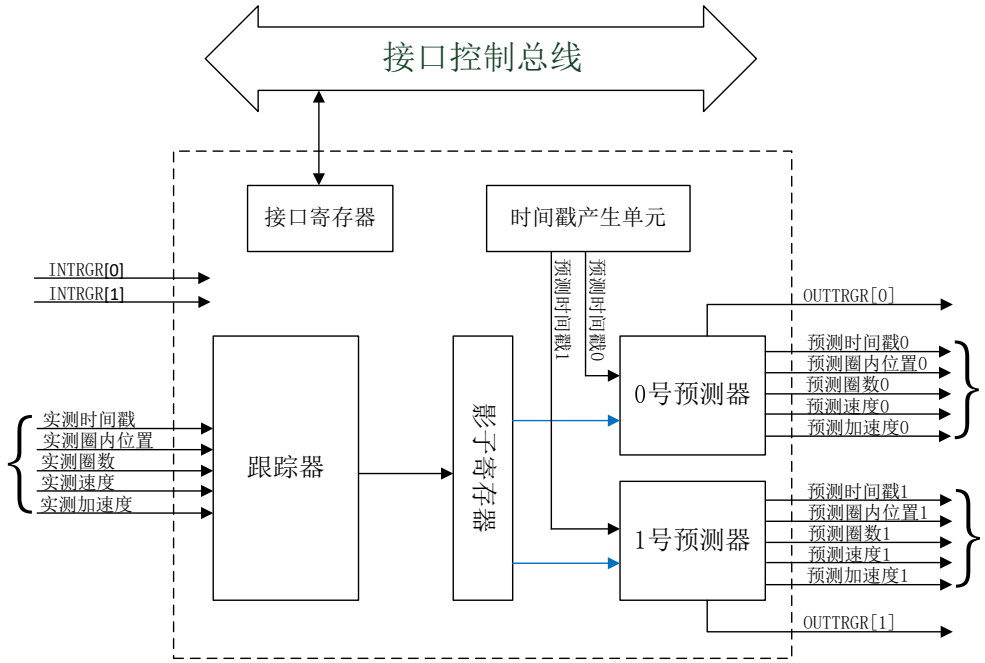


Figure 50: MMC Structure Block Diagram

- Blue curved arrow is controlled by register BR [n] [BR_TRG_PERIOD]
- Red curved arrow is controlled by register BR [n] [BR_TIMEOFF]
- Black arrow is controlled by register BR [n] [BR_TRG_F_TIME]
- Yellow arrow indicates the arrival time of the trigger signal
- Blue small box indicates the time consumed by MMC calculation

When BR [n] [BR_CTRL [PRED_MODE]] = 0, BR [n] [BR_CTRL [F_TRG_TYPE]] = 0 and BR [n] [BR_CTRL [NF_TRG_TYPE]] = 0, periodic prediction will be generated. The timestamp generation mode is shown in Figure 51. This diagram shows that the first time to be predicted t_0 is configured in BR [n] [BR_TRG_F_TIME], and the prediction calculation starts at t_0' ($t_0' = t_0 - BR[n][BR_TIMEOFF]$); the next predicted time is $t_1 = t_0 + BR[n][BR_TRG_PERIOD]$; subsequent calculations follow the same rule.

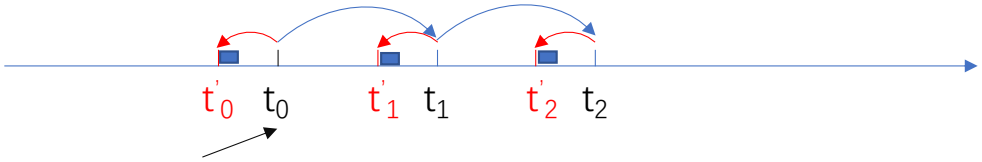


Figure 51: MMC Timestamp Generation Mode 0

When BR [n] [BR_CTRL [PRED_MODE]] = 0, BR [n] [BR_CTRL [F_TRG_TYPE]] = 1 and BR [n] [BR_CTRL [NF_TRG_TYPE]] = 1, trigger signal based prediction will be generated. The timestamp generation mode is shown in Figure 52. This diagram shows that the first time to be predicted t_1 is trigger time $t_t1 + BR[n][BR_TRG_PERIOD]$, and the prediction calculation starts at t_1' ($t_1' = t_1 - BR[n][BR_TIMEOFF]$); the next predicted time t_2 is determined by the trigger time t_t2; subsequent calculations follow the same rule.

When BR [n] [BR_CTRL [PRED_MODE]] = 1, continuous prediction will be generated.

- When BR [n] [BR_CTRL [F_TRG_TYPE]] = 0, the first time to be predicted t_0 is configured in BR [n] [BR_TRG_F_TIME], and the prediction calculation starts at t_0' ($t_0' = t_0 - BR[n][BR_TIMEOFF]$);

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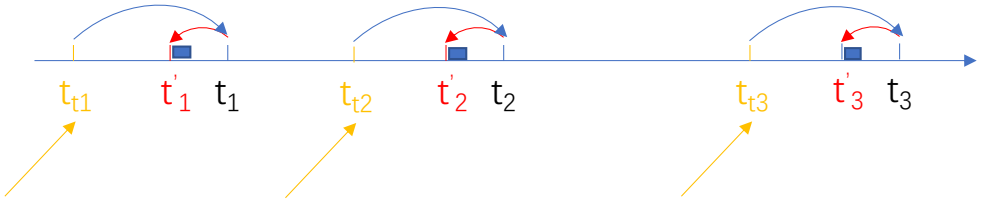


Figure 52: MMC Timestamp Generation Mode 1

the next predicted time is t_1 = calculation completion time + BR [n] [BR_TRG_PERIOD]; subsequent calculations follow the same rule.

- When BR [n] [BR_CTRL [F_TRG_TYPE]] = 1, the first time to be predicted t_1 is trigger time $t_t1 + BR[n][BR_TRG_PERIOD]$, and the prediction calculation starts at t_1' ($t_1' = t_1 - BR[n][BR_TIMEOFF]$); the next predicted time is t_2 = calculation completion time + BR [n] [BR_TRG_PERIOD]; subsequent calculations follow the same rule.

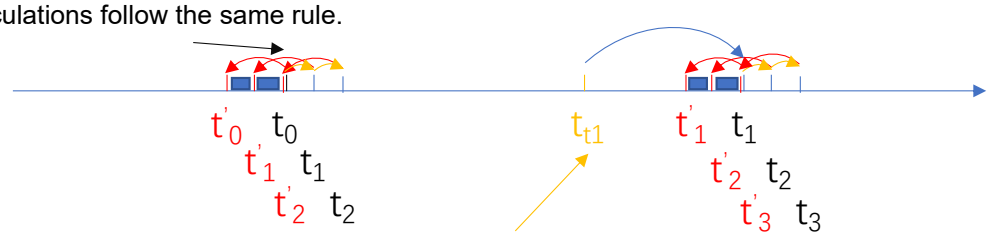


Figure 53: MMC Timestamp Generation Mode 2

The above are all multi-prediction modes. When BR [n] [BR_CTRL [PRED_MODE]] = 2, only one prediction will be generated, and other aspects are the same as the above modes.

38.3.3 MMC Data Flow Diagram

Its specific data flow diagram is shown in Figure 54.

This is the diagram of the tracker, with inputs of "measured position" and "$\triangle_T$"; the predictor is similar to the tracker, except that it only needs to input "$\triangle_T$" to calculate "predicted position" and "biased predicted position", and other relevant parameters are copied from the tracker through the shadow register. Among them, α is also called P coefficient; β is also called I coefficient; $\gamma/2$ is also called A coefficient.

38.3.4 MMC Prediction Output

Note: This section describes the configuration and impacts of prediction output. To clearly distinguish the differences between different configurations, the effects in the diagrams are only schematic and exaggerated for easy understanding, and do not represent real data.

Prediction output can be generated in the following two ways:

38.3.4.1 Prediction Output when CR[ADJOP]=0

An example of MMC prediction effect when CR[ADJOP]=0 is shown in Figure 55. In the diagram, the horizontal axis represents time, the vertical axis represents position, the red line represents real position, and the yellow line represents predicted position. The mark Tn on the horizontal axis represents the timestamp of the observed measured position, which is not uniform.

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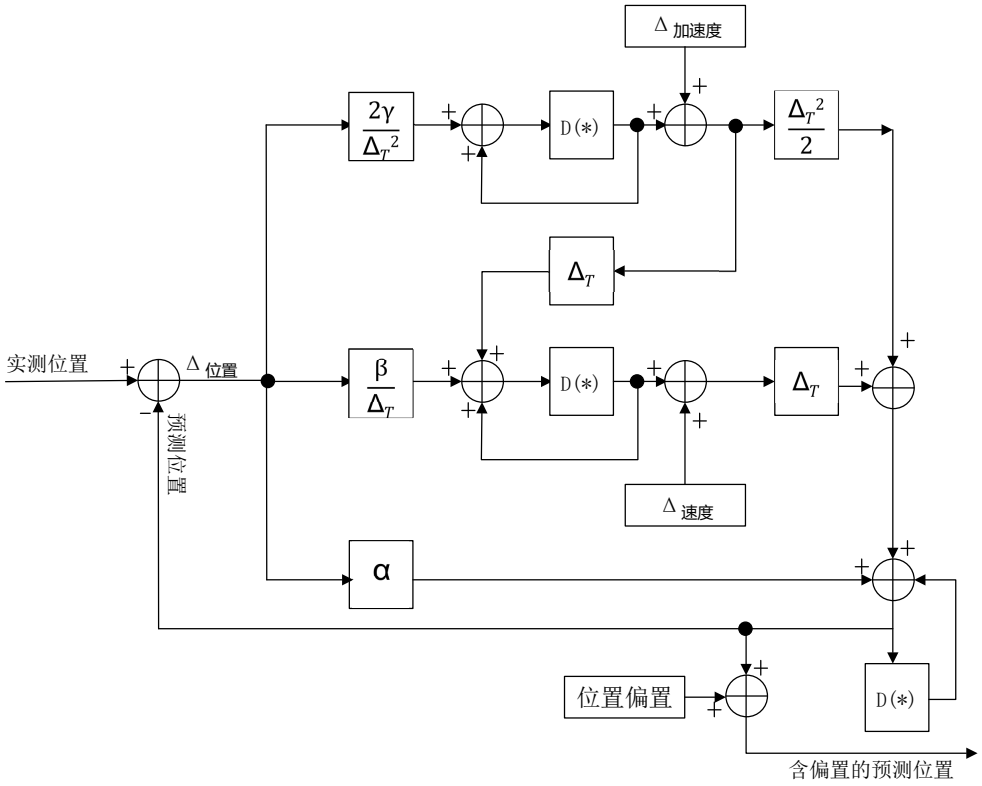


Figure 54: MMC Structure Block Diagram

Tracking and prediction are both continuous; each time a new measured position arrives, the tracked parameters will be updated, and the predicted value will be pulled closer to the measured position.

38.3.4.2 Prediction Output when CR[ADJOP]=1

An example of MMC prediction effect when CR[ADJOP]=1 is shown in Figure 56. In the diagram, the black line represents predicted position.

In this mode, each time a new measured position arrives, while updating the tracked parameters, the prediction will start from the measured position and use the tracked parameters for prediction.

When CR[ADJOP]=1, an out-of-step readjustment mechanism is also provided. The out-of-step threshold needs to be configured to an appropriate value in register OOSYNC_THETA_THR. When the absolute value of $\triangle$_position in Figure 54 is greater than OOSYNC_THETA_THR, it is considered that the current tracked data deviates greatly from the actual value, and the tracked coefficients will be forcibly reinitialized: clear the estimated acceleration to 0, and set the estimated speed to

$$\frac{(\text{Newmeasuredposition} - \text{Lastmeasuredposition})}{(\text{Newmeasuredtime} - \text{Lastmeasuredtime})}$$

. An example of MMC prediction effect at this time is shown in Figure 57. In the diagram, the solid blue line represents predicted position, and the dashed blue line represents the basis for speed estimation, which shows that the solid blue line is actually an extension of the dashed blue line.

When CR[ADJOP]=1, an upper and lower threshold mechanism is also provided. The threshold needs to be configured to an appropriate value in register DISCRETECFG1 [INV_POSMAX]. When the absolute value

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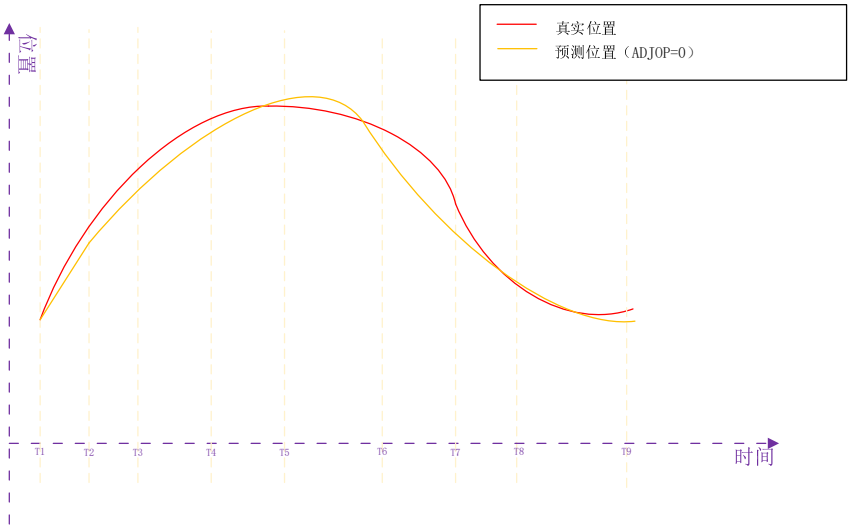


Figure 55: MMC Prediction Waveform (CR[ADJOP]=0)

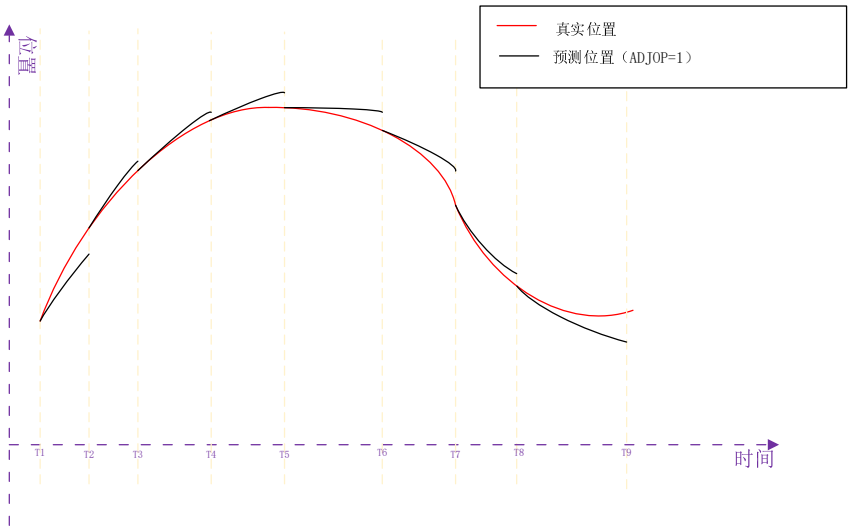


Figure 56: MMC Prediction Waveform (CR[ADJOP]=1)

of (Predictedposition − Lastmeasuredposition) in Figure 54 is greater than DISCRETECFG1 [INV_POSMAX], it is considered that the current predicted data deviates greatly from the actual value, and the predicted data will be forcibly locked at this threshold. An example of MMC prediction effect at this time is shown in Figure 58. In the diagram, only the situation where the threshold is reached after time T8 is marked. With threshold limitation, the predicted value follows the black line; without threshold limitation, the predicted value follows the green line. For clarity, the black line and green line are drawn separately, but they are actually the same immediately after time T8 and then diverge.

38.3.5 Open-loop and Closed-loop

Open-loop mode refers to continuously outputting new predicted values over time starting from a specific position with a given speed and acceleration. Open-loop mode can be implemented only with the prediction module.

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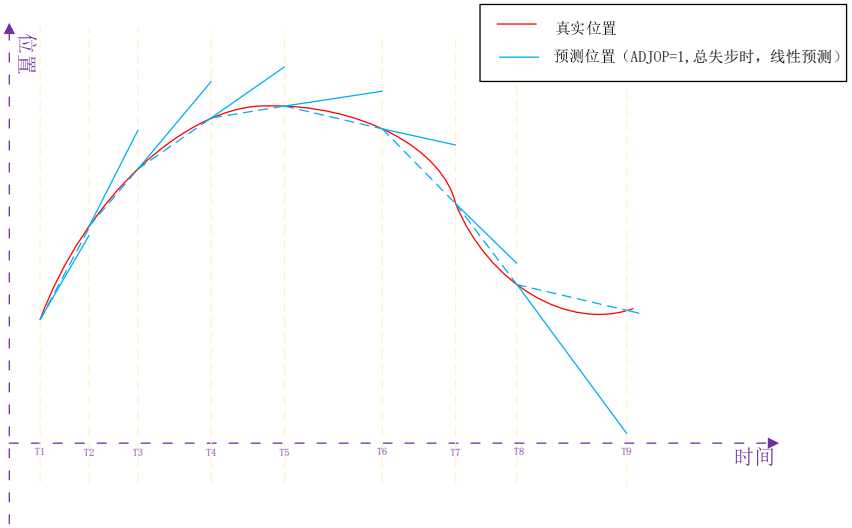


Figure 57: MMC Prediction Waveform (CR[ADJOP]=1, continuous out-of-step)

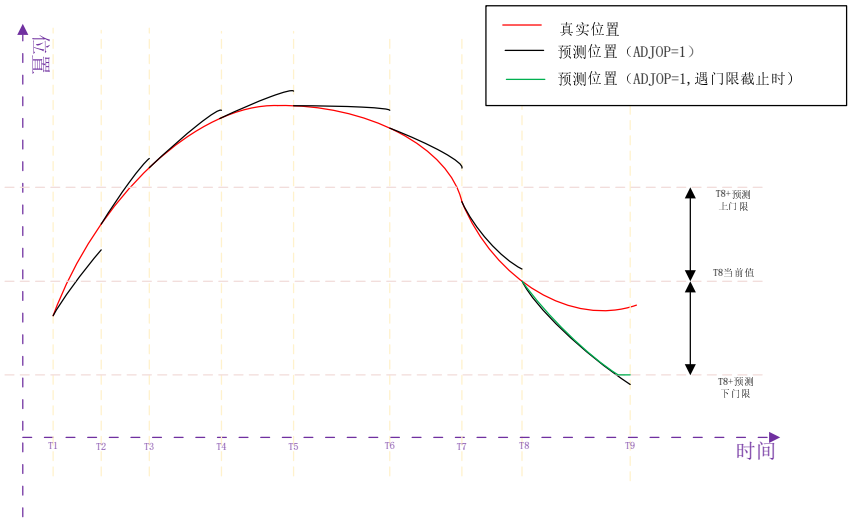


Figure 58: MMC Prediction Waveform (CR[ADJOP]=1, threshold cutoff after time T8)

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Closed-loop mode refers to that the module estimates speed and acceleration according to input time and position, and predicts the new position of the motor in a short period of time. This new position can be used to generate more accurate motor drive waveforms. Closed-loop mode requires both the tracking module and the prediction module to work simultaneously.

For specific configuration method, see Section 38.6.6.

38.3.5.1 Switching from Open-loop to Closed-loop

This is a method to simulate and drive the motor from stationary to moving. By setting a certain acceleration, a series of assumed positions can be generated, so that the subsequent circuit modules can forcibly drive the motor according to these positions; at the same time, the tracking module continuously evaluates the current position, speed and acceleration.

Generally, after the tracking module tracks to a certain speed, you can exit open-loop mode and enter closed-loop mode, and use the tracked position, speed, acceleration and other information to drive the motor.

38.3.6 Initialization of Position, Speed and Acceleration

If the relevant position, speed and acceleration are known before the MMC module runs, they can be configured in advance to reduce convergence time. For specific configuration method, see Section 38.6.2.

38.3.7 Initialization of Increments of Position, Speed and Acceleration (i.e. Position Offset, △_speed, △_acceleration)

Since parameter convergence always takes a certain time, faster convergence speed may be required in some scenarios. In actual driving, you always know whether to accelerate or decelerate, and even can predict the value of speed and/or acceleration increments in some cases. In this case, you can pre-configure the speed and acceleration increments according to the actual situation, which will speed up system convergence. Therefore, speed and acceleration increments are generally configured in the tracker.

Sometimes there is a small fixed deviation between the electrical angle and physical angle of the motor, you can configure the position increment as this deviation to output a more accurate (biased) position. Therefore, position increment is generally configured in the predictor.

For specific configuration method, see Section 38.6.3.

38.3.8 Trigger Mechanism

The trigger mechanism is mainly used to set the time to configure position, speed, acceleration, position increment, speed increment, acceleration increment, etc. Trigger sources include time, position, speed and external trigger signals.

For specific configuration method, see Section 38.6.5.

38.3.9 Internal Data Monitoring

The following registers can be used to monitor internal information:

- Tracker Status
  - The timestamp of estimated data is displayed in the ESTM_TIM register.

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Toreadthisdata,firstsetCR[SHADOW_RD_REQ]=1,thenreadafterCR[SHADOW_RD_REQ]becomes0.

- The position of estimated data is displayed in the ESTM_POS and ESTM_REV registers. Toreadthisdata,firstsetCR[SHADOW_RD_REQ]=1,thenreadafterCR[SHADOW_RD_REQ]becomes0.
- The speed of estimated data is displayed in the ESTM_SPEED register. Toreadthisdata,firstsetCR[SHADOW_RD_REQ]=1,thenreadafterCR[SHADOW_RD_REQ]becomes0.
- The acceleration of estimated data is displayed in the ESTM_ACCEL register. Toreadthisdata,firstsetCR[SHADOW_RD_REQ]=1,thenreadafterCR[SHADOW_RD_REQ]becomes0.
- The acceleration of estimated data is displayed in the CUR_PCOEF register. Toreadthisdata,firstsetCR[SHADOW_RD_REQ]=1,thenreadafterCR[SHADOW_RD_REQ]becomes0.
- The acceleration of estimated data is displayed in the CUR_ICOEF register. Toreadthisdata,firstsetCR[SHADOW_RD_REQ]=1,thenreadafterCR[SHADOW_RD_REQ]becomes0.
- The acceleration of estimated data is displayed in the CUR_ACOEF register. Toreadthisdata,firstsetCR[SHADOW_RD_REQ]=1,thenreadafterCR[SHADOW_RD_REQ]becomes0.
- The latest input of the tracker is displayed in real time in BK0_TIMESTAMP, BK0_POSITION, BK0_REVOLUTION, BK0_SPEED, BK0_ACCELERATOR.
- The second latest input of the tracker is displayed in real time in BK1_TIMESTAMP, BK1_POSITION, BK1_REVOLUTION, BK1_SPEED, BK1_ACCELERATOR.

- Predictor Status
  - The timestamp from the tracking module is displayed in the BR [n] [BR_CUR_POS_TIME] register. Direct reading is supported, no shadow register.
  - The position from the tracking module is displayed in the BR [n] [BR_CUR_POS] and BR [n] [BR_CUR_REV] registers. Direct reading is supported, no shadow register.
  - The speed from the tracking module is displayed in the BR [n] [BR_CUR_SPEED] register. Direct reading is supported, no shadow register.
  - The acceleration from the tracking module is displayed in the BR [n] [BR_CUR_ACCEL] register. Direct reading is supported, no shadow register.

38.3.10 Single and Multiple P, I, A Parameters

In closed-loop mode, the loop convergence speed depends on P, I, A parameters. Too small parameters will lead to too slow convergence and even frequent out-of-step; too large parameters will converge quickly but may cause larger noise and overshoot. For specific configuration method of single parameters, see Section 38.6.4. When the parameter values increase, the impacts are generally shown in Table 178.

Since it is difficult to configure these parameters as universal parameters, it is better to automatically select different coefficients according to different convergence conditions. Therefore, this module also supports multiple P, I, A parameter configuration, see Section 38.6.7 for details.

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Table 178: Impacts of P, I, A Parameters

Response	Rise Time	Overshoot	Settling Time	Steady-state Error
P	Decrease	Increase	Uncertain	Decrease
I	Decrease	Increase	Increase	Eliminate
A	Decrease	Increase	Uncertain	Uncertain

38.3.11 Interrupt

This module supports configuration completion interrupt, trigger interrupt and exception interrupt. The specific description of relevant interrupt sources is shown in Table 179.

Interrupt Event	Event Flag	Enable Control Bit
Tracker Out-of-step	STA[OOSYNC]	INT_EN[OOSYNC_IE]
Tracker Speed Trigger	STA[SPEED_TRG_VALID]	INT_EN[SPEED_TRG_VLD_IE]
Tracker Position Trigger	STA[POS_TRG_VALID]	INT_EN[POS_TRG_VLD_IE]
Tracker Coefficient Initialization Command Completed	STA[INI_COEFS_CMD_DONE]	INT_EN[INI_COEFS_CMD_DONE_IE]
Tracker Position Increment Initialization Command Completed	STA[INI_DELTA_POS_REQ_CMD_DONE]	INT_EN[INI_DELTA_POS_REQ_CMD_DONE_IE]
Tracker Position Initialization Command Completed	STA[INI_POS_REQ_CMD_DONE]	INT_EN[INI_POS_REQ_CMD_DONE_IE]
Predictor 0 Speed Trigger	BR[0][BR_ST[SPEED_TRG_VLD]]	BR[0][BR_CTRL[SPEED_TRG_VLD_IE]]
Predictor 0 Position Trigger	BR[0][BR_ST[POS_TRG_VLD]]	BR[0][BR_CTRL[POS_TRG_VLD_IE]]
Predictor 0 Position Increment Initialization Command Completed	BR[0][BR_ST[INI_DELTA_POS_DONE]]	BR[0][BR_CTRL[INI_DELTA_POS_DONE_IE]]
Predictor 0 Position Initialization Command Completed	STA[INI_BR0_POS_REQ_CMD_DONE]	INT_EN[INI_BR0_POS_REQ_CMD_DONE_IE]
Predictor 1 Speed Trigger	BR[1][BR_ST[SPEED_TRG_VLD]]	BR[1][BR_CTRL[SPEED_TRG_VLD_IE]]
Predictor 1 Position Trigger	BR[1][BR_ST[POS_TRG_VLD]]	BR[1][BR_CTRL[POS_TRG_VLD_IE]]

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Interrupt Event	Event Flag	Enable Control Bit
Predictor 1 Position Increment Initialization Command Completed	BR[1][BR_ST[INI_DELTA_POS_DONE]]	BR[1][BR_CTRL[INI_DELTA_POS_DONE_IE]]
Predictor 1 Position Initialization Command Completed	STA[INI_BR1_POS_REQ_CMD_DONE]	INT_EN[INI_BR1_POS_REQ_CMD_DONE_IE]

Table 179: MMC Interrupt Requests

38.4 MMC Register List

The MMC register list is as follows:

MMC0 base address: 0xF0310000

MMC1 base address: 0xF0314000

Offset	Register Name	Description	Reset Value
0x0000	CR	Control Register	0x00000000
0x0004	STA	Status Register	0x00000020
0x0008	INT_EN	Interrupt Enable Register	0x00000000
0x000C	SYSCLK_FREQ	System Clock Frequency Register	0x00000000
0x0010	SYSCLK_PERIOD	System Clock Period Register	0x00000000
0x0014	OOSYNC_THETA_THR	Position Out-Of-Sync Threshold Regster	0x00000000
0x0018	DISCRETECFG0	Discrete Mode Configuration 0 Register	0x00000000
0x001C	DISCRETECFG1	Discrete Mode Configuration 1 Register	0x00000000
0x0020	CONTCFG0	Continuous Mode Configuration 0 Register	0x00000000
0x0024	INI_POS_TIME	The destined timestamp register for position initialization	0x00000000
0x0028	INI_POS	The destined position register for position initialization	0x00000000
0x002C	INI_REV	The destined revolution register for position initialization	0x00000000
0x0030	INI_SPEED	The destined speed register for position initialization	0x00000000
0x0034	INI_ACCEL	The destined accelerator register for position initialization	0x00000000

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Offset	Register Name	Description	Reset Value
0x0038	INI_COEF_TIME	The destined timestamp register for coefficients initialization	0x00000000
0x003C	INI_PCOEF	The destined coefficient P register for coefficients initialization	0x00000000
0x0040	INI_ICOEF	The destined coefficient I register for coefficients initialization	0x00000000
0x0044	INI_ACOEF	The destined coefficient A register for coefficients initialization	0x00000000
0x0048	ESTM_TIM	The timestamp register for internal estimation	0x00000000
0x004C	ESTM_POS	The position register for the internal estimation	0x00000000
0x0050	ESTM_REV	The revolution register for the internal estimation	0x00000000
0x0054	ESTM_SPEED	The speed register for the internal estimation	0x00000000
0x0058	ESTM_ACCEL	The accelerator register for theinternal estimation	0x00000000
0x005C	CUR_PCOEF	The coefficient P register for the internal estimation	0x00000000
0x0060	CUR_ICOEF	The coefficient I register for the internal estimation	0x00000000
0x0064	CUR_ACOEF	The coefficient A register for the internal estimation	0x00000000
0x0068	INI_DELTA_POS_TIME	The destined timestamp register for delta position initialization	0x00000000
0x006C	INI_DELTA_POS	The destined delta position register for delta position initialization	0x00000000
0x0070	INI_DELTA_REV	The destined delta revolution register for delta position initialization	0x00000000
0x0074	INI_DELTA_SPEED	The destined delta speed register for delta position initialization	0x00000000
0x0078	INI_DELTA_ACCEL	The destined delta accelerator register for delta position initialization	0x00000000

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Offset	Register Name	Description	Reset Value
0x0080	POS_TRG_CFG	Tracking Configuration pos trigger cfg	0x00000000
0x0084	POS_TRG_POS_THR	Tracking Configuration position threshold	0x00000000
0x0088	POS_TRG_REV_THR	Tracking Configuration revolution threshold	0x00000000
0x008C	SPEED_TRG_CFG	Tracking Configuration speed trigger cfg	0x00000000
0x0090	SPEED_TRG_THR	Tracking Configuration speed threshold	0x00000000
0x00A0	COEF_TRG_CFG[0][ERR_THR]	Tracking Configuration coef trigger cfg0	0x00000000
0x00A4	COEF_TRG_CFG[0][P]	Tracking Configuration coef trigger cfg0 P	0x00000000
0x00A8	COEF_TRG_CFG[0][I]	Tracking Configuration coef trigger cfg0 I	0x00000000
0x00AC	COEF_TRG_CFG[0][A]	Tracking Configuration coef trigger cfg0 A	0x00000000
0x00B0	COEF_TRG_CFG[0][TIME]	Tracking Configuration coef trigger cfg0 time	0x00000000
0x00B4	COEF_TRG_CFG[1][ERR_THR]	Tracking Configuration coef trigger cfg1	0x00000000
0x00B8	COEF_TRG_CFG[1][P]	Tracking Configuration coef trigger cfg1 P	0x00000000
0x00BC	COEF_TRG_CFG[1][I]	Tracking Configuration coef trigger cfg1 I	0x00000000
0x00C0	COEF_TRG_CFG[1][A]	Tracking Configuration coef trigger cfg1 A	0x00000000
0x00C4	COEF_TRG_CFG[1][TIME]	Tracking Configuration coef trigger cfg1 time	0x00000000
0x00C8	COEF_TRG_CFG[2][ERR_THR]	Tracking Configuration coef trigger cfg2	0x00000000
0x00CC	COEF_TRG_CFG[2][P]	Tracking Configuration coef trigger cfg2 P	0x00000000
0x00D0	COEF_TRG_CFG[2][I]	Tracking Configuration coef trigger cfg2 I	0x00000000
0x00D4	COEF_TRG_CFG[2][A]	Tracking Configuration coef trigger cfg2 A	0x00000000
0x00D8	COEF_TRG_CFG[2][TIME]	Tracking Configuration coef trigger cfg2 time	0x00000000
0x0100	BR[0][BR_CTRL]	Prediction Control Register	0x00000000

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Offset	Register Name	Description	Reset Value
0x0104	BR[0][BR_TIMEOFF]	Prediction Timing Offset Register	0x00000000
0x0108	BR[0][BR_TRG_PERIOD]	Prediction Triggering Period Offset Register	0x00000000
0x010C	BR[0][BR_TRG_F_TIME]	Prediction Triggering First Offset Register	0x00000000
0x0110	BR[0][BR_ST]	Prediction Status Register	0x00000000
0x0140	BR[0][BR_TRG_POS_CFG]	Prediction Configuration postion trigger cfg	0x00000000
0x0144	BR[0][BR_TRG_POS_THR]	Prediction Configuration postion threshold	0x00000000
0x0148	BR[0][BR_TRG_REV_THR]	Prediction Configuration revolutiom threshold	0x00000000
0x014C	BR[0][BR_TRG_SPEED_CFG]	Prediction Configuration speed trigger cfg	0x00000000
0x0150	BR[0][BR_TRG_SPEED_THR]	Prediction Configuration speed threshold	0x00000000
0x01C0	BR[0][BR_INI_POS_TIME]	Initialization timestamp for open-loop mode	0x00000000
0x01C4	BR[0][BR_INI_POS]	Initialization position for open-loop mode	0x00000000
0x01C8	BR[0][BR_INI_REV]	Initialization revolution for open-loop mode	0x00000000
0x01CC	BR[0][BR_INI_SPEED]	Initialization speed for open-loop mode	0x00000000
0x01D0	BR[0][BR_INI_ACCEL]	Initialization acceleration for open-loop mode	0x00000000
0x01D4	BR[0][BR_INI_DELTA_POS_TIME]	Initialization timestamp for delta mode in prediction mode	0x00000000
0x01D8	BR[0][BR_INI_DELTA_POS]	Initialization delta position for delta mode in prediction mode	0x00000000
0x01DC	BR[0][BR_INI_DELTA_REV]	Initialization delta revolution for delta mode in prediction mode	0x00000000
0x01E0	BR[0][BR_INI_DELTA_SPEED]	Initialization delta speed for delta mode in prediction mode	0x00000000
0x01E4	BR[0][BR_INI_DELTA_ACCEL]	Initialization delta acceleration for delta mode in prediction mode	0x00000000
0x01EC	BR[0][BR_CUR_POS_TIME]	Monitor of the output timestamp	0x00000000
0x01F0	BR[0][BR_CUR_POS]	Monitor of the output position	0x00000000
0x01F4	BR[0][BR_CUR_REV]	Monitor of the output revolution	0x00000000

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Offset	Register Name	Description	Reset Value
0x01F8	BR[0][BR_CUR_SPEED]	Monitor of the output speed	0x00000000
0x01FC	BR[0][BR_CUR_ACCEL]	Monitor of the output acceleration	0x00000000
0x0200	BR[1][BR_CTRL]	Prediction Control Register	0x00000000
0x0204	BR[1][BR_TIMEOFF]	Prediction Timing Offset Register	0x00000000
0x0208	BR[1][BR_TRG_PERIOD]	Prediction Triggering Period Offset Register	0x00000000
0x020C	BR[1][BR_TRG_F_TIME]	Prediction Triggering First Offset Register	0x00000000
0x0210	BR[1][BR_ST]	Prediction Status Register	0x00000000
0x0240	BR[1][BR_TRG_POS_CFG]	Prediction Configuration postion trigger cfg	0x00000000
0x0244	BR[1][BR_TRG_POS_THR]	Prediction Configuration postion threshold	0x00000000
0x0248	BR[1][BR_TRG_REV_THR]	Prediction Configuration revolutiom threshold	0x00000000
0x024C	BR[1][BR_TRG_SPEED_CFG]	Prediction Configuration speed trigger cfg	0x00000000
0x0250	BR[1][BR_TRG_SPEED_THR]	Prediction Configuration speed threshold	0x00000000
0x02C0	BR[1][BR_INI_POS_TIME]	Initialization timestamp for open-loop mode	0x00000000
0x02C4	BR[1][BR_INI_POS]	Initialization position for open-loop mode	0x00000000
0x02C8	BR[1][BR_INI_REV]	Initialization revolution for open-loop mode	0x00000000
0x02CC	BR[1][BR_INI_SPEED]	Initialization speed for open-loop mode	0x00000000
0x02D0	BR[1][BR_INI_ACCEL]	Initialization acceleration for open-loop mode	0x00000000
0x02D4	BR[1][BR_INI_DELTA_POS_TIME]	Initialization timestamp for delta mode in prediction mode	0x00000000
0x02D8	BR[1][BR_INI_DELTA_POS]	Initialization delta position for delta mode in prediction mode	0x00000000
0x02DC	BR[1][BR_INI_DELTA_REV]	Initialization delta revolution for delta mode in prediction mode	0x00000000
0x02E0	BR[1][BR_INI_DELTA_SPEED]	Initialization delta speed for delta mode in prediction mode	0x00000000

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Offset	Register Name	Description	Reset Value
0x02E4	BR[1][BR_INI_DELTA_ACCEL]	Initialization delta acceleration for delta mode in prediction mode	0x00000000
0x02EC	BR[1][BR_CUR_POS_TIME]	Monitor of the output timestamp	0x00000000
0x02F0	BR[1][BR_CUR_POS]	Monitor of the output position	0x00000000
0x02F4	BR[1][BR_CUR_REV]	Monitor of the output revolution	0x00000000
0x02F8	BR[1][BR_CUR_SPEED]	Monitor of the output speed	0x00000000
0x02FC	BR[1][BR_CUR_ACCEL]	Monitor of the output acceleration	0x00000000
0x0300	BK0_TIMESTAMP	Monitor of the just received input timestamp for tracing logic	0x00000000
0x0304	BK0_POSITION	Monitor of the just received input position for tracing logic	0x00000000
0x0308	BK0_REVOLUTION	Monitor of the just received input revolution for tracing logic	0x00000000
0x030C	BK0_SPEED	Monitor of the just received input speed for tracing logic	0x00000000
0x0310	BK0_ACCELERATOR	Monitor of the just received input acceleration for tracing logic	0x00000000
0x0320	BK1_TIMESTAMP	Monitor of the previous received input timestamp for tracing logic	0x00000000
0x0324	BK1_POSITION	Monitor of the previous received input position for tracing logic	0x00000000
0x0328	BK1_REVOLUTION	Monitor of the previous received input revolution for tracing logic	0x00000000
0x032C	BK1_SPEED	Monitor of the previous received input speed for tracing logic	0x00000000
0x0330	BK1_ACCELERATOR	Monitor of the previous received input acceleration for tracing logic	0x00000000

Table 180: MMC Register List

38.5 MMC Register Details

The detailed description of MMC registers is as follows:

38.5.1 CR (0x0)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SFTRST	RSVD	INI_BR0_POS_REQ	INI_BR1_POS_REQ	FRCACCELZERO	MS_COEF_EN	INI_DELTA_POS_TRG_TYPE			INI_POS_TRG_TYPE			INI_DELTA_POS_CMD_MSK				INI_DELTA_POS_REQ	OPEN_LOOP_MODE	POS_TYPE	INI_POS_CMD_MSK				INI_POS_REQ	INI_COEFS_CMD_MSK			INI_COEFS_CMD	SHADOW_RD_REQ	ADJOP	DISCRETETRC	MOD_EN
RW	N/A	RW	RW	RW	RW	RW			RW			RW				RW	RW	RW	RW				RW	RW			RW	RW	RW	RW	RW
0	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CR [31:0]

Bits	Name	Description
31	SFTRST	Software reset, high active. When write 1 ,all internal logical will be reset. 0b - No action 1b - All MMC internal registers are forced into their reset state. Interface registers are not affected.
29	INI_BR0_POS_REQ	Auto clear. Only effective in open_loop mode.
28	INI_BR1_POS_REQ	Auto clear. Only effective in open_loop mode.
27	FRCACCELZERO	Zeroise the accelerator calculation.
26	MS_COEF_EN	Multiple Coefficients Enable
25-23	INI_DELTA_POS_TRG_TYPE	0: Time Stamp in the configuration 1: Risedge of In Trg[0] 2: Risedge of In Trg[1] 3: Risedge of out trg[0] 4: Risedge of out trg[1] 5: triggered by self position trigger 6: triggered by self speed trigger Otherser: no function
22-20	INI_POS_TRG_TYPE	0: Time Stamp in the configuration 1: Risedge of In Trg[0] 2: Risedge of In Trg[1] 3: Risedge of out trg[0] 4: Risedge of out trg[1] 5: triggered by self position trigger 6: triggered by self speed trigger Otherser: no function

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Bits	Name	Description
19-16	INI_DELTA_POS_CMD_MSK	1: change 0: won't change bit 3: for delta accel bit 2: for delta speed bit 1: for delta revolution bit 0: for delta position
15	INI_DELTA_POS_REQ	1: Command to reload the delta pos. Auto clear 0:
14	OPEN_LOOP_MODE	1: in open loop mode 0: not in open loop mode
13	POS_TYPE	1: 32-bit for rev+pos, with each element occupying 16 bits 0: 32-bit for rev, and 32 bit for pos When CR[MANUAL_IO]==1, 1: means that the INI_POS is acting as INI_POS cmds 0: means that the INI_POS is simulating the input of iposition and itimestamp
12-9	INI_POS_CMD_MSK	1: change 0: won't change bit 3: for accel bit 2: for speed bit 1: for revolution bit 0: for position
8	INI_POS_REQ	1: Command to reload the positions. Auto clear 0:
7-5	INI_COEFS_CMD_MSK	1: change 0: won't change bit 2: for ACOEF bit 1: for ICOEF bit 0: for PCOEF
4	INI_COEFS_CMD	1: Command to reload the coefs. Auto clear 0:
3	SHADOW_RD_REQ	1: Shadow Request for read of tracking parameters. Auto clear 0:
2	ADJOP	1: use the input iposition whenever a new iposition comes, and force the predicted output stop at the boundaries. 0: Continuous tracking mode, without any boundary check
1	DISCRETETRC	1: Discrete position input 0: Continuous position input
0	MOD_EN	Module Enable

CR Bits

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38.5.2 STA (0x4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ERR_ID				RSVD																	SPEED_TRG_VALID	POS_TRG_VALID	INI_DELTA_POS_REQ_CMD_DONE	INI_BR0_POS_REQ_CMD_DONE	INI_BR1_POS_REQ_CMD_DONE	IDLE	OOSYNC	RSVD	INI_POS_REQ_CMD_DONE	INI_COEFS_CMD_DONE	SHADOW_RD_DONE
RO				N/A																	W1C	W1C	W1C	W1C	W1C	RO	W1C	N/A	W1C	W1C	RO
0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	1	0	x	0	0	0

STA [31:0]

Bits	Name	Description
31-28	ERR_ID	Tracking ERR_ID
10	SPEED_TRG_VALID	W1C
9	POS_TRG_VALID	W1C
8	INI_DELTA_POS_REQ_CMD_DONE	W1C
7	INI_BR0_POS_REQ_CMD_DONE	W1C
6	INI_BR1_POS_REQ_CMD_DONE	W1C
5	IDLE	Tracking Module in Idle status
4	OOSYNC	Tracking module out-of sync. W1C
2	INI_POS_REQ_CMD_DONE	W1C
1	INI_COEFS_CMD_DONE	W1C
0	SHADOW_RD_DONE	Shadow ready for read. Auto cleared by setting CR[SHADOW_RD_REQ] as 1

STA Bits

38.5.3 INT_EN (0x8)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																					SPEED_TRG_VLD_IE	POS_TRG_VLD_IE	INI_DELTA_POS_REQ_CMD_DONE_IE	INI_BR0_POS_REQ_CMD_DONE_IE	INI_BR1_POS_REQ_CMD_DONE_IE	RSVD	OOSYNC_IE	RSVD	INI_POS_REQ_CMD_DONE_IE	INI_COEFS_CMD_DONE_IE	SHADOW_RD_DONE_IE
N/A																					RW	RW	RW	RW	RW	N/A	RW	N/A	RW	RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	x	0	x	0	0	0

INT_EN [31:0]

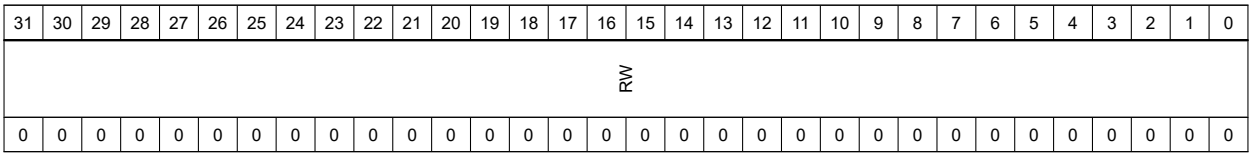
Bits	Name	Description
10	SPEED_TRG_VLD_IE	Interrupt Enable for SPEED_TRG_VALID
9	POS_TRG_VLD_IE	Interrupt Enable for POS_TRG_VALID
8	INI_DELTA_POS_REQ_CMD_DONE_IE	Interrupt Enable for INI_DELTA_POS_REQ_CMD_DONE
7	INI_BR0_POS_REQ_CMD_DONE_IE	Interrupt Enable for INI_BR0_POS_REQ_CMD_DONE
6	INI_BR1_POS_REQ_CMD_DONE_IE	Interrupt Enable for INI_BR1_POS_REQ_CMD_DONE
4	OOSYNC_IE	Interrupt Enable for OOSYNC
2	INI_POS_REQ_CMD_DONE_IE	Interrupt Enable for INI_POS_REQ_CMD_DONE
1	INI_COEFS_CMD_DONE_IE	Interrupt Enable for INI_COEFS_CMD_DONE
0	SHADOW_RD_DONE_IE	Interrupt Enable for SHADOW_RD_DONE

INT_EN Bits

38.5.4 SYSCLK_FREQ (0xC)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															

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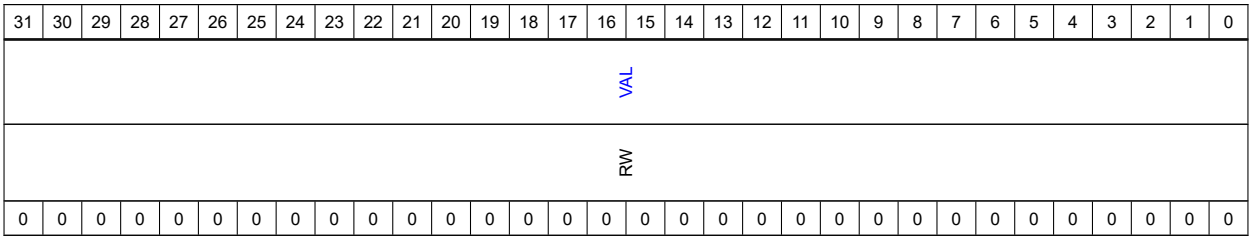


SYSCLK_FREQ [31:0]

Bits	Name	Description
31-0	VAL	system clock frequency, ufix<32, 0>

SYSCLK_FREQ Bits

38.5.5 SYSCLK_PERIOD (0x10)

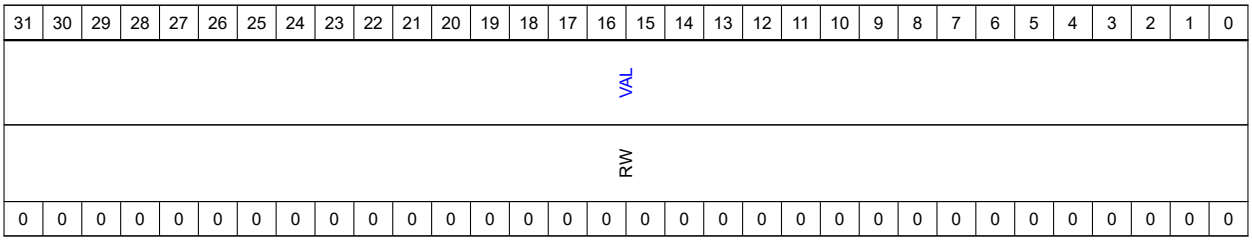


SYSCLK_PERIOD [31:0]

Bits	Name	Description
31-0	VAL	round(the value of clock period * (2^24)*(2^20)), ufix<32, 0>

SYSCLK_PERIOD Bits

38.5.6 OOSYNC_THETA_THR (0x14)



OOSYNC_THETA_THR [31:0]

Bits	Name	Description
31-0	VAL	the threshold of theta difference between actual and prediction for out-of-sync determination，ufix<32, 32>

OOSYNC_THETA_THR Bits

38.5.7 DISCRETECFG0 (0x18)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD												POSMAX																			
N/A												RW																			
x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DISCRETECFG0 [31:0]

Bits	Name	Description
19-0	POSMAX	Max ID Of Lines. For example-1, for 512 lines, it is 511. ufix<32, 0>

DISCRETECFG0 Bits

38.5.8 DISCRETECFG1 (0x1C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INV_POSMAX																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DISCRETECFG1 [31:0]

Bits	Name	Description
31-0	INV_POSMAX	discrete mode: ufix<32, 0> of 1/(Number Of Lines) continuous mode: the max delta for tracking from the last received position, ufix<32, 32>

DISCRETECFG1 Bits

38.5.9 CONTCFG0 (0x20)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HALF_CIRC_THETA																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CONTCFG0 [31:0]

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Bits	Name	Description
31-0	HALF_CIRC_THETA	the theta for cal the clockwise or anticlockwise rotation between two adjacent inputs, ufix<32, 32>

CONTCFG0 Bits

38.5.10 INI_POS_TIME (0x24)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

INI_POS_TIME [31:0]

Bits	Name	Description
31-0	VAL	indicate the time to change the values. 0: instant change

INI_POS_TIME Bits

38.5.11 INI_POS (0x28)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

INI_POS [31:0]

Bits	Name	Description
31-0	VAL	the value； continuous mode: ufix<32, 32>

INI_POS Bits

38.5.12 INI_REV (0x2C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

INI_REV [31:0]

Bits	Name	Description
31-0	VAL	the value； continuous mode: ufix<32, 0>

INI_REV Bits

38.5.13 INI_SPEED (0x30)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

INI_SPEED [31:0]

Bits	Name	Description
31-0	VAL	the value; continuous mode: fix<32, 19>

INI_SPEED Bits

38.5.14 INI_ACCEL (0x34)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

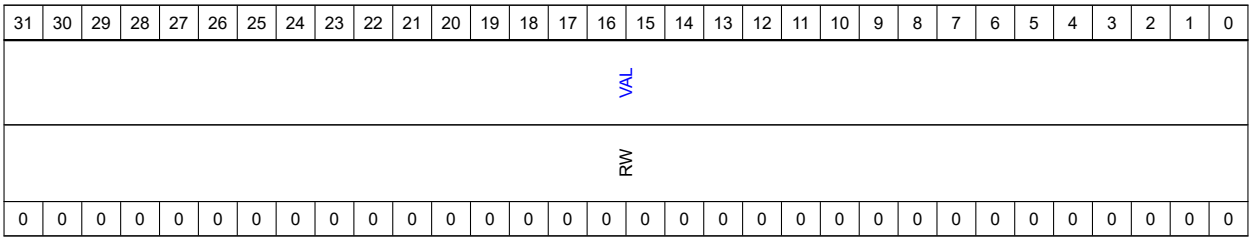
INI_ACCEL [31:0]

Bits	Name	Description
31-0	VAL	the value continuous mode: fix<32, 19>

INI_ACCEL Bits

38.5.15 INI_COEF_TIME (0x38)

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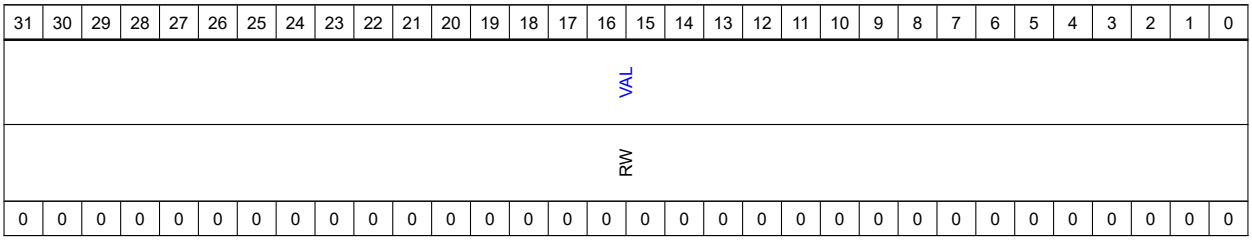


INI_COEF_TIME [31:0]

Bits	Name	Description
31-0	VAL	indicate the time to change the values. 0: instant change

INI_COEF_TIME Bits

38.5.16 INI_PCOEF (0x3C)

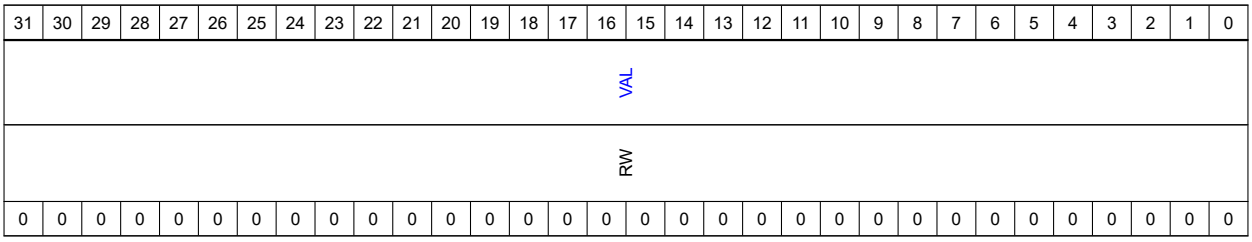


INI_PCOEF [31:0]

Bits	Name	Description
31-0	VAL	the value, fix<32, 15>

INI_PCOEF Bits

38.5.17 INI_ICOEF (0x40)



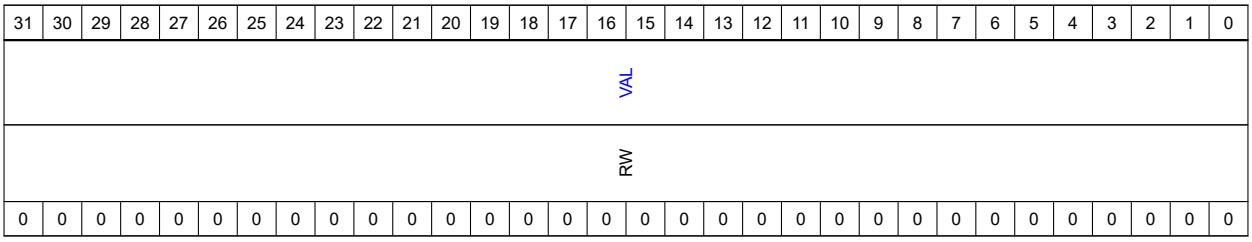
INI_ICOEF [31:0]

Bits	Name	Description
31-0	VAL	the value, fix<32, 21>

INI_ICOEF Bits

38.5.18 INI_ACOEF (0x44)

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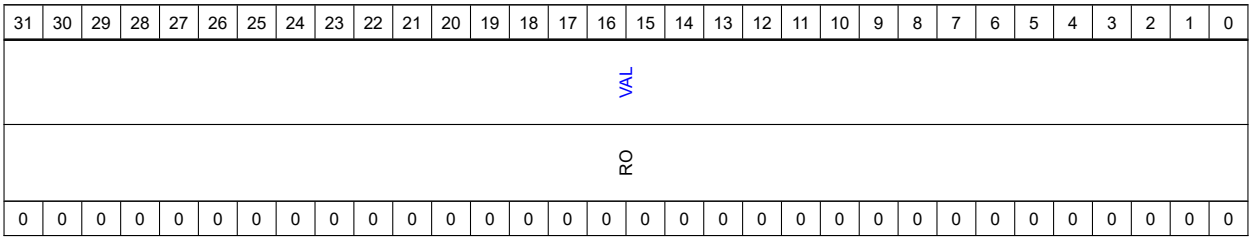


INI_ACOEF [31:0]

Bits	Name	Description
31-0	VAL	the value, fix<32, 19>

INI_ACOEF Bits

38.5.19 ESTM_TIM (0x48)

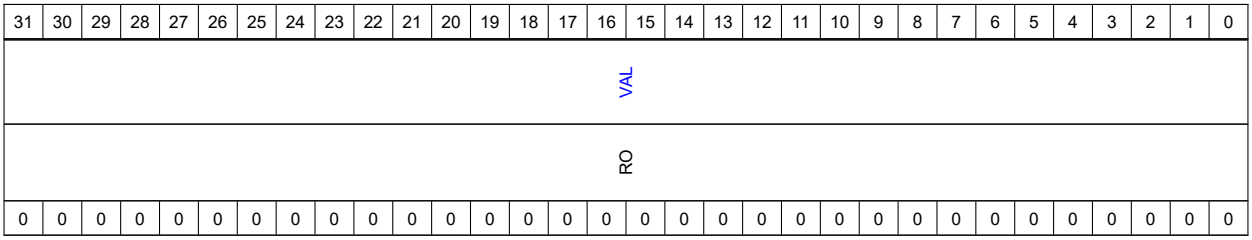


ESTM_TIM [31:0]

Bits	Name	Description
31-0	VAL	the value

ESTM_TIM Bits

38.5.20 ESTM_POS (0x4C)



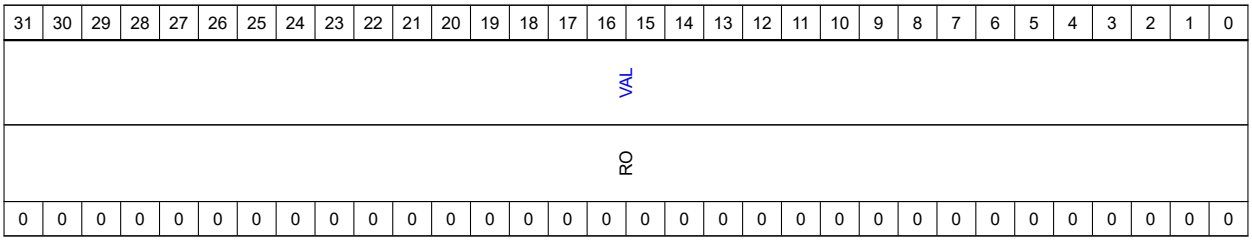
ESTM_POS [31:0]

Bits	Name	Description
31-0	VAL	the value

ESTM_POS Bits

38.5.21 ESTM_REV (0x50)

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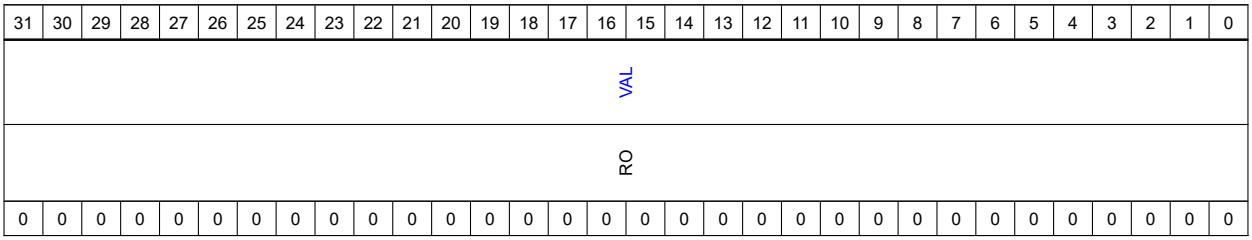


ESTM_REV [31:0]

Bits	Name	Description
31-0	VAL	the value

ESTM_REV Bits

38.5.22 ESTM_SPEED (0x54)

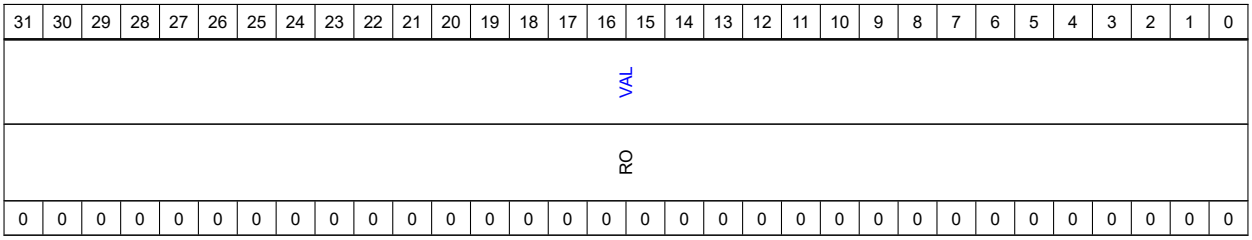


ESTM_SPEED [31:0]

Bits	Name	Description
31-0	VAL	the value

ESTM_SPEED Bits

38.5.23 ESTM_ACCEL (0x58)



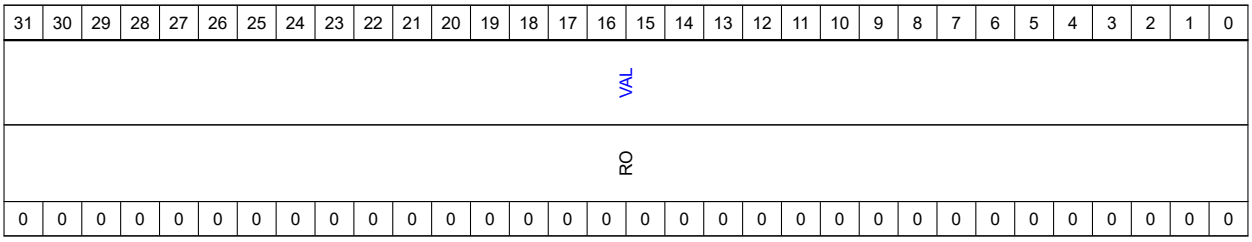
ESTM_ACCEL [31:0]

Bits	Name	Description
31-0	VAL	the value

ESTM_ACCEL Bits

38.5.24 CUR_PCOEF (0x5C)

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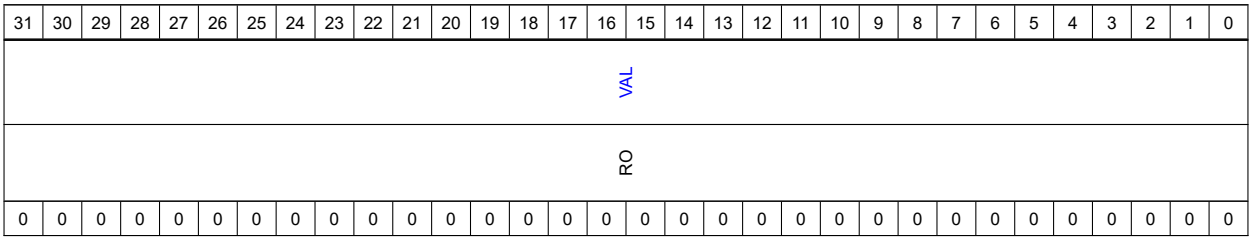


CUR_PCOEF [31:0]

Bits	Name	Description
31-0	VAL	the value

CUR_PCOEF Bits

38.5.25 CUR_ICOEF (0x60)

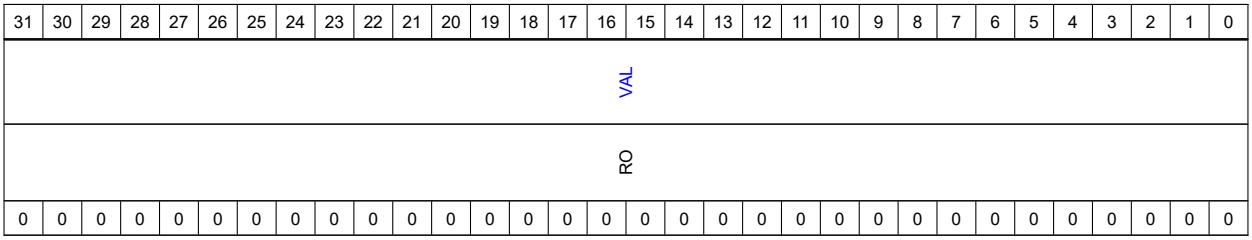


CUR_ICOEF [31:0]

Bits	Name	Description
31-0	VAL	the value

CUR_ICOEF Bits

38.5.26 CUR_ACOEF (0x64)



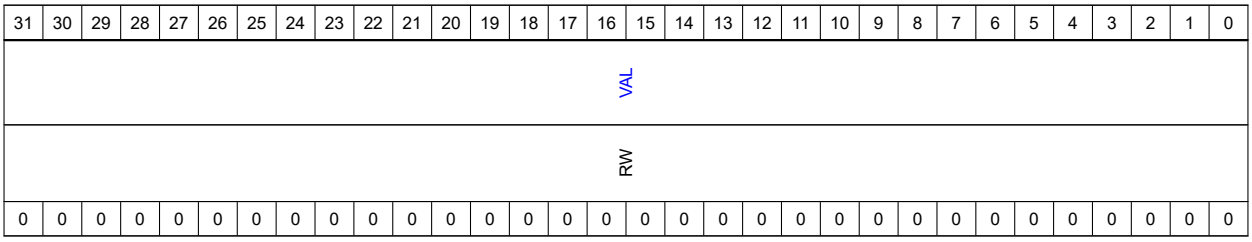
CUR_ACOEF [31:0]

Bits	Name	Description
31-0	VAL	the value

CUR_ACOEF Bits

38.5.27 INI_DELTA_POS_TIME (0x68)

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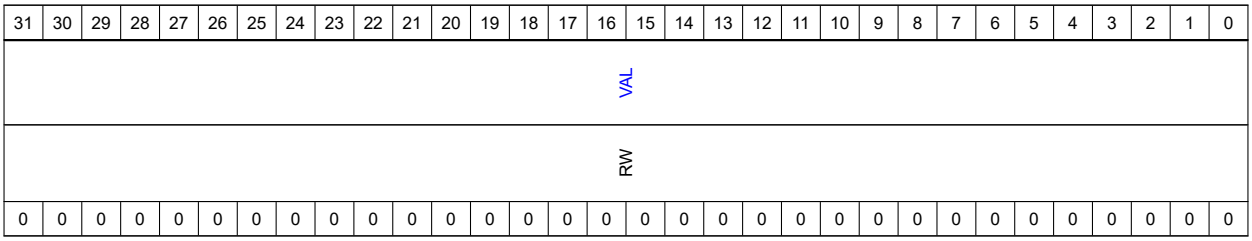


INI_DELTA_POS_TIME [31:0]

Bits	Name	Description
31-0	VAL	indicate the time to change the values. 0: instant change

INI_DELTA_POS_TIME Bits

38.5.28 INI_DELTA_POS (0x6C)

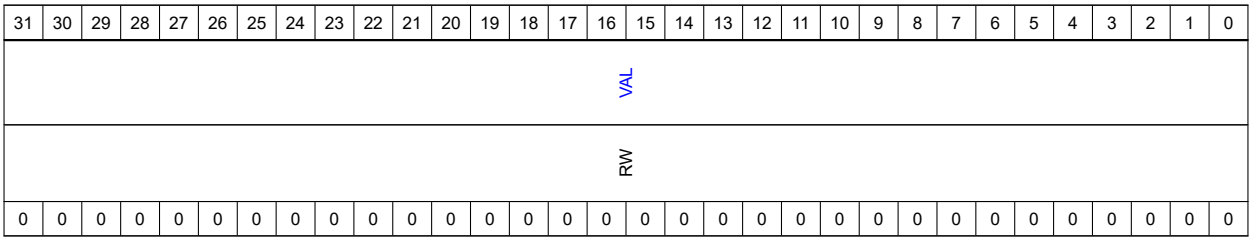


INI_DELTA_POS [31:0]

Bits	Name	Description
31-0	VAL	the value continuous mode: ufix <32, 32>

INI_DELTA_POS Bits

38.5.29 INI_DELTA_REV (0x70)



INI_DELTA_REV [31:0]

Bits	Name	Description
31-0	VAL	the value continuous mode: fix<32, 0>

INI_DELTA_REV Bits

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38.5.30 INI_DELTA_SPEED (0x74)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

INI_DELTA_SPEED [31:0]

Bits	Name	Description
31-0	VAL	the value； continuous mode: fix<32, 19>

INI_DELTA_SPEED Bits

38.5.31 INI_DELTA_ACCEL (0x78)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

INI_DELTA_ACCEL [31:0]

Bits	Name	Description
31-0	VAL	the value continuous mode: fix<32, 19>

INI_DELTA_ACCEL Bits

38.5.32 POS_TRG_CFG (0x80)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																														EDGE	EN
N/A																														RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0

POS_TRG_CFG [31:0]

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Bits	Name	Description
1	EDGE	0: (rising edge) pos inc greater than, 1: (falling edge) pos dec less than
0	EN	1-trigger valid; 0-Trigger not valid”

POS_TRG_CFG Bits

38.5.33 POS_TRG_POS_THR (0x84)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

POS_TRG_POS_THR [31:0]

Bits	Name	Description
31-0	VAL	For pos out trigger (pos). ufix<32, 32>

POS_TRG_POS_THR Bits

38.5.34 POS_TRG_REV_THR (0x88)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

POS_TRG_REV_THR [31:0]

Bits	Name	Description
31-0	VAL	For pos out trigger (rev) fix<32, 0>

POS_TRG_REV_THR Bits

38.5.35 SPEED_TRG_CFG (0x8C)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																													COMP_TYPE	EDGE	EN
N/A																													RW	RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0

SPEED_TRG_CFG [31:0]

Bits	Name	Description
2	COMP_TYPE	1: Use abs value for comparion. 0: Use the speed with direction info (so not the abs value)
1	EDGE	0: (rising edge) speed inc greater than, 1: (falling edge) speed dec less than
0	EN	1-trigger valid; 0-Trigger not valid Normally it means either the max pos speed, or the min negative speed.

SPEED_TRG_CFG Bits

38.5.36 SPEED_TRG_THR (0x90)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

SPEED_TRG_THR [31:0]

Bits	Name	Description
31-0	VAL	For speed trigger. continuous mode: fix<32, 19>

SPEED_TRG_THR Bits

38.5.37 COEF_TRG_CFG[ERR_THR] (0xA0 + 0x14 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

COEF_TRG_CFG[ERR_THR] [31:0]

Bits	Name	Description
31-0	VAL	ErrThr0: Error Threshold 0, (abs(tracking error)>= will choose the coefs as below) Note: ErrThr0>ErrThr1>ErrThr2 ufix<31, 28>

COEF_TRG_CFG[ERR_THR] Bits

38.5.38 COEF_TRG_CFG[P] (0xA4 + 0x14 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

COEF_TRG_CFG[P] [31:0]

Bits	Name	Description
31-0	VAL	P0_Coef, fix<32, 15>

COEF_TRG_CFG[P] Bits

38.5.39 COEF_TRG_CFG[I] (0xA8 + 0x14 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

COEF_TRG_CFG[I] [31:0]

Bits	Name	Description
31-0	VAL	I0_Coef, fix<32, 21>

COEF_TRG_CFG[I] Bits

38.5.40 COEF_TRG_CFG[A] (0xAC + 0x14 * n)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

COEF_TRG_CFG[A] [31:0]

Bits	Name	Description
31-0	VAL	A0_Coef，fix<32, 19>

COEF_TRG_CFG[A] Bits

38.5.41 COEF_TRG_CFG[TIME] (0xB0 + 0x14 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

COEF_TRG_CFG[TIME] [31:0]

Bits	Name	Description
31-0	VAL	CoefTime0: Time Stayed using this coefs (counted in input samples). Ideal value of tracing cycles should +1. ufix<32,0>

COEF_TRG_CFG[TIME] Bits

38.5.42 BR[BR_CTRL] (0x100 + 0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD	SPEED_TRG_VALID_IE	POS_TRG_VALID_IE	RSVD			INI_POS_TRG_TYPE			RSVD	INI_POS_CMD_MSK				RSVD	INI_DELTA_POS_TRG_TYPE			INI_DELTA_POS_DONE_IE	INI_DELTA_POS_CMD_MSK				INI_DELTA_POS_REQ	OPEN_LOOP_MODE	RSVD		PRED_MODE		RSVD	NF_TRG_TYPE	F_TRG_TYPE	BR_EN
N/A	RW	RW	N/A			RW			N/A	RW				N/A	RW			RW	RW				RW	RW	N/A		RW		N/A	RW	RW	RW
x	0	0	x	x	x	0	0	0	x	0	0	0	0	x	0	0	0	0	0	0	0	0	0	0	x	0	0	x	0	0	0

BR[BR_CTRL] [31:0]

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Bits	Name	Description
30	SPEED_TRG_VALID_IE	Interrupt Enable for SPEED_TRG_VALID
29	POS_TRG_VALID_IE	Interrupt Enable for POS_TRG_VALID
25-23	INI_POS_TRG_TYPE	0: Time Stamp in the configuration 1: Risedge of In Trg[0] 2: Risedge of In Trg[1] 3: Risedge of out trg[0] 4: Risedge of out trg[1] 5: Risedge of self pos trigger 6: Risedge of self speed trigger Others: no function
21-18	INI_POS_CMD_MSK	1: change 0: won't change bit 3: for accel bit 2: for speed bit 1: for revolution bit 0: for position
16-14	INI_DELTA_POS_TRG_TYPE	0: Time Stamp in the configuration 1: Risedge of In Trg[0] 2: Risedge of In Trg[1] 3: Risedge of out trg[0] 4: Risedge of out trg[1] 5: Risedge of self pos trigger 6: Risedge of self speed trigger Others: no function
13	INI_DELTA_POS_DONE_IE	Interrupt Enable for INI_DELTA_POS_DONE
12-9	INI_DELTA_POS_CMD_MSK	1: change 0: won't change bit 3: for delta accel bit 2: for delta speed bit 1: for delta revolution bit 0: for delta position
8	INI_DELTA_POS_REQ	1: Command to reload the delta pos. Auto clear 0:
7	OPEN_LOOP_MODE	1: in open loop mode 0: not in open loop mode
5-4	PRED_MODE	1:continuously repeat pred, 0:cal the pred based on a definite time-stamp offset, 2:programed one-shot prediction mode

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Bits	Name	Description
2	NF_TRG_TYPE	1. Each non-first trigger by external trigger pin 0. Each non-first trigger by the timer
1	F_TRG_TYPE	1. First trigger by external trigger pin 0. First trigger by the timer When in CR[MANUAL_IO]=1 mode, it is the prediction trigger
0	BR_EN	Branch Enable

BR[BR_CTRL] Bits

38.5.43 BR[BR_TIMEOFF] (0x104 + 0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

BR[BR_TIMEOFF] [31:0]

Bits	Name	Description
31-0	VAL	ufix<32, 0> time offset incycles from the trigger time

BR[BR_TIMEOFF] Bits

38.5.44 BR[BR_TRG_PERIOD] (0x108 + 0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

BR[BR_TRG_PERIOD] [31:0]

Bits	Name	Description
31-0	VAL	uifx<32, 0>, time offset incycles between each trigger time

BR[BR_TRG_PERIOD] Bits

38.5.45 BR[BR_TRG_F_TIME] (0x10C + 0x100 * n)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

BR[BR_TRG_F_TIME] [31:0]

Bits	Name	Description
31-0	VAL	uifx<32, 0> the time for the first trigger

BR[BR_TRG_F_TIME] Bits

38.5.46 BR[BR_ST] (0x110 + 0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																					OPEN_LOOP_ST	SPEED_TRG_VLD	POS_TRG_VLD	RSVD	INI_DELTA_POS_DONE	IDLE	RSVD	ERR_ID			
N/A																					RO	W1C	W1C	N/A	W1C	RO	N/A	RO			
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	x	0	1	x	0	0	0	0

BR[BR_ST] [31:0]

Bits	Name	Description
10	OPEN_LOOP_ST	1：in open loop mode 0：in closed loop mode
9	SPEED_TRG_VLD	1：self speed trigger event found 0：self speed trigger event not found yet
8	POS_TRG_VLD	1：self position trigger event found 0：self position trigger event not found yet
6	INI_DELTA_POS_DONE	1: the initialization of delta position command is done 0: the initialization of delta position command is not done
5	IDLE	1: The prediction module is idle. 0: The prediction module is not idle.
3-0	ERR_ID	The module’s error ID output

BR[BR_ST] Bits

38.5.47 BR[BR_TRG_POS_CFG] (0x140 + 0x100 * n)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																														EDGE	EN
N/A																														RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0

BR[BR_TRG_POS_CFG] [31:0]

Bits	Name	Description
1	EDGE	bit1: 0: (rising edge) pos inc greater than, 1: (falling edge) pos dec less than
0	EN	1-trigger valid; 0-Trigger not valid

BR[BR_TRG_POS_CFG] Bits

38.5.48 BR[BR_TRG_POS_THR] (0x144 + 0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

BR[BR_TRG_POS_THR] [31:0]

Bits	Name	Description
31-0	VAL	For pos out trigger (pos). ufix<32, 32>

BR[BR_TRG_POS_THR] Bits

38.5.49 BR[BR_TRG_REV_THR] (0x148 + 0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

BR[BR_TRG_REV_THR] [31:0]

Bits	Name	Description
31-0	VAL	For pos out trigger (rev) ufix<32, 0>

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Bits	Name	Description

BR[BR_TRG_REV_THR] Bits

38.5.50 BR[BR_TRG_SPEED_CFG] (0x14C + 0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																													COMP_TYPE	EDGE_SEL	EN
N/A																													RW	RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0

BR[BR_TRG_SPEED_CFG] [31:0]

Bits	Name	Description
2	COMP_TYPE	Use abs value for comparion. 0: Use the speed with direction info (so not the abs value)
1	EDGE_SEL	0: (rising edge) speed inc greater than, 1: (falling edge) speed dec less than
0	EN	1-trigger valid; 0-Trigger not valid Normally it means either the max pos speed, or the min negative speed.

BR[BR_TRG_SPEED_CFG] Bits

38.5.51 BR[BR_TRG_SPEED_THR] (0x150 + 0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

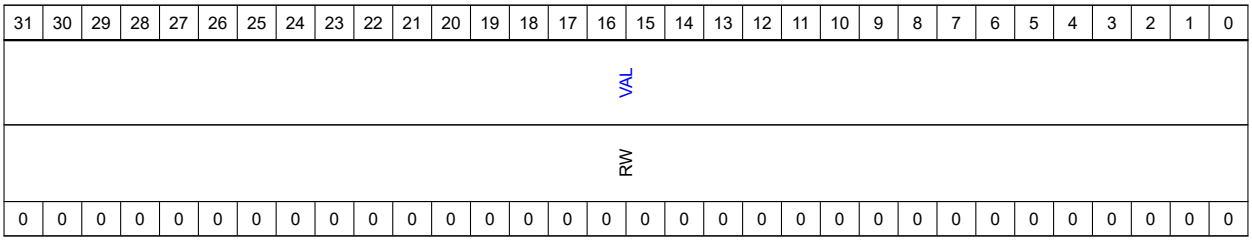
BR[BR_TRG_SPEED_THR] [31:0]

Bits	Name	Description
31-0	VAL	For speed trigger. continuous mode: fix<32, 19>

BR[BR_TRG_SPEED_THR] Bits

38.5.52 BR[BR_INI_POS_TIME] (0x1C0 + 0x100 * n)

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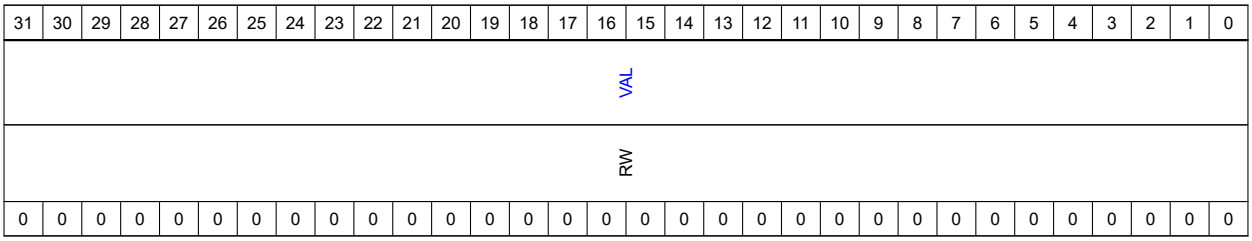


BR[BR_INI_POS_TIME] [31:0]

Bits	Name	Description
31-0	VAL	indicate the time to change the values. 0: instant change

BR[BR_INI_POS_TIME] Bits

38.5.53 BR[BR_INI_POS] (0x1C4 + 0x100 * n)

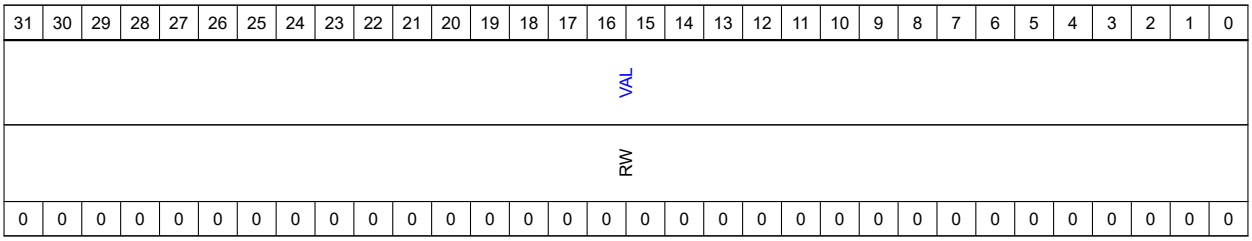


BR[BR_INI_POS] [31:0]

Bits	Name	Description
31-0	VAL	the value ufix<32, 32>

BR[BR_INI_POS] Bits

38.5.54 BR[BR_INI_REV] (0x1C8 + 0x100 * n)



BR[BR_INI_REV] [31:0]

Bits	Name	Description
31-0	VAL	the value ufix<32, 0>

BR[BR_INI_REV] Bits

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38.5.55 BR[BR_INI_SPEED] (0x1CC + 0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

BR[BR_INI_SPEED] [31:0]

Bits	Name	Description
31-0	VAL	the value fix<32, 19>

BR[BR_INI_SPEED] Bits

38.5.56 BR[BR_INI_ACCEL] (0x1D0 + 0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

BR[BR_INI_ACCEL] [31:0]

Bits	Name	Description
31-0	VAL	the value continuous mode: fix<32, 19>

BR[BR_INI_ACCEL] Bits

38.5.57 BR[BR_INI_DELTA_POS_TIME] (0x1D4 + 0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

BR[BR_INI_DELTA_POS_TIME] [31:0]

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Bits	Name	Description
31-0	VAL	indicate the time to change the values. 0: instant change

BR[BR_INI_DELTA_POS_TIME] Bits

38.5.58 BR[BR_INI_DELTA_POS] (0x1D8 + 0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

BR[BR_INI_DELTA_POS] [31:0]

Bits	Name	Description
31-0	VAL	the value continuous mode: ufix<32, 32>

BR[BR_INI_DELTA_POS] Bits

38.5.59 BR[BR_INI_DELTA_REV] (0x1DC + 0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

BR[BR_INI_DELTA_REV] [31:0]

Bits	Name	Description
31-0	VAL	the value continuous mode: fix<32, 0>

BR[BR_INI_DELTA_REV] Bits

38.5.60 BR[BR_INI_DELTA_SPEED] (0x1E0 + 0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

BR[BR_INI_DELTA_SPEED] [31:0]

Bits	Name	Description
31-0	VAL	the value continuous mode: fix<32, 19>

BR[BR_INI_DELTA_SPEED] Bits

38.5.61 BR[BR_INI_DELTA_ACCEL] (0x1E4 + 0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

BR[BR_INI_DELTA_ACCEL] [31:0]

Bits	Name	Description
31-0	VAL	the value continuous mode: fix<32, 19>

BR[BR_INI_DELTA_ACCEL] Bits

38.5.62 BR[BR_CUR_POS_TIME] (0x1EC + 0x100 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

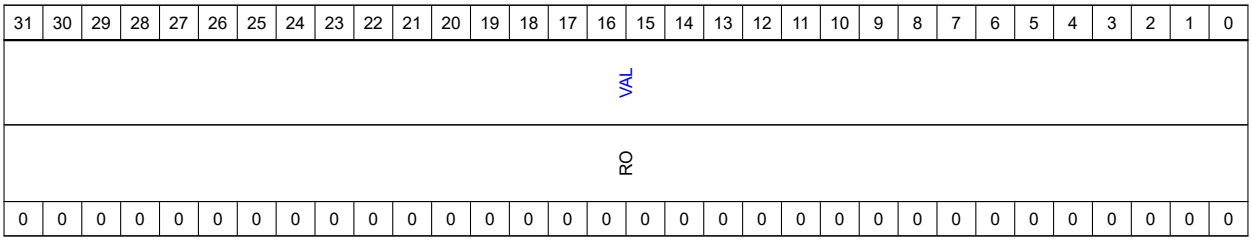
BR[BR_CUR_POS_TIME] [31:0]

Bits	Name	Description
31-0	VAL	the value

BR[BR_CUR_POS_TIME] Bits

38.5.63 BR[BR_CUR_POS] (0x1F0 + 0x100 * n)

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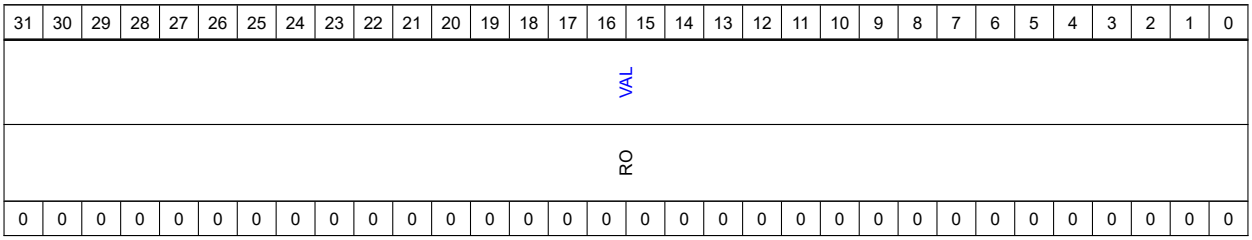


BR[BR_CUR_POS] [31:0]

Bits	Name	Description
31-0	VAL	the value

BR[BR_CUR_POS] Bits

38.5.64 BR[BR_CUR_REV] (0x1F4 + 0x100 * n)

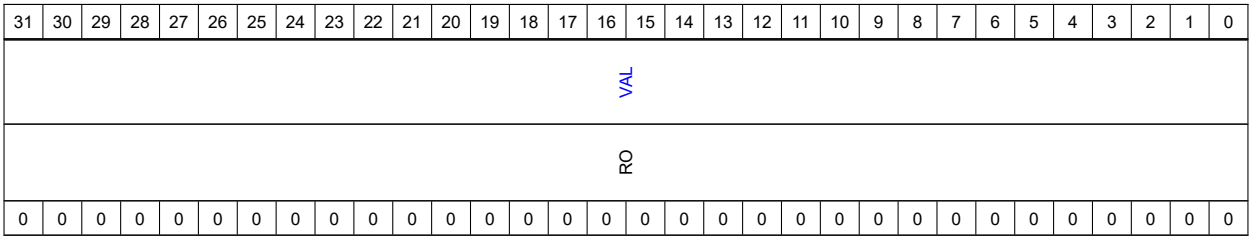


BR[BR_CUR_REV] [31:0]

Bits	Name	Description
31-0	VAL	the value

BR[BR_CUR_REV] Bits

38.5.65 BR[BR_CUR_SPEED] (0x1F8 + 0x100 * n)



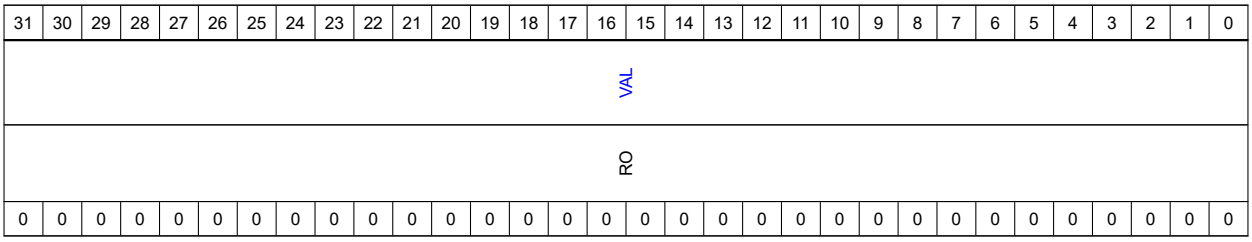
BR[BR_CUR_SPEED] [31:0]

Bits	Name	Description
31-0	VAL	the value

BR[BR_CUR_SPEED] Bits

38.5.66 BR[BR_CUR_ACCEL] (0x1FC + 0x100 * n)

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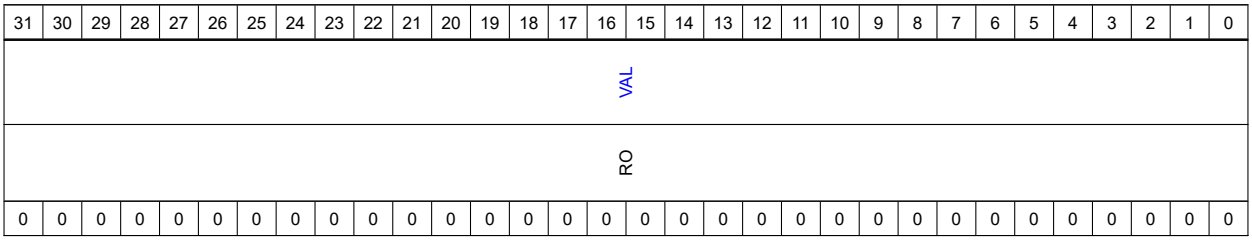


BR[BR_CUR_ACCEL] [31:0]

Bits	Name	Description
31-0	VAL	the value

BR[BR_CUR_ACCEL] Bits

38.5.67 BK0_TIMESTAMP (0x300)

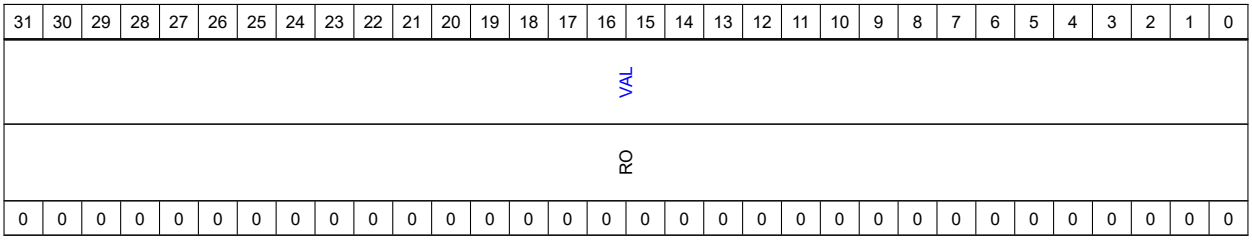


BK0_TIMESTAMP [31:0]

Bits	Name	Description
31-0	VAL	the value

BK0_TIMESTAMP Bits

38.5.68 BK0_POSITION (0x304)



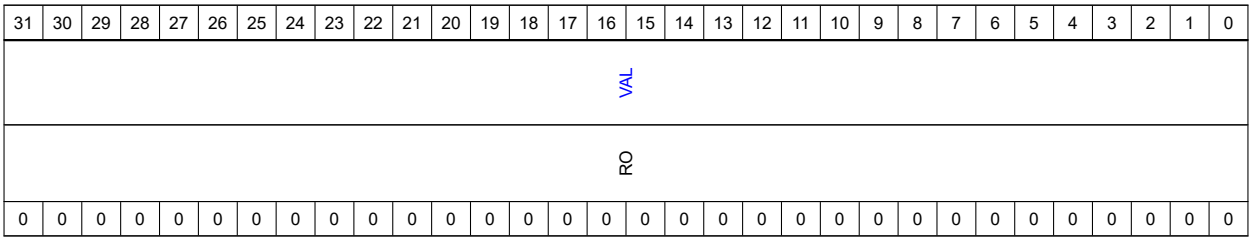
BK0_POSITION [31:0]

Bits	Name	Description
31-0	VAL	the value

BK0_POSITION Bits

38.5.69 BK0_REVOLUTION (0x308)

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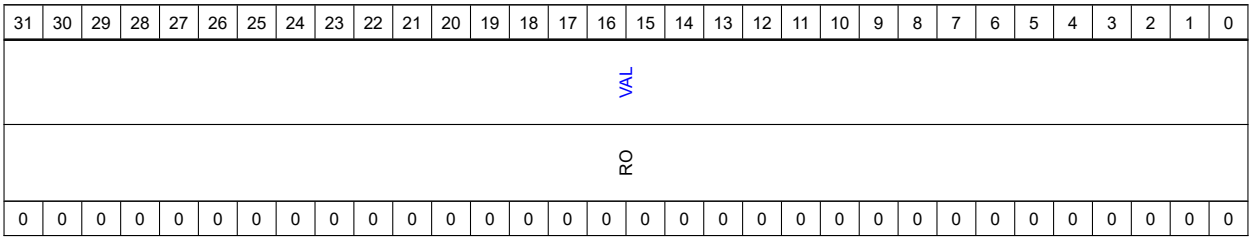


BK0_REVOLUTION [31:0]

Bits	Name	Description
31-0	VAL	the value

BK0_REVOLUTION Bits

38.5.70 BK0_SPEED (0x30C)

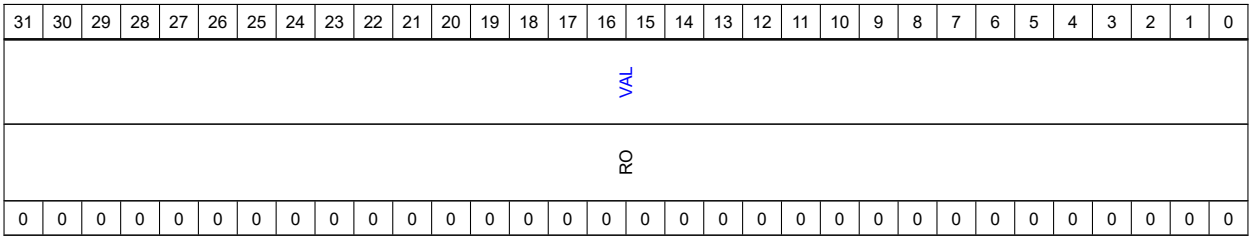


BK0_SPEED [31:0]

Bits	Name	Description
31-0	VAL	the value

BK0_SPEED Bits

38.5.71 BK0_ACCELERATOR (0x310)



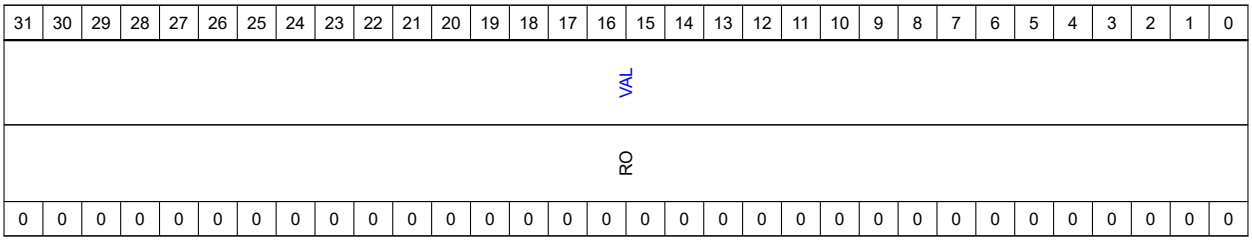
BK0_ACCELERATOR [31:0]

Bits	Name	Description
31-0	VAL	the value

BK0_ACCELERATOR Bits

38.5.72 BK1_TIMESTAMP (0x320)

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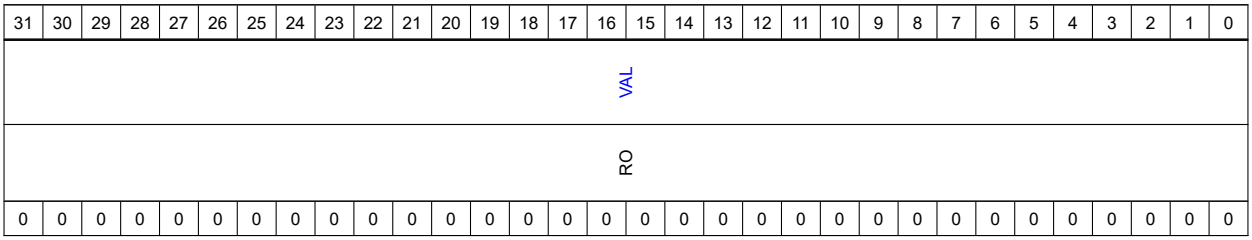


BK1_TIMESTAMP [31:0]

Bits	Name	Description
31-0	VAL	the value

BK1_TIMESTAMP Bits

38.5.73 BK1_POSITION (0x324)

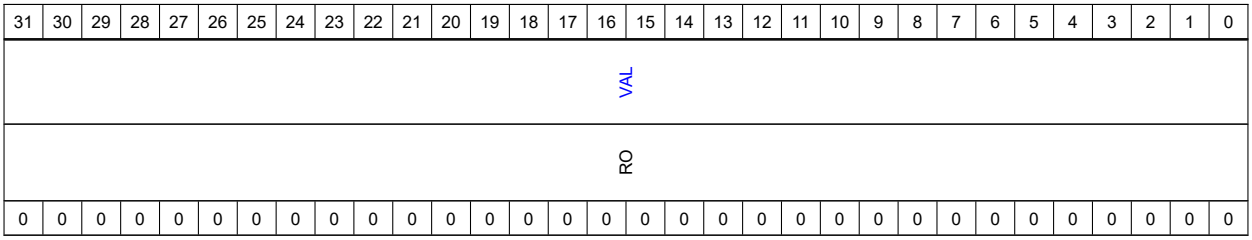


BK1_POSITION [31:0]

Bits	Name	Description
31-0	VAL	the value

BK1_POSITION Bits

38.5.74 BK1_REVOLUTION (0x328)



BK1_REVOLUTION [31:0]

Bits	Name	Description
31-0	VAL	the value

BK1_REVOLUTION Bits

38.5.75 BK1_SPEED (0x32C)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

BK1_SPEED [31:0]

Bits	Name	Description
31-0	VAL	the value

BK1_SPEED Bits

38.5.76 BK1_ACCELERATOR (0x330)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

BK1_ACCELERATOR [31:0]

Bits	Name	Description
31-0	VAL	the value

BK1_ACCELERATOR Bits

38.6 Simple Guide for MMC Configuration and Usage

38.6.1 Software Reset

Set CR[SFTRST] to 1 to perform reset, then set CR[SFTRST] to 0 to release reset.

38.6.2 Configure Position, Speed and Acceleration

There are multiple trigger modes, take time trigger mode as an example first.

The steps to configure the tracker are as follows:

1. Configure register INI_POS_TIME to the time when parameters such as position are to be updated.
2. Configure registers INI_POS, INI_REV, INI_SPEED, INI_ACCEL to the target values. Unneeded parameters can be left unconfigured.
3. Configure register CR[INI_POS_CMD_MSK] as required.
4. Set register CR[INI_POS_TRG_TYPE] to 0 for time trigger mode.
5. Set register CR[INI_POS_REQ] to 1.

In this way, parameters such as the position of the tracker can be modified at the specified time.

The steps to configure predictor 0 are as follows:

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1. Configure register BR [0] [BR_INI_POS_TIME] to the time when parameters such as position are to be updated.
2. Configure registers BR [0] [BR_INI_POS], BR [0] [BR_INI_REV], BR [0] [BR_INI_SPEED], BR [0] [BR_INI_ACCEL] to the target values. Unneeded parameters can be left unconfigured.
3. Configure register BR [0] [BR_CTRL [INI_POS_CMD_MSK]] as required.
4. Set register BR [0] [BR_CTRL [INI_POS_TRG_TYPE]] to 0 for time trigger mode.
5. Set register CR[INI_BR0_POS_REQ] to 1.

In this way, parameters such as the position of predictor 0 can be modified at the specified time.

The steps to configure predictor 1 are as follows:

1. Configure register BR [1] [BR_INI_POS_TIME] to the time when parameters such as position are to be updated.
2. Configure registers BR [1] [BR_INI_POS], BR [1] [BR_INI_REV], BR [1] [BR_INI_SPEED], BR [1] [BR_INI_ACCEL] to the target values. Unneeded parameters can be left unconfigured.
3. Configure register BR [1] [BR_CTRL [INI_POS_CMD_MSK]] as required.
4. Set register BR [1] [BR_CTRL [INI_POS_TRG_TYPE]] to 0 for time trigger mode.
5. Set register CR[INI_BR1_POS_REQ] to 1.

In this way, parameters such as the position of predictor 1 can be modified at the specified time.

38.6.3 Configure Increments of Position, Speed and Acceleration

There are multiple trigger modes, take time trigger mode as an example first.

The steps to configure the tracker are as follows:

1. Configure register INI_DELTA_POS_TIME to the time when parameters such as position increments are to be updated.
2. Configure registers INI_DELTA_POS, INI_DELTA_REV, INI_DELTA_SPEED, INI_DELTA_ACCEL to the target values. Unneeded parameters can be left unconfigured.
3. Configure register CR[INI_DELTA_POS_CMD_MSK] as required.
4. Set register CR[INI_DELTA_POS_TRG_TYPE] to 0 for time trigger mode.
5. Set register CR[INI_DELTA_POS_REQ] to 1.

In this way, parameters such as the position increments of the tracker can be modified at the specified time.

The steps to configure predictor 0 are as follows:

1. Configure register BR [0] [BR_INI_DELTA_POS_TIME] to the time when parameters such as position increments are to be updated.
2. Configure registers BR [0] [BR_INI_DELTA_POS], BR [0] [BR_INI_DELTA_REV], BR [0] [BR_INI_DELTA_SPEED] BR [0] [BR_INI_DELTA_ACCEL] to the target values. Unneeded parameters can be left unconfigured.
3. Configure register BR [0] [BR_CTRL [INI_DELTA_POS_CMD_MSK]] as required.
4. Set register BR [0] [BR_CTRL [INI_DELTA_POS_TRG_TYPE]] to 0 for time trigger mode.
5. Set register BR [0] [BR_CTRL [INI_DELTA_POS_REQ]] to 1.

In this way, parameters such as the position increments of predictor 0 can be modified at the specified time.

The steps to configure predictor 1 are as follows:

1. Configure register BR [1] [BR_INI_DELTA_POS_TIME] to the time when parameters such as position

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increments are to be updated.

2. Configure registers BR [1] [BR_INI_DELTA_POS], BR [1] [BR_INI_DELTA_REV], BR [1] [BR_INI_DELTA_SPEED] BR [1] [BR_INI_DELTA_ACCEL] to the target values. Unneeded parameters can be left unconfigured.
3. Configure register BR [1] [BR_CTRL [INI_DELTA_POS_CMD_MSK]] as required.
4. Set register BR [1] [BR_CTRL [INI_DELTA_POS_TRG_TYPE]] to 0 for time trigger mode.
5. Set register BR [1] [BR_CTRL [INI_DELTA_POS_REQ]] to 1.

In this way, parameters such as the position increments of predictor 1 can be modified at the specified time.

38.6.4 Configure P, I, A Parameters

The steps are as follows:

1. Configure register INI_COEF_TIME to the time when P, I, A parameters are to be updated.
2. Configure registers INI_PCOEF, INI_ICOEF, INI_ACOEF to the target values. Unneeded parameters can be left unconfigured.
3. Configure register CR[INI_COEFS_CMD_MSK] as required.
4. Set register CR[INI_COEFS_REQ] to 1.

In this way, the P, I, A parameters of the tracker can be modified at the specified time.

38.6.5 Configure Trigger Mechanism

This module supports multiple trigger sources, including time, position, speed, input INTRGR[0], input INTRGR[1], output OUTTRGR[0], output OUTTRGR[1].

Take configuring predictor 1 position parameters with predictor 1 speed trigger as an example:

1. Configure registers BR [1] [BR_INI_POS], BR [1] [BR_INI_REV], BR [1] [BR_INI_SPEED], BR [1] [BR_INI_ACCEL] to the target values. Unneeded parameters can be left unconfigured.
2. Configure register BR [1] [BR_CTRL [INI_POS_CMD_MSK]] as required.
3. Configure register BR [1] [BR_TRG_SPEED_THR] to the speed threshold.
4. Configure register BR [1] [BR_TRG_SPEED_CFG] as required.
5. Set register BR [1] [BR_CTRL [INI_DELTA_POS_TRG_TYPE]] to 6 for speed trigger mode.
6. Set register CR[INI_BR1_POS_REQ] to 1.

In this way, when the speed observed inside predictor 1 meets the threshold, parameters such as the position of predictor 1 can be modified.

Take configuring predictor 0 position parameters with predictor 0 position trigger as an example:

1. Configure registers BR [0] [BR_INI_POS], BR [0] [BR_INI_REV], BR [0] [BR_INI_SPEED], BR [0] [BR_INI_ACCEL] to the target values. Unneeded parameters can be left unconfigured.
2. Configure register BR [0] [BR_CTRL [INI_POS_CMD_MSK]] as required.
3. Configure registers BR [0] [BR_TRG_POS_THR], BR [0] [BR_TRG_REV_THR] to the position threshold.
4. Configure register BR [0] [BR_TRG_POS_CFG] as required.
5. Set register BR [0] [BR_CTRL [INI_DELTA_POS_TRG_TYPE]] to 5 for position trigger mode.
6. Set register CR[INI_BR0_POS_REQ] to 1.

In this way, when the position observed inside predictor 0 meets the threshold, parameters such as the

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position of predictor 0 can be modified.

The trigger source can also be OUTTRGR[0] or OUTTRGR[1].

OUTTRGR[n] is a one-cycle signal, which can only be generated by predictor n. The generation logic is: (valid edge of the predictor's own position trigger signal | valid edge of the speed trigger signal).

38.6.6 Configure Open-loop to Closed-loop Switching

38.6.6.1 From Open-loop to Closed-loop

Clear register BR [n] [BR_CTRL [OPEN_LOOP_MODE]] of predictor n to 0 to select closed-loop mode. The predictor will actually enter closed-loop mode only after the tracker has finished calculating the first input sampling point.

38.6.6.2 From Closed-loop to Open-loop

Set register BR [n] [BR_CTRL [OPEN_LOOP_MODE]] of predictor n to 1 to select open-loop mode. You need to stop the predictor first, configure relevant registers such as position and position increment, and then re-enable the predictor.

38.6.7 Configure Multiple P, I, A Parameters

To achieve a more flexible trade-off between convergence speed and convergence accuracy, this module supports 4 sets of P, I, A parameter settings:

- Parameter set 0: fast convergence speed but large convergence error, applicable to scenarios with the largest tracking error;
- Parameter set 1: slower convergence speed than set 0 but smaller convergence error, applicable to scenarios with the second largest tracking error;
- Parameter set 2: slower convergence speed than set 1 but smaller convergence error, applicable to scenarios with smaller tracking error;
- Parameter set 3: slowest convergence speed but smallest convergence error, applicable to scenarios with the smallest tracking error.

For parameter set n (n=0,1,2), the error threshold is configured in register COEF_TRG_CFG [n] [ERR_THR], the minimum number of consecutive sampling points to stay in this set is configured in register COEF_TRG_CFG [n] [TIME], and the parameters used are COEF_TRG_CFG [n] [P], COEF_TRG_CFG [n] [I], COEF_TRG_CFG [n] [A]. As n increases, COEF_TRG_CFG [n] [ERR_THR] decreases.

For parameter set 3 (n=3), there is no error threshold and no minimum stay sampling point requirement, and the parameters used are INI_PCOEF, INI_ICOEF, INI_ACOEF.

In this way, when the tracking error is large, parameters with fast convergence speed but poor accuracy will be automatically selected; as the error decreases, parameter sets with smaller convergence error will be automatically selected step by step.

During configuration, first configure the above relevant registers, then set CR[MS_COEF_EN]=1 to enable automatic switching and tracking of multiple P, I, A parameters.

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38.6.8 MMC Enable Procedure

1. Configure register SYSCLK_FREQ to the required value. For example, for 200MHz main frequency, 200e+6 Hz = 0x0BEBC200 Hz, so set SYSCLK_FREQ = 0xBEBC200.
2. Configure register SYSCLK_PERIOD to the required value. For example, for 200MHz main frequency, the corresponding clock period is 5e-9 s, round(5e-9 * (2^{24}) * (2^{20})) = 0x15799, so set SYSCLK_PERIOD = 0x15799.
3. Configure register OOSYNC_THETA_THR to the required value. For example, if the maximum acceptable tracking error is (0.5/270) cycle, set this register to round(0.5 / 270 * 2^{32}) = 0x795CEB.
4. Set register DISCRETECFG0 to 0.
5. Configure register DISCRETECFG1 to the required value. For example, if the maximum acceptable error between two consecutive tracking values is (1/270) cycle (predicted values exceeding this limit will be locked at the threshold), set this register to round(1 / 270 * 2^{32}) = 0xF2B9D6.
6. Set register CONTCFG0 to 0x01000000. In this case (since CONTCFG0[0]==0), if the in-cycle phase of the current external input is more than 220 degrees larger than that of the previous input (calculated as unsigned number), it is considered as reverse rotation; otherwise, it is forward rotation.
7. If you need to configure position registers, follow the steps in Section 38.6.2 (temporarily do not set the CR[INI_POS_REQ] bit). If you skip this step, the module will automatically extract relevant information from external inputs.
8. Follow the steps in Section 38.6.4 (temporarily do not set the CR[INI_COEFS_REQ] bit) to configure P, I, A coefficient registers.
9. Set CR[INI_POS_REQ] (if position registers are configured) and CR[INI_COEFS_REQ] to 1 in the same configuration command.
10. Read register STA[INI_COEFS_CMD_DONE] repeatedly until it becomes 1, then write 1 to this bit to clear it.
11. Configure registers BR [0] [BR_CTRL [INI_DELTA_POS_TRG_TYPE]] and/or BR [1] [BR_CTRL [INI_DELTA_POS_TRG_TYPE]] according to the selected trigger type, and set BR [0] [BR_CTRL [BR_EN]]=1 and/or BR [1] [BR_CTRL [BR_EN]]=1 to enable the corresponding predictor(s).
12. Set register CR[MOD_EN] to 1, and the module will start working.

38.7 MMC Usage Notes

The following precautions should be noted:

- The timestamp generated by the predictor is valid only after the tracker has generated at least one tracking event.

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39 Programmable Logic Block (PLB)

This section describes the main functions and features of the Programmable Logic Block (PLB).

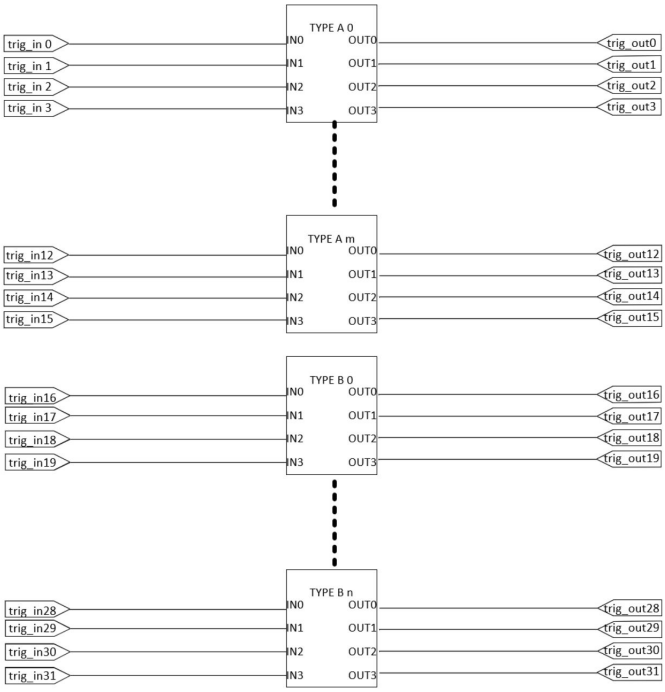
39.1 Feature Summary

This section describes the main features of the PLB:

- Two programmable types
- TYPE_A supports programming of both combinational logic and sequential logic
- TYPE_B supports complex operations such as counting, shifting, filtering, etc.

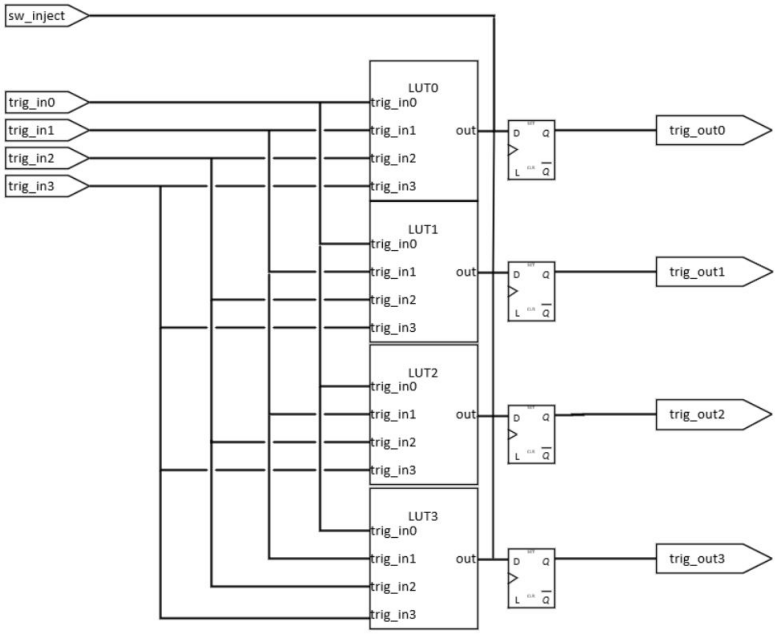
39.2 Architecture Diagram

All PLB inputs and outputs are connected to the TRIG_MUX. The overall structure is shown in the figure below:

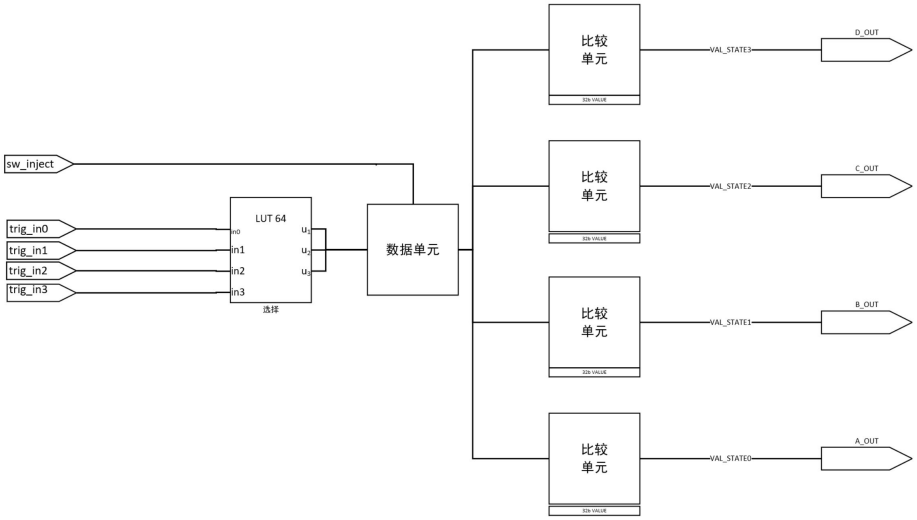


The structure of TYPE A is shown in the figure below:

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The structure of TYPE B is shown in the figure below:



39.3 Function Description

39.3.1 TYPE A Function Description

The PLB has 4 built-in TYPE A channels, each with 4 trig_in inputs and 4 trig_out outputs. Each trig_out output corresponds to a Look-Up Table (LUT) unit.

All 4 LUTs use the 4-bit trig_in as the index, and the content of the LUT can be regarded as the truth table of the 4-bit trig_in. These 4 LUTs share the same trig_in, and the trig_in of different TYPE A modules increases sequentially. For example, TYPE A 0 uses trig_in[3:0], and TYPE A 1 uses trig_in[7:4].

The output of the LUT is output to trig_out through a register.

Software can configure TYPE_A[SW_INJECT] to inject values directly to trig_out. The injected value will last for one system clock cycle, after which the trig_out value will continue to be determined by trig_in and the LUT.

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39.3.2 TYPE B Function Description

The PLB has 4 built-in TYPE B channels, each with 4 trig_in inputs and 4 trig_out outputs.

Each TYPE B channel has a 64-bit LUT, which is formed by concatenating TYPE_B[LUT][0] and TYPE_B[LUT][1] into a 64-bit double word: lut0 occupies bits 31:0, and lut1 occupies bits 63:32. The 64-bit value is divided into 16 4-bit slice units, where slice0 corresponds to bits 3:0, slice1 corresponds to bits 7:4, and so on, with slice15 corresponding to bits 63:60. Each TYPE B channel is assigned 4 trig_in inputs, and this 4-bit trig_in is used as the slice index, so that each channel generates a corresponding 4-bit operation selection, which determines the operation of the TYPE B data unit.

0: The data unit retains the value from the previous cycle.

1: The data unit uses the value of the cmp[0] register of this channel as the value for the next cycle.

2: The data unit uses the value of the cmp[1] register of this channel as the value for the next cycle.

3: The data unit uses the value of the cmp[2] register of this channel as the value for the next cycle.

4: When MODE.OPT_SEL=0, the value of the data unit in the next cycle is the current value plus 1; when MODE.OPT_SEL=1, the value of the data unit in the next cycle is the current value shifted left by 1 bit, with 0 filled in the low bit.

5: When MODE.OPT_SEL=0, the value of the data unit in the next cycle is the current value plus 2; when MODE.OPT_SEL=1, the value of the data unit in the next cycle is the current value shifted left by 1 bit, with 1 filled in the low bit.

6: When MODE.OPT_SEL=0, the value of the data unit in the next cycle is the current value minus 1; when MODE.OPT_SEL=1, the value of the data unit in the next cycle is the current value shifted right by 1 bit, with 0 filled in the high bit.

7: When MODE.OPT_SEL=0, the value of the data unit in the next cycle is the current value minus 2; when MODE.OPT_SEL=1, the value of the data unit in the next cycle is the current value shifted right by 1 bit, with 1 filled in the high bit.

Other values: The value of the data unit remains unchanged.

Each TYPE B channel has 4 comparison units, and the output of each comparison unit corresponds to the respective trig_out.

The operation mode of the comparison unit is determined by TYPE_B[MODE].OUTn_SEL.

0: Output 0

1: Output 1

2: Check whether the data of the data unit is greater than the value of TYPE_B[CMP]. Output 1 if yes, 0 otherwise.

3: Check whether the data of the data unit is less than the value of TYPE_B[CMP]. Output 1 if yes, 0 otherwise.

4: Check whether the data of the data unit is equal to the value of TYPE_B[CMP]. Output 1 if yes, 0 otherwise.

5: Check whether the data of the data unit is not equal to the value of TYPE_B[CMP]. Output 1 if yes, 0 otherwise.

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6: Check whether the data of the data unit is greater than or equal to the value of TYPE_B[CMP]. Output 1 if yes, 0 otherwise.

7: Check whether the data of the data unit is less than or equal to the value of TYPE_B[CMP]. Output 1 if yes, 0 otherwise.

8: Output 0

9: Output 1

10: Perform AND operation on the bits of the data unit where the corresponding bit in TYPE_B[CMP] is 1.

11: Perform OR operation on the bits of the data unit where the corresponding bit in TYPE_B[CMP] is 1.

12: Perform XOR operation on the bits of the data unit where the corresponding bit in TYPE_B[CMP] is 1.

13: Perform NAND operation on the bits of the data unit where the corresponding bit in TYPE_B[CMP] is 1.

14: Perform NOR operation on the bits of the data unit where the corresponding bit in TYPE_B[CMP] is 1.

15: Perform XNOR operation on the bits of the data unit where the corresponding bit in TYPE_B[CMP] is 1.

39.4 PLB Register List

The PLB register list is as follows:

PLB base address: 0xF0324000

Offset	Register Name	Description	Reset Value
0x0000	TYPE_A[0][LOOKUP_TABLE][0]	TYPE A CHN0 lookup_table 0	0x00000000
0x0004	TYPE_A[0][LOOKUP_TABLE][1]	TYPE A CHN0 lookup_table 1	0x00000000
0x0008	TYPE_A[0][LOOKUP_TABLE][2]	TYPE A CHN0 lookup_table 2	0x00000000
0x000C	TYPE_A[0][LOOKUP_TABLE][3]	TYPE A CHN0 lookup_table 3	0x00000000
0x0010	TYPE_A[0][SW_INJECT]	TYPE A CHN0 software inject	0x00000000
0x0020	TYPE_A[1][LOOKUP_TABLE][0]	TYPE A CHN1 lookup_table 0	0x00000000
0x0024	TYPE_A[1][LOOKUP_TABLE][1]	TYPE A CHN1 lookup_table 1	0x00000000
0x0028	TYPE_A[1][LOOKUP_TABLE][2]	TYPE A CHN1 lookup_table 2	0x00000000
0x002C	TYPE_A[1][LOOKUP_TABLE][3]	TYPE A CHN1 lookup_table 3	0x00000000
0x0030	TYPE_A[1][SW_INJECT]	TYPE A CHN1 software inject	0x00000000
0x0040	TYPE_A[2][LOOKUP_TABLE][0]	TYPE A CHN2 lookup_table 0	0x00000000
0x0044	TYPE_A[2][LOOKUP_TABLE][1]	TYPE A CHN2 lookup_table 1	0x00000000
0x0048	TYPE_A[2][LOOKUP_TABLE][2]	TYPE A CHN2 lookup_table 2	0x00000000
0x004C	TYPE_A[2][LOOKUP_TABLE][3]	TYPE A CHN2 lookup_table 3	0x00000000
0x0050	TYPE_A[2][SW_INJECT]	TYPE A CHN2 software inject	0x00000000
0x0060	TYPE_A[3][LOOKUP_TABLE][0]	TYPE A CHN3 lookup_table 0	0x00000000
0x0064	TYPE_A[3][LOOKUP_TABLE][1]	TYPE A CHN3 lookup_table 1	0x00000000
0x0068	TYPE_A[3][LOOKUP_TABLE][2]	TYPE A CHN3 lookup_table 2	0x00000000
0x006C	TYPE_A[3][LOOKUP_TABLE][3]	TYPE A CHN3 lookup_table 3	0x00000000
0x0070	TYPE_A[3][SW_INJECT]	TYPE A CHN3 software inject	0x00000000
0x0400	TYPE_B[0][LUT][0]	TYPE B CHN0 lookup table 0	0x00000000

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Offset	Register Name	Description	Reset Value
0x0404	TYPE_B[0][LUT][1]	TYPE B CHN0 lookup table 1	0x00000000
0x0408	TYPE_B[0][CMP][0]	TYPE B CHN0 data unit cmp0	0x00000000
0x040C	TYPE_B[0][CMP][1]	TYPE B CHN0 data unit cmp1	0x00000000
0x0410	TYPE_B[0][CMP][2]	TYPE B CHN0 data unit cmp2	0x00000000
0x0414	TYPE_B[0][CMP][3]	TYPE B CHN0 data unit cmp3	0x00000000
0x0418	TYPE_B[0][MODE]	TYPE B CHN0 mode ctrl	0x00000000
0x041C	TYPE_B[0][SW_INJECT]	TYPE B CHN0 software inject	0x00000000
0x0420	TYPE_B[1][LUT][0]	TYPE B CHN1 lookup table 0	0x00000000
0x0424	TYPE_B[1][LUT][1]	TYPE B CHN1 lookup table 1	0x00000000
0x0428	TYPE_B[1][CMP][0]	TYPE B CHN1 data unit cmp0	0x00000000
0x042C	TYPE_B[1][CMP][1]	TYPE B CHN1 data unit cmp1	0x00000000
0x0430	TYPE_B[1][CMP][2]	TYPE B CHN1 data unit cmp2	0x00000000
0x0434	TYPE_B[1][CMP][3]	TYPE B CHN1 data unit cmp3	0x00000000
0x0438	TYPE_B[1][MODE]	TYPE B CHN1 mode ctrl	0x00000000
0x043C	TYPE_B[1][SW_INJECT]	TYPE B CHN1 software inject	0x00000000
0x0440	TYPE_B[2][LUT][0]	TYPE B CHN2 lookup table 0	0x00000000
0x0444	TYPE_B[2][LUT][1]	TYPE B CHN2 lookup table 1	0x00000000
0x0448	TYPE_B[2][CMP][0]	TYPE B CHN2 data unit cmp0	0x00000000
0x044C	TYPE_B[2][CMP][1]	TYPE B CHN2 data unit cmp1	0x00000000
0x0450	TYPE_B[2][CMP][2]	TYPE B CHN2 data unit cmp2	0x00000000
0x0454	TYPE_B[2][CMP][3]	TYPE B CHN2 data unit cmp3	0x00000000
0x0458	TYPE_B[2][MODE]	TYPE B CHN2 mode ctrl	0x00000000
0x045C	TYPE_B[2][SW_INJECT]	TYPE B CHN2 software inject	0x00000000
0x0460	TYPE_B[3][LUT][0]	TYPE B CHN3 lookup table 0	0x00000000
0x0464	TYPE_B[3][LUT][1]	TYPE B CHN3 lookup table 1	0x00000000
0x0468	TYPE_B[3][CMP][0]	TYPE B CHN3 data unit cmp0	0x00000000
0x046C	TYPE_B[3][CMP][1]	TYPE B CHN3 data unit cmp1	0x00000000
0x0470	TYPE_B[3][CMP][2]	TYPE B CHN3 data unit cmp2	0x00000000
0x0474	TYPE_B[3][CMP][3]	TYPE B CHN3 data unit cmp3	0x00000000
0x0478	TYPE_B[3][MODE]	TYPE B CHN3 mode ctrl	0x00000000
0x047C	TYPE_B[3][SW_INJECT]	TYPE B CHN3 software inject	0x00000000

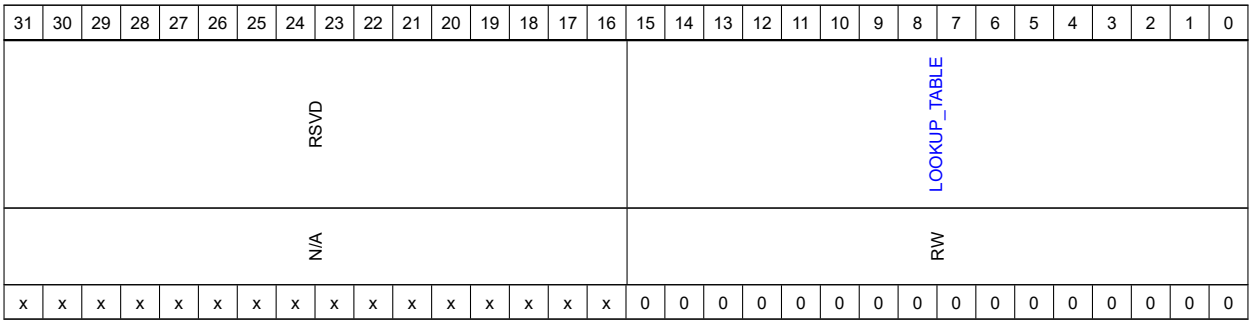
Table 181: PLB Register List

39.5 PLB Register Details

The detailed description of PLB registers is as follows:

39.5.1 TYPE_A[LOOKUP_TABLE] (0x0 + 0x20 * n + 0x4 * m)

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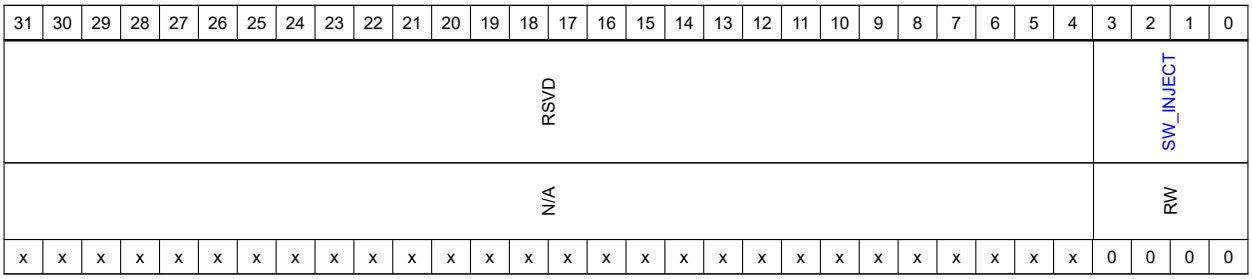


TYPE_A[LOOKUP_TABLE] [31:0]

Bits	Name	Description
15-0	LOOKUP_TABLE	using 4 bit trig_in as lookup index. software can program this register as trig_in's true table.

TYPE_A[LOOKUP_TABLE] Bits

39.5.2 TYPE_A[SW_INJECT] (0x10 + 0x20 * n)

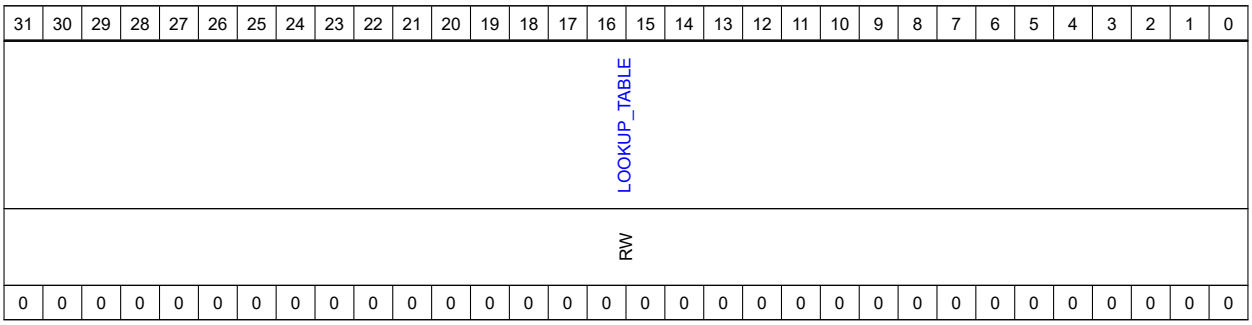


TYPE_A[SW_INJECT] [31:0]

Bits	Name	Description
3-0	SW_INJECT	software can inject value to TYPEA's output

TYPE_A[SW_INJECT] Bits

39.5.3 TYPE_B[LUT] (0x400 + 0x20 * n + 0x4 * m)



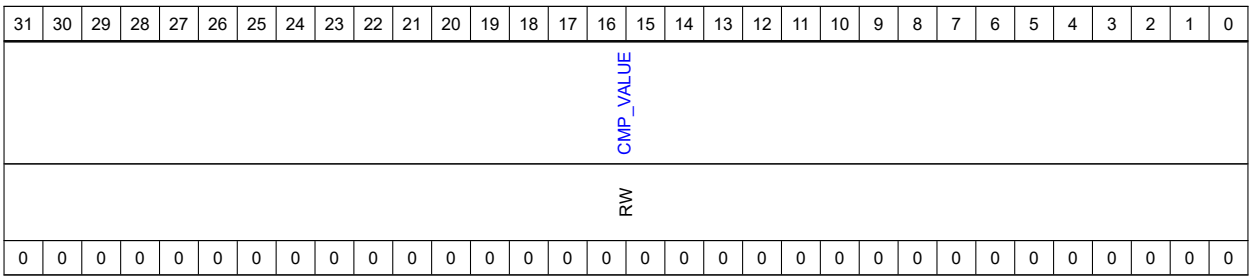
TYPE_B[LUT] [31:0]

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Bits	Name	Description
31-0	LOOKUP_TABLE	lut0 and lut1 union as 64bit, consider each 4bit as one slice. then, total 16 slice. slice0 as bit3:0, slice1 as bit7:4...etc. using 4bit trig in as index of slice. the operate sel in data unit of type B channle is decided by which slice value choosed by trig_in

TYPE_B[LUT] Bits

39.5.4 TYPE_B[CMP] (0x408 + 0x20 * n + 0x4 * m)

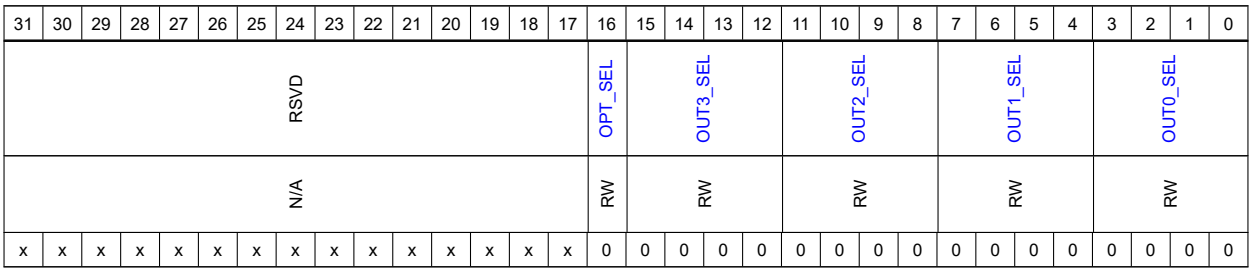


TYPE_B[CMP] [31:0]

Bits	Name	Description
31-0	CMP_VALUE	cmp value, using as data unit operation

TYPE_B[CMP] Bits

39.5.5 TYPE_B[MODE] (0x418 + 0x20 * n)



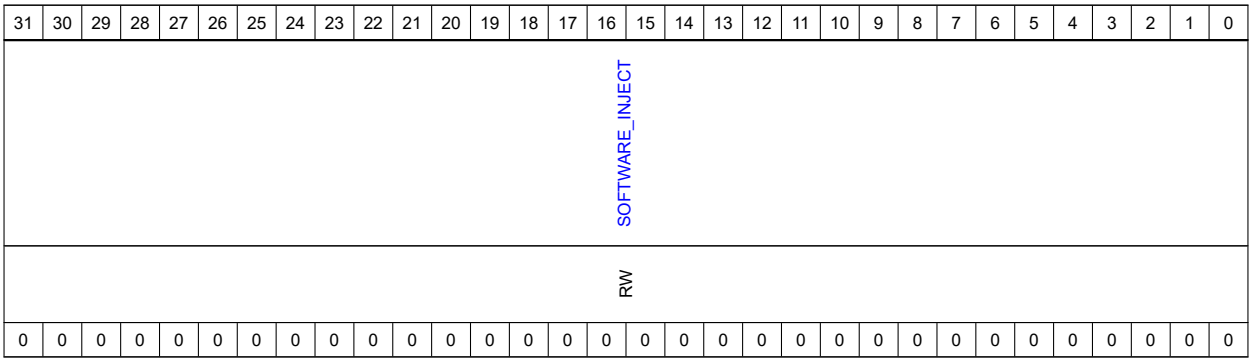
TYPE_B[MODE] [31:0]

Bits	Name	Description
16	OPT_SEL	operation selection in data unit.
15-12	OUT3_SEL	trig out 3 output type in current channel
11-8	OUT2_SEL	trig out 2 output type in current channel
7-4	OUT1_SEL	trig out 1 output type in current channel
3-0	OUT0_SEL	trig out 0 output type in current channel

TYPE_B[MODE] Bits

39.5.6 TYPE_B[SW_INJECT] (0x41C + 0x20 * n)

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TYPE_B[SW_INJECT] [31:0]

Bits	Name	Description
31-0	SOFTWARE_INJECT	data unit value can be changed if program this register

TYPE_B[SW_INJECT] Bits

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40 Rotary Encoder Interface RDC

This chapter introduces the main functions and features of the rotary encoder interface RDC.

40.1 Overview

The RDC module can generate excitation signals and perform accumulation preprocessing on the induction signals of the resolver. The currently supported features are as follows:

Function	Main Features
Excitation Signal Generation	· Single DAC output · Two-way differential DAC output · PWM output, NMOS+PMOS · PWM output, PMOS+NMOS · Support amplitude and bias adjustment · Support dead zone configuration
ADC Sampling Control	· Single ADC alternate sampling · Dual ADC synchronous sampling · Continuous sampling · Sample once at the aligned high point of each excitation signal cycle · Sample once aligned to the midpoint of PWM low level · Sample once aligned to the midpoint of PWM high level · Sample once aligned to the midpoint of PWM high/low level · Record maximum and minimum values of ADC input
Rectification Control	· Reference point based on 0/90/180/270 phase of excitation signal · Reference point supports external synchronization signal input · Zero-crossing delay measurement of ADC sampling data · Zero-crossing jitter filtering of ADC sampling data
Position Pre-processing	· Allow setting the number of accumulation cycles · Allow dynamic configuration of the number of accumulation cycles · Allow transparent transmission of ADC sampling results to QEI · Allow software reading of ADC data accumulation results · Maximum value recording of accumulation results · Minimum value recording of accumulation results · Accumulation modulus over-limit alarm

Table 182: Main Features of RDC

40.2 RDC Architecture Diagram

The structure diagram of this module is shown in Figure 59.

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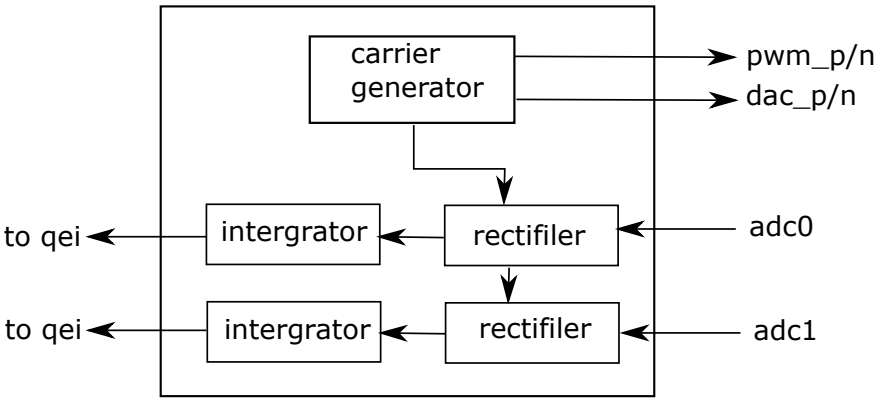


Figure 59: RDC Structure Block Diagram

The carrier generator in the RDC module is an excitation signal generator, which can generate sine waves with a frequency of 10k-20kHz. The sine wave can be output in the form of two PWM signals, or converted to analog signal through DAC and then output. The rectifier module can realize rectification of ADC sampling data, and the rectified data is periodically accumulated in the integrator module according to user configuration. The accumulation result can be understood as the envelope of the amplitude modulation of the excitation signal caused by motor rotation, and this result can be output to the subsequent QEI module for further processing.

40.3 Pin Description

Table 183: RDC Pin Description

Pin	Direction	Function Description
pwm_p	Output	Output PWM signal
pwm_n	Output	Output complementary PWM signal
dac_p	Output	Signal output to DAC
dac_n	Output	Signal output to DAC
adc_0	Input	Data after ADC conversion
adc_1	Input	Data after ADC conversion

40.4 Function Description

40.4.1 Excitation Generation

RDC has a built-in trigonometric function table with a depth of 256, which can generate sine waves of specified frequency and points according to user configuration. Assuming the frequency of the sine wave is f_sin and the operating clock is f_soc, the frequency of the sine wave conforms to the following formula:

$$f_sin = \frac{f_soc}{a * 2^b * 4}$$

Where a is the value of register exc_timming.smp_rate, b is the value of register exc_timming.smp_num.

There are two trigger modes for excitation signal: software trigger and hardware trigger. For software trigger mode, the software can trigger the generation of excitation signal by setting register rdc_ctl.exc_start to 1, and this

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register will automatically clear to 0 after writing 1. For hardware trigger mode, the trigger of excitation signal is controlled by the output signal of trig_mux module. When the signal jumps, the excitation signal can be triggered after a certain delay, which is controlled by register exc_sync_dly.delay. This delay can be used to adjust the phase of the excitation signal.

The excitation signal can be directly output as analog signal through DAC, or as digital signal through PWM interface. The digital signal can be restored to sine wave after being shaped by external power stage circuit.

When the excitation signal is output through DAC, the swing amplitude and offset of the sine wave can be adjusted, as shown in the following formula:

$$v_dac = \left(\frac{\sin\theta * 4095 * 4096 * m}{2^d} + s \right) * 256$$

Where m is the value of register exc_scaling.amp_man; d is the value of register exc_scaling.ampl_exp; s is the value of register exc_offset.ampl_offset; v_dac is the data output to DAC, and DAC converts the digital signal to analog voltage according to this data.

When the excitation signal is output through DAC, there can also be an inverted signal, whose amplitude and offset are the same as the first DAC output, only the phase is opposite, as shown in the following formula:

$$v_dac = \left(\frac{-\sin\theta * 4095 * 4096 * m}{2^d} + s \right) * 256$$

When the excitation signal is output through PWM, the duty cycle of PWM wave in one PWM cycle is used to simulate the value of trigonometric function. By default, one PWM cycle is consistent with the time difference between two points of DAC output, and the PWM cycle can also be wider, as shown in the following formula:

$$f_pwm = \frac{f_soc}{a * (p + 1)}$$

Where p is the value of register exc_timming.pwm_prd.

Since the duty cycle of PWM output waveform corresponds to the value of trigonometric function, its swing amplitude and offset can also be adjusted, as shown in the following formula:

$$d_pwm = \frac{\sin\theta * pm}{2^{pd}} + \frac{ps}{a}$$

Where pm is the value of register pwm_scaling.ampl_man; pd is the value of register pwm_scaling.ampl_exp; ps is the value of register pwm_offset.ampl_offset.

In one PWM cycle, the duration of high level is used to represent the value of sin. If a PWM cycle is a (sampling period), the value of d_pwm represents a threshold. A counter increases from 0 to a, when it exceeds d_pwm, PWM outputs 1, otherwise outputs 0.

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PWM output signals include PWM_P and PWM_N, PWM_N is the inverted output of PWM_P. In addition, PWM output supports dead zone configuration, which is disabled by default, and the dead zone function can be configured through register pwm_dz.

In general, the steps to enable excitation output are as follows:

1. Configure excitation signal timing control register exc_timming to appropriate values according to system requirements;
2. Configure excitation signal amplitude control registers exc_scaling, pwm_scaling, exc_offset and pwm_offset to appropriate values according to system requirements;
3. Set rdc_ctl.exc_en to 1;
4. Configure software or hardware trigger mode. For hardware trigger, configure exc_sync_dly register, and the excitation signal will be output after the trigger occurs;

Along with the output of excitation signal, RDC module can also generate corresponding trigger output signals for guiding ADC sampling or other events. The supported trigger outputs of RDC include:

- Output in the middle of PWM high level, and support front and rear offset adjustment controlled by register trig_out1_cfg;
- Output in the middle of PWM low level, and support front and rear offset adjustment controlled by register trig_out1_cfg;
- Output at the maximum value of sin function;
- Output at the minimum value of sin function;

The above four output formats support any combination.

40.4.2 Delay Measurement

After the excitation signal is induced by the rotary encoder, the two channels of signals are converted by ADC and input to RDC module. RDC module can assist to measure the phase delay between excitation signal and induction signal, and the delay measurement of the two channels can be carried out independently. The basic principle of measurement is shown in the following figure:

In the figure:

- The green exc_sin represents the output excitation signal;
- sign represents the sign bit of the excitation signal;
- sdcount is the 32-bit internal counter of RDC;
- adc_in is the data obtained from ADC conversion, which is also a sin function;
- sdcap is the 32-bit internal register rise_delay_i/q of RDC, which is used to save the value of adcount when adc_in crosses zero;
- sign_dly is the delayed rectification signal generated inside RDC;

The delay measurement process is:

1. Set rdc_ctl.exc_en and rdc_ctl.acc_en to 1, set rdc_ctl.rectify_sel to 0, select excitation signal as synchronization source;
2. Read the values of registers max_i and max_q to obtain the maximum value v_max and minimum

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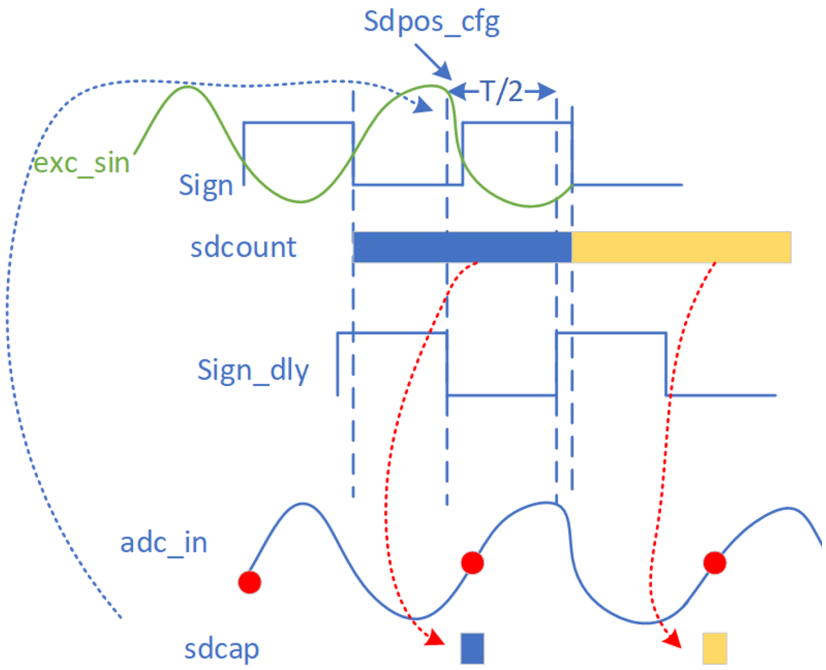


Figure 60: Schematic Diagram of Delay Measurement

value v_min of ADC input;

3. Calculate the offset of ADC input signal, and configure the result to thrs_i and thrs_q, the calculation formula is as follows:

$$v_thrs = \frac{v_max - v_min}{2} + v_min - 0x800000$$

4. Configure the value of edg_det_ctl.filter as required to enable the jitter filtering function for sampling zero crossing point;
5. Read the value of register rise_delay_i/q for multiple times and calculate the average, which corresponds to the specific value of sdcap;
6. If the delay measurement result exceeds half of the sin function clock cycle, set rdc_ctl.rectify_sel to 2 and repeat the above measurement process;

After the delay measurement is completed, the measured delay value needs to be configured to register sync_delay_i/q, and the sin function period needs to be configured to register exc_period. RDC will regenerate the delayed rectification signal internally, the generation method of rectification signal is:

- When the value of sdcount is equal to the value of sync_delay_i/q, the rectification signal sign_dly changes from 1 to 0;
- When the value of sdcount is equal to the value of sync_delay_i/q plus half of the sin function clock cycle, the rectification signal sign_dly changes from 0 to 1;

It should be noted that the delay measurement of the two ADC input channels of RDC can be carried out independently, and the measurement results of the two delays are also different. In actual use, the configuration values of the two delays should be as close as possible.

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40.4.3 Carrier Accumulation

The data output from RDC to QEI is the data accumulated periodically at a certain time interval, and the accumulation process can be expressed by the following formula:

$$acc = \sum \left(\frac{v_adc}{256} - 0x800000 \right) * sign_dly$$

Where acc is the accumulation result output to QEI, v_adc is the value obtained from ADC conversion, sign_dly is the rectification signal generated inside RDC.

Periodic accumulation at a certain time interval means that the accumulation result is output once every one or more sin function cycles from the ADC input data, and this time is controlled by register rdc_ctl.acc_len. Generally, when the motor speed is high, this time interval needs to be smaller to realize fast detection of motor position; when the motor speed is low, this time interval can be larger to realize more accurate position detection.

RDC assumes that the ADC input data is 32-bit unsigned number, the accumulation result will become 40-bit signed number after actual accumulation, while the actual accumulation result output from RDC is 32-bit. Register acc_scaling.acc_shift can control how the accumulation result is output.

The interval between adjacent accumulation results output by RDC is relatively long. For the cooperation between RDC and QEI, RDC needs to give QEI a sampling time indication before the next accumulation result is output. This indication signal can be given at the beginning, middle or end of an accumulation cycle, controlled by register rdc_ctl.ts_sel.

RDC module also has modulus anomaly detection function. The basic principle of detection is to first calculate the modulus of the two channels of accumulation results, and then detect whether the modulus is within the given range. The formula for calculating modulus m is as follows:

$$m = \left| max(acc_i, acc_q) + \frac{min(acc_i, acc_q)}{2} \right|$$

Where acc_i represents the accumulation value of the first ADC input data; acc_q represents the accumulation value of the second ADC input data.

The legal range of modulus is controlled by register amp_max and register amp_min.

40.4.4 Interrupt

The internal supported interrupt sources of RDC include the following:

- Two channels of accumulation result update interrupt;
- Two channels of ADC input signal rising zero-crossing interrupt;
- Two channels of ADC input signal falling zero-crossing interrupt;
- Two channels of interrupt for sampling ADC input value at the rising edge of delayed rectification signal sign_dly;
- Two channels of interrupt for sampling ADC input value at the falling edge of delayed rectification signal sign_dly;
- Two channels of accumulation value overflow interrupt;

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- Two channels of accumulation value underflow interrupt;
- Accumulation result modulus overflow interrupt;
- Accumulation result modulus underflow interrupt;

40.5 RDC Register List

The register list of RDC is as follows:

RDC base address: 0xF0320000

Offset	Register Name	Description	Reset Value
0x0000	RDC_CTL	rdc control	0x00000000
0x0004	ACC_I	accumulate result of i_channel	0x00000000
0x0008	ACC_Q	accumulate result of q_channel	0x00000000
0x000C	IN_CTL	input channel selection	0x00000000
0x0010	OUT_CTL	output channel selection	0x00000000
0x0034	EXC_TIMMING	excitation signal timming setting	0x000400C8
0x0038	EXC_SCALING	amplitude scaling for excitation	0x00000011
0x003C	EXC_OFFSET	amplitude offset setting	0x00800000
0x0040	PWM_SCALING	amplitude scaling for excitation	0x00000111
0x0044	PWM_OFFSET	amplitude offset setting	0x00000064
0x0048	TRIG_OUT0_CFG	Configuration for trigger out 0 in clock cycle	0x00100019
0x004C	TRIG_OUT1_CFG	Configuration for trigger out 1 in clock cycle	0x0010004B
0x0050	PWM_DZ	pwm dead zone control in clock cycle	0x00000000
0x0054	SYNC_OUT_CTRL	synchronize output signal control	0x00000000
0x0058	EXC_SYNC_DLY	trigger in delay timming in soc bus cycle	0x01000001
0x0070	MAX_I	max value of i_channel	0x00000000
0x0074	MIN_I	min value of i_channel	0x00000000
0x0078	MAX_Q	max value of q_channel	0x00000000
0x007C	MIN_Q	min value of q_channel	0x00000000
0x0080	THRS_I	the offset setting for edge detection of the i_channel	0x00000000
0x0084	THRS_Q	the offset setting for edge detection of the q_channel	0x00000000
0x0088	EDG_DET_CTL	the control for edge detection	0x00000080
0x008C	ACC_SCALING	scaling for accumulation result	0x00000000
0x0090	EXC_PERIOD	period of excitation	0x00001770
0x00A0	SYNC_DELAY_I	delay setting in clock cycle for synchronous signal	0x00000008

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Offset	Register Name	Description	Reset Value
0x00A8	RISE_DELAY_I	delay in clock cycle between excitation synchrnous signal and rising edge of i_channel data	0x00000000
0x00AC	FALL_DELAY_I	delay in clock cycle between excitation synchrnous signal and falling edge of i_channel data	0x00000000
0x00B0	SAMPLE_RISE_I	sample value on rising edge of rectify signal	0x00000000
0x00B4	SAMPLE_FALL_I	sample value on falling edge of rectify signal	0x00000000
0x00B8	ACC_CNT_I	number of accumulation	0x00000000
0x00BC	SIGN_CNT_I	sample counter of opposite sign with rectify signal	0x00000000
0x00C0	SYNC_DELAY_Q	delay setting in clock cycle for synchronous signal	0x00000008
0x00C8	RISE_DELAY_Q	delay in clock cycle between excitation synchrnous signal and rising edge of q_channel data	0x00000000
0x00CC	FALL_DELAY_Q	delay in clock cycle between excitation synchrnous signal and falling edge of q_channel data	0x00000000
0x00D0	SAMPLE_RISE_Q	sample value on rising edge of rectify signal	0x00000000
0x00D4	SAMPLE_FALL_Q	sample value on falling edge of rectify signal	0x00000000
0x00D8	ACC_CNT_Q	number of accumulation	0x00000000
0x00DC	SIGN_CNT_Q	sample counter of opposite sign with rectify signal	0x00000000
0x00E0	AMP_MAX	the maximum of acc amplitude	0x01000000
0x00E4	AMP_MIN	the minimum of acc amplitude	0x00400000
0x00E8	INT_EN	the interrupt mask control	0x00000000
0x00EC	ADC_INT_STATE	the interrupt state	0x00000000

Table 184: RDC Register List

40.6 RDC Register Detailed Information

The detailed description of RDC registers is as follows:

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40.6.1 RDC_CTL (0x0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD										TS_SEL		ACC_LEN								RSVD					RECTIFY_SEL			RSVD	ACC_EN	EXC_START	EXC_EN
N/A										RW		RW								N/A					RW			N/A	RW	RW1C	RW
x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	0	0	0	x	0	0	0

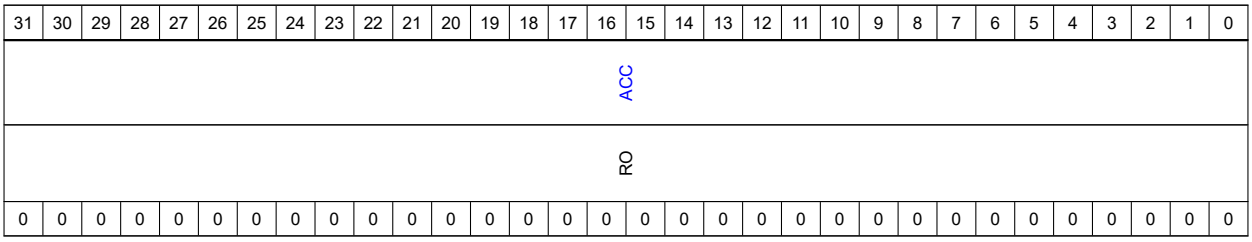
RDC_CTL [31:0]

Bits	Name	Description
21-20	TS_SEL	Time stamp selection for accumulation 0: end of accumulation 1: start of accumulation 2: center of accumulation
19-12	ACC_LEN	Accumulate time, support on the fly change 0：1 cycle 1：2 cycles … 255: 256 cycles
6-4	RECTIFY_SEL	Select reference point of rectify signal 0: 0 phase of internal exciting signal 1: 90 phase of internal exciting signal 2: 180 phase of internal exciting signal 3: 270 phase of internal exciting signal 4: use value on external pin 5: use invert value on external pin
2	ACC_EN	Enable rdc accumulate 0: rdc disable 1: rdc enable
1	EXC_START	Write 1 start excite signal, always read 0 0: no effect 1: start excite signal
0	EXC_EN	Enable rdc excite signal 0: rdc disable 1: rdc enable

RDC_CTL Bits

40.6.2 ACC_I (0x4)

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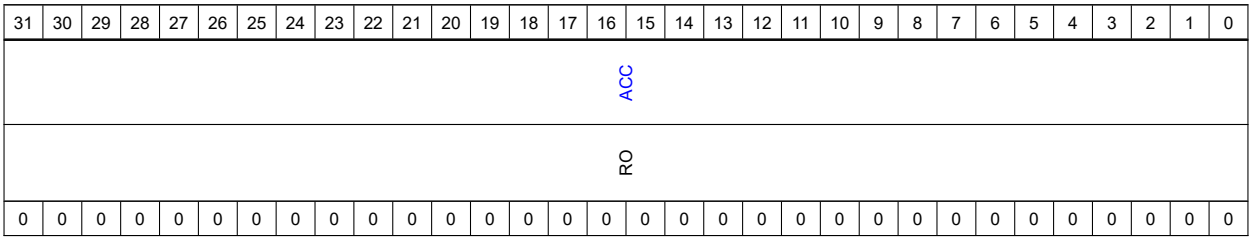


ACC_I [31:0]

Bits	Name	Description
31-0	ACC	accumulate result of i_channel, this is a signed number

ACC_I Bits

40.6.3 ACC_Q (0x8)

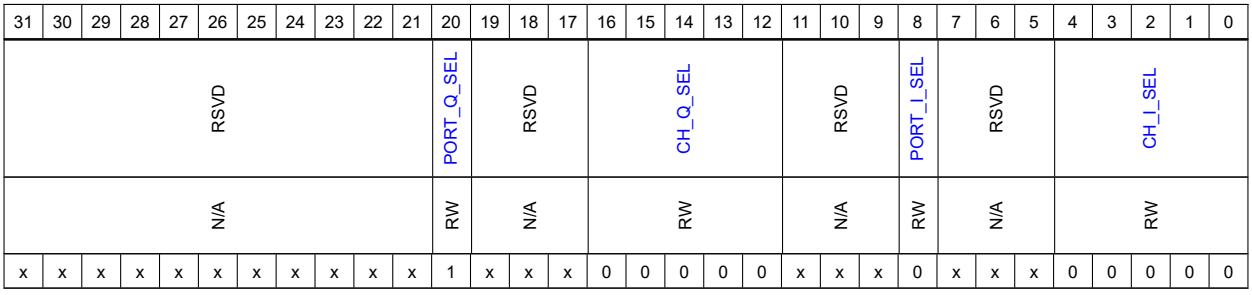


ACC_Q [31:0]

Bits	Name	Description
31-0	ACC	accumulate result of q_channel, this is a signed number

ACC_Q Bits

40.6.4 IN_CTL (0xC)



IN_CTL [31:0]

Bits	Name	Description
20	PORT_Q_SEL	Input port selection for q_channel, 0:sel port0 1:sel port1

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Bits	Name	Description
16-12	CH_Q_SEL	Input channel selection for q_channel 0: channel 0 selected 1: channel 1 selected … 31: channel 31 selected
8	PORT_I_SEL	Input port selection for i_channel, 0:sel port0 1:sel port1
4-0	CH_I_SEL	Input channel selection for i_channel 0: channel 0 selected 1: channel 1 selected … 31: channel 31 selected

IN_CTL Bits

40.6.5 OUT_CTL (0x10)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																			CH_Q_SEL					RSVD			CH_I_SEL				
N/A																			RW					N/A			RW				
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	x	x	x	0	0	0	0	0

OUT_CTL [31:0]

Bits	Name	Description
12-8	CH_Q_SEL	Output channel selection for q_channel
4-0	CH_I_SEL	Output channel selection for i_channel

OUT_CTL Bits

40.6.6 EXC_TIMMING (0x34)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD							SWAP	PWM_PRD				SMP_NUM				SMP_RATE																
N/A							RW	RW				RW				RW																
x	x	x	x	x	x	x	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	1	0	0	1	0	0	0

EXC_TIMMING [31:0]

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Bits	Name	Description
24	SWAP	Swap output of PWM and DAC 0: disable swap 1: swap output
23-20	PWM_PRD	Pwm period in samples， 0：1 sample period 1: 2 sample period ... 15: 16 sample period
19-16	SMP_NUM	Number of sample every excitation period 0: 4 point 1: 8 point … 8: 1024 point
15-0	SMP_RATE	The period for excitation sample in clock cycle， 0: not allowed 1: 1 cycle 2: 2 cycles … 65535 : 65535 cycles

EXC_TIMMING Bits

40.6.7 EXC_SCALING (0x38)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																								AMP_EXP				AMP_MAN			
N/A																								RW				RW			
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	1	0	0	0	1

EXC_SCALING [31:0]

Bits	Name	Description
7-4	AMP_EXP	Amplitude scaling for excitation, amplitude = [table value] x man / 2^exp
3-0	AMP_MAN	Amplitude scaling for excitation, amplitude = [table value] x man / 2^exp

EXC_SCALING Bits

40.6.8 EXC_OFFSET (0x3C)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD								AMP_OFFSET																							
N/A								RW																							
x	x	x	x	x	x	x	x	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

EXC_OFFSET [31:0]

Bits	Name	Description
23-0	AMP_OFFSET	Offset for excitation

EXC_OFFSET Bits

40.6.9 PWM_SCALING (0x40)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																		N_POL	P_POL	RSVD			DITHER	AMP_EXP				AMP_MAN			
N/A																		RW	RW	N/A			RW	RW				RW			
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	x	x	x	1	0	0	0	1	0	0	0	1

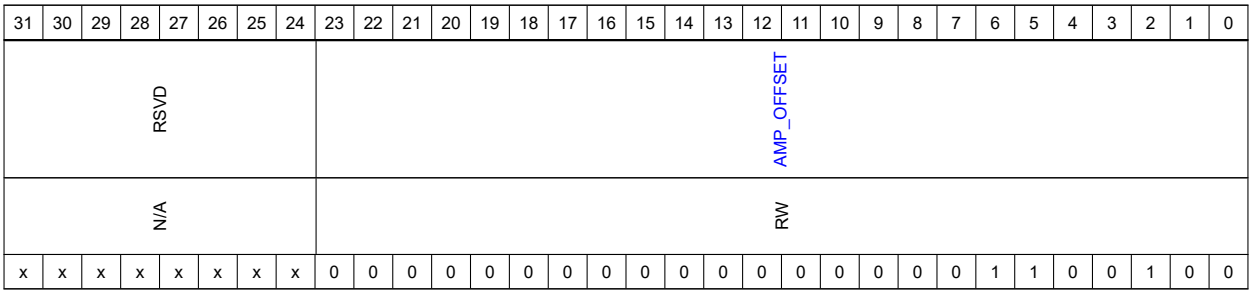
PWM_SCALING [31:0]

Bits	Name	Description
13	N_POL	Polarity of exc_n signal 0: high active 1: low active
12	P_POL	Polarity of exc_p signal 0: high active 1: low active
8	DITHER	Enable dither of pwm 0: disable 1: enable
7-4	AMP_EXP	Amplitude scaling for excitation, amplitude = [table value] x man / 2^exp
3-0	AMP_MAN	Amplitude scaling for excitation, amplitude = [table value] x man / 2^exp

PWM_SCALING Bits

40.6.10 PWM_OFFSET (0x44)

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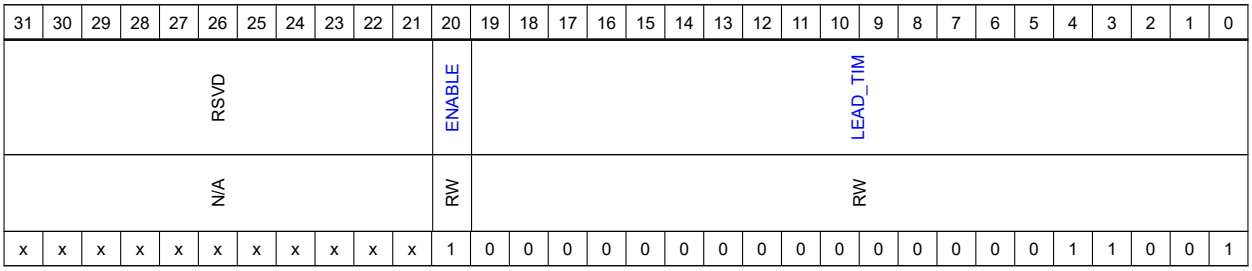


PWM_OFFSET [31:0]

Bits	Name	Description
23-0	AMP_OFFSET	Offset for excitation

PWM_OFFSET Bits

40.6.11 TRIG_OUT0_CFG (0x48)



TRIG_OUT0_CFG [31:0]

Bits	Name	Description
20	ENABLE	Enable trigger out0 0: disable 1: enable
19-0	LEAD_TIM	Lead time for trigger out0 from center of low level , this is a signed value ... 2: 2 cycle befor center of low level 1: 1 cycle before center of low level 0: center of low level -1: 1cycle after center of low level -2: 2cycle after center of low level

TRIG_OUT0_CFG Bits

40.6.12 TRIG_OUT1_CFG (0x4C)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD											ENABLE	LEAD_TIM																			
N/A											RW	RW																			
x	x	x	x	x	x	x	x	x	x	x	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	1	1

TRIG_OUT1_CFG [31:0]

Bits	Name	Description
20	ENABLE	Enable trigger out1 0: disable 1: enable
19-0	LEAD_TIM	Lead time for trigger out0 from center of hight level , this is a signed value ... 2: 2 cycle befor center of hight level 1: 1 cycle before center of hight level 0: center of hight level -1: 1cycle after center of hight level -2: 2cycle after center of hight level

TRIG_OUT1_CFG Bits

40.6.13 PWM_DZ (0x50)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																DZ_N								DZ_P							
N/A																RW								RW							
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PWM_DZ [31:0]

Bits	Name	Description
15-8	DZ_N	Exc_n dead zone in clock cycle before swap 0: no dead zone 1: 1 cycle dead zone 2: 2 cycle dead zone ...

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Bits	Name	Description
7-0	DZ_P	Exc_p dead zone in clock cycle before swap 0: no dead zone 1: 1 cycle dead zone 2: 2 cycle dead zone ...

PWM_DZ Bits

40.6.14 SYNC_OUT_CTRL (0x54)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PWM_OUT_DLY																RSVD										MIN2TRIG_EN	MAX2TRIG_EN	RSVD		SYNC_OUT_SEL	
RO																N/A										RW	RW	N/A		RW	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x	x	x	0	0	x	x	0	0

SYNC_OUT_CTRL [31:0]

Bits	Name	Description
31-16	PWM_OUT_DLY	Delay bettween the delyed trigger and the first pwm pulse in clock cycle 1: 1 cycle 2: 2 cycle ...
5	MIN2TRIG_EN	Enable trigger out from the min point of exciting signal 1: enable 0: disable
4	MAX2TRIG_EN	Enable trigger out from the max point of exciting signal 1: enable 0: disable
1-0	SYNC_OUT_SEL	Select output synchornize signal 0: 0 phase of internal exciting signal 1: 90 phase of internal exciting signal 2: 180 phase of internal exciting signal 3: 270 phase of internal exciting signal

SYNC_OUT_CTRL Bits

40.6.15 EXC_SYNC_DLY (0x58)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD							DISABLE	DELAY																							
N/A							RW	RW																							
x	x	x	x	x	x	x	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

EXC_SYNC_DLY [31:0]

Bits	Name	Description
24	DISABLE	Disable hardware trigger input 0: enable 1: disable
23-0	DELAY	Trigger in delay timming in bus cycle from rising edge of trigger signal 0: 1 cycle 1: 2 cycle ... 0xffffff: 2^24 cycle

EXC_SYNC_DLY Bits

40.6.16 MAX_I (0x70)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MAX																								RSVD							VALID
RWC																								N/A							RWC
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	0

MAX_I [31:0]

Bits	Name	Description
31-8	MAX	Max value of i_channel, write clear
0	VALID	Max value valid, write clear 0: max value is not valid 1: max value is valid

MAX_I Bits

40.6.17 MIN_I (0x74)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MIN																								RSVD							VALID
RWC																								N/A							RWC
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	0

MIN_I [31:0]

Bits	Name	Description
31-8	MIN	Min value of i_channel, write clear
0	VALID	Min value valid, write clear 0: min value is not valid 1: min value is valid

MIN_I Bits

40.6.18 MAX_Q (0x78)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MAX																								RSVD							VALID
RWC																								N/A							RWC
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	0

MAX_Q [31:0]

Bits	Name	Description
31-8	MAX	Max value of q_channel, write clear
0	VALID	Max value valid, write clear 0: max value is not valid 1: max value is valid

MAX_Q Bits

40.6.19 MIN_Q (0x7C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MIN																								RSVD							VALID
RWC																								N/A							RWC
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	0

MIN_Q [31:0]

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Bits	Name	Description
31-8	MIN	Min value of q_channel, write clear
0	VALID	Min value valid, write clear 0: min value is not valid 1: min value is valid

MIN_Q Bits

40.6.20 THRS_I (0x80)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
THRS																								RSVD							
RW																								N/A							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x

THRS_I [31:0]

Bits	Name	Description
31-8	THRS	The offset setting for edge detection of the i_channel, signed number … 2: the offset is 0x800000+2 1: the offset is 0x800000+1 0: the offset is 0x800000 -1: the offset is 0x800000-1 -2: the offset is 0x800000-2 …

THRS_I Bits

40.6.21 THRS_Q (0x84)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
THRS																								RSVD							
RW																								N/A							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x

THRS_Q [31:0]

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Bits	Name	Description
31-8	THRS	The offset setting for edge detection of the q_channel, signed number … 2: the offset is 0x800000+2 1: the offset is 0x800000+1 0: the offset is 0x800000 -1: the offset is 0x800000-1 -2: the offset is 0x800000-2 …

THRS_Q Bits

40.6.22 EDG_DET_CTL (0x88)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																						HOLD						RSVD	FILTER		
N/A																						RW						N/A	RW		
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	1	0	0	0	x	0	0	0

EDG_DET_CTL [31:0]

Bits	Name	Description
9-4	HOLD	The minimum edge distance in sample 0:1 sample 1:2 sample 2:3 samples … 63:64 samples
2-0	FILTER	The continuous positive or negative number for edge detection 0: 1 1: 2 … 7: 8

EDG_DET_CTL Bits

40.6.23 ACC_SCALING (0x8C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																							TOXIC_LK	RSVD				ACC_SHIFT			

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
N/A																							RW	N/A				RW			
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	x	0	0	0	0

ACC_SCALING [31:0]

Bits	Name	Description
8	TOXIC_LK	Toxic accumulation data be removed control 1: enable 0: disable
3-0	ACC_SHIFT	Accumulation value shift control, this is a sign number. 0: {acc[39],acc[38:8]} 1: {acc[39],acc[37:7]} 2: {acc[39],acc[36:6]} … 7: {acc[39],acc[31:1]} 8: {acc[39],acc[30:0]} 9: acc/2^9 10: acc/2^10 … 15:acc/2^15

ACC_SCALING Bits

40.6.24 EXC_PERIOD (0x90)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EXC_PERIOD																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	1	0	1	1	1	0	0	0	0

EXC_PERIOD [31:0]

Bits	Name	Description
31-0	EXC_PERIOD	The num in clock cycle for period of excitation 0: invalid value 1:1 cycle 2:2 cycles …

EXC_PERIOD Bits

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40.6.25 SYNC_DELAY_I (0xA0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DELAY																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0

SYNC_DELAY_I [31:0]

Bits	Name	Description
31-0	DELAY	Delay in clock cycle for synchronous signal, the value shoud less than half of exc_period.exc_period. 0: invalid value 1: 1 cycles 2: 2 cycles ...

SYNC_DELAY_I Bits

40.6.26 RISE_DELAY_I (0xA8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RISE_DELAY																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

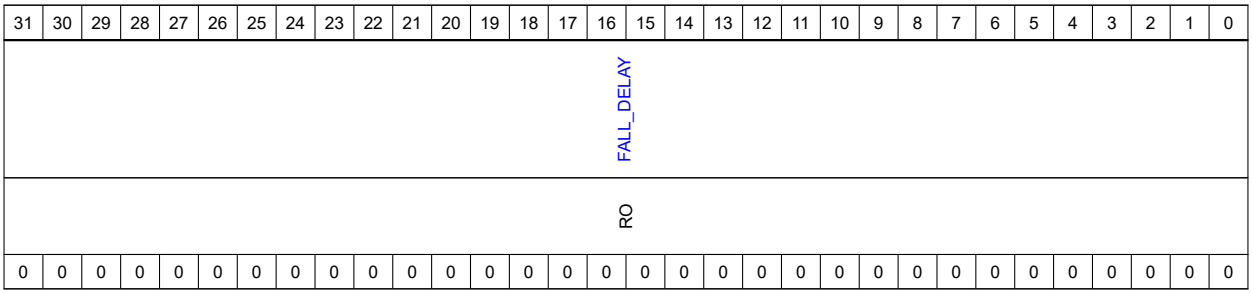
RISE_DELAY_I [31:0]

Bits	Name	Description
31-0	RISE_DELAY	Delay value on rising edge of i_channel data 0: 1 cycle 1: 2 cycles ...

RISE_DELAY_I Bits

40.6.27 FALL_DELAY_I (0xAC)

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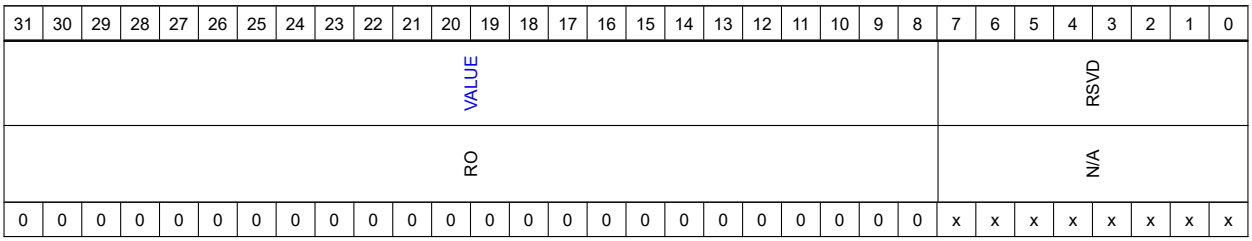


FALL_DELAY_I [31:0]

Bits	Name	Description
31-0	FALL_DELAY	Delay value on falling edge of i_channel data 0: 1 cycle 1: 2 cycles ...

FALL_DELAY_I Bits

40.6.28 SAMPLE_RISE_I (0xB0)

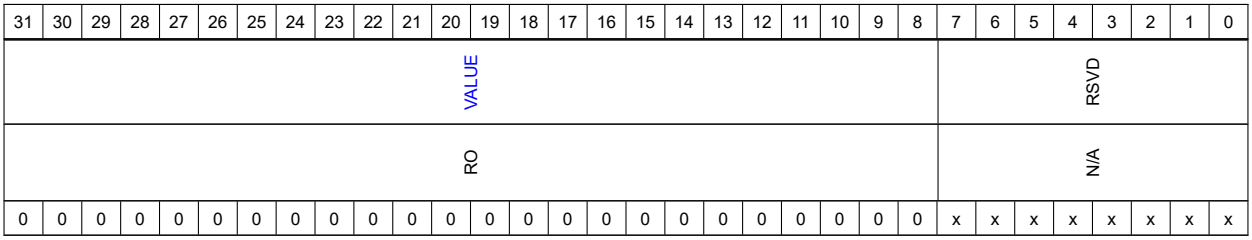


SAMPLE_RISE_I [31:0]

Bits	Name	Description
31-8	VALUE	sample value on rising edge of rectify signal

SAMPLE_RISE_I Bits

40.6.29 SAMPLE_FALL_I (0xB4)



SAMPLE_FALL_I [31:0]

Bits	Name	Description
31-8	VALUE	sample value on falling edge of rectify signal

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Bits	Name	Description

SAMPLE_FALL_I Bits

40.6.30 ACC_CNT_I (0xB8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT_NEG																CNT_POS															
RO																RO															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ACC_CNT_I [31:0]

Bits	Name	Description
31-16	CNT_NEG	sample number during the negtive of rectify signal 1: 1 2: 2 ...
15-0	CNT_POS	sample number during the positive of rectify signal 1: 1 2: 2 ...

ACC_CNT_I Bits

40.6.31 SIGN_CNT_I (0xBC)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT_NEG																CNT_POS															
RO																RO															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

SIGN_CNT_I [31:0]

Bits	Name	Description
31-16	CNT_NEG	Positive sample counter during negative rectify signal
15-0	CNT_POS	Negative sample counter during positive rectify signal

SIGN_CNT_I Bits

40.6.32 SYNC_DELAY_Q (0xC0)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DELAY																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0

SYNC_DELAY_Q [31:0]

Bits	Name	Description
31-0	DELAY	Delay in clock cycle for synchronous signal, the value shoud less than half of exc_period.exc_period. 0: invalid value 1: 1 cycles 2: 2 cycles ...

SYNC_DELAY_Q Bits

40.6.33 RISE_DELAY_Q (0xC8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RISE_DELAY																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

RISE_DELAY_Q [31:0]

Bits	Name	Description
31-0	RISE_DELAY	Delay value on rising edge of q_channel data 0: 1 cycle 1: 2 cycles ...

RISE_DELAY_Q Bits

40.6.34 FALL_DELAY_Q (0xCC)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FALL_DELAY																															

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

FALL_DELAY_Q [31:0]

Bits	Name	Description
31-0	FALL_DELAY	Delay value on falling edge of q_channel data 0: 1 cycle 1: 2 cycles ...

FALL_DELAY_Q Bits

40.6.35 SAMPLE_RISE_Q (0xD0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VALUE																								RSVD							
RO																								N/A							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x

SAMPLE_RISE_Q [31:0]

Bits	Name	Description
31-8	VALUE	sample value on rising edge of rectify signal

SAMPLE_RISE_Q Bits

40.6.36 SAMPLE_FALL_Q (0xD4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VALUE																								RSVD							
RO																								N/A							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x

SAMPLE_FALL_Q [31:0]

Bits	Name	Description
31-8	VALUE	sample value on falling edge of rectify signal

SAMPLE_FALL_Q Bits

40.6.37 ACC_CNT_Q (0xD8)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT_NEG																CNT_POS															
RO																RO															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ACC_CNT_Q [31:0]

Bits	Name	Description
31-16	CNT_NEG	sample number during the negtive of rectify signal 1: 1 2: 2 ...
15-0	CNT_POS	sample number during the positive of rectify signal 1: 1 2: 2 ...

ACC_CNT_Q Bits

40.6.38 SIGN_CNT_Q (0xDC)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT_NEG																CNT_POS															
RO																RO															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

SIGN_CNT_Q [31:0]

Bits	Name	Description
31-16	CNT_NEG	Positive sample counter during negative rectify signal
15-0	CNT_POS	Negative sample counter during positive rectify signal

SIGN_CNT_Q Bits

40.6.39 AMP_MAX (0xE0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MAX																															
RW																															
0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AMP_MAX [31:0]																															

Bits	Name	Description
31-0	MAX	the maximum of acc amplitude

AMP_MAX Bits

40.6.40 AMP_MIN (0xE4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MIN																															
RW																															
0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

AMP_MIN [31:0]

Bits	Name	Description
31-0	MIN	the minimum of acc amplitude

AMP_MIN Bits

40.6.41 INT_EN (0xE8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INT_EN	RSVD															ACC_VLD_I_EN	ACC_VLD_Q_EN	RISING_DELAY_I_EN	FALLING_DELAY_I_EN	RISING_DELAY_Q_EN	FALLING_DELAY_Q_EN	SAMPLE_RISING_I_EN	SAMPLE_FALLING_I_EN	SAMPLE_RISING_Q_EN	SAMPLE_FALLING_Q_EN	ACC_VLD_I_OVH_EN	ACC_VLD_Q_OVH_EN	ACC_VLD_I_OVL_EN	ACC_VLD_Q_OVL_EN	ACC_AMP_OVH_EN	ACC_AMP_OVL_EN
RW	N/A															RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW
0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

INT_EN [31:0]

Bits	Name	Description
31	INT_EN	enable interrupt output
15	ACC_VLD_I_EN	i_channel accumulate valid interrupt enable for i_channel
14	ACC_VLD_Q_EN	q_channel accumulate valid interrupt enable for i_channel
13	RISING_DELAY_I_EN	i_channel delayed rectify signal rising edge interrupt enable

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Bits	Name	Description
12	FALLING_DELAY_I_EN	i_channel delayed rectify signal falling edge interrupt enable
11	RISING_DELAY_Q_EN	q_channel delayed rectify signal rising edge interrupt enable
10	FALLING_DELAY_Q_EN	q_channel delayed rectify signal falling edge interrupt enable
9	SAMPLE_RISING_I_EN	i_channel rising edge interrupt enable
8	SAMPLE_FALLING_I_EN	i_channel falling edge interrupt enable
7	SAMPLE_RISING_Q_EN	q_channel rising edge interrupt enable
6	SAMPLE_FALLING_Q_EN	q_channel falling edge interrupt enable
5	ACC_VLD_I_OVH_EN	i_channel accumulate overflow interrupt enable
4	ACC_VLD_Q_OVH_EN	q_channel accumulate overflow interrupt enable
3	ACC_VLD_I_OVL_EN	i_channel accumulate underflow interrupt enable
2	ACC_VLD_Q_OVL_EN	q_channel accumulate underflow interrupt enable
1	ACC_AMP_OVH_EN	accumulate ample overflow interrupt enable
0	ACC_AMP_OVL_EN	accumulate ample underflow interrupt enable

INT_EN Bits

40.6.42 ADC_INT_STATE (0xEC)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																ACC_VLD_I_STA	ACC_VLD_Q_STA	RISING_DELAY_I_STA	FALLING_DELAY_I_STA	RISING_DELAY_Q_STA	FALLING_DELAY_Q_STA	SAMPLE_RISING_I_STA	SAMPLE_FALLING_I_STA	SAMPLE_RISING_Q_STA	SAMPLE_FALLING_Q_STA	ACC_VLD_I_OVH_STA	ACC_VLD_Q_OVH_STA	ACC_VLD_I_OVL_STA	ACC_VLD_Q_OVL_STA	ACC_AMP_OVH_STA	ACC_AMP_OVL_STA
N/A																W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

ADC_INT_STATE [31:0]

Bits	Name	Description
15	ACC_VLD_I_STA	i_channel accumulate valid interrupt status for i_channel
14	ACC_VLD_Q_STA	q_channel accumulate valid interrupt status for i_channel
13	RISING_DELAY_I_STA	i_channel delayed rectify signal rising edge interrupt status
12	FALLING_DELAY_I_STA	i_channel delayed rectify signal falling edge interrupt status
11	RISING_DELAY_Q_STA	q_channel delayed rectify signal rising edge interrupt status
10	FALLING_DELAY_Q_STA	q_channel delayed rectify signal falling edge interrupt status
9	SAMPLE_RISING_I_STA	i_channel rising edge interrupt status
8	SAMPLE_FALLING_I_STA	i_channel falling edge interrupt status
7	SAMPLE_RISING_Q_STA	q_channel rising edge interrupt status
6	SAMPLE_FALLING_Q_STA	q_channel falling edge interrupt status
5	ACC_VLD_I_OVH_STA	i_channel accumulate overflow interrupt status
4	ACC_VLD_Q_OVH_STA	q_channel accumulate overflow interrupt status
3	ACC_VLD_I_OVL_STA	i_channel accumulate underflow interrupt status
2	ACC_VLD_Q_OVL_STA	q_channel accumulate underflow interrupt status
1	ACC_AMP_OVH_STA	accumulate ample overflow interrupt status
0	ACC_AMP_OVL_STA	accumulate ample underflow interrupt status

ADC_INT_STATE Bits

40.7 RDC Configuration and Usage Instructions

40.7.1 Excitation Generation Configuration Instructions

The excitation signal output supports DAC output and PWM output, the steps are as follows:

1. Configure the period of excitation signal, set registers exc_timming.smp_rate and exc_timming.smp_num

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to required values;

2. If the excitation signal uses DAC output mode, configure registers exc_scaling.amp_man, exc_scaling.ampl_exp and exc_offset.ampl_offset to appropriate values;
3. If the excitation signal uses PWM output mode, configure registers pwm_scaling.ampl_man, pwm_scaling.ampl_exp and pwm_offset.ampl_offset to appropriate values; if you need to accumulate the duty cycle decimal part of the previous PWM cycle to the next PWM cycle, you also need to set register pwm_scaling.dither to 1;
4. Set register rdc_ctl.exc_en to 1 to enable excitation signal generation;
5. If you use software to trigger excitation signal, write register rdc_ctl.exc_start to 1 to trigger excitation signal output;
6. If you use hardware event to trigger excitation signal, you need to configure register exc_sync_dly.dly to appropriate value to control the delay from hardware event to excitation signal generation, then set exc_sync_dly.disable to 0, the excitation signal will be generated when the hardware event is triggered.

40.7.2 Delay Measurement Configuration Instructions

There are multiple methods for delay measurement. The following describes the steps to measure the phase delay from excitation signal generation to induction signal reception with the assistance of RDC module:

1. Set rdc_ctl.exc_en and rdc_ctl.acc_en to 1, set rdc_ctl.rectify_sel to 0, select excitation signal generator as synchronization source;
2. Configure in_ctl to appropriate value, select appropriate ADC input and channel number;
3. Read the values of max_i/q to obtain the maximum and minimum values of ADC input; calculate the intermediate value for multiple times and take the average, then configure the result to thrs_i/q;
4. If you need to enable jitter filtering function, configure edg_det_ctl.filter to appropriate value;
5. Read the value of rise_delay_i/q and calculate the average for multiple times, this value is the phase delay value. When this value exceeds half of the excitation signal period, set register rdc_ctl.rectify_sel to 2 and repeat the above steps;
6. Configure the phase delay value to register sync_delay_i/q, and configure the excitation signal period to register exc_period at the same time.

40.7.3 Carrier Accumulation Configuration Instructions

The configuration method of carrier accumulation is:

1. Set rdc_ctl.acc_en to 1, configure RDC_ctl.acc_len to required value;
2. Configure in_ctl to appropriate value, select correct ADC input source;
3. Wait for interrupt generation, read the values of acc_i/q to obtain the accumulation result;

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41 Serial Encoder Interface (SEI)

This chapter describes the main functions and features of SEI.

41.1 Feature Summary

This section describes the main features of SEI:

- 2 SEI controllers
- 9 data register groups
- Supports up to 64 instructions
- Supports synchronous and asynchronous communication
- Supports master mode and slave mode (acting as an encoder)
- Supports RS-485 and RS-422 interfaces
- Each SEI controller supports 3 trigger modes
  - External trigger, with 8 channels each for trigger input and trigger output
  - Periodic trigger
  - Software trigger
- Supports precise control of the timing for SAMPLE or UPDATE of position information and timestamps
- Supports command matching and instruction jump
- Supports automatic CRC check
- Supports automatic parity check
- Supports WatchDog
- Supports TIMEOUT and transceiving CDM/CDS

The block diagram of SEI is shown in Figure 61.

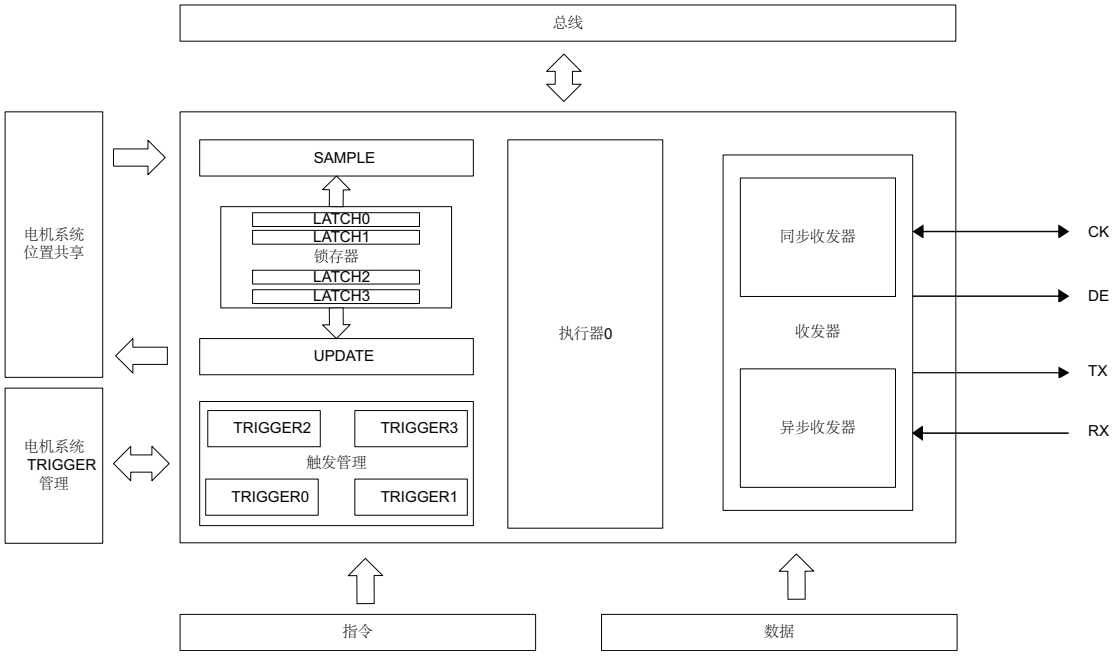


Figure 61: SEI Block Diagram

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41.2 Functional Description

41.2.1 Data Transmission and Reception

Each SEI group has 4 pins: DE, CK, TX and RX, supporting RS-485 and RS-422 interfaces.

When working as a master, SEI is mainly used to obtain position information from the outside (such as rotor position, number of turns, rotation speed, motor operating status, etc.). When working as a slave, SEI is mainly used to send position information obtained from the motor system to the outside, acting as an encoder.

41.2.1.1 Asynchronous Mode

When the transceiver works in asynchronous mode, configure the BAUD_DIV bit field in the CTRL[n] [XCVR] [BAUD_CFG] register to set the baud rate for data transceiving, and adjust the data transceiving timing by setting the RXD_POINT and TXD_POINT bit fields in the CTRL[n] [XCVR] [DATA_CFG] register.

In addition, it is necessary to configure the CTRL[n] [XCVR] [TYPE_CFG] register according to specific protocol requirements to set the status of DE, TX and RX pins, and control the line status of the data line and clock line in idle state. The data length during transceiving is set through the DATA_LEN bit field in the CTRL[n] [XCVR] [TYPE_CFG] register, supporting up to 32 bits. The transceiver will automatically insert start bit, stop bit and parity bit during communication according to this setting. Select whether to enable parity check and choose odd or even parity through PAR_EN and PAR_POL in the CTRL[n] [XCVR] [TYPE_CFG] register.

41.2.1.2 Synchronous Mode

In synchronous mode, select whether the transceiver works in master mode or slave mode as required. When it is necessary to obtain position information from an external encoder, configure SEI as master mode to use SEI as the host; when it is necessary to send the position information inside the motor system to the outside, configure SEI as slave mode, and SEI acts as an encoder at this time.

By configuring the BAUD_DIV bit field in the CTRL[n] [XCVR] [BAUD_CFG] register, you can set the clock frequency for data transceiving, and configure the CTRL[n] [XCVR] [CLK_CFG] register to set the clock transition points CK0_POINT and CK1_POINT. Configure the RXD_POINT and TXD_POINT bit fields in the CTRL[n] [XCVR] [DATA_CFG] register to adjust the data transceiving timing. In addition, configure the CTRL[n] [XCVR] [TYPE_CFG] register according to specific protocol requirements to set the status of CK, DE, TX and RX pins, and control the line status of the data line and clock line in idle state.

41.2.2 Instructions

SEI has 8 different operations in total, and performs data transceiving related operations according to the configuration of the instruction register, supporting up to 64 instructions. Each instruction consists of operation, clock, CRC register, DATA register, and length fields. Users configure each instruction in sequence according to the application scenario and timing sequence, and combine multiple instructions to form a timing that meets the target communication protocol. The description of each field is as follows:

- Operation field: Indicates the operation performed by this instruction. The description of each operation type in the operation field is as follows:

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Op-code	Operation	Function Description
000	Stop/Wait	The transceiver stops for a period of time or ends operation, used to generate delay or waiting time
001	Jump	Used for instruction jump: Can jump to the corresponding instruction to start execution according to the result of command matching; Can directly jump to the instruction in POINTER_WDOG to start execution; Can directly jump to the instruction in POINTER_INIT to start execution;
010	Timed Send	Send a certain number of data bits, with time limit
011	Untimed Send	Send a certain number of data bits, no time limit
100	Timed Wait	Wait for the expected level on the data line, used for synchronization between master and slave, with time limit
101	Untimed Wait	Wait for the expected level on the data line, used for synchronization between master and slave, no time limit
110	Timed Receive	Receive a certain number of data bits, with time limit
111	Untimed Receive	Receive a certain number of data bits, no time limit

Table 185: Operation Types and Function Descriptions

- Clock field: Configure the clock line status when this instruction is executed.
- CRC register field: Specify the data register used for CRC calculation.
- DATA register field: Used to select the data register used for data transceiving of this instruction.
- Operand field: The length of data transceived by this instruction, or the duration of instruction execution.

41.2.3 Data

SEI has a total of 9 groups of shared data registers, including 1 group of ignore data registers, and the rest are general-purpose data registers. Each group of data registers is independent of each other. All SEIs share these 9 groups of data registers, which can be allocated and used according to the requirements of each SEI during use. In addition, each SEI has a group of private command registers.

No.	Purpose	Function Description
DATA[0]	Ignore Data Register	Used to receive meaningless data or data that users do not care about during communication. The data received by this register will be discarded, and users cannot perform other processing on the data.

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No.	Purpose	Function Description
CMD	Command Register	Used to receive or send commands in communication. If the trigger function is enabled, the hardware will load the command in the corresponding trigger command register to the command register for transceiving after the trigger is generated.
DATA[9:2]	General-purpose Data Register	When working in data mode, used to receive or send data; When working in check mode, used to check and compare the received data with the expected data; When working in CRC mode, used to calculate CRC or CHECKSUM;

Table 186: Purpose and Description of Data Register Groups

In addition, the DAT field in SEI instructions also has constant 0 and constant 1 that can be used as the data content for reception or transmission. During use, you can flexibly configure to send 0 or 1 of a certain length to the data line, or wait until 0 or 1 of a certain length is received on the data line as required.

It should be noted that when the same instruction using the command register is executed by SEI[n] and SEI[m] respectively, the internal command registers of SEI[n] and SEI[m] will be automatically used, no additional configuration is required.

Data Register Working Modes

- Data Mode
  When working in data mode, you can configure the data format of the data register for data transceiving through the DATA[n] [MODE] register, including data length, word order, bit order, and whether it is a signed number, etc. At the same time, you also need to configure the data filling method and filling order in the DATA[n] [IDX] register. After the configuration is completed, you need to set the reset pointer of the data register to update the initial pointer in the data register.
- Check Mode
  When working in check mode, the software writes the expected value to the DATA[n] [GOLD] register in advance, and other configurations are similar to those in data mode. When using this data register to receive data, the received data will be checked and compared with the expected data bit by bit, and a transmission error will be generated if they are inconsistent.
- CRC Mode
  When working in CRC mode, fill the CRC polynomial and CRC initial value into the DATA[n] [CRCPOLY] register and DATA[n] [CRCINIT] register respectively, and configure the CRC length, shift mode and whether to enable CRC inversion in the DATA[n] [MODE] register as required. By configuring operation instructions, the CRC calculation result can be sent as data, or the CRC calculation result can be used as the expected value to check and compare with the received CRC. Each time the CRC calculation result is sent to the line, or the data received on the line is compared with the CRC calculation value,

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the CRC calculation is considered completed; when CRC calculation is performed on the received data or sent data again, a new CRC calculation will automatically start again.
It should be noted that when SAMPLE or UPDATE occurs, the CRC calculation result in the data register will also be cleared and automatically reset to the initial value. In addition, due to configurations such as transceiving order and whether to invert bit by bit, the value directly read by the software from the CRC register will not be directly equal to the data on the communication line.

Command Register and Command Matching

The basic functions and configuration methods of the command register are the same as those of the general-purpose data register, but the command register can match the received command with the pre-filled command in the command table, and jump to the corresponding instruction to perform subsequent operations according to the matching result.

After receiving CMD, use the jump command to start command matching and jump to the matching result. SEI will automatically compare from CMD_TABLE[0] to CMD_TABLE[6] one by one according to the matching conditions. If none are matched, it will unconditionally jump to the last one, that is, the PTR[n] instruction of CMD_TABLE[7]. The matching conditions are:

- CMD_TABLE[n] [MSK]: Select the bits that need to be compared. When all zeros, it means that the value of any bit field is not cared, that is, this CMD_TABLE is unconditionally satisfied
- CMD_TABLE[n] [MIN]: The minimum value of CMD, the received CMD needs to be greater than or equal to the value of this register
- CMD_TABLE[n] [MAX]: The maximum value of CMD, the received CMD needs to be less than or equal to the value of this register

If the matching conditions are met, jump to the PTR[n] instruction of the CMD_TABLE to start execution (the value of PTR[n] uses absolute index), where n is selected by the OPR[3:0] field of the jump instruction.

41.2.4 Trigger Management

SEI supports direct execution of program instructions, and also supports controlling the execution of program instructions through trigger signals. SEI supports 3 trigger modes: external trigger, periodic trigger and software trigger, where external trigger is TRIGGER[0] and TRIGGER[1], periodic trigger is TRIGGER[2], and software trigger is TRIGGER[3].

There are 4 trigger command registers inside SEI (namely TRG_TABLE[CMD] [n]), which correspond to TRIGGER[0] ~TRIGGER[3] one by one. When using the trigger function, the user needs to write the command to the corresponding trigger command register in advance, so that when the trigger is generated, the hardware will automatically fill the command in the trigger command register corresponding to the trigger mode into the command register, and send the command when executing the instruction of sending the command register. In addition, when a trigger is generated, the time when the trigger is generated will be recorded, and the user can obtain the timestamp of each trigger generation time by reading the corresponding register, so as to estimate the transmission time.

41.2.4.1 External Trigger

- Trigger Input

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Each SEI supports 8 channels of external trigger signal input. Users can select any input signal from the 8 channels as the trigger signal for TRIGGER[0] and TRIGGER[1]. The source of the external trigger signal can be selected by configuring TRIGM, usually from other modules in the motor system, or from outside the chip through the GPIO of TRIGM, so as to trigger SEI to start executing instructions.

- Trigger Output
  Each SEI supports 8 channels of trigger signal output, and the trigger signals of the 8 channels come from the 4 latches LATCH[3:0] inside SEI. Users can configure the OUTx_SEL bit field in the CTRL[n] [TRG] [OUT_CFG] register to output the output signal of the corresponding latch LATCH[x] to TRGM from any channel, so as to output to other modules in the motor system or outside the chip.

41.2.4.2 Periodic Trigger

SEI has a 32-bit counter inside to realize periodic triggering of program instructions. By configuring the trigger period, program instructions can be executed periodically. Users can view the status of periodic triggers and the number of triggers from related registers. Periodic trigger can choose to trigger directly or wait for the trigger source before starting to trigger:

- If configured as direct trigger, the first execution will be triggered immediately after the periodic trigger is enabled, and timing will restart. When the trigger period is reached, a new trigger signal will be generated, and the second execution will start, and so on.
- If configured to wait for the trigger source to generate the trigger, after enabling the periodic trigger, the first execution and timing will not start immediately, and SEI will start the periodic trigger only after the external trigger source arrives. According to requirements, you can also choose whether to synchronize the external trigger source with the periodic trigger signal. If synchronized, a trigger will be generated each time the external trigger source arrives, followed by periodic triggering.

It should be noted that the operation of enabling periodic trigger can only be performed after all other configurations are completed, to avoid SEI starting to work before the configuration is completed. And the configured trigger period needs to be longer than the time for a single execution of the program instruction, otherwise the new trigger signal will interrupt the ongoing communication.

41.2.4.3 Software Trigger

The software can trigger SEI to directly start executing instructions by setting the CTRL[n] [TRG] [SW] register. The software-triggered instruction is executed only once. If multiple executions or repeated executions are required, a jump instruction can be added to the instruction. After executing the jump instruction, the program can jump to a specific instruction to start execution.

41.2.5 SAMPLE and UPDATE of Position Information

Through the SAMPLE and UPDATE functions, SEI can not only obtain position information from external encoders with communication protocols, but also send position information generated by other modules inside the motor system to the outside through communication protocols.

41.2.5.1 LATCH State Machine

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The implementation of SEI's SAMPLE and UPDATE is mainly based on state machine transitions. Each SEI has 4 LATCH state machines available for selection to control SAMPLE or UPDATE. Each LATCH state machine has 4 states, and these 4 states can only flow in one direction, that is, state 0 to state 1, state 1 to state 2, state 2 to state 3, state 3 to state 0. By configuring registers, specific time points in the communication process can be used as state transition conditions. The optional conditions for state machine transition are as follows:

- Transceiving timeout, such as when status is 1, status is 0, rising edge
- In master mode, clock line status, such as when status is 0, status is 1, rising edge, falling edge
- Transmit data line TX status, such as when status is 0, status is 1, rising edge, falling edge
- Pointer, such as when pointer matches, pointer does not match, pointer starts, pointer ends

In actual configuration, it is not necessary to set transition conditions for each state of the state machine. When 4 states are not required, the intermediate state can be configured as unconditional jump. By configuring the LATCH delay, the occurrence timing of SAMPLE or UPDATE can be delayed.

41.2.5.2 SAMPLE Function

When working as a master, the SAMPLE function is mainly used for timestamp sampling, providing timestamps for the position information obtained by each communication, so that other modules inside the motor system can perform subsequent analysis and processing, such as prediction and following of position information. When working as a slave, the SAMPLE function samples the position information generated by other modules of the motor system to the specified data register for sending to the outside, and the timestamp is also sampled at the same time. For the selection of SAMPLE time point, please refer to the relevant protocol.

According to the requirements of the application scenario, any one or more of position, number of turns, speed and acceleration can be used as the sampled data when SAMPLE occurs. You only need to specify the data register used for transceiving the data and enable the corresponding control bit. In addition, you can also configure the sampling window and sampling method when SAMPLE occurs.

41.2.5.3 UPDATE Function

When working as a master, the UPDATE function is mainly used to update the obtained position information and timestamp to the inside of the motor system for use by other modules after the transmission is completed.

According to the requirements of the application scenario, any one or more of position, number of turns, speed and acceleration can be used as the updated data when UPDATE occurs. You only need to specify the data register used for transceiving the data and enable the corresponding control bit. You can also configure whether to use the received value or the overwrite value filled by internal software when UPDATE occurs. In addition, when transmission errors occur, you can also choose whether to still pass the data to the next level through UPDATE.

41.2.6 Interrupt and DMA

SEI supports generating the following interrupts:

- TRG_ERR[3:0]: When an error occurs in TRIGGER[0]~TRIGGER[3];
- TRIGER[3:0]: When TRIGGER[0]~TRIGGER[3] generates a trigger signal;
- LATCH[3:0]: When LATCH[0]~LATCH[3] state transitions occur;

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- SMP_ERR: When an error occurs in sampling position information;
- TIMEOUT: When a timeout occurs during communication;
- TRX_ERR: When a transmission error occurs during transmission, including CRC check result error, data check comparison error, etc.;
- INSTR0/1: At the start or end of instruction 0 or instruction 1, you need to configure the CTRL[n] [IRQ] [INSTR0] or CTRL[n] [IRQ] [INSTR1] register to specify any instruction as instruction 0 or instruction 1, so an interrupt can be triggered when any specific instruction is executed;
- PTR0/1: At the start or end of pointer 0 or pointer 1, you need to configure CTRL[n] [IRQ] [POINTER0] or CTRL[n] [IRQ] [POINTER1] to specify any pointer as pointer 0 or pointer 1, so an interrupt can be triggered when any specific pointer is executed;
- WDOG: When the set watchdog is triggered;
- EXCEPT: When an abnormal event occurs; in this product, it only indicates whether the interpretation of WDOG timeout as an exception has been successfully enabled. WDOG timeout errors or communication abnormalities are checked by WDOG interrupt and TRX_ERR interrupt.
- STALL: When the communication of the transceiver stops;

SEI supports generating the following DMA requests:

- TRG_ERR[3:0]: When an error occurs in TRIGGER[0]~TRIGGER[3];
- TRIGER[3:0]: When TRIGGER[0]~TRIGGER[3] generates a trigger signal;
- LATCH[3:0]: When LATCH[0]~LATCH[3] state transitions occur;
- SMP_ERR: When an error occurs in sampling position information;
- TIMEOUT: When a timeout occurs during communication;
- TRX_ERR: When a transmission error occurs during transmission, including CRC check result error, data check comparison error, etc.;
- INSTR0/1: At the start or end of instruction 0 or instruction 1, you need to configure the CTRL[n] [IRQ] [INSTR0] or CTRL[n] [IRQ] [INSTR1] register to specify any instruction as instruction 0 or instruction 1, so a DMA request can be generated when any specific instruction is executed;
- PTR0/1: At the start or end of pointer 0 or pointer 1, you need to configure CTRL[n] [IRQ] [POINTER0] or CTRL[n] [IRQ] [POINTER1] to specify any pointer as pointer 0 or pointer 1, so a DMA request can be generated when any specific pointer is executed;
- WDOG: When the set watchdog is triggered;
- EXCEPT: When an abnormal event occurs; in this product, it only indicates whether the interpretation of WDOG timeout as an exception has been successfully enabled. WDOG timeout errors or communication abnormalities are checked by WDOG interrupt and TRX_ERR interrupt.
- STALL: When the communication of the transceiver stops;

41.2.7 Protocol Support

SEI can implement a variety of communication protocols through programming, such as Tamagawa encoder protocol, HIPERFACE®, Nikon encoder protocol, SSI, BiSS-C and EnDat2.1/2.2 protocols, etc. Users can also support more protocols through custom programming. Although the supported protocols are very rich, the overall configuration process and idea of SEI are basically the same. Taking the Tamagawa encoder communication protocol as an example, when SEI acts as the host to periodically read the encoder data through the DataID 3

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command, the configuration process is as follows:

- Configure control registers
  - Configure the initial program pointer position and data register address offset;
  - Configure SEI working mode, asynchronous transceiver baud rate and line status in idle state;
  - Configure periodic trigger;
  - Fill the Data ID 3 command into the periodic trigger command register CTRL[0] [TRG_TABLE] [CMD] [2];
- Configure operation instructions
  - Use the command register to send an 8-bit read position command, that is, send the Control Field (CF);
  - Use data register 2 to receive the 8-bit read position command CF data, and pass the received data to data register 7 to calculate CRC;
  - Use data register 3 to receive 8-bit Status Field (SF) data, and use data register 7 to calculate CRC;
  - Use data register 4 to receive 24-bit position information (ABS0~ABS2, Absolute data in one revolution) in the Data Field (DF), and use data register 7 to calculate CRC;
  - Use data register 0 to receive the 8-bit Encoder ID (ENID) in the data field, and use data register 7 to calculate CRC;
  - Use data register 5 to receive 24-bit turn number information (ABM0~ABM2, Multi-turn data) in the data field, and use data register 7 to calculate CRC;
  - Use data register 6 to receive 8-bit encoder error information (ALMC) in the data field, and use data register 7 to calculate CRC;
  - Use data register 7 to receive CRC check bit (CRC Field) data and perform CRC check;
- Configure the data registers to be used
  - Configure the data mode register corresponding to the command register (CTRL[0] [CMD] [MODE]) and data index register (CTRL[0] [CMD] [IDX]). MODE is configured as data mode.
  - Configure the data mode registers corresponding to data registers 2~6 (DATA[n] [MODE]) and data index registers (DATA[n] [IDX]), where MODE is configured as data mode;
  - Configure data register 7 as CRC mode, configure CRC polynomial, initial value, etc.;
- Enable periodic trigger
  - Enable ENGINE and periodic trigger, and the instructions start to execute.

41.2.7.1 Asynchronous Communication Protocol

Tamagawa instruction configuration reference:

INSTR	OP Field	CK Field	CRC Field	DAT Field	LEN Field	Register Configuration	Configuration Description
0	Untimed Send	Any	Do not calculate CRC	Command register	8 bits	0x0C000107	Use command register to send CF command

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INSTR	OP Field	CK Field	CRC Field	DAT Field	LEN Field	Register Configuration	Configuration Description
1	Untimed Receive	Any	Data register 7	Data register 2	8 bits	0x1C070207	Use data register 2 to receive CF data, and use data register 7 to calculate CRC
2	Untimed Receive	Any	Data register 7	Data register 3	8 bits	0x1C070307	Use data register 3 to receive SF, and calculate CRC
3	Untimed Receive	Any	Data register 7	Data register 4	24 bits	0x1C070417	Use data register 4 to receive ABS0~ABS2, and calculate CRC
4	Untimed Receive	Any	Data register 7	Data register 0	8 bits	0x1C070007	Use data register 0 to receive ENID, and calculate CRC
5	Untimed Receive	Any	Data register 7	Data register 5	24 bits	0x1C070517	Use data register 5 to receive ABM0~ABM2, and calculate CRC
6	Untimed Receive	Any	Data register 7	Data register 6	8 bits	0x1C070607	Use data register 6 to receive ALMC, and calculate CRC
7	Untimed Receive	Any	Do not calculate CRC	Data register 7	8 bits	0x1C000707	Use data register 7 to receive and compare CRC

Table 187: SEI Instruction Reference When Working as Tamagawa Host

HIPERFACE® instruction configuration reference (taking 42H command as an example):

INSTR	OP Field	CK Field	CRC Field	DAT Field	LEN Field	Register Configuration	Configuration Description
0	Untimed Send	Any	Data register 7	Data register 2	8 bits	0x0C070207	Use data register 2 to send address data, and use data register 7 to calculate checksum
1	Untimed Send	Any	Data register 7	Command register	8 bits	0x0C070107	Use command register to send command, and calculate checksum

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INSTR	OP Field	CK Field	CRC Field	DAT Field	LEN Field	Register Configuration	Configuration Description
2	Untimed Send	Any	Do not calculate CRC	Data register 7	8 bits	0x0C000707	Send the checksum in data register 7
3	Untimed Receive	Any	Data register 7	Data register 3	8 bits	0x1C070307	Use data register 3 to receive address data returned by the encoder, and use data register 7 to calculate checksum
4	Untimed Receive	Any	Data register 7	Data register 4	8 bits	0x1C070407	Use data register 4 to receive command data returned by the encoder, and calculate checksum
5	Untimed Receive	Any	Data register 7	Data register 5	8 bits	0x1C070507	Use data register 5 to receive Position[31:24] in position data, and calculate checksum
6	Untimed Receive	Any	Data register 7	Data register 5	8 bits	0x1C070507	Use data register 5 to receive Position[23:16] in position data, and calculate checksum
7	Untimed Receive	Any	Data register 7	Data register 5	8 bits	0x1C070507	Use data register 5 to receive Position[15:8] in position data, and calculate checksum
8	Untimed Receive	Any	Data register 7	Data register 5	8 bits	0x1C070507	Use data register 5 to receive Position[7:0] in position data, and calculate checksum
9	Untimed Receive	Any	Do not calculate CRC	Data register 7	8 bits	0x1C000707	Use data register 7 to receive and compare checksum

Table 188: SEI Instruction Reference When Working as HIPERFACE® Host

Nikon instruction configuration reference (taking CDF0 command as an example):

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INSTR	OP Field	CK Field	CRC Field	DAT Field	LEN Field	Register Configuration	Configuration Description
0	Untimed Send	Any	Data register 5	Command register	13 bits	0x0C05010C	Send the Sink Code, Frame Code, EA and CC parts of the CDF0 command in the command register, and use data register 5 to calculate CRC
1	Untimed Send	Any	Do not calculate CRC	Data register 5	3 bits	0x0C000502	Send the CRC data in data register 5, which is the CRC calculation result of the data in the command register
2	Untimed Receive	Any	Data register 5	Data register 2	16 bits	0x1C05020F	Use data register 2 to receive Information Field (IF), and use data register 5 to calculate CRC
3	Untimed Receive	Any	Data register 5	Data register 3	16 bits	0x1C05030F	Use data register 3 to receive Data Field 0 (DF0), and calculate CRC
4	Untimed Receive	Any	Data register 5	Data register 3	16 bits	0x1C05030F	Use data register 3 to receive Data Field 1 (DF1), and calculate CRC
5	Untimed Receive	Any	Data register 5	Data register 4	8 bits	0x1C050407	Use data register 4 to receive the non-CRC part of Data Field 2 (DF2), and calculate CRC
6	Untimed Receive	Any	Do not calculate CRC	Data register 5	8 bits	0x1C000507	Use data register 5 to receive the CRC part of Data Field 2 (DF2), and compare CRC

Table 189: SEI Instruction Reference When Working as Nikon Host

41.2.7.2 Synchronous Communication Protocol

SSI instruction configuration reference (taking POSITAL's SSI protocol description as an example):

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INSTR	OP Field	CK Field	CRC Field	DAT Field	LEN Field	Register Configuration	Configuration Description
0	Stop/Wait	Clock line high	Do not calculate CRC	Ignore data	2 bits	0x03000001	Pull the clock line high and wait for 2 clock cycles
1	Untimed Receive	Clock line rising-falling	Do not calculate CRC	Constant 1	1 bits	0x1D001F00	Send the clock and receive 1 bit of 1 in the idle state of the data line at the first falling edge
2	Untimed Receive	Clock line rising-falling	Do not calculate CRC	Data register 3	1 bit	0x1D000300	Use data register 3 to receive 1 bit of 1
3	Untimed Receive	Clock line rising-falling	Do not calculate CRC	Data register 8	12 bits	0x1D00080B	Use data register 8 to receive 12 bits of multi-turn value (MT)
4	Untimed Receive	Clock line rising-falling	Do not calculate CRC	Data register 9	12 bits	0x1D00090B	Use data register 9 to receive 12 bits of single-turn value (ST)
5	Untimed Wait	Clock line high	Do not calculate CRC	Constant 1	1 bit	0x17001F00	Pull the clock line high, wait until 1 bit of 1 is received, wait for WatchDog to generate TIMEOUT

Table 190: Master Mode

INSTR	OP Field	CK Field	CRC Field	DAT Field	LEN Field	Register Configuration	Configuration Description
0	Untimed Send	Clock line falling-rising	Do not calculate CRC	Data register 6	1 bit	0x0E000600	Send 1 bit of 1
1	Untimed Send	Clock line falling-rising	Do not calculate CRC	Data register 6	12 bits	0x0E00040B	Use data register 6 to send 12 bits of multi-turn value (MT)
2	Untimed Send	Clock line falling-rising	Do not calculate CRC	Data register 5	12 bits	0x0E00050B	Use data register 5 to send 12 bits of single-turn value (ST)

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INSTR	OP Field	CK Field	CRC Field	DAT Field	LEN Field	Register Configuration	Configuration Description
3	Untimed Send	Clock line falling-rising	Do not calculate CRC	Constant 0	2 bits	0x0E001E00	Send 2 bits of 0 as TIMEOUT signal, pull the data line low until WatchDog generates TIMEOUT

Table 191: Slave Mode

BiSS-C instruction configuration reference:

INSTR	OP Field	CK Field	CRC Field	DAT Field	LEN Field	Register Configuration	Configuration Description
0	Stop/Wait	Clock line high	Do not calculate CRC	Ignore data	2 bits	0x03000001	Pull the clock line high and wait for 2 clock cycles
1	Untimed Wait	Clock line rising-falling	Do not calculate CRC	Constant 1	1 bit	0x15001F00	Wait until 1 bit of 1 is received
2	Untimed Wait	Clock line rising-falling	Do not calculate CRC	Constant 1	1 bit	0x15001F00	Wait until 1 bit of 1 is received as the start bit
3	Untimed Receive	Clock line rising-falling	Do not calculate CRC	Data register 4	1 bit	0x1D000400	Use data register 4 to receive 1 bit of CDS
4	Untimed Receive	Clock line rising-falling	Data register 7	Data register 8	12 bits	0x1D07080B	Use data register 8 to receive 12 bits of multi-turn position, and use data register 7 to calculate CRC
5	Untimed Receive	Clock line rising-falling	Data register 7	Data register 9	12 bits	0x1D07090B	Use data register 9 to receive 12 bits of single-turn position, and calculate CRC
6	Untimed Receive	Clock line rising-falling	Data register 7	Constant 0	1 bit	0x1D071E00	Receive error status bit, and calculate CRC

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INSTR	OP Field	CK Field	CRC Field	DAT Field	LEN Field	Register Configuration	Configuration Description
7	Untimed Receive	Clock line rising-falling	Data register 7	Constant 1	1 bit	0x1D071F00	Receive warning status bit, and calculate CRC
8	Untimed Receive	Clock line rising-falling	Do not calculate CRC	Data register 7	6 bits	0x1D000705	Use data register 7 to receive and compare CRC

Table 192: Master Mode

INSTR	OP Field	CK Field	CRC Field	DAT Field	LEN Field	Register Configuration	Configuration Description
0	Untimed Send	Clock line falling-rising	Do not calculate CRC	Constant 1	1 bit	0x0E001F00	Send 1 bit of 1
1	Untimed Send	Clock line falling-rising	Do not calculate CRC	Data register 3	5 bits	0x0E000304	Use data register 3 to send header
2	Untimed Send	Clock line falling-rising	Data register 6	Data register 4	12 bits	0x0E06040B	Use data register 4 to send 12 bits of multi-turn position, and use data register 6 to calculate CRC
3	Untimed Send	Clock line falling-rising	Data register 6	Data register 5	12 bits	0x0E06050B	Use data register 5 to send 12 bits of single-turn position, and calculate CRC
4	Untimed Send	Clock line falling-rising	Data register 6	Data register 3	1 bit	0x0E060300	Use data register 3 to send error status bit, and calculate CRC
5	Untimed Send	Clock line falling-rising	Data register 6	Data register 3	1 bit	0x0E060300	Use data register 3 to send warning status bit, and calculate CRC
6	Untimed Send	Clock line falling-rising	Do not calculate CRC	Data register 6	6 bits	0x0E000605	Send the CRC calculation result in data register 6

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INSTR	OP Field	CK Field	CRC Field	DAT Field	LEN Field	Register Configuration	Configuration Description
7	Untimed Send	Clock line high	Do not calculate CRC	Constant 0	2 bits	0x0F001E01	Send 2 bits of 0 as TIMEOUT signal, pull the data line low until WatchDog generates TIMEOUT

Table 193: Slave Mode

EnDat instruction configuration reference (taking 110001 command as an example):

INSTR	OP Field	CK Field	CRC Field	DAT Field	LEN Field	Register Configuration	Configuration Description
0	Untimed Wait	Clock line low	Do not calculate CRC	Constant 1	1 bit	0x14001F00	Pull the clock line low until 1 bit of 1 is received
1	Untimed Send	Clock line falling-rising	Do not calculate CRC	Constant 1	2 bits	0x0E001F01	Send 1 bit of 1
2	Untimed Send	Clock line falling-rising	Do not calculate CRC	Command register	6 bits	0x0E000105	Use command register to send CMD10
3	Untimed Send	Clock line falling-rising	Do not calculate CRC	Data register 2	8 bits	0x0E000207	Use data register 2 to send address
4	Untimed Send	Clock line falling-rising	Do not calculate CRC	Data register 3	16 bits	0x0E00030F	Use data register 3 to send 16 bits of data content
5	Untimed Send	Clock line falling-rising	Do not calculate CRC	Constant 1	1 bit	0x0E001F00	Send 1 bit of 1
6	Untimed Wait	Clock line falling-rising	Do not calculate CRC	Constant 0	1 bit	0x16001E00	Wait until 1 bit of 0 is received
7	Untimed Wait	Clock line falling-rising	Do not calculate CRC	Constant 1	1 bit	0x16001F00	Wait until start bit is received

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INSTR	OP Field	CK Field	CRC Field	DAT Field	LEN Field	Register Configuration	Configuration Description
8	Timed Receive	Clock line falling-rising	Data register 7	Data register 4	8 bits	0x1A070407	Use data register 4 to receive address, and use data register 7 to calculate CRC
9	Timed Receive	Clock line falling-rising	Data register 7	Data register 5	16 bits	0x1A07050F	Use data register to receive 16 bits of data content, and calculate CRC
10	Timed Receive	Clock line falling-rising	Do not calculate CRC	Data register 7	5 bits	0x1A000704	Use data register 7 to receive and compare CRC
11	Timed Receive	Clock line falling-rising	Do not calculate CRC	Ignore data	1 bit	0x1A000000	Receive 1 bit of 1
12	Untimed Wait	Clock line high	Do not calculate CRC	Constant 0	1 bit	0x17001E00	Pull the clock line high and wait until 1 bit of 0 is received, wait for WatchDog to trigger TIMEOUT

Table 194: Master Mode

INSTR	OP Field	CK Field	CRC Field	DAT Field	LEN Field	Register Configuration	Configuration Description
0	Timed Receive	Clock line rising-falling	Do not calculate CRC	Ignore data	2 bits	0x19000001	Receive 2 bits of data
1	Timed Receive	Clock line rising-falling	Do not calculate CRC	Command register	6 bits	0x19000105	Use command register to receive command
2	Timed Receive	Clock line rising-falling	Do not calculate CRC	Data register 4	8 bits	0x19000407	Use data register 4 to receive 8 bits of data content
3	Timed Receive	Clock line rising-falling	Do not calculate CRC	Data register 5	16 bits	0x1900050F	Use data register 5 to receive 16 bits of data content

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INSTR	OP Field	CK Field	CRC Field	DAT Field	LEN Field	Register Configuration	Configuration Description
4	Timed Receive	Clock line rising-falling	Do not calculate CRC	Ignore data	1 bit	0x19000000	Receive 1 bit of data
5	Untimed Send	Clock line rising-falling	Do not calculate CRC	Constant 0	1 bit	0x0D001E00	Send 1 bit of constant 0
6	Untimed Send	Clock line falling-rising	Do not calculate CRC	Constant 0	3 bits	0x0E001E02	Send 3 bits of constant 0
7	Untimed Send	Clock line falling-rising	Do not calculate CRC	Constant 1	1 bit	0x0E001F00	Send 1 bit of constant 1
8	Untimed Send	Clock line falling-rising	Data register 6	Data register 7	8 bits	0x0E060707	Use data register 7 to send address data, and use data register 6 to calculate CRC
9	Untimed Send	Clock line falling-rising	Data register 6	Data register 8	16 bits	0x0E06080F	Use data register 8 to send 16 bits of data content, and calculate CRC
10	Untimed Send	Clock line falling-rising	Do not calculate CRC	Data register 6	5 bits	0x0E000604	Send the CRC calculation result in data register 6
11	Untimed Send	Clock line falling-rising	Do not calculate CRC	Constant 1	2 bits	0x0E001F01	Send 2 bits of 1, wait for WatchDog to trigger TIMEOUT

Table 195: Slave Mode

41.3 SEI Register List

The SEI register list is as follows:

SEI base address: 0xF032C000

Offset	Register Name	Description	Reset Value
0x0000	CTRL[0][ENGINE][CTRL]	Engine control register	0x00000000
0x0004	CTRL[0][ENGINE][PTR_CFG]	Pointer configuration register	0x00000000
0x0008	CTRL[0][ENGINE][WDG_CFG]	Watch dog configuration register	0x00000000
0x0010	CTRL[0][ENGINE][EXE_STA]	Execution status	0x00000000
0x0014	CTRL[0][ENGINE][EXE_PTR]	Execution pointer	0x00000000

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Offset	Register Name	Description	Reset Value
0x0018	CTRL[0][ENGINE][EXE_INST]	Execution instruction	0x00000000
0x001C	CTRL[0][ENGINE][WDG_STA]	Watch dog status	0x00000000
0x0020	CTRL[0][XCVR][CTRL]	Transceiver control register	0x00000000
0x0024	CTRL[0][XCVR][TYPE_CFG]	Transceiver configuration register	0x00000000
0x0028	CTRL[0][XCVR][BAUD_CFG]	Transceiver baud rate register	0x00000000
0x002C	CTRL[0][XCVR][DATA_CFG]	Transceiver data timing configuration	0x00000000
0x0030	CTRL[0][XCVR][CLK_CFG]	Transceiver clock timing configuration	0x00000000
0x0038	CTRL[0][XCVR][PIN]	Transceiver pin status	0x00000000
0x003C	CTRL[0][XCVR][STATE]	FSM of asynchronous	0x00000000
0x0040	CTRL[0][TRG][IN_CFG]	Trigger input configuration	0x00000000
0x0044	CTRL[0][TRG][SW]	Software trigger	0x00000000
0x0048	CTRL[0][TRG][PRD_CFG]	Period trigger configuration	0x00000000
0x004C	CTRL[0][TRG][PRD]	Trigger period	0x00000000
0x0050	CTRL[0][TRG][OUT_CFG]	Trigger output configuration	0x00000000
0x0060	CTRL[0][TRG][PRD_STS]	Period trigger status	0x00000000
0x0064	CTRL[0][TRG][PRD_CNT]	Period trigger counter	0x00000000
0x0080	CTRL[0][TRG_TABLE][CMD][0]	Trigger command	0x00000000
0x0084	CTRL[0][TRG_TABLE][CMD][1]	Trigger command	0x00000000
0x0088	CTRL[0][TRG_TABLE][CMD][2]	Trigger command	0x00000000
0x008C	CTRL[0][TRG_TABLE][CMD][3]	Trigger command	0x00000000
0x00A0	CTRL[0][TRG_TABLE][TIME][0]	Trigger Time	0x00000000
0x00A4	CTRL[0][TRG_TABLE][TIME][1]	Trigger Time	0x00000000
0x00A8	CTRL[0][TRG_TABLE][TIME][2]	Trigger Time	0x00000000
0x00AC	CTRL[0][TRG_TABLE][TIME][3]	Trigger Time	0x00000000
0x00C0	CTRL[0][CMD][MODE]	command register mode	0x00000000
0x00C4	CTRL[0][CMD][IDX]	command register configuration	0x00000000
0x00E0	CTRL[0][CMD][CMD]	command	0x00000000
0x00E4	CTRL[0][CMD][SET]	command bit set register	0x00000000
0x00E8	CTRL[0][CMD][CLR]	command bit clear register	0x00000000
0x00EC	CTRL[0][CMD][INV]	command bit invert register	0x00000000
0x00F0	CTRL[0][CMD][IN]	Commad input	0x00000000
0x00F4	CTRL[0][CMD][OUT]	Command output	0x00000000
0x00F8	CTRL[0][CMD][STS]	Command status	0x00000000
0x0100	CTRL[0][CMD_TABLE][0][MIN]	command start value	0x00000000
0x0104	CTRL[0][CMD_TABLE][0][MAX]	command end value	0x00000000
0x0108	CTRL[0][CMD_TABLE][0][MSK]	command compare bit enable	0x00000000
0x0110	CTRL[0][CMD_TABLE][0][PTA]	command pointer 0 - 3	0x00000000

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Offset	Register Name	Description	Reset Value
0x0114	CTRL[0][CMD_TABLE][0][PTB]	command pointer 4 - 7	0x00000000
0x0120	CTRL[0][CMD_TABLE][1][MIN]	command start value	0x00000000
0x0124	CTRL[0][CMD_TABLE][1][MAX]	command end value	0x00000000
0x0128	CTRL[0][CMD_TABLE][1][MSK]	command compare bit enable	0x00000000
0x0130	CTRL[0][CMD_TABLE][1][PTA]	command pointer 0 - 3	0x00000000
0x0134	CTRL[0][CMD_TABLE][1][PTB]	command pointer 4 - 7	0x00000000
0x0140	CTRL[0][CMD_TABLE][2][MIN]	command start value	0x00000000
0x0144	CTRL[0][CMD_TABLE][2][MAX]	command end value	0x00000000
0x0148	CTRL[0][CMD_TABLE][2][MSK]	command compare bit enable	0x00000000
0x0150	CTRL[0][CMD_TABLE][2][PTA]	command pointer 0 - 3	0x00000000
0x0154	CTRL[0][CMD_TABLE][2][PTB]	command pointer 4 - 7	0x00000000
0x0160	CTRL[0][CMD_TABLE][3][MIN]	command start value	0x00000000
0x0164	CTRL[0][CMD_TABLE][3][MAX]	command end value	0x00000000
0x0168	CTRL[0][CMD_TABLE][3][MSK]	command compare bit enable	0x00000000
0x0170	CTRL[0][CMD_TABLE][3][PTA]	command pointer 0 - 3	0x00000000
0x0174	CTRL[0][CMD_TABLE][3][PTB]	command pointer 4 - 7	0x00000000
0x0180	CTRL[0][CMD_TABLE][4][MIN]	command start value	0x00000000
0x0184	CTRL[0][CMD_TABLE][4][MAX]	command end value	0x00000000
0x0188	CTRL[0][CMD_TABLE][4][MSK]	command compare bit enable	0x00000000
0x0190	CTRL[0][CMD_TABLE][4][PTA]	command pointer 0 - 3	0x00000000
0x0194	CTRL[0][CMD_TABLE][4][PTB]	command pointer 4 - 7	0x00000000
0x01A0	CTRL[0][CMD_TABLE][5][MIN]	command start value	0x00000000
0x01A4	CTRL[0][CMD_TABLE][5][MAX]	command end value	0x00000000
0x01A8	CTRL[0][CMD_TABLE][5][MSK]	command compare bit enable	0x00000000
0x01B0	CTRL[0][CMD_TABLE][5][PTA]	command pointer 0 - 3	0x00000000
0x01B4	CTRL[0][CMD_TABLE][5][PTB]	command pointer 4 - 7	0x00000000
0x01C0	CTRL[0][CMD_TABLE][6][MIN]	command start value	0x00000000
0x01C4	CTRL[0][CMD_TABLE][6][MAX]	command end value	0x00000000
0x01C8	CTRL[0][CMD_TABLE][6][MSK]	command compare bit enable	0x00000000
0x01D0	CTRL[0][CMD_TABLE][6][PTA]	command pointer 0 - 3	0x00000000
0x01D4	CTRL[0][CMD_TABLE][6][PTB]	command pointer 4 - 7	0x00000000
0x01E0	CTRL[0][CMD_TABLE][7][MIN]	command start value	0x00000000
0x01E4	CTRL[0][CMD_TABLE][7][MAX]	command end value	0x00000000
0x01E8	CTRL[0][CMD_TABLE][7][MSK]	command compare bit enable	0x00000000
0x01F0	CTRL[0][CMD_TABLE][7][PTA]	command pointer 0 - 3	0x00000000
0x01F4	CTRL[0][CMD_TABLE][7][PTB]	command pointer 4 - 7	0x00000000
0x0200	CTRL[0][LATCH][0][TRAN][0_1]	Latch state transition configuration	0x00000000
0x0204	CTRL[0][LATCH][0][TRAN][1_2]	Latch state transition configuration	0x00000000

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Offset	Register Name	Description	Reset Value
0x0208	CTRL[0][LATCH][0][TRAN][2_3]	Latch state transition configuration	0x00000000
0x020C	CTRL[0][LATCH][0][TRAN][3_0]	Latch state transition configuration	0x00000000
0x0210	CTRL[0][LATCH][0][CFG]	Latch configuration	0x00000000
0x0218	CTRL[0][LATCH][0][TIME]	Latch time	0x00000000
0x021C	CTRL[0][LATCH][0][STS]	Latch status	0x00000000
0x0220	CTRL[0][LATCH][1][TRAN][0_1]	Latch state transition configuration	0x00000000
0x0224	CTRL[0][LATCH][1][TRAN][1_2]	Latch state transition configuration	0x00000000
0x0228	CTRL[0][LATCH][1][TRAN][2_3]	Latch state transition configuration	0x00000000
0x022C	CTRL[0][LATCH][1][TRAN][3_0]	Latch state transition configuration	0x00000000
0x0230	CTRL[0][LATCH][1][CFG]	Latch configuration	0x00000000
0x0238	CTRL[0][LATCH][1][TIME]	Latch time	0x00000000
0x023C	CTRL[0][LATCH][1][STS]	Latch status	0x00000000
0x0240	CTRL[0][LATCH][2][TRAN][0_1]	Latch state transition configuration	0x00000000
0x0244	CTRL[0][LATCH][2][TRAN][1_2]	Latch state transition configuration	0x00000000
0x0248	CTRL[0][LATCH][2][TRAN][2_3]	Latch state transition configuration	0x00000000
0x024C	CTRL[0][LATCH][2][TRAN][3_0]	Latch state transition configuration	0x00000000
0x0250	CTRL[0][LATCH][2][CFG]	Latch configuration	0x00000000
0x0258	CTRL[0][LATCH][2][TIME]	Latch time	0x00000000
0x025C	CTRL[0][LATCH][2][STS]	Latch status	0x00000000
0x0260	CTRL[0][LATCH][3][TRAN][0_1]	Latch state transition configuration	0x00000000
0x0264	CTRL[0][LATCH][3][TRAN][1_2]	Latch state transition configuration	0x00000000
0x0268	CTRL[0][LATCH][3][TRAN][2_3]	Latch state transition configuration	0x00000000
0x026C	CTRL[0][LATCH][3][TRAN][3_0]	Latch state transition configuration	0x00000000
0x0270	CTRL[0][LATCH][3][CFG]	Latch configuration	0x00000000
0x0278	CTRL[0][LATCH][3][TIME]	Latch time	0x00000000
0x027C	CTRL[0][LATCH][3][STS]	Latch status	0x00000000
0x0280	CTRL[0][POS][SMP_EN]	Sample selection register	0x00000000

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Offset	Register Name	Description	Reset Value
0x0284	CTRL[0][POS][SMP_CFG]	Sample configuration	0x00000000
0x0288	CTRL[0][POS][SMP_DAT]	Sample data	0x00000000
0x0290	CTRL[0][POS][SMP_POS]	Sample override position	0x00000000
0x0294	CTRL[0][POS][SMP_REV]	Sample override revolution	0x00000000
0x0298	CTRL[0][POS][SMP_SPD]	Sample override speed	0x00000000
0x029C	CTRL[0][POS][SMP_ACC]	Sample override accelerate	0x00000000
0x02A0	CTRL[0][POS][UPD_EN]	Update configuration	0x00000000
0x02A4	CTRL[0][POS][UPD_CFG]	Update configuration	0x00000000
0x02A8	CTRL[0][POS][UPD_DAT]	Update data	0x00000000
0x02AC	CTRL[0][POS][UPD_TIME]	Update overide time	0x00000000
0x02B0	CTRL[0][POS][UPD_POS]	Update override position	0x00000000
0x02B4	CTRL[0][POS][UPD_REV]	Update override revolution	0x00000000
0x02B8	CTRL[0][POS][UPD_SPD]	Update override speed	0x00000000
0x02BC	CTRL[0][POS][UPD_ACC]	Update override accelerate	0x00000000
0x02C0	CTRL[0][POS][SMP_VAL]	Sample valid	0x00000000
0x02C4	CTRL[0][POS][SMP_STS]	Sample status	0x00000000
0x02CC	CTRL[0][POS][TIME_IN]	input time	0x00000000
0x02D0	CTRL[0][POS][POS_IN]	Input position	0x00000000
0x02D4	CTRL[0][POS][REV_IN]	Input revolution	0x00000000
0x02D8	CTRL[0][POS][SPD_IN]	Input speed	0x00000000
0x02DC	CTRL[0][POS][ACC_IN]	Input accelerate	0x00000000
0x02E4	CTRL[0][POS][UPD_STS]	Update status	0x00000000
0x0300	CTRL[0][IRQ][INT_EN]	Interrupt Enable	0x00000000
0x0304	CTRL[0][IRQ][INT_FLAG]	Interrupt flag	0x00000000
0x0308	CTRL[0][IRQ][INT_STS]	Interrupt status	0x00000000
0x0310	CTRL[0][IRQ][POINTER0]	Match pointer 0	0x00000000
0x0314	CTRL[0][IRQ][POINTER1]	Match pointer 1	0x00000000
0x0318	CTRL[0][IRQ][INSTR0]	Match instruction 0	0x00000000
0x031C	CTRL[0][IRQ][INSTR1]	Match instruction 1	0x00000000
0x0320	CTRL[0][DMA_EN]	DMA Enable	0x00000000
0x0400	CTRL[1][ENGINE][CTRL]	Engine control register	0x00000000
0x0404	CTRL[1][ENGINE][PTR_CFG]	Pointer configuration register	0x00000000
0x0408	CTRL[1][ENGINE][WDG_CFG]	Watch dog configuration register	0x00000000
0x0410	CTRL[1][ENGINE][EXE_STA]	Execution status	0x00000000
0x0414	CTRL[1][ENGINE][EXE_PTR]	Execution pointer	0x00000000
0x0418	CTRL[1][ENGINE][EXE_INST]	Execution instruction	0x00000000
0x041C	CTRL[1][ENGINE][WDG_STA]	Watch dog status	0x00000000
0x0400	CTRL[1][XCVR][CTRL]	Transceiver control register	0x00000000
0x0404	CTRL[1][XCVR][TYPE_CFG]	Transceiver configuration register	0x00000000

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Offset	Register Name	Description	Reset Value
0x0408	CTRL[1][XCVR][BAUD_CFG]	Transceiver baud rate register	0x00000000
0x040C	CTRL[1][XCVR][DATA_CFG]	Transceiver data timing configuration	0x00000000
0x0410	CTRL[1][XCVR][CLK_CFG]	Transceiver clock timing configuration	0x00000000
0x0418	CTRL[1][XCVR][PIN]	Transceiver pin status	0x00000000
0x041C	CTRL[1][XCVR][STATE]	FSM of asynchronous	0x00000000
0x0400	CTRL[1][TRG][IN_CFG]	Trigger input configuration	0x00000000
0x0404	CTRL[1][TRG][SW]	Software trigger	0x00000000
0x0408	CTRL[1][TRG][PRD_CFG]	Period trigger configuration	0x00000000
0x040C	CTRL[1][TRG][PRD]	Trigger period	0x00000000
0x0410	CTRL[1][TRG][OUT_CFG]	Trigger output configuration	0x00000000
0x0420	CTRL[1][TRG][PRD_STS]	Period trigger status	0x00000000
0x0424	CTRL[1][TRG][PRD_CNT]	Period trigger counter	0x00000000
0x0400	CTRL[1][TRG_TABLE][CMD][0]	Trigger command	0x00000000
0x0404	CTRL[1][TRG_TABLE][CMD][1]	Trigger command	0x00000000
0x0408	CTRL[1][TRG_TABLE][CMD][2]	Trigger command	0x00000000
0x040C	CTRL[1][TRG_TABLE][CMD][3]	Trigger command	0x00000000
0x0420	CTRL[1][TRG_TABLE][TIME][0]	Trigger Time	0x00000000
0x0424	CTRL[1][TRG_TABLE][TIME][1]	Trigger Time	0x00000000
0x0428	CTRL[1][TRG_TABLE][TIME][2]	Trigger Time	0x00000000
0x042C	CTRL[1][TRG_TABLE][TIME][3]	Trigger Time	0x00000000
0x0400	CTRL[1][CMD][MODE]	command register mode	0x00000000
0x0404	CTRL[1][CMD][IDX]	command register configuration	0x00000000
0x0420	CTRL[1][CMD][CMD]	command	0x00000000
0x0424	CTRL[1][CMD][SET]	command bit set register	0x00000000
0x0428	CTRL[1][CMD][CLR]	command bit clear register	0x00000000
0x042C	CTRL[1][CMD][INV]	command bit invert register	0x00000000
0x0430	CTRL[1][CMD][IN]	Commad input	0x00000000
0x0434	CTRL[1][CMD][OUT]	Command output	0x00000000
0x0438	CTRL[1][CMD][STS]	Command status	0x00000000
0x0400	CTRL[1][CMD_TABLE][0][MIN]	command start value	0x00000000
0x0404	CTRL[1][CMD_TABLE][0][MAX]	command end value	0x00000000
0x0408	CTRL[1][CMD_TABLE][0][MSK]	command compare bit enable	0x00000000
0x0410	CTRL[1][CMD_TABLE][0][PTA]	command pointer 0 - 3	0x00000000
0x0414	CTRL[1][CMD_TABLE][0][PTB]	command pointer 4 - 7	0x00000000
0x0420	CTRL[1][CMD_TABLE][1][MIN]	command start value	0x00000000
0x0424	CTRL[1][CMD_TABLE][1][MAX]	command end value	0x00000000
0x0428	CTRL[1][CMD_TABLE][1][MSK]	command compare bit enable	0x00000000
0x0430	CTRL[1][CMD_TABLE][1][PTA]	command pointer 0 - 3	0x00000000

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Offset	Register Name	Description	Reset Value
0x0434	CTRL[1][CMD_TABLE][1][PTB]	command pointer 4 - 7	0x00000000
0x0440	CTRL[1][CMD_TABLE][2][MIN]	command start value	0x00000000
0x0444	CTRL[1][CMD_TABLE][2][MAX]	command end value	0x00000000
0x0448	CTRL[1][CMD_TABLE][2][MSK]	command compare bit enable	0x00000000
0x0450	CTRL[1][CMD_TABLE][2][PTA]	command pointer 0 - 3	0x00000000
0x0454	CTRL[1][CMD_TABLE][2][PTB]	command pointer 4 - 7	0x00000000
0x0460	CTRL[1][CMD_TABLE][3][MIN]	command start value	0x00000000
0x0464	CTRL[1][CMD_TABLE][3][MAX]	command end value	0x00000000
0x0468	CTRL[1][CMD_TABLE][3][MSK]	command compare bit enable	0x00000000
0x0470	CTRL[1][CMD_TABLE][3][PTA]	command pointer 0 - 3	0x00000000
0x0474	CTRL[1][CMD_TABLE][3][PTB]	command pointer 4 - 7	0x00000000
0x0480	CTRL[1][CMD_TABLE][4][MIN]	command start value	0x00000000
0x0484	CTRL[1][CMD_TABLE][4][MAX]	command end value	0x00000000
0x0488	CTRL[1][CMD_TABLE][4][MSK]	command compare bit enable	0x00000000
0x0490	CTRL[1][CMD_TABLE][4][PTA]	command pointer 0 - 3	0x00000000
0x0494	CTRL[1][CMD_TABLE][4][PTB]	command pointer 4 - 7	0x00000000
0x04A0	CTRL[1][CMD_TABLE][5][MIN]	command start value	0x00000000
0x04A4	CTRL[1][CMD_TABLE][5][MAX]	command end value	0x00000000
0x04A8	CTRL[1][CMD_TABLE][5][MSK]	command compare bit enable	0x00000000
0x04B0	CTRL[1][CMD_TABLE][5][PTA]	command pointer 0 - 3	0x00000000
0x04B4	CTRL[1][CMD_TABLE][5][PTB]	command pointer 4 - 7	0x00000000
0x04C0	CTRL[1][CMD_TABLE][6][MIN]	command start value	0x00000000
0x04C4	CTRL[1][CMD_TABLE][6][MAX]	command end value	0x00000000
0x04C8	CTRL[1][CMD_TABLE][6][MSK]	command compare bit enable	0x00000000
0x04D0	CTRL[1][CMD_TABLE][6][PTA]	command pointer 0 - 3	0x00000000
0x04D4	CTRL[1][CMD_TABLE][6][PTB]	command pointer 4 - 7	0x00000000
0x04E0	CTRL[1][CMD_TABLE][7][MIN]	command start value	0x00000000
0x04E4	CTRL[1][CMD_TABLE][7][MAX]	command end value	0x00000000
0x04E8	CTRL[1][CMD_TABLE][7][MSK]	command compare bit enable	0x00000000
0x04F0	CTRL[1][CMD_TABLE][7][PTA]	command pointer 0 - 3	0x00000000
0x04F4	CTRL[1][CMD_TABLE][7][PTB]	command pointer 4 - 7	0x00000000
0x0400	CTRL[1][LATCH][0][TRAN][0_1]	Latch state transition configuration	0x00000000
0x0404	CTRL[1][LATCH][0][TRAN][1_2]	Latch state transition configuration	0x00000000
0x0408	CTRL[1][LATCH][0][TRAN][2_3]	Latch state transition configuration	0x00000000
0x040C	CTRL[1][LATCH][0][TRAN][3_0]	Latch state transition configuration	0x00000000
0x0410	CTRL[1][LATCH][0][CFG]	Latch configuration	0x00000000

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Offset	Register Name	Description	Reset Value
0x0418	CTRL[1][LATCH][0][TIME]	Latch time	0x00000000
0x041C	CTRL[1][LATCH][0][STS]	Latch status	0x00000000
0x0420	CTRL[1][LATCH][1][TRAN][0_1]	Latch state transition configuration	0x00000000
0x0424	CTRL[1][LATCH][1][TRAN][1_2]	Latch state transition configuration	0x00000000
0x0428	CTRL[1][LATCH][1][TRAN][2_3]	Latch state transition configuration	0x00000000
0x042C	CTRL[1][LATCH][1][TRAN][3_0]	Latch state transition configuration	0x00000000
0x0430	CTRL[1][LATCH][1][CFG]	Latch configuration	0x00000000
0x0438	CTRL[1][LATCH][1][TIME]	Latch time	0x00000000
0x043C	CTRL[1][LATCH][1][STS]	Latch status	0x00000000
0x0440	CTRL[1][LATCH][2][TRAN][0_1]	Latch state transition configuration	0x00000000
0x0444	CTRL[1][LATCH][2][TRAN][1_2]	Latch state transition configuration	0x00000000
0x0448	CTRL[1][LATCH][2][TRAN][2_3]	Latch state transition configuration	0x00000000
0x044C	CTRL[1][LATCH][2][TRAN][3_0]	Latch state transition configuration	0x00000000
0x0450	CTRL[1][LATCH][2][CFG]	Latch configuration	0x00000000
0x0458	CTRL[1][LATCH][2][TIME]	Latch time	0x00000000
0x045C	CTRL[1][LATCH][2][STS]	Latch status	0x00000000
0x0460	CTRL[1][LATCH][3][TRAN][0_1]	Latch state transition configuration	0x00000000
0x0464	CTRL[1][LATCH][3][TRAN][1_2]	Latch state transition configuration	0x00000000
0x0468	CTRL[1][LATCH][3][TRAN][2_3]	Latch state transition configuration	0x00000000
0x046C	CTRL[1][LATCH][3][TRAN][3_0]	Latch state transition configuration	0x00000000
0x0470	CTRL[1][LATCH][3][CFG]	Latch configuration	0x00000000
0x0478	CTRL[1][LATCH][3][TIME]	Latch time	0x00000000
0x047C	CTRL[1][LATCH][3][STS]	Latch status	0x00000000
0x0400	CTRL[1][POS][SMP_EN]	Sample selection register	0x00000000
0x0404	CTRL[1][POS][SMP_CFG]	Sample configuration	0x00000000
0x0408	CTRL[1][POS][SMP_DAT]	Sample data	0x00000000
0x0410	CTRL[1][POS][SMP_POS]	Sample override position	0x00000000
0x0414	CTRL[1][POS][SMP_REV]	Sample override revolution	0x00000000
0x0418	CTRL[1][POS][SMP_SPD]	Sample override speed	0x00000000

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Offset	Register Name	Description	Reset Value
0x041C	CTRL[1][POS][SMP_ACC]	Sample override accelerate	0x00000000
0x0420	CTRL[1][POS][UPD_EN]	Update configuration	0x00000000
0x0424	CTRL[1][POS][UPD_CFG]	Update configuration	0x00000000
0x0428	CTRL[1][POS][UPD_DAT]	Update data	0x00000000
0x042C	CTRL[1][POS][UPD_TIME]	Update overide time	0x00000000
0x0430	CTRL[1][POS][UPD_POS]	Update override position	0x00000000
0x0434	CTRL[1][POS][UPD_REV]	Update override revolution	0x00000000
0x0438	CTRL[1][POS][UPD_SPD]	Update override speed	0x00000000
0x043C	CTRL[1][POS][UPD_ACC]	Update override accelerate	0x00000000
0x0440	CTRL[1][POS][SMP_VAL]	Sample valid	0x00000000
0x0444	CTRL[1][POS][SMP_STS]	Sample status	0x00000000
0x044C	CTRL[1][POS][TIME_IN]	input time	0x00000000
0x0450	CTRL[1][POS][POS_IN]	Input position	0x00000000
0x0454	CTRL[1][POS][REV_IN]	Input revolution	0x00000000
0x0458	CTRL[1][POS][SPD_IN]	Input speed	0x00000000
0x045C	CTRL[1][POS][ACC_IN]	Input accelerate	0x00000000
0x0464	CTRL[1][POS][UPD_STS]	Update status	0x00000000
0x0400	CTRL[1][IRQ][INT_EN]	Interrupt Enable	0x00000000
0x0404	CTRL[1][IRQ][INT_FLAG]	Interrupt flag	0x00000000
0x0408	CTRL[1][IRQ][INT_STS]	Interrupt status	0x00000000
0x0410	CTRL[1][IRQ][POINTER0]	Match pointer 0	0x00000000
0x0414	CTRL[1][IRQ][POINTER1]	Match pointer 1	0x00000000
0x0418	CTRL[1][IRQ][INSTR0]	Match instruction 0	0x00000000
0x041C	CTRL[1][IRQ][INSTR1]	Match instruction 1	0x00000000
0x0720	CTRL[1][DMA_EN]	DMA Enable	0x00000000
0x3400	INSTR[0]	Instructions	0x00000000
0x3404	INSTR[1]	Instructions	0x00000000
0x3408	INSTR[2]	Instructions	0x00000000
0x340C	INSTR[3]	Instructions	0x00000000
0x3410	INSTR[4]	Instructions	0x00000000
0x3414	INSTR[5]	Instructions	0x00000000
0x3418	INSTR[6]	Instructions	0x00000000
0x341C	INSTR[7]	Instructions	0x00000000
0x3420	INSTR[8]	Instructions	0x00000000
0x3424	INSTR[9]	Instructions	0x00000000
0x3428	INSTR[10]	Instructions	0x00000000
0x342C	INSTR[11]	Instructions	0x00000000
0x3430	INSTR[12]	Instructions	0x00000000
0x3434	INSTR[13]	Instructions	0x00000000
0x3438	INSTR[14]	Instructions	0x00000000

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Offset	Register Name	Description	Reset Value
0x343C	INSTR[15]	Instructions	0x00000000
0x3440	INSTR[16]	Instructions	0x00000000
0x3444	INSTR[17]	Instructions	0x00000000
0x3448	INSTR[18]	Instructions	0x00000000
0x344C	INSTR[19]	Instructions	0x00000000
0x3450	INSTR[20]	Instructions	0x00000000
0x3454	INSTR[21]	Instructions	0x00000000
0x3458	INSTR[22]	Instructions	0x00000000
0x345C	INSTR[23]	Instructions	0x00000000
0x3460	INSTR[24]	Instructions	0x00000000
0x3464	INSTR[25]	Instructions	0x00000000
0x3468	INSTR[26]	Instructions	0x00000000
0x346C	INSTR[27]	Instructions	0x00000000
0x3470	INSTR[28]	Instructions	0x00000000
0x3474	INSTR[29]	Instructions	0x00000000
0x3478	INSTR[30]	Instructions	0x00000000
0x347C	INSTR[31]	Instructions	0x00000000
0x3480	INSTR[32]	Instructions	0x00000000
0x3484	INSTR[33]	Instructions	0x00000000
0x3488	INSTR[34]	Instructions	0x00000000
0x348C	INSTR[35]	Instructions	0x00000000
0x3490	INSTR[36]	Instructions	0x00000000
0x3494	INSTR[37]	Instructions	0x00000000
0x3498	INSTR[38]	Instructions	0x00000000
0x349C	INSTR[39]	Instructions	0x00000000
0x34A0	INSTR[40]	Instructions	0x00000000
0x34A4	INSTR[41]	Instructions	0x00000000
0x34A8	INSTR[42]	Instructions	0x00000000
0x34AC	INSTR[43]	Instructions	0x00000000
0x34B0	INSTR[44]	Instructions	0x00000000
0x34B4	INSTR[45]	Instructions	0x00000000
0x34B8	INSTR[46]	Instructions	0x00000000
0x34BC	INSTR[47]	Instructions	0x00000000
0x34C0	INSTR[48]	Instructions	0x00000000
0x34C4	INSTR[49]	Instructions	0x00000000
0x34C8	INSTR[50]	Instructions	0x00000000
0x34CC	INSTR[51]	Instructions	0x00000000
0x34D0	INSTR[52]	Instructions	0x00000000
0x34D4	INSTR[53]	Instructions	0x00000000
0x34D8	INSTR[54]	Instructions	0x00000000

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Offset	Register Name	Description	Reset Value
0x34DC	INSTR[55]	Instructions	0x00000000
0x34E0	INSTR[56]	Instructions	0x00000000
0x34E4	INSTR[57]	Instructions	0x00000000
0x34E8	INSTR[58]	Instructions	0x00000000
0x34EC	INSTR[59]	Instructions	0x00000000
0x34F0	INSTR[60]	Instructions	0x00000000
0x34F4	INSTR[61]	Instructions	0x00000000
0x34F8	INSTR[62]	Instructions	0x00000000
0x34FC	INSTR[63]	Instructions	0x00000000
0x3800	DAT[0][MODE]		0x00000000
0x3804	DAT[0][IDX]	Data register bit index	0x00000000
0x3808	DAT[0][GOLD]	Gold data for data check	0x00000000
0x380C	DAT[0][CRCINIT]	CRC calculation initial vector	0x00000000
0x3810	DAT[0][CRCPOLY]	CRC calculation polynomial	0x00000000
0x3820	DAT[0][DATA]	Data value	0x00000000
0x3824	DAT[0][SET]	Data bit set	0x00000000
0x3828	DAT[0][CLR]	Data bit clear	0x00000000
0x382C	DAT[0][INV]	Data bit invert	0x00000000
0x3830	DAT[0][IN]	Data input	0x00000000
0x3834	DAT[0][OUT]	Data output	0x00000000
0x3838	DAT[0][STS]	Data status	0x00000000
0x3840	DAT[1][MODE]		0x00000000
0x3844	DAT[1][IDX]	Data register bit index	0x00000000
0x3848	DAT[1][GOLD]	Gold data for data check	0x00000000
0x384C	DAT[1][CRCINIT]	CRC calculation initial vector	0x00000000
0x3850	DAT[1][CRCPOLY]	CRC calculation polynomial	0x00000000
0x3860	DAT[1][DATA]	Data value	0x00000000
0x3864	DAT[1][SET]	Data bit set	0x00000000
0x3868	DAT[1][CLR]	Data bit clear	0x00000000
0x386C	DAT[1][INV]	Data bit invert	0x00000000
0x3870	DAT[1][IN]	Data input	0x00000000
0x3874	DAT[1][OUT]	Data output	0x00000000
0x3878	DAT[1][STS]	Data status	0x00000000
0x3880	DAT[2][MODE]		0x00000000
0x3884	DAT[2][IDX]	Data register bit index	0x00000000
0x3888	DAT[2][GOLD]	Gold data for data check	0x00000000
0x388C	DAT[2][CRCINIT]	CRC calculation initial vector	0x00000000
0x3890	DAT[2][CRCPOLY]	CRC calculation polynomial	0x00000000
0x38A0	DAT[2][DATA]	Data value	0x00000000
0x38A4	DAT[2][SET]	Data bit set	0x00000000

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Offset	Register Name	Description	Reset Value
0x38A8	DAT[2][CLR]	Data bit clear	0x00000000
0x38AC	DAT[2][INV]	Data bit invert	0x00000000
0x38B0	DAT[2][IN]	Data input	0x00000000
0x38B4	DAT[2][OUT]	Data output	0x00000000
0x38B8	DAT[2][STS]	Data status	0x00000000
0x38C0	DAT[3][MODE]		0x00000000
0x38C4	DAT[3][IDX]	Data register bit index	0x00000000
0x38C8	DAT[3][GOLD]	Gold data for data check	0x00000000
0x38CC	DAT[3][CRCINIT]	CRC calculation initial vector	0x00000000
0x38D0	DAT[3][CRCPOLY]	CRC calculation polynomial	0x00000000
0x38E0	DAT[3][DATA]	Data value	0x00000000
0x38E4	DAT[3][SET]	Data bit set	0x00000000
0x38E8	DAT[3][CLR]	Data bit clear	0x00000000
0x38EC	DAT[3][INV]	Data bit invert	0x00000000
0x38F0	DAT[3][IN]	Data input	0x00000000
0x38F4	DAT[3][OUT]	Data output	0x00000000
0x38F8	DAT[3][STS]	Data status	0x00000000
0x3900	DAT[4][MODE]		0x00000000
0x3904	DAT[4][IDX]	Data register bit index	0x00000000
0x3908	DAT[4][GOLD]	Gold data for data check	0x00000000
0x390C	DAT[4][CRCINIT]	CRC calculation initial vector	0x00000000
0x3910	DAT[4][CRCPOLY]	CRC calculation polynomial	0x00000000
0x3920	DAT[4][DATA]	Data value	0x00000000
0x3924	DAT[4][SET]	Data bit set	0x00000000
0x3928	DAT[4][CLR]	Data bit clear	0x00000000
0x392C	DAT[4][INV]	Data bit invert	0x00000000
0x3930	DAT[4][IN]	Data input	0x00000000
0x3934	DAT[4][OUT]	Data output	0x00000000
0x3938	DAT[4][STS]	Data status	0x00000000
0x3940	DAT[5][MODE]		0x00000000
0x3944	DAT[5][IDX]	Data register bit index	0x00000000
0x3948	DAT[5][GOLD]	Gold data for data check	0x00000000
0x394C	DAT[5][CRCINIT]	CRC calculation initial vector	0x00000000
0x3950	DAT[5][CRCPOLY]	CRC calculation polynomial	0x00000000
0x3960	DAT[5][DATA]	Data value	0x00000000
0x3964	DAT[5][SET]	Data bit set	0x00000000
0x3968	DAT[5][CLR]	Data bit clear	0x00000000
0x396C	DAT[5][INV]	Data bit invert	0x00000000
0x3970	DAT[5][IN]	Data input	0x00000000
0x3974	DAT[5][OUT]	Data output	0x00000000

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Offset	Register Name	Description	Reset Value
0x3978	DAT[5][STS]	Data status	0x00000000
0x3980	DAT[6][MODE]		0x00000000
0x3984	DAT[6][IDX]	Data register bit index	0x00000000
0x3988	DAT[6][GOLD]	Gold data for data check	0x00000000
0x398C	DAT[6][CRCINIT]	CRC calculation initial vector	0x00000000
0x3990	DAT[6][CRCPOLY]	CRC calculation polynomial	0x00000000
0x39A0	DAT[6][DATA]	Data value	0x00000000
0x39A4	DAT[6][SET]	Data bit set	0x00000000
0x39A8	DAT[6][CLR]	Data bit clear	0x00000000
0x39AC	DAT[6][INV]	Data bit invert	0x00000000
0x39B0	DAT[6][IN]	Data input	0x00000000
0x39B4	DAT[6][OUT]	Data output	0x00000000
0x39B8	DAT[6][STS]	Data status	0x00000000
0x39C0	DAT[7][MODE]		0x00000000
0x39C4	DAT[7][IDX]	Data register bit index	0x00000000
0x39C8	DAT[7][GOLD]	Gold data for data check	0x00000000
0x39CC	DAT[7][CRCINIT]	CRC calculation initial vector	0x00000000
0x39D0	DAT[7][CRCPOLY]	CRC calculation polynomial	0x00000000
0x39E0	DAT[7][DATA]	Data value	0x00000000
0x39E4	DAT[7][SET]	Data bit set	0x00000000
0x39E8	DAT[7][CLR]	Data bit clear	0x00000000
0x39EC	DAT[7][INV]	Data bit invert	0x00000000
0x39F0	DAT[7][IN]	Data input	0x00000000
0x39F4	DAT[7][OUT]	Data output	0x00000000
0x39F8	DAT[7][STS]	Data status	0x00000000
0x3A00	DAT[8][MODE]		0x00000000
0x3A04	DAT[8][IDX]	Data register bit index	0x00000000
0x3A08	DAT[8][GOLD]	Gold data for data check	0x00000000
0x3A0C	DAT[8][CRCINIT]	CRC calculation initial vector	0x00000000
0x3A10	DAT[8][CRCPOLY]	CRC calculation polynomial	0x00000000
0x3A20	DAT[8][DATA]	Data value	0x00000000
0x3A24	DAT[8][SET]	Data bit set	0x00000000
0x3A28	DAT[8][CLR]	Data bit clear	0x00000000
0x3A2C	DAT[8][INV]	Data bit invert	0x00000000
0x3A30	DAT[8][IN]	Data input	0x00000000
0x3A34	DAT[8][OUT]	Data output	0x00000000
0x3A38	DAT[8][STS]	Data status	0x00000000
0x3A40	DAT[9][MODE]		0x00000000
0x3A44	DAT[9][IDX]	Data register bit index	0x00000000
0x3A48	DAT[9][GOLD]	Gold data for data check	0x00000000

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Offset	Register Name	Description	Reset Value
0x3A4C	DAT[9][CRCINIT]	CRC calculation initial vector	0x00000000
0x3A50	DAT[9][CRCPOLY]	CRC calculation polynomial	0x00000000
0x3A60	DAT[9][DATA]	Data value	0x00000000
0x3A64	DAT[9][SET]	Data bit set	0x00000000
0x3A68	DAT[9][CLR]	Data bit clear	0x00000000
0x3A6C	DAT[9][INV]	Data bit invert	0x00000000
0x3A70	DAT[9][IN]	Data input	0x00000000
0x3A74	DAT[9][OUT]	Data output	0x00000000
0x3A78	DAT[9][STS]	Data status	0x00000000

Table 196: SEI Register List

41.4 SEI Register Detailed Information

The detailed description of SEI registers is as follows:

41.4.1 CTRL[ENGINE][CTRL] (0x0 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD							WATCH	RSVD							ARMING	RSVD							EXCEPT	RSVD			REWIND	RSVD			ENABLE
N/A							RW	N/A							RW	N/A							RW	N/A			RW	N/A			RW
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[ENGINE][CTRL] [31:0]

Bits	Name	Description
24	WATCH	Enable watch dog 0: Watch dog disabled 1: Watch dog enabled
16	ARMING	Wait for trigger before excuting 0: Execute on enable 1: Wait trigger before exection after enabled
8	EXCEPT	Explain timout as exception 0: when timeout, pointer move to next instruction 1: when timeout, pointer jump to timeout vector
4	REWIND	Rewind execution pointer 0: run 1: clean status and rewind
0	ENABLE	Enable 0: disable 1: enable

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Bits	Name	Description

CTRL[ENGINE][CTRL] Bits

41.4.2 CTRL[ENGINE][PTR_CFG] (0x4 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD			DAT_CDM					RSVD			DAT_BASE					POINTER_WDOG								POINTER_INIT							
N/A			RW					N/A			RW					RW								RW							
0	0	0	0	0	0	0	0	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[ENGINE][PTR_CFG] [31:0]

Bits	Name	Description
28-24	DAT_CDM	Select DATA register to receive CDM bit in BiSSC slave mode 0: ignore 1: command 2: data register 2 3: data register 3 ... 29:data register 29 30: value 0 when send, ignore in receive 31: value1 when send, ignore in receive
20-16	DAT_BASE	Bias for data register access, if calculated index bigger than 32, index will wrap around 0: real data index 1: access index is 1 greater than instruction address 2: access index is 2 greater than instruction address ... 31: access index is 31 greater than instruction address
15-8	POINTER_WDOG	Pointer to the instruction that the program starts executing after the instruction timeout. The timeout is WDOG_TIME
7-0	POINTER_INIT	Initial execute pointer

CTRL[ENGINE][PTR_CFG] Bits

41.4.3 CTRL[ENGINE][WDG_CFG] (0x8 + 0x400 * n)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																WDOG_TIME															
N/A																RW															
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[ENGINE][WDG_CFG] [31:0]

Bits	Name	Description
15-0	WDOG_TIME	Time out count for each instruction, counter in bit time.

CTRL[ENGINE][WDG_CFG] Bits

41.4.4 CTRL[ENGINE][EXE_STA] (0x10 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD											TRIGERED	RSVD			ARMED	RSVD							EXPIRE	RSVD							STALL
N/A											RO	N/A			RO	N/A							RO	N/A							RO
x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	0	x	x	x	x	x	x	x	0	x	x	x	x	x	x	x	0

CTRL[ENGINE][EXE_STA] [31:0]

Bits	Name	Description
20	TRIGERED	Execution has been triggered 0: Execution not triggered 1: Execution triggered
16	ARMED	Waiting for trigger for execution 0: Not in waiting status 1: In waiting status
8	EXPIRE	Watchdog timer expired 0: Not expired 1: Expired
0	STALL	Program finished 0: Program is executing 1: Program finished

CTRL[ENGINE][EXE_STA] Bits

41.4.5 CTRL[ENGINE][EXE_PTR] (0x14 + 0x400 * n)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD			HALT_CNT					RSVD			BIT_CNT					RSVD								POINTER							
N/A			RO					N/A			RO					N/A								RO							
x	x	x	0	0	0	0	0	x	x	x	0	0	0	0	0	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0

CTRL[ENGINE][EXE_PTR] [31:0]

Bits	Name	Description
28-24	HALT_CNT	Halt count in halt instrution
20-16	BIT_CNT	Bit count in send and receive instruction execution
7-0	POINTER	Current program pointer

CTRL[ENGINE][EXE_PTR] Bits

41.4.6 CTRL[ENGINE][EXE_INST] (0x18 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INST																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[ENGINE][EXE_INST] [31:0]

Bits	Name	Description
31-0	INST	Current instruction

CTRL[ENGINE][EXE_INST] Bits

41.4.7 CTRL[ENGINE][WDG_STA] (0x1C + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																WDOG_CNT															
N/A																RO															
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[ENGINE][WDG_STA] [31:0]

Bits	Name	Description
15-0	WDOG_CNT	Current watch dog counter value

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Bits	Name	Description

CTRL[ENGINE][WDG_STA] Bits

41.4.8 CTRL[XCVR][CTRL] (0x20 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																			TRISMP	RSVD			PAR_CLR	RSVD			RESTART	RSVD		MODE	
N/A																			RW	N/A			WC	N/A			WC	N/A		RW	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[XCVR][CTRL] [31:0]

Bits	Name	Description
12	TRISMP	Tipple sampe 0: sample 1 time for data transmission 1: sample 3 times for data transmission and take the same value twice as the sampling result
8	PAR_CLR	Clear parity error, this is a self clear bit 0: no effect 1: clear parity error
4	RESTART	Restart transceiver, this is a self clear bit 0: no effect 1: reset transceiver
1-0	MODE	Transceiver mode 0: synchronous master 1: synchronous slave 2: asynchronous mode 3: asynchronous mode

CTRL[XCVR][CTRL] Bits

41.4.9 CTRL[XCVR][TYPE_CFG] (0x24 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WAIT_LEN								RSVD			DATA_LEN					RSVD						PAR_POL	PAR_EN	RSVD				DA_IDLEZ	CK_IDLEZ	DA_IDLEV	CK_IDLEV
RW								N/A			RW					N/A						RW	RW	N/A				RW	RW	RW	RW
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[XCVR][TYPE_CFG] [31:0]

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Bits	Name	Description
31-24	WAIT_LEN	Number of extra stop bit for asynchronous mode 0: 1 bit 1: 2 bit ... 255: 256 bit
20-16	DATA_LEN	Number of data bit for asynchronous mode 0: 1 bit 1: 2 bit ... 31: 32 bit
9	PAR_POL	Polarity of parity for asynchronous mode 0: even 1: odd
8	PAR_EN	enable parity check for asynchronous mode 0: disable 1: enable
3	DA_IDLEZ	Idle state driver of data line 0: output 1: high-Z
2	CK_IDLEZ	Idle state driver of clock line 0: output 1: high-Z
1	DA_IDLEV	Idle state value of data line 0: data'0' 1: data'1'
0	CK_IDLEV	Idle state value of clock line 0: data'0' 1: data'1'

CTRL[XCVR][TYPE_CFG] Bits

41.4.10 CTRL[XCVR][BAUD_CFG] (0x28 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SYNC_POINT																BAUD_DIV															
RW																RW															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[XCVR][BAUD_CFG] [31:0]

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Bits	Name	Description
31-16	SYNC_POINT	Baud synchronous time, minmum bit time
15-0	BAUD_DIV	Baud rate, bit time in system clock cycle

CTRL[XCVR][BAUD_CFG] Bits

41.4.11 CTRL[XCVR][DATA_CFG] (0x2C + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TXD_POINT																RXD_POINT															
RW																RW															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[XCVR][DATA_CFG] [31:0]

Bits	Name	Description
31-16	TXD_POINT	data transmit point in system clcok cycle
15-0	RXD_POINT	data receive point in system clcok cycle

CTRL[XCVR][DATA_CFG] Bits

41.4.12 CTRL[XCVR][CLK_CFG] (0x30 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CK1_POINT																CK0_POINT															
RW																RW															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[XCVR][CLK_CFG] [31:0]

Bits	Name	Description
31-16	CK1_POINT	clock point 1 in system clcok cycle
15-0	CK0_POINT	clock point 0 in system clcok cycle

CTRL[XCVR][CLK_CFG] Bits

41.4.13 CTRL[XCVR][PIN] (0x38 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD					OE_CK	DI_CK	DO_CK	RSVD					OE_RX	DI_RX	DO_RX	RSVD					OE_DE	DI_DE	DO_DE	RSVD					OE_TX	DI_TX	DO_TX

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
N/A					RO	RO	RO	N/A					RO	RO	RO	N/A					RO	RO	RO	N/A					RO	RO	RO
x	x	x	x	x	0	0	0	x	x	x	x	x	0	0	0	x	x	x	x	x	0	0	0	x	x	x	x	x	0	0	0

CTRL[XCVR][PIN] [31:0]

Bits	Name	Description
26	OE_CK	CK drive state 0: input 1: output
25	DI_CK	CK state 0: data 0 1: data 1
24	DO_CK	CK output 0: data 0 1: data 1
18	OE_RX	RX drive state 0: input 1: output
17	DI_RX	RX state 0: data 0 1: data 1
16	DO_RX	RX output 0: data 0 1: data 1
10	OE_DE	DE drive state 0: input 1: output
9	DI_DE	DE state 0: data 0 1: data 1
8	DO_DE	DE output 0: data 0 1: data 1
2	OE_TX	TX drive state 0: input 1: output
1	DI_TX	TX state 0: data 0 1: data 1

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Bits	Name	Description
0	DO_TX	TX output 0: data 0 1: data 1

CTRL[XCVR][PIN] Bits

41.4.14 CTRL[XCVR][STATE] (0x3C + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD					RECV_STATE			RSVD					SEND_STATE			RSVD															
N/A					RO			N/A					RO			N/A															
x	x	x	x	x	0	0	0	x	x	x	x	x	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

CTRL[XCVR][STATE] [31:0]

Bits	Name	Description
26-24	RECV_STATE	FSM of asynchronous receive
18-16	SEND_STATE	FSM of asynchronous transmit

CTRL[XCVR][STATE] Bits

41.4.15 CTRL[TRG][IN_CFG] (0x40 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD								PRD_EN	RSVD				SYNC_SEL			IN1_EN	RSVD				IN1_SEL			IN0_EN	RSVD				IN0_SEL		
N/A								RW	N/A				RW			RW	N/A				RW			RW	N/A				RW		
x	x	x	x	x	x	x	x	0	x	x	x	x	0	0	0	0	x	x	x	x	0	0	0	0	x	x	x	x	0	0	0

CTRL[TRG][IN_CFG] [31:0]

Bits	Name	Description
23	PRD_EN	Enable period trigger (tigger 2) 0: periodical trigger disabled 1: periodical trigger enabled
18-16	SYNC_SEL	Synchronize sigal selection (tigger 2) 0: trigger in 0 1: trigger in 1 ... 7: trigger in 7

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Bits	Name	Description
15	IN1_EN	Enable trigger 1 0: disable trigger 1 1: enable trigger 1
10-8	IN1_SEL	Trigger 1 sigal selection 0: trigger in 0 1: trigger in 1 ... 7: trigger in 7
7	IN0_EN	Enable trigger 0 0: disable trigger 1 1: enable trigger 1
2-0	IN0_SEL	Trigger 0 sigal selection 0: trigger in 0 1: trigger in 1 ... 7: trigger in 7

CTRL[TRG][IN_CFG] Bits

41.4.16 CTRL[TRG][SW] (0x44 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																															SOFT
N/A																															WC
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

CTRL[TRG][SW] [31:0]

Bits	Name	Description
0	SOFT	Software trigger (tigger 3). this bit is self-clear 0: trigger source disabled 1: trigger source enabled

CTRL[TRG][SW] Bits

41.4.17 CTRL[TRG][PRD_CFG] (0x48 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD															ARMING	RSVD															SYNC

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
N/A															RW	N/A															RW
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

CTRL[TRG][PRD_CFG] [31:0]

Bits	Name	Description
16	ARMING	Wait for trigger synchronous before trigger 0: Trigger directly 1: Wait trigger source before period trigger
0	SYNC	Synchronous 0: Not synchronous 1: Synchronous every trigger source

CTRL[TRG][PRD_CFG] Bits

41.4.18 CTRL[TRG][PRD] (0x4C + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PERIOD																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[TRG][PRD] [31:0]

Bits	Name	Description
31-0	PERIOD	Trigger period

CTRL[TRG][PRD] Bits

41.4.19 CTRL[TRG][OUT_CFG] (0x50 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OUT3_EN	RSVD				OUT3_SEL			OUT2_EN	RSVD				OUT2_SEL			OUT1_EN	RSVD				OUT1_SEL			OUT0_EN	RSVD				OUT0_SEL		
RW	N/A				RW			RW	N/A				RW			RW	N/A				RW			RW	N/A				RW		
0	x	x	x	x	0	0	0	0	x	x	x	x	0	0	0	0	x	x	x	x	0	0	0	0	x	x	x	x	0	0	0

CTRL[TRG][OUT_CFG] [31:0]

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Bits	Name	Description
31	OUT3_EN	Enable trigger 3 0: disable trigger 3 1: enable trigger 3
26-24	OUT3_SEL	Trigger 3 sigal selection 0: trigger out 0 1: trigger out 1 ... 7: trigger out 7
23	OUT2_EN	Enable trigger 2 0: disable trigger 2 1: enable trigger 2
18-16	OUT2_SEL	Trigger 2 sigal selection 0: trigger out 0 1: trigger out 1 ... 7: trigger out 7
15	OUT1_EN	Enable trigger 1 0: disable trigger 1 1: enable trigger 1
10-8	OUT1_SEL	Trigger 1 sigal selection 0: trigger out 0 1: trigger out 1 ... 7: trigger out 7
7	OUT0_EN	Enable trigger 0 0: disable trigger 1 1: enable trigger 1
2-0	OUT0_SEL	Trigger 0 sigal selection 0: trigger out 0 1: trigger out 1 ... 7: trigger out 7

CTRL[TRG][OUT_CFG] Bits

41.4.20 CTRL[TRG][PRD_STS] (0x60 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD											TRIGERED	RSVD			ARMED	RSVD															

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
N/A											RO	N/A			RO	N/A															
x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

CTRL[TRG][PRD_STS] [31:0]

Bits	Name	Description
20	TRIGERED	Period has been triggered 0: Not triggered 1: Triggered
16	ARMED	Waiting for trigger 0: Not in waiting status 1: In waiting status

CTRL[TRG][PRD_STS] Bits

41.4.21 CTRL[TRG][PRD_CNT] (0x64 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PERIOD_CNT																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[TRG][PRD_CNT] [31:0]

Bits	Name	Description
31-0	PERIOD_CNT	Trigger period counter

CTRL[TRG][PRD_CNT] Bits

41.4.22 CTRL[TRG_TABLE][CMD] (0x80 + 0x400 * n + 0x4 * x)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMD_TRIGGER0																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

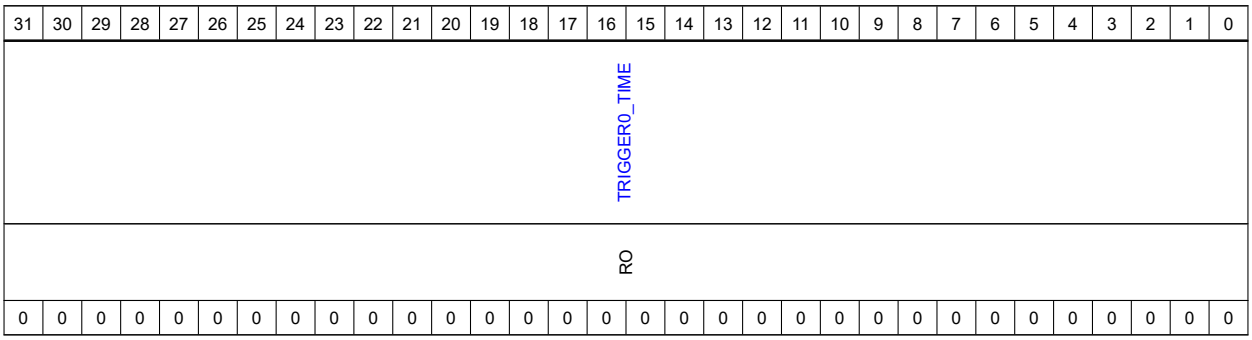
CTRL[TRG_TABLE][CMD] [31:0]

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Bits	Name	Description
31-0	CMD_TRIGGER0	Trigger command

CTRL[TRG_TABLE][CMD] Bits

41.4.23 CTRL[TRG_TABLE][TIME] (0xA0 + 0x400 * n + 0x4 * x)

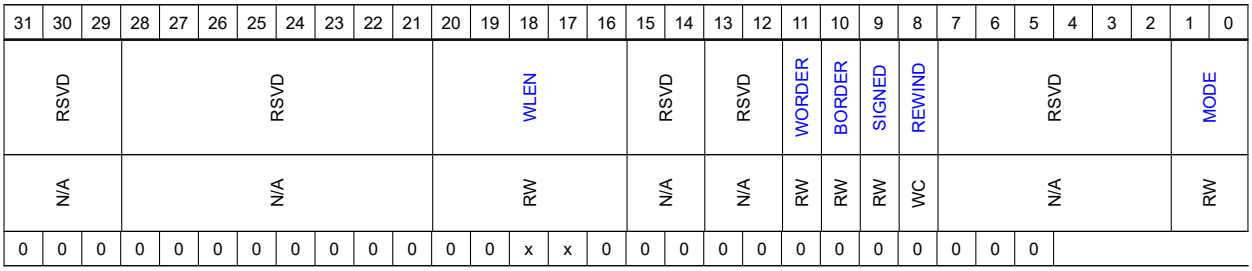


CTRL[TRG_TABLE][TIME] [31:0]

Bits	Name	Description
31-0	TRIGGER0_TIME	Trigger time

CTRL[TRG_TABLE][TIME] Bits

41.4.24 CTRL[CMD][MODE] (0xC0 + 0x400 * n)



CTRL[CMD][MODE] [31:0]

Bits	Name	Description
20-16	WLEN	word length 0: 1 bit 1: 2 bit ... 31: 32 bit
11	WORDER	word order 0: sample as bit order 1: different from bit order

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Bits	Name	Description
10	BORDER	bit order 0: LSB first 1: MSB first
9	SIGNED	Signed 0: unsigned value 1: signed value
8	REWIND	Write 1 to rewind read/write pointer, this is a self clear bit
1-0	MODE	Data mode(CMD register only support data mode) 0: data mode 1: check mode 2: CRC mode

CTRL[CMD][MODE] Bits

41.4.25 CTRL[CMD][IDX] (0xC4 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD			LAST_BIT					RSVD			FIRST_BIT					RSVD			MAX_BIT					RSVD			MIN_BIT				
N/A			RW					N/A			RW					N/A			RW					N/A			RW				
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[CMD][IDX] [31:0]

Bits	Name	Description
28-24	LAST_BIT	Last bit index for tranceive
20-16	FIRST_BIT	First bit index for tranceive
12-8	MAX_BIT	Highest bit index
4-0	MIN_BIT	Lowest bit index

CTRL[CMD][IDX] Bits

41.4.26 CTRL[CMD][CMD] (0xE0 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

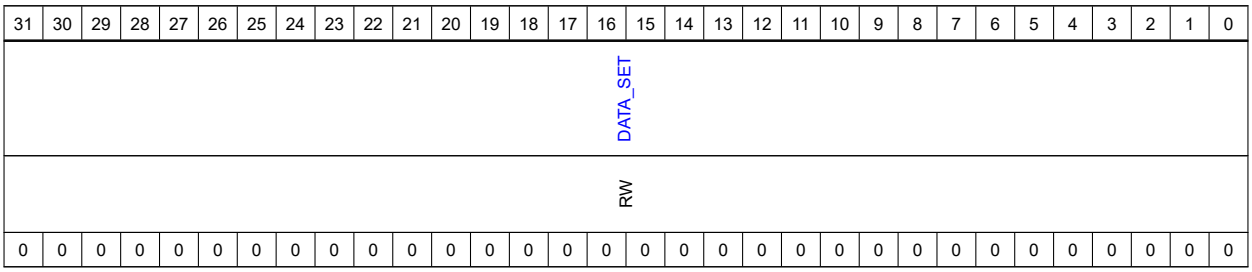
CTRL[CMD][CMD] [31:0]

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Bits	Name	Description
31-0	DATA	DATA

CTRL[CMD][CMD] Bits

41.4.27 CTRL[CMD][SET] (0xE4 + 0x400 * n)

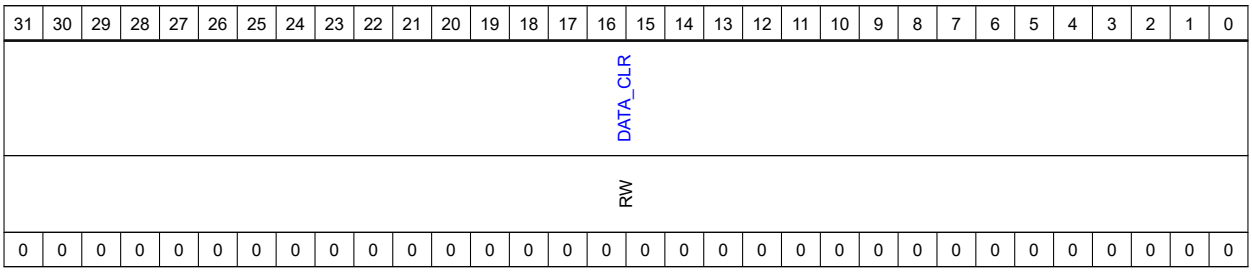


CTRL[CMD][SET] [31:0]

Bits	Name	Description
31-0	DATA_SET	DATA bit set

CTRL[CMD][SET] Bits

41.4.28 CTRL[CMD][CLR] (0xE8 + 0x400 * n)

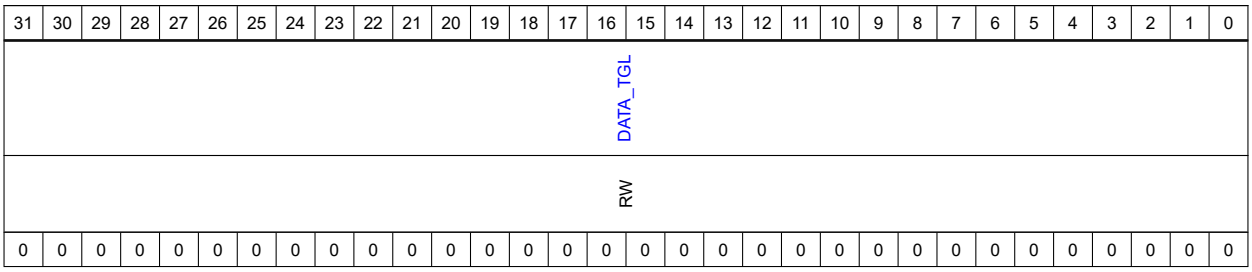


CTRL[CMD][CLR] [31:0]

Bits	Name	Description
31-0	DATA_CLR	DATA bit clear

CTRL[CMD][CLR] Bits

41.4.29 CTRL[CMD][INV] (0xEC + 0x400 * n)



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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

CTRL[CMD][INV] [31:0]

Bits	Name	Description
31-0	DATA_TGL	DATA bit toggle

CTRL[CMD][INV] Bits

41.4.30 CTRL[CMD][IN] (0xF0 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA_IN																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[CMD][IN] [31:0]

Bits	Name	Description
31-0	DATA_IN	Commad input

CTRL[CMD][IN] Bits

41.4.31 CTRL[CMD][OUT] (0xF4 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA_OUT																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[CMD][OUT] [31:0]

Bits	Name	Description
31-0	DATA_OUT	Command output

CTRL[CMD][OUT] Bits

41.4.32 CTRL[CMD][STS] (0xF8 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD											WORD_IDX					RSVD					WORD_CNT					RSVD					BIT_IDX				

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
N/A			N/A								RO					N/A			RO					N/A				RO				
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0					

CTRL[CMD][STS] [31:0]

Bits	Name	Description
20-16	WORD_IDX	Word index
12-8	WORD_CNT	Word counter
4-0	BIT_IDX	Bit index

CTRL[CMD][STS] Bits

41.4.33 CTRL[CMD_TABLE][MIN] (0x100 + 0x400 * n + 0x20 * m)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMD_MIN																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[CMD_TABLE][MIN] [31:0]

Bits	Name	Description
31-0	CMD_MIN	minimum command value

CTRL[CMD_TABLE][MIN] Bits

41.4.34 CTRL[CMD_TABLE][MAX] (0x104 + 0x400 * n + 0x20 * m)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMD_MAX																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

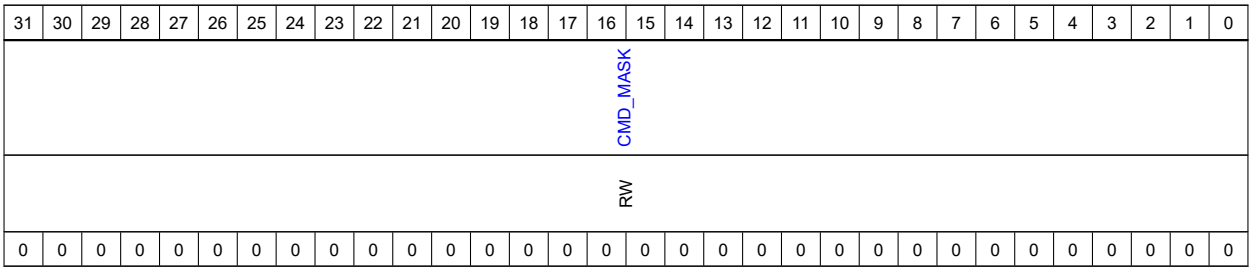
CTRL[CMD_TABLE][MAX] [31:0]

Bits	Name	Description
31-0	CMD_MAX	maximum command value

CTRL[CMD_TABLE][MAX] Bits

41.4.35 CTRL[CMD_TABLE][MSK] (0x108 + 0x400 * n + 0x20 * m)

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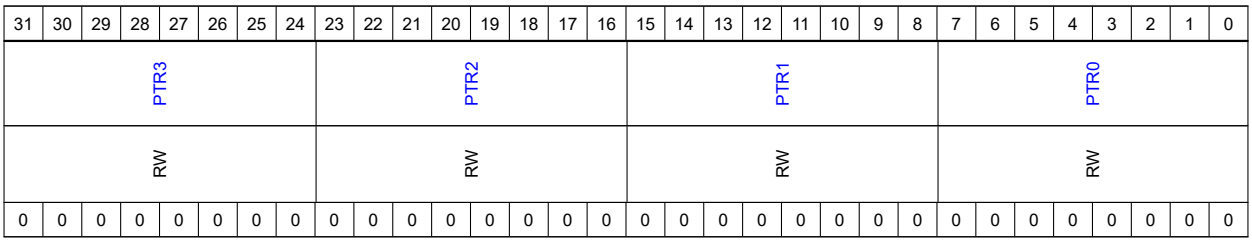


CTRL[CMD_TABLE][MSK] [31:0]

Bits	Name	Description
31-0	CMD_MASK	compare mask

CTRL[CMD_TABLE][MSK] Bits

41.4.36 CTRL[CMD_TABLE][PTA] (0x110 + 0x400 * n + 0x20 * m)

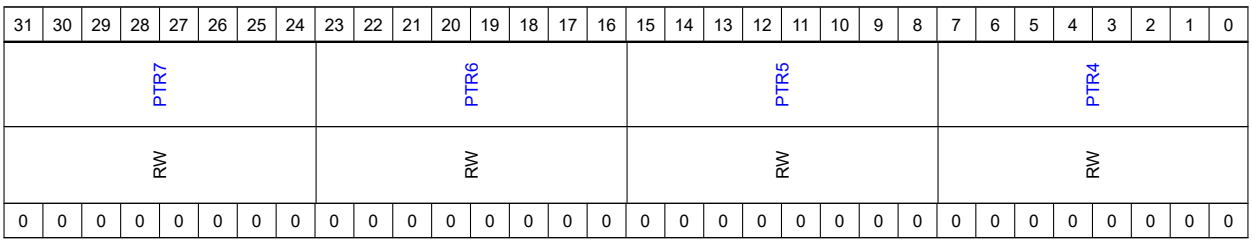


CTRL[CMD_TABLE][PTA] [31:0]

Bits	Name	Description
31-24	PTR3	pointer3
23-16	PTR2	pointer2
15-8	PTR1	pointer1
7-0	PTR0	pointer0

CTRL[CMD_TABLE][PTA] Bits

41.4.37 CTRL[CMD_TABLE][PTB] (0x114 + 0x400 * n + 0x20 * m)



CTRL[CMD_TABLE][PTB] [31:0]

Bits	Name	Description
31-24	PTR7	pointer7

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Bits	Name	Description
23-16	PTR6	pointer6
15-8	PTR5	pointer5
7-0	PTR4	pointer4

CTRL[CMD_TABLE][PTB] Bits

41.4.38 CTRL[LATCH][TRAN] (0x200 + 0x400 * n + 0x20 * m + 0x4 * x)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
POINTER								RSVD						CFG_TM		RSVD		CFG_TXD		CFG_CLK		CFG_PTR		RSVD			OV_TM	RSVD	OV_TXD	OV_CLK	OV_PTR
RW								N/A						RW		N/A		RW		RW		RW		N/A			RW	N/A	RW	RW	RW
0	0	0	0	0	0	0	0	0	0	0	0	x	x	0	0	0	0	0	0	0	0	0	0	x	0	0	0				

CTRL[LATCH][TRAN] [31:0]

Bits	Name	Description
31-24	POINTER	pointer
17-16	CFG_TM	timeout 0: high 1: low 2: rise 3: fall
13-12	CFG_TXD	data send 0: high 1: low 2: rise 3: fall
11-10	CFG_CLK	clock(only support master mode) 0: high 1: low 2: rise 3: fall
9-8	CFG_PTR	pointer 0: match 1: not match 2:entry 3:leave
4	OV_TM	override timeout check
2	OV_TXD	override TX data check
1	OV_CLK	override clock check

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Bits	Name	Description
0	OV_PTR	override pointer check

CTRL[LATCH][TRAN] Bits

41.4.39 CTRL[LATCH][CFG] (0x210 + 0x400 * n + 0x20 * m)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EN	RSVD				SELECT			RSVD								DELAY																
RW	N/A				RW			N/A								RW															
0	x	x	x	x	0	0	0	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[LATCH][CFG] [31:0]

Bits	Name	Description
31	EN	Enable latch 0: disable 1: enable
26-24	SELECT	Output select 0: state0-state1 1: state1-state2 2: state2-state3 3: state3-state0
15-0	DELAY	Delay in system clock cycle, for state transition

CTRL[LATCH][CFG] Bits

41.4.40 CTRL[LATCH][TIME] (0x218 + 0x400 * n + 0x20 * m)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LAT_TIME																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[LATCH][TIME] [31:0]

Bits	Name	Description
31-0	LAT_TIME	Latch time

CTRL[LATCH][TIME] Bits

41.4.41 CTRL[LATCH][STS] (0x21C + 0x400 * n + 0x20 * m)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD					STATE			RSVD								LAT_CNT																
N/A					RO			N/A								RO																
x	x	x	x	x	0	0	0	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[LATCH][STS] [31:0]

Bits	Name	Description
26-24	STATE	State
15-0	LAT_CNT	Latch counter

CTRL[LATCH][STS] Bits

41.4.42 CTRL[POS][SMP_EN] (0x280 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ACC_EN	RSVD		ACC_SEL					SPD_EN	RSVD		SPD_SEL					REV_EN	RSVD		REV_SEL					POS_EN	RSVD		POS_SEL				
RW	N/A		RW					RW	N/A		RW					RW	N/A		RW					RW	N/A		RW				
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[POS][SMP_EN] [31:0]

Bits	Name	Description
31	ACC_EN	Position include acceleration 0: use acceleration from sample override acceleration register 1: use acceleration from motor group
28-24	ACC_SEL	Data register for acceleration transfer
23	SPD_EN	Position include speed 0: use speed from sample override speed register 1: use speed from motor group
20-16	SPD_SEL	Data register for speed transfer
15	REV_EN	Position include revolution 0: use revolution from sample override revolution register 1: use revolution from motor group
12-8	REV_SEL	Data register for revolution transfer
7	POS_EN	Position include position 0: use position from sample override position register 1: use position from motor group
4-0	POS_SEL	Data register for position transfer

CTRL[POS][SMP_EN] Bits

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41.4.43 CTRL[POS][SMP_CFG] (0x284 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD							ONCE	RSVD						LAT_SEL		WINDOW																
N/A							RW	N/A						RW		RW																
x	x	x	x	x	x	x	0	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[POS][SMP_CFG] [31:0]

Bits	Name	Description
24	ONCE	Sample one time 0: Sample during windows time 1: Close sample window after first sample
17-16	LAT_SEL	Latch selection 0: latch 0 1: latch 1 2: latch 2 3: latch 3
15-0	WINDOW	Sample window, in clock cycle

CTRL[POS][SMP_CFG] Bits

41.4.44 CTRL[POS][SMP_DAT] (0x288 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DAT_SEL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

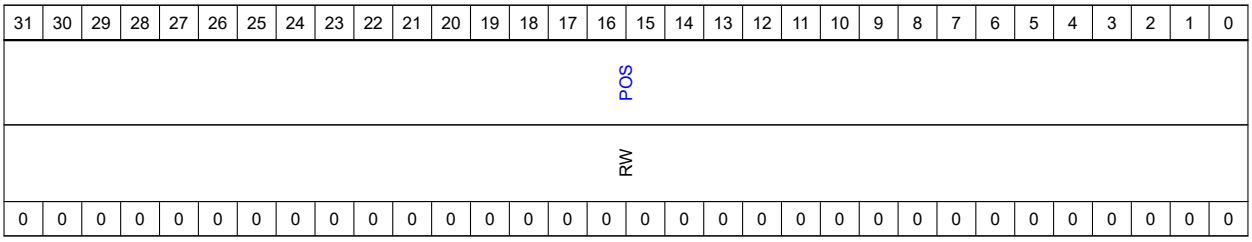
CTRL[POS][SMP_DAT] [31:0]

Bits	Name	Description
31-0	DAT_SEL	Data register sampled, each bit represent a data register, select the DATA registers need to be updated when SAMPLE happens. Note: CRC register will be cleared automatically by SAMPLE if select the DATA register used for CRC.

CTRL[POS][SMP_DAT] Bits

41.4.45 CTRL[POS][SMP_POS] (0x290 + 0x400 * n)

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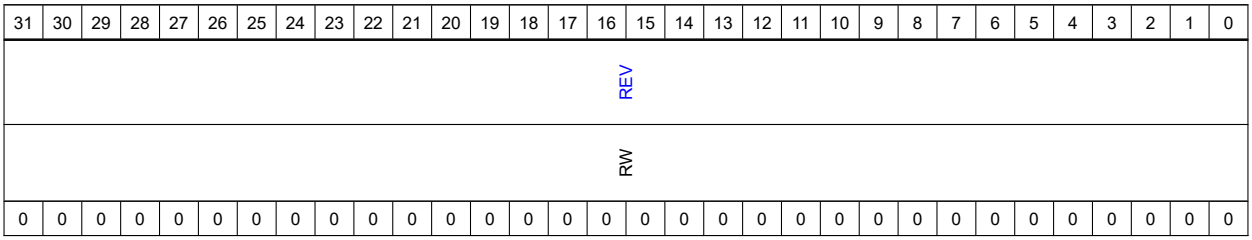


CTRL[POS][SMP_POS] [31:0]

Bits	Name	Description
31-0	POS	Sample override position

CTRL[POS][SMP_POS] Bits

41.4.46 CTRL[POS][SMP_REV] (0x294 + 0x400 * n)

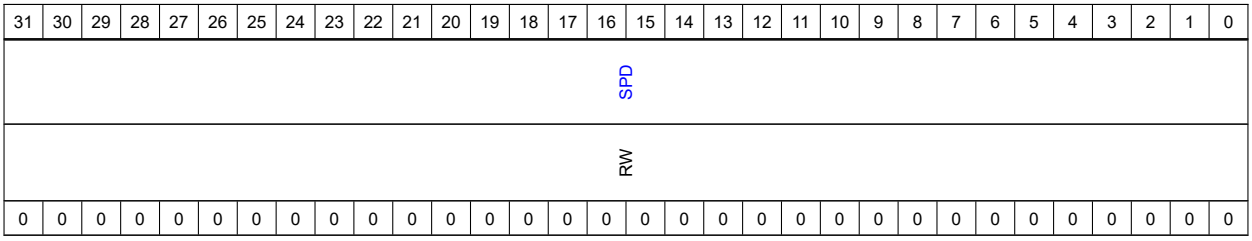


CTRL[POS][SMP_REV] [31:0]

Bits	Name	Description
31-0	REV	Sample override revolution

CTRL[POS][SMP_REV] Bits

41.4.47 CTRL[POS][SMP_SPD] (0x298 + 0x400 * n)



CTRL[POS][SMP_SPD] [31:0]

Bits	Name	Description
31-0	SPD	Sample override speed

CTRL[POS][SMP_SPD] Bits

41.4.48 CTRL[POS][SMP_ACC] (0x29C + 0x400 * n)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ACC																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[POS][SMP_ACC] [31:0]

Bits	Name	Description
31-0	ACC	Sample override accelerate

CTRL[POS][SMP_ACC] Bits

41.4.49 CTRL[POS][UPD_EN] (0x2A0 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ACC_EN	RSVD			ACC_SEL				SPD_EN	RSVD		SPD_SEL					REV_EN	RSVD		REV_SEL					POS_EN	RSVD		POS_SEL				
RW	N/A			RW				RW	N/A		RW					RW	N/A		RW					RW	N/A		RW				
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[POS][UPD_EN] [31:0]

Bits	Name	Description
31	ACC_EN	Position include acceleration 0: use acceleration from update override acceleration register 1: use acceleration from data register
28-24	ACC_SEL	Data register for acceleration transfer
23	SPD_EN	Position include speed 0: use speed from update override speed register 1: use speed from data register
20-16	SPD_SEL	Data register for speed transfer
15	REV_EN	Position include revolution 0: use revolution from update override revolution register 1: use revolution from data register
12-8	REV_SEL	Data register for revolution transfer
7	POS_EN	Position include position 0: use position from update override position register 1: use position from data register
4-0	POS_SEL	Data register for position transfer

CTRL[POS][UPD_EN] Bits

41.4.50 CTRL[POS][UPD_CFG] (0x2A4 + 0x400 * n)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIME_OVRD	RSVD						ONERR	RSVD						LAT_SEL		RSVD																
RW	N/A						RW	N/A						RW		N/A																
0	x	x	x	x	x	x	0	x	x	x	x	x	x	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

CTRL[POS][UPD_CFG] [31:0]

Bits	Name	Description
31	TIME_OVRD	Use override time 0: use time sample from motor group 1: use override time
24	ONERR	Sample one time 0: Sample during windows time 1: Close sample window after first sample
17-16	LAT_SEL	Latch selection 0: latch 0 1: latch 1 2: latch 2 3: latch 3

CTRL[POS][UPD_CFG] Bits

41.4.51 CTRL[POS][UPD_DAT] (0x2A8 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DAT_SEL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

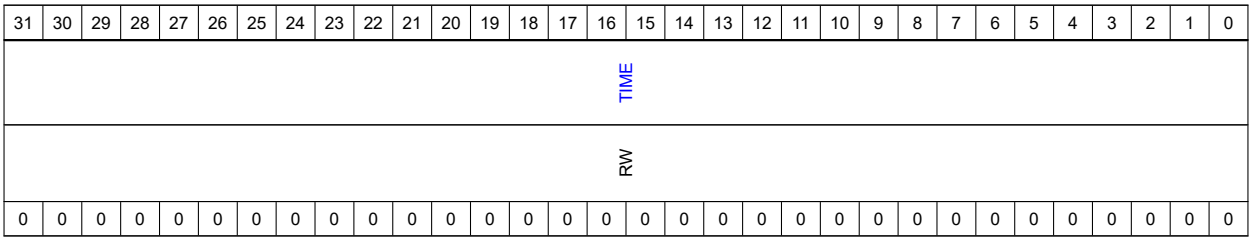
CTRL[POS][UPD_DAT] [31:0]

Bits	Name	Description
31-0	DAT_SEL	Data register sampled, each bit represent a data register, select the DATA registers need to be updated when UPDATE happen. Note: CRC register will be cleared automatically by UPDATE if select the DATA register used for CRC.

CTRL[POS][UPD_DAT] Bits

41.4.52 CTRL[POS][UPD_TIME] (0x2AC + 0x400 * n)

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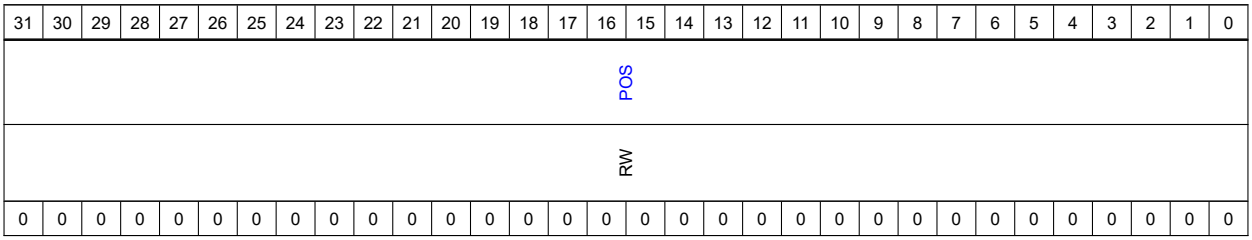


CTRL[POS][UPD_TIME] [31:0]

Bits	Name	Description
31-0	TIME	Update override time

CTRL[POS][UPD_TIME] Bits

41.4.53 CTRL[POS][UPD_POS] (0x2B0 + 0x400 * n)

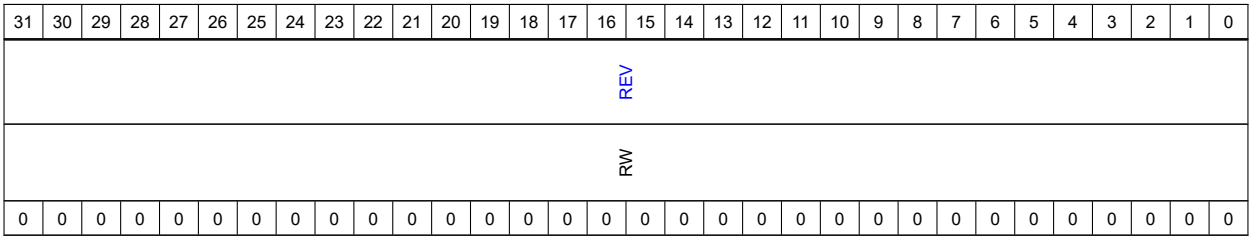


CTRL[POS][UPD_POS] [31:0]

Bits	Name	Description
31-0	POS	Update override position

CTRL[POS][UPD_POS] Bits

41.4.54 CTRL[POS][UPD_REV] (0x2B4 + 0x400 * n)



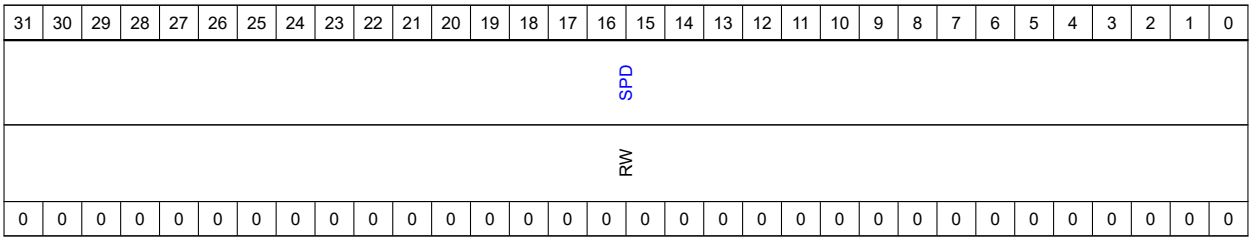
CTRL[POS][UPD_REV] [31:0]

Bits	Name	Description
31-0	REV	Update override revolution

CTRL[POS][UPD_REV] Bits

41.4.55 CTRL[POS][UPD_SPD] (0x2B8 + 0x400 * n)

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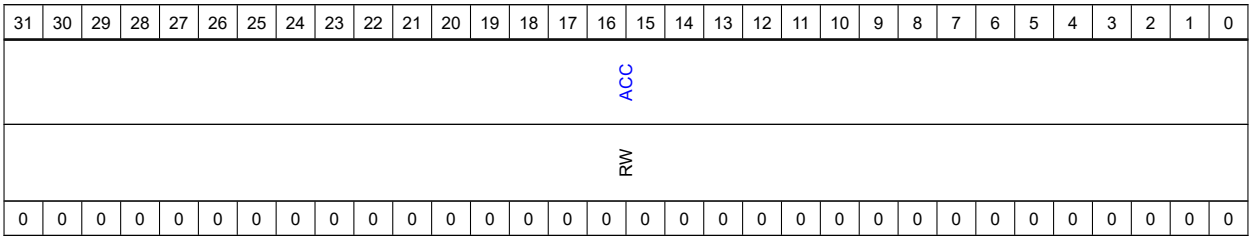


CTRL[POS][UPD_SPD] [31:0]

Bits	Name	Description
31-0	SPD	Update override speed

CTRL[POS][UPD_SPD] Bits

41.4.56 CTRL[POS][UPD_ACC] (0x2BC + 0x400 * n)

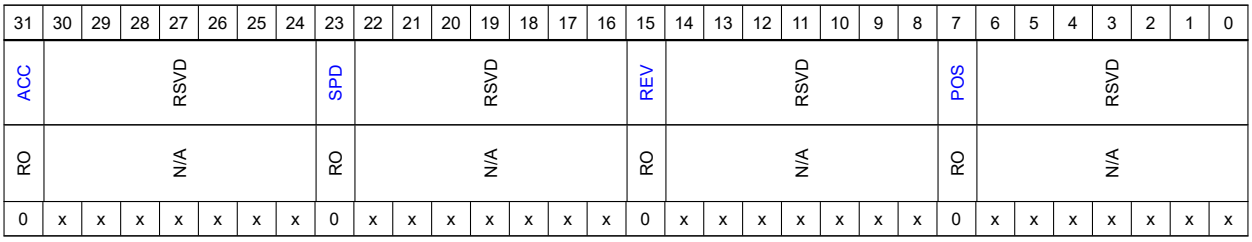


CTRL[POS][UPD_ACC] [31:0]

Bits	Name	Description
31-0	ACC	Update override accelerate

CTRL[POS][UPD_ACC] Bits

41.4.57 CTRL[POS][SMP_VAL] (0x2C0 + 0x400 * n)



CTRL[POS][SMP_VAL] [31:0]

Bits	Name	Description
31	ACC	Position include acceleration
23	SPD	Position include speed
15	REV	Position include revolution
7	POS	Position include position

CTRL[POS][SMP_VAL] Bits

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41.4.58 CTRL[POS][SMP_STS] (0x2C4 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD							OCCUR	RSVD								WIN_CNT															
N/A							RO	N/A								RO															
x	x	x	x	x	x	x	0	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[POS][SMP_STS] [31:0]

Bits	Name	Description
24	OCCUR	Sample occured 0: Sample not happened 1: Sample occured
15-0	WIN_CNT	Sample window counter

CTRL[POS][SMP_STS] Bits

41.4.59 CTRL[POS][TIME_IN] (0x2CC + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIME																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[POS][TIME_IN] [31:0]

Bits	Name	Description
31-0	TIME	input time

CTRL[POS][TIME_IN] Bits

41.4.60 CTRL[POS][POS_IN] (0x2D0 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
POS																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[POS][POS_IN] [31:0]

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Bits	Name	Description
31-0	POS	Input position

CTRL[POS][POS_IN] Bits

41.4.61 CTRL[POS][REV_IN] (0x2D4 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
REV																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[POS][REV_IN] [31:0]

Bits	Name	Description
31-0	REV	Input revolution

CTRL[POS][REV_IN] Bits

41.4.62 CTRL[POS][SPD_IN] (0x2D8 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SPD																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[POS][SPD_IN] [31:0]

Bits	Name	Description
31-0	SPD	Input speed

CTRL[POS][SPD_IN] Bits

41.4.63 CTRL[POS][ACC_IN] (0x2DC + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ACC																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[POS][ACC_IN] [31:0]

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Bits	Name	Description
31-0	ACC	Input accelerate

CTRL[POS][ACC_IN] Bits

41.4.64 CTRL[POS][UPD_STS] (0x2E4 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD							UPD_ERR	RSVD																							
N/A							RO	N/A																							
x	x	x	x	x	x	x	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

CTRL[POS][UPD_STS] [31:0]

Bits	Name	Description
24	UPD_ERR	Update error 0: data receive normally 1: data receive error

CTRL[POS][UPD_STS] Bits

41.4.65 CTRL[IRQ][INT_EN] (0x300 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TRG_ERR3	TRG_ERR2	TRG_ERR1	TRG_ERR0	TRIGER3	TRIGER2	TRIGER1	TRIGER0	RSVD			SMP_ERR	LATCH3	LATCH2	LATCH1	LATCH0	RSVD		TIMEOUT	TRX_ERR	INSTR1_END	INSTR0_END	PTR1_END	PTR0_END	INSTR1_ST	INSTR0_ST	PTR1_ST	PTR0_ST	RSVD	WDOG	EXCEPT	STALL
RW	RW	RW	RW	RW	RW	RW	RW	N/A			RW	RW	RW	RW	RW	N/A		RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	N/A	RW	RW	RW
0	0	0	0	0	0	0	0	x	x	x	0	0	0	0	0	x	x	0	0	0	0	0	0	0	0	0	0	x	0	0	0

CTRL[IRQ][INT_EN] [31:0]

Bits	Name	Description
31	TRG_ERR3	Trigger3 failed
30	TRG_ERR2	Trigger2 failed
29	TRG_ERR1	Trigger1 failed
28	TRG_ERR0	Trigger0 failed
27	TRIGER3	Trigger3
26	TRIGER2	Trigger2
25	TRIGER1	Trigger1
24	TRIGER0	Trigger0
20	SMP_ERR	Sample error

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Bits	Name	Description
19	LATCH3	Latch3
18	LATCH2	Latch2
17	LATCH1	Latch1
16	LATCH0	Latch0
13	TIMEOUT	Timeout
12	TRX_ERR	Transfer error
11	INSTR1_END	Instruction 1 end
10	INSTR0_END	Instruction 0 end
9	PTR1_END	Pointer 1 end
8	PTR0_END	Pointer 0 end
7	INSTR1_ST	Instruction 1 start
6	INSTR0_ST	Instruction 0 start
5	PTR1_ST	Pointer 1 start
4	PTR0_ST	Pointer 0 start
2	WDOG	Watch dog
1	EXCEPT	Exception
0	STALL	Stall

CTRL[IRQ][INT_EN] Bits

41.4.66 CTRL[IRQ][INT_FLAG] (0x304 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TRG_ERR3	TRG_ERR2	TRG_ERR1	TRG_ERR0	TRIGER3	TRIGER2	TRIGER1	TRIGER0	RSVD			SMP_ERR	LATCH3	LATCH2	LATCH1	LATCH0	RSVD		TIMEOUT	TRX_ERR	INSTR1_END	INSTR0_END	PTR1_END	PTR0_END	INSTR1_ST	INSTR0_ST	PTR1_ST	PTR0_ST	RSVD	WDOG	EXCEPT	STALL
W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	N/A			W1C	W1C	W1C	W1C	W1C	N/A		W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	N/A	W1C	W1C	W1C
0	0	0	0	0	0	0	0	x	x	x	0	0	0	0	0	x	x	0	0	0	0	0	0	0	0	0	0	x	0	0	0

CTRL[IRQ][INT_FLAG] [31:0]

Bits	Name	Description
31	TRG_ERR3	Trigger3 failed
30	TRG_ERR2	Trigger2 failed
29	TRG_ERR1	Trigger1 failed
28	TRG_ERR0	Trigger0 failed
27	TRIGER3	Trigger3
26	TRIGER2	Trigger2
25	TRIGER1	Trigger1
24	TRIGER0	Trigger0
20	SMP_ERR	Sample error

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Bits	Name	Description
19	LATCH3	Latch3
18	LATCH2	Latch2
17	LATCH1	Latch1
16	LATCH0	Latch0
13	TIMEOUT	Timeout
12	TRX_ERR	Transfer error
11	INSTR1_END	Instruction 1 end
10	INSTR0_END	Instruction 0 end
9	PTR1_END	Pointer 1 end
8	PTR0_END	Pointer 0 end
7	INSTR1_ST	Instruction 1 start
6	INSTR0_ST	Instruction 0 start
5	PTR1_ST	Pointer 1 start
4	PTR0_ST	Pointer 0 start
2	WDOG	Watch dog
1	EXCEPT	Exception
0	STALL	Stall

CTRL[IRQ][INT_FLAG] Bits

41.4.67 CTRL[IRQ][INT_STS] (0x308 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TRG_ERR3	TRG_ERR2	TRG_ERR1	TRG_ERR0	TRIGER3	TRIGER2	TRIGER1	TRIGER0	RSVD			SMP_ERR	LATCH3	LATCH2	LATCH1	LATCH0	RSVD		TIMEOUT	TRX_ERR	INSTR1_END	INSTR0_END	PTR1_END	PTR0_END	INSTR1_ST	INSTR0_ST	PTR1_ST	PTR0_ST	RSVD	WDOG	EXCEPT	STALL
RO	RO	RO	RO	RO	RO	RO	RO	N/A			RO	RO	RO	RO	RO	N/A		RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	N/A	RO	RO	RO
0	0	0	0	0	0	0	0	x	x	x	0	0	0	0	0	x	x	0	0	0	0	0	0	0	0	0	0	x	0	0	0

CTRL[IRQ][INT_STS] [31:0]

Bits	Name	Description
31	TRG_ERR3	Trigger3 failed
30	TRG_ERR2	Trigger2 failed
29	TRG_ERR1	Trigger1 failed
28	TRG_ERR0	Trigger0 failed
27	TRIGER3	Trigger3
26	TRIGER2	Trigger2
25	TRIGER1	Trigger1
24	TRIGER0	Trigger0
20	SMP_ERR	Sample error

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Bits	Name	Description
19	LATCH3	Latch3
18	LATCH2	Latch2
17	LATCH1	Latch1
16	LATCH0	Latch0
13	TIMEOUT	Timeout
12	TRX_ERR	Transfer error
11	INSTR1_END	Instruction 1 end
10	INSTR0_END	Instruction 0 end
9	PTR1_END	Pointer 1 end
8	PTR0_END	Pointer 0 end
7	INSTR1_ST	Instruction 1 start
6	INSTR0_ST	Instruction 0 start
5	PTR1_ST	Pointer 1 start
4	PTR0_ST	Pointer 0 start
2	WDOG	Watch dog
1	EXCEPT	Exception
0	STALL	Stall

CTRL[IRQ][INT_STS] Bits

41.4.68 CTRL[IRQ][POINTER0] (0x310 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																								POINTER							
N/A																								RW							
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0

CTRL[IRQ][POINTER0] [31:0]

Bits	Name	Description
7-0	POINTER	Match pointer 0

CTRL[IRQ][POINTER0] Bits

41.4.69 CTRL[IRQ][POINTER1] (0x314 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																								POINTER							
N/A																								RW							

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0

CTRL[IRQ][POINTER1] [31:0]

Bits	Name	Description
7-0	POINTER	Match pointer 1

CTRL[IRQ][POINTER1] Bits

41.4.70 CTRL[IRQ][INSTR0] (0x318 + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INSTR																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[IRQ][INSTR0] [31:0]

Bits	Name	Description
31-0	INSTR	Match instruction 0

CTRL[IRQ][INSTR0] Bits

41.4.71 CTRL[IRQ][INSTR1] (0x31C + 0x400 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
INSTR																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL[IRQ][INSTR1] [31:0]

Bits	Name	Description
31-0	INSTR	Match instruction 1

CTRL[IRQ][INSTR1] Bits

41.4.72 CTRL[DMA_EN] (0x320 + 0x400 * n)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TRG_ERR3	TRG_ERR2	TRG_ERR1	TRG_ERR0	TRIGER3	TRIGER2	TRIGER1	TRIGER0	RSVD			SMP_ERR	LATCH3	LATCH2	LATCH1	LATCH0	RSVD		TIMEOUT	TRX_ERR	INSTR1_END	INSTR0_END	PTR1_END	PTR0_END	INSTR1_ST	INSTR0_ST	PTR1_ST	PTR0_ST	RSVD	WDOG	EXCEPT	STALL
RW	RW	RW	RW	RW	RW	RW	RW	N/A			RW	RW	RW	RW	RW	N/A		RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	N/A	RW	RW	RW
0	0	0	0	0	0	0	0	x	x	x	0	0	0	0	0	x	x	0	0	0	0	0	0	0	0	0	0	x	0	0	0

CTRL[DMA_EN] [31:0]

Bits	Name	Description
31	TRG_ERR3	Trigger3 failed
30	TRG_ERR2	Trigger2 failed
29	TRG_ERR1	Trigger1 failed
28	TRG_ERR0	Trigger0 failed
27	TRIGER3	Trigger3
26	TRIGER2	Trigger2
25	TRIGER1	Trigger1
24	TRIGER0	Trigger0
20	SMP_ERR	Sample error
19	LATCH3	Latch3
18	LATCH2	Latch2
17	LATCH1	Latch1
16	LATCH0	Latch0
13	TIMEOUT	Timeout
12	TRX_ERR	Transfer error
11	INSTR1_END	Instruction 1 end
10	INSTR0_END	Instruction 0 end
9	PTR1_END	Pointer 1 end
8	PTR0_END	Pointer 0 end
7	INSTR1_ST	Instruction 1 start
6	INSTR0_ST	Instruction 0 start
5	PTR1_ST	Pointer 1 start
4	PTR0_ST	Pointer 0 start
2	WDOG	Watch dog
1	EXCEPT	Exception
0	STALL	Stall

CTRL[DMA_EN] Bits

41.4.73 INSTR (0x3400 + 0x4 * n)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD			OP			CK		RSVD			CRC					RSVD			DAT					RSVD			OPR				
N/A			RW			RW		N/A			RW					N/A			RW					N/A			RW				
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

INSTR [31:0]

Bits	Name	Description
28-26	OP	operation 0: halt 1: jump 2: send with timeout check 3: send without timout check 4: wait with timeout check 5: wait without timout check 6: receive with timeout check 7: receive without timout check
25-24	CK	clock state configure a. In synchronous master mode: 0: low 1: rise-fall 2: fall-rise 3: high b. In synchronous slave mode: 0：Use TX_POINT and RX_POINT as the timing for data transmission and reception. Disable the TIMEOUT function in the communication protocol(this is not WDG). 1：Switch the timing for data transmission and reception (e.g., switching edges for receive/transmit in an EnDat Encoder communication cycle). Disable the TIMEOUT function in the communication protocol. 2：Use TX_POINT and RX_POINT as the timing for data transmission and reception；Enable the TIMEOUT function in the communication protocol. 3：Switch the timing for data transmission and reception. Enable the TIMEOUT function in the communication protocol. c. In asynchronous mode: please keep 0.

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Bits	Name	Description
20-16	CRC	CRC register 0: don't calculate CRC 1: do not set this value 2: data register 2 3: data register 3 ... 29: data register 29 30: do not set this value 31: do not set this value
12-8	DAT	DATA register 0: ignore data 1: command 2: data register 2 3: data register 3 ... 29: data register 29 30: value 0 when send, wait 0 in receive 31: value1 when send, wait 1 in receive
4-0	OPR	a. When OP is 0, this area is the halt time in baudrate, 0 represents infinite time. b. When OP is 1, this area is the the pointer to the command table. OPR[4]=1, OPR[3:0] value is CMD_TABLE instruct pointer; OPR[4]=0, OPR[3:0]=0 is INIT_POINTER; OPR[4]=0, OPR[3:0]=1 is WDG_POINTER. c. When OP is 2-7, this area is the data length as fellow: 0: 1 bit 1: 2 bit ... 31: 32 bit

INSTR Bits

41.4.74 DAT[MODE] (0x3800 + 0x40 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD			CRC_LEN					RSVD			WLEN					RSVD		CRC_SHIFT	CRC_INV	WORDER	BORDER	SIGNED	REWIND	RSVD						MODE	
N/A			RW					N/A			RW					N/A		RW	RW	RW	RW	RW	RW	N/A						RW	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

DAT[MODE] [31:0]

Bits	Name	Description
28-24	CRC_LEN	CRC length 0: 1 bit 1: 2 bit ... 31: 32 bit
20-16	WLEN	word length 0: 1 bit 1: 2 bit ... 31: 32 bit
13	CRC_SHIFT	CRC shift mode, this mode is used to perform repeat code check 0: CRC 1: shift mode
12	CRC_INV	CRC invert 0: use CRC 1: use inverted CRC
11	WORDER	word order 0: sample as bit order 1: different from bit order
10	BORDER	bit order 0: LSB first 1: MSB first
9	SIGNED	Signed 0: unsigned value 1: signed value
8	REWIND	Write 1 to rewind read/write pointer, this is a self clear bit
1-0	MODE	Data mode 0: data mode 1: check mode 2: CRC mode

DAT[MODE] Bits

41.4.75 DAT[IDX] (0x3804 + 0x40 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD			LAST_BIT					RSVD			FIRST_BIT					RSVD			MAX_BIT					RSVD				MIN_BIT			

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
N/A			RW					N/A			RW					N/A			RW					N/A				RW			
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DAT[IDX] [31:0]

Bits	Name	Description
28-24	LAST_BIT	Last bit index for tranceive
20-16	FIRST_BIT	First bit index for tranceive
12-8	MAX_BIT	Highest bit index
4-0	MIN_BIT	Lowest bit index

DAT[IDX] Bits

41.4.76 DAT[GOLD] (0x3808 + 0x40 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GOLD_VALUE																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DAT[GOLD] [31:0]

Bits	Name	Description
31-0	GOLD_VALUE	Gold value for check mode

DAT[GOLD] Bits

41.4.77 DAT[CRCINIT] (0x380C + 0x40 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CRC_INIT																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

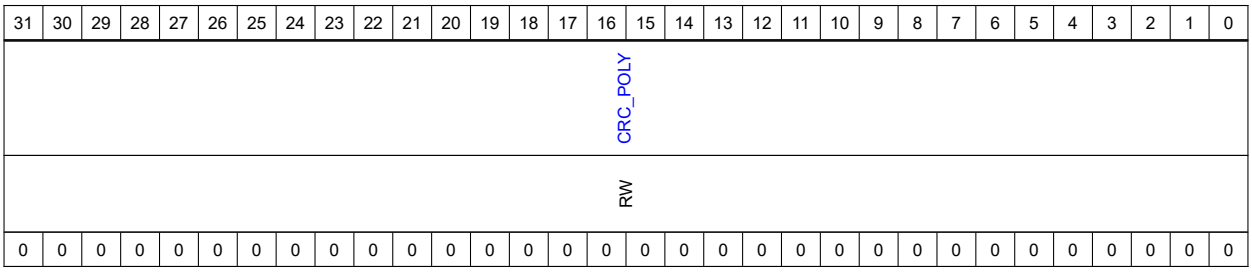
DAT[CRCINIT] [31:0]

Bits	Name	Description
31-0	CRC_INIT	CRC initial value

DAT[CRCINIT] Bits

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41.4.78 DAT[CRCPOLY] (0x3810 + 0x40 * n)

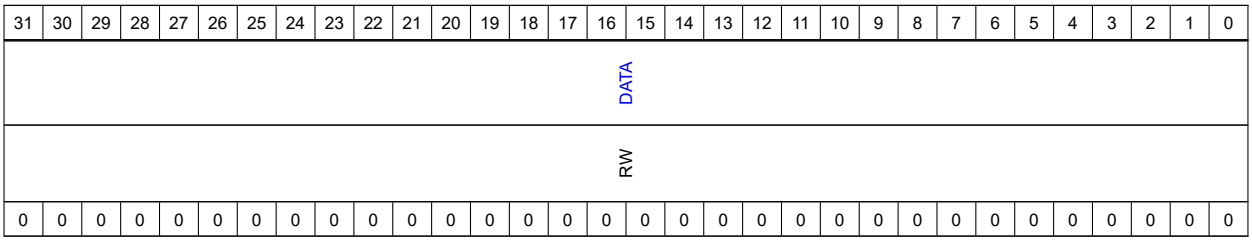


DAT[CRCPOLY] [31:0]

Bits	Name	Description
31-0	CRC_POLY	CRC polymonial

DAT[CRCPOLY] Bits

41.4.79 DAT[DATA] (0x3820 + 0x40 * n)

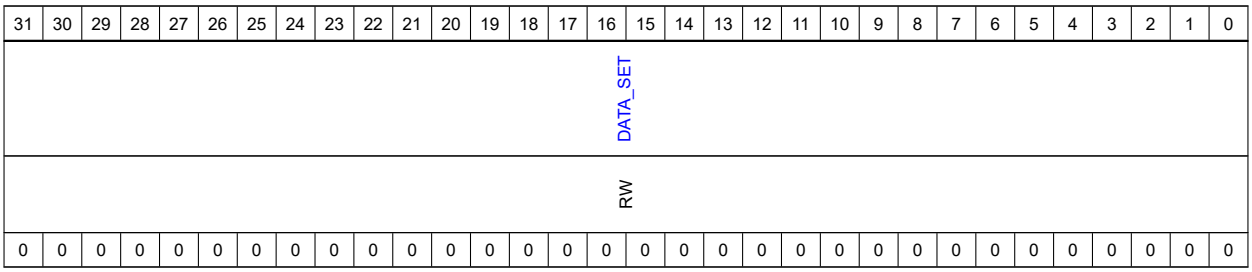


DAT[DATA] [31:0]

Bits	Name	Description
31-0	DATA	DATA

DAT[DATA] Bits

41.4.80 DAT[SET] (0x3824 + 0x40 * n)



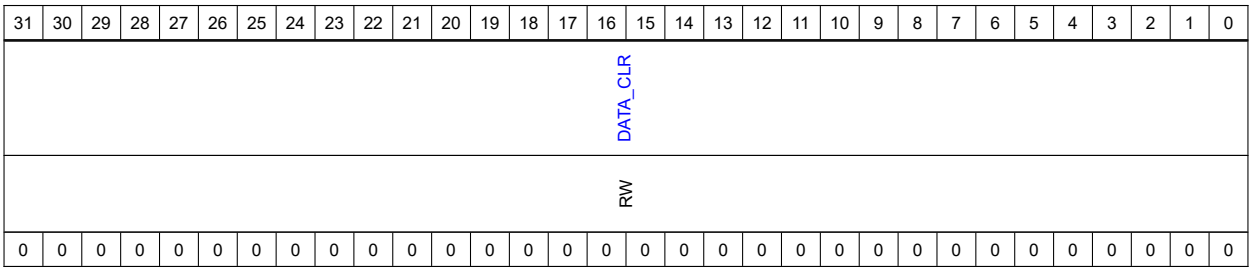
DAT[SET] [31:0]

Bits	Name	Description
31-0	DATA_SET	DATA bit set

DAT[SET] Bits

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41.4.81 DAT[CLR] (0x3828 + 0x40 * n)

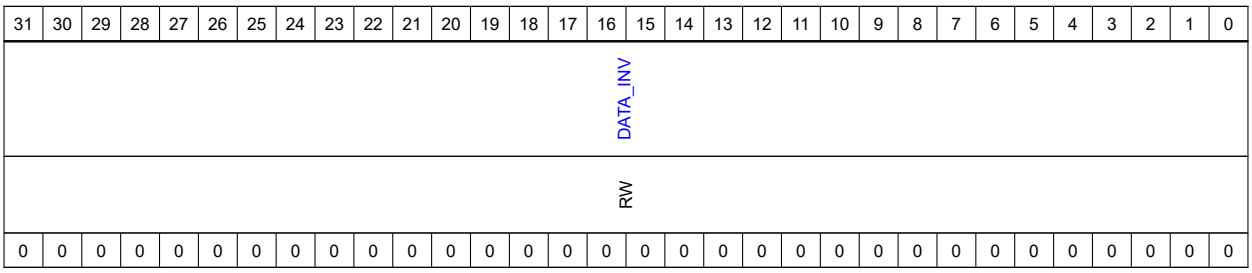


DAT[CLR] [31:0]

Bits	Name	Description
31-0	DATA_CLR	DATA bit clear

DAT[CLR] Bits

41.4.82 DAT[INV] (0x382C + 0x40 * n)

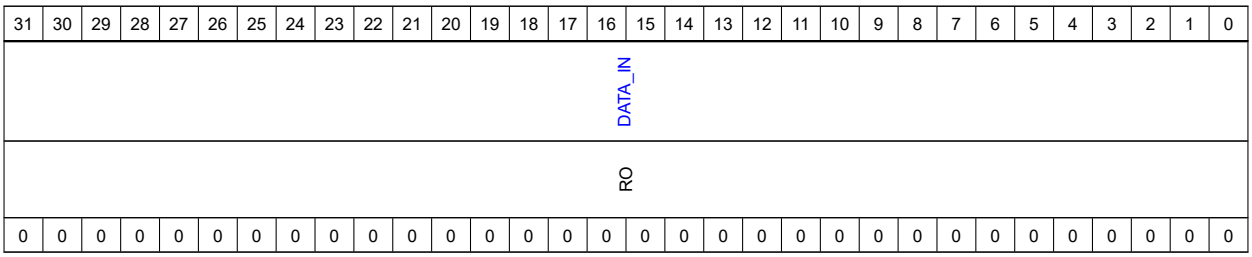


DAT[INV] [31:0]

Bits	Name	Description
31-0	DATA_INV	DATA bit toggle

DAT[INV] Bits

41.4.83 DAT[IN] (0x3830 + 0x40 * n)



DAT[IN] [31:0]

Bits	Name	Description
31-0	DATA_IN	Data input

DAT[IN] Bits

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41.4.84 DAT[OUT] (0x3834 + 0x40 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA_OUT																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DAT[OUT] [31:0]

Bits	Name	Description
31-0	DATA_OUT	Data output

DAT[OUT] Bits

41.4.85 DAT[STS] (0x3838 + 0x40 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD			CRC_IDX					RSVD			WORD_IDX					RSVD			WORD_CNT					RSVD			BIT_IDX				
N/A			RO					N/A			RO					N/A			RO					N/A			RO				
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DAT[STS] [31:0]

Bits	Name	Description
28-24	CRC_IDX	CRC index
20-16	WORD_IDX	Word index
12-8	WORD_CNT	Word counter
4-0	BIT_IDX	Bit index

DAT[STS] Bits

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42 Timer Overview

This chapter introduces the timers of this product. The timers on this product include general-purpose timer and watchdog timer.

42.1 General-Purpose Timers GPTMR, PTMR

This product supports 5 general-purpose timers, each of which supports 4 channels. Each channel is equipped with a 32-bit counter, a reload register and an input capture/output compare channel, supporting general timing, input capture, output compare, PWM generation, as well as interrupt and DMA request generation.

- 4 timers located in the system power domain are called general-purpose timers GPTMR0~3
- 1 timer located in the power management domain is called power management domain timer PTMR.

On this product, among the 4 input capture/output compare channels of GPTMR0~3, in addition to being connected to IOs, channel 2 and channel 3 are connected to each interconnection manager of the motor control unit. The counter synchronization input SYNCI of GPTMR0~3 comes from the interconnection manager.

For details of the internal signal interconnection of GPTMR0~3, please refer to Section 32.8.

On this product, the 4 input capture/output compare channels of PTMR are connected to the power management domain IOs (GPIO port Y). PTMR can keep working when the system power domain is powered down. PTMR interrupt can wake up the system from low-power modes. PTMR does not support DMA request generation.

42.2 Watchdog Timers WDG, PWDG

This product supports 3 watchdog timers, which can prevent the system from locking up due to software or hardware errors. The watchdog timer supports generating system reset when software feeding times out.

2 of them are located in the system power domain called WDG0~1, the clock of WDGx is CLK_TOP_AHB, and the external clock of the watchdog counter comes from CLK_32K.

1 is located in the power management domain called PWDG. PWDG can keep working when the system power domain is powered down. The clock of PWDG is CLK_24M, and the external clock of the watchdog counter is CLK_32K. PWDG supports outputting reset signals from the power management domain IOs.

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43 Timer TMR

This chapter describes the main functions and features of the Timer TMR.

43.1 Feature Summary

This section lists the main features of Timer TMR:

- The timer consists of 4 channels
- Each channel is configured with a 32-bit resolution counter supporting count-up mode
- Counters of all channels support synchronization
- Each channel supports a configurable 32-bit reload register
- Channels can be used for both input capture and output compare simultaneously
- Each channel supports 2 output comparators
- Supports rising edge and/or falling edge capture, and supports PWM period and duty cycle measurement
- Supports DMA and interrupt request generation

The block diagram of the timer is shown in Figure 62.

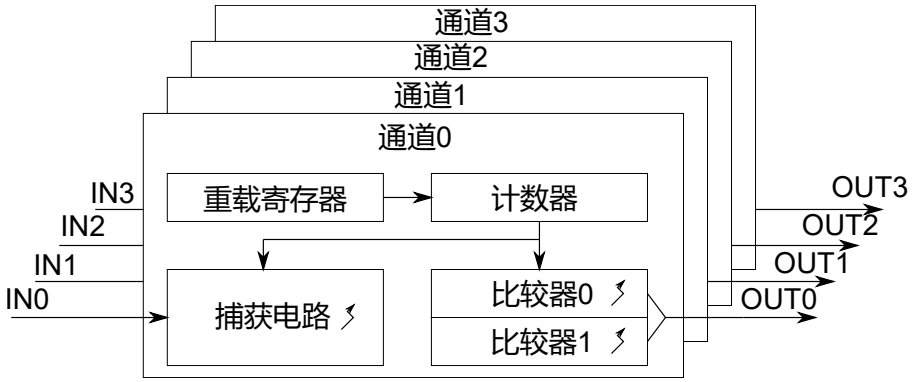


Figure 62: Timer Block Diagram

43.2 Functional Description

This section describes the functions of each sub-module of the timer.

43.2.1 Timer Time Base

The time base module of each timer channel is used to determine the operating time and period of the timer.

It consists of the following parts:

- 32-bit counter
- 32-bit reload register

The timer supports count-up mode.

Users can set the CHxCR [CEN] bit to enable the counter. After enabling, the counter always starts counting from 0. When the counter value reaches the reload value, the reload flag bit is set to 1, and the counter resets to 0 to continue counting.

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Users can reset the counter at any time through the CHxCR [CNTRST] bit. Setting CHxCR[CNTRST] to 1 resets the counter to 0. The counter will restart counting only after CHxCR[CNTRST] is set to 0 again. Resetting the counter through the CHxCR[CNTRST] bit also resets all input capture value registers, and resets the output compare output to the initial level defined by the CHxCR[CMPINIT] bit.

The counter register is read-only. Users can update the CHxCNT counter value to the expected value through the following steps:

- Write the expected value to the CHxCNTUPTVAL register
- Set CHxCR [CNTUPT] to 1 to load the CHxCNTNEW value into the CHxCNT counter

Users can also use the timer's synchronization trigger input (SYNCI) to synchronize the start timing of the counter. The timer supports synchronization mechanisms including software synchronization and hardware synchronization:

- After setting CHxCR [SWSYNCIEN] to 1, software synchronization can be triggered by writing to the GCR[SWSYNCTx] bit, and each channel has an independent software trigger bit. Users can set the software trigger bits of multiple channels to 1 at the same time to realize synchronous counting start of these channels.
- The hardware synchronization signal comes from outside the timer, i.e. SYNCI. Users can select whether synchronization occurs on the rising or falling edge of SYNCI by setting CHxCR [SYNCIREN] or CHxCR [SYNCIFEN] to 1.
- In addition, for channels with a channel number greater than 0, users can set CHxCR [SYNCFLW] to 1, so that the counter of this channel will be synchronized with the counter of the previous channel. For example, the counter of channel 1 will be synchronized with the counter of channel 0.

If the counter is reset to 0 through the synchronization trigger input, the reload flag bit RLDF will also be set to 1.

43.2.2 Output Compare

General-purpose timer channels support the output compare function.

The configuration procedure for using the channel output compare function is as follows:

- Configure the initial output state through the CHxCR [CMPINIT] bit. When this bit is set to 0, the initial output state is logic 0 when the counter starts counting; otherwise, the initial output state is logic 1 when the counter starts counting. In addition, when the counter is reloaded (CNT == RLD), the output will also be reset to the initial state.
- Configure the two comparators of the channel, CMP0 and CMP1. Whenever the counter reaches the value of either comparator, the channel output toggles.
  If CMP0 and CMP1 are configured to be equal, the output will not change when the counter reaches the comparator value.
- Enable the output compare function by setting CHxCR [CMPEN] to 1
- Enable the counter by setting CHxCR [CEN] to 1

If users need to output edge-aligned PWM, they can set the PWM duty cycle through CMP0, and set CMP1 equal to the reload register value (CMP1 = RLD).

If users need to output center-aligned PWM, they can set $0 < CMP0 < CMP1 < RLD$. In this case, the PWM

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period is the RLD value, and the duty cycle is (CMP1 –CMP0).

Output compare supports fast output of logic 0 and logic 1: when CMP0 is configured as 32'hFFFFFFFF, the channel output is always logic 0; when CMP1 is configured as 32'hFFFFFFFF and CMP0 is not all 1s, the channel output is always logic 1.

Note: The initial values of CMP0 and CMP1 are both 32'hFFFFFFF0. Users need to configure both values when using the compare output function.

43.2.3 Input Capture

General-purpose timer channels support input capture.

Users can configure the CHxCR[CAPMODE] bit to select different capture modes:

- 3'b000: Disable input capture
- 3'b001: Capture rising edge. Whenever the input changes from logic 0 to logic 1, the current counter value is saved to the CAPPOS register, and the CHxCAPF flag bit is set to 1
- 3'b010: Capture falling edge. Whenever the input changes from logic 1 to logic 0, the current counter value is saved to the CAPNEG register, and the CHxCAPF flag bit is set to 1
- 3'b011: Capture both rising and falling edges. Whenever the input toggles, the current counter value is saved to the CAPPOS and CAPNEG registers, and the CHxCAPF flag bit is set to 1
- 3'b100: PWM measurement mode. In this mode, the timer automatically measures the PWM period and duty cycle. The timer saves the count difference between two rising edges of the input signal (i.e. PWM period) in the CAPPRD register, and saves the count difference between the rising edge and falling edge of the signal (i.e. PWM duty cycle) in the CAPDTY register. The CAPPOS register still saves the counter value at the last rising edge of the signal. In this mode, the CHxCAPF flag bit is set to 1 when a rising edge of the input signal is captured.

Note that the values of CAPPOS register, CAPNEG register, CAPPRD register and CAPDTY register will be refreshed when a new edge is captured according to different configurations, and the previously captured values will be lost.

Note that in PWM measurement mode, users should pay attention to the CNT counter value. If the counter is reloaded and reset to 0 between adjacent signal edge transitions, the values of CAPPRD register and CAPDTY register will be inaccurate.

43.2.4 Interrupt and DMA

The timer supports generating the following interrupts:

- CHxRLDF flag bit is set to 1, which means the counter counts to the reload register value, or the counter is reloaded triggered by the SYNCI synchronization trigger input
- CHxCMPxF flag bit is set to 1, which means when the output compare function is enabled, the counter counts to the value of CMP0 or CMP1
- CHxCAPF flag bit is set to 1, which means when the input capture function is enabled, the specified edge transition is captured on the input channel

The above interrupt requests can be enabled or disabled by setting the corresponding bits of the CHxIRQEN register.

Users can query all flag bits by reading the CHxSR register, and writing 1 to the corresponding flag bit of the SR register can clear the flag bit.

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The timer supports generating the following DMA requests:

- CHxRLDF flag bit is set to 1, which means the counter counts to the reload register value, or the counter is reloaded triggered by the SYNCI synchronization trigger input
- CHxCMPxF flag bit is set to 1, which means when the output compare function is enabled, the counter counts to the value of CMP0 or CMP1
- CHxCAPF flag bit is set to 1, which means when the input capture function is enabled, the specified edge transition is captured on the input channel

One of the above requests can be selected as the DMA request output by setting the CHxCR[DMASEL] register bit.

DMA requests can be enabled or disabled by setting the CHxCR [DMAEN] register bit.

43.2.5 Debug Mode Support

The general-purpose timer supports the debug function, that is, when the core enters debug mode, the counter pauses counting. This function takes effect when users set the CHxCR[DBGPAUSE] bit to 1.

43.3 Timer Registers

43.3.1 Register Description

The list of timer registers is as follows:

GPTMR0 base address: 0xF0000000

GPTMR1 base address: 0xF0004000

GPTMR2 base address: 0xF0008000

GPTMR3 base address: 0xF000C000

PTMR base address: 0xF4120000

Offset	Register Name	Description	Reset Value
0x0000	CHANNEL[CH0][CR]	Control Register	0x00000000
0x0004	CHANNEL[CH0][CMP][CMP0]	Comparator register 0	0xFFFFFFF0
0x0008	CHANNEL[CH0][CMP][CMP1]	Comparator register 1	0xFFFFFFF0
0x000C	CHANNEL[CH0][RLD]	Reload register	0xFFFFFFFF
0x0010	CHANNEL[CH0][CNTUPTVAL]	Counter update value register	0x00000000
0x0020	CHANNEL[CH0][CAPPOS]	Capture rising edge register	0x00000000
0x0024	CHANNEL[CH0][CAPNEG]	Capture falling edge register	0x00000000
0x0028	CHANNEL[CH0][CAPPRD]	PWM period measure register	0x00000000
0x002C	CHANNEL[CH0][CAPDTY]	PWM duty cycle measure register	0x00000000
0x0030	CHANNEL[CH0][CNT]	Counter	0x00000000

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Offset	Register Name	Description	Reset Value
0x0040	CHANNEL[CH1][CR]	Control Register	0x00000000
0x0044	CHANNEL[CH1][CMP][CMP0]	Comparator register 0	0xFFFFFFF0
0x0048	CHANNEL[CH1][CMP][CMP1]	Comparator register 1	0xFFFFFFF0
0x004C	CHANNEL[CH1][RLD]	Reload register	0xFFFFFFFF
0x0050	CHANNEL[CH1][CNTUPTVAL]	Counter update value register	0x00000000
0x0060	CHANNEL[CH1][CAPPOS]	Capture rising edge register	0x00000000
0x0064	CHANNEL[CH1][CAPNEG]	Capture falling edge register	0x00000000
0x0068	CHANNEL[CH1][CAPPRD]	PWM period measure register	0x00000000
0x006C	CHANNEL[CH1][CAPDTY]	PWM duty cycle measure register	0x00000000
0x0070	CHANNEL[CH1][CNT]	Counter	0x00000000
0x0080	CHANNEL[CH2][CR]	Control Register	0x00000000
0x0084	CHANNEL[CH2][CMP][CMP0]	Comparator register 0	0xFFFFFFF0
0x0088	CHANNEL[CH2][CMP][CMP1]	Comparator register 1	0xFFFFFFF0
0x008C	CHANNEL[CH2][RLD]	Reload register	0xFFFFFFFF
0x0090	CHANNEL[CH2][CNTUPTVAL]	Counter update value register	0x00000000
0x00A0	CHANNEL[CH2][CAPPOS]	Capture rising edge register	0x00000000
0x00A4	CHANNEL[CH2][CAPNEG]	Capture falling edge register	0x00000000
0x00A8	CHANNEL[CH2][CAPPRD]	PWM period measure register	0x00000000
0x00AC	CHANNEL[CH2][CAPDTY]	PWM duty cycle measure register	0x00000000
0x00B0	CHANNEL[CH2][CNT]	Counter	0x00000000
0x00C0	CHANNEL[CH3][CR]	Control Register	0x00000000
0x00C4	CHANNEL[CH3][CMP][CMP0]	Comparator register 0	0xFFFFFFF0
0x00C8	CHANNEL[CH3][CMP][CMP1]	Comparator register 1	0xFFFFFFF0
0x00CC	CHANNEL[CH3][RLD]	Reload register	0xFFFFFFFF
0x00D0	CHANNEL[CH3][CNTUPTVAL]	Counter update value register	0x00000000
0x00E0	CHANNEL[CH3][CAPPOS]	Capture rising edge register	0x00000000
0x00E4	CHANNEL[CH3][CAPNEG]	Capture falling edge register	0x00000000
0x00E8	CHANNEL[CH3][CAPPRD]	PWM period measure register	0x00000000
0x00EC	CHANNEL[CH3][CAPDTY]	PWM duty cycle measure register	0x00000000

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Offset	Register Name	Description	Reset Value
0x00F0	CHANNEL[CH3][CNT]	Counter	0x00000000
0x0200	SR	Status register	0x00000000
0x0204	IRQEN	Interrupt request enable register	0x00000000
0x0208	GCR	Global control register	0x00000000

Table 197: GPTMR Register List

The detailed description of timer registers is as follows:

43.3.2 CHANNEL[CR] (0x0 + 0x40 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNTUPT	RSVD													OPMODE	MONITOR_SEL	MONITOR_EN	CNTRST	SYNCFLW	SYNCIFEN	SYNCIREN	CEN	CMPINIT	CMPEN	DMASEL		DMAEN	SWSYNCIEN	DBGPAUSE	CAPMODE		
WO	N/A													RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW		RW	RW	RW	RW		
0	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CHANNEL[CR] [31:0]

Bits	Name	Description
31	CNTUPT	1- update counter to new value as CNTUPTVAL This bit will be auto cleared after 1 cycle
17	OPMODE	0: round mode 1: one-shot mode, timer will stopped at reload point.user need clear CEN and set it to start timer agian. NOTE: reload irq will be always set at one-shot mode at end
16	MONITOR_SEL	set to monitor input signal high level time(chan_meas_high) clr to monitor input signal period(chan_meas_prd)
15	MONITOR_EN	set to monitor input signal period or high level time. When this bit is set, if detected period less than val_0 or more than val_1, will set related irq_sts only can be used when trig_mode is selected as measure mode(100) the time may not correct after reload, so monitor is disabled after reload point, and enabled again after two continul posedge. if no posedge after reload for more than val_1, will also assert irq_capt
14	CNTRST	1- reset counter
13	SYNCFLW	1- enable this channel to reset counter to reload(RLD) together with its previous channel. This bit is not valid for channel 0.

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Bits	Name	Description
12	SYNCIFEN	1- SYNCI is valid on its falling edge
11	SYNCIREN	1- SYNCI is valid on its rising edge
10	CEN	1- counter enable
9	CMPINIT	Output compare initial poliarity 1- The channel output initial level is high 0- The channel output initial level is low User should set this bit before set CMPEN to 1.
8	CMPEN	1- Enable the channel output compare function. The output signal can be generated per comparator (CMPx) settings.
7-6	DMASEL	select one of DMA request: 00- CMP0 flag 01- CMP1 flag 10- Input signal toggle captured 11- RLD flag, counter reload;
5	DMAEN	1- enable dma
4	SWSYNCIEN	1- enable software sync. When this bit is set, counter will reset to RLD when swsynct bit is set
3	DBGPAUSE	1- counter will pause if chip is in debug mode
2-0	CAPMODE	This bitfield define the input capture mode 100: width measure mode, timer will calculate the input signal period and duty cycle 011: capture at both rising edge and falling edge 010: capture at falling edge 001: capture at rising edge 000: No capture

CHANNEL[CR] Bits

43.3.3 CHANNEL[CMP] (0x4 + 0x40 * n + 0x4 * m)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CMP																															
RW																															
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0

CHANNEL[CMP] [31:0]

Bits	Name	Description
31-0	CMP	compare value 0

CHANNEL[CMP] Bits

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43.3.4 CHANNEL[RLD] (0xC + 0x40 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RLD																															
RW																															
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

CHANNEL[RLD] [31:0]

Bits	Name	Description
31-0	RLD	reload value

CHANNEL[RLD] Bits

43.3.5 CHANNEL[CNTUPTVAL] (0x10 + 0x40 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNTUPTVAL																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CHANNEL[CNTUPTVAL] [31:0]

Bits	Name	Description
31-0	CNTUPTVAL	counter will be set to this value when software write cntupt bit in CR

CHANNEL[CNTUPTVAL] Bits

43.3.6 CHANNEL[CAPPOS] (0x20 + 0x40 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CAPPOS																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CHANNEL[CAPPOS] [31:0]

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Bits	Name	Description
31-0	CAPPOS	This register contains the counter value captured at input signal rising edge

CHANNEL[CAPPOS] Bits

43.3.7 CHANNEL[CAPNEG] (0x24 + 0x40 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CAPNEG																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CHANNEL[CAPNEG] [31:0]

Bits	Name	Description
31-0	CAPNEG	This register contains the counter value captured at input signal falling edge

CHANNEL[CAPNEG] Bits

43.3.8 CHANNEL[CAPPRD] (0x28 + 0x40 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CAPPRD																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CHANNEL[CAPPRD] [31:0]

Bits	Name	Description
31-0	CAPPRD	This register contains the input signal period when channel is configured to input capture measure mode.

CHANNEL[CAPPRD] Bits

43.3.9 CHANNEL[CAPDTY] (0x2C + 0x40 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MEAS_HIGH																															

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CHANNEL[CAPDTY] [31:0]

Bits	Name	Description
31-0	MEAS_HIGH	This register contains the input signal duty cycle when channel is configured to input capture measure mode.

CHANNEL[CAPDTY] Bits

43.3.10 CHANNEL[CNT] (0x30 + 0x40 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
COUNTER																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CHANNEL[CNT] [31:0]

Bits	Name	Description
31-0	COUNTER	32 bit counter value

CHANNEL[CNT] Bits

43.3.11 SR (0x200)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																CH3CMP1F	CH3CMP0F	CH3CAPF	CH3RLDF	CH2CMP1F	CH2CMP0F	CH2CAPF	CH2RLDF	CH1CMP1F	CH1CMP0F	CH1CAPF	CH1RLDF	CH0CMP1F	CH0CMP0F	CH0CAPF	CH0RLDF
N/A																W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C	W1C
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

SR [31:0]

Bits	Name	Description
15	CH3CMP1F	channel 3 compare value 1 match flag
14	CH3CMP0F	channel 3 compare value 1 match flag
13	CH3CAPF	channel 3 capture flag, the flag will be set at the valid capture edge per CAPMODE setting. If the capture channel is set to measure mode, the flag will be set at rising edge.

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Bits	Name	Description
12	CH3RLDF	channel 3 counter reload flag
11	CH2CMP1F	channel 2 compare value 1 match flag
10	CH2CMP0F	channel 2 compare value 1 match flag
9	CH2CAPF	channel 2 capture flag, the flag will be set at the valid capture edge per CAPMODE setting. If the capture channel is set to measure mode, the flag will be set at rising edge.
8	CH2RLDF	channel 2 counter reload flag
7	CH1CMP1F	channel 1 compare value 1 match flag
6	CH1CMP0F	channel 1 compare value 1 match flag
5	CH1CAPF	channel 1 capture flag, the flag will be set at the valid capture edge per CAPMODE setting. If the capture channel is set to measure mode, the flag will be set at rising edge.
4	CH1RLDF	channel 1 counter reload flag
3	CH0CMP1F	channel 1 compare value 1 match flag
2	CH0CMP0F	channel 1 compare value 1 match flag
1	CH0CAPF	channel 1 capture flag, the flag will be set at the valid capture edge per CAPMODE setting. If the capture channel is set to measure mode, the flag will be set at rising edge.
0	CH0RLDF	channel 1 counter reload flag

SR Bits

43.3.12 IRQEN (0x204)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																CH3CMP1EN	CH3CMP0EN	CH3CAPEN	CH3RLDEN	CH2CMP1EN	CH2CMP0EN	CH2CAPEN	CH2RLDEN	CH1CMP1EN	CH1CMP0EN	CH1CAPEN	CH1RLDEN	CH0CMP1EN	CH0CMP0EN	CH0CAPEN	CH0RLDEN
N/A																RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

IRQEN [31:0]

Bits	Name	Description
15	CH3CMP1EN	1- generate interrupt request when ch3cmp1f flag is set
14	CH3CMP0EN	1- generate interrupt request when ch3cmp0f flag is set
13	CH3CAPEN	1- generate interrupt request when ch3capf flag is set
12	CH3RLDEN	1- generate interrupt request when ch3rldf flag is set
11	CH2CMP1EN	1- generate interrupt request when ch2cmp1f flag is set
10	CH2CMP0EN	1- generate interrupt request when ch2cmp0f flag is set
9	CH2CAPEN	1- generate interrupt request when ch2capf flag is set
8	CH2RLDEN	1- generate interrupt request when ch2rldf flag is set

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Bits	Name	Description
7	CH1CMP1EN	1- generate interrupt request when ch1cmp1f flag is set
6	CH1CMP0EN	1- generate interrupt request when ch1cmp0f flag is set
5	CH1CAPEN	1- generate interrupt request when ch1capf flag is set
4	CH1RLDEN	1- generate interrupt request when ch1rldf flag is set
3	CH0CMP1EN	1- generate interrupt request when ch0cmp1f flag is set
2	CH0CMP0EN	1- generate interrupt request when ch0cmp0f flag is set
1	CH0CAPEN	1- generate interrupt request when ch0capf flag is set
0	CH0RLDEN	1- generate interrupt request when ch0rldf flag is set

IRQEN Bits

43.3.13 GCR (0x208)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																												SWSYNCT			
N/A																												W1C			
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0

GCR [31:0]

Bits	Name	Description
3-0	SWSYNCT	set this bitfield to trigger software counter sync event

GCR Bits

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44 Watchdog EWDG

This chapter introduces the functions and features of the watchdog EWDG.

44.1 Feature Summary

The main features of the watchdog EWDG include:

- Supports feeding the watchdog within a specific time window
- Supports unlock protection before watchdog feeding, with multiple configurable unlock modes
- Supports locking of key control registers, which require unlocking before being updated
- Configurable watchdog operating clock source, can still operate under 32K clock when the bus clock stops
- Key control registers are equipped with parity check function
- The watchdog can be configured to automatically stop or continue working when the chip enters debug mode
- The watchdog can be configured to automatically stop or continue working when the chip enters low-power mode

44.2 EWDG Architecture Diagram

The block diagram of the watchdog is shown in Figure 63.

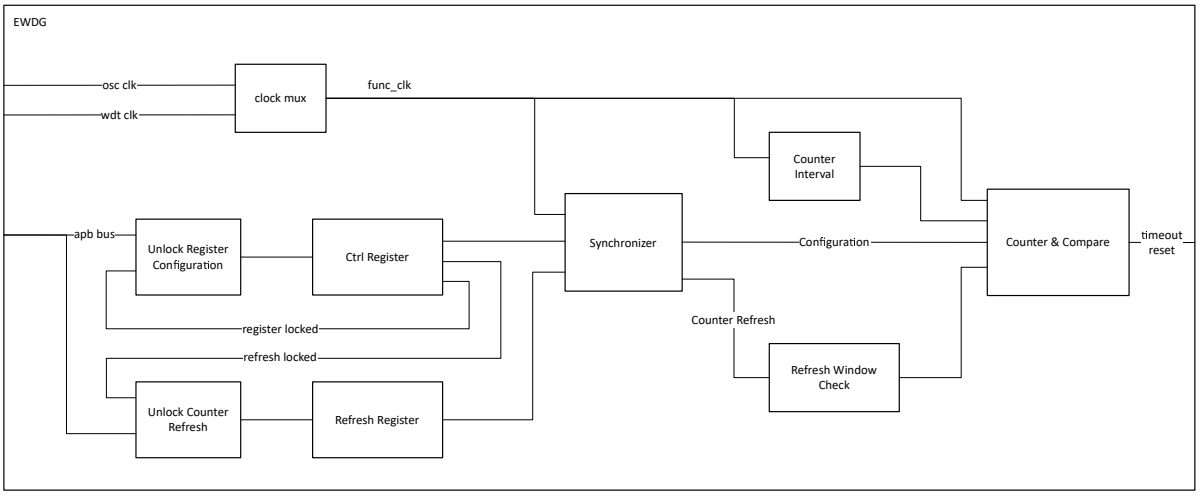


Figure 63: Watchdog Block Diagram

44.3 Functional Description

44.3.1 Function Overview

EWDG has a built-in 16-bit timer, whose value increases after a configurable number of operating clock cycles (up to 32). If no watchdog feeding is performed within the specified time, a reset operation will be triggered when the timer value exceeds the value specified in the reset value register. The watchdog feeding operation is to write a specific value 32'h5A45_524F to the watchdog feeding register.

For security considerations, EWDG can require separate unlock operations before modifying key registers

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and performing watchdog feeding. Unlock failure or no unlock operation will generate a violation, triggering corresponding interrupt or reset. In addition, window mode is supported: watchdog feeding is only allowed within a specific time window, and feeding the watchdog outside the window will generate a violation and trigger interrupt or reset.

The reset signal generated by EWDG can also be output to IO pins to reset other devices in the system at the same time. Please refer to the PINMUX configuration information to confirm whether this function is supported in the current project.

44.3.2 Clock Selection

EWDG has three clock inputs, namely bus configuration clock, external operating clock 1 and external operating clock 2. The bus operating clock is used for EWDG register configuration and watchdog feeding operations. External operating clock 1 and external operating clock 2 serve as the main operating clocks of EWDG, which can be selected through register configuration.

> In general, SoC connects external operating clock 1 to the bus configuration clock to realize fast EWDG response, and connects external operating clock 2 to 32K clock to realize longer watchdog feeding interval and keep EWDG operating when the bus clock stops.

The timer of EWDG can be configured to increment after a certain number of operating clock cycles. It will be cleared to 0 after receiving a valid watchdog feeding operation, and generate a reset operation when the counter reaches the pre-configured timeout value.

The calculation formula of EWDG timeout period is:

$$T_timeout = T_clk_period * (1 << counter_interval) * counter_rst_value$$

44.3.3 Register Configuration Unlock and Parity Check Requirements

EWDG control register 0 and configuration register 1 are equipped with parity check. When writing to configuration register 0 and configuration register 1, the check bit should be filled in BIT31 of the corresponding register to ensure that the number of 1s in the register (including the check bit) is always even. Otherwise, the register configuration will fail, and an interrupt or reset operation will be generated.

After register protection enable is enabled, once EWDG starts working by configuring the EWDG enable register, the register protection function is activated. Subsequent updates to control register 0, configuration register 1, EWDG reset value and other registers require an unlock operation first. Registers in the protected state can only be modified after correct unlock. Updating registers without correct unlock will cause update failure and trigger interrupt or reset operation.

After the register protection function takes effect, you can unlock the register update operation by writing the correct password to the configuration protection unlock register. The password should be written to the configuration protection unlock register before the register protection enable is turned on.

> Note: Once the register protection enable bit in control register 0 is turned on, the configuration protection unlock register cannot be modified any more, so modification of this register should be completed before the register protection enable bit in control register 0 is enabled.

After the register configuration protection is unlocked, you can modify EWDG control register 0/1, reset value and other registers through software. EWDG will automatically re-enable the register configuration protection

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function under the following circumstances:

- If EWDG is currently in stopped state, it will automatically enter the protection state after EWDG restarts
- If EWDG is currently in operating state, it will automatically enter the protection state after a configurable number of bus clock cycles

44.3.4 Watchdog Feeding Unlock Operation

EWDG supports watchdog feeding protection function. If this function is enabled, you need to unlock the watchdog feeding operation before each feeding, otherwise the current feeding will fail and trigger corresponding interrupt or reset function.

44.3.4.1 Watchdog Feeding Protection Enable

If the watchdog feeding protection enable bit in EWDG control register 0 is configured, EWDG will enter watchdog feeding protection mode after it starts working.

44.3.4.2 Unlock Watchdog Feeding Operation

After EWDG watchdog feeding protection takes effect, you need to perform an unlock operation before feeding the watchdog. The unlock operation can be configured as one of the following modes:

- Fixed configuration password: Write the initial password to the watchdog feeding protection unlock register in the initial stage, and then write the same password to this register to unlock
- Circular left shift password: Write the initial password to the watchdog feeding protection unlock register in the initial stage, then write the circular left shift result of the password to this register to unlock, and the password will be automatically updated to the circular left shift result
- LSFR password: Write the initial password to the watchdog feeding protection unlock register in the initial stage, then write the LSFR result calculated according to the formula $X^{15} + 1$ to this register to unlock, and the password will be updated to this LSFR result
- Fixed value unlock: No initial password configuration is required, write the fixed value 16'h55AA to the watchdog feeding protection unlock register during the unlock phase to unlock

Note: The watchdog feeding protection unlock register can only be configured before the watchdog feeding protection enable bit in control register 0 is enabled. Once the watchdog feeding protection enable bit is turned on, the value of the watchdog feeding protection unlock register cannot be modified.

After each watchdog feeding protection is unlocked, EWDG will automatically re-enable the protection function under the following circumstances:

- The protection function will be re-enabled immediately after the watchdog feeding operation is completed, and re-unlocking is required before the next feeding
- If the unlock operation is not completed within a configurable specific time after unlocking, the current unlock will time out, the system will re-enter the watchdog feeding protection state, and a watchdog feeding violation flag will be generated at the same time

44.3.5 Window Mode

EWDG can be configured to window mode. When window mode is enabled, EWDG only allows watchdog feeding operations within a specific timer window. Feeding the watchdog outside the window time will lead to

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feeding failure and trigger corresponding interrupt or reset operation.

The lower limit of the window can be configured as 4/8, 5/8, 6/8 or 7/8 of the watchdog reset value register. The upper limit of the window is calculated by dividing the reset value register into sixteen equal parts, and adding several one-sixteenth of the reset value register to the lower limit.

$$window_lower = reset_counter_value * (1/2 + win_lower_reg * 1/8)$$

$$window_upper = window_lower + reset_counter_value/16 * win_upper_reg$$

Note: If the upper limit register is not configured, or the upper limit exceeds the value of the reset value register, the value of the reset value register will be used as the upper limit of watchdog feeding, and you only need to complete the feeding operation before the timeout reset.

44.3.6 Working State in Debug Mode

EWDG pauses working by default when the chip is in debug mode, and resumes working after the chip exits debug mode. You can configure EWDG to work normally in debug mode through registers.

44.3.7 Working State in Low-Power Mode

EWDG pauses working by default when the chip is in low-power mode, and resumes working after the chip exits low-power mode. You can configure EWDG to work normally in low-power mode through registers.

44.4 EWDG Register List

The EWDG register list is as follows:

EWDG0 base address: 0xF00B0000

EWDG1 base address: 0xF00B4000

PEWDG base address: 0xF4128000

Offset	Register Name	Description	Reset Value
0x0000	CTRL0	wdog ctrl register 0 Note: Parity check is required once writing to this register. The result should be zero by modular two addition of all bits	0x00000000
0x0004	CTRL1	wdog ctrl register 1 Note: Parity check is required once writing to this register. The result should be zero by modular two addition of all bits	0x00000000
0x0008	OT_INT_VAL	wdog timeout interrupt counter value	0x00000000

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Offset	Register Name	Description	Reset Value
0x000C	OT_RST_VAL	wdog timeout reset counter value	0x00000000
0x0010	WDT_REFRESH_REG	wdog refresh register	0x00000000
0x0014	WDT_STATUS	wdog status register	0x00000000
0x0018	CFG_PROT	ctrl register protection register	0x00000000
0x001C	REF_PROT	refresh protection register	0x00000000
0x0020	WDT_EN	Wdog enable	0x00000000
0x0024	REF_TIME	Refresh period value	0x00000000

Table 198: EWDG Register List

44.5 EWDG Register Description

The detailed description of EWDG registers is as follows:

44.5.1 CTRL0 (0x0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD		CLK_SEL	RSVD	DIV_VALUE			WIN_EN	WIN_LOWER		CFG_LOCK	RSVD					REF_OT_REQ	WIN_UPPER			RSVD						REF_LOCK	REF_UNLOCK_MEC		EN_DBG	EN_LP	
N/A		RW	N/A	RW			RW	RW		RW	N/A					RW	RW			N/A						RW	RW		RW	RW	
x	x	0	x	0	0	0	0	0	0	0	x	x	x	x	x	0	0	0	0	x	x	x	x	x	x	0	0	0	0	0	0

CTRL0 [31:0]

Bits	Name	Description
29	CLK_SEL	clock select 0：bus clock 1：ext clock
27-25	DIV_VALUE	clock divider, the clock divider works as 2 ^ div_value for wdt counter
24	WIN_EN	window mode enable
23-22	WIN_LOWER	Once window mode is opened, the lower counter value to refresh wdt 00: 4/8 overtime value 01: 5/8 of overtime value 10: 6/8 of overtime value 11: 7/8 of overtime value

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Bits	Name	Description
21	CFG_LOCK	The register is locked and unlock is needed before re-config registers Once the lock mechanism takes effect, the CTRL0, CTRL1, timeout int register, timeout rst register, needs unlock before re-config them. The register update needs to be finished in the required period defined by UPD_OT_TIME register
15	REF_OT_REQ	If refresh event has to be limited into a period after refresh unlocked. Note: the refresh overtime counter works in bus clock domain, not in wdt function clock domain. The wdt divider doesn't take effect for refresh counter
14-12	WIN_UPPER	The upper threshold of window value The window period upper limit is: lower_limit + (overtime_rst_value / 16) * upper_reg_value If this register value is zero, then no upper level limitation
5	REF_LOCK	WDT refresh has to be unlocked firstly once refresh lock is enable.
4-3	REF_UNLOCK_MEC	Unlock refresh mechanism 00: the required unlock password is the same with refresh_psd_register 01: the required unlock password is a ring shift left value of refresh_psd_register 10: the required unlock password is always 16'h55AA, no matter what refresh_psd_register is 11: the required unlock password is a LSFR result of refresh_psd_register, the characteristic polynomial is X^15 + 1
2	EN_DBG	WTD enable or not in debug mode
1-0	EN_LP	WDT enable or not in low power mode 2'b00: wdt is halted once in low power mode 2'b01: wdt will work with 1/4 normal clock freq in low power mode 2'b10: wdt will work with 1/2 normal clock freq in low power mode 2'b11: wdt will work with normal clock freq in low power mode

CTRL0 Bits

44.5.2 CTRL1 (0x4)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD								REF_FAIL_RST_EN	REF_FAIL_INT_EN	UNL_REF_FAIL_RST_EN	UNL_REF_FAIL_INT_EN	RSVD		OT_RST_EN	OT_INT_EN	RSVD								CTL_VIO_RST_EN	CTL_VIO_INT_EN	UNL_CTL_FAIL_RST_EN	UNL_CTL_FAIL_INT_EN	PARITY_FAIL_RST_EN	PARITY_FAIL_INT_EN	RSVD	
N/A								RW	RW	RW	RW	N/A		RW	RW	N/A								RW	RW	RW	RW	RW	RW	N/A	
x	x	x	x	x	x	x	x	0	0	0	0	x	x	0	0	x	x	x	x	x	x	x	x	0	0	0	0	0	0	x	x

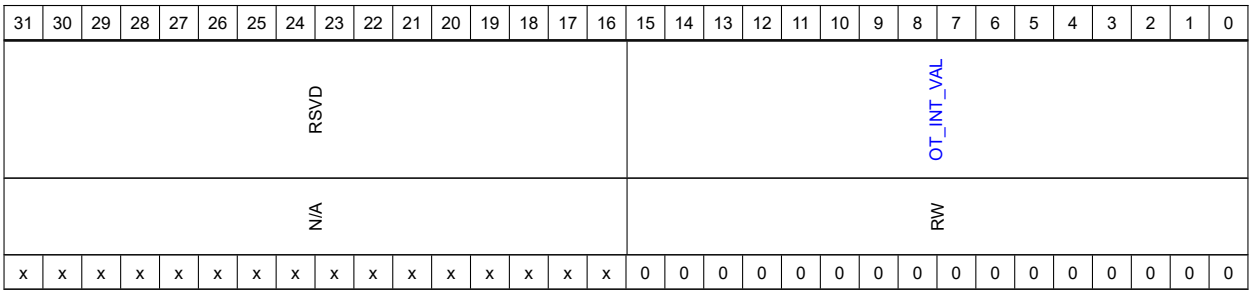
CTRL1 [31:0]

Bits	Name	Description
23	REF_FAIL_RST_EN	Refresh violation will trigger an reset. These event will be taken as a refresh violation: 1) Not refresh in the window once window mode is enabled 2) Not unlock refresh firstly if unlock is required 3) Not refresh in the required time after unlock, once refresh unlock overtime is enabled. 4) Not write the required word to refresh wdt.
22	REF_FAIL_INT_EN	Refresh violation will trigger an interrupt
21	UNL_REF_FAIL_RST_EN	Refresh unlock fail will trigger a reset
20	UNL_REF_FAIL_INT_EN	Refresh unlock fail will trigger a interrupt
17	OT_RST_EN	WDT overtime will generate a reset
16	OT_INT_EN	WDT can generate an interrupt warning before timeout
7	CTL_VIO_RST_EN	Ctrl update violation will trigger a reset The violation event is to try updating the locked register before unlock them
6	CTL_VIO_INT_EN	Ctrl update violation will trigger a interrupt
5	UNL_CTL_FAIL_RST_EN	Unlock register update failure will trigger a reset
4	UNL_CTL_FAIL_INT_EN	Unlock register update failure will trigger a interrupt
3	PARITY_FAIL_RST_EN	Parity error will trigger a reset A parity check is required once writing to ctrl0 and ctrl1 register. The result should be zero by modular two addition of all bits
2	PARITY_FAIL_INT_EN	Parity error will trigger a interrupt

CTRL1 Bits

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44.5.3 OT_INT_VAL (0x8)

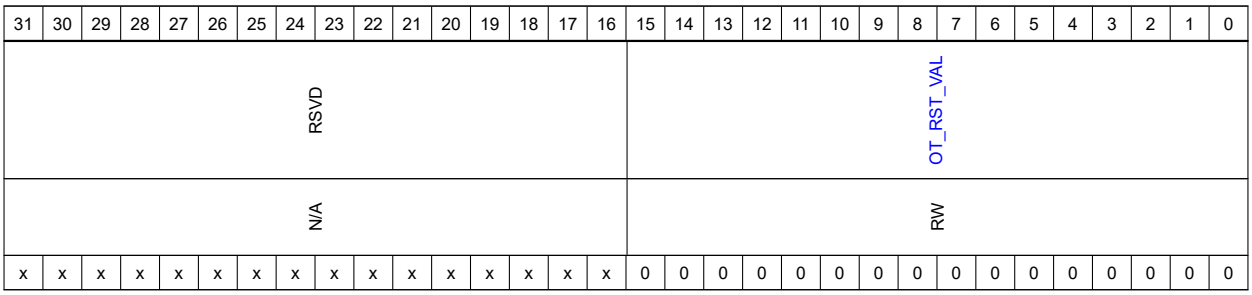


OT_INT_VAL [31:0]

Bits	Name	Description
15-0	OT_INT_VAL	WDT timeout interrupt value

OT_INT_VAL Bits

44.5.4 OT_RST_VAL (0xC)

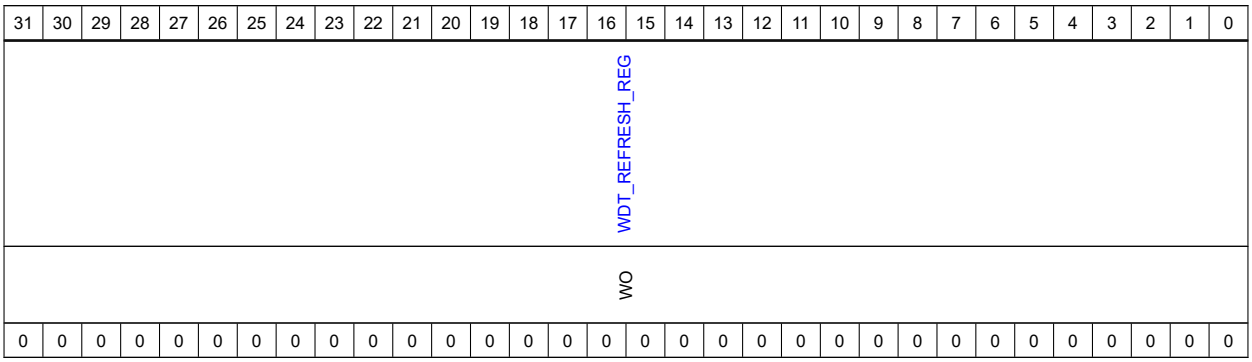


OT_RST_VAL [31:0]

Bits	Name	Description
15-0	OT_RST_VAL	WDT timeout reset value

OT_RST_VAL Bits

44.5.5 WDT_REFRESH_REG (0x10)



WDT_REFRESH_REG [31:0]

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Bits	Name	Description
31-0	WDT_REFRESH_REG	Write this register by 32'h5A45_524F to refresh wdog Note: Reading this register can read back wdt real time counter value, while it is only used by debug purpose

WDT_REFRESH_REG Bits

44.5.6 WDT_STATUS (0x14)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																									PARITY_ERROR	OT_RST	OT_INT	CTL_UNL_FAIL	CTL_VIO	REF_UNL_FAIL	REF_VIO
N/A																									W1C	W1C	W1C	W1C	W1C	W1C	W1C
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0

WDT_STATUS [31:0]

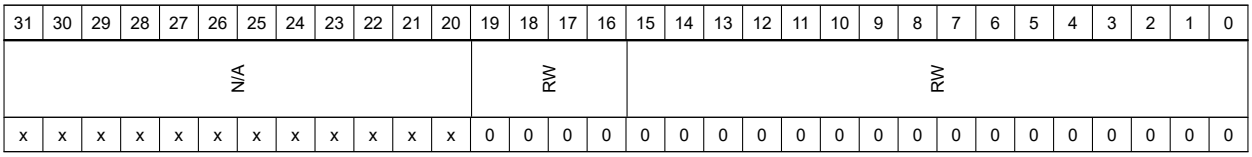
Bits	Name	Description
6	PARITY_ERROR	parity error Write one to clear the bit
5	OT_RST	Timeout happens, a reset will happen once enable bit set This bit can be cleared only by refreshing wdt or reset
4	OT_INT	Timeout happens, a interrupt will happen once enable bit set This bit can be cleared only by refreshing wdt or reset
3	CTL_UNL_FAIL	Unlock ctrl reg update protection fail Write one to clear the bit
2	CTL_VIO	Violate register update protection mechanism Write one to clear the bit
1	REF_UNL_FAIL	Refresh unlock fail Write one to clear the bit
0	REF_VIO	Refresh fail Write one to clear the bit

WDT_STATUS Bits

44.5.7 CFG_PROT (0x18)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD												UPD_OT_TIME				UPD_PSD															

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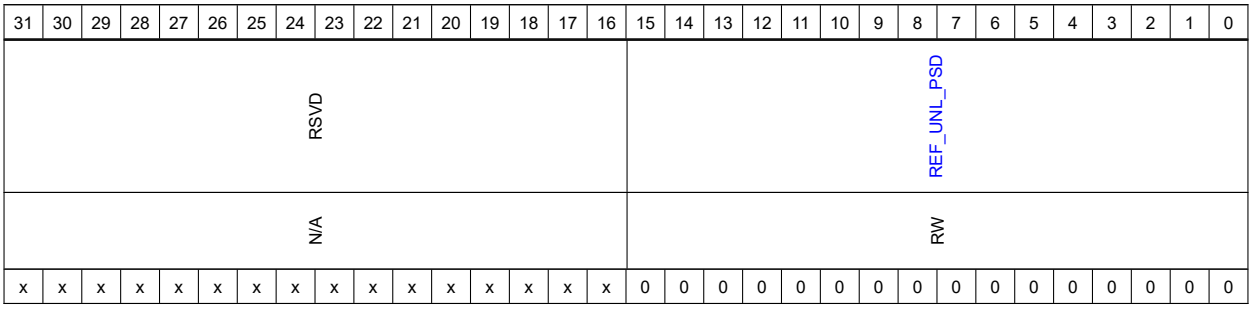


CFG_PROT [31:0]

Bits	Name	Description
19-16	UPD_OT_TIME	The period in which register update has to be in after unlock The required period is less than：128 * 2 ^ UPD_OT_TIME * bus_clock_cycle
15-0	UPD_PSD	The password of unlocking register update

CFG_PROT Bits

44.5.8 REF_PROT (0x1C)

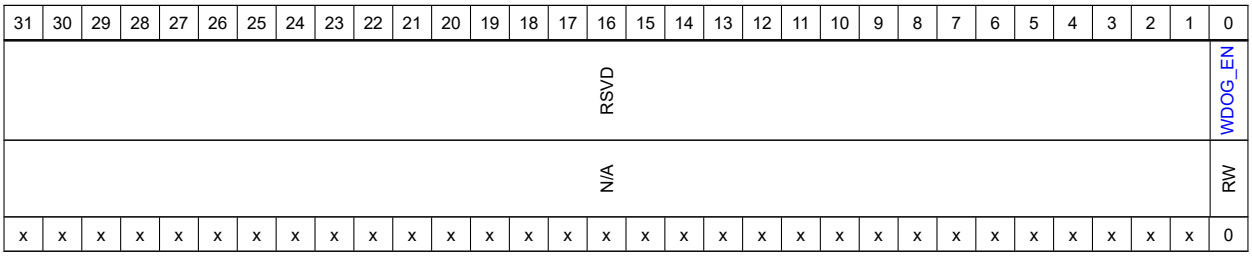


REF_PROT [31:0]

Bits	Name	Description
15-0	REF_UNL_PSD	The password to unlock refreshing

REF_PROT Bits

44.5.9 WDT_EN (0x20)



WDT_EN [31:0]

Bits	Name	Description
0	WDOG_EN	Wdog is enabled, the re-written of this register is impacted by enable lock function

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Bits	Name	Description

WDT_EN Bits

44.5.10 REF_TIME (0x24)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																REFRESH_PERIOD															
N/A																RW															
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

REF_TIME [31:0]

Bits	Name	Description
15-0	REFRESH_PERIOD	The refresh period after refresh unlocked Note: the refresh overtime counter works in bus clock domain, not in wdt function clock domain. The wdt divider doesn't take effect for refresh counter

REF_TIME Bits

44.6 Watchdog Configuration and Usage Instructions

44.6.1 Configuration Reference: Window Working Mode, Watchdog Feeding Protection Enabled

Requirements: EWDG selects operating clock 2 (OSC32K), works in window mode, allows watchdog feeding in the 4/8 to 7/8 phase of the timer, and requires unlock before feeding, with the unlock password adopting circular left shift mode.

Reference configuration:

- Configure the watchdog feeding protection register (REF_PROT), write the initial password PSD. If you expect to complete the feeding operation within 512 bus clock cycles after unlocking, configure the watchdog feeding time limit register (REF_TIME) to 512.
- Configure the watchdog feeding protection mode unlock method (REF_UNLOCK_MEC) of the control register to 01, select circular left shift mode. Configure the watchdog feeding protection mode enable (REF_LOCK).
- Configure CLK_SEL of control register 0 to 1, select OSC 32K.
- Configure WIN_EN of control register 0 to enable window mode, set the lower limit WIN_LOWER to 00 (select 4/8 of the timer timeout period), set the upper limit to be 3/8 of the timeout period higher than the lower limit, so configure WIN_UPPER to 6 (6/16 + lower limit value).

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- If you expect EWDG to generate a reset signal after timeout, configure the timeout reset enable (OT_RST_EN) of control register 1. If you want to generate an interrupt when the watchdog is not fed as required, configure the watchdog feeding operation violation interrupt enable (REF_FAIL_INT_EN) and watchdog feeding protection unlock failure interrupt enable (UNL_REF_FAIL_INT_EN) of control register 1, so that an interrupt will be generated when feeding fails or feeding unlock fails.
- Note that when updating configuration register 0 and configuration register 1, ensure that the parity check passes, i.e. the number of bits with value 1 is even.
- Configure the enable register WDT_EN, write 1 to WDOG_EN, and the watchdog starts working.
- When the timer enters the allowed feeding window and you are ready to feed the watchdog, first unlock the feeding protection: write the circular left shift result of the initial password in the watchdog feeding protection register (REF_PROT) to the REF_PROT register to unlock the feeding operation, and the password in the feeding protection register will be updated to the circular left shift result at the same time.
- Write the reserved word 32'h5A45524F to the watchdog feeding register (WDT_REFRESH_REG) within 512 bus clock cycles to complete the feeding operation.

44.6.2 Configuration Reference: Register Configuration Lock and Unlock

Scenario: After the configuration register is locked, you need to update configuration register 1 to change the unlock failure behavior from interrupt to reset.

- Configure the configuration update time (UPD_OD_TIME) of the configuration protection register to 2, so that the configuration can be completed within 512 cycles ($128 * 2^2$) after configuration unlock. Write the initial password PSD to UPD_PSD.
- Enable the register protection enable (CFG_LOCK) of control register 0.
- After EWDG is enabled (write 1 to the enable register), the configuration protection function is activated. Updating the control register and reset value register without unlock will generate a violation.
- When you need to unlock: first write the password to the configuration protection register to unlock configuration protection. Write 0x0 to the enable register bit to stop EWDG (recommended to prevent unexpected accidents in critical states). Then update control register 1 within 512 system clock cycles to enable the watchdog feeding operation violation reset enable (REF_FAIL_RST_EN), configure the enable register to 0x01 to restart EWDG, and the configuration protection function will be automatically re-enabled after EWDG starts working.

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45 Overview of Communication Peripherals

This chapter introduces the communication peripherals of this product.

45.1 Universal Asynchronous Receiver/Transmitter UART, PUART

This product supports 9 universal asynchronous receivers/transmitters, which support two-way asynchronous communication with external devices.

8 universal asynchronous receivers/transmitters are located in the system power domain, called UART0~7.

1 universal asynchronous receiver/transmitter is located in the power management domain, called PUART. PUART supports maintaining operation when the system power domain is powered down, and the generated interrupt can be used for low-power wake-up. PUART does not support generating DMA requests.

45.2 Serial Peripheral Bus SPI

This product supports 4 serial peripheral buses SPI, which support full-duplex, synchronous communication with external devices. SPI supports the classic unidirectional 2-data-line mode, namely MISO and MOSI, and also supports bidirectional 2-line and 4-line data modes, namely Dual-SPI and Quad-SPI.

45.3 Inter-Integrated Circuit Bus I2C

This product supports 4 inter-integrated circuit buses I2C. I2C supports a maximum of 100 Kbps in standard mode, 400 Kbps in fast mode and 1 Mbps in fast+ mode.

45.4 Controller Area Network MCAN

This product supports 4 controller area networks MCAN0~4. MCAN complies with CAN bus protocol 2.0A and 2.0B, and supports CAN-FD protocol.

On this product, MCAN supports time-triggered CAN communication, and MCAN can directly load timestamp information through the timestamp output port of the Precision Time Protocol module PTPC.

45.5 Precision Time Protocol Module PTPC

This product supports 1 Precision Time Protocol module PTPC. PTPC supports 2 sets of timestamps, and can generate precise timestamp information with resolution of seconds and nanoseconds for clock synchronization between different nodes in the network.

PTPC supports 2 timestamp output ports, which are respectively connected to CAN0~1, and the CAN module can directly load timestamp information from these ports.

PTPC supports 2 timestamp input capture signals CAP0 and CAP1, and supports capturing the current timestamp according to the input capture signals. CAP0 and CAP1 come from the interconnection manager TRGM0~1. PTPC supports outputting 2 timestamp output comparison CMP0 and CMP1 when the timestamp matches. CMP0 and CMP1 are output to the interconnection manager TRGM0~1. Please refer to Section 32.8 for specific connections.

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45.6 Universal Serial Bus USB

This product supports 1 universal serial bus USB. USB integrates HS PHY and supports USB 2.0 low-speed, full-speed and high-speed transmission. USB supports Host, Device and OTG modes.

On this product, the USB controller supports outputting the USB SOF (Start of Frame) signal, which can be connected to other modules on the chip through the interconnection manager TRGM.

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46 Universal Asynchronous Receiver/Transmitter (UART)

This chapter describes the functions and features of the Universal Asynchronous Receiver/Transmitter (UART).

46.1 Feature Summary

This chapter describes the main features of UART:

- Supports 5~9 bits data length and address matching
- Configurable stop bits: 1 bit, 1.5 bits or 2 bits
- Configurable parity bits: odd parity, even parity, sticky parity
- Supports DMA data transmission
- Supports configurable baud rate and independent baud rate generation clock
- Supports hardware flow control
- Supports error detection for parity error, data FIFO overflow, etc.
- 16-byte TXFIFO and RXFIFO
- Supports RX IDLE status flag
- Cooperates with TRIGGER MUX to support hardware-triggered transmission
- Supports various types of interrupts

The block diagram of UART is shown in Figure 64.

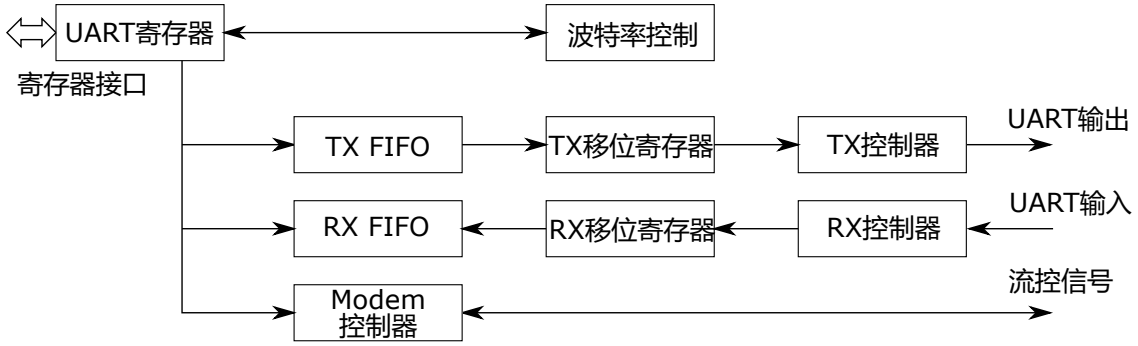


Figure 64: UART Block Diagram

46.2 Function Description

This section describes the functions of UART.

46.2.1 UART Transmission

The UART transmitter consists of TX FIFO, TX shift register and TX controller.

TX FIFO stores the data to be transmitted and transfers the data to the TX shift register.

The TX shift register is a parallel-to-serial converter, which converts the transmitted data into a serial bit stream.

When transmitting a piece of data, the TX controller will generate a START bit, a configurable parity bit (PARITY bit), and a stop bit with configurable length. Users can configure the parity bit and stop bit through the

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LCR register (Line Control Register).

The write entry of TXFIFO is the THR register (Transmitter Holding Register). Users need to set the FIFOE bit of the FCR register (FIFO Control Register) to 1 to enable TXFIFO.

46.2.2 UART Reception

The UART receiver consists of RX FIFO, RX shift register and RX controller.

The RX controller uses the oversampling clock generated by the baud rate control module to sample each input bit, and shifts each received bit into the RX shift register.

The RX shift register performs serial-to-parallel conversion on the data and stores the data in the RX FIFO.

The read entry of RX FIFO is the RBR register (Receiver Buffer Register). Users need to set the FIFOE bit of the FCR register (FIFO Control Register) to 1 to enable RXFIFO.

The RX controller supports detecting errors during data reception, such as parity error, stop bit error, RX FIFO overflow, etc.

46.2.3 Idle Detection

UART idle detection is divided into transmit idle detection and receive idle detection.

Taking receive idle detection as an example: idle detection is used to detect the idle duration of UART, which can be divided into pin idle and internal UART state machine idle, distinguished by the $TX_IDLE_CONDbitofIDLE_CFG.Thisfuncti$

When the idle time reaches the value set by $RX_IDLE_THRofIDLE_CFG(countedinbitunits), aninterruptcanbetriggered.Whenthe$

46.2.4 Address Matching

When address matching is enabled, the highest bit can be used to distinguish between address and data. Take 9-bit mode as an example. For non-9-bit modes, the address is one bit less than the number of characters, that is, in 8-bit mode, the valid address bit is 7 bits, and the highest bit distinguishes between address and data.

During transmission: each character is sent with 9 bits, the 9th bit being 1 indicates address, and 0 indicates data. When transmitting, set the 9th bit of the first character to 1 and send it as an address. The specific address is configured by software.

During reception: in 9-bit mode, check whether the 9th bit of the first received character is 1. If it is 1, address matching will be performed. ADDR0 and ADDR1 are set in $ADDR_CFGformatching, whichcanbeselectedasrequired.$

46.2.5 Baud Rate Control

The baud rate control module divides the UART clock to obtain the baud rate clock. The divider is 16 bits long and is located in two registers, each storing an 8-bit frequency division value. The MSB of the frequency division value is located in the DLM register (Divisor Latch MSB), and the LSB is located in the DLL register (Divisor Latch LSB).

The ratio of UART clock to baud rate is the oversampling rate. The oversampling rate OSC is stored in the OSCR register (Over Sample Control Register). The default value of OSC is 16.

The formula of the frequency division value is shown in (46.2.5):

$$Divisor = \frac{f_{uart_clock}}{Baudrate \times OSR}$$

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The RX controller uses the oversampling clock to sample the input data. Assuming that the oversampling rate OSC is 16, when the first falling edge of the input signal (start START bit) is detected, the counter starts counting from 1 to 16. When the count reaches 8, the RX controller samples the input data. After counting to 16, the counter will reset to 1, so as to sample the next bit of data, and cycle until the stop STOP bit. The TX controller also uses the oversampling clock to generate the output data stream.

46.2.6 Modem Controller

The Modem controller provides Modem control functions and also supports automatic flow control, which can further reduce software control overhead.

The transmit and receive flow control of UART is realized through RTS / CTS handshake. Flow control can avoid overrun errors caused by the data transmission rate exceeding the data processing rate.

UART supports automatic flow control, including auto-RTS and auto-CTS functions. The former is used for data reception, and the latter for data transmission.

When the auto-RTS function is enabled, the RTS output of the UART needs to be connected to the CTS signal of the peer UART.

- N is the trigger threshold of RX FIFO
- Bn is the number of bytes received by RX FIFO
- When $B_n \geq N$, RTS is set low
- When $B_n < N$, RTS will be automatically set high, indicating that the peer UART can send data

When the auto-CTS function is enabled, the CTS input of the UART is connected to the RTS signal of the peer UART. The UART will only send data when CTS detects a low input. CTS must be pulled high before the stop STOP bit of the currently transmitted data is sent, so that the UART will not send the next data.

DE flow control pin: it controls the direction of data, which is suitable for half-duplex UART communication. When the DE pin is at high level, it indicates that it is currently in the data transmission mode, and the external device receives data. The DE pin signal can be used for external UART devices to switch between transmit and receive modes.

46.2.7 Loopback Mode

UART supports Loopback mode.

When Loopback mode is enabled, the TX of UART is connected to RX, and RTS is connected to CTS. The data sent through the THR register can be received from the RBR register.

46.2.8 Interrupt

UART supported interrupts mainly include: receive idle interrupt, transmit idle interrupt, address matching interrupt, data loss interrupt, flow control interrupt, receive status interrupt, transmit status interrupt, receive data valid interrupt, character timeout interrupt, etc.

By configuring the interrupt enable register IER (0x24), UART can support the above interrupts. The specific interrupt types and detailed introduction can be distinguished through IIR2 (0x0C) (Interrupt Identification Register 2). It is recommended to use IIR2 to operate interrupts instead of IIR.

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46.2.9 DMA

UART supports transmit DMA requests and receive DMA requests.

The user configures the trigger threshold RFIFOT of RX FIFO through the FCR register. When the data received by RX FIFO exceeds the threshold, a DMA receive request is generated.

=1 When the user configures TIME_RXDMA_DIS of the FCR register to 1, DMA only moves data when the trigger threshold is reached, and TIMEOUT will not move data.

When TIME_RXDMA_DIS is configured to 0, if the data in RX FIFO does not reach the threshold and there is no new data for 4 bytes, a DMA receive request will also be generated to send the remaining data in FIFO, avoiding the situation that data smaller than the threshold cannot be received.

The user configures the trigger threshold TFIFOT of TX FIFO through the FCR register. When the data to be transmitted in TX FIFO is less than the threshold, a DMA transmit request is generated.

46.3 Register Description

The list of UART registers is as follows:

UART0 base address: 0xF0040000

UART1 base address: 0xF0044000

UART2 base address: 0xF0048000

UART3 base address: 0xF004C000

UART4 base address: 0xF0050000

UART5 base address: 0xF0054000

UART6 base address: 0xF0058000

UART7 base address: 0xF005C000

PUART base address: 0xF4124000

Offset	Register Name	Description	Reset Value
0x0004	IDLE_CFG	Idle Configuration Register	0x00000000
0x0008	ADDR_CFG	address match config register	0x00000000
0x000C	IIR2	Interrupt Identification Register2	0x00000001
0x0010	CFG	Configuration Register	0x00000000
0x0014	OSCR	Over Sample Control Register	0x00000010
0x0018	FCRR	FIFO Control Register config	0x00000000
0x001C	MOTO_CFG	moto system control register	0x00000000
0x0020	RBR	Receiver Buffer Register (when DLAB = 0)	0x00000000
0x0020	THR	Transmitter Holding Register (when DLAB = 0)	0x00000000
0x0020	DLL	Divisor Latch LSB (when DLAB = 1)	0x00000001

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Offset	Register Name	Description	Reset Value
0x0024	IER	Interrupt Enable Register (when DLAB = 0)	0x00000000
0x0024	DLM	Divisor Latch MSB (when DLAB = 1)	0x00000000
0x0028	IIR	Interrupt Identification Register	0x00000001
0x0028	FCR	FIFO Control Register	0x00000000
0x002C	LCR	Line Control Register	0x00000000
0x0030	MCR	Modem Control Register (	0x00000000
0x0034	LSR	Line Status Register	0x00000000
0x0038	MSR	Modem Status Register	0x00000000
0x003C	GPR	GPR Register	0x00000000

Table 199: UART Register List

46.4 Detailed Register Information

The detailed description of UART registers is as follows:

46.4.1 IDLE_CFG (0x4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD						TX_IDLE_COND	TX_IDLE_EN	TX_IDLE_THR								RSVD				RXEN	RSVD	RX_IDLE_COND	RX_IDLE_EN	RX_IDLE_THR							
N/A						RW	RW	RW								N/A				RW	N/A	RW	RW	RW							
x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	x	x	x	x	0	x	0	0	0	0	0	0	0	0	0	0

IDLE_CFG [31:0]

Bits	Name	Description
25	TX_IDLE_COND	IDLE Detection Condition 0 - Treat as idle if TX pin is logic one 1 - Treat as idle if UART state machine state is idle
24	TX_IDLE_EN	UART TX Idle Detect Enable 0 - Disable 1 - Enable
23-16	TX_IDLE_THR	Threshold for UART transmit Idle detection (in terms of bits)
11	RXEN	UART receive enable. 0 - hold RX input to high, avoide wrong data input when config pinmux 1 - bypass RX input from PIN software should set it after config pinmux

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Bits	Name	Description
9	RX_IDLE_COND	IDLE Detection Condition 0 - Treat as idle if RX pin is logic one 1 - Treat as idle if UART state machine state is idle
8	RX_IDLE_EN	UART Idle Detect Enable 0 - Disable 1 - Enable it should be enabled if enable address match feature
7-0	RX_IDLE_THR	Threshold for UART Receive Idle detection (in terms of bits)

IDLE_CFG Bits

46.4.2 ADDR_CFG (0x8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD											TXEN_9BIT	RXEN_ADDR_MSB	RXEN_9BIT	A1_EN	A0_EN	ADDR1								ADDR0								
N/A											RW	RW	RW	RW	RW	RW								RW								
x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ADDR_CFG [31:0]

Bits	Name	Description
20	TXEN_9BIT	set to use 9bit mode for transmitter, will set the MSB for the first character as address flag, keep 0 for others.
19	RXEN_ADDR_MSB	set to use MSB as address flag at receiver(actually this is done by software set correct MSB in addr0/addr1). Clr to use first character as address. Only needed if enable address match feature
18	RXEN_9BIT	set to use 9bit mode for receiver, only valid if rxen_addr_msb is set
17	A1_EN	enable addr1 compare for the first character. If a1_en OR a0_en, then do not receive data if address not match. If ~a1_en AND ~a0_en, the receive all data like before. NOTE: should set idle_tmout_en if enable address match feature
16	A0_EN	enable addr0 compare for the first character

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Bits	Name	Description
15-8	ADDR1	address 1 fileld. in 9bit mode, this is the full address byte. For other mode(8/7/6/5bit), MSB should be set for address flag. If want address==0 to be matched at 8bit mode, should set addr1=0x80
7-0	ADDR0	address 0 field.

ADDR_CFG Bits

46.4.3 IIR2 (0xC)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RXIDLE_FLAG	TXIDLE_FLAG	ADDR_MATCH	ADDR_MATCH_IDLE	DATA_LOST	RSVD																			FIFOED		RSVD		INTRID			
W1C	W1C	W1C	W1C	W1C	N/A																			RO		N/A		RO			
0	0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	x	x	0	0	0	1

IIR2 [31:0]

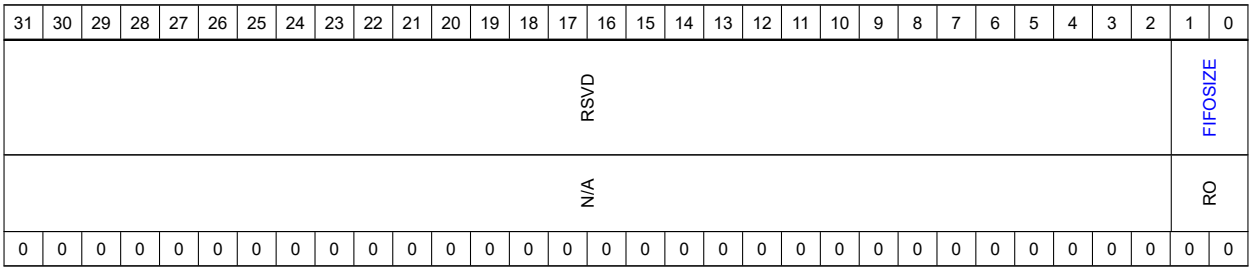
Bits	Name	Description
31	RXIDLE_FLAG	UART RX IDLE Flag, assert after rxd high and then rx idle timeout, write one clear 0 - UART RX is busy 1 - UART RX is idle
30	TXIDLE_FLAG	UART TX IDLE Flag, assert after txd high and then tx idle timeout, write one clear 0 - UART TX is busy 1 - UART TX is idle
29	ADDR_MATCH	address match irq status, assert if either address match(and enabled). Write one clear NOTE: the address byte may not moved by DMA at this point. User can wait next addr_match_idle irq for the whole data include address
28	ADDR_MATCH_IDLE	address match and idle irq status, assert at rx bus idle if address match event triggered. Write one clear;
27	DATA_LOST	assert if data lost before address match status, write one clear; It will not assert if no address match occurs

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Bits	Name	Description
7-6	FIFOED	FIFOs enabled These two bits are 1 when bit 0 of the FIFO Control Register (FIFOE) is set to 1.
3-0	INTRID	Interrupt ID, see IIR2 for detail decoding

IIR2 Bits

46.4.4 CFG (0x10)

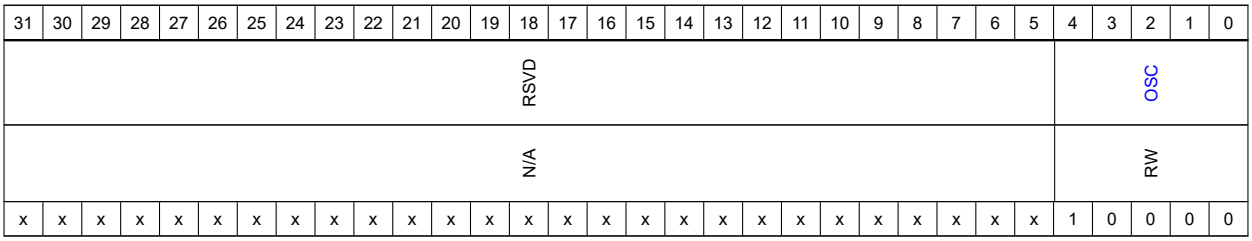


CFG [31:0]

Bits	Name	Description
1-0	FIFOSIZE	The depth of RXFIFO and TXFIFO 0: 16-byte FIFO 1: 32-byte FIFO 2: 64-byte FIFO 3: 128-byte FIFO

CFG Bits

46.4.5 OSCR (0x14)



OSCR [31:0]

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Bits	Name	Description
4-0	OSC	Over-sample control The value must be an even number; any odd value writes to this field will be converted to an even value. OSC=0: reserved OSC<=8: The over-sample ratio is 8 8 < OSC< 32: The over sample ratio is OSC

OSCR Bits

46.4.6 FCRR (0x18)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD								FIFOT4EN	RSVD		TFIFOT4					RSVD			RFIFOT4					RFIFOT		TFIFOT		DMAE	TFIFORST	RFIFORST	FIFOE
N/A								RW	N/A		RW					N/A			RW					RW		RW		RW	WO	WO	RW
x	x	x	x	x	x	x	x	0	x	x	0	0	0	0	0	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0

FCRR [31:0]

Bits	Name	Description
23	FIFOT4EN	set to use new 4bit fifo threshold(TFIFOT4 and RFIFOT4) clr to use 2bit(TFIFOT and RFIFOT)
20-16	TFIFOT4	txfifo threshold(0 for 1byte, 0xF for 32bytes), uart will send tx_dma_req when data in fifo is less than threshold.
12-8	RFIFOT4	rxfifo threshold(0 for 1byte, 0xF for 32bytes). Uart will send rx_dma_req if data in fifo reachs the threshold, also will set the rxdata irq if enabled
7-6	RFIFOT	Receiver FIFO trigger level
5-4	TFIFOT	Transmitter FIFO trigger level
3	DMAE	DMA enable 0: Disable 1: Enable
2	TFIFORST	Transmitter FIFO reset Write 1 to clear all bytes in the TXFIFO and resets its counter. The Transmitter Shift Register is not cleared. This bit will automatically be cleared.
1	RFIFORST	Receiver FIFO reset Write 1 to clear all bytes in the RXFIFO and resets its counter. The Receiver Shift Register is not cleared. This bit will automatically be cleared.

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Bits	Name	Description
0	FIFOE	FIFO enable Write 1 to enable both the transmitter and receiver FIFOs. The FIFOs are reset when the value of this bit toggles.

FCRR Bits

46.4.7 MOTO_CFG (0x1C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SWTRG	RSVD															TXSTP_BITS								HWTRG_EN	TRG_MODE	TRG_CLR_RFIFO	TXSTOP_INSERT	RSVD			
WO	N/A															RW								RW	RW	RW	RW	N/A			
0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x

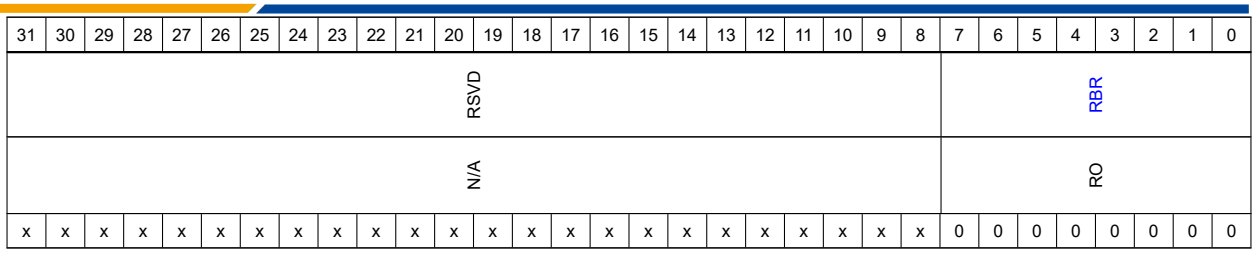
MOTO_CFG [31:0]

Bits	Name	Description
31	SWTRG	software trigger. User should avoid use sw/hw trigger at same time, otherwise result unknown. Hardware auto reset.
15-8	TXSTP_BITS	if TXSTOP_INSERT is enabled, the STOP bits to be inserted between each byte. 0 for 1 bit; 0xFF for 256bits
7	HWTRG_EN	set to enable hardware trigger(trigger from moto is shared by other UART)
6	TRG_MODE	set to enable trigger mode. software should push needed data into txbuffer frist, uart will not start transmission at this time. User should send trigger signal(by hw or sw), uart will send all data in txfifo till empty NOTE: the hw_trigger should be pulse signal from trig mux.
5	TRG_CLR_RFIFO	set to enable the feature that, clear rxfifo at tx trigger(sw or hw), avoid unexpected data in rxfifo.
4	TXSTOP_INSERT	set to insert STOP bits between each tx byte till tx fifo empty. NOTE: there will be no 1.5/2 STOP bits if enabled this feature, LCR.STB should be set to 0 if this bit is set

MOTO_CFG Bits

46.4.8 RBR (0x20)

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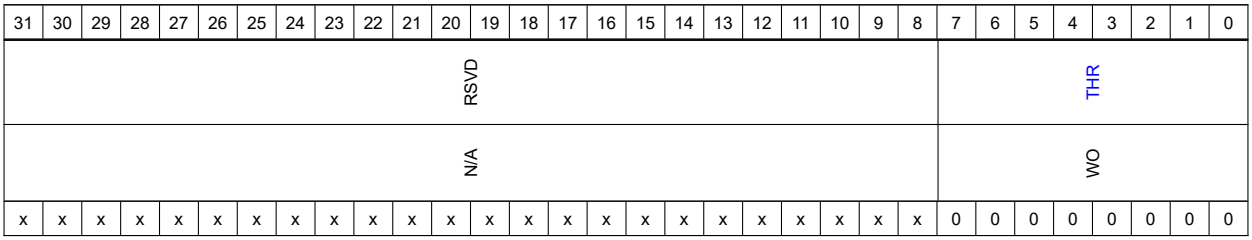


RBR [31:0]

Bits	Name	Description
7-0	RBR	Receive data read port

RBR Bits

46.4.9 THR (0x20)

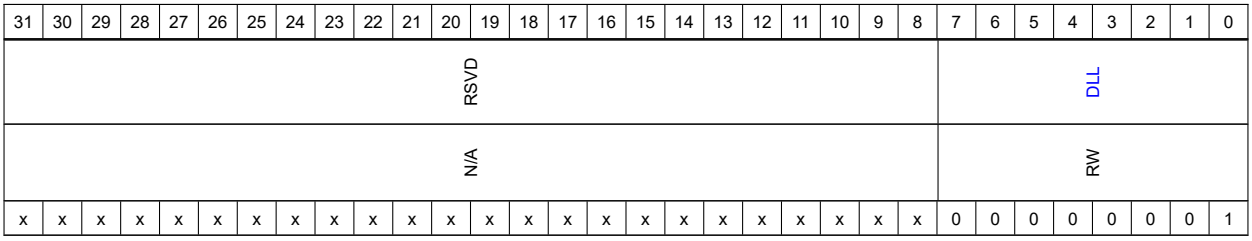


THR [31:0]

Bits	Name	Description
7-0	THR	Transmit data write port

THR Bits

46.4.10 DLL (0x20)



DLL [31:0]

Bits	Name	Description
7-0	DLL	Least significant byte of the Divisor Latch

DLL Bits

46.4.11 IER (0x24)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ERXIDLE	ETXIDLE	EADDRM	EADDRM_IDLE	EDATLOST	RSVD																							EMSI	ELSI	ETHEI	ERBI
RW	RW	RW	RW	RW	N/A																							RW	RW	RW	RW
0	0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0

IER [31:0]

Bits	Name	Description
31	ERXIDLE	Enable Receive Idle interrupt 0 - Disable Idle interrupt 1 - Enable Idle interrupt
30	ETXIDLE	enable transmit idle interrupt
29	EADDRM	enable ADDR_MATCH interrupt
28	EADDRM_IDLE	enable ADDR_MATCH_IDLE interrupt
27	EDATLOST	enable DATA_LOST interrupt
3	EMSI	Enable modem status interrupt The interrupt asserts when the status of one of the following occurs: The status of modem_rin, modem_dcdn, modem_dsrn or modem_ctsn (If the auto-cts mode is disabled) has been changed. If the auto-cts mode is enabled (MCR bit4 (AFE) = 1), modem_ctsn would be used to control the transmitter.
2	ELSI	Enable receiver line status interrupt
1	ETHEI	Enable transmitter holding register interrupt
0	ERBI	Enable received data available interrupt and the character timeout interrupt 0: Disable 1: Enable

IER Bits

46.4.12 DLM (0x24)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																								DLM							
N/A																								RW							
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0

DLM [31:0]

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Bits	Name	Description
7-0	DLM	Most significant byte of the Divisor Latch

DLM Bits

46.4.13 IIR (0x28)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RXIDLE_FLAG	RSVD																							FIFOED		RSVD		INTRID			
W1C	N/A																							RO		N/A		RO			
0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	x	x	0	0	0	1

IIR [31:0]

Bits	Name	Description
31	RXIDLE_FLAG	UART IDLE Flag 0 - UART is busy 1 - UART is idle NOTE: when write one to clear this bit, avoid changging FCR register since it's same address as IIR
7-6	FIFOED	FIFOs enabled These two bits are 1 when bit 0 of the FIFO Control Register (FIFOE) is set to 1.
3-0	INTRID	Interrupt ID, see IIR2 for detail decoding

IIR Bits

46.4.14 FCR (0x28)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																								RFIFOT		TFIFOT		DMAE	TFIFORST	RFIFORST	FIFOE
N/A																								WO		WO		WO	WO	WO	WO
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0

FCR [31:0]

Bits	Name	Description
7-6	RFIFOT	Receiver FIFO trigger level(0 for 1byte, 0x3 for 4bytes). Uart will send rx_dma_req if data in fifo reachs the threshold

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Bits	Name	Description
5-4	TFIFOT	Transmitter FIFO trigger level(0 for 1byte, 0x3 for 4bytes), uart will send tx_dma_req when data in fifo is less than threshold.
3	DMAE	DMA enable 0: Disable 1: Enable
2	TFIFORST	Transmitter FIFO reset Write 1 to clear all bytes in the TXFIFO and resets its counter. The Transmitter Shift Register is not cleared. This bit will automatically be cleared.
1	RFIFORST	Receiver FIFO reset Write 1 to clear all bytes in the RXFIFO and resets its counter. The Receiver Shift Register is not cleared. This bit will automatically be cleared.
0	FIFOE	FIFO enable Write 1 to enable both the transmitter and receiver FIFOs. The FIFOs are reset when the value of this bit toggles.

FCR Bits

46.4.15 LCR (0x2C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																								DLAB	BC	SPS	EPS	PEN	STB	WLS	
N/A																								RW	RW	RW	RW	RW	RW	RW	
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0

LCR [31:0]

Bits	Name	Description
7	DLAB	Divisor latch access bit
6	BC	Break control
5	SPS	Stick parity 1: Parity bit is constant 0 or 1, depending on bit4 (EPS). 0: Disable the sticky bit parity.
4	EPS	Even parity select 1: Even parity (an even number of logic-1 is in the data and parity bits) 0: Old parity.

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Bits	Name	Description
3	PEN	Parity enable When this bit is set, a parity bit is generated in transmitted data before the first STOP bit and the parity bit would be checked for the received data.
2	STB	Number of STOP bits 0: 1 bits 1: The number of STOP bit is based on the WLS setting When WLS = 0, STOP bit is 1.5 bits When WLS = 1, 2, 3, STOP bit is 2 bits
1-0	WLS	Word length setting 0: 5 bits 1: 6 bits 2: 7 bits 3: 8 bits

LCR Bits

46.4.16 MCR (0x30)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																										AFE	LOOP	RSVD		RTS	RSVD
N/A																										RW	RW	N/A		RW	N/A
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	x	x	0	x

MCR [31:0]

Bits	Name	Description
5	AFE	Auto flow control enable 0: Disable 1: The auto-CTS and auto-RTS setting is based on the RTS bit setting: When RTS = 0, auto-CTS only When RTS = 1, auto-CTS and auto-RTS
4	LOOP	Enable loopback mode 0: Disable 1: Enable
1	RTS	Request to send This bit controls the modem_rtsn output. 0: The modem_rtsn output signal will be driven HIGH 1: The modem_rtsn output signal will be driven LOW

MCR Bits

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46.4.17 LSR (0x34)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RXIDLE	TXIDLE	RSVD									RFIFO_NUM					RSVD			TFIFO_NUM					ERRF	TEMT	THRE	LBREAK	FE	PE	OE	DR
RO	RO	N/A									RO					N/A			RO					RO	RO	RO	RO	RO	RO	RO	RO
0	0	x	x	x	x	x	x	x	x	x	0	0	0	0	0	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0

LSR [31:0]

Bits	Name	Description
31	RXIDLE	rxidle after timeout, clear after rx idle condition not match
30	TXIDLE	txidle after timeout, clear after tx idle condition not match
20-16	RFIFO_NUM	data bytes in rxfifo not read
12-8	TFIFO_NUM	data bytes in txfifo not sent
7	ERRF	Error in RXFIFO In the FIFO mode, this bit is set when there is at least one parity error, framing error, or line break associated with data in the RXFIFO. It is cleared when this register is read and there is no more error for the rest of data in the RXFIFO.
6	TEMT	Transmitter empty This bit is 1 when the THR (TXFIFO in the FIFO mode) and the Transmitter Shift Register (TSR) are both empty. Otherwise, it is zero.
5	THRE	Transmitter Holding Register empty This bit is 1 when the THR (TXFIFO in the FIFO mode) is empty. Otherwise, it is zero. If the THRE interrupt is enabled, an interrupt is triggered when THRE becomes 1.
4	LBREAK	Line break This bit is set when the uart_sin input signal was held LOWfor longer than the time for a full-word transmission. A full-word transmission is the transmission of the START, data, parity, and STOP bits. It is cleared when this register is read. In the FIFO mode, this bit indicates the line break for the received data at the top of the RXFIFO.

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Bits	Name	Description
3	FE	Framing error This bit is set when the received STOP bit is not HIGH. It is cleared when this register is read. In the FIFO mode, this bit indicates the framing error for the received data at the top of the RXFIFO.
2	PE	Parity error This bit is set when the received parity does not match with the parity selected in the LCR[5:4]. It is cleared when this register is read. In the FIFO mode, this bit indicates the parity error for the received data at the top of the RXFIFO.
1	OE	Overrun error This bit indicates that data in the Receiver Buffer Register (RBR) is overrun.
0	DR	Data ready. This bit is set when there are incoming received data in the Receiver Buffer Register (RBR). It is cleared when all of the received data are read.

LSR Bits

46.4.18 MSR (0x38)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																											CTS	RSVD			DCTS
N/A																											RO	N/A			RC
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	0

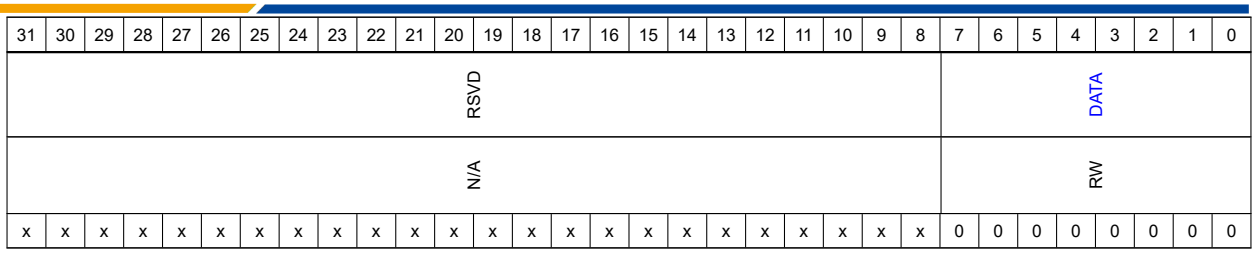
MSR [31:0]

Bits	Name	Description
4	CTS	Clear to send 0: The modem_ctsn input signal is HIGH. 1: The modem_ctsn input signal is LOW.
0	DCTS	Delta clear to send This bit is set when the state of the modem_ctsn input signal has been changed since the last time this register is read.

MSR Bits

46.4.19 GPR (0x3C)

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GPR [31:0]

Bits	Name	Description
7-0	DATA	A one-byte storage register

GPR Bits

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47 Serial Peripheral Bus SPI

Serial Peripheral Bus (SPI) supports master mode and slave mode. In master mode, it can control multiple peripheral devices, while in slave mode, it can respond to master requests to complete data exchange.

47.1 Module Functions

- Supports both master mode and slave mode;
- Configurable MSB-first or LSB-first transmission order;
- Configurable clock frequency;
- Supports DMA data transmission;
- Supports DSPI and QSPI modes;
- Configurable clock polarity and phase;
- Both Tx_FIFO and Rx_FIFO have a depth of 8 words;

47.2 Functional Description

This subsection is divided into 4 parts, describing master mode, slave mode, DSPI mode and QSPI mode respectively. The master and slave mode sections mainly introduce initialization and transmission formats, while the DSPI and QSPI mode sections mainly explain their differences from normal mode.

The architecture diagram of this module is shown below. SPI connects to external devices through six pins:

- SCLK: Serial clock line, generated by the master and provided to slave devices;
- CS: Slave device selection line, which enables master-slave data exchange after the target slave device is selected;
- MOSI: Master Output, Slave Input line;
- MISO: Master Input, Slave Output line;
- DATA2: Data line 2 for QSPI mode;
- DATA3: Data line 3 for QSPI mode;

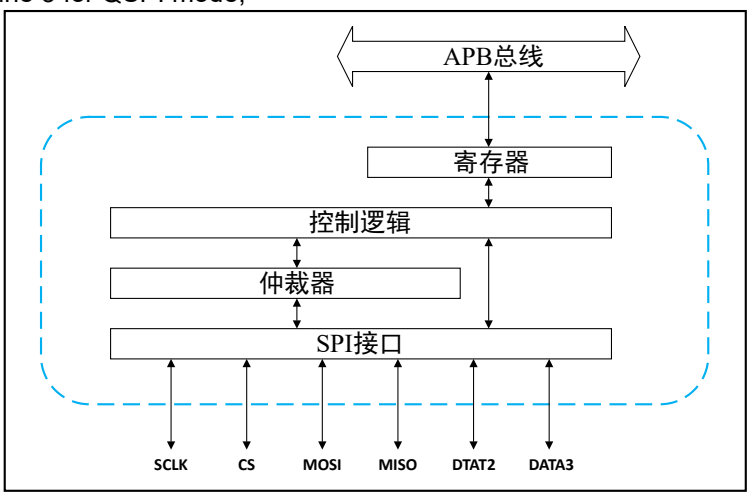


Figure 65: SPI Structure Diagram

In normal mode, i.e. when TRANSFMT[MOSIBIDIR] = 0, the transmission direction of MOSI is from master to slave, and the transmission direction of MISO is from slave to master. In master mode, the polarity and phase of SPI transmission can be set through TRANSFMT[CPOL] and TRANSFMT[CPHA]. SPI sends data on the

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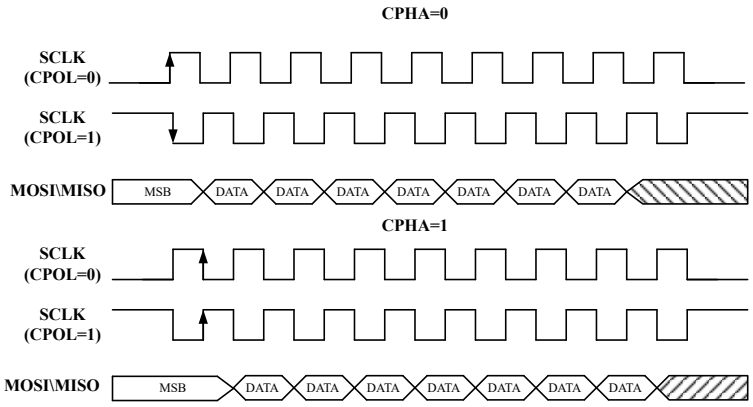


Figure 66: SPI Data Timing Diagram

rising or falling edge of the SCLK line, and receives data on the opposite valid clock edge.

47.2.1 Master Mode

In master mode, the SPI initiates transmission control. In this mode, SPI transmission format, transmission control and interface timing are configured through the configuration bus.

The transmission format includes command, address and data fields. The command and address fields can also be disabled to implement pure data transmission only.

Interface timing includes SCLK frequency, the minimum time from CS valid to SCLK valid edge, and the minimum high-level duration of CS. In master mode, the interface timing register TIMING needs to be configured.

Transmission/reception in master mode can be completed via register access, or via DMA after thresholds are set in CTRL[TXTHRES] / CTRL[RXTHRES]. The SPI controller supports transmit/receive buffer threshold interrupts to reduce the software load of flow control. The controller also supports transmission completion interrupt and other interrupt types.

In addition, SPI allows software to directly control pin signals, which can be configured in the DIRECTIO register to implement special timings.

The SPI module supports up to 4 slave chip selects. The chip select signals can be controlled via the [CS_EN] bit field of the CTRL register, with each bit controlling one CS pin. For specific slave data parameters, please refer to the corresponding packaging information.

47.2.2 Slave Mode

When SPI transmission format register TRANSFMT[SLVMODE] = 1, the SPI is configured as slave mode. In slave mode, there is no need to configure the SPI interface timing register TIMING. Besides, when SPI acts as a slave, the SCLK frequency is required to be no higher than 1/4 of the SPI clock frequency.

In slave mode, the controller receives signals on the interface and interprets the signals as commands, dummy cycles and data. The command and dummy fields are fixed at 8 bits, and the length of the data segment is determined by the received command and register settings.

In slave mode, built-in fixed commands are available for direct use, and this mode also supports user-defined arbitrary 8-bit commands in registers. For user-defined instructions, the data segment format is configured via registers. For example, if the transmission mode is set to (dummy, write), the dummy cycle is (dummy-

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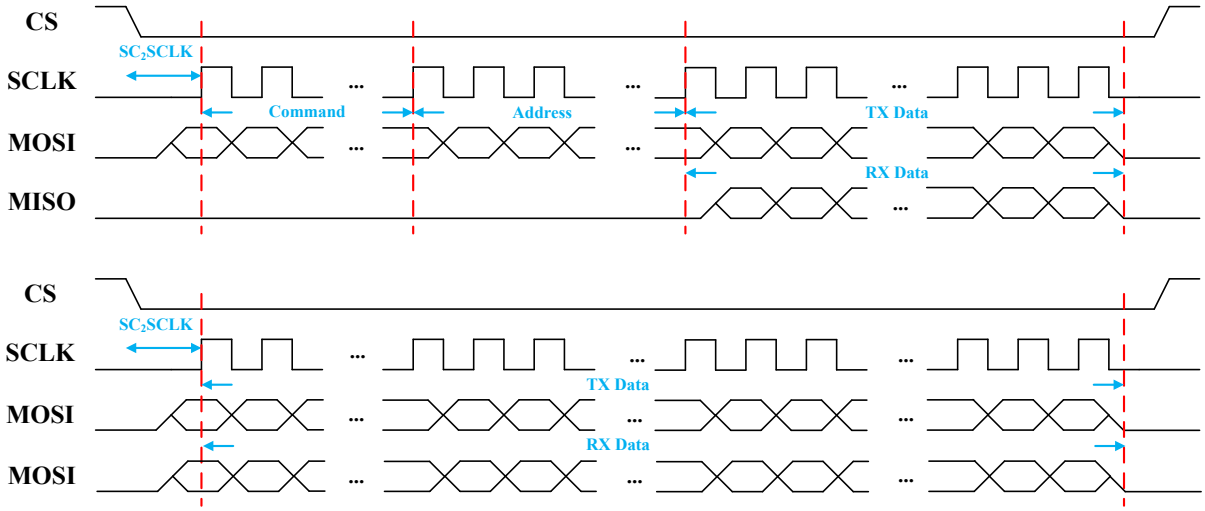


Figure 67: Two SPI Transmission Formats

Cnt+1)*((DataLen+1)/IO width), the dummy phase field is discarded, and the subsequent data is saved to the data register.

47.2.3 DSPI Mode

When register TRANSCTRL[DUALQUAD] = 1, the SPI is configured as DSPI mode. In this mode, set TRANSFMT[MOSIBIDIR] = 1, and MOSI and MISO work in half-duplex transmission mode, thus doubling the transmission bandwidth.

DSPI mode provides two transmission methods: one uses both MOSI and MISO lines for transmission of both address and data; the other uses MOSI and MISO lines for data transmission only. The transmission method can be configured via the TRANSCTRL register.

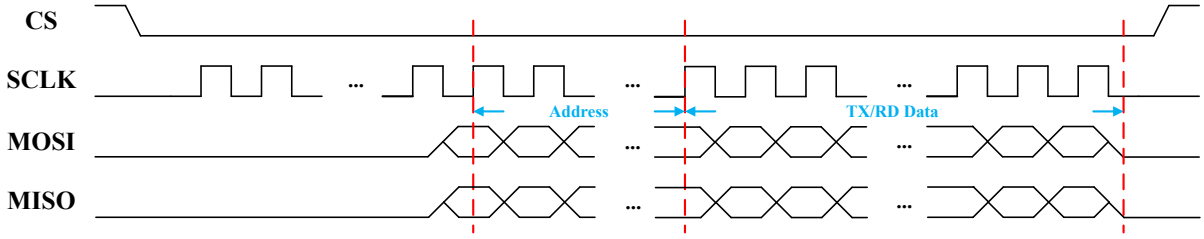


Figure 68: DSPI Address Two-wire Transmission Format Diagram

47.2.4 QSPI Mode

When register TRANSCTRL[DUALQUAD] = 2, the SPI is configured as QSPI mode. In this mode, set TRANSFMT[MOSIBIDIR] = 1, and MOSI, MISO, DATA2 and DATA3 work in half-duplex transmission mode, thus quadrupling the transmission bandwidth.

QSPI mode provides two transmission methods: one uses MOSI, MISO, DATA2 and DATA3 four lines for transmission of both address and data; the other uses the four lines for data transmission only. The transmission method can be configured via the TRANSCTRL register.

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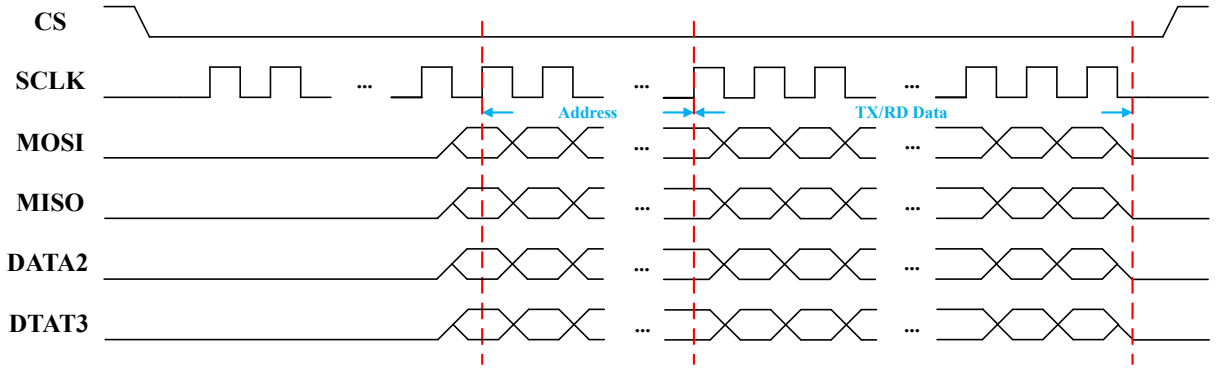


Figure 69: QSPI Address Four-wire Transmission Format Diagram

47.2.5 Transmission Initiation and Command Values

When the SPI module works in master mode, transmission (read or write operation) starts after the corresponding instruction is written to the command register (CMD). For example, if the SPI module is set to master mode, and both command segment and address segment are disabled in the transmission control register (i.e. the master works in pure data transmission mode), after writing 0xFF to the command register, the master will start transmission (the SPI will not send the instruction value in this scenario). Another example: the user sets the SPI module to master mode, uses DMA to send data at the specified address to the slave in pure data format. After completing DMA configuration, write 0xFF to the command register, and the SPI module will obtain data via DMA and send it out. To perform another transmission, just write 0xFF to the command register again.

If the SPI module works in master mode and the command segment is enabled in the transmission control register, write the corresponding instruction value to the command register, the SPI module will start transmission and send out the instruction value.

When the SPI module works as a slave, it supports multiple instruction operations. For example, when receiving the command value 0x05, the SPI module will return the value of the slave status register (SLVST), with a return length of 32 bits. The following table lists the supported operations when SPI is in slave mode.

Instruction Value	Operation Content
0x05	Read 32-bit SPI slave status register value in SPI mode
0x15	Read 32-bit SPI slave status register value in DSPI mode
0x25	Read 32-bit SPI slave status register value in QSPI mode
0x0B	Read FIFO value in SPI slave in SPI mode
0x0C	Read FIFO value in SPI slave in DSPI mode
0x0E	Read FIFO value in SPI slave in QSPI mode
0x51	Write data to FIFO in SPI slave in SPI mode
0x52	Write data to FIFO in SPI slave in DSPI mode
0x54	Write data to FIFO in SPI slave in QSPI mode

Table 200: Supported Command Values in SPI Slave Mode

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47.2.6 FIFO Threshold, Interrupt and DMA

Both Tx_FIFO and Rx_FIFO of the SPI module have a depth of 8 words, which should be noted when configuring FIFO thresholds. The FIFO depth can be obtained from the [TXFIFOSIZE] and [RXFIFOSIZE] bit fields in the CONFIG register. FIFO threshold settings will affect the interrupt and DMA functions of the SPI module.

Users can enable the interrupt function of the SPI module in the INTREN register according to requirements. The SPI module supports the following interrupts in master mode:

- Transmission completion interrupt;
- Tx_FIFO threshold interrupt (triggered when valid data in Tx_FIFO is less than or equal to the transmit threshold);
- Rx_FIFO threshold interrupt (triggered when valid data in Rx_FIFO is greater than or equal to the receive threshold);

SPI supports the following interrupts in slave mode:

- Command received interrupt (valid when the command field is enabled);
- Tx_FIFO threshold interrupt (triggered when valid data in Tx_FIFO is less than or equal to the transmit threshold);
- Rx_FIFO threshold interrupt (triggered when valid data in Rx_FIFO is greater than or equal to the receive threshold);
- Tx_FIFO underrun interrupt;
- Rx_FIFO overflow interrupt;

The DMA function of the SPI module is enabled via the [TXDMAEN] and [RXDMAEN] bits in the CTRL register. The FIFO thresholds that trigger DMA transmission are set in the [TXTHRES] and [RXTHRES] bit fields.

47.3 Brief SPI Configuration Instructions

47.3.1 Command-Address-Data Transmission with DMA in Master Mode

1. Confirm that SPIACTIVE bit in the SPI status register (0x034 STATUS) is 0, to ensure the previous SPI transmission is completed;
2. Configure the SPI transmission format register (0x010 TRANSFMT) according to requirements;
3. Configure the SPI transmission control register (0x020 TRANSCTRL) according to requirements;
4. Set the transmit FIFO threshold in the SPI control register (0x030 CTRL) according to requirements;
5. Configure interface timing in the SPI TIMING register (0x040 TIMING) according to requirements;
6. Set interrupts as required, for example, to enable transmission completion interrupt: set INTREN[ENDINTEN] = 1 in the SPI interrupt enable register (0x038 INTREN);
7. Configure the DMA controller to ensure data is moved from memory to the SPI data register (0x02c DATA);
8. Write the target address to the SPI address register (0x028 ADDR) as required;
9. Write the command to the SPI command register (0x024 CMD);

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10. Wait for the transmission completion interrupt, and read the SPI interrupt status register (0x03c INTRST) to check status bits.

47.3.2 Pure Data Transmission without DMA in Master Mode

1. Confirm that SPIACTIVE bit in the SPI status register (0x034 STATUS) is 0, to ensure the previous SPI transmission is completed;

2. Configure the SPI transmission format register (0x010 TRANSFMT) according to requirements;

3. Configure the SPI transmission control register (0x020 TRANSCTRL) according to requirements, disable command and address fields by setting TRANSFMT[CMDEN] = 0 and TRANSFMT[ADDREN] = 0;

4. Configure interface timing in the SPI TIMING register (0x040 TIMING) according to requirements;

5. Set interrupts as required, for example, to enable transmission completion interrupt: set INTREN[ENDINTEN] = 1 in the SPI interrupt enable register (0x038 INTREN);

6. Write data to the SPI data register (0x02c DATA);

7. Write 0xFF to the CMD register to start data transmission;

8. Wait for the transmission completion interrupt, and read the SPI interrupt status register (0x03c INTRST) to check status bits.

47.4 SPI Register List

SPI0 base address: 0xF0070000

SPI1 base address: 0xF0074000

SPI2 base address: 0xF0078000

SPI3 base address: 0xF007C000

Offset	Register Name	Description	Reset Value
0x0004	WR_TRANS_CNT	Transfer count for write data	0x00000000
0x0008	RD_TRANS_CNT	Transfer count for read data	0x00000000
0x0010	TRANSFMT	Transfer Format Register	0x00020780
0x0014	DIRECTIO	Direct IO Control Register	0x00003100
0x0020	TRANSCTRL	Transfer Control Register	0x00000000
0x0024	CMD	Command Register	0x00000000
0x0028	ADDR	Address Register	0x00000000
0x002C	DATA	Data Register	0x00000000
0x0030	CTRL	Control Register	0x00000000
0x0034	STATUS	Status Register	0x00000000
0x0038	INTREN	Interrupt Enable Register	0x00000000
0x003C	INTRST	Interrupt Status Register	0x00000000
0x0040	TIMING	Interface Timing Register	0x00000000
0x0060	SLVST	Slave Status Register	0x00000000
0x0064	SLVDATACNT	Slave Data Count Register	0x00000000

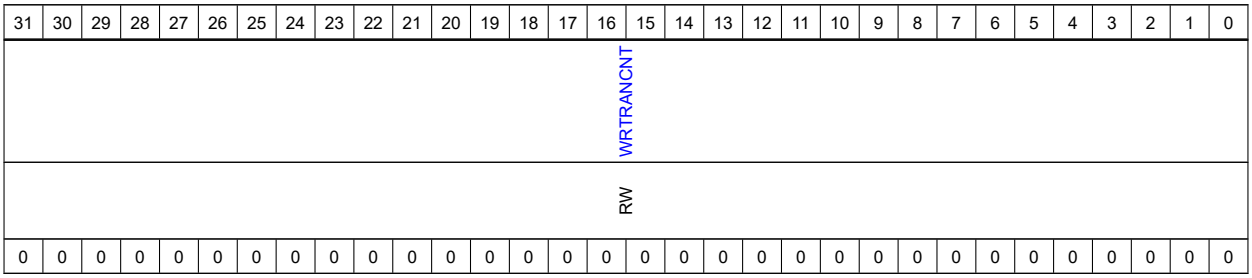
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Offset	Register Name	Description	Reset Value
0x0068	SLVDATAWCNT	WCnt	0x00000000
0x006C	SLVDATARCNT	RCnt	0x00000000
0x007C	CONFIG	Configuration Register	0x00004311

Table 201: SPI Register List

47.5 SPI Register Description

47.5.1 WR_TRANS_CNT (0x4)

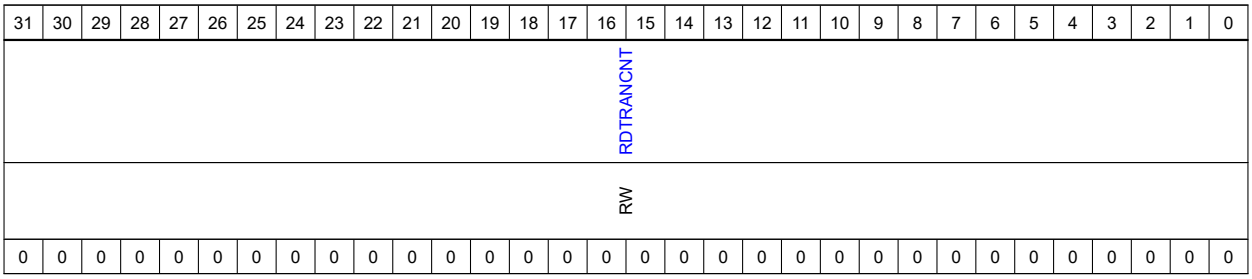


WR_TRANS_CNT [31:0]

Bits	Name	Description
31-0	WRTRANCNT	Transfer count for write data WrTranCnt indicates the number of units of data to be transmitted to the SPI bus from the Data Register. The actual transfer count is (WrTranCnt+1). WrTranCnt only takes effect when TransMode is 0, 1, 3, 4, 5, 6 or 8. The size (bit-width) of a data unit is defined by the DataLen field of the Transfer Format Register. For TransMode 0, WrTranCnt must be equal to RdTranCnt.

WR_TRANS_CNT Bits

47.5.2 RD_TRANS_CNT (0x8)



RD_TRANS_CNT [31:0]

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Bits	Name	Description
31-0	RDTRANCNT	Transfer count for read data RdTranCnt indicates the number of units of data to be received from SPI bus and stored to the Data Register. The actual received count is (RdTranCnt+1). RdTransCnt only takes effect when TransMode is 0, 2, 3, 4, 5, 6 or 9. The size (bit-width) of a data unit is defined by the DataLen field of the Transfer Format Register. For TransMode 0, WrTranCnt must equal RdTranCnt.

RD_TRANS_CNT Bits

47.5.3 TRANSFMT (0x10)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD														ADDRLEN		RSVD			DATALEN					DATAMERGE	RSVD		MOSIBIDIR	LSB	SLVMODE	CPOL	CPHA
N/A														RW		N/A			RW					RW	N/A		RW	RW	RW	RW	RW
0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	x	x	x	0	0	1	1	1	1	x	x	0	0	0	0	0

TRANSFMT [31:0]

Bits	Name	Description
17-16	ADDRLEN	Address length in bytes 0x0: 1 byte 0x1: 2 bytes 0x2: 3 bytes 0x3: 4 bytes
12-8	DATALEN	The length of each data unit in bits The actual bit number of a data unit is (DataLen + 1)
7	DATAMERGE	Enable Data Merge mode, which does automatic data split on write and data coalescing on read. This bit only takes effect when DataLen = 0x7. Under Data Merge mode, each write to the Data Register will transmit all fourbytes of the write data; each read from the Data Register will retrieve four bytes of received data as a single word data. When Data Merge mode is disabled, only the least (DataLen+1) significient bits of the Data Register are valid for read/write operations; no automatic data split/coalescing will be performed.

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Bits	Name	Description
4	MOSIBIDIR	Bi-directional MOSI in regular (single) mode 0x0: MOSI is uni-directional signal in regular mode. 0x1: MOSI is bi-directional signal in regular mode. This bi-directional signal replaces the two
3	LSB	Transfer data with the least significant bit first 0x0: Most significant bit first 0x1: Least significant bit first
2	SLVMODE	SPI Master/Slave mode selection 0x0: Master mode 0x1: Slave mode
1	CPOL	SPI Clock Polarity 0x0: SCLK is LOW in the idle states 0x1: SCLK is HIGH in the idle states
0	CPHA	SPI Clock Phase 0x0: Sampling data at odd SCLK edges 0x1: Sampling data at even SCLK edges

TRANSFMT Bits

47.5.4 DIRECTIO (0x14)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD							DIRECTIOEN	RSVD		HOLD_OE	WP_OE	MISO_OE	MOSI_OE	SCLK_OE	CS_OE	RSVD		HOLD_O	WP_O	MISO_O	MOSI_O	SCLK_O	CS_O	RSVD		HOLD_I	WP_I	MISO_I	MOSI_I	SCLK_I	CS_I
N/A							RW	N/A		RW	RW	RW	RW	RW	RW	N/A		RW	RW	RW	RW	RW	RW	N/A		RO	RO	RO	RO	RO	RO
x	x	x	x	x	x	x	0	x	x	0	0	0	0	0	0	x	x	1	1	0	0	0	1	x	x	0	0	0	0	0	0

DIRECTIO [31:0]

Bits	Name	Description
24	DIRECTIOEN	Enable Direct IO 0x0: Disable 0x1: Enable
21	HOLD_OE	Output enable for the SPI Flash hold signal
20	WP_OE	Output enable for the SPI Flash write protect signal
19	MISO_OE	Output enable fo the SPI MISO signal
18	MOSI_OE	Output enable for the SPI MOSI signal
17	SCLK_OE	Output enable for the SPI SCLK signal
16	CS_OE	Output enable for SPI CS (chip select) signal
13	HOLD_O	Output value for the SPI Flash hold signal

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Bits	Name	Description
12	WP_O	Output value for the SPI Flash write protect signal
11	MISO_O	Output value for the SPI MISO signal
10	MOSI_O	Output value for the SPI MOSI signal
9	SCLK_O	Output value for the SPI SCLK signal
8	CS_O	Output value for the SPI CS (chip select) signal
5	HOLD_I	Status of the SPI Flash hold signal
4	WP_I	Status of the SPI Flash write protect signal
3	MISO_I	Status of the SPI MISO signal
2	MOSI_I	Status of the SPI MOSI signal
1	SCLK_I	Status of the SPI SCLK signal
0	CS_I	Status of the SPI CS (chip select) signal

DIRECTIO Bits

47.5.5 TRANSCTRL (0x20)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SLVDATAONLY	CMDEN	ADDREN	ADDRFMT	TRANSMODE				DUALQUAD		TOKENEN	WRTRANCNT									TOKENVALUE	DUMMYCNT		RDTRANCNT										
RW	RW	RW	RW	RW				RW		RW	RW									RW	RW		RW										
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TRANSCTRL [31:0]

Bits	Name	Description
31	SLVDATAONLY	Data-only mode (slave mode only) 0x0: Disable the data-only mode 0x1: Enable the data-only mode Note: This mode only works in the uni-directional regular (single) mode so MOSIBiDir, DualQuad and TransMode should be set to 0.
30	CMDEN	SPI command phase enable (Master mode only) 0x0: Disable the command phase 0x1: Enable the command phase
29	ADDREN	SPI address phase enable (Master mode only) 0x0: Disable the address phase 0x1: Enable the address phase

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Bits	Name	Description
28	ADDRFMT	SPI address phase format (Master mode only) 0x0: Address phase is the regular (single) mode 0x1: The format of the address phase is the same as the data phase (DualQuad).
27-24	TRANSMODE	Transfer mode The transfer sequence could be 0x0: Write and read at the same time 0x1: Write only 0x2: Read only 0x3: Write, Read 0x4: Read, Write 0x5: Write, Dummy, Read 0x6: Read, Dummy, Write 0x7: None Data (must enable CmdEn or AddrEn in master mode) 0x8: Dummy, Write 0x9: Dummy, Read 0xa~0xf: Reserved
23-22	DUALQUAD	SPI data phase format 0x0: Regular (Single) mode 0x1: Dual I/O mode 0x2: Quad I/O mode 0x3: Reserved
21	TOKENEN	Token transfer enable (Master mode only) Append a one-byte special token following the address phase for SPI read transfers. The value of the special token should be selected in TokenValue. 0x0: Disable the one-byte special token 0x1: Enable the one-byte special token
20-12	WRTRANCNT	Transfer count for write data WrTranCnt indicates the number of units of data to be transmitted to the SPI bus from the Data Register. The actual transfer count is (WrTranCnt+1). WrTranCnt only takes effect when TransMode is 0, 1, 3, 4, 5, 6 or 8. The size (bit-width) of a data unit is defined by the DataLen field of the Transfer Format Register. For TransMode 0, WrTranCnt must be equal to RdTranCnt.

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Bits	Name	Description
11	TOKENVALUE	Token value (Master mode only) The value of the one-byte special token following the address phase for SPI read transfers. 0x0: token value = 0x00 0x1: token value = 0x69
10-9	DUMMYCNT	Dummy data count. The actual dummy count is (DummyCnt +1). The number of dummy cycles on the SPI interface will be (DummyCnt+1)* ((DataLen+1)/SPI IO width) The Data pins are put into the high impedance during the dummy data phase. DummyCnt is only used for TransMode 5, 6, 8 and 9, which has dummy data phases.
8-0	RDTRANCNT	Transfer count for read data RdTranCnt indicates the number of units of data to be received from SPI bus and stored to the Data Register. The actual received count is (RdTranCnt+1). RdTransCnt only takes effect when TransMode is 0, 2, 3, 4, 5, 6 or 9. The size (bit-width) of a data unit is defined by the DataLen field of the Transfer Format Register. For TransMode 0, WrTranCnt must equal RdTranCnt.

TRANSCTRL Bits

47.5.6 CMD (0x24)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																								CMD							
N/A																								RW							
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0

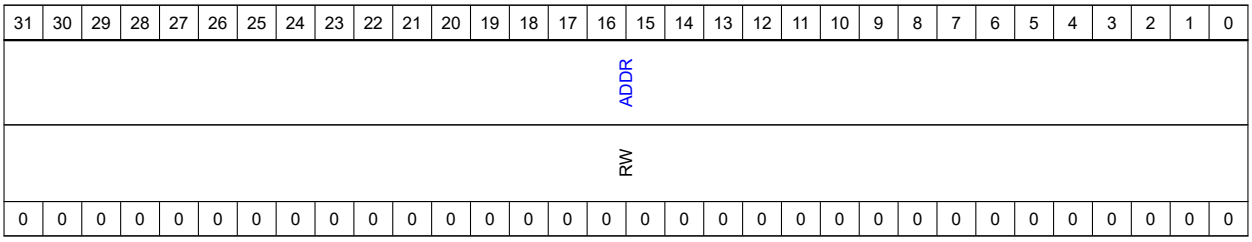
CMD [31:0]

Bits	Name	Description
7-0	CMD	SPI Command

CMD Bits

47.5.7 ADDR (0x28)

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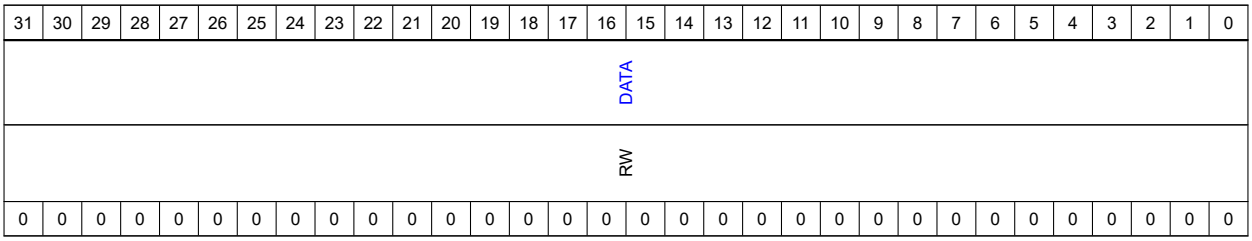


ADDR [31:0]

Bits	Name	Description
31-0	ADDR	SPI Address (Master mode only)

ADDR Bits

47.5.8 DATA (0x2C)



DATA [31:0]

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Bits	Name	Description
31-0	DATA	Data to transmit or the received data For writes, data is enqueued to the TX FIFO. The least significant byte is always transmitted first. If the TX FIFO is full and the SPIActive bit of the status register is 1, the ready signal hready/pready will be deasserted to insert wait states to the transfer. For reads, data is read and dequeued from the RX FIFO. The least significant byte is the first received byte. If the RX FIFO is empty and the SPIActive bit of the status register is 1, the ready signal hready/pready will be deasserted to insert wait states to the transfer. The FIFOs decouple the speed of the SPI transfers and the software 鈥樉 generation/consumption of data. When the TX FIFO is empty, SPI transfers will hold until more data is written to the TX FIFO; when the RX FIFO is full, SPI transfers will hold until there is more room in the RX FIFO. If more data is written to the TX FIFO than the write transfer count (WrTranCnt), the remaining data will stay in the TX FIFO for the next transfer or until the TX FIFO is reset.

DATA Bits

47.5.9 CTRL (0x30)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD				CS_EN				TXTHRES								RXTHRES								RSVD			TXDMAEN	RXDMAEN	TXFIFORST	RXFIFORST	SPIRST
N/A				RW				RW								RW								N/A			RW	RW	RW	RW	RW
x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	0	0	0	0	0

CTRL [31:0]

Bits	Name	Description
27-24	CS_EN	Enable the selection of slave units. Each bit controls a slave selection signal.
23-16	TXTHRES	Transmit (TX) FIFO Threshold The TXFIFOInt interrupt or DMA request would be issued to replenish the TX FIFO when the TX data count is less than or equal to the TX FIFO threshold.

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Bits	Name	Description
15-8	RXTHRES	Receive (RX) FIFO Threshold The RXFIFOInt interrupt or DMA request would be issued for consuming the RX FIFO when the RX data count is more than or equal to the RX FIFO threshold.
4	TXDMAEN	TX DMA enable
3	RXDMAEN	RX DMA enable
2	TXFIFORST	Transmit FIFO reset Write 1 to reset. It is automatically cleared to 0 after the reset operation completes. When this bit is set to 1, it is necessary to first poll this bit to ensure it is 0 before proceeding with the operation. Otherwise, subsequent operations may encounter abnormalities.
1	RXFIFORST	Receive FIFO reset Write 1 to reset. It is automatically cleared to 0 after the reset operation completes. When this bit is set to 1, it is necessary to first poll this bit to ensure it is 0 before proceeding with the operation. Otherwise, subsequent operations may encounter abnormalities.
0	SPIRST	SPI reset Write 1 to reset. It is automatically cleared to 0 after the reset operation completes. When this bit is set to 1, it is necessary to first poll this bit to ensure it is 0 before proceeding with the operation. Otherwise, subsequent operations may encounter abnormalities.

CTRL Bits

47.5.10 STATUS (0x34)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD		TXNUM_7_6		RSVD		RXNUM_7_6		TXFULL	TXEMPTY	TXNUM_5_0						RXFULL	RXEMPTY	RXNUM_5_0						RSVD							SPIACTIVE
N/A		RO		N/A		RO		RO	RO	RO						RO	RO	RO						N/A							RO
x	x	0	0	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	0

STATUS [31:0]

Bits	Name	Description
29-28	TXNUM_7_6	Number of valid entries in the Transmit FIFO
25-24	RXNUM_7_6	Number of valid entries in the Receive FIFO
23	TXFULL	Transmit FIFO Full flag

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Bits	Name	Description
22	TXEMPTY	Transmit FIFO Empty flag
21-16	TXNUM_5_0	Number of valid entries in the Transmit FIFO
15	RXFULL	Receive FIFO Full flag
14	RXEMPTY	Receive FIFO Empty flag
13-8	RXNUM_5_0	Number of valid entries in the Receive FIFO
0	SPIACTIVE	SPI register programming is in progress. In master mode, SPIActive becomes 1 after the SPI command register is written and becomes 0 after the transfer is finished. In slave mode, SPIActive becomes 1 after the SPI CS signal is asserted and becomes 0 after the SPI CS signal is deasserted. Note that due to clock synchronization, it may take at most two spi_clock cycles for SPIActive to change when the corresponding condition happens. Note this bit stays 0 when Direct IO Control or the memory-mapped interface is used.

STATUS Bits

47.5.11 INTREN (0x38)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																										SLVCMDEN	ENDINTEN	TXFIFOINTEN	RXFIFOINTEN	TXFIFOURINTEN	RXFIFOORINTEN
N/A																										RW	RW	RW	RW	RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0

INTREN [31:0]

Bits	Name	Description
5	SLVCMDEN	Enable the Slave Command Interrupt. Control whether interrupts are triggered whenever slave commands are received. (Slave mode only)
4	ENDINTEN	Enable the End of SPI Transfer interrupt. Control whether interrupts are triggered when SPI transfers end. (In slave mode, end of read status transaction doesn 鈥 trigger this interrupt.)
3	TXFIFOINTEN	Enable the SPI Transmit FIFO Threshold interrupt. Control whether interrupts are triggered when the valid entries are less than or equal to the TX FIFO threshold.

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Bits	Name	Description
2	RXFIFOINTEN	Enable the SPI Receive FIFO Threshold interrupt. Control whether interrupts are triggered when the valid entries are greater than or equal to the RX FIFO threshold.
1	TXFIFOURINTEN	Enable the SPI Transmit FIFO Underrun interrupt. Control whether interrupts are triggered when the Transmit FIFO run out of data. (Slave mode only)
0	RXFIFOORINTEN	Enable the SPI Receive FIFO Overrun interrupt. Control whether interrupts are triggered when the Receive FIFO overflows. (Slave mode only)

INTREN Bits

47.5.12 INTRST (0x3C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																										SLVCMDINT	ENDINT	TXFIFOINT	RXFIFOINT	TXFIFOURINT	RXFIFOORINT
N/A																										W1C	W1C	W1C	W1C	W1C	W1C
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0

INTRST [31:0]

Bits	Name	Description
5	SLVCMDINT	Slave Command Interrupt. This bit is set when Slave Command interrupts occur. (Slave mode only)
4	ENDINT	End of SPI Transfer interrupt. This bit is set when End of SPI Transfer interrupts occur.
3	TXFIFOINT	TX FIFO Threshold interrupt. This bit is set when TX FIFO Threshold interrupts occur.
2	RXFIFOINT	RX FIFO Threshold interrupt. This bit is set when RX FIFO Threshold interrupts occur.
1	TXFIFOURINT	TX FIFO Underrun interrupt. This bit is set when TX FIFO Underrun interrupts occur. (Slave mode only)
0	RXFIFOORINT	RX FIFO Overrun interrupt. This bit is set when RX FIFO Overrun interrupts occur. (Slave mode only)

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Bits	Name	Description

INTRST Bits

47.5.13 TIMING (0x40)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																		CS2SCLK		CSHT				SCLK_DIV							
N/A																		RW		RW				RW							
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TIMING [31:0]

Bits	Name	Description
13-12	CS2SCLK	The minimum time between the edges of SPI CS and the edges of SCLK. SCLK_period * (CS2SCLK + 1) / 2
11-8	CSHT	The minimum time that SPI CS should stay HIGH. SCLK_period * (CSHT + 1) / 2
7-0	SCLK_DIV	The clock frequency ratio between the clock source and SPI interface SCLK. SCLK_period = ((SCLK_DIV + 1) * 2) * (Period of the SPI clock source) The SCLK_DIV value 0xff is a special value which indicates that the SCLK frequency should be the same as the spi_clock frequency.

TIMING Bits

47.5.14 SLVST (0x60)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD													UNDERRUN	OVERRUN	READY	USR_STATUS															
N/A													W1C	RW	RW	RW															
x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

SLVST [31:0]

Bits	Name	Description
18	UNDERRUN	Data underrun occurs in the last transaction

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Bits	Name	Description
17	OVERRUN	Data overrun occurs in the last transaction
16	READY	Set this bit to indicate that the ATCSPI200 is ready for data transaction. When an SPI transaction other than slave status-reading command ends, this bit will be cleared to 0.
15-0	USR_STATUS	User defined status flags

SLVST Bits

47.5.15 SLVDATACNT (0x64)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD						WCNT										RSVD						RCNT									
N/A						RO										N/A						RO									
x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0

SLVDATACNT [31:0]

Bits	Name	Description
25-16	WCNT	Slave transmitted data count
9-0	RCNT	Slave received data count

SLVDATACNT Bits

47.5.16 SLVDATAWCNT (0x68)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VAL																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

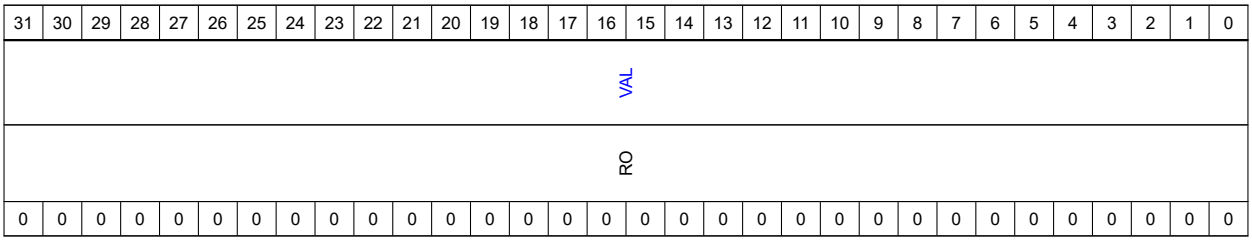
SLVDATAWCNT [31:0]

Bits	Name	Description
31-0	VAL	

SLVDATAWCNT Bits

47.5.17 SLVDATARCNT (0x6C)

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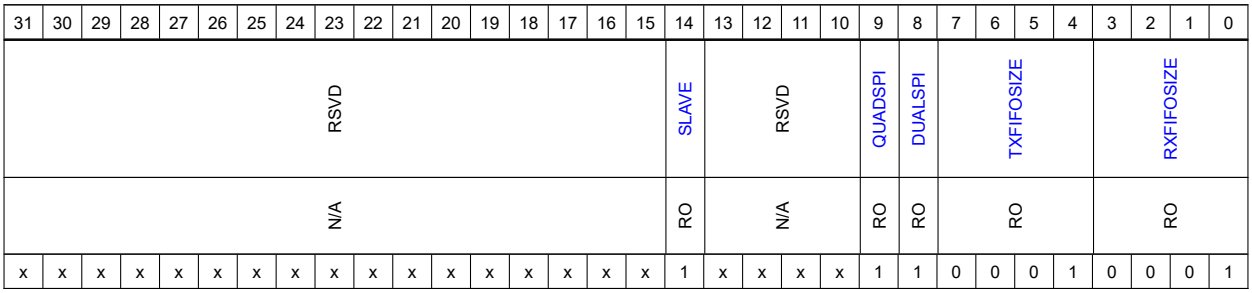


SLVDATARCNT [31:0]

Bits	Name	Description
31-0	VAL	

SLVDATARCNT Bits

47.5.18 CONFIG (0x7C)



CONFIG [31:0]

Bits	Name	Description
14	SLAVE	Support for SPI Slave mode
9	QUADSPI	Support for Quad I/O SPI
8	DUALSPI	Support for Dual I/O SPI
7-4	TXFIFOSIZE	Depth of TX FIFO 0x0: 2 words 0x1: 4 words 0x2: 8 words 0x3: 16 words 0x4: 32 words 0x5: 64 words 0x6: 128 words

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Bits	Name	Description
3-0	RXFIFOSIZE	Depth of RX FIFO 0x0: 2 words 0x1: 4 words 0x2: 8 words 0x3: 16 words 0x4: 32 words 0x5: 64 words 0x6: 128 words

CONFIG Bits

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48 Inter-Integrated Circuit Bus (I2C)

This chapter introduces the functions and features of I2C.

48.1 Feature Summary

This section describes the main features of I2C:

- Supports Standard mode (100 Kb/s), Fast mode (400 Kb/s) and Fast mode Plus (1 Mb/s)
- Configurable master/slave mode
- Supports 7-bit and 10-bit address modes
- Supports general call address
- Automatic clock stretching
- Configurable clock/data timing
- Supports Direct Memory Access (DMA)
- 4-byte FIFO

The I2C block diagram is shown in Figure 70.

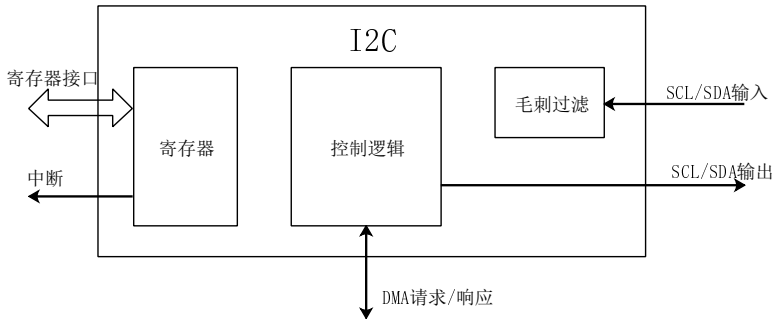


Figure 70: I2C Block Diagram

48.2 Functional Description

This section describes the functions of I2C.

48.2.1 Main Functions

The I2C can be configured to operate in master or slave mode via the MASTER bit in the SETUP register. As an I2C master, the controller can initiate transmissions efficiently. Each transmission consists of 4 phases: START, address, data and STOP. A START operation is generated in the START phase, the address is sent in the address phase, one or more data bytes are transmitted in the data phase, and a STOP operation is generated in the STOP phase. Execution of each phase can be controlled independently.

As an I2C slave, the controller is selected when the address transmitted on I2C matches the address in the ADDR register. The ADDRHIT bit in the INTEN register can be configured to make the controller generate an interrupt to notify the software to prepare for subsequent operations. If a general call address is received, the controller responds with an ACK and sets the GENCALL bit in the STATUS register to 1.

When the software is not ready for the next byte of transmit data, or the FIFO is full during data reception, the controller automatically stretches the I2C bus clock to suspend bus transmission. Automatic clock stretching

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is supported in both master and slave modes.

The controller enables automatic response by default, that is, it automatically sends an ACK for each received byte except the last byte. The software can disable the automatic response function by enabling the ByteReceive interrupt, and decide whether to send an ACK response after each byte is received.

48.2.2 Timing Configuration

I2C timing parameters are configured in the SETUP register, and all parameters are in units of clock cycles of the I2C module function clock (CLK_TOP_I2Cx). Due to the limited bit width of these configurable parameters, when the function clock frequency is very high, the bit width of the timing parameters may not meet the configuration requirements. Therefore, the controller provides the TPM register to set the timing parameter multiplier, which can extend the range of timing parameters. The TPM register should be modified when the I2C bus is idle and the I2C controller is disabled (the IICEN bit of the SETUP register is 0).

The following uses CLK_TOP_I2C frequency of 40 MHz and TPM equal to 0 as an example to illustrate the timing parameter configuration.

Glitch filtering: The pulse width that can be filtered out on the SCL and SDA input signals is defined by the T_SP bit of the SETUP register. The calculation formula is shown in (2):

$$PulseWidth = T_SP \times Clk_Period \times (TPM + 1) \tag{2}$$

For Fast mode and Fast mode Plus, glitches shorter than 50 ns must be filtered out, and T_SP should be set to 2 according to calculation.

Data Setup Time: The data setup time defines the time that SDA should remain stable before the rising edge of SCL, which is set by the T_SUDAT bit of the SETUP register. The calculation formula is shown in (3):

$$SetupTime = (2 \times Clk_Period) + (2 + T_SP + T_SUDAT) \times Clk_Period \times (TPM + 1) \tag{3}$$

For Standard mode, when the data setup time requirement is 250 ns, T_SUDAT should be set to 4.

Data Hold Time: The data hold time defines the time that SDA should remain stable after the falling edge of SCL, which is set by the T_HDDAT bit of the SETUP register. The calculation formula is shown in (4):

$$HoldTime = (2 \times Clk_Period) + (2 + T_SP + T_HDDAT) \times Clk_Period \times (TPM + 1) \tag{4}$$

For Standard mode, when the data hold time requirement is 300 ns, T_HDDAT should be set to 6.

Bus clock frequency: The I2C bus clock frequency is defined by the T_SCLHI bit and T_SCLRATIO bit of the SETUP register. The calculation formulas are shown in (5) and (6):

$$SCL\ HighPeriod = (2 \times Clk_Period) + (2 + T_SP + T_SCLHI) \times Clk_Period \times (TPM + 1) \tag{5}$$

$$SCL\ LowPeriod = (2 \times Clk_Period) + (2 + T_SP + T_SCLHI \times T_SCLRATIO) \times Clk_Period \times (TPM + 1) \tag{6}$$

48.2.3 Master Mode

Data transmission without DMA:

1. Set timing parameters
2. Configure master mode (MASTER=1) and enable the controller (IICEN=1)

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3. Set data length DATACNT, transmission direction DIR, and transmission phase selection PHASE_START / ADDR / DATA / STOP
4. Configure target slave address ADDRESS
5. Set CMPL to generate an interrupt when transmission is complete, set FIFOEMPTY to generate an interrupt when FIFO is empty
6. Write 0x1 to the COMMAND register to start transmission
7. Wait for interrupt: If the FIFO is empty, write data to the DATA register; if all data has been stored in the FIFO, disable the FIFO empty interrupt, otherwise repeat step 7; if the transmission is complete, check ADDRHIT to ensure that the target slave has correctly received the transmission, write 1 to the CMPL register to clear the interrupt, and go to step 8.
8. Disable all interrupts, check DATACNT to ensure all data transmission is completed

Data reception without DMA:

1. Set timing parameters
2. Configure master mode (MASTER=1) and enable the controller (IICEN=1)
3. Set data length DATACNT, transmission direction DIR, and transmission phase selection PHASE_START / ADDR / DATA / STOP
4. Configure target slave address ADDRESS
5. Set CMPL to generate an interrupt when transmission is complete, set FIFOFULL to generate an interrupt when FIFO is full
6. Write 0x1 to the COMMAND register to start transmission
7. Wait for interrupt: If the FIFO is full, read data from the DATA register; if all data has been stored in the FIFO, disable the FIFO empty interrupt, otherwise repeat step 7; if the transmission is complete, check ADDRHIT to ensure that the target slave has correctly received the transmission, write 1 to the CMPL register to clear the interrupt, and go to step 8.
8. Disable all interrupts, check DATACNT to ensure all data transmission is completed

Data transmission with DMA:

1. Set timing parameters
2. Configure master mode (MASTER=1), enable the controller (IICEN=1), and enable DMA function (DMAEn=1)
3. Configure DMA controller IP
4. Set data length DATACNT, transmission direction DIR, and transmission phase selection PHASE_START / ADDR / DATA / STOP
5. Configure target slave address ADDRESS
6. Set CMPL to generate an interrupt when transmission is complete
7. Write 0x1 to the COMMAND register to start transmission
8. Wait for the interrupt, check DATACNT to ensure all data transmission is completed

48.2.4 Slave Mode

Data transmission without DMA:

1. Set slave address
2. Set timing parameters

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3. Configure slave mode (MASTER=0) and enable the controller (IICEN=1)
4. Enable address hit interrupt ADDRHIT and transmission complete interrupt CMPL
5. Wait for address hit interrupt: Read transmission direction DIR, check GENCALL to confirm whether it is a general call. If DIR is slave receive, enable FIFO full interrupt and go to step 6; if DIR is slave transmit, enable FIFO empty interrupt and go to step 7
6. Slave receive: Wait for FIFO full interrupt, read data until transmission complete interrupt is received
7. Slave transmit: Wait for FIFO empty interrupt, write data until transmission complete interrupt is received
8. Check the DATACNT register to confirm the amount of transmitted data, and clear the transmission complete interrupt

Data transmission with DMA:

1. Set slave address
2. Set timing parameters
3. Configure slave mode (MASTER=0) and enable the controller (IICEN=1)
4. Enable address hit interrupt ADDRHIT and transmission complete interrupt CMPL
5. Wait for address hit interrupt: Read transmission direction DIR, if it is not the expected transmission direction, go to the data transmission process without DMA. Check GENCALL to confirm whether it is a general call
6. Slave receive: Wait for FIFO full interrupt, read data until transmission complete interrupt is received
7. Slave transmit: Wait for FIFO empty interrupt, write data until transmission complete interrupt is received
8. Check the DATACNT register to confirm the amount of transmitted data, and clear the transmission complete interrupt

48.2.5 Interrupts

I2C supports the following interrupt types: transmission complete interrupt, byte receive interrupt, byte transmit interrupt, START interrupt, STOP interrupt, bus arbitration loss interrupt, address hit interrupt, FIFO half-empty/half-full interrupt, FIFO full interrupt, FIFO empty interrupt, etc.

By configuring the interrupt enable register INTEN (0x14), I2C can support the above interrupts. The specific interrupt type and detailed information can be distinguished via the STATUS (0x18) register.

48.2.6 DMA

I2C supports DMA function, and supports DMA operation in both master and slave modes. The relevant DMA configuration required is described in detail in the introduction of master mode and slave mode in this chapter for DMA-based data transmission.

48.3 I2C Registers

48.3.1 Register Description

The list of I2C registers is as follows:

I2C0 base address: 0xF0060000

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I2C1 base address: 0xF0064000

I2C2 base address: 0xF0068000

I2C3 base address: 0xF006C000

Offset	Register Name	Description	Reset Value
0x0010	CFG	Configuration Register	0x00000001
0x0014	INTEN	Interrupt Enable Register	0x00000000
0x0018	STATUS	Status Register	0x00000001
0x001C	ADDR	Address Register	0x00000000
0x0020	DATA	Data Register	0x00000000
0x0024	CTRL	Control Register	0x00905E00
0x0028	CMD	Command Register	0x00000000
0x002C	SETUP	Setup Register	0x05252100
0x0030	TPM	I2C Timing Paramater Multiplier	0x00000000

Table 202: I2C Register List

48.3.2 Detailed Register Information

The detailed description of I2C registers is as follows:

48.3.3 CFG (0x10)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																														FIFOSIZE	
N/A																														RO	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

CFG [31:0]

Bits	Name	Description
1-0	FIFOSIZE	FIFO Size: 0: 2 bytes 1: 4 bytes 2: 8 bytes 3: 16 bytes

CFG Bits

48.3.4 INTEN (0x14)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																						CMPL	BYTERECV	BYTETRANS	START	STOP	ARBLOSE	ADDRHIT	FIFOHALF	FIFOFULL	FIFOEMPTY
N/A																						RW	RW	RW	RW	RW	RW	RW	RW	RW	RW
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

INTEN [31:0]

Bits	Name	Description
9	CMPL	Set to enable the Completion Interrupt. Master: interrupts when a transaction is issued from this master and completed without losing the bus arbitration. Slave: interrupts when a transaction addressing the controller is completed.
8	BYTERECV	Set to enable the Byte Receive Interrupt. Interrupts when a byte of data is received Auto-ACK will be disabled if this interrupt is enabled, that is, the software needs to ACK/NACK the received byte manually.
7	BYTETRANS	Set to enable the Byte Transmit Interrupt. Interrupts when a byte of data is transmitted.
6	START	Set to enable the START Condition Interrupt. Interrupts when a START condition/repeated START condition is detected.
5	STOP	Set to enable the STOP Condition Interrupt Interrupts when a STOP condition is detected.
4	ARBLOSE	Set to enable the Arbitration Lose Interrupt. Master: interrupts when the controller loses the bus arbitration Slave: not available in this mode.
3	ADDRHIT	Set to enable the Address Hit Interrupt. Master: interrupts when the addressed slave returned an ACK. Slave: interrupts when the controller is addressed.
2	FIFOHALF	Set to enable the FIFO Half Interrupt. Receiver: Interrupts when the FIFO is half-empty, i.e, there is >= 1/2 entries in the FIFO. Transmitter: Interrupts when the FIFO is half-empty, i.e. there is <= 1/2 entries in the FIFO. This interrupt depends on the transaction direction; don’t enable this interrupt unless the transfer direction is determined, otherwise unintended interrupts may be triggered.
1	FIFOFULL	Set to enable the FIFO Full Interrupt. Interrupts when the FIFO is full.

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Bits	Name	Description
0	FIFOEMPTY	Set to enabled the FIFO Empty Interrupt Interrupts when the FIFO is empty.

INTEN Bits

48.3.5 STATUS (0x18)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																	LINESDA	LINESCL	GENCALL	BUSBUSY	ACK	CMPL	BYTERECV	BYTETRANS	START	STOP	ARBLOSE	ADDRHIT	FIFOHALF	FIFOFULL	FIFOEMPTY
N/A																	RO	RO	RO	RO	RO	W1C	W1C	W1C	W1C	W1C	W1C	W1C	RO	RO	RO
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

STATUS [31:0]

Bits	Name	Description
14	LINESDA	Indicates the current status of the SDA line on the bus 1: high 0: low
13	LINESCL	Indicates the current status of the SCL line on the bus 1: high 0: low
12	GENCALL	Indicates that the address of the current transaction is a general call address: 1: General call 0: Not general call
11	BUSBUSY	Indicates that the bus is busy The bus is busy when a START condition is on bus and it ends when a STOP condition is seen on bus 1: Busy 0: Not busy
10	ACK	Indicates the type of the last received/transmitted acknowledgement bit: 1: ACK 0: NACK

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Bits	Name	Description
9	CMPL	Transaction Completion Master: Indicates that a transaction has been issued from this master and completed without losing the bus arbitration Slave: Indicates that a transaction addressing the controller has been completed. This status bit must be cleared to receive the next transaction; otherwise, the next incoming transaction will be blocked.
8	BYTERECV	Indicates that a byte of data has been received.
7	BYTETRANS	Indicates that a byte of data has been transmitted.
6	START	Indicates that a START Condition or a repeated START condition has been transmitted/received.
5	STOP	Indicates that a STOP Condition has been transmitted/received.
4	ARBLOSE	Indicates that the controller has lost the bus arbitration.
3	ADDRHIT	Master: indicates that a slave has responded to the transaction. Slave: indicates that a transaction is targeting the controller (including the General Call).
2	FIFOHALF	Transmitter: Indicates that the FIFO is half-empty.
1	FIFOFULL	Indicates that the FIFO is full.
0	FIFOEMPTY	Indicates that the FIFO is empty.

STATUS Bits

48.3.6 ADDR (0x1C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																						ADDR									
N/A																						RW									
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ADDR [31:0]

Bits	Name	Description
9-0	ADDR	The slave address. For 7-bit addressing mode, the most significant 3 bits are ignored and only the least-significant 7 bits of Addr are valid

ADDR Bits

48.3.7 DATA (0x20)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																								DATA							
N/A																								RW							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DATA [31:0]

Bits	Name	Description
7-0	DATA	Write this register to put one byte of data to the FIFO. Read this register to get one byte of data from the FIFO.

DATA Bits

48.3.8 CTRL (0x24)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATACNT_HIGH								RESET_LEN				RSVD					RESET_HOLD_SCKIN	RESET_ON	PHASE_START	PHASE_ADDR	PHASE_DATA	PHASE_STOP	DIR	DATACNT							
RW								RW				N/A					RW	RW	RW	RW	RW	RW	RW	RW							
0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	0	1	0	1	1	1	1	0	0	0	0	0	0	0	0	0

CTRL [31:0]

Bits	Name	Description
31-24	DATACNT_HIGH	Data counts in bytes. Master: The number of bytes to transmit/receive. 0 means max length. DataCnt will be decreased by one for each byte transmitted/received. Slave: the meaning of DataCnt depends on the DMA mode: If DMA is not enabled, DataCnt is the number of bytes transmitted/received from the bus master. It is reset to 0 when the controller is addressed and then increased by one for each byte of data transmitted/received. If DMA is enabled, DataCnt is the number of bytes to transmit/receive. It will not be reset to 0 when the slave is addressed and it will be decreased by one for each byte of data transmitted/received.

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Bits	Name	Description
23-20	RESET_LEN	reset clock cycles. the clock high/low time is defined by Setup.T_SCLHi, 50% duty cycle.
14	RESET_HOLD_SCKIN	set to hold input clock to high when reset is active
13	RESET_ON	set to send reset signals(just toggle clock bus defined by reset_len). this register is clered when reset is end, can't be cleared by software
12	PHASE_START	Enable this bit to send a START condition at the beginning of transaction. Master mode only.
11	PHASE_ADDR	Enable this bit to send the address after START condition. Master mode only.
10	PHASE_DATA	Enable this bit to send the data after Address phase. Master mode only.
9	PHASE_STOP	Enable this bit to send a STOP condition at the end of a transaction. Master mode only.
8	DIR	Transaction direction Master: Set this bit to determine the direction for the next transaction. 0: Transmitter 1: Receiver Slave: The direction of the last received transaction. 0: Receiver 1: Transmitter
7-0	DATACNT	Data counts in bytes. Master: The number of bytes to transmit/receive. 0 means max length. DataCnt will be decreased by one for each byte transmitted/received. Slave: the meaning of DataCnt depends on the DMA mode: If DMA is not enabled, DataCnt is the number of bytes transmitted/received from the bus master. It is reset to 0 when the controller is addressed and then increased by one for each byte of data transmitted/received. If DMA is enabled, DataCnt is the number of bytes to transmit/receive. It will not be reset to 0 when the slave is addressed and it will be decreased by one for each byte of data transmitted/received.

CTRL Bits

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48.3.9 CMD (0x28)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																													CMD		
N/A																													RW		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CMD [31:0]

Bits	Name	Description
2-0	CMD	Write this register with the following values to perform the corresponding actions: 0x0: no action 0x1: issue a data transaction (Master only) 0x2: respond with an ACK to the received byte 0x3: respond with a NACK to the received byte 0x4: clear the FIFO 0x5: reset the I2C controller (abort current transaction, set the SDA and SCL line to the open-drain mode, reset the Status Register and the Interrupt Enable Register, and empty the FIFO) When issuing a data transaction by writing 0x1 to this register, the CMD field stays at 0x1 for the duration of the entire transaction, and it is only cleared to 0x0 after when the transaction has completed or when the controller loses the arbitration. Note: No transaction will be issued by the controller when all phases (Start, Address, Data and Stop) are disabled.

CMD Bits

48.3.10 SETUP (0x2C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD			T_SUDAT					T_SP			T_HDDAT					RSVD		T_SCLRADIO	T_SCLHI									DMAEN	MASTER	ADDRESSING	IICEN
N/A			RW					RW			RW					N/A		RW	RW									RW	RW	RW	RW
0	0	0	0	0	1	0	1	0	0	1	0	0	1	0	1	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0

SETUP [31:0]

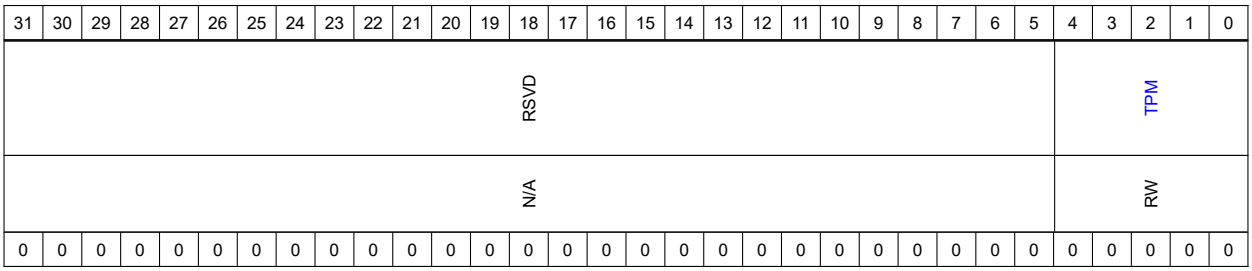
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Bits	Name	Description
28-24	T_SUDAT	T_SUDAT defines the data setup time before releasing the SCL. Setup time = (2 * tpclk) + (2 + T_SP + T_SUDAT) * tpclk* (TPM+1) tpclk = PCLK period TPM = The multiplier value in Timing Parameter Multiplier Register
23-21	T_SP	T_SP defines the pulse width of spikes that must be suppressed by the input filter. Pulse width = T_SP * tpclk* (TPM+1)
20-16	T_HDDAT	T_HDDAT defines the data hold time after SCL goes LOW Hold time = (2 * tpclk) + (2 + T_SP + T_HDDAT) * tpclk* (TPM+1)
13	T_SCLRADIO	The LOW period of the generated SCL clock is defined by the combination of T_SCLRatio and T_SCLHi values. When T_SCLRatio = 0, the LOW period is equal to HIGH period. When T_SCLRatio = 1, the LOW period is roughly two times of HIGH period. SCL LOW period = (2 * tpclk) + (2 + T_SP + T_SCLHi * ratio) * tpclk * (TPM+1) 1: ratio = 2 0: ratio = 1 This field is only valid when the controller is in the master mode.
12-4	T_SCLHI	The HIGH period of generated SCL clock is defined by T_SCLHi. SCL HIGH period = (2 * tpclk) + (2 + T_SP + T_SCLHi) * tpclk* (TPM+1) The T_SCLHi value must be greater than T_SP and T_HDDAT values. This field is only valid when the controller is in the master mode.
3	DMAEN	Enable the direct memory access mode data transfer. 1: Enable 0: Disable
2	MASTER	Configure this device as a master or a slave. 1: Master mode 0: Slave mode
1	ADDRESSING	I2C addressing mode: 1: 10-bit addressing mode 0: 7-bit addressing mode
0	IICEN	Enable the I2C controller. 1: Enable 0: Disable

SETUP Bits

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48.3.11 TPM (0x30)



TPM [31:0]

Bits	Name	Description
4-0	TPM	A multiplication value for I2C timing parameters. All the timing parameters in the Setup Register are multiplied by (TPM+1).

TPM Bits

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49 Controller Area Network (MCAN)

This chapter introduces the functions and features of the Controller Area Network (MCAN).

49.1 Overview

The features of MCAN are as follows:

- Supports CAN 2.0B protocol, supports data payload up to 8 bytes, data rate up to 1 Mbit/s;
- Supports CAN FD protocol, supports data payload up to 64 bytes, data rate up to 8 Mbit/s;
- Supports baud rate prescaler from 1 to 1/16, which can be flexibly configured with the system CAN clock divider;
- Multiple MCAN modules share HRAM (AHB SRAM, 32 KB) as message memory
  - CPU can access HRAM via AHB bus, different MCAN modules share the same HRAM by configuring the message RAM to different addresses (only lower 16-bit address is required)
  - Can be flexibly configured as different transmit buffers, receive buffers, filters, and event buffers.
- Programmable ID CODE bits and MASK bits;
- Both PTB/STB support single-shot transmission mode;
- Supports silent mode;
- Supports loopback mode;
- Supports standby mode;
- Supports capturing transmission error types and locating arbitration failure positions;
- Programmable error warning threshold;
- Compliant with ISO 11898-1:2015
- Supports AUTOSAR
- Supports SAE J1939
- Supports debugging
- Supports DMA
- Supports hardware timestamp according to CiA 603
  - 16 groups of 32-bit timestamps, or 8 groups of 64-bit timestamps
  - TSU can use its own timestamp, or use external timestamp
  - Can output timestamp for other MCAN modules in the system

49.2 CAN Architecture Diagram

The structure diagram of this module is shown in Figure 71. The functions of each main unit are as follows:

Transmit processing unit: Controls the transmit transmission from the message storage to the CAN core, can be configured with up to 32 transmit buffers.

Receive processing unit: Controls which matching messages can be received, can be configured with up to 64 receive buffers.

CAN core: The CAN core controls the transmit and receive shift registers, handles all functions of the ISO 11898-1:2015 protocol, and supports 11-bit and 29-bit identifiers.

Interrupt control: Handles interrupt generation and clearing.

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Message storage interface and message storage: Stores transmitted and received messages.

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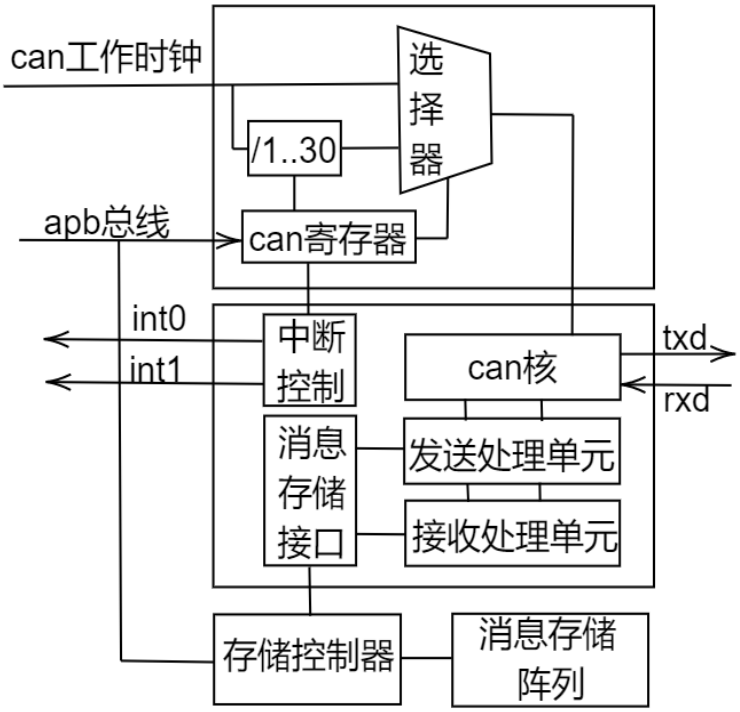


Figure 71: CAN Block Diagram

49.3 Pin Description

Table 203: CAN Pin Description

Pin	Direction	Function Description
CANn[1]_TXD	Output	CAN bus data output
CANn_RXD	Input	CAN bus data input
CANn_STANDBY	Output	Indicates that the CAN controller is in standby mode

[1] n=0,..,3

49.4 Function Description

49.4.1 General CAN Description

CAN pin functions

- CANn_TXD: CAN bus output data.
- CANn_RXD: CAN bus input data.
- CANn_STANDBY: When this signal is high, it indicates that the CAN controller is in standby mode, and the CAN clock can be turned off to save power at this time.

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49.4.2 Bit Timing

Bit time refers to the period T of each bit on the CAN bus, T=1/baud rate, which consists of three parts as shown in Figure 72:

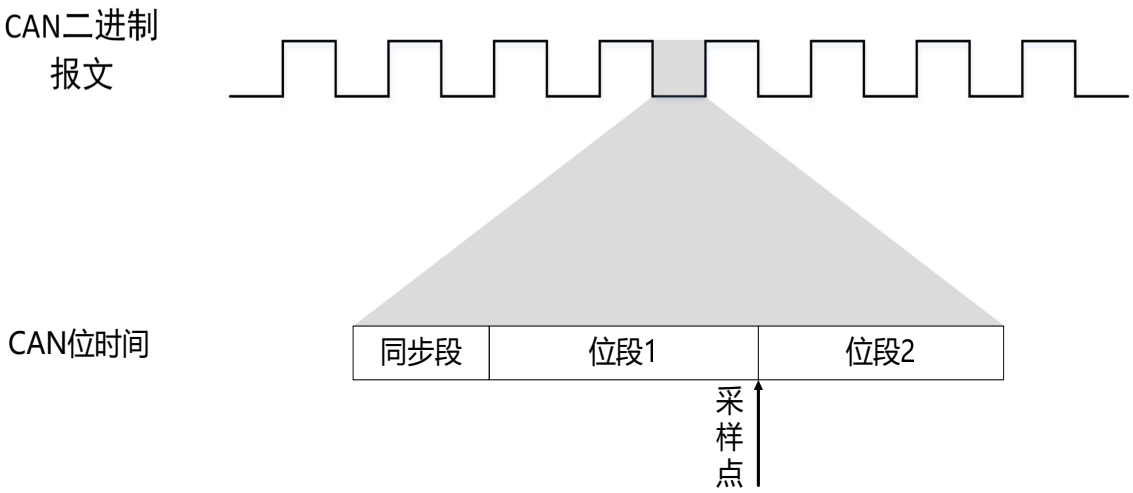


Figure 72: CAN Bit Timing

Sync Segment: Refers to the time required from when the value changes until it stabilizes, which is a fixed 1 time quantum here.

Bit Segment 1: Defines the sampling point, including the propagation segment and phase buffer segment 1 defined in the CAN standard. Its period is configurable, with a maximum of 16 time quanta and a minimum of 1 time quantum.

Bit Segment 2: Defines the position of the transmission point, which refers to the phase buffer segment 2 in the CAN standard. Its period is configurable, with a maximum of 8 time quanta and a minimum of 1 time quantum.

Resynchronization Jump Width: The resynchronization jump width (SJW) defines the extension or shortening of the bit segment, and its value is configurable, with a maximum of 4 time quanta and a minimum of 1 time quantum.

Therefore, the time calculation for each segment is as follows:

Time quantum tq = (CAN_DBTP.DBRP[4:0] + 1) * can_tq_clk_period

Sync Segment time = 1 * tq

Bit Segment 1 time = tq * (CAN_DBTP.DTSEG1[4:0] + 1)

Bit Segment 2 time = tq * (CAN_DBTP.DTSEG2[3:0] + 1)

For example, if the CAN clock frequency is set to 80 MHz (i.e. 12.5 ns), and you want to configure the baud rate to 1 M, you can configure the time quantum tq to 8 clock cycles, that is, configure the CAN_DBTP.DBRP[4:0] register to 7. At this time, one time quantum is configured to 100 ns, the sync segment time is 1*tq=100 ns, the Bit Segment 1 time can be set according to the sampling point, for example, set the CAN_DBTP.DTSEG1[4:0] register to 4, that is, Bit Segment 1 is 5tq = 500 ns. The Bit Segment 2 time can be determined according to the baud rate to be set, for example, to set a baud rate of 1 M, Bit Segment 2 needs to be configured as 1000 ns-100 ns-500 ns=400 ns, that is, the CAN_DBTP.DTSEG2[3:0] register is set to 3. In summary, to set a baud rate of 1 M when the CAN clock frequency is 80 M, you can configure the CAN_DBTP.DBRP[4:0] register to 7, the CAN_DBTP.DTSEG1[4:0] register to 4, and the CAN_DBTP.DTSEG2[3:0] register to 3.

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49.4.3 Normal Operation Mode

After CAN initialization is completed, the init bit of the CCCR register can be set to 0, and the CAN controller will be ready for communication. When the message address received from the CAN bus matches the rules set by the CAN controller, the received data is stored in the receive buffer or receive FIFO 0/1. The data placed in the transmit buffer or transmit FIFO 0/1 will also automatically be ready for transmission to the CAN bus after the register TXBAR is configured.

49.4.4 CAN FD Operation Mode

Compared with the normal mode, the FD mode has two differences: first, the length of the data part is greatly expanded, and the DLC supports up to 64 bytes. Second, the control part, data part and check part of the CAN FD frame are at a higher bit rate.

CAN FD can be enabled by setting the FDOE bit of CCCR. When CAN FD is enabled, FD frames can be transmitted and received, as well as normal frames. When the FDF bit in the transmit buffer or receive buffer is 0, the CAN controller will parse the frame as a normal frame; when the FDF bit is 1 and FDOE is enabled, the CAN controller will parse it as an FD frame. In actual use, after the system starts, all nodes transmit normal frame formats. Only when it is confirmed that all nodes in the system support CAN FD frame format, all nodes should switch to FD operation together.

In CAN FD format, the meaning of DLC is slightly different from that of normal CAN format. When the DLC value is 0 to 8, the meaning is the same as that of normal CAN. The meaning of values 9 to 15 is shown in Table 204:

Table 204: DLC Meaning

DLC Value	9	10	11	12	13	14	15
Corresponding Number of Bytes	12	16	20	24	32	48	64

49.4.5 Silent Mode

This mode can be enabled by setting the MON bit of CCCR to 1. In silent mode (ISO 11898-1:2015), the CAN controller only receives data frames and does not send data frames. The advantage of this mode is that it can be used to observe the data flow of the CAN bus without affecting the CAN bus, and its structure is shown in Figure 73.

49.4.6 Standby Mode

This mode can be enabled by setting the CSR bit of CCCR to 1. When the CSR bit of CCCR is set to 1, the CAN controller will wait for all tasks in the transmit buffer to complete, and then the INIT bit of CCCR is automatically set to 1 to prevent users from adding new tasks. When successfully entering standby mode, the CSA of CCCR will become 1, then you can configure the register to set the STANDBY output to high, and finally turn off the input clock to save power. When you want to exit standby mode, first turn on the clock of the CAN module, then write 0 to the CSR bit of CCCR, and finally the software clears the INIT bit of CCCR to wake up.

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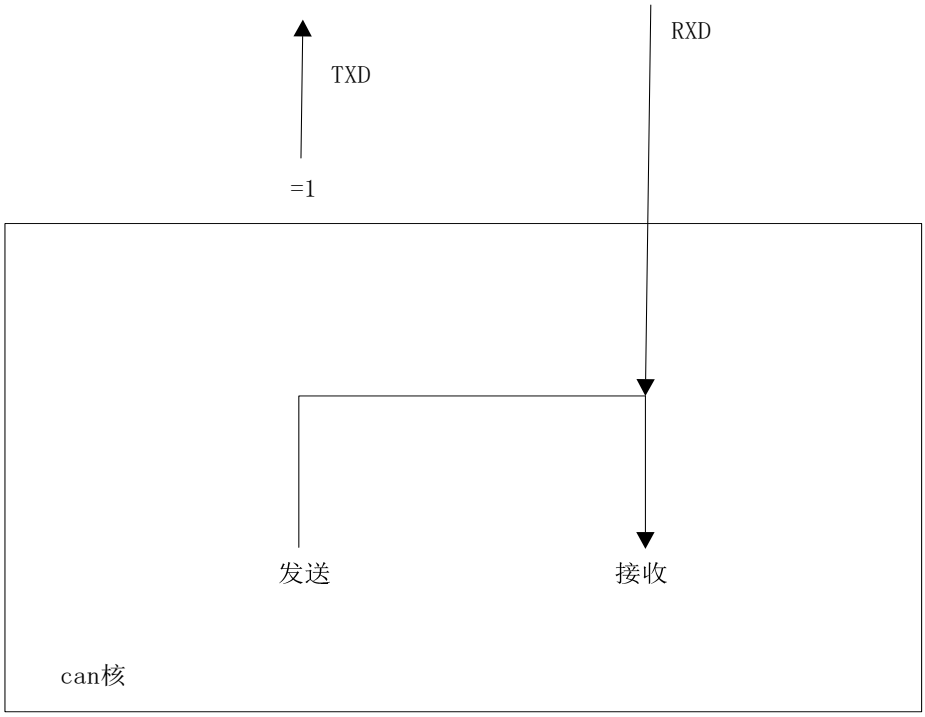


Figure 73: CAN Silent Mode

49.4.7 External Loopback Mode

This mode can be enabled by setting the LBCK bit of the TEST register. Before setting the TEST register, you need to set the TEST bit of CCCR to 1. In this loopback mode, CAN places the output data frame directly to the input port and stores the data frame in the receive buffer or receive FIFO, and the input RXD is disconnected from the external port. As can be seen in Figure 74, the relationship between the transmit port and the receive port of external loopback is shown. External loopback is used for hardware self-test.

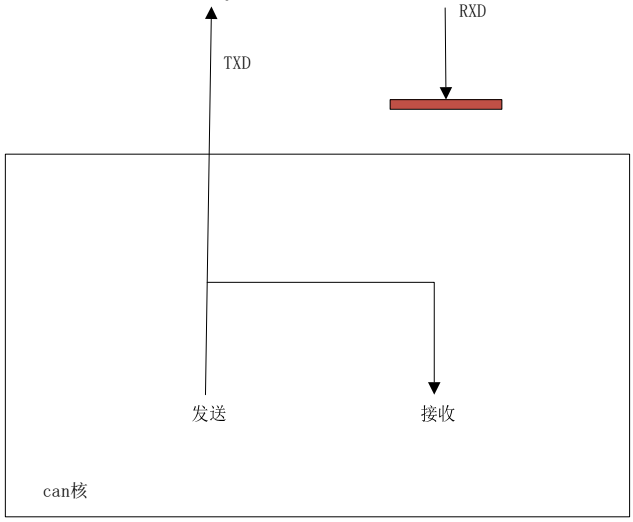


Figure 74: CAN External Loopback

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49.4.8 Internal Loopback Mode

This mode can be enabled by setting the LBCK bit of the TEST register and the MON bit of the CCCR register. Before setting the TEST register, you need to set the TEST bit of CCCR to 1. This mode can be used for hot self-test, which can perform self-test on CAN without affecting the entire running CAN system. In this case, the TXD of CAN is set to 1, and RXD is disconnected from the outside, as shown in Figure 75.

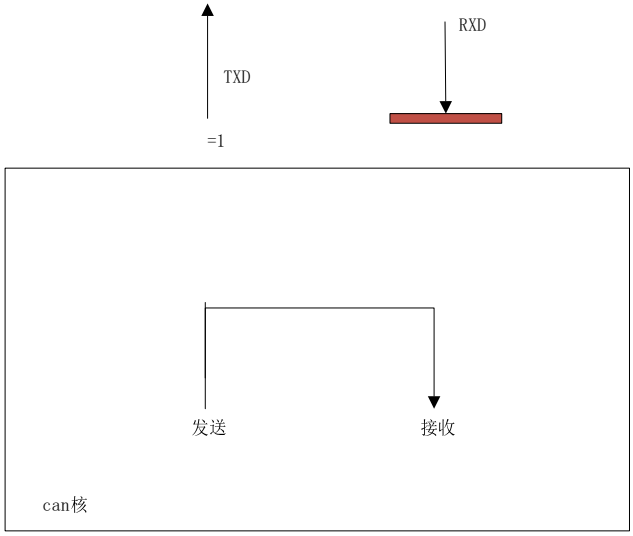


Figure 75: CAN Internal Loopback

49.4.9 Timestamp Generation

The CAN controller provides 4 64-bit loop counters internally. When CAN is used, the time source can be selected through the tsu_tbin_sel of register GLB_CTRL. The Time Stamp Unit (TSU) can store 8 64-bit times or 16 32-bit times. Before use, you need to configure the UTSU bit of register CCCR to 1 first. When using timestamps to receive messages, at the end of receiving a frame, the Time Stamp Unit will be triggered to record the value of the currently selected counter, and according to the value of the Timestamp Pointer (TSP), the counter value is stored in the corresponding timestamp record register, while the timestamp pointer is incremented by 1. And the value of the timestamp pointer (TSP) is placed in RXTSP in R1B of the receive buffer format.

49.4.10 Receive Buffer Data Format

The data format of the receive buffer is shown in Figure 76, where:

R1A: When the Time Stamp Unit (TSU) is not used (UTSU bit of CCCR is 0), RXTS[15:0] of R1A saves the 16-bit timestamp of the CAN controller itself at this time.

R1B: When the Time Stamp Unit (TSU) is used (UTSU bit of CCCR is 1) and the SSYNC/ESYNC bit in the filter is set, the TSC bit of R1B is 1 indicating that the timestamp capture is successful, and RXTSP of R1B saves the TSU timestamp register number, and the timestamp register of the corresponding number saves a 32-bit or 64-bit timestamp.

- ESI (Bit 31 of R0): Error State Indicator.
  - 0: Transmission node error
  - 1: No error in transmission node
- XTD (Bit 30 of R0): Extended Identifier.

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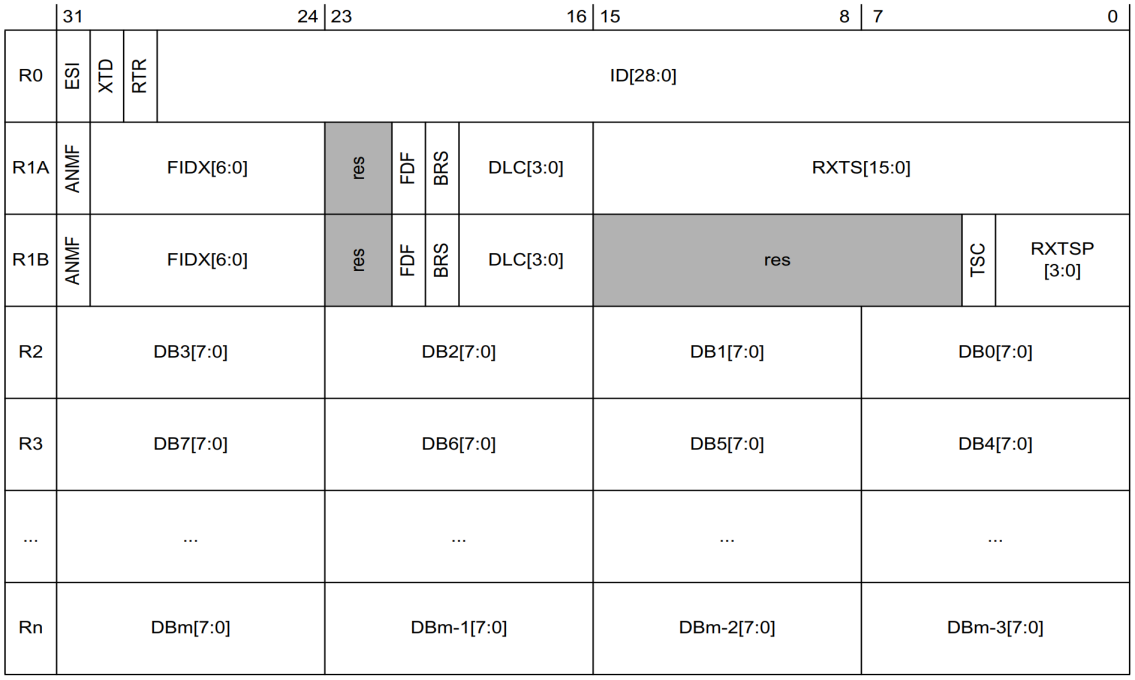


Figure 76: CAN Receive Buffer Data Format

  - 0: 11-bit standard identifier
  - 1: 29-bit extended identifier
- RTR (Bit 29 of R0): Remote Transmission Request.
  - 0: Received frame is a data frame
  - 1: Received frame is a remote frame
- ID (Bits 28 to 0 of R0): Identifier.
  - Valid bits of standard identifier are ID[28:18], valid bits of extended identifier are ID[28:0]
- ANMF (Bit 31 of R1): Accept Non-Matching Frame.
  - 0: Received frame matches the receive filter
  - 1: Received frame does not match the receive filter
- FIDX[6:0] (Bits 30 to 24 of R1): Filter Index.
  - Range from 0 to 127, indicating the index of the matched receive filter
- FDF (Bit 21 of R1): FD Frame Format.
  - 0: Indicates traditional CAN frame format
  - 1: Indicates FD frame format
- BRS (Bit 20 of R1): Bit Rate Switch.
  - 0: Received frame has no bit rate switch
  - 1: Received frame has bit rate switch
- DLC[3:0] (Bits 19 to 16 of R1): Data Length Code.
  - 0 to 8: Indicates that the data length of the received frame is 0 to 8 bytes
  - 9 to 15: For normal frames, it indicates that the received frame has 8 bytes; for FD frames, it indicates 12/16/20/24/32/48/64 bytes respectively
- RXTS[15:0] (Bits 15 to 0 of R1A): Receive Timestamp.

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  - Timestamp counter value captured at the start of frame reception.
- TSC (Bit 4 of R1B): Timestamp Capture.
  - 0: No timestamp captured
  - 1: Timestamp captured, and the timestamp register number is in RXTSP position of R1B
- RXTSP[3:0] (Bits 3 to 0 of R1B): Receive Timestamp Pointer.
  - Indicates which timestamp register the timestamp at reception is stored in, i.e. the number of the timestamp register.
- DB3[7:0] (Bits 31 to 24 of R2): Data Byte 3.
- DB2[7:0] (Bits 23 to 16 of R2): Data Byte 2.
- DB1[7:0] (Bits 15 to 8 of R2): Data Byte 1.
- DB0[7:0] (Bits 7 to 0 of R2): Data Byte 0.
- DB3[7:0] (Bits 31 to 24 of R3): Data Byte 7.
- DB2[7:0] (Bits 23 to 16 of R3): Data Byte 6.
- DB1[7:0] (Bits 15 to 8 of R3): Data Byte 5.
- DB0[7:0] (Bits 7 to 0 of R3): Data Byte 4.
- ... ... : ...
- DB3[7:0] (Bits 31 to 24 of Rn): Data Byte m.
- DB2[7:0] (Bits 23 to 16 of Rn): Data Byte m-1.
- DB1[7:0] (Bits 15 to 8 of Rn): Data Byte m-2.
- DB0[7:0] (Bits 7 to 0 of Rn): Data Byte m-3.

49.4.11 Transmit Buffer Data Format

The data format of the transmit buffer is shown in Figure 77

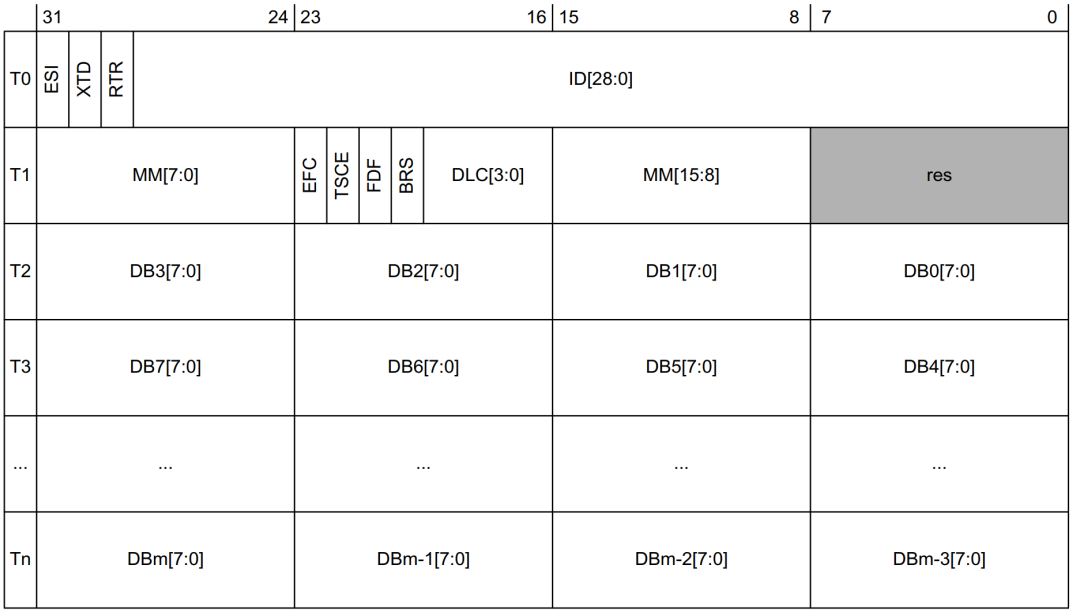


Figure 77: CAN Transmit Buffer Data Format

- ESI (Bit 31 of T0): Error State Indicator.
  - 0: Transmission node error
  - 1: No error in transmission node

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- XTD (Bit 30 of T0): Extended Identifier.
  - 0: 11-bit standard identifier
  - 1: 29-bit extended identifier
- RTR (Bit 29 of T0): Remote Transmission Request.
  - 0: Transmitted frame is a data frame
  - 1: Transmitted frame is a remote frame
- ID (Bits 28 to 0 of T0): Identifier.
  - Valid bits of standard identifier are ID[28:18], valid bits of extended identifier are ID[28:0]
- MM[7:0] (Bits 31 to 24 of T1): Lower 8 bits of Message Marker.
  - Written by CPU when configuring transmit buffer data, then copied to the transmit time FIFO for identification and judgment of transmitted messages
- EFC (Bit 23 of T1): Event FIFO Control.
  - 0: Do not save transmission behavior events
  - 1: Save transmission behavior events
- BTSCE (Bit 22 of T1): TSU Timestamp Capture Enable, only valid when the UTSU bit of CCCR register is set
  - 0: Timestamp capture disabled
  - 1: Timestamp capture enabled
- FDF (Bit 21 of T1): FD Frame Format.
  - 0: Indicates that the transmitted frame is in normal frame format
  - 1: Indicates that the transmitted frame is in CAN FD frame format
- BRS (Bit 20 of T1): Bit Rate Switch.
  - 0: Transmitted frame has no bit rate switch
  - 1: Transmitted frame has bit rate switch
- DLC[3:0] (Bits 19 to 16 of T1): Data Length Code.
  - 0 to 8: Indicates that the data length of the transmitted frame is 0 to 8 bytes
  - 9 to 15: For normal frames, it indicates that the transmitted frame has 8 bytes; for FD frames, it indicates 12/16/20/24/32/48/64 bytes respectively
- MM[15:8] (Bits 15 to 8 of T1): Higher 8 bits of Message Marker.
  - Written by CPU when configuring transmit buffer data, then copied to the transmit time FIFO for identification and judgment of transmitted messages.
- DB3[7:0] (Bits 31 to 24 of T2): Data Byte 3.
- DB2[7:0] (Bits 23 to 16 of T2): Data Byte 2.
- DB1[7:0] (Bits 15 to 8 of T2): Data Byte 1.
- DB0[7:0] (Bits 7 to 0 of T2): Data Byte 0.
- DB3[7:0] (Bits 31 to 24 of T3): Data Byte 7.
- DB2[7:0] (Bits 23 to 16 of T3): Data Byte 6.
- DB1[7:0] (Bits 15 to 8 of T3): Data Byte 5.
- DB0[7:0] (Bits 7 to 0 of T3): Data Byte 4.
- ... ... : ...
- DB3[7:0] (Bits 31 to 24 of Tn): Data Byte m.
- DB2[7:0] (Bits 23 to 16 of Tn): Data Byte m-1.

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- DB1[7:0] (Bits 15 to 8 of Tn): Data Byte m-2.
- DB0[7:0] (Bits 7 to 0 of Tn): Data Byte m-3.

49.4.12 Transmit Event Data Format

The transmit event data saves information about the transmission. The CPU can read the transmit event FIFO in the order of transmission. The status of the transmit event FIFO can be read through the register TXEFS. The meaning of the data in the FIFO is shown in Figure 78. Among them:

E1A: When the Time Stamp Unit (TSU) is not used (UTSU bit of CCCR is 0) and the WMM of CCCR register is 0, TXTS[15:0] of E1A saves the 16-bit timestamp of the CAN controller itself at this time.

E1B: When the Time Stamp Unit (TSU) is used (UTSU bit of CCCR is 1) or the 16-bit message marker bit is enabled (WMM of CCCR is configured to 1), the TSC bit of E1B is 1 indicating that the timestamp capture is successful, and TXTSP of T1B saves the TSU timestamp register number, and the timestamp register of the corresponding number saves a 32-bit or 64-bit timestamp.

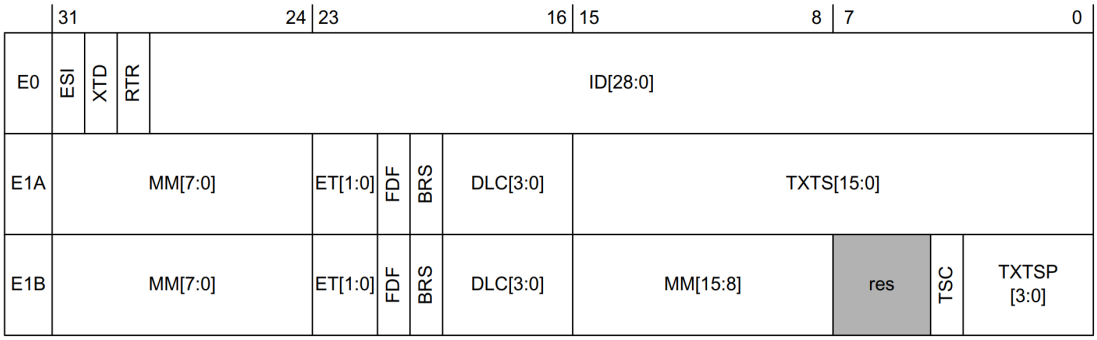


Figure 78: CAN Transmit Event Data Format

- ESI (Bit 31 of E0): Error State Indicator.
  - 0: Transmission node error
  - 1: No error in transmission node
- XTD (Bit 30 of E0): Extended Identifier.
  - 0: 11-bit standard identifier
  - 1: 29-bit extended identifier
- RTR (Bit 29 of E0): Remote Transmission Request.
  - 0: Transmitted frame is a data frame
  - 1: Transmitted frame is a remote frame
- ID (Bits 28 to 0 of E0): Identifier.
  - Valid bits of standard identifier are ID[28:18], valid bits of extended identifier are ID[28:0]
- MM[7:0] (Bits 31 to 24 of E1): Filter Index.
  - Range from 0 to 127, indicating the index of the matched receive filter
- ET[1:0] (Bits 23 to 22 of E1): Event Type.
  - 00: Reserved
  - 01: Transmit event
  - 10: Transmit after cancellation
  - 11: Reserved
- FDF (Bit 21 of E1): FD Frame Format.

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  - 0: Indicates traditional CAN frame format
  - 1: Indicates FD frame format
- BRS (Bit 20 of E1): Bit Rate Switch.
  - 0: Transmitted frame has no bit rate switch
  - 1: Transmitted frame has bit rate switch
- DLC[3:0] (Bits 19 to 16 of E1): Data Length Code.
  - 0 to 8: Indicates that the data length of the transmitted frame is 0 to 8 bytes
  - 9 to 15: For normal frames, it indicates that the transmitted frame has 8 bytes; for FD frames, it indicates 12/16/20/24/32/48/64 bytes respectively
- TXTS[15:0] (Bits 15 to 0 of E1A): Transmit Timestamp.
  - Timestamp counter value captured at the start of frame transmission.
- MM[15:8] (Bits 15 to 8 of E1B): Higher 8 bits of Message Marker.
  - Written by CPU when configuring transmit buffer data, then copied to the transmit time FIFO for identification and judgment of transmitted messages.
- TSC (Bit 4 of E1B): Timestamp Capture.
  - 0: No timestamp captured
  - 1: Timestamp captured, and the timestamp register number is in RXTSP position of R1B
- TXTSP[3:0] (Bits 3 to 0 of E1B): Transmit Timestamp Pointer.
  - Indicates which timestamp register the timestamp at transmission is stored in, i.e. the number of the timestamp register.

49.4.13 Standard Message Identifier Filtering

Up to 128 filters can be configured for 11-bit identifiers. When accessing the filter, its address is the value of the filter start address register SIDFC plus the filter index offset. The meaning of the filter content is shown in Figure 79

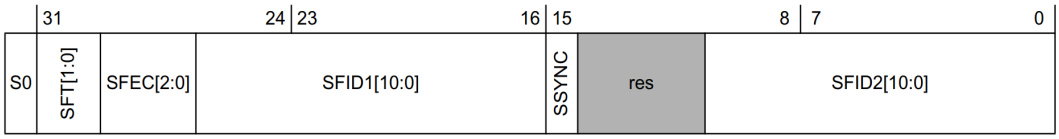


Figure 79: CAN Standard Message Identifier Filter

- SFT[1:0] (Bits 31 to 30 of S0): Standard Filter Type.
  - 00: Identifier range filter, from SFID1 to SFID2, where SFID2>=SFID1
  - 01: Identifier value filter, equal to SFID1 or SFID2
  - 10: Classic mask filter, SFID1 is the filter value, SFID2 is the corresponding mask. When the mask bit is 0, the corresponding bits of SFID1 and ID can be ignored, and the bit is considered matching. When the mask bit is 1, the corresponding bit of ID must be strictly equal to the corresponding bit of SFID1. For example, when all bits of SFID2 are 1, it means that the ID value must be exactly equal to SFID1; when all bits of SFID2 are 0, it means that all IDs match SFID1.
  - 11: Filter disabled
- SFEC[2:0] (Bits 29 to 27 of S0): Standard Filter Configuration.
  - 000: Filter disabled

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  - 001: Store data to receive FIFO 0 after filter match
  - 010: Store data to receive FIFO 1 after filter match
  - 011: Do not receive data after filter match
  - 100: Only set priority after filter match, no storage
  - 101: Set priority and store data to receive FIFO 0 after filter match
  - 110: Set priority and store data to receive FIFO 1 after filter match
  - 111: Store data to receive buffer (Rx buffer) after filter match
- SFID1[10:0] (Bits 26 to 16 of S0): Standard Identifier Filter 1.
  - The first standard identifier filter, used for filtering in different scenarios
- SSYNC (Bit 15 of S0): Standard Synchronization Message. Only valid when the UTSU bit of CCCR is 1, triggers the Time Stamp Unit (TSU) to record the timestamp
  - 0: Disable triggering TSU to record timestamp
  - 1: Enable triggering TSU to record timestamp
- SFID2[10:0] (Bits 10 to 0 of S0): Standard Identifier Filter 2. It has different meanings according to different SFEC values.

  When SFEC value is 000 to 110, it represents the second part of the standard filter.

  When SFEC value is 111:
  - SFID2[10:9] defines whether the received message is stored in the buffer or treated as a debug message.
    - 00: Store to receive buffer (Rx buffer)
    - 01: Debug message A
    - 10: Debug message B
    - 11: Debug message C
  - SFID2[8:6] defines the filter event pins used to control the extended interface
  - SFID2[5:0] defines the offset relative to the storage start address where the received data is stored

49.4.14 Extended Message Identifier Filtering

Up to 64 filters can be configured for 29-bit identifiers. When accessing the filter, its address is the value of the filter start address register XIDFC plus twice the filter index offset. The meaning of the filter content is shown in Figure 80

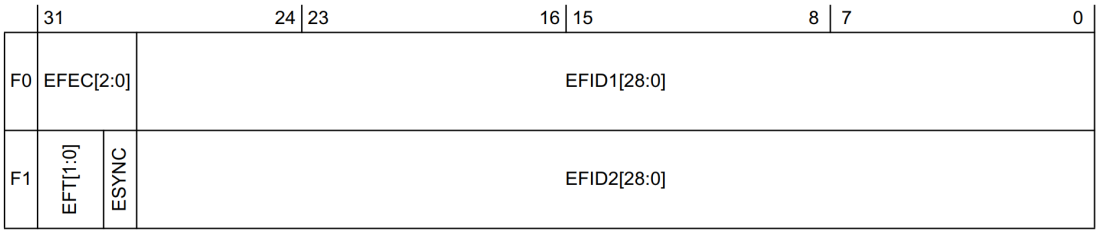


Figure 80: CAN Extended Message Identifier Filter

- EFEC[2:0] (Bits 31 to 29 of F0): Extended Filter Configuration.
  - 000: Filter disabled
  - 001: Store data to receive FIFO 0 after filter match

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  - 010: Store data to receive FIFO 1 after filter match
  - 011: Do not receive data after filter match
  - 100: Only set priority after filter match, no storage
  - 101: Set priority and store data to receive FIFO 0 after filter match
  - 110: Set priority and store data to receive FIFO 1 after filter match
  - 111: Store data to receive buffer (Rx buffer) after filter match
- EFID1[28:0] (Bits 28 to 0 of F0): Extended Identifier Filter 1.
  - The first extended identifier filter, used for filtering in different scenarios
- EFT[1:0] (Bits 31 to 30 of F1): Extended Filter Type.
  - 00: Identifier range filter, from EFID1 to EFID2, where EFID2>=EFID1
  - 01: Identifier value filter, equal to EFID1 or EFID2
  - 10: Classic mask filter, EFID1 is the filter value, EFID2 is the corresponding mask
  - 11: Filter disabled
- ESYNC (Bit 29 of F1): Extended Synchronization Message. Only valid when the UTSU bit of CCCR is 1, triggers the Time Stamp Unit (TSU) to record the timestamp
  - 0: Disable triggering TSU to record timestamp
  - 1: Enable triggering TSU to record timestamp
- EFID2[28:0] (Bits 28 to 0 of F1): Extended Identifier Filter 2. It has different meanings according to different EFEC values.
  When EFEC value is 000 to 110, it represents the second part of the extended filter.
  When EFEC value is 111:
  - EFID2[10:9] defines whether the received message is stored in the buffer or treated as a debug message.
    * 00: Store to receive buffer (Rx buffer)
    * 01: Debug message A
    * 10: Debug message B
    * 11: Debug message C
  - EFID2[8:6] defines the filter event pins used to control the extended interface
  - EFID2[5:0] defines the offset relative to the storage start address where the received data is stored

49.4.15 Receive FIFO

Receive FIFO 0 and Receive FIFO 1 can be configured as 64 entries through registers RXF0C and RXF1C respectively. Received messages can be selected to be stored in the receive FIFO after passing through the receive filter. To avoid receive FIFO overflow, the receive FIFO watermark threshold indicator register RXFnC.FnWM can be used. When the number of entries in the FIFO is greater than or equal to the threshold, the interrupt flag IR.RFnW is issued.

49.4.15.1 Receive FIFO Blocking Mode

The receive FIFO blocking mode is controlled by register RXFnC.FnOM. When this bit is configured to 0 (default value is 0), it indicates that the receive FIFO is in blocking mode. When the receive FIFO is full, if you write a message to the FIFO again, the write will fail, and the written message may be lost. At the same time,

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flag bit RXFns.RFnL is set to 1, and interrupt flag bit IR.RFnL is set.

49.4.15.2 Receive FIFO Overwrite Mode

The receive FIFO overwrite mode is controlled by register RXFnC.FnOM. When this bit is configured to 1 (default value is 0), it indicates that the receive FIFO is in overwrite mode. When the receive FIFO is full, if you write a message to the FIFO again, the earliest message will be overwritten, and the earliest message will be lost. In this case, flag bit RXFns.RFnL will not be set to 1.

49.4.16 Interrupt

This module provides two groups of interrupts. Interrupt sources can be connected to interrupt 0 or interrupt 1 through register configuration. By default, all interrupts are connected to interrupt 0. Interrupt enabling or disabling can be controlled through configuration registers ILE.EINT0 and ILE.EINT1 respectively. At the upper layer, interrupt 0 and interrupt 1 are both connected to one interrupt source, which means that as long as either interrupt 0 or interrupt 1 generates an interrupt, the interrupt will be output. Therefore, interrupt 0 and interrupt 1 can be considered the same, and you can choose either one to use. There are 30 types of interrupts. When the corresponding interrupt bit is enabled and an interrupt is generated for the corresponding bit, the interrupt will be sent. For detailed interrupt types, please refer to the interrupt register IR.

49.4.17 Message Memory Diagram

The message storage width is 32 bits, and the maximum space of each part is shown in Figure 81. There is no strict order between each part.

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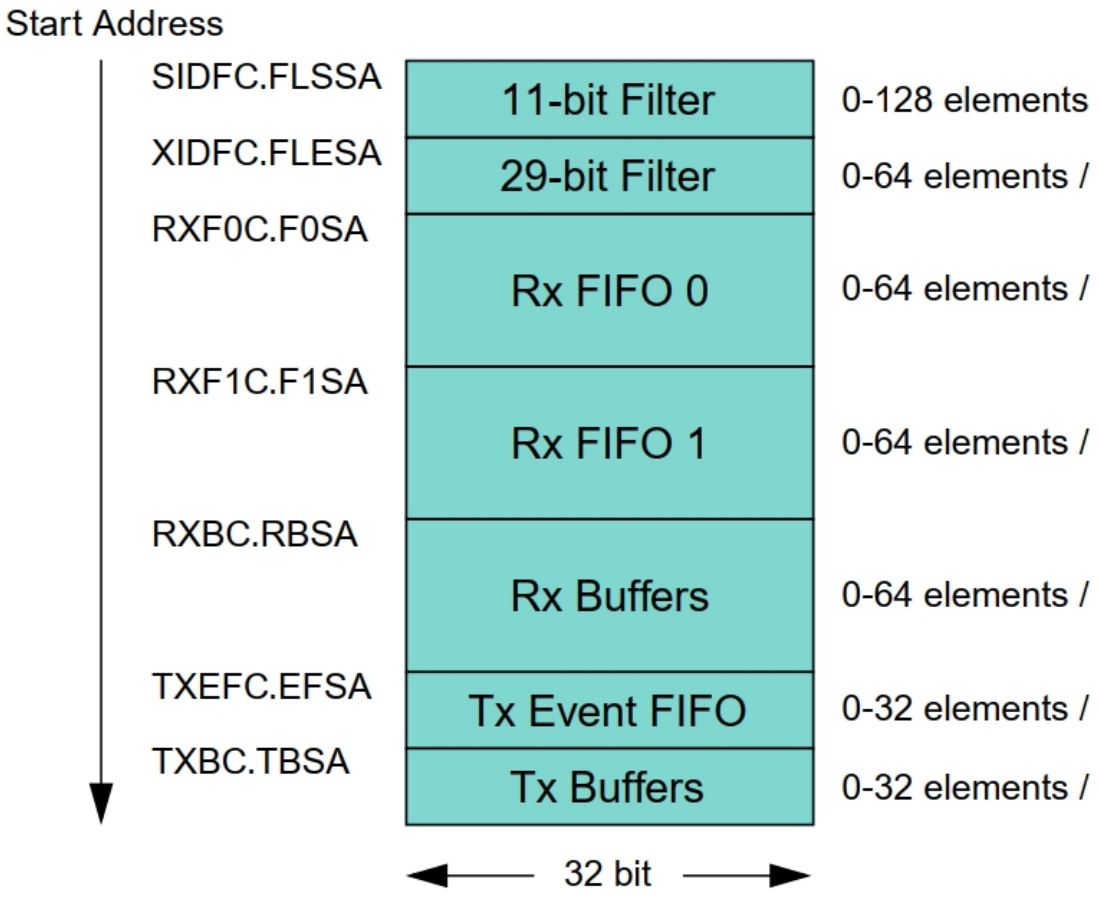


Figure 81: Message Memory Diagram

49.5 MCAN Registers

49.5.1 MCAN Register Overview

The MCAN register list is as follows:

MCAN0 base address: 0xF0280000

MCAN1 base address: 0xF0284000

MCAN2 base address: 0xF0288000

MCAN3 base address: 0xF028C000

Offset	Register Name	Description	Reset Value
0x0004	ENDN	endian register	0x87654321
0x000C	DBTP	data bit timing and prescaler, writeable when CCCR.CCE and CCCR.INT are set	0x00000A33
0x0010	TEST	test register	0x00000000
0x0014	RWD	ram watchdog	0x00000000
0x0018	CCCR	CC control register	0x00000001

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Offset	Register Name	Description	Reset Value
0x001C	NBTP	nominal bit timing and prescaler register	0x06000A03
0x0020	TSCC	timestamp counter configuration	0x00000000
0x0024	TSCV	timestamp counter value	0x00000000
0x0028	TOCC	timeout counter configuration	0xFFFF0000
0x002C	TOCV	timeout counter value	0x0000FFFF
0x0040	ECR	error counter register	0x00000000
0x0044	PSR	protocol status register	0x00000707
0x0048	TDCR	transmitter delay compensation	0x00000000
0x0050	IR	interrupt register	0x00000000
0x0054	IE	interrupt enable	0x00000000
0x0058	ILS	interrupt line select	0x00000000
0x005C	ILE	interrupt line enable	0x00000000
0x0080	GFC	global filter configuration	0x00000000
0x0084	SIDFC	standard ID filter configuration	0x00000000
0x0088	XIDFC	extended ID filter configuration	0x00000000
0x0090	XIDAM	extended id and mask	0x1FFFFFFF
0x0094	HPMS	high priority message status	0x00000000
0x0098	NDAT1	new data1	0x00000000
0x009C	NDAT2	new data2	0x00000000
0x00A0	RXF0C	rx fifo 0 configuration	0x00000000
0x00A4	RXF0S	rx fifo 0 status	0x00000000
0x00A8	RXF0A	rx fifo0 acknowledge	0x00000000
0x00AC	RXBC	rx buffer configuration	0x00000000
0x00B0	RXF1C	rx fifo1 configuration	0x00000000
0x00B4	RXF1S	rx fifo1 status	0x00000000
0x00B8	RXF1A	rx fifo 1 acknowledge	0x00000000
0x00BC	RXESC	rx buffer/fifo element size configuration	0x00000000
0x00C0	TXBC	tx buffer configuration	0x00000000
0x00C4	TXFQS	tx fifo/queue status	0x00000000
0x00C8	TXESC	tx buffer element size configuration	0x00000000
0x00CC	TXBRP	tx buffer request pending	0x00000000
0x00D0	TXBAR	tx buffer add request	0x00000000
0x00D4	TXBCR	tx buffer cancellation request	0x00000000
0x00D8	TXBTO	tx buffer transmission occurred	0x00000000

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Offset	Register Name	Description	Reset Value
0x00DC	TXBCF	tx buffer cancellation finished	0x00000000
0x00E0	TXBTIE	tx buffer transmission interrupt enable	0x00000000
0x00E4	TXBCIE	tx buffer cancellation finished interrupt enable	0x00000000
0x00F0	TXEFC	tx event fifo configuration	0x00000000
0x00F4	TXEFS	tx event fifo status	0x00000000
0x00F8	TXEFA	tx event fifo acknowledge	0x00000000
0x0200	TS_SEL[TS_SEL0]	timestamp 0	0x00000000
0x0204	TS_SEL[TS_SEL1]	timestamp 1	0x00000000
0x0208	TS_SEL[TS_SEL2]	timestamp 2	0x00000000
0x020C	TS_SEL[TS_SEL3]	timestamp 3	0x00000000
0x0210	TS_SEL[TS_SEL4]	timestamp 4	0x00000000
0x0214	TS_SEL[TS_SEL5]	timestamp 5	0x00000000
0x0218	TS_SEL[TS_SEL6]	timestamp 6	0x00000000
0x021C	TS_SEL[TS_SEL7]	timestamp 7	0x00000000
0x0220	TS_SEL[TS_SEL8]	timestamp 8	0x00000000
0x0224	TS_SEL[TS_SEL9]	timestamp 9	0x00000000
0x0228	TS_SEL[TS_SEL10]	timestamp 10	0x00000000
0x022C	TS_SEL[TS_SEL11]	timestamp 11	0x00000000
0x0230	TS_SEL[TS_SEL12]	timestamp 12	0x00000000
0x0234	TS_SEL[TS_SEL13]	timestamp 13	0x00000000
0x0238	TS_SEL[TS_SEL14]	timestamp 14	0x00000000
0x023C	TS_SEL[TS_SEL15]	timestamp 15	0x00000000
0x0240	CREL	core release register	0x00000000
0x0244	TSCFG	timestamp configuration	0x00000000
0x0248	TSS1	timestamp status1	0x00000000
0x024C	TSS2	timestamp status2	0x00000000
0x0250	ATB	actual timebase	0x00000000
0x0254	ATBH	actual timebase high	0x00000000
0x0400	GLB_CTL	global control	0x00000000
0x0404	GLB_STATUS	global status	0x00000000

Table 205: MCAN Register List

49.5.2 Detailed Register Information

The detailed description of MCAN registers is as follows:

49.5.3 ENDN (0x4)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EVT																															
R																															
1	0	0	0	0	1	1	1	0	1	1	0	0	1	0	1	0	1	0	0	0	0	1	1	0	0	1	0	0	0	0	1

ENDN [31:0]

Bits	Name	Description
31-0	EVT	Endianness Test Value The endianness test value is 0x87654321.

ENDN Bits

49.5.4 DBTP (0xC)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD								TDC	RSVD		DBRP					RSVD			DTSEG1					DTSEG2				DSJW			
N/A								RW	N/A		RW					N/A			RW					RW				RW			
x	x	x	x	x	x	x	x	0	x	x	0	0	0	0	0	x	x	x	0	1	0	1	0	0	0	1	1	0	0	1	1

DBTP [31:0]

Bits	Name	Description
23	TDC	transmitter delay compensation enable 0= Transmitter Delay Compensation disabled 1= Transmitter Delay Compensation enabled
20-16	DBRP	Data Bit Rate Prescaler The value by which the oscillator frequency is divided for generating the bit time quanta. The bit time is built up from a multiple of this quanta. Valid values for the Bit Rate Prescaler are 0 to 31. When TDC = ‘1’, the range is limited to 0,1. The actual interpretation by the hardware of this value is such that one more than the value programmed here is used.
12-8	DTSEG1	Data time segment before sample point Valid values are 0 to 31. The actual interpretation by the hardware of this value is such that one more than the programmed value is used.

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Bits	Name	Description
7-4	DTSEG2	Data time segment after sample point Valid values are 0 to 15. The actual interpretation by the hardware of this value is such that one more than the programmed value is used.
3-0	DSJW	Data (Re)Synchronization Jump Width Valid values are 0 to 15. The actual interpretation by the hardware of this value is such that one more than the value programmed here is used.

DBTP Bits

49.5.5 TEST (0x10)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD										SVAL	TXBNS					RSVD		PVAL	TXBNP					RX	TX		LBCK	RSVD			
N/A										R	R					N/A		R	R					R	RW		RW	N/A			
x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	x	x	0	0	0	0	0	0	0	0	0	0	x	x	x	x

TEST [31:0]

Bits	Name	Description
21	SVAL	Started Valid 0= Value of TXBNS not valid 1= Value of TXBNS valid
20-16	TXBNS	Tx Buffer Number Started Tx Buffer number of message whose transmission was started last. Valid when SVAL is set. Valid values are 0 to 31.
13	PVAL	Prepared Valid 0= Value of TXBNP not valid 1= Value of TXBNP valid
12-8	TXBNP	Tx Buffer Number Prepared Tx Buffer number of message that is ready for transmission. Valid when PVAL is set.Valid values are 0 to 31.
7	RX	Receive Pin Monitors the actual value of pin m_can_rx 0= The CAN bus is dominant (m_can_rx = ‘0’) 1= The CAN bus is recessive (m_can_rx = ‘1’)

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Bits	Name	Description
6-5	TX	Control of Transmit Pin 00 Reset value, m_can_tx controlled by the CAN Core, updated at the end of the CAN bit time 01 Sample Point can be monitored at pin m_can_tx 10 Dominant (‘0’) level at pin m_can_tx 11 Recessive (‘1’) at pin m_can_tx
4	LBCK	Loop Back Mode 0= Reset value, Loop Back Mode is disabled 1= Loop Back Mode is enabled

TEST Bits

49.5.6 RWD (0x14)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																WDV								WDC							
N/A																R								RW							
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

RWD [31:0]

Bits	Name	Description
15-8	WDV	Watchdog Value Actual Message RAM Watchdog Counter Value.
7-0	WDC	Watchdog Configuration Start value of the Message RAM Watchdog Counter. With the reset value of “00” the counter is disabled.

RWD Bits

49.5.7 CCCR (0x18)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																NISO	TXP	EFBI	PXHD	WMM	UTSU	BRSE	FDOE	TEST	DAR	MON	CSR	CSA	ASM	CCE	INIT
N/A																RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	R	RW	RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

CCCR [31:0]

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Bits	Name	Description
15	NISO	Non ISO Operation If this bit is set, the M_CAN uses the CAN FD frame format as specified by the Bosch CAN FD Specification V1.0. 0= CAN FD frame format according to ISO 11898-1:2015 1= CAN FD frame format according to Bosch CAN FD Specification V1.0 Note: When the generic parameter iso_only_g is set to ‘1’ in hardware synthesis, this bit becomes reserved and is read as ‘0’. The M_CAN always operates with the CAN FD frame format according to ISO 11898-1:2015.
14	TXP	Transmit Pause If this bit is set, the M_CAN pauses for two CAN bit times before starting the next transmission after itself has successfully transmitted a frame (see Section 3.5). 0= Transmit pause disabled 1= Transmit pause enabled
13	EFBI	Edge Filtering during Bus Integration 0= Edge filtering disabled 1= Two consecutive dominant tq required to detect an edge for hard synchronization
12	PXHD	Protocol Exception Handling Disable 0= Protocol exception handling enabled 1= Protocol exception handling disabled Note: When protocol exception handling is disabled, the M_CAN will transmit an error frame when it detects a protocol exception condition.
11	WMM	Wide Message Marker Enables the use of 16-bit Wide Message Markers. When 16-bit Wide Message Markers are used (WMM = ‘1’), 16-bit internal timestamping is disabled for the Tx Event FIFO. 0= 8-bit Message Marker used 1= 16-bit Message Marker used, replacing 16-bit timestamps in Tx Event FIFO

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Bits	Name	Description
10	UTSU	Use Timestamping Unit When UTSU is set, 16-bit Wide Message Markers are also enabled regardless of the value of WMM. 0= Internal time stamping 1= External time stamping by TSU Note: When generic parameter connected_tsu_g = ‘0’, there is no TSU connected to the M_CAN. In this case bit UTSU is fixed to zero by synthesis.
9	BRSE	Bit Rate Switch Enable 0= Bit rate switching for transmissions disabled 1= Bit rate switching for transmissions enabled Note: When CAN FD operation is disabled FDOE = ‘0’, BRSE is not evaluated.
8	FDOE	FD Operation Enable 0= FD operation disabled 1= FD operation enabled
7	TEST	Test Mode Enable 0= Normal operation, register TEST holds reset values 1= Test Mode, write access to register TEST enabled
6	DAR	Disable Automatic Retransmission 0= Automatic retransmission of messages not transmitted successfully enabled 1= Automatic retransmission disabled
5	MON	Bus Monitoring Mode Bit MON can only be set by the Host when both CCE and INIT are set to ‘1’. The bit can be reset by the Host at any time. 0= Bus Monitoring Mode is disabled 1= Bus Monitoring Mode is enabled
4	CSR	Clock Stop Request 0= No clock stop is requested 1= Clock stop requested. When clock stop is requested, first INIT and then CSA will be set after all pending transfer requests have been completed and the CAN bus reached idle.
3	CSA	Clock Stop Acknowledge 0= No clock stop acknowledged 1= M_CAN may be set in power down by stopping m_can_hclk and m_can_cclk

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Bits	Name	Description
2	ASM	Restricted Operation Mode Bit ASM can only be set by the Host when both CCE and INIT are set to ‘1’. The bit can be reset by the Host at any time. For a description of the Restricted Operation Mode see Section 3.1.5. 0= Normal CAN operation 1= Restricted Operation Mode active
1	CCE	Configuration Change Enable 0= The CPU has no write access to the protected configuration registers 1= The CPU has write access to the protected configuration registers (while CCCR.INIT = ‘1’)
0	INIT	Initialization 0= Normal Operation 1= Initialization is started

CCCR Bits

49.5.8 NBTP (0x1C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
NSJW							NBRP									NTSEG1								RSVD	NTSEG2						
RW							RW									RW								N/A	RW						
0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	x	0	0	0	0	0	1	1

NBTP [31:0]

Bits	Name	Description
31-25	NSJW	Nominal (Re)Synchronization Jump Width Valid values are 0 to 127. The actual interpretation by the hardware of this value is such that one more than the value programmed here is used.
24-16	NBRP	Nominal Bit Rate Prescaler The value by which the oscillator frequency is divided for generating the bit time quanta. The bit time is built up from a multiple of this quanta. Valid values for the Bit Rate Prescaler are 0 to 511. The actual interpretation by the hardware of this value is such that one more than the value programmed here is used.

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Bits	Name	Description
15-8	NTSEG1	Nominal Time segment before sample point Valid values are 1 to 255. The actual interpretation by the hardware of this value is such that one more than the programmed value is used.
6-0	NTSEG2	Nominal Time segment after sample point Valid values are 1 to 127. The actual interpretation by the hardware of this value is such that one more than the programmed value is used.

NBTP Bits

49.5.9 TSCC (0x20)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD												TCP				RSVD														TSS	
N/A												RW				N/A														RW	
x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0

TSCC [31:0]

Bits	Name	Description
19-16	TCP	Timestamp Counter Prescaler Configures the timestamp and timeout counters time unit in multiples of CAN bit times [1…16]. The actual interpretation by the hardware of this value is such that one more than the value programmed here is used.
1-0	TSS	timestamp Select 00= Timestamp counter value always 0x0000 01= Timestamp counter value incremented according to TCP 10= External timestamp counter value used 11= Same as “00”

TSCC Bits

49.5.10 TSCV (0x24)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																TSC															
N/A																RC															
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TSCV [31:0]																															

Bits	Name	Description
15-0	TSC	Timestamp Counter The internal/external Timestamp Counter value is captured on start of frame (both Rx and Tx).When TSCC.TSS = “01”, the Timestamp Counter is incremented in multiples of CAN bit times [1…16] depending on the configuration of TSCC.TCP. A wrap around sets interrupt flag IR.TSW. Write access resets the counter to zero. When TSCC.TSS = “10”, TSC reflects the external Timestamp Counter value. A write access has no impact.

TSCV Bits

49.5.11 TOCC (0x28)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TOP																RSVD													TOS		RP
RW																N/A													RW		RW
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0
TOCC [31:0]																															

Bits	Name	Description
31-16	TOP	Timeout Period Start value of the Timeout Counter (down-counter). Configures the Timeout Period.
2-1	TOS	Timeout Select When operating in Continuous mode, a write to TOCV presets the counter to the value configured by TOCC.TOP and continues down-counting. When the Timeout Counter is controlled by one of the FIFOs, an empty FIFO presets the counter to the value configured by TOCC.TOP. Down-counting is started when the first FIFO element is stored. 00= Continuous operation 01= Timeout controlled by Tx Event FIFO 10= Timeout controlled by Rx FIFO 0 11= Timeout controlled by Rx FIFO 1

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Bits	Name	Description
0	RP	Enable Timeout Counter 0= Timeout Counter disabled 1= Timeout Counter enabled

TOCC Bits

49.5.12 TOCV (0x2C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																TOC															
N/A																RC															
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

TOCV [31:0]

Bits	Name	Description
15-0	TOC	Timeout Counter The Timeout Counter is decremented in multiples of CAN bit times [1…16] depending on the configuration of TSCC.TCP. When decremented to zero, interrupt flag IR.TOO is set and the Timeout Counter is stopped. Start and reset/restart conditions are configured via TOCC.TOS. Note: Byte access: when TOCC.TOS = “00，writing one of the register bytes 3/2/1/0 will preset the Timeout Counter.

TOCV Bits

49.5.13 ECR (0x40)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD								CEL								RP	REC							TEC							
N/A								X								R	R							R							
x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ECR [31:0]

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Bits	Name	Description
23-16	CEL	CAN Error Logging The counter is incremented each time when a CAN protocol error causes the 8-bit Transmit Error Counter TEC or the 7-bit Receive Error Counter REC to be incremented. The counter is also incremented when the Bus_Off limit is reached. It is not incremented when only RP is set without changing REC. The increment of CEL follows after the increment of REC or TEC. The counter is reset by read access to CEL. The counter stops at 0xFF; the next increment of TEC or REC sets interrupt flag IR.ELO. Note: Byte access: Reading byte 2 will reset CEL to zero, reading bytes 3/1/0 has no impact.
15	RP	Receive Error Passive 0= The Receive Error Counter is below the error passive level of 128 1= The Receive Error Counter has reached the error passive level of 128
14-8	REC	Receive Error Counter Actual state of the Receive Error Counter, values between 0 and 127
7-0	TEC	Transmit Error Counter Actual state of the Transmit Error Counter, values between 0 and 255 Note: When CCCR.ASM is set, the CAN protocol controller does not increment TEC and REC when a CAN protocol error is detected, but CEL is still incremented.

ECR Bits

49.5.14 PSR (0x44)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD									TDCV							RSVD	PXE	RFDF	RBRS	RESI	DLEC			BO	EW	EP	ACT			LEC		
N/A									R							N/A	X	X	X	X	S			R	R	R	R			S		
x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	x	0	0	0	0	1	1	1	0	0	0	0	0	1	1	1

PSR [31:0]

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Bits	Name	Description
22-16	TDCV	Transmitter Delay Compensation Value Position of the secondary sample point, defined by the sum of the measured delay from m_can_tx to m_can_rx and TDCR.TDCO. The SSP position is, in the data phase, the number of mtq between the start of the transmitted bit and the secondary sample point. Valid values are 0 to 127 mtq.
14	PXE	Protocol Exception Event 0= No protocol exception event occurred since last read access 1= Protocol exception event occurred Note: Byte access: Reading byte 0 will reset PXE, reading bytes 3/2/1 has no impact.
13	RFDF	Received a CAN FD Message This bit is set independent of acceptance filtering. 0= Since this bit was reset by the CPU, no CAN FD message has been received 1= Message in CAN FD format with FDF flag set has been received Note: Byte access: Reading byte 0 will reset RFDF, reading bytes 3/2/1 has no impact.
12	RBRS	BRS flag of last received CAN FD Message This bit is set together with RFDF, independent of acceptance filtering. 0= Last received CAN FD message did not have its BRS flag set 1= Last received CAN FD message had its BRS flag set Note: Byte access: Reading byte 0 will reset RBRS, reading bytes 3/2/1 has no impact.
11	RESI	ESI flag of last received CAN FD Message This bit is set together with RFDF, independent of acceptance filtering. 0= Last received CAN FD message did not have its ESI flag set 1= Last received CAN FD message had its ESI flag set Note: Byte access: Reading byte 0 will reset RESI, reading bytes 3/2/1 has no impact.

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Bits	Name	Description
10-8	DLEC	Data Phase Last Error Code Type of last error that occurred in the data phase of a CAN FD format frame with its BRS flag set.Coding is the same as for LEC. This field will be cleared to zero when a CAN FD format frame with its BRS flag set has been transferred (reception or transmission) without error. Note: Byte access: Reading byte 0 will set DLEC to “111”, reading bytes 3/2/1 has no impact.
7	BO	Bus_Off Status 0= The M_CAN is not Bus_Off 1= The M_CAN is in Bus_Off state
6	EW	Warning Status 0= Both error counters are below the Error_Warning limit of 96 1= At least one of error counter has reached the Error_Warning limit of 96
5	EP	Error Passive 0= The M_CAN is in the Error_Active state. It normally takes part in bus communication and sends an active error flag when an error has been detected 1= The M_CAN is in the Error_Passive state
4-3	ACT	Activity Monitors the module’s CAN communication state. 00= Synchronizing - node is synchronizing on CAN communication 01= Idle - node is neither receiver nor transmitter 10= Receiver - node is operating as receiver 11= Transmitter - node is operating as transmitter Note: ACT is set to “00” by a Protocol Exception Event.

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Bits	Name	Description
2-0	LEC	Last Error Code The LEC indicates the type of the last error to occur on the CAN bus. This field will be cleared to ‘0’ when a message has been transferred (reception or transmission) without error. 0= No Error: No error occurred since LEC has been reset by successful reception or transmission. 1= Stuff Error: More than 5 equal bits in a sequence have occurred in a part of a received message where this is not allowed. 2= Form Error: A fixed format part of a received frame has the wrong format. 3= AckError: The message transmitted by the M_CAN was not acknowledged by another node. 4= Bit1Error: During the transmission of a message (with the exception of the arbitration field), the device wanted to send a recessive level (bit of logical value ‘1’), but the monitored bus value was dominant. 5= Bit0Error: During the transmission of a message (or acknowledge bit, or active error flag, or overload flag), the device wanted to send a dominant level (data or identifier bit logical value ‘0’), but the monitored bus value was recessive. During Bus_Off recovery this status is set each time a sequence of 11 recessive bits has been monitored. This enables the CPU to monitor the proceeding of the Bus_Off recovery sequence (indicating the bus is not stuck at dominant or continuously disturbed). 6= CRCError: The CRC check sum of a received message was incorrect. The CRC of an incoming message does not match with the CRC calculated from the received data. 7= NoChange: Any read access to the Protocol Status Register re-initializes the LEC to ‘7’. When the LEC shows the value ‘7’, no CAN bus event was detected since the last CPU read access to the Protocol Status Register. Note: When a frame in CAN FD format has reached the data phase with BRS flag set, the next CAN event (error or valid frame) will be shown in DLEC instead of LEC. An error in a fixed stuff bit of a CAN FD CRC sequence will be shown as a Form Error, not Stuff Error. Note: The Bus_Off recovery sequence (see ISO 11898-1:2015) cannot be shortened by setting or resetting CCCR.INIT. If the device goes Bus_Off, it will set CCCR.INIT of its own accord,stopping all bus activities. Once CCCR.INIT has been cleared by the CPU, the device will then wait for 129 occurrences of Bus Idle (129 * 11 consecutive recessive bits) before resuming normal operation. At the end of the Bus_Off recovery sequence, the Error

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Bits	Name	Description

PSR Bits

49.5.15 TDCR (0x48)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																	TDCO							RSVD	TDCF						
N/A																	RW							N/A	RW						
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	x	0	0	0	0	0	0	0

TDCR [31:0]

Bits	Name	Description
14-8	TDCO	Transmitter Delay Compensation SSP Offset Offset value defining the distance between the measured delay from m_can_tx to m_can_rx and the secondary sample point. Valid values are 0 to 127 mtq.
6-0	TDCF	Transmitter Delay Compensation Filter Window Length Defines the minimum value for the SSP position, dominant edges on m_can_rx that would result in an earlier SSP position are ignored for transmitter delay measurement. The feature is enabled when TDCF is configured to a value greater than TDCO. Valid values are 0 to 127 mtq.

TDCR Bits

49.5.16 IR (0x50)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD		ARA	PED	PEA	WDI	BO	EW	EP	ELO	BEU	BEC	DRX	TOO	MRAF	TSW	TEFL	TEFF	TEFW	TEFN	TFE	TCF	TC	HPM	RF1L	RF1F	RF1W	RF1N	RF0L	RF0F	RF0W	RF0N
N/A		RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW
x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

IR [31:0]

Bits	Name	Description
29	ARA	Access to Reserved Address 0= No access to reserved address occurred 1= Access to reserved address occurred

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Bits	Name	Description
28	PED	Protocol Error in Data Phase (Data Bit Time is used) 0= No protocol error in data phase 1= Protocol error in data phase detected (PSR.DLEC ≠ 0,7)
27	PEA	Protocol Error in Arbitration Phase (Nominal Bit Time is used) 0= No protocol error in arbitration phase 1= Protocol error in arbitration phase detected (PSR.LEC ≠ 0,7)
26	WDI	Watchdog Interrupt 0= No Message RAM Watchdog event occurred 1= Message RAM Watchdog event due to missing READY
25	BO	Bus_Off Status 0= Bus_Off status unchanged 1= Bus_Off status changed
24	EW	Warning Status 0= Error_Warning status unchanged 1= Error_Warning status changed
23	EP	Error Passive 0= Error_Passive status unchanged 1= Error_Passive status changed
22	ELO	Error Logging Overflow 0= CAN Error Logging Counter did not overflow 1= Overflow of CAN Error Logging Counter occurred
21	BEU	Bit Error Uncorrected Message RAM bit error detected, uncorrected. Controlled by input signal m_can_aeim_berr[1] generated by an optional external parity / ECC logic attached to the Message RAM. An uncorrected Message RAM bit error sets CCCR.INIT to ‘1’. This is done to avoid transmission of corrupted data. 0= No bit error detected when reading from Message RAM 1= Bit error detected, uncorrected (e.g. parity logic)
20	BEC	Bit Error Corrected Message RAM bit error detected and corrected. Controlled by input signal m_can_aeim_berr[0] generated by an optional external parity / ECC logic attached to the Message RAM. 0= No bit error detected when reading from Message RAM 1= Bit error detected and corrected (e.g. ECC)
19	DRX	Message stored to Dedicated Rx Buffer The flag is set whenever a received message has been stored into a dedicated Rx Buffer. 0= No Rx Buffer updated 1= At least one received message stored into an Rx Buffer

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Bits	Name	Description
18	TOO	Timeout Occurred 0= No timeout 1= Timeout reached
17	MRAF	Message RAM Access Failure The flag is set, when the Rx Handler .has not completed acceptance filtering or storage of an accepted message until the arbitration field of the following message has been received. In this case acceptance filtering or message storage is aborted and the Rx Handler starts processing of the following message. .was not able to write a message to the Message RAM. In this case message storage is aborted. In both cases the FIFO put index is not updated resp. the New Data flag for a dedicated Rx Buffer is not set, a partly stored message is overwritten when the next message is stored to this location. The flag is also set when the Tx Handler was not able to read a message from the Message RAM in time. In this case message transmission is aborted. In case of a Tx Handler access failure the M_CAN is switched into Restricted Operation Mode (see Section 3.1.5). To leave Restricted Operation Mode, the Host CPU has to reset CCCR.ASM. 0= No Message RAM access failure occurred 1= Message RAM access failure occurred
16	TSW	Timestamp Wraparound 0= No timestamp counter wrap-around 1= Timestamp counter wrapped around
15	TEFL	Tx Event FIFO Element Lost 0= No Tx Event FIFO element lost 1= Tx Event FIFO element lost, also set after write attempt to Tx Event FIFO of size zero
14	TEFF	Tx Event FIFO Full 0= Tx Event FIFO not full 1= Tx Event FIFO full
13	TEFW	Tx Event FIFO Watermark Reached 0= Tx Event FIFO fill level below watermark 1= Tx Event FIFO fill level reached watermark

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Bits	Name	Description
12	TEFN	Tx Event FIFO New Entry 0= Tx Event FIFO unchanged 1= Tx Handler wrote Tx Event FIFO element
11	TFE	Tx FIFO Empty 0= Tx FIFO non-empty 1= Tx FIFO empty
10	TCF	Transmission Cancellation Finished 0= No transmission cancellation finished 1= Transmission cancellation finished
9	TC	Transmission Completed 0= No transmission completed 1= Transmission completed
8	HPM	High Priority Message 0= No high priority message received 1= High priority message received
7	RF1L	Rx FIFO 1 Message Lost 0= No Rx FIFO 1 message lost 1= Rx FIFO 1 message lost, also set after write attempt to Rx FIFO 1 of size zero
6	RF1F	Rx FIFO 1 Full 0= Rx FIFO 1 not full 1= Rx FIFO 1 full
5	RF1W	Rx FIFO 1 Watermark Reached 0= Rx FIFO 1 fill level below watermark 1= Rx FIFO 1 fill level reached watermark
4	RF1N	Rx FIFO 1 New Message 0= No new message written to Rx FIFO 1 1= New message written to Rx FIFO 1
3	RF0L	Rx FIFO 0 Message Lost 0= No Rx FIFO 0 message lost 1= Rx FIFO 0 message lost, also set after write attempt to Rx FIFO 0 of size zero
2	RF0F	Rx FIFO 0 Full 0= Rx FIFO 0 not full 1= Rx FIFO 0 full
1	RF0W	Rx FIFO 0 Watermark Reached 0= Rx FIFO 0 fill level below watermark 1= Rx FIFO 0 fill level reached watermark
0	RF0N	Rx FIFO 0 New Message 0= No new message written to Rx FIFO 0 1= New message written to Rx FIFO 0

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Bits	Name	Description

IR Bits

49.5.17 IE (0x54)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD		ARAE	PEDE	PEAE	WDIE	BOE	EWE	EPE	ELOE	BEUE	BECE	DRXE	TOOE	MRAFE	TSWE	TEFLE	TEFFE	TEFWE	TEFNE	TFEE	TCFE	TCE	HPME	RF1LE	RF1FE	RF1WE	RF1NE	RF0LE	RF0FE	RF0WE	RF0NE
N/A		RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW
x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

IE [31:0]

Bits	Name	Description
29	ARAE	Access to Reserved Address Enable
28	PEDE	Protocol Error in Data Phase Enable
27	PEAE	Protocol Error in Arbitration Phase Enable
26	WDIE	Watchdog Interrupt Enable
25	BOE	Bus_Off Status Interrupt Enable
24	EWE	Warning Status Interrupt Enable
23	EPE	Error Passive Interrupt Enable
22	ELOE	Error Logging Overflow Interrupt Enable
21	BEUE	Bit Error Uncorrected Interrupt Enable
20	BECE	Bit Error Corrected Interrupt Enable
19	DRXE	Message stored to Dedicated Rx Buffer Interrupt Enable
18	TOOE	Timeout Occurred Interrupt Enable
17	MRAFE	Message RAM Access Failure Interrupt Enable
16	TSWE	Timestamp Wraparound Interrupt Enable
15	TEFLE	Tx Event FIFO Event Lost Interrupt Enable
14	TEFFE	Tx Event FIFO Full Interrupt Enable
13	TEFWE	Tx Event FIFO Watermark Reached Interrupt Enable
12	TEFNE	Tx Event FIFO New Entry Interrupt Enable
11	TFEE	Tx FIFO Empty Interrupt Enable
10	TCFE	Transmission Cancellation Finished Interrupt Enable
9	TCE	Transmission Completed Interrupt Enable
8	HPME	High Priority Message Interrupt Enable
7	RF1LE	Rx FIFO 1 Message Lost Interrupt Enable
6	RF1FE	Rx FIFO 1 Full Interrupt Enable
5	RF1WE	Rx FIFO 1 Watermark Reached Interrupt Enable
4	RF1NE	Rx FIFO 1 New Message Interrupt Enable
3	RF0LE	Rx FIFO 0 Message Lost Interrupt Enable

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Bits	Name	Description
2	RF0FE	Rx FIFO 0 Full Interrupt Enable
1	RF0WE	Rx FIFO 0 Watermark Reached Interrupt Enable
0	RF0NE	Rx FIFO 0 New Message Interrupt Enable

IE Bits

49.5.18 ILS (0x58)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD		ARAL	PEDL	PEAL	WDIL	BOL	EWL	EPL	ELOL	BEUL	BECL	DRXL	TOOL	MRAFL	TSWL	TEFLL	TEFFL	TEFWL	TEFNL	TFEL	TCFL	TCL	HPML	RF1LL	RF1FL	RF1WL	RF1NL	RF0LL	RF0FL	RF0WL	RF0NL
N/A		RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW
x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ILS [31:0]

Bits	Name	Description
29	ARAL	Access to Reserved Address Line
28	PEDL	Protocol Error in Data Phase Line
27	PEAL	Protocol Error in Arbitration Phase Line
26	WDIL	Watchdog Interrupt Line
25	BOL	Bus_Off Status Interrupt Line
24	EWL	Warning Status Interrupt Line
23	EPL	Error Passive Interrupt Line
22	ELOL	Error Logging Overflow Interrupt Line
21	BEUL	Bit Error Uncorrected Interrupt Line
20	BECL	Bit Error Corrected Interrupt Line
19	DRXL	Message stored to Dedicated Rx Buffer Interrupt Line
18	TOOL	Timeout Occurred Interrupt Line
17	MRAFL	Message RAM Access Failure Interrupt Line
16	TSWL	Timestamp Wraparound Interrupt Line
15	TEFLL	Tx Event FIFO Event Lost Interrupt Line
14	TEFFL	Tx Event FIFO Full Interrupt Line
13	TEFWL	Tx Event FIFO Watermark Reached Interrupt Line
12	TEFNL	Tx Event FIFO New Entry Interrupt Line
11	TFEL	Tx FIFO Empty Interrupt Line
10	TCFL	Transmission Cancellation Finished Interrupt Line
9	TCL	Transmission Completed Interrupt Line
8	HPML	High Priority Message Interrupt Line
7	RF1LL	Rx FIFO 1 Message Lost Interrupt Line
6	RF1FL	Rx FIFO 1 Full Interrupt Line

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Bits	Name	Description
5	RF1WL	Rx FIFO 1 Watermark Reached Interrupt Line
4	RF1NL	Rx FIFO 1 New Message Interrupt Line
3	RF0LL	Rx FIFO 0 Message Lost Interrupt Line
2	RF0FL	Rx FIFO 0 Full Interrupt Line
1	RF0WL	Rx FIFO 0 Watermark Reached Interrupt Line
0	RF0NL	Rx FIFO 0 New Message Interrupt Line

ILS Bits

49.5.19 ILE (0x5C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																														EINT1	EINT0
N/A																														RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0

ILE [31:0]

Bits	Name	Description
1	EINT1	Enable Interrupt Line 1 0= Interrupt line m_can_int1 disabled 1= Interrupt line m_can_int1 enabled
0	EINT0	Enable Interrupt Line 0 0= Interrupt line m_can_int0 disabled 1= Interrupt line m_can_int0 enabled

ILE Bits

49.5.20 GFC (0x80)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																										ANFS		ANFE		RRFS	RRFE
N/A																										RW		RW		RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0

GFC [31:0]

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Bits	Name	Description
5-4	ANFS	Accept Non-matching Frames Standard Defines how received messages with 11-bit IDs that do not match any element of the filter list are treated. 00= Accept in Rx FIFO 0 01= Accept in Rx FIFO 1 10= Reject 11= Reject
3-2	ANFE	Accept Non-matching Frames Extended Defines how received messages with 29-bit IDs that do not match any element of the filter list are treated. 00= Accept in Rx FIFO 0 01= Accept in Rx FIFO 1 10= Reject 11= Reject
1	RRFS	Reject Remote Frames Standard 0= Filter remote frames with 11-bit standard IDs 1= Reject all remote frames with 11-bit standard IDs
0	RRFE	Reject Remote Frames Extended 0= Filter remote frames with 29-bit extended IDs 1= Reject all remote frames with 29-bit extended IDs

GFC Bits

49.5.21 SIDFC (0x84)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD								LSS								FLSSA														RSVD	
N/A								RW								RW														N/A	
x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x

SIDFC [31:0]

Bits	Name	Description
23-16	LSS	List Size Standard 0= No standard Message ID filter 1-128= Number of standard Message ID filter elements >128= Values greater than 128 are interpreted as 128
15-2	FLSSA	Filter List Standard Start Address Start address of standard Message ID filter list (32-bit word address)

SIDFC Bits

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49.5.22 XIDFC (0x88)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD									LSE							FLESA														RSVD	
N/A									RW							RW														N/A	
x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x

XIDFC [31:0]

Bits	Name	Description
22-16	LSE	List Size Extended 0= No extended Message ID filter 1-64= Number of extended Message ID filter elements >64= Values greater than 64 are interpreted as 64
15-2	FLESA	Filter List Extended Start Address Start address of extended Message ID filter list (32-bit word address).

XIDFC Bits

49.5.23 XIDAM (0x90)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD			EIDM																												
N/A			RW																												
x	x	x	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

XIDAM [31:0]

Bits	Name	Description
28-0	EIDM	Extended ID Mask For acceptance filtering of extended frames the Extended ID AND Mask is ANDed with the Message ID of a received frame. Intended for masking of 29-bit IDs in SAE J1939. With the reset value of all bits set to one the mask is not active.

XIDAM Bits

49.5.24 HPMS (0x94)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																FLST	FIDX							MSI		BIDX					
N/A																R	R							R		R					
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

HPMS [31:0]

Bits	Name	Description
15	FLST	Filter List Indicates the filter list of the matching filter element. 0= Standard Filter List 1= Extended Filter List
14-8	FIDX	Filter Index Index of matching filter element. Range is 0 to SIDFC.LSS - 1 resp. XIDFC.LSE - 1.
7-6	MSI	Message Storage Indicator 00= No FIFO selected 01= FIFO message lost 10= Message stored in FIFO 0 11= Message stored in FIFO 1
5-0	BIDX	Buffer Index Index of Rx FIFO element to which the message was stored. Only valid when MSI[1] = ‘1’.

HPMS Bits

49.5.25 NDAT1 (0x98)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ND1																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

NDAT1 [31:0]

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Bits	Name	Description
31-0	ND1	New Data[31:0] The register holds the New Data flags of Rx Buffers 0 to 31. The flags are set when the respective Rx Buffer has been updated from a received frame. The flags remain set until the Host clears them.A flag is cleared by writing a ’ 1’ to the corresponding bit position. Writing a ’ 0’ has no effect. A hard reset will clear the register. 0= Rx Buffer not updated 1= Rx Buffer updated from new message

NDAT1 Bits

49.5.26 NDAT2 (0x9C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ND2																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

NDAT2 [31:0]

Bits	Name	Description
31-0	ND2	New Data[63:32] The register holds the New Data flags of Rx Buffers 32 to 63. The flags are set when the respective Rx Buffer has been updated from a received frame. The flags remain set until the Host clears them. A flag is cleared by writing a ’ 1’ to the corresponding bit position. Writing a ’ 0’ has no effect. A hard reset will clear the register. 0= Rx Buffer not updated 1= Rx Buffer updated from new message

NDAT2 Bits

49.5.27 RXF0C (0xA0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
F0OM	F0WM							RSVD	F0S							F0SA														RSVD	
RW	RW							N/A	RW							RW														N/A	
0	0	0	0	0	0	0	0	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

RXF0C [31:0]

Bits	Name	Description
31	F0OM	FIFO 0 Operation Mode FIFO 0 can be operated in blocking or in overwrite mode (see Section 3.4.2). 0= FIFO 0 blocking mode 1= FIFO 0 overwrite mode
30-24	F0WM	Rx FIFO 0 Watermark 0= Watermark interrupt disabled 1-64= Level for Rx FIFO 0 watermark interrupt (IR.RF0W) >64= Watermark interrupt disabled
22-16	F0S	Rx FIFO 0 Size 0= No Rx FIFO 0 1-64= Number of Rx FIFO 0 elements >64= Values greater than 64 are interpreted as 64 The Rx FIFO 0 elements are indexed from 0 to F0S-1
15-2	F0SA	Rx FIFO 0 Start Address Start address of Rx FIFO 0 in Message RAM (32-bit word address)

RXF0C Bits

49.5.28 RXF0S (0xA4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD						RF0L	F0F	RSVD		F0PI						RSVD		F0GI						RSVD	F0FL						
N/A						R	R	N/A		R						N/A		R						N/A	R						
x	x	x	x	x	x	0	0	x	x	0	0	0	0	0	0	x	x	0	0	0	0	0	0	x	0	0	0	0	0	0	0

RXF0S [31:0]

Bits	Name	Description
25	RF0L	Rx FIFO 0 Message Lost This bit is a copy of interrupt flag IR.RF0L. When IR.RF0L is reset, this bit is also reset. 0= No Rx FIFO 0 message lost 1= Rx FIFO 0 message lost, also set after write attempt to Rx FIFO 0 of size zero Note: Overwriting the oldest message when RXF0C.F0OM = ‘1’ will not set this flag.

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Bits	Name	Description
24	F0F	Rx FIFO 0 Full 0= Rx FIFO 0 not full 1= Rx FIFO 0 full
21-16	F0PI	Rx FIFO 0 Put Index Rx FIFO 0 write index pointer, range 0 to 63.
13-8	F0GI	Rx FIFO 0 Get Index Rx FIFO 0 read index pointer, range 0 to 63.
6-0	F0FL	Rx FIFO 0 Fill Level Number of elements stored in Rx FIFO 0, range 0 to 64.

RXF0S Bits

49.5.29 RXF0A (0xA8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																										F0AI					
N/A																										RW					
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0

RXF0A [31:0]

Bits	Name	Description
5-0	F0AI	Rx FIFO 0 Acknowledge Index After the Host has read a message or a sequence of messages from Rx FIFO 0 it has to write the buffer index of the last element read from Rx FIFO 0 to F0AI. This will set the Rx FIFO 0 Get Index RXF0S.F0GI to F0AI + 1 and update the FIFO 0 Fill Level RXF0S.F0FL.

RXF0A Bits

49.5.30 RXBC (0xAC)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																RBSA														RSVD	
N/A																RW														N/A	
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x

RXBC [31:0]

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Bits	Name	Description
15-2	RBSA	Rx Buffer Start Address Configures the start address of the Rx Buffers section in the Message RAM (32-bit word address).Also used to reference debug messages A,B,C.

RXBC Bits

49.5.31 RXF1C (0xB0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
F1OM	F1WM							RSVD	F1S							F1SA														RSVD	
RW	RW							N/A	RW							RW														N/A	
0	0	0	0	0	0	0	0	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x

RXF1C [31:0]

Bits	Name	Description
31	F1OM	FIFO 1 Operation Mode FIFO 1 can be operated in blocking or in overwrite mode (see Section 3.4.2). 0= FIFO 1 blocking mode 1= FIFO 1 overwrite mode
30-24	F1WM	Rx FIFO 1 Watermark 0= Watermark interrupt disabled 1-64= Level for Rx FIFO 1 watermark interrupt (IR.RF1W) >64= Watermark interrupt disabled
22-16	F1S	Rx FIFO 1 Size 0= No Rx FIFO 1 1-64= Number of Rx FIFO 1 elements >64= Values greater than 64 are interpreted as 64 The Rx FIFO 1 elements are indexed from 0 to F1S - 1
15-2	F1SA	Rx FIFO 1 Start Address Start address of Rx FIFO 1 in Message RAM (32-bit word address)

RXF1C Bits

49.5.32 RXF1S (0xB4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DMS	RSVD					RF1L	F1F	RSVD		F1PI						RSVD		F1GI						RSVD	F1FL						

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R		N/A				R	R	N/A		R						N/A		R						N/A		R					
0	0	x	x	x	x	0	0	x	x	0	0	0	0	0	0	x	x	0	0	0	0	0	0	x	0	0	0	0	0	0	0

RXF1S [31:0]

Bits	Name	Description
31-30	DMS	Debug Message Status 00= Idle state, wait for reception of debug messages, DMA request is cleared 01= Debug message A received 10= Debug messages A, B received 11= Debug messages A, B, C received, DMA request is set
25	RF1L	Rx FIFO 1 Message Lost This bit is a copy of interrupt flag IR.RF1L. When IR.RF1L is reset, this bit is also reset. 0= No Rx FIFO 1 message lost 1= Rx FIFO 1 message lost, also set after write attempt to Rx FIFO 1 of size zero Note: Overwriting the oldest message when RXF1C.F1OM = ‘1’ will not set this flag.
24	F1F	Rx FIFO 1 Full 0= Rx FIFO 1 not full 1= Rx FIFO 1 full
21-16	F1PI	Rx FIFO 1 Put Index Rx FIFO 1 write index pointer, range 0 to 63.
13-8	F1GI	Rx FIFO 1 Get Index Rx FIFO 1 read index pointer, range 0 to 63.
6-0	F1FL	Rx FIFO 1 Fill Level Number of elements stored in Rx FIFO 1, range 0 to 64.

RXF1S Bits

49.5.33 RXF1A (0xB8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																										F1AI					
N/A																										RW					
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0

RXF1A [31:0]

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Bits	Name	Description
5-0	F1AI	Rx FIFO 1 Acknowledge Index After the Host has read a message or a sequence of messages from Rx FIFO 1 it has to write the buffer index of the last element read from Rx FIFO 1 to F1AI. This will set the Rx FIFO 1 Get Index RXF1S.F1GI to F1AI + 1 and update the FIFO 1 Fill Level RXF1S.F1FL.

RXF1A Bits

49.5.34 RXESC (0xBC)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																					RBDS			RSVD	F1DS			RSVD	F0DS		
N/A																					RW			N/A	RW			N/A	RW		
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	x	0	0	0	x	0	0	0

RXESC [31:0]

Bits	Name	Description
10-8	RBDS	Rx Buffer Data Field Size 000= 8 byte data field 001= 12 byte data field 010= 16 byte data field 011= 20 byte data field 100= 24 byte data field 101= 32 byte data field 110= 48 byte data field 111= 64 byte data field
6-4	F1DS	Rx FIFO 1 Data Field Size 000= 8 byte data field 001= 12 byte data field 010= 16 byte data field 011= 20 byte data field 100= 24 byte data field 101= 32 byte data field 110= 48 byte data field 111= 64 byte data field

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Bits	Name	Description
2-0	F0DS	Rx FIFO 0 Data Field Size 000= 8 byte data field 001= 12 byte data field 010= 16 byte data field 011= 20 byte data field 100= 24 byte data field 101= 32 byte data field 110= 48 byte data field 111= 64 byte data field Note: In case the data field size of an accepted CAN frame exceeds the data field size configured for the matching Rx Buffer or Rx FIFO, only the number of bytes as configured by RXESC are stored to the Rx Buffer resp. Rx FIFO element. The rest of the frame’s data field is ignored.

RXESC Bits

49.5.35 TXBC (0xC0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD	TFQM	TFQS						RSVD		NDTB						TBSA														RSVD	
N/A	RW	RW						N/A		RW						RW														N/A	
x	0	0	0	0	0	0	0	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x

TXBC [31:0]

Bits	Name	Description
30	TFQM	Tx FIFO/Queue Mode 0= Tx FIFO operation 1= Tx Queue operation
29-24	TFQS	Transmit FIFO/Queue Size 0= No Tx FIFO/Queue 1-32= Number of Tx Buffers used for Tx FIFO/Queue >32= Values greater than 32 are interpreted as 32
21-16	NDTB	Number of Dedicated Transmit Buffers 0= No Dedicated Tx Buffers 1-32= Number of Dedicated Tx Buffers >32= Values greater than 32 are interpreted as 32

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Bits	Name	Description
15-2	TBSA	Tx Buffers Start Address Start address of Tx Buffers section in Message RAM (32-bit word address, see Figure 2). Note: Be aware that the sum of TFQS and NDTB may be not greater than 32. There is no check for erroneous configurations. The Tx Buffers section in the Message RAM starts with the dedicated Tx Buffers.

TXBC Bits

49.5.36 TXFQS (0xC4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD										TFQF	TFQPI					RSVD			TFGI					RSVD		TFFL					
N/A										R	R					N/A			R					N/A		R					
x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	x	x	x	0	0	0	0	0	x	x	0	0	0	0	0	0

TXFQS [31:0]

Bits	Name	Description
21	TFQF	Tx FIFO/Queue Full 0= Tx FIFO/Queue not full 1= Tx FIFO/Queue full
20-16	TFQPI	Tx FIFO/Queue Put Index Tx FIFO/Queue write index pointer, range 0 to 31.
12-8	TFGI	Tx FIFO Get Index Tx FIFO read index pointer, range 0 to 31. Read as zero when Tx Queue operation is configured (TXBC.TFQM = ‘1’).
5-0	TFFL	Tx FIFO Free Level Number of consecutive free Tx FIFO elements starting from TFGI, range 0 to 32. Read as zero when Tx Queue operation is configured (TXBC.TFQM = ‘1’) Note: In case of mixed configurations where dedicated Tx Buffers are combined with a Tx FIFO or a Tx Queue, the Put and Get Indices indicate the number of the Tx Buffer starting with the first dedicated Tx Buffers. Example: For a configuration of 12 dedicated Tx Buffers and a Tx FIFO of 20 Buffers a Put Index of 15 points to the fourth buffer of the Tx FIFO.

TXFQS Bits

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49.5.37 TXESC (0xC8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																													TBDS		
N/A																													RW		
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0

TXESC [31:0]

Bits	Name	Description
2-0	TBDS	Tx Buffer Data Field Size 000= 8 byte data field 001= 12 byte data field 010= 16 byte data field 011= 20 byte data field 100= 24 byte data field 101= 32 byte data field 110= 48 byte data field 111= 64 byte data field Note: In case the data length code DLC of a Tx Buffer element is configured to a value higher than the Tx Buffer data field size TXESC.TBDS, the bytes not defined by the Tx Buffer are transmitted as “0xCC” (padding bytes).

TXESC Bits

49.5.38 TXBRP (0xCC)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TRP																															
R																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TXBRP [31:0]

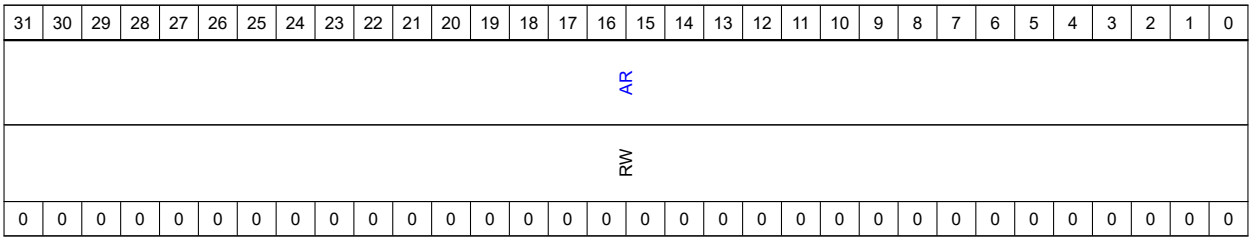
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Bits	Name	Description
31-0	TRP	Transmission Request Pending Each Tx Buffer has its own Transmission Request Pending bit. The bits are set via register TXBAR.The bits are reset after a requested transmission has completed or has been cancelled via register TXBCR. TXBRP bits are set only for those Tx Buffers configured via TXBC. After a TXBRP bit has been set, a Tx scan (see Section 3.5, Tx Handling) is started to check for the pending Tx request with the highest priority (Tx Buffer with lowest Message ID). A cancellation request resets the corresponding transmission request pending bit of register TXBRP. In case a transmission has already been started when a cancellation is requested, this is done at the end of the transmission, regardless whether the transmission was successful or not. The cancellation request bits are reset directly after the corresponding TXBRP bit has been reset. After a cancellation has been requested, a finished cancellation is signalled via TXBCF ? after successful transmission together with the corresponding TXBTO bit ? when the transmission has not yet been started at the point of cancellation ? when the transmission has been aborted due to lost arbitration ? when an error occurred during frame transmission In DAR mode all transmissions are automatically cancelled if they are not successful. The corresponding TXBCF bit is set for all unsuccessful transmissions. 0= No transmission request pending 1= Transmission request pending Note: TXBRP bits which are set while a Tx scan is in progress are not considered during this particular Tx scan. In case a cancellation is requested for such a Tx Buffer, this Add Request is cancelled immediately, the corresponding TXBRP bit is reset.

TXBRP Bits

49.5.39 TXBAR (0xD0)

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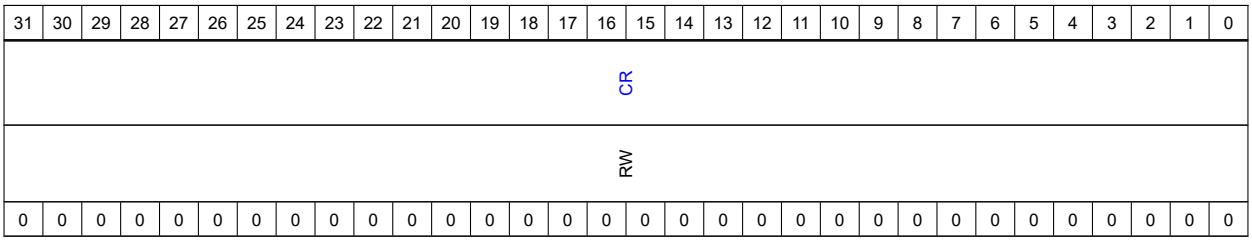


TXBAR [31:0]

Bits	Name	Description
31-0	AR	Add Request Each Tx Buffer has its own Add Request bit. Writing a ‘1’ will set the corresponding Add Request bit; writing a ‘0’ has no impact. This enables the Host to set transmission requests for multiple Tx Buffers with one write to TXBAR. TXBAR bits are set only for those Tx Buffers configured via TXBC. When no Tx scan is running, the bits are reset immediately, else the bits remain set until the Tx scan process has completed. 0= No transmission request added 1= Transmission requested added Note: If an add request is applied for a Tx Buffer with pending transmission request (corresponding TXBRP bit already set), this add request is ignored.

TXBAR Bits

49.5.40 TXBCR (0xD4)



TXBCR [31:0]

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Bits	Name	Description
31-0	CR	Cancellation Request Each Tx Buffer has its own Cancellation Request bit. Writing a '1' will set the corresponding Cancellation Request bit; writing a '0' has no impact. This enables the Host to set cancellation requests for multiple Tx Buffers with one write to TXBCR. TXBCR bits are set only for those Tx Buffers configured via TXBC. The bits remain set until the corresponding bit of TXBRP is reset. 0= No cancellation pending 1= Cancellation pending

TXBCR Bits

49.5.41 TXBTO (0xD8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TO																															
R																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TXBTO [31:0]

Bits	Name	Description
31-0	TO	Transmission Occurred Each Tx Buffer has its own Transmission Occurred bit. The bits are set when the corresponding TXBRP bit is cleared after a successful transmission. The bits are reset when a new transmission is requested by writing a '1' to the corresponding bit of register TXBAR. 0= No transmission occurred 1= Transmission occurred

TXBTO Bits

49.5.42 TXBCF (0xDC)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CF																															
R																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TXBCF [31:0]																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Name	Description
31-0	CF	Cancellation Finished Each Tx Buffer has its own Cancellation Finished bit. The bits are set when the corresponding TXBRP bit is cleared after a cancellation was requested via TXBCR. In case the corresponding TXBRP bit was not set at the point of cancellation, CF is set immediately. The bits are reset when a new transmission is requested by writing a ‘1’ to the corresponding bit of register TXBAR. 0= No transmit buffer cancellation 1= Transmit buffer cancellation finished

TXBCF Bits

49.5.43 TXBTIE (0xE0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TIE																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TXBTIE [31:0]

Bits	Name	Description
31-0	TIE	Transmission Interrupt Enable Each Tx Buffer has its own Transmission Interrupt Enable bit. 0= Transmission interrupt disabled 1= Transmission interrupt enable

TXBTIE Bits

49.5.44 TXBCIE (0xE4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CFIE																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TXBCIE [31:0]

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Bits	Name	Description
31-0	CFIE	Cancellation Finished Interrupt Enable Each Tx Buffer has its own Cancellation Finished Interrupt Enable bit. 0= Cancellation finished interrupt disabled 1= Cancellation finished interrupt enabled

TXBCIE Bits

49.5.45 TXEFC (0xF0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD		EFWM						RSVD		EFS						EFSA														RSVD	
N/A		RW						N/A		RW						RW														N/A	
x	x	0	0	0	0	0	0	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x

TXEFC [31:0]

Bits	Name	Description
29-24	EFWM	Event FIFO Watermark 0= Watermark interrupt disabled 1-32= Level for Tx Event FIFO watermark interrupt (IR.TEFW) >32= Watermark interrupt disabled
21-16	EFS	Event FIFO Size 0= Tx Event FIFO disabled 1-32= Number of Tx Event FIFO elements >32= Values greater than 32 are interpreted as 32 The Tx Event FIFO elements are indexed from 0 to EFS - 1
15-2	EFSA	Event FIFO Start Address Start address of Tx Event FIFO in Message RAM (32-bit word address)

TXEFC Bits

49.5.46 TXEFS (0xF4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD						TEFL	EFF	RSVD			EFPI					RSVD			EFGI					RSVD		EFFL					
N/A						R	R	N/A			R					N/A			R					N/A		R					
x	x	x	x	x	x	0	0	x	x	x	0	0	0	0	0	x	x	x	0	0	0	0	0	x	x	0	0	0	0	0	0

TXEFS [31:0]

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Bits	Name	Description
25	TEFL	Tx Event FIFO Element Lost This bit is a copy of interrupt flag IR.TEFL. When IR.TEFL is reset, this bit is also reset. 0= No Tx Event FIFO element lost 1= Tx Event FIFO element lost, also set after write attempt to Tx Event FIFO of size zero.
24	EFF	Event FIFO Full 0= Tx Event FIFO not full 1= Tx Event FIFO full
20-16	EFPI	Event FIFO Put Index Tx Event FIFO write index pointer, range 0 to 31.
12-8	EFGI	Event FIFO Get Index Tx Event FIFO read index pointer, range 0 to 31.
5-0	EFFL	Event FIFO Fill Level Number of elements stored in Tx Event FIFO, range 0 to 32.

TXEFS Bits

49.5.47 TXEFA (0xF8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																											EFAI					
N/A																											RW					
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0

TXEFA [31:0]

Bits	Name	Description
4-0	EFAI	Event FIFO Acknowledge Index After the Host has read an element or a sequence of elements from the Tx Event FIFO it has to write the index of the last element read from Tx Event FIFO to EFAI. This will set the Tx Event FIFO Get Index TXEFS.EFGI to EFAI + 1 and update the Event FIFO Fill Level TXEFS.EFFL.

TXEFA Bits

49.5.48 TS_SEL (0x200 + 0x4 * n)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TS																															
R																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TS_SEL [31:0]

Bits	Name	Description
31-0	TS	Timestamp Word TS default can save 16 timestamps with 32bit; if ts64_en is set, then work at 64bit mode, can save 8 timestamps with 01/23/45….

TS_SEL Bits

49.5.49 CREL (0x240)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
REL				STEP				SUBSTEP				YEAR				MON								DAY							
R				R				R				R				R								R							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CREL [31:0]

Bits	Name	Description
31-28	REL	Core Release One digit, BCD-coded
27-24	STEP	Step of Core Release One digit, BCD-coded.
23-20	SUBSTEP	Sub-step of Core Release One digit, BCD-coded
19-16	YEAR	Timestamp Year One digit, BCD-coded. This field is set by generic parameter on synthesis.
15-8	MON	Timestamp Month Two digits, BCD-coded. This field is set by generic parameter on synthesis.
7-0	DAY	Timestamp Day Two digits, BCD-coded. This field is set by generic parameter on synthesis.

CREL Bits

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49.5.50 TSCFG (0x244)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																TBPRE								RSVD				EN64	SCP	TBCS	TSUE
N/A																RW								N/A				RW	RW	RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	x	x	x	x	0	0	0	0

TSCFG [31:0]

Bits	Name	Description
15-8	TBPRE	Timebase Prescaler 0x00 to 0xFF The value by which the oscillator frequency is divided for generating the timebase counter clock. Valid values for the Timebase Prescaler are 0 to 255. The actual interpretation by the hardware of this value is such that one more than the value programmed here is used. Affects only the TSU internal timebase. When the internal timebase is excluded by synthesis, TBPRE[7:0] is fixed to 0x00, the Timestamp Prescaler is not used.
3	EN64	set to use 64bit timestamp. when enabled, tsu can save up to 8 different timestamps, TS(k) and TS(k+1) are used for one 64bit timestamp, k is 0~7. TSP can be used to select different one
2	SCP	Select Capturing Position 0: Capture Timestamp at EOF 1: Capture Timestamp at SOF
1	TBCS	Timebase Counter Select When the internal timebase is excluded by synthesis, TBCS is fixed to ‘1’. 0: Timestamp value captured from internal timebase counter, ATB.TB[31:0] is the internal timbase counter 1: Timestamp value captured from input tsu_tbin[31:0],ATB.TB[31:0] is tsu_tbin[31:0]
0	TSUE	Timestamp Unit Enable 0: TSU disabled 1: TSU enabled

TSCFG Bits

49.5.51 TSS1 (0x248)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TSL																TSN															
R																R															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TSS1 [31:0]

Bits	Name	Description
31-16	TSL	Timestamp Lost Each Timestamp register (TS0-TS15) is assigned one bit. The bits are set when the timestamp stored in the related Timestamp register was overwritten before it was read. Reading a Timestamp register resets the related bit.
15-0	TSN	Timestamp New Each Timestamp register (TS0-TS15) is assigned one bit. The bits are set when a timestamp was stored in the related Timestamp register. Reading a Timestamp register resets the related bit.

TSS1 Bits

49.5.52 TSS2 (0x24C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																												TSP			
N/A																												R			
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0

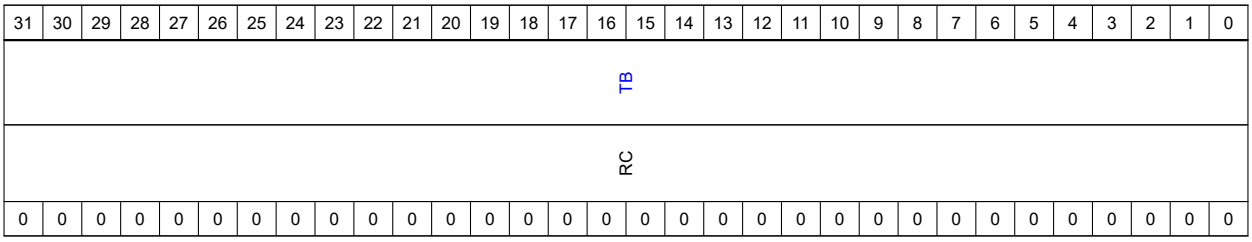
TSS2 [31:0]

Bits	Name	Description
3-0	TSP	Timestamp Pointer The Timestamp Pointer is incremented by one each time a timestamp is captured. From its maximum value (3, 7, or 15 depending on number_ts_g), it is incremented to 0. Value also signalled on output m_can_tsp[3:0].

TSS2 Bits

49.5.53 ATB (0x250)

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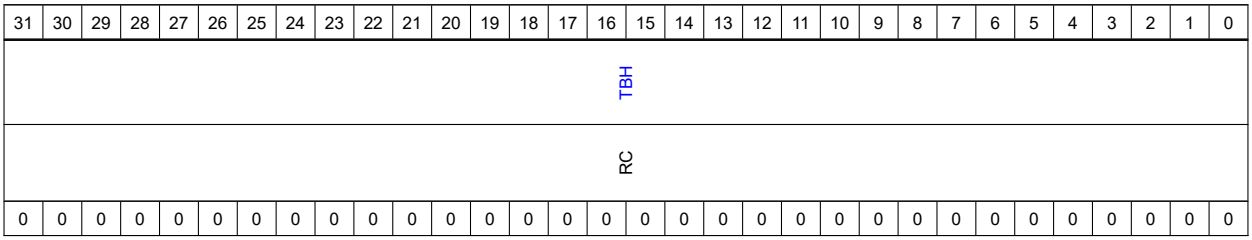


ATB [31:0]

Bits	Name	Description
31-0	TB	timebase for timestamp generation 31-0

ATB Bits

49.5.54 ATBH (0x254)

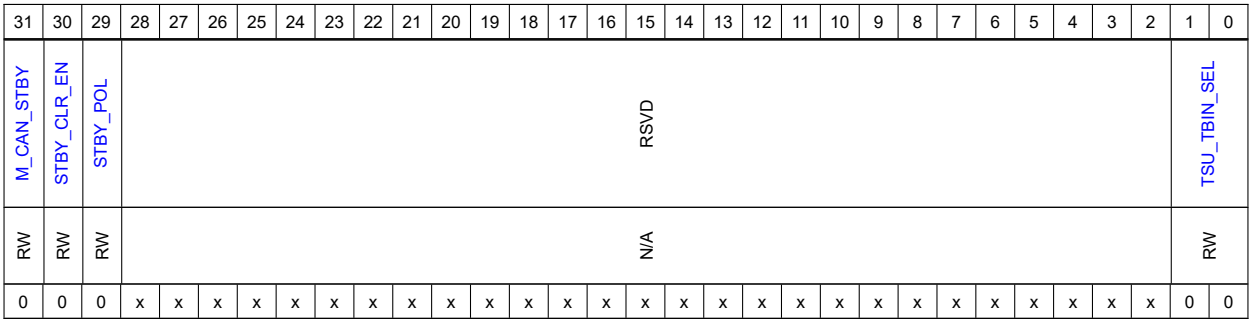


ATBH [31:0]

Bits	Name	Description
31-0	TBH	timebase for timestamp generation 63-32

ATBH Bits

49.5.55 GLB_CTL (0x400)



GLB_CTL [31:0]

Bits	Name	Description
31	M_CAN_STBY	m_can standby control

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Bits	Name	Description
30	STBY_CLR_EN	m_can standby clear control 0:controlled by software by standby bit[bit31] 1:auto clear standby by hardware when rx data is 0
29	STBY_POL	standby polarity selection
1-0	TSU_TBIN_SEL	external timestamp select. each CAN block has 4 timestamp input, this register is used to select one of them as timestame if TSCFG.TBCS is set to 1

GLB_CTL Bits

49.5.56 GLB_STATUS (0x404)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																												M_CAN_INT1	M_CAN_INT0	RSVD	
N/A																												R	R	N/A	
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	x	x

GLB_STATUS [31:0]

Bits	Name	Description
3	M_CAN_INT1	m_can interrupt status1
2	M_CAN_INT0	m_can interrupt status0

GLB_STATUS Bits

49.6 CAN Configuration and Usage Quick Guide

49.6.1 Software Initialization

CAN software can be initialized by setting the init bit of the CCCR register. When the init bit of the CCCR register is set to 1, the CAN bus stops receiving and transmitting data, and the output TXD is always 1. At this time, it can control the read and write permissions of the register together with the cce bit of CCCR. When there is read and write permission, the software can initialize the register. Note that the cce bit can only be configured when the INIT bit is 1. When INIT is cleared to 0, CCE will also be automatically cleared to 0. The following registers will be reset when CCE is set:

- HPMS: High Priority Message Status
- RXF0S: Receive FIFO 0 Status
- RXF1S: Receive FIFO 1 Status
- TXFQS: Transmit FIFO/Queue Status
- TXBRP: Transmit Request Pending
- TXBTO: Transmission Occurred

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- TXBCF: Transmit Buffer Cancellation Finished
- TXEFS: Transmit Event FIFO Status

And the following two registers can only be written after CCE is cleared to 0:

- TXBAR: Transmit Buffer Add Request
- TXBCR: Transmit Buffer Cancellation Request

49.6.2 Receive Processing

The CAN controller provides two types of receive filters, one using standard identifiers and the other using extended identifiers. These filters can be assigned to receive buffers or receive FIFO 0/1. For receive filters, each filter starts from element 0 until a matching end is encountered, and elements after the matched element are not queried for matching. The main functions of the receive filter are:

- Each filter can be configured as:
  - Range filter: The message identifier matches one or two identifiers in registers SF1ID/SF2ID or EF1ID/EFID2
  - Filter for one or two specific identifiers: Configure registers SF1ID=SF2ID or EF1ID=EFID2
  - Classic mask filter: The received identifier is ANDed with the bit mask, SF1ID/EF1ID is used as the message filter, and SF2ID/EF2ID is used as the filter mask.
- Each filter can be configured to accept or reject received data after matching
- Each filter can be enabled and disabled individually

According to the filter configuration (SFEC/EFEC), one of the following operations will be triggered after matching:

- Place the received frame into FIFO0 or FIFO1
- Place the received frame into the receive buffer
- Reject the received data frame
- Set the HPM bit of the high priority message interrupt flag register IR
- Set the HPM bit of the high priority message interrupt flag register IR and place the received frame into FIFO0 or FIFO1

The receive filter starts filtering after receiving the complete identifier. After filtering is completed, if it is a matching identifier, the received data frame is stored in the receive buffer or FIFO in 32-bit format. If the data frame is not accepted or an error is detected in the received data frame (such as CRC check error), the data will be discarded.

49.6.3 Transmit Processing

The transmit processing handles transmit requests from transmit buffers, transmit FIFOs and transmit queues. Up to 32 transmit buffers can be set for transmitting data, and each transmit buffer can be individually configured as normal CAN bus mode or CAN FD mode. The specific configuration is shown in the table below.

When the register TXBRP is updated, the transmit buffer data will be automatically scanned to transmit the highest priority transmit request. The hardware automatically updates the value of register TXBRP when transmission is completed or the software adds or cancels a transmit task.

49.6.3.1 Transmit Buffer

Each transmit buffer has a configurable message identifier. If multiple transmit buffers are configured with

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Table 206: Frame Transmission Configuration

Register CCCR		Transmit Buffer		Frame Transmission
BRSE	FDOE	FDF	BRS	
Ignore	0	Ignore	Ignore	Normal CAN format transmission
0	1	0	Ignore	Normal CAN format transmission
0	1	1	Ignore	CAN FD format transmission without bit rate switch
1	1	0	Ignore	Normal CAN format transmission
1	1	1	0	CAN FD format transmission without bit rate switch
1	1	1	1	CAN FD format transmission with bit rate switch

the same message identifier, the transmit buffer with the smaller buffer table number is transmitted first. After updating the transmitted data part, you can add a transmit request by configuring register TXBRP. At this time, the transmit request is arbitrated with the transmit FIFO or transmit queue internally according to the message identifier, and the one with higher priority is transmitted first.

49.6.3.2 Transmit FIFO

The transmit FIFO is implemented by configuring the TFQM bit of register TXBC to 0. When adding a message to the transmit FIFO, write 1 to the corresponding bit of register TXBAR to enable the transmit operation. For each message transmitted in the transmit FIFO, the TFGI of the corresponding register TXFQS is incremented by 1 until the transmit FIFO is empty.

49.6.3.3 Transmit Queue

The transmit queue is implemented by configuring the TFQM bit of register TXBC to 1. Messages with lower message identifiers in the transmit queue have higher priority. New messages should be placed in the transmit buffer corresponding to TFQPI of TXFQS.

49.6.3.4 Transmit Cancellation

The CAN controller supports canceling transmit operations, which is very suitable for gateway applications and AUTOSAR. If you want to cancel a transmit request, you only need to set the bit corresponding to the transmit request in register TXBCR to 1. The function of canceling transmit requests only supports those in transmit buffers or transmit queues, and does not support canceling transmit requests in transmit FIFOs for the time being. After successfully canceling the transmit request, the corresponding bit of register TXBCF will be set to 1.

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50 Precision Time Protocol Module PTPC

This chapter describes the main functions and features of the Precision Time Protocol module PTPC.

50.1 Feature Summary

This chapter describes the main features of the Precision Time Protocol module PTPC:

- Supports 2 sets of timestamps
- Timestamps support 63-bit counters
- The counter is divided into a 32-bit second counter and a 31-bit nanosecond counter
- Supports input capture timestamps
- Supports generating interrupts when timestamp matching occurs
- Supports sending data to the CAN module from the timestamp read interface

50.2 Functional Description

This chapter describes the functions of the Precision Time Protocol module PTPC.

50.2.1 Structure Diagram of Relationship with CAN

The relationship between the Precision Time Protocol module PTPC and the CAN module is shown in Figure 82.

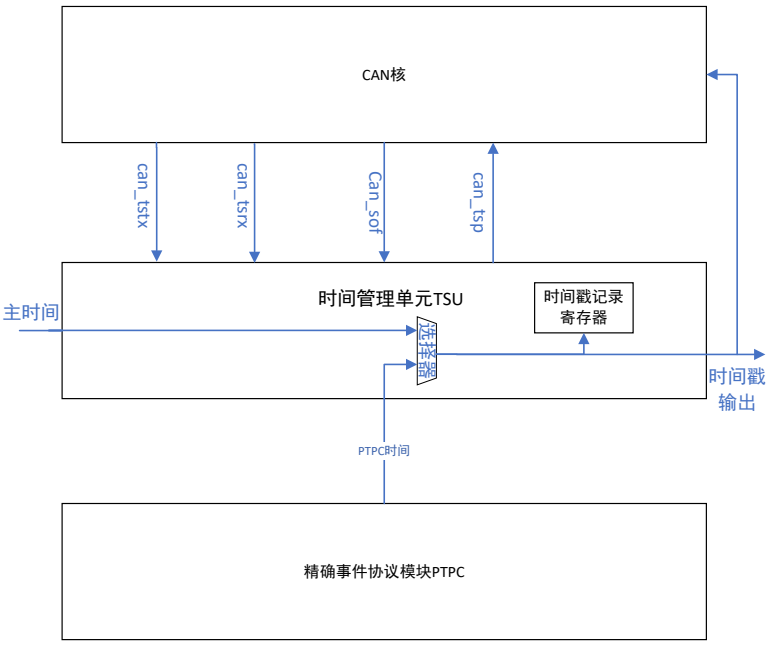


Figure 82: Relationship between PTPC and CAN

50.2.2 Timestamp Module

The Precision Time Protocol module PTPC contains 2 timestamp modules, each with a 63-bit counter.

The counter is divided into a 32-bit second counter TIMEH and a 31-bit nanosecond counter TIMEL

When TIMEL is updated each time, it does not simply add 1 like a traditional counter, but adds the value of

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CTRL1[SS_INCR]:

$$TIMEL = TIMEL + SS_INCR$$

Assume that the PTPC clock is 100 MHz, and each clock cycle is 10 ns. At this time, CTRL1 [SS_INCR] needs to be configured to 8'd10, that is, TIMEL accumulates in units of 10 nanoseconds each time.

The counter is divided into 31-bit TIMEL and 32-bit TIMEH, and TIMEH accumulates whenever TIMEL overflows. The overflow point of TIMEL supports the following 2 types:

- Counter overflow: the lower 31-bit TIMEL overflows when counting to 32'h7FFFFFFF. When TIMEL overflows, TIMEH increases by 1
- Second count overflow: the lower 31-bit TIMEL overflows when counting to 1000000000, i.e. 32'h3B9ACA00. When TIMEL overflows, TIMEH increases by 1. TIMEL counts in nanoseconds, and TIMEH counts in seconds.

When the user sets CTRL0 [SUBSEC_DIGITAL_ROLLOVER] to 0, the counter is configured in counter mode. When set to 1, the counter is configured in second count overflow mode.

It is recommended that the user set this bit to 1, so that when reading the timestamp counter, the real time of TIMEH seconds plus TIMEL nanoseconds can be obtained intuitively.

Before using PTPC, you need to complete the PTPC initialization first. The steps to initialize the timestamp counters TIMEL and TIMEH are as follows:

- Set the CTRL0[TIMER_ENABLE] bit to 1
- Write the target values to the TS_UPDTH and TS_UPDTL registers. Note that if TIMEL is set to second count overflow, the value of TS_UPDTL should not exceed 1000000000, i.e. 32'h3B9ACA00
- Set the CTRL0[INIT_TIMER] bit to 1

After PTPC starts timing, users can update the timer as needed. The steps for PTPC to update the timestamps TIMEL and TIMEH are as follows:

- Write the target values to the TS_UPDTH and TS_UPDTL registers.
- Configure the TS_UPDTL[ADD_SUB] bit: when set to 0, TIMEL/TIMEH will add TS_UPDTL and TS_UPDTH to the original value; when set to 1, TIMEL/TIMEH will subtract TS_UPDTL and TS_UPDTH from the original value
- Set the CTRL0[UPDATE_TIMER] bit to 1

PTPC supports fine-tuning the timing speed of its timestamp counter by configuring the ADDEND register.

As a nanosecond counter, TIMEL accumulates CTRL1 [SS_INCR] every PTPC clock cycle. When the PTPC clock is 100 MHz (i.e. the cycle is 10 ns), SS_INCR should be configured to 10, that is, TIMEL accumulates 10 every PTPC clock cycle. When the clock synchronization software performs clock synchronization based on the PTPC timer, it may find that the error between the local time and the network clock Master is caused by the error of the PTPC clock. At this time, equivalent fine adjustment can be achieved by configuring the ADDEND register.

During fine adjustment, there is an accumulator that adds ADDEND for each PTPC clock, and TIMEL adds SS_INCR when the accumulator overflows. Therefore, if you use the fine adjustment mode, it is recommended to set SS_INCR to 20, when ADDEND=0x80000000, the effect is the same as the original, that is, add 20 in two clock cycles.

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50.2.3 Timestamp Capture and Comparison

PTPC supports input capture function. Each timestamp supports one capture input signal, and the valid edge of the CAP0 or CAP1 signal can be configured. When the rising or falling edge of CAP0 or CAP1 occurs, the values of the timestamps TIMEL and TIMEH are captured and stored in the registers CAPT_SNAPL and CAPT_SNAPH. When capture occurs, the capture flag bit will be set to 1. If the corresponding interrupt enable bit is set to 1 in the interrupt enable register, a timestamp capture interrupt request will be generated.

PTPC supports timestamp matching function. The user writes the target timestamp value to the TARH and TARL registers and sets CTRL0[COMP_EN] to 1. When the timestamp counters TIMEH and TIMEL are greater than or equal to TARH and TARL, a matching event occurs, and COMP_EN will be automatically cleared after the matching event occurs.

At this time, the matching flag bit will be set to 1. If the corresponding interrupt enable bit is set to 1 in the interrupt enable register, a timestamp matching interrupt request will be generated.

PTPC supports generating timestamp matching output comparison, and outputs CMP0 or CMP1 to other modules of the system when the timestamp matches.

50.2.4 Timestamp Output Port

PTPC supports 4 timestamp output ports, from which other modules on the system can directly load the values of TIMEL and TIMEH. Users can directly configure the source of timestamp information on port x through the TIME_SEL register. When the corresponding selection bit is set to 0, the timestamp of module 0 is output. When the selection bit is set to 1, the timestamp of module 1 is output.

50.3 PTPC Register List

The register list of PTPC is as follows:

PTPC base address: 0xF02FC000

Offset	Register Name	Description	Reset Value
0x0000	PTPC[0][CTRL0]	Control Register 0	0x00000000
0x0004	PTPC[0][CTRL1]	Control Register 1	0x00000000
0x0008	PTPC[0][TIMEH]	timestamp high	0x00000000
0x000C	PTPC[0][TIMEL]	timestamp low	0x00000000
0x0010	PTPC[0][TS_UPDTH]	timestamp update high	0x00000000
0x0014	PTPC[0][TS_UPDTL]	timestamp update low	0x00000000
0x0018	PTPC[0][ADDEND]		0x00000000
0x001C	PTPC[0][TARH]		0x00000000
0x0020	PTPC[0][TARL]		0x00000000
0x002C	PTPC[0][PPS_CTRL]		0x00000000
0x0030	PTPC[0][CAPT_SNAPH]		0x00000000
0x0034	PTPC[0][CAPT_SNAPL]		0x00000000
0x1000	PTPC[1][CTRL0]	Control Register 0	0x00000000
0x1004	PTPC[1][CTRL1]	Control Register 1	0x00000000
0x1008	PTPC[1][TIMEH]	timestamp high	0x00000000

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Offset	Register Name	Description	Reset Value
0x100C	PTPC[1][TIMEL]	timestamp low	0x00000000
0x1010	PTPC[1][TS_UPDTH]	timestamp update high	0x00000000
0x1014	PTPC[1][TS_UPDTL]	timestamp update low	0x00000000
0x1018	PTPC[1][ADDEND]		0x00000000
0x101C	PTPC[1][TARH]		0x00000000
0x1020	PTPC[1][TARL]		0x00000000
0x102C	PTPC[1][PPS_CTRL]		0x00000000
0x1030	PTPC[1][CAPT_SNAPH]		0x00000000
0x1034	PTPC[1][CAPT_SNAPL]		0x00000000
0x2000	TIME_SEL		0x00000000
0x2004	INT_STS		0x00000000
0x2008	INT_EN		0x00000000
0x3000	PTPC_CAN_TS_SEL		0x00000000

Table 207: PTPC Register List

50.4 PTPC Register Description

The detailed description of PTPC registers is as follows:

50.4.1 PTPC[CTRL0] (0x0 + 0x1000 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																						SUBSEC_DIGITAL_ROLLOVER	CAPT_SNAP_KEEP	CAPT_SNAP_POS_EN	CAPT_SNAP_NEG_EN	RSVD	COMP_EN	UPDATE_TIMER	INIT_TIMER	FINE_COARSE_SEL	TIMER_ENABLE
N/A																						RW	RW	RW	RW	N/A	RW	WO	WO	RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0

PTPC[CTRL0] [31:0]

Bits	Name	Description
9	SUBSEC_DIGITAL_ROLLOVER	Format for ns counter rollover, 1-digital, overflow time 1000000000/0x3B9ACA00 0-binary, overflow time 0x7FFFFFFF

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Bits	Name	Description
8	CAPT_SNAP_KEEP	set will keep capture snap till software read capt_snapl. If this bit is set, software should read capt_snaph first to avoid wrong result. If this bit is cleared, capture result will be updated at each capture event
7	CAPT_SNAP_POS_EN	set will use posege of input capture signal to latch timestamp value
6	CAPT_SNAP_NEG_EN	
4	COMP_EN	set to enable compare, will be cleared by HW when compare event triggered
3	UPDATE_TIMER	update timer with +/- ts_updt, pulse, clear after set
2	INIT_TIMER	initial timer with ts_updt, pulse, clear after set
1	FINE_COARSE_SEL	0: coarse update, ns counter add ss_incr[7:0] each clk 1: fine update, ns counter add ss_incr[7:0] each time addend counter overflow
0	TIMER_ENABLE	

PTPC[CTRL0] Bits

50.4.2 PTPC[CTRL1] (0x4 + 0x1000 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																								SS_INCR							
N/A																								RW							
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0

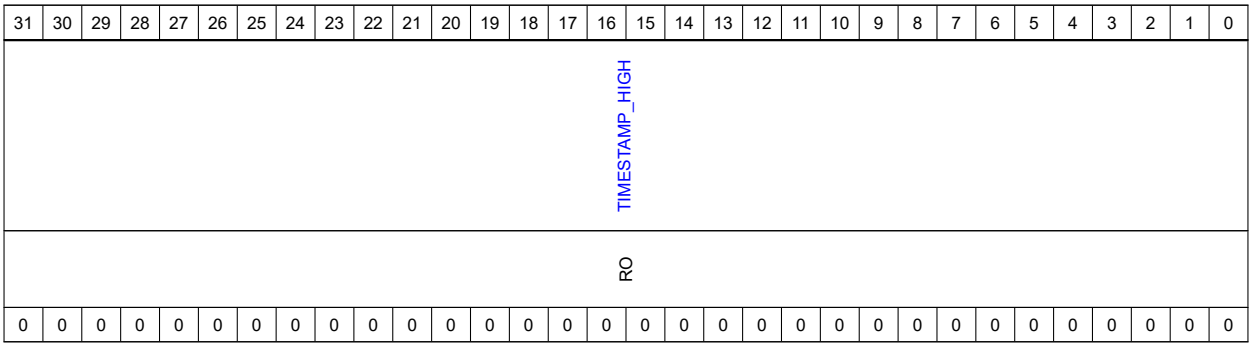
PTPC[CTRL1] [31:0]

Bits	Name	Description
7-0	SS_INCR	constant value used to add ns counter; such as for 50MHz timer clock, set it to 8'd20

PTPC[CTRL1] Bits

50.4.3 PTPC[TIMEH] (0x8 + 0x1000 * n)

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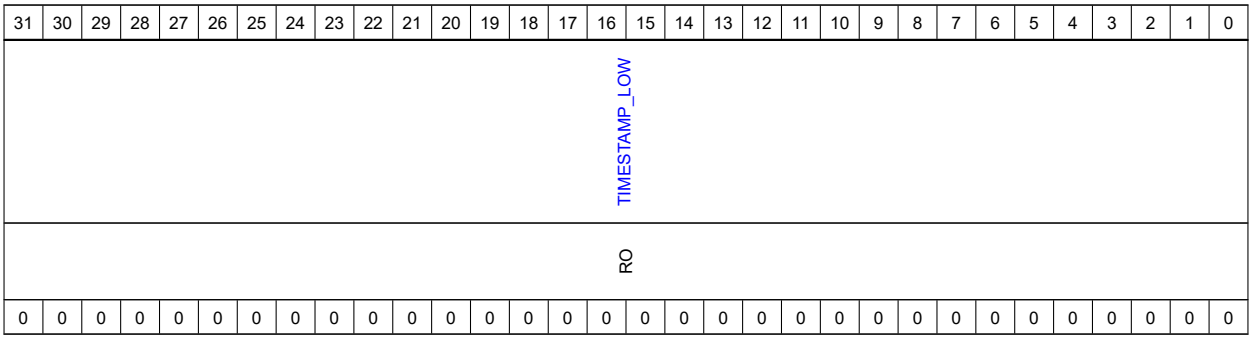


PTPC[TIMEH] [31:0]

Bits	Name	Description
31-0	TIMESTAMP_HIGH	

PTPC[TIMEH] Bits

50.4.4 PTPC[TIMEL] (0xC + 0x1000 * n)

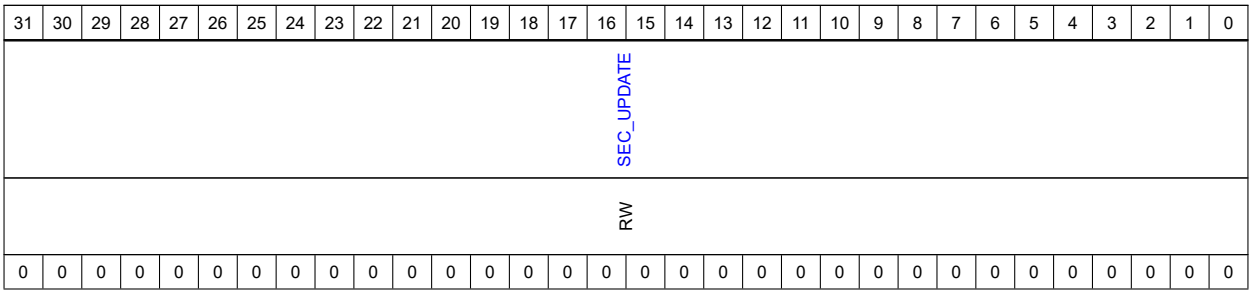


PTPC[TIMEL] [31:0]

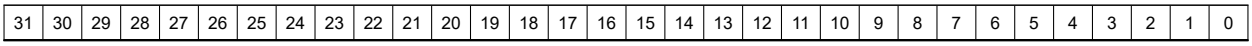
Bits	Name	Description
31-0	TIMESTAMP_LOW	

PTPC[TIMEL] Bits

50.4.5 PTPC[TS_UPDTH] (0x10 + 0x1000 * n)



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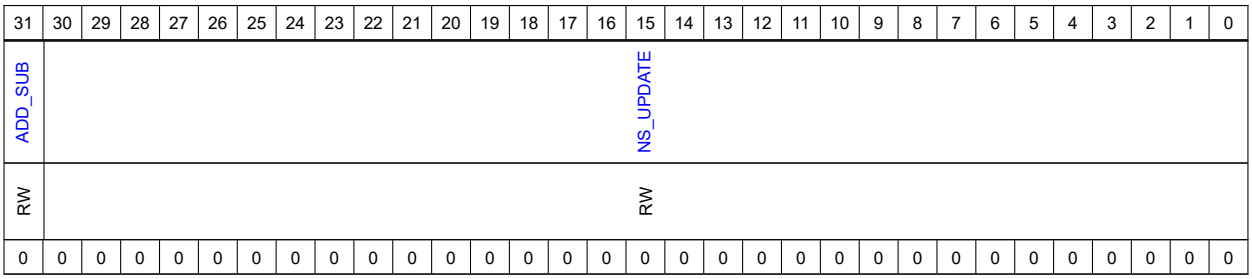


PTPC[TS_UPDTH] [31:0]

Bits	Name	Description
31-0	SEC_UPDATE	together with ts_updtl, used to initial or update timestamp

PTPC[TS_UPDTH] Bits

50.4.6 PTPC[TS_UPDTL] (0x14 + 0x1000 * n)

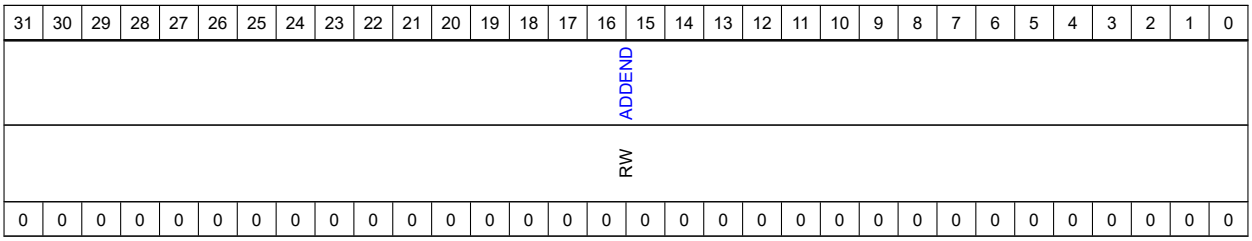


PTPC[TS_UPDTL] [31:0]

Bits	Name	Description
31	ADD_SUB	1 for sub; 0 for add, used only at update
30-0	NS_UPDATE	

PTPC[TS_UPDTL] Bits

50.4.7 PTPC[ADDEND] (0x18 + 0x1000 * n)



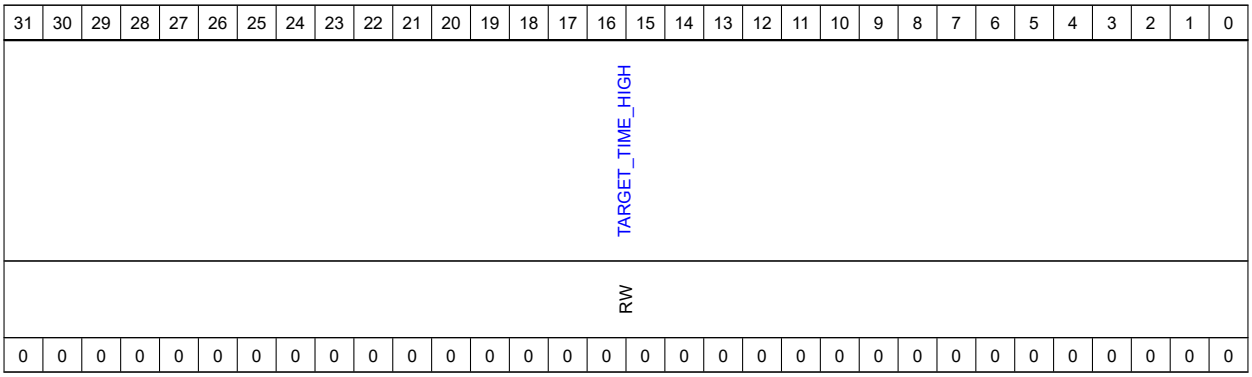
PTPC[ADDEND] [31:0]

Bits	Name	Description
31-0	ADDEND	used in fine update mode only

PTPC[ADDEND] Bits

50.4.8 PTPC[TARH] (0x1C + 0x1000 * n)

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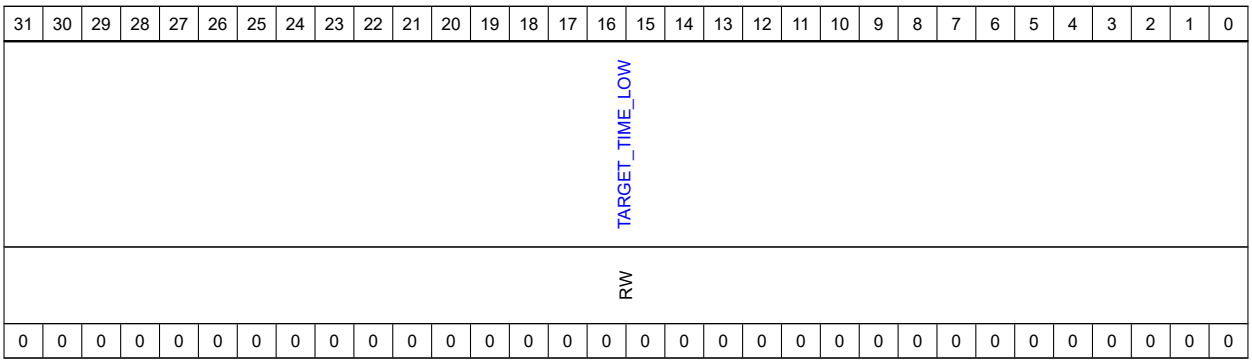


PTPC[TARH] [31:0]

Bits	Name	Description
31-0	TARGET_TIME_HIGH	used for generate compare signal if enabled

PTPC[TARH] Bits

50.4.9 PTPC[TARL] (0x20 + 0x1000 * n)

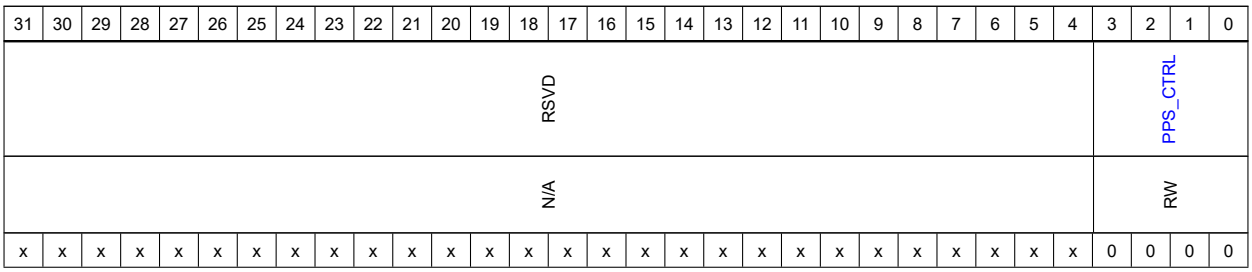


PTPC[TARL] [31:0]

Bits	Name	Description
31-0	TARGET_TIME_LOW	

PTPC[TARL] Bits

50.4.10 PTPC[PPS_CTRL] (0x2C + 0x1000 * n)



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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

PTPC[PPS_CTRL] [31:0]

Bits	Name	Description
3-0	PPS_CTRL	

PTPC[PPS_CTRL] Bits

50.4.11 PTPC[CAPT_SNAPH] (0x30 + 0x1000 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CAPT_SNAP_HIGH																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PTPC[CAPT_SNAPH] [31:0]

Bits	Name	Description
31-0	CAPT_SNAP_HIGH	take snapshot for input capture signal, at pos or neg or both; the result can be kept or updated at each event according to cfg0.bit8

PTPC[CAPT_SNAPH] Bits

50.4.12 PTPC[CAPT_SNAPL] (0x34 + 0x1000 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CAPT_SNAP_LOW																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PTPC[CAPT_SNAPL] [31:0]

Bits	Name	Description
31-0	CAPT_SNAP_LOW	

PTPC[CAPT_SNAPL] Bits

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50.4.13 TIME_SEL (0x2000)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																												CAN3_TIME_SEL	CAN2_TIME_SEL	CAN1_TIME_SEL	CAN0_TIME_SEL
N/A																												RW	RW	RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0

TIME_SEL [31:0]

Bits	Name	Description
3	CAN3_TIME_SEL	
2	CAN2_TIME_SEL	
1	CAN1_TIME_SEL	
0	CAN0_TIME_SEL	set to use ptpc1 for canx clr to use ptpc0 for canx

TIME_SEL Bits

50.4.14 INT_STS (0x2004)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD													COMP_INT_STS1	CAPTURE_INT_STS1	PPS_INT_STS1	RSVD													COMP_INT_STS0	CAPTURE_INT_STS0	PPS_INT_STS0
N/A													W1C	W1C	W1C	N/A													W1C	W1C	W1C
x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0

INT_STS [31:0]

Bits	Name	Description
18	COMP_INT_STS1	
17	CAPTURE_INT_STS1	
16	PPS_INT_STS1	
2	COMP_INT_STS0	
1	CAPTURE_INT_STS0	
0	PPS_INT_STS0	

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Bits	Name	Description

INT_STS Bits

50.4.15 INT_EN (0x2008)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD													COMP_INT_STS1	CAPTURE_INT_STS1	PPS_INT_STS1	RSVD													COMP_INT_STS0	CAPTURE_INT_STS0	PPS_INT_STS0
N/A													RW	RW	RW	N/A													RW	RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0

INT_EN [31:0]

Bits	Name	Description
18	COMP_INT_STS1	
17	CAPTURE_INT_STS1	
16	PPS_INT_STS1	
2	COMP_INT_STS0	
1	CAPTURE_INT_STS0	
0	PPS_INT_STS0	

INT_EN Bits

50.4.16 PTPC_CAN_TS_SEL (0x3000)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TSU_TBIN3_SEL						TSU_TBIN2_SEL						TSU_TBIN1_SEL						TSU_TBIN0_SEL						RSVD							
RW						RW						RW						RW						N/A							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x

PTPC_CAN_TS_SEL [31:0]

Bits	Name	Description
31-26	TSU_TBIN3_SEL	

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Bits	Name	Description
25-20	TSU_TBIN2_SEL	
19-14	TSU_TBIN1_SEL	
13-8	TSU_TBIN0_SEL	

PTPC_CAN_TS_SEL Bits

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51 Universal Serial Bus (USB)

51.1 Function Introduction

Provides USB communication solutions, fully compatible with the USB 2.0 protocol.

- Supports OTG (configurable as host mode or device mode)
- Host mode supports high-speed (480Mbps), full-speed (12Mbps) and low-speed (1.5Mbps)
- Device mode supports high-speed (480Mbps) and full-speed (12Mbps)
- Device mode supports 16 bidirectional endpoints (16IN + 16OUT endpoints)
- Built-in on-chip high-speed USB PHY (supports full-speed and low-speed)
- Built-in DMA, completes data interaction between USB and system memory without CPU participation via descriptors
- Host mode software architecture is fully compatible with the EHCI 1.0 (Enhanced Host Controller Interface Specification for Universal Serial Bus) protocol
- Low-power mode: when all on-chip PLLs are disabled, wake-up can be completed via the on-chip OSC clock

51.2 Pin Description

Table 208: USB Pin Description

Pin	Direction	Function Description
USB_DP	Bidirectional	Positive end of differential signal.
USB_DN	Bidirectional	Negative end of differential signal.

51.3 Working Process

51.3.1 USB PHY Initialization

Exit reset state: OTG_CTRL0.OTG_UTMI_RESRET_SW=0

Exit suspend state: OTG_CTRL0.OTG_UTMI_SUSPENDM_SW=1

Detect USB PHY clock: PHY_STATUS.UTMI_CLK_VALID=1

Wait for USB PHY clock to stabilize: while(PHY_STATUS.UTMI_CLK_VALID==1)

51.3.2 Working Mode Configuration

After system power-on, the USB module is in OTG mode. If you need to change the mode, perform USB reset first (USBCMD.RST=1).

Users can configure it as host or device mode according to requirements.

You can also keep OTG mode and configure it according to external connection status.

For micro-AB interface, keep ID high (configure pull-up via GPIO) and keep VBUS low

- If ID is detected low, a device is connected (or an OTG cable is inserted), set to host mode (USBMODE.CM=3)
- If VBUS is detected high, a host is connected, set to device mode (USBMODE.CM=2)

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For Type-C interface, please refer to Type-C protocol to set host or device mode.

51.3.3 Host Initialization Process

Enable VBUS (generally controlled via GPIO to the on-board power supply chip).

Enable port: PORTSC1.PP=1

Wait for device connection. After receiving port change interrupt (USBSTS.PCI), check port connection status (PORTSC1.CCS), 1 means device is connected.

Check linestate: PHY_STATUS.LINE_STATE

Value 2 means low-speed device is connected, serial mode is required (set PORTSC1.STS=1)

Value 1 means full-speed or high-speed device is connected, parallel mode is used (set PORTSC1.STS=0)

Start USB: USBCMD.RS=1

Reset device: PORTSC1.PR=1

Wait for bus reset completion: while(PORTSC1.PR==1), this process takes about 55ms, a port change interrupt (USBSTS.PCI) will be triggered during this period.

Determine device connection speed: PORTSC1.PSPD

Enumerate device (refer to EHCI protocol for detailed process)

51.3.4 Device Initialization Process

Set parallel mode (device mode does not support low speed, all can work in parallel mode): PORTSC1.STS=0

Prepare receive descriptor for endpoint 0, write its address to ENDPTLISTADDR.

Wait for VBUS (OTGSC.AVV) to be 1.

Start USB: USBCMD.RS=1

Wait for host to send bus reset (USBSTS.URI)

Wait for port change interrupt USBSTS.PCI.

Determine host connection speed: PORTSC1.PSPD

Wait for host to initiate enumeration and respond.

51.4 Data Structure

51.4.1 Host Data Structure

For software development convenience, host data structure is fully compatible with EHCI 1.0 protocol, refer to the protocol for details.

Here is a brief introduction to the asynchronous queue used for BULK/CTRL transmission, all figures are from EHCI protocol.

The asynchronous queue is a circular linked list, as shown in Figure 83. Each item (also called qhead) in the linked list corresponds to one endpoint (EP). A host can connect to multiple devices, and one device can have multiple endpoints.

Hardware will poll this linked list for pending data transmission. One qhead is marked with H=1 as the linked

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list head. When no data is transmitted between two consecutive polls to the linked list head, hardware will stop polling. Software shall write the address of the linked list head to ASYNCLISTADDR register during initialization, hardware will update ASYNCLISTADDR in real time to the address of the currently executing qhead during polling. Each qhead can link 0 or more qTDs. qTD is used for data transmission, including packet address, transmission status, and next qTD address.

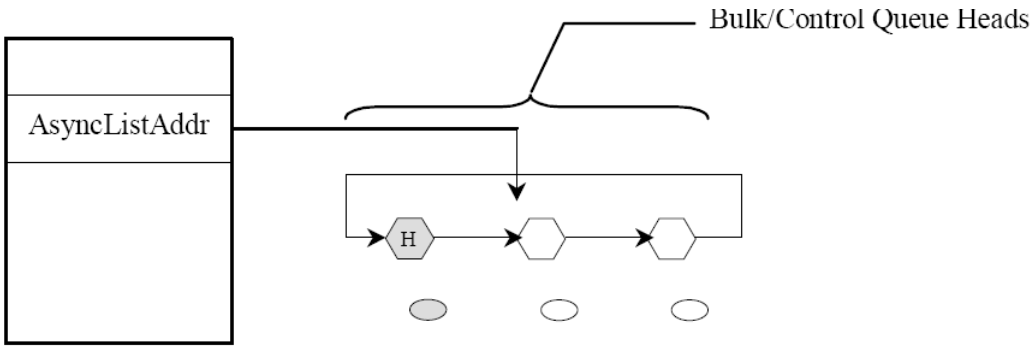
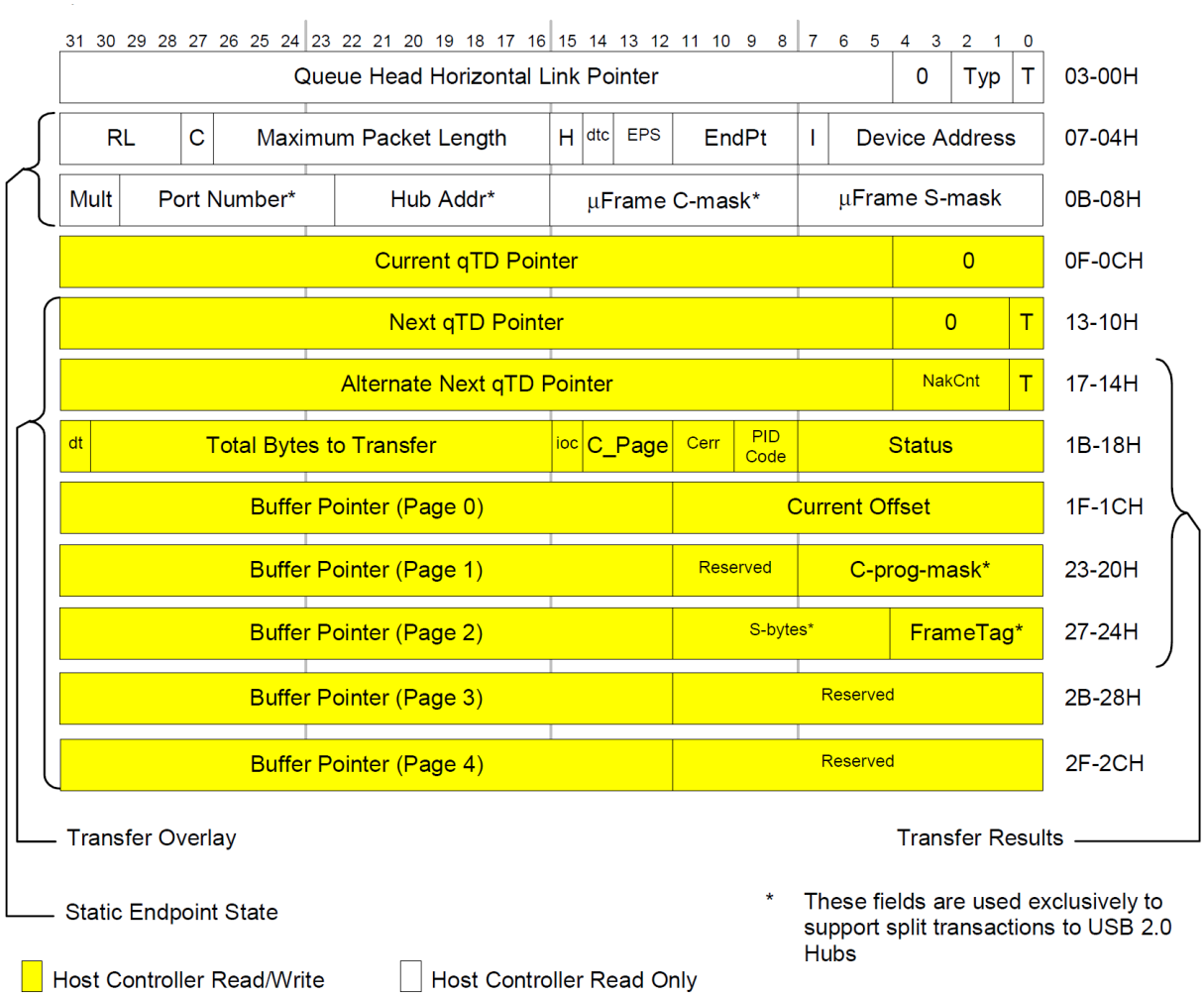


Figure 83: Asynchronous Queue Diagram

qhead structure is shown in Figure 84,



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Figure 84: qhead Diagram

The following introduces the parts required for BULK/CTRL transmission, unmentioned parts refer to EHCI protocol.

The part after byte 0x10 of qhead is also called Transfer Overlay. Hardware will write the currently transmitted qTD to this part of qhead, update the status in qhead before current qTD transmission is completed, and write back to qTD after transmission is completed.

Byte	Bit	Description
0x00	31:5	Queue Head Horizontal Link Pointer Address of next qhead, qhead data structure must be 32-byte aligned. If only one endpoint of one device is used in the application, this address can point to the qhead itself.
0x00	2:1	Typ: qhead type, shall be set to 01 for BULK and CTRL transmission.
0x00	0	T: short for Terminate, set to 1 means current address is invalid. For BULK/CTRL qhead, T bit of Queue Head Horizontal Link Pointer shall be 0.
0x04	31:28	RL: set to 0
0x04	27	C: indicates CTRL endpoint. Set to 1 when qhead.EPS is high-speed device and current endpoint is CTRL endpoint, otherwise set to 0
0x04	26:16	Maximum Packet Length: maximum packet length of current endpoint, can be obtained from device during host enumeration, set to 8 before acquisition.
0x04	15	H: indicates current qhead is the first entry in the linked list. Exactly one qhead in the entire qhead linked list shall be set to H=1
0x04	14	dtc: data toggle control. Set to 1 means use DT bit of qtd as data toggle bit for current transmission; 0 means hardware maintains data toggle. Special rules apply for CTRL transmission (see USB 2.0 protocol 8.5.3 for details)
0x04	13:12	EPS: endpoint speed, full speed 00, low speed 01, high speed 10, 11 is reserved
0x04	11:8	EndPt: endpoint number corresponding to current qhead
0x04	7	I: used for isochronous queues, set to 0 for BULK/CTRL transmission
0x04	6:0	DeviceAddress: device address corresponding to current qhead
0x08	31:0	Used for isochronous transmission and hub related functions, set to 0 for simple applications
0x0C	31:5	Current qTD Pointer: maintained by hardware for the address of currently transmitted qTD, software initializes it to 0
0x10	31:5	Next qTD Pointer: address of next qTD
0x10	0	T: short for Terminate, set to 0 means current address is valid. When initializing qhead, if there is no pending transmission, software shall set T to 1
0x14	31:0	Set to 1 for simple applications (or T=1, other bits are arbitrary, e.g. 0xbadbadff)

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Byte	Bit	Description
0x18	31	dt: data toggle. When qhead.dtc is 1, hardware uses this bit as the starting data toggle for current transmission, only used for CTRL transmission in simple applications
0x18	30:16	Total Bytes Transfer: data length to be transmitted by current dTD, maximum 20Kbyte supported, recommended maximum 16Kbyte. For OUT: software initializes it to the length of data to send, hardware decrements it after each packet transmission (packet length is Maximum Packet Length except the last packet), it shall be 0 after transmission completes; For IN: software initializes it to the size of receive buffer, hardware updates this value after each packet transmission. Transmission completes when value is 0 or short packet (packet length less than Maximum Packet Length) is received, the final value may not be 0
0x18	15	ioc: Interrupt On Complete. Set to 1 means USBSTS.UI will be set after current qTD transmission completes to trigger interrupt. For multi-qTD transmission, software can only set ioc bit for the last qTD to reduce interrupt count and improve transmission efficiency
0x18	14:12	CPage: data buffer pointer, software initializes it to 0, hardware updates it in real time to indicate which buffer the current data is in (valid value 0 4, corresponding to Buffer Pointer Page 0 4 in qTD)
0x18	11:10	Cerr: transmission error counter, software initializes it to 3. It decrements by 1 each time transmission error occurs, current qTD ends and USBSTS.UEI is set to trigger error interrupt when it reaches 0
0x18	9:8	PID Code: PID of current qTD transmission, valid values: OUT-00, IN-01, SETUP-10
0x18	7:0	Status: indicates current qTD status, software initializes it to 0x80. Bit7: Active: indicates whether current qTD is valid. Hardware only processes qTD when Active=1, clears Active after transmission completes; Bit6: Halt: hardware sets Halt to 1 and clears Active when serious error occurs (babble, Cerr count reaches 0, STALL received); Bit5: Data Buffer Error: set to 1 when overrun or underrun occurs; Bit4: Babble: set to 1 when hardware detects babble (USB 2.0 protocol 8.7.4); Bit3: XactErr: indicates transmission error, see EHCI protocol 4.15.1.1 for specific error types; Bits 2:0 can be ignored in simple applications.
0x1C	31:12	Buffer Pointer (Page 0): start address of data packet, specified by buffer0 and current offset. Hardware switches to buffer1 after buffer0 transmission completes. 5 buffers can be consecutive or non-consecutive, only need 4Kbyte alignment

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Byte	Bit	Description
0x1C	11:0	Current Offset: software initializes it to the start offset of data packet in buffer0, hardware updates it in real time to the offset of current buffer (specified by cpage) during transmission, indicates transmitted data length
0x20 0x2F	31:12	Remaining part of qhead, only need to configure data buffer pointer address (page1 4) for simple applications

Table 209: qhead Structure

qtd structure is shown in Figure 85, its content is the same as the overlay part of qhead. When there is pending data transmission, software shall initialize one or more qtds, and write the address of the first qtd to qhead.Next qTD Pointer.

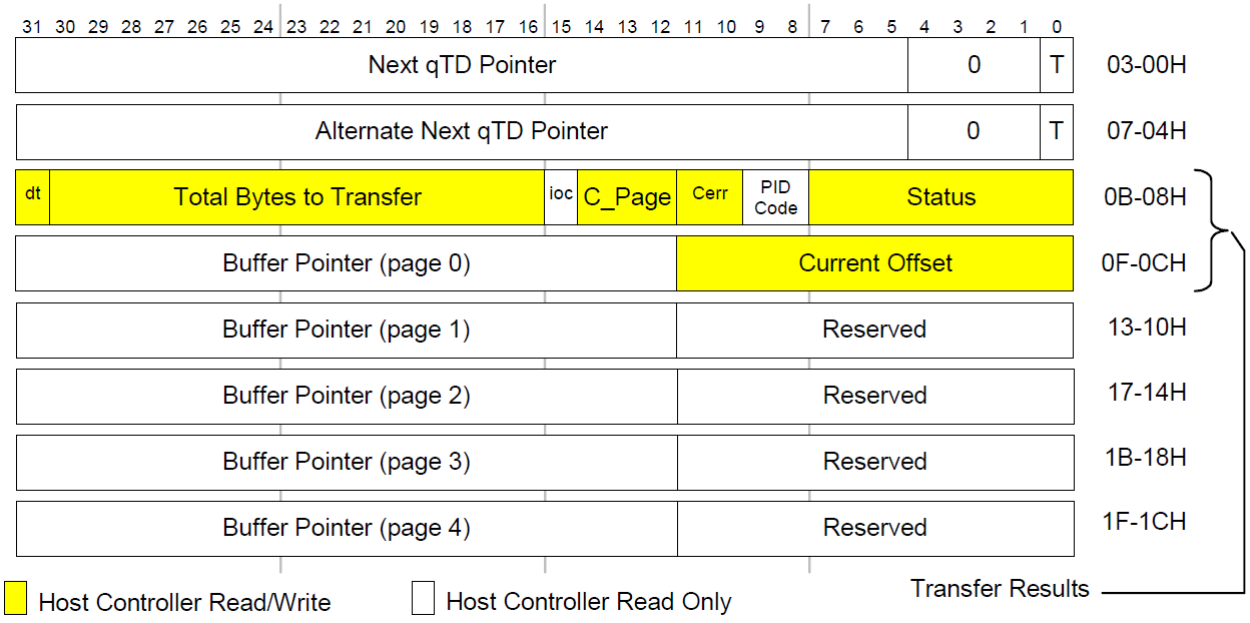


Figure 85: qtd Diagram

51.4.2 Device Data Structure

Device data structure is a space allocated by software in storage, consisting of the following parts:

- dqhlist: stores all dqhs, each endpoint corresponds to two dqhs (IN and OUT), each dqh occupies 64 bytes, start address of dqhlist must be 2Kbyte aligned.
- dqh: configures endpoint parameters, including maximum packet length, current and next transmission descriptor addresses, and saves current transmission descriptor content, as shown in Figure 87. See Table 210 for details.
- dtd: transmission descriptor, used for data transmission, including data length, start address, status and other information, as shown in Figure 88

As shown in Figure 86, register ENDPTLISTADDR points to the start address of dqhlist. Software shall prepare at least endpoint 0 receive dqh (EP0 OUT dqh, located at the start address of dqhlist) before starting USB

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(USBCMD.RS=1), to receive the first SETUP packet in enumeration information sent by host, then prepare dqhs and dtds for corresponding endpoints according to received data.

Take receiving the first SETUP packet of enumeration as an example, the software process is as follows:

- Allocate a 2Kbyte aligned space for dqhlist, write the start address to ENDPTLISTADDR, the size is (maximum endpoint number required by device) * 128 bytes. For example, a mass storage device needs endpoint 1 as IN, endpoint 2 and 3 as OUT besides endpoint 0, so dqhlist needs 4*128=512 bytes; In extreme case, a device only needs endpoint 7 as IN, dqhlist needs 8*128=1Kbyte;
- Prepare endpoint 0 receive dqh (address: dqhlist+0x00), set Maximum Packet Length to the maximum packet length supported by the device, generally 8 for low speed, 64 for full/high speed; set ios to 1 to trigger interrupt after SETUP is received; T=1; Status=0; other parts are arbitrary;
- Prepare endpoint 0 send dqh (address: dqhlist+0x40), set Maximum Packet Length to the maximum packet length supported by the device, Status=0, other parts are arbitrary;
- Set USBINTR.UI=1, wait for interrupt;
- After receiving UI interrupt, the 8-byte SETUP packet content is stored in the Setup Buffer of endpoint 0 receive dqh, prepare response according to SETUP content;
- If the received command is GET DESC, prepare device information to return to host, software needs to prepare an IN dtd for returning data, and a 0-byte OUT dtd as the end of Status phase.
- For IN dtd: T=1; TotalBytes is the length of device information; Status=0x80; write data start address to buffer0 and Current Offset, prepare buffer1 if device information crosses 4K address boundary, other parts are 0;
- For OUT dtd: T=1, TotalBytes=0, Status=0x80, ioc=1, other parts are 0;
- Write IN dtd address to Next dTD Pointer of endpoint 0 IN dqh and clear T, set bit16 of ENDPTPRIME, wait for bit16 to be cleared automatically by hardware (hardware will read corresponding content from IN0 dqh, load data to be sent into internal storage, and clear corresponding bit of ENDPTPRIME after completion); this process is called PRIME EP0 IN.
- Perform the same operation to PRIME EP0 OUT
- Wait for interrupt; if completed normally, EP0 OUT interrupt will be received, because only OUT dtd ioc is set in the above process, IN dtd ioc is not set.

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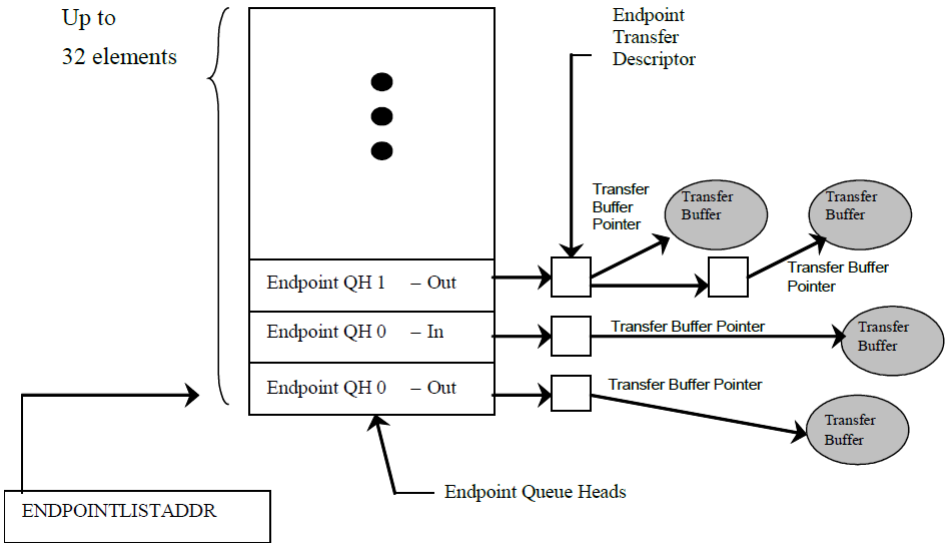


Figure 86: dqhlist Diagram

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	09	08	07	06	05	04	03	02	01	00	word
mult		zlt		0	max_packet_size											ios	0															03-00h
cur_dtd_addr																												0				07-04h
nxt_dtd_addr																												0			T	0B-08h
0	total_bytes															ioc	c_page				multo		0		status							0F-0Ch
buffer(page 0)																					current_offset											13-10h
buffer(page 1)																					reserved											17-14h
buffer(page 2)																					reserved											1B-18h
buffer(page 3)																					reserved											1F-1Ch
buffer(page 4)																					reserved											23-20h
reserved																																27-24h
setup buffer bytes 3..0																																2B-28h
setup buffer bytes 7..4																																2F-2Ch

Figure 87: dqh Diagram

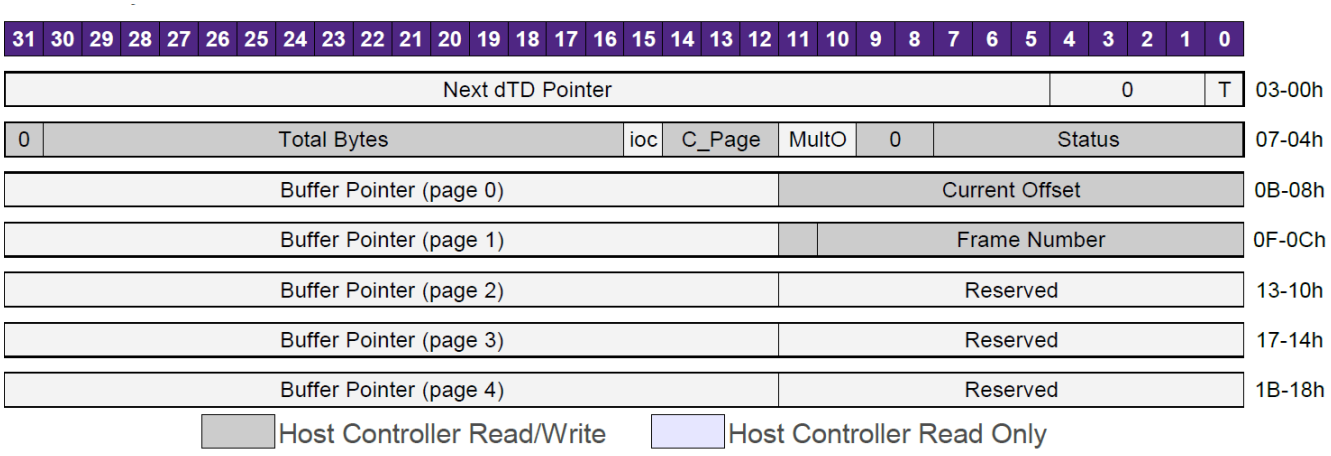


Figure 88: dtd Diagram

Similar to the overlay part of host qhead, device dtd will be fully copied to dqh by hardware before transmis-

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sion, so part of dqh (0x08 0x23) is exactly the same as dtd.

Detailed description of device data structure dqh is as follows:

Byte	Bit	Description
0x00	31:30	mult: for ISO endpoints, must be set to 1/2/3, indicating how many packets each dTD can transmit For non-ISO endpoints, must be set to 0.
0x00	29	zlt: set to 0 means when transmission length is integer multiple of maximum packet length, a 0-length packet is added at the end as transmission end flag. Only applicable for non-ISO endpoints
0x00	26:16	max_packet_size: maximum packet length of the endpoint.
0x00	15	ios: interrupt on setup, set to 1 means USBSTS.UI will be set when CTRL endpoint receives SETUP packet. dqh of CTRL endpoint shall set this bit
0x04	31:5	cur_dtd_addr: physical address of dtd being transmitted in current dqh, 32-byte aligned. When endpoint is primed, or after a dtd transmission completes, hardware copies nxt_dtd_addr to this position
0x08	31:5	nxt_dtd_addr: physical address of next dtd to be processed, 32-byte aligned
0x08	0	T(Terminate): indicates whether nxt_dtd_addr is valid, 0 means valid, 1 means current dtd is the last one, no more valid dtd in queue
0x0C	30:16	total_bytes: length to be transmitted by current dtd. For IN: length of data packet to send to host; For OUT: maximum length that can be received from host, actual received length may be less than this value.
0x0C	15	ioc: Interrupt On Complete, set to 1 means USBSTS.UI will be set after current dtd transmission completes
0x0C	14:12	CPage: current buffer, software initializes it to 0, hardware updates it in real time during transmission
0x0C	11:10	multo: used for dtd to overwrite mult part of dqh. For example: dqh.mult=3, max_packet_size=512. When 1023byte data needs to be transmitted, software shall set dtd.multo to 2, so 2 packets will be transmitted: 512+511; If multo is not set (value 0), 3 packets will be transmitted: 512+511+0.
0x0C	7:0	Status: indicates current dtd status, software initializes it to 0x80. Bit7: Active: indicates current dtd is valid. Software shall set this bit to 1 when preparing dtd, hardware clears this bit after transmission completes. Bit6: Halted: indicates current endpoint is in Halt state Bit5: data buffer error: indicates buffer underrun (IN) or overrun (OUT) Bit3: transaction error: indicates data transmission error Bits 2:0 can be ignored in simple applications.

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Byte	Bit	Description
0x10	31:12	Buffer Pointer (Page 0): each dtd has 5 buffers, each 4Kbyte, can be consecutive physical addresses (buffer[n]=buffer[n-1]+0x1000) or discrete addresses. Start address of Buffer page0 is specified by current_offset, other buffers must be 4Kbyte aligned. Maximum transmission length is 20Kbyte, if start address is not 4Kbyte aligned, recommended maximum length is 16Kbyte
0x10	11:0	Current Offset: software initializes it to the start offset of data packet in buffer0, hardware updates it in real time to the offset of current buffer (specified by cpage) during transmission, indicates transmitted data length
0x14 0x23	31:12	Data buffer pointer address (page1 4)
0x28 0x2F	31:0	setup buffer: last 8 bytes of dqh, used to store SETUP packets. After receiving SETUP packet, 8-byte SETUP data is stored in OUT dqh of endpoint 0

Table 210: dqh Structure

51.5 USB Register List

USB0 base address: 0xF300C000

Offset	Register Name	Description	Reset Value
0x0080	GPTIMER0LD	General Purpose Timer #0 Load Register	0x00000000
0x0084	GPTIMER0CTRL	General Purpose Timer #0 Controller Register	0x00000000
0x0088	GPTIMER1LD	General Purpose Timer #1 Load Register	0x00000000
0x008C	GPTIMER1CTRL	General Purpose Timer #1 Controller Register	0x00000000
0x0090	SBUSCFG	System Bus Config Register	0x00000000
0x0140	USBCMD	USB Command Register	0x00080000
0x0144	USBSTS	USB Status Register	0x00000000
0x0148	USBINTR	Interrupt Enable Register	0x00000000
0x014C	FRINDEX	USB Frame Index Register	0x00000000
0x0154	DEVICEADDR	Device Address Register	0x00000000
0x0154	PERIODICLISTBASE	Frame List Base Address Register	0x00000000
0x0158	ASYNCLISTADDR	Next Asynch. Address Register	0x00000000
0x0158	ENDPTLISTADDR	Endpoint List Address Register	0x00000000
0x0160	BURSTSIZE	Programmable Burst Size Register	0x00000000

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Offset	Register Name	Description	Reset Value
0x0164	TXFILLTUNING	TX FIFO Fill Tuning Register	0x00000000
0x0178	ENDPTNAK	Endpoint NAK Register	0x00000000
0x017C	ENDPTNAKEN	Endpoint NAK Enable Register	0x00000000
0x0184	PORTSC1	Port Status & Control	0x00000000
0x01A4	OTGSC	On-The-Go Status & control Register	0x00000000
0x01A8	USBMODE	USB Device Mode Register	0x00000000
0x01AC	ENDPTSETUPSTAT	Endpoint Setup Status Register	0x00000000
0x01B0	ENDPTPRIME	Endpoint Prime Register	0x00000000
0x01B4	ENDPTFLUSH	Endpoint Flush Register	0x00000000
0x01B8	ENDPTSTAT	Endpoint Status Register	0x00000000
0x01BC	ENDPTCOMPLETE	Endpoint Complete Register	0x00000000
0x01C0	ENDPTCTRL[ENDPTCTRL0]	Endpoint Control0 Register... Endpoint Control7 Register	0x00000000
0x01C4	ENDPTCTRL[ENDPTCTRL1]	Endpoint Control0 Register... Endpoint Control7 Register	0x00000000
0x01C8	ENDPTCTRL[ENDPTCTRL2]	Endpoint Control0 Register... Endpoint Control7 Register	0x00000000
0x01CC	ENDPTCTRL[ENDPTCTRL3]	Endpoint Control0 Register... Endpoint Control7 Register	0x00000000
0x01D0	ENDPTCTRL[ENDPTCTRL4]	Endpoint Control0 Register... Endpoint Control7 Register	0x00000000
0x01D4	ENDPTCTRL[ENDPTCTRL5]	Endpoint Control0 Register... Endpoint Control7 Register	0x00000000
0x01D8	ENDPTCTRL[ENDPTCTRL6]	Endpoint Control0 Register... Endpoint Control7 Register	0x00000000
0x01DC	ENDPTCTRL[ENDPTCTRL7]	Endpoint Control0 Register... Endpoint Control7 Register	0x00000000
0x01E0	ENDPTCTRL[ENDPTCTRL8]	Endpoint Control0 Register... Endpoint Control7 Register	0x00000000
0x01E4	ENDPTCTRL[ENDPTCTRL9]	Endpoint Control0 Register... Endpoint Control7 Register	0x00000000
0x01E8	ENDPTCTRL[ENDPTCTRL10]	Endpoint Control0 Register... Endpoint Control7 Register	0x00000000
0x01EC	ENDPTCTRL[ENDPTCTRL11]	Endpoint Control0 Register... Endpoint Control7 Register	0x00000000
0x01F0	ENDPTCTRL[ENDPTCTRL12]	Endpoint Control0 Register... Endpoint Control7 Register	0x00000000
0x01F4	ENDPTCTRL[ENDPTCTRL13]	Endpoint Control0 Register... Endpoint Control7 Register	0x00000000

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Offset	Register Name	Description	Reset Value
0x01F8	ENDPTCTRL[ENDPTCTRL14]	Endpoint Control0 Register... Endpoint Control7 Register	0x00000000
0x01FC	ENDPTCTRL[ENDPTCTRL15]	Endpoint Control0 Register... Endpoint Control7 Register	0x00000000
0x0200	OTG_CTRL0		0x00000000
0x0210	PHY_CTRL0		0x00000000
0x0214	PHY_CTRL1		0x00000000
0x0220	TOP_STATUS		0x00000000
0x0224	PHY_STATUS		0x00000000

Table 211: USB Register List

51.6 USB Register Description

51.6.1 GPTIMER0LD (0x80)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD								GPTLD																							
N/A								RW																							
x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

GPTIMER0LD [31:0]

Bits	Name	Description
23-0	GPTLD	GPTLD General Purpose Timer Load Value These bit fields are loaded to GPTCNT bits when GPTRST bit is set ’1b’. This value represents the time in microseconds minus 1 for the timer duration. Example: for a one millisecond timer, load 1000-1=999 or 0x0003E7. NOTE: Max value is 0xFFFFFF or 16.777215 seconds.

GPTIMER0LD Bits

51.6.2 GPTIMER0CTRL (0x84)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GPTRUN	GPTRST	RSVD					GPTMODE	GPTCNT																							

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RW	WO	N/A					RW	RO																							
0	0	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

GPTIMER0CTRL [31:0]

Bits	Name	Description
31	GPTRUN	GPTRUN General Purpose Timer Run GPTCNT bits are not effected when setting or clearing this bit. 0 - Stop counting 1 - Run
30	GPTRST	GPTRST General Purpose Timer Reset 0 - No action 1 - Load counter value from GPTLD bits in n_GPTIMER0LD
24	GPTMODE	GPTMODE General Purpose Timer Mode In one shot mode, the timer will count down to zero, generate an interrupt, and stop until the counter is reset by software; In repeat mode, the timer will count down to zero, generate an interrupt and automatically reload the counter value from GPTLD bits to start again. 0 - One Shot Mode 1 - Repeat Mode
23-0	GPTCNT	GPTCNT General Purpose Timer Counter. This field is the count value of the countdown timer.

GPTIMER0CTRL Bits

51.6.3 GPTIMER1LD (0x88)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD								GPTLD																							
N/A								RW																							
x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

GPTIMER1LD [31:0]

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Bits	Name	Description
23-0	GPTLD	GPTLD General Purpose Timer Load Value These bit fields are loaded to GPTCNT bits when GPTRST bit is set ’1b’. This value represents the time in microseconds minus 1 for the timer duration. Example: for a one millisecond timer, load 1000-1=999 or 0x0003E7. NOTE: Max value is 0xFFFFFF or 16.777215 seconds.

GPTIMER1LD Bits

51.6.4 GPTIMER1CTRL (0x8C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
GPTRUN	GPTRST	RSVD					GPTMODE	GPTCNT																							
RW	WO	N/A					RW	RO																							
0	0	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

GPTIMER1CTRL [31:0]

Bits	Name	Description
31	GPTRUN	GPTRUN General Purpose Timer Run GPTCNT bits are not effected when setting or clearing this bit. 0 - Stop counting 1 - Run
30	GPTRST	GPTRST General Purpose Timer Reset 0 - No action 1 - Load counter value from GPTLD bits in USB_n_GPTIMER1LD

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Bits	Name	Description
24	GPTMODE	GPTMODE General Purpose Timer Mode In one shot mode, the timer will count down to zero, generate an interrupt, and stop until the counter is reset by software. In repeat mode, the timer will count down to zero, generate an interrupt and automatically reload the counter value from GPTLD bits to start again. 0 - One Shot Mode 1 - Repeat Mode
23-0	GPTCNT	GPTCNT General Purpose Timer Counter. This field is the count value of the countdown timer.

GPTIMER1CTRL Bits

51.6.5 SBUSCFG (0x90)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																														AHBBRST	
N/A																														RW	
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0

SBUSCFG [31:0]

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Bits	Name	Description
2-0	AHBBRST	AHBBRST AHB master interface Burst configuration These bits control AHB master transfer type sequence (or priority). NOTE: This register overrides n_BURSTSIZE register when its value is not zero. 000 - Incremental burst of unspecified length only 001 - INCR4 burst, then single transfer 010 - INCR8 burst, INCR4 burst, then single transfer 011 - INCR16 burst, INCR8 burst, INCR4 burst, then single transfer 100 - Reserved, don't use 101 - INCR4 burst, then incremental burst of unspecified length 110 - INCR8 burst, INCR4 burst, then incremental burst of unspecified length 111 - INCR16 burst, INCR8 burst, INCR4 burst, then incremental burst of unspecified length

SBUSCFG Bits

51.6.6 USBCMD (0x140)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD								ITC								FS_2	ATDTW	SUTW	PRM	ASPE	RSVD	ASP		RSVD	IAA	ASE	PSE	FS_1		RST	RS
N/A								RW								RW	RW	RW	WO	RW	N/A	RW		N/A	RW	RW	RW	RW		RW	RW
x	x	x	x	x	x	x	x	0	0	0	0	1	0	0	0	0	0	0	0	0	x	0	0	x	0	0	0	0	0	0	0

USBCMD [31:0]

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Bits	Name	Description
23-16	ITC	ITC Interrupt Threshold Control -Read/Write. The system software uses this field to set the maximum rate at which the host/device controller will issue interrupts. ITC contains the maximum interrupt interval measured in micro-frames. Valid values are shown below. Value Maximum Interrupt Interval 00000000 - Immediate (no threshold) 00000001 - 1 micro-frame 00000010 - 2 micro-frames 00000100 - 4 micro-frames 00001000 - 8 micro-frames 00010000 - 16 micro-frames 00100000 - 32 micro-frames 01000000 - 64 micro-frames
15	FS_2	FS_2 Frame List Size - (Read/Write or Read Only). [host mode only] This field is Read/Write only if Programmable Frame List Flag in the HCCPARAMS registers is set to one. This field specifies the size of the frame list that controls which bits in the Frame Index Register should be used for the Frame List Current index. NOTE: This field is made up from USBCMD bits 15, 3 and 2. Value Meaning 0b000 - 1024 elements (4096 bytes) Default value 0b001 - 512 elements (2048 bytes) 0b010 - 256 elements (1024 bytes) 0b011 - 128 elements (512 bytes) 0b100 - 64 elements (256 bytes) 0b101 - 32 elements (128 bytes) 0b110 - 16 elements (64 bytes) 0b111 - 8 elements (32 bytes)
14	ATDTW	ATDTW Add dTD TripWire - Read/Write. [device mode only] This bit is used as a semaphore to ensure proper addition of a new dTD to an active (primed) endpoint's linked list. This bit is set and cleared by software. This bit would also be cleared by hardware when state machine is hazard region for which adding a dTD to a primed endpoint may go unrecognized.

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Bits	Name	Description
13	SUTW	SUTW Setup TripWire - Read/Write. [device mode only] This bit is used as a semaphore to ensure that the setup data payload of 8 bytes is extracted from a QH by the DCD without being corrupted. If the setup lockout mode is off (SLOM bit in USB core register n_USBMODE, see USBMODE) then there is a hazard when new setup data arrives while the DCD is copying the setup data payload from the QH for a previous setup packet. This bit is set and cleared by software. This bit would also be cleared by hardware when a hazard detected.
12	PRM	Asynchronous Schedule start- Write only，host mode only。 this bit is used to notify hostcontroller to start async schedule immediately.
11	ASPE	ASPE Asynchronous Schedule Park Mode Enable - Read/Write. If the Asynchronous Park Capability bit in the HCCPARAMS register is a one, then this bit defaults to a 1h and is R/W. Otherwise the bit must be a zero and is RO. Software uses this bit to enable or disable Park mode. When this bit is one, Park mode is enabled. When this bit is a zero, Park mode is disabled. NOTE: ASPE bit reset value: ’0b’ for OTG controller .
9-8	ASP	ASP Asynchronous Schedule Park Mode Count - Read/Write. If the Asynchronous Park Capability bit in the HCCPARAMS register is a one, then this field defaults to 3h and is R/W. Otherwise it defaults to zero and is Read-Only. It contains a count of the number of successive transactions the host controller is allowed to execute from a high-speed queue head on the Asynchronous schedule before continuing traversal of the Asynchronous schedule. Valid values are 1h to 3h. Software must not write a zero to this bit when Park Mode Enable is a one as this will result in undefined behavior. This field is set to 3h in all controller core.

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Bits	Name	Description
6	IAA	IAA Interrupt on Async Advance Doorbell - Read/Write. This bit is used as a doorbell by software to tell the host controller to issue an interrupt the next time it advances asynchronous schedule. Software must write a 1 to this bit to ring the doorbell. When the host controller has evicted all appropriate cached schedule states, it sets the Interrupt on Async Advance status bit in the USBSTS register. If the Interrupt on Sync Advance Enable bit in the USBINTR register is one, then the host controller will assert an interrupt at the next interrupt threshold. The host controller sets this bit to zero after it has set the Interrupt on Sync Advance status bit in the USBSTS register to one. Software should not write a one to this bit when the asynchronous schedule is inactive. Doing so will yield undefined results. This bit is only used in host mode. Writing a one to this bit when device mode is selected will have undefined results.
5	ASE	ASE Asynchronous Schedule Enable - Read/Write. Default 0b. This bit controls whether the host controller skips processing the Asynchronous Schedule. Only the host controller uses this bit. Values Meaning 0 - Do not process the Asynchronous Schedule. 1 - Use the ASYNCLISTADDR register to access the Asynchronous Schedule.
4	PSE	PSE Periodic Schedule Enable- Read/Write. Default 0b. This bit controls whether the host controller skips processing the Periodic Schedule. Only the host controller uses this bit. Values Meaning 0 - Do not process the Periodic Schedule 1 - Use the PERIODICLISTBASE register to access the Periodic Schedule.
3-2	FS_1	FS_1 See description at bit 15

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Bits	Name	Description
1	RST	RST Controller Reset (RESET) - Read/Write. Software uses this bit to reset the controller. This bit is set to zero by the Host/Device Controller when the reset process is complete. Software cannot terminate the reset process early by writing a zero to this register. Host operation mode: When software writes a one to this bit, the Controller resets its internal pipelines, timers, counters, state machines etc. to their initial value. Any transaction currently in progress on USB is immediately terminated. A USB reset is not driven on downstream ports. Software should not set this bit to a one when the HCHalted bit in the USBSTS register is a zero. Attempting to reset an actively running host controller will result in undefined behavior. Device operation mode: When software writes a one to this bit, the Controller resets its internal pipelines, timers, counters, state machines etc. to their initial value. Writing a one to this bit when the device is in the attached state is not recommended, because the effect on an attached host is undefined. In order to ensure that the device is not in an attached state before initiating a device controller reset, all primed endpoints should be flushed and the USBCMD Run/Stop bit should be set to 0.

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Bits	Name	Description
0	RS	RS Run/Stop (RS) - Read/Write. Default 0b. 1=Run. 0=Stop. Host operation mode: When set to '1b', the Controller proceeds with the execution of the schedule. The Controller continues execution as long as this bit is set to a one. When this bit is set to 0, the Host Controller completes the current transaction on the USB and then halts. The HC Halted bit in the status register indicates when the Controller has finished the transaction and has entered the stopped state. Software should not write a one to this field unless the controller is in the Halted state (that is, HCHalted in the USBSTS register is a one). Device operation mode: Writing a one to this bit will cause the controller to enable a pull-up on D+ and initiate an attach event. This control bit is not directly connected to the pull-up enable, as the pull-up will become disabled upon transitioning into high-speed mode. Software should use this bit to prevent an attach event before the controller has been properly initialized. Writing a 0 to this will cause a detach event.

USBCMD Bits

51.6.7 USBSTS (0x144)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD						TI1	TI0	RSVD				UPI	UAI	RSVD	NAKI	AS	PS	RCL	HCH	RSVD			SLI	SRI	URI	AAI	SEI	FRI	PCI	UEI	UI
N/A						RWC	RWC	N/A				RWC	RWC	N/A	RO	RO	RO	RO	RO	N/A			RWC	RWC	RWC	RWC	RWC	RWC	RWC	RWC	RWC
x	x	x	x	x	x	0	0	x	x	x	x	0	0	x	0	0	0	0	0	x	x	x	0	0	0	0	0	0	0	0	0

USBSTS [31:0]

Bits	Name	Description
25	TI1	TI1 General Purpose Timer Interrupt 1(GPTINT1)–R/WC. This bit is set when the counter in the GPTIMER1CTRL register transitions to zero, writing a one to this bit will clear it.

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Bits	Name	Description
24	TI0	TI0 General Purpose Timer Interrupt 0(GPTINT0)–R/WC. This bit is set when the counter in the GPTIMER0CTRL register transitions to zero, writing a one to this bit clears it.
19	UPI	USB Host Periodic Interrupt –RWC. Default = 0b. This bit is set by the Host Controller when the cause of an interrupt is a completion of a USB transaction where the Transfer Descriptor (TD) has an interrupt on complete (IOC) bit set and the TD was from the periodic schedule. This bit is also set by the Host Controller when a short packet is detected and the packet is on the periodic schedule. A short packet is when the actual number of bytes received was less than expected. This bit is not used by the device controller and will always be zero.
18	UAI	USB Host Asynchronous Interrupt –RWC. Default = 0b. This bit is set by the Host Controller when the cause of an interrupt is a completion of a USB transaction where the Transfer Descriptor (TD) has an interrupt on complete (IOC) bit set AND the TD was from the asynchronous schedule. This bit is also set by the Host when a short packet is detected and the packet is on the asynchronous schedule. A short packet is when the actual number of bytes received was less than expected. This bit is not used by the device controller and will always be zero
16	NAKI	NAKI NAK Interrupt Bit–RO. This bit is set by hardware when for a particular endpoint both the TX/RX Endpoint NAK bit and corresponding TX/RX Endpoint NAK Enable bit are set. This bit is automatically cleared by hardware when all Enabled TX/RX Endpoint NAK bits are cleared.

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Bits	Name	Description
15	AS	AS Asynchronous Schedule Status - Read Only. This bit reports the current real status of the Asynchronous Schedule. When set to zero the asynchronous schedule status is disabled and if set to one the status is enabled. The Host Controller is not required to immediately disable or enable the Asynchronous Schedule when software transitions the Asynchronous Schedule Enable bit in the USBCMD register. When this bit and the Asynchronous Schedule Enable bit are the same value, the Asynchronous Schedule is either enabled (1) or disabled (0). Only used in the host operation mode.
14	PS	PS Periodic Schedule Status - Read Only. This bit reports the current real status of the Periodic Schedule. When set to zero the periodic schedule is disabled, and if set to one the status is enabled. The Host Controller is not required to immediately disable or enable the Periodic Schedule when software transitions the Periodic Schedule Enable bit in the USBCMD register. When this bit and the Periodic Schedule Enable bit are the same value, the Periodic Schedule is either enabled (1) or disabled (0). Only used in the host operation mode.
13	RCL	RCL Reclamation - Read Only. This is a read-only status bit used to detect an empty asynchronous schedule. Only used in the host operation mode.

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Bits	Name	Description
12	HCH	HCH HCHalted - Read Only. This bit is a zero whenever the Run/Stop bit is a one. The Controller sets this bit to one after it has stopped executing because of the Run/Stop bit being set to 0, either by software or by the Controller hardware (for example, an internal error). Only used in the host operation mode. Default value is ’0b’ for OTG core . This is because OTG core is not operating as host in default. Please see CM bit in USB_n_USBMODE register. NOTE: HCH bit reset value: ’0b’ for OTG controller core .
8	SLI	SLI DCSuspend - R/WC. When a controller enters a suspend state from an active state, this bit will be set to a one. The device controller clears the bit upon exiting from a suspend state. Only used in device operation mode.
7	SRI	SRI SOF Received - R/WC. When the device controller detects a Start Of (micro) Frame, this bit will be set to a one. When a SOF is extremely late, the device controller will automatically set this bit to indicate that an SOF was expected. Therefore, this bit will be set roughly every 1ms in device FS mode and every 125ms in HS mode and will be synchronized to the actual SOF that is received. Because the device controller is initialized to FS before connect, this bit will be set at an interval of 1ms during the prelude to connect and chirp. In host mode, this bit will be set every 125us and can be used by host controller driver as a time base. Software writes a 1 to this bit to clear it.
6	URI	URI USB Reset Received - R/WC. When the device controller detects a USB Reset and enters the default state, this bit will be set to a one. Software can write a 1 to this bit to clear the USB Reset Received status bit. Only used in device operation mode.

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Bits	Name	Description
5	AAI	AAI Interrupt on Async Advance - R/WC. System software can force the host controller to issue an interrupt the next time the host controller advances the asynchronous schedule by writing a one to the Interrupt on Async Advance Doorbell bit in the n_USBCMD register. This status bit indicates the assertion of that interrupt source. Only used in host operation mode.
4	SEI	System Error –RWC. Default = 0b. In the BVCI implementation of the USBHS core, this bit is not used, and will always be cleared to ’0b’. In the AMBA implementation, this bit will be set to ’1b’ when an Error response is seen by the master interface (HRESP[1:0]=ERROR)
3	FRI	FRI Frame List Rollover - R/WC. The Host Controller sets this bit to a one when the Frame List Index rolls over from its maximum value to zero. The exact value at which the rollover occurs depends on the frame list size. For example. If the frame list size (as programmed in the Frame List Size field of the USB_n_USBCMD register) is 1024, the Frame Index Register rolls over every time FRINDEX [13] toggles. Similarly, if the size is 512, the Host Controller sets this bit to a one every time FHINDEX [12] toggles. Only used in host operation mode.
2	PCI	PCI Port Change Detect - R/WC. The Host Controller sets this bit to a one when on any port a Connect Status occurs, a Port Enable/Disable Change occurs, or the Force Port Resume bit is set as the result of a J-K transition on the suspended port. The Device Controller sets this bit to a one when the port controller enters the full or high-speed operational state. When the port controller exits the full or high-speed operation states due to Reset or Suspend events, the notification mechanisms are the USB Reset Received bit and the DCSuspend bits Respectively.

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Bits	Name	Description
1	UEI	UEI USB Error Interrupt (USBERRINT) - R/WC. When completion of a USB transaction results in an error condition, this bit is set by the Host/Device Controller. This bit is set along with the USBINT bit, if the TD on which the error interrupt occurred also had its interrupt on complete (IOC) bit set.
0	UI	UI USB Interrupt (USBINT) - R/WC. This bit is set by the Host/Device Controller when the cause of an interrupt is a completion of a USB transaction where the Transfer Descriptor (TD) has an interrupt on complete (IOC) bit set. This bit is also set by the Host/Device Controller when a short packet is detected. A short packet is when the actual number of bytes received was less than the expected number of bytes.

USBSTS Bits

51.6.8 USBINTR (0x148)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD						TIE1	TIE0	RSVD				UPIE	UAIE	RSVD	NAKE	RSVD							SLE	SRE	URE	AAE	SEE	FRE	PCE	UEE	UE
N/A						RW	RW	N/A				RW	RW	N/A	RO	N/A							RW	RW	RW	RW	RW	RW	RW	RWC	RW
x	x	x	x	x	x	0	0	x	x	x	x	0	0	x	0	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0

USBINTR [31:0]

Bits	Name	Description
25	TIE1	TIE1 General Purpose Timer #1 Interrupt Enable When this bit is one and the TI1 bit in n_USBSTS register is a one the controller will issue an interrupt.
24	TIE0	TIE0 General Purpose Timer #0 Interrupt Enable When this bit is one and the TI0 bit in n_USBSTS register is a one the controller will issue an interrupt.

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Bits	Name	Description
19	UPIE	UPIE USB Host Periodic Interrupt Enable When this bit is one, and the UPI bit in the n_USBSTS register is one, host controller will issue an interrupt at the next interrupt threshold.
18	UAIE	UAIE USB Host Asynchronous Interrupt Enable When this bit is one, and the UAI bit in the n_USBSTS register is one, host controller will issue an interrupt at the next interrupt threshold.
16	NAKE	NAKE NAK Interrupt Enable When this bit is one and the NAKI bit in n_USBSTS register is a one the controller will issue an interrupt.
8	SLE	SLE Sleep Interrupt Enable When this bit is one and the SLI bit in n_n_USBSTS register is a one the controller will issue an interrupt. Only used in device operation mode.
7	SRE	SRE SOF Received Interrupt Enable When this bit is one and the SRI bit in n_USBSTS register is a one the controller will issue an interrupt.
6	URE	URE USB Reset Interrupt Enable When this bit is one and the URI bit in n_USBSTS register is a one the controller will issue an interrupt. Only used in device operation mode.
5	AAE	AAE Async Advance Interrupt Enable When this bit is one and the AAI bit in n_USBSTS register is a one the controller will issue an interrupt. Only used in host operation mode.
4	SEE	SEE System Error Interrupt Enable When this bit is one and the SEI bit in n_USBSTS register is a one the controller will issue an interrupt. Only used in host operation mode.

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Bits	Name	Description
3	FRE	FRE Frame List Rollover Interrupt Enable When this bit is one and the FRI bit in n_USBSTS register is a one the controller will issue an interrupt. Only used in host operation mode.
2	PCE	PCE Port Change Detect Interrupt Enable When this bit is one and the PCI bit in n_USBSTS register is a one the controller will issue an interrupt.
1	UEE	UEE USB Error Interrupt Enable When this bit is one and the UEI bit in n_USBSTS register is a one the controller will issue an interrupt.
0	UE	UE USB Interrupt Enable When this bit is one and the UI bit in n_USBSTS register is a one the controller will issue an interrupt.

USBINTR Bits

51.6.9 FRINDEX (0x14C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																		FRINDEX													
N/A																		RW													
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0

FRINDEX [31:0]

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Bits	Name	Description
13-0	FRINDEX	FRINDEX Frame Index. The value, in this register, increments at the end of each time frame (micro-frame). Bits [N: 3] are used for the Frame List current index. This means that each location of the frame list is accessed 8 times (frames or micro-frames) before moving to the next index. The following illustrates values of N based on the value of the Frame List Size field in the USBCMD register, when used in host mode. USBCMD [Frame List Size] Number Elements N In device mode the value is the current frame number of the last frame transmitted. It is not used as an index. In either mode bits 2:0 indicate the current microframe. The bit field values description below is represented as (Frame List Size) Number Elements N. 00000000000000 - (1024) 12 00000000000001 - (512) 11 00000000000010 - (256) 10 00000000000011 - (128) 9 00000000000100 - (64) 8 00000000000101 - (32) 7 00000000000110 - (16) 6 00000000000111 - (8) 5

FRINDEX Bits

51.6.10 DEVICEADDR (0x154)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
USBADR							USBADRA	RSVD																							
RW							RW	N/A																							
0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

DEVICEADDR [31:0]

Bits	Name	Description
31-25	USBADR	USBADR Device Address. These bits correspond to the USB device address

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Bits	Name	Description
24	USBADRA	USBADRA Device Address Advance. Default=0. When this bit is '0', any writes to USBADR are instantaneous. When this bit is written to a '1' at the same time or before USBADR is written, the write to the USBADR field is staged and held in a hidden register. After an IN occurs on endpoint 0 and is ACKed, USBADR will be loaded from the holding register. Hardware will automatically clear this bit on the following conditions: 1) IN is ACKed to endpoint 0. (USBADR is updated from staging register). 2) OUT/SETUP occur to endpoint 0. (USBADR is not updated). 3) Device Reset occurs (USBADR is reset to 0). NOTE: After the status phase of the SET_ADDRESS descriptor, the DCD has 2 ms to program the USBADR field. This mechanism will ensure this specification is met when the DCD can not write of the device address within 2ms from the SET_ADDRESS status phase. If the DCD writes the USBADR with USBADRA=1 after the SET_ADDRESS data phase (before the prime of the status phase), the USBADR will be programmed instantly at the correct time and meet the 2ms USB requirement.

DEVICEADDR Bits

51.6.11 PERIODICLISTBASE (0x154)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BASEADR																				RSVD											
RW																				N/A											
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x

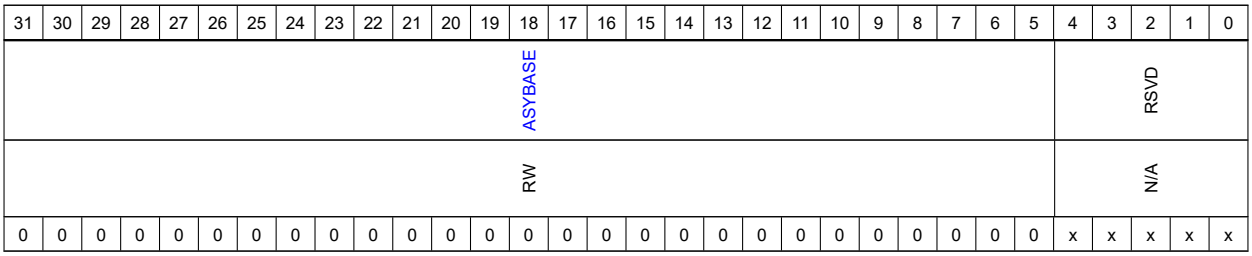
PERIODICLISTBASE [31:0]

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Bits	Name	Description
31-12	BASEADR	BASEADR Base Address (Low). These bits correspond to memory address signals [31:12], respectively. Only used by the host controller.

PERIODICLISTBASE Bits

51.6.12 ASYNCLISTADDR (0x158)

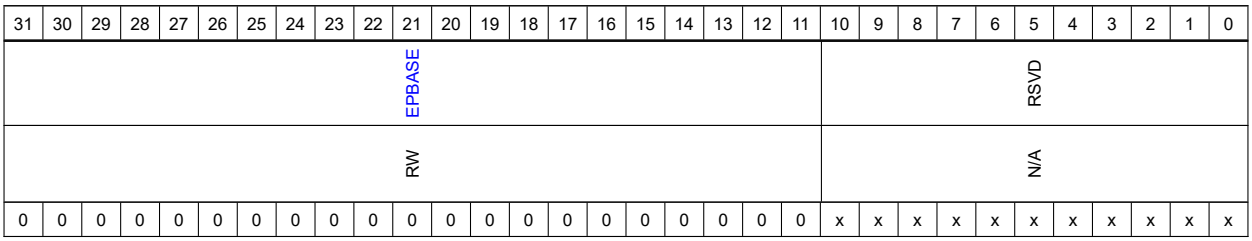


ASYNCLISTADDR [31:0]

Bits	Name	Description
31-5	ASYBASE	ASYBASE Link Pointer Low (LPL). These bits correspond to memory address signals [31:5], respectively. This field may only reference a Queue Head (QH). Only used by the host controller.

ASYNCLISTADDR Bits

51.6.13 ENDPTLISTADDR (0x158)



ENDPTLISTADDR [31:0]

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Bits	Name	Description
31-11	EPBASE	EPBASE Endpoint List Pointer(Low). These bits correspond to memory address signals [31:11], respectively. This field will reference a list of up to 32 Queue Head (QH) (that is, one queue head per endpoint & direction).

ENDPTLISTADDR Bits

51.6.14 BURSTSIZE (0x160)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																TXPBURST								RXPBURST							
N/A																RW								RW							
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

BURSTSIZE [31:0]

Bits	Name	Description
15-8	TXPBURST	TXPBURST Programmable TX Burst Size. Default value is determined by TXBURST bits in n_HWTXBUF. This register represents the maximum length of a the burst in 32-bit words while moving data from system memory to the USB bus.
7-0	RXPBURST	RXPBURST Programmable RX Burst Size. Default value is determined by TXBURST bits in n_HWRXBUF. This register represents the maximum length of a the burst in 32-bit words while moving data from the USB bus to system memory.

BURSTSIZE Bits

51.6.15 TXFILLTUNING (0x164)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD										TXFIFOTHRES						RSVD			TXSCHHEALTH					RSVD	TXSCHOH						
N/A										RW						N/A			RWC					N/A	RW						

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	x	x	x	0	0	0	0	0	x	0	0	0	0	0	0	0

TXFILLTUNING [31:0]

Bits	Name	Description
21-16	TXFIFOTHRES	TXFIFOTHRES FIFO Burst Threshold. (Read/Write) This register controls the number of data bursts that are posted to the TX latency FIFO in host mode before the packet begins on to the bus. The minimum value is 2 and this value should be a low as possible to maximize USB performance. A higher value can be used in systems with unpredictable latency and/or insufficient bandwidth where the FIFO may underrun because the data transferred from the latency FIFO to USB occurs before it can be replenished from system memory. This value is ignored if the Stream Disable bit in USB_n_USBMODE register is set.
12-8	TXSCHHEALTH	TXSCHHEALTH Scheduler Health Counter. (Read/Write To Clear) Table continues on the next page This register increments when the host controller fails to fill the TX latency FIFO to the level programmed by TXFIFOTHRES before running out of time to send the packet before the next Start-Of-Frame. This health counter measures the number of times this occurs to provide feedback to selecting a proper TXSCHOH. Writing to this register will clear the counter and this counter will max. at 31.

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Bits	Name	Description
6-0	TXSCHOH	TXSCHOH Scheduler Overhead. (Read/Write) [Default = 0] This register adds an additional fixed offset to the schedule time estimator described above as Tff. As an approximation, the value chosen for this register should limit the number of back-off events captured in the TXSCHHEALTH to less than 10 per second in a highly utilized bus. Choosing a value that is too high for this register is not desired as it can needlessly reduce USB utilization. The time unit represented in this register is 1.267us when a device is connected in High-Speed Mode. The time unit represented in this register is 6.333us when a device is connected in Low/Full Speed Mode. Default value is ’08h’ for OTG controller core .

TXFILLTUNING Bits

51.6.16 ENDPTNAK (0x178)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EPTN																EPRN															
RWC																RWC															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENDPTNAK [31:0]

Bits	Name	Description
31-16	EPTN	EPTN TX Endpoint NAK - R/WC. Each TX endpoint has 1 bit in this field. The bit is set when the device sends a NAK handshake on a received IN token for the corresponding endpoint. Bit [N] - Endpoint #[N], N is 0-7
15-0	EPRN	EPRN RX Endpoint NAK - R/WC. Each RX endpoint has 1 bit in this field. The bit is set when the device sends a NAK handshake on a received OUT or PING token for the corresponding endpoint. Bit [N] - Endpoint #[N], N is 0-7

ENDPTNAK Bits

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51.6.17 ENDPTNAKEN (0x17C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EPTNE																EPRNE															
RW																RW															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENDPTNAKEN [31:0]

Bits	Name	Description
31-16	EPTNE	EPTNE TX Endpoint NAK Enable - R/W. Each bit is an enable bit for the corresponding TX Endpoint NAK bit. If this bit is set and the corresponding TX Endpoint NAK bit is set, the NAK Interrupt bit is set. Bit [N] - Endpoint #[N], N is 0-7
15-0	EPRNE	EPRNE RX Endpoint NAK Enable - R/W. Each bit is an enable bit for the corresponding RX Endpoint NAK bit. If this bit is set and the corresponding RX Endpoint NAK bit is set, the NAK Interrupt bit is set. Bit [N] - Endpoint #[N], N is 0-7

ENDPTNAKEN Bits

51.6.18 PORTSC1 (0x184)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD		STS	PTW	PSPD		RSVD	PFSC	PHCD	WKOC	WKDC	WKCN	PTC				RSVD			PP	LS		HSP	PR	SUSP	FPR	OCC	OCA	PEC	PE	CSC	CCS
N/A		RW	RW	RO		N/A	RW	RW	RW	RW	RW	RW				N/A			RW	RO		RO	RW	RW	RW	RW	RO	RWC	RWC	RWC	RWC
x	x	0	0	0	0	x	0	0	0	0	0	0	0	0	0	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0

PORTSC1 [31:0]

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Bits	Name	Description
29	STS	STS Serial Transceiver Select 1 Serial Interface Engine is selected 0 Parallel Interface signals is selected Serial Interface Engine can be used in combination with UTMI+/ULPI physical interface to provide FS/LS signaling instead of the parallel interface signals. When this bit is set '1b', serial interface engine will be used instead of parallel interface signals.
28	PTW	PTW Parallel Transceiver Width This bit has no effect if serial interface engine is used. 0 - Select the 8-bit UTMI interface [60MHz] 1 - Select the 16-bit UTMI interface [30MHz]
27-26	PSPD	PSPD Port Speed - Read Only. This register field indicates the speed at which the port is operating. 00 - Full Speed 01 - Low Speed 10 - High Speed 11 - Undefined
24	PFSC	PFSC Port Force Full Speed Connect - Read/Write. Default = 0b. When this bit is set to '1b', the port will be forced to only connect at Full Speed, It disables the chirp sequence that allows the port to identify itself as High Speed. 0 - Normal operation 1 - Forced to full speed

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Bits	Name	Description
23	PHCD	PHCD PHY Low Power Suspend - Clock Disable (PLPSCD) - Read/Write. Default = 0b. When this bit is set to '1b', the PHY clock is disabled. Reading this bit will indicate the status of the PHY clock. NOTE: The PHY clock cannot be disabled if it is being used as the system clock. In device mode, The PHY can be put into Low Power Suspend when the device is not running (USBCMD Run/Stop=0b) or the host has signalled suspend (PORTSC1 SUSPEND=1b). PHY Low power suspend will be cleared automatically when the host initials resume. Before forcing a resume from the device, the device controller driver must clear this bit. In host mode, the PHY can be put into Low Power Suspend when the downstream device has been put into suspend mode or when no downstream device is connected. Low power suspend is completely under the control of software. 0 - Enable PHY clock 1 - Disable PHY clock
22	WKOC	WKOC Wake on Over-current Enable (WKOC_E) - Read/Write. Default = 0b. Writing this bit to a one enables the port to be sensitive to over-current conditions as wake-up events. This field is zero if Port Power(PORTSC1) is zero.
21	WKDC	WKDC Wake on Disconnect Enable (WKDSCNNT_E) - Read/Write. Default=0b. Writing this bit to a one enables the port to be sensitive to device disconnects as wake-up events. This field is zero if Port Power(PORTSC1) is zero or in device mode.
20	WKCN	WKCN Wake on Connect Enable (WKCNNT_E) - Read/Write. Default=0b. Writing this bit to a one enables the port to be sensitive to device connects as wake-up events. This field is zero if Port Power(PORTSC1) is zero or in device mode.

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Bits	Name	Description
19-16	PTC	PTC Port Test Control - Read/Write. Default = 0000b. Refer to Port Test Mode for the operational model for using these test modes and the USB Specification Revision 2.0, Chapter 7 for details on each test mode. The FORCE_ENABLE_FS and FORCE ENABLE_LS are extensions to the test mode support specified in the EHCI specification. Writing the PTC field to any of the FORCE_ENABLE_{HS/FS/LS} values will force the port into the connected and enabled state at the selected speed. Writing the PTC field back to TEST_MODE_DISABLE will allow the port state machines to progress normally from that point. NOTE: Low speed operations are not supported as a peripheral device. Any other value than zero indicates that the port is operating in test mode. Value Specific Test 0000 - TEST_MODE_DISABLE 0001 - J_STATE 0010 - K_STATE 0011 - SE0 (host) / NAK (device) 0100 - Packet 0101 - FORCE_ENABLE_HS 0110 - FORCE_ENABLE_FS 0111 - FORCE_ENABLE_LS 1000-1111 - Reserved

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Bits	Name	Description
12	PP	PP Port Power (PP)-Read/Write or Read Only. The function of this bit depends on the value of the Port Power Switching (PPC) field in the HCSPARAMS register. The behavior is as follows: PPC PP Operation 0 1b Read Only - Host controller does not have port power control switches. Each port is hard-wired to power. 1 1b/0b - Read/Write. OTG controller requires port power control switches. This bit represents the current setting of the switch (0=off, 1=on). When power is not available on a port (that is, PP equals a 0), the port is non-functional and will not report attaches, detaches, etc. When an over-current condition is detected on a powered port and PPC is a one, the PP bit in each affected port may be transitional by the host controller driver from a one to a zero (removing power from the port). This feature is implemented in all controller cores (PPC = 1).
11-10	LS	LS Line Status-Read Only. These bits reflect the current logical levels of the D+ (bit 11) and D- (bit 10) signal lines. In host mode, the use of linestate by the host controller driver is not necessary (unlike EHCI), because the port controller state machine and the port routing manage the connection of LS and FS. In device mode, the use of linestate by the device controller driver is not necessary. The encoding of the bits are: Bits [11:10] Meaning 00 - SE0 01 - K-state 10 - J-state 11 - Undefined

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Bits	Name	Description
9	HSP	HSP High-Speed Port - Read Only. Default = 0b. When the bit is one, the host/device connected to the port is in high-speed mode and if set to zero, the host/device connected to the port is not in a high-speed mode. NOTE: HSP is redundant with PSPD(bit 27, 26) but remained for compatibility.
8	PR	PR Port Reset - Read/Write or Read Only. Default = 0b. In Host Mode: Read/Write. 1=Port is in Reset. 0=Port is not in Reset. Default 0. When software writes a one to this bit the bus-reset sequence as defined in the USB Specification Revision 2.0 is started. This bit will automatically change to zero after the reset sequence is complete. This behavior is different from EHCI where the host controller driver is required to set this bit to a zero after the reset duration is timed in the driver. In Device Mode: This bit is a read only status bit. Device reset from the USB bus is also indicated in the USBSTS register.

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Bits	Name	Description
7	SUSP	SUSP Suspend - Read/Write or Read Only. Default = 0b. 1=Port in suspend state. 0=Port not in suspend state. In Host Mode: Read/Write. Port Enabled Bit and Suspend bit of this register define the port states as follows: Bits [Port Enabled, Suspend] Port State 0x Disable 10 Enable 11 Suspend When in suspend state, downstream propagation of data is blocked on this port, except for port reset. The blocking occurs at the end of the current transaction if a transaction was in progress when this bit was written to 1. In the suspend state, the port is sensitive to resume detection. Note that the bit status does not change until the port is suspended and that there may be a delay in suspending a port if there is a transaction currently in progress on the USB. The host controller will unconditionally set this bit to zero when software sets the Force Port Resume bit to zero. The host controller ignores a write of zero to this bit. If host software sets this bit to a one when the port is not enabled (that is, Port enabled bit is a zero) the results are undefined. This field is zero if Port Power(PORTSC1) is zero in host mode. In Device Mode: Read Only. In device mode this bit is a read only status bit.

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Bits	Name	Description
6	FPR	FPR Force Port Resume -Read/Write. 1= Resume detected/driven on port. 0=No resume (K-state) detected driven on port. Default = 0. In Host Mode: Software sets this bit to one to drive resume signaling. The Host Controller sets this bit to one if a J-to-K transition is detected while the port is in the Suspend state. When this bit transitions to a one because a J-to-K transition is detected, the Port Change Detect bit in the USBSTS register is also set to one. This bit will automatically change to zero after the resume sequence is complete. This behavior is different from EHCI where the host controller driver is required to set this bit to a zero after the resume duration is timed in the driver. Note that when the Host controller owns the port, the resume sequence follows the defined sequence documented in the USB Specification Revision 2.0. The resume signaling (Full-speed 'K') is driven on the port as long as this bit remains a one. This bit will remain a one until the port has switched to the high-speed idle. Writing a zero has no effect because the port controller will time the resume operation, clear the bit the port control state switches to HS or FS idle. This field is zero if Port Power(PORTSC1) is zero in host mode. This bit is not-EHCI compatible. In Device mode: After the device has been in Suspend State for 5ms or more, software must set this bit to one to drive resume signaling before clearing. The Device Controller will set this bit to one if a J-to-K transition is detected while the port is in the Suspend state. The bit will be cleared when the device returns to normal operation. Also, when this bit wil be cleared because a K-to-J transition detected, the Port Change Detect bit in the USBSTS register is also set to one.

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Bits	Name	Description
5	OCC	OCC Over-current Change-R/WC. Default=0. This bit is set '1b' by hardware when there is a change to Over-current Active. Software can clear this bit by writing a one to this bit position.
4	OCA	OCA Over-current Active-Read Only. Default 0. This bit will automatically transition from one to zero when the over current condition is removed. 0 - This port does not have an over-current condition. 1 - This port currently has an over-current condition
3	PEC	PEC Port Enable/Disable Change-R/WC. 1=Port enabled/disabled status has changed. 0=No change. Default = 0. In Host Mode: For the root hub, this bit is set to a one only when a port is disabled due to disconnect on the port or due to the appropriate conditions existing at the EOF2 point (See Chapter 11 of the USB Specification). Software clears this by writing a one to it. This field is zero if Port Power(PORTSC1) is zero. In Device mode: The device port is always enabled, so this bit is always '0b'.
2	PE	PE Port Enabled/Disabled-Read/Write. 1=Enable. 0=Disable. Default 0. In Host Mode: Ports can only be enabled by the host controller as a part of the reset and enable. Software cannot enable a port by writing a one to this field. Ports can be disabled by either a fault condition (disconnect event or other fault condition) or by the host software. Note that the bit status does not change until the port state actually changes. There may be a delay in disabling or enabling a port due to other host controller and bus events. When the port is disabled, (0b) downstream propagation of data is blocked except for reset. This field is zero if Port Power(PORTSC1) is zero in host mode. In Device Mode: The device port is always enabled, so this bit is always '1b'.

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Bits	Name	Description
1	CSC	CSC Connect Status Change-R/WC. 1 =Change in Current Connect Status. 0=No change. Default 0. In Host Mode: Indicates a change has occurred in the port's Current Connect Status. The host/device controller sets this bit for all changes to the port device connect status, even if system software has not cleared an existing connect status change. For example, the insertion status changes twice before system software has cleared the changed condition, hub hardware will be 'setting' an already-set bit (that is, the bit will remain set). Software clears this bit by writing a one to it. This field is zero if Port Power(PORTSC1) is zero in host mode. In Device Mode: This bit is undefined in device controller mode.
0	CCS	CCS Current Connect Status-Read Only. In Host Mode: 1=Device is present on port. 0=No device is present. Default = 0. This value reflects the current state of the port, and may not correspond directly to the event that caused the Connect Status Change bit (Bit 1) to be set. This field is zero if Port Power(PORTSC1) is zero in host mode. In Device Mode: 1=Attached. 0=Not Attached. Default=0. A one indicates that the device successfully attached and is operating in either high speed or full speed as indicated by the High Speed Port bit in this register. A zero indicates that the device did not attach successfully or was forcibly disconnected by the software writing a zero to the Run bit in the USBCMD register. It does not state the device being disconnected or Suspended.

PORTSC1 Bits

51.6.19 OTGSC (0x1A4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD					ASVIE	AVVIE	IDIE	RSVD					ASVIS	AVVIS	IDIS	RSVD					ASV	AVV	ID	RSVD		IDPU	RSVD			VC	VD

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
N/A					RW	RW	RW	N/A					RWC	RWC	RWC	N/A					RO	RO	RO	N/A		RW	N/A			RW	RW
x	x	x	x	x	0	0	0	x	x	x	x	x	0	0	0	x	x	x	x	x	0	0	0	x	x	0	x	x	x	0	0

OTGSC [31:0]

Bits	Name	Description
26	ASVIE	ASVIE A Session Valid Interrupt Enable - Read/Write.
25	AVVIE	AVVIE A VBus Valid Interrupt Enable - Read/Write. Setting this bit enables the A VBus valid interrupt.
24	IDIE	IDIE USB ID Interrupt Enable - Read/Write. Setting this bit enables the USB ID interrupt.
18	ASVIS	ASVIS A Session Valid Interrupt Status - Read/Write to Clear. This bit is set when VBus has either risen above or fallen below the A session valid threshold. Software must write a one to clear this bit.
17	AVVIS	AVVIS A VBus Valid Interrupt Status - Read/Write to Clear. This bit is set when VBus has either risen above or fallen below the VBus valid threshold on an A device. Software must write a one to clear this bit.
16	IDIS	IDIS USB ID Interrupt Status - Read/Write. This bit is set when a change on the ID input has been detected. Software must write a one to clear this bit.
10	ASV	ASV A Session Valid - Read Only. Indicates VBus is above the A session valid threshold.
9	AVV	AVV A VBus Valid - Read Only. Indicates VBus is above the A VBus valid threshold.
8	ID	ID USB ID - Read Only. 0 = A device, 1 = B device

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Bits	Name	Description
5	IDPU	IDPU ID Pullup - Read/Write This bit provide control over the ID pull-up resistor; 0 = off, 1 = on [default]. When this bit is 0, the ID input will not be sampled.
1	VC	VC VBUS Charge - Read/Write. Setting this bit causes the VBus line to be charged. This is used for VBus pulsing during SRP.
0	VD	VD VBUS_Discharge - Read/Write. Setting this bit causes VBus to discharge through a resistor.

OTGSC Bits

51.6.20 USBMODE (0x1A8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																											SDIS	SLOM	ES	CM	
N/A																											RW	RW	RW	RW	
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0

USBMODE [31:0]

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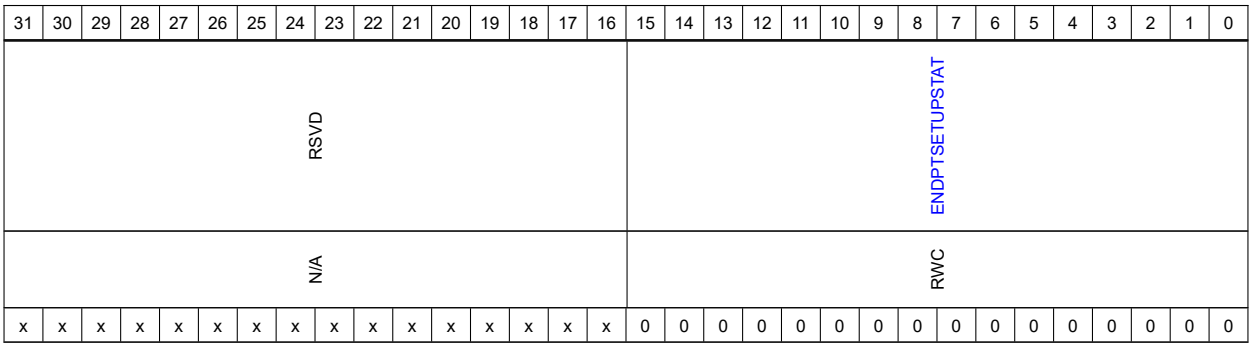
Bits	Name	Description
4	SDIS	SDIS Stream Disable Mode. (0 - Inactive [default]; 1 - Active) Device Mode: Setting to a '1' disables double priming on both RX and TX for low bandwidth systems. This mode ensures that when the RX and TX buffers are sufficient to contain an entire packet that the standard double buffering scheme is disabled to prevent overruns/underruns in bandwidth limited systems. Note: In High Speed Mode, all packets received are responded to with a NYET handshake when stream disable is active. Host Mode: Setting to a '1' ensures that overruns/underruns of the latency FIFO are eliminated for low bandwidth systems where the RX and TX buffers are sufficient to contain the entire packet. Enabling stream disable also has the effect of ensuring the TX latency is filled to capacity before the packet is launched onto the USB. NOTE: Time duration to pre-fill the FIFO becomes significant when stream disable is active. See TXFILLTUNING and TXTTFILLTUNING [MPH Only] to characterize the adjustments needed for the scheduler when using this feature. NOTE: The use of this feature substantially limits of the overall USB performance that can be achieved.
3	SLOM	SLOM Setup Lockout Mode. In device mode, this bit controls behavior of the setup lock mechanism. See Control Endpoint Operation Model . 0 - Setup Lockouts On (default); 1 - Setup Lockouts Off. DCD requires use of Setup Data Buffer Tripwire in USBCMD.
2	ES	ES Endian Select - Read/Write. This bit can change the byte alignment of the transfer buffers to match the host microprocessor. The bit fields in the microprocessor interface and the data structures are unaffected by the value of this bit because they are based upon the 32-bit word. Bit Meaning 0 - Little Endian [Default] 1 - Big Endian

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Bits	Name	Description
1-0	CM	CM Controller Mode - R/WO. Controller mode is defaulted to the proper mode for host only and device only implementations. For those designs that contain both host & device capability, the controller defaults to an idle state and needs to be initialized to the desired operating mode after reset. For combination host/ device controllers, this register can only be written once after reset. If it is necessary to switch modes, software must reset the controller by writing to the RESET bit in the USBCMD register before reprogramming this register. For OTG controller core, reset value is ’00b’. 00 - Idle [Default for combination host/device] 01 - Reserved 10 - Device Controller [Default for device only controller] 11 - Host Controller [Default for host only controller]

USBMODE Bits

51.6.21 ENDPTSETUPSTAT (0x1AC)



ENDPTSETUPSTAT [31:0]

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Bits	Name	Description
15-0	ENDPTSETUPSTAT	ENDPTSETUPSTAT Setup Endpoint Status. For every setup transaction that is received, a corresponding bit in this register is set to one. Software must clear or acknowledge the setup transfer by writing a one to a respective bit after it has read the setup data from Queue head. The response to a setup packet as in the order of operations and total response time is crucial to limit bus time outs while the setup lock out mechanism is engaged. This register is only used in device mode.

ENDPTSETUPSTAT Bits

51.6.22 ENDPTPRIME (0x1B0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PETB																PERB															
RWS																RWS															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENDPTPRIME [31:0]

Bits	Name	Description
31-16	PETB	PETB Prime Endpoint Transmit Buffer - R/WS. For each endpoint a corresponding bit is used to request that a buffer is prepared for a transmit operation in order to respond to a USB IN/INTERRUPT transaction. Software should write a one to the corresponding bit when posting a new transfer descriptor to an endpoint queue head. Hardware automatically uses this bit to begin parsing for a new transfer descriptor from the queue head and prepare a transmit buffer. Hardware clears this bit when the associated endpoint(s) is (are) successfully primed. NOTE: These bits are momentarily set by hardware during hardware re-priming operations when a dTD is retired, and the dQH is updated. PETB[N] - Endpoint #N, N is in 0..7

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Bits	Name	Description
15-0	PERB	PERB Prime Endpoint Receive Buffer - R/WS. For each endpoint, a corresponding bit is used to request a buffer prepare for a receive operation for when a USB host initiates a USB OUT transaction. Software should write a one to the corresponding bit whenever posting a new transfer descriptor to an endpoint queue head. Hardware automatically uses this bit to begin parsing for a new transfer descriptor from the queue head and prepare a receive buffer. Hardware clears this bit when the associated endpoint(s) is (are) successfully primed. NOTE: These bits are momentarily set by hardware during hardware re-priming operations when a dTD is retired, and the dQH is updated. PERB[N] - Endpoint #N, N is in 0..7

ENDPTPRIME Bits

51.6.23 ENDPTFLUSH (0x1B4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FETB																FERB															
RWS																RWS															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENDPTFLUSH [31:0]

Bits	Name	Description
31-16	FETB	FETB Flush Endpoint Transmit Buffer - R/WS. Writing one to a bit(s) in this register causes the associated endpoint(s) to clear any primed buffers. If a packet is in progress for one of the associated endpoints, then that transfer continues until completion. Hardware clears this register after the endpoint flush operation is successful. FETB[N] - Endpoint #N, N is in 0..7

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Bits	Name	Description
15-0	FERB	FERB Flush Endpoint Receive Buffer - R/WS. Writing one to a bit(s) causes the associated endpoint(s) to clear any primed buffers. If a packet is in progress for one of the associated endpoints, then that transfer continues until completion. Hardware clears this register after the endpoint flush operation is successful. FERB[N] - Endpoint #N, N is in 0..7

ENDPTFLUSH Bits

51.6.24 ENDPTSTAT (0x1B8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETBR																ERBR															
RO																RO															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENDPTSTAT [31:0]

Bits	Name	Description
31-16	ETBR	ETBR Endpoint Transmit Buffer Ready – Read Only. One bit for each endpoint indicates status of the respective endpoint buffer. This bit is set to one by the hardware as a response to receiving a command from a corresponding bit in the ENDPTPRIME register. There is always a delay between setting a bit in the ENDPTPRIME register and endpoint indicating ready. This delay time varies based upon the current USB traffic and the number of bits set in the ENDPRIME register. Buffer ready is cleared by USB reset, by the USB DMA system, or through the ENDPTFLUSH register. NOTE: These bits are momentarily cleared by hardware during hardware endpoint re-priming operations when a dTD is retired, and the dQH is updated. ETBR[N] - Endpoint #N, N is in 0..7

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Bits	Name	Description
15-0	ERBR	ERBR Endpoint Receive Buffer Ready – Read Only. One bit for each endpoint indicates status of the respective endpoint buffer. This bit is set to a one by the hardware as a response to receiving a command from a corresponding bit in the ENDPRIME register. There is always a delay between setting a bit in the ENDPRIME register and endpoint indicating ready. This delay time varies based upon the current USB traffic and the number of bits set in the ENDPRIME register. Buffer ready is cleared by USB reset, by the USB DMA system, or through the ENDPTFLUSH register. NOTE: These bits are momentarily cleared by hardware during hardware endpoint re-priming operations when a dTD is retired, and the dQH is updated. ERBR[N] - Endpoint #N, N is in 0..7

ENDPTSTAT Bits

51.6.25 ENDPTCOMPLETE (0x1BC)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ETCE																ERCE															
RWC																RWC															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ENDPTCOMPLETE [31:0]

Bits	Name	Description
31-16	ETCE	ETCE Endpoint Transmit Complete Event - R/WC. Each bit indicates a transmit event (IN/INTERRUPT) occurred, and software should read the corresponding endpoint queue to determine the endpoint status. If the corresponding IOC bit is set in the Transfer Descriptor, then this bit is set simultaneously with the USBINT . Writing one clears the corresponding bit in this register. ETCE[N] - Endpoint #N, N is in 0..7

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Bits	Name	Description
15-0	ERCE	ERCE Endpoint Receive Complete Event - RW/C. Each bit indicates a received event (OUT/SETUP) occurred and software should read the corresponding endpoint queue to determine the transfer status. If the corresponding IOC bit is set in the Transfer Descriptor, then this bit is set simultaneously with the USBINT . Writing one clears the corresponding bit in this register. ERCE[N] - Endpoint #N, N is in 0..7

ENDPTCOMPLETE Bits

51.6.26 ENDPTCTRL (0x1C0 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD								TXE	TXR	RSVD		TXT		RSVD	TXS	RSVD								RXE	RXR	RSVD		RXT		RSVD	RXS
N/A								RW	WS	N/A		RW		N/A	RW	N/A								RW	WS	N/A		RW		N/A	RW
x	x	x	x	x	x	x	x	0	0	x	x	0	0	x	0	x	x	x	x	x	x	x	x	0	0	x	x	0	0	x	0

ENDPTCTRL [31:0]

Bits	Name	Description
23	TXE	TXE TX Endpoint Enable 0 Disabled [Default] 1 Enabled An Endpoint should be enabled only after it has been configured.
22	TXR	TXR TX Data Toggle Reset (WS) Write 1 - Reset PID Sequence Whenever a configuration event is received for this Endpoint, software must write a one to this bit in order to synchronize the data PID’s between the Host and device.

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Bits	Name	Description
19-18	TXT	TXT TX Endpoint Type - Read/Write 00 Control 01 Isochronous 10 Bulk 11 Interrupt NOTE: when this TX endpoint type is set to none-Control type, must set RXT in same register to none-Control type also, even it's not enabled. (control endpoint must be existed with pair)
16	TXS	TXS TX Endpoint Stall - Read/Write 0 End Point OK 1 End Point Stalled This bit will be cleared automatically upon receipt of a SETUP request if this Endpoint is configured as a Control Endpoint and this bit will continue to be cleared by hardware until the associated ENDPTSETUPSTAT bit is cleared. Software can write a one to this bit to force the endpoint to return a STALL handshake to the Host. This control will continue to STALL until this bit is either cleared by software or automatically cleared as above for control endpoints. NOTE: [CONTROL ENDPOINT TYPES ONLY]: there is a slight delay (50 clocks max) between the ENDPTSETUPSTAT begin cleared and hardware continuing to clear this bit. In most systems, it is unlikely the DCD software will observe this delay. However, should the DCD observe that the stall bit is not set after writing a one to it then follow this procedure: continually write this stall bit until it is set or until a new setup has been received by checking the associated endptsetupstat Bit.
7	RXE	RXE RX Endpoint Enable 0 Disabled [Default] 1 Enabled An Endpoint should be enabled only after it has been configured.

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Bits	Name	Description
6	RXR	RXR RX Data Toggle Reset (WS) Write 1 - Reset PID Sequence Whenever a configuration event is received for this Endpoint, software must write a one to this bit in order to synchronize the data PID's between the host and device.
3-2	RXT	RXT RX Endpoint Type - Read/Write 00 Control(default) 01 Isochronous 10 Bulk 11 Interrupt NOTE: when this RX endpoint type is set to none-Control type, must set TXT in same register to none-Control type also, even it's not enabled. (control endpoint must be existed with pair)
0	RXS	RXS RX Endpoint Stall - Read/Write 0 End Point OK. [Default] 1 End Point Stalled This bit is set automatically upon receipt of a SETUP request if this Endpoint is configured as a Control Endpointand this bit will continue to be cleared by hardware until the associated ENDPTSETUPSTAT bit is cleared. Software can write a one to this bit to force the endpoint to return a STALL handshake to the Host. This control will continue to STALL until this bit is either cleared by software or automatically cleared as above for control endpoints. NOTE: [CONTROL ENDPOINT TYPES ONLY]: there is a slight delay (50 clocks max) between the ENDPTSETUPSTAT begin cleared and hardware continuing to clear this bit. In most systems, it is unlikely the DCD software will observe this delay. However, should the DCD observe that the stall bit is not set after writing a one to it then follow this procedure: continually write this stall bit until it is set or until a new setup has been received by checking the associated endptsetupstat Bit.

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Bits	Name	Description

ENDPTCTRL Bits

51.6.27 OTG_CTRL0 (0x200)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD						OTG_WKDPDMCHG_EN	RSVD					AUTORESUME_EN	RSVD	OTG_VBUS_WAKEUP_EN	OTG_ID_WAKEUP_EN	RSVD		OTG_VBUS_SOURCE_SEL	OTG_UTMI_SUSPENDM_SW	OTG_UTMI_RESET_SW	OTG_WAKEUP_INT_ENABLE	OTG_POWER_MASK	OTG_OVER_CUR_POL	OTG_OVER_CUR_DIS	RSVD		SER_MODE_SUSPEND_EN	RSVD			
N/A						RW	N/A					RW	N/A	RW	RW	N/A		RW	RW	RW	RW	RW	RW	RW	N/A		RW	N/A			
x	x	x	x	x	x	0	x	x	x	x	x	0	x	0	0	x	x	0	0	0	0	0	0	0	x	x	0	x	x	x	x

OTG_CTRL0 [31:0]

Bits	Name	Description
25	OTG_WKDPDMCHG_EN	DP/DM wakeup enable when USB enters into low power mode(suspend), setting this bit to high, or VBUS is valid(after selected by otg_vbus_source_sel), will wakeup USB(exit suspend), and send wakeup interrupt if otg_wakeup_int_enable is set to high. this bit is only used in device mode. use WKDC/WKCN in PORTSC1 to enable DP/DM wakeup at host mode
19	AUTORESUME_EN	auturesume enable used in host mode only, for response to device remote wakeup if PLL is not enabled in time.
17	OTG_VBUS_WAKEUP_EN	vbus wakeup enable
16	OTG_ID_WAKEUP_EN	ID wakeup enable
13	OTG_VBUS_SOURCE_SEL	VBUS wakeup source select. 0: use vbus valid as wakeup source(>4.2V); 1: use session valid as wakeup source(> about 2.4V);

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Bits	Name	Description
12	OTG_UTMI_SUSPENDM_SW	software usbphy suspend control. default 0 will suspend usbphy. user need to set this bit to 1 during usbphy initiallization NOTE: make sure at least 1us between setting this bit and clear otg_utmi_reset_sw
11	OTG_UTMI_RESET_SW	software usbphy reset control. default 1 will let usbphy in reset state. user need to clr this bit to 0 during usbphy initiallization NOTE: before clearing this bit, make sure at least 1us after setting otg_utmi_suspendm_sw.
10	OTG_WAKEUP_INT_ENABLE	wakeup interrupt enable. when USB in suspend state(PORTSC1.PHCD is 1), user can close all PLL and external crystal, keep internal 24MHz osc active as wakeup clock. when receive wakeup event and related enable is set, an interrupt to CPU will be asserted if this bit is set. the interrupt status bit is located at TOP_STATUS.wakeup_int_status，user need clear this bit to 0 to clear the wakeup interrupt.
9	OTG_POWER_MASK	USB*_PWR polarity control during host mode, user can config IOMUX to output USB*_PWR pin(which is from PORTSC1.PP), to control external VBUS on/off. this bit can control the polarity of external VBUS control logic active high or active low(default 0). user can also use a GPIO to control.
8	OTG_OVER_CUR_POL	overcurrent polarity control. during host mode, user can config IOMUX to input USB*_OC as overcurrent status indication(generally come from VBUS control logic). this bit can control the polarity of overcurrent. 0 for active high; 1 for active low.
7	OTG_OVER_CUR_DIS	overcurrent disable. setting this bit to 1 can disable overcurrent function.
4	SER_MODE_SUSPEND_EN	setting this bit to 1.

OTG_CTRL0 Bits

51.6.28 PHY_CTRL0 (0x210)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD						GPIO_ID_SEL_N	RSVD										ID_DIG_OVERRIDE	SESS_VALID_OVERRIDE	VBUS_VALID_OVERRIDE	OP_MODE_SUSPENDM_ENJ	RSVD								ID_DIG_OVERRIDE_EN	SESS_VALID_OVERRIDE_EN	VBUS_VALID_OVERRIDE_EN
N/A						RW	N/A										RW	RW	RW	RW	N/A								RW	RW	RW
x	x	x	x	x	x	0	x	x	x	x	x	x	x	x	x	x	0	0	0	0	x	x	x	x	x	x	x	x	0	0	0

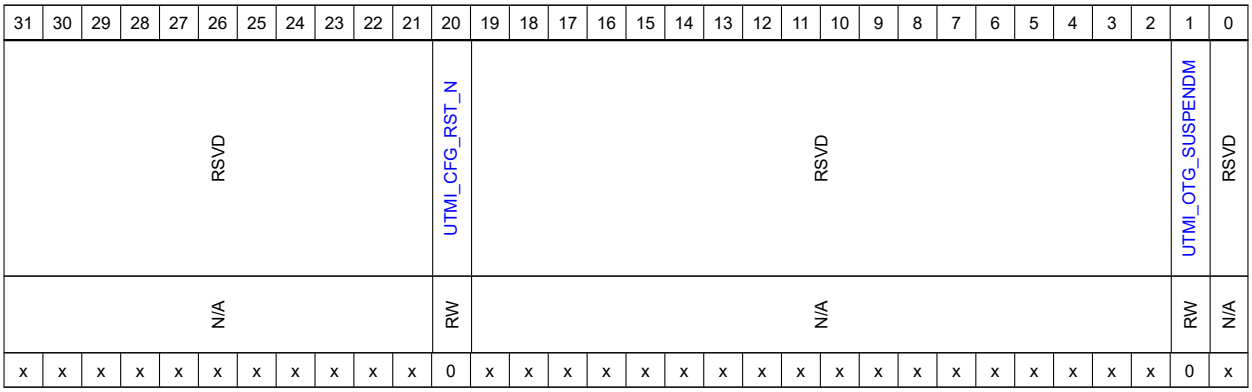
PHY_CTRL0 [31:0]

Bits	Name	Description
25	GPIO_ID_SEL_N	ID selection USBPHY has ID pin, but it maybe not exist in some small package. user can use some GPIO(see pinmux for detail) as ID in these package if need ID functon. 0: use usbphy ID. 1: use GPIO ID(need config pinmux)
14	ID_DIG_OVERRIDE	ID override value. use this register as ID value if id_dig_override_en is set to 1. this feature is useful when external ID is not used.
13	SESS_VALID_OVERRIDE	sess_valid override value. use this register as session valid value if sess_valid_override_enis set to 1.
12	VBUS_VALID_OVERRIDE	vbus_valid override value. use this register as vbus valid value if vbus_valid_override_en is set to 1.
11	OP_MODE_SUSPENDM_ENJ	set op_mode to 2’b01 in suspend, for naneng usbphy
2	ID_DIG_OVERRIDE_EN	ID override enable
1	SESS_VALID_OVERRIDE_EN	SESS_VALID override enable
0	VBUS_VALID_OVERRIDE_EN	VBUS_VALID override enable

PHY_CTRL0 Bits

51.6.29 PHY_CTRL1 (0x214)

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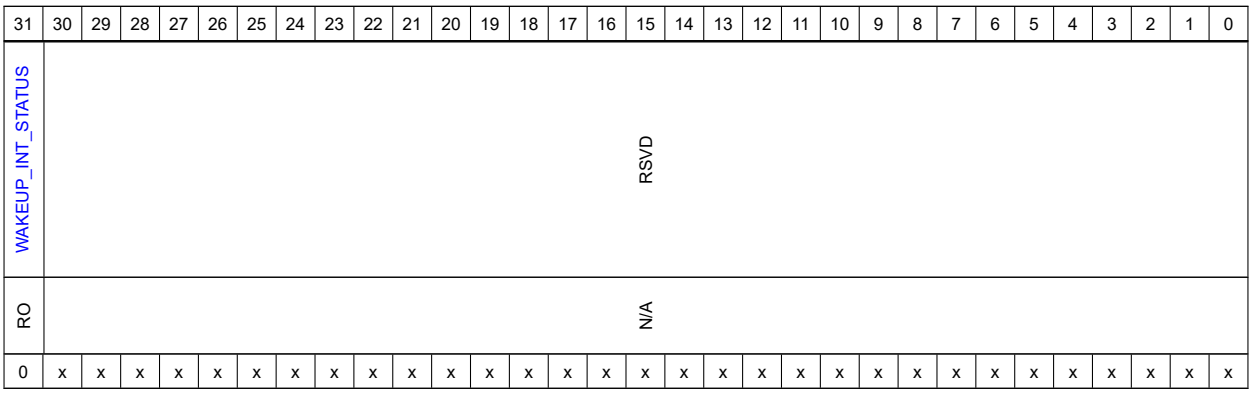


PHY_CTRL1 [31:0]

Bits	Name	Description
20	UTMI_CFG_RST_N	usbphy config reset defult 0 for usbphy config in reset state. user need set it to 1 during usbphy initialization.
1	UTMI_OTG_SUSPENDM	OTG suspend, not utmi_suspendm. setting this bit also enable host high-speed disconnect detection logic(check DP/DM voltage at end of SOF, to determine whether 50Ohm are still there on DP/DM) should clear this bit before entering into suspend state(setting portsc.phcd), avoide unwanted glitch on vbus/sess_vld/sess_end

PHY_CTRL1 Bits

51.6.30 TOP_STATUS (0x220)



TOP_STATUS [31:0]

Bits	Name	Description
31	WAKEUP_INT_STATUS	wakeup interrupt status bit. NOTE: this bit is not write 1 clear. user need clear OTG_CTRL0.otg_wakeup_int_enable to clear wakeup interrupt.

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Bits	Name	Description

TOP_STATUS Bits

51.6.31 PHY_STATUS (0x224)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
UTMI_CLK_VALID	RSVD																							LINE_STATE		HOST_DISCONNECT	ID_DIG	RSVD	UTMI_SESS_VALID	RSVD	VBUS_VALID
RW	N/A																							RW		RW	RW	N/A	RW	N/A	RW
0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	x	0	x	0

PHY_STATUS [31:0]

Bits	Name	Description
31	UTMI_CLK_VALID	usbphy clock valid status bit. this bit is used to check whether usbphy is outputing valid UTMI clock. user need write 1 to clear it first, and read to check whether UTMI clock is valid. 0 means no valid UTMI clock; 1 means there's valid UTMI clock.
7-6	LINE_STATE	USB bus state. during full speed or low speed, bit7 is DM state, bit6 is DP state. during high speed, this filed only appeared whether in data transfer(Squelch), 00 means no data transfer, 01 means active data transfer
5	HOST_DISCONNECT	host disconnection status this bit is used only for host mode working at high speed. when high speed device connected, usb host can't use DP/DM status to detect device disconnection. usbphy will detect device disconnection at end of SOF(32-bit time), by check signal level(400mv or 800mv). this bit set to 1 means usbphy detect high speed device disconection.
4	ID_DIG	usbphy ID pin state.
2	UTMI_SESS_VALID	usbphy SESS_VALID state.
0	VBUS_VALID	usbphy VBUS_VALID state.

PHY_STATUS Bits

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52 Analog Peripherals Overview

This chapter introduces the analog peripherals of this product.

52.1 16-bit Analog-to-Digital Converter ADC16

This product supports 2 16-bit analog-to-digital converters ADC16. ADC16 is a successive approximation analog-to-digital converter that can convert analog signals from external pins and inside the chip. When the conversion accuracy of ADC16 is set to 16 bits, the maximum sampling rate is 2MSPS, and when the conversion accuracy is set to 12 bits, the maximum sampling rate is 4MSPS.

On this product, the two 16-bit ADCs are called ADC0~1.

On this product, the reference voltage inputs of the two 16-bit ADCs are VREFH and VREFL.

Note that when using the built-in DCDC, the accuracy of the 16-bit ADC will decrease to a certain extent; for precautions for using the on-chip ADC, please refer to the ADC-related application documents at https://www.hpmicro.com/design-resources/chip-information. For the power supply structure without using the built-in DCDC, please refer to Figure 5

Note: On this product, ADC16 related registers (registers starting with ADC16_, with offset greater than or equal to 0x1400) need to configure the following registers to be written normally:

- Set CLK_DIVIDER=1 in the CONV_CFG1 register;
- Set ADC_CLK_ON=1 in the ANA_CTRL0 register;

52.1.1 ADC0 Input Channel Allocation

The 16-bit analog-to-digital converter ADC0 supports input channels ina0~15, all from IO.

For the specific connection between ADC0 input channels and IO, please refer to Chapter 20.

52.1.2 ADC1 Input Channel Allocation

The 16-bit analog-to-digital converter ADC1 supports input channels ina0~15, all from IO.

For the specific connection between ADC1 input channels and IO, please refer to Chapter 20.

52.1.3 ADC Conversion Trigger Signal Connection

The 16-bit analog-to-digital converter ADC16 on this product supports 2 types of queue conversion: sequence conversion and preemption conversion:

- The maximum sequence conversion length can reach 16, supporting software or hardware triggering. The hardware trigger signal STRIGI comes from the interconnection manager TRGM. Other on-chip modules can trigger ADC sequence conversion through this signal.
- The maximum preemption conversion length can reach 4. Each ADC supports 4 groups, 3 per group, a total of 12 preemption conversion queues. Each preemption queue supports software or hardware triggering, 12 hardware trigger signals come from the interconnection manager, and other on-chip modules can trigger ADC preemption conversion through these signals.
  - PTRGI0A, PTRGI0B, PTRGI0C come from the interconnection manager TRGM
  - PTRGI1A, PTRGI1B, PTRGI1C come from the interconnection manager TRGM
  - PTRGI2A, PTRGI2B, PTRGI2C come from the interconnection manager TRGM
  - PTRGI3A, PTRGI3B, PTRGI3C come from the interconnection manager TRGM

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– The preemption conversion trigger signals with the same serial number of each ADC on the chip are short-circuited, so the same trigger signal can trigger the preemption conversion sequences with the same serial number of all ADCs on the chip at the same time.

For the specific connection of ADC16 trigger signals, please refer to Section 32.8.

Note: Before receiving the trigger signal, the ADC must be initialized first, otherwise it may lead to abnormal operation.

52.2 Comparator ACMP

This product supports 2 analog comparators ACMP. ACMP can compare two analog voltage inputs (positive terminal voltage INP) and negative terminal voltage (INN), and output the comparison result. ACMP supports an internal 8-bit digital-to-analog converter DAC, and supports comparing two external analog signals, or comparing an external analog signal with a reference signal generated by the internal DAC.

On this product, the reference voltage of the internal DAC of ACMP comes from VREFH.

On this product, ACMP supports 8 positive terminal voltage INP input channels and 8 negative terminal voltage INN input channels. INP0 and INN0 come from the output of the built-in DAC of ACMP. The remaining input channels come from IO. For the specific connection between each ACMP input channel and IO on this product, please refer to Chapter 20.

On this product, the outputs of the 2 ACMPs are connected to the IO and also to the interconnection manager of the motor system. For specific connections, please refer to Chapter 20 and Section 34.3.5.

52.3 Temperature Sensor

On this product, a temperature sensor is integrated, which can be used to measure the internal temperature of the chip.

52.4 Digital-to-Analog Converter DAC

On this product, 2 12-bit digital-to-analog converters DAC are integrated, which can convert digital signals into analog signal outputs.

52.5 Operational Amplifier OPAMP

On this product, 2 operational amplifiers OPAMP are integrated.

The output OPA0_OUT of OPAMP0 and the input channel IN15 of ADC0/1 are connected to PB00, The output OPA1_OUT of OPAMP1 and the input channel IN14 of ADC0/1 are connected to PB01, which is convenient for the ADC to directly convert the output signal of the operational amplifier.

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53 16-bit Analog-to-Digital Converter ADC16

This chapter describes the main functions and features of the 16-bit analog-to-digital converter ADC16:

53.1 Feature Summary

This chapter describes the main features of the 16-bit analog-to-digital converter ADC16:

- 16-bit Successive Approximation Register (SAR) ADC
- Up to 2MHz sampling rate
- Supports single-ended input
- Independent ADC conversion clock
- Supports arbitrarily configurable AD conversion resolution
- Configurable number of sampling cycles
- Built-in DMA can directly write ADC conversion results to memory
- Supports read conversion mode
  - Reading the result register directly triggers the conversion
- Supports periodic conversion mode
  - Built-in timer performs conversion periodically
  - Supports hardware threshold comparison, alarms for out-of-range conversion results
- Supports sequence conversion mode
  - Can be triggered by software or hardware
  - Maximum sequence length up to 16
  - Supports single conversion or continuous conversion
  - Supports sequence loop
- Supports preemptive conversion mode
  - Can be triggered by software or hardware
  - Maximum continuous conversion sequence length up to 4
- Supports generation of various interrupts

53.2 Functional Description

This chapter describes the functions of the 16-bit analog-to-digital converter ADC16.

53.2.1 ADC Clock

The ADC clock includes the ADC controller clock and the ADC conversion clock.

The ADC conversion clock is the clock that controls the timing of the ADC successive approximation analog conversion.

It is obtained by prescaling the ADC controller clock, which is configured by the system control module and defaults to the system bus clock of 200MHz (synchronized with the motor system clock).

Users can configure the prescaler value of the ADC conversion clock relative to the ADC controller clock by setting the CONVCFG1[CONVCLKDIV] register bit. When CONVCFG1[CONVCLKDIV] is 0, the prescaler value is /1; when it is 1, the prescaler value is /2, with a maximum value of /16.

The maximum frequency of the ADC16 conversion clock is 50MHz.

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53.2.2 ADC Input Channel Configuration

Each input channel of the ADC can be configured individually through the SAMPLE_CFGx register.

Users can set the sampling time of the channel through the SAMPLE_CFGx [SAMPLE_CLOCK_NUMBER] and [SAMPLE_CLOCK_NUMBER_SHIFT] bits.

The sampling time is $SAMPLE_CLOCK_NUMBER \times 2^{SAMPLE_CLOCK_NUMBER_SHIFT}$ clock cycles. The minimum sampling time is 1 clock cycle.

In addition, users need to configure CONV_CFG1[CONVERT_CLOCK_NUMBER] to 21, that is, the conversion time is 21 clock cycles, to achieve 16-bit conversion accuracy.

If users do not need 16-bit accuracy but want to get results faster, they can do so by shortening the conversion time.

Users need to change CONV_CFG1[CONVERT_CLOCK_NUMBER] and ADC16_CONFIG1[COV_END_CNT] at the same time.

COV_END_CNT = 21-CONVERT_CLOCK_NUMBER; When CONVERT_CLOCK_NUMBER=21, ADC16 operates in full 16-bit mode;

When CONVERT_CLOCK_NUMBER=14, ADC16 operates in approximately 12-bit mode;

When CONVERT_CLOCK_NUMBER=11, ADC16 operates in approximately 10-bit mode;

When CONVERT_CLOCK_NUMBER=9, ADC16 operates in approximately 8-bit mode;

Other values are also supported, and users can freely choose according to accuracy and speed requirements.

The conversion time for each channel is the sum of the sampling time and the conversion time.

53.2.3 Read Conversion Mode

Users can intuitively read the conversion result of a specific ADC input channel through the read conversion mode.

This mode is suitable for scenarios where users need to obtain the result of a channel as soon as possible and have no other tasks to perform before the result is available.

When the user reads the BUS_RESULTx register, it triggers a conversion of ADC input x. After the conversion is completed, the result is returned directly.

The time taken to return the conversion result varies according to different ADC channel configurations.

Before the ADC completes the conversion, it will block access to the relevant peripheral bus, and the blocking time caused by the conversion may be relatively long. Users can set the BUF_CFG0 [WAITDIS] bit to 1 to disable bus blocking, at which point reading the BUS_RESULTx register will directly return the result of the last conversion. When the BUS_RESULTx [VALID] bit is set to 1, it indicates that the ADC conversion is completed, and the BUS_RESULTx register holds the latest conversion result.

53.2.4 Periodic Conversion Mode

Users can require one or more input channels of the ADC to perform periodic conversion through the periodic conversion mode.

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This mode is suitable for scenarios with timed conversion requirements, or for monitoring whether a channel exceeds a threshold.

Users can set the cycle length through the PRD_CFGx [PRD_CNT] and [PRESCALE] register bits, and the cycle length is specifically $PRD_CNT \times 2^{PRESCALE}$, The counter decrements by 1 every ADC clock cycle. When the counter counts to 0, the conversion starts and the counter is reloaded.

Note: Clearing the PRD_CFGx [PRD_CNT] bit to 0 disables the periodic conversion of this input.

The result of periodic conversion is stored in the PRD_RESULTx register, and the software can read the latest conversion result of input channel x from PRD_RESULTx.

Note: PRD_RESULTx always saves the latest conversion result of ADC input channel x. PRD_RESULTx stores not only the results of periodic conversion, but also the conversion results of other conversion modes.

Users can monitor the conversion result of input channel x by setting PRD_THSHD_CFGx[THSHDH] and PRD_THSHD_CFGx[THSHDL]. Once the conversion result is out of range (Result is greater than THSHDHx or less than THSHDLx), an alarm will be triggered, the corresponding flag bit INT_STS[ADWDGx] will be set to 1, and an interrupt will be generated if the corresponding interrupt is enabled.

Note: ADC conversion result detection is not limited to periodic conversion mode, but is valid for all read modes. If the threshold is set incorrectly, for example, setting THSHDHx to 0, an alarm will be triggered every time the channel conversion is completed.

If multiple channels have their timers expire at the same time and need to start conversion, the input channel with the smaller serial number will be converted first.

53.2.5 Sequence Conversion Mode

The ADC supports sequence conversion mode, in which the ADC can convert specified input channels one by one according to a pre-programmed sequence.

This mode is suitable for scenarios where continuous conversion of one or several channels is required with a large amount of data, such as testing signal-to-noise ratio or ENOB (Effective Number of Bits).

Users can specify which input channel is the conversion target of the sequence by configuring the SEQ_QUEx registers in sequence. The SEQ_QUE0 [CHAN_NUM] bit field configures the channel number of the first AD conversion after the sequence conversion is triggered; the SEQ_QUE1 [CHNUM] bit field configures the channel number of the second AD conversion, and so on, the SEQ_QUE15 [CHNUM] bit configures the channel number of the 16th AD conversion.

In addition, users need to configure the SEQ_CFG0 [SEQ_LEN] bit to specify the length of the conversion sequence, and the maximum length of the ADC conversion sequence is 16.

The start of sequence conversion can be triggered by software or hardware:

- The user first sets the SEQ_CFG0 [SW_TRIG_EN] bit to 1 through software, then sets the SEQ_CFG0 [SW_TRIG] bit to 1 to trigger the sequence conversion.
- After the SEQ_CFG0 [HW_TRIG_EN] bit is set to 1, a rising edge captured on STRGI triggers the sequence conversion

Once triggered, the ADC will start converting the input channel according to the configuration of SEQ_QUE0 in sequence. After the conversion is completed, the flag bit INT_STS [SEQ_CVC] (conversion complete) will be set to 1. At this time:

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- If the SEQ_CFG0 [CONT_EN] bit is set to 1, the ADC will automatically start the next AD conversion according to the configuration of SEQ_QUE1 until the specified sequence length SEQ_QuEn is reached.
- If the SEQ_CFG0 [CONT_EN] bit is cleared to 0, the ADC will wait for the next software or hardware trigger before starting the conversion of the next channel.

When the ADC completes all conversions of the sequence (the length is set by the SEQ_CFG0 [SEQ_LEN] bit), the flag bit INT_STS [SEQ_CMPT] (Sequence Complete) will be set to 1. At this time:

- If SEQ_CFG0 [RESTART_EN] bit is set to 1 and SEQ_CFG0 [CONT_EN] bit is set to 1, the ADC will automatically start continuous conversion sequentially from SEQ_QUE0 according to the configuration of SEQ_QUEx
- If SEQ_CFG0 [RESTART_EN] bit is cleared to 0 and SEQ_CFG0 [CONT_EN] bit is set to 1, the ADC will continuously convert until the entire sequence is completed after the next software or hardware trigger
- If the SEQ_CFG0 [CONT_EN] bit is cleared to 0, regardless of whether the SEQ_CFG0 [RESTART_EN] bit is set to 1 or 0, the ADC will restart conversion according to the configuration of SEQ_QUEx only after the next software or hardware trigger.

If a new sequence conversion trigger signal is received during the ADC sequence conversion, the ADC will ignore this trigger. If it is a software-triggered sequence conversion, the INT_STS [SEQ_SW_CFLCT] bit will be set to 1. If it is a hardware-triggered sequence conversion caused by STRGI, the INT_STS [SEQ_HW_CFLCT] bit will be set to 1.

53.2.6 DMA for Sequence Conversion Mode

The ADC sequence conversion mode supports built-in DMA, which can directly write conversion results to the buffer specified by the user in memory. Users can configure the target address through the SEQ_DMA_ADDR register, and set the data area length through SEQ_DMA_CFG [BUF_LEN]. This configures a circular buffer for the ADC sequence conversion results, that is:

After the first AD conversion is completed, the DMA will write the conversion result to SEQ_DMA_ADDR, and the result of the next AD conversion will be written to SEQ_DMA_ADDR + 4. After each subsequent conversion is completed, the pointer address will continue to accumulate. If the address exceeds SEQ_DMA_ADDR + BUF_LEN, the ADC will rewrite the next AD conversion result to the memory address unit pointed to by SEQ_DMA_ADDR.

Note: DMA writes 32-bit (4-byte) data to memory each time, where the 16-bit AD conversion result is located in [15:0]. Bits [31:16] also include flag data such as cycle, sequence number, and AD input channel number, which facilitates software interpretation of AD conversion information.

Figure 89 shows the data format written to memory by DMA in sequence conversion mode.

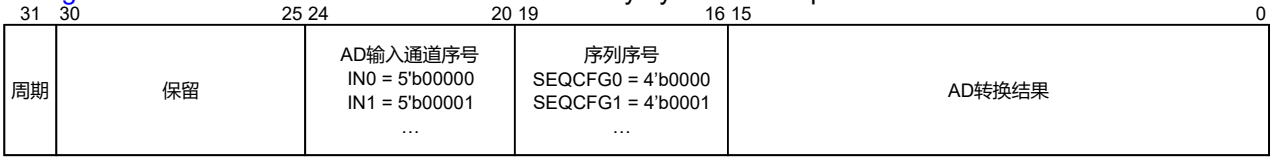


Figure 89: Data Format of Sequence Conversion Mode

Users can use the Cycle bit [31] to assist in judging the real-time performance of AD conversion results. When DMA writes the conversion result to the memory buffer for the first time, this bit will be set to 1. When the

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entire buffer is full, the DMA pointer loops back to the starting position, and this bit will be set to 0 in this round. When DMA fills the buffer again and the pointer returns to the starting position for the third time, the Cycle bit will be set to 1 again in this round.

To prevent DMA from overwriting data in the buffer that has not been read in time, users can set the SEQ_DMA_CFG [STOP_EN] bit to 1 and set the SEQ_DMA_CFG [STOP_POS] bit to mark the position where the software last read the conversion result. After the pointer reaches SEQ_DMA_ADDR + STOP_POS, DMA will no longer write data to the buffer. The results of this and subsequent AD conversions will be lost, the flag bit INT_STS [SEQ_DMAABT] will be set to 1, and the ADC will generate an interrupt alarm if the corresponding interrupt is enabled.

Users can reset the entire conversion sequence by setting the SEQ_DMA_CFG [DMA_RST] bit to 1 and then clearing it to 0. After reset, the ADC will restart the sequence conversion according to the configuration of SEQ_QUE0 after the next software or hardware trigger. This bit also resets the DMA pointer to the address contained in the SEQ_DMA_ADDR register. The Cycle bit is also reset to 1.

It is recommended that after setting the SEQ_DMA_CFG [DMA_RST] bit to 1, the software initializes and clears the buffer in memory, and then clears the SEQ_DMA_CFG [DMA_RST] bit to 0.

53.2.7 Preemptive Conversion Mode

The ADC supports preemptive conversion mode, which is the conversion mode with the highest priority.

This mode is suitable for motor control systems and other scenarios with high real-time requirements that require conversion of specified channels at specified times.

The ADC supports 12 groups of preemptive conversion sequences, with trigger signals coming from the on-chip interconnection module of the chip, totaling 12 preemptive trigger sources.

Users can configure these preemptive conversion sequences separately through the CONFIGx registers.

Preemptive conversion is triggered by hardware:

- A rising edge captured on PTRGIxA, PTRGIxB or PTRGIxC triggers the preemptive sequence xA, xB or xC

Users first need to configure the length of the preemptive conversion sequence through the [TRIG_LEN] bit of the CONFIGx register. The maximum sequence length is 4, that is, each trigger can trigger a conversion of up to length 4.

Users can configure the conversion order of preemptive conversion through the CONFIGx register. The [CHAN0], [CHAN1], [CHAN2], and [CHAN3] bit fields of the configuration register can sequentially configure the AD input channel numbers for the 1st, 2nd, 3rd, and 4th conversions after triggering.

Once the preemptive conversion starts, it will continuously convert the entire preemptive sequence according to the configuration of [TRIG_LEN]. After the xth conversion of the preemptive sequence is completed, if the [INTENx] bit of the CONFIGx register is set to 1, the INT_STS [TRIG_CMPT] flag bit will be set to 1, and if the corresponding interrupt control bit is also set to 1, the ADC will generate an interrupt.

If a new preemptive conversion trigger signal is received during the ADC preemptive conversion, the ADC will not respond to the new trigger. However, depending on whether the trigger source is software or hardware, the INT_STS [TRIG_SW_CFLICT] flag bit or INT_STS [TRIG_HW_CFLICT] flag bit will be set to 1, indicating that a preemptive conversion trigger conflict has occurred. If the corresponding interrupt control bit is set to 1,

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the ADC will generate an interrupt alarm.

If several different preemptive conversions are triggered at the same time, the ADC will respond preferentially in the following order:

Preemptive conversion with a smaller group number, for example, PTRGI0A has higher priority than PTRGI1A. Within the same group, xA has higher priority than xB, and xB has higher priority than xC. For example, PTRGI0A has higher priority than PTRGI0B. At the same time, the INT_STS [PTCHWCFLICT] flag bit is set to 1.

53.2.8 DMA for Preemptive Conversion Mode

The ADC preemptive conversion mode supports built-in DMA, which can directly write conversion results to the buffer specified by the user in memory.

Users can configure the target address through the TRG_DMA_ADDR register. Using this address as the base address, the ADC will occupy a 192-byte area as the buffer for preemptive conversion results. The buffer allocation is shown in Table 212:

Serial No.	Address	Description
0	TRG_DMA_ADDR + 0x0	First AD conversion result and additional information triggered by PTRGI0A
1	TRG_DMA_ADDR + 0x4	Second AD conversion result and additional information triggered by PTRGI0A
2	TRG_DMA_ADDR + 0x8	Third AD conversion result and additional information triggered by PTRGI0A
3	TRG_DMA_ADDR + 0xC	Fourth AD conversion result and additional information triggered by PTRGI0A
4	TRG_DMA_ADDR + 0x10	First AD conversion result and additional information triggered by PTRGI0B
5	TRG_DMA_ADDR + 0x14	Second AD conversion result and additional information triggered by PTRGI0B
6	TRG_DMA_ADDR + 0x18	Third AD conversion result and additional information triggered by PTRGI0B
7	TRG_DMA_ADDR + 0x1C	Fourth AD conversion result and additional information triggered by PTRGI0B
8	TRG_DMA_ADDR + 0x20	First AD conversion result and additional information triggered by PTRGI0C
9	TRG_DMA_ADDR + 0x24	Second AD conversion result and additional information triggered by PTRGI0C
10	TRG_DMA_ADDR + 0x28	Third AD conversion result and additional information triggered by PTRGI0C
11	TRG_DMA_ADDR + 0x2C	Fourth AD conversion result and additional information triggered by PTRGI0C

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Serial No.	Address	Description
12	TRG_DMA_ADDR + 0x30	First AD conversion result and additional information triggered by PTRGI1A
13	TRG_DMA_ADDR + 0x34	Second AD conversion result and additional information triggered by PTRGI1A
14	TRG_DMA_ADDR + 0x38	Third AD conversion result and additional information triggered by PTRGI1A
15	TRG_DMA_ADDR + 0x3C	Fourth AD conversion result and additional information triggered by PTRGI1A
16	TRG_DMA_ADDR + 0x40	First AD conversion result and additional information triggered by PTRGI1B
17	TRG_DMA_ADDR + 0x44	Second AD conversion result and additional information triggered by PTRGI1B
18	TRG_DMA_ADDR + 0x48	Third AD conversion result and additional information triggered by PTRGI1B
19	TRG_DMA_ADDR + 0x4C	Fourth AD conversion result and additional information triggered by PTRGI1B
20	TRG_DMA_ADDR + 0x50	First AD conversion result and additional information triggered by PTRGI1C
21	TRG_DMA_ADDR + 0x54	Second AD conversion result and additional information triggered by PTRGI1C
22	TRG_DMA_ADDR + 0x58	Third AD conversion result and additional information triggered by PTRGI1C
23	TRG_DMA_ADDR + 0x5C	Fourth AD conversion result and additional information triggered by PTRGI1C
24	TRG_DMA_ADDR + 0x60	First AD conversion result and additional information triggered by PTRGI2A
25	TRG_DMA_ADDR + 0x64	Second AD conversion result and additional information triggered by PTRGI2A
26	TRG_DMA_ADDR + 0x68	Third AD conversion result and additional information triggered by PTRGI2A
27	TRG_DMA_ADDR + 0x6C	Fourth AD conversion result and additional information triggered by PTRGI2A
28	TRG_DMA_ADDR + 0x70	First AD conversion result and additional information triggered by PTRGI2B
29	TRG_DMA_ADDR + 0x74	Second AD conversion result and additional information triggered by PTRGI2B
30	TRG_DMA_ADDR + 0x78	Third AD conversion result and additional information triggered by PTRGI2B

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Serial No.	Address	Description
31	TRG_DMA_ADDR + 0x7C	Fourth AD conversion result and additional information triggered by PTRGI2B
32	TRG_DMA_ADDR + 0x80	First AD conversion result and additional information triggered by PTRGI2C
33	TRG_DMA_ADDR + 0x84	Second AD conversion result and additional information triggered by PTRGI2C
34	TRG_DMA_ADDR + 0x88	Third AD conversion result and additional information triggered by PTRGI2C
35	TRG_DMA_ADDR + 0x8C	Fourth AD conversion result and additional information triggered by PTRGI2C
36	TRG_DMA_ADDR + 0x90	First AD conversion result and additional information triggered by PTRGI3A
37	TRG_DMA_ADDR + 0x94	Second AD conversion result and additional information triggered by PTRGI3A
38	TRG_DMA_ADDR + 0x98	Third AD conversion result and additional information triggered by PTRGI3A
39	TRG_DMA_ADDR + 0x9C	Fourth AD conversion result and additional information triggered by PTRGI3A
40	TRG_DMA_ADDR + 0xA0	First AD conversion result and additional information triggered by PTRGI3B
41	TRG_DMA_ADDR + 0xA4	Second AD conversion result and additional information triggered by PTRGI3B
42	TRG_DMA_ADDR + 0xA8	Third AD conversion result and additional information triggered by PTRGI3B
43	TRG_DMA_ADDR + 0xAC	Fourth AD conversion result and additional information triggered by PTRGI3B
44	TRG_DMA_ADDR + 0xB0	First AD conversion result and additional information triggered by PTRGI3C
45	TRG_DMA_ADDR + 0xB4	Second AD conversion result and additional information triggered by PTRGI3C
46	TRG_DMA_ADDR + 0xB8	Third AD conversion result and additional information triggered by PTRGI3C
47	TRG_DMA_ADDR + 0xBC	Fourth AD conversion result and additional information triggered by PTRGI3C

Table 212: ADC Preemptive Conversion DMA Data Buffer Allocation

After the preemptive conversion is triggered, DMA will write the conversion result to the corresponding position in the buffer according to the trigger source and the serial number in the preemptive conversion sequence.

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Note: DMA writes 32-bit (4-byte) data to memory each time, where the 16-bit AD conversion result is located in [15:0]. Bits [31:16] also include flag data such as cycle, sequence number, and AD input channel number, which facilitates software interpretation of AD conversion information.

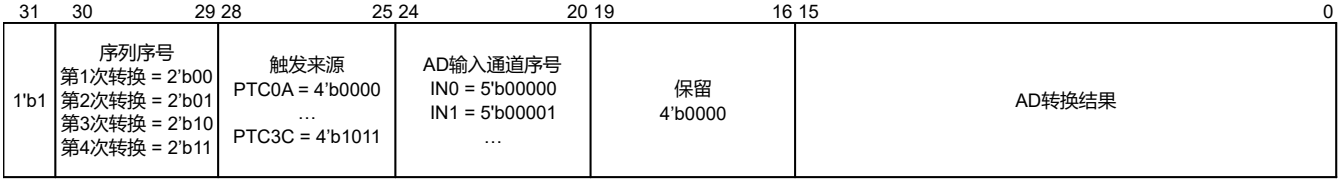


Figure 90: Data Format of Preemptive Conversion Mode

Bit 31 is used for interaction with the software. The software can clear this bit to 0 when initializing the buffer. When reading the result after receiving the conversion completion interrupt, it needs to check that this bit is high to ensure that the ADC has written the new result into the buffer.

53.2.9 ADC Interrupts

The ADC supports the following interrupts:

- In sequence conversion mode, when a certain AD conversion in the sequence is completed, the INT_STS [SEQ_CVC] bit is set to 1
- In sequence conversion mode, when the entire sequence conversion is completed, the INT_STS [SEQ_CMPT] bit is set to 1
- When a software trigger for sequence conversion occurs during ongoing ADC sequence conversion, the INT_STS [SEQ_SW_CFLCT] bit is set to 1
- When a hardware trigger for sequence conversion occurs during ongoing ADC sequence conversion, the INT_STS [SEQ_HW_CFLCT] bit is set to 1
- In sequence conversion mode, when SEQ_DMA_CFG [STOP_EN] bit is set to 1 and SEQ_DMA_CFG [STOP_POS] bit is set, DMA will stop writing data to the buffer after the pointer reaches SEQ_DMA_ADDR + STOP_POS. The results of this and subsequent AD conversions will be lost, and the flag bit INT_STS [SEQ_DMAABT] will be set to 1
- In preemptive conversion mode, after the xth conversion of the preemptive sequence is completed, if the [INTENx] bit of the CONFIGx register is set to 1, the INT_STS [TRIG_CMPT] flag bit will be set to 1
- When a software trigger for preemptive conversion occurs during ongoing ADC preemptive conversion, the INT_STS [TRIG_SW_CFLICT] bit is set to 1
- When a new preemptive trigger signal is received during ongoing ADC preemptive conversion, or multiple preemptive trigger signals arrive at the same time, the INT_STS [TRIG_HW_CFLICT] bit is set to 1
- When ADC read conversion is in progress and BUF_CFG0 [WAITDIS] bit has been set to 1, if the software reads another BUS_RESULT register, the INT_STS [READ_CFLCT] bit is set to 1
- When the AD conversion result of input channel x exceeds the range set by the THSHDx register, an alarm will be triggered and the INT_STS [WDOG] bit is set to 1
- When the DMA FIFO for sequence conversion and preemptive conversion overflows, the INT_STS [DMA_FIFO_FULL] bit is set to 1
- When DMA for sequence conversion and preemptive conversion fails to write ADC conversion results to memory, the INT_STS [AHB_ERR] bit is set to 1

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Users can clear each flag bit in the INT_STS register by writing 1 to it.

The ADC can manage these interrupts through the INT_EN register. If the corresponding interrupt enable bit is set to 1, an interrupt is generated when the flag bit is set to 1.

53.3 ADC16 Register List

The register list of ADC16 is as follows:

ADC0 base address: 0xF3080000

ADC1 base address: 0xF3084000

Offset	Register Name	Description	Reset Value
0x0000	CONFIG[TRG0A]		0x00000000
0x0004	CONFIG[TRG0B]		0x00000000
0x0008	CONFIG[TRG0C]		0x00000000
0x000C	CONFIG[TRG1A]		0x00000000
0x0010	CONFIG[TRG1B]		0x00000000
0x0014	CONFIG[TRG1C]		0x00000000
0x0018	CONFIG[TRG2A]		0x00000000
0x001C	CONFIG[TRG2B]		0x00000000
0x0020	CONFIG[TRG2C]		0x00000000
0x0024	CONFIG[TRG3A]		0x00000000
0x0028	CONFIG[TRG3B]		0x00000000
0x002C	CONFIG[TRG3C]		0x00000000
0x0030	TRG_DMA_ADDR		0x00000000
0x0034	TRG_SW_STA		0x00000000
0x0400	BUS_RESULT[CHN0]		0x00000000
0x0404	BUS_RESULT[CHN1]		0x00000000
0x0408	BUS_RESULT[CHN2]		0x00000000
0x040C	BUS_RESULT[CHN3]		0x00000000
0x0410	BUS_RESULT[CHN4]		0x00000000
0x0414	BUS_RESULT[CHN5]		0x00000000
0x0418	BUS_RESULT[CHN6]		0x00000000
0x041C	BUS_RESULT[CHN7]		0x00000000
0x0420	BUS_RESULT[CHN8]		0x00000000
0x0424	BUS_RESULT[CHN9]		0x00000000
0x0428	BUS_RESULT[CHN10]		0x00000000
0x042C	BUS_RESULT[CHN11]		0x00000000
0x0430	BUS_RESULT[CHN12]		0x00000000
0x0434	BUS_RESULT[CHN13]		0x00000000
0x0438	BUS_RESULT[CHN14]		0x00000000
0x043C	BUS_RESULT[CHN15]		0x00000000
0x0500	BUF_CFG0		0x00000000

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Offset	Register Name	Description	Reset Value
0x0800	SEQ_CFG0		0x00000000
0x0804	SEQ_DMA_ADDR		0x00000000
0x0808	SEQ_WR_ADDR		0x00000000
0x080C	SEQ_DMA_CFG		0x00000000
0x0810	SEQ_QUE[CFG0]		0x00000000
0x0814	SEQ_QUE[CFG1]		0x00000000
0x0818	SEQ_QUE[CFG2]		0x00000000
0x081C	SEQ_QUE[CFG3]		0x00000000
0x0820	SEQ_QUE[CFG4]		0x00000000
0x0824	SEQ_QUE[CFG5]		0x00000000
0x0828	SEQ_QUE[CFG6]		0x00000000
0x082C	SEQ_QUE[CFG7]		0x00000000
0x0830	SEQ_QUE[CFG8]		0x00000000
0x0834	SEQ_QUE[CFG9]		0x00000000
0x0838	SEQ_QUE[CFG10]		0x00000000
0x083C	SEQ_QUE[CFG11]		0x00000000
0x0840	SEQ_QUE[CFG12]		0x00000000
0x0844	SEQ_QUE[CFG13]		0x00000000
0x0848	SEQ_QUE[CFG14]		0x00000000
0x084C	SEQ_QUE[CFG15]		0x00000000
0x0850	SEQ_HIGH_CFG		0x00000000
0x0C00	PRD_CFG[CHN0][PRD_CFG]		0x00000000
0x0C04	PRD_CFG[CHN0][PRD_THSHD_CFG]		0x00000000
0x0C08	PRD_CFG[CHN0][PRD_RESULT]		0x00000000
0x0C10	PRD_CFG[CHN1][PRD_CFG]		0x00000000
0x0C14	PRD_CFG[CHN1][PRD_THSHD_CFG]		0x00000000
0x0C18	PRD_CFG[CHN1][PRD_RESULT]		0x00000000
0x0C20	PRD_CFG[CHN2][PRD_CFG]		0x00000000
0x0C24	PRD_CFG[CHN2][PRD_THSHD_CFG]		0x00000000
0x0C28	PRD_CFG[CHN2][PRD_RESULT]		0x00000000
0x0C30	PRD_CFG[CHN3][PRD_CFG]		0x00000000
0x0C34	PRD_CFG[CHN3][PRD_THSHD_CFG]		0x00000000
0x0C38	PRD_CFG[CHN3][PRD_RESULT]		0x00000000
0x0C40	PRD_CFG[CHN4][PRD_CFG]		0x00000000
0x0C44	PRD_CFG[CHN4][PRD_THSHD_CFG]		0x00000000
0x0C48	PRD_CFG[CHN4][PRD_RESULT]		0x00000000
0x0C50	PRD_CFG[CHN5][PRD_CFG]		0x00000000
0x0C54	PRD_CFG[CHN5][PRD_THSHD_CFG]		0x00000000
0x0C58	PRD_CFG[CHN5][PRD_RESULT]		0x00000000
0x0C60	PRD_CFG[CHN6][PRD_CFG]		0x00000000

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Offset	Register Name	Description	Reset Value
0x0C64	PRD_CFG[CHN6][PRD_THSHD_CFG]		0x00000000
0x0C68	PRD_CFG[CHN6][PRD_RESULT]		0x00000000
0x0C70	PRD_CFG[CHN7][PRD_CFG]		0x00000000
0x0C74	PRD_CFG[CHN7][PRD_THSHD_CFG]		0x00000000
0x0C78	PRD_CFG[CHN7][PRD_RESULT]		0x00000000
0x0C80	PRD_CFG[CHN8][PRD_CFG]		0x00000000
0x0C84	PRD_CFG[CHN8][PRD_THSHD_CFG]		0x00000000
0x0C88	PRD_CFG[CHN8][PRD_RESULT]		0x00000000
0x0C90	PRD_CFG[CHN9][PRD_CFG]		0x00000000
0x0C94	PRD_CFG[CHN9][PRD_THSHD_CFG]		0x00000000
0x0C98	PRD_CFG[CHN9][PRD_RESULT]		0x00000000
0x0CA0	PRD_CFG[CHN10][PRD_CFG]		0x00000000
0x0CA4	PRD_CFG[CHN10][PRD_THSHD_CFG]		0x00000000
0x0CA8	PRD_CFG[CHN10][PRD_RESULT]		0x00000000
0x0CB0	PRD_CFG[CHN11][PRD_CFG]		0x00000000
0x0CB4	PRD_CFG[CHN11][PRD_THSHD_CFG]		0x00000000
0x0CB8	PRD_CFG[CHN11][PRD_RESULT]		0x00000000
0x0CC0	PRD_CFG[CHN12][PRD_CFG]		0x00000000
0x0CC4	PRD_CFG[CHN12][PRD_THSHD_CFG]		0x00000000
0x0CC8	PRD_CFG[CHN12][PRD_RESULT]		0x00000000
0x0CD0	PRD_CFG[CHN13][PRD_CFG]		0x00000000
0x0CD4	PRD_CFG[CHN13][PRD_THSHD_CFG]		0x00000000
0x0CD8	PRD_CFG[CHN13][PRD_RESULT]		0x00000000
0x0CE0	PRD_CFG[CHN14][PRD_CFG]		0x00000000
0x0CE4	PRD_CFG[CHN14][PRD_THSHD_CFG]		0x00000000
0x0CE8	PRD_CFG[CHN14][PRD_RESULT]		0x00000000
0x0CF0	PRD_CFG[CHN15][PRD_CFG]		0x00000000
0x0CF4	PRD_CFG[CHN15][PRD_THSHD_CFG]		0x00000000
0x0CF8	PRD_CFG[CHN15][PRD_RESULT]		0x00000000
0x1000	SAMPLE_CFG[CHN0]		0x00000001
0x1004	SAMPLE_CFG[CHN1]		0x00000001
0x1008	SAMPLE_CFG[CHN2]		0x00000001
0x100C	SAMPLE_CFG[CHN3]		0x00000001
0x1010	SAMPLE_CFG[CHN4]		0x00000001
0x1014	SAMPLE_CFG[CHN5]		0x00000001
0x1018	SAMPLE_CFG[CHN6]		0x00000001
0x101C	SAMPLE_CFG[CHN7]		0x00000001
0x1020	SAMPLE_CFG[CHN8]		0x00000001
0x1024	SAMPLE_CFG[CHN9]		0x00000001
0x1028	SAMPLE_CFG[CHN10]		0x00000001

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Offset	Register Name	Description	Reset Value
0x102C	SAMPLE_CFG[CHN11]		0x00000001
0x1030	SAMPLE_CFG[CHN12]		0x00000001
0x1034	SAMPLE_CFG[CHN13]		0x00000001
0x1038	SAMPLE_CFG[CHN14]		0x00000001
0x103C	SAMPLE_CFG[CHN15]		0x00000001
0x1104	CONV_CFG1		0x00000000
0x1108	ADC_CFG0		0x00000000
0x1110	INT_STS		0x00000000
0x1114	INT_EN		0x00000000
0x1200	ANA_CTRL0		0x00000000
0x1210	ANA_STATUS		0x00000000
0x1400	ADC16_PARAMS[ADC16_PARA00]		0x0000
0x1402	ADC16_PARAMS[ADC16_PARA01]		0x0000
0x1404	ADC16_PARAMS[ADC16_PARA02]		0x0000
0x1406	ADC16_PARAMS[ADC16_PARA03]		0x0000
0x1408	ADC16_PARAMS[ADC16_PARA04]		0x0000
0x140A	ADC16_PARAMS[ADC16_PARA05]		0x0000
0x140C	ADC16_PARAMS[ADC16_PARA06]		0x0000
0x140E	ADC16_PARAMS[ADC16_PARA07]		0x0000
0x1410	ADC16_PARAMS[ADC16_PARA08]		0x0000
0x1412	ADC16_PARAMS[ADC16_PARA09]		0x0000
0x1414	ADC16_PARAMS[ADC16_PARA10]		0x0000
0x1416	ADC16_PARAMS[ADC16_PARA11]		0x0000
0x1418	ADC16_PARAMS[ADC16_PARA12]		0x0000
0x141A	ADC16_PARAMS[ADC16_PARA13]		0x0000
0x141C	ADC16_PARAMS[ADC16_PARA14]		0x0000
0x141E	ADC16_PARAMS[ADC16_PARA15]		0x0000
0x1420	ADC16_PARAMS[ADC16_PARA16]		0x0000
0x1422	ADC16_PARAMS[ADC16_PARA17]		0x0000
0x1424	ADC16_PARAMS[ADC16_PARA18]		0x0000
0x1426	ADC16_PARAMS[ADC16_PARA19]		0x0000
0x1428	ADC16_PARAMS[ADC16_PARA20]		0x0000
0x142A	ADC16_PARAMS[ADC16_PARA21]		0x0000
0x142C	ADC16_PARAMS[ADC16_PARA22]		0x0000
0x142E	ADC16_PARAMS[ADC16_PARA23]		0x0000
0x1430	ADC16_PARAMS[ADC16_PARA24]		0x0000
0x1432	ADC16_PARAMS[ADC16_PARA25]		0x0000
0x1434	ADC16_PARAMS[ADC16_PARA26]		0x0000
0x1436	ADC16_PARAMS[ADC16_PARA27]		0x0000
0x1438	ADC16_PARAMS[ADC16_PARA28]		0x0000

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Offset	Register Name	Description	Reset Value
0x143A	ADC16_PARAMS[ADC16_PARA29]		0x0000
0x143C	ADC16_PARAMS[ADC16_PARA30]		0x0000
0x143E	ADC16_PARAMS[ADC16_PARA31]		0x0000
0x1440	ADC16_PARAMS[ADC16_PARA32]		0x0000
0x1442	ADC16_PARAMS[ADC16_PARA33]		0x0000
0x1444	ADC16_CONFIG0		0x00000000
0x1460	ADC16_CONFIG1		0x00000100

Table 213: ADC16 Register List

53.4 ADC16 Register Description

The detailed description of ADC16 registers is as follows:

53.4.1 CONFIG (0x0 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TRIG_LEN		INTEN3	CHAN3					RSVD		INTEN2	CHAN2					RSVD		INTEN1	CHAN1					RSVD	QUEUE_EN	INTEN0	CHAN0				
WO		RW	RW					N/A		RW	RW					N/A		RW	RW					N/A	RW	RW	RW				
0	0	0	0	0	0	0	0	x	x	0	0	0	0	0	0	x	x	0	0	0	0	0	0	x	0	0	0	0	0	0	0

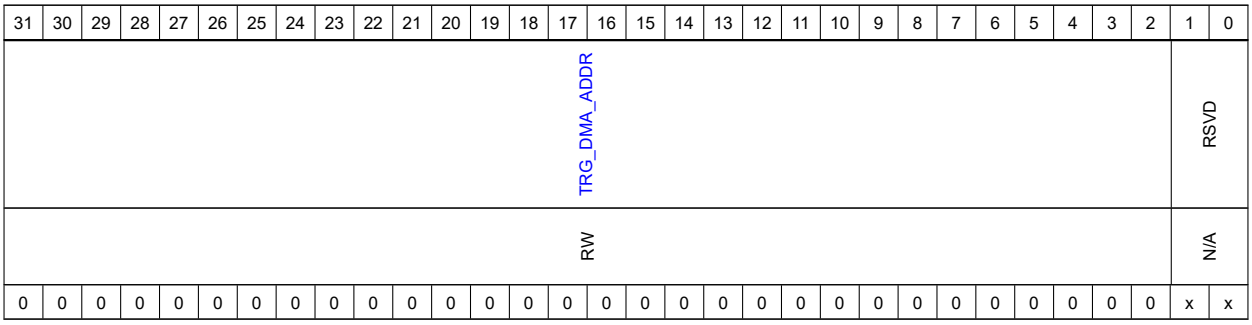
CONFIG [31:0]

Bits	Name	Description
31-30	TRIG_LEN	length for current trigger, can up to 4 conversions for one trigger, from 0 to 3
29	INTEN3	interrupt enable for 4th conversion
28-24	CHAN3	channel number for 4th conversion
21	INTEN2	interrupt enable for 3rd conversion
20-16	CHAN2	channel number for 3rd conversion
13	INTEN1	interrupt enable for 2nd conversion
12-8	CHAN1	channel number for 2nd conversion
6	QUEUE_EN	preemption queue enable control
5	INTEN0	interrupt enable for 1st conversion
4-0	CHAN0	channel number for 1st conversion

CONFIG Bits

53.4.2 TRG_DMA_ADDR (0x30)

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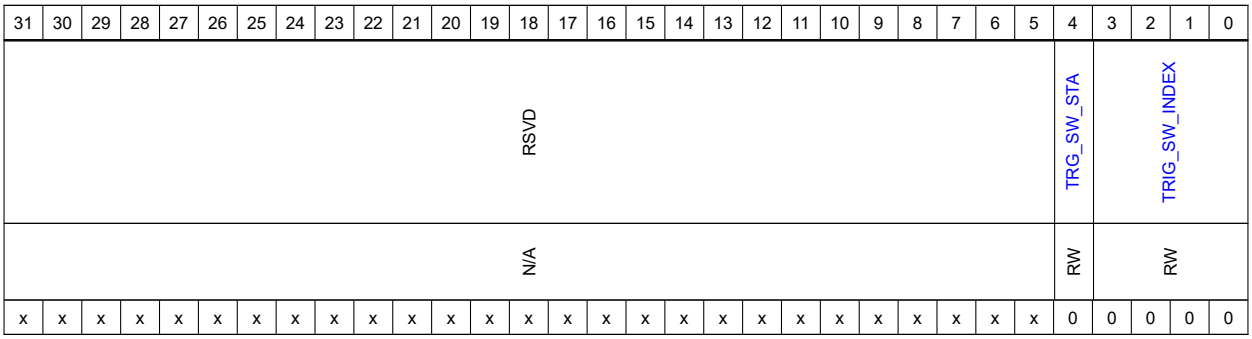


TRG_DMA_ADDR [31:0]

Bits	Name	Description
31-2	TRG_DMA_ADDR	buffer start address for trigger queue, 192byte total, 16 bytes for each trigger (4 bytes for each conversion)

TRG_DMA_ADDR Bits

53.4.3 TRG_SW_STA (0x34)

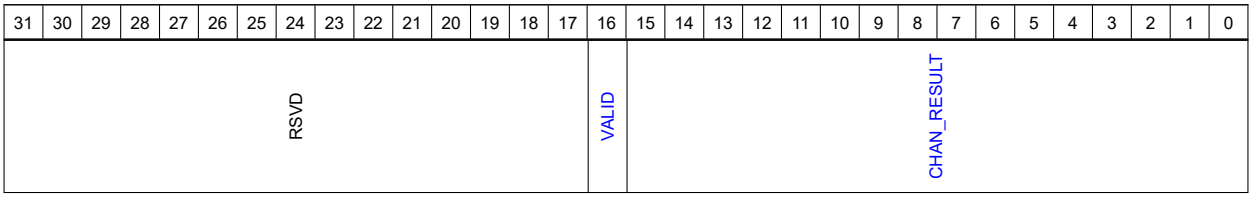


TRG_SW_STA [31:0]

Bits	Name	Description
4	TRG_SW_STA	SW trigger start bit, HW will clear it after all conversions(up to 4) finished. SW should make sure it's 0 before set it.
3-0	TRIG_SW_INDEX	which trigger for the SW trigger 0 for trig0a, 1 for trig0b… 3 for trig1a, …11 for trig3c

TRG_SW_STA Bits

53.4.4 BUS_RESULT (0x400 + 0x4 * n)



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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
N/A															RO	RO															
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

BUS_RESULT [31:0]

Bits	Name	Description
16	VALID	set after conversion finished if wait_dis is set, cleared after software read. The first time read with 0 will trigger one new conversion. If SW read other channel when one channel conversion is in progress, it will not trigger new conversion at other channel, and will get old result with valid 0, also with read_cflct interrupt status bit set. the result may not realtime if software read once and wait long time to read again
15-0	CHAN_RESULT	read this register will trigger one adc conversion. If wait_dis bit is set, SW will get the latest conversion result(not current one) with valid bit is 0, SW need polling valid bit till it's set to get current result If wait_dis bit is 0, SW can get the current conversion result with holding the bus, valid bit is always set at this mode. this is not recommended if channel sample time is too long

BUS_RESULT Bits

53.4.5 BUF_CFG0 (0x500)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																														BUS_MODE_EN	WAIT_DIS
N/A																														RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0

BUF_CFG0 [31:0]

Bits	Name	Description
1	BUS_MODE_EN	bus mode enable
0	WAIT_DIS	set to disable read waiting, get result immediately but maybe not current conversion result.

BUF_CFG0 Bits

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53.4.6 SEQ_CFG0 (0x800)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CYCLE	RSVD																			SEQ_LEN				RSVD			RESTART_EN	CONT_EN	SW_TRIG	SW_TRIG_EN	HW_TRIG_EN
RO	N/A																			RW				N/A			RW	RW	WO	RW	RW
0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	x	x	x	0	0	0	0	0

SEQ_CFG0 [31:0]

Bits	Name	Description
31	CYCLE	current dma write cycle bit
11-8	SEQ_LEN	sequence queue length, 0 for one, 0xF for 16
4	RESTART_EN	if set together with cont_en, HW will continue process the whole queue after trigger once. If cont_en is 0, this bit is not used
3	CONT_EN	if set, HW will continue process the queue till end(seq_len) after trigger once
2	SW_TRIG	SW trigger, pulse signal, cleared by HW one cycle later
1	SW_TRIG_EN	set to enable SW trigger
0	HW_TRIG_EN	set to enable external HW trigger, only trigger on posedge

SEQ_CFG0 Bits

53.4.7 SEQ_DMA_ADDR (0x804)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TAR_ADDR																														RSVD	
RW																														N/A	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x

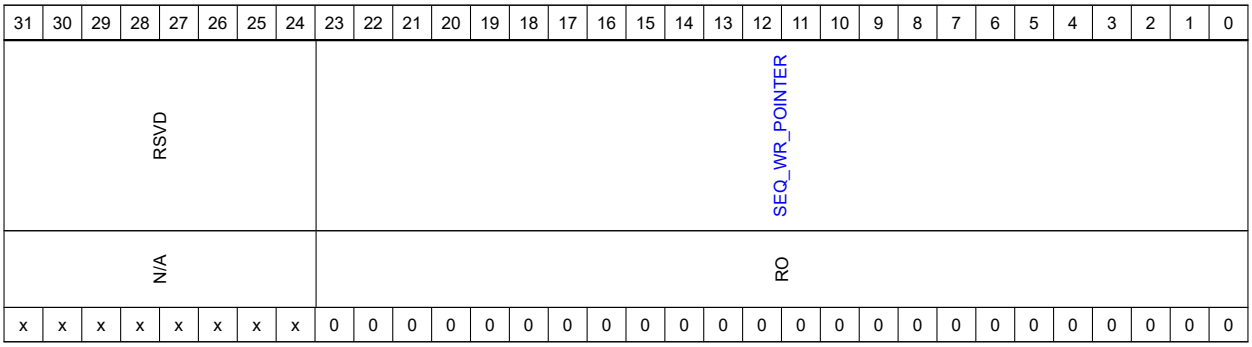
SEQ_DMA_ADDR [31:0]

Bits	Name	Description
31-2	TAR_ADDR	dma target address, should be 4-byte aligned

SEQ_DMA_ADDR Bits

53.4.8 SEQ_WR_ADDR (0x808)

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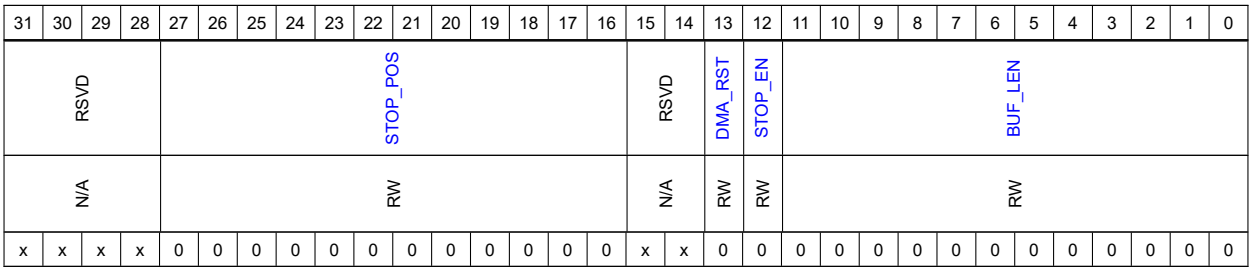


SEQ_WR_ADDR [31:0]

Bits	Name	Description
23-0	SEQ_WR_POINTER	HW update this field after each dma write, it indicate the next dma write pointer. dma write address is (tar_addr+seq_wr_pointer)*4

SEQ_WR_ADDR Bits

53.4.9 SEQ_DMA_CFG (0x80C)



SEQ_DMA_CFG [31:0]

Bits	Name	Description
27-16	STOP_POS	if stop_en is set, SW is responsible to update this field to the next read point, HW should not write data to this point since it's not read out by SW yet
13	DMA_RST	set this bit will reset HW dma write pointer to seq_dma_addr, and set HW cycle bit to 1. dma is halted if this bit is set. SW should clear all cycle bit in buffer to 0 before clear dma_rst
12	STOP_EN	set to stop dma if reach the stop_pos
11-0	BUF_LEN	dma buffer length, after write to (tar_addr[31:2]+buf_len)*4, the next dma address will be tar_addr[31:2]*4 0 for 4byte; 0xFFF for 16kbyte.

SEQ_DMA_CFG Bits

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53.4.10 SEQ_QUE (0x810 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																										SEQ_INT_EN	CHAN_NUM_4_0				
N/A																										RW	RW				
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0

SEQ_QUE [31:0]

Bits	Name	Description
5	SEQ_INT_EN	interrupt enable for current conversion
4-0	CHAN_NUM_4_0	channel number for current conversion

SEQ_QUE Bits

53.4.11 SEQ_HIGH_CFG (0x850)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD								STOP_POS_HIGH												BUF_LEN_HIGH											
N/A								RW												RW											
x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

SEQ_HIGH_CFG [31:0]

Bits	Name	Description
23-12	STOP_POS_HIGH	
11-0	BUF_LEN_HIGH	

SEQ_HIGH_CFG Bits

53.4.12 PRD_CFG[PRD_CFG] (0xC00 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																				PRESCALE				PRD							
N/A																				RW				RW							
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

PRD_CFG[PRD_CFG] [31:0]

Bits	Name	Description
12-8	PRESCALE	0: 1xclock, 1: 2x, 2: 4x, 3: 8x,⋯,15: 32768x,⋯,31: 2Gx
7-0	PRD	conver period, with prescale. Set to 0 means disable current channel

PRD_CFG[PRD_CFG] Bits

53.4.13 PRD_CFG[PRD_THSHD_CFG] (0xC04 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
THSHDH																THSHDL															
RW																RW															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PRD_CFG[PRD_THSHD_CFG] [31:0]

Bits	Name	Description
31-16	THSHDH	threshold high, assert interrupt(if enabled) if result exceed high or low.
15-0	THSHDL	threshold low

PRD_CFG[PRD_THSHD_CFG] Bits

53.4.14 PRD_CFG[PRD_RESULT] (0xC08 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																CHAN_RESULT															
N/A																RO															
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PRD_CFG[PRD_RESULT] [31:0]

Bits	Name	Description
15-0	CHAN_RESULT	adc convert result, update after each valid conversion. it may be updated period according to config, also may be updated due to other queue convert the same channel

PRD_CFG[PRD_RESULT] Bits

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53.4.15 SAMPLE_CFG (0x1000 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																				SAMPLE_CLOCK_NUMBER_SHIFT			SAMPLE_CLOCK_NUMBER						
N/A																				RW			RW								
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	1

SAMPLE_CFG [31:0]

Bits	Name	Description
11-9	SAMPLE_CLOCK_NUMBER_SHIFT	shift for sample clock number
8-0	SAMPLE_CLOCK_NUMBER	sample clock number, base on clock_period, default one period

SAMPLE_CFG Bits

53.4.16 CONV_CFG1 (0x1104)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																									CONVERT_CLOCK_NUMBER				CLOCK_DIVIDER			
N/A																							RW					RW			
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0

CONV_CFG1 [31:0]

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Bits	Name	Description
8-4	CONVERT_CLOCK_NUMBER	convert clock numbers, set to 21 (0x15) for 16bit mode, which means convert need 21 adc clock cycles(based on clock after divider); user can use small value to get faster conversion, but less accuracy, need to config cov_end_cnt at adc16_config1 also. Ex: use 200MHz bus clock for adc, set sample_clock_number to 4, sample_clock_number_shift to 0, covert_clk_number to 21 for 16bit mode, clock_divder to 3, then each ADC conversion(plus sample) need 25 cycles(50MHz).
3-0	CLOCK_DIVIDER	clock_period, N half clock cycle per half adc cycle 0 for same adc_clk and bus_clk, 1 for 1:2, 2 for 1:3, ... 15 for 1:16 Note: set to 2 can genenerate 66.7MHz adc_clk at 200MHz bus_clk

CONV_CFG1 Bits

53.4.17 ADC_CFG0 (0x1108)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SEL_SYNC_AHB	RSVD	ADC_AHB_EN	RSVD																												PORT3_REALTIME
RW	N/A	RW	N/A																												RW
0	x	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

ADC_CFG0 [31:0]

Bits	Name	Description
31	SEL_SYNC_AHB	set to 1 will enable sync AHB bus, to get better bus performance. Adc_clk must to be set to same as bus clock at this mode
29	ADC_AHB_EN	set to 1 to enable ADC DMA to write data to soc memory bus, for trig queue and seq queue;
0	PORT3_REALTIME	set to enable trg queue stop other queues

ADC_CFG0 Bits

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53.4.18 INT_STS (0x1110)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TRIG_CMPT	TRIG_SW_CFLCT	TRIG_HW_CFLCT	READ_CFLCT	SEQ_SW_CFLCT	SEQ_HW_CFLCT	SEQ_DMAABT	SEQ_CMPT	SEQ_CVC	DMA_FIFO_FULL	AHB_ERR	RSVD					WDOG															
RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	N/A					RW1C															
0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

INT_STS [31:0]

Bits	Name	Description
31	TRIG_CMPT	interrupt for one trigger conversion complete if enabled
30	TRIG_SW_CFLCT	
29	TRIG_HW_CFLCT	
28	READ_CFLCT	read conflict interrupt, set if wait_dis is set, one conversion is in progress, SW read another channel
27	SEQ_SW_CFLCT	sequence queue conflict interrupt, set if HW or SW trigger received during conversion
26	SEQ_HW_CFLCT	
25	SEQ_DMAABT	dma abort interrupt, set if seqence dma write pointer reachs sw read pointer if stop_en is set
24	SEQ_CMPT	the whole sequence complete interrupt
23	SEQ_CVC	one conversion complete in seq_queue if related seq_int_en is set
22	DMA_FIFO_FULL	DMA fifo full interrupt, user need to check clock frequency if it's set.
21	AHB_ERR	set if got hresp=1, generally caused by wrong trg_dma_addr or seq_dma_addr
15-0	WDOG	set if one chanel watch dog event triggered

INT_STS Bits

53.4.19 INT_EN (0x1114)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TRIG_CMPT	TRIG_SW_CFLCT	TRIG_HW_CFLCT	READ_CFLCT	SEQ_SW_CFLCT	SEQ_HW_CFLCT	SEQ_DMAABT	SEQ_CMPT	SEQ_CVC	DMA_FIFO_FULL	AHB_ERR	RSVD					WDOG															
RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	RW	N/A					RW															

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

INT_EN [31:0]

Bits	Name	Description
31	TRIG_CMPT	interrupt for one trigger conversion complete if enabled
30	TRIG_SW_CFLCT	
29	TRIG_HW_CFLCT	
28	READ_CFLCT	read conflict interrupt, set if wait_dis is set, one conversion is in progress, SW read another channel
27	SEQ_SW_CFLCT	sequence queue conflict interrupt, set if HW or SW trigger received during conversion
26	SEQ_HW_CFLCT	
25	SEQ_DMAABT	dma abort interrupt, set if seqence dma write pointer reachs sw read pointer if stop_en is set
24	SEQ_CMPT	the whole sequence complete interrupt
23	SEQ_CVC	one conversion complete in seq_queue if related seq_int_en is set
22	DMA_FIFO_FULL	DMA fifo full interrupt, user need to check clock frequency if it's set.
21	AHB_ERR	set if got hresp=1, generally caused by wrong trg_dma_addr or seq_dma_addr
15-0	WDOG	set if one chanel watch dog event triggered

INT_EN Bits

53.4.20 ANA_CTRL0 (0x1200)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MOTO_EN	RSVD																		ADC_CLK_ON	RSVD									STARTCAL	RSVD	
RW	N/A																		RW	N/A									RW	N/A	
0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	x	x	x	x	x	x	0	x	x

ANA_CTRL0 [31:0]

Bits	Name	Description
31	MOTO_EN	"set to enable moto_soc and moto_valid. Should use AHB clock for adc, this bit can be used avoid async output"

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Bits	Name	Description
12	ADC_CLK_ON	set to enable adc clock to analog, Software should set this bit before access to any adc16_* register. MUST set clock_period to 0 or 1 for adc16 reg access
2	STARTCAL	set to start the offset calibration cycle (Active H). user need to clear it after setting it.

ANA_CTRL0 Bits

53.4.21 ANA_STATUS (0x1210)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																								CALON	RSVD						
N/A																								RW	N/A						
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	x	x	x	x

ANA_STATUS [31:0]

Bits	Name	Description
7	CALON	Indicates if the ADC is in calibration mode (Active H).

ANA_STATUS Bits

53.4.22 ADC16_PARAMS (0x1400 + 0x2 * n)

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PARAM_VAL															
RW															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ADC16_PARAMS [15:0]

Bits	Name	Description
15-0	PARAM_VAL	

ADC16_PARAMS Bits

53.4.23 ADC16_CONFIG0 (0x1444)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD							REG_EN	BANDGAP_EN	CAL_AVG_CFG			RSVD						CONV_PARAM													
N/A							RW	RW	RW			N/A						RW													
x	x	x	x	x	x	x	0	0	0	0	0	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ADC16_CONFIG0 [31:0]

Bits	Name	Description
24	REG_EN	set to enable regulator
23	BANDGAP_EN	set to enable bandgap. user should set reg_en and bandgap_en before use adc16.
22-20	CAL_AVG_CFG	for average the calibration result. 0- 1 loop; 1- 2 loops; 2- 4 loops; 3- 8 loops; 4- 16 loops; 5-32 loops; others reserved
13-0	CONV_PARAM	conversion parameter

ADC16_CONFIG0 Bits

53.4.24 ADC16_CONFIG1 (0x1460)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																			COV_END_CNT					RSVD							
N/A																			RW					N/A							
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	1	x	x	x	x	x	x	x	x

ADC16_CONFIG1 [31:0]

Bits	Name	Description
12-8	COV_END_CNT	used for faster conversion, user can change it to get higher convert speed(but less accuracy). should set to (21-convert_clock_number+1).

ADC16_CONFIG1 Bits

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54 Analog Comparator (ACMP)

This chapter describes the functions and features of the Analog Comparator (ACMP).

54.1 Feature Summary

The main features of the ACMP are as follows:

- Operating voltage from 3.0 V to 3.6 V
- Supports rail-to-rail input
- Programmable hysteresis
- Supports window mode output
- Supports digital filtering on output
- Supports interrupt generation
- Supports DMA
- Built-in 8-bit DAC

54.2 Function Description

This chapter describes the functions of the ACMP.

The comparator is used to compare the two analog input signals PIN and MIN. When the voltage of PIN is higher than that of MIN, the comparator outputs logic 1. Conversely, when the voltage of PIN is lower than that of MIN, the comparator outputs logic 0.

54.2.1 ACMP Input Configuration

Users can select the input source of PIN by configuring the CFG[PINSEL] bit of the comparator:

- INP0, permanently connected to DAC_OUT
- INP1
- INP2
- INP3
- INP4
- INP5
- INP6
- INP7

Users can select the input source of MIN by configuring the CFG[MINSEL] bit of the comparator:

- INN0, permanently connected to DAC_OUT
- INN1
- INN2
- INN3
- INN4
- INN5
- INN6
- INN7

Users can set the hysteresis of the comparator by configuring the CFG[HYST] bit. The comparator supports

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4 different levels of hysteresis.

54.2.2 ACMP Output Control

The output OUT of the ACMP reflects the voltage magnitude relationship between the two inputs. Users can also intervene the output to a certain extent through the comparator control logic.

Users can invert the output through the CFG[OPOL] bit.

Users can apply digital filtering to the comparator output. When CFG[FLTBYPS] is set to 0, the digital filter is enabled; when set to 1, the digital filter is disabled.

When the digital filter is enabled, users can specify the working mode of the filter by configuring CFG[FLTMODE]:

- 3'b000: Bypass mode, digital filter is turned off
- 3'b100: Glitch filtering mode. After the filter input flips, the output will also flip immediately, and then ignore the filter input for a certain period of time. In this mode, the filter output follows the input closely while avoiding glitches in the output signal.
- 3'b101: Delay filter mode. The filter input needs to remain flipped for a certain period of time before the filter output flips.
- 3'b110: Peak filtering mode. After the filter input is set to logic 0, it needs to remain for a certain period of time before the filter output is set to logic 0; when the filter input is set to 1, the filter output will be set to 1 immediately. This mode is designed to filter out input peaks that are not long enough.
- 3'b111: Valley filtering mode. After the filter input is set to logic 1, it needs to remain for a certain period of time before the filter output is set to logic 1; when the filter input is set to 0, the filter output will be set to 0 immediately. This mode is designed to filter out input valleys that are not long enough.

Users can configure the filter length through the CFG[FLTLEN] bit.

Users can enable the output window mode through the CFG[WINEN] bit. In window mode, that is, when CFG[WINEN] is set to 1, the comparator always outputs logic 0 when the WIN input of the ACMP is logic 0; when the WIN input is logic 1, the comparator outputs normally.

54.2.3 ACMP Working Modes

The ACMP supports two working modes: high-performance mode and normal mode. Users can set the ACMP to high-performance mode by setting the CFG[HPMODE] bit to 1. In high-performance mode, the output delay of the ACMP is smaller than that in normal mode, but the power consumption is higher.

54.2.4 Interrupt and DMA

The ACMP can generate interrupts on the rising edge, falling edge, or double edges of the output. Users can enable or disable the corresponding interrupt requests by configuring the IRQEN[REDGEN] and IRQEN[FEDGEN] bits.

Similarly, the ACMP can generate DMA requests on the rising edge, falling edge, or double edges of the output. Users can enable or disable the corresponding DMA requests by configuring the DMAEN[REDGEN] and IRQEN[FEDGEN] bits.

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54.3 ACMP Register List

The ACMP register list is as follows:

54.4 ACMP Register Description

The detailed description of ACMP registers is as follows:

54.4.1 CHANNEL[CFG] (0x0 + 0x20 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HYST		DACEN	HPMODE	CMPEN	MINSEL			DAC_TRIG_EN	PINSEL			CMPOEN	FLTBYPS	WINEN	OPOL	FLTMODE			RSVD	FLTLEN											
RW		RW	RW	RW	RW			RW	RW			RW	RW	RW	RW	RW			N/A	RW											
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	0	0	0	0	0	0	1	0	0	0	0	0

CHANNEL[CFG] [31:0]

Bits	Name	Description
31-30	HYST	This bitfield configure the comparator hysteresis. 0: Hysteresis about 30mV 1: Hysteresis about 20mV 2: Hysteresis about 10mV 3: Disable hysteresis
29	DACEN	This bit enable the comparator internal DAC 0: DAC disabled 1: DAC enabled
28	HPMODE	This bit enable the comparator high performance mode. 0: HP mode disabled 1: HP mode enabled
27	CMPEN	This bit enable the comparator. 0: ACMP disabled 1: ACMP enabled
26-24	MINSEL	PIN select, from pad_ai_acmp[7:1] and dac_out
23	DAC_TRIG_EN	if set, the dac value is from moto system when valid if clr, use dac_cfg value
22-20	PINSEL	MIN select, from pad_ai_acmp[7:1] and dac_out
19	CMPOEN	This bit enable the comparator output on pad. 0: ACMP output disabled 1: ACMP output enabled
18	FLTBYPS	This bit bypass the comparator output digital filter. 0: The ACMP output need pass digital filter 1: The ACMP output digital filter is bypassed.

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Bits	Name	Description
17	WINEN	This bit enable the comparator window mode. 0: Window mode is disabled 1: Window mode is enabled
16	OPOL	The output polarity control bit. 0: The ACMP output remain un-changed. 1: The ACMP output is inverted.
15-13	FLTMODE	This bitfield define the ACMP output digital filter mode: 000-bypass 100-change immediately; 101-change after filter; 110-stalbe low; 111-stable high
11-0	FLTLEN	This bitfield define the ACMP output digital filter length. The unit is ACMP clock cycle.

CHANNEL[CFG] Bits

54.4.2 CHANNEL[DACCFG] (0x4 + 0x20 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																								DACCFG							
N/A																								RW							
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0

CHANNEL[DACCFG] [31:0]

Bits	Name	Description
7-0	DACCFG	8bit DAC digital value

CHANNEL[DACCFG] Bits

54.4.3 CHANNEL[SR] (0x10 + 0x20 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																														FEDGF	REDGF
N/A																														RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0

CHANNEL[SR] [31:0]

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Bits	Name	Description
1	FEDGF	Output falling edge flag. Write 1 to clear this flag.
0	REDGF	Output rising edge flag. Write 1 to clear this flag.

CHANNEL[SR] Bits

54.4.4 CHANNEL[IRQEN] (0x14 + 0x20 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																														FEDGEN	REDGEN
N/A																														RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0

CHANNEL[IRQEN] [31:0]

Bits	Name	Description
1	FEDGEN	Output falling edge flag interrupt enable bit.
0	REDGEN	Output rising edge flag interrupt enable bit.

CHANNEL[IRQEN] Bits

54.4.5 CHANNEL[DMAEN] (0x18 + 0x20 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																														FEDGEN	REDGEN
N/A																														RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0

CHANNEL[DMAEN] [31:0]

Bits	Name	Description
1	FEDGEN	Output falling edge flag DMA request enable bit.
0	REDGEN	Output rising edge flag DMA request enable bit.

CHANNEL[DMAEN] Bits

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55 Digital-to-Analog Converter (DAC)

This chapter describes the functions and features of the Digital-to-Analog Converter (DAC).

55.1 Feature Summary

The main features of the DAC are as follows:

- Up to 1MHz output frequency, 12-bit resolution
- Two clock domains: AHB, DAC
- Built-in 16-bit divider, which can divide the DAC clock to a data output frequency of no more than 1MHz
- Supports direct mode, step mode and memory mode
- Built-in DMA controller
- Supports interrupt and DMA requests

55.2 Function Description

This section describes the functions of the DAC.

55.2.1 Direct Mode

Users can directly output the required voltage by writing to the 12-bit register.

55.2.2 Step Mode

The DAC output voltage ranges from the start value to the end value.

- Supports increment or decrement direction;
- Each step value is configurable;
- After reaching the end value, it can be configured to stop at the end value or return to the start value to continue operation;
- Supports 4 groups of step configurations, each group can be independently configured with start/end values, step value, cycle mode, increment/decrement direction, etc.;

55.2.3 Memory Mode

The built-in DMA of the DAC can read a segment of data from the specified storage space and output it through the DAC;

The DMA reads 32-bit data from the storage space, and each 32-bit data can be configured as one or two DAC output data;

Users can configure two storage spaces, each with a maximum size of 1MB, supporting up to 512K points. After reading data from one storage space, the DAC controller automatically reads data from the other one.

55.2.4 Trigger Control

Both step mode and memory mode can be enabled by software or hardware trigger. After enabling, data is output according to the configured DAC output frequency;

- Software trigger: Triggered via software register;
- Hardware trigger: Hardware trigger is generated by configuring the interconnection manager;

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55.2.5 Interrupt and DMA Requests

In memory mode, it supports sending an interrupt or DMA request when the read operation of each storage space is completed.

55.2.6 Clock Control

The DAC has two input clocks: APB clock and DAC clock. The APB clock is used for CPU to read and write configuration registers, and for the internal DMA to read data via AHB in memory mode;

The DAC clock is used for analog circuit control and output data frequency control;

Users need to configure dac_cfg1.div_cfg to set the DAC output data frequency;

The analog circuit requires a clock to latch the DAC output data. In direct mode, users need to set dac_cfg1.ana_clk_en to enable this clock; no configuration is required in step mode and memory mode.

55.3 DAC Registers

The list of DAC registers is as follows:

DAC0 base address: 0xF3090000

DAC1 base address: 0xF3094000

Offset	Register Name	Description	Reset Value
0x0000	CFG0		0x00000000
0x0004	CFG1		0x00010000
0x0008	CFG2		0x00000000
0x0010	STEP_CFG[STEP0]		0x00000000
0x0014	STEP_CFG[STEP1]		0x00000000
0x0018	STEP_CFG[STEP2]		0x00000000
0x001C	STEP_CFG[STEP3]		0x00000000
0x0020	BUF_ADDR[BUF0]		0x00000000
0x0024	BUF_ADDR[BUF1]		0x00000000
0x0028	BUF_LENGTH		0x00000000
0x0030	IRQ_STS		0x00000000
0x0034	IRQ_EN		0x00000000
0x0038	DMA_EN		0x00000000
0x0040	ANA_CFG0		0x00000030
0x0044	CFG0_BAK		0x00000000
0x0048	STATUS0		0x00000000

Table 214: DAC Register List

55.4 DAC Register Details

The detailed description of DAC registers is as follows:

55.4.1 CFG0 (0x0)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD				SW_DAC_DATA												RSVD						DMA_AHB_EN	SYNC_MODE	RSVD	HW_TRIG_EN	DAC_MODE		BUF_DATA_MODE	HBURST_CFG		
N/A				WO												N/A						WO	WO	N/A	WO	WO		WO	WO		
x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	0	0	x	0	0	0	0	0	0	0

CFG0 [31:0]

Bits	Name	Description
27-16	SW_DAC_DATA	dac data used in direct mode(dac_mode==2'b10)
9	DMA_AHB_EN	set to enable internal DMA, it will read one burst if enough space in FIFO. Should only be used in buffer mode.
8	SYNC_MODE	1: sync dac clock and ahb clock. all HW trigger signals are pulse in sync mode, can get faster response; 0: async dac clock and ahb_clock all HW trigger signals should be level and should be more than one dac clock cycle, used to get accurate output frequency(which may not be divided from AHB clock)
6	HW_TRIG_EN	set to use trigger signal from trigger_mux, user should config it to pulse in single mode, and level in continual mode
5-4	DAC_MODE	00: direct mode, DAC output the fixed configured data(from sw_dac_data) 01: step mode, DAC output from start_point to end point, with configured step, can step up or step down 10: buffer mode, read data from buffer, then output to analog, internal DMA will load next burst if enough space in local FIFO; 11: trigger mode, DAC output from external trigger signals Note: Trigger mode is not supported in hpm63xx and hpm62xx families.
3	BUF_DATA_MODE	data structure for buffer mode, 0: each 32-bit data contains 2 points, b11:0 for first, b27:16 for second. 1: each 32-bit data contains 1 point, b11:0 for first
2-0	HBURST_CFG	DAC support following fixed burst only 000-SINGLE; 011-INCR4; 101: INCR8 others are reserved

CFG0 Bits

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55.4.2 CFG1 (0x4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD													ANA_CLK_EN	ANA_DIV_CFG		DIV_CFG																
N/A													RW	RW		RW																
x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CFG1 [31:0]

Bits	Name	Description
18	ANA_CLK_EN	set to enable analog clock(divided by ana_div_cfg) need to be set in direct mode and trigger mode
17-16	ANA_DIV_CFG	clock divider config for ana_clk to dac analog; 00: div2 01: div4 10: div6 11: div8
15-0	DIV_CFG	step mode and buffer mode: defines how many clk_dac cycles to change data to analog, should configured to less than 1MHz data rate. Direct mode and trigger mode: defines how many clk_dac cycles to accpet the input data, dac will not accept new written data or trigger data before the clock cycles passed. should configured to less than 1MHz. Note: For direct mode and trigger mode, this config is not supported in hpm63xx and hpm62xx families.

CFG1 Bits

55.4.3 CFG2 (0x8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																								DMA_RST1	DMA_RST0	FIFO_CLR	BUF_SW_TRIG	STEP_SW_TRIG3	STEP_SW_TRIG2	STEP_SW_TRIG1	STEP_SW_TRIG0
N/A																								WO	WO	WO	RW	RW	RW	RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0

CFG2 [31:0]

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Bits	Name	Description
7	DMA_RST1	set to reset dma read pointer to buf1_start_addr; if set both dma_rst0&dma_rst1, will set to buf0_start_addr user can set fifo_clr bit when use dma_rst*
6	DMA_RST0	set to reset dma read pointer to buf0_start_addr
5	FIFO_CLR	set to clear FIFO content(set both read/write pointer to 0)
4	BUF_SW_TRIG	software trigger for buffer mode, W1C in single mode. RW in continual mode
3	STEP_SW_TRIG3	
2	STEP_SW_TRIG2	
1	STEP_SW_TRIG1	
0	STEP_SW_TRIG0	software trigger0 for step mode, W1C in single mode. RW in continual mode

CFG2 Bits

55.4.4 STEP_CFG (0x10 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD		ROUND_MODE	UP_DOWN	END_POINT												STEP_NUM				START_POINT											
N/A		RW	RW	RW												RW				RW											
x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

STEP_CFG [31:0]

Bits	Name	Description
29	ROUND_MODE	0: stop at end point; 1: reload start point, step again
28	UP_DOWN	0 for up, 1 for down
27-16	END_POINT	
15-12	STEP_NUM	output data change step_num each DAC clock cycle. Ex: if step_num=3, output data sequence is 0,3,6,9 NOTE: user should make sure end_point can be reached if step_num is not 1 if step_num is 0, output data will always at start point
11-0	START_POINT	

STEP_CFG Bits

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55.4.5 BUF_ADDR (0x20 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BUF_START_ADDR																														RSVD	BUF_STOP
RW																														N/A	RW
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	0

BUF_ADDR [31:0]

Bits	Name	Description
31-2	BUF_START_ADDR	buffer start address, should be 4-byte aligned AHB burst can't cross 1K-byte boundary, user should config the address/length/burst to avoid such issue.
0	BUF_STOP	set to stop read point at end of bufffer0

BUF_ADDR Bits

55.4.6 BUF_LENGTH (0x28)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BUF1_LEN																BUF0_LEN															
RW																RW															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

BUF_LENGTH [31:0]

Bits	Name	Description
31-16	BUF1_LEN	buffer length, 1 indicate one 32bit date, 256K-byte max for one buffer
15-0	BUF0_LEN	

BUF_LENGTH Bits

55.4.7 IRQ_STS (0x30)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																											STEP_CMPT	AHB_ERROR	FIFO_EMPTY	BUF1_CMPT	BUF0_CMPT

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
N/A																											W1C	W1C	W1C	W1C	W1C
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0

IRQ_STS [31:0]

Bits	Name	Description
4	STEP_CMPT	
3	AHB_ERROR	set if hresp==2'b01(ERROR)
2	FIFO_EMPTY	
1	BUF1_CMPT	
0	BUF0_CMPT	

IRQ_STS Bits

55.4.8 IRQ_EN (0x34)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																											STEP_CMPT	AHB_ERROR	FIFO_EMPTY	BUF1_CMPT	BUF0_CMPT
N/A																											RW	RW	RW	RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0

IRQ_EN [31:0]

Bits	Name	Description
4	STEP_CMPT	
3	AHB_ERROR	
2	FIFO_EMPTY	
1	BUF1_CMPT	
0	BUF0_CMPT	

IRQ_EN Bits

55.4.9 DMA_EN (0x38)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																											STEP_CMPT	RSVD		BUF1_CMPT	BUF0_CMPT
N/A																											RW	N/A		RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	0	0

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

DMA_EN [31:0]

Bits	Name	Description
4	STEP_CMPT	
1	BUF1_CMPT	
0	BUF0_CMPT	

DMA_EN Bits

55.4.10 ANA_CFG0 (0x40)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																							DAC12BIT_LP_MODE	DAC_CONFIG				CALI_DELTA_V_CFG		BYPASS_CALI_GM	DAC12BIT_EN
N/A																							RW	RW				RW		RW	RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	1	1	0	0	0	0

ANA_CFG0 [31:0]

Bits	Name	Description
8	DAC12BIT_LP_MODE	
7-4	DAC_CONFIG	
3-2	CALI_DELTA_V_CFG	
1	BYPASS_CALI_GM	
0	DAC12BIT_EN	

ANA_CFG0 Bits

55.4.11 CFG0_BAK (0x44)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD				SW_DAC_DATA												RSVD						DMA_AHB_EN	SYNC_MODE	TRIG_MODE	HW_TRIG_EN	DAC_MODE		BUF_DATA_MODE		HBURST_CFG	

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
N/A				RW												N/A						RW	RW	RW	RW	RW		RW	RW		
x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0

CFG0_BAK [31:0]

Bits	Name	Description
27-16	SW_DAC_DATA	dac data used in direct mode(dac_mode==2'b10)
9	DMA_AHB_EN	set to enable internal DMA, it will read one burst if enough space in FIFO. Should only be used in buffer mode.
8	SYNC_MODE	1: sync dac clock and ahb clock. all HW trigger signals are pulse in sync mode, can get faster response; 0: async dac clock and ahb_clock all HW trigger signals should be level and should be more than one dac clock cycle, used to get accurate output frequency(which may not be divided from AHB clock)
7	TRIG_MODE	0: single mode, one trigger pulse will send one 12bit data to DAC analog; 1: continual mode, if trigger signal(either or HW) is set, DAC will send data if FIFO is not empty, if trigger signal is clear, DAC will stop send data.
6	HW_TRIG_EN	set to use trigger signal from trigger_mux, user should config it to pulse in single mode, and level in continual mode
5-4	DAC_MODE	00: direct mode, DAC output the fixed configured data(from sw_dac_data) 01: step mode, DAC output from start_point to end point, with configured step, can step up or step down 10: buffer mode, read data from buffer, then output to analog, internal DMA will load next burst if enough space in local FIFO;
3	BUF_DATA_MODE	data structure for buffer mode, 0: each 32-bit data contains 2 points, b11:0 for first, b27:16 for second. 1: each 32-bit data contains 1 point, b11:0 for first
2-0	HBURST_CFG	DAC support following fixed burst only 000-SINGLE; 011-INCR4; 101: INCR8 others are reserved

CFG0_BAK Bits

55.4.12 STATUS0 (0x48)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD								CUR_BUF_OFFSET																CUR_BUF_INDEX	RSVD						
N/A								RW																RW	N/A						
x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x

STATUS0 [31:0]

Bits	Name	Description
23-8	CUR_BUF_OFFSET	
7	CUR_BUF_INDEX	

STATUS0 Bits

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56 Temperature Sensor TSNS

This chapter describes the functions and features of the TSNS temperature sensor.

56.1 Feature Summary

This product integrates a temperature sensor to measure the die temperature. Its main features are as follows:

- Single measurement
- Continuous measurement
- Averaging of multiple measurements
- Temperature upper and lower limit recording
- Temperature range comparison
- Interrupt generation when temperature is out of range
- Chip reset when temperature is out of range

56.2 Function Description

TSNS converts the signal of the built-in temperature-sensitive diode through the internal ADC to obtain the die temperature. The software needs to enable its bias circuit before temperature acquisition.

During temperature acquisition, the ADC samples the internal temperature sensor multiple times and averages the results to achieve higher accuracy. The number of averages is configured by the AVERAGE field, supporting up to 128 averages.

AGE register: The temperature sensor module provides the time elapsed since the last temperature acquisition, and the time value provided by this register is counted in 24M clock cycles.

The temperature sensor internally records the highest and lowest temperatures ever detected. They can be read through the TMAX and TMIN registers, and can be cleared via RECORD_MAX_CLR and RECORD_MIN_CLR bits for re-recording.

56.2.1 Temperature Acquisition Mode

Conversion can be triggered by software or performed periodically in continuous acquisition mode.

In single acquisition mode, the software initiates temperature measurement by writing to the TRIGGER bit. After the measurement is completed, the VALID bit is set to 1, indicating that the temperature value is valid.

Continuous mode is configured via the CONTINOUS field. In continuous mode, users set the temperature sampling interval through the VALIDITY field, and the interval is counted in 24M clock cycles.

56.2.2 Temperature Comparison Function

The temperature sensor supports interrupt generation and chip reset when the temperature is out of range. This function can be enabled by setting COMPARE_MIN_EN and COMPARE_MAX_EN bits to allow low temperature and high temperature comparison respectively.

The temperature ranges for interrupt and reset are configured independently.

The temperature thresholds for interrupt comparison are set in the UPPER_LIM_IRQ and LOWER_LIM_IRQ

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registers.

The temperature thresholds for reset comparison are set in the UPPER_LIM_RST and LOWER_LIM_RST registers.

When the measured die temperature exceeds the set range, the temperature sensor module sets the corresponding flag bit. If the interrupt enable bit IRQ_EN is set to 1, an interrupt will be generated. If the reset enable bit RST_EN is set to 1, a chip reset will be triggered.

56.3 TSNS Registers

The register list of TSNS is as follows:

TSNS base address: 0xF0090000

Offset	Register Name	Description	Reset Value
0x0000	T	Temperature	0x00000000
0x0004	TMAX	Maximum Temperature	0xFF800000
0x0008	TMIN	Minimum Temperature	0x007FFFFF
0x000C	AGE	Sample age	0x00000000
0x0010	STATUS	Status	0x00000000
0x0014	CONFIG	Configuration	0x00600300
0x0018	VALIDITY	Sample validity	0x016E3600
0x001C	FLAG	Temperature flag	0x00000000
0x0020	UPPER_LIM_IRQ	Maximum temperature to interrupt	0x00000000
0x0024	LOWER_LIM_IRQ	Minimum temperature to interrupt	0x00000000
0x0028	UPPER_LIM_RST	Maximum temperature to reset	0x00000000
0x002C	LOWER_LIM_RST	Minimum temperature to reset	0x00000000
0x0030	ASYNC	Configuration in asynchronous mode	0x00000000
0x0038	ADVAN	Advance configuration	0x00000000

Table 215: TSNS Register List

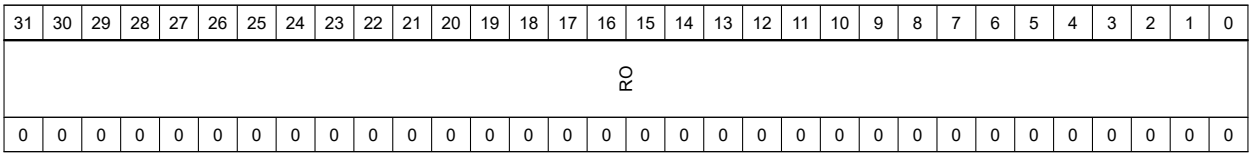
56.4 TSNS Register Details

The detailed description of TSNS registers is as follows:

56.4.1 T (0x0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
T																															

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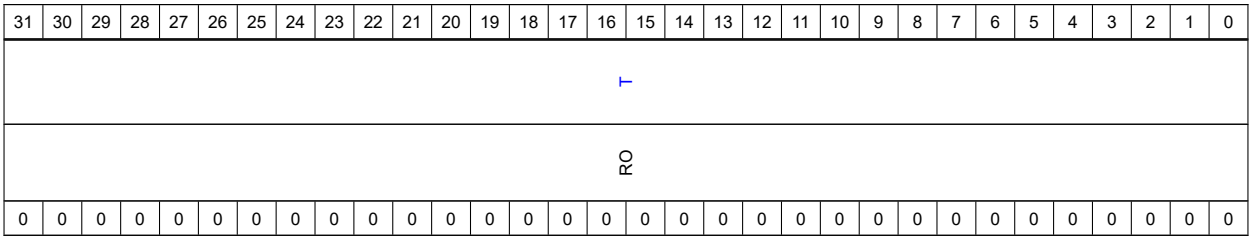


T [31:0]

Bits	Name	Description
31-0	T	Signed number of temperature in 256 x celsius degree

T Bits

56.4.2 TMAX (0x4)

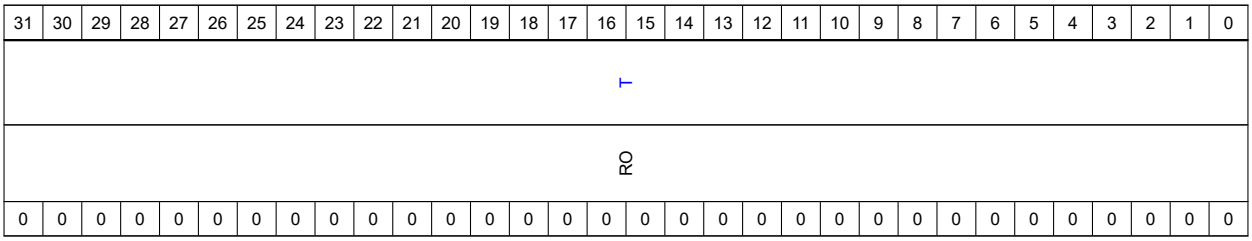


TMAX [31:0]

Bits	Name	Description
31-0	T	maximum temperature ever found

TMAX Bits

56.4.3 TMIN (0x8)



TMIN [31:0]

Bits	Name	Description
31-0	T	minimum temperature ever found

TMIN Bits

56.4.4 AGE (0xC)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AGE																															
RO																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

AGE [31:0]

Bits	Name	Description
31-0	AGE	age of T register in 24MHz clock cycles

AGE Bits

56.4.5 STATUS (0x10)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VALID	RSVD																														TRIGGER
RO	N/A																														W1C
0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

STATUS [31:0]

Bits	Name	Description
31	VALID	indicate value in T is valid or not 0: not valid 1:valid
0	TRIGGER	Software trigger for sensing in trigger mode, trigger will be ignored if in sensing or other mode

STATUS Bits

56.4.6 CONFIG (0x14)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IRQ_EN	RST_EN	RSVD				COMPARE_MIN_EN	COMPARE_MAX_EN	SPEED								RSVD					AVERAGE			RSVD			CONTINUOUS	RSVD		ASYNC	ENABLE
RW	RW	N/A				RW	RW	RW								N/A					RW			N/A			RW	N/A		RW	RW
0	0	x	x	x	x	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	0	0	0	x	x	x	0	x	x	0	0

CONFIG [31:0]

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Bits	Name	Description
31	IRQ_EN	Enable interrupt
30	RST_EN	Enable reset
25	COMPARE_MIN_EN	Enable compare for minimum temperature
24	COMPARE_MAX_EN	Enable compare for maximum temperature
23-16	SPEED	cycles of a progressive step in 24M clock, valid from 24-255, default 96 24: 24 cycle for a step 25: 25 cycle for a step 26: 26 cycle for a step ... 255: 255 cycle for a step
10-8	AVERAGE	Average time, default in 3 0: measure and return 1: twice and average 2: 4 times and average . . . 7: 128 times and average
4	CONTINUOUS	continuous mode that keep sampling temperature peridically 0: trigger mode 1: continuous mode
1	ASYNC	Acynchronous mode, this mode can work without clock, only available function ios compare to certain ADC value 0: active mode 1: Async mode
0	ENABLE	Enable temperature 0: disable, temperature sensor is shut down 1: enable. Temperature sensor enabled

CONFIG Bits

56.4.7 VALIDITY (0x18)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VALIDITY																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

VALIDITY [31:0]

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Bits	Name	Description
31-0	VALIDITY	time for temperature values to expire in 24M clock cycles

VALIDITY Bits

56.4.8 FLAG (0x1C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD										RECORD_MIN_CLR	RECORD_MAX_CLR	RSVD		UNDER_TEMP	OVER_TEMP	RSVD															IRQ
N/A										RW	RW	N/A		RW	RW	N/A															RW
x	x	x	x	x	x	x	x	x	x	0	0	x	x	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

FLAG [31:0]

Bits	Name	Description
21	RECORD_MIN_CLR	Clear minimum recorder of temerature, write 1 to clear
20	RECORD_MAX_CLR	Clear maximum recorder of temerature, write 1 to clear
17	UNDER_TEMP	Clear under temperature status, write 1 to clear
16	OVER_TEMP	Clear over temperature status, write 1 to clear
0	IRQ	IRQ flag, write 1 to clear

FLAG Bits

56.4.9 UPPER_LIM_IRQ (0x20)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
T																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

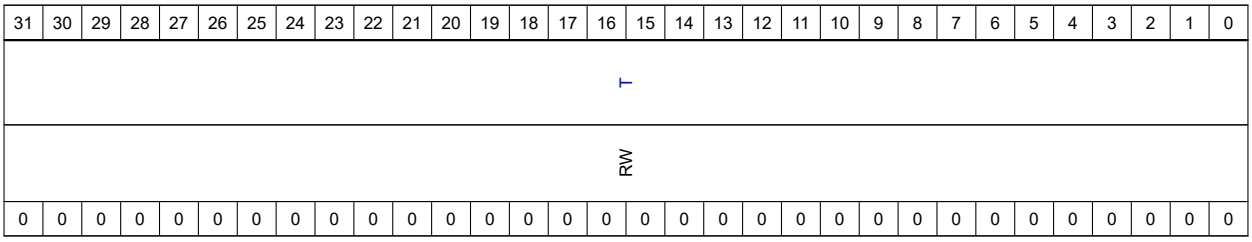
UPPER_LIM_IRQ [31:0]

Bits	Name	Description
31-0	T	Maximum temperature for compare

UPPER_LIM_IRQ Bits

56.4.10 LOWER_LIM_IRQ (0x24)

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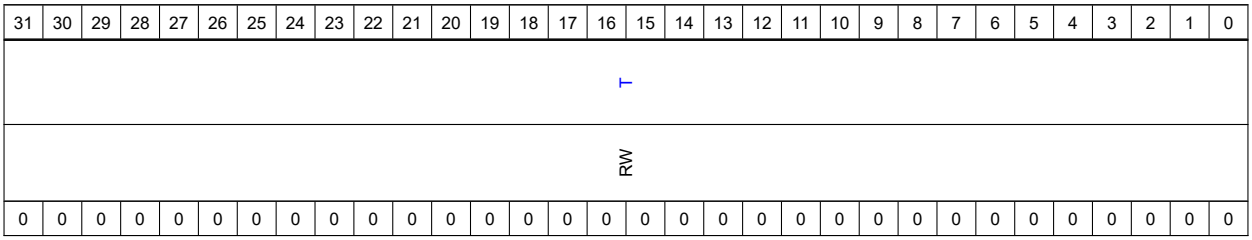


LOWER_LIM_IRQ [31:0]

Bits	Name	Description
31-0	T	Minimum temperature for compare

LOWER_LIM_IRQ Bits

56.4.11 UPPER_LIM_RST (0x28)

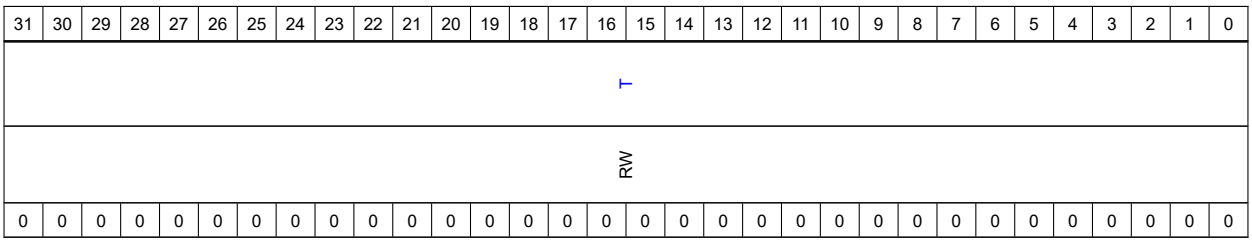


UPPER_LIM_RST [31:0]

Bits	Name	Description
31-0	T	Maximum temperature for compare

UPPER_LIM_RST Bits

56.4.12 LOWER_LIM_RST (0x2C)



LOWER_LIM_RST [31:0]

Bits	Name	Description
31-0	T	Minimum temperature for compare

LOWER_LIM_RST Bits

56.4.13 ASYNC (0x30)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD							ASYNC_TYPE	RSVD							POLARITY	RSVD					VALUE										
N/A							RW	N/A							RW	N/A					RW										
x	x	x	x	x	x	x	0	x	x	x	x	x	x	x	0	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0

ASYNC [31:0]

Bits	Name	Description
24	ASYNC_TYPE	Compare hotter than or colder than in asynchoronous mode 0: hotter than 1: colder than
16	POLARITY	Polarity of internal comparator
10-0	VALUE	Value of async mode to compare

ASYNC Bits

56.4.14 ADVAN (0x38)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD						ASYNC_IRQ	ACTIVE_IRQ	RSVD							SAMPLING	RSVD														NEG_ONLY	POS_ONLY
N/A						RO	RO	N/A							RO	N/A														RW	RW
x	x	x	x	x	x	0	0	x	x	x	x	x	x	x	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0

ADVAN [31:0]

Bits	Name	Description
25	ASYNC_IRQ	interrupt status of asynchronous mode
24	ACTIVE_IRQ	interrupt status of active mode
16	SAMPLING	temperature sampling is working
1	NEG_ONLY	use negative compare polarity only
0	POS_ONLY	use positive compare polarity only

ADVAN Bits

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57 Operational Amplifier (OPAMP)

57.1 Overview

The operational amplifier (OPAMP) is an analog module used to amplify input differential signals to obtain corresponding output voltages, which can expand the analog application range of the MCU.

- Functions including voltage follower, non-inverting amplification, inverting amplification and custom mode can be realized by configuring the internal feedback network and using external components.
- Supports rail-to-rail input and output range, and the open-loop gain of the OPAMP remains relatively consistent within the rail-to-rail input range.
- Adaptable to a wide range of capacitive loads (≤50pF) and resistive loads (≥1KOhm), and the OPAMP response speed can be optimized according to different capacitive loads.
- The reference voltage can be provided by internal reference circuit, external excitation source or DAC output voltage.
- The output can be connected to the input port of the ADC.
- Equipped with 4-channel positive input terminals and 4-channel negative input terminals.
- Supports direct configuration of working mode via software, or selection of preset working mode via hardware trigger.
- Provides 1 set of direct configuration parameter registers and 8 sets of preset parameter registers.

57.2 Pin Description

Table 216: OPAMP Pin Description

Pin	Direction	Function Description
OPAn[1]_OUT	Output	Amplifier output.
OPAn_EXT	Input	Auxiliary function.
OPAn_INP0	Input	Differential input positive terminal.
OPAn_INN0	Input	Differential input negative terminal.
OPAn_INP1	Input	Differential input 1 positive terminal.
OPAn_INN1	Input	Differential input 1 negative terminal.
OPAn_INP2	Input	Differential input 2 positive terminal, from PB08, system DAC0.
OPAn_INN2	Input	Differential input 2 negative terminal, from PB09, system DAC1.
OPAn_INP3	Input	Differential input 3 positive terminal.
OPAn_INN3	Input	Differential input 3 negative terminal.

[1] n=0,1

57.3 Function Description

57.3.1 Follower Mode

Vout=Vin

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Where Vin is the input voltage, Vout is the output voltage.

Pin OPAn_INP0/1/2/3 can be selected as the input terminal, pin OPAn_OUT is the output terminal.

57.3.2 Non-Inverting Amplification Mode

Vout=(Vin-Vref)*Av+Vref

Av=2,4,8,16,32,64,128

Where Vin is the input voltage, Vref is the reference voltage, Vout is the output voltage, and Av is the amplification factor.

 Use internal resistor feedback network, with internal GND as reference voltage.

Pin OPAn_INP0/1/2/3 can be selected as the input terminal, pin OPAn_OUT is the output terminal.

 Use internal resistor feedback network and external reference voltage.

Pin OPAn_INP0/1/2/3 can be selected as the input terminal, pin OPAn_INN0/1/2/3 can be selected to connect external reference voltage, pin OPAn_OUT is the output terminal.

 Use internal resistor feedback network, with internal GND as reference voltage, and external filter capacitor.

Pin OPAn_INP0/1/2/3 can be selected as the input terminal, connect the filter capacitor between pin OPAn_INN0/1/2/3 and OPAn_OUT, pin OPAn_OUT is the output terminal.

 Use internal resistor feedback network, external reference voltage, and external filter capacitor.

Pin OPAn_INP0/1/2/3 can be selected as the input terminal, pin OPAn_INN0/1/2/3 can be selected to connect external reference voltage, connect the filter capacitor between pin OPAn_EXT and OPAn_OUT, pin OPAn_OUT is the output terminal.

 Use internal resistor feedback network, with DAC output as reference voltage.

Pin OPAn_INP0/1/2/3 can be selected as the input terminal, select DAC output as reference voltage through corresponding configuration, pin OPAn_OUT is the output terminal.

57.3.3 Inverting Amplification Mode

Vout=(Vin-Vref)*Av+Vref

Av=-2,-4,-8,-16,-32,-64,-128

Where Vin is the input voltage, Vref is the reference voltage, Vout is the output voltage, and Av is the amplification factor.

 Use internal resistor feedback network and internal reference voltage.

Pin OPAn_INN0/1/2/3 can be selected as the input terminal, pin OPAn_OUT is the output terminal.

 Use internal resistor feedback network and external reference voltage.

Pin OPAn_INN0/1/2/3 can be selected as the input terminal, pin OPAn_INP0/1/2/3 can be selected to connect external reference voltage, pin OPAn_OUT is the output terminal.

 Use internal resistor feedback network, external reference voltage, and external filter capacitor.

Pin OPAn_INN0/1/2/3 can be selected as the input terminal, pin OPAn_INP0/1/2/3 can be selected to connect

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external reference voltage, connect the filter capacitor between pin OPAn_EXT and OPAn_OUT, pin OPAn_OUT is the output terminal.

 Use internal resistor feedback network, with DAC output as reference voltage.

Pin OPAn_INN0/1/2/3 can be selected as the input terminal, select DAC output as reference voltage through corresponding configuration, pin OPAn_OUT is the output terminal.

57.3.4 Custom Mode

Freely build external circuits for the operational amplifier to realize multiple functions.

Pin OPAn_INP0/1/2/3 can be selected as the positive input terminal of the OPAMP, pin OPAn_INN0/1/2/3 can be selected as the negative input terminal of the OPAMP, pin OPAn_OUT is the output terminal.

57.4 Function Table

Except for ISO which refers to voltage ISO_LV_VDD33, all other abbreviations in the first row of the function table come from registers, which can be either direct configuration parameters or preset parameters selected by trigger:

ISO	EN	BP	VW	PS	MS	P7	P2	IS	C3	C2	X1	X0	Function
3.3	x	x	x	x	x	x	x	x	x	x	x	x	isolation
0	0	x	x	x	x	x	x	x	x	x	x	x	Turn off OPAMP.
0	1	x	x	0	x	x	x	x	x	x	x	x	OPAMP positive input terminal is gated to pin OPAn_INP0.
0	1	x	x	1	x	x	x	x	x	x	x	x	OPAMP positive input terminal is gated to pin OPAn_INP1.
0	1	x	x	2	x	x	x	x	x	x	x	x	OPAMP positive input terminal is gated to pin OPAn_INP2.
0	1	x	x	3	x	x	x	x	x	x	x	x	OPAMP positive input terminal is gated to pin OPAn_INP3.
0	1	x	x	4	x	x	x	x	x	x	x	x	OPAMP positive input terminal is gated to internal reference voltage 0.25*Vsupply.
0	1	x	x	5	x	x	x	x	x	x	x	x	OPAMP positive input terminal is gated to internal reference voltage 0.5*Vsupply.
0	1	x	x	6	x	x	x	x	x	x	x	x	OPAMP positive input terminal is gated to internal reference voltage 0.75*Vsupply.
0	1	x	x	7	x	x	x	x	x	x	x	x	OPAMP positive input terminal is floating.
0	1	x	x	x	0	x	x	x	x	x	x	x	OPAMP negative input terminal is gated to pin OPAn_INN0.
0	1	x	x	x	1	x	x	x	x	x	x	x	OPAMP negative input terminal is gated to pin OPAn_INN1.
0	1	x	x	x	2	x	x	x	x	x	x	x	OPAMP negative input terminal is gated to pin OPAn_INN2.

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ISO	EN	BP	VW	PS	MS	P7	P2	IS	C3	C2	X1	X0	Function
0	1	x	x	x	3	x	x	x	x	x	x	x	OPAMP negative input terminal is gated to pin OPAn_INN3.
0	1	x	x	x	4	x	x	x	x	x	x	x	OPAMP negative input terminal is floating.
0	1	x	x	x	x	x	x	x	x	6	x	x	Select 5 sets of compensation capacitors.
0	1	x	x	x	x	x	x	x	x	7	x	x	Select 7 sets of compensation capacitors.
0	1	x	x	x	x	x	x	x	x	0	x	x	Select 9 sets of compensation capacitors.
0	1	x	x	x	x	x	x	x	x	1	x	x	Select 11 sets of compensation capacitors.
0	1	x	x	x	x	x	x	x	x	2	x	x	Select 14 sets of compensation capacitors.
0	1	x	x	x	x	x	x	x	x	3	x	x	Select 18 sets of compensation capacitors.
0	1	x	x	x	x	x	x	x	x	4	x	x	Select 23 sets of compensation capacitors.
0	1	x	x	x	x	x	x	x	x	5	x	x	Select 30 sets of compensation capacitors.
0	1	x	x	x	x	x	x	x	x	0	x	x	(Example) Low load: Cload≤5pF, Rload≥1Kohm or No Rload. (Select 9 sets of compensation capacitors).
0	1	x	x	x	x	x	x	x	x	3	x	x	(Example) High load: Cload≤50pF, Rload≥1Kohm or No Rload. (Select 18 sets of compensation capacitors).
0	1	x	x	x	x	x	0	x	x	x	x	x	Select 2x amplification factor.
0	1	x	x	x	x	x	1	x	x	x	x	x	Select 4x amplification factor.
0	1	x	x	x	x	x	2	x	x	x	x	x	Select 8x amplification factor.
0	1	x	x	x	x	x	3	x	x	x	x	x	Select 16x amplification factor.
0	1	x	x	x	x	x	4	x	x	x	x	x	Select 32x amplification factor.
0	1	x	x	x	x	x	5	x	x	x	x	x	Select 64x amplification factor.
0	1	x	x	x	x	x	6	x	x	x	x	x	Select 128x amplification factor.
0	1	x	x	x	x	x	7	x	x	x	x	x	Select 128x amplification factor.
0	1	1	0	<4	4	6	x	0	1	0	0	0	Function: Follower mode. Pin OPAn_INP0/1/2/3 can be selected as the input terminal, pin OPAn_OUT is the output terminal.
0	1	0	0	5	<4	8	x	0	0	x	0	0	Function: Inverting amplification mode (use internal resistor feedback network and internal reference voltage). Pin OPAn_INN0/1/2/3 can be selected as the input terminal, pin OPAn_OUT is the output terminal.
0	1	0	0	<4	<4	8	x	0	0	x	0	0	Function: Inverting amplification mode (use internal resistor feedback network and external reference voltage). Pin OPAn_INN0/1/2/3 can be selected as the input terminal, pin OPAn_INP0/1/2/3 can be selected to connect external reference voltage, pin OPAn_OUT is the output terminal.

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ISO	EN	BP	VW	PS	MS	P7	P2	IS	C3	C2	X1	X0	Function
0	1	0	0	<4	<4	18	x	0	1	x	0	0	Function: Inverting amplification mode (use internal resistor feedback network, external reference voltage, and external filter capacitor). Pin OPAn_INN0/1/2/3 can be selected as the input terminal, pin OPAn_INP0/1/2/3 can be selected to connect external reference voltage, connect the filter capacitor between pin OPAn_EXT and OPAn_OUT, pin OPAn_OUT is the output terminal.
0	1	0	0	2	<4	8	x	0	0	x	0	0	Function: Inverting amplification mode (use internal resistor feedback network, with DAC output as reference voltage). Pin OPAn_INN0/1/2/3 can be selected as the input terminal, select DAC output as reference voltage through corresponding configuration, pin OPAn_OUT is the output terminal.
0	1	0	0	<4	4	1	x	0	0	x	0	0	Function: Non-inverting amplification mode (use internal resistor feedback network, with internal GND as reference voltage). Pin OPAn_INP0/1/2/3 can be selected as the input terminal, pin OPAn_OUT is the output terminal.
0	1	0	0	<4	<4	9	x	0	0	x	0	0	Function: Non-inverting amplification mode (use internal resistor feedback network and external reference voltage). Pin OPAn_INP0/1/2/3 can be selected as the input terminal, pin OPAn_INN0/1/2/3 can be selected to connect external reference voltage, pin OPAn_OUT is the output terminal.
0	1	0	0	<4	<4	11	x	0	1	x	0	0	Function: Non-inverting amplification mode (use internal resistor feedback network, with internal GND as reference voltage, and external filter capacitor). Pin OPAn_INP0/1/2/3 can be selected as the input terminal, connect the filter capacitor between pin OPAn_INN0/1/2/3 and OPAn_OUT, pin OPAn_OUT is the output terminal.
0	1	0	0	<4	<4	19	x	0	1	x	0	0	Function: Non-inverting amplification mode (use internal resistor feedback network, external reference voltage, and external filter capacitor). Pin OPAn_INP0/1/2/3 can be selected as the input terminal, pin OPAn_INN0/1/2/3 can be selected to connect external reference voltage, connect the filter capacitor between pin OPAn_EXT and OPAn_OUT, pin OPAn_OUT is the output terminal.

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ISO	EN	BP	VW	PS	MS	P7	P2	IS	C3	C2	X1	X0	Function
0	1	0	0	<4	2	9	x	0	0	x	0	0	Function: Non-inverting amplification mode (use internal resistor feedback network, with DAC output as reference voltage). Pin OPAn_INP0/1/2/3 can be selected as the input terminal, select DAC output as reference voltage through corresponding configuration, pin OPAn_OUT is the output terminal.
0	1	1	0	<4	<4	4	x	0	1	x	0	0	Function: Custom mode (freely build external circuits for the operational amplifier to realize multiple functions). Pin OPAn_INP0/1/2/3 can be selected as the positive input terminal of the OPAMP, pin OPAn_INN0/1/2/3 can be selected as the negative input terminal of the OPAMP, pin OPAn_OUT is the output terminal.

Table 218: Function Table Abbreviations

Abbreviation	Parameter Name	Bit Width
BP	VBYPASS	1
EN	EN_LV	1
VW	VSWITCH_SEL	3
PS	VIP_SEL	3
MS	VIM_SEL	3
P7	MODE	5
P2	GAIN_SEL	3
IS	ISEL	2
C3	DISABLE_PM_CAP	1
C2	MILLER_SEL	3
X1	OPAOUT_SEL[1]	1
X0	OPAOUT_SEL[0]	1

57.5 OPAMP Registers

The list of OPAMP registers is as follows:

OPAMP0 base address: 0xF30A0000

OPAMP1 base address: 0xF30A4000

Offset	Register Name	Description	Reset Value
0x0000	CTRL0	control reg	0x00000000
0x0004	STATUS	status reg	0x00000000
0x0008	CTRL1	control reg1	0x00000000
0x0010	CFG[PRESET0][CFG0]		0x00000000

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Offset	Register Name	Description	Reset Value
0x0014	CFG[PRESET0][CFG1]		0x00000000
0x0018	CFG[PRESET0][CFG2]		0x00000000
0x0020	CFG[PRESET1][CFG0]		0x00000000
0x0024	CFG[PRESET1][CFG1]		0x00000000
0x0028	CFG[PRESET1][CFG2]		0x00000000
0x0030	CFG[PRESET2][CFG0]		0x00000000
0x0034	CFG[PRESET2][CFG1]		0x00000000
0x0038	CFG[PRESET2][CFG2]		0x00000000
0x0040	CFG[PRESET3][CFG0]		0x00000000
0x0044	CFG[PRESET3][CFG1]		0x00000000
0x0048	CFG[PRESET3][CFG2]		0x00000000
0x0050	CFG[PRESET4][CFG0]		0x00000000
0x0054	CFG[PRESET4][CFG1]		0x00000000
0x0058	CFG[PRESET4][CFG2]		0x00000000
0x0060	CFG[PRESET5][CFG0]		0x00000000
0x0064	CFG[PRESET5][CFG1]		0x00000000
0x0068	CFG[PRESET5][CFG2]		0x00000000
0x0070	CFG[PRESET6][CFG0]		0x00000000
0x0074	CFG[PRESET6][CFG1]		0x00000000
0x0078	CFG[PRESET6][CFG2]		0x00000000
0x0080	CFG[PRESET7][CFG0]		0x00000000
0x0084	CFG[PRESET7][CFG1]		0x00000000
0x0088	CFG[PRESET7][CFG2]		0x00000000

Table 219: OPAMP Register List

57.6 OPAMP Register Details

The detailed description of OPAMP registers is as follows:

57.6.1 CTRL0 (0x0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD					EN_LV	RSVD							VIM_SEL			MODE					GAIN_SEL			DISABLE_PM_CAP	MILLER_SEL			VBYPASS	VIP_SEL		
N/A					RW	N/A							RW			RW					RW			RW	RW			RW	RW		
x	x	x	x	x	0	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL0 [31:0]

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Bits	Name	Description
26	EN_LV	
18-16	VIM_SEL	
15-11	MODE	
10-8	GAIN_SEL	
7	DISABLE_PM_CAP	
6-4	MILLER_SEL	
3	VBYPASS	
2-0	VIP_SEL	

CTRL0 Bits

57.6.2 STATUS (0x4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD				TRIG_CONFLICT								PRESET_ACT	CUR_PRESET			RSVD															
N/A				RWC								RO	RO			N/A															
x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

STATUS [31:0]

Bits	Name	Description
27-20	TRIG_CONFLICT	if more than one hardware trigger is set, will put all trigger input here; write any value to clear
19	PRESET_ACT	1 for preset active; one of cur_preset is selected for OPAMP; 0 for no preset, OPAMP use cfg0 parameters
18-16	CUR_PRESET	current selected preset

STATUS Bits

57.6.3 CTRL1 (0x8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SW_PRESET	RSVD																												SW_SEL		
RW	N/A																												RW		
0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

CTRL1 [31:0]

Bits	Name	Description
31	SW_PRESET	set to use preset defined by sw_sel. NOTE: when set, the hardware trigger will not be used
2-0	SW_SEL	

CTRL1 Bits

57.6.4 CFG[CFG0] (0x10 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD				DISABLE_PM_CAP	MILLER_SEL			RSVD													VIM_SEL			RSVD					VIP_SEL		
N/A				RW	RW			N/A													RW			N/A					RW		
x	x	x	x	0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	x	x	x	x	x	0	0	0

CFG[CFG0] [31:0]

Bits	Name	Description
27	DISABLE_PM_CAP	
26-24	MILLER_SEL	
10-8	VIM_SEL	
2-0	VIP_SEL	

CFG[CFG0] Bits

57.6.5 CFG[CFG1] (0x14 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HW_TRIG_EN	EN_LV	VBYPASS_LV	RSVD																					MODE					GAIN_SEL		
RW	RW	RW	N/A																					RW					RW		
0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0

CFG[CFG1] [31:0]

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Bits	Name	Description
31	HW_TRIG_EN	set to enable hardware trigger from moto system. NOTE: when sw_preset is enabled, this bit will not take effert
30	EN_LV	
29	VBYPASS_LV	
7-3	MODE	
2-0	GAIN_SEL	

CFG[CFG1] Bits

57.6.6 CFG[CFG2] (0x18 + 0x10 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD					CHANNEL			RSVD																							
N/A					RW			N/A																							
x	x	x	x	x	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

CFG[CFG2] [31:0]

Bits	Name	Description
26-24	CHANNEL	

CFG[CFG2] Bits

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58 Overview of Information Security Module

This chapter introduces the information security module of this product.

The information security modules of this product are mainly divided into the following categories:

- Encryption and decryption engine
  - Secure Data Processor SDP
  - Inline Decryption Module EXIP
- Key management for encryption and decryption engines
  - Key Manager KEYM
  - Key area in OTP
- True Random Number Generator
- System Security Status Monitor
  - Based on product lifecycle management
  - Monitor JTAG debug interface
  - Monitor processor entering Debug mode
- BOOT ROM secure boot mechanism to ensure safe and trusted loading of user program images

58.1 Secure Data Processor SDP

This product contains a secure data processor SDP, which supports AES-128/256 encryption and decryption operations, supports AES ECB and CBC encryption modes, and supports Chinese national cryptographic SM4 algorithm. SDP supports SHA-1/256 hash operations, CRC32 and Chinese national cryptographic SM3 algorithm.

SDP supports loading keys from the key manager KEY. On this product, the following keys of SDP come from the key manager KEYM.

- MK[0:255], from the MK key output of the key manager
- SK0[0:255], from the SK0 key output of the key manager
- SK1[0:255], from the SK1 key output of the key manager
- SK2[0:255], from the SK2 key output of the key manager
- SK3[0:255], from the SK3 key output of the key manager

58.2 Inline Decryption Engine EXIP

This product includes 1 inline decryption engine EXIP0, which is tightly coupled with XPI0. Users can use AES-128 CTR mode to encrypt data stored in external NOR Flash according to specifications. When this product is connected to encrypted external NOR Flash via XPI, EXIP performs real-time inline decryption with zero wait cycles, restores the encrypted code to plaintext for the processor to execute directly online.

EXIP supports using the EXIP KEK (Key Encryption Key) stored in OTP to restore DEK (Data Encryption Key) from Key Blob, so as to realize DEK protection.

For the key management of this product, please refer to Section 58.4.

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58.3 Key Manager

This product supports key manager KEYM. The key manager supports loading keys from the key area of OTP through an independent data path, and after obfuscation processing, transmits keys to the secure data processor SDP through an independent data path. It realizes key loading without processor participation, ensuring that keys are not exposed to any software.

On this product, the key manager supports loading key information from the following modules:

- SMKRNG[0:255], from Random Number Generator RNG
- FMK[0:255], from the key area of OTP

On this product, KEYM sends key information to SDP through an independent data path:

- MK[0:255]
- SK0[0:255]
- SK1[0:255]
- SK2[0:255]
- SK3[0:255]

For the key management of this product, please refer to Section 58.4.

58.4 Key Management Summary

For this product, the key management summary is shown in Figure 91:

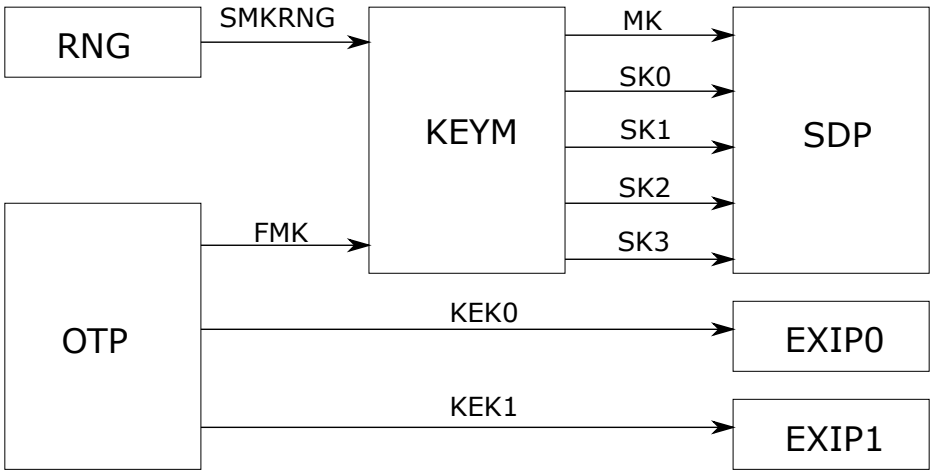


Figure 91: Key Management Summary

58.5 One-Time Programmable Memory OTP

The OTP of this product supports storing various data related to security functions:

- 128-bit Debug Key: This product supports JTAG port locking. After locking, the debugger needs to provide a debug password to unlock the JTAG interface for debugging.
- 128-bit UUID, the unique identifier of the chip
- Public key HASH for secure boot, which is the 256-bit SHA-256 hash value of the public key
- 256-bit KEK for EXIP to decrypt the Key Blob containing DEK, EXIP0 KEK

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- 256-bit FMK for the key manager

For specific information about these security-related data, please refer to the relevant chapters of OTP and other security modules.

58.6 True Random Number Generator RNG

This product contains a true random number generator. The entropy source is an internal analog noise source, which is very sensitive to on-chip environments such as temperature, voltage and clock, and generates 512-bit entropy. RNG supports error detection.

58.7 Security Manager PSEC

This product contains a security manager PSEC.

The main function of the security manager PSEC is to configure the chip security status according to the current lifecycle of the chip.

The security manager PSEC can formulate security rules and detect various security events. When a security violation event occurs, different measures are taken to protect users' sensitive information, such as keys.

The security manager supports monitoring the following events:

- OSC24M clock failure
- CPU0 enters debug mode
- VPMC power glitch or OSC24M clock glitch, from power domain monitor PMON
- VPMC brownout
- JTAG port connection
- JTAG TCK pin activity

When a security event occurs, the chip security status will switch to FAIL, which will lock each key data path of the key manager KEYM, and lock the key and sensitive data areas of OTP and their shadow registers to be unreadable.

58.8 Security Monitor PMON

This product contains a security monitor PMON. The function of the security monitor is to monitor the power and clock of the power domain. When the security monitor detects abnormal power glitches on VPMC, clock glitches on OSC24M, or OSC24M stops working, it will alarm the security manager PSEC.

58.9 BOOT ROM

This product integrates BOOT ROM, and the BOOT ROM code is always executed first after system reset. The physical characteristics of read-only memory ensure that the BOOT ROM code is almost impossible to tamper with, thus constructing the initial trusted secure execution environment.

BOOT ROM supports the following security functions:

- Execute code directly from encrypted serial NOR Flash through EXIP and XPI.
- Secure boot: Before jumping to the user software image, it can first verify that the source of the image is legal and trusted. Support ECDSA or SM2 signature verification
- Support loading encrypted software images, support AES or SM4 encryption algorithms

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- Flash loader supports burning encrypted firmware

For details of the security functions supported by BOOT ROM, please refer to the relevant chapters of ROM.

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59 Secure Data Processor (SDP)

This chapter describes the main features and functions of the Secure Data Processor (SDP).

59.1 Feature Summary

The main features of the SDP are as follows:

- Supports AES-128/256, ECB and CBC modes
- Supports SM4
- Supports SHA-1/256
- Supports CRC-32
- Supports SM3
- Built-in DMA, supports data copy, block copy and data filling
- Supports loading protected keys from the key manager, or using keys written by users to the SDP key memory
- Read protection for key registers

59.2 Function Description

The Secure Data Processor SDP is a security operation engine that implements data encryption/decryption and data hash value calculation. As a bus master, SDP can independently access on-chip storage peripherals; it can independently read command descriptors, fetch data to be processed, and write back processed data.

59.2.1 Command Descriptor

The operation of SDP is controlled by command descriptors, which can be prepared by users in memory. SDP independently reads command descriptors from memory, executes tasks according to the descriptor configuration, and feeds back results.

Tasks that SDP can execute via command descriptors are summarized as follows:

- AES-128/256 ECB mode: encrypt plaintext of specified length or decrypt ciphertext of specified length
- AES-128/256 CBC mode: encrypt plaintext of specified length or decrypt ciphertext of specified length
- SHA-1/256: calculate Hash value for data of specified length
- CRC-32: calculate CRC32 value for data of specified length
- Copy data of specified length from source address to destination address
- Fill the target address with data of specified length and specified pattern

Users can specify the address of the command descriptor in memory through the CMDPTR register.

The format of the command descriptor is shown in Table 220:

Address	Name	Description
CMDPTR + 0x00	NXTCMD	Next command descriptor pointer
CMDPTR + 0x04	PKTCTL	Command control word, 32-bit
CMDPTR + 0x08	PKTSRC	Source data address, 32-bit
CMDPTR + 0x0C	PKTDST	Destination data address, 32-bit

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Address	Name	Description
CMDPTR + 0x10	BUFSIZE	Data length (in bytes), 32-bit
CMDPTR + 0x14	RESERVED	Reserved
CMDPTR + 0x18	RESERVED	Reserved
CMDPTR + 0x1C	RESERVED	Reserved

Table 220: SDP Command Descriptor Format

NXTCMD points to the next command descriptor. If PKTCTL[3] (CHAIN bit) is set to 1, SDP will continue to execute the command descriptor pointed to by NXTCMD after completing the current command.

PKTCTL is the control word of the command descriptor, which configures the SDP task, described as follows:

- PKTCTL[31:24], PKTTAG: packet tag, users can assign different tags to each packet;
- PKTCTL[6], CIPHIV: set to 1 when using AES-CBC mode, prompting AES to load the Initial Vector;
- PKTCTL[5], HASFNL: set to 1 to prompt the HASH engine that this is the end of the data to be processed;
- PKTCTL[4], HASINI: set to 1 to prompt the HASH engine that this is the start of the data to be processed;
- PKTCTL[3], CHAIN: set to 1 to prompt SDP to fetch the next command descriptor from NXTCMD after executing the current command;
- PKTCTL[2], DCRSEMA: set to 1 to prompt SDP to decrement PKTCNT [CNTVAL] by 1 after executing the current command. When PKTCNT [CNTVAL] = 0, SDP will stop working;
- PKTCTL[1], PKTINT: when set to 1, SDP will generate an interrupt request after executing the current command;

PKTSRC is the source data address, i.e., the read address of the data to be processed by SDP. PKTSRC should point to the plaintext to be encrypted, the ciphertext to be decrypted, or the data to be compressed by the HASH function. When SDP performs data copy, PKTSRC points to the data to be copied.

PKTDST is the destination data address, i.e., the storage address of the data processed by SDP. PKTDST should point to the encrypted ciphertext or decrypted plaintext. When SDP performs data copy, PKTDST points to the target address of the copy.

BUFSIZE is the data length (in bytes), i.e., the length of plaintext to be encrypted by SDP, the length of ciphertext to be decrypted, the length of the data block for which Hash value or CRC value needs to be calculated, or the length of the data block for data copy.

Before using SDP for operation, users need to:

- Configure SDPCR and MODCTRL registers as required
- Prepare command descriptors, note that:
  - Set PKTCTL[2] (DCRSEMA) to 1 to prompt PKTCNT [CNTVAL] to decrement by 1 after the command is executed
  - If multiple commands need to be executed, set PKTCTL[3] (CHAIN bit) to 1 for all command descriptors except the last one, and fill NXTCMD with the address of the next command
  - If an interrupt request is required after the command is executed, set PKTCTL[1] (PKTINT) of the command descriptor to 1
- Write the address of the command descriptor to the CMDPTR register
- Write the number of command descriptors to be executed to PKTCNT [CNTINCR]. Writing to PKTCNT

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[CNTINCR] will update PKTCNT [CNTVAL] = PKTCNT [CNTVAL] + PKTCNT [CNTINCR], and PKTCNT [CNTVAL] represents the number of commands to be executed

- Write 1 if only one command needs to be executed
- Write n if n consecutive commands need to be executed

59.2.2 AES Encryption/Decryption Engine

SDP supports AES encryption and decryption engine. AES is the abbreviation of Advanced Encryption Standard, which was published in FIPS PUB 197 by the National Institute of Standards and Technology (NIST) of the United States on November 26, 2001. Users can refer to official documents for details of AES.

Users can enable the AES encryption/decryption engine by setting SDPCR[CIPHEN] bit to 1.

The AES engine supports AES-128 and AES-256, i.e., AES key lengths of 128 bits and 256 bits.

Users can configure the AES key length through the MODCTRL[AESALG] bit:

- 4'b0000: AES-128, key length is 128 bits
- 4'b0001: AES-256, key length is 256 bits

The AES engine supports encryption, which encrypts plaintext into ciphertext through a key, and decryption, which restores ciphertext to plaintext through a key. Users can configure the AES engine to perform encryption or decryption through the MODCTRL [AESDIR] bit:

- MODCTRL [AESDIR] = 1'b0: AES engine performs encryption operation
- MODCTRL [AESDIR] = 1'b1: AES engine performs decryption operation

The AES engine supports ECB and CBC operating modes.

ECB (Electronic Codebook) mode is the most basic AES encryption mode. Before encryption, all data is divided into several blocks according to the data block size (AES data block length is 16 bytes, i.e., 128 bits), and each block is encrypted separately with the same key, the same applies to decryption. It is usually not recommended for users to use ECB mode to encrypt data longer than one AES data block.

CBC (Cipher-block chaining) mode: For each data block to be encrypted, it is XORed with the ciphertext of the previous data block before being encrypted by the encryptor. The first plaintext block is XORed with a data block called Initial Vector. The decryption process is similar: after the first ciphertext block is decrypted with the key, it is XORed with the initial vector data block to recover the first plaintext block, and subsequent ciphertext blocks are decrypted and XORed with the ciphertext of the previous data block to recover the plaintext.

Users can configure the AES operating mode through the MODCTRL[AESMOD] bit:

- MODCTRL[AESMOD] = 4'b0000: AES operating mode is ECB mode
- MODCTRL[AESMOD] = 4'b0001: AES operating mode is CBC mode

When using AES-CBC mode, users need to set PKTCTL[6] (CIPHIV bit) of the first command descriptor to 1 to prompt the AES engine to load the Initial Vector, and at the same time, write the Initial Vector to the CIPHIV0 CIPHIV3 registers.

59.2.3 AES Key Configuration

As a publicly released standard algorithm, the data processing steps of AES encryption and decryption are publicly known. The security of AES algorithm does not depend on the algorithm itself, but on the security of the key.

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The AES engine of SDP adopts a series of designs to protect the AES encryption and decryption keys from leakage.

The AES engine key has the following options:

- Load key from SDP internal memory KEYRAM: KEYRAM has a capacity of 256 bytes, which can store 16 128-bit AES-128 keys or 8 256-bit AES-256 keys
- Load key from the dedicated key path of the key manager KEYM: available keys include MK, SK0 SK3

The internal memory KEYRAM of SDP is write-only and unreadable. Users can initialize the internal memory KEYRAM of SDP through the KEYADDR and KEYDAT registers as follows:

- Specify the internal address index of KEYRAM by writing the KEYADDR [INDEX] bit field.
  - KEYADDR [INDEX] = 0: points to the first 128-bit key of KEYRAM
  - KEYADDR [INDEX] = 1: points to the second 128-bit key of KEYRAM
  - ……
  - KEYADDR [INDEX] = 15: points to the 16th 128-bit key of KEYRAM
- Write the key through KEYDAT, which is a 32-bit register
  - First write to KEYDAT: write key[0:31]
  - Second write to KEYDAT: write key[32:63]
  - Third write to KEYDAT: write key[64:95]
  - Fourth write to KEYDAT: write key[96:127]

Obviously, if users want to use a 256-bit AES-256 key, they need to first set KEYADDR [INDEX] to an even number between 0 15, write the first 128 bits of the key according to the above steps, then add 1 to the current KEYADDR [INDEX], and write to the KEYDAT register 4 more times to write the last 128 bits of the key (key[128:255])

After the key initialization is completed, users can select the key through the MODCTRL[AESKS] bit field:

When the AES engine is set to AES-128:

- MODCTRL[AESKS] = 6'h00: select the first 128-bit key of KEYRAM, i.e., KEYRAM [0:127]
- MODCTRL[AESKS] = 6'h01: select the second 128-bit key of KEYRAM, i.e., KEYRAM [128:255]
- ……
- MODCTRL[AESKS] = 6'h0E: select the 15th 128-bit key of KEYRAM, i.e., KEYRAM [1792:1919]
- MODCTRL[AESKS] = 6'h0F: select the 16th 128-bit key of KEYRAM, i.e., KEYRAM [1920:2047]
- MODCTRL[AESKS] = 6'h20: select SK0[0:127] from KEYM
- MODCTRL[AESKS] = 6'h21: select SK0[128:255] from KEYM
- MODCTRL[AESKS] = 6'h22: select SK1[0:127] from KEYM
- MODCTRL[AESKS] = 6'h23: select SK1[128:255] from KEYM
- MODCTRL[AESKS] = 6'h24: select SK2[0:127] from KEYM
- MODCTRL[AESKS] = 6'h25: select SK2[128:255] from KEYM
- MODCTRL[AESKS] = 6'h26: select SK3[0:127] from KEYM
- MODCTRL[AESKS] = 6'h27: select SK3[128:255] from KEYM
- MODCTRL[AESKS] = 6'h2E: select MK[0:127] from KEYM
- MODCTRL[AESKS] = 6'h2F: select MK[128:255] from KEYM

When the AES engine is set to AES-256:

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- MODCTRL[AESKS] = 6'h00: select the first 256-bit key of KEYRAM, i.e., KEYRAM [0:255]
- MODCTRL[AESKS] = 6'h01: Invalid
- ……
- MODCTRL[AESKS] = 6'h0E: select the 8th 256-bit key of KEYRAM, i.e., KEYRAM [1792:2048]
- MODCTRL[AESKS] = 6'h0F: Invalid
- MODCTRL[AESKS] = 6'h20: select SK0[0:255] from KEYM
- MODCTRL[AESKS] = 6'h21: Invalid
- MODCTRL[AESKS] = 6'h22: select SK1[0:255] from KEYM
- MODCTRL[AESKS] = 6'h23: Invalid
- MODCTRL[AESKS] = 6'h24: select SK2[0:255] from KEYM
- MODCTRL[AESKS] = 6'h25: Invalid
- MODCTRL[AESKS] = 6'h26: select SK3[0:255] from KEYM
- MODCTRL[AESKS] = 6'h27: Invalid
- MODCTRL[AESKS] = 6'h2E: select MK[0:255] from KEYM
- MODCTRL[AESKS] = 6'h2F: Invalid

59.2.4 SM4 Encryption/Decryption Engine

SDP supports the national cryptographic SM4 block encryption/decryption algorithm, which complies with the international standard ISO/IEC 18033-3:2010/AMD1:2021 *Information technology —Security techniques — Encryption algorithms —Part 3: Block ciphers —Amendment 1: SM4*.

59.2.5 HASH Module

SDP supports a HASH module.

HASH is a function that converts input data of any length into a fixed-length output through a HASH algorithm, and the output is called a HASH value. This conversion is a compression mapping, that is, the space of HASH values is usually much smaller than the input space, and different inputs may be hashed into the same output, so it is theoretically impossible to determine the unique input value from the hash value. Simply put, it is a function that compresses a message of any length into a message digest of a fixed length, so the HASH value of a piece of information is usually also called the digest of the information.

Although theoretically there is no one-to-one correspondence between a piece of data and its HASH value, that is, there are always different data with the same HASH value, which is called a collision. However, cryptography theory holds that modern HASH algorithms have the following characteristics:

- The original data cannot be derived reversely from the HASH value, i.e., unidirectionality
- Very sensitive to input data, the slightest modification of the original data will lead to a completely different HASH value
- Although the probability of collision is not theoretically 0, mathematically, it is considered to be approximately 0.

Therefore, the data digest (HASH value) obtained by the HASH algorithm can be used to verify the integrity of data, and many applications have been derived in the field of information security.

The HASH module of SDP supports SHA-1, SHA-256 and SM3.

SHA-1 (Secure Hash Algorithm 1) is designed by the National Security Agency of the United States and

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published as a Federal Information Processing Standard (FIPS) by the National Institute of Standards and Technology (NIST). SHA-1 supports input data with a length not exceeding $(2^{64} - 1)$ bits, and can generate a 160-bit (20-byte) HASH value.

SHA-2 (Secure Hash Algorithm 2) is developed by the National Security Agency of the United States and published by NIST in 2001, it is the successor algorithm of SHA-1. SHA-256 is one of the SHA-2 algorithms, which supports input data with a length not exceeding $(2^{64} - 1)$ bits, and can generate a 256-bit (32-byte) HASH value.

SM3 cryptographic hash algorithm is a Chinese commercial cryptographic hash algorithm standard issued by the State Cryptography Administration of China, which is applicable to digital signature and verification in commercial cryptographic applications.

Users can enable the HASH module by setting SDPCR[HASHEN] bit to 1.

Users can select the HASH algorithm through the MODCTRL[HASALG] bit:

- MODCTRL[HASALG] = 4'b0000: select SHA-1
- MODCTRL[HASALG] = 4'b0010: select SHA-256

When using the HASH module, users must configure the PKTCTL word of the command descriptor as required:

- If the data processed by this command descriptor includes the end of the HASH algorithm input data, set PKTCTL[5] (HASFNL) to 1
- If the data processed by this command descriptor includes the start of the HASH algorithm input data, set PKTCTL[4] (HASINI) to 1
- If performing HASH algorithm on continuous data in a memory segment, users can set both PKTCTL[5] and PKTCTL[4] to 1, and calculate the HASH value of a piece of continuous data with one command descriptor.

59.2.6 Data Copy and Data Filling

SDP supports data copy and data filling functions for memory segments.

Users can enable the data copy function by setting SDPCR[MCPEN] bit to 1. In this case, the SDP function is similar to DMA, and by configuring the command descriptor, data of any length can be copied from the memory segment specified by the source address to the memory segment specified by the destination address.

Users can enable the data filling function by setting SDPCR[CONFEN] bit to 1. In this case, users configure the command descriptor, write the data to be filled into the source address word PKTSRC, and SDP will fill the area starting from the destination address word PKTDST with a length of BUFSIZE with the value of PKTSRC.

59.2.7 Data Reordering

SDP supports data reordering for input data read from the source address, output data written to the destination address, and keys, to adapt to the data ordering requirements of different systems.

The MODCTRL[DINSWP] bit field configures the reordering method of input data:

- 2'b00: The bytes of the input 32-bit data are [Byte3, Byte2, Byte1, Byte0] from high to low, remain unchanged
- 2'b01: The bytes of the input 32-bit data are [Byte3, Byte2, Byte1, Byte0] from high to low, reordered to

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[Byte2, Byte3, Byte0, Byte1]

- 2'b10: The bytes of the input 32-bit data are [Byte3, Byte2, Byte1, Byte0] from high to low, reordered to [Byte1, Byte0, Byte3, Byte2]
- 2'b11: The bytes of the input 32-bit data are [Byte3, Byte2, Byte1, Byte0] from high to low, reordered to [Byte0, Byte1, Byte2, Byte3]

The MODCTRL[DOUTSWP] bit field configures the reordering method of output data:

- 2'b00: The bytes of the output 32-bit data are [Byte3, Byte2, Byte1, Byte0] from high to low, remain unchanged
- 2'b01: The bytes of the output 32-bit data are [Byte3, Byte2, Byte1, Byte0] from high to low, reordered to [Byte2, Byte3, Byte0, Byte1]
- 2'b10: The bytes of the output 32-bit data are [Byte3, Byte2, Byte1, Byte0] from high to low, reordered to [Byte1, Byte0, Byte3, Byte2]
- 2'b11: The bytes of the output 32-bit data are [Byte3, Byte2, Byte1, Byte0] from high to low, reordered to [Byte0, Byte1, Byte2, Byte3]

The MODCTRL[KEYSWP] bit field configures the reordering method of key data:

- 2'b00: The bytes of the key 32-bit data are [Byte3, Byte2, Byte1, Byte0] from high to low, remain unchanged
- 2'b01: The bytes of the key 32-bit data are [Byte3, Byte2, Byte1, Byte0] from high to low, reordered to [Byte2, Byte3, Byte0, Byte1]
- 2'b10: The bytes of the key 32-bit data are [Byte3, Byte2, Byte1, Byte0] from high to low, reordered to [Byte1, Byte0, Byte3, Byte2]
- 2'b11: The bytes of the key 32-bit data are [Byte3, Byte2, Byte1, Byte0] from high to low, reordered to [Byte0, Byte1, Byte2, Byte3]

59.3 SDP Registers

59.3.1 Register Overview

The register list of SDP is as follows:

SDP base address: 0xF3040000

Offset	Register Name	Description	Reset Value
0x0000	SDPCR	SDP control register	0x30000000
0x0004	MODCTRL	Mod control register.	0x00000000
0x0008	PKTCNT	packet counter registers.	0x00000000
0x000C	STA	Status Registers	0x00000000
0x0010	KEYADDR	Key Address	0x00000040
0x0014	KEYDAT	Key Data	0x00000030
0x0018	CIPHIV[CIPHIV0]	Cipher Initializtion Vector 0	0x00000000
0x001C	CIPHIV[CIPHIV1]	Cipher Initializtion Vector 1	0x00000000
0x0020	CIPHIV[CIPHIV2]	Cipher Initializtion Vector 2	0x00000000
0x0024	CIPHIV[CIPHIV3]	Cipher Initializtion Vector 3	0x00000000

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Offset	Register Name	Description	Reset Value
0x0028	HASWRD[HASWRD0]	Hash Data Word 0	0x00000030
0x002C	HASWRD[HASWRD1]	Hash Data Word 1	0x00000030
0x0030	HASWRD[HASWRD2]	Hash Data Word 2	0x00000030
0x0034	HASWRD[HASWRD3]	Hash Data Word 3	0x00000030
0x0038	HASWRD[HASWRD4]	Hash Data Word 4	0x00000030
0x003C	HASWRD[HASWRD5]	Hash Data Word 5	0x00000030
0x0040	HASWRD[HASWRD6]	Hash Data Word 6	0x00000030
0x0044	HASWRD[HASWRD7]	Hash Data Word 7	0x00000030
0x0048	CMDPTR	Command Pointer	0x00000000
0x004C	NPKTPTR	Next Packet Address Pointer	0x00000000
0x0050	PKTCTL	Packet Control Registers	0x00000000
0x0054	PKTSRC	Packet Memory Source Address	0x00000000
0x0058	PKTDST	Packet Memory Destination Address	0x00000000
0x005C	PKTBUF	Packet buffer size.	0x00000000

Table 221: SDP Register List

The detailed description of SDP registers is as follows:

59.3.2 SDPCR (0x0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SFTRST	CLKGAT	CIPDIS	HASDIS	RSVD				CIPHEN	HASHEN	MCPEN	CONFEN	DCRPDI	RSVD	TSTPKT0IRQ	RSVD								RDSCEN	RSVD							INTEN
RW	RW	RO	RO	N/A				RW	RW	RW	RW	RW	N/A	RW	N/A								RW	N/A							RW
0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x	0	x	x	x	x	x	x	x	0

SDPCR [31:0]

Bits	Name	Description
31	SFTRST	soft reset. Write 1 then 0, to reset the SDP block.
30	CLKGAT	Clock Gate for the SDP main logic. Write to 1 will clock gate for most logic of the SDP block, dynamic power saving when not use SDP block.
29	CIPDIS	Cipher Disable, read the info, whether the CIPHER features is besing disable in this chip or not. 1, Cipher is disabled in this chip. 0, Cipher is enabled in this chip.

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Bits	Name	Description
28	HASDIS	HASH Disable, read the info, whether the HASH features is besing disable in this chip or not. 1, HASH is disabled in this chip. 0, HASH is enabled in this chip.
23	CIPHEN	Cipher Enablement, controlled by SW. 1, Cipher is Enabled. 0, Cipher is Disabled.
22	HASHEN	HASH Enablement, controlled by SW. 1, HASH is Enabled. 0, HASH is Disabled.
21	MCPEN	Memory Copy Enablement, controlled by SW. 1, Memory copy is Enabled. 0, Memory copy is Disabled.
20	CONFEN	Constant Fill to memory, controlled by SW. 1, Constant fill is Enabled. 0, Constant fill is Disabled.
19	DCRPDI	Decryption Disable bit, Write to 1 to disable the decryption.
17	TSTPKT0IRQ	Test purpose for interrupt when Packet counter reachs ”0”, but CHAIN=1 in the current packet.
8	RDSCEN	when set to ”1”, the 1st data packet descriptor loacted in the register(CMDPTR, NPKTPTR, ...) when set to ”0”, the 1st data packet descriptor loacted in the memeory(pointed by CMDPTR)
0	INTEN	Interrupt Enablement, controlled by SW. 1, SDP interrupt is enabled. 0, SDP interrupt is disabled.

SDPCR Bits

59.3.3 MODCTRL (0x4)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
AESALG				AESMOD				AESKS						RSVD	AESDIR	HASALG				RSVD	HASCHK	HASOUT	RSVD	RSVD		DINSWP		DOUTSWP		KEYSWP	
RW				RW				RW						N/A	RW	RW				N/A	RW	RW	N/A	N/A		RW		RW		RW	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	0	0	0	0	0	0	0	0	0	0	0

MODCTRL [31:0]

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Bits	Name	Description
31-28	AESALG	AES algorithem selection. 0x0 = AES 128; 0x1 = AES 256; 0x8 = SM4； Others, reserved.
27-24	AESMOD	AES mode selection. 0x0 = ECB; 0x1 = CBC; Others, reserved.

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Bits	Name	Description
23-18	AESKS	AES Key Selection. These regisgers are being used to select the AES key that stored in the 16x128 key ram of the SDP, or select the key from the OTP. Detail as following: 0x00: key from the 16x128, this is the key read address, valid for AES128; AES256 will use 128 bit from this address and 128 bit key from next address as 256 bit AES key. 0x01: key from the 16x128, this is the key read address, valid for AES128, not valid for AES286. .... 0x0E: key from the 16x128, this is the key read address, valid for AES128; AES256 will use 128 from this add and 128 from next add for the AES key. 0x0F: key from the 16x128, this is the key read address, valid for AES128, not valid for AES286. 0x20: kman_sk0[127:0] from the key manager for AES128; AES256 will use kman_sk0[255:0] as AES key. 0x21: kman_sk0[255:128] from the key manager for AES128; not valid for AES256. 0x22: kman_sk1[127:0] from the key manager for AES128; AES256 will use kman_sk1[255:0] as AES key. 0x23: kman_sk1[255:128] from the key manager for AES128; not valid for AES256. 0x24: kman_sk2[127:0] from the key manager for AES128; AES256 will use kman_sk2[255:0] as AES key. 0x25: kman_sk2[255:128] from the key manager for AES128; not valid for AES256. 0x26: kman_sk3[127:0] from the key manager for AES128; AES256 will use kman_sk3[255:0] as AES key. 0x27: kman_sk3[255:128] from the key manager for AES128; not valid for AES256. 0x30: exip0_key[127:0] from OTP for AES128; AES256 will use exip0_key[255:0] as AES key. 0x31: exip0_key[255:128] from OTP for AES128; not valid for AES256. 0x32: exip1_key[127:0] from OTP for AES128; AES256 will use exip1_key[255:0] as AES key. 0x33: exip1_key[255:128] from OTP for AES128; not valid for AES256. Other values, reserved.

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Bits	Name	Description
16	AESDIR	AES direction 1x1, AES Decryption 1x0, AES Encryption.
15-12	HASALG	HASH Algorithem selection. 0x0 SHA1 — 0x1 CRC32 — 0x2 SHA256 —
10	HASCHK	HASH Check Enable Bit. 1x1, HASH check need, hash result will compare with the HASHRSLT 0-7 registers; 1x0, HASH check is not enabled, HASHRSLT0-7 store the HASH result. For SHA1, will use HASHRSLT0-3 words, and HASH 256 will use HASH0-7 words.
9	HASOUT	When hashing is enabled, this bit controls the input or output data of the AES engine is hashed. 0 INPUT HASH 1 OUTPUT HASH
5-4	DINSWP	Decide whether the SDP byteswaps the input data (big-endian data); When all bits are set, the data is assumed to be in the big-endian format
3-2	DOUTSWP	Decide whether the SDP byteswaps the output data (big-endian data); When all bits are set, the data is assumed to be in the big-endian format
1-0	KEYSWP	Decide whether the SDP byteswaps the Key (big-endian data). When all bits are set, the data is assumed to be in the big-endian format

MODCTRL Bits

59.3.4 PKTCNT (0x8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD	RSVD	RSVD						CNTVAL								RSVD								CNTINCR							
N/A	N/A	N/A						RO								N/A								RW							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PKTCNT [31:0]

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Bits	Name	Description
23-16	CNTVAL	This read-only field shows the current (instantaneous) value of the packet counter
7-0	CNTINCR	The value written to this field is added to the spacket count.

PKTCNT Bits

59.3.5 STA (0xC)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TAG								IRQ	RSVD		CHN1PKT0	AESBSY	HASBSY	PKTCNT0	PKTDON	RSVD										ERRSET	ERRPKT	ERRSRC	ERRDST	ERRHAS	ERRCHAIN
RO								W1C	N/A		W1C	RO	RO	W1C	W1C	N/A										W1C	W1C	W1C	W1C	W1C	W1C
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

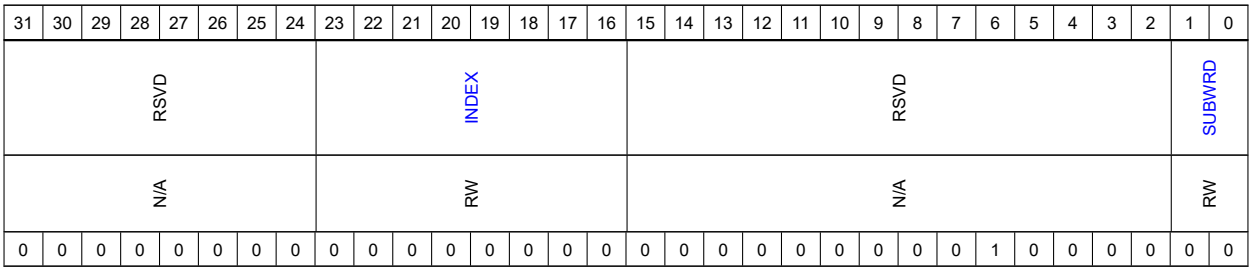
STA [31:0]

Bits	Name	Description
31-24	TAG	packet tag.
23	IRQ	interrupt Request, requested when error happen, or when packet processing done, packet counter reach to zero.
20	CHN1PKT0	the chain buffer ”chain” bit is ”1”, while packet counter is ”0”, now, waiting for new buffer data.
19	AESBSY	AES Busy
18	HASBSY	Hashing Busy
17	PKTCNT0	Packet Counter registers reachs to ZERO now.
16	PKTDON	Packet processing done, will trigger this itnerrrupt when the ”PKTINT” bit set in the packet control word.
5	ERRSET	Working mode setup error.
4	ERRPKT	Packet head access error, or status update error.
3	ERRSRC	Source Buffer Access Error
2	ERRDST	Destination Buffer Error
1	ERRHAS	Hashing Check Error
0	ERRCHAIN	buffer chain error happen when packet’s CHAIN bit=0, but the Packet counter is still not zero.

STA Bits

59.3.6 KEYADDR (0x10)

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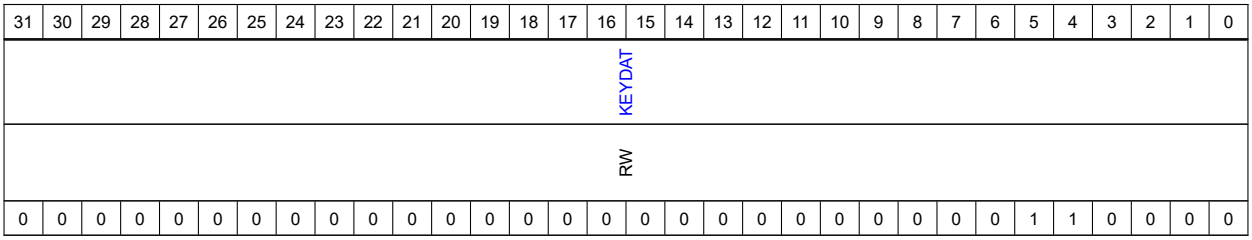


KEYADDR [31:0]

Bits	Name	Description
23-16	INDEX	To write a key to the SDP KEY RAM, the software must first write the desired key index/subword to this register. Key index pointer. The valid indices are 0-[number_keys]. In the SDP, there is a 16x128 key ram can store 16 AES128 keys or 8 AES 256 Keys; this index is for addressing the 16 128-bit key addresses.
1-0	SUBWRD	Key subword pointer. The valid indices are 0-3. After each write to the key data register, this field increments; To write a key, the software must first write the desired key index/subword to this register.

KEYADDR Bits

59.3.7 KEYDAT (0x14)



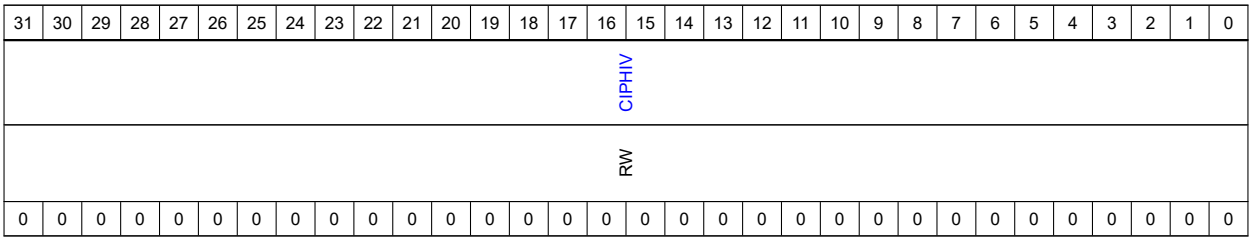
KEYDAT [31:0]

Bits	Name	Description
31-0	KEYDAT	This register provides the write access to the key/key subword specified by the key index register. Writing this location updates the selected subword for the key located at the index specified by the key index register. The write also triggers the SUBWORD field of the KEY register to increment to the next higher word in the key

KEYDAT Bits

59.3.8 CIPHIV (0x18 + 0x4 * n)

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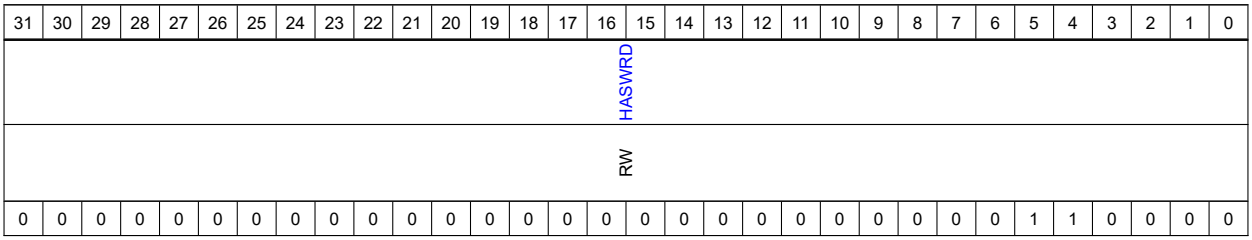


CIPHIV [31:0]

Bits	Name	Description
31-0	CIPHIV	cipher initialization vector.

CIPHIV Bits

59.3.9 HASWRD (0x28 + 0x4 * n)

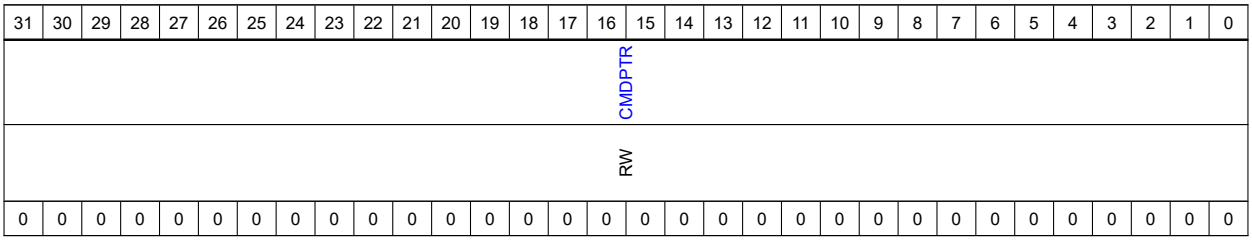


HASWRD [31:0]

Bits	Name	Description
31-0	HASWRD	Hash Data Word x - HASH result bit; will store the expected hash result bit if hash check enabled; when hash check is not enabled, the hash engine will store the final hash result[31:0] here. If CRC mode enabled, this work store the CRC expected result if the check enabled, or store the final calcuated CRC result.

HASWRD Bits

59.3.10 CMDPTR (0x48)



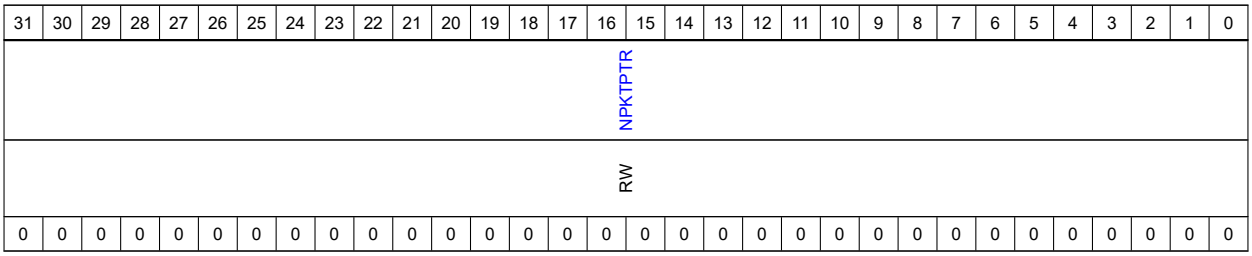
CMDPTR [31:0]

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Bits	Name	Description
31-0	CMDPTR	current command addresses the register points to the multiword descriptor that is to be executed (or is currently being executed)

CMDPTR Bits

59.3.11 NPKTPTR (0x4C)

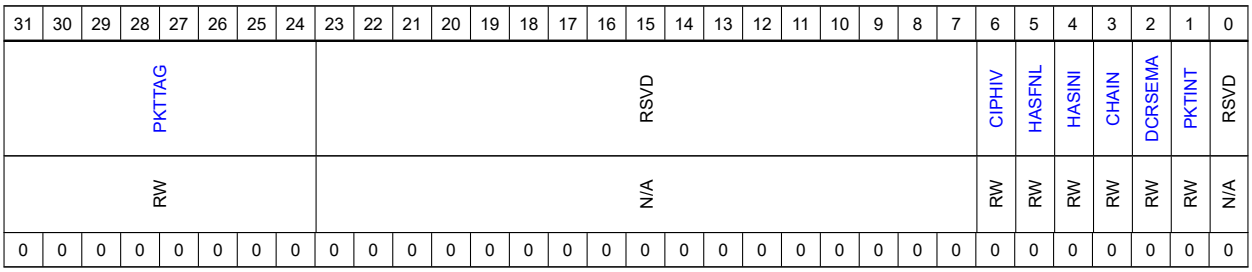


NPKTPTR [31:0]

Bits	Name	Description
31-0	NPKTPTR	Next Packet Address Pointer

NPKTPTR Bits

59.3.12 PKTCTL (0x50)



PKTCTL [31:0]

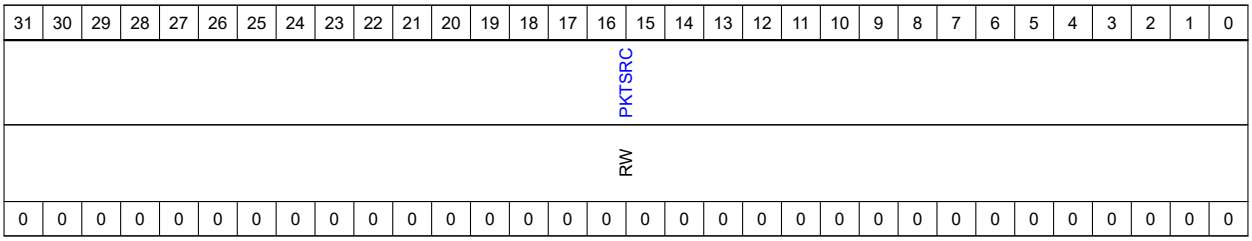
Bits	Name	Description
31-24	PKTTAG	packet tag
6	CIPHIV	Load Initial Vector for the AES in this packet.
5	HASFNL	Hash Termination packet
4	HASINI	Hash Initialization packat
3	CHAIN	whether the next command pointer register must be loaded into the channel's current descriptor pointer.
2	DCRSEMA	whether the channel's semaphore must be decremented at the end of the current operation. When the semaphore reaches a value of zero, no more operations are issued from the channel.

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Bits	Name	Description
1	PKTINT	Reflects whether the channel must issue an interrupt upon the completion of the packet

PKTCTL Bits

59.3.13 PKTSRC (0x54)

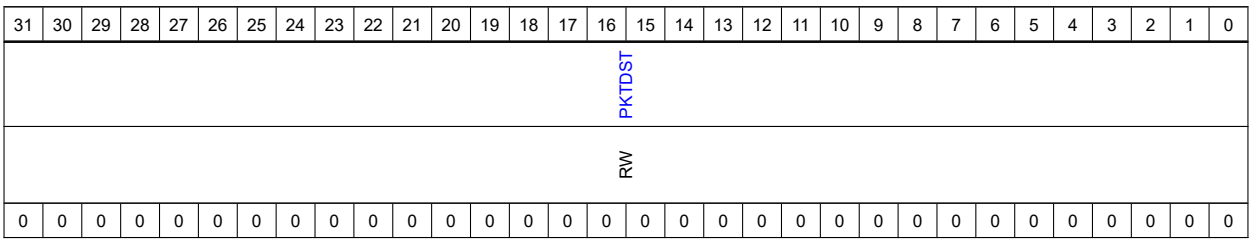


PKTSRC [31:0]

Bits	Name	Description
31-0	PKTSRC	Packet Memory Source Address

PKTSRC Bits

59.3.14 PKTDST (0x58)

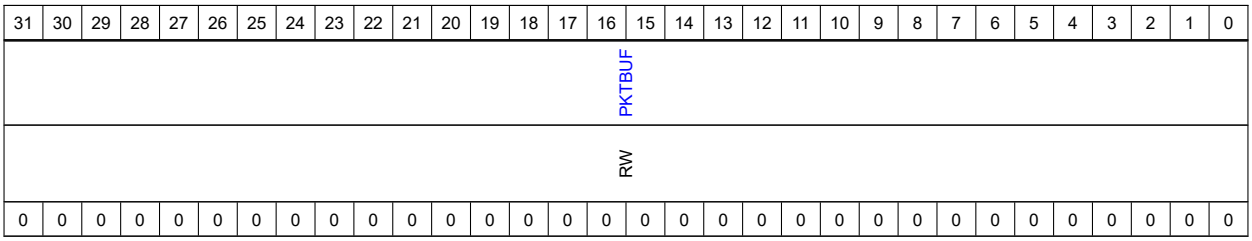


PKTDST [31:0]

Bits	Name	Description
31-0	PKTDST	Packet Memory Destination Address

PKTDST Bits

59.3.15 PKTBUF (0x5C)



PKTBUF [31:0]

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Bits	Name	Description
31-0	PKTBUF	

PKTBUF Bits

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60 Online Decryption Engine EXIP

This chapter describes the main features and functions of the online decryption engine EXIP.

60.1 Feature Summary

The online decryption engine EXIP can perform real-time decryption on encrypted external NOR Flash. The main features of EXIP are as follows:

- Supports AES-128 CTR mode decryption
- Supports 4 regions, each can be encrypted with independent keys
- Supports encapsulating sensitive information such as data encryption keys via Key Blob (compliant with RFC3394)
- Supports hardware Key Blob unwrapping (compliant with RFC3394)
- Supports reading KEK (Key Encryption Key) for Key Blob unwrapping from OTP

60.2 Functional Description

On this product, users can connect serial NOR Flash of different brands and capacities via the XPI controller. The XPI controller supports direct mapping of external memory to the system address space, so master devices such as processors and DMA can directly access external memory, load data or execute code just like accessing on-chip memory. This access method that does not require copying external memory data to RAM before execution is called Execution In Place (XIP). The online decryption engine EXIP is tightly coupled with the XPI controller, supporting real-time decryption of data while executing in place from external serial NOR Flash. Therefore, users can achieve data and code protection as long as they encrypt the data and code on external NOR Flash with the AES-128 CTR algorithm.

60.2.1 EXIP Regions, Keys and Counters

EXIP supports 4 regions, each can be configured with independent keys and counters.

Each region can be configured with a start address and an end address, the area between the two addresses is the encrypted region. EXIP is tightly coupled with the XPI controller: when a bus master (processor, DMA, etc.) initiates an access to external memory via XPI, EXIP will monitor the accessed system address. If the address falls within an encrypted region, decryption will be performed.

RGNx_DSCR_W0 [START] stores the start address of region x, and RGNx_DSCR_W1 [END] stores the end address of region x. EXIP requires the encrypted region length to be in 1KB units, and the start address must be 1KB aligned. If the encrypted region length is x KB, the end address should be START + x*1024 - 1. Therefore, only the upper 22 bits of START are valid, the lower 10 bits should be 0; only the upper 22 bits of END are valid, the lower 10 bits should be 1.

EXIP supports AES-128 CTR mode decryption with a 128-bit key, which is stored in registers RGNx_KEY0, RGNx_KEY1, RGNx_KEY2 and RGNx_KEY3.

The counter length is the same as the AES data block length, which is 128 bits. AES CTR mode requires that the counter value corresponding to each data block must not be repeated for data encrypted with the same key. For each 128-bit (16-byte) data block of EXIP, its counter consists of the 32-bit system address of the data block and a 64-bit NONCE, which is stored in RGNx_CTR0 and RGNx_CTR1. The counter composition is as

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follows:

- Counter[127:96] = CTR0[31:0]
- Counter[95:64] = CTR1[31:0]
- Counter[63:32] = CTR0[31:0] XOR CTR1[31:0]
- Counter[31:0] = Sysaddr[31:0] AND 0xFFFFFFF0

Where:

- CTR0 is the lower 32 bits of the 64-bit NONCE
- CTR1 is the upper 32 bits of the 64-bit NONCE
- SysAddr is the system mapped address of the ciphertext block. The ciphertext block length is 128 bits / 16 bytes, one ciphertext block corresponds to one counter value, so only the upper 28 bits of the address are used for the counter value, with the lower 4 bits of the address set to 0.

60.2.2 EXIP Key Wrap and Key Unwrap

EXIP uses the key wrap and key unwrap algorithms defined in RFC3394 to protect EXIP decryption keys and related sensitive data. For distinction, the key used by EXIP for key wrap/unwrap is called KEK (Key-Encryption Key), and the key used for decrypting data/code on external memory is called DEK (Data Encryption Key).

For each EXIP region, the DEK and related data to be wrapped include 5 64-bit data blocks, totaling 40 bytes, as summarized below:

Address Offset	Description
0x00	DEKx[31:0], partial key stored in RGNx_KEY0
0x04	DEKx[63:32], partial key stored in RGNx_KEY1
0x08	DEKx[95:64], partial key stored in RGNx_KEY2
0x0C	DEKx[127:96], partial key stored in RGNx_KEY3
0x10	NONCE[31:0], partial NONCE stored in RGNx_CTR0
0x14	NONCE[63:32], partial NONCE stored in RGNx_CTR1
0x18	START, i.e. RGNx_RGN_SA[31:10], start address of the encrypted region, other bits set to 0
0x1C	END, i.e. RGNx_RGN_EA[31:10], end address of the encrypted region RO, i.e. RGNx_RGN_EA[2], when set to 1, RGNx configuration registers are read-protected DECEN, i.e. RGNx_RGN_EA[1], when set to 1, EXIP performs decryption on this region VALID, i.e. RGNx_RGN_EA[0], when set to 1, region x is valid Other bits set to 1
0x20	Padding data, all set to 0
0x24	Padding data, all set to 0

Table 222: EXIP Key Wrap Data Summary

After these data go through the Key Wrap process, 6 64-bit ciphertext blocks are obtained, totaling 48 bytes.

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When the BE bit and KBPE bit of the CFG register are set to 1, EXIP will start the hardware key unwrapping process. EXIP will directly fetch ciphertext blocks from the start address of external memory via the XPI controller. The locations of the ciphertext blocks in external memory are as follows:

Address Offset	Description
0x00	Ciphertext package of DEK, NONCE and region address information for region 0 (RGN0)
0x40	Ciphertext package of DEK, NONCE and region address information for region 1 (RGN1)
0x80	Ciphertext package of DEK, NONCE and region address information for region 2 (RGN2)
0xC0	Ciphertext package of DEK, NONCE and region address information for region 3 (RGN3)

Table 223: EXIP Key Wrap Data Summary

Once EXIP starts unwrapping ciphertext packages, it will automatically perform the Key Unwrap process for all 4 regions' ciphertext packages. After unwrapping is completed, the recovered DEK, NONCE, START, END and other information will be loaded into the corresponding RGNx registers. If the initial value obtained after unwrapping does not match the defined IV (0xA6A6A6A6A6A6A6A6), an error will be reported: the RGNx_DESC_W1[0] VALID bit will be set to 1, and the STA[KBERR] flag bit will also be set to 1.

If the number of regions actually used by users is less than 4, it is still required to generate ciphertext packages for all 4 regions according to the key wrap steps and program them to the corresponding addresses of Flash. For unused regions, just set both the start address and end address to 0.

The KEK used by EXIP for key unwrapping is directly loaded from the dedicated data path of on-chip OTP. Once KEK is programmed to the specified OTP location and configured with appropriate read/write protection, it will no longer be visible to software. For OTP programming and read/write protection configuration, please refer to the OTP related chapters.

On this product, EXIP enabling and key unwrapping are managed by BOOT ROM. After the ENCRYPT_XIP field in OTP is set to 1, the ROM will forcibly enable EXIP during XPI NOR Flash boot.

In summary, to implement online decryption execution with EXIP, users need to:

- Generate KEK, and perform Key Wrap on DEK, NONCE, START, END and other data with KEK to generate ciphertext packages. Note that ciphertext packages for all 4 EXIP regions need to be generated
- Program the ciphertext packages to the specified addresses of the external serial NOR Flash
- Encrypt the firmware image with DEK and NONCE, then program the encrypted image to the external serial NOR Flash
- Program KEK to the specified OTP location, and configure read/write protection
- Program the ENCRYPT_XIP field in OTP to 1
- After reset, the chip will support booting from the encrypted external serial NOR Flash

60.3 Appendix

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60.3.1 Introduction to RFC3394

EXIP supports AES-128 CTR mode decryption of encrypted data and code on external NOR Flash. Obviously, the security of these data and code depends on the security of EXIP decryption keys. EXIP supports a key encapsulation algorithm called Key Wrap to protect EXIP decryption keys and other related sensitive data.

The Key Wrap algorithm used by EXIP complies with RFC3394 "Advanced Encryption Standard (AES) Key Wrap Algorithm". RFC3394 defines a key wrap algorithm based on the AES standard, which can encrypt and protect data of any length and provide data integrity check. RFC3394 defines the key used to encapsulate protected data as KEK (Key-encryption Key), which can be AES-128, AES-192 or AES-256 keys supported by the AES standard. The KEK used by EXIP is 128 bits long.

As defined in RFC3394, protected data can be divided into multiple 64-bit data blocks, with a minimum of 2 blocks and no upper limit on the maximum number. After n plaintext data blocks go through the key wrap process, n+1 ciphertext data blocks are obtained; the n+1 ciphertext data blocks can be restored to n plaintext data blocks after the key unwrap process.

The Key Wrap steps of RFC3394 are quoted as follows:

Input: Plaintext, n 64-bit data blocks {P1, P2, ..., Pn} and key K (i.e., KEK).
Output: Ciphertext, (n+1) 64-bit ciphertext blocks {C0, C1, ..., Cn}.

Step 1: Initialization
Set A to initial value IV
For i = 1 to n
  R[i] = P[i]

Step 2: Calculate intermediate values
For j = 0 to 5
  For i=1 to n
    B = AES(K, A | R[i])
    A = MSB(64, B) ^ t where t = (n*j)+i
    R[i] = LSB(64, B)

Step 3: Output results
C[0] = A
For i = 1 to n
  C[i] = R[i]

The Key Unwrap steps of RFC3394 are quoted as follows:

Input: Ciphertext, (n+1) 64-bit ciphertext blocks {C0, C1, ..., Cn}, and key K (i.e., KEK).
Output: Plaintext, n 64-bit data blocks {P1, P2, ..., Pn}.

Step 1: Initialization
A = C[0]
For i = 1 to n

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R[i] = C[i]

Step 2: Calculate intermediate values

For j = 5 to 0

For i = n to 1

B = AES-1(K, (A ^ t) | R[i]) where t = n*j+i

A = MSB(64, B)

R[i] = LSB(64, B)

Step 3: Output results

If A matches the initial value IV:

For i = 1 to n

P[i] = R[i]

Else:

Error occurred

Descriptions of expressions used in the above key wrap and unwrap steps are as follows:

AES(K, W) Encrypt data W with key K using AES algorithm

AES-1(K, W) Decrypt data W with key K using AES algorithm

MSB(j, W) Return j MSBs of data W, i.e., the highest j bits

LSB(j, W) Return j LSBs of data W, i.e., the lowest j bits

B1 ^ B2 Bitwise XOR of data B1 and B2

B1 | B2 Concatenate data B1 and B2

K i.e. KEK, Key Encryption Key

n Number of 64-bit data blocks

s Number of key wrap steps s = 6n

P[i] The i-th 64-bit plaintext data block

C[i] The i-th 64-bit ciphertext data block

A 64-bit register storing integrity check data

R[i] A set of 64-bit register arrays, where i = 0, 1, ..., n

IV 64-bit initial value during wrap, used for comparison with A during unwrap. EXIP uses the default initial value defined in RF

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61 Random Number Generator (RNG)

This chapter introduces the main features and functions of the Random Number Generator (RNG).

61.1 Feature Summary

The features of RNG are as follows:

- 256-bit hardware True Random Number Generator (TRNG) with multiple entropy sources to generate seeds (SEED)
- NIST-compliant 160-bit hardware Pseudo Random Number Generator (PRNG), with a 256-bit seed for pseudo-random numbers;
- Supports the key generation algorithm defined in the Digital Signature Standard (DSS);
- Supports a random number transfer interface invisible to software;

61.2 Function Description

This chapter describes the functions of the Random Number Generator (RNG).

61.2.1 RNG Initialization

The RNG module needs to be initialized before use, and the initialization process is as follows:

1. Set the CLRERR bit of the CMD register to 1 to clear all errors and interrupts
2. Set the GENSD bit of the CMD register to 1 to start generating seed (SEED)
3. Wait until the FSTSDDN flag bit of the STA register is set to 1, which indicates that the SEED is ready

61.2.2 RNG SEED Generation

After the RNG is initialized, it can start generating SEED.

RNG supports both automatic SEED generation and software-triggered SEED generation. When the AUTRSD bit of the CTRL register is set to 1, RNG can automatically regenerate SEED. If the AUTRSD bit is set to 0, the user needs to set the GENSD bit of the CMD register to 1 to instruct RNG to generate a new SEED. When the SEED generation is completed, the NEWSDDN flag bit of the STA register will be set to 1.

Whenever the FIFO storing random numbers (five words totaling 160 bits) is empty, RNG will quickly generate 160-bit random numbers based on the seed and fill them into the FIFO; After the current seed is used to generate a certain number of random numbers, RNG will indicate that a new seed needs to be generated through the status register. At this time, if the automatic seed generation mode (AUTRSD is 1) is enabled, the seed will be generated automatically; otherwise, the software needs to set the CMD.GENSD bit to trigger the generation of a new seed.

You can read the random numbers stored in the FIFO through the FO2B register. You can judge the number of random numbers stored in the current FIFO through the FRNNU flag bit of the STA register.

RNG supports a direct random number output port, which is directly transmitted to other modules through hardware, and the random number is invisible to the software at this time; the random number transfer steps are as follows:

1. Judge that there is valid data in the SEED FIFO through the FLVL flag bit of the STA register
2. The software reads registers R2SK[0] ~ R2SK[7] in sequence. After reading R2SK[7] is completed,

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the 256-bit SEED data on the SEED port becomes valid.

Note: When the software reads registers R2SK[0] ~ R2SK[7], the return value is always 0. The real valid random number will be output to other modules from the output port.

61.2.3 RNG Self-Test

RNG supports self-test to ensure that the relevant circuits work normally.

When the user sets the SLFCHK bit of the CMD register to 1, the RNG module starts the self-test process. When the self-test is completed, the SLFCHKDN flag bit of the STA register will be set to 1. If there is a problem in the self-test, the SLFCHNPF flag bit will be set to 1.

61.2.4 Interrupt

RNG supports generating the following interrupt requests:

- When RNG finishes generating SEED or completes self-test
- When an error occurs in RNG

When the MIRQDN bit and MIRQERR bit of the CTRL register are cleared to 0, the corresponding interrupt generation is supported.

61.3 RNG Register List

The RNG register list is as follows:

RNG base address: 0xF304C000

Offset	Register Name	Description	Reset Value
0x0000	CMD	Command Register	0x00000000
0x0004	CTRL	Control Register	0x00000000
0x0008	STA	Status Register	0x00000000
0x000C	ERR	Error Registers	0x00000000
0x0010	FO2B	FIFO out to bus/cpu	0x00000000
0x0020	R2SK[FO2S0]	FIFO out to SDP as AES engine key	0x00000000
0x0024	R2SK[FO2S1]	FIFO out to SDP as AES engine key	0x00000000
0x0028	R2SK[FO2S2]	FIFO out to SDP as AES engine key	0x00000000
0x002C	R2SK[FO2S3]	FIFO out to SDP as AES engine key	0x00000000
0x0030	R2SK[FO2S4]	FIFO out to SDP as AES engine key	0x00000000
0x0034	R2SK[FO2S5]	FIFO out to SDP as AES engine key	0x00000000
0x0038	R2SK[FO2S6]	FIFO out to SDP as AES engine key	0x00000000

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Offset	Register Name	Description	Reset Value
0x003C	R2SK[FO2S7]	FIFO out to SDP as AES engine key	0x00000000

Table 224: RNG Register List

61.3.1 RNG Register Description

The detailed description of RNG registers is as follows:

61.3.2 CMD (0x0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																									SFTRST	CLRERR	CLRINT	RSVD		GENSD	SLFCHK
N/A																									RW	RW	RW	N/A		RW	RW
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CMD [31:0]

Bits	Name	Description
6	SFTRST	Soft Reset, Perform a software reset of the RNG This bit is self-clearing. 0 Do not perform a software reset. 1 Software reset
5	CLRERR	Clear the Error, clear the errors in the ESR register and the RNG interrupt. This bit is self-clearing. 0 Do not clear the errors and the interrupt. 1 Clear the errors and the interrupt.
4	CLRINT	Clear the Interrupt, clear the RNG interrupt if an error is not present. This bit is self-clearing. 0 Do not clear the interrupt. 1 Clear the interrupt
1	GENSD	Generate Seed, when both ST and GS triggered, ST first and GS next.
0	SLFCHK	Self Test, when both ST and GS triggered, ST first and GS next.

CMD Bits

61.3.3 CTRL (0x4)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																									MIRQERR	MIRQDN	AUTRSD	RSVD		FUFMOD	
N/A																									RW	RW	RW	N/A		RW	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

CTRL [31:0]

Bits	Name	Description
6	MIRQERR	Mask Interrupt Request for Error
5	MIRQDN	Mask Interrupt Request for Done Event, asks the interrupts generated upon the completion of the seed and self-test modes. The status of these jobs can be viewed by: • Reading the STA and viewing the seed done and the self-test done bits (STA[SDN, STDN]). • Viewing the RNG_CMD for the generate-seed or the self-test bits (CMD[GS,ST]) being set, indicating that the operation is still taking place.
4	AUTRSD	Auto Reseed
1-0	FUFMOD	FIFO underflow response mode 00 Return all zeros and set the ESR[FUFE]. 01 Return all zeros and set the ESR[FUFE]. 10 Generate the bus transfer error 11 Generate the interrupt and return all zeros (overrides the CTRL[MASKERR]).

CTRL Bits

61.3.4 STA (0x8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD								SCPF			RSVD				FUNCERR	FSIZE				FRNNU				RSVD	NSDDN	FSDDN	SCDN	RSDREQ	IDLE	BUSY	RSVD
N/A								RO			N/A				RO	RO				RO				N/A	RO	RO	RO	RO	RO	RO	N/A
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

STA [31:0]

Bits	Name	Description
23-21	SCPF	Self Check Pass Fail
16	FUNCERR	Error was detected, check ESR register for details
15-12	FSIZE	Fifo Size, it is 5 in this design.

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Bits	Name	Description
11-8	FRNNU	Fifo Level, Indicates the number of random words currently in the output FIFO
6	NSDDN	New seed done.
5	FSDDN	1st Seed done When ”1”, Indicates that the RNG generated the first seed.
4	SCDN	Self Check Done Indicates whether Self Test is done or not. Can be cleared by the hardware reset or a new self test is initiated by setting the CMD[ST]. 0 Self test not completed 1 Completed a self test since the last reset.
3	RSDREQ	Reseed needed Indicates that the RNG needs to be reseeded. This is done by setting the CMD[GS], or automatically if the CTRL[ARS] is set.
2	IDLE	Idle, the RNG is in the idle mode, and internal clocks are disabled, in this mode, access to the FIFO is allowed. Once the FIFO is empty, the RNGB fills the FIFO and then enters idle mode again.
1	BUSY	when 1, means the RNG engine is busy for seeding or random number generation, self test and so on.

STA Bits

61.3.5 ERR (0xC)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD								RSVD								RSVD								RSVD		FUFE	RSVD	SCKERR	RSVD	RSVD	RSVD
N/A								N/A								N/A								N/A		RO	N/A	RO	N/A	N/A	N/A
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	0	0	0	0	0	0

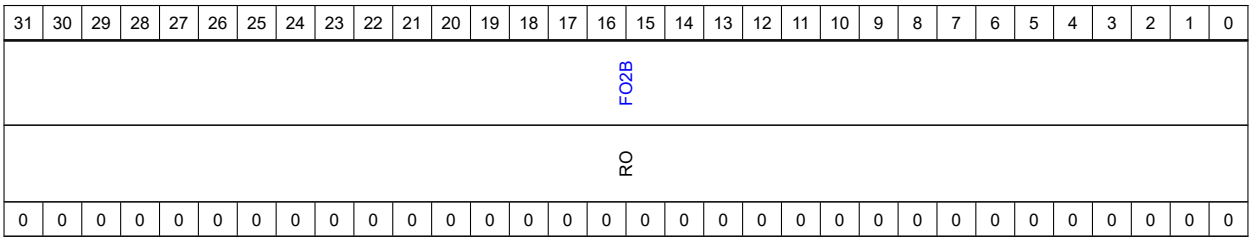
ERR [31:0]

Bits	Name	Description
5	FUFE	FIFO access error(underflow)
3	SCKERR	Self-test error Indicates that the RNG failed the most recent self test. This bit is sticky and can only be reset by a hardware reset or by writing 1 to the CMD[CE]

ERR Bits

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61.3.6 FO2B (0x10)

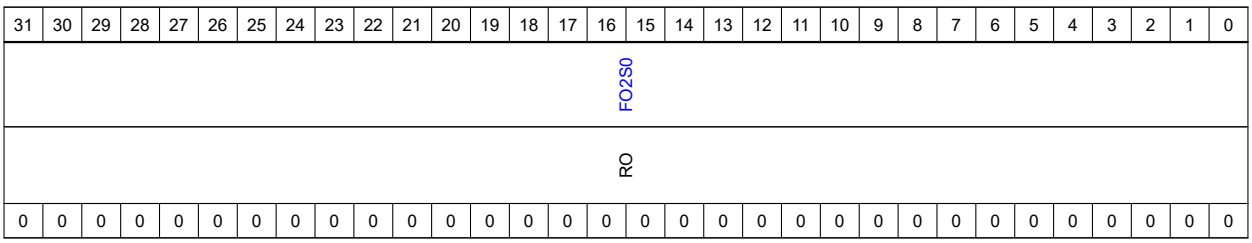


FO2B [31:0]

Bits	Name	Description
31-0	FO2B	SW read the FIFO output.

FO2B Bits

61.3.7 R2SK (0x20 + 0x4 * n)



R2SK [31:0]

Bits	Name	Description
31-0	FO2S0	FIFO out to KMAN, will be SDP engine key.

R2SK Bits

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62 Key Manager KEYM

This chapter describes the main features and functions of the key manager KEYM. The main function of the key manager is to deliver keys to the encryption and decryption engine, implement secure key management, and ensure that keys are not exposed to any software.

62.1 Feature Summary

Main features of the key manager KEYM:

- Load keys from other modules
- Support scrambling of input keys
- Deliver keys to the encryption engine
- Support software key SMK
- Support generating multiple SK (session key)
- Support synthesizing one key from multiple keys

62.2 Function Description

This chapter describes the functions of the key manager KEYM.

62.2.1 FMK Key

FMK key is a 256-bit key. FMK is transmitted by the key manager from OTP. Each chip has a unique OTP root key.

Users can choose between two scrambled versions when using FMK, and the original key cannot be used directly.

Note: To realize key isolation under different system security states, the key manager supports separately configuring whether to scramble the key in the two different states of SEC and NSC, and different scrambling algorithms are used in different modes. That is, for the same original data, the scrambled output is also different. Please refer to the relevant chapters of the system security manager for detailed information about security states.

Setting SEC_KEY_SEL[FMK_SEL] bit to 0 means that when the system security state is SEC, the scrambled FMK is used; otherwise, another scrambled version of FMK is used.

Setting NSC_KEY_SEL[FMK_SEL] bit to 0 means that when the system security state is NSC, the scrambled FMK is used; otherwise, another scrambled version of FMK is used.

62.2.2 SMK Key

SMK key is a 256-bit key. SMK can be written to SFK0 SFK7 registers by software.

SMK supports direct loading from SMKRNG port, and loading options can be configured by configuring KEYM_RNG register:

- RNG[RNG_XOR] = 0’b0, SMK(KEYM_SFKx) is directly filled with the random number generated by RNG
- RNG[RNG_XOR] = 0’b1, SMK(KEYM_SFKx) is updated to the bitwise XOR of the original value and the random number generated by RNG

Users can set KEYM_RNG[BLOCK_RNG_XOR] bit to 1 to lock the value of KEYM_RNG[RNG_XOR] bit,

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which cannot be modified later.

The key manager supports setting read protection for the SFK0 SFK7 registers storing SMK keys. Setting READ_CONTROL [BLOCK_SMK_READ] bit to 1 will prevent software from reading SFK0 SFK7.

Users can choose to use the original data or the scrambled version when using SMK.

Note: To realize key isolation under different system security states, the key manager supports separately configuring whether to scramble the key in the two different states of SEC and NSC, and different scrambling algorithms are used in different modes. That is, for the same original data, the scrambled output is also different. Please refer to the relevant chapters of the system security manager for detailed information about security states.

Setting SEC_KEY_SEL[SMK_SEL] bit to 1 means that when the system security state is SEC, the scrambled SMK is used; otherwise, the original SMK data is used.

Setting NSC_KEY_SEL[SMK_SEL] bit to 1 means that when the system security state is NSC, the scrambled SMK is used; otherwise, the original SMK data is used. If the secure mode is set to use the scrambled version, the non-secure mode is fixed to use the scrambled version.

62.2.3 MK Key

MK can be selected from FMK and SMK, or synthesized from multiple or all of them. Key synthesis enables decentralized management of root keys and reduces risks.

Note: To realize key isolation under different system security states, the key manager supports separately configuring MK synthesis in the two different states of SEC and NSC.

The specific configuration method of MK synthesis is as follows:

- Setting KEY_SEL[0] of SEC_KEY_SEL register to 1 means that when the system security state is SEC, FMK participates in key synthesis, otherwise the output key is irrelevant to FMK
- Setting KEY_SEL[2] of SEC_KEY_SEL register to 1 means that when the system security state is SEC, SMK participates in key synthesis, otherwise the output key is irrelevant to SMK
- Setting KEY_SEL[0] of NSC_KEY_SEL register to 1 means that when the system security state is NSC, FMK participates in key synthesis, otherwise the output key is irrelevant to FMK
- Setting KEY_SEL[2] of NSC_KEY_SEL register to 1 means that when the system security state is NSC, SMK participates in key synthesis, otherwise the output key is irrelevant to SMK

Suppose the user sets KEY_SEL[0] of SEC_KEY_SEL register to 1 and KEY_SEL[2] to 0, then when the security state is SEC, the key manager will output FMK as MK.

If both synthesis factors are 0, or any of the selected factors input invalid value, the all-zero key is output. You can detect whether the key is set successfully by comparing the encrypted data with the ciphertext of the all-zero key.

62.2.4 SK Key

SK key is also called session key. The key manager supports outputting SK0, SK1, SK2, SK3 to the encryption and decryption engines on the system.

Note: To realize key isolation under different system security states, the key manager supports generating different SKs in the two different states of SEC and NSC.

Setting SEC_KEY_SEL[SK_VAL] bit to 1 means that when the system security state is SEC, SK generation is valid; otherwise, the output of SKx is all 0.

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Setting NSC_KEY_SEL[SK_VAL] bit to 1 means that when the system security state is NSC, SK generation is valid; otherwise, the output of SKx is all 0.

The SK key generation of the key manager is realized by the key generation module. The input of the key generation module is SMK, each chip has different key generation logic, and the key generation module logic is also different between different SKx.

Therefore, on the same chip, when SMK = a, SK0 = SK0a, SK1 = SK1a, SK2 = SK2a, SK3 = SK3a are generated. As long as the value of a is recorded, setting SMK = a again can restore SK0a, SK1a, SK2a, SK3a. But on another chip, inputting SMK = a cannot get the same SK.

Note: If session Key is used, SMK should not be introduced as a synthesis factor in MK.

62.3 KEYM Register List

The register list of KEYM is as follows:

KEYM base address: 0xF3054000

Offset	Register Name	Description	Reset Value
0x0000	SOFTMKEY[SFK0]	software set symmetric key	0x00000000
0x0004	SOFTMKEY[SFK1]	software set symmetric key	0x00000000
0x0008	SOFTMKEY[SFK2]	software set symmetric key	0x00000000
0x000C	SOFTMKEY[SFK3]	software set symmetric key	0x00000000
0x0010	SOFTMKEY[SFK4]	software set symmetric key	0x00000000
0x0014	SOFTMKEY[SFK5]	software set symmetric key	0x00000000
0x0018	SOFTMKEY[SFK6]	software set symmetric key	0x00000000
0x001C	SOFTMKEY[SFK7]	software set symmetric key	0x00000000
0x0020	SOFTPKEY[SPK0]	system asymmetric key	0x00000000
0x0024	SOFTPKEY[SPK1]	system asymmetric key	0x00000000
0x0028	SOFTPKEY[SPK2]	system asymmetric key	0x00000000
0x002C	SOFTPKEY[SPK3]	system asymmetric key	0x00000000
0x0030	SOFTPKEY[SPK4]	system asymmetric key	0x00000000
0x0034	SOFTPKEY[SPK5]	system asymmetric key	0x00000000
0x0038	SOFTPKEY[SPK6]	system asymmetric key	0x00000000
0x003C	SOFTPKEY[SPK7]	system asymmetric key	0x00000000
0x0040	SEC_KEY_CTL	secure key generation	0x00000000
0x0044	NSC_KEY_CTL	non-secure key generation	0x00000000
0x0048	RNG	Random number interface behavior	0x00000000
0x004C	READ_CONTROL	key read out control	0x00000000

Table 225: KEYM Register List

62.4 Register Description

The detailed description of KEYM registers is as follows:

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62.4.1 SOFTMKEY (0x0 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
KEY																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

SOFTMKEY [31:0]

Bits	Name	Description
31-0	KEY	software symmetric key key will be scambled to 4 variants for software to use, and replicable on same chip. scramble keys are chip different, and not replicable on different chip must be write sequencely from 0 - 7, otherwise key value will be treated as all 0

SOFTMKEY Bits

62.4.2 SOFTPKEY (0x20 + 0x4 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
KEY																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

SOFTPKEY [31:0]

Bits	Name	Description
31-0	KEY	software asymmetric key key is derived from scrambles of fuse private key, software input key, SRK, and system security status. This key os read once, sencondary read will read out 0

SOFTPKEY Bits

62.4.3 SEC_KEY_CTL (0x40)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LOCK_SEC_CTL	RSVD														SK_VAL	RSVD			SMK_SEL	RSVD			ZMK_SEL	RSVD			FMK_SEL	RSVD	KEY_SEL		
RW	N/A														RO	N/A			RW	N/A			RW	N/A			RW	N/A	RW		
0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	0	x	x	x	0	x	x	x	0	x	0	0	0

SEC_KEY_CTL [31:0]

Bits	Name	Description
31	LOCK_SEC_CTL	block secure state key setting being changed
16	SK_VAL	session key valid 0: session key is all 0's and not usable 1: session key is valid
12	SMK_SEL	software symmetric key selection 0: use origin value in software symmetric key 1: use scramble version of software symmetric key
8	ZMK_SEL	batt symmetric key selection 0: use scramble version of software symmetric key 1: use origin value in software symmetric key
4	FMK_SEL	fuse symmetric key selection 0: use scramble version of fuse symmetric key 1: use alnertave scramble of fuse symmetric key
2-0	KEY_SEL	secure symmtric key synthesize setting, key is a XOR of following bit0: fuse mk, 0: not selected, 1:selected bit1: zmk from batt, 0: not selected, 1:selected bit2: software key 0: not selected, 1:selected

SEC_KEY_CTL Bits

62.4.4 NSC_KEY_CTL (0x44)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LOCK_NSC_CTL	RSVD														SK_VAL	RSVD			SMK_SEL	RSVD			ZMK_SEL	RSVD			FMK_SEL	RSVD	KEY_SEL		
RW	N/A														RO	N/A			RW	N/A			RW	N/A			RW	N/A	RW		
0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	0	x	x	x	0	x	x	x	0	x	0	0	0

NSC_KEY_CTL [31:0]

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Bits	Name	Description
31	LOCK_NSC_CTL	block non-secure state key setting being changed
16	SK_VAL	session key valid 0: session key is all 0's and not usable 1: session key is valid
12	SMK_SEL	software symmetric key selection 0: use scramble version of software symmetric key 1: use origin value in software symmetric key
8	ZMK_SEL	batt symmetric key selection 0: use scramble version of software symmetric key 1: use origin value in software symmetric key
4	FMK_SEL	fuse symmetric key selection 0: use scramble version of fuse symmetric key 1: use origin value in fuse symmetric key
2-0	KEY_SEL	non-secure symmtric key synthesize setting, key is a XOR of following bit0: fuse mk, 0: not selected, 1:selected bit1: zmk from batt, 0: not selected, 1:selected bit2: software key 0: not selected, 1:selected

NSC_KEY_CTL Bits

62.4.5 RNG (0x48)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD															BLOCK_RNG_XOR	RSVD															RNG_XOR
N/A															RW	N/A															RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

RNG [31:0]

Bits	Name	Description
16	BLOCK_RNG_XOR	block RNG_XOR bit from changing, if this bit is written to 1, it will hold 1 until next reset 0: RNG_XOR can be changed by software 1: RNG_XOR ignore software change from software
0	RNG_XOR	control how SFK is accepted from random number generator 0: SFK value replaced by random number input 1: SFK value exclusive or with random number input,this help generate random number using 2 rings inside RNG

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Bits	Name	Description

RNG Bits

62.4.6 READ_CONTROL (0x4C)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD															BLOCK_PK_READ	RSVD															BLOCK_SMK_READ
N/A															RW	N/A															RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

READ_CONTROL [31:0]

Bits	Name	Description
16	BLOCK_PK_READ	asymmetric key readout control, if this bit is written to 1, it will hold 1 until next reset 0: key can be read out 1: key cannot be read out
0	BLOCK_SMK_READ	symmetric key readout control, if this bit is written to 1, it will hold 1 until next reset 0: key can be read out 1: key cannot be read out

READ_CONTROL Bits

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63 Security Manager PSEC

This chapter describes the features and functions of the security manager PSEC.

63.1 Feature Summary

Main features of the security manager PSEC:

- Manages the security lifecycle of the chip
- Manages the security status of the chip
- Detects chip security events
- Takes measures to protect sensitive information when an exception occurs

63.2 Functional Description

This section describes the functions of the security manager PSEC.

63.2.1 Chip Lifecycle

Read and write permissions of internal chip information are managed according to different stages of the chip lifecycle. The 4-bit security lifecycle LIFECYCLE is redundantly configured in OTP, with two independent copies A and B.

The 4 bits of chip lifecycle are described as follows:

- LIFECYCLE = 4’b0000, CREATE, reserved.
- LIFECYCLE = 4’b0001, NONSEC: The state when the chip passes testing and is delivered to users, where users carry out development work.
  - Chip debug ports, JTAG and Debug ports are open.
  - Test functions are enabled.
  - BOOT ROM does not perform secure boot, nor verify the signature of user code image.
  - The key manager and its associated modules use NONSEC keys.
- LIFECYCLE = 4’b0011, SECURE: Users inject sensitive information and code into the system, and protect the system from functional damage.
  - Chip debug ports, JTAG and Debug ports are closed, and will be opened only after debug authentication succeeds.
  - Test functions are disabled.
  - BOOT ROM performs secure boot and verifies the signature of the image. If verification fails, the image will not run.
  - The key manager and its associated modules use SEC keys.
- LIFECYCLE = 4’b0111, RETURN: This state re-enables test functions on the premise of ensuring no leakage of user information.
  - Chip debug ports, JTAG and Debug ports are re-opened.
  - Test functions are enabled.
  - BOOT ROM does not perform secure boot, nor verify the signature of user code image.
  - No keys are loaded from OTP any more.
- LIFECYCLE = 4’b1011, NONRET: Factory return test is prohibited. The security protection functions in this state are the same as those in SECURE state, but test functions are permanently disabled.

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  - On the basis of SECURE lifecycle, the chip no longer supports configuration of factory return mode bit.
- LIFECYCLE = 4’b1111, SCRIBE: The chip is discarded, sensitive information is destroyed, and the chip cannot work normally in this state.
  - The key manager and its associated modules can no longer load keys from OTP, and keys in the battery domain will also be erased.
  - ROM stops booting the system.

Note: If users force to close the debug port and JTAG port by programming the DEBUG_DISABLE and JTAG_DISABLE bits in OTP, these ports cannot be re-opened no matter which lifecycle the chip is in.

In addition to reading OTP, users can read the current lifecycle state of the chip from the LIFECYCLE register of the system security monitor:

- LIFECYCLE[0]: Set to 1 indicates unknown lifecycle.
- LIFECYCLE[1]: Set to 1 indicates the lifecycle is CREATE, reserved.
- LIFECYCLE[2]: Set to 1 indicates the lifecycle is NONSEC, the chip is open.
- LIFECYCLE[3]: Set to 1 indicates the lifecycle is SECURE, the chip is in secure state.
- LIFECYCLE[4]: Set to 1 indicates the lifecycle is RETURN, the chip is in factory return state.
- LIFECYCLE[5]: Set to 1 indicates the lifecycle is NONRET, factory return of the chip is prohibited.
- LIFECYCLE[6]: Set to 1 indicates the lifecycle is SCRIBE, the chip is discarded.
- LIFECYCLE[7]: Set to 1 indicates the lifecycle is DEBATE, redundancy state conflict occurs.

63.2.2 Security State Management

The power management domain security manager can configure and monitor the security state of the chip. The following security states are supported:

- INSPECT: Used for self-test of system security related components. The key is fixed to 1F1E1D1C1B1A1918171615
in this state.
- NONSEC: This state mainly protects basic security information of HPM such as pre-installed keys and identification numbers.
- SECURE: This state mainly protects customer assets such as user-installed keys.
- FAIL: This state disables the hardware encryption and decryption engine.

When the chip is released from reset, the security state is initially in INSPECT state, and BOOT ROM will configure the chip to NONSEC or SECURE state according to the LIFECYCLE bit settings in OTP.

- If the current chip lifecycle is NONSEC, set the security state to NONSEC;
- If the current chip lifecycle is SECURE, set the security state to SECURE;
- If the current chip lifecycle is RETURN, set the security state to NONSEC;
- If the current chip lifecycle is NONRET, set the security state to SECURE;
- If the current chip lifecycle is in other states, set the security state to FAIL;

On-chip key modules such as the key manager provide two sets of keys: NONSEC or SECURE according to the chip security state. In other security states, the key manager will block key output.

Users can query and configure the security state through the SECURE_STATE register.

The security manager monitors several security related events on the system. For the specific event list, please refer to the relevant chapters of system security configuration.

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Note: The security manager can monitor and respond to security events in both SECURE state and NONSEC state, and will set the security state to FAIL when a security event occurs.

Users can configure whether the monitor responds to detected security events through the VIOLATION_CONFIG register:

- The NSC_VIO_CFG bit field contains response enable bits for each security event. 0 means the corresponding security event is not regarded as a security rule violation in NONSEC state, 1 means the corresponding security event is regarded as a security rule violation in NONSEC state, and the security state will switch to FAIL accordingly.
- LOCK_NSC bit is the NONSEC configuration lock bit, set to 1 to lock the NSC_VIO_CFG bit field.
- The SEC_VIO_CFG bit field contains response enable bits for each security event. 0 means the corresponding security event is not regarded as a security rule violation in SECURE state, 1 means the corresponding security event is regarded as a security rule violation in SECURE state, and the security state will switch to FAIL accordingly.
- LOCK_SEC bit is the SECURE configuration lock bit, set to 1 to lock the SEC_VIO_CFG bit field.

Users can query whether a security event is detected through each flag bit of the EVENT register.

When the security state is FAIL, PSEC will disable access to sensitive information and disable the encryption and decryption engine.

63.2.3 Security Event Notification

Users can configure whether the security monitor reports security events to the battery domain (if battery domain exists) security manager through the ESCALATE_CONFIG register, allowing the internal security manager of the battery domain to take corresponding protection measures.

- The NSC_VIO_CFG bit field contains notification enable bits for each security event. When set to 1, the battery domain will respond to the corresponding security event when it is in NONSEC mode.
- The SEC_VIO_CFG bit field contains notification enable bits for each security event. When set to 1, the battery domain will respond to the corresponding security event when it is in SECURE mode.

Users can query whether a security event notification occurs through each flag bit of the EVENT register.

63.3 PSEC Register List

The PSEC register list is as follows:

SEC base address: 0xF3044000

Offset	Register Name	Description	Reset Value
0x0000	SECURE_STATE	Secure state	0x00000000
0x0004	SECURE_STATE_CONFIG	secure state configuration	0x00000000
0x0008	VIOLATION_CONFIG	Security violation config	0x00000000
0x000C	ESCALATE_CONFIG	Escalate behavior on security event	0x00000000
0x0010	EVENT	Event and escalate status	0x00000000
0x0014	LIFECYCLE	Lifecycle	0x00000000

Table 226: SEC Register List

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63.4 PSEC Register Description

The detailed description of PSEC registers is as follows:

63.4.1 SECURE_STATE (0x0)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD														ALLOW_NSC	ALLOW_SEC	RSVD								PMIC_FAIL	PMIC_NSC	PMIC_SEC	PMIC_INS	RSVD			
N/A														RO	RO	N/A								RW	RW	RW	RW	N/A			
x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	x	x	x	x	x	x	x	x	0	0	0	0	x	x	x	x

SECURE_STATE [31:0]

Bits	Name	Description
17	ALLOW_NSC	Non-secure state allow 0: system is not healthy to enter non-secure state, request to enter non-secure state will cause a fail state 1: system is healthy to enter non-secure state
16	ALLOW_SEC	Secure state allow 0: system is not healthy to enter secure state, request to enter non-secure state will cause a fail state 1: system is healthy to enter secure state
7	PMIC_FAIL	PMIC secure state one hot indicator 0: secure state is not in fail state 1: secure state is in fail state
6	PMIC_NSC	PMIC secure state one hot indicator 0: secure state is not in non-secure state 1: secure state is in non-secure state
5	PMIC_SEC	PMIC secure state one hot indicator 0: secure state is not in secure state 1: secure state is in secure state
4	PMIC_INS	PMIC secure state one hot indicator 0: secure state is not in inspect state 1: secure state is in inspect state

SECURE_STATE Bits

63.4.2 SECURE_STATE_CONFIG (0x4)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																												LOCK	RSVD		ALLOW_RESTART
N/A																												RW	N/A		RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	0

SECURE_STATE_CONFIG [31:0]

Bits	Name	Description
3	LOCK	Lock bit of allow restart setting, once locked, lock bit itself and configuration register will keep value until next reset 0: not locked, register can be modified 1: register locked, write access to the register is ignored
0	ALLOW_RESTART	allow secure state restart from fail state 0: restart is not allowed, only hardware reset can recover secure state 1: software is allowed to switch to inspect state from fail state

SECURE_STATE_CONFIG Bits

63.4.3 VIOLATION_CONFIG (0x8)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LOCK_NSC								NSC_VIO_CFG								LOCK_SEC								SEC_VIO_CFG							
RW								RW								RW								RW							
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

VIOLATION_CONFIG [31:0]

Bits	Name	Description
31	LOCK_NSC	Lock bit non-secure violation setting, once locked, lock bit itself and configuration will keep value until next reset 0: not locked, configuration can be modified 1: register locked, write access to the configuration is ignored
30-16	NSC_VIO_CFG	configuration of non-secure state violations, each bit represents one security event 0: event is not a security violation 1: event is a security violation

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Bits	Name	Description
15	LOCK_SEC	Lock bit secure violation setting, once locked, lock bit itself and configuration will keep value until next reset 0: not locked, configuration can be modified 1: register locked, write access to the configuration is ignored
14-0	SEC_VIO_CFG	configuration of secure state violations, each bit represents one security event 0: event is not a security violation 1: event is a security violation

VIOLATION_CONFIG Bits

63.4.4 ESCALATE_CONFIG (0xC)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
LOCK_NSC	NSC_VIO_CFG															LOCK_SEC	SEC_VIO_CFG														
RW	RW															RW	RW														
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

ESCALATE_CONFIG [31:0]

Bits	Name	Description
31	LOCK_NSC	Lock bit non-secure escalate setting, once locked, lock bit itself and configuration will keep value until next reset 0: not locked, configuration can be modified 1: register locked, write access to the configuration is ignored
30-16	NSC_VIO_CFG	configuration of non-secure state escalates, each bit represents one security event 0: event is not a security escalate 1: event is a security escalate
15	LOCK_SEC	Lock bit secure escalate setting, once locked, lock bit itself and configuration will keep value until next reset 0: not locked, configuration can be modified 1: register locked, write access to the configuration is ignored
14-0	SEC_VIO_CFG	configuration of secure state escalates, each bit represents one security event 0: event is not a security escalate 1: event is a security escalate

ESCALATE_CONFIG Bits

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63.4.5 EVENT (0x10)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EVENT																RSVD												PMIC_ESC_NSC	PMIC_ESC_SEC	RSVD	
RO																N/A												RO	RO	N/A	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	0	0	x	x

EVENT [31:0]

Bits	Name	Description
31-16	EVENT	local event statue, each bit represents one security event
3	PMIC_ESC_NSC	PMIC is escalating non-secure event
2	PMIC_ESC_SEC	PMIC is escalting secure event

EVENT Bits

63.4.6 LIFECYCLE (0x14)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																								LIFECYCLE							
N/A																								RO							
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0

LIFECYCLE [31:0]

Bits	Name	Description
7-0	LIFECYCLE	lifecycle status, bit7: lifecycle_debate, bit6: lifecycle_scribe, bit5: lifecycle_no_ret, bit4: lifecycle_return, bit3: lifecycle_secure, bit2: lifecycle_nonsec, bit1: lifecycle_create, bit0: lifecycle_unknow

LIFECYCLE Bits

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64 Security Monitor PMON

This chapter describes the main features and functions of the Security Monitor PMON.

64.1 Feature Summary

The Security Monitor PMON is part of the security manager, which can be used to configure power supply and clock monitoring circuits. Its main features are as follows:

- 2 glitch detection circuits
- 2 clock fault detection circuits
- Supports interrupt generation

64.2 Functional Description

This chapter describes the functions of the Security Monitor PMON.

64.2.1 Chip Lifecycle

The security monitor includes power supply and clock fault detection circuits, which are used to monitor unexpectedly injected glitches on the power supply and clock, and notify the security manager.

The security monitor includes 2 independent power glitch detection modules and 2 independent clock fault detection modules.

- The glitch detection module detects the power input on the VPMC pin and the clock noise of OSC24M
- The clock fault detection module detects the external clock OSC24M or internal RC24M oscillator

MONITOR.GLITCH0 and MONITOR.GLITCH1 correspond to the 2 power glitch detection modules respectively, which are configured through their respective CONTROL registers:

- When CONTROL[ENABLE] is set to 0, the detection module is disabled; when set to 1, the detection module performs monitoring.
- When CONTROL[ACTIVE] is set to 0, it is set to passive detection; when set to 1, it is set to active detection

MONITOR.CLOKC0 and MONITOR.CLOKC1 correspond to the 2 clock fault detection modules respectively, which are configured through their respective CONTROL registers:

- When CONTROL[ENABLE] is set to 0, the detection module is disabled; when set to 1, the detection module performs monitoring.

Each detection module can independently detect power glitches or clock faults, and when an abnormality is found, it notifies the security manager PSEC as a security event, and PSEC will respond according to the configuration.

64.2.2 Interrupt

The security monitor can also be configured to generate an interrupt request when a power glitch or clock fault is detected.

The IRQ_ENABLE register contains the corresponding interrupt enable bits.

The interrupt flag bits can be queried through the IRQ_FLAG register.

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64.3 PMON Register List

The PMON register list is as follows:

MON base address: 0xF3048000

Offset	Register Name	Description	Reset Value
0x0000	MONITOR[GLITCH0][CONTROL]	Glitch and clock monitor control	0x00000000
0x0004	MONITOR[GLITCH0][STATUS]	Glitch and clock monitor status	0x00000000
0x0008	MONITOR[GLITCH1][CONTROL]	Glitch and clock monitor control	0x00000000
0x000C	MONITOR[GLITCH1][STATUS]	Glitch and clock monitor status	0x00000000
0x0010	MONITOR[CLOCK0][CONTROL]	Glitch and clock monitor control	0x00000000
0x0014	MONITOR[CLOCK0][STATUS]	Glitch and clock monitor status	0x00000000
0x0018	MONITOR[CLOCK1][CONTROL]	Glitch and clock monitor control	0x00000000
0x001C	MONITOR[CLOCK1][STATUS]	Glitch and clock monitor status	0x00000000
0x0040	IRQ_FLAG		0x00000000
0x0044	IRQ_ENABLE		0x00000000

Table 227: MON Register List

64.4 PMON Register Description

The detailed description of PMON registers is as follows:

64.4.1 MONITOR[CONTROL] (0x0 + 0x8 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																											ACTIVE	RSVD				ENABLE
N/A																											RW	N/A				RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	x	x	x	0

MONITOR[CONTROL] [31:0]

Bits	Name	Description
4	ACTIVE	select glitch works in active mode or passve mode. 0: passive mode, depends on power glitch destroy DFF value 1: active mode, check glitch by DFF chain
0	ENABLE	enable glitch detector 0: detector disabled 1: detector enabled

MONITOR[CONTROL] Bits

64.4.2 MONITOR[STATUS] (0x4 + 0x8 * n)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																															FLAG
N/A																															RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

MONITOR[STATUS] [31:0]

Bits	Name	Description
0	FLAG	flag for glitch detected, write 1 to clear this flag 0: glitch not detected 1: glitch detected

MONITOR[STATUS] Bits

64.4.3 IRQ_FLAG (0x40)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																												FLAG			
N/A																												RW			
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0

IRQ_FLAG [31:0]

Bits	Name	Description
3-0	FLAG	interrupt flag, each bit represents for one monitor, write 1 to clear interrupt flag 0: no monitor interrupt 1: monitor interrupt happened

IRQ_FLAG Bits

64.4.4 IRQ_ENABLE (0x44)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																												ENABLE			
N/A																												RW			
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0

IRQ_ENABLE [31:0]

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Bits	Name	Description
3-0	ENABLE	interrupt enable, each bit represents for one monitor 0: monitor interrupt disabled 1: monitor interrupt enabled

IRQ_ENABLE Bits

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65 Overview of System Debug

This chapter introduces the debug module of this product.

The Debug system complies with The RISC-V Debug Specification, Version 0.13. The Debug system consists of 2 parts: JTAG interface conversion module (DTM) and debug module (DM). DTM connects to an external debugger through a standard JTAG interface, and can convert the debug instructions received on JTAG into read and write access to the DM module. The debug module DM integrates debug functions, which can pause or resume CPU operation, generate reset, and access on-chip resources.

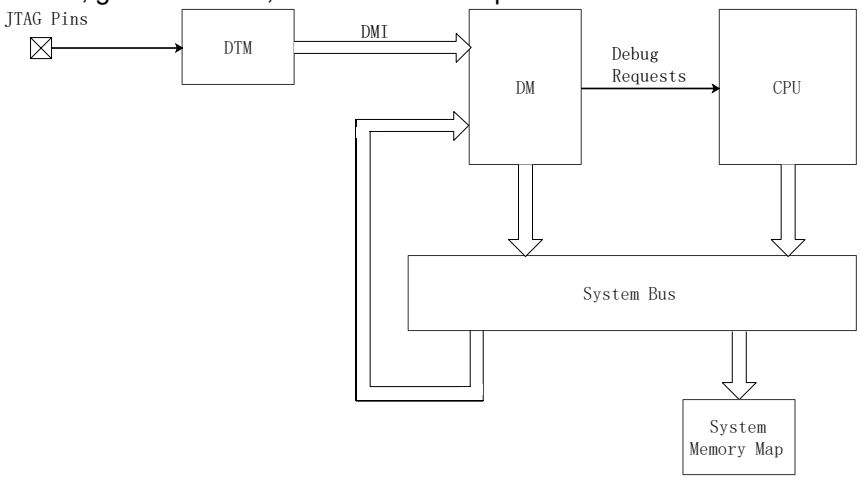


Figure 92: Debug System Block Diagram

The features of the Debug system are as follows:

- Supports 4-wire JTAG interface, compliant with IEEE Std 1149.1
- Supports off-chip debugger to connect to the Debug module through JTAG interface conversion module
- Supports processor to access the Debug module through system address mapping
- Supports 2 hardware breakpoints
- Supports single-step debugging
- Supports System Bus Access, the debug interface can act as a bus master, and can directly access on-chip resources without interrupting kernel operation
- Supports Debug security mechanism, which can enable, disable or lock the Debug interface. When Debug is locked, users need to unlock it with a security code.

The JTAG ID of this product is 0x1000563d.

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66 Debug Transport Module (DTM)

Debug Transport Module (DTM)

This module complies with the definition of RISC-V Version 0.13 specification, and supports the JTAG interface defined in IEEE 1149.1. The instruction length is 5 bits.

66.1 DTM Instructions

The list of INSTRUCTIONs is as follows:

Value	Name	Description	Data Width
0x00	USERCODE	Device identification code	32
0x01	IDCODE	Device identification code	32
0x08	CLAMP	CLAMP instruction	404
0x09	EXTEST	EXTEST instruction	404
0x0A	SAMPLE/PRELOAD	SAMPLE/PRELOAD instruction	404
0x0B	HIGHZ	HIGHZ instruction	1
0x0F	AUTHEN	Debug authentication instruction	32
0x10	DTMCS	DTM Control and Status	32
0x11	DMI	DMI interface	41
0x1F	BYPASS	BYPASS	1

Table 228: DTM Instruction List

66.2 DTM Instruction Description

The detailed description of INSTRUCTIONs is as follows:

66.2.1 USERCODE (0x0)

Bits	Name	Access	Description
31-0	USERCODE	RO	

USERCODE Bits

66.2.2 IDCODE (0x1)

Bits	Name	Access	Description
31-0	IDCODE	RO	

IDCODE Bits

66.2.3 CLAMP (0x8)

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Bits	Name	Access	Description
403-0	BOUNDRY	RO	Refer to IEEE 1149.1 for detals.

CLAMP Bits

66.2.4 EXTEST (0x9)

Bits	Name	Access	Description
403-0	BOUNDRY	RO	Refer to IEEE 1149.1 for detals.

EXTEST Bits

66.2.5 SAMPLE/PRELOAD (0xA)

Bits	Name	Access	Description
403-0	BOUNDRY	RO	Refer to IEEE 1149.1 for detals.

SAMPLE/PRELOAD Bits

66.2.6 HIGHZ (0xB)

Bits	Name	Access	Description
0	BYPASS	RO	Refer to IEEE 1149.1 for detals.

HIGHZ Bits

66.2.7 AUTHEN (0xF)

Bits	Name	Access	Description
31-0	AUTH	RW	Debug authentication, perform 8 DR shift to authenticate debugger, first 4 shift shift out 128 bit die trace information. following 4 shift accept 128 bit authentication password in OTP.

AUTHEN Bits

66.2.8 DTMCS (0x10)

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Bits	Name	Access	Description
17	DMIHARDRESET	W1	Writing 1 to this bit does a hard reset of the DTM, causing the DTM to forget about any outstanding DMI transactions. In general, this should only be used when the debugger has reasons to expect that the outstanding DMI transaction will never complete (e.g., a reset condition causes an inflight DMI transaction to be canceled).
16	DMIRESET	W1	Writing 1 to this bit clears the sticky error state and allows the DTM to retry or complete the previous transaction.
14-12	IDLE	RO	This is a hint to the debugger of the minimum number of cycles a debugger should spend in RunTest/Idle after every DMI scan to avoid a busy return code (dmistat of 3).
11-10	DMISTAT	RO	State of DMI 0: No error 1: Reserved 2: An operation failed (resulted in op of 2) 3: An operation was attempted while a DMI access was still in progress (resulted in op of 3)
9-4	ABITS	RO	Bit width of DMI address is 7
3-0	VERSION	RO	Version of the implemented DTM. 0x1 indicates that the current implementation conforms to RISC-V External Debug Support (TD003) V0.13.

DTMCS Bits

66.2.9 DMI (0x11)

Bits	Name	Access	Description
40-34	ADDRESS	RW	Address used for DMI access. In Update-DR this value is used to access the DM over the DMI.
33-2	DATA	RW	The data to send to the DM over the DMI during Update-DR, and the data returned from the DM as a result of the previous operation.

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Bits	Name	Access	Description
1-0	OP	RW	Write operation: 0: Ignore data and address. (nop) 1: Read from address. (read) 2: Write data to address. (write) 3: Reserved Read operation: 0: The previous operation completed successfully. 1: Reserved 2: A previous operation failed. 3: An operation was attempted while a DMI request is still in progress. The data scanned into dmi in this access will be ignored.

DMI Bits

66.2.10 BYPASS (0x1F)

Bits	Name	Access	Description
0	BYPASS	RO	Refer to IEEE 1149.1 for detals.

BYPASS Bits

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67 Debug Module (DM)

This module complies with the RISC-V Debug Specification Version 0.13, and is equipped with a 16×32-bit program buffer.

Registers in the debug module support two access methods: access by the CPU via the system bus, or access by the debugger via the DTM. The two access methods correspond to different memory mappings.

67.1 System Memory Map

地址偏移	名称	描述
0x0000 - 0x007F	DEBUGROM	Debug ROM
0x0080 - 0x00BF	PROGBUF	Program Buffer 0-15
0x00C0 - 0x00CF	DATA	Abstract Data 0-3
0x00D0 - 0x01FF	RESERVED	Reserved for internal use

Table 229: DM 系统内存映射

67.2 DMI Memory Map

Offset	Register Name	Description	Reset Value
0x0004	DATA[DATA0]	Abstract Data 0-3	0x00000000
0x0005	DATA[DATA1]	Abstract Data 0-3	0x00000000
0x0006	DATA[DATA2]	Abstract Data 0-3	0x00000000
0x0007	DATA[DATA3]	Abstract Data 0-3	0x00000000
0x0010	DMCONTROL	Debug Module Control	0x00000000
0x0011	DMSTATUS	Debug Module Status	0x004000A2
0x0012	HARTINFO	Hart Info	0x002140C0
0x0013	HALTSUM	Halt Summary	0x00000000
0x0014	HAWINDOWSEL	Hart Array Window Select	0x00000000
0x0015	HAWINDOW	Hart Array Window	0x00000000
0x0016	ABSTRACTCS	Abstract Control and Status	0x08000004
0x0017	COMMAND	Abstract Command	0x00000000
0x0018	ABSTRACTAUTO	Abstract Command Autoexec	0x00000000
0x0020	PROGBUF[PROGBUF0]	Program Buffer 0-15	0x00000000
0x0021	PROGBUF[PROGBUF1]	Program Buffer 0-15	0x00000000
0x0022	PROGBUF[PROGBUF2]	Program Buffer 0-15	0x00000000
0x0023	PROGBUF[PROGBUF3]	Program Buffer 0-15	0x00000000
0x0024	PROGBUF[PROGBUF4]	Program Buffer 0-15	0x00000000
0x0025	PROGBUF[PROGBUF5]	Program Buffer 0-15	0x00000000
0x0026	PROGBUF[PROGBUF6]	Program Buffer 0-15	0x00000000
0x0027	PROGBUF[PROGBUF7]	Program Buffer 0-15	0x00000000
0x0038	SBCS	System Bus Access Control and Status	0x20040407

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Offset	Register Name	Description	Reset Value
0x0039	SBADDRESS[SBADDRESS0]	System Bus Address	0x00000000
0x003A	SBADDRESS[SBADDRESS1]	System Bus Address	0x00000000
0x003B	SBADDRESS[SBADDRESS2]	System Bus Address	0x00000000
0x003C	SBDATA[SBDATA0]	System Bus Data	0x00000000
0x003D	SBDATA[SBDATA1]	System Bus Data	0x00000000
0x003E	SBDATA[SBDATA2]	System Bus Data	0x00000000
0x003F	SBDATA[SBDATA3]	System Bus Data	0x00000000

Table 230: DM Register List

67.3 DM Register Description

67.3.1 DATA (0x4 + 0x1 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DATA [31:0]

Bits	Name	Description
31-0	DATA	The registers are accessible from both the DMI interface and the system bus interface to support data exchanges between the external debugger and the processor

DATA Bits

67.3.2 DMCONTROL (0x10)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HALTREQ	RESUMEREQ	RSVD	ACKHAVERESET	RSVD	HASEL	HARTSEL										RSVD												SETRESETHALTREQ	CLRRESETHALTREQ	NDMRESET	DMACTIVE
WO	WO	N/A	RW	N/A	RW	RW										N/A												W1S	W1C	RW	RW
0	0	x	0	x	0	0	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0

DMCONTROL [31:0]

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Bits	Name	Description
31	HALTREQ	Writes the halt request bit for all currently selected harts. When set to 1, each selected hart will halt if it is not currently halted. Writing 1 or 0 has no effect on a hart which is already halted, but the bit must be cleared to 0 before the hart is resumed. Harts are selected based on the new value of hartsel and hasel being written.
30	RESUMEREQ	Writes the resume request bit for all currently selected harts. When set to 1, each selected hart will resume if it is currently halted. The resume request bit is ignored while the halt request bit is set. Harts are selected based on the new value of hartsel and hasel being written.
28	ACKHAVERESET	Writing 1 to this bit clears the havereset bits for any selected harts. Harts are selected based on the new value of hartsel and hasel being written.
26	HASEL	Selects the definition of currently selected harts.
25-16	HARTSEL	Selecting the target hart to be debugged.
3	SETRESETHALTREQ	This optional field writes the halt-on-reset request bit for all currently selected harts, unless clrresethaltreq is simultaneously set to 1. When set to 1, each selected hart will halt upon the next deassertion of its reset. The halt-on-reset request bit is not automatically cleared. The debugger must write to clrresethaltreq to clear it. Writes apply to the new value of hartsel and hasel. If hasresethaltreq is 0, this field is not implemented.
2	CLRRESETHALTREQ	This optional field clears the halt-on-reset request bit for all currently selected harts. Writes apply to the new value of hartsel and hasel.
1	NDMRESET	Controlling reset signal from the Debug Module to the rest of the system.
0	DMACTIVE	Controlling reset signal for Debug Module itself.

DMCONTROL Bits

67.3.3 DMSTATUS (0x11)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD									IMPEBREAK	RSVD		ALLHAVERESET	ANYHAVERESET	ALLRESUMEACK	ANYRESUMEACK	ALLNONEXISTENT	ANYNONEXISTENT	ALLUNAVAIL	ANYUNAVAIL	ALLRUNNING	ANYRUNNING	ALLHALTED	ANYHALTED	AUTHENTICATED	AUTHBUSY	HASRESETHALTREQ	DEVTREEVALID	VERSION			
N/A									RO	N/A		RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO	RO			

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
x	x	x	x	x	x	x	x	x	1	x	x	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	0

DMSTATUS [31:0]

Bits	Name	Description
22	IMPEBREAK	Indicates whether there is an implicit EBREAK instruction at the nonexistent word immediately after the program buffer.
19	ALLHAVERESET	Indicates whether all currently selected harts have been reset but the reset has not been acknowledged
18	ANYHAVERESET	Indicates whether any currently selected hart has been reset but the reset has not been acknowledged
17	ALLRESUMEACK	Indicates whether all currently selected harts have acknowledged the previous resume request.
16	ANYRESUMEACK	Indicates whether any currently selected hart has acknowledged the previous resume request.
15	ALLNONEXISTENT	Indicates whether all currently selected harts do not exist in this system.
14	ANYNONEXISTENT	Indicates whether any currently selected hart does not exist in this system.
13	ALLUNAVAIL	Indicates whether all currently selected harts are unavailable.
12	ANYUNAVAIL	Indicates whether any currently selected hart is unavailable.
11	ALLRUNNING	Indicates whether all currently selected harts are running.
10	ANYRUNNING	Indicates whether any currently selected hart is running.
9	ALLHALTED	Indicates whether all currently selected harts are halted.
8	ANYHALTED	Indicates whether any currently selected hart is halted.
7	AUTHENTICATED	Indicates whether authentication check has passed.
6	AUTHBUSY	Indicates whether authentication module is busy.
5	HASRESETHALTREQ	This bit indicates the supported status of resethaltreq.
4	DEVTREEVALID	Whether the information in devtreeaddr0 - devtreeaddr3 registers holds the address of the Device Tree.
3-0	VERSION	Version of the implemented RISC-V External Debug Support. 0x2 indicates that the current implemented version is 0.13.

DMSTATUS Bits

67.3.4 HARTINFO (0x12)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD								NSCRATCH				RSVD			DATAACCESS	DATASIZE				DATAADDR												

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
N/A								RO				N/A			RO	RO				RO											
x	x	x	x	x	x	x	x	0	0	1	0	x	x	x	1	0	1	0	0	0	0	0	0	1	1	0	0	0	0	0	0

HARTINFO [31:0]

Bits	Name	Description
23-20	NSCRATCH	Number of dscratch registers available for the debugger to use during program buffer execution, starting from dscratch0.
16	DATAACCESS	The method for accessing the shadowed data registers. The value of this field is 0x1, indicating that the data registers are shadowed in the memory map of the selected hart under Debug Mode. 0: The data registers are shadowed in the hart by CSR registers. 1: The data registers are shadowed in the hart's memory map. Each register takes up 4 bytes in the memory map.
15-12	DATASIZE	Number of 32-bit words in the memory map dedicated to shadowing the data registers.
11-0	DATAADDR	Signed offset for accessing the shadowed data registers by the processor, to be used as offsets for load/store instructions with the zero register as the base register in Debug Mode.

HARTINFO Bits

67.3.5 HALTSUM (0x13)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																															HALT
N/A																															RW
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

HALTSUM [31:0]

Bits	Name	Description
0	HALT	Bit 0 contains the logical OR of 32 halt bits for hart 0 - hart 31.

HALTSUM Bits

67.3.6 HAWINDOWSEL (0x14)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD																	HAWINDOWSEL														
N/A																	RW														
x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

HAWINDOWSEL [31:0]

Bits	Name	Description
14-0	HAWINDOWSEL	This register selects which of the 32-bit portion of the hart array mask register is accessible in hawindow.

HAWINDOWSEL Bits

67.3.7 HAWINDOW (0x15)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
HAWINDOW																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

HAWINDOW [31:0]

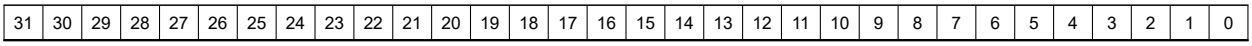
Bits	Name	Description
31-0	HAWINDOW	This register provides R/W accesses to a 32-bit portion of the hart array mask register. The position of the window is determined by hawindowsel. That is, bit 0 refers to hart (hawindowsel * 32), while bit 31 refers to hart (hawindowsel * 32 + 31).

HAWINDOW Bits

67.3.8 ABSTRACTCS (0x16)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD			PROGBUFSIZE					RSVD											BUSY	RSVD	CMDERR			RSVD			DATACOUNT				
N/A			RO					N/A											RO	N/A	RW1C			N/A			RO				
x	x	x	0	1	0	0	0	x	x	x	x	x	x	x	x	x	x	x	0	x	0	0	0	x	x	x	0	0	1	0	0

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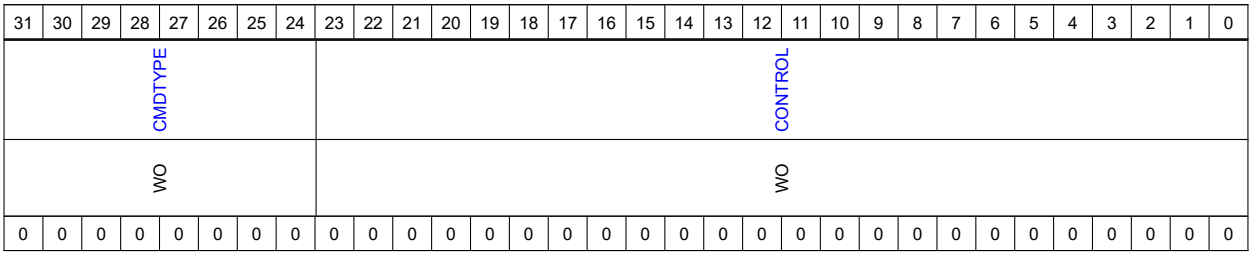


ABSTRACTCS [31:0]

Bits	Name	Description
28-24	PROGBUFSIZE	Size of the program buffer, in 32-bit words.
12	BUSY	Flag indicating an abstract command is currently being executed.
10-8	CMDERR	Error code indicating that an abstract command fails. The bits in this field remain set until they are cleared by writing 1 to them. No abstract command is started until the value is reset to 0 0: None: no error 1: Busy: an abstract command was executing while command, abstractcs, or abstractauto was written, or when one of the data or progbuf registers was read or written. 2: Not supported: the requested commandis not supported. 3: Exception: an exception occurred while executing thecommand. 4: Halt/resume: an abstract command cannot execute because the hart wasn't in the expected state (running/halted).
4-0	DATACOUNT	Number of data registers that are implemented.

ABSTRACTCS Bits

67.3.9 COMMAND (0x17)



COMMAND [31:0]

Bits	Name	Description
31-24	CMDTYPE	Controlling the overall functionality of this abstract command.
23-0	CONTROL	The field is interpreted in a command-specific manner.

COMMAND Bits

67.3.10 ABSTRACTAUTO (0x18)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RSVD								AUTOEXECPROGBUF								RSVD												AUTOEXECDATA			
N/A								RW								N/A												RW			
x	x	x	x	x	x	x	x	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0

ABSTRACTAUTO [31:0]

Bits	Name	Description
23-16	AUTOEXECPROGBUF	When a bit in this field is 1, read or write accesses to the corresponding progbuf word cause the command in command to be executed again.
3-0	AUTOEXECDATA	When a bit in this field is 1, read or write accesses to the corresponding data word cause the command in command to be executed again.

ABSTRACTAUTO Bits

67.3.11 PROGBUF (0x20 + 0x1 * n)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DATA																															
RW																															
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PROGBUF [31:0]

Bits	Name	Description
31-0	DATA	The progbuf registers provide read/write accesses to the program buffer. DM supports program buffer 0-7 only. Program buffer 8-155 are hardwired to 0x00100073 (the EBREAK instruction).

PROGBUF Bits

67.3.12 SBCS (0x38)

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31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SBVERSION			RSVD						SBBUSYERROR	SBBUSY	SBREADONADDR	SBACCESS			SBAUTOINCREMENT	SBREADONDATA	SBERROR			SBASIZE							SBACCESS128	SBACCESS64	SBACCESS32	SBACCESS16	SBACCESS8	
RO			N/A						RW1C	RO	RW	RW			RW	RW	RW1C			RO							RO	RO	RO	RO	RO	
0	0	1	x	x	x	x	x	x	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1	1

SBCS [31:0]

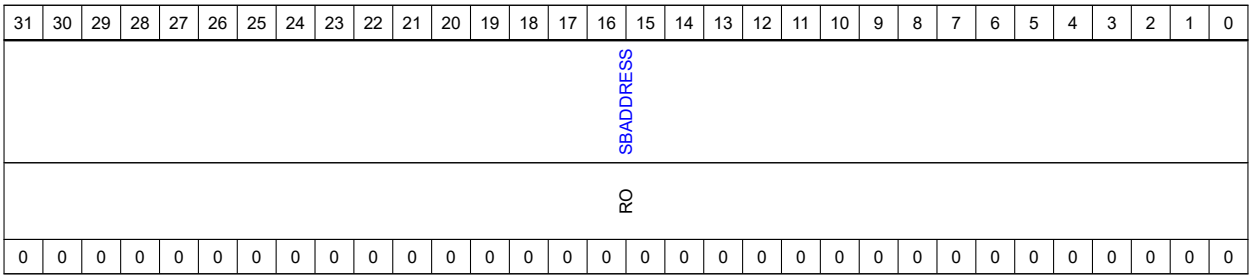
Bits	Name	Description
31-29	SBVERSION	The version of the supported System Bus Access version. It is currently 1 for v0.13 of the RISC-V External Debug Support specification.
22	SBBUSYERROR	Indicate the debugger attempts to execute system bus access before sbbusy is cleared.
21	SBBUSY	Indicate the system bus master is busy.
20	SBREADONADDR	Every write to sbaddress0 automatically triggers a system bus read at the new address.
19-17	SBACCESS	Access size. 0: 8-bit 1: 16-bit 2: 32-bit 3: 64-bit 4: 128-bit
16	SBAUTOINCREMENT	sbaddress is incremented by the size specified in sbaccess_ after every system bus access.
15	SBREADONDATA	Every read from sbdata0 automatically triggers a system bus read at the (possibly auto-incremented) address.
14-12	SBERROR	Error code indicating the failure type of the system bus accesses. Write 1 to clear the status. 0: No bus error 1: Timeout error 2: Bad address 3: Alignment error 4: Unsupported size was requested 7: Others
11-5	SBASIZE	Address width of System bus addresses in bits. This field is 0 if there is no bus access support. Otherwise, it is the value of the SYS_ADDR_WIDTH parameter.

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Bits	Name	Description
4	SBACCESS128	This bit indicates the supported status of 128-bit system bus data accesses.
3	SBACCESS64	This bit indicates the supported status of 64-bit system bus data accesses. This bit is 1 if SYS_DATA_WIDTH == 64.
2	SBACCESS32	This bit indicates the supported status of 32-bit system bus data accesses.
1	SBACCESS16	This bit indicates the supported status of 16-bit system bus data accesses.
0	SBACCESS8	This bit indicates the supported status of 8-bit system bus data accesses.

SBCS Bits

67.3.13 SBADDRESS (0x39 + 0x1 * n)

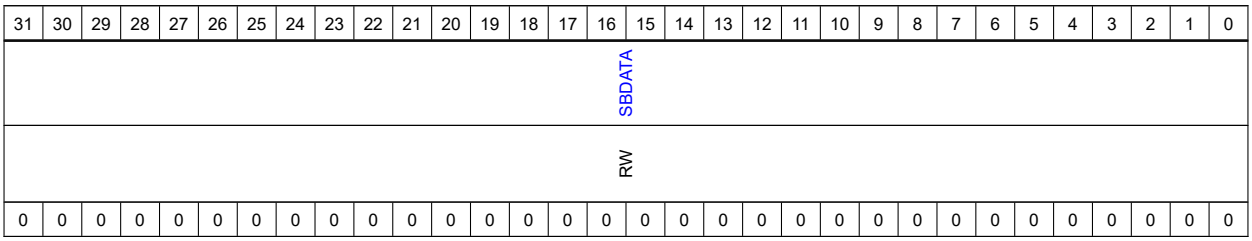


SBADDRESS [31:0]

Bits	Name	Description
31-0	SBADDRESS	bit of the address in sbaddress

SBADDRESS Bits

67.3.14 SBDATA (0x3C + 0x1 * n)



SBDATA [31:0]

Bits	Name	Description
31-0	SBDATA	bit of the address in sbdata

SBDATA Bits

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68 Revision Information

Date	Revision	Description
Rev0.0	2023/05/28	Internal version Rev 0.0 released.
Rev0.1	2023/06/26	Internal version Rev 0.1 released. Added function description for each module of the motion control system. Updated the description of the function overview module.
Rev0.2	2023/07/28	Internal version Rev 0.2 released. Updated product introduction description and system block diagram. Added SYNT and TRGM chapters. Updated CPU core description. Corrected relevant function descriptions in the overview chapter of each module. Fixed register descriptions in each chapter.
Rev0.3	2023/08/07	Internal version Rev 0.3 released. Supplemented register information in the QEO chapter.
Rev0.4	2023/08/15	Internal version Rev 0.4 released. Updated product model information on the cover.
Rev0.5	2023/09/15	Internal version Rev 0.5 released. Updated the analog peripherals overview chapter. Corrected part of the description in the RDC chapter. Updated the SEI chapter. Updated OPAMP register information.
Rev0.6	2023/10/27	Internal version Rev 0.6 released. Updated product model information on the cover. Updated information in the TSNS chapter. Updated part of the information in the BOOT ROM chapter. Updated IO reset status, added description of IO power supply pins.
Rev0.7	2024/03/08	Internal version Rev 0.7 released. Removed the LIN module. Updated part of the description information in the ADC chapter. Updated DMA-related description information in the TRGM chapter. Corrected the description of the motion control system part in the product introduction chapter. Added description of the VPMC undervoltage detection BOR function.

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Date	Revision	Description
Rev0.8	2024/11/27	Internal version Rev 0.8 released. Updated part of the description information in the QEIV2 chapter Updated part of the description information in the OPAMP chapter Updated part of the description information in the BOOTROM chapter Updated part of the description information in the UART chapter Updated part of the description information in the I2C chapter Updated part of the description information in the EWDG chapter Updated part of the description information in the clock system chapter Updated part of the description information in the interconnection manager output assignment chapter Updated part of the description information in the SEI chapter Updated part of the description information in the SYNT chapter Updated part of the description information in the MCAN chapter Updated part of the description information in the MCHTMR chapter Updated part of the description information in the PLIC chapter Updated part of the description information in the MBX chapter Updated part of the description information in the PWM chapter
Rev0.9	2024/12/25	Internal version Rev 0.9 released. Updated part of the description information in the ACMP chapter Updated part of the description information in the TRGM chapter Updated part of the description information in the EWDG chapter Updated part of the description information in the UART chapter Updated part of the description information in the QEI chapter
Rev1.0	2025/04/15	Internal version Rev 1.0 released. Updated part of the description information in the SEI chapter Updated part of the description information in the UART chapter Updated part of the description information in the ADC chapter Updated part of the description information in the ACMP chapter Updated part of the description information in the TRGM chapter
Rev1.1	2025/11/20	Internal version Rev 1.1 released. Added power supply block diagram for DCDC BYPASS scenarios

Table 231: Revision Information

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